Guide Statistics · Evaluation · Training

Frontier Model Field Guide

How large language models are measured and how they are made, explained with pictures you can poke at. Every topic opens in plain words with an everyday example, then switches to what a leader needs to decide, then goes under the hood to the math and code.

48	3	95	⌘K
chapters in 6 parts	depths per topic	interactive widgets	search anything

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Introduction

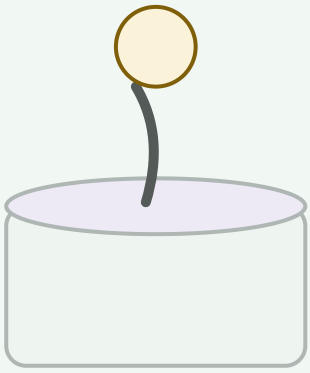
How to use this guide

This guide follows one idea from end to end: you cannot improve what you cannot measure, and you cannot trust a measurement you do not understand. Part I gives you the statistics. Part II turns that into model evaluation. Part III explains what a neural network and an LLM actually are. Part IV explains how models are trained, always connecting back to how you would measure whether the training worked. Part V is about leading that work, and Part VI closes with experimental design: how to set up a measurement so the answer can be trusted at all.

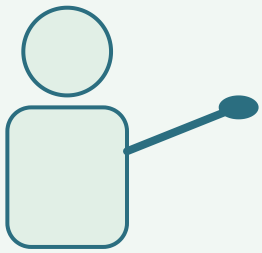
Diagram

The whole guide in one kitchen

Five diagrams. Use Next or the step numbers.

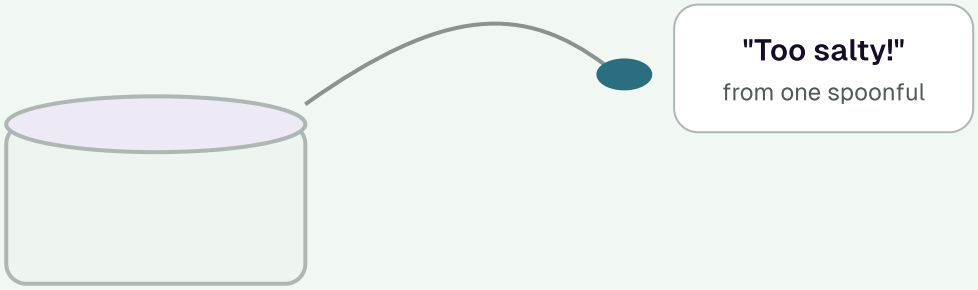


The cook = training



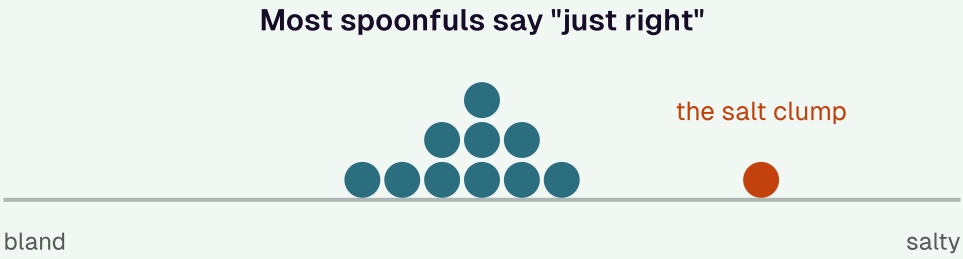
The taster = evaluation

Making a model is cooking: you choose ingredients (data), a recipe (the training method), and how long to cook (compute). Checking the model is tasting. Part III explains the kitchen, Part IV is the cooking, and Part II is the tasting.

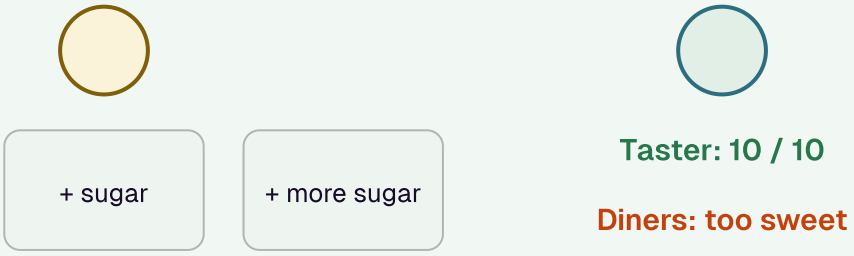


One spoonful might have landed on a salt clump.

One taste is one sample. It might be unlucky. Part I is about how much a handful of samples can and cannot tell you.

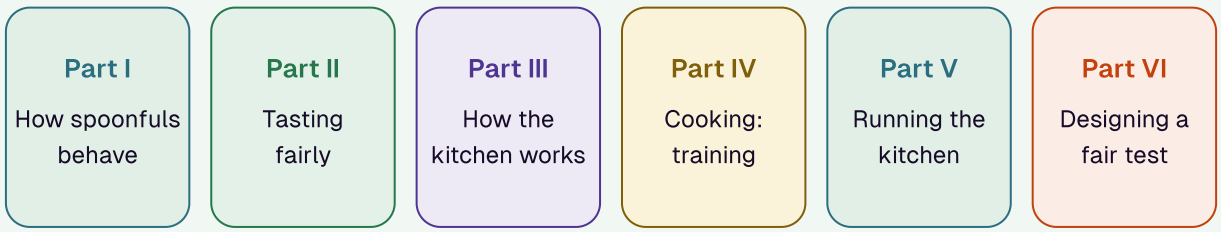


Taste twenty spoonfuls and a shape appears. The shape is a distribution. With it you can say "just right, give or take a little", which is what a confidence interval says about a model's score.



When a score becomes the target, it stops measuring what you wanted.

If the cook learns that one taster loves sugar, every dish gets sweeter and the taster's score climbs while real diners suffer. In model training this is reward hacking, and it is why evaluation and training have to be designed together (Part V).



Each chapter picks its own everyday example, like this one.

That is the map. Every chapter opens with a picture like this, using whatever everyday example fits best: mail sorting, exams, sports, shopping, maps.

Three layers in every chapter

Plain words

An analogy and the core idea, with no jargon and no formulas. Read this first, even if you think you know the topic.

For leaders

What decision the idea informs, what it costs, how it fails, and the questions a senior reviewer asks. This is the layer for staff, principal, and partner conversations.

Under the hood

The math, the derivations, and short runnable code. Enough to check someone's work, not just follow it.

Each topic shows its three depths as tabs. The **Plain / Leader / Deep** switch in the top bar flips every topic in the guide at once; **All** stacks them. A good first pass is Plain for everything, then Leader, then Deep for the chapters that matter to your work. Press `⌘K` (or `/`) to search chapters, topics, widgets, and glossary terms.

Plain words

What a layer looks like

Imagine tasting soup to decide if it needs salt. One spoonful tells you something, but not everything. Evaluation is tasting the model. Training is cooking. This guide teaches both, and how to tell when one spoonful is lying to you.

For leaders

Why both halves matter at senior levels

At senior levels in a frontier lab, decisions rarely turn on a single technique. They turn on whether a result is real, what it cost, and what it will break. Training choices (data, objective, RL method) determine what can go wrong. Evaluation choices determine whether you notice. Leaders who understand both can push back on a launch, scope a research bet, or kill a project with evidence instead of instinct.

Under the hood

How the math and code are shown

Formulas are typeset, for example the cross-entropy loss that almost every chapter in Parts III and IV builds on:

$$\mathcal{L}(\theta) = -\frac{1}{T}\sum_{t=1}^{T} \log p_\theta(x_t \mid x_{<t})$$

The average negative log probability the model assigns to each correct next token x_t, given everything before it.

python

```python
import math

def cross_entropy(probs_of_correct_token):
    # average surprise, in nats, at the token that actually came next
    return -sum(math.log(p) for p in probs_of_correct_token) / len(probs_of_correct_token)

print(round(cross_entropy([0.9, 0.5, 0.25]), 3))  # 0.728
print(round(math.exp(0.728), 2))                  # perplexity 2.07
```

rust

```rust
fn cross_entropy(probs_of_correct_token: &[f64]) -> f64 {
    // average surprise, in nats, at the token that actually came next
    -probs_of_correct_token.iter().map(|p| p.ln()).sum::<f64>() / probs_of_correct_token.len() as f64
}

fn main() {
    println!("{:.3}", cross_entropy(&[0.9, 0.5, 0.25])); // 0.728
    println!("{:.2}", 0.728_f64.exp());                  // perplexity 2.07
}
```

Map of the guide

Part	What you get	Start here if
Reading the math	Every symbol used in the guide: subscripts, bars and hats, Σ and Π, Greek letters, probability notation, with a sum builder to practise	Formulas look like pictures rather than sentences
I. Statistical foundations	Distributions, sampling, standard error, intervals, p values, reading a shape, heavy tails	It has been years since you did statistics
II. Evaluating models	Pass-rate intervals, paired comparisons, power, multiple comparisons, classification metrics through AUC, LLM judges, the offline to online gap, contamination, harness engineering, regression triage	You need to defend or challenge an eval result
III. How neural networks and LLMs work	Neurons and layers, gradient descent and local minima, why not infinite layers, the family tree from CNNs to transformers and GNNs, next-token prediction, how an assistant is made, reasoning modes vs temperature, one-pass decision models, and how framing a decision changes the answer	You want to know what is actually inside the model
IV. Training models	The pipeline, pretraining, optimisation, scaling laws, SFT, LoRA, reward models, RLHF and PPO, DPO and its relatives, GRPO and RL with verifiable rewards, distillation, inference-time compute, safety training	You need to talk credibly about how models are built
V. Leading the work	How training and evaluation connect, review checklists, the questions principals ask, and a glossary	You are preparing for a senior conversation tomorrow
VI. Experimental design	Correlation vs causation, who is really in your sample, random assignment and its alternatives, data types and the right formulas, A/B tests end to end, metrics and logging, and designing experiments for AI systems	You need to design a test whose answer you can trust

> **Note**
> Interactive panels run in the web version. The PDF shows each one at its default setting.

Introduction

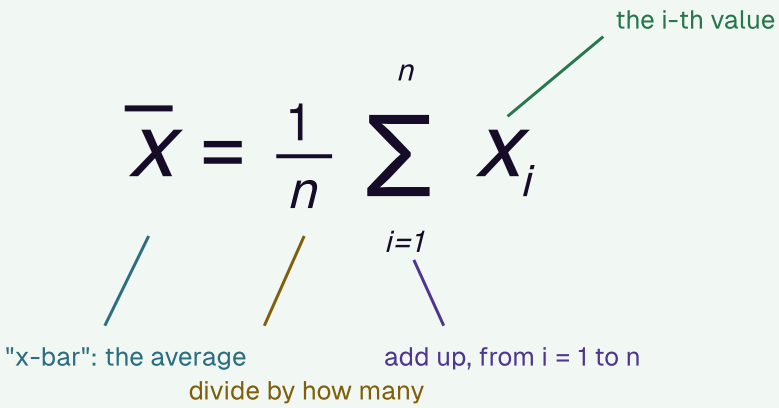
Reading the math: symbols, subscripts, and sums

Every formula in this guide is built from a small set of symbols. Once you can read them aloud, a formula stops being a picture and becomes a sentence. This chapter is the decoder ring. Come back to it whenever a symbol looks unfamiliar; the ⌘K search also finds every symbol by name.

Diagram

Decoding one formula: the average

Five commutes, one formula, read piece by piece.



Read it aloud: "x-bar equals one over n times the sum, from i equals 1 to n, of x sub i." In plain words: add up all the values and divide by how many there are. Four symbols, one familiar idea.

Five commutes, in minutes. The subscript is just a seat number.

n = 5 values. x_3 = 31. "x sub i" means "whichever one we are looking at".

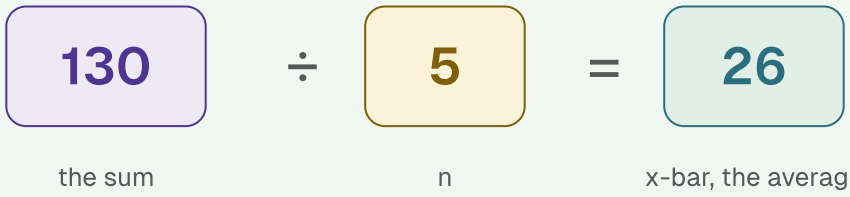
x_i (say "x sub i") is the value in position i. Here $x_1 = 22$ and $x_3 = 31$. The letter n is how many values there are, so $n = 5$. Code calls this `x[i]`, except most code starts counting at 0 and most math starts at 1.

$\sum$ "add up everything that follows"

$$22 + 25 + 31 + 24 + 28 = 130$$

The bottom of Σ says where to start (i = 1); the top says where to stop (n = 5).
In code: sum(x), or a for-loop with a running total.

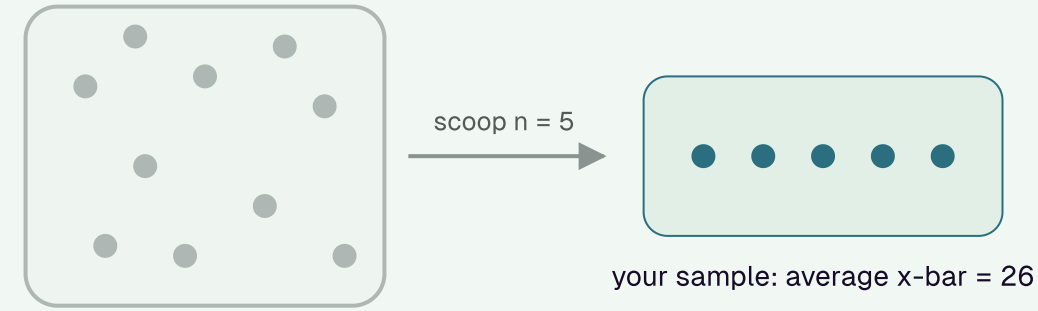
$\sum_{i=1}^{n} x_i$ is a for-loop written on one line: start with $i = 1$, add x_i, move to the next i, stop after $i = n$. Here that is $22 + 25 + 31 + 24 + 28 = 130$.



Multiplying by 1/n is the same as dividing by n.

The $\frac{1}{n}$ out front divides the total by the count: $130/5 = 26$. So $\bar{x} = 26$ minutes. A bar over a letter almost always means "the average of these values".

Greek letters (μ, σ) describe the whole jar. Latin letters (x-bar, s) describe your scoop.



One more convention that runs through the whole guide: Greek letters usually name the true, unknown quantity (μ, "mu", the true average), and Latin letters or hats name what you computed from a sample ($\bar{x}$, or $\hat{\mu}$, "mu-hat"). Statistics is mostly about how close $\bar{x}$ is likely to be to μ.

The symbols, one table at a time

Each row gives the symbol, how to say it, what it means, and a tiny example. Everything here is used somewhere later in the guide.

Values, positions, and counts

Symbol	Say it	Meaning	Example
x	"x"	A value, or a whole list of values	commute times
x_i	"x sub i"	The value in position i. The subscript is a seat number, not multiplication	$x_3 = 31$
$i,\ j,\ k,\ t$	"i", "j"…	Index variables: counters that run over positions. t is often a time step or token position	$i = 1, \ldots, n$
$n,\ N$	"n"	How many values. N is also used for model parameter count in scaling laws	$n = 5$
x_{ij}	"x sub i j"	Two indices: row i, column j (a table or matrix)	score of model i on task j
$x_{<t}$	"x before t"	Everything up to, but not including, position t	the tokens before token t
x^2	"x squared"	A superscript usually means a power	$5^2 = 25$
$x^{(k)}$	"x superscript k"	Parentheses mean a label, not a power: the k-th version or sample	bootstrap replicate k
$\mathbf{x},\ \mathbf{W}$	"bold x"	Bold lowercase is a vector (a list); bold uppercase is a matrix (a grid)	$\mathbf{W}$ a weight matrix
$\mathbf{W}^\top$	"W transpose"	Flip a matrix so rows become columns	used in LoRA and attention

Accents: bars, hats, tildes, stars

Symbol	Say it	Meaning	Example
$\bar{x}$	"x-bar"	The average (mean) of the values	$\bar{x} = 26$
$\hat{\theta}$, $\hat{p}$	"theta-hat", "p-hat"	An estimate computed from data, of the true θ or p	$\hat{p} = 124/200 = 0.62$ pass rate
$\tilde{x}$	"x-tilde"	Often the median, or a modified version of x	$\tilde{x} = 25$
x^*, π^*	"x-star"	The best or optimal value	π^* the optimal policy (DPO)
x'	"x-prime"	Another value of the same kind, or a changed version	a perturbed eval item

Greek letters and what they usually mean here

Letter	Name	Usual meaning in this guide
μ	mu	True mean of a population
σ, σ^2	sigma, sigma squared	True standard deviation and variance (lowercase sigma)
Σ	capital sigma	Sum (as an operator). Also a covariance matrix in some papers
Π, $\prod$	capital pi	Product: multiply everything that follows
π	pi	3.14159..., or in reinforcement learning a **policy** (the model): $\pi_\theta(y \mid x)$
θ	theta	The model's parameters (its weights), or any unknown quantity being estimated
α	alpha	Significance level (0.05), a scaling constant (LoRA), or a learning rate
β	beta	Type II error rate, the KL strength in DPO and RLHF, or Adam's decay rates
η	eta	Learning rate
ϵ, ε	epsilon	A small number: noise, an error term, or PPO's clip range
λ	lambda	A rate (Poisson), a penalty weight, or GAE's mixing factor
ρ	rho	Correlation, such as the intra-cluster correlation
γ	gamma	Discount factor in RL
δ, Δ	delta	A difference or change: Δ = "change in"
∇	nabla, "grad"	Gradient: the direction of steepest increase (optimisation)
Φ, ϕ	phi	Standard normal CDF and density
χ^2	chi squared	A distribution used in tests of counts

Operators and relations

Symbol	Say it	Meaning	Example
$\sum_{i=1}^{n} x_i$	"sum of x sub i, i from 1 to n"	Add x_1 through x_n	130
$\prod_{i=1}^{n} p_i$	"product of p sub i"	Multiply p_1 through p_n	$0.9 \times 0.5 \times 0.25 = 0.1125$
$\sqrt{x}$	"square root"	The number that squared gives x	$\sqrt{12.5} = 3.54$
$\lvert x \rvert$	"absolute value"	Distance from zero, always non-negative	$\lvert -4 \rvert = 4$
$\lVert \mathbf{x} \rVert$	"norm of x"	Length of a vector	gradient clipping by norm
$\log x,\ \ln x$	"log", "natural log"	The inverse of e^x. In ML papers $\log$ almost always means natural log	$\ln 0.1125 = -2.18$
$e^x,\ \exp(x)$	"e to the x"	Exponential, $e \approx 2.718$	perplexity $= e^{\text{loss}}$
$\max,\ \min$	"max", "min"	Largest or smallest value	$\max x_i = 31$
$\arg\max_i x_i$	"arg max"	The *position* of the largest value, not the value	$\arg\max = 3$ (the third commute)
$=,\ \approx,\ \equiv,\ :=$	equals, approximately, identical, defined as	$:=$ introduces a new name	$\text{SE} := s/\sqrt{n}$
$\propto$	"proportional to"	Equal up to a constant multiplier	$\text{SE} \propto 1/\sqrt{n}$
$\in,\ \notin$	"in", "not in"	Membership in a set	$p \in [0, 1]$

Symbol	Say it	Meaning	Example
$\{\dots\},\ [a,b],\ (a,b)$	set, closed and open interval	Square brackets include the ends, round ones exclude them	a 95% CI $[0.55, 0.68]$
$\mathbb{R}$	"the reals"	All real numbers; $\mathbb{R}^d$ is a list of d of them	$\mathbf{x} \in \mathbb{R}^{4096}$
$\to,\ \Rightarrow$	"tends to", "implies"	Approaches a limit; one statement leads to another	as $n \to \infty$
$n!\ ,\ \binom{n}{k}$	"n factorial", "n choose k"	$1 \times 2 \times \cdots \times n$; number of ways to pick k of n	$\binom{5}{2} = 10$

Probability and expectation

Symbol	Say it	Meaning	Example
$P(A)$	"probability of A"	A number from 0 to 1	$P(\text{pass}) = 0.62$
$P(A \mid B)$	"probability of A given B"	Probability of A when you already know B happened. The bar is "given"	$P(\text{spam} \mid \text{flagged})$ = precision
$p_\theta(x_t \mid x_{<t})$	"p theta of x t given x before t"	The model's probability for the next token, given the previous ones	the language model
$X \sim \mathcal{N}(\mu, \sigma^2)$	"X is distributed normal"	X is a random quantity with that distribution	$\sim$ means "is drawn from"
$\mathbb{E}[X]$	"expected value of X"	The long-run average of a random quantity	$\mathbb{E}[\text{die}] = 3.5$
$\mathbb{E}_{x \sim D}[f(x)]$	"expectation over x drawn from D"	Average f over inputs drawn from D. In practice: sample and average	training losses
$\text{Var}(X),\ \text{Cov}(X, Y)$	variance, covariance	Spread of one quantity; how two move together	see Distributions
$\text{KL}(p \,\|\, q)$	"KL divergence of p from q"	How different two distributions are (not symmetric)	RLHF's leash to the reference model

Build a sum yourself

Interactive **Sum builder** (shown at its default setting)

Type some numbers, pick an expression, and see every term written out. Each expression is one you will meet in the guide.

$$\sum_{i=1}^{n} x_i$$

+ + + + =

The sum of the 5 values is **130**. In code: `sum(x)`.

Three habits that make any formula readable

Plain words

1. **Find the big operator first.** A Σ means "add up", a Π means "multiply", an $\mathbb{E}$ means "average over". That tells you what kind of answer comes out.
2. **Find the index.** The letter under the operator (usually i or t) is what changes from term to term. Everything with that letter in it changes; everything without it stays fixed.
3. **Plug in tiny numbers.** Write out two or three terms with real values, as the sum builder does. Most formulas become obvious after one worked line.

Reading a formula in a paper or a review

For leaders

You do not need to derive every equation to evaluate a result. Four questions get most of the value:

- **What is summed or averaged over?** Tokens, sequences, prompts, users? A loss averaged per token weights long answers differently from one averaged per sequence, which is the whole argument in the Dr. GRPO paper (GRPO).
- **What has a hat?** Hatted quantities are estimates and carry uncertainty. If a decision rests on one, ask for its standard error.
- **What is held fixed?** Subscripts like π_{ref} or θ_{old} mark frozen copies. Knowing which model is frozen tells you what the method is anchoring to.

- **Which constants were tuned?** Greek letters such as β, ϵ, λ, η are usually knobs. Ask how sensitive the result is to each.

Under the hood

Sum rules, and the symbols as code

A handful of rules let you rearrange any sum. Each is just arithmetic on the written-out terms.

$$\sum_{i=1}^{n} c\,x_i = c\sum_{i=1}^{n} x_i \qquad \sum_{i=1}^{n}(x_i + y_i) = \sum_{i=1}^{n} x_i + \sum_{i=1}^{n} y_i \qquad \sum_{i=1}^{n} c = n\,c \qquad \sum_{i=1}^{n}(x_i - \bar{x}) =$$

Constants pull out; sums split; adding a constant n times gives nc; and deviations from the mean always cancel, which is why variance squares them.

$$\ln \prod_{i} p_i = \sum_{i} \ln p_i \qquad \mathbb{E}[X] = \sum_{x} x\,P(X = x) \qquad \sum_{i=1}^{m}\sum_{j=1}^{n} x_{ij} = \text{add every c}$$

Logs turn products into sums, which is why likelihoods are computed as log-likelihoods. Expectation is a probability-weighted sum. Two sums in a row are nested loops.

The log rule connects straight to training. The probability a model assigns to a sentence is the product of its token probabilities, and the loss is the negative log of that, averaged per token:

$$0.9 \times 0.5 \times 0.25 = 0.1125, \qquad -\tfrac{1}{3}\left(\ln 0.9 + \ln 0.5 + \ln 0.25\right) = -\tfrac{1}{3}\ln 0.1125 = 0.728$$

That 0.728 is the cross-entropy loss computed in the introduction.

python

```python
import math

x = [22, 25, 31, 24, 28]
n = len(x)

total = sum(x)                                  # Σ x_i
x_bar = total / n                               # <span class="ovl">x</span> = (1/n) Σ x_i
ss = sum((xi - x_bar) ** 2 for xi in x)         # Σ (x_i − <span class="ovl">x</span>)²
s2 = ss / (n - 1)                               # s²
i_max = max(range(n), key=lambda i: x[i]) + 1   # arg max, counting from 1 like the math
```

```
p = [0.9, 0.5, 0.25]
prod = math.prod(p)                                  # Π p_i
log_sum = sum(math.log(pi) for pi in p)              # Σ ln p_i = ln Π p_i

print(total, x_bar, ss, s2)                          # 130 26.0 50.0 12.5
print(max(x), i_max)                                 # 31 3
print(round(prod, 4), round(log_sum, 4), round(-log_sum / len(p), 3))  # 0.1125 -2.1848 0.728
```

rust

```
fn main() {
    let x = [22.0_f64, 25.0, 31.0, 24.0, 28.0];
    let n = x.len() as f64;

    let total: f64 = x.iter().sum();                                    // Σ x_i
    let x_bar = total / n;                                              // <span class="ovl">x</span> = (1/n) Σ
x_i
    let ss: f64 = x.iter().map(|xi| (xi - x_bar).powi(2)).sum();        // Σ (x_i − <span class="ovl">x</span>)²
    let s2 = ss / (n - 1.0);                                            // s²
    let (i_max, max) = x.iter().enumerate()
        .fold((0, f64::MIN), |best, (i, &v)| if v > best.1 { (i, v) } else { best });

    let p = [0.9_f64, 0.5, 0.25];
    let prod: f64 = p.iter().product();                                 // Π p_i
    let log_sum: f64 = p.iter().map(|pi| pi.ln()).sum();                // Σ ln p_i = ln Π p_i

    println!("{} {} {} {}", total, x_bar, ss, s2);                      // 130 26 50 12.5
    println!("{} {}", max, i_max + 1);                                  // 31 3 (arg max, counting from 1)
    println!("{:.4} {:.4} {:.3}", prod, log_sum, -log_sum / p.len() as f64); // 0.1125 -2.1848 0.728
}
```

Practice: read four formulas from later chapters

Open each one and try reading it aloud before looking at the translation.

Cross-entropy loss (pretraining)

Sample variance (statistics)

Group-relative advantage (GRPO)

Expected value (everywhere)

Part I

Statistical foundations

Distributions, sampling, intervals, and how to read the shape of your data. Everything later leans on this.

Part I · Chapter 3

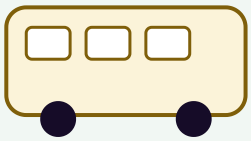
Distributions: centre, spread, and shape

Every number an eval produces, whether a pass rate, a latency, a token count, or a judge score, is a summary of a spread of measurements. This chapter gives you the vocabulary for describing that spread and the reflexes for spotting when a summary is lying.

Diagram

A hundred stopwatches on one bus route

What a distribution is, and the three words that describe it: centre, spread, shape.

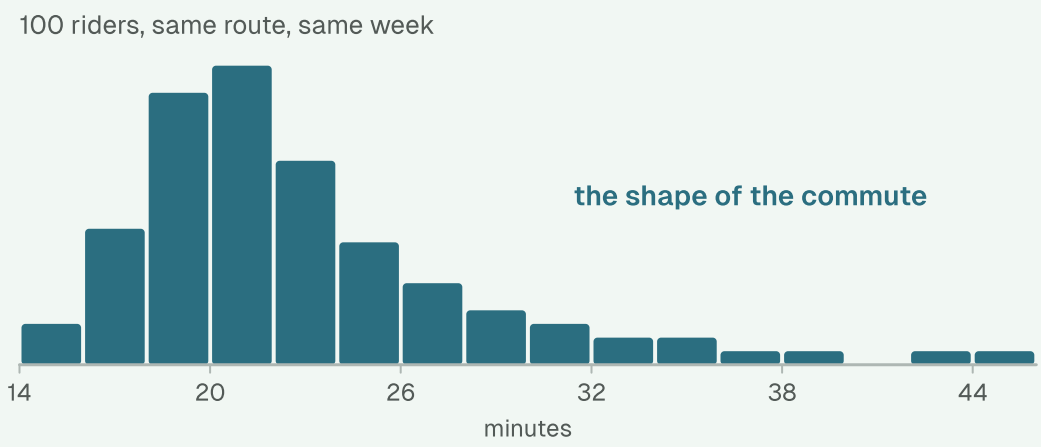


Route 12, Tuesday 8:05

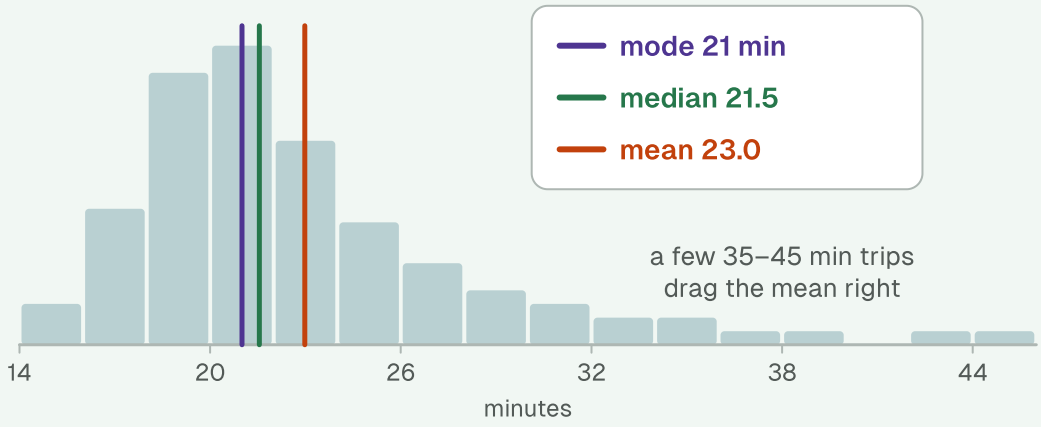
20 minutes

One trip gives one number. Is 20 minutes typical?

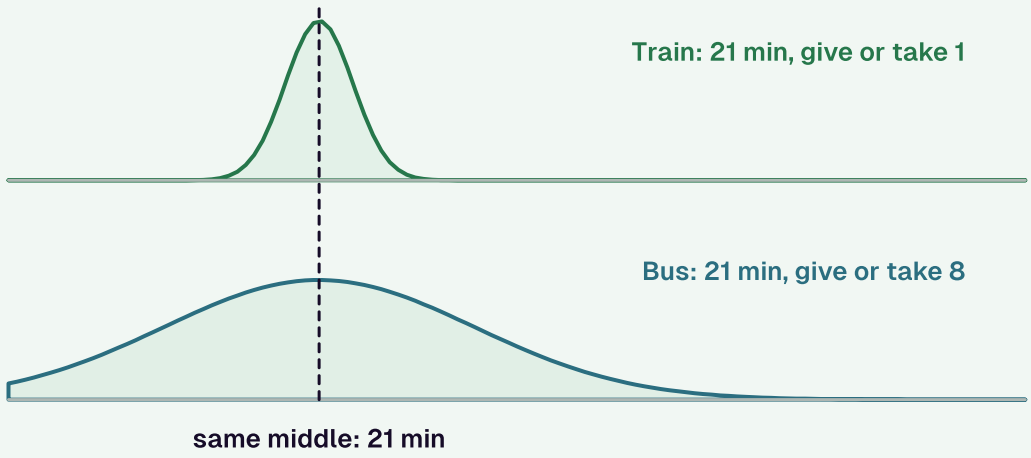
You time your bus ride once: 20 minutes. One measurement is a single dot. It cannot tell you whether 20 is normal, lucky, or slow.



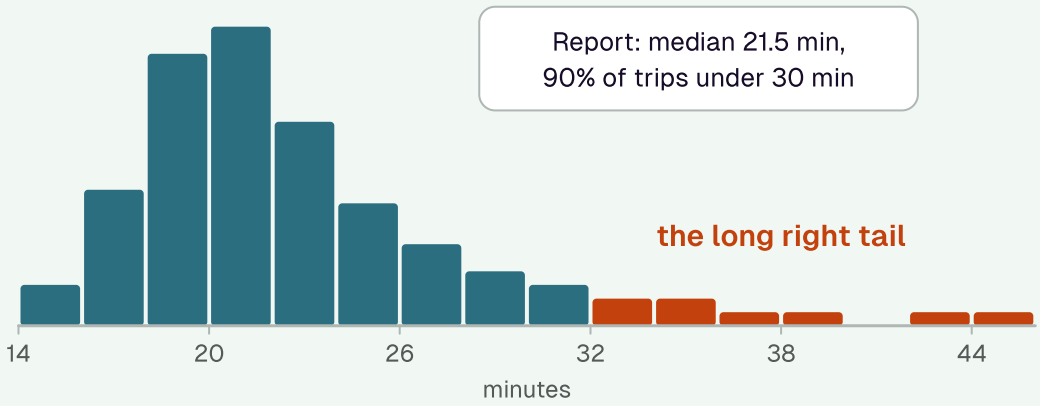
Ask 100 riders on the same route and stack their times into bars. Most rides take 18 to 24 minutes, a few take much longer. That outline is the **distribution**.



"Typical" has three answers. The **mode** (most common) is about 21, the **median** (half above, half below) is 21.5, and the **mean** is 23.0. The handful of 35 to 45 minute trips pull the mean to the right.



Two routes can share a middle and feel completely different. The train is 21 minutes almost every day; the bus is 21 on average but swings by 8. That swing is the **spread**, and it is what makes you late.



The **shape** leans right: a long tail of bad days. When a distribution leans, a single average misleads. Say what riders feel: the median plus a high percentile. Eval latency, token counts, and loop lengths all lean like this.

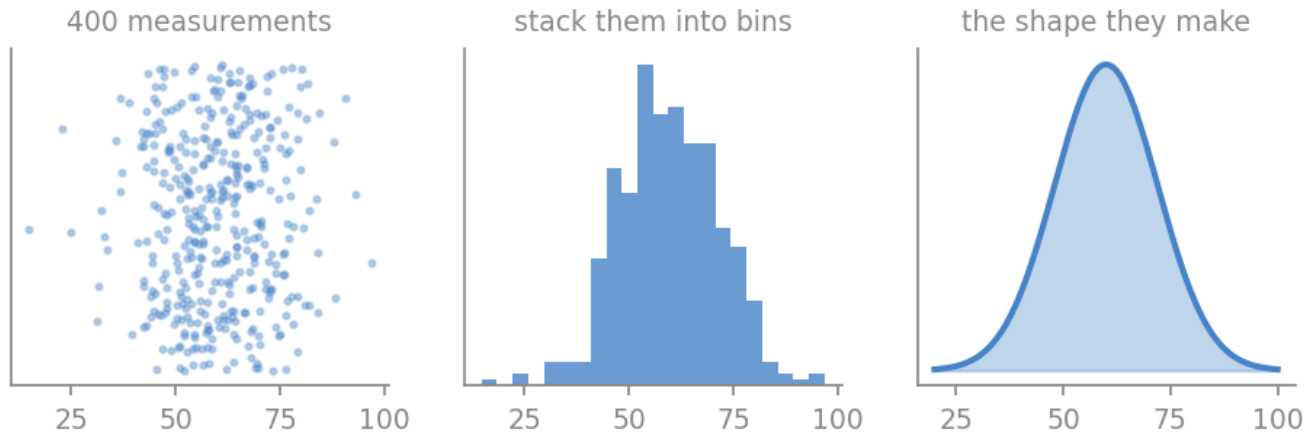
Plain words

A distribution is the shape a spread of answers makes

Measure the same thing many times and you do not get one answer, you get a spread of answers. A **distribution** is the shape that spread makes. That is the whole idea. Everything else in this chapter is vocabulary for describing shapes: where the middle is, how wide it is, and which way it leans.

Think of a hundred people timing the same bus route. Most say about twenty minutes, a few say fifteen, and one person who hit every red light says forty-five. If you stack their answers into a bar chart, the shape you see is the distribution of the commute.

What a distribution actually is



Left: 400 individual measurements. Middle: stack them into bins. Right: the shape they make. That shape is the distribution.

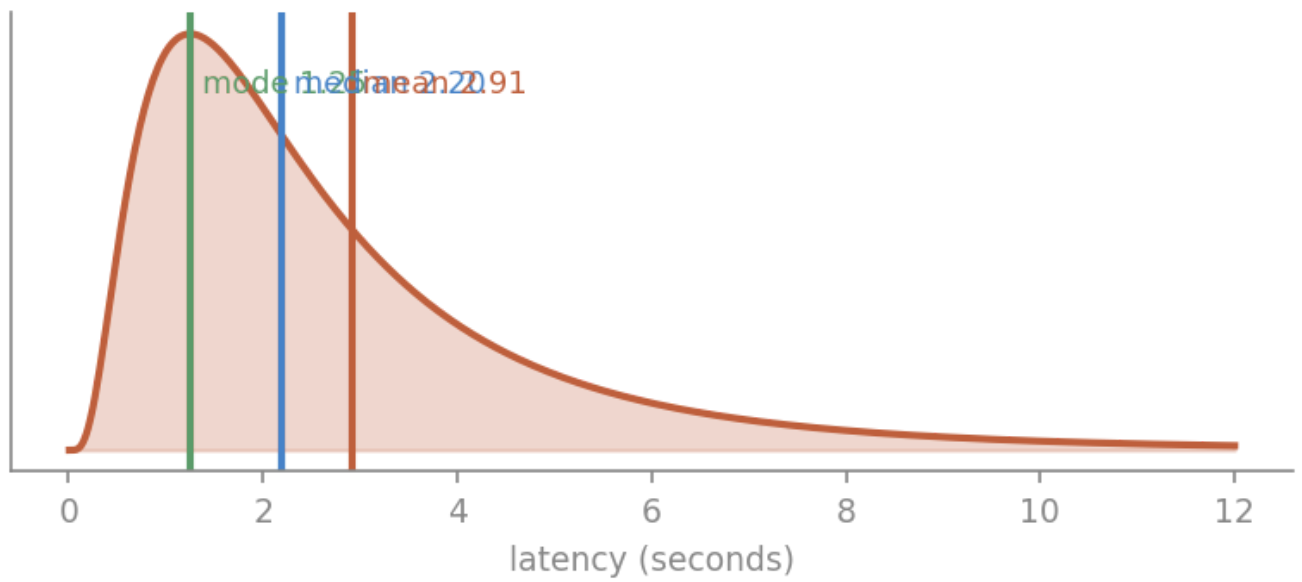
Three words get used loosely and mean different things:

- **Population** is every possible measurement you could ever take. Usually imaginary, always unavailable.
- **Sample** is the measurements you actually have. 200 tasks, 5000 sessions.
- **Distribution** is the shape. A population has one, and your sample gives you a blurry picture of it.

Eval lens

Your eval suite is a sample. Production is closer to the population. The offline to online gap is, in one sentence, the sample not looking like the population.

Centre: three different answers to "what is typical"



The same data has three different centres. On a skewed distribution they are far apart, and picking the wrong one is the most common reporting error there is.

Measure	What it is	Use when	Breaks when
Mean	Add everything, divide by n	Symmetric data. Also anything where the total matters, like compute cost	The tail is long. One 400 second request drags it and no user experiences it
Median	The middle value when sorted	Skewed data. Almost all latency and token reporting	You need totals, or the data is bimodal (the middle can fall in the empty valley)
Mode	The most common value	Categorical data, spotting bimodality	Continuous data, where no value repeats exactly

Key idea

On a right skewed distribution the order is always mode < median < mean. If someone reports a mean latency well above the median, they are reporting a number no user experienced.

Under the hood

Mean and median, defined

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \qquad \text{median} = x_{((n+1)/2)} \text{ of the sorted values}$$

For even n, the median is the average of the two middle sorted values.

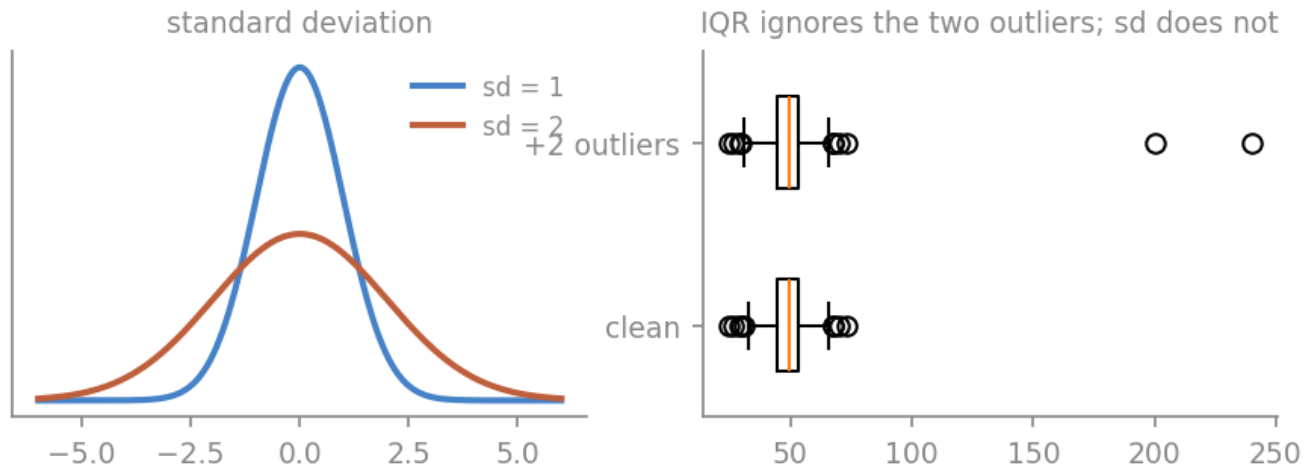
python

```python
import numpy as np

lat = [1.9, 2.1, 2.0, 2.4, 2.2, 1.8, 2.3, 2.0, 41.0]   # one slow request

# THE MEAN: add everything, divide by n. One outlier drags it a long way.
print(sum(lat) / len(lat))              # 6.41 - no request took anywhere near this

# THE MEDIAN: sort, take the middle. The outlier moves it by nothing.
s = sorted(lat)
n = len(s)
median = s[n // 2] if n % 2 else (s[n // 2 - 1] + s[n // 2]) / 2
print(median)                           # 2.10 - describes a typical request

# THE RATIO: the fastest diagnostic in this whole part of the guide.
print((sum(lat) / len(lat)) / median)   # 3.05 -> heavily right skewed
```

rust

```rust
fn main() {
    let lat = [1.9, 2.1, 2.0, 2.4, 2.2, 1.8, 2.3, 2.0, 41.0]; // one slow request

    // THE MEAN: add everything, divide by n. One outlier drags it a long way.
    let mean = lat.iter().sum::<f64>() / lat.len() as f64;
    println!("{}", mean);          // 6.41 - no request took anywhere near this

    // THE MEDIAN: sort, take the middle. The outlier moves it by nothing.
    let mut s = lat.to_vec();
    s.sort_by(|a, b| a.total_cmp(b));
    let n = s.len();
    let median = if n % 2 == 1 { s[n / 2] } else { (s[n / 2 - 1] + s[n / 2]) / 2.0 };
    println!("{}", median);        // 2.10 - describes a typical request

    // THE RATIO: the fastest diagnostic in this whole part of the guide.
    println!("{}", mean / median); // 3.05 -> heavily right skewed
}
```

Spread: how much things vary



Left: standard deviation controls the width. Right: two outliers barely move the box (which is built from quartiles) but would move the standard deviation a lot.

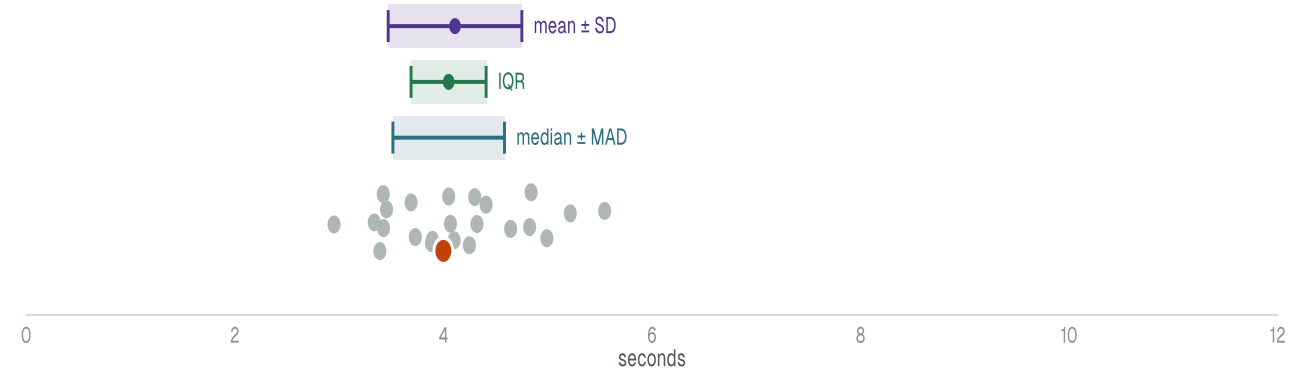
Measure	Definition in words	Strength	Weakness
Variance	Average squared distance from the mean	Adds up nicely across independent sources, which is why the maths uses it	Wrong units. Variance of seconds is in seconds squared
Standard deviation	Square root of variance, back in the original units	Interpretable. Roughly 68% of normal data sits within 1 sd	One extreme value can dominate it
IQR	The range covering the middle 50%, p75 minus p25	Outliers cannot touch it	Ignores the tails entirely, which may be the thing you care about
MAD	Median of the absolute distances from the median	Very robust	Less familiar, and needs a scaling constant to compare with sd

The centre tells you what is typical. Spread tells you how far from typical things wander. There are four measures you will meet everywhere, and they disagree in useful ways: two are built from squared distances

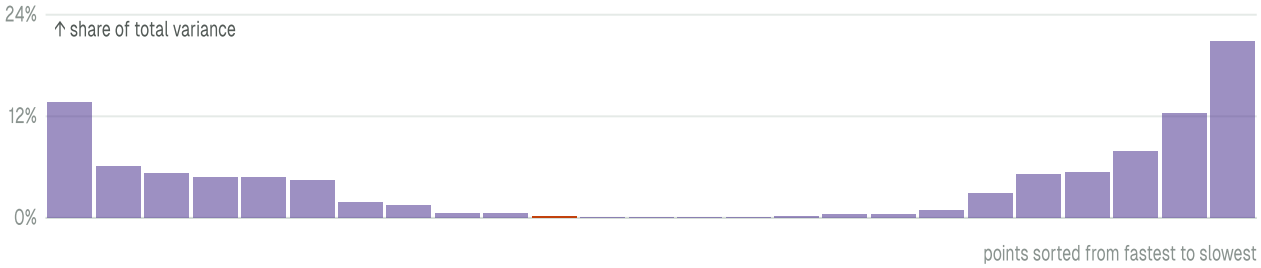
and react strongly to extremes, two are built from ranks and barely notice them. Try it below, then read each one in depth.

Interactive **Spread explorer: one dataset, four rulers** (shown at its default setting)

Twenty-four response times (seconds) plus one request you control. Drag it out to the right and watch which measures react. The lower chart shows how much of the total variance each point owns.



mean ± 1 SD · median and IQR (middle 50%) · median ± scaled MAD · the request you control



Variance (s^2)	Standard deviation	IQR	MAD (scaled ×1.4826)
0.41	**0.64 s**	**0.72 s**	**0.54 s**

Your request alone owns **0%** of the variance. Compared with a typical 4 s request, SD is **1.0×** larger, IQR **1.00×**, MAD **1.00×**. SD and scaled MAD agree: no single point is dominating.

Plain words **Four ways to say "how much does my commute vary?"**

Picture a month of commutes. Most days take about 25 minutes, give or take a few. One day a crash makes it 95.

- **Variance** asks: on average, how big is the surprise, squared? Squaring makes the crash day count enormously, and the answer comes out in "minutes squared", which nobody can picture.
- **Standard deviation** takes the square root of that, back to minutes: "a typical day is about this far from average". It is still pulled up by the crash.
- **IQR** ignores the best and worst quarter of days and reports how wide the middle half is: "half my commutes fall in this 6-minute window". The crash does not change it.
- **MAD** asks: on a typical day, how far am I from my usual commute? It looks at the middle of the distances, so one disaster cannot move it.

None is "right". Variance and SD describe the whole month including the disaster. IQR and MAD describe a normal day. Which one you want depends on whether the disaster is the point.

For leaders

Which spread to ask for, and when

- **Error bars and significance run on variance.** Every standard error, confidence interval, power calculation, and sample-size estimate in Part II is built from variance. If someone shows an interval, the variance behind it is the thing to question.
- **For skewed operational metrics, ask for IQR or percentiles, not SD.** Latency, cost per request, and tokens per response are long tailed. A mean plus or minus SD on them describes no real user and can imply negative latencies.
- **For monitoring and anomaly alerts, MAD is the safe default.** A single incident inflates the SD, which then hides the next incident. MAD-based thresholds stay put.
- **When SD and scaled MAD disagree by more than about 1.5 times, the data has heavy tails or outliers.** That gap is a finding. Ask what is in the tail before choosing which number to report.

Principal asks

Is that standard deviation of the users, or standard error of the average? What does the spread look like with the top 1% removed? Are we alerting on SD-based thresholds, and what happens to them the day after an incident?

Under the hood

Four measures of spread

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2 \qquad s = \sqrt{s^2}$$

Sample variance and standard deviation. Dividing by $n - 1$ rather than n corrects for having used the same data to estimate the mean.

$$\mathrm{IQR} = Q_3 - Q_1 = p_{75} - p_{25} \qquad \mathrm{MAD} = \mathrm{median}\left(|x_i - \mathrm{median}(x)|\right)$$

The two robust measures. Multiply MAD by 1.4826 to compare it with the sd on normal data.

python

```python
import numpy as np

def spread(x):
    x = np.asarray(x, dtype=float)
    n = len(x)
    m = x.mean()

    # VARIANCE: mean squared distance from the mean.
    # n-1, not n. Dividing by n underestimates, because you used the same
    # data to compute the mean you are measuring distance from.
    var = ((x - m) ** 2).sum() / (n - 1)

    q1, q3 = np.quantile(x, [0.25, 0.75])          # IQR: middle 50%
    med = np.median(x)
    mad = np.median(np.abs(x - med))               # robust alternative to sd

    return {"sd": var ** 0.5, "var": var, "iqr": q3 - q1,
            "mad": mad, "mad_scaled": 1.4826 * mad}   # scaled to match sd on normal data

clean = [1.9, 2.1, 2.0, 2.4, 2.2, 1.8, 2.3, 2.0]
print(spread(clean))                    # sd 0.203, iqr 0.250, mad 0.150
print(spread(clean + [41.0]))           # sd 13.0 (!), iqr 0.300, mad 0.200
# One outlier multiplied the sd by 64x and barely touched the IQR or MAD.
```

rust

```rust
#[derive(Debug)]
struct Spread {
    sd: f64,
    var: f64,
    iqr: f64,
    mad: f64,
    mad_scaled: f64, // scaled to match sd on normal data
}

/// Sum the way numpy does: eight running totals combined in pairs.
/// More accurate than one running total, and it reproduces numpy's last digit.
```

```rust
fn pairwise_sum(x: &[f64]) -> f64 {
    if x.len() < 8 {
        return x.iter().sum();
    }
    if x.len() > 128 {
        let half = x.len() / 2 / 8 * 8;
        return pairwise_sum(&x[..half]) + pairwise_sum(&x[half..]);
    }
    let mut r = [0.0; 8];
    let full = x.len() / 8 * 8;
    for chunk in x[..full].chunks(8) {
        for j in 0..8 {
            r[j] += chunk[j];
        }
    }
    let mut total = ((r[0] + r[1]) + (r[2] + r[3])) + ((r[4] + r[5]) + (r[6] + r[7]));
    for v in &x[full..] {
        total += v;
    }
    total
}

fn mean(x: &[f64]) -> f64 {
    pairwise_sum(x) / x.len() as f64
}

/// Linear-interpolation quantile of SORTED data (numpy's default method).
fn quantile(sorted: &[f64], q: f64) -> f64 {
    let h = (sorted.len() - 1) as f64 * q;
    let lo = h.floor() as usize;
    let hi = (lo + 1).min(sorted.len() - 1);
    let (a, b, t) = (sorted[lo], sorted[hi], h - lo as f64);
    // interpolate from the nearer end, as numpy does
    if t >= 0.5 { b - (b - a) * (1.0 - t) } else { a + (b - a) * t }
}

fn median(x: &[f64]) -> f64 {
    let mut s = x.to_vec();
    s.sort_by(|a, b| a.total_cmp(b));
    let n = s.len();
    if n % 2 == 1 { s[n / 2] } else { (s[n / 2 - 1] + s[n / 2]) / 2.0 }
}

fn spread(x: &[f64]) -> Spread {
    let n = x.len() as f64;
    let m = mean(x);

    // VARIANCE: mean squared distance from the mean.
    // n-1, not n. Dividing by n underestimates, because you used the same
    // data to compute the mean you are measuring distance from.
    let sq: Vec<f64> = x.iter().map(|v| (v - m).powi(2)).collect();
    let var = pairwise_sum(&sq) / (n - 1.0);

    let mut s = x.to_vec();
    s.sort_by(|a, b| a.total_cmp(b));
    let (q1, q3) = (quantile(&s, 0.25), quantile(&s, 0.75)); // IQR: middle 50%
    let med = median(x);
    let dev: Vec<f64> = x.iter().map(|v| (v - med).abs()).collect();
    let mad = median(&dev); // robust alternative to sd
```

```
    Spread { sd: var.sqrt(), var, iqr: q3 - q1, mad, mad_scaled: 1.4826 * mad }
}

fn main() {
    let clean = vec![1.9, 2.1, 2.0, 2.4, 2.2, 1.8, 2.3, 2.0];
    println!("{:?}", spread(&clean)); // sd 0.203, iqr 0.250, mad 0.150
    let mut dirty = clean.clone();
    dirty.push(41.0);
    println!("{:?}", spread(&dirty)); // sd 13.0 (!), iqr 0.300, mad 0.200
    // One outlier multiplied the sd by 64x and barely touched the IQR or MAD.
}
```

Variance in depth

$$\sigma^2 = \mathbb{E}\big[(X-\mu)^2\big] \qquad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})^2$$

Left: the population variance, a property of the process that generates the data. Right: the sample variance, your estimate of it from n observations.

By hand

Data 2, 4, 4, 4, 5, 5, 7, 9. Mean 5. Squared distances 9, 1, 1, 1, 0, 0, 4, 16, sum 32. Divide by $n-1=7$: $s^2 = 4.57$. Divide by $n=8$ instead and you get 4.00, which underestimates.

Why squares

Squares make variance **add up**: for independent sources, $\operatorname{Var}(X+Y) = \operatorname{Var}X + \operatorname{Var}Y$. Scaling multiplies it: $\operatorname{Var}(aX) = a^2 \operatorname{Var}X$. These two rules are why all of inference is built on variance.

Use it for

The arithmetic underneath standard errors, power, bootstrap, ANOVA, the design effect, and in training: Adam's second moment, advantage normalisation, LayerNorm.

Watch out

Units are squared and one point can own most of it. Add a 60 to the data above and that single point contributes 88% of the total variance.

The $n - 1$ (Bessel's correction) exists because the deviations are measured from $\bar{x}$, which was fitted to the same data and sits closer to the points than the true μ does. Dividing by $n - 1$ makes s^2 unbiased for σ^2. For large n the difference vanishes; for a 5-seed ablation it is 25%.

The general addition rule includes a covariance term: $\operatorname{Var}(X + Y) = \operatorname{Var} X + \operatorname{Var} Y + 2\operatorname{Cov}(X, Y)$. That term is why paired comparisons are so powerful: for a difference $X - Y$ the sign flips, and positively correlated scores on shared tasks *subtract* variance (see paired comparisons). It is also why clustered data needs care: correlated items add variance a naive formula ignores (see clustering).

Standard deviation in depth

$$\sigma = \sqrt{\sigma^2} \qquad s = \sqrt{s^2} \qquad z = \frac{x - \bar{x}}{s}$$

SD is variance brought back to the original units. The z-score expresses any value as "how many SDs from the mean".

By hand

Same data: $s = \sqrt{4.57} = 2.14$. A value of 9 has $z = (9 - 5)/2.14 = 1.87$: almost two typical distances above the mean.

Normal benchmarks

For normal data, about 68%, 95%, and 99.7% of values fall within 1, 2, and 3 SDs. Only for normal data: for skewed data these percentages are wrong.

Use it for

Describing variability in readable units, z-scores, effect sizes such as Cohen's $d = (\bar{x}_1 - \bar{x}_2)/s$, seed-to-seed noise in training runs, and as the numerator of the standard error.

Watch out

SD does not add. $\sigma_{X+Y} = \sqrt{\sigma_X^2 + \sigma_Y^2}$, not $\sigma_X + \sigma_Y$. And SD is as fragile as variance: with the 60 added, SD jumps from 2.14 to 18.4.

Interquartile range in depth

$$\mathrm{IQR} = Q_3 - Q_1 = F^{-1}(0.75) - F^{-1}(0.25) \qquad \text{fences: } \left[Q_1 - 1.5\,\mathrm{IQR},\ Q_3 + 1.5\,\mathrm{IQR}\right]$$

The width of the middle half of the data, from the quantile function F^{-1}. The Tukey fences on the right are the usual box-plot rule for flagging outliers.

By hand

Sorted 2, 4, 4, 4, 5, 5, 7, 9. With linear interpolation (numpy's default) $Q_1 = 4$, $Q_3 = 5.5$, IQR $= 1.5$. Add the 60 and IQR becomes 3.0. Quantile conventions differ slightly between tools, so state which one you use.

Normal benchmark

On normal data $\mathrm{IQR} = 1.349\,\sigma$, so $\mathrm{IQR}/1.349$ is a robust stand-in for SD.

Use it for

Box plots, outlier fences, describing latency and token counts, and robust scaling of features. It pairs naturally with the median, the same way SD pairs with the mean.

Watch out

It ignores the outer half of the data by design. If the tail is the product (p99 latency, rare catastrophic answers), IQR hides exactly what matters. Report a tail percentile alongside it.

Median absolute deviation in depth

$$\mathrm{MAD} = \mathrm{median}_i\left(|x_i - \mathrm{median}(x)|\right) \qquad \hat{\sigma}_{\mathrm{MAD}} = 1.4826 \cdot \mathrm{MAD} \qquad M_i = \frac{0.6745\,(x_i - \tilde{x})}{\mathrm{MAD}}$$

The middle of the distances from the middle. The constant $1.4826 = 1/\Phi^{-1}(0.75)$ rescales MAD so it estimates σ on normal data. M_i is the robust (modified) z-score; $|M_i| > 3.5$ is a common outlier flag.

By hand

Median of the data is 4.5. Distances 2.5, 0.5, 0.5, 0.5, 0.5, 0.5, 2.5, 4.5; their median is 0.5. So MAD $= 0.5$, scaled $= 0.74$. Five of the eight values sit within half a unit of the median, and MAD reports exactly that.

Breakdown point

Up to half the data can be replaced by garbage before MAD can be pushed anywhere. Add the 60: MAD moves from 0.5 to 1.0. SD moved from 2.14 to 18.4.

Use it for

Anomaly detection and alert thresholds, cleaning before fitting, flagging flaky eval tasks, and any dashboard where one bad day should not redefine "normal".

Watch out

It is less precise than SD when data really is normal (about 37% efficiency), and it can be zero when more than half the values are identical, as with pass/fail scores. Use it on continuous metrics.

The four side by side

Property	Variance	Standard deviation	IQR	MAD
Units	Squared	Original	Original	Original
Built from	Squared distances from the mean	Same, square-rooted	Ranks (quartiles)	Ranks (median of distances)
Breakdown point (share of bad data it survives)	0%	0%	25%	50%
On normal data equals	σ^2	σ	$1.349\,\sigma$	$0.6745\,\sigma$
Efficiency on normal data	100%	100%	≈37%	≈37%
Adds across independent parts	Yes	No (add variances, then root)	No	No
Natural partner	Mean	Mean	Median	Median

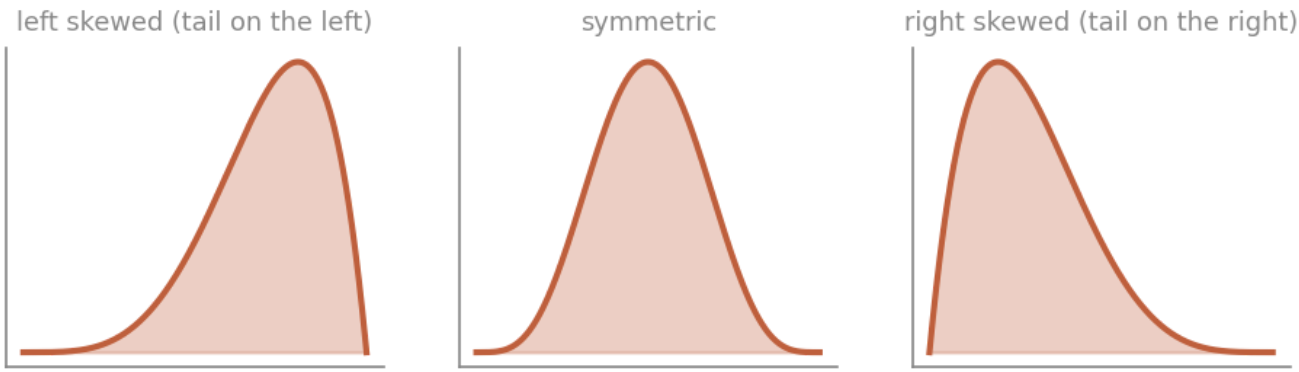
Where each one shows up in the rest of the guide

Measure	Where you will meet it
Variance	Standard errors and confidence intervals (the sampling chapter), paired tests and power (eval statistics), the design effect for clustered agent traces (intervals), Adam's per-parameter running variance of gradients (optimisation), advantage normalisation in PPO and GRPO (RLHF, GRPO), the bias–variance view of overfitting (SFT)
Standard deviation	Seed-to-seed noise and how many seeds you need (leading), z-scores and effect sizes, GRPO's division by the group SD and why Dr. GRPO removes it (GRPO), flaky-task quarantine thresholds (harness)
IQR	Box plots and outlier fences (long tails), latency and token-count reporting (shape to method), robust comparisons when data is skewed
MAD	Robust outlier detection (long tails), regression alerts that survive incidents (triage), cleaning noisy offline metrics before backtesting (offline to online)

Key idea

Spread of the data is not precision of an estimate. Everything above describes how individual values vary. The **standard error** describes how much an *estimate*, such as the mean of those values, would vary if you repeated the whole measurement. Formally, the standard error of an estimator $\hat{\theta}$ is the standard deviation of its sampling distribution, $\mathrm{SE}(\hat{\theta}) = \sqrt{\mathrm{Var}(\hat{\theta})}$. For the mean of n independent values it is $\sigma/\sqrt{n}$. Variance stays put as you collect more data; standard error shrinks. The full definition, the derivation, and an interactive comparison are in the sampling chapter.

Shape: skew and modality



Skew is named for the side the tail is on, not the side the bulk is on. This trips up almost everyone at first.

- **Right skewed** (positive skew). Long tail to the right. Latency, token counts, tool calls, repo sizes. By far the most common shape in eval work.
- **Left skewed**. Long tail to the left. Rarer. Shows up in scores near a ceiling, like a pass rate close to 100%.
- **Symmetric**. Both tails alike. Test statistics, averages of many things.
- **Bimodal**. Two humps. Almost always two populations mixed together, not one distribution being odd.

Key idea

The quickest skew check you can do in your head: **divide the mean by the median**. Above about 1.2 means right skewed enough that you should stop reporting the mean. Below about 0.85 means left skewed. This one ratio catches most bad reporting.

Under the hood

Skewness, formally and quickly

$$g_1 = \frac{\frac{1}{n}\sum(x_i - \bar{x})^3}{s^3}$$

Formal skewness. Positive means a right tail. In practice the mean/median ratio below is quicker and just as useful.

$$\frac{\bar{x}}{\text{median}(x)} > 1.2 \Rightarrow \text{right skewed}$$

Density, cumulative, and quantiles

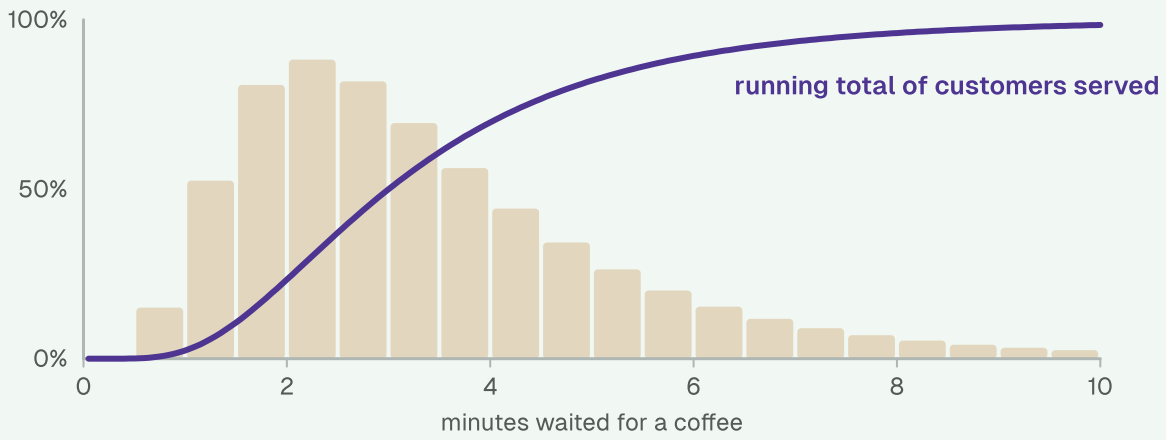
Three ways of describing the same shape. You need all three because different questions want different ones.

Diagram

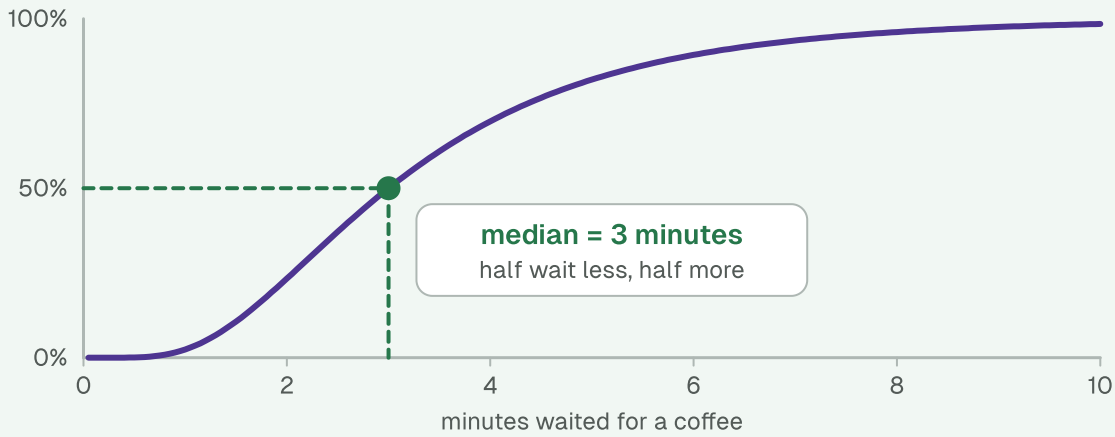
Reading a café queue as a cumulative curve

How a CDF turns a pile of waiting times into answers like "half of customers wait under 3 minutes".

Two hundred coffee orders, timed. The histogram shows where waits pile up, but it is hard to read off a question like "how long do most people wait?"

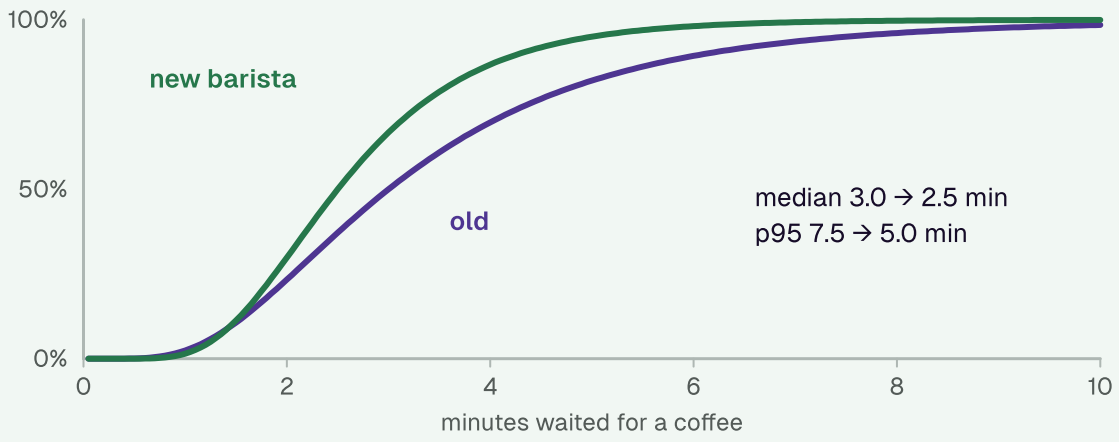


Walk left to right and keep a running total of the share of customers already served. That rising curve is the **cumulative distribution** (CDF). It always climbs from 0% to 100%.



To read the **median**, go across from 50% until you hit the curve, then drop down: 3 minutes. Half the customers waited less.

The same move at 95% gives the **p95**: 7.5 minutes. Nineteen customers in twenty waited less. Latency targets in model serving are written exactly like this.



Compare two curves at a glance: a curve that sits to the left is faster everywhere, from the median to the tail. Overlaying CDFs is the quickest honest way to compare two latency or token-count distributions.

- **Density (PDF)** answers "how common is a value around here". This is the humped curve you picture when someone says distribution.
- **Cumulative (CDF)** answers "what share of values are at or below here". Always rises from 0 to 1.
- **Quantiles** are the CDF read backwards: "what value has 95% of the data below it". p95 is a quantile. So is the median, which is p50.

> **Note**
>
> Percentile and quantile are the same thing at different scales. p95, the 95th percentile, and the 0.95 quantile are three names for one number.

Under the hood CDF and quantile function

$$F(x) = P(X \le x) \qquad Q(p) = F^{-1}(p)$$

The CDF gives the share at or below a value. The quantile function is its inverse: give it 0.95 and it hands back p95.

python

```python
import numpy as np

lat = np.array([1.2, 1.5, 1.9, 2.1, 2.4, 3.0, 4.5, 8.0, 15.0, 40.0])

# QUANTILE: what value has p of the data below it?
print(np.quantile(lat, 0.50))     # 2.70  -> the median
print(np.quantile(lat, 0.95))     # 28.7  -> p95

# CDF: run it the other way. What share is at or below 4.5?
print((lat <= 4.5).mean())        # 0.70

# These are inverses of each other. Same information, two directions.
```

rust

```rust
/// Linear-interpolation quantile of SORTED data (numpy's default method).
fn quantile(sorted: &[f64], q: f64) -> f64 {
    let h = (sorted.len() - 1) as f64 * q;
    let lo = h.floor() as usize;
    let hi = (lo + 1).min(sorted.len() - 1);
    let (a, b, t) = (sorted[lo], sorted[hi], h - lo as f64);
    // interpolate from the nearer end, as numpy does
    if t >= 0.5 { b - (b - a) * (1.0 - t) } else { a + (b - a) * t }
}

fn main() {
    let lat = [1.2, 1.5, 1.9, 2.1, 2.4, 3.0, 4.5, 8.0, 15.0, 40.0]; // already sorted

    // QUANTILE: what value has p of the data below it?
    println!("{}", quantile(&lat, 0.50)); // 2.70  -> the median
    println!("{}", quantile(&lat, 0.95)); // 28.7  -> p95

    // CDF: run it the other way. What share is at or below 4.5?
    let share = lat.iter().filter(|&&v| v <= 4.5).count() as f64 / lat.len() as f64;
    println!("{}", share);              // 0.70
```

```
    // These are inverses of each other. Same information, two directions.
}
```

For leaders

Reading a dashboard number without being fooled

Most bad decisions built on eval data are not caused by exotic statistics. They come from a single summary number standing in for a shape nobody looked at. The questions below take thirty seconds and catch most of it.

- **"Is that a mean or a median, and what is the other one?"** For latency, cost per task, or tokens per response, a mean far above the median tells you a tail is driving the number. A mean is right for capacity and budget (totals); a median or p95 is right for user experience.
- **"What is the spread, and in what units?"** A 3 point win means little if per-task scores swing by 20 points. Ask for an IQR or a percentile band, not just an average.
- **"Has anyone plotted it?"** A bimodal distribution, such as cache hits versus cold starts or easy versus hard tasks, means there are two stories. Averaging them produces a number that describes neither and hides which one moved.
- **"Which population is this a sample of?"** An eval suite is a sample chosen by people. If it over-represents easy or old tasks, every summary inherits that.

Principal asks

"The mean latency improved by 12%. What happened to p50 and p95?" If the mean improved because a handful of pathological requests got fixed, that is good engineering but says nothing about the typical user. If p95 got worse while the mean improved, the launch may hurt the users who already suffer most.

Part I · Chapter 4

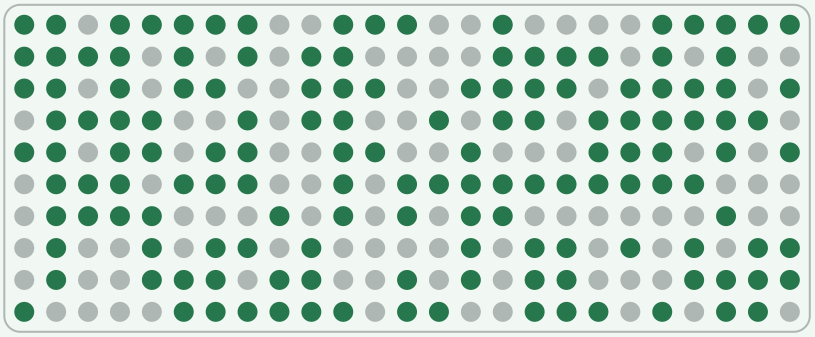
Two big ideas: the sampling distribution and the standard error

If you take only two things from Part I, take these. Nearly every statistical mistake in eval work is a misunderstanding of one of them, and every confidence interval and p value later in the guide is built on them.

Diagram

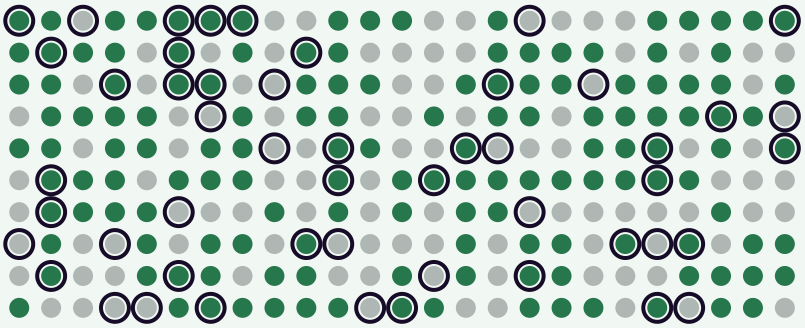
Polling a neighbourhood about a new park

The sampling distribution and the standard error, with dots you can count.



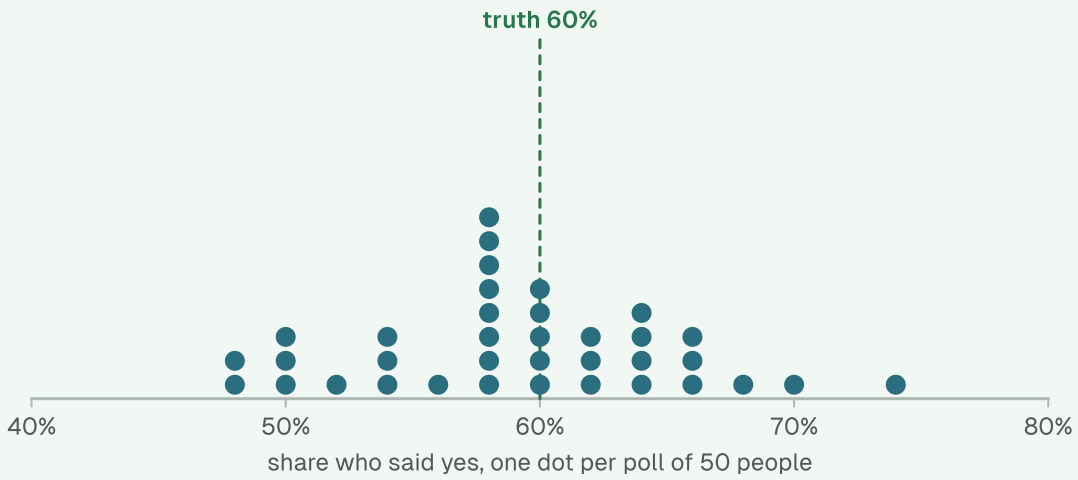
likes the new park does not **true share: 60%**

A town of 10,000 people (each dot stands for 40). Exactly 60% like the plan for a new park. In real life you never see this picture; you only get to ask a few people.

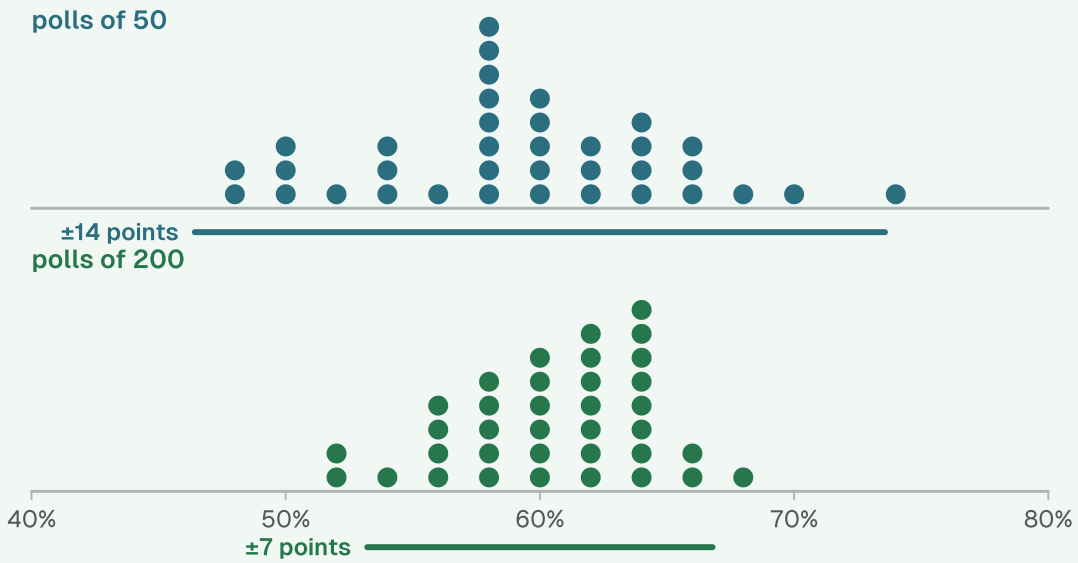


You ask 50 people
31 say yes: 62%

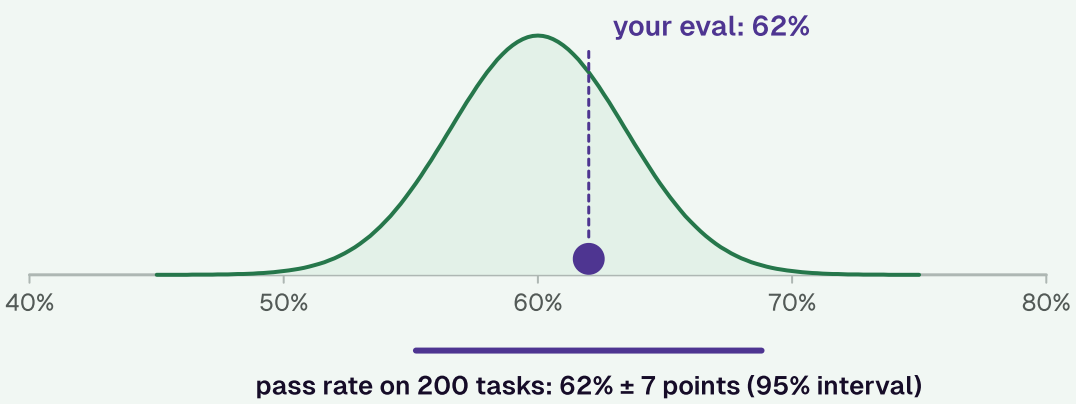
You stop 50 people at random. 31 say yes, so your poll says 62%. Close to the truth, but not equal to it, and you have no way to know how close.



Imagine 36 different pollsters, each asking 50 different people. Their answers pile up around 60%, some as low as 48%, some as high as 72%. This pile is the **sampling distribution**.



Ask 200 people per poll and the pile tightens to about half the width. Four times the people buys half the uncertainty. The width of the pile is the **standard error**.



An eval score is one poll of tasks. A 62% pass rate on 200 tasks is one dot from a pile about ±7 points wide. That is why "62% vs 60%" on 200 tasks is not a result yet.

Plain words

Your score is one draw from a lottery of possible scores

You ran 200 tasks and got 62%. If you had run a *different* 200 tasks you would have got something else, maybe 59%, maybe 65%. The spread of all those hypothetical alternative answers is the **sampling distribution**. Every confidence interval and every p value is a statement about it, not about your data.

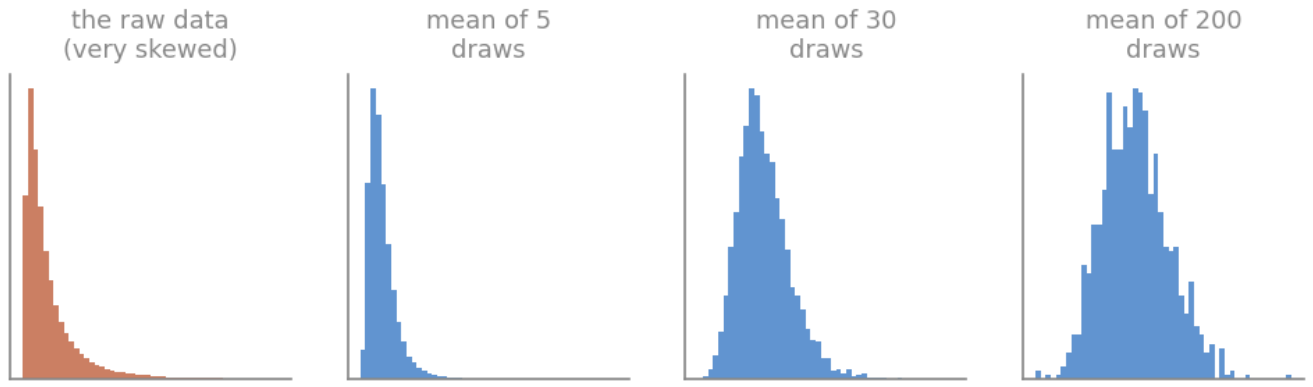
The **standard error** is how wide that lottery is. It shrinks as you test more tasks, but slowly: to cut the uncertainty in half you need four times as many tasks.

The sampling distribution, and why the bell curve is everywhere

Here is the part that surprises people. The raw data can be any shape at all. But the *average* of many draws is always roughly a bell, and it gets more bell-like as you average more. That is the central limit theorem.

$$\bar{X}_n \longrightarrow \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right) \quad \text{as } n \text{ grows}$$

The average of n draws tends toward a normal shape centred on the true mean, with variance shrinking as one over n.



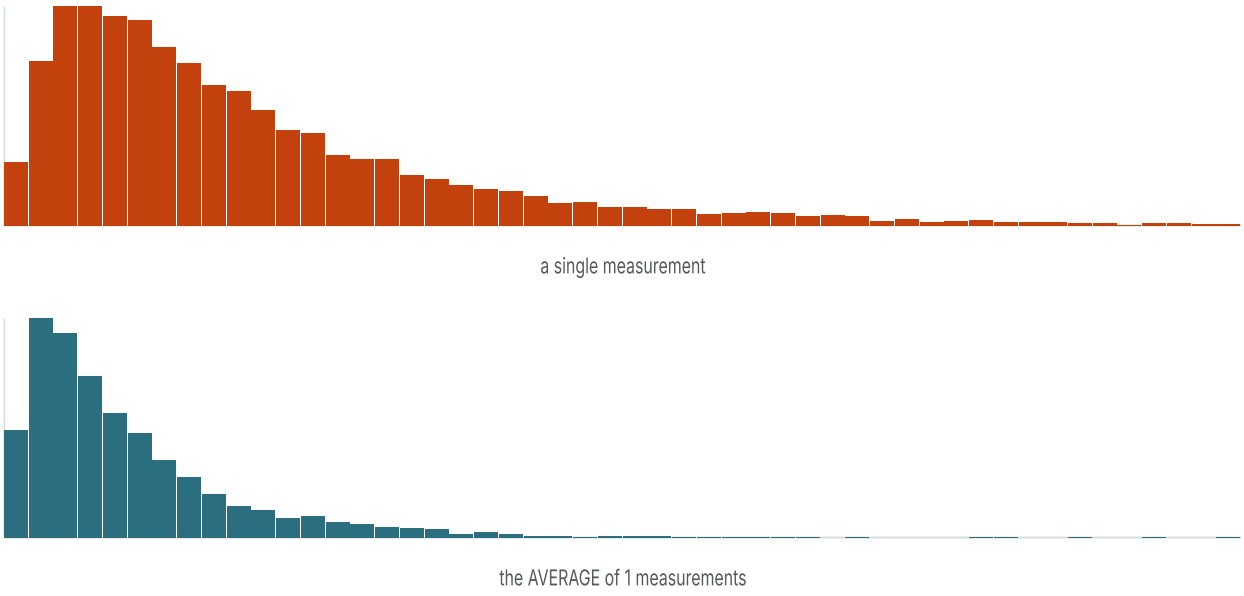
The raw data on the left is wildly skewed. Average 5 of them and it is still skewed. Average 30 and it is nearly a bell. Average 200 and it is a bell.

Key idea

This is why so much of statistics assumes normality even though almost no real data is normal. The assumption is not about your measurements, it is about their average. Your latency data is lognormal and always will be; the mean of 500 latencies is close to normal.

Interactive **Watch the central limit theorem happen** (shown at its default setting)

Pick any starting shape, however ugly. The top panel is a single measurement. The bottom is the average of n measurements. Push n up and watch the bottom become a bell regardless of what the top looks like.



Raw data: mean 4.11, median 3.03, skew ratio 1.35
Averages of 1: mean 4.00, median 2.94, skew ratio 1.36, sd 3.68
The top shape can be anything at all. The bottom one becomes a bell as n grows, and its spread shrinks like 1/sqrt(n). That is the central limit theorem, and it is why so much of statistics assumes normality even when the raw data is nothing like normal.

Try it

Set the starting shape to Pareto and push n to 200. It becomes a bell far more slowly than the others. That is what a genuinely heavy tail does: the central limit theorem still applies, but it needs much more data before you can lean on it.

Standard error vs variance (and standard deviation)

This is the single most common confusion in applied statistics, and the two get swapped in dashboards constantly. They answer two different questions.

Variance s^2 (and SD s)

Question: how much do individual values differ from each other?

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2, \qquad s = \sqrt{}$$

Describes the data: users, tasks, latencies.
Units squared (variance) or original (SD).
Depends on the data only.
With more data it settles to its true value σ^2. It does not shrink.

Standard error SE

Question: how much would my *estimate* change if I repeated the whole study?

$$\mathrm{SE}(\hat{\theta}) = \sqrt{\mathrm{Var}(\hat{\theta})}, \qquad \mathrm{SE}(\bar{x}) = \frac{s}{\sqrt{n}}$$

Describes an estimate: a mean, a pass rate, a difference.
Units original.
Depends on the data's variance, the sample size n, the estimator, and whether items are independent.
With more data it shrinks like $1/\sqrt{n}$.

Formal definition

The **standard error** of an estimator $\hat{\theta}$ is the standard deviation of its **sampling distribution**: the spread of the values $\hat{\theta}$ would take across many repeats of the same study. Variance is a property of the data; standard error is a property of a number you computed from the data. For the mean of n independent values, $\mathrm{Var}(\bar{x}) = \sigma^2/n$, so $\mathrm{SE} = \sigma/\sqrt{n}$. The derivation is in the **Under the hood** tab below.

The same five commutes, both numbers

Quantity	Formula	Value for 22, 25, 31, 24, 28 (n = 5)	If n were 20, same spread
Sample variance	s^2	12.5 min²	≈ 12.5 min²
Standard deviation	s	3.54 min	≈ 3.54 min
Variance of the mean	s^2/n	2.5 min²	0.625 min²
Standard error of the mean	$s/\sqrt{n}$	1.58 min	0.79 min
95% interval for the true average	$\bar{x} \pm 1.96\ \mathrm{SE}$	26 ± 3.1 min	26 ± 1.5 min

Four times the data left the variance and SD where they were and cut the standard error in half. (With only five values a t-based interval is a little wider than ±1.96 SE; see intervals.)

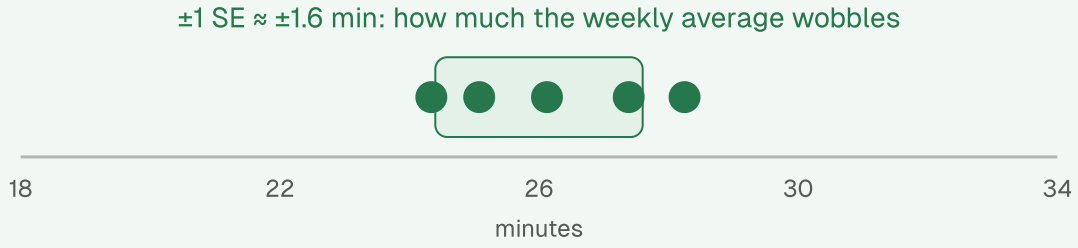
Diagram

Repeat the week: variance stays, standard error shrinks

Same commute, measured week after week. Numbers are illustrative but consistent with σ ≈ 3.5 minutes.

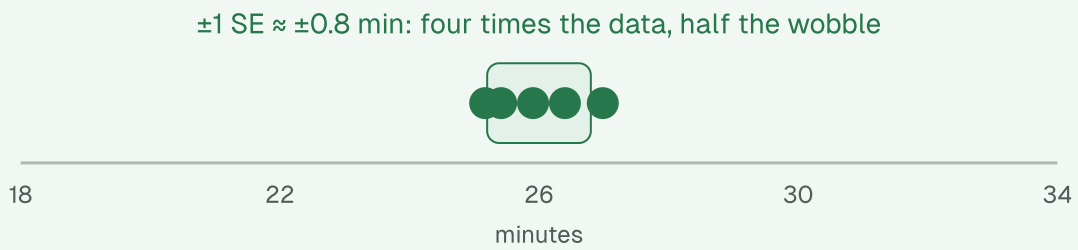
One week, five commutes. They spread about 3.5 minutes either side of their average. That spread is the standard deviation, and its square (12.5) is the variance. It describes **days**.

Five weeks: one average per week (n = 5 each)

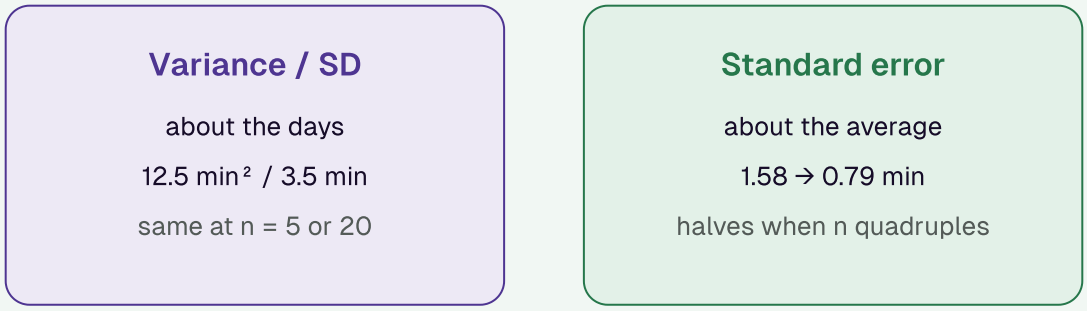


Repeat the week five times and plot only each week's **average**: 26.0, 24.2, 28.1, 25.0, 27.3. The averages wobble about 1.6 minutes, close to $3.54/\sqrt{5} = 1.58$. That wobble is the standard error. It describes the **estimate**, not the days.

Twenty commutes per week (n = 20)



Log twenty commutes a week instead of five. Single days are exactly as varied as before (SD still about 3.5), but the weekly averages now land within about 0.8 minutes of each other, close to $3.54/\sqrt{20} = 0.79$.



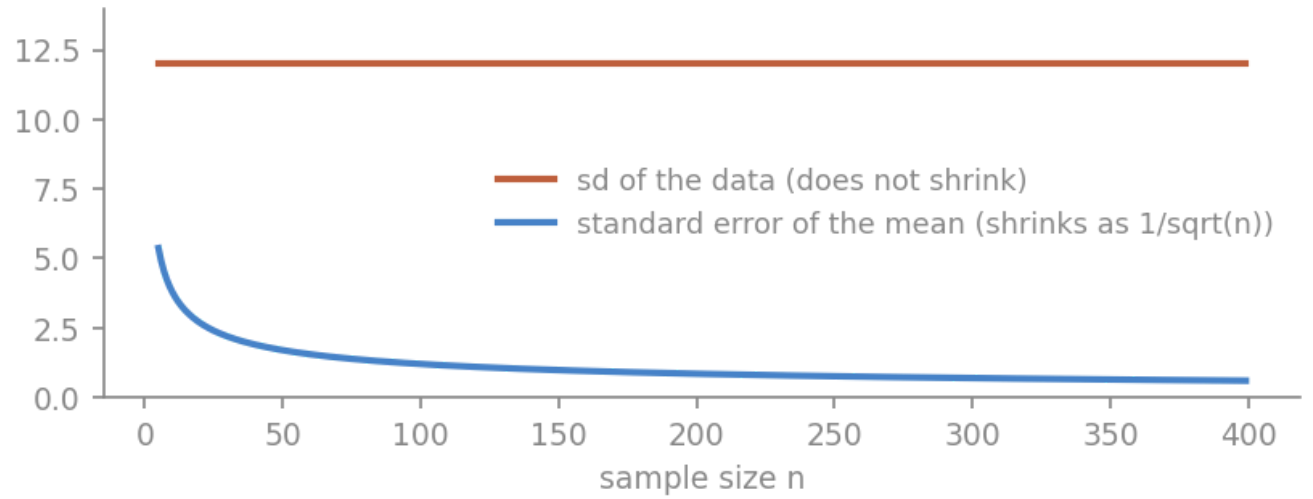
Use SD to describe variability. Use SE for error bars on an average.

Variance and SD describe how varied the things you measured are. Standard error describes how precisely you have pinned down an average of them. An eval report needs both, labelled.

Four quantities that get confused

Quantity	Symbol	Describes	Units	As n grows
Population variance	σ^2	How much individual values vary in the process	Squared	Fixed: a property of the world
Sample variance	s^2	Your estimate of σ^2	Squared	Settles toward σ^2
Variance of the mean	σ^2/n	How much the average would vary across repeats	Squared	Shrinks like $1/n$
Standard error of the mean	$\sigma/\sqrt{n}$	The same, in original units	Original	Shrinks like $1/\sqrt{n}$

So the difference between variance and standard error is not just a square root. Variance (and SD) describe the *data*; standard error describes an *estimate*, and it depends on three things the data's variance does not: the sample size, the estimator, and the dependence between observations.



The standard deviation of your data does not shrink as you collect more. The standard error of the mean does, as one over the square root of n.

	Standard deviation	Standard error
Describes	How spread out the individual measurements are	How precisely you know the average
Formula	s	$s/\sqrt{n}$
As n grows	Stays put. More data does not make users more alike	Shrinks, but slowly
Use for	Describing variability, setting SLOs	Error bars on a mean, confidence intervals

Key idea

Because standard error shrinks as one over the square root of n, **quadrupling your sample halves your error bar**. Not doubling, quadrupling. That single fact explains why every sample size answer feels so unsatisfying and why chasing small effects gets expensive so fast.

Plain words

Scattered arrows, wobbling averages

An archer shoots a volley of 20 arrows. The arrows land all over the target: that scatter is the **standard deviation**, and practising more volleys will not make a single arrow land closer to the others.

Now mark the average landing spot of each volley. Shoot another volley and the average lands somewhere slightly different. How much that average wobbles from volley to volley is the **standard error**. Shoot 80 arrows per volley instead of 20 and the average wobbles half as much, even though each arrow is just as scattered as before.

Variance is the standard deviation squared. It describes the arrows. The standard error describes your *estimate* built from the arrows.

For leaders

Reading error bars correctly

- **Ask which one the bars are.** SD bars show how varied the users or tasks are and never shrink. SE bars (or a confidence interval, about ±2 SE) show how sure we are of the average and shrink with more data. Decks swap them constantly.
- **Precision is bought with the square root.** Halving an error bar costs four times the data. That is why small effects get expensive and why reducing variance (paired designs, cleaner tasks) is often cheaper than collecting more.
- **Standard errors assume independent items.** Multi-turn conversations, several calls per agent session, or many tasks from one repository are correlated. The honest SE is larger, sometimes several times larger.

Principal asks

Are those SD or SE bars? What is n, and are the items independent? If we doubled the eval set, how much would the interval shrink, and is that cheaper than a paired design?

Under the hood

Standard error, formally, and the square-root law in code

Definition. An estimator $\hat{\theta}$ (a mean, a pass rate, a median, a difference between two models) is computed from a random sample, so it is itself a random variable with its own distribution across hypothetical repeats of the study: its *sampling distribution*. The **standard error** is the standard deviation of that sampling distribution:

$$\mathrm{SE}(\hat{\theta}) \;=\; \sqrt{\mathrm{Var}(\hat{\theta})} \;=\; \sqrt{\mathbb{E}\Big[\big(\hat{\theta} - \mathbb{E}[\hat{\theta}]\big)^2\Big]}$$

Same formula as a standard deviation, applied to the estimate instead of to individual observations. In practice it is unknown and estimated from the data, written $\widehat{\mathrm{SE}}$.

Derivation for the mean. Let $X_1, \dots, X_n$ be independent with variance σ^2. Two variance rules from the distributions chapter do all the work: variances of independent terms add, and scaling by a multiplies variance by a^2.

$$\mathrm{Var}(\bar{X}) = \mathrm{Var}\left(\tfrac{1}{n}\sum_{i=1}^{n} X_i\right) = \tfrac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(X_i) = \tfrac{1}{n^2}\cdot n\sigma^2 = \frac{\sigma^2}{n} \qquad \Longrightarrow \qquad \mathrm{SE}(\bar{X}) =$$

The $1/n^2$ from averaging beats the n from adding, leaving $1/n$ for the variance and $1/\sqrt{n}$ for the standard error. Independence is used in the second step; drop it and covariance terms appear.

Standard errors you will actually use

Estimate	Standard error	Note
Mean	$s/\sqrt{n}$	The workhorse
Pass rate $\hat{p}$	$\sqrt{\hat{p}(1-\hat{p})/n}$	124 of 200 passing: 0.0343, so about ±6.7 points at 95%. Use Wilson near 0 or 1 (eval statistics)
Difference, independent samples	$\sqrt{s_1^2/n_1 + s_2^2/n_2}$	Variances add, so the difference is noisier than either part
Difference, paired on the same items	$s_d/\sqrt{n}$, with s_d the SD of per-item differences	Usually far smaller, because shared item difficulty cancels
Median (normal data, large n)	$\approx 1.2533\,\sigma/\sqrt{n}$	The price of robustness: 25% wider than the mean's
Mean of clustered data	$\frac{s}{\sqrt{n}}\sqrt{1+(m-1)\rho}$	m items per cluster, intra-cluster correlation ρ (clustering)
Anything else	SD of bootstrap replicates	When no formula exists (bootstrap)

python

```python
import math

def se_mean(sd, n):
    return sd / math.sqrt(n)

def se_proportion(p, n):
    return math.sqrt(p * (1 - p) / n)

def se_diff_independent(s1, n1, s2, n2):
    # variances add for independent estimates; standard errors do not
    return math.sqrt(s1 ** 2 / n1 + s2 ** 2 / n2)

def se_clustered(se_naive, m, icc):
    # design effect: m items per cluster, intra-cluster correlation icc
    return se_naive * math.sqrt(1 + (m - 1) * icc)

print(round(se_mean(12.0, 400), 3))                        # 0.6 seconds
print(round(se_proportion(124 / 200, 200), 4))             # 0.0343 -> about +/- 6.7 points at 95%
print(round(se_diff_independent(0.49, 200, 0.48, 200), 4)) # 0.0485 two models on different task sets
print(round(se_clustered(0.0343, 12, 0.6), 4))             # 0.0946 same pass rate, 12 calls per session
```

rust

```rust
fn se_mean(sd: f64, n: f64) -> f64 {
    sd / n.sqrt()
}

fn se_proportion(p: f64, n: f64) -> f64 {
    (p * (1.0 - p) / n).sqrt()
}

fn se_diff_independent(s1: f64, n1: f64, s2: f64, n2: f64) -> f64 {
    // variances add for independent estimates; standard errors do not
    (s1.powi(2) / n1 + s2.powi(2) / n2).sqrt()
}

fn se_clustered(se_naive: f64, m: f64, icc: f64) -> f64 {
    // design effect: m items per cluster, intra-cluster correlation icc
    se_naive * (1.0 + (m - 1.0) * icc).sqrt()
}

fn round(x: f64, digits: i32) -> f64 {
    let f = 10f64.powi(digits);
    (x * f).round() / f
}

fn main() {
    println!("{}", round(se_mean(12.0, 400.0), 3));                      // 0.6 seconds
    println!("{}", round(se_proportion(124.0 / 200.0, 200.0), 4));       // 0.0343 -> about +/- 6.7 points at
95%
    println!("{}", round(se_diff_independent(0.49, 200.0, 0.48, 200.0), 4)); // 0.0485 two models on different
task sets
    println!("{}", round(se_clustered(0.0343, 12.0, 0.6), 4));           // 0.0946 same pass rate, 12 calls per
session
}
```

The square-root law, simulated

$$\mathrm{SE} = \frac{s}{\sqrt{n}} \qquad \mathrm{SE} \propto n^{-1/2} \;\Rightarrow\; 4n \text{ gives half the error bar}$$

python

```python
import numpy as np

rng = np.random.default_rng(0)
population_sd = 12.0

for n in (25, 100, 400, 1600):
    sample = rng.normal(60, population_sd, n)

    sd = sample.std(ddof=1)          # spread of the MEASUREMENTS
    se = sd / np.sqrt(n)             # spread of the ESTIMATE of their mean
```

```
    print(n, "sd:", round(sd, 2), " se:", round(se, 3),
          " 95% CI half width:", round(1.96 * se, 3))

# n=25   sd 10.47  se 2.095   CI half width 4.106
# n=100  sd 11.68  se 1.168   CI half width 2.29
# n=400  sd 12.35  se 0.617   CI half width 1.21
# n=1600 sd 11.97  se 0.299   CI half width 0.587
#
# The sd column does not move. More data does not make users more alike.
# The se column halves every time n quadruples. That is the whole story.
```

rust

```
// A tiny inline PRNG (SplitMix64) so the snippet needs no crates.
// Its draws differ from numpy's, so the random numbers below differ from the Python's.
struct Rng(u64);

impl Rng {
    fn new(seed: u64) -> Self {
        Rng(seed)
    }
    fn next_u64(&mut self) -> u64 {
        self.0 = self.0.wrapping_add(0x9E37_79B9_7F4A_7C15);
        let mut z = self.0;
        z = (z ^ (z >> 30)).wrapping_mul(0xBF58_476D_1CE4_E5B9);
        z = (z ^ (z >> 27)).wrapping_mul(0x94D0_49BB_1331_11EB);
        z ^ (z >> 31)
    }
    /// Uniform on [0, 1).
    fn random(&mut self) -> f64 {
        (self.next_u64() >> 11) as f64 / (1u64 << 53) as f64
    }
    /// Normal draw by Box-Muller.
    fn normal(&mut self, mean: f64, sd: f64) -> f64 {
        let u1 = 1.0 - self.random(); // in (0, 1], safe for ln
        let u2 = self.random();
        mean + sd * (-2.0 * u1.ln()).sqrt() * (std::f64::consts::TAU * u2).cos()
    }
}

fn round(x: f64, digits: i32) -> f64 {
    let f = 10f64.powi(digits);
    (x * f).round() / f
}

/// Sample standard deviation (ddof = 1).
fn sample_sd(x: &[f64]) -> f64 {
    let n = x.len() as f64;
    let m = x.iter().sum::<f64>() / n;
    (x.iter().map(|v| (v - m).powi(2)).sum::<f64>() / (n - 1.0)).sqrt()
}

fn main() {
    let mut rng = Rng::new(0);
    let population_sd = 12.0;

    for n in [25, 100, 400, 1600] {
        let sample: Vec<f64> = (0..n).map(|_| rng.normal(60.0, population_sd)).collect();

        let sd = sample_sd(&sample);    // spread of the MEASUREMENTS
```

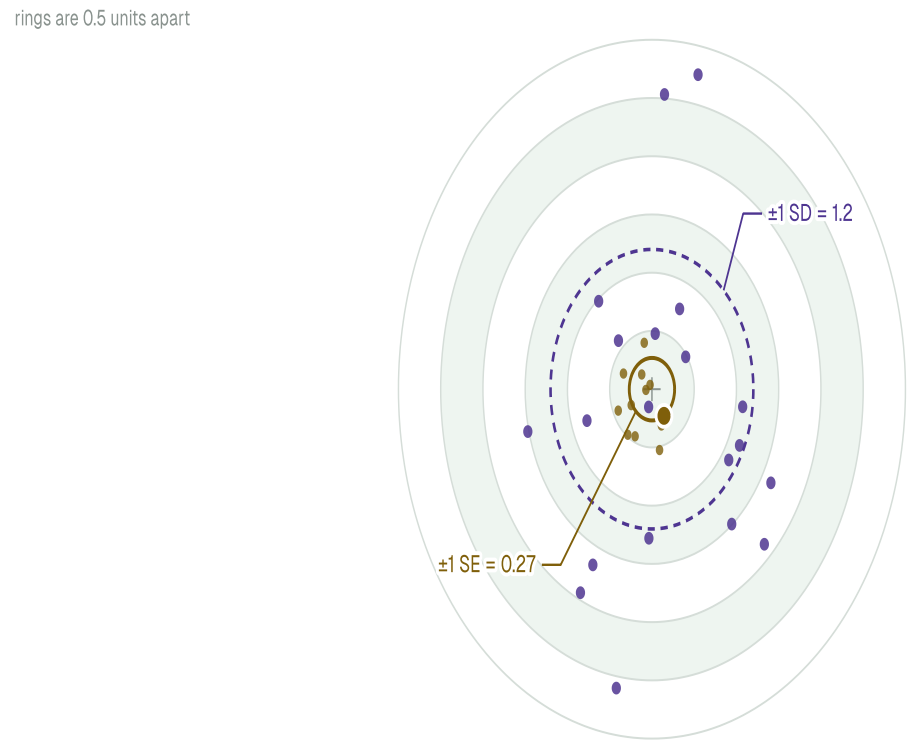
```
        let se = sd / (n as f64).sqrt(); // spread of the ESTIMATE of their mean

        println!("{} sd: {}  se: {}  95% CI half width: {}",
                 n, round(sd, 2), round(se, 3), round(1.96 * se, 3));
    }

    // SplitMix64 draws, so the numbers differ from numpy's; the pattern is the same:
    // n=25   sd 11.56  se 2.313  CI half width 4.533
    // n=100  sd 11.12  se 1.112  CI half width 2.179
    // n=400  sd 11.85  se 0.592  CI half width 1.161
    // n=1600 sd 12.02  se 0.3    CI half width 0.589
    //
    // The sd column does not move. More data does not make users more alike.
    // The se column halves every time n quadruples. That is the whole story.
}
```

Here is the archer from the plain-words tab, ready to shoot. Each arrow lands with the same scatter, but the average of a volley wobbles much less, and you can watch by how much.

Interactive

Archer's target: scatter vs wobble (shown at its default setting)

Shoot 20 volleys, then move "arrows per volley" from 20 to 80: the wide circle for single arrows stays put while the cloud of volley averages, and its circle, shrinks by half.



arrows of the latest volley, and the ±1 SD circle (σ) volley averages, and the ±1 SE circle (σ/√n)

Arrow spread SD σ	SD within the latest volley	SE = σ/√n	Observed spread of averages
1.20	1.09	0.268	0.229

Volleys shot

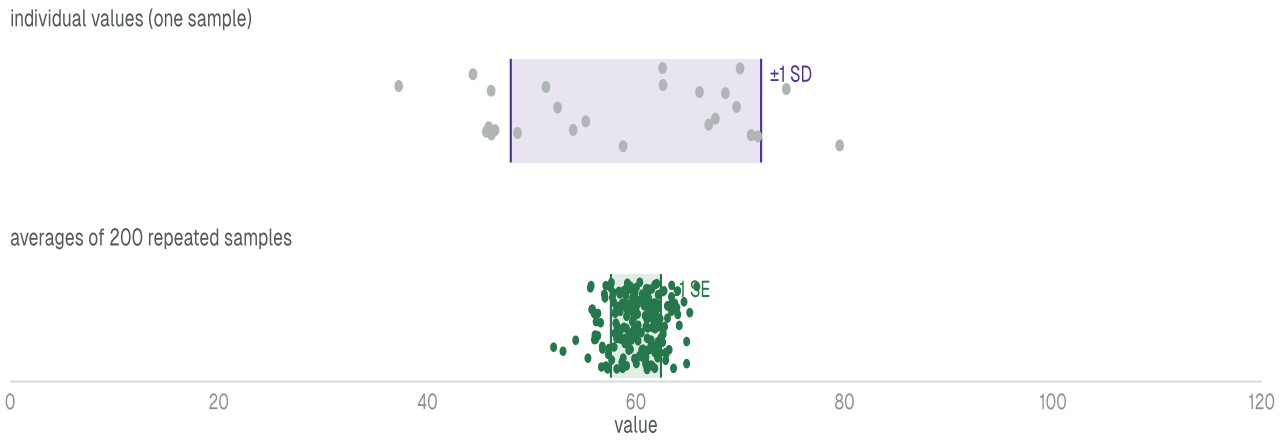
12

Each arrow misses the bullseye by about **σ = 1.2** units in each direction (dashed circle, first card); the latest volley's own scatter is 1.09 (second card). The average of n = 20 arrows wobbles only about **σ/√n = 0.27** units (solid circle, third card). Across 12 volleys the averages actually spread 0.23 (fourth card), close to the formula. Quadruple the arrows to 80 and the SE would halve to

0.13 while σ stays 1.2. (In two dimensions only about 4 arrows in 10 fall inside a 1 SD circle; compare the two circles, not the counts.)

Interactive **Standard deviation vs standard error** (shown at its default setting)

Top: one sample of n values. Their spread is the SD. Bottom: the averages of 200 repeated samples of the same size. Their spread is the SE. Raise n and watch only the bottom row tighten.



SD of the sample	SE by formula σ/√n	SD of the 200 averages	Variance of the mean σ²/n
11.67	**2.400**	**2.329**	**5.760**

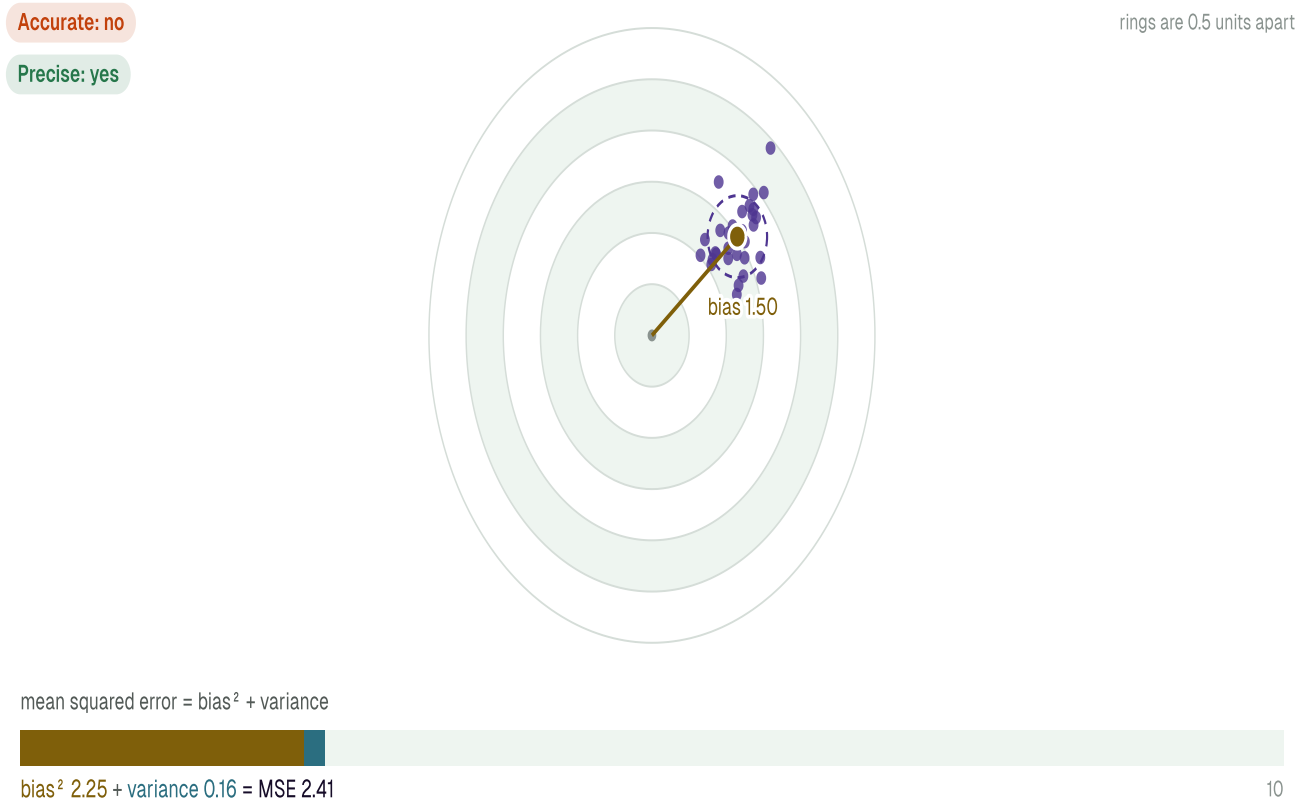
With n = 25, the values still spread about **11.7** around their mean, close to σ = 12. Their average is pinned to within about **±4.70** (95%). Quadruple n to 100 and that window halves to ±2.35; the SD does not move.

A small standard error means your estimate is **precise**: repeat the study and you land in nearly the same place. It does not mean the estimate is **accurate**, because a measurement can be precisely wrong. Statisticians call the systematic miss **bias** and the scatter **variance**, and the average squared error is exactly the sum of the two: $\mathbf{MSE} = \mathbf{bias}^2 + \mathbf{variance}$. Here "precision" is the measurement sense (how repeatable a result is), not the classification metric of the same name in the classification chapter.

Interactive

Accuracy vs precision (bias vs variance) (shown at its default setting)

Press the four corner presets, then drag bias up with spread low: the throws stay tight but the whole cluster drifts, and the bias part of the error bar takes over.



Bias (distance)	Bias²	Variance (spread²)	Mean squared error
1.50	2.25	0.16	2.41

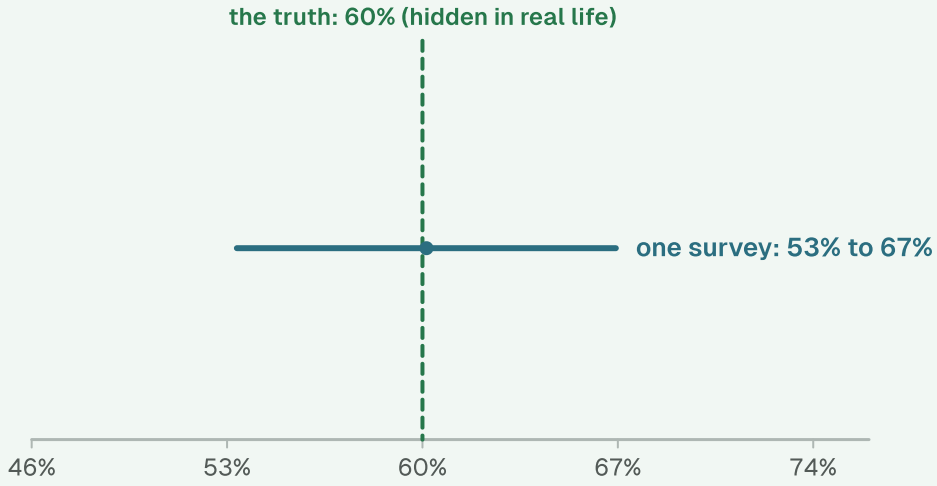
Precise, not accurate. The 30 throws sit about 0.40 units from their own centre (spread), and that centre sits 1.50 units from the bullseye (bias). Average squared miss = $1.50^2 + 0.40^2 = 2.25 + 0.16 = \mathbf{2.41}$; bias is 93% of it. Throwing more darts would pin down the mean point more tightly but leave it just as far from the bullseye: only fixing the aim removes bias. In eval terms, more tasks shrink the standard error, but a leaked or mis-graded benchmark stays wrong. (The labels use illustrative cut-offs: accurate means bias ≤ 0.5 units, precise means spread ≤ 0.6 units.)

What a confidence interval actually means

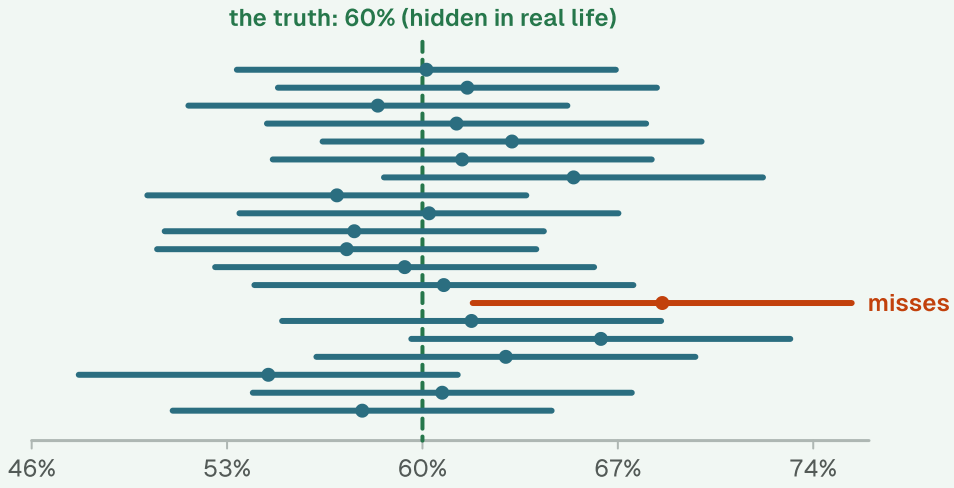
Diagram

What "95% confident" really means

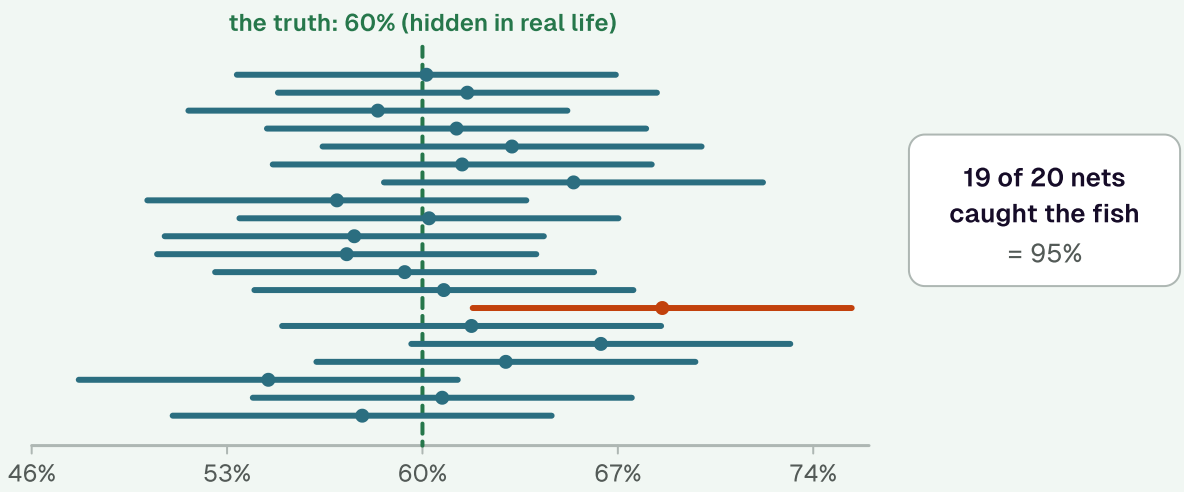
Twenty surveys, twenty nets thrown at a fish you cannot see.



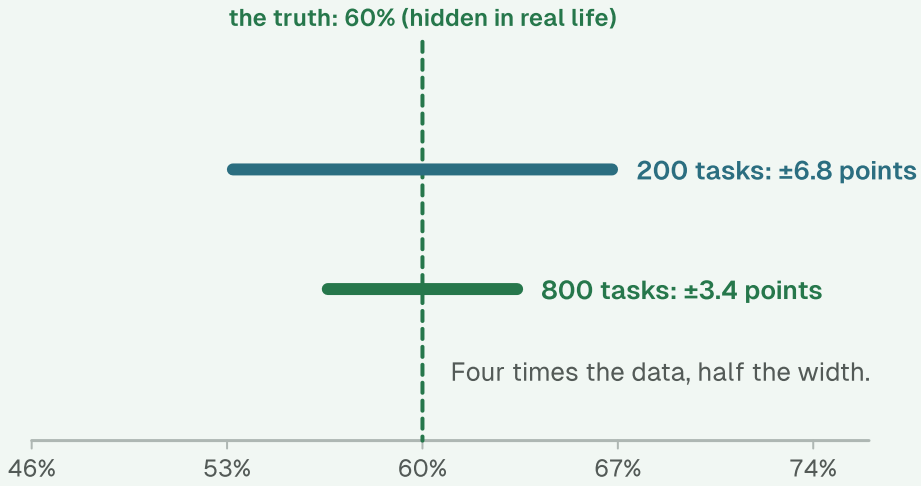
The true pass rate is 60%, but you never get to see it. You run one survey of 200 tasks and build an interval around your result. Think of the interval as a net thrown at a hidden fish.



Now repeat the whole survey 20 times with fresh samples. Each one throws its own net. Most land across the truth. One, in red, misses.



That is all "95%" means: the **method** catches the truth in about 19 of 20 throws. It does not mean a 95% chance the truth is inside *your* particular net; that net either caught it or did not.



To make the net smaller without missing more often, you need more data. Quadrupling the tasks halves the width, the same square-root law as the polls.

☆ **In plain words**

A 95% confidence interval is produced by a procedure that captures the true value 95 times out of 100. It is **not** a 95% chance the truth is inside the one interval in front of you. The truth is a fixed number; the interval is the thing that moves from experiment to experiment.

Under the hood

The textbook interval and the Wilson interval

$$\bar{x} \pm 1.96 \cdot \frac{s}{\sqrt{n}} \qquad \frac{\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}$$

Left: the textbook interval for a mean. Right: Wilson, for a proportion $\hat{p} = k/n$. Use Wilson for pass rates: the textbook version breaks near 0 and 1, and can hand you an upper bound above 100 percent.

python

```python
import math

def wilson(k: int, n: int, z: float = 1.96):
    """Interval for a pass rate. k passed out of n."""
    if n == 0:
        return (0.0, 1.0)
    p = k / n
    den = 1 + z * z / n                       # shrink factor, -> 1 as n grows

    # The centre is NOT p. It is pulled toward 0.5 by z^2/(2n), and that pull
    # is what keeps the interval inside [0, 1] at extreme rates.
    centre = (p + z * z / (2 * n)) / den

    half = z * math.sqrt(p * (1 - p) / n + z * z / (4 * n * n)) / den
    return (max(0.0, centre - half), min(1.0, centre + half))


def naive(k, n, z=1.96):
    p = k / n
    h = z * math.sqrt(p * (1 - p) / n)
    return (p - h, p + h)                     # note: no clamping

print(wilson(49, 50))    # (0.8950, 0.9965)  stays inside [0, 1]
print(naive(49, 50))     # (0.9412, 1.0188)  claims a rate above 100%
```

rust

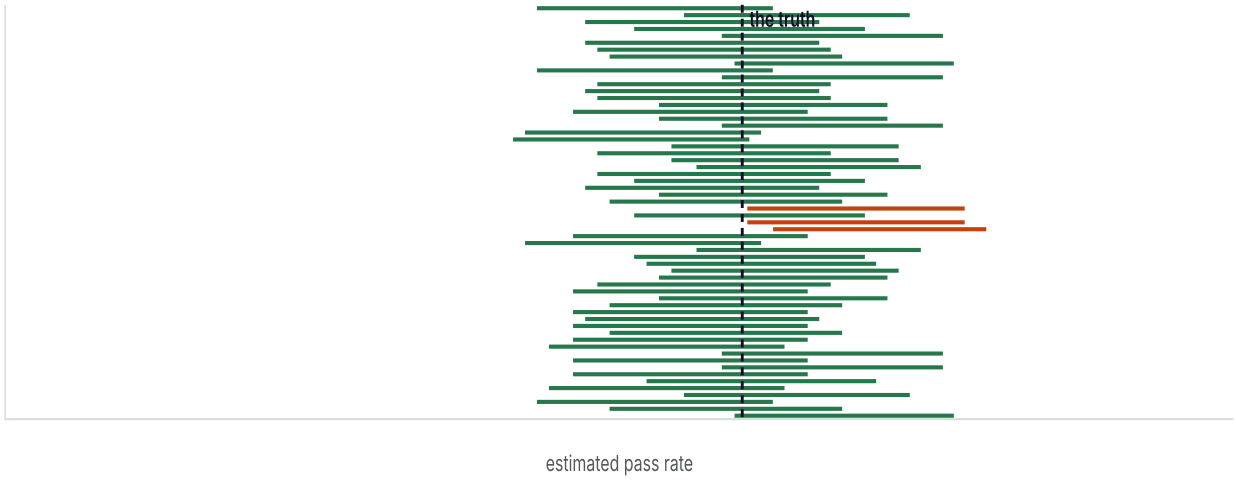
```rust
/// Interval for a pass rate. k passed out of n.
fn wilson(k: u32, n: u32, z: f64) -> (f64, f64) {
    if n == 0 {
        return (0.0, 1.0);
    }
    let (k, n) = (k as f64, n as f64);
    let p = k / n;
    let den = 1.0 + z * z / n; // shrink factor, -> 1 as n grows

    // The centre is NOT p. It is pulled toward 0.5 by z^2/(2n), and that pull
    // is what keeps the interval inside [0, 1] at extreme rates.
    let centre = (p + z * z / (2.0 * n)) / den;
```

```
    let half = z * (p * (1.0 - p) / n + z * z / (4.0 * n * n)).sqrt() / den;
    ((centre - half).max(0.0), (centre + half).min(1.0))
}

fn naive(k: u32, n: u32, z: f64) -> (f64, f64) {
    let p = k as f64 / n as f64;
    let h = z * (p * (1.0 - p) / n as f64).sqrt();
    (p - h, p + h) // note: no clamping
}

fn main() {
    println!("{:?}", wilson(49, 50, 1.96)); // (0.8950, 0.9965)  stays inside [0, 1]
    println!("{:?}", naive(49, 50, 1.96));  // (0.9412, 1.0188)  claims a rate above 100%
}
```

Interactive

Sixty experiments, sixty intervals (shown at its default setting)

Each bar is one experiment on the same underlying truth. Green bars caught it, orange missed. The truth never moves; the bars do. Shrink n and watch the bars get wider but the hit rate stay near 95%.



57 of 60 intervals (95%) contain the true rate. Each orange bar is an experiment that got unlucky.

This is what "95% confident" actually means: the PROCEDURE captures the truth 95 times in 100. It is not a 95% chance that the truth is inside the one interval you happen to be looking at. The truth is fixed; the interval is what moves.

What a p value is, and the four things it is not

A p value answers exactly one question: **if there were genuinely no difference, how often would I see a result at least this extreme by chance alone?** Small p means your result would be surprising in a world with no real effect.

$$p = P\big(T \geq t_{\text{obs}} \;\big|\; H_0 \text{ true}\big)$$

The probability of a result at least this extreme, assuming there is no real effect. Note which side of the bar the assumption sits on.

- It is **not** the probability that the null hypothesis is true.
- It is **not** the probability your result was a fluke.
- It is **not** a measure of effect size. A tiny, useless difference gets a tiny p value with enough data.
- p = 0.06 is **not** "no effect". It is slightly weaker evidence than p = 0.04. The 0.05 line is a convention, not a law of nature.

Key idea

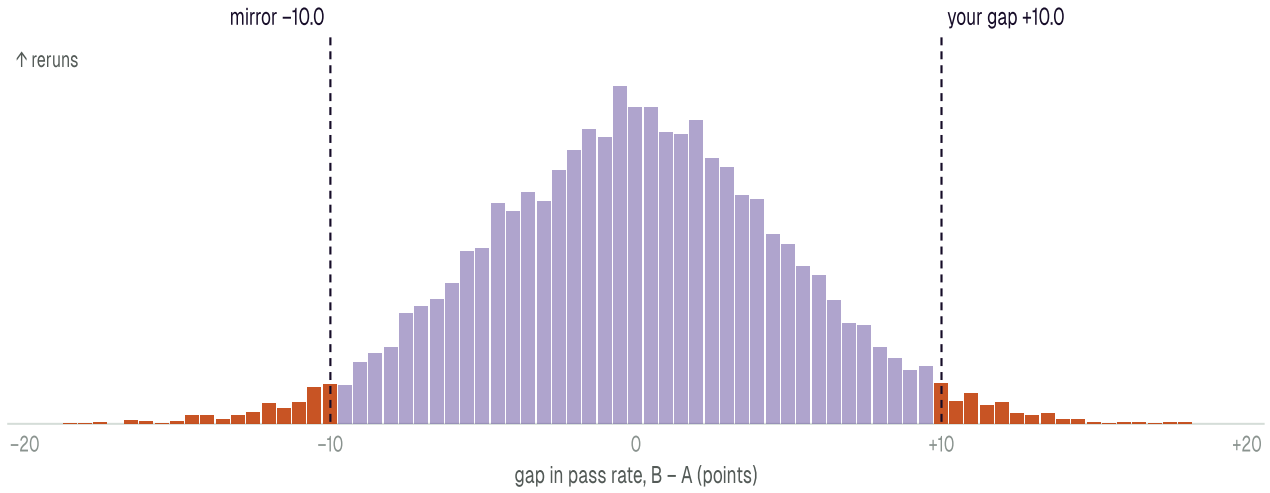
Always report the effect size next to the p value. "Model B is better, p = 0.003" is nearly useless on its own. "Model B passes 4.2 points more, 95% CI [1.1, 7.3], p = 0.003" tells someone whether to ship.

The p value is easiest to understand by building one. Pretend the two models are identical, rerun the eval thousands of times on paper, and count how often chance alone produces a gap as big as yours.

Interactive

Where a p value comes from (shown at its default setting)

Leave the gap at 10 points and drag tasks per model from 200 to 800: the same gap moves from "borderline" to "very surprising under no difference", because the no-difference histogram narrows. Then press "Add a true effect" to see how often a real edge actually reaches p < 0.05.



- gaps from 10,000 reruns where both models truly pass 60%
- reruns at least as extreme as yours, either direction: their share is p

p value (two-sided)	Reruns as extreme	Gap needed for p < 0.05	Power at p < 0.05
0.048	**484 / 10,000**	**±10.0**	**off**

If A and B were truly identical (both pass 60%) and each ran n = 200 tasks, a gap of 10.0 points or more in *either* direction turned up in **484 of 10,000** simulated reruns (orange). That share is the p value: **p ≈ 0.048**. It is **not** the probability that the models are identical, not the chance your result is a fluke, and it says nothing about whether 10.0 points matters. At this n, any gap of ±10.0 points or more would give p < 0.05. Press **Add a true effect** to see how often a real edge clears that bar.

Under the hood

A useless effect becomes "significant" with enough data

python

```python
import numpy as np
from scipy import stats

rng = np.random.default_rng(0)

# A tiny, useless difference. 0.5 percentage points.
def experiment(n):
    a = rng.random(n) < 0.605
    b = rng.random(n) < 0.600
    return stats.fisher_exact([[a.sum(), n - a.sum()],
                               [b.sum(), n - b.sum()]]).pvalue

for n in (200, 2_000, 20_000, 200_000):
    print(n, round(experiment(n), 4))

# The effect never changes. Only n does. With enough data a difference
# nobody could notice becomes "statistically significant".
# THIS is why a p value alone tells you nothing about whether to ship.
```

rust

```rust
// A tiny inline PRNG (SplitMix64) so the snippet needs no crates.
// Its draws differ from numpy's, so the random numbers below differ from the Python's.
struct Rng(u64);

impl Rng {
    fn new(seed: u64) -> Self {
        Rng(seed)
    }
    fn next_u64(&mut self) -> u64 {
        self.0 = self.0.wrapping_add(0x9E37_79B9_7F4A_7C15);
        let mut z = self.0;
        z = (z ^ (z >> 30)).wrapping_mul(0xBF58_476D_1CE4_E5B9);
        z = (z ^ (z >> 27)).wrapping_mul(0x94D0_49BB_1331_11EB);
        z ^ (z >> 31)
    }
    /// Uniform on [0, 1).
    fn random(&mut self) -> f64 {
        (self.next_u64() >> 11) as f64 / (1u64 << 53) as f64
    }
}

/// ln Gamma(x) by the Lanczos approximation (g = 7, 9 terms).
fn ln_gamma(x: f64) -> f64 {
    const G: [f64; 9] = [
        0.999_999_999_999_809_93, 676.520_368_121_885_1, -1259.139_216_722_402_8,
        771.323_428_777_653_13, -176.615_029_162_140_59, 12.507_343_278_686_905,
        -0.138_571_095_265_720_12, 9.984_369_578_019_571_6e-6, 1.505_632_735_149_311_6e-7,
    ];
    let pi = std::f64::consts::PI;
    if x < 0.5 {
        return (pi / (pi * x).sin()).ln() - ln_gamma(1.0 - x); // reflection
    }
    let x = x - 1.0;
    let t = x + 7.5;
    let a = G[0] + (1..9).map(|i| G[i] / (x + i as f64)).sum::<f64>();
    0.5 * (2.0 * pi).ln() + (x + 0.5) * t.ln() - t + a.ln()
}
```

```rust
fn ln_choose(n: u64, k: u64) -> f64 {
    ln_gamma(n as f64 + 1.0) - ln_gamma(k as f64 + 1.0) - ln_gamma((n - k) as f64 + 1.0)
}

/// Two-sided Fisher exact test on [[a, b], [c, d]]: add up every table with the
/// same margins that is no more likely than the one observed.
fn fisher_exact(t: [[u64; 2]; 2]) -> f64 {
    let [[a, b], [c, d]] = t;
    let (row1, col1, n) = (a + b, a + c, a + b + c + d);
    let ln_pmf = |x: u64| ln_choose(row1, x) + ln_choose(n - row1, col1 - x) - ln_choose(n, col1);
    let cutoff = ln_pmf(a) + 1e-7_f64.ln_1p(); // small tolerance for rounding
    let lo = col1.saturating_sub(n - row1);
    let hi = row1.min(col1);
    let p: f64 = (lo..=hi).map(ln_pmf).filter(|&l| l <= cutoff).map(f64::exp).sum();
    p.min(1.0)
}

fn round(x: f64, digits: i32) -> f64 {
    let f = 10f64.powi(digits);
    (x * f).round() / f
}

// A tiny, useless difference. 0.5 percentage points.
fn experiment(rng: &mut Rng, n: u64) -> f64 {
    let a = (0..n).filter(|_| rng.random() < 0.605).count() as u64;
    let b = (0..n).filter(|_| rng.random() < 0.600).count() as u64;
    fisher_exact([[a, n - a], [b, n - b]])
}

fn main() {
    let mut rng = Rng::new(0);
    for n in [200, 2_000, 20_000, 200_000] {
        println!("{} {}", n, round(experiment(&mut rng, n), 4));
    }
    // 200 0.2518 | 2000 0.3487 | 20000 0.0521 | 200000 0.0029  (SplitMix64 draws;
    // numpy's give 0.6869, 0.1832, 0.2365, 0.0001). Small samples wander at random;
    // at 200,000 per arm a 0.5 point gap is about 3 standard errors, so p < 0.05
    // is the typical outcome.

    // The effect never changes. Only n does. With enough data a difference
    // nobody could notice becomes "statistically significant".
    // THIS is why a p value alone tells you nothing about whether to ship.
}
```

For leaders

What these two ideas let you demand

The sampling distribution is the reason an eval number is never just a number. It comes with a margin, and that margin decides whether a result can carry a decision.

- **No interval, no decision.** A pass rate without a confidence interval cannot support a ship call. On 200 tasks at around 60%, the 95% interval is roughly ±7 points. Two models 3 points apart on

that suite are usually indistinguishable unless the comparison is paired (Part II covers how pairing shrinks the margin).

- **Budget with the square-root law.** Halving the error bar costs four times the eval compute and four times the labelling. Detecting a 1 point change typically needs thousands of tasks. Put that in planning documents before someone promises "we'll know by Friday".
- **Separate "real" from "matters".** A p value says whether a difference is likely real. It says nothing about whether it is big enough to care about. Ask for the effect size and its interval, and agree in advance on the smallest difference worth acting on.
- **Watch for SD and SE swaps.** Error bars drawn with the standard deviation look alarmingly wide; error bars drawn with the standard error look reassuringly tight. Both are legitimate for different questions, and dashboards frequently mix them up.

Principal asks

"If we reran this eval on a fresh draw of tasks, how much would the headline move?" A team that can answer with a number, and knows where it came from, understands its own result. A team that cannot has a point estimate and a hope.

Part I · Chapter 5

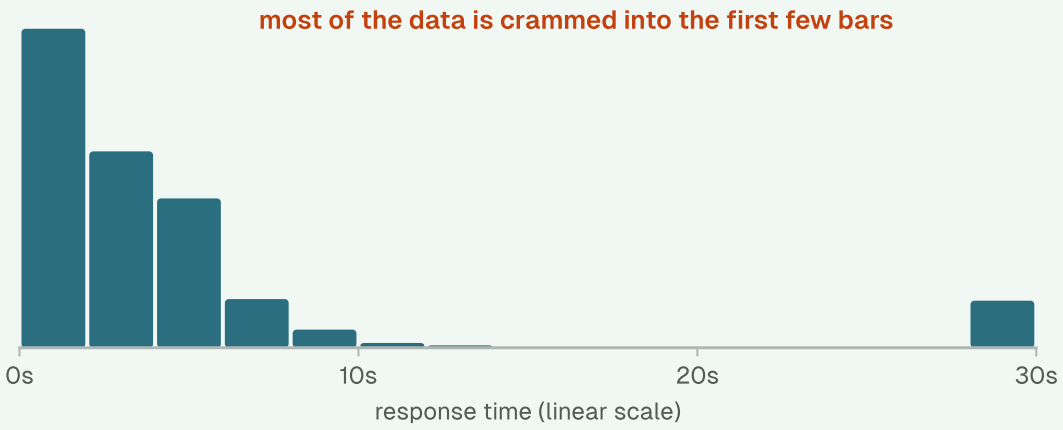
Reading a shape: work out what you have before choosing a method

This is the practical skill: look at data and work out what you have. The checklist takes about two minutes and catches nearly everything that would otherwise invalidate the analysis that follows.

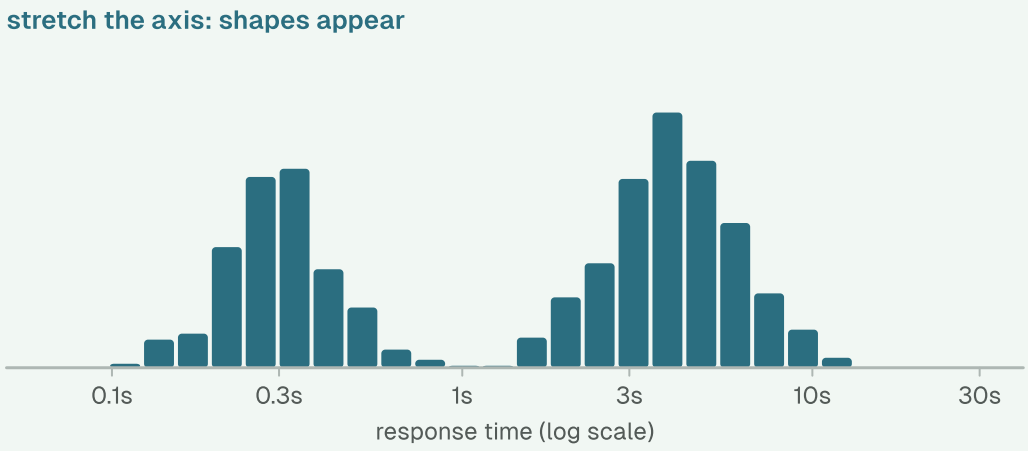
Diagram

A two-minute checkup for a chatbot's response times

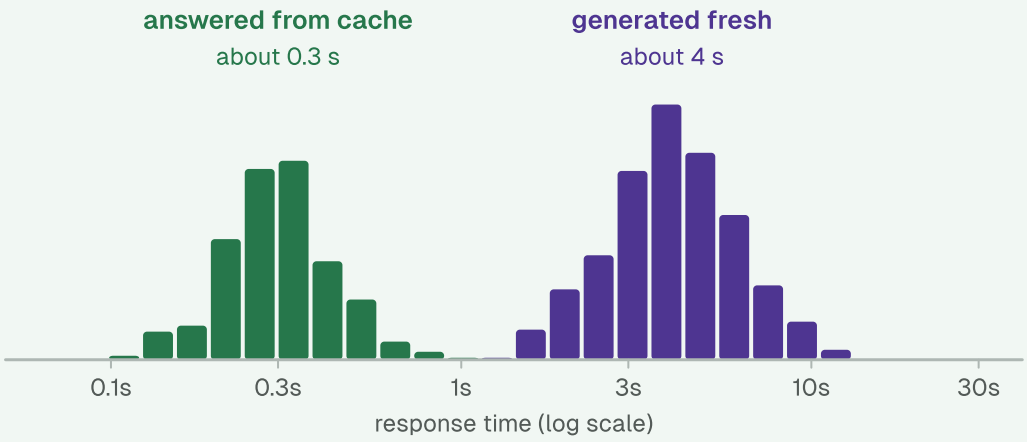
Four looks that tell you what kind of data you have before you summarise it.



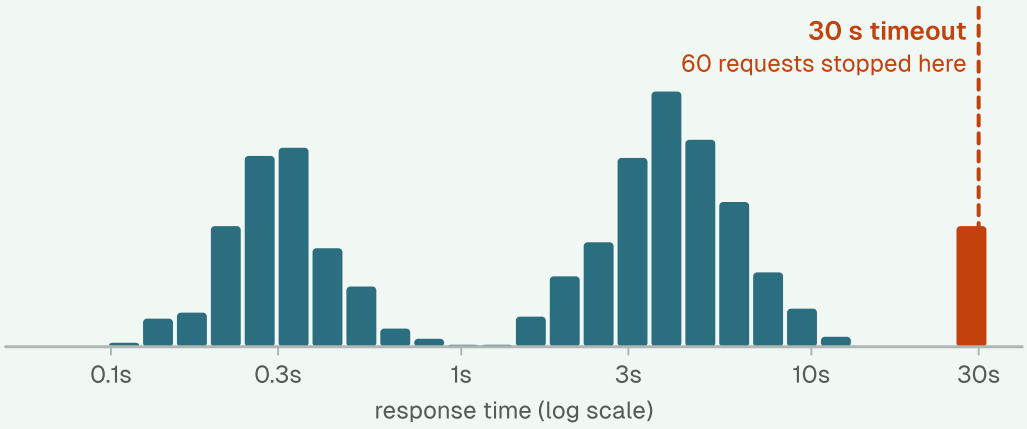
A thousand response times on a normal axis. A few tall bars on the left and a long empty stretch. You cannot see anything useful, which is itself a clue: the data is heavily skewed.



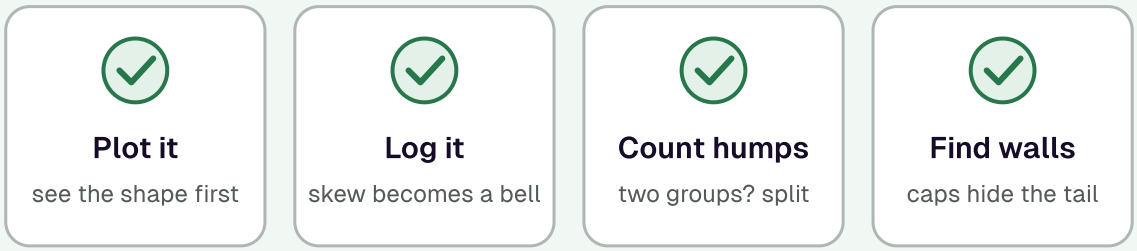
Put time on a log scale, where each tick is about three times the last. Skewed timing data usually turns into tidy bumps. If it looks like a bell on a log axis, it is roughly lognormal.



Here there are two bumps: answers served from a cache in about 0.3 seconds, and answers generated fresh in about 4 seconds. Two humps mean two populations. Summarise each separately, or the average describes nobody.



Last look: a spike at a wall. The 60 requests that hit the 30 second timeout were recorded as exactly 30. The real tail is longer than the data shows. That is **censoring**.



Two minutes of looking, before any statistics.

The checkup in four moves. Each one changes which summary and which test are honest, which is what the rest of this chapter and the lookup table are for.

Plain words

Look before you measure

A doctor looks at a patient before prescribing. A summary statistic is a prescription. Before you choose one, look at the data: draw a picture, check whether it leans, check whether there are two groups hiding in it, and check whether something chopped off the top. Most mistakes come from skipping the look.

The two-minute checklist

1. **Plot a histogram.** Always. Before any number. Summary statistics hide bimodality, censoring and outliers with total reliability.
2. **Divide mean by median.** Above 1.2 is right skewed, below 0.85 is left skewed.
3. **Divide variance by mean** for counts. Near 1 means Poisson is plausible. Well above 1 means overdispersed.
4. **Check whether values are whole numbers**, and whether they are bounded (at 0, at 1, at n).
5. **Try a log scale.** If the shape turns symmetric, you have something lognormal.
6. **Look for a spike at a round number.** That is censoring, not data.
7. **Look for two humps.** That is two populations.

$$\frac{s^2}{\bar{x}} \approx 1 \;\Rightarrow\; \text{Poisson;} \qquad \frac{s^2}{\bar{x}} \gg 1 \;\Rightarrow\; \text{overdispersed}$$

The dispersion check for count data: variance divided by mean.

Under the hood

The checklist as a function

python

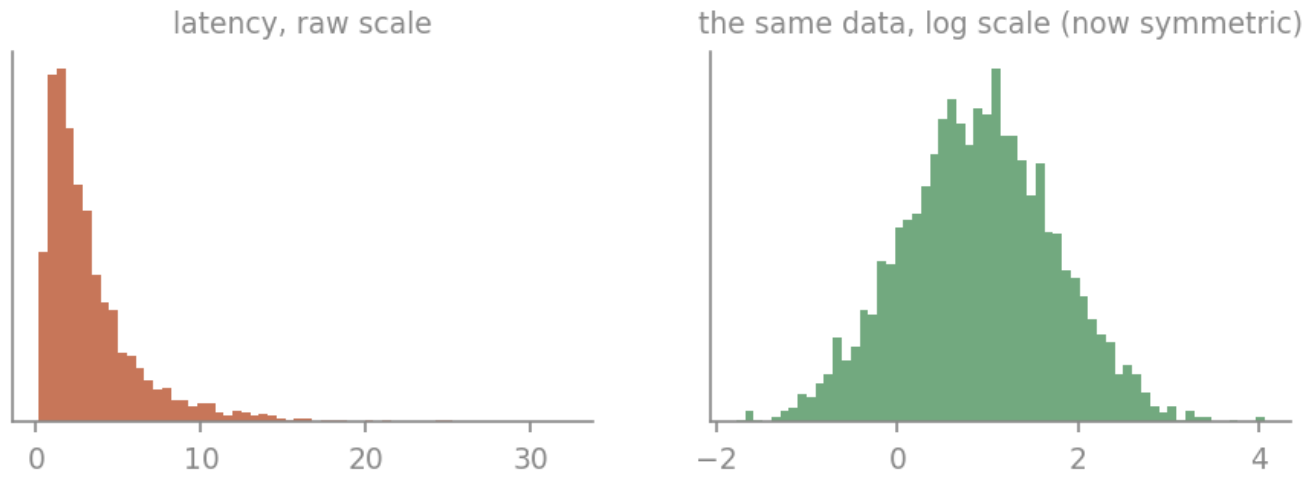
```python
import numpy as np
from scipy import stats

def diagnose(x):
    """Run this before choosing any method. Takes two seconds, catches most errors."""
    x = np.asarray(x, dtype=float)
    n = len(x)
    m, med = x.mean(), np.median(x)
    var = x.var(ddof=1)

    out = {
        "n": n,
        "mean": m,
        "median": med,
        "skew_ratio": m / med if med else float("inf"),   # >1.2 = right skewed
        "var_over_mean": var / m if m else float("inf"),  # ~1 = Poisson-like
        "whole_numbers": bool(np.all(np.abs(x - np.round(x)) < 1e-9)),
        "min": x.min(), "max": x.max(),
        "zeros": int((x == 0).sum()),                     # blocks a log transform
    }

    # Does a log transform fix the skew? Only meaningful for positive data.
    if x.min() > 0:
        out["skew_raw"] = float(stats.skew(x))
        out["skew_logged"] = float(stats.skew(np.log(x)))
        out["log_helps"] = abs(out["skew_logged"]) < abs(out["skew_raw"]) / 2

    # CENSORING CHECK: is there a pile-up at a single value near the max?
    top = x[x >= np.quantile(x, 0.98)]
    if len(top):
        common = stats.mode(np.round(top, 6), keepdims=False)
        out["censoring_suspect"] = bool(common.count > 0.01 * n)
        out["suspect_value"] = float(common.mode)

    return out


lat = stats.lognorm.rvs(0.9, scale=3, size=4000, random_state=0)
for k, v in diagnose(lat).items():
    print(k, v)
# skew_ratio 1.49 -> right skewed, do not report the mean
# log_helps True  -> it is lognormal, analyse on the log scale
```

rust

```rust
// A tiny inline PRNG (SplitMix64) so the snippet needs no crates.
// Its draws differ from numpy's, so the random numbers below differ from the Python's.
struct Rng(u64);

impl Rng {
    fn new(seed: u64) -> Self {
        Rng(seed)
    }
    fn next_u64(&mut self) -> u64 {
        self.0 = self.0.wrapping_add(0x9E37_79B9_7F4A_7C15);
        let mut z = self.0;
        z = (z ^ (z >> 30)).wrapping_mul(0xBF58_476D_1CE4_E5B9);
        z = (z ^ (z >> 27)).wrapping_mul(0x94D0_49BB_1331_11EB);
        z ^ (z >> 31)
    }
    /// Uniform on [0, 1).
    fn random(&mut self) -> f64 {
        (self.next_u64() >> 11) as f64 / (1u64 << 53) as f64
    }
    /// Normal draw by Box-Muller.
    fn normal(&mut self, mean: f64, sd: f64) -> f64 {
        let u1 = 1.0 - self.random(); // in (0, 1], safe for ln
        let u2 = self.random();
        mean + sd * (-2.0 * u1.ln()).sqrt() * (std::f64::consts::TAU * u2).cos()
    }
}

fn mean(x: &[f64]) -> f64 {
    x.iter().sum::<f64>() / x.len() as f64
}

fn sorted(x: &[f64]) -> Vec<f64> {
    let mut s = x.to_vec();
    s.sort_by(|a, b| a.total_cmp(b));
    s
}

/// Linear-interpolation quantile (numpy's default method).
fn quantile(x: &[f64], q: f64) -> f64 {
    let s = sorted(x);
    let h = (s.len() - 1) as f64 * q;
    let lo = h.floor() as usize;
    let hi = (lo + 1).min(s.len() - 1);
    let (a, b, t) = (s[lo], s[hi], h - lo as f64);
    if t >= 0.5 { b - (b - a) * (1.0 - t) } else { a + (b - a) * t }
}

/// Sample skewness, as scipy.stats.skew computes it: m3 / m2^1.5.
fn skew(x: &[f64]) -> f64 {
    let m = mean(x);
    let m2 = x.iter().map(|v| (v - m).powi(2)).sum::<f64>() / x.len() as f64;
    let m3 = x.iter().map(|v| (v - m).powi(3)).sum::<f64>() / x.len() as f64;
    m3 / m2.powf(1.5)
}

/// Run this before choosing any method. Takes two seconds, catches most errors.
fn diagnose(x: &[f64]) -> Vec<(&'static str, String)> {
    let n = x.len();
```

```rust
    let (m, med) = (mean(x), quantile(x, 0.5));
    let var = x.iter().map(|v| (v - m).powi(2)).sum::<f64>() / (n as f64 - 1.0);
    let min = x.iter().cloned().fold(f64::INFINITY, f64::min);
    let max = x.iter().cloned().fold(f64::NEG_INFINITY, f64::max);

    let mut out: Vec<(&'static str, String)> = vec![
        ("n", n.to_string()),
        ("mean", m.to_string()),
        ("median", med.to_string()),
        ("skew_ratio", (if med != 0.0 { m / med } else { f64::INFINITY }).to_string()), // >1.2 = right skewed
        ("var_over_mean", (if m != 0.0 { var / m } else { f64::INFINITY }).to_string()), // ~1 = Poisson-like
        ("whole_numbers", x.iter().all(|v| (v - v.round()).abs() < 1e-9).to_string()),
        ("min", min.to_string()),
        ("max", max.to_string()),
        ("zeros", x.iter().filter(|&&v| v == 0.0).count().to_string()), // blocks a log transform
    ];

    // Does a log transform fix the skew? Only meaningful for positive data.
    if min > 0.0 {
        let logs: Vec<f64> = x.iter().map(|v| v.ln()).collect();
        let (skew_raw, skew_logged) = (skew(x), skew(&logs));
        out.push(("skew_raw", skew_raw.to_string()));
        out.push(("skew_logged", skew_logged.to_string()));
        out.push(("log_helps", (skew_logged.abs() < skew_raw.abs() / 2.0).to_string()));
    }

    // CENSORING CHECK: is there a pile-up at a single value near the max?
    let q98 = quantile(x, 0.98);
    let top: Vec<f64> = x.iter().filter(|&&v| v >= q98).map(|v| (v * 1e6).round() / 1e6).collect();
    if !top.is_empty() {
        // the most common value; ties go to the smallest, as scipy.stats.mode does
        let top = sorted(&top);
        let (mut mode, mut count) = (top[0], 0);
        for run in top.chunk_by(|a, b| a == b) {
            if run.len() > count {
                (mode, count) = (run[0], run.len());
            }
        }
        out.push(("censoring_suspect", (count as f64 > 0.01 * n as f64).to_string()));
        out.push(("suspect_value", mode.to_string()));
    }

    out
}

fn main() {
    // lognormal(sigma 0.9, median 3): exp of a normal
    let mut rng = Rng::new(0);
    let lat: Vec<f64> = (0..4000).map(|_| 3.0 * rng.normal(0.0, 0.9).exp()).collect();
    for (k, v) in diagnose(&lat) {
        println!("{} {}", k, v);
    }
    // skew_ratio 1.47 -> right skewed, do not report the mean
    // log_helps true -> it is lognormal, analyse on the log scale
}
```

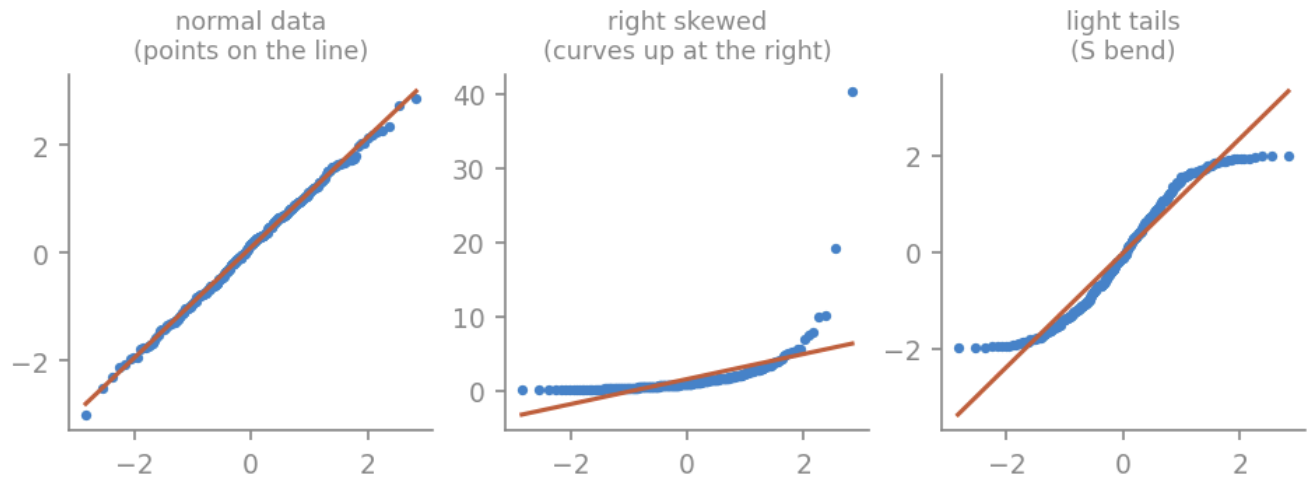
The log scale check



Same data, two scales. If the log version looks symmetric, the raw version is lognormal, and the log scale is where the analysis should happen.

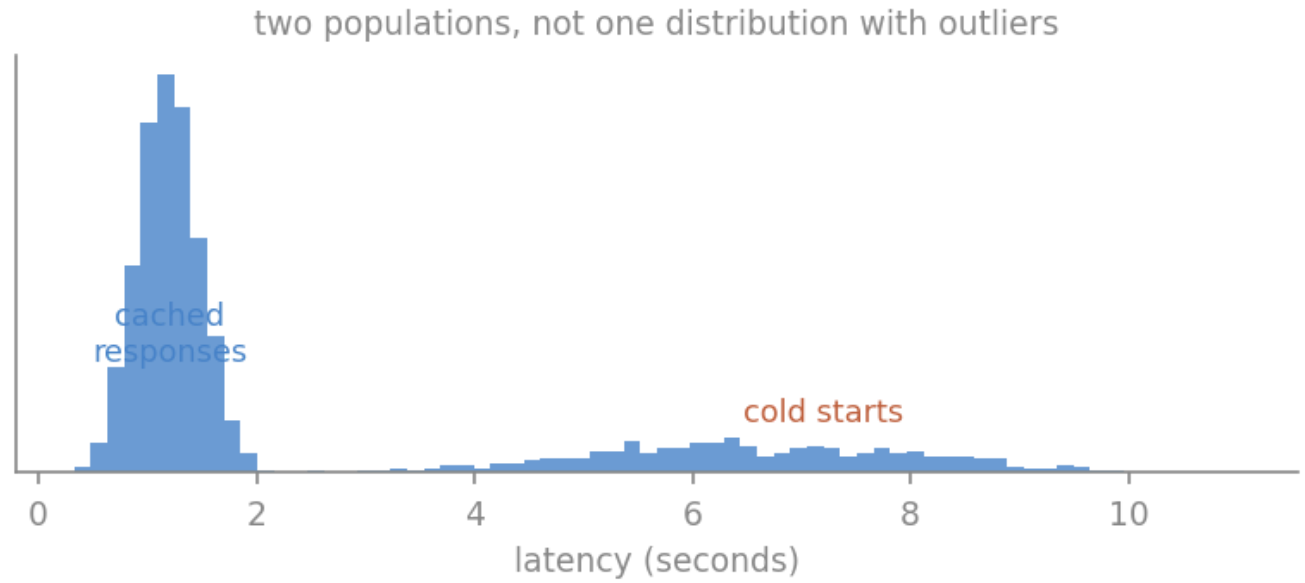
QQ plots, once you want more than eyeballing

A QQ plot compares your data's quantiles against a reference distribution's. A straight line means a good match. The way it bends tells you how it differs.



Left: normal data sits on the line. Middle: right skew curves up at the right end. Right: light tails make an S bend.

Two humps means two populations



This is not one distribution with outliers. It is cached responses and cold starts sharing an axis.

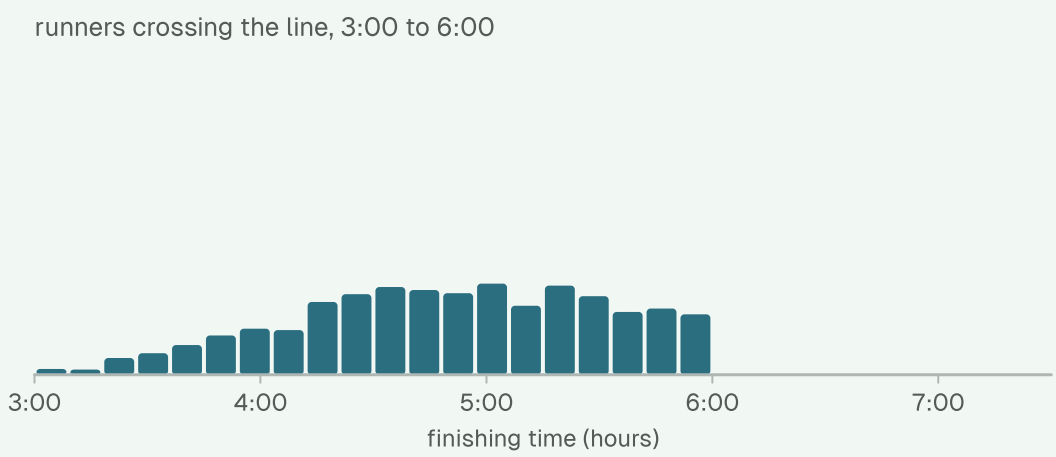
Eval lens

When you see this, **stop reaching for statistics and go find the segmenting variable**. Split by cache hit, retry status, repo size, task category. Two clean distributions beat one messy one every time, and the split is usually the actual finding.

Censoring: the one that hides

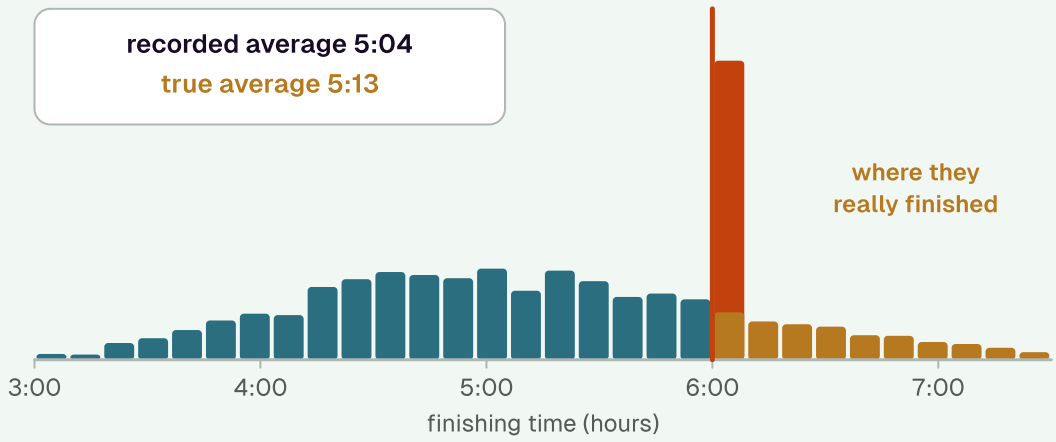
Diagram **The marathon that closed at six hours**

Censoring: when the data stops before the story does.

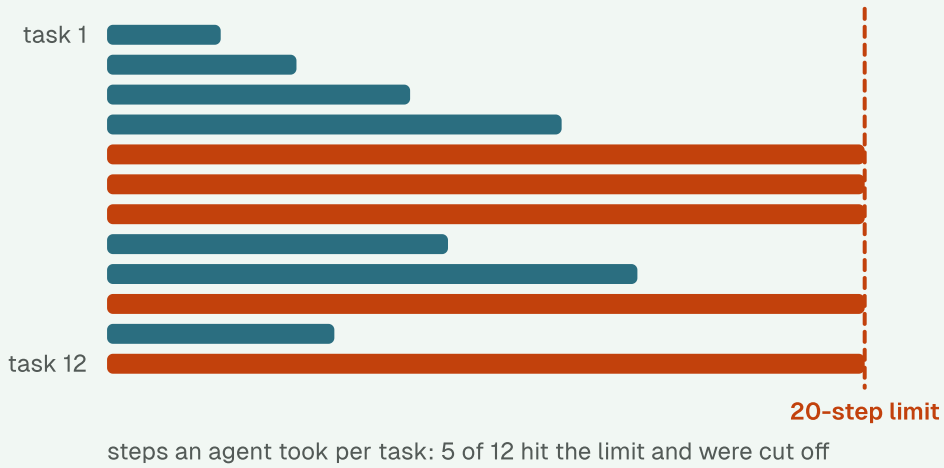


Finishing times for 3,000 marathon runners. Most finish around five hours. So far, an ordinary bell-shaped pile.

But the course closes at 6:00. The 21% of runners still out there get written down as "6:00". All of them pile into one spike at the wall.

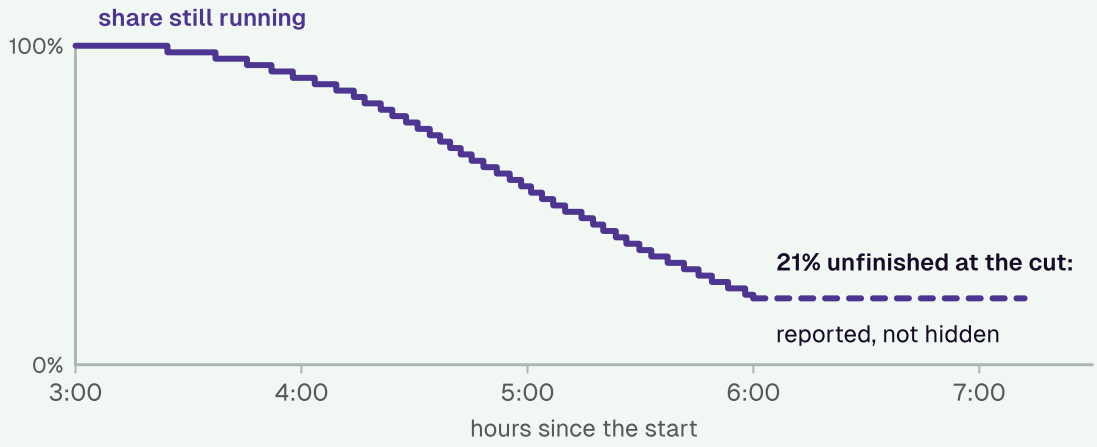


Their real times were later, so the recorded average (5:04) is too optimistic; the true average is 5:13. Dropping them would be worse: you would be averaging only the faster runners.

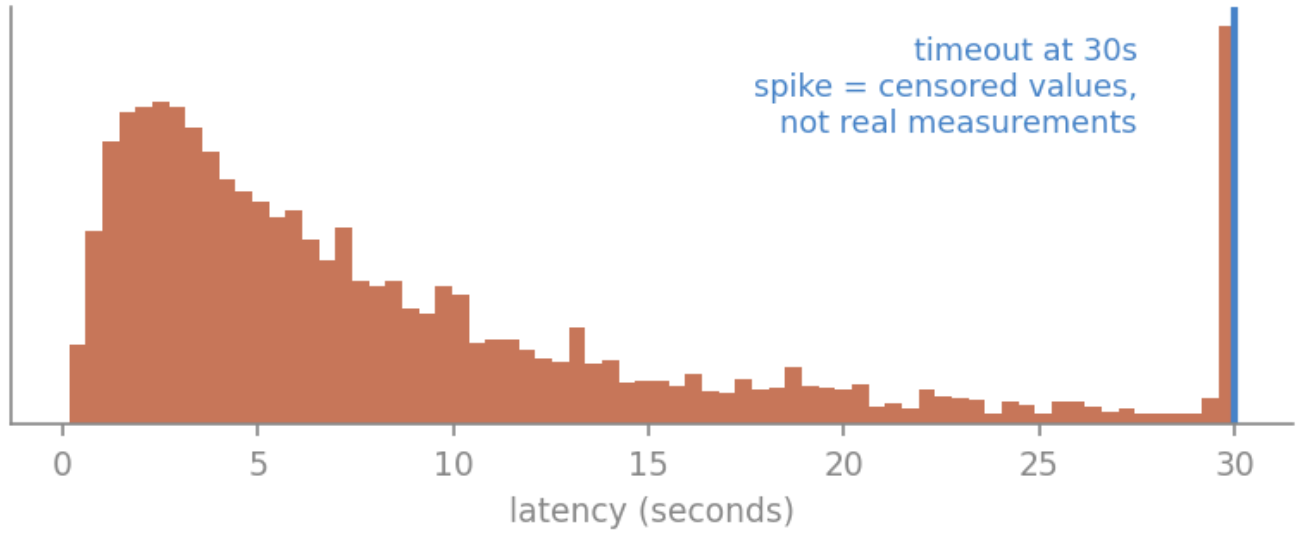


steps an agent took per task: 5 of 12 hit the limit and were cut off

Agent evals have the same wall. A task stopped at a step limit or timeout is recorded at the cap. Average "steps to solve" then looks better than reality, and the hardest tasks vanish into the spike.



The fix: say how many hit the wall, and summarise with a survival curve (Kaplan-Meier) that treats "still running at the cut" as information, not as a finish time.



A spike sitting exactly at your timeout is not a measurement. Those requests were 30 seconds or more and you will never know which.

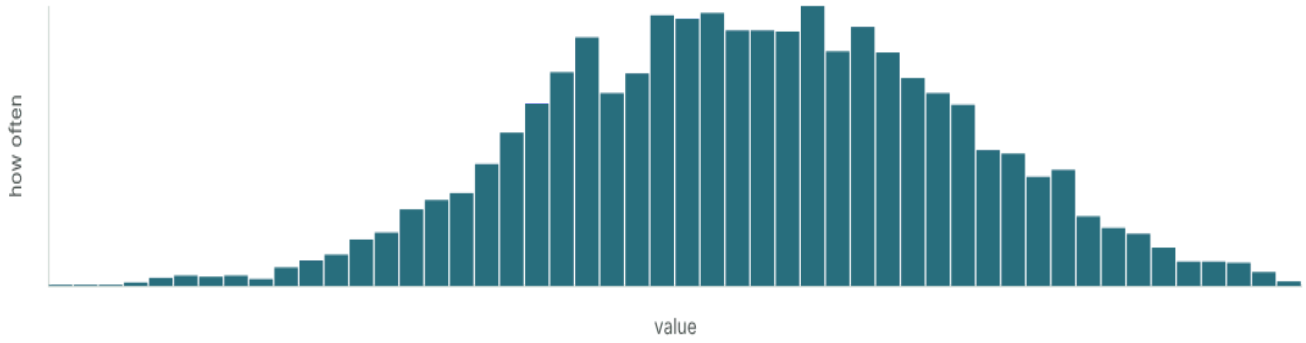
Watch out

Timeouts do not create outliers, they create **censored data**. Every value at the cap is really "at least this much". Your mean, p95 and p99 are all biased downward by an amount you cannot see from the data. Report the censoring rate next to any percentile, or use survival methods. Check for this before reaching for any outlier technique.

Interactive

Name that distribution (shown at its default setting)

A random sample with its diagnostics. Work through the checklist above, make a guess, and get the reasoning either way. Do ten of these and the skill sticks.



- Lognormal
- Normal
- Poisson
- Negative binomial
- Exponential
- Binomial
- Mixture (two populations)
- Uniform
- Pareto (power law)
- Geometric

Diagnostics: mean 5.03 median 5.01 mean/median 1.00 variance/mean 0.44 min -1.10 max 10.52 whole numbers: no

For leaders

Make "show me the histogram" a habit

The cheapest review step in evaluation is asking to see the raw distribution before anyone argues about the summary. It surfaces three failure modes that no amount of downstream rigour can fix.

Hidden segments

Two humps mean two populations. A change that helps one and hurts the other can net to zero on average. The useful question becomes "which segment moved?"

Silent censoring

A spike at a timeout or token cap means the worst cases were truncated. A model that times out more often can look faster. Insist the censoring rate is reported beside every percentile.

Wrong scale

Latency, cost and token counts are multiplicative. Comparing them on a linear scale overweights the tail. Ratios and log-scale comparisons are usually the honest framing.

Principal asks

"What fraction of runs hit the timeout or the max-token limit, before and after?" If that fraction changed, every latency and length metric in the report is comparing different things, and the headline needs to be redone.

Part I · Chapter 6

The distribution zoo: seventeen shapes and the stories that make them

Seventeen distributions. For each one: the picture, the story that generates it, what it looks like in eval work, and when not to use it. The generating story is the most useful part. If you can match the story, you have found your distribution.

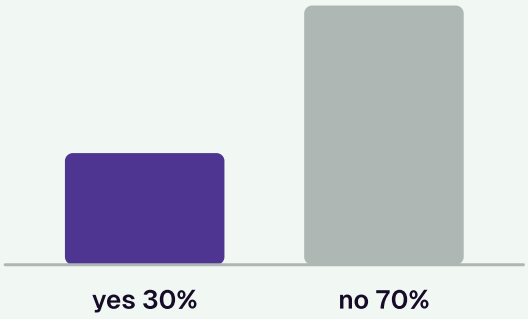
Diagram

One café, six distributions

Every shape in the zoo comes from a story about how the numbers were made. One morning at a café tells six of them.

Bernoulli

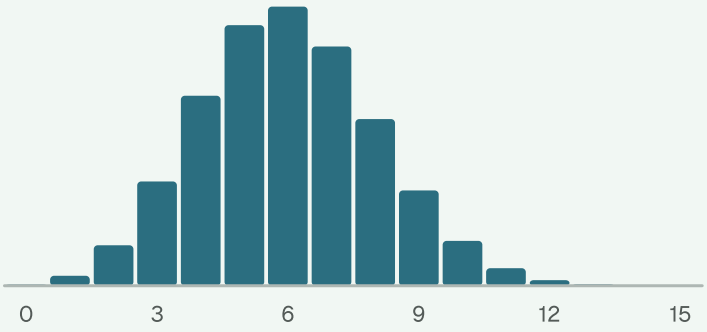
One customer:
pastry, yes or no?



One customer either buys a pastry or does not. A single yes/no with a fixed chance (here 30%) is a **Bernoulli**. In evals: one task, pass or fail.

Binomial

Out of 20 customers,
how many buy one?

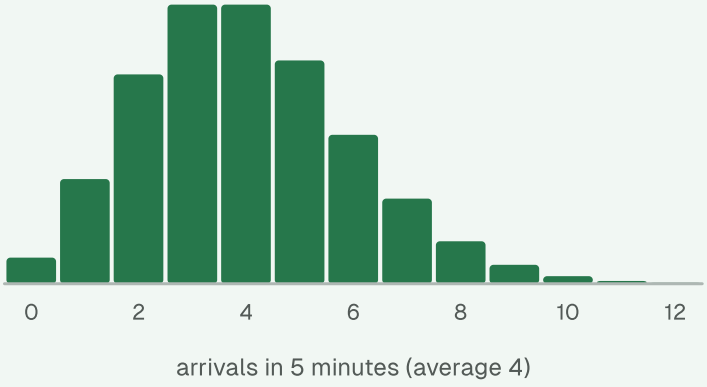


pastries sold per 20 customers (peak at 6)

Count the yeses among 20 customers and you get a **binomial**: usually around 6, sometimes 2, sometimes 11. In evals: passes out of n tasks.

Poisson

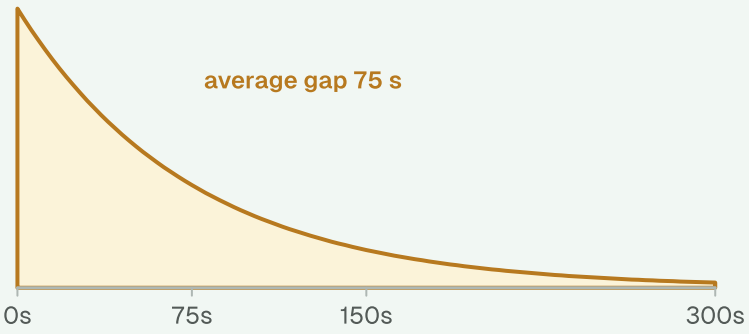
How many customers walk in per 5 minutes?



Count how many people walk in during any 5 minutes, when arrivals are independent and the average is 4: that count is **Poisson**. In evals: errors per thousand requests.

Exponential

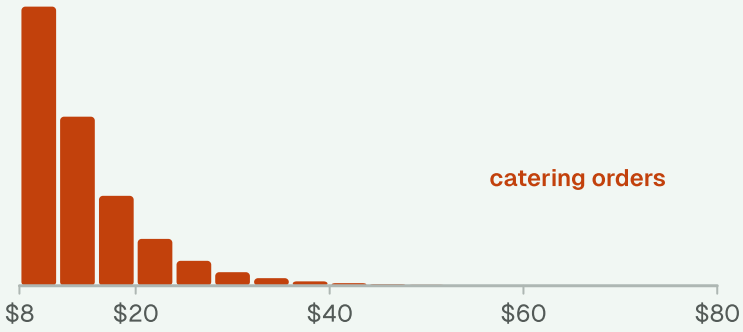
Seconds between one arrival and the next



Measure the gap between one arrival and the next: lots of short gaps, a few long ones, falling away smoothly. That is **exponential**, with an average of 75 seconds here. In evals: time between failures.

Lognormal

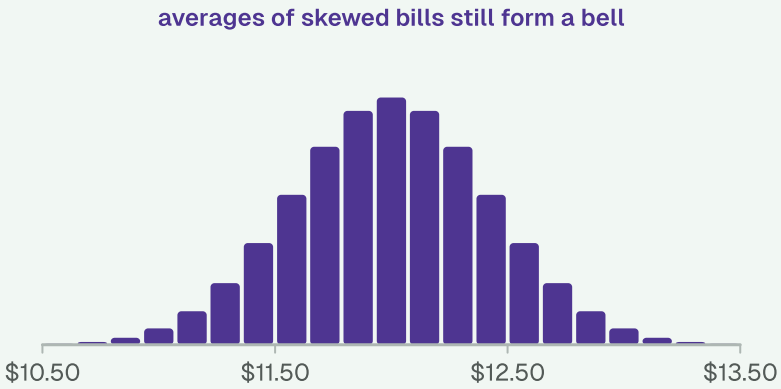
How big is each bill?
mostly $8–15, a few $70



Bill sizes multiply out of several factors (people, items, extras), which makes a right-skewed **lognormal**: mostly $8 to $15, occasionally a $70 catering order. In evals: latency and token counts.

Normal

Average bill per day,
over 100 days



Average each day's bills and plot 100 days of averages: a symmetric bell, the **normal**, even though single bills were skewed. That is the central limit theorem from the previous chapter, and why intervals on averages work.

Plain words

Match the story, find the shape

Each distribution is the fingerprint of a process. Flip one coin and you get a Bernoulli. Count heads in 200 flips and you get a binomial. Add up many small nudges and you get a bell curve. Multiply many small nudges, like a slow network times a big payload times a cold cache, and you get a lopsided

lognormal. You do not need to memorise formulas. You need to ask "what process made these numbers?" and match it to a story below.

For leaders

Why the choice of shape changes decisions

Picking a distribution sounds academic until it changes a ship decision. The common failure is assuming a simple shape when the real process is messier, which makes uncertainty look smaller than it is.

- **Assuming independence that is not there.** Treating tool calls inside one session as independent coin flips (binomial) understates uncertainty, often by a factor of two or more. Results look significant that are not.
- **Assuming Poisson for agent counts.** Tool calls, turns and retries vary with task difficulty, so variance runs several times the mean. A Poisson model will flag a routine long session as a one-in-a-million event and send engineers chasing noise.
- **Reporting a mean for a heavy tail.** For lognormal data the mean describes nobody; for a genuine power law it may not even settle as data grows. Percentiles are the defensible summary.
- **Fitting one shape to a mixture.** Two populations on one axis need to be split, not modelled.

Principal asks

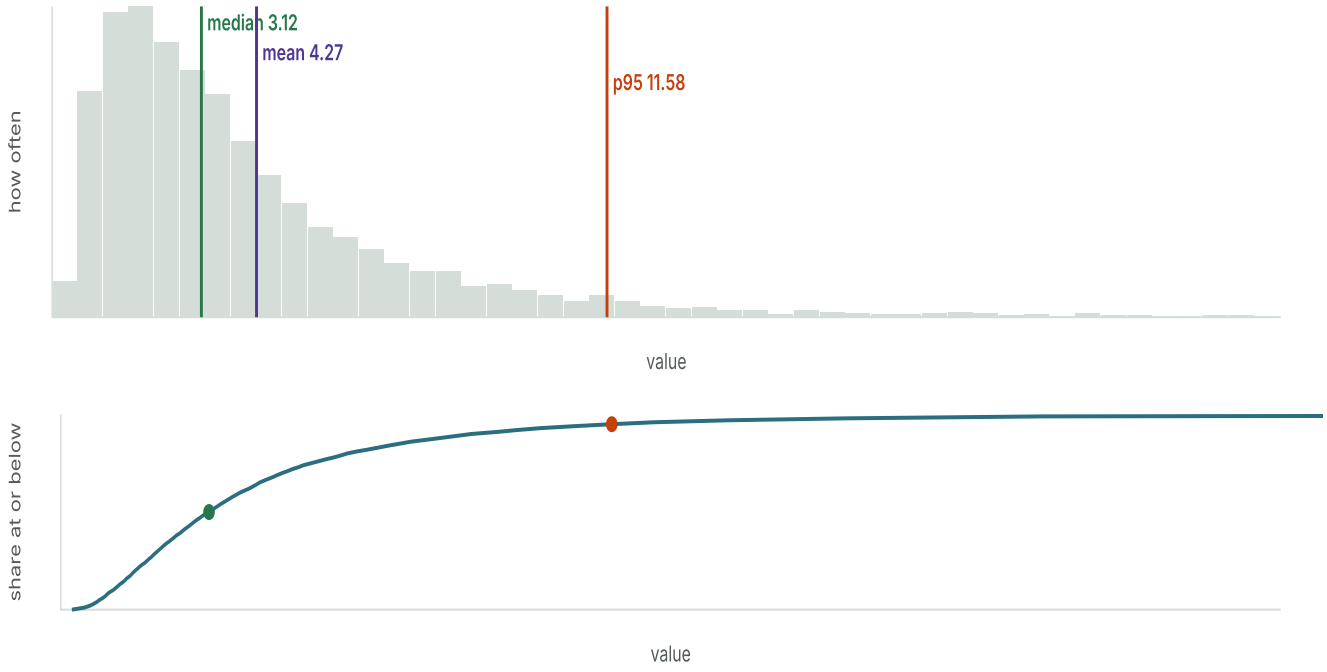
"What is the variance-to-mean ratio of that count metric, and what model did the interval assume?" If the answer is Poisson and the ratio is 4, the error bars on the dashboard are about half as wide as they should be.

Quick reference

Distribution	The story	Typical eval quantity
Bernoulli	One yes/no trial	One task result
Binomial	Successes in n independent trials	Tasks passed out of 200
Uniform	Every value equally likely	p values under a true null
Normal	Sum of many small independent effects	Averages, test statistics
Student's t	Normal mean with estimated spread	Small-sample means
Lognormal	Product of many small effects	Latency, token counts
Exponential	Waiting time, constant rate	Time between failures
Gamma	Sum of several waits	Multi-stage pipeline latency
Poisson	Events in a window, constant rate	Requests per minute
Negative binomial	Counts when the rate varies	Tool calls per session
Geometric	Attempts until first success	Retry counts
Beta	Belief about a rate	Uncertainty about a pass rate
Chi square	Sum of squared normal deviations	Test statistics
F	Ratio of two variances	ANOVA
Pareto / power law	Scale free, no typical size	Repo and file sizes
Categorical / multinomial	One draw picks one class	Intent labels, failure codes
Mixture	Several populations on one axis	Cache hits and cold starts

Interactive **Distribution explorer** (shown at its default setting)

Every distribution below, live. The top panel is the density, the bottom is the cumulative view with the median and p95 marked. Move the parameters and watch the mean detach from the median as skew grows.

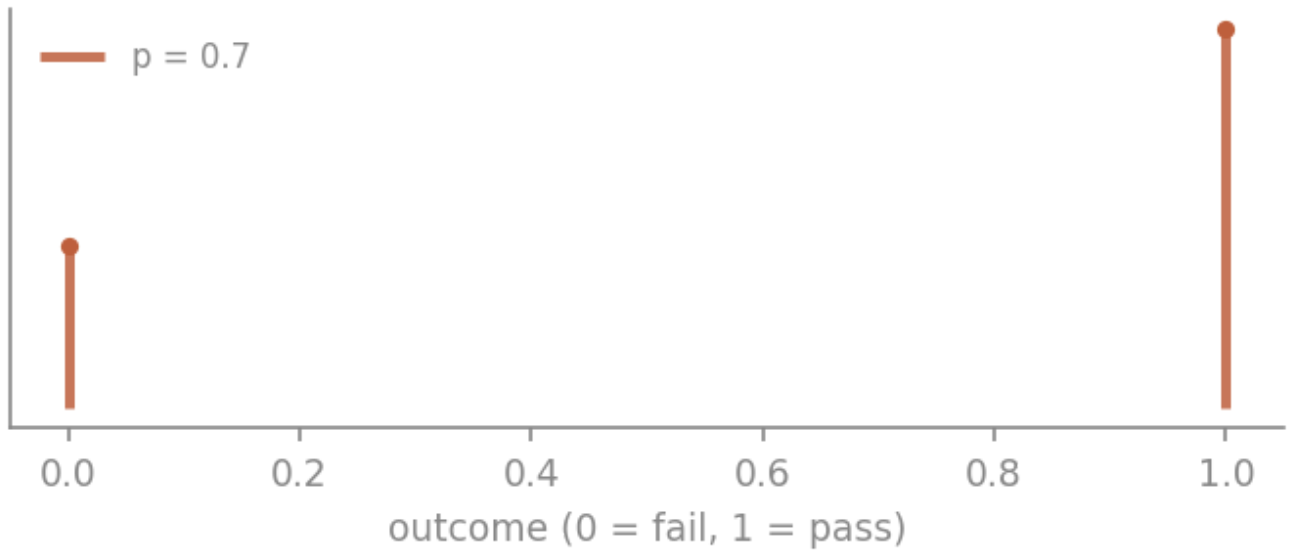


mean 4.27 **median** 3.12 **sd** 4.02 **p95** 11.58 **max** 49.22

Right skewed. mean/median = 1.37. Report the median and percentiles, not the mean.

Bernoulli

One yes/no trial.



Bernoulli: one yes/no trial.

The story	A single task either passes or fails. That is it. The building block everything else in pass/fail land is made from.
In eval work	One task result. One tool call succeeding. One thumbs up.
Use when	Use for a single binary outcome. It has one parameter, p. Its variance is p(1-p), which is largest at p = 0.5 and shrinks to nothing at the extremes, which is why rates near 0 or 1 are easier to pin down.
Avoid when	There is nothing to misuse. The mistakes come one level up, in how you aggregate them.

Under the hood

Bernoulli: formulas and code

$$P(X = k) = p^k (1-p)^{1-k}, \quad k \in \{0, 1\}$$

$$E[X] = p \qquad \mathrm{Var}(X) = p(1-p)$$

python

```python
import random

# GENERATE: one task result. This is the atom everything else is built from.
def one_task(p_pass: float, rng) -> int:
    return 1 if rng.random() < p_pass else 0        # 1 = passed, 0 = failed

rng = random.Random(0)
results = [one_task(0.7, rng) for _ in range(200)]  # 200 independent tasks

# SUMMARISE
n = len(results)
k = sum(results)                                    # how many passed
print(k / n)                                        # the pass rate, e.g. 0.695

# NOTE the variance: p(1-p) is largest at p = 0.5 and shrinks toward the edges.
# That is WHY a model at 50% needs more tasks to pin down than one at 95%.
for p in (0.5, 0.7, 0.95):
    print(p, "variance per task:", round(p * (1 - p), 4))
```

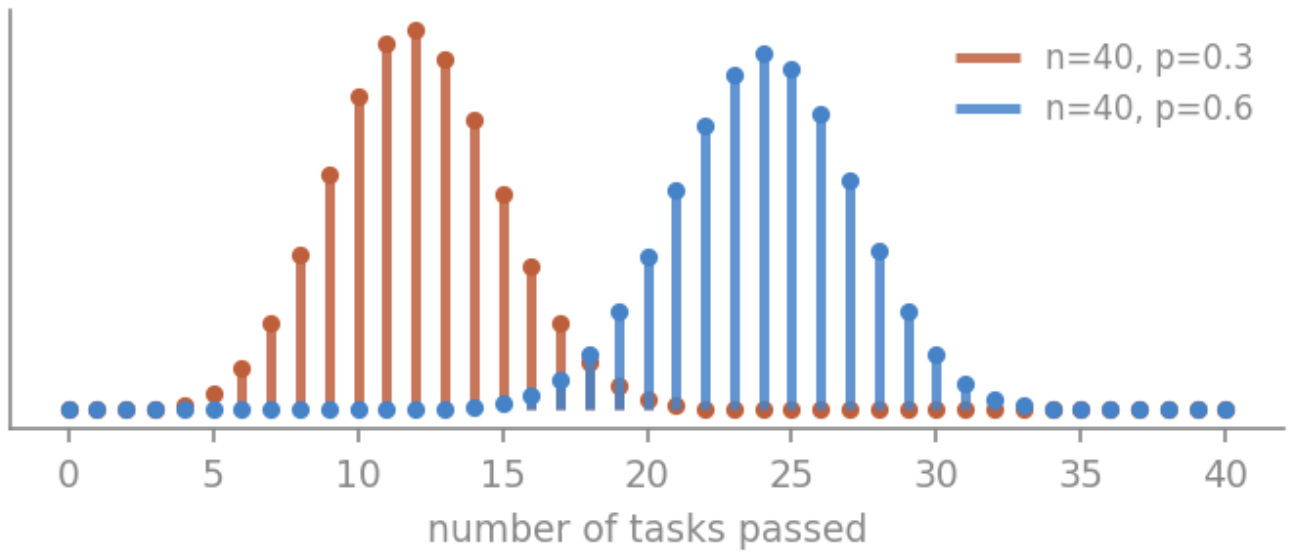
rust

```rust
// A tiny inline PRNG (SplitMix64) so the snippet needs no crates.
// Its draws differ from numpy's, so the random numbers below differ from the Python's.
struct Rng(u64);

impl Rng {
    fn new(seed: u64) -> Self {
        Rng(seed)
    }
    fn next_u64(&mut self) -> u64 {
        self.0 = self.0.wrapping_add(0x9E37_79B9_7F4A_7C15);
        let mut z = self.0;
        z = (z ^ (z >> 30)).wrapping_mul(0xBF58_476D_1CE4_E5B9);
        z = (z ^ (z >> 27)).wrapping_mul(0x94D0_49BB_1331_11EB);
        z ^ (z >> 31)
    }
    /// Uniform on [0, 1).
    fn random(&mut self) -> f64 {
        (self.next_u64() >> 11) as f64 / (1u64 << 53) as f64
    }
}

fn round(x: f64, digits: i32) -> f64 {
    let f = 10f64.powi(digits);
    (x * f).round() / f
}

// GENERATE: one task result. This is the atom everything else is built from.
fn one_task(p_pass: f64, rng: &mut Rng) -> u32 {
    if rng.random() < p_pass { 1 } else { 0 } // 1 = passed, 0 = failed
}

fn main() {
    let mut rng = Rng::new(0);
    let results: Vec<u32> = (0..200).map(|_| one_task(0.7, &mut rng)).collect(); // 200 independent tasks

    // SUMMARISE
    let n = results.len();
```

```
    let k: u32 = results.iter().sum(); // how many passed
    println!("{}", k as f64 / n as f64); // the pass rate, e.g. 0.68 (Python's draws give 0.695)

    // NOTE the variance: p(1-p) is largest at p = 0.5 and shrinks toward the edges.
    // That is WHY a model at 50% needs more tasks to pin down than one at 95%.
    for p in [0.5, 0.7, 0.95] {
        println!("{} variance per task: {}", p, round(p * (1.0 - p), 4));
    }
}
```

Binomial

Count of successes in n independent trials.



Binomial: count of successes in n independent trials.

The story	Run n tasks, each passing with probability p, count the passes. The distribution of that count is binomial.
In eval work	Number of tasks passed out of 200. Number of sessions with a successful tool call, IF sessions are independent.
Use when	Use when trials are independent and share the same p. Wilson intervals are built on this.
Avoid when	**Do not use when trials are not independent**, which is the big one. Multiple tool calls inside one session are not independent, and treating them as binomial understates your uncertainty, sometimes by a factor of two or more.

Under the hood

Binomial: formulas and code

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$E[X] = np \qquad \mathrm{Var}(X) = np(1-p)$$

python

```python
from scipy import stats
import math

# The count of passes out of n tasks IS binomial, if the tasks are independent.
n, p = 200, 0.62
print("expected passes:", n * p)                        # 124.0
print("sd of that count:", math.sqrt(n * p * (1 - p))) # 6.86

# How likely is exactly 130 passes? And 130 or more?
print(stats.binom.pmf(130, n, p))     # exactly     -> 0.0401
print(stats.binom.sf(129, n, p))      # 130 or more -> 0.2121

# The interval you should report for an observed rate.
def wilson(k, n, z=1.96):
    p_hat = k / n
    den = 1 + z * z / n
    centre = (p_hat + z * z / (2 * n)) / den
    half = z * math.sqrt(p_hat * (1 - p_hat) / n + z * z / (4 * n * n)) / den
    return max(0.0, centre - half), min(1.0, centre + half)
```

```
print(wilson(124, 200))   # (0.5511, 0.6844)

# THE TRAP: this is only valid if tasks are independent.
# 200 tool calls from 25 sessions are NOT 200 independent trials.
```

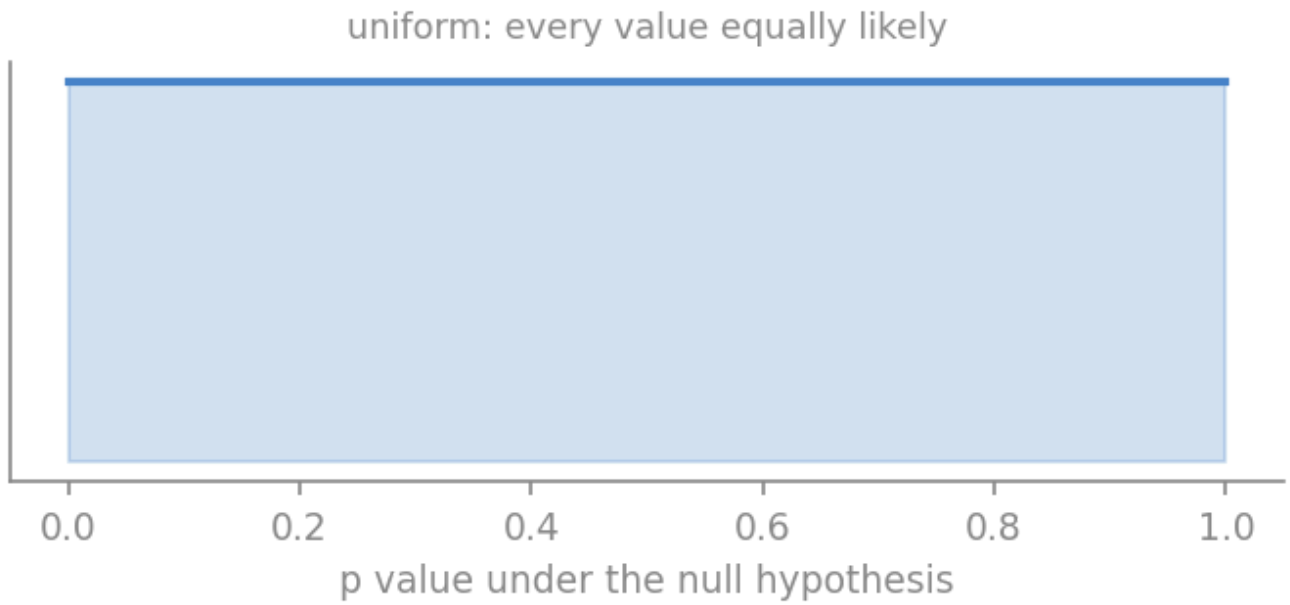
rust

```
/// ln Gamma(x) by the Lanczos approximation (g = 7, 9 terms).
fn ln_gamma(x: f64) -> f64 {
    const G: [f64; 9] = [
        0.999_999_999_999_809_93, 676.520_368_121_885_1, -1259.139_216_722_402_8,
        771.323_428_777_653_13, -176.615_029_162_140_59, 12.507_343_278_686_905,
        -0.138_571_095_265_720_12, 9.984_369_578_019_571_6e-6, 1.505_632_735_149_311_6e-7,
    ];
    let pi = std::f64::consts::PI;
    if x < 0.5 {
        return (pi / (pi * x).sin()).ln() - ln_gamma(1.0 - x); // reflection
    }
    let x = x - 1.0;
    let t = x + 7.5;
    let a = G[0] + (1..9).map(|i| G[i] / (x + i as f64)).sum::<f64>();
    0.5 * (2.0 * pi).ln() + (x + 0.5) * t.ln() - t + a.ln()
}

fn ln_choose(n: u64, k: u64) -> f64 {
    ln_gamma(n as f64 + 1.0) - ln_gamma(k as f64 + 1.0) - ln_gamma((n - k) as f64 + 1.0)
}

fn binom_pmf(k: u64, n: u64, p: f64) -> f64 {
    (ln_choose(n, k) + k as f64 * p.ln() + (n - k) as f64 * (1.0 - p).ln()).exp()
}

/// Survival function P(X > k): add up the upper tail.
fn binom_sf(k: u64, n: u64, p: f64) -> f64 {
    (k + 1..=n).map(|j| binom_pmf(j, n, p)).sum()
}

// The interval you should report for an observed rate.
fn wilson(k: u64, n: u64, z: f64) -> (f64, f64) {
    let (k, n) = (k as f64, n as f64);
    let p_hat = k / n;
    let den = 1.0 + z * z / n;
    let centre = (p_hat + z * z / (2.0 * n)) / den;
    let half = z * (p_hat * (1.0 - p_hat) / n + z * z / (4.0 * n * n)).sqrt() / den;
    ((centre - half).max(0.0), (centre + half).min(1.0))
}

fn main() {
    // The count of passes out of n tasks IS binomial, if the tasks are independent.
    let (n, p) = (200u64, 0.62);
    println!("expected passes: {:?}", n as f64 * p);                          // 124.0
    println!("sd of that count: {}", (n as f64 * p * (1.0 - p)).sqrt());      // 6.86

    // How likely is exactly 130 passes? And 130 or more?
    println!("{}", binom_pmf(130, n, p)); // exactly     -> 0.0401
    println!("{}", binom_sf(129, n, p));  // 130 or more -> 0.2121

    println!("{:?}", wilson(124, 200, 1.96)); // (0.5511, 0.6844)

    // THE TRAP: this is only valid if tasks are independent.
```

```
    // 200 tool calls from 25 sessions are NOT 200 independent trials.
}
```

Uniform

Every value in a range equally likely.



Uniform: every value in a range equally likely.

The story	Nothing favours any part of the range. Flat.
In eval work	P values under a true null hypothesis are uniform. That fact is what makes the multiple comparisons problem work the way it does: run 40 benchmarks with no real effect and you get 40 uniform draws, so about 2 land below 0.05.
Use when	Use as a null model, for random baselines, and for reasoning about p values.
Avoid when	Almost no real measurement is uniform. If your data looks flat, suspect a rounding or bucketing artefact.

Under the hood

Uniform: formulas and code

$$f(x) = \frac{1}{b-a}, \quad a \le x \le b$$

$$E[X] = \frac{a+b}{2} \qquad \mathrm{Var}(X) = \frac{(b-a)^2}{12}$$

python

```python
import random

# The single most useful fact about the uniform distribution in eval work:
# p values from a TRUE NULL are uniform. That is the whole multiple
# comparisons problem in one line.
rng = random.Random(1)
suite = 40                                  # benchmarks in your standing suite
p_values = [rng.random() for _ in range(suite)]   # all nulls, no real effect

false_alarms = sum(1 for p in p_values if p < 0.05)
print(false_alarms)      # 6 with this seed; 2 on average, purely by luck

# Expected number of spurious "significant" results:
print(suite * 0.05)      # 2.0
```

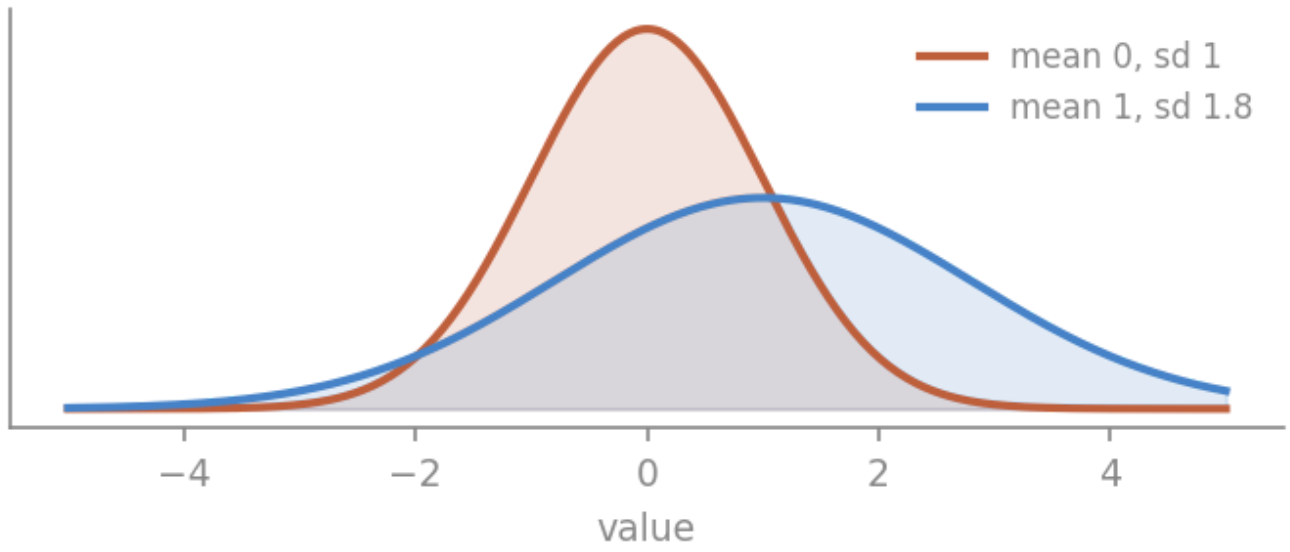
rust

```rust
// A tiny inline PRNG (SplitMix64) so the snippet needs no crates.
// Its draws differ from numpy's, so the random numbers below differ from the Python's.
struct Rng(u64);

impl Rng {
    fn new(seed: u64) -> Self {
        Rng(seed)
    }
    fn next_u64(&mut self) -> u64 {
        self.0 = self.0.wrapping_add(0x9E37_79B9_7F4A_7C15);
        let mut z = self.0;
        z = (z ^ (z >> 30)).wrapping_mul(0xBF58_476D_1CE4_E5B9);
        z = (z ^ (z >> 27)).wrapping_mul(0x94D0_49BB_1331_11EB);
        z ^ (z >> 31)
    }
    /// Uniform on [0, 1).
    fn random(&mut self) -> f64 {
        (self.next_u64() >> 11) as f64 / (1u64 << 53) as f64
    }
}

fn main() {
    // The single most useful fact about the uniform distribution in eval work:
```

```rust
    // p values from a TRUE NULL are uniform. That is the whole multiple
    // comparisons problem in one line.
    let mut rng = Rng::new(1);
    let suite = 40; // benchmarks in your standing suite
    let p_values: Vec<f64> = (0..suite).map(|_| rng.random()).collect(); // all nulls, no real effect

    let false_alarms = p_values.iter().filter(|&&p| p < 0.05).count();
    println!("{}", false_alarms);            // 2 with these draws (Python's: 6); 2 on average, purely by luck

    // Expected number of spurious "significant" results:
    println!("{:?}", suite as f64 * 0.05);   // 2.0
}
```

Normal (Gaussian)

The sum of many small independent effects.



Normal (Gaussian): the sum of many small independent effects.

The story	When lots of small independent things add up, the total is normal. That is the only story it tells, and it is why normality shows up in averages and test statistics rather than in raw measurements.
In eval work	Averages of large samples. Test statistics. Almost never your raw latency or token counts.
Use when	Use for averages, for differences of averages, and as the reference for QQ plots and z scores.
Avoid when	**Do not assume your raw data is normal.** Latency is not. Token counts are not. Counts are not. Assuming it and then using a t test on skewed data is the classic error.

Under the hood

Normal (Gaussian): formulas and code

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

$$E[X] = \mu \qquad \mathrm{Var}(X) = \sigma^2$$

python

```python
from scipy import stats
import numpy as np

# Raw eval data is almost never normal. AVERAGES of it get close.
latency = stats.lognorm.rvs(0.9, scale=3, size=5000, random_state=0)  # skewed
print(stats.skew(latency))                    # ~4.4, strongly right skewed

means = latency[:5000].reshape(-1, 50).mean(axis=1)   # averages of 50
print(stats.skew(means))                      # ~0.27 with this seed (theory ~0.67), far closer to symmetric

# The 68 / 95 / 99.7 rule, for genuinely normal data:
for k in (1, 2, 3):
    print(k, "sd covers", round(100 * (stats.norm.cdf(k) - stats.norm.cdf(-k)), 1), "%")

# Test normality rather than assuming it. Small p = not normal.
print(stats.shapiro(latency[:500]).pvalue)    # ~0 -> definitely not normal
print(stats.shapiro(means[:80]).pvalue)       # 0.35 with this seed, but across seeds this rejects
                                              # about half the time: closer to normal, not exactly normal
```

rust

```rust
// A tiny inline PRNG (SplitMix64) so the snippet needs no crates.
// Its draws differ from numpy's, so the random numbers below differ from the Python's.
struct Rng(u64);

impl Rng {
    fn new(seed: u64) -> Self {
        Rng(seed)
    }
    fn next_u64(&mut self) -> u64 {
        self.0 = self.0.wrapping_add(0x9E37_79B9_7F4A_7C15);
        let mut z = self.0;
        z = (z ^ (z >> 30)).wrapping_mul(0xBF58_476D_1CE4_E5B9);
        z = (z ^ (z >> 27)).wrapping_mul(0x94D0_49BB_1331_11EB);
        z ^ (z >> 31)
    }
    /// Uniform on [0, 1).
    fn random(&mut self) -> f64 {
        (self.next_u64() >> 11) as f64 / (1u64 << 53) as f64
    }
    /// Normal draw by Box-Muller.
    fn normal(&mut self, mean: f64, sd: f64) -> f64 {
        let u1 = 1.0 - self.random(); // in (0, 1], safe for ln
        let u2 = self.random();
        mean + sd * (-2.0 * u1.ln()).sqrt() * (std::f64::consts::TAU * u2).cos()
    }
}

/// ln Gamma(x) by the Lanczos approximation (g = 7, 9 terms).
fn ln_gamma(x: f64) -> f64 {
    const G: [f64; 9] = [
        0.999_999_999_999_809_93, 676.520_368_121_885_1, -1259.139_216_722_402_8,
        771.323_428_777_653_13, -176.615_029_162_140_59, 12.507_343_278_686_905,
        -0.138_571_095_265_720_12, 9.984_369_578_019_571_6e-6, 1.505_632_735_149_311_6e-7,
    ];
    let pi = std::f64::consts::PI;
    if x < 0.5 {
        return (pi / (pi * x).sin()).ln() - ln_gamma(1.0 - x); // reflection
    }
    let x = x - 1.0;
    let t = x + 7.5;
    let a = G[0] + (1..9).map(|i| G[i] / (x + i as f64)).sum::<f64>();
    0.5 * (2.0 * pi).ln() + (x + 0.5) * t.ln() - t + a.ln()
}

/// Regularized incomplete gamma functions (P, Q) = (lower, upper), P + Q = 1.
fn gamma_pq(a: f64, x: f64) -> (f64, f64) {
    if x <= 0.0 {
        return (0.0, 1.0);
    }
    let front = (-x + a * x.ln() - ln_gamma(a)).exp();
    if x < a + 1.0 {
        // power series converges fast here
        let (mut ap, mut term) = (a, 1.0 / a);
        let mut sum = term;
        while term.abs() > sum.abs() * 1e-16 {
            ap += 1.0;
            term *= x / ap;
            sum += term;
```

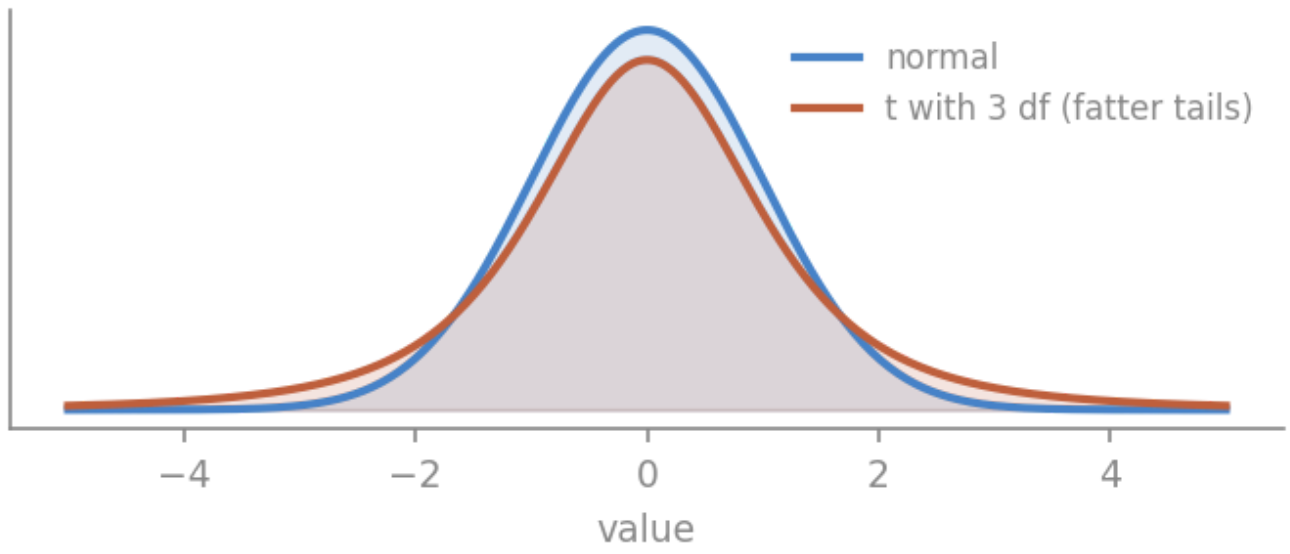
```
        }
        let p = sum * front;
        (p, 1.0 - p)
    } else {
        // continued fraction (modified Lentz) converges fast here
        let tiny = 1e-300;
        let (mut b, mut c) = (x + 1.0 - a, 1.0 / tiny);
        let mut d = 1.0 / b;
        let mut h = d;
        for i in 1..1000 {
            let an = -(i as f64) * (i as f64 - a);
            b += 2.0;
            d = an * d + b;
            if d.abs() < tiny { d = tiny; }
            c = b + an / c;
            if c.abs() < tiny { c = tiny; }
            d = 1.0 / d;
            h *= d * c;
            if (d * c - 1.0).abs() < 1e-16 { break; }
        }
        let q = front * h;
        (1.0 - q, q)
    }
}

/// Standard normal CDF, via the incomplete gamma function (erf(x) = P(1/2, x^2)).
fn norm_cdf(z: f64) -> f64 {
    let tail = 0.5 * gamma_pq(0.5, z * z / 2.0).1; // P(Z > |z|), accurate far out
    if z >= 0.0 { 1.0 - tail } else { tail }
}

fn mean(x: &[f64]) -> f64 {
    x.iter().sum::<f64>() / x.len() as f64
}

/// Sample skewness, as scipy.stats.skew computes it: m3 / m2^1.5.
fn skew(x: &[f64]) -> f64 {
    let m = mean(x);
    let m2 = x.iter().map(|v| (v - m).powi(2)).sum::<f64>() / x.len() as f64;
    let m3 = x.iter().map(|v| (v - m).powi(3)).sum::<f64>() / x.len() as f64;
    m3 / m2.powf(1.5)
}

fn round(x: f64, digits: i32) -> f64 {
    let f = 10f64.powi(digits);
    (x * f).round() / f
}

/// Inverse normal CDF: Acklam's rational approximation, polished by one Newton step.
fn norm_ppf(p: f64) -> f64 {
    let a = [-3.969683028665376e1, 2.209460984245205e2, -2.759285104469687e2,
             1.383577518672690e2, -3.066479806614716e1, 2.506628277459239];
    let b = [-5.447609879822406e1, 1.615858368580409e2, -1.556989798598866e2,
             6.680131188771972e1, -1.328068155288572e1];
    let c = [-7.784894002430293e-3, -3.223964580411365e-1, -2.400758277161838,
             -2.549732539343734, 4.374664141464968, 2.938163982698783];
    let d = [7.784695709041462e-3, 3.224671290700398e-1, 2.445134137142996, 3.754408661907416];
    let tail = |q: f64| (((((c[0] * q + c[1]) * q + c[2]) * q + c[3]) * q + c[4]) * q + c[5])
        / ((((d[0] * q + d[1]) * q + d[2]) * q + d[3]) * q + 1.0);
    let x = if p < 0.02425 {
```

```
        tail((-2.0 * p.ln()).sqrt())
    } else if p > 1.0 - 0.02425 {
        -tail((-2.0 * (1.0 - p).ln()).sqrt())
    } else {
        let q = p - 0.5;
        let r = q * q;
        (((((a[0] * r + a[1]) * r + a[2]) * r + a[3]) * r + a[4]) * r + a[5]) * q
            / (((((b[0] * r + b[1]) * r + b[2]) * r + b[3]) * r + b[4]) * r + 1.0)
    };
    let pdf = (-x * x / 2.0).exp() / (2.0 * std::f64::consts::PI).sqrt();
    x - (norm_cdf(x) - p) / pdf
}

fn poly(c: &[f64], x: f64) -> f64 {
    c.iter().rev().fold(0.0, |acc, &ci| acc * x + ci)
}

/// Shapiro-Wilk test (Royston's 1995 algorithm, as scipy uses), for n >= 12.
/// Returns the p value: small means "not normal".
fn shapiro_pvalue(x: &[f64]) -> f64 {
    let n = x.len();
    let nf = n as f64;
    let mut s = x.to_vec();
    s.sort_by(|a, b| a.total_cmp(b));

    // expected normal order statistics, and the weights built from them
    let m: Vec<f64> = (1..=n).map(|i| norm_ppf((i as f64 - 0.375) / (nf + 0.25))).collect();
    let mm: f64 = m.iter().map(|v| v * v).sum();
    let u = 1.0 / nf.sqrt();
    let an = m[n - 1] / mm.sqrt() + poly(&[0.0, 0.221157, -0.147981, -2.071190, 4.434685, -2.706056], u);
    let an1 = m[n - 2] / mm.sqrt() + poly(&[0.0, 0.042981, -0.293762, -1.752461, 5.682633, -3.582633], u);
    let phi = (mm - 2.0 * m[n - 1].powi(2) - 2.0 * m[n - 2].powi(2))
        / (1.0 - 2.0 * an * an - 2.0 * an1 * an1);
    let mut a: Vec<f64> = m.iter().map(|v| v / phi.sqrt()).collect();
    (a[0], a[1], a[n - 2], a[n - 1]) = (-an, -an1, an1, an);

    // W = squared correlation between the sorted data and the weights
    let xbar = mean(&s);
    let sax: f64 = a.iter().zip(&s).map(|(ai, xi)| ai * (xi - xbar)).sum();
    let ssa: f64 = a.iter().map(|v| v * v).sum();
    let ssx: f64 = s.iter().map(|v| (v - xbar).powi(2)).sum();
    let w = sax * sax / (ssa * ssx);

    // ln(1 - W) is close to normal; its mean and sd depend on ln n
    let ln_n = nf.ln();
    let mu = poly(&[-1.5861, -0.31082, -0.083751, 0.0038915], ln_n);
    let sigma = poly(&[-0.4803, -0.082676, 0.0030302], ln_n).exp();
    norm_cdf(-((1.0 - w).ln() - mu) / sigma) // upper tail
}

fn main() {
    // Raw eval data is almost never normal. AVERAGES of it are.
    let mut rng = Rng::new(0);
    let latency: Vec<f64> = (0..5000).map(|_| 3.0 * rng.normal(0.0, 0.9).exp()).collect(); // skewed (lognormal)
    println!("{}", skew(&latency)); // 4.51, strongly right skewed

    let means: Vec<f64> = latency.chunks(50).map(mean).collect(); // averages of 50
    println!("{}", skew(&means));   // 0.46, much closer to symmetric

    // The 68 / 95 / 99.7 rule, for genuinely normal data:
```

```
    for k in [1.0, 2.0, 3.0] {
        println!("{} sd covers {} %", k, round(100.0 * (norm_cdf(k) - norm_cdf(-k)), 1));
    }

    // Test normality rather than assuming it. Small p = not normal.
    println!("{:e}", shapiro_pvalue(&latency[..500])); // 8.3e-35 -> definitely not normal
    println!("{}", shapiro_pvalue(&means[..80]));      // 0.025 with these draws: closer to normal, not there yet
    // (theory: a mean of 50 still has skew 4.75 / sqrt(50) = 0.67)
}
```

Student's t

A normal mean when you had to estimate the spread too.



Student's t: a normal mean when you had to estimate the spread too.

The story	Same bell shape but with fatter tails, because estimating the standard deviation from a small sample adds uncertainty. As n grows it converges on the normal.
In eval work	Small sample comparisons of means, under about 30 observations.
Use when	Use for means of small samples where the data is roughly symmetric.
Avoid when	The fatter tails do not rescue you from skew. On lognormal latency, t is still the wrong tool. Bootstrap or work on the log scale instead.

Under the hood

Student's t: formulas and code

$$f(x) \propto \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}$$

$$E[X] = 0 \ (\nu > 1) \qquad \mathrm{Var}(X) = \frac{\nu}{\nu - 2} \ (\nu > 2)$$

python

```python
from scipy import stats

# t is what you use when you estimated the spread from the same small sample.
# Compare the critical value at 95% as the sample grows:
for df in (5, 10, 30, 100, 1000):
    print(df, round(stats.t.ppf(0.975, df), 3))
# 5 -> 2.571,  30 -> 2.042,  1000 -> 1.962.  It converges on the normal 1.96.

# A paired t test on per task score differences:
import numpy as np
rng = np.random.default_rng(0)
model_a = rng.normal(0.62, 0.20, 40)
model_b = rng.normal(0.58, 0.20, 40)
print(stats.ttest_rel(model_a, model_b))

# BUT: check the differences are roughly symmetric first. On skewed data
# this is the wrong tool no matter how fat the t tails are.
```

rust

```rust
// A tiny inline PRNG (SplitMix64) so the snippet needs no crates.
// Its draws differ from numpy's, so the random numbers below differ from the Python's.
```

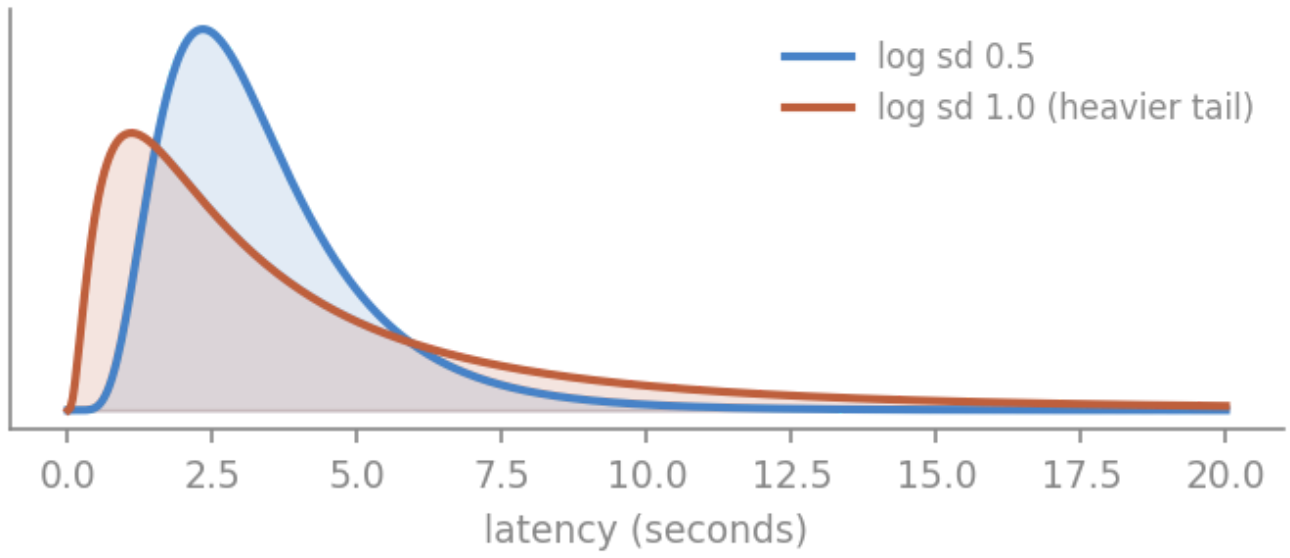
```rust
struct Rng(u64);

impl Rng {
    fn new(seed: u64) -> Self {
        Rng(seed)
    }
    fn next_u64(&mut self) -> u64 {
        self.0 = self.0.wrapping_add(0x9E37_79B9_7F4A_7C15);
        let mut z = self.0;
        z = (z ^ (z >> 30)).wrapping_mul(0xBF58_476D_1CE4_E5B9);
        z = (z ^ (z >> 27)).wrapping_mul(0x94D0_49BB_1331_11EB);
        z ^ (z >> 31)
    }
    /// Uniform on [0, 1).
    fn random(&mut self) -> f64 {
        (self.next_u64() >> 11) as f64 / (1u64 << 53) as f64
    }
    /// Normal draw by Box-Muller.
    fn normal(&mut self, mean: f64, sd: f64) -> f64 {
        let u1 = 1.0 - self.random(); // in (0, 1], safe for ln
        let u2 = self.random();
        mean + sd * (-2.0 * u1.ln()).sqrt() * (std::f64::consts::TAU * u2).cos()
    }
}

/// ln Gamma(x) by the Lanczos approximation (g = 7, 9 terms).
fn ln_gamma(x: f64) -> f64 {
    const G: [f64; 9] = [
        0.999_999_999_999_809_93, 676.520_368_121_885_1, -1259.139_216_722_402_8,
        771.323_428_777_653_13, -176.615_029_162_140_59, 12.507_343_278_686_905,
        -0.138_571_095_265_720_12, 9.984_369_578_019_571_6e-6, 1.505_632_735_149_311_6e-7,
    ];
    let pi = std::f64::consts::PI;
    if x < 0.5 {
        return (pi / (pi * x).sin()).ln() - ln_gamma(1.0 - x); // reflection
    }
    let x = x - 1.0;
    let t = x + 7.5;
    let a = G[0] + (1..9).map(|i| G[i] / (x + i as f64)).sum::<f64>();
    0.5 * (2.0 * pi).ln() + (x + 0.5) * t.ln() - t + a.ln()
}

/// Regularized incomplete beta I_x(a, b), by continued fraction (modified Lentz).
fn beta_inc(a: f64, b: f64, x: f64) -> f64 {
    if x <= 0.0 {
        return 0.0;
    }
    if x >= 1.0 {
        return 1.0;
    }
    let front = (ln_gamma(a + b) - ln_gamma(a) - ln_gamma(b) + a * x.ln() + b * (1.0 - x).ln()).exp();
    // the fraction converges fast on one side of the mean; use symmetry for the other
    if x < (a + 1.0) / (a + b + 2.0) {
        front * beta_cf(a, b, x) / a
    } else {
        1.0 - front * beta_cf(b, a, 1.0 - x) / b
    }
}

fn beta_cf(a: f64, b: f64, x: f64) -> f64 {
```

```
    let tiny = 1e-300;
    let clamp = |v: f64| if v.abs() < tiny { tiny } else { v };
    let mut c = 1.0;
    let mut d = 1.0 / clamp(1.0 - (a + b) * x / (a + 1.0));
    let mut h = d;
    for m in 1..1000 {
        let m = m as f64;
        let even = m * (b - m) * x / ((a + 2.0 * m - 1.0) * (a + 2.0 * m));
        d = 1.0 / clamp(1.0 + even * d);
        c = clamp(1.0 + even / c);
        h *= d * c;
        let odd = -(a + m) * (a + b + m) * x / ((a + 2.0 * m) * (a + 2.0 * m + 1.0));
        d = 1.0 / clamp(1.0 + odd * d);
        c = clamp(1.0 + odd / c);
        h *= d * c;
        if (d * c - 1.0).abs() < 1e-16 {
            break;
        }
    }
    h
}

fn round(x: f64, digits: i32) -> f64 {
    let f = 10f64.powi(digits);
    (x * f).round() / f
}

/// Student t CDF with df degrees of freedom, via the incomplete beta function.
fn t_cdf(t: f64, df: f64) -> f64 {
    let tail = 0.5 * beta_inc(df / 2.0, 0.5, df / (df + t * t)); // P(T > |t|)
    if t >= 0.0 { 1.0 - tail } else { tail }
}

/// Inverse t CDF by bisection: the CDF only goes up, so halve the bracket.
fn t_ppf(p: f64, df: f64) -> f64 {
    let (mut lo, mut hi) = (-100.0, 100.0);
    for _ in 0..200 {
        let mid = 0.5 * (lo + hi);
        if t_cdf(mid, df) < p { lo = mid } else { hi = mid }
    }
    0.5 * (lo + hi)
}

#[derive(Debug)]
struct TtestResult {
    statistic: f64,
    pvalue: f64,
    df: usize,
}

/// Paired t test: a one-sample t test on the per-task differences.
fn ttest_rel(a: &[f64], b: &[f64]) -> TtestResult {
    let d: Vec<f64> = a.iter().zip(b).map(|(x, y)| x - y).collect();
    let n = d.len() as f64;
    let m = d.iter().sum::<f64>() / n;
    let sd = (d.iter().map(|v| (v - m).powi(2)).sum::<f64>() / (n - 1.0)).sqrt();
    let t = m / (sd / n.sqrt());
    let pvalue = 2.0 * (1.0 - t_cdf(t.abs(), n - 1.0)); // two sided
    TtestResult { statistic: t, pvalue, df: d.len() - 1 }
}
```

```rust
fn main() {
    // t is what you use when you estimated the spread from the same small sample.
    // Compare the critical value at 95% as the sample grows:
    for df in [5, 10, 30, 100, 1000] {
        println!("{} {}", df, round(t_ppf(0.975, df as f64), 3));
    }
    // 5 -> 2.571,  30 -> 2.042,  1000 -> 1.962.  It converges on the normal 1.96.

    // A paired t test on per task score differences:
    let mut rng = Rng::new(0);
    let model_a: Vec<f64> = (0..40).map(|_| rng.normal(0.62, 0.20)).collect();
    let model_b: Vec<f64> = (0..40).map(|_| rng.normal(0.58, 0.20)).collect();
    println!("{:?}", ttest_rel(&model_a, &model_b));
    // with these draws: statistic 1.398, pvalue 0.170, df 39

    // BUT: check the differences are roughly symmetric first. On skewed data
    // this is the wrong tool no matter how fat the t tails are.
}
```

Lognormal

The PRODUCT of many small independent effects.



Lognormal: the PRODUCT of many small independent effects.

The story	Where normal comes from adding, lognormal comes from multiplying. Latency is multiplicative: a slow network multiplies with a big payload multiplies with a cold cache. So the log of latency is normal, and latency itself is lognormal.
In eval work	**Latency. Token counts. Session durations. Repo sizes.** This is the workhorse shape of eval engineering.
Use when	Use for anything strictly positive with a long right tail. Analyse on the log scale, report percentiles in the original units. The geometric mean is its natural centre.
Avoid when	Do not report the arithmetic mean unless you need a total. Do not run a t test on it raw. And distinguish it from a genuine power law: lognormal has a finite, well behaved mean, a power law may not.

Under the hood

Lognormal: formulas and code

$$f(x) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln x - \mu)^2}{2\sigma^2}\right)$$

$$E[X] = e^{\mu+\sigma^2/2} \qquad \text{median} = e^{\mu} \qquad \text{Var}(X) = \left(e^{\sigma^2} - 1\right) e^{2\mu+\sigma^2}$$

python

```python
import numpy as np
from scipy import stats

lat = stats.lognorm.rvs(0.85, scale=2.5, size=5000, random_state=3)

# WRONG: the arithmetic mean sits between the median and p95 and
# describes nobody's actual experience.
print("mean  ", lat.mean())                    # 3.48
print("median", np.median(lat))                # 2.42
print("p95   ", np.quantile(lat, 0.95))        # 9.90

# RIGHT: work on the log scale, where it is symmetric.
logs = np.log(lat)
print("log data is symmetric:", round(stats.skew(logs), 3))   # 0.037

# The geometric mean is the natural centre, and it recovers the median.
```

```
geo = np.exp(logs.mean())
print("geometric mean:", geo)                  # 2.42, matches the median closely

# Fit and read off the parameters:
shape, loc, scale = stats.lognorm.fit(lat, floc=0)
print("sigma:", shape, " median:", scale)      # 0.846, 2.42 - recovered
```

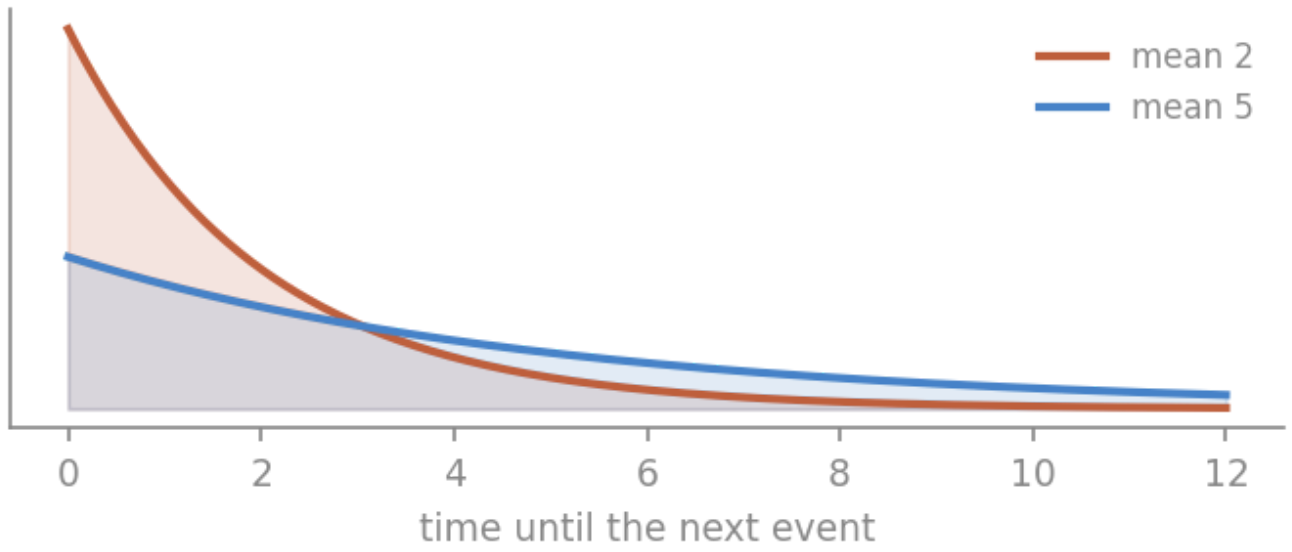
rust

```
// A tiny inline PRNG (SplitMix64) so the snippet needs no crates.
// Its draws differ from numpy's, so the random numbers below differ from the Python's.
struct Rng(u64);

impl Rng {
    fn new(seed: u64) -> Self {
        Rng(seed)
    }
    fn next_u64(&mut self) -> u64 {
        self.0 = self.0.wrapping_add(0x9E37_79B9_7F4A_7C15);
        let mut z = self.0;
        z = (z ^ (z >> 30)).wrapping_mul(0xBF58_476D_1CE4_E5B9);
        z = (z ^ (z >> 27)).wrapping_mul(0x94D0_49BB_1331_11EB);
        z ^ (z >> 31)
    }
    /// Uniform on [0, 1).
    fn random(&mut self) -> f64 {
        (self.next_u64() >> 11) as f64 / (1u64 << 53) as f64
    }
    /// Normal draw by Box-Muller.
    fn normal(&mut self, mean: f64, sd: f64) -> f64 {
        let u1 = 1.0 - self.random(); // in (0, 1], safe for ln
        let u2 = self.random();
        mean + sd * (-2.0 * u1.ln()).sqrt() * (std::f64::consts::TAU * u2).cos()
    }
}

fn mean(x: &[f64]) -> f64 {
    x.iter().sum::<f64>() / x.len() as f64
}

/// Linear-interpolation quantile of SORTED data (numpy's default method).
fn quantile(sorted: &[f64], q: f64) -> f64 {
    let h = (sorted.len() - 1) as f64 * q;
    let lo = h.floor() as usize;
    let hi = (lo + 1).min(sorted.len() - 1);
    let (a, b, t) = (sorted[lo], sorted[hi], h - lo as f64);
    if t >= 0.5 { b - (b - a) * (1.0 - t) } else { a + (b - a) * t }
}

/// Sample skewness, as scipy.stats.skew computes it: m3 / m2^1.5.
fn skew(x: &[f64]) -> f64 {
    let m = mean(x);
    let m2 = x.iter().map(|v| (v - m).powi(2)).sum::<f64>() / x.len() as f64;
    let m3 = x.iter().map(|v| (v - m).powi(3)).sum::<f64>() / x.len() as f64;
    m3 / m2.powf(1.5)
}

fn round(x: f64, digits: i32) -> f64 {
    let f = 10f64.powi(digits);
    (x * f).round() / f
```

```
}

fn main() {
    // lognormal(sigma 0.85, median 2.5): exp of a normal
    let mut rng = Rng::new(3);
    let lat: Vec<f64> = (0..5000).map(|_| 2.5 * rng.normal(0.0, 0.85).exp()).collect();
    let mut sorted = lat.clone();
    sorted.sort_by(|a, b| a.total_cmp(b));

    // WRONG: the arithmetic mean sits between the median and p95 and
    // describes nobody's actual experience.
    println!("mean   {}", mean(&lat));                  // 3.52
    println!("median {}", quantile(&sorted, 0.5));      // 2.53
    println!("p95    {}", quantile(&sorted, 0.95));     // 9.75

    // RIGHT: work on the log scale, where it is symmetric.
    let logs: Vec<f64> = lat.iter().map(|v| v.ln()).collect();
    println!("log data is symmetric: {}", round(skew(&logs), 3)); // -0.021

    // The geometric mean is the natural centre, and it recovers the median.
    let geo = mean(&logs).exp();
    println!("geometric mean: {}", geo);                // 2.47, close to the median

    // Fit and read off the parameters. With loc fixed at 0 the maximum likelihood
    // fit is closed form: sigma = sd of the logs (dividing by n), median = geo.
    let mu = mean(&logs);
    let shape = (logs.iter().map(|v| (v - mu).powi(2)).sum::<f64>() / logs.len() as f64).sqrt();
    let scale = mu.exp();
    println!("sigma: {}  median: {}", shape, scale);    // 0.845, 2.47 - recovered
}
```

Exponential

Waiting time when the event rate is constant.



Exponential: waiting time when the event rate is constant.

The story	How long until the next event, when the chance of it happening in the next instant never changes. It is memoryless: having waited 10 seconds tells you nothing about how much longer you will wait.
In eval work	Time between independent failures. Idle time between requests, roughly.
Use when	Use for waiting times with a constant hazard rate.
Avoid when	Real systems are rarely memoryless. If a request has already run 60 seconds it is probably stuck, and its remaining time is not the same as a fresh request's. Use gamma or lognormal when the hazard changes.

Under the hood

Exponential: formulas and code

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

$$E[X] = \frac{1}{\lambda} \qquad \mathrm{Var}(X) = \frac{1}{\lambda^2}$$

python

```python
from scipy import stats
import numpy as np

# Waiting time with a CONSTANT hazard rate.
gaps = stats.expon.rvs(scale=4.0, size=4000, random_state=1)  # mean 4 seconds

# Memorylessness, demonstrated. Among requests that already waited 5 seconds,
# the REMAINING wait still averages 4 seconds. Waiting tells you nothing.
already = gaps[gaps > 5] - 5
print(gaps.mean(), already.mean())     # 3.98  vs  3.91

# THE TEST: is your data actually memoryless? Compare the mean and sd.
# For an exponential they are equal.
print(gaps.mean(), gaps.std())         # 3.98, 3.90 -> consistent

lat = stats.lognorm.rvs(0.9, scale=3, size=4000, random_state=1)
print(lat.mean(), lat.std())           # 4.56, 5.28 -> not exponential
```

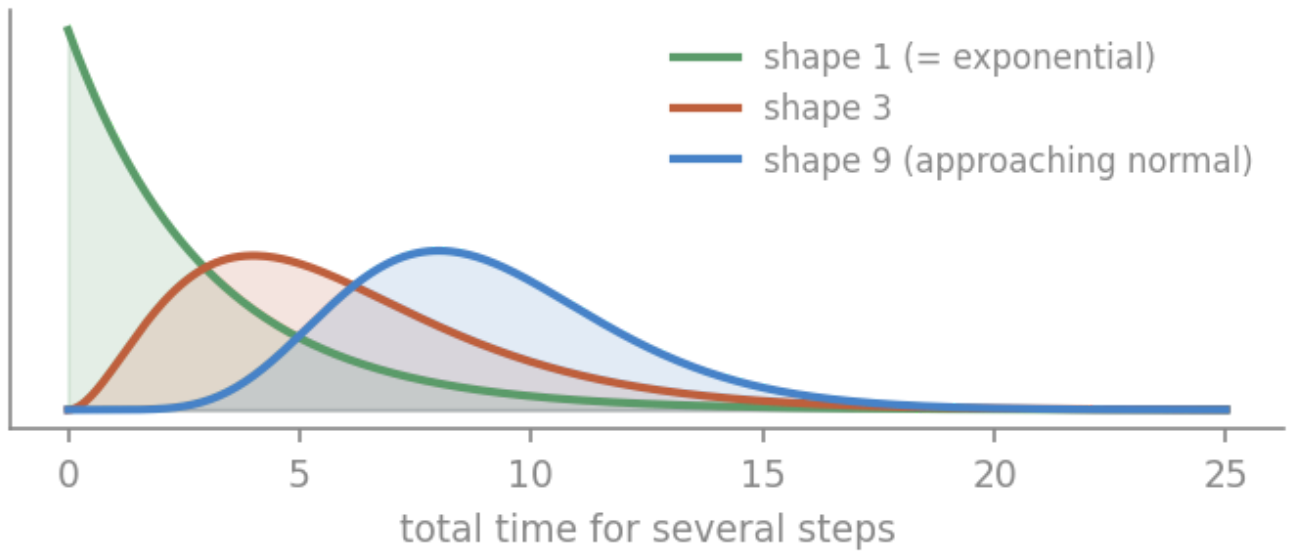
rust

```rust
// A tiny inline PRNG (SplitMix64) so the snippet needs no crates.
// Its draws differ from numpy's, so the random numbers below differ from the Python's.
struct Rng(u64);

impl Rng {
    fn new(seed: u64) -> Self {
        Rng(seed)
    }
    fn next_u64(&mut self) -> u64 {
        self.0 = self.0.wrapping_add(0x9E37_79B9_7F4A_7C15);
        let mut z = self.0;
        z = (z ^ (z >> 30)).wrapping_mul(0xBF58_476D_1CE4_E5B9);
        z = (z ^ (z >> 27)).wrapping_mul(0x94D0_49BB_1331_11EB);
        z ^ (z >> 31)
    }
    /// Uniform on [0, 1).
    fn random(&mut self) -> f64 {
        (self.next_u64() >> 11) as f64 / (1u64 << 53) as f64
    }
    /// Normal draw by Box-Muller.
    fn normal(&mut self, mean: f64, sd: f64) -> f64 {
        let u1 = 1.0 - self.random(); // in (0, 1], safe for ln
        let u2 = self.random();
        mean + sd * (-2.0 * u1.ln()).sqrt() * (std::f64::consts::TAU * u2).cos()
    }
}

fn mean(x: &[f64]) -> f64 {
    x.iter().sum::<f64>() / x.len() as f64
}

/// Population standard deviation (ddof = 0, numpy's default).
fn std(x: &[f64]) -> f64 {
    let m = mean(x);
    (x.iter().map(|v| (v - m).powi(2)).sum::<f64>() / x.len() as f64).sqrt()
}

fn main() {
```

```rust
    // Waiting time with a CONSTANT hazard rate.
    let mut rng = Rng::new(1);
    let gaps: Vec<f64> = (0..4000).map(|_| -4.0 * (1.0 - rng.random()).ln()).collect(); // mean 4 seconds

    // Memorylessness, demonstrated. Among requests that already waited 5 seconds,
    // the REMAINING wait still averages 4 seconds. Waiting tells you nothing.
    let already: Vec<f64> = gaps.iter().filter(|&&g| g > 5.0).map(|g| g - 5.0).collect();
    println!("{} {}", mean(&gaps), mean(&already)); // 3.91  vs  4.13

    // THE TEST: is your data actually memoryless? Compare the mean and sd.
    // For an exponential they are equal.
    println!("{} {}", mean(&gaps), std(&gaps));     // 3.91, 4.03 -> consistent

    let lat: Vec<f64> = (0..4000).map(|_| 3.0 * rng.normal(0.0, 0.9).exp()).collect(); // lognormal
    println!("{} {}", mean(&lat), std(&lat));       // 4.44, 5.04 -> not exponential
}
```

Gamma

Total time for several sequential steps.



Gamma: total time for several sequential steps.

The story	Add up several exponential waits and you get a gamma. Shape parameter = how many steps.
In eval work	End to end latency through a multi stage pipeline. Total time for an agent loop with several tool calls.
Use when	Use for sums of waiting times. With shape 1 it IS exponential; with large shape it approaches normal.
Avoid when	Needs the steps to be independent and similarly scaled. A pipeline with one dominant slow stage is not gamma, it is basically that stage's own distribution.

Under the hood

Gamma: formulas and code

$$f(x) = \frac{\lambda^k x^{k-1} e^{-\lambda x}}{\Gamma(k)}$$

$$E[X] = \frac{k}{\lambda} \qquad \mathrm{Var}(X) = \frac{k}{\lambda^2}$$

python

```python
from scipy import stats
import numpy as np

# Total time through a pipeline of k independent stages.
# Sum of k exponentials with the same rate IS a gamma with shape k.
rng = np.random.default_rng(0)
stages = rng.exponential(2.0, size=(5000, 3))     # 3 stages, mean 2s each
total = stages.sum(axis=1)

print(total.mean(), 3 * 2.0)                      # 6.0 - matches k/lambda
print(stats.gamma.fit(total, floc=0)[0])          # recovers shape ~3

# Shape 1 IS exponential; large shape approaches normal.
for k in (1, 3, 20):
    print(k, round(stats.skew(stats.gamma.rvs(k, size=20000, random_state=1)), 2))
# 1 -> 1.93 (very skewed),  3 -> 1.18,  20 -> 0.45 (nearly symmetric)
# theory, 2 / sqrt(k): 2.0, 1.15, 0.45
```

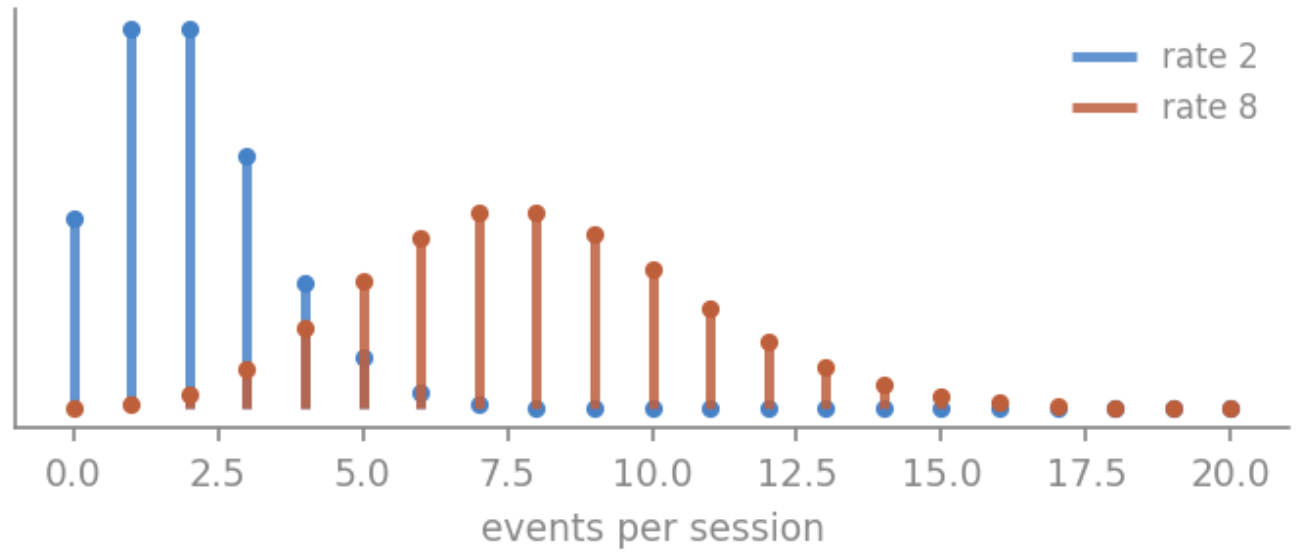
rust

```rust
// A tiny inline PRNG (SplitMix64) so the snippet needs no crates.
// Its draws differ from numpy's, so the random numbers below differ from the Python's.
struct Rng(u64);

impl Rng {
    fn new(seed: u64) -> Self {
        Rng(seed)
    }
    fn next_u64(&mut self) -> u64 {
        self.0 = self.0.wrapping_add(0x9E37_79B9_7F4A_7C15);
        let mut z = self.0;
        z = (z ^ (z >> 30)).wrapping_mul(0xBF58_476D_1CE4_E5B9);
        z = (z ^ (z >> 27)).wrapping_mul(0x94D0_49BB_1331_11EB);
        z ^ (z >> 31)
    }
    /// Uniform on [0, 1).
    fn random(&mut self) -> f64 {
        (self.next_u64() >> 11) as f64 / (1u64 << 53) as f64
    }
    /// Normal draw by Box-Muller.
    fn normal(&mut self, mean: f64, sd: f64) -> f64 {
        let u1 = 1.0 - self.random(); // in (0, 1], safe for ln
        let u2 = self.random();
        mean + sd * (-2.0 * u1.ln()).sqrt() * (std::f64::consts::TAU * u2).cos()
    }
}

impl Rng {
    /// Gamma(shape, scale) draw, Marsaglia-Tsang method (shape >= 1).
    fn gamma(&mut self, shape: f64, scale: f64) -> f64 {
        let d = shape - 1.0 / 3.0;
        let c = 1.0 / (9.0 * d).sqrt();
        loop {
            let z = self.normal(0.0, 1.0);
            let v = (1.0 + c * z).powi(3);
            if v > 0.0 && self.random().ln() < 0.5 * z * z + d - d * v + d * v.ln() {
                return d * v * scale;
            }
        }
    }
}

fn mean(x: &[f64]) -> f64 {
    x.iter().sum::<f64>() / x.len() as f64
}

/// Sample skewness, as scipy.stats.skew computes it: m3 / m2^1.5.
fn skew(x: &[f64]) -> f64 {
    let m = mean(x);
    let m2 = x.iter().map(|v| (v - m).powi(2)).sum::<f64>() / x.len() as f64;
    let m3 = x.iter().map(|v| (v - m).powi(3)).sum::<f64>() / x.len() as f64;
    m3 / m2.powf(1.5)
}

fn round(x: f64, digits: i32) -> f64 {
    let f = 10f64.powi(digits);
    (x * f).round() / f
}
```

```rust
fn digamma(mut x: f64) -> f64 {
    let mut r = 0.0;
    while x < 6.0 {
        r -= 1.0 / x; // psi(x) = psi(x + 1) - 1/x
        x += 1.0;
    }
    let f = 1.0 / (x * x); // asymptotic series for large x
    r + x.ln() - 0.5 / x - f * (1.0 / 12.0 - f * (1.0 / 120.0 - f * (1.0 / 252.0 - f / 240.0)))
}

/// Maximum likelihood gamma shape with loc fixed at 0:
/// solve ln(k) - digamma(k) = ln(mean) - mean(ln x). The left side falls as k grows.
fn gamma_fit_shape(x: &[f64]) -> f64 {
    let s = mean(x).ln() - x.iter().map(|v| v.ln()).sum::<f64>() / x.len() as f64;
    let (mut lo, mut hi) = (1e-3_f64, 1e3_f64);
    for _ in 0..200 {
        let mid = (lo * hi).sqrt();
        if mid.ln() - digamma(mid) > s { lo = mid } else { hi = mid }
    }
    (lo * hi).sqrt()
}

fn main() {
    // Total time through a pipeline of k independent stages.
    // Sum of k exponentials with the same rate IS a gamma with shape k.
    let mut rng = Rng::new(0);
    let total: Vec<f64> = (0..5000)
        .map(|_| (0..3).map(|_| -2.0 * (1.0 - rng.random()).ln()).sum()) // 3 stages, mean 2s each
        .collect();

    println!("{} {:?}", mean(&total), 3.0 * 2.0); // 5.97 - matches k/lambda
    println!("{}", gamma_fit_shape(&total));      // recovers shape ~3 (3.04)

    // Shape 1 IS exponential; large shape approaches normal.
    let mut rng = Rng::new(1);
    for k in [1, 3, 20] {
        let draws: Vec<f64> = (0..20000).map(|_| rng.gamma(k as f64, 1.0)).collect();
        println!("{} {}", k, round(skew(&draws), 2));
    }
    // 1 -> 1.93 (very skewed),  3 -> 1.13,  20 -> 0.48 (nearly symmetric)
    // theory, 2 / sqrt(k): 2.0, 1.15, 0.45
}
```

Poisson

Count of events in a fixed window, at a constant rate.



Poisson: count of events in a fixed window, at a constant rate.

The story	Events happen independently at a steady average rate; you count how many landed in one window. Its defining property: **variance equals the mean**.
In eval work	Requests per minute during steady traffic. Errors per hour when nothing unusual is happening.
Use when	Use when your variance/mean ratio comes out near 1.
Avoid when	**Check that ratio first, because it usually is not near 1.** Tool calls per session run 3 to 6, not 1. Assuming Poisson there makes your intervals roughly sqrt(ratio) times too narrow, and you will call noise a change.

● Under the hood

Poisson: formulas and code

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

$$E[X] = \lambda \qquad \mathrm{Var}(X) = \lambda \qquad \frac{\mathrm{Var}}{E} = 1$$

python

```python
from scipy import stats
import numpy as np

# THE defining property: variance equals the mean. Test this FIRST,
# before you use any Poisson based interval.
counts = stats.poisson.rvs(6, size=3000, random_state=0)
print(counts.mean(), counts.var(ddof=1))          # 5.90, 6.01 -> ratio 1.02

def dispersion_test(counts):
    m = np.mean(counts)
    v = np.var(counts, ddof=1)
    ratio = v / m if m else float("inf")
    return {"mean": m, "var": v, "ratio": ratio,
            "verdict": "Poisson ok" if ratio < 1.5 else "OVERDISPERSED"}

print(dispersion_test(counts))                    # Poisson ok

# Real agent data almost never passes this.
real = np.concatenate([stats.poisson.rvs(2, size=1000, random_state=1),
                       stats.poisson.rvs(15, size=1000, random_state=2)])
print(dispersion_test(real))                      # ratio ~5.9, OVERDISPERSED
```

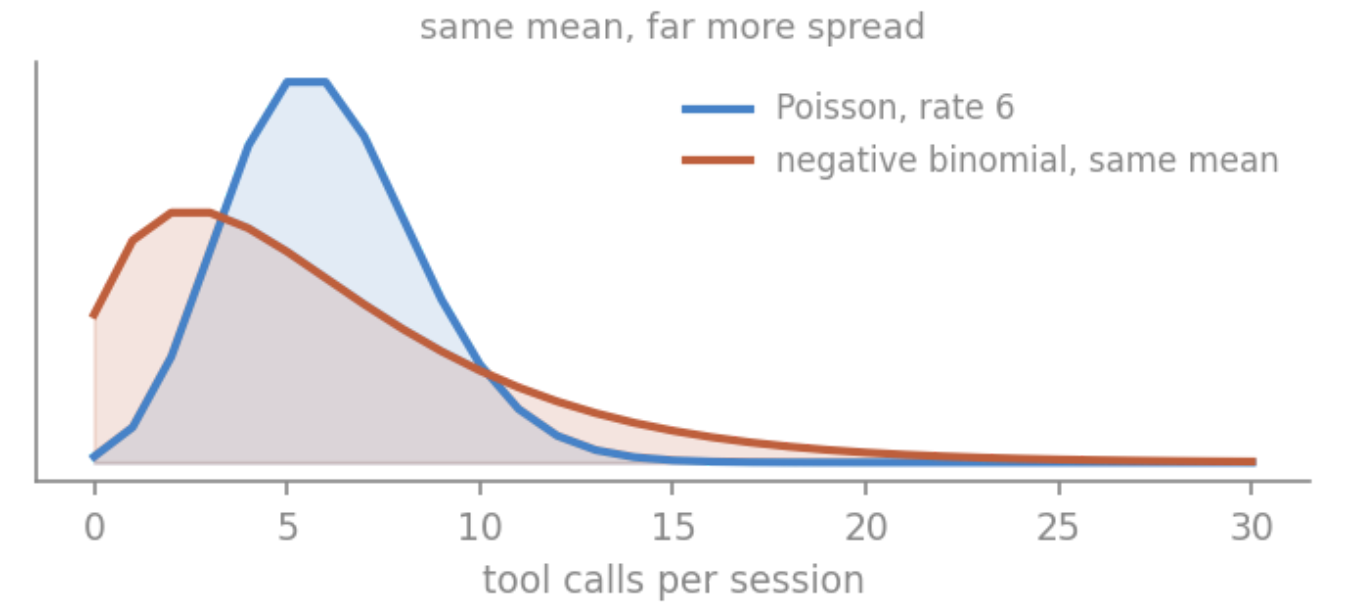
rust

```rust
// A tiny inline PRNG (SplitMix64) so the snippet needs no crates.
// Its draws differ from numpy's, so the random numbers below differ from the Python's.
struct Rng(u64);

impl Rng {
    fn new(seed: u64) -> Self {
        Rng(seed)
    }
    fn next_u64(&mut self) -> u64 {
        self.0 = self.0.wrapping_add(0x9E37_79B9_7F4A_7C15);
        let mut z = self.0;
        z = (z ^ (z >> 30)).wrapping_mul(0xBF58_476D_1CE4_E5B9);
        z = (z ^ (z >> 27)).wrapping_mul(0x94D0_49BB_1331_11EB);
        z ^ (z >> 31)
    }
    /// Uniform on [0, 1).
    fn random(&mut self) -> f64 {
        (self.next_u64() >> 11) as f64 / (1u64 << 53) as f64
    }
}
```

```rust
impl Rng {
    /// Poisson draw: multiply uniforms until the product drops below e^-lambda.
    fn poisson(&mut self, lambda: f64) -> u64 {
        let limit = (-lambda).exp();
        let (mut k, mut prod) = (0, self.random());
        while prod > limit {
            k += 1;
            prod *= self.random();
        }
        k
    }
}

#[derive(Debug)]
struct Dispersion {
    mean: f64,
    var: f64,
    ratio: f64,
    verdict: &'static str,
}

fn dispersion_test(counts: &[u64]) -> Dispersion {
    let n = counts.len() as f64;
    let m = counts.iter().sum::<u64>() as f64 / n;
    let v = counts.iter().map(|&c| (c as f64 - m).powi(2)).sum::<f64>() / (n - 1.0);
    let ratio = if m != 0.0 { v / m } else { f64::INFINITY };
    Dispersion { mean: m, var: v, ratio, verdict: if ratio < 1.5 { "Poisson ok" } else { "OVERDISPERSED" } }
}

fn main() {
    // THE defining property: variance equals the mean. Test this FIRST,
    // before you use any Poisson based interval.
    let mut rng = Rng::new(0);
    let counts: Vec<u64> = (0..3000).map(|_| rng.poisson(6.0)).collect();
    let d = dispersion_test(&counts);
    println!("{} {}", d.mean, d.var);         // 5.97, 5.73 -> ratio 0.96

    println!("{:?}", d);                      // Poisson ok

    // Real agent data almost never passes this.
    let mut real: Vec<u64> = (0..1000).map(|_| rng.poisson(2.0)).collect();
    real.extend((0..1000).map(|_| rng.poisson(15.0)));
    println!("{:?}", dispersion_test(&real)); // ratio ~6.0, OVERDISPERSED
}
```

Negative binomial

Counts when the rate itself varies.



Negative binomial: counts when the rate itself varies.

The story	A mixture of Poissons with different rates is a negative binomial. Easy tasks take two tool calls, hard ones take twenty. The rate is not constant across tasks, so the counts spread far wider than Poisson allows.
In eval work	**Tool calls per session. Turns per conversation. Retries per task.** Nearly every count in agent evaluation.
Use when	Use whenever variance/mean is well above 1, which for agent counts is essentially always.
Avoid when	It has one more parameter than Poisson, so it needs a bit more data. If you only need an interval rather than a model, bootstrapping quantiles is simpler and just as defensible.

Under the hood

Negative binomial: formulas and code

$$P(X = k) = \binom{k + r - 1}{k} (1 - p)^k p^r$$

$$E[X] = \mu \qquad \mathrm{Var}(X) = \mu + \frac{\mu^2}{r} \; > \; \mu$$

python

```python
from scipy import stats
import numpy as np

# A mixture of Poissons with DIFFERENT rates is a negative binomial.
# That is exactly what varied task difficulty produces.
rng = np.random.default_rng(0)
rates = rng.gamma(shape=2.0, scale=3.0, size=4000)   # difficulty varies
calls = rng.poisson(rates)                           # tool calls per session

m, v = calls.mean(), calls.var(ddof=1)
print(m, v, v / m)                    # 6.15, 25.6, ratio 4.16 -> not Poisson

# Fit: parameterise by mean and dispersion r. Var = mu + mu^2 / r.
mu = m
r = mu ** 2 / (v - mu)                # method of moments estimate
print("dispersion r:", r)             # 1.94, recovering the gamma shape of 2.0

# Compare what each model says about seeing 20+ calls in a session:
print("Poisson says: ", stats.poisson.sf(19, mu))            # 7.3e-06 (absurd)
print("NegBinom says:", stats.nbinom.sf(19, r, r / (r + mu)))  # 0.022 (real)
# The Poisson model would flag a routine session as a once-in-a-million event.
```

rust

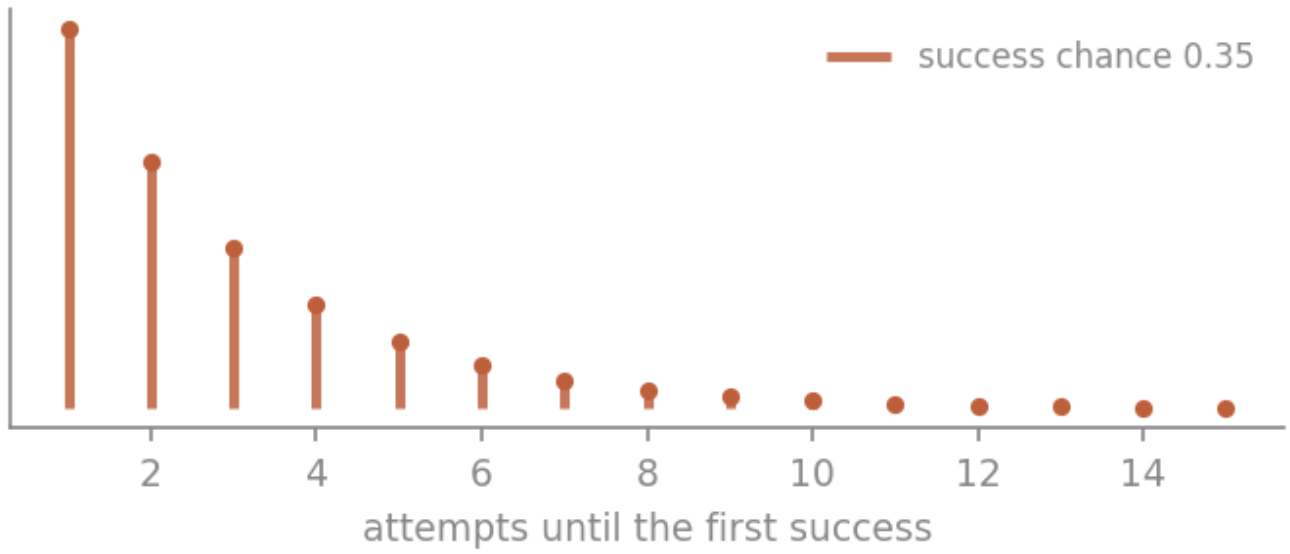
```rust
// A tiny inline PRNG (SplitMix64) so the snippet needs no crates.
// Its draws differ from numpy's, so the random numbers below differ from the Python's.
struct Rng(u64);

impl Rng {
    fn new(seed: u64) -> Self {
        Rng(seed)
    }
    fn next_u64(&mut self) -> u64 {
        self.0 = self.0.wrapping_add(0x9E37_79B9_7F4A_7C15);
        let mut z = self.0;
        z = (z ^ (z >> 30)).wrapping_mul(0xBF58_476D_1CE4_E5B9);
        z = (z ^ (z >> 27)).wrapping_mul(0x94D0_49BB_1331_11EB);
        z ^ (z >> 31)
    }
    /// Uniform on [0, 1).
    fn random(&mut self) -> f64 {
        (self.next_u64() >> 11) as f64 / (1u64 << 53) as f64
    }
    /// Normal draw by Box-Muller.
```

```
    fn normal(&mut self, mean: f64, sd: f64) -> f64 {
        let u1 = 1.0 - self.random(); // in (0, 1], safe for ln
        let u2 = self.random();
        mean + sd * (-2.0 * u1.ln()).sqrt() * (std::f64::consts::TAU * u2).cos()
    }
}

impl Rng {
    /// Gamma(shape, scale) draw, Marsaglia-Tsang method (shape >= 1).
    fn gamma(&mut self, shape: f64, scale: f64) -> f64 {
        let d = shape - 1.0 / 3.0;
        let c = 1.0 / (9.0 * d).sqrt();
        loop {
            let z = self.normal(0.0, 1.0);
            let v = (1.0 + c * z).powi(3);
            if v > 0.0 && self.random().ln() < 0.5 * z * z + d - d * v + d * v.ln() {
                return d * v * scale;
            }
        }
    }
}

impl Rng {
    /// Poisson draw: multiply uniforms until the product drops below e^-lambda.
    fn poisson(&mut self, lambda: f64) -> u64 {
        let limit = (-lambda).exp();
        let (mut k, mut prod) = (0, self.random());
        while prod > limit {
            k += 1;
            prod *= self.random();
        }
        k
    }
}

/// ln Gamma(x) by the Lanczos approximation (g = 7, 9 terms).
fn ln_gamma(x: f64) -> f64 {
    const G: [f64; 9] = [
        0.999_999_999_999_809_93, 676.520_368_121_885_1, -1259.139_216_722_402_8,
        771.323_428_777_653_13, -176.615_029_162_140_59, 12.507_343_278_686_905,
        -0.138_571_095_265_720_12, 9.984_369_578_019_571_6e-6, 1.505_632_735_149_311_6e-7,
    ];
    let pi = std::f64::consts::PI;
    if x < 0.5 {
        return (pi / (pi * x).sin()).ln() - ln_gamma(1.0 - x); // reflection
    }
    let x = x - 1.0;
    let t = x + 7.5;
    let a = G[0] + (1..9).map(|i| G[i] / (x + i as f64)).sum::<f64>();
    0.5 * (2.0 * pi).ln() + (x + 0.5) * t.ln() - t + a.ln()
}

fn poisson_pmf(k: u64, mu: f64) -> f64 {
    (k as f64 * mu.ln() - mu - ln_gamma(k as f64 + 1.0)).exp()
}

/// Negative binomial pmf: failures before the r-th success, success chance p.
fn nbinom_pmf(k: u64, r: f64, p: f64) -> f64 {
    let k = k as f64;
    (ln_gamma(k + r) - ln_gamma(r) - ln_gamma(k + 1.0) + r * p.ln() + k * (1.0 - p).ln()).exp()
```

```rust
}

/// Survival function P(X > k): add up the upper tail (far enough that the rest is negligible).
fn sf(pmf: impl Fn(u64) -> f64, k: u64) -> f64 {
    (k + 1..k + 2000).map(pmf).sum()
}

fn main() {
    // A mixture of Poissons with DIFFERENT rates is a negative binomial.
    // That is exactly what varied task difficulty produces.
    let mut rng = Rng::new(0);
    let rates: Vec<f64> = (0..4000).map(|_| rng.gamma(2.0, 3.0)).collect();  // difficulty varies
    let calls: Vec<f64> = rates.iter().map(|&r| rng.poisson(r) as f64).collect(); // tool calls per session

    let n = calls.len() as f64;
    let m = calls.iter().sum::<f64>() / n;
    let v = calls.iter().map(|c| (c - m).powi(2)).sum::<f64>() / (n - 1.0);
    println!("{} {} {}", m, v, v / m); // 5.94, 23.5, ratio 3.95 -> not Poisson

    // Fit: parameterise by mean and dispersion r. Var = mu + mu^2 / r.
    let mu = m;
    let r = mu * mu / (v - mu);        // method of moments estimate
    println!("dispersion r: {}", r);   // 2.01, recovering the gamma shape of 2.0

    // Compare what each model says about seeing 20+ calls in a session:
    println!("Poisson says:  {:e}", sf(|k| poisson_pmf(k, mu), 19));           // 4.4e-6 (absurd)
    println!("NegBinom says: {}", sf(|k| nbinom_pmf(k, r, r / (r + mu)), 19)); // 0.018 (real)
    // The Poisson model would flag a routine session as a once-in-a-million event.
}
```

Geometric

How many attempts until the first success.



Geometric: how many attempts until the first success.

The story	Keep retrying until something works; count the attempts. The discrete cousin of the exponential, and also memoryless.
In eval work	**Retry counts.** Attempts before a flaky tool call succeeds.
Use when	Use for attempts-until-success with a constant success chance.
Avoid when	Assumes the success chance does not change between attempts. If a retry is more likely to succeed because a cache warmed up, or less likely because the task is genuinely broken, this is the wrong model.

Under the hood

Geometric: formulas and code

$$P(X = k) = (1-p)^{k-1}p, \quad k = 1, 2, \ldots$$

$$E[X] = \frac{1}{p} \qquad \mathrm{Var}(X) = \frac{1-p}{p^2}$$

python

```python
from scipy import stats
import numpy as np

# Attempts until the first success. Directly models harness retries.
p_success = 0.7
attempts = stats.geom.rvs(p_success, size=5000, random_state=0)

print("mean attempts:", attempts.mean(), " theory:", 1 / p_success)   # 1.43

# What fraction of tasks need more than one attempt?
print((attempts > 1).mean())                    # 0.30, i.e. 1 - p

# Retry budget: what share of tasks are still unfinished after k attempts?
for k in (1, 2, 3, 4):
    print(k, "attempts leaves", round((1 - p_success) ** k, 4), "unfinished")
# 4 attempts leaves 0.8% - that residue is your silently dropped tasks.

# ASSUMPTION CHECK: this needs a constant success chance per attempt.
# If a retry succeeds because a cache warmed up, the model does not hold.
```

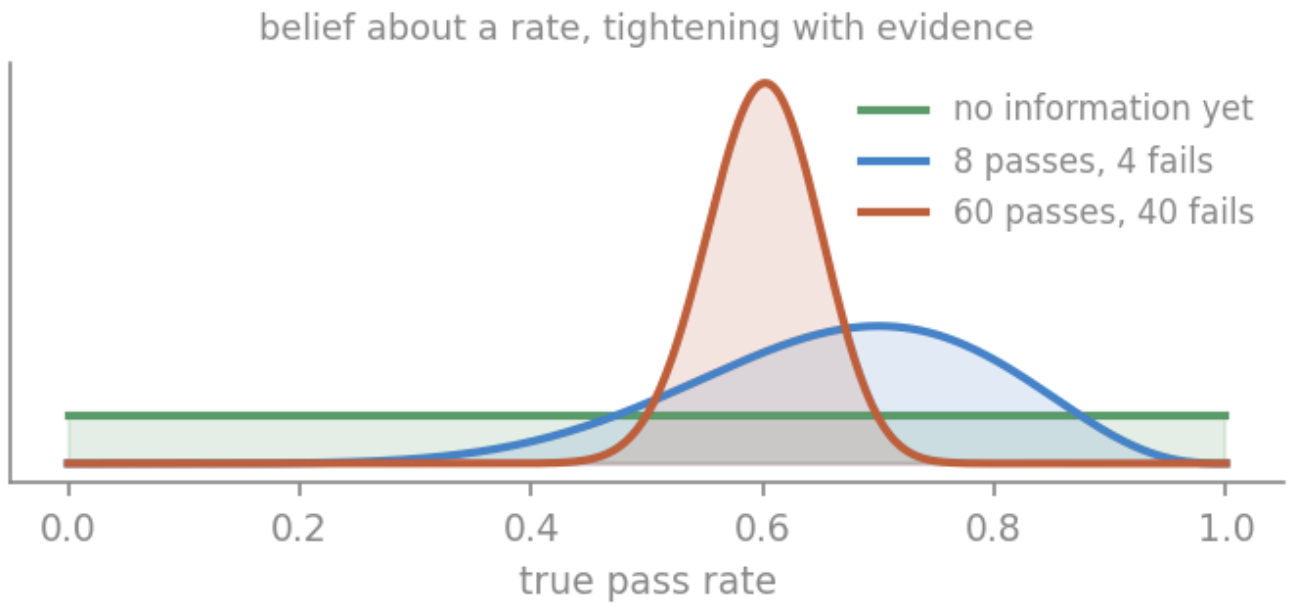
rust

```rust
// A tiny inline PRNG (SplitMix64) so the snippet needs no crates.
// Its draws differ from numpy's, so the random numbers below differ from the Python's.
struct Rng(u64);

impl Rng {
    fn new(seed: u64) -> Self {
        Rng(seed)
    }
    fn next_u64(&mut self) -> u64 {
        self.0 = self.0.wrapping_add(0x9E37_79B9_7F4A_7C15);
        let mut z = self.0;
        z = (z ^ (z >> 30)).wrapping_mul(0xBF58_476D_1CE4_E5B9);
        z = (z ^ (z >> 27)).wrapping_mul(0x94D0_49BB_1331_11EB);
        z ^ (z >> 31)
    }
    /// Uniform on [0, 1).
    fn random(&mut self) -> f64 {
        (self.next_u64() >> 11) as f64 / (1u64 << 53) as f64
    }
}

fn round(x: f64, digits: i32) -> f64 {
    let f = 10f64.powi(digits);
    (x * f).round() / f
}

fn main() {
    // Attempts until the first success. Directly models harness retries.
    let p_success = 0.7;
    let mut rng = Rng::new(0);
    let attempts: Vec<u32> = (0..5000)
        .map(|_| {
            let mut k = 1;
            while rng.random() >= p_success {
                k += 1;
            }
```

```
            k
        })
        .collect();

    let n = attempts.len() as f64;
    let mean = attempts.iter().sum::<u32>() as f64 / n;
    println!("mean attempts: {}  theory: {}", mean, 1.0 / p_success); // 1.428 vs 1.43

    // What fraction of tasks need more than one attempt?
    println!("{}", attempts.iter().filter(|&&a| a > 1).count() as f64 / n); // 0.30, i.e. 1 - p

    // Retry budget: what share of tasks are still unfinished after k attempts?
    for k in 1..=4 {
        println!("{} attempts leaves {} unfinished", k, round((1.0 - p_success).powi(k), 4));
    }
    // 4 attempts leaves 0.8% - that residue is your silently dropped tasks.

    // ASSUMPTION CHECK: this needs a constant success chance per attempt.
    // If a retry succeeds because a cache warmed up, the model does not hold.
}
```

Beta

A distribution over a RATE you are uncertain about.



Beta: a distribution over a RATE you are uncertain about.

The story	Everything else here describes data. Beta describes your belief about a probability. After seeing 8 passes and 4 fails, Beta(8,4) is the shape of your uncertainty about the true pass rate. More evidence, tighter shape.
In eval work	Uncertainty about a pass rate. Priors when you have very few tasks. Smoothing a rate estimated from a handful of observations.
Use when	Use for reasoning about rates, especially with small samples. It is the natural partner to the binomial.
Avoid when	It is a belief, not a measurement. Do not report a Beta as if it described your data, and be explicit about any prior you chose, because with small n the prior is doing real work.

Under the hood

Beta: formulas and code

$$f(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}, \quad 0 \le x \le 1$$

$$E[X] = \frac{\alpha}{\alpha+\beta} \qquad \mathrm{Var}(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

python

```python
from scipy import stats

# Beta describes your BELIEF about a rate, not the data itself.
# Start with no information, then add evidence.
passes, fails = 8, 4
post = stats.beta(1 + passes, 1 + fails)          # uniform prior + evidence

print("best guess:", post.mean())                 # 0.643
print("95% credible interval:", post.interval(0.95))   # (0.386, 0.861)

# With only 12 tasks the interval is enormous. Watch it tighten:
for n in (12, 50, 200, 1000):
    a = 1 + round(0.65 * n)
    b = 1 + n - round(0.65 * n)
    lo, hi = stats.beta(a, b).interval(0.95)
    print(n, "tasks ->", round(hi - lo, 3), "wide")
# 12 -> 0.476,  200 -> 0.131,  1000 -> 0.059
```

```
# Probability model A truly beats a 0.60 baseline, given the data:
print(1 - stats.beta(1 + 8, 1 + 4).cdf(0.60))     # 0.647 - genuinely unsure
```

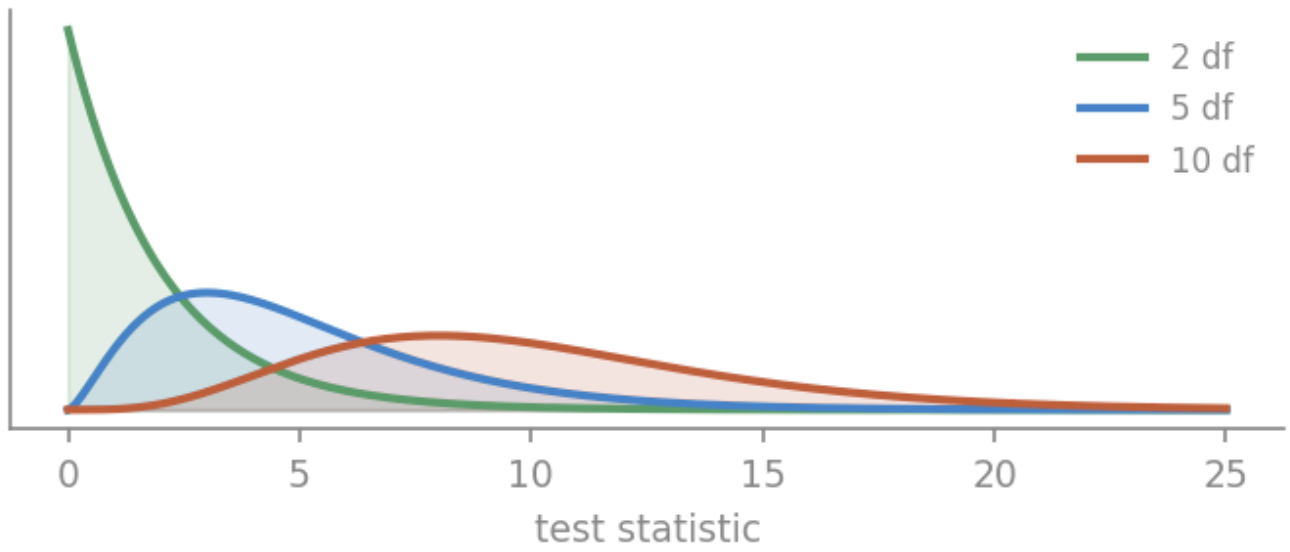
rust

```
/// ln Gamma(x) by the Lanczos approximation (g = 7, 9 terms).
fn ln_gamma(x: f64) -> f64 {
    const G: [f64; 9] = [
        0.999_999_999_999_809_93, 676.520_368_121_885_1, -1259.139_216_722_402_8,
        771.323_428_777_653_13, -176.615_029_162_140_59, 12.507_343_278_686_905,
        -0.138_571_095_265_720_12, 9.984_369_578_019_571_6e-6, 1.505_632_735_149_311_6e-7,
    ];
    let pi = std::f64::consts::PI;
    if x < 0.5 {
        return (pi / (pi * x).sin()).ln() - ln_gamma(1.0 - x); // reflection
    }
    let x = x - 1.0;
    let t = x + 7.5;
    let a = G[0] + (1..9).map(|i| G[i] / (x + i as f64)).sum::<f64>();
    0.5 * (2.0 * pi).ln() + (x + 0.5) * t.ln() - t + a.ln()
}

/// Regularized incomplete beta I_x(a, b), by continued fraction (modified Lentz).
fn beta_inc(a: f64, b: f64, x: f64) -> f64 {
    if x <= 0.0 {
        return 0.0;
    }
    if x >= 1.0 {
        return 1.0;
    }
    let front = (ln_gamma(a + b) - ln_gamma(a) - ln_gamma(b) + a * x.ln() + b * (1.0 - x).ln()).exp();
    // the fraction converges fast on one side of the mean; use symmetry for the other
    if x < (a + 1.0) / (a + b + 2.0) {
        front * beta_cf(a, b, x) / a
    } else {
        1.0 - front * beta_cf(b, a, 1.0 - x) / b
    }
}

fn beta_cf(a: f64, b: f64, x: f64) -> f64 {
    let tiny = 1e-300;
    let clamp = |v: f64| if v.abs() < tiny { tiny } else { v };
    let mut c = 1.0;
    let mut d = 1.0 / clamp(1.0 - (a + b) * x / (a + 1.0));
    let mut h = d;
    for m in 1..1000 {
        let m = m as f64;
        let even = m * (b - m) * x / ((a + 2.0 * m - 1.0) * (a + 2.0 * m));
        d = 1.0 / clamp(1.0 + even * d);
        c = clamp(1.0 + even / c);
        h *= d * c;
        let odd = -(a + m) * (a + b + m) * x / ((a + 2.0 * m) * (a + 2.0 * m + 1.0));
        d = 1.0 / clamp(1.0 + odd * d);
        c = clamp(1.0 + odd / c);
        h *= d * c;
        if (d * c - 1.0).abs() < 1e-16 {
            break;
        }
    }
```

```rust
    h
}

fn round(x: f64, digits: i32) -> f64 {
    let f = 10f64.powi(digits);
    (x * f).round() / f
}

struct Beta {
    a: f64,
    b: f64,
}

impl Beta {
    fn mean(&self) -> f64 {
        self.a / (self.a + self.b)
    }
    fn cdf(&self, x: f64) -> f64 {
        beta_inc(self.a, self.b, x)
    }
    /// Inverse CDF by bisection: the CDF only goes up, so halve the bracket.
    fn ppf(&self, p: f64) -> f64 {
        let (mut lo, mut hi) = (0.0, 1.0);
        for _ in 0..100 {
            let mid = 0.5 * (lo + hi);
            if self.cdf(mid) < p { lo = mid } else { hi = mid }
        }
        0.5 * (lo + hi)
    }
    /// Central interval holding `level` of the probability.
    fn interval(&self, level: f64) -> (f64, f64) {
        (self.ppf((1.0 - level) / 2.0), self.ppf((1.0 + level) / 2.0))
    }
}

fn main() {
    // Beta describes your BELIEF about a rate, not the data itself.
    // Start with no information, then add evidence.
    let (passes, fails) = (8.0, 4.0);
    let post = Beta { a: 1.0 + passes, b: 1.0 + fails }; // uniform prior + evidence

    println!("best guess: {}", post.mean());                        // 0.643
    println!("95% credible interval: {:?}", post.interval(0.95));  // (0.386, 0.861)

    // With only 12 tasks the interval is enormous. Watch it tighten:
    for n in [12, 50, 200, 1000] {
        let k = (0.65 * n as f64).round_ties_even(); // Python's round() sends halves to even
        let (lo, hi) = Beta { a: 1.0 + k, b: 1.0 + n as f64 - k }.interval(0.95);
        println!("{} tasks -> {} wide", n, round(hi - lo, 3));
    }
    // 12 -> 0.476,  200 -> 0.131,  1000 -> 0.059

    // Probability model A truly beats a 0.60 baseline, given the data:
    println!("{}", 1.0 - Beta { a: 1.0 + 8.0, b: 1.0 + 4.0 }.cdf(0.60)); // 0.647 - genuinely unsure
}
```

Chi square

Sum of squared normal deviations.



Chi square: sum of squared normal deviations.

The story	Add up squared standardised differences. You rarely measure something chi square distributed; you compute a test statistic that follows it.
In eval work	The distribution behind chi square tests of independence, Cochran's Q, and variance tests.
Use when	Use as the reference distribution for those tests.
Avoid when	Its approximations get unreliable when expected cell counts fall below about 5. On small eval sets prefer exact tests or a permutation test, which is why the exact McNemar and permutation approaches are worth having.

Under the hood

Chi square: formulas and code

$$X = \sum_{i=1}^{k} Z_i^2, \quad Z_i \sim \mathcal{N}(0,1)$$

$$E[X] = k \qquad \mathrm{Var}(X) = 2k$$

python

```python
from scipy import stats
import numpy as np

# You rarely MEASURE something chi square. You compute a statistic that follows it.
# Example: are failure reasons distributed the same across two models?
observed = np.array([[42, 31, 27],       # model A: timeout / wrong / crash
                     [18, 55, 29]])      # model B
chi2, p, dof, expected = stats.chi2_contingency(observed)
print(chi2, p, dof)                      # 16.4, 2.8e-04, 2

# THE ASSUMPTION nobody checks: every expected count should exceed ~5.
print(expected.min())                    # 27.7 -> fine here
# If it is below 5, the chi square approximation is unreliable.
# Use an exact test or a permutation test instead:
print(stats.fisher_exact(observed[:, :2]))
```

rust

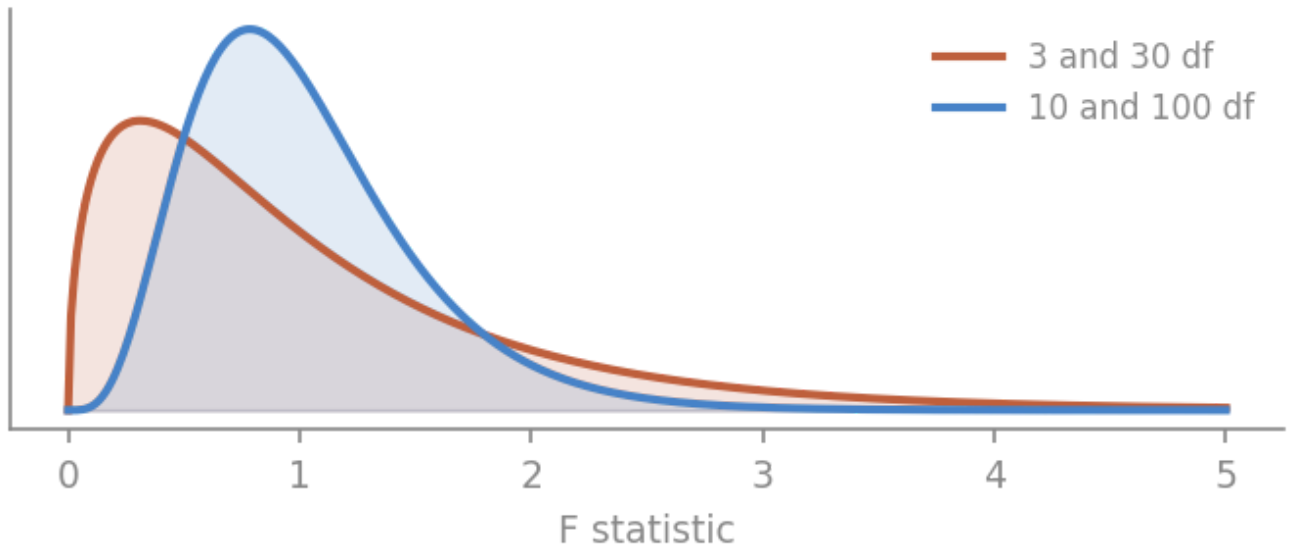
```rust
/// ln Gamma(x) by the Lanczos approximation (g = 7, 9 terms).
fn ln_gamma(x: f64) -> f64 {
    const G: [f64; 9] = [
        0.999_999_999_999_809_93, 676.520_368_121_885_1, -1259.139_216_722_402_8,
        771.323_428_777_653_13, -176.615_029_162_140_59, 12.507_343_278_686_905,
        -0.138_571_095_265_720_12, 9.984_369_578_019_571_6e-6, 1.505_632_735_149_311_6e-7,
    ];
    let pi = std::f64::consts::PI;
    if x < 0.5 {
        return (pi / (pi * x).sin()).ln() - ln_gamma(1.0 - x); // reflection
    }
    let x = x - 1.0;
    let t = x + 7.5;
    let a = G[0] + (1..9).map(|i| G[i] / (x + i as f64)).sum::<f64>();
    0.5 * (2.0 * pi).ln() + (x + 0.5) * t.ln() - t + a.ln()
}

/// Regularized incomplete gamma functions (P, Q) = (lower, upper), P + Q = 1.
fn gamma_pq(a: f64, x: f64) -> (f64, f64) {
    if x <= 0.0 {
        return (0.0, 1.0);
    }
    let front = (-x + a * x.ln() - ln_gamma(a)).exp();
    if x < a + 1.0 {
        // power series converges fast here
        let (mut ap, mut term) = (a, 1.0 / a);
```

```
        let mut sum = term;
        while term.abs() > sum.abs() * 1e-16 {
            ap += 1.0;
            term *= x / ap;
            sum += term;
        }
        let p = sum * front;
        (p, 1.0 - p)
    } else {
        // continued fraction (modified Lentz) converges fast here
        let tiny = 1e-300;
        let (mut b, mut c) = (x + 1.0 - a, 1.0 / tiny);
        let mut d = 1.0 / b;
        let mut h = d;
        for i in 1..1000 {
            let an = -(i as f64) * (i as f64 - a);
            b += 2.0;
            d = an * d + b;
            if d.abs() < tiny { d = tiny; }
            c = b + an / c;
            if c.abs() < tiny { c = tiny; }
            d = 1.0 / d;
            h *= d * c;
            if (d * c - 1.0).abs() < 1e-16 { break; }
        }
        let q = front * h;
        (1.0 - q, q)
    }
}

fn ln_choose(n: u64, k: u64) -> f64 {
    ln_gamma(n as f64 + 1.0) - ln_gamma(k as f64 + 1.0) - ln_gamma((n - k) as f64 + 1.0)
}

/// Chi square test of independence on an r x c table of counts.
/// Returns (statistic, p value, degrees of freedom, expected counts).
fn chi2_contingency(observed: &[Vec<f64>]) -> (f64, f64, usize, Vec<Vec<f64>>) {
    let row: Vec<f64> = observed.iter().map(|r| r.iter().sum()).collect();
    let col: Vec<f64> = (0..observed[0].len()).map(|j| observed.iter().map(|r| r[j]).sum()).collect();
    let total: f64 = row.iter().sum();
    let expected: Vec<Vec<f64>> = row.iter().map(|ri| col.iter().map(|cj| ri * cj / total).collect()).collect();
    let chi2: f64 = observed.iter().flatten().zip(expected.iter().flatten())
        .map(|(o, e)| (o - e).powi(2) / e)
        .sum();
    let dof = (row.len() - 1) * (col.len() - 1);
    let p = gamma_pq(dof as f64 / 2.0, chi2 / 2.0).1; // chi square upper tail
    (chi2, p, dof, expected)
}

#[derive(Debug)]
struct SignificanceResult {
    statistic: f64, // the sample odds ratio
    pvalue: f64,
}

/// Two-sided Fisher exact test on [[a, b], [c, d]]: add up every table with the
/// same margins that is no more likely than the one observed.
fn fisher_exact(t: [[u64; 2]; 2]) -> SignificanceResult {
    let [[a, b], [c, d]] = t;
    let (row1, col1, n) = (a + b, a + c, a + b + c + d);
```

```
    let ln_pmf = |x: u64| ln_choose(row1, x) + ln_choose(n - row1, col1 - x) - ln_choose(n, col1);
    let cutoff = ln_pmf(a) + 1e-7_f64.ln_1p(); // small tolerance for rounding
    let lo = col1.saturating_sub(n - row1);
    let hi = row1.min(col1);
    let p: f64 = (lo..=hi).map(ln_pmf).filter(|&l| l <= cutoff).map(f64::exp).sum();
    SignificanceResult { statistic: (a * d) as f64 / (b * c) as f64, pvalue: p.min(1.0) }
}

fn main() {
    // You rarely MEASURE something chi square. You compute a statistic that follows it.
    // Example: are failure reasons distributed the same across two models?
    let observed = vec![vec![42.0, 31.0, 27.0],  // model A: timeout / wrong / crash
                        vec![18.0, 55.0, 29.0]]; // model B
    let (chi2, p, dof, expected) = chi2_contingency(&observed);
    println!("{} {} {}", chi2, p, dof); // 16.4, 2.8e-04, 2

    // THE ASSUMPTION nobody checks: every expected count should exceed ~5.
    let min_expected = expected.iter().flatten().cloned().fold(f64::INFINITY, f64::min);
    println!("{}", min_expected);       // 27.7 -> fine here
    // If it is below 5, the chi square approximation is unreliable.
    // Use an exact test or a permutation test instead:
    println!("{:?}", fisher_exact([[42, 31], [18, 55]]));
}
```

F

Ratio of two variances.



F: ratio of two variances.

The story	Signal variance over noise variance. Large F means the differences between groups are big relative to the scatter inside them.
In eval work	The distribution behind ANOVA.
Use when	Use as the reference for ANOVA.
Avoid when	Sensitive to non-normality and unequal variances. On skewed eval data, prefer Friedman or a permutation test.

Under the hood

F: formulas and code

$$F = \frac{\chi^2_{d_1}/d_1}{\chi^2_{d_2}/d_2} = \frac{\mathrm{MS}_{\text{between}}}{\mathrm{MS}_{\text{within}}}$$

$$E[F] = \frac{d_2}{d_2 - 2} \quad (d_2 > 2)$$

python

```python
from scipy import stats
import numpy as np

# F is a ratio of variances: between-group signal over within-group noise.
rng = np.random.default_rng(0)
config_a = rng.normal(2.0, 0.4, 60)      # latency under three serving configs
config_b = rng.normal(2.1, 0.4, 60)
config_c = rng.normal(2.6, 0.4, 60)

F, p = stats.f_oneway(config_a, config_b, config_c)
print(F, p)                              # 32.8, 7.5e-13

# ALWAYS report the effect size next to F. Eta squared = share of variance explained.
groups = [config_a, config_b, config_c]
allv = np.concatenate(groups)
ss_between = sum(len(g) * (g.mean() - allv.mean()) ** 2 for g in groups)
ss_total = ((allv - allv.mean()) ** 2).sum()
print("eta squared:", ss_between / ss_total)     # 0.27

# On skewed data (latency usually is), prefer the rank based version:
print(stats.kruskal(config_a, config_b, config_c))
```

rust

```rust
// A tiny inline PRNG (SplitMix64) so the snippet needs no crates.
// Its draws differ from numpy's, so the random numbers below differ from the Python's.
struct Rng(u64);

impl Rng {
    fn new(seed: u64) -> Self {
        Rng(seed)
    }
    fn next_u64(&mut self) -> u64 {
        self.0 = self.0.wrapping_add(0x9E37_79B9_7F4A_7C15);
        let mut z = self.0;
        z = (z ^ (z >> 30)).wrapping_mul(0xBF58_476D_1CE4_E5B9);
        z = (z ^ (z >> 27)).wrapping_mul(0x94D0_49BB_1331_11EB);
        z ^ (z >> 31)
    }
    /// Uniform on [0, 1).
    fn random(&mut self) -> f64 {
        (self.next_u64() >> 11) as f64 / (1u64 << 53) as f64
    }
    /// Normal draw by Box-Muller.
    fn normal(&mut self, mean: f64, sd: f64) -> f64 {
        let u1 = 1.0 - self.random(); // in (0, 1], safe for ln
        let u2 = self.random();
        mean + sd * (-2.0 * u1.ln()).sqrt() * (std::f64::consts::TAU * u2).cos()
    }
}

/// ln Gamma(x) by the Lanczos approximation (g = 7, 9 terms).
fn ln_gamma(x: f64) -> f64 {
    const G: [f64; 9] = [
        0.999_999_999_999_809_93, 676.520_368_121_885_1, -1259.139_216_722_402_8,
        771.323_428_777_653_13, -176.615_029_162_140_59, 12.507_343_278_686_905,
        -0.138_571_095_265_720_12, 9.984_369_578_019_571_6e-6, 1.505_632_735_149_311_6e-7,
    ];
    let pi = std::f64::consts::PI;
    if x < 0.5 {
        return (pi / (pi * x).sin()).ln() - ln_gamma(1.0 - x); // reflection
    }
    let x = x - 1.0;
    let t = x + 7.5;
    let a = G[0] + (1..9).map(|i| G[i] / (x + i as f64)).sum::<f64>();
    0.5 * (2.0 * pi).ln() + (x + 0.5) * t.ln() - t + a.ln()
}

/// Regularized incomplete gamma functions (P, Q) = (lower, upper), P + Q = 1.
fn gamma_pq(a: f64, x: f64) -> (f64, f64) {
    if x <= 0.0 {
        return (0.0, 1.0);
    }
    let front = (-x + a * x.ln() - ln_gamma(a)).exp();
    if x < a + 1.0 {
        // power series converges fast here
        let (mut ap, mut term) = (a, 1.0 / a);
        let mut sum = term;
        while term.abs() > sum.abs() * 1e-16 {
            ap += 1.0;
            term *= x / ap;
            sum += term;
```

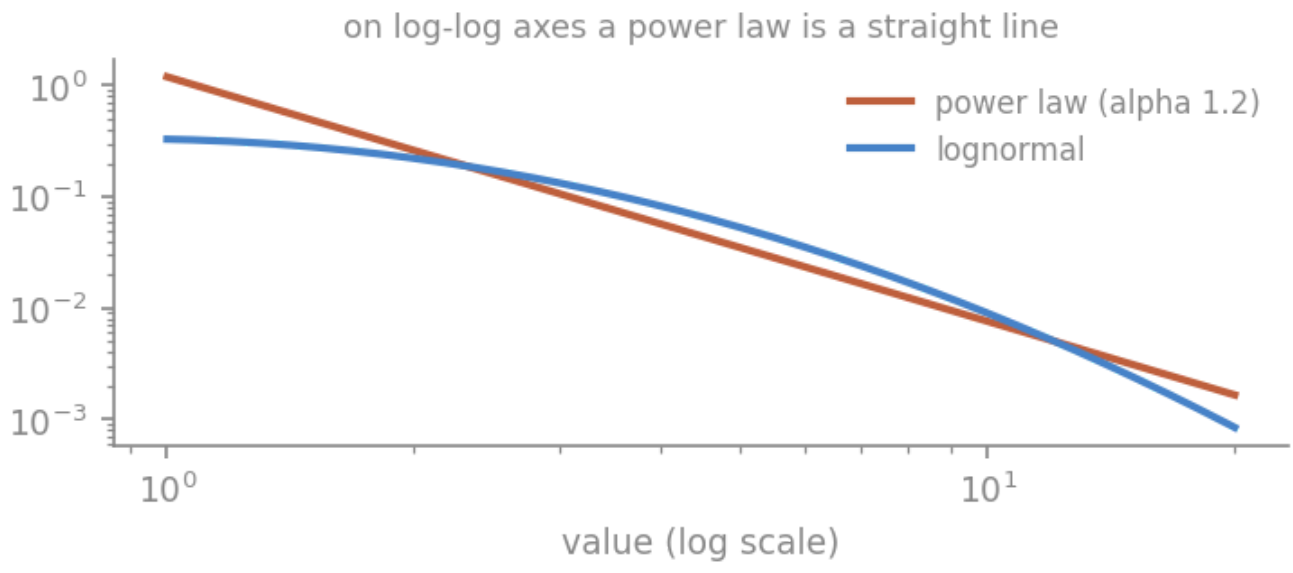
```
        }
        let p = sum * front;
        (p, 1.0 - p)
    } else {
        // continued fraction (modified Lentz) converges fast here
        let tiny = 1e-300;
        let (mut b, mut c) = (x + 1.0 - a, 1.0 / tiny);
        let mut d = 1.0 / b;
        let mut h = d;
        for i in 1..1000 {
            let an = -(i as f64) * (i as f64 - a);
            b += 2.0;
            d = an * d + b;
            if d.abs() < tiny { d = tiny; }
            c = b + an / c;
            if c.abs() < tiny { c = tiny; }
            d = 1.0 / d;
            h *= d * c;
            if (d * c - 1.0).abs() < 1e-16 { break; }
        }
        let q = front * h;
        (1.0 - q, q)
    }
}

/// Regularized incomplete beta I_x(a, b), by continued fraction (modified Lentz).
fn beta_inc(a: f64, b: f64, x: f64) -> f64 {
    if x <= 0.0 {
        return 0.0;
    }
    if x >= 1.0 {
        return 1.0;
    }
    let front = (ln_gamma(a + b) - ln_gamma(a) - ln_gamma(b) + a * x.ln() + b * (1.0 - x).ln()).exp();
    // the fraction converges fast on one side of the mean; use symmetry for the other
    if x < (a + 1.0) / (a + b + 2.0) {
        front * beta_cf(a, b, x) / a
    } else {
        1.0 - front * beta_cf(b, a, 1.0 - x) / b
    }
}

fn beta_cf(a: f64, b: f64, x: f64) -> f64 {
    let tiny = 1e-300;
    let clamp = |v: f64| if v.abs() < tiny { tiny } else { v };
    let mut c = 1.0;
    let mut d = 1.0 / clamp(1.0 - (a + b) * x / (a + 1.0));
    let mut h = d;
    for m in 1..1000 {
        let m = m as f64;
        let even = m * (b - m) * x / ((a + 2.0 * m - 1.0) * (a + 2.0 * m));
        d = 1.0 / clamp(1.0 + even * d);
        c = clamp(1.0 + even / c);
        h *= d * c;
        let odd = -(a + m) * (a + b + m) * x / ((a + 2.0 * m) * (a + 2.0 * m + 1.0));
        d = 1.0 / clamp(1.0 + odd * d);
        c = clamp(1.0 + odd / c);
        h *= d * c;
        if (d * c - 1.0).abs() < 1e-16 {
            break;
```

```
        }
    }
    h
}

fn mean(x: &[f64]) -> f64 {
    x.iter().sum::<f64>() / x.len() as f64
}

/// One-way ANOVA. Returns (F, p value).
fn f_oneway(groups: &[&[f64]]) -> (f64, f64) {
    let all: Vec<f64> = groups.concat();
    let grand = mean(&all);
    let ss_between: f64 = groups.iter().map(|g| g.len() as f64 * (mean(g) - grand).powi(2)).sum();
    let ss_within: f64 = groups.iter()
        .map(|g| { let m = mean(g); g.iter().map(|v| (v - m).powi(2)).sum::<f64>() })
        .sum();
    let (df_b, df_w) = ((groups.len() - 1) as f64, (all.len() - groups.len()) as f64);
    let f = (ss_between / df_b) / (ss_within / df_w);
    let p = beta_inc(df_w / 2.0, df_b / 2.0, df_w / (df_w + df_b * f)); // F upper tail
    (f, p)
}

#[derive(Debug)]
struct KruskalResult {
    statistic: f64,
    pvalue: f64,
}

/// Kruskal-Wallis H test: ANOVA on ranks, so skew and outliers do not matter.
fn kruskal(groups: &[&[f64]]) -> KruskalResult {
    // pool, sort, and give tied values their average rank
    let mut pooled: Vec<(f64, usize)> = groups.iter().enumerate()
        .flat_map(|(gi, g)| g.iter().map(move |&v| (v, gi)))
        .collect();
    pooled.sort_by(|a, b| a.0.total_cmp(&b.0));
    let n = pooled.len() as f64;
    let mut rank_sum = vec![0.0; groups.len()];
    let mut ties = 0.0; // for the tie correction
    let mut i = 0;
    while i < pooled.len() {
        let mut j = i;
        while j + 1 < pooled.len() && pooled[j + 1].0 == pooled[i].0 {
            j += 1;
        }
        let avg_rank = (i + j) as f64 / 2.0 + 1.0;
        for item in &pooled[i..=j] {
            rank_sum[item.1] += avg_rank;
        }
        let t = (j - i + 1) as f64;
        ties += t * t * t - t;
        i = j + 1;
    }
    let h: f64 = 12.0 / (n * (n + 1.0))
        * groups.iter().zip(&rank_sum).map(|(g, r)| r * r / g.len() as f64).sum::<f64>()
        - 3.0 * (n + 1.0);
    let h = h / (1.0 - ties / (n * n * n - n));
    let df = (groups.len() - 1) as f64;
    KruskalResult { statistic: h, pvalue: gamma_pq(df / 2.0, h / 2.0).1 } // chi square upper tail
}
```

```rust
fn main() {
    // F is a ratio of variances: between-group signal over within-group noise.
    let mut rng = Rng::new(0);
    let config_a: Vec<f64> = (0..60).map(|_| rng.normal(2.0, 0.4)).collect(); // latency under three serving
configs
    let config_b: Vec<f64> = (0..60).map(|_| rng.normal(2.1, 0.4)).collect();
    let config_c: Vec<f64> = (0..60).map(|_| rng.normal(2.6, 0.4)).collect();
    let groups: [&[f64]; 3] = [&config_a, &config_b, &config_c];

    let (f, p) = f_oneway(&groups);
    println!("{} {:e}", f, p); // 38.9, 1e-14 (Python's draws: 32.8, 7.5e-13)

    // ALWAYS report the effect size next to F. Eta squared = share of variance explained.
    let allv: Vec<f64> = groups.concat();
    let grand = mean(&allv);
    let ss_between: f64 = groups.iter().map(|g| g.len() as f64 * (mean(g) - grand).powi(2)).sum();
    let ss_total: f64 = allv.iter().map(|v| (v - grand).powi(2)).sum();
    println!("eta squared: {}", ss_between / ss_total); // 0.31

    // On skewed data (latency usually is), prefer the rank based version:
    println!("{:?}", kruskal(&groups));
    // with these draws: statistic 53.9, pvalue 2.0e-12
}
```

Pareto / power law

Scale free: no typical size.



Pareto / power law: scale free: no typical size.

The story	The heaviest tail you will meet in practice. Doubling the value divides its frequency by a constant, which produces a straight line on log-log axes. Depending on the tail index, the mean and even the variance may not converge.
In eval work	Repo sizes. File sizes. Occasionally session lengths in a long tailed product.
Use when	Use when a log-log plot is straight and the tail refuses to settle down.
Avoid when	**Never report a mean for a genuine power law.** It may not exist, and your sample mean will keep drifting as you collect more data. Percentiles and log scales only. Distinguish carefully from lognormal, which looks similar but is much better behaved.

Under the hood

Pareto / power law: formulas and code

$$P(X > x) = \left(\frac{x_m}{x}\right)^{\alpha}$$

$$E[X] = \frac{\alpha x_m}{\alpha - 1} \quad (\alpha > 1), \quad \text{else undefined}$$

python

```python
import numpy as np
from scipy import stats

# The test that separates a power law from a lognormal: does the MEAN settle?
heavy = stats.pareto.rvs(1.2, size=200000, random_state=0)     # power law
mild = stats.lognorm.rvs(1.0, scale=3, size=200000, random_state=0)

for n in (1000, 10000, 100000, 200000):
    print(n, "power law mean:", round(heavy[:n].mean(), 1),
             " lognormal mean:", round(mild[:n].mean(), 2))
# lognormal: 4.67 -> 4.81 -> 4.94 -> 4.95, settling down.
# power law: 5.7 -> 6.7 -> 5.4 -> 5.9, still wandering at 200,000 samples.

# Estimate the tail index. alpha <= 1 means the mean does not exist at all.
def hill_estimator(x, k=500):
    s = np.sort(x)[::-1][:k + 1]
    return 1.0 / np.mean(np.log(s[:k] / s[k]))
```

```
print("alpha:", hill_estimator(heavy))     # 1.135, close to the true 1.2
# alpha < 2 -> infinite variance. Never report a mean or an sd.
```

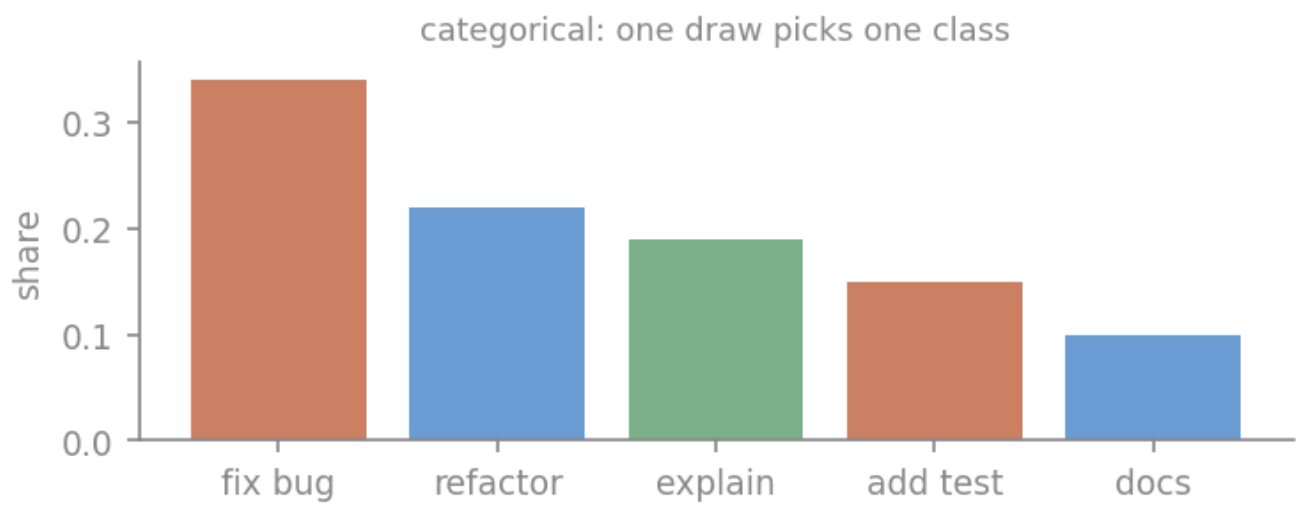
rust

```
// A tiny inline PRNG (SplitMix64) so the snippet needs no crates.
// Its draws differ from numpy's, so the random numbers below differ from the Python's.
struct Rng(u64);

impl Rng {
    fn new(seed: u64) -> Self {
        Rng(seed)
    }
    fn next_u64(&mut self) -> u64 {
        self.0 = self.0.wrapping_add(0x9E37_79B9_7F4A_7C15);
        let mut z = self.0;
        z = (z ^ (z >> 30)).wrapping_mul(0xBF58_476D_1CE4_E5B9);
        z = (z ^ (z >> 27)).wrapping_mul(0x94D0_49BB_1331_11EB);
        z ^ (z >> 31)
    }
    /// Uniform on [0, 1).
    fn random(&mut self) -> f64 {
        (self.next_u64() >> 11) as f64 / (1u64 << 53) as f64
    }
    /// Normal draw by Box-Muller.
    fn normal(&mut self, mean: f64, sd: f64) -> f64 {
        let u1 = 1.0 - self.random(); // in (0, 1], safe for ln
        let u2 = self.random();
        mean + sd * (-2.0 * u1.ln()).sqrt() * (std::f64::consts::TAU * u2).cos()
    }
}

fn mean(x: &[f64]) -> f64 {
    x.iter().sum::<f64>() / x.len() as f64
}

fn round(x: f64, digits: i32) -> f64 {
    let f = 10f64.powi(digits);
    (x * f).round() / f
}

// Estimate the tail index. alpha <= 1 means the mean does not exist at all.
fn hill_estimator(x: &[f64], k: usize) -> f64 {
    let mut s = x.to_vec();
    s.sort_by(|a, b| b.total_cmp(a)); // largest first
    1.0 / (s[..k].iter().map(|v| (v / s[k]).ln()).sum::<f64>() / k as f64)
}

fn main() {
    // The test that separates a power law from a lognormal: does the MEAN settle?
    let mut rng = Rng::new(0);
    let heavy: Vec<f64> = (0..200_000).map(|_| (1.0 - rng.random()).powf(-1.0 / 1.2)).collect(); // power law
(Pareto, alpha 1.2)
    let mild: Vec<f64> = (0..200_000).map(|_| 3.0 * rng.normal(0.0, 1.0).exp()).collect();      // lognormal

    for n in [1000, 10000, 100000, 200000] {
        println!("{} power law mean: {:?}  lognormal mean: {:?}",
                 n, round(mean(&heavy[..n]), 1), round(mean(&mild[..n]), 2));
    }
    // lognormal: 4.82 -> 5.08 -> 4.96 -> 4.95, settling down.
```

```
    // power law: 4.0 -> 5.2 -> 5.2 -> 5.3, still drifting (the true mean is 6).

    println!("alpha: {}", hill_estimator(&heavy, 500)); // 1.23, close to the true 1.2
    // alpha < 2 -> infinite variance. Never report a mean or an sd.
}
```

Categorical / multinomial

One draw picks one class.



Categorical / multinomial: one draw picks one class.

The story	Each observation lands in exactly one of k buckets, with a probability per bucket. Multinomial is the count version across many draws.
In eval work	**Intent classification.** Task categories. Failure reason codes.
Use when	Use for any labelled class assignment. The counts across classes are multinomial.
Avoid when	Class shares are not independent: they must sum to 1, so pushing one up pulls the others down. Treating each class as its own separate binomial ignores that constraint.

Under the hood

Categorical / multinomial: formulas and code

$$P(X = i) = p_i, \qquad \sum_{i=1}^{k} p_i = 1$$

$$E[\text{count}_i] = np_i \qquad \text{Cov}(n_i, n_j) = -np_ip_j$$

python

```python
from collections import Counter
from scipy import stats
import numpy as np

# Intent classification: each session gets exactly one label.
labels = ["fix_bug", "refactor", "explain", "add_test", "docs"]
counts = np.array([340, 220, 190, 150, 100])
n = counts.sum()

for lab, c in zip(labels, counts):
    print(lab, round(c / n, 3))

# THE CONSTRAINT people forget: shares must sum to 1, so they are NOT
# independent. Pushing one class up pulls the others down, and the
# covariance between two class counts is negative.
p = counts / n
print("cov(fix_bug, refactor):", -n * p[0] * p[1])    # negative, by construction

# Did the intent mix shift between two releases?
before = np.array([340, 220, 190, 150, 100])
after = np.array([290, 260, 185, 165, 100])
print(stats.chisquare(after, f_exp=before * after.sum() / before.sum()))
```

rust

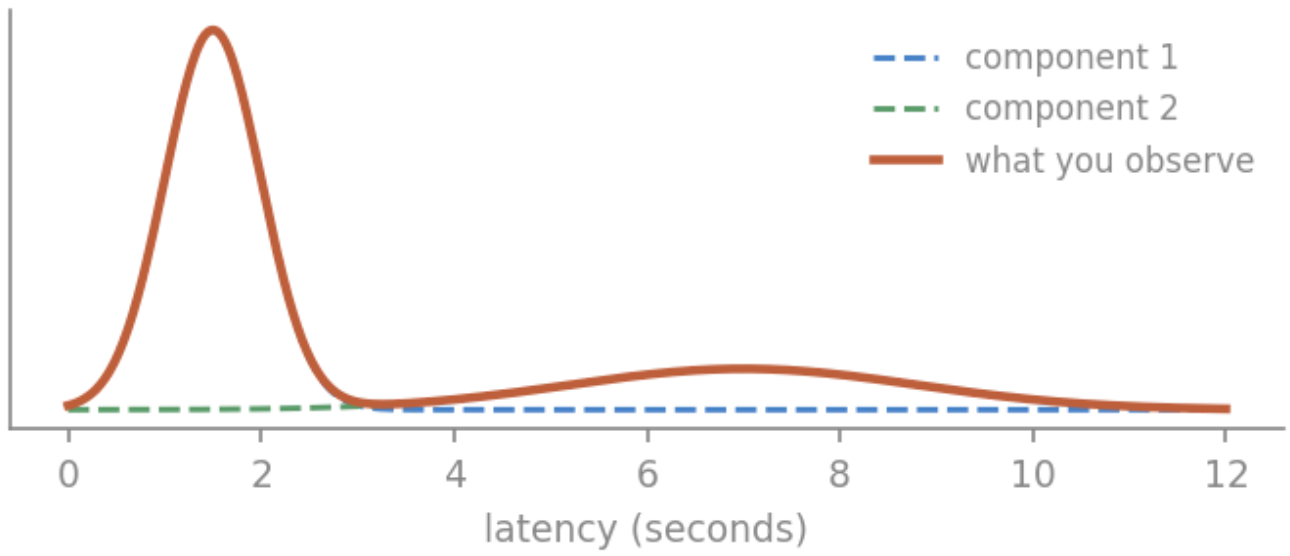
```rust
/// ln Gamma(x) by the Lanczos approximation (g = 7, 9 terms).
fn ln_gamma(x: f64) -> f64 {
    const G: [f64; 9] = [
        0.999_999_999_999_809_93, 676.520_368_121_885_1, -1259.139_216_722_402_8,
        771.323_428_777_653_13, -176.615_029_162_140_59, 12.507_343_278_686_905,
        -0.138_571_095_265_720_12, 9.984_369_578_019_571_6e-6, 1.505_632_735_149_311_6e-7,
    ];
    let pi = std::f64::consts::PI;
    if x < 0.5 {
        return (pi / (pi * x).sin()).ln() - ln_gamma(1.0 - x); // reflection
    }
    let x = x - 1.0;
    let t = x + 7.5;
    let a = G[0] + (1..9).map(|i| G[i] / (x + i as f64)).sum::<f64>();
    0.5 * (2.0 * pi).ln() + (x + 0.5) * t.ln() - t + a.ln()
```

```rust
}

/// Regularized incomplete gamma functions (P, Q) = (lower, upper), P + Q = 1.
fn gamma_pq(a: f64, x: f64) -> (f64, f64) {
    if x <= 0.0 {
        return (0.0, 1.0);
    }
    let front = (-x + a * x.ln() - ln_gamma(a)).exp();
    if x < a + 1.0 {
        // power series converges fast here
        let (mut ap, mut term) = (a, 1.0 / a);
        let mut sum = term;
        while term.abs() > sum.abs() * 1e-16 {
            ap += 1.0;
            term *= x / ap;
            sum += term;
        }
        let p = sum * front;
        (p, 1.0 - p)
    } else {
        // continued fraction (modified Lentz) converges fast here
        let tiny = 1e-300;
        let (mut b, mut c) = (x + 1.0 - a, 1.0 / tiny);
        let mut d = 1.0 / b;
        let mut h = d;
        for i in 1..1000 {
            let an = -(i as f64) * (i as f64 - a);
            b += 2.0;
            d = an * d + b;
            if d.abs() < tiny { d = tiny; }
            c = b + an / c;
            if c.abs() < tiny { c = tiny; }
            d = 1.0 / d;
            h *= d * c;
            if (d * c - 1.0).abs() < 1e-16 { break; }
        }
        let q = front * h;
        (1.0 - q, q)
    }
}

fn round(x: f64, digits: i32) -> f64 {
    let f = 10f64.powi(digits);
    (x * f).round() / f
}

#[derive(Debug)]
struct PowerDivergenceResult {
    statistic: f64,
    pvalue: f64,
}

/// Chi square goodness of fit: observed counts against expected counts.
fn chisquare(observed: &[f64], expected: &[f64]) -> PowerDivergenceResult {
    let statistic: f64 = observed.iter().zip(expected).map(|(o, e)| (o - e).powi(2) / e).sum();
    let dof = (observed.len() - 1) as f64;
    PowerDivergenceResult { statistic, pvalue: gamma_pq(dof / 2.0, statistic / 2.0).1 }
}

fn main() {
```

```
    // Intent classification: each session gets exactly one label.
    let labels = ["fix_bug", "refactor", "explain", "add_test", "docs"];
    let counts = [340.0, 220.0, 190.0, 150.0, 100.0];
    let n: f64 = counts.iter().sum();

    for (lab, c) in labels.iter().zip(&counts) {
        println!("{} {}", lab, round(c / n, 3));
    }

    // THE CONSTRAINT people forget: shares must sum to 1, so they are NOT
    // independent. Pushing one class up pulls the others down, and the
    // covariance between two class counts is negative.
    let p: Vec<f64> = counts.iter().map(|c| c / n).collect();
    println!("cov(fix_bug, refactor): {}", -n * p[0] * p[1]); // negative, by construction

    // Did the intent mix shift between two releases?
    let before = [340.0, 220.0, 190.0, 150.0, 100.0];
    let after = [290.0, 260.0, 185.0, 165.0, 100.0];
    let scale = after.iter().sum::<f64>() / before.iter().sum::<f64>();
    let f_exp: Vec<f64> = before.iter().map(|b| b * scale).collect();
    println!("{:?}", chisquare(&after, &f_exp));
}
```

Mixture

Two or more populations sharing an axis.



Mixture: two or more populations sharing an axis.

The story	Not a distribution family so much as a warning. Any two distributions can be blended, and the result often looks like one distribution with strange outliers.
In eval work	**Cache hits and cold starts. Retried and first-try requests. Easy and hard task categories.** Very common and very often missed.
Use when	Use it as a diagnosis, not a model. When you see two humps, go find the variable that separates them.
Avoid when	Do not fit a single distribution to a mixture and then report its mean or median. The median can land in the empty valley between the humps, describing a case that never occurs.

Under the hood

Mixture: formulas and code

$$f(x) = \sum_j w_j f_j(x), \qquad \sum_j w_j = 1$$

$$E[X] = \sum_j w_j \mu_j \quad \text{(but the mean may describe nobody)}$$

python

```python
import numpy as np
from sklearn.mixture import GaussianMixture   # or fit by hand

lat = np.concatenate([
    np.random.default_rng(0).normal(1.2, 0.3, 2200),   # cached
    np.random.default_rng(1).normal(6.5, 1.5, 800),    # cold start
])

# DIAGNOSIS, not a model. Two components fit far better than one?
logs = np.log(lat).reshape(-1, 1)
one = GaussianMixture(1, random_state=0).fit(logs)
two = GaussianMixture(2, random_state=0).fit(logs)
print("BIC one component:", round(one.bic(logs)))     # higher = worse
print("BIC two components:", round(two.bic(logs)))    # much lower -> mixture

# Why the median is dangerous here:
print("median:", np.median(lat))          # 1.33 - describes the fast group only
print("mean:  ", lat.mean())              # 2.58 - describes nobody
```

```
# THE RIGHT MOVE: stop modelling and go find the variable that splits them.
# Then report each segment separately, with its own share of traffic.
```

rust

```rust
// A tiny inline PRNG (SplitMix64) so the snippet needs no crates.
// Its draws differ from numpy's, so the random numbers below differ from the Python's.
struct Rng(u64);

impl Rng {
    fn new(seed: u64) -> Self {
        Rng(seed)
    }
    fn next_u64(&mut self) -> u64 {
        self.0 = self.0.wrapping_add(0x9E37_79B9_7F4A_7C15);
        let mut z = self.0;
        z = (z ^ (z >> 30)).wrapping_mul(0xBF58_476D_1CE4_E5B9);
        z = (z ^ (z >> 27)).wrapping_mul(0x94D0_49BB_1331_11EB);
        z ^ (z >> 31)
    }
    /// Uniform on [0, 1).
    fn random(&mut self) -> f64 {
        (self.next_u64() >> 11) as f64 / (1u64 << 53) as f64
    }
    /// Normal draw by Box-Muller.
    fn normal(&mut self, mean: f64, sd: f64) -> f64 {
        let u1 = 1.0 - self.random(); // in (0, 1], safe for ln
        let u2 = self.random();
        mean + sd * (-2.0 * u1.ln()).sqrt() * (std::f64::consts::TAU * u2).cos()
    }
}

fn mean(x: &[f64]) -> f64 {
    x.iter().sum::<f64>() / x.len() as f64
}

fn median(x: &[f64]) -> f64 {
    let mut s = x.to_vec();
    s.sort_by(|a, b| a.total_cmp(b));
    let n = s.len();
    if n % 2 == 1 { s[n / 2] } else { (s[n / 2 - 1] + s[n / 2]) / 2.0 }
}

fn normal_pdf(x: f64, mu: f64, var: f64) -> f64 {
    (-(x - mu).powi(2) / (2.0 * var)).exp() / (2.0 * std::f64::consts::PI * var).sqrt()
}

/// Fit a 1-D Gaussian mixture with k components by EM (fitting by hand).
/// Returns the BIC: -2 log likelihood + parameters * ln n. Lower is better.
fn gmm_bic(x: &[f64], k: usize) -> f64 {
    let n = x.len();
    let mut s = x.to_vec();
    s.sort_by(|a, b| a.total_cmp(b));
    // start the means spread across the data, each component with the overall variance
    let mut mu: Vec<f64> = (0..k).map(|j| s[(2 * j + 1) * n / (2 * k)]).collect();
    let m = mean(x);
    let mut var = vec![x.iter().map(|v| (v - m).powi(2)).sum::<f64>() / n as f64; k];
    let mut w = vec![1.0 / k as f64; k];
    let mut resp = vec![vec![0.0; k]; n];
    let mut loglik = f64::NEG_INFINITY;
```

```
    for _ in 0..500 {
        // E step: how much each component "owns" each point
        let mut ll = 0.0;
        for (i, &xi) in x.iter().enumerate() {
            let dens: Vec<f64> = (0..k).map(|j| w[j] * normal_pdf(xi, mu[j], var[j])).collect();
            let total: f64 = dens.iter().sum();
            ll += total.ln();
            for j in 0..k {
                resp[i][j] = dens[j] / total;
            }
        }
        // M step: refit each component to the points it owns
        for j in 0..k {
            let nj: f64 = resp.iter().map(|r| r[j]).sum();
            w[j] = nj / n as f64;
            mu[j] = x.iter().zip(&resp).map(|(xi, r)| r[j] * xi).sum::<f64>() / nj;
            var[j] = x.iter().zip(&resp).map(|(xi, r)| r[j] * (xi - mu[j]).powi(2)).sum::<f64>() / nj + 1e-6;
        }
        let done = (ll - loglik).abs() / (n as f64) < 1e-6; // average log likelihood has settled
        loglik = ll;
        if done {
            break;
        }
    }
    let params = (3 * k - 1) as f64; // k means, k variances, k - 1 free weights
    -2.0 * loglik + params * (n as f64).ln()
}

fn main() {
    let mut rng0 = Rng::new(0);
    let mut rng1 = Rng::new(1);
    let mut lat: Vec<f64> = (0..2200).map(|_| rng0.normal(1.2, 0.3)).collect(); // cached
    lat.extend((0..800).map(|_| rng1.normal(6.5, 1.5)));                        // cold start

    // DIAGNOSIS, not a model. Two components fit far better than one?
    let logs: Vec<f64> = lat.iter().map(|v| v.ln()).collect();
    println!("BIC one component: {}", gmm_bic(&logs, 1).round());  // higher = worse
    println!("BIC two components: {}", gmm_bic(&logs, 2).round()); // much lower -> mixture

    // Why the median is dangerous here:
    println!("median: {}", median(&lat)); // 1.34 - describes the fast group only
    println!("mean:   {}", mean(&lat));   // 2.62 - describes nobody

    // THE RIGHT MOVE: stop modelling and go find the variable that splits them.
    // Then report each segment separately, with its own share of traffic.
}
```

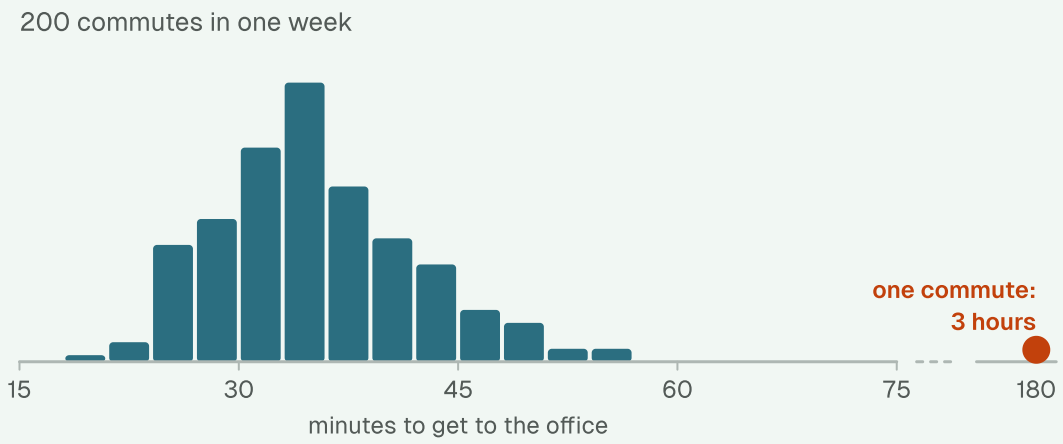
Long tails and outliers: error or product?

Latency, token counts, and agent loop lengths all have long tails, and every team eventually argues about what to do with the extreme values. The first question is never which technique. It is whether the outlier is an **error** or the **product**.

Diagram

The three-hour commute

An outlier is either a broken measurement or a real person. Treat them differently.



A week of office commutes: mostly 25 to 45 minutes. Then one trip of three hours, far off the chart. The instinct is to delete it so the chart looks normal.

An error

The badge reader logged a commute of −4 minutes.

Fix it or drop it.

The product

Her train broke down. She really waited three hours.

Keep it. It happened.

The first question is never "which technique?"

Ask first: is it an error or the product? A negative commute is a broken sensor; fix or drop it. A three-hour wait after a breakdown really happened to someone. In eval data, a 40-second response is a user who waited 40 seconds.

average with her

36.0 min

→

average after deleting

35.2 min

The number got better. Nobody got to work sooner.

Deleting the real one moves the average from 36.0 to 35.2 minutes. The dashboard improves; the commute does not. Teams do this to latency charts all the time.

median	p95	over 90 min
34 min	**48 min**	**0.5%**
what a typical person feels	1 in 20 trips is worse	1 trip in 200, kept

Change the summary, not the data.

Keep the data and change the summary: the median for the typical trip, a high percentile for the bad days, and a count of extreme cases. All three are honest and none hides the person who waited.

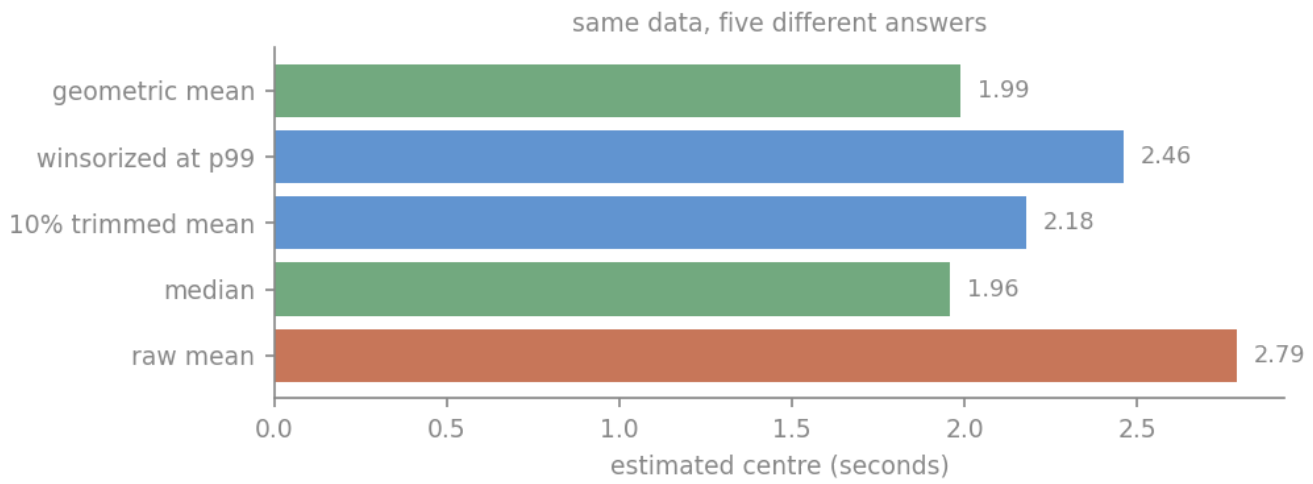
Plain words

The slow request is a real customer

If a restaurant serves most meals in ten minutes but one table waits an hour, that hour is not a typo. It is a customer who left angry. Deleting it from the spreadsheet makes the average look better and the restaurant no better. Only remove a value if you can show the measurement itself was broken, such as a clock error or a logging bug. Otherwise, change how you summarise, not what you measured.

Key idea

In eval data most long tails are signal. A 40 second response is something a real user actually waited through. Winsorizing it makes your chart prettier and your product no better. The honest default is to change the summary statistic, not the data.



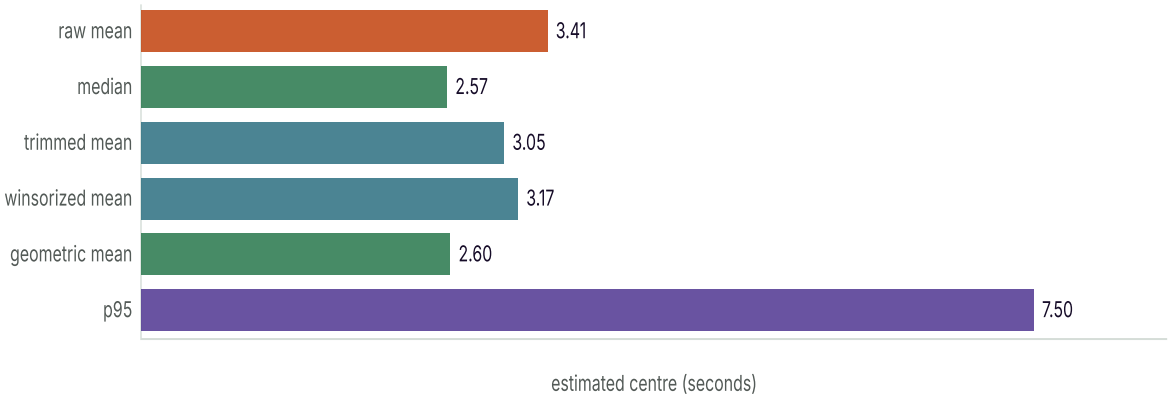
The same 1000 measurements, five different answers for "the centre". The spread between them is entirely a choice someone made.

Interactive

Outlier handling, side by side

(shown at its default setting)

Add outliers and watch which estimators move. The median barely notices. The raw mean tracks the tail directly. Move the winsorization percentile and see how much the answer depends on a threshold somebody picked arbitrarily.



8 outliers out of 1000 spread the "centre" estimates by 1.33x.

Notice the median barely moves no matter what you do to the tail, while the raw mean tracks it directly. Before you pick a method, decide whether those 8 points are broken instrumentation (remove them and say so) or slow requests real users actually waited through (keep them and report percentiles).

The options, in the order to try them

Option	What it does	Cons	Use when
1. Change the statistic	Report p50, p95, p99 with bootstrap intervals instead of a mean	No single headline number; p99 unstable at small n	The tail is real user experience. Almost always, for latency
2. Segment	Find the variable that separates the tail into its own population	Needs a plausible mechanism; you can fish until something separates	You can name a candidate first. **Highest value option and the most skipped**
3. Log transform	Analyse on the log scale, report back in original units	Geometric mean is not the arithmetic mean, so it cannot answer capacity or cost questions. Breaks on zeros	Data is roughly lognormal and you want to model it
4. Winsorize	Cap values above a percentile at that percentile	Biases the mean by an unknown amount; the cap point is arbitrary and two teams will pick differently	You have **confirmed instrumentation errors**, not slow requests
5. Trimmed mean	Drop a fixed share from each end, average the rest	Symmetric trimming on an asymmetric tail removes real data from one side and nothing from the other	Rarely right for a one sided tail
6. Robust estimators	Huber, Hodges-Lehmann: downweight rather than delete	Harder to explain; tuning constants are another arbitrary choice	You need a mean-like number that resists contamination
7. Rank based tests	Mann-Whitney instead of a t test	No longer a statement about means; effect sizes less intuitive	Comparing two groups where outliers would flip a t test
8. Model the tail	Fit a generalized Pareto above a threshold	Real machinery, real assumptions	The tail itself is the deliverable

> ⚠ **Watch out**
>
> **Check for censoring before any of this.** A spike of values sitting exactly at your timeout means those are not measurements, they are lower bounds. No outlier technique fixes that, and every one of them will quietly make the bias worse.

Per metric

- **Latency:** quantiles, never a mean. Segment by cold start and retry status first.
- **Token counts:** log scale for analysis, quantiles for reporting. Watch for a bimodal split between short answers and agentic loops, which is segmentation, not outliers.
- **Tool calls per session:** the tail is task difficulty, so it is signal. Negative binomial or quantiles. Never trim.
- **Pass rates:** bounded at 0 and 1, so no tail. Different problem entirely.

Under the hood

Winsorized, trimmed, and geometric means

$$\tilde{x}_i = \min\big(x_i,\ Q(0.99)\big) \qquad \bar{x}_\alpha = \frac{1}{n - 2\lfloor \alpha n \rfloor} \sum_{i=\lfloor \alpha n \rfloor + 1}^{n - \lfloor \alpha n \rfloor} x_{(i)}$$

Left: winsorizing caps each value at the 99th percentile. Right: the α-trimmed mean averages the sorted values $x_{(i)}$ after dropping a share α from each end.

$$\mathrm{GM} = \exp\left(\frac{1}{n}\sum \ln x_i\right) = \left(\prod x_i\right)^{1/n}$$

The geometric mean: average on the log scale, then come back. On lognormal data it lands on the median.

python

```python
import numpy as np
from scipy import stats

lat = np.concatenate([stats.lognorm.rvs(0.7, scale=2, size=995, random_state=7),
                      [45, 52, 61, 80, 120]])          # five real slow requests
```

```
def centres(x):
    x = np.asarray(x, dtype=float)
    cap = np.quantile(x, 0.99)
    return {
        "raw mean":        x.mean(),                  # moves with every outlier
        "median":          np.median(x),              # ignores them entirely
        "10% trimmed":     stats.trim_mean(x, 0.10),  # drops both tails
        "winsorized p99":  np.minimum(x, cap).mean(), # caps rather than drops
        "geometric mean":  np.exp(np.log(x).mean()),  # centre on the log scale
        "p95":             np.quantile(x, 0.95),      # when the tail IS the point
    }

for k, v in centres(lat).items():
    print(k.ljust(16), round(v, 3))

# CENSORING CHECK FIRST. A pile-up at one value near the top means those
# are lower bounds, not measurements, and every method above is then biased.
top = lat[lat >= np.quantile(lat, 0.98)]
print("distinct values in the top 2%:", len(np.unique(np.round(top, 3))))
# If that comes back as 1 or 2, you have a timeout, not a tail.
```

rust

```
// numpy and scipy pieces written out by hand. The 995 lognormal draws come
// from a small SplitMix64 + Box-Muller generator, so they differ from scipy's
// random_state=7 draws and the numbers below are this program's output.
struct Rng(u64);

impl Rng {
    fn next_u64(&mut self) -> u64 {
        self.0 = self.0.wrapping_add(0x9E37_79B9_7F4A_7C15);
        let mut z = self.0;
        z = (z ^ (z >> 30)).wrapping_mul(0xBF58_476D_1CE4_E5B9);
        z = (z ^ (z >> 27)).wrapping_mul(0x94D0_49BB_1331_11EB);
        z ^ (z >> 31)
    }
    fn uniform(&mut self) -> f64 {
        (self.next_u64() >> 11) as f64 / (1u64 << 53) as f64
    }
    fn normal(&mut self) -> f64 {
        // Box-Muller: two uniforms in, one standard normal out
        let u1 = 1.0 - self.uniform(); // in (0, 1], so ln is finite
        let u2 = self.uniform();
        (-2.0 * u1.ln()).sqrt() * (2.0 * std::f64::consts::PI * u2).cos()
    }
}

fn mean(x: &[f64]) -> f64 {
    x.iter().sum::<f64>() / x.len() as f64
}

// np.quantile's default: linear interpolation between order statistics
fn quantile(sorted: &[f64], q: f64) -> f64 {
    let pos = q * (sorted.len() - 1) as f64;
    let lo = pos.floor() as usize;
    let hi = (lo + 1).min(sorted.len() - 1);
    sorted[lo] + (pos - lo as f64) * (sorted[hi] - sorted[lo])
}

// stats.trim_mean: cut int(prop * n) points from EACH end, average the rest
```

```rust
fn trim_mean(sorted: &[f64], prop: f64) -> f64 {
    let cut = (prop * sorted.len() as f64) as usize;
    mean(&sorted[cut..sorted.len() - cut])
}

fn centres(x: &[f64]) -> Vec<(&'static str, f64)> {
    let mut s = x.to_vec();
    s.sort_by(f64::total_cmp);
    let cap = quantile(&s, 0.99);
    let capped: Vec<f64> = x.iter().map(|&v| v.min(cap)).collect();
    let logs: Vec<f64> = x.iter().map(|v| v.ln()).collect();
    vec![
        ("raw mean", mean(x)),                    // moves with every outlier
        ("median", quantile(&s, 0.5)),            // ignores them entirely
        ("10% trimmed", trim_mean(&s, 0.10)),     // drops both tails
        ("winsorized p99", mean(&capped)),        // caps rather than drops
        ("geometric mean", mean(&logs).exp()),    // centre on the log scale
        ("p95", quantile(&s, 0.95)),              // when the tail IS the point
    ]
}

fn round_to(v: f64, digits: i32) -> f64 {
    let f = 10f64.powi(digits);
    (v * f).round() / f
}

fn main() {
    let mut rng = Rng(7);
    // lognorm(0.7, scale=2) is 2 * exp(0.7 * Z)
    let mut lat: Vec<f64> = (0..995).map(|_| 2.0 * (0.7 * rng.normal()).exp()).collect();
    lat.extend([45.0, 52.0, 61.0, 80.0, 120.0]);       // five real slow requests

    for (k, v) in centres(&lat) {
        println!("{:<16} {:?}", k, round_to(v, 3));
    }
    // This program's draws (they differ from scipy's):
    // raw mean         2.941
    // median           1.998
    // 10% trimmed      2.25
    // winsorized p99   2.611
    // geometric mean   2.068
    // p95              6.574

    // CENSORING CHECK FIRST. A pile-up at one value near the top means those
    // are lower bounds, not measurements, and every method above is then biased.
    let mut s = lat.clone();
    s.sort_by(f64::total_cmp);
    let cut = quantile(&s, 0.98);
    let mut top: Vec<i64> = lat.iter().filter(|&&v| v >= cut).map(|&v| (v * 1000.0).round() as i64).collect();
    top.sort();
    top.dedup();
    println!("distinct values in the top 2%: {}", top.len()); // 20
    // If that comes back as 1 or 2, you have a timeout, not a tail.
}
```

For leaders

Outlier policy is a product decision

How a team handles extreme values decides which users the metric speaks for. That choice should be explicit, written down, and fixed before results come in.

- **Pre-register the rule.** If the winsorization threshold or exclusion criterion is chosen after seeing the data, it can be tuned (often unconsciously) to produce the desired answer. Two teams picking different cut-offs on the same data can report opposite conclusions.
- **Treat "we removed outliers" as a red flag until explained.** Ask what was removed, how many, and what evidence shows each was a measurement error rather than a slow or failed run. In agent evals, the tail is often exactly the failures you most need to see.
- **Ask for the segment, not the trim.** The highest-value response to a long tail is finding the variable that explains it: a task category, a tool, a context length bucket. That turns a statistics argument into a fixable engineering problem.
- **Separate cost questions from experience questions.** Budgets need means (totals). Users experience percentiles. A report that uses one summary for both is answering one of the two questions wrongly.

> **Principal asks**
>
> "If we had not excluded anything, would the conclusion change?" Ask for both versions side by side. If the conclusion flips, the exclusion rule is carrying the result, and it needs much stronger justification than a footnote.

Part I · Chapter 8

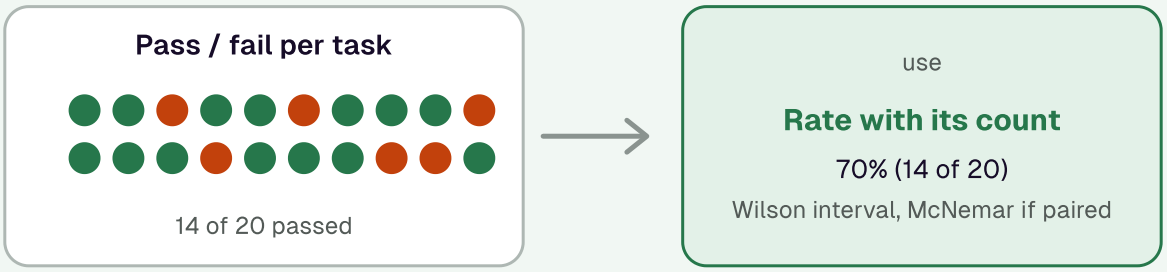
From shape to method: a lookup table

Given what you have, use this. Every row assumes you have already plotted a histogram and checked for censoring and bimodality, as in the reading-a-shape checklist.

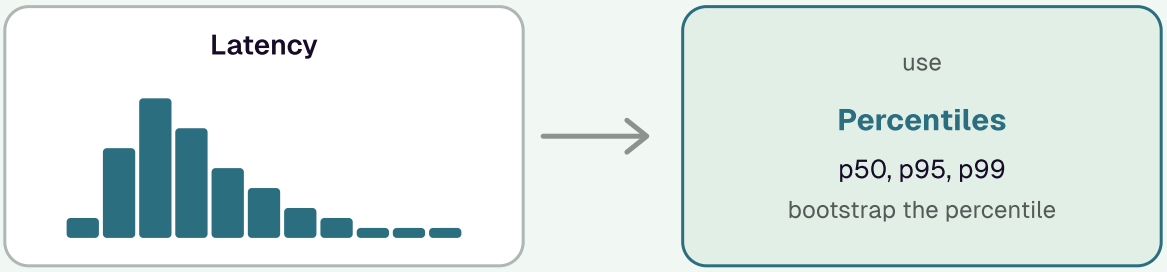
Diagram

Match the data to the tool

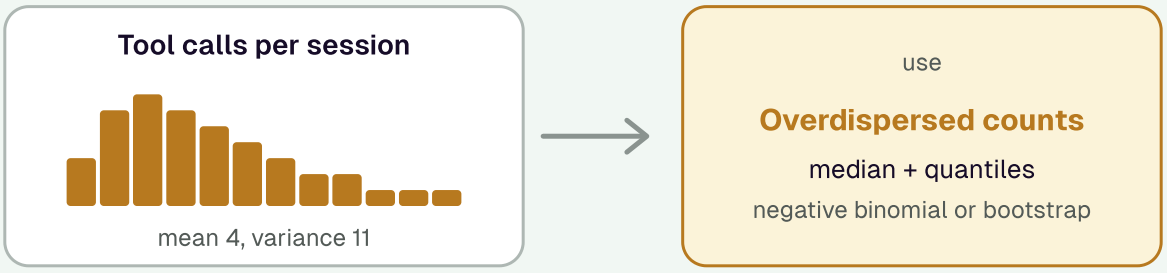
You would not weigh flour with a thermometer. Five kinds of eval data, five toolboxes.



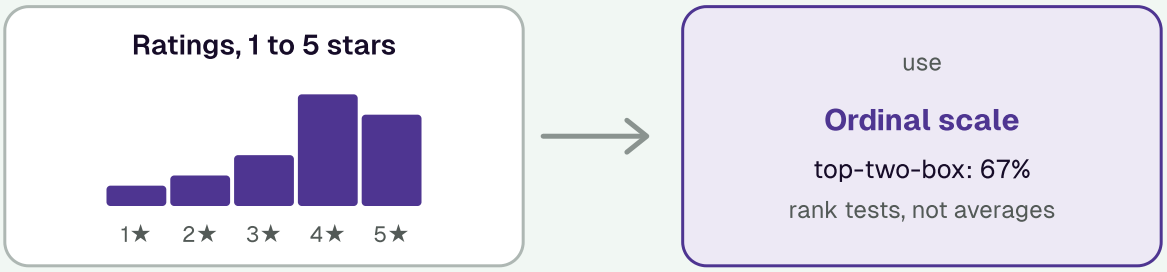
Pass or fail on each task is a pile of coin flips. Report the rate *with* its count, give it a Wilson interval, and compare paired runs with McNemar.



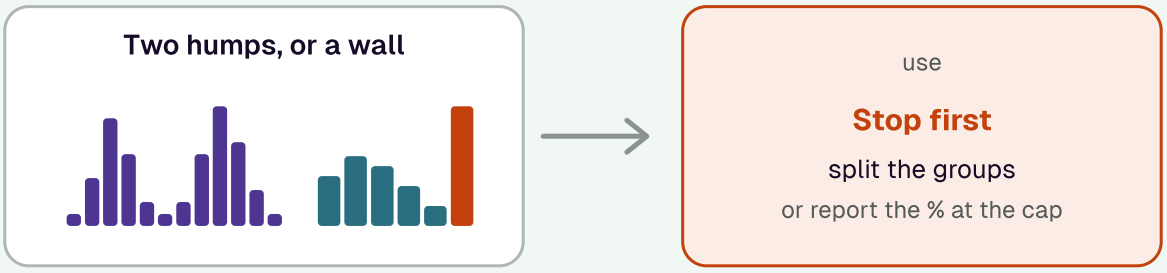
Latency leans right with a long tail. Averages hide the bad days, so report percentiles and get their uncertainty from the bootstrap.



Counts like tool calls per session usually vary far more than their average would suggest (variance 11 against a mean of 4). Poisson tools will claim false precision; use quantiles or a negative binomial.



Star ratings are ordered labels, not measurements: the gap from 2 to 3 stars is not the gap from 4 to 5. Report the share of 4s and 5s and compare with rank tests.



Two humps or a spike at a cap means stop before you summarise. Split the populations, or report how many hit the wall. The table below lists every row.

You have	Shape	Summarise with	Compare with	Interval from
Pass / fail per task	Bernoulli / binomial	Rate + count	Exact McNemar (paired), Cochran's Q (3+)	Wilson
Pass / fail nested in sessions	Clustered binomial	Rate + effective n	Cluster-level comparison	Cluster bootstrap
Latency	Lognormal, right skewed	p50, p95, p99	Mann-Whitney, or t test on logs	Bootstrap on the percentile
Token counts	Right skewed, overdispersed	Median + quantiles	Mann-Whitney	Bootstrap
Tool calls per session	Negative binomial	Median + quantiles, report var/mean	Negative binomial model, or bootstrap	Bootstrap
Retry counts	Geometric	Median, share needing >1 attempt	Proportion test on "needed a retry"	Wilson on that proportion
Likert 1-5	Ordinal	Top-two-box share	Mann-Whitney + effect size	Wilson on top-two-box
Judge score 0-10	Ordinal, often compressed	Median + distribution	Friedman (3+), Mann-Whitney (2)	Bootstrap
A rate you barely have data for	Beta	Posterior mean + interval	Overlap of posteriors	Beta credible interval
Anything with two humps	Mixture	**Split it first**	Compare within segments	Per segment
Anything with a spike at a cap	Censored	**Report the censoring rate**	Survival methods	Kaplan-Meier

> **Note**
>
> If a row makes you reach for the bootstrap, remember its limits: it fails at very small n, on extreme quantiles, on maxima, and on clustered data unless you resample the clusters. Part II covers each of these in detail.

Plain words

Pick the tool that fits the data

You would not measure flour with a thermometer. Each kind of eval data, yes/no results, waiting times, counts, star ratings, has tools that fit it and tools that quietly give wrong answers. This table matches each kind of data to a way of summarising it, a way of comparing two models, and a way of drawing an error bar.

For leaders

Use the table as a review checklist

You do not need to run these methods yourself to use this table well. When a result lands in front of you, find the row that matches the metric and check the analysis against it.

1. **Was the comparison paired?** If both models ran the same tasks, the comparison should use the pairing (McNemar, paired bootstrap). An unpaired test throws away most of the statistical power and can miss real wins.
2. **Does the interval method match the data?** Wilson for rates, bootstrap for percentiles, cluster bootstrap for anything nested in sessions or conversations. A normal-approximation interval on a 98% pass rate or a session-level metric is a warning sign.
3. **Was the data split or flagged first?** Mixtures need segmenting and censored data needs its censoring rate reported. If neither check is mentioned, ask.
4. **Is the summary the one users feel?** Percentiles for latency, top-two-box for satisfaction, rates with counts for pass/fail.

> **Principal asks**
>
> "Which row of the table is this metric, and does the method match?" A team that can answer instantly has thought about its measurement. A pause usually means the default tool was used without checking whether it fits.

Under the hood

The decision, compressed

The table encodes four questions asked in order:

1. **Is it binary, count, continuous, or ordinal?** This fixes the family: binomial and beta for binary, Poisson or negative binomial for counts, lognormal or quantile methods for positive continuous data, rank methods for ordinal scores.
2. **Are observations independent?** If they are nested (calls in sessions, turns in conversations), the effective sample size is closer to the number of clusters, and intervals must resample clusters.
3. **Are the comparisons paired?** Shared tasks call for paired tests, which use within-task differences and remove between-task difficulty from the noise.
4. **Is the statistic smooth?** Means and medians bootstrap well at moderate n. Extreme quantiles and maxima do not, and need more data, a parametric tail model, or an honest "insufficient data".

Part II

Evaluating models

Turning model outputs into numbers you can defend: eval statistics, classification metrics, judges, the offline to online gap, and the harness that produces it all.

Part II · Chapter 9

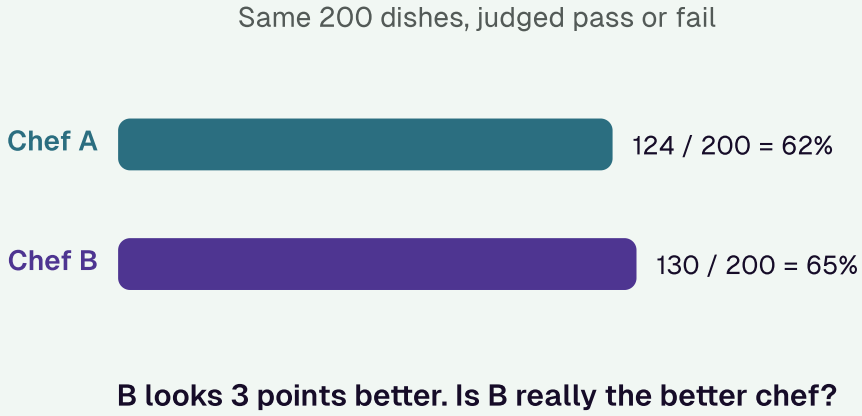
Statistics for eval sets: is the difference real?

Every eval number is an estimate from a finite sample of tasks. This chapter covers the handful of methods that decide whether a result can be trusted: an interval on one pass rate, a paired comparison of two models, power and sample size, many benchmarks at once, pass@k, and comparisons across more than two models. It assumes the ideas from sampling distributions and standard error; here they are applied to the numbers an eval team actually reports.

Diagram

Two chefs, the same 200 dishes

Why comparing two models on shared tasks only counts the disagreements.



Two chefs cook the same 200 dishes and a judge marks each pass or fail. Chef B scores 130, chef A 124. Product wants to promote B. Comparing the two totals is the obvious move, and the wrong one.

Line the dishes up side by side instead of counting totals.

Line the dishes up one by one. On 112 dishes both chefs passed, and on 58 both failed. Only 30 dishes split them: B alone passed 18, A alone passed 12.

170 dishes where the chefs tied say nothing about who is better.

The 170 dishes where both chefs got the same result cannot tell you who is better, so set them aside. The whole comparison rests on the 30 dishes where they differed. This is a paired comparison.

A fair coin splits 18 to 12 or worse 36% of the time (p = 0.36).

Treat each disagreement as a coin toss. If the chefs were equally good, a split of 18 to 12 or more lopsided would happen about 36% of the time. That is not evidence. The exact McNemar test gives p = 0.36.

Both kitchens: 124 vs 130, the same 3-point headline.

Same 124 versus 130 headline, different story. If the chefs had disagreed on only 6 dishes and B won all 6, the result would be convincing (p = 0.03). The totals cannot tell these kitchens apart. The disagreements can.

Key idea

Most eval mistakes are not exotic. They are a missing interval, an ignored pairing, an underpowered suite, or forty benchmarks read as forty independent findings. Get those four right and you are ahead of most results decks.

Missing interval

A score with no range. Fix: Wilson interval on every pass rate (section 1).

Ignored pairing

Comparing totals on shared tasks. Fix: count only the disagreements (section 2).

Underpowered suite

Too few tasks to see the effect you care about. Fix: size the suite first (sections 3 and 4).

Forty "findings"

Many benchmarks read one at a time. Fix: correct for multiple comparisons (section 5).

1. The uncertainty on a single pass rate

Plain words

One score is one spoonful

If a model passes 49 of 50 tasks, its "true" pass rate is probably high, but you only tasted 50 spoonfuls. A different 50 tasks would give a slightly different score. A **confidence interval** is the range of true pass rates that could reasonably have produced what you saw. Fewer tasks means a wider range.

The most common shortcut for building that range breaks when scores are near 0% or 100%, which is exactly where good models live. For 49 of 50 it claims the true rate might be 102%, which is impossible. A better recipe, the **Wilson interval**, never does that.

For leaders

What to require in every report

- **No pass rate without n and an interval.** "62%" is not a result. "62% (124/200, 95% CI 55 to 68)" is.
- **Near the ceiling, intervals get lopsided.** On saturated benchmarks the honest interval is asymmetric, and small suites cannot separate 96% from 99%. That is a sign the benchmark has stopped being informative, not that the models are equal.

- **Red flag:** symmetric "±" error bars on a score above 90% or below 10%. They come from the normal approximation and are wrong in a predictable direction.

Under the hood

The Wilson score interval

With k passes out of n, $\hat{p} = k/n$, and $z = 1.96$ for 95%:

$$\frac{\hat{p} + \frac{z^2}{2n}}{1 + \frac{z^2}{n}} \pm \frac{z}{1 + \frac{z^2}{n}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}$$

The centre is pulled toward 0.5 by $z^2/2n$, and the width gets a small-sample correction $z^2/4n^2$. Wilson comes from inverting the score test, so the interval is asymmetric in the right direction near a boundary.

python

```python
import math

def wilson(k: int, n: int, z: float = 1.96) -> tuple[float, float]:
    """Wilson score interval for a binomial proportion."""
    # With no data every proportion is possible; avoids dividing by zero below.
    if n == 0:
        return (0.0, 1.0)

    p = k / n                       # observed pass rate, the point estimate
    denom = 1.0 + z * z / n         # shrink factor; tends to 1 as n grows

    # The CENTRE is not p. Wilson pulls it toward 0.5 by z^2/(2n), which is what
    # stops the interval running past 0 or 1 at extreme rates.
    center = (p + z * z / (2 * n)) / denom

    # Half width: binomial variance p(1-p)/n plus the small-sample term z^2/(4n^2).
    margin = z * math.sqrt(p * (1 - p) / n + z * z / (4 * n * n)) / denom

    # Clamp against floating point nudging a bound past a boundary.
    return (max(0.0, center - margin), min(1.0, center + margin))


def normal_approx(k: int, n: int, z: float = 1.96) -> tuple[float, float]:
    """The default most people reach for. Shown so you can see why it fails."""
    p = k / n
    margin = z * math.sqrt(p * (1 - p) / n)    # symmetric, no correction
    return (p - margin, p + margin)            # no clamping: can exceed 1


print(wilson(49, 50))         # (0.8950, 0.9965)  stays inside [0, 1]
print(normal_approx(49, 50))  # (0.9412, 1.0188)  claims a rate above 100%
```

rust

```rust
/// Wilson score interval for a binomial proportion.
fn wilson(k: u64, n: u64, z: f64) -> (f64, f64) {
    // With no data every proportion is possible; avoids dividing by zero below.
    if n == 0 {
        return (0.0, 1.0);
    }
    let (kf, nf) = (k as f64, n as f64);

    let p = kf / nf;                    // observed pass rate, the point estimate
    let denom = 1.0 + z * z / nf;       // shrink factor; tends to 1 as n grows

    // The CENTRE is not p. Wilson pulls it toward 0.5 by z^2/(2n), which is what
    // stops the interval running past 0 or 1 at extreme rates.
    let center = (p + z * z / (2.0 * nf)) / denom;

    // Half width: binomial variance p(1-p)/n plus the small-sample term z^2/(4n^2).
    let margin = z * (p * (1.0 - p) / nf + z * z / (4.0 * nf * nf)).sqrt() / denom;

    // Clamp against floating point nudging a bound past a boundary.
    ((center - margin).max(0.0), (center + margin).min(1.0))
}

/// The default most people reach for. Shown so you can see why it fails.
fn normal_approx(k: u64, n: u64, z: f64) -> (f64, f64) {
    let p = k as f64 / n as f64;
    let margin = z * (p * (1.0 - p) / n as f64).sqrt();  // symmetric, no correction
    (p - margin, p + margin)                             // no clamping: can exceed 1
}

fn main() {
    println!("{:?}", wilson(49, 50, 1.96));         // (0.8950, 0.9965)  stays inside [0, 1]
    println!("{:?}", normal_approx(49, 50, 1.96));  // (0.9412, 1.0188)  claims a rate above 100%
}
```

The normal approximation fails because it assumes the sampling distribution of $\hat{p}$ is symmetric around $\hat{p}$. Near a boundary it is not. Exact coverage at a true rate of 0.95 with $n = 30$: Wilson 0.94, normal approximation 0.78. At $n = 50$: 0.96 versus 0.92.

Interactive **Wilson vs the normal approximation** (shown at its default setting)

Both claim to be 95 percent intervals. The readout runs 3000 simulated experiments at your settings and reports what fraction actually contained the true rate. Drag the rate toward 0.95 and shrink n to watch the normal approximation quietly stop being a 95 percent interval.

Actual coverage over 3000 simulated runs

Wilson 0.932 Normal 0.781

Nominal is 0.950. The normal approximation is not delivering a 95% interval here.

Normal upper bound exceeds 1.0, which is impossible.

⚠ **Watch out**

At a true rate of 0.95 with n = 30, the normal approximation covers about 78 percent of the time and its upper bound can exceed 1.0. Extreme rates on small sets are where eval suites live.

2. Comparing two models on the same tasks

Plain words **Only the disagreements count**

Two students take the same 200-question exam. On most questions they both get it right, or both get it wrong. Those questions tell you nothing about who is better. Only the questions where one was right and the other wrong carry information. If there are 30 such questions and one student wins 18 of them, that could easily be luck. If they win 22 of 30, it probably is not.

Evals work the same way. When two models run on the same tasks, you compare them task by task. This is called a **paired** comparison, and it is far more sensitive than comparing two separate averages.

For leaders

What to say when product wants to ship

- **Overlapping intervals is not a test.** Two intervals can overlap heavily while the paired difference is clearly significant, and the reverse. Ask for the paired result.
- **The headline delta is not enough.** The same 3-point gap can be noise or real depending on how the disagreements split. Ask: "How many tasks did they disagree on, and how did those split?"
- **Frame the answer as a decision.** "The 3-point gain rests on 30 disagreements split 18 to 12. A coin flip does that about a third of the time. I can't call this an improvement yet. Here is what it would take to know."

Under the hood

Paired bootstrap and exact McNemar

Let b be the tasks A passed and B failed, and c the reverse. Under the null hypothesis that the models are equally good, each discordant task is a fair coin flip, so $b \sim \mathrm{Binomial}(b + c, \tfrac{1}{2})$. The exact two-sided p value is

$$p = \min\left(1,\ 2 \sum_{i=0}^{\min(b,c)} \binom{n_d}{i} 2^{-n_d}\right), \qquad n_d = b + c$$

Tasks both models passed or both failed never enter the formula. That is why the test has power on small suites. Use the exact version rather than the chi-squared approximation when n_d is small.

python

```python
from math import comb

def mcnemar_exact(a: list[int], b: list[int]) -> dict:
    """Exact McNemar test on paired binary outcomes."""
    b_count = sum(1 for x, y in zip(a, b) if x == 1 and y == 0)   # A won, B lost
    c_count = sum(1 for x, y in zip(a, b) if x == 0 and y == 1)   # B won, A lost
    n = b_count + c_count            # discordant pairs: the effective sample
    if n == 0:
        return {"b": 0, "c": 0, "p": 1.0}
    k = min(b_count, c_count)
    tail = sum(comb(n, i) for i in range(0, k + 1)) / (2 ** n)
    return {"b": b_count, "c": c_count, "n_discordant": n, "p": min(1.0, 2 * tail)}
```

```
# 200 tasks, the models agree on 188. A wins 9 of the 12 disagreements.
print(mcnemar_exact([1]*9 + [0]*3 + [1]*94 + [0]*94,
                    [0]*9 + [1]*3 + [1]*94 + [0]*94))
# {'b': 9, 'c': 3, 'n_discordant': 12, 'p': 0.146}   nine to three is not enough
```

rust

```
struct McNemar {
    b: usize,
    c: usize,
    n_discordant: usize,
    p: f64,
}

/// P(X <= k) for X ~ Binomial(n, 1/2): sum of C(n, i) over 2^n.
fn binom_half_cdf(n: usize, k: usize) -> f64 {
    if n <= 120 {
        // Exact integers, like Python's math.comb; u128 holds C(120, 60) * 61.
        let (mut c, mut total) = (1u128, 1u128);
        for i in 1..=k {
            c = c * (n - i + 1) as u128 / i as u128;
            total += c;
        }
        total as f64 / 2f64.powi(n as i32)
    } else {
        // Large n: add the terms in log space so nothing overflows.
        let mut log_term = -(n as f64) * std::f64::consts::LN_2;
        let mut total = log_term.exp();
        for i in 1..=k {
            log_term += ((n - i + 1) as f64 / i as f64).ln();
            total += log_term.exp();
        }
        total
    }
}

/// Exact McNemar test on paired binary outcomes.
fn mcnemar_exact(a: &[u8], b: &[u8]) -> McNemar {
    let b_count = a.iter().zip(b).filter(|&(&x, &y)| x == 1 && y == 0).count(); // A won, B lost
    let c_count = a.iter().zip(b).filter(|&(&x, &y)| x == 0 && y == 1).count(); // B won, A lost
    let n = b_count + c_count;           // discordant pairs: the effective sample
    if n == 0 {
        return McNemar { b: 0, c: 0, n_discordant: 0, p: 1.0 };
    }
    let k = b_count.min(c_count);
    let tail = binom_half_cdf(n, k);
    McNemar { b: b_count, c: c_count, n_discordant: n, p: (2.0 * tail).min(1.0) }
}

fn main() {
    // 200 tasks, the models agree on 188. A wins 9 of the 12 disagreements.
    let a: Vec<u8> = [vec![1; 9], vec![0; 3], vec![1; 94], vec![0; 94]].concat();
    let b: Vec<u8> = [vec![0; 9], vec![1; 3], vec![1; 94], vec![0; 94]].concat();
    let r = mcnemar_exact(&a, &b);
    println!("{{'b': {}, 'c': {}, 'n_discordant': {}, 'p': {:?}}}", r.b, r.c, r.n_discordant, r.p);
    // {'b': 9, 'c': 3, 'n_discordant': 12, 'p': 0.146}   nine to three is not enough
}
```

The paired bootstrap gives an interval on the difference itself. The one line that matters is resampling *task indices* and reading both models at each index.

python

```python
import random

def paired_bootstrap(a: list[int], b: list[int], iters: int = 10_000, seed: int = 0) -> dict:
    """CI for the difference in pass rate between two models on the SAME tasks."""
    assert len(a) == len(b)          # misaligned lists silently produce garbage
    n = len(a)
    rng = random.Random(seed)
    observed = sum(a) / n - sum(b) / n

    diffs = []
    for _ in range(iters):
        # Resample TASK INDICES, then read BOTH models at each index.
        # Sampling a[] and b[] independently would destroy the pairing.
        sample = [rng.randrange(n) for _ in range(n)]
        diffs.append(sum(a[i] for i in sample) / n - sum(b[i] for i in sample) / n)
    diffs.sort()
    lo, hi = diffs[int(0.025 * iters)], diffs[int(0.975 * iters)]
    p_val = 2 * min(sum(d <= 0 for d in diffs), sum(d >= 0 for d in diffs)) / iters
    return {"observed": observed, "ci95": (lo, hi), "p": min(1.0, p_val)}
```

rust

```rust
// Small inline PRNG (SplitMix64); its draws differ from Python's random module.
struct Rng(u64);

impl Rng {
    fn next_u64(&mut self) -> u64 {
        self.0 = self.0.wrapping_add(0x9E37_79B9_7F4A_7C15);
        let mut z = self.0;
        z = (z ^ (z >> 30)).wrapping_mul(0xBF58_476D_1CE4_E5B9);
        z = (z ^ (z >> 27)).wrapping_mul(0x94D0_49BB_1331_11EB);
        z ^ (z >> 31)
    }
    fn randrange(&mut self, n: usize) -> usize {
        ((self.next_u64() >> 11) as f64 / (1u64 << 53) as f64 * n as f64) as usize
    }
}

struct PairedBootstrap {
    observed: f64,
    ci95: (f64, f64),
    p: f64,
}

/// CI for the difference in pass rate between two models on the SAME tasks.
fn paired_bootstrap(a: &[u8], b: &[u8], iters: usize, seed: u64) -> PairedBootstrap {
    assert_eq!(a.len(), b.len());        // misaligned lists silently produce garbage
    let n = a.len();
    let nf = n as f64;
    let mut rng = Rng(seed);
    let rate = |x: &[u8]| x.iter().map(|&v| v as f64).sum::<f64>() / nf;
    let observed = rate(a) - rate(b);

    let mut diffs = Vec::with_capacity(iters);
```

```
    for _ in 0..iters {
        // Resample TASK INDICES, then read BOTH models at each index.
        // Sampling a[] and b[] independently would destroy the pairing.
        let sample: Vec<usize> = (0..n).map(|_| rng.randrange(n)).collect();
        let sa: f64 = sample.iter().map(|&i| a[i] as f64).sum();
        let sb: f64 = sample.iter().map(|&i| b[i] as f64).sum();
        diffs.push(sa / nf - sb / nf);
    }
    diffs.sort_by(f64::total_cmp);
    let (lo, hi) = (diffs[(0.025 * iters as f64) as usize], diffs[(0.975 * iters as f64) as usize]);
    let below = diffs.iter().filter(|&&d| d <= 0.0).count();
    let above = diffs.iter().filter(|&&d| d >= 0.0).count();
    let p_val = 2.0 * below.min(above) as f64 / iters as f64;
    PairedBootstrap { observed, ci95: (lo, hi), p: p_val.min(1.0) }
}
```

Worked problem

Two models, 200 shared tasks. Ship or not? Model A passed 124 of 200. Model B passed 130 of 200. Product wants to ship B.

Wilson intervals: A is 0.551 to 0.684, B is 0.582 to 0.713. They overlap heavily, and the instinct is "no significant difference". That instinct uses the wrong method. The tasks were shared, so treating the models as independent throws away most of the power.

The right analysis looks at the discordant pairs. Suppose the models agreed on 170 tasks and split the other 30.

python

```python
from math import comb

def mcnemar_exact(b, c):
    """b = tasks B won and A lost. c = the reverse."""
    n = b + c
    if n == 0:
        return 1.0
    k = min(b, c)
    tail = sum(comb(n, i) for i in range(k + 1)) / (2 ** n)
    return min(1.0, 2 * tail)

print(mcnemar_exact(18, 12))   # 0.3616  not significant
print(mcnemar_exact(6, 0))     # 0.0312  significant
```

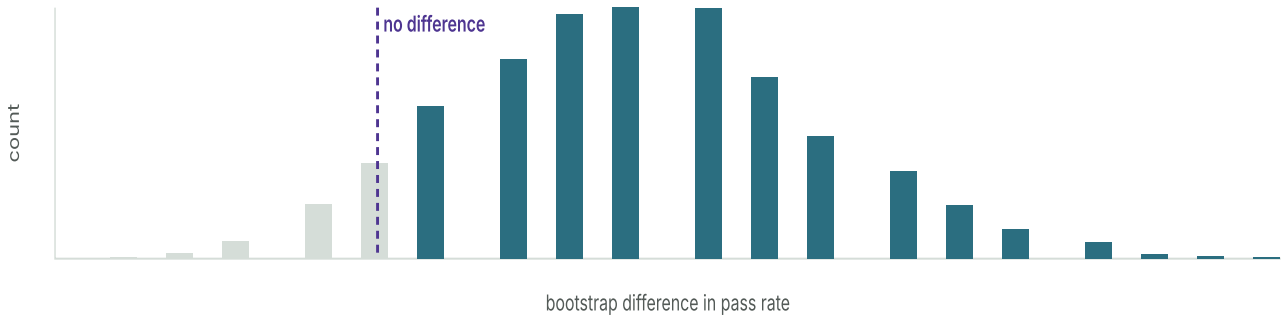
rust

```rust
/// b = tasks B won and A lost. c = the reverse.
fn mcnemar_exact(b: u32, c: u32) -> f64 {
    let n = b + c;
    if n == 0 {
        return 1.0;
    }
    let k = b.min(c);
    // sum of C(n, i) for i in 0..=k, in exact integers (fine up to n = 120)
    let (mut comb, mut sum) = (1u128, 1u128);
    for i in 1..=k {
        comb = comb * (n - i + 1) as u128 / i as u128;
        sum += comb;
    }
    let tail = sum as f64 / 2f64.powi(n as i32);
    (2.0 * tail).min(1.0)
}

fn main() {
    println!("{:?}", mcnemar_exact(18, 12));   // 0.3616  not significant
    println!("{:?}", mcnemar_exact(6, 0));     // 0.0312  significant
}
```

Now suppose instead the models disagreed on only 6 tasks and B won all 6. B still beats A by 6 tasks, so the headline is the same 124 versus 130, but now p = 0.03. Same 3-point headline, opposite conclusions: 18 to 12 is noise, 6 to 0 is real. The pass rates alone cannot tell you which world you are in.

Interactive **Paired comparison, live** (shown at its default setting)

Two models on the same tasks. Only the disagreements carry information. Set agreement to 0.94 and the effect to 0.03 and watch a convincing looking win count fail to reach significance.



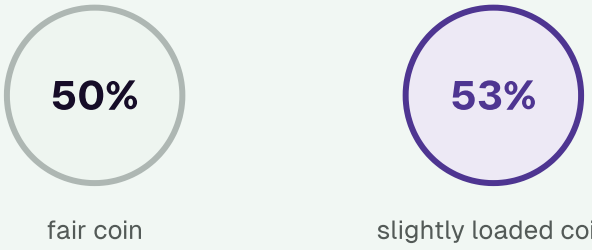
A wins (b)	B wins (c)
6	2
both pass	both fail
106	86

Discordant pairs: 8 of 200 tasks. Everything else carries no information.
Exact McNemar p = 0.2891 **bootstrap 95% CI** [-0.0050, 0.0500]
Observed difference 0.0200. **Not significant.** This difference is consistent with noise, however convincing the raw win count looks.

3. Power: the smallest difference you can see

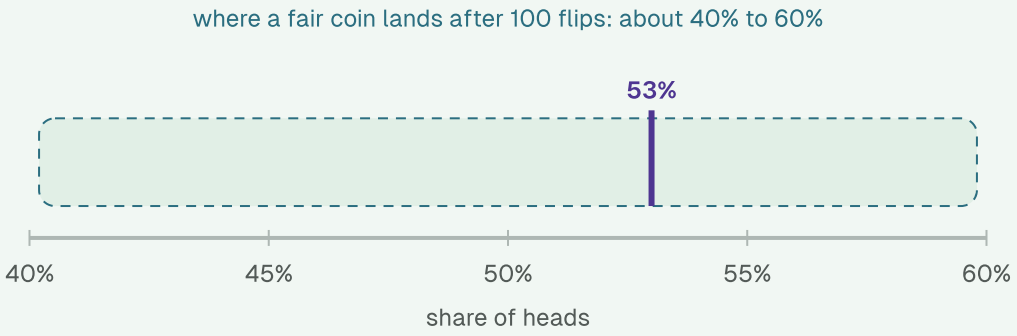
Diagram **How many flips to catch a loaded coin**

Power and sample size, with a coin that is only slightly unfair.

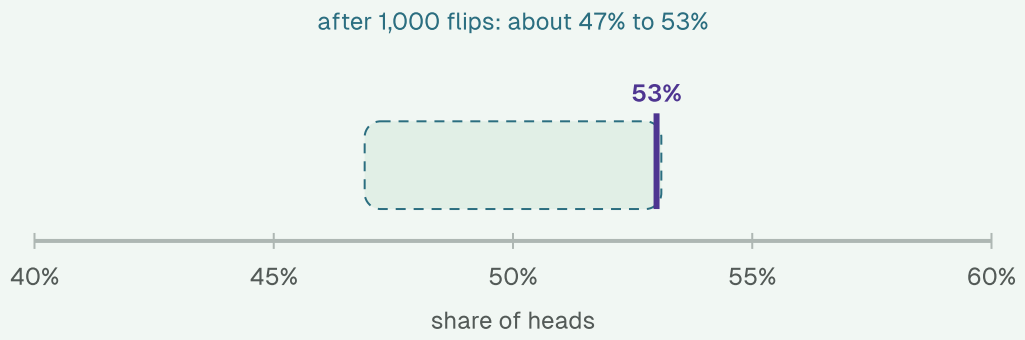


They look identical. How many flips before you can tell?

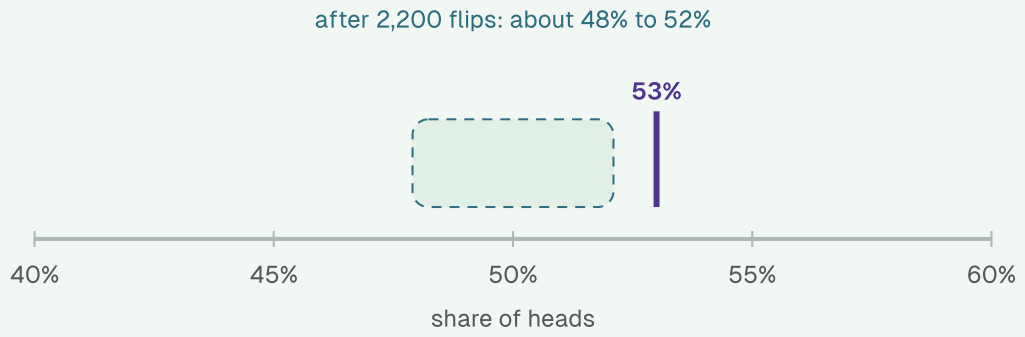
A fair coin lands heads 50% of the time. A slightly loaded one lands heads 53%. That 3-point gap is the size of a typical model improvement. The question is how many flips it takes to see it.



After 100 flips, a fair coin can easily land anywhere from about 40% to 60% heads. The loaded coin's 53% sits deep inside that band, so 100 flips cannot separate them. Neither can 100 eval tasks.

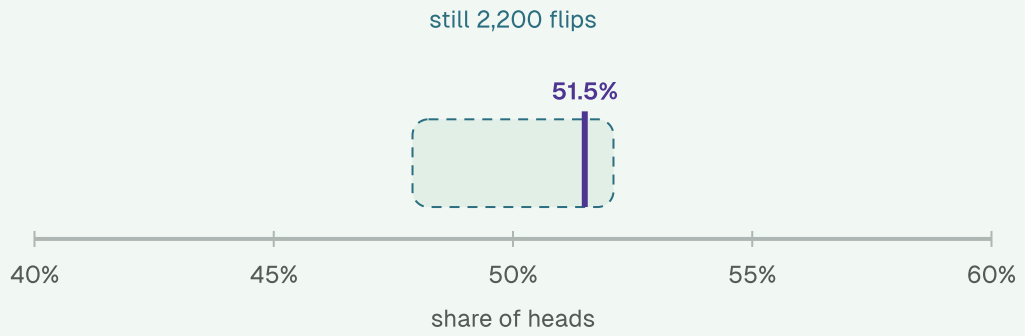


After 1,000 flips the band shrinks to about 47% to 53%. The loaded coin sits right at the edge. Sometimes you would catch it, often you would not.



Caught about 80% of the time: that is "80% power".

At about 2,200 flips the band is roughly 48% to 52%, and a 53% coin lands outside it about 80% of the time. That is what "80% power" means: the chance your test notices a difference that is really there.



Halve the gap and you need four times the flips: about 8,700.

Make the coin only 51.5% loaded and it hides inside the band again. Halving the gap you want to detect quadruples the flips needed, to about 8,700. Small improvements are expensive to prove. Pairing (the chefs story above) is the main way evals cut that cost.

Plain words

A blurry camera

A small eval set is a blurry camera. Big differences show up; small ones disappear into the blur. **Power** is the chance your eval detects a real improvement of a given size. The **minimum detectable effect** is the smallest improvement you could reliably see. If your camera can only see 5-point changes, it will report "no difference" for a real 2-point gain.

For leaders

Decide the effect size before you run

- **Ask what size of change matters, then size the suite.** If the team cares about 2 points, a 200-task suite cannot answer the question. Rule of thumb: a 2-point difference near a 50% base rate needs on the order of 1,000 to 4,000 paired tasks, depending on how often the models disagree.
- **Underpowered evals fail in two ways.** They miss real gains, and the gains they do "find" are inflated, because only lucky draws clear the bar. That second effect is the winner's curse, and it is why a surprising win on a small suite often shrinks on re-run.
- **"No significant difference" from an underpowered test is not evidence of no difference.** Ask for the minimum detectable effect alongside every null result.

Under the hood

Power by simulation

Simulation is the most defensible way to answer "how many tasks?", because the only assumptions are the ones you write into the simulator: base rate, effect, and how correlated the two models are per task.

python

```python
import random

def mde_by_simulation(base_rate: float, n_tasks: int, correlation: float = 0.8,
                      alpha: float = 0.05, power_target: float = 0.8,
                      trials: int = 2000, seed: int = 0) -> float:
    """Smallest true effect detectable at 80% power on n_tasks paired examples.

    correlation approximates how much the two models agree per task. Higher
    correlation gives a tighter paired comparison and a smaller detectable effect.
    """
    rng = random.Random(seed)

    def power_at(effect: float) -> float:
        hits = 0
        for _ in range(trials):
            a, b = [], []
            for _ in range(n_tasks):
                pa = base_rate + effect
                if rng.random() < correlation:          # shared difficulty draw
                    u = rng.random()
                    a.append(1 if u < pa else 0)
                    b.append(1 if u < base_rate else 0)
                else:                                   # independent draws
                    a.append(1 if rng.random() < pa else 0)
                    b.append(1 if rng.random() < base_rate else 0)
            if mcnemar_exact(a, b)["p"] < alpha:
                hits += 1
        return hits / trials

    lo, hi = 0.0, 0.30
    for _ in range(12):                         # binary search on the effect size
        mid = (lo + hi) / 2
        if power_at(mid) >= power_target:
            hi = mid
        else:
            lo = mid
    return hi
```

rust

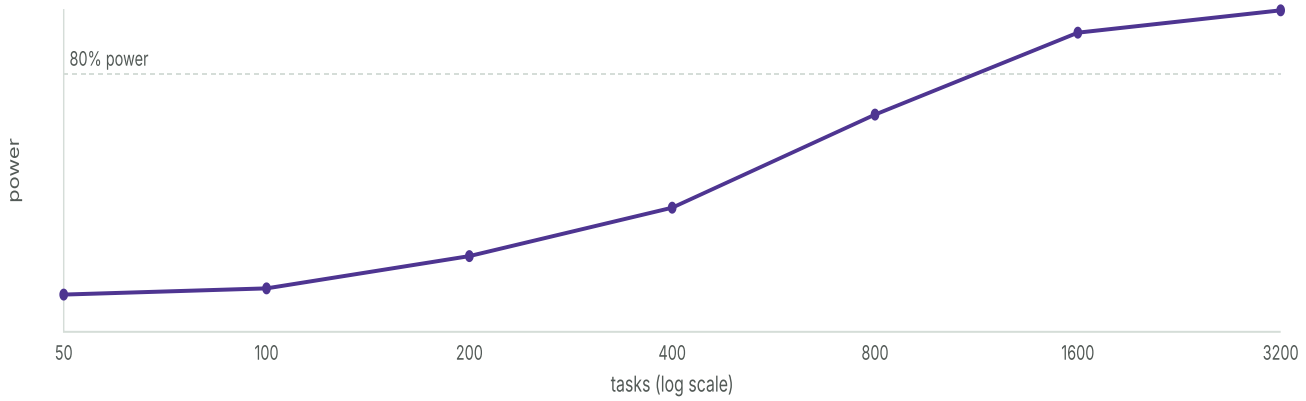
```rust
// Small inline PRNG (SplitMix64); its draws differ from Python's random module,
// so the effect this finds can differ from the Python version in the third digit.
struct Rng(u64);

impl Rng {
    fn next_u64(&mut self) -> u64 {
        self.0 = self.0.wrapping_add(0x9E37_79B9_7F4A_7C15);
        let mut z = self.0;
        z = (z ^ (z >> 30)).wrapping_mul(0xBF58_476D_1CE4_E5B9);
        z = (z ^ (z >> 27)).wrapping_mul(0x94D0_49BB_1331_11EB);
        z ^ (z >> 31)
    }
```

```
    fn random(&mut self) -> f64 {
        (self.next_u64() >> 11) as f64 / (1u64 << 53) as f64
    }
}

// The exact McNemar p value from the paired-test snippet above, repeated so
// this one compiles on its own (summed in log space, so any n works).
fn mcnemar_exact_p(a: &[u8], b: &[u8]) -> f64 {
    let b_count = a.iter().zip(b).filter(|&(&x, &y)| x == 1 && y == 0).count();
    let c_count = a.iter().zip(b).filter(|&(&x, &y)| x == 0 && y == 1).count();
    let n = b_count + c_count;
    if n == 0 {
        return 1.0;
    }
    // tail = P(X <= min(b, c)) for X ~ Binomial(n, 1/2), summed in log space
    let mut log_term = -(n as f64) * std::f64::consts::LN_2;
    let mut tail = log_term.exp();
    for i in 1..=b_count.min(c_count) {
        log_term += ((n - i + 1) as f64 / i as f64).ln();
        tail += log_term.exp();
    }
    (2.0 * tail).min(1.0)
}

/// Smallest true effect detectable at 80% power on n_tasks paired examples.
///
/// correlation approximates how much the two models agree per task. Higher
/// correlation gives a tighter paired comparison and a smaller detectable effect.
fn mde_by_simulation(base_rate: f64, n_tasks: usize, correlation: f64,
                     alpha: f64, power_target: f64, trials: usize, seed: u64) -> f64 {
    let mut rng = Rng(seed);

    let mut power_at = |effect: f64| -> f64 {
        let mut hits = 0;
        for _ in 0..trials {
            let (mut a, mut b) = (Vec::with_capacity(n_tasks), Vec::with_capacity(n_tasks));
            for _ in 0..n_tasks {
                let pa = base_rate + effect;
                if rng.random() < correlation {          // shared difficulty draw
                    let u = rng.random();
                    a.push((u < pa) as u8);
                    b.push((u < base_rate) as u8);
                } else {                                 // independent draws
                    a.push((rng.random() < pa) as u8);
                    b.push((rng.random() < base_rate) as u8);
                }
            }
            if mcnemar_exact_p(&a, &b) < alpha {
                hits += 1;
            }
        }
        hits as f64 / trials as f64
    };

    let (mut lo, mut hi) = (0.0, 0.30);
    for _ in 0..12 {                         // binary search on the effect size
        let mid = (lo + hi) / 2.0;
        if power_at(mid) >= power_target {
            hi = mid;
        } else {
```

```
            lo = mid;
        }
    }
    hi
}
```

Interactive **Power and sample size** (shown at its default setting)

How many tasks do you actually need. Model agreement is the slider people forget: the more two models agree, the fewer discordant pairs carry information, and the more tasks you need to separate them. Drag it up and watch the curve collapse.



Tasks needed for 80% power: ≈ 1600

Raise model agreement and the pairing helps you. Shrink the effect and cost explodes: power scales with the square of the effect.

Principal asks

Set the effect to 0.02 and agreement to 0.9. The curve does not reach 80 percent power even at 3,200 tasks. That is the quantitative case for generating tasks where models actually differ instead of growing a static suite.

4. Sample size, stated plainly

For leaders **Why converging models get expensive to compare**

A paired test only learns from the tasks where models disagree. As frontier models converge on a fixed suite, they disagree on fewer tasks, and the suite has to grow just to keep the same resolution. At a fixed win share among disagreements, halving the disagreement rate doubles the tasks you need. This is the economic argument for dynamic benchmark generation: build tasks where current models differ, instead of adding more tasks they all pass.

Use the closed form below for a quick answer in a meeting, and the simulation in the previous section when the number has to be defended.

Under the hood

Closed-form sample size for a paired binary comparison

Let p_d be the fraction of tasks where the models disagree and π the share of those disagreements the better model wins. Testing $\pi = \frac{1}{2}$ on the discordant pairs needs

$$n_d = \frac{\left(z_{1-\alpha/2} \cdot \frac{1}{2} + z_{1-\beta}\sqrt{\pi(1-\pi)}\right)^2}{\left(\pi - \frac{1}{2}\right)^2}, \qquad n = \frac{n_d}{p_d}$$

The difference in pass rates is $p_d(2\pi - 1)$. For 95% confidence and 80% power, $z_{1-\alpha/2} = 1.96$ and $z_{1-\beta} = 0.84$.

python

```python
import math
from statistics import NormalDist

def paired_sample_size(p_discordant: float, win_share: float = 0.7,
                       alpha: float = 0.05, power: float = 0.8) -> int:
    """Tasks needed for a paired binary comparison.

    p_discordant: fraction of tasks where the two models disagree.
    win_share:    of those disagreements, the fraction the better model wins.
    """
    z_a = NormalDist().inv_cdf(1 - alpha / 2)
    z_b = NormalDist().inv_cdf(power)
    d = win_share - 0.5
    if d <= 0:
        return 10 ** 9
    n_discordant = (0.5 * z_a + z_b * math.sqrt(win_share * (1 - win_share))) ** 2 / d ** 2
    return math.ceil(n_discordant / p_discordant)


print(paired_sample_size(0.12, 0.70))   # 389   models disagree on 12% of tasks
print(paired_sample_size(0.04, 0.70))   # 1166  same models after they converge
```

rust

```rust
// Horner evaluation, highest coefficient first.
fn poly(coeffs: &[f64], r: f64) -> f64 {
    coeffs.iter().fold(0.0, |acc, &c| acc * r + c)
}

// Inverse standard normal CDF: Wichura's AS241, the algorithm behind
// Python's NormalDist().inv_cdf, so the results agree to the last bit.
fn inv_cdf(p: f64) -> f64 {
    let q = p - 0.5;
    if q.abs() <= 0.425 {
        let r = 0.180625 - q * q;
        let num = poly(&[2.5090809287301226727e+3, 3.3430575583588128105e+4, 6.7265770927008700853e+4,
                         4.5921953931549871457e+4, 1.3731693765509461125e+4, 1.9715909503065514427e+3,
                         1.3314166789178437745e+2, 3.3871328727963666080e+0], r) * q;
        let den = poly(&[5.2264952788528545610e+3, 2.8729085735721942674e+4, 3.9307895800092710610e+4,
                         2.1213794301586595867e+4, 5.3941960214247511077e+3, 6.8718700749205790830e+2,
                         4.2313330701600911252e+1, 1.0], r);
        return num / den;
    }
    let r = if q <= 0.0 { p } else { 1.0 - p };
    let r = (-r.ln()).sqrt();
    let (num, den) = if r <= 5.0 {
        let r = r - 1.6;
        (poly(&[7.7454501427834140764e-4, 2.2723844989269184583e-2, 2.4178072517745061177e-1,
                1.2704582524523683826e+0, 3.6478483247632046050e+0, 5.7694972214606914055e+0,
                4.6303378461565452959e+0, 1.4234371107496835773e+0], r),
         poly(&[1.0507500716444168432e-9, 5.4759380849953449460e-4, 1.5198666563616457197e-2,
                1.4810397642748007459e-1, 6.8976733498510000455e-1, 1.6763848301838038494e+0,
                2.0531916266377588219e+0, 1.0], r))
    } else {
        let r = r - 5.0;
        (poly(&[2.0103343992922881327e-7, 2.7115555687434875782e-5, 1.2426609473880784386e-3,
                2.6532189526576123093e-2, 2.9656057182850489123e-1, 1.7848265399172913358e+0,
                5.4637849111641143699e+0, 6.6579046435011037772e+0], r),
         poly(&[2.0442631033899397856e-15, 1.4215117583164458887e-7, 1.8463183175100546818e-5,
                7.8686913114561325910e-4, 1.4875361290850614853e-2, 1.3692988092273580531e-1,
                5.9983220655588793769e-1, 1.0], r))
    };
    if q < 0.0 { -num / den } else { num / den }
}

/// Tasks needed for a paired binary comparison.
///
/// p_discordant: fraction of tasks where the two models disagree.
/// win_share:    of those disagreements, the fraction the better model wins.
fn paired_sample_size(p_discordant: f64, win_share: f64, alpha: f64, power: f64) -> u64 {
    let z_a = inv_cdf(1.0 - alpha / 2.0);
    let z_b = inv_cdf(power);
    let d = win_share - 0.5;
    if d <= 0.0 {
        return 1_000_000_000;
    }
    let n_discordant = (0.5 * z_a + z_b * (win_share * (1.0 - win_share)).sqrt()).powi(2) / d.powi(2);
    (n_discordant / p_discordant).ceil() as u64
}

fn main() {
    println!("{}", paired_sample_size(0.12, 0.70, 0.05, 0.8));   // 389   models disagree on 12% of tasks
```

```
    println!("{}", paired_sample_size(0.04, 0.70, 0.05, 0.8));   // 1166  same models after they converge
}
```

Worked problem

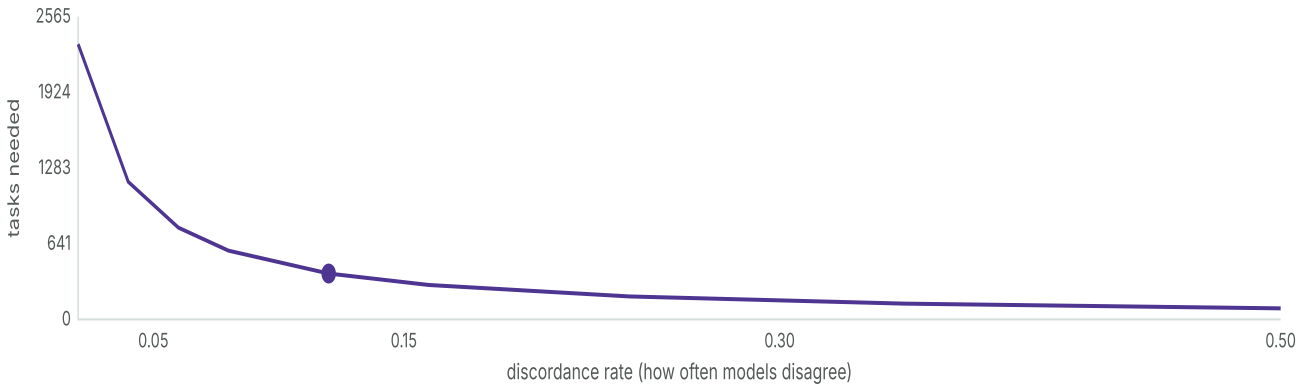
How many tasks do I need? A product team wants to detect an improvement of about 2 to 3 points.

Disagreement rate	Win share	Pass-rate gap	Tasks needed
0.12	0.62	2.9 pts	**1,116**
0.12	0.58	1.9 pts	**2,536**
0.06 (models have converged)	0.62	1.4 pts	**2,232**

A 2-point move needs over a thousand tasks, not two hundred. That is the honest answer to most sample size questions. And at a fixed win share, halving the disagreement rate doubles the requirement.

Interactive **Sample size calculator** (shown at its default setting)

The reciprocal curve that makes converging models so expensive to compare. Drag the discordance rate down toward 0.04 and watch the required task count run away.



Tasks needed: 389

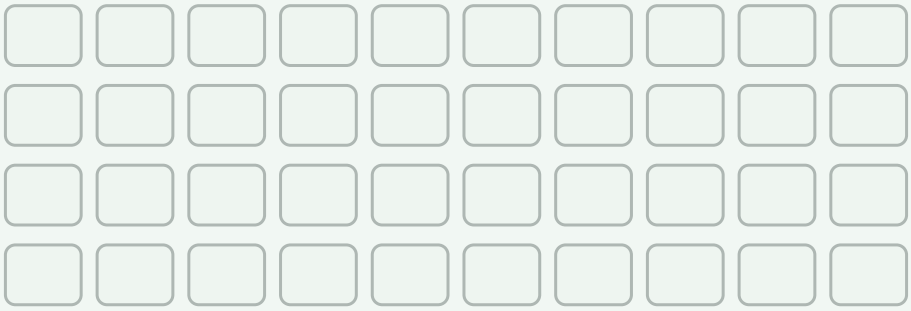
The curve is a reciprocal: halving the discordance rate doubles the tasks required. As frontier models converge on a fixed task set, discordance falls and this number runs away from you. Generating tasks where models actually differ is the only thing that pushes back.

5. Many benchmarks at once

Diagram **Forty scratch cards**

Why a big benchmark suite always shows a few "significant" wins.

40 benchmarks, and the new model changed none of them



Each card: did this benchmark "improve significantly"?

A new model runs on a standing suite of 40 benchmarks. Suppose, for now, that it is no better or worse on any of them. Each benchmark is a scratch card that might say "significant win".

With a 0.05 threshold, each card has a 1 in 20 chance of a fake win

40 × 0.05 = 2 fake wins on average, from pure luck.

Testing at p < 0.05 means each card has a 1 in 20 chance of a fake win even when nothing changed. Across 40 cards, expect about 2 fake wins every time you run the suite.

87%

chance that at least one of 40 null benchmarks lights up

$1 - 0.95^{40} \approx 0.87$

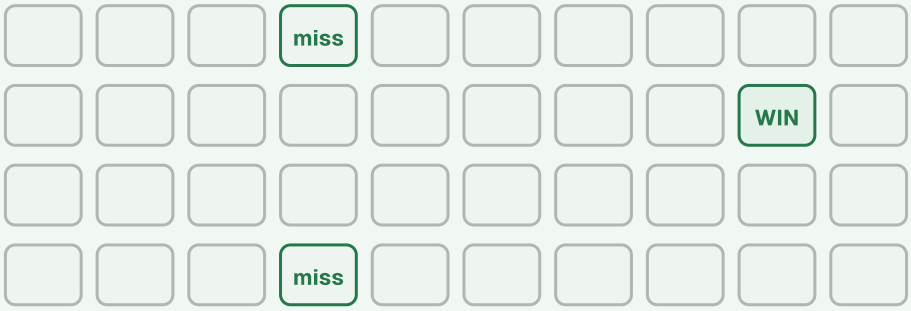
The chance that at least one card lights up is about 87%. A big standing suite will almost always show something "significant", which is why a list of wins with no correction is a red flag.

Now 3 benchmarks really improved (green) among 37 that did not

Six wins. From the outside, the real ones and the lucky ones look the same.

Now suppose 3 benchmarks genuinely improved. Without correction you see 6 wins: the 3 real ones and 3 lucky ones. This matches the worked problem below. Nothing on the card tells you which kind it is.

A correction raises the bar: Bonferroni tests each card at 0.05 / 40



No fake wins left, but two real wins were missed too.

A correction raises the bar for every card. The lucky wins disappear, and so do two of the real ones. There is no free option, only a choice about which mistake costs less: for regression monitoring, a missed regression usually costs more than a false alarm.

Plain words

Forty coin flips

Flip 40 fair coins five times each and a couple will land heads every time, by luck. Run 40 benchmarks on a new model and a couple will "improve significantly" by luck too. The more things you test, the more false wins you get. A **multiple comparisons correction** raises the bar to account for how many things you looked at.

For leaders

Pick which error you would rather make

- At a 0.05 threshold, 30 benchmarks with no real change produce about 1.5 false "significant" results on average. A big standing suite will always have something lighting up.
- Corrections are not free. They cut false alarms and also miss real effects. There is no option without a cost, only a choice about which error is cheaper.
- **The asymmetry to say out loud:** for regression monitoring, a missed regression ships a worse model while a false alarm costs an engineer a day. So monitoring deliberately accepts more false alarms than a publication claim would. For a launch announcement, the reverse.
- **Red flag:** a deck that lists "6 of 40 benchmarks improved" with no correction and no note of how many got worse.

Under the hood

Bonferroni and Benjamini–Hochberg

Bonferroni controls the chance of any false positive (the family-wise error rate) by testing each of m hypotheses at α/m. It is simple and very conservative on large suites.

Benjamini–Hochberg controls the false discovery rate, the expected share of discoveries that are false. Sort the p values $p_{(1)} \le \cdots \le p_{(m)}$ and find

$$k^* = \max\left\{ k : p_{(k)} \le \frac{k}{m}\, q \right\}$$

Reject the hypotheses with the k^* smallest p values, including any below k^* that individually failed the check. The threshold loosens as rank grows, which keeps power on a large suite.

python

```python
def benjamini_hochberg(p_values: list[float], fdr: float = 0.05) -> list[bool]:
    """Control the false discovery rate across a suite of benchmarks."""
    m = len(p_values)
    order = sorted(range(m), key=lambda i: p_values[i])   # indices, ascending p

    # PASS ONE: the largest rank whose p value clears rank/m * fdr.
    # Scan ALL ranks; a later rank can pass after an earlier one fails.
    max_rank = -1
    for rank, i in enumerate(order, start=1):
        if p_values[i] <= fdr * rank / m:
            max_rank = rank

    # PASS TWO: everything up to that rank is a discovery (the step-up rule,
    # and the part people most often implement wrong).
    keep = [False] * m
    for rank, i in enumerate(order, start=1):
        if rank <= max_rank:
            keep[i] = True
    return keep
```

rust

```rust
/// Control the false discovery rate across a suite of benchmarks.
fn benjamini_hochberg(p_values: &[f64], fdr: f64) -> Vec<bool> {
    let m = p_values.len();
    let mut order: Vec<usize> = (0..m).collect();          // indices, ascending p
    order.sort_by(|&i, &j| p_values[i].total_cmp(&p_values[j]));

    // PASS ONE: the largest rank whose p value clears rank/m * fdr.
    // Scan ALL ranks; a later rank can pass after an earlier one fails.
    let mut max_rank = 0;
    for (rank, &i) in (1..).zip(&order) {
        if p_values[i] <= fdr * rank as f64 / m as f64 {
            max_rank = rank;
        }
    }

    // PASS TWO: everything up to that rank is a discovery (the step-up rule,
    // and the part people most often implement wrong).
```

```
    let mut keep = vec![false; m];
    for (rank, &i) in (1..).zip(&order) {
        if rank <= max_rank {
            keep[i] = true;
        }
    }
    keep
}
```

Worked problem

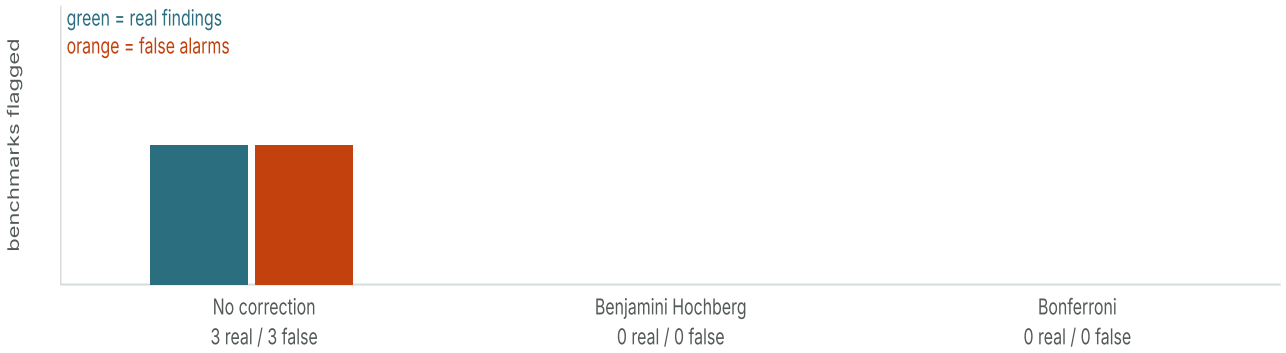
Forty benchmarks, six moved. Which do you believe? A new model lands, the standing suite of 40 benchmarks runs, and six come back with p below 0.05. One illustrative simulation where exactly three benchmarks genuinely moved:

Method	Real found	False alarms	Real missed
No correction	3	**3**	0
Benjamini–Hochberg	1	0	**2**
Bonferroni	1	0	**2**

Half the uncorrected "discoveries" were noise. Both corrections removed them and also missed two real effects. The right choice depends on the loss function, which is a product decision as much as a statistical one.

Interactive **Multiple comparisons, live** (shown at its default setting)

Run a whole suite against a new model and see how many benchmarks light up by luck. Compare no correction, Benjamini Hochberg and Bonferroni on the same draw.



Run 40 benchmarks where only 3 truly moved.

No correction: 3 real, **3 false alarms**, 0 missed.

Benjamini Hochberg: 0 real, 0 false, 3 missed.

Bonferroni: 0 real, 0 false, **3 missed**.

Bonferroni controls the chance of ANY false alarm and pays for it in missed regressions. For a standing suite that is the wrong trade: a missed regression ships, a false alarm costs an engineer a day.

6. pass@k and sampling variance

Diagram **Best of three free throws**

pass@1, pass@k, and pass^k with one basketball player.

A player who makes 40% of free throws

One shot each time: success 40% of the time

pass@1 = 0.40

A player makes 40% of free throws. Give them one shot and they succeed 40% of the time. For a model, that is pass@1: how often the first answer is right.

Three shots, and you only need one to go in

All three miss: 0.6 × 0.6 × 0.6 = 0.216

pass@3 = 1 – 0.216 ≈ 0.78

Now give them three shots and count it a success if any go in. They only fail if all three miss, which happens 21.6% of the time, so pass@3 is about 78%. This is pass@k, and it only matters if something can pick out the good attempt, like unit tests for code.

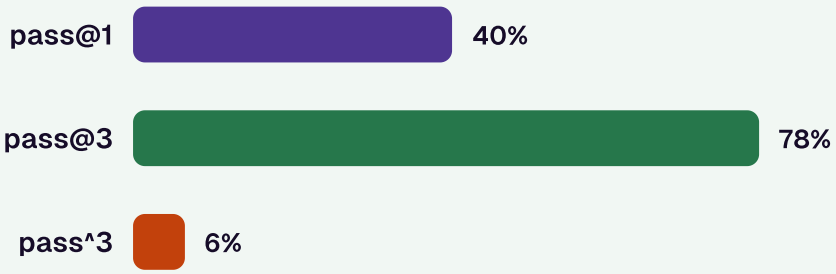
Three shots, and all three must go in

0.4 × 0.4 × 0.4 = 0.064

pass^3 ≈ 0.06

Flip the rule: all three must go in. That happens only 6.4% of the time. This is pass^k, a reliability measure. An agent that succeeds once and fails the next time is a problem pass@1 hides.

Same player, three different numbers



Ask which number matches how the product is used.

One player, one skill level, three honest numbers: 40%, 78% and 6%. A pass@k quoted next to a competitor's pass@1 is not a comparison. Always ask how many attempts, at what temperature, and whether the product can actually pick the best one.

Plain words

How many tries does the model get?

A model with some randomness can answer the same question differently each time. **pass@1** asks how often its first try is right. **pass@k** asks how often at least one of k tries is right, which matters

when something can check the answers (like running unit tests on code). The opposite question, how often *all* k tries are right, measures reliability.

For leaders

Match the metric to the product

- **pass@1** is what a user sees when they get one answer. **pass@k** is relevant only if the product samples several times and has a way to pick the right one, such as tests or a verifier.
- **pass^k** (all k attempts succeed, popularised by the τ-bench agent benchmark, Yao et al. 2024) measures consistency. Agents that pass once and fail the next time are a reliability problem pass@1 hides.
- Ask what temperature and how many samples produced a number. A pass@k quoted next to a competitor's pass@1 is not a comparison.

Under the hood

The unbiased pass@k estimator

Sample $n \geq k$ completions per task and count the c that pass. The unbiased estimator from the Codex paper (Chen et al. 2021) is the probability that a random subset of k contains at least one pass, averaged over tasks:

$$\widehat{\text{pass@}k} = \mathbb{E}_{\text{tasks}}\left[1 - \frac{\binom{n-c}{k}}{\binom{n}{k}}\right]$$

Sampling exactly k and checking is unbiased but high variance; sampling n and plugging c/n into $1-(1-c/n)^k$ is biased. The combinatorial form uses all n samples without bias.

python

```python
import numpy as np

def pass_at_k(n: int, c: int, k: int) -> float:
    """Unbiased estimate of pass@k given n samples of which c passed."""
    # Fewer than k failures: any k samples include a pass.
    if n - c < k:
        return 1.0
    # 1 - C(n-c, k)/C(n, k), computed as a product so large n does not overflow.
    return 1.0 - float(np.prod(1.0 - k / np.arange(n - c + 1, n + 1)))


def suite_pass_at_k(results: dict[str, tuple[int, int]], k: int) -> float:
    """results maps task_id -> (n_samples, n_passed). Average PER TASK,
    so one heavily sampled task cannot dominate the number."""
    return sum(pass_at_k(n, c, k) for n, c in results.values()) / len(results)
```

rust

```rust
use std::collections::HashMap;

/// Unbiased estimate of pass@k given n samples of which c passed.
fn pass_at_k(n: u64, c: u64, k: u64) -> f64 {
    // Fewer than k failures: any k samples include a pass.
    if n - c < k {
        return 1.0;
    }
    // 1 - C(n-c, k)/C(n, k), computed as a product so large n does not overflow.
    1.0 - (n - c + 1..=n).map(|i| 1.0 - k as f64 / i as f64).product::<f64>()
}

/// results maps task_id -> (n_samples, n_passed). Average PER TASK,
/// so one heavily sampled task cannot dominate the number.
fn suite_pass_at_k(results: &HashMap<String, (u64, u64)>, k: u64) -> f64 {
    results.values().map(|&(n, c)| pass_at_k(n, c, k)).sum::<f64>() / results.len() as f64
}
```

7. More than two models, and where the noise comes from

Plain words

A tournament, then a referee check

With three or more models, first ask whether they differ at all, then which pairs differ. Separately, it helps to know where the wobble in your scores comes from: are the tasks themselves very different in difficulty, or does the same task give different results each time you run it? That answer tells you whether to buy more tasks or more runs.

For leaders

More tasks or more samples?

This is a budgeting question that teams often guess at. Variance components answer it with a number. In one illustrative measurement on 60 tasks with 5 runs each, doubling the tasks reduced the standard error from 0.041 to 0.029, while doubling the runs per task only got it to 0.037, for the same compute. Task variance is divided only by the number of tasks; run-to-run noise is divided by tasks times runs. So once you have a few runs per task, new tasks are almost always the better buy.

Plain one-way ANOVA is usually the wrong tool for evals, because outcomes are binary and the design is paired. Ask for the paired version from the table below.

Under the hood

Cochran's Q, ANOVA, and variance components

Situation	Test	Why
2 models, binary, same tasks	McNemar exact	Paired, exact on small discordant counts
3+ models, binary, same tasks	Cochran's Q	The paired multi-group extension of McNemar
3+ models, continuous or ordinal scores	Friedman	Ranks within task, no distribution assumption
3+ models, continuous, independent groups	One-way ANOVA	Only when tasks are not shared, which is rare in evals
Score by model and task category	Two-way ANOVA	Separates model effect from task difficulty and their interaction
How much noise comes from where	Variance components	Tells you whether to add tasks or add samples

Cochran's Q

With k models, G_j the passes for model j, and L_i the passes on task i:

$$Q = (k-1)\,\frac{k\sum_j G_j^2 - \left(\sum_j G_j\right)^2}{k\sum_i L_i - \sum_i L_i^2} \;\sim\; \chi^2_{k-1} \text{ under } H_0$$

Tasks every model passed ($L_i = k$) or failed ($L_i = 0$) contribute nothing to the denominator: the same "only disagreement counts" logic as McNemar.

python

```python
import random

def cochran_q(X: list[list[int]]) -> float:
    """X[i][j] = 1 if model j passed task i. Paired across k models."""
    n, k = len(X), len(X[0])
    G = [sum(X[i][j] for i in range(n)) for j in range(k)]   # per model totals
    L = [sum(row) for row in X]                             # per task totals
    num = (k - 1) * (k * sum(g * g for g in G) - sum(G) ** 2)
    den = k * sum(L) - sum(l * l for l in L)
    return num / den if den else 0.0    # den == 0 when every task is unanimous
```

```python
def permutation_p(X, stat_fn, iters: int = 5000, seed: int = 0) -> tuple[float, float]:
    """p value with no distribution assumptions."""
    rng = random.Random(seed)
    observed = stat_fn(X)
    hits = 0
    for _ in range(iters):
        shuffled = []
        for row in X:
            r = list(row)
            rng.shuffle(r)      # shuffle WITHIN a task: models are exchangeable
            shuffled.append(r)  # under H0, but tasks differ in difficulty
        if stat_fn(shuffled) >= observed:
            hits += 1
    return observed, (hits + 1) / (iters + 1)   # +1 avoids ever reporting p = 0
```

rust

```rust
// Small inline PRNG (SplitMix64); its draws differ from Python's random module.
struct Rng(u64);

impl Rng {
    fn next_u64(&mut self) -> u64 {
        self.0 = self.0.wrapping_add(0x9E37_79B9_7F4A_7C15);
        let mut z = self.0;
        z = (z ^ (z >> 30)).wrapping_mul(0xBF58_476D_1CE4_E5B9);
        z = (z ^ (z >> 27)).wrapping_mul(0x94D0_49BB_1331_11EB);
        z ^ (z >> 31)
    }
    fn randrange(&mut self, n: usize) -> usize {
        ((self.next_u64() >> 11) as f64 / (1u64 << 53) as f64 * n as f64) as usize
    }
    // Fisher-Yates, walking down from the end like random.shuffle
    fn shuffle<T>(&mut self, v: &mut [T]) {
        for i in (1..v.len()).rev() {
            v.swap(i, self.randrange(i + 1));
        }
    }
}

/// X[i][j] = 1 if model j passed task i. Paired across k models.
fn cochran_q(x: &[Vec<u8>]) -> f64 {
    let (n, k) = (x.len(), x[0].len() as i64);
    let g: Vec<i64> = (0..x[0].len()).map(|j| (0..n).map(|i| x[i][j] as i64).sum()).collect(); // per model
totals
    let l: Vec<i64> = x.iter().map(|row| row.iter().map(|&v| v as i64).sum()).collect();       // per task totals
    let g_sum: i64 = g.iter().sum();
    let num = (k - 1) * (k * g.iter().map(|v| v * v).sum::<i64>() - g_sum * g_sum);
    let den = k * l.iter().sum::<i64>() - l.iter().map(|v| v * v).sum::<i64>();
    if den != 0 { num as f64 / den as f64 } else { 0.0 }   // den == 0 when every task is unanimous
}

/// p value with no distribution assumptions.
fn permutation_p(x: &[Vec<u8>], stat_fn: impl Fn(&[Vec<u8>]) -> f64,
                 iters: usize, seed: u64) -> (f64, f64) {
    let mut rng = Rng(seed);
    let observed = stat_fn(x);
    let mut hits = 0;
    for _ in 0..iters {
        let mut shuffled = Vec::with_capacity(x.len());
```

```
        for row in x {
            let mut r = row.clone();
            rng.shuffle(&mut r);   // shuffle WITHIN a task: models are exchangeable
            shuffled.push(r);      // under H0, but tasks differ in difficulty
        }
        if stat_fn(&shuffled) >= observed {
            hits += 1;
        }
    }
    (observed, (hits + 1) as f64 / (iters + 1) as f64)   // +1 avoids ever reporting p = 0
}
```

> **Watch out**
>
> Cochran's Q answers "are these models different at all". It does not say which. Running it and then reporting the winner is the multiple comparisons mistake in a different form. Follow a significant Q with pairwise McNemar tests and Benjamini–Hochberg across the pairs.

One-way ANOVA, for continuous scores

$$F = \frac{SS_{\text{between}}/(g-1)}{SS_{\text{within}}/(N-g)}, \qquad \eta^2 = \frac{SS_{\text{between}}}{SS_{\text{between}} + SS_{\text{within}}}$$

F is signal over noise. η^2 is the effect size: the share of variance explained by group membership. Always report η^2, because F grows with sample size and can be large for a difference nobody would notice.

python

```
def one_way_anova(groups: list[list[float]]) -> dict:
    g = len(groups)
    N = sum(len(x) for x in groups)
    grand = sum(sum(x) for x in groups) / N
    ss_between = sum(len(x) * (sum(x) / len(x) - grand) ** 2 for x in groups)   # signal
    ss_within = sum((v - sum(x) / len(x)) ** 2 for x in groups for v in x)       # noise
    df_b, df_w = g - 1, N - g
    ms_b, ms_w = ss_between / df_b, ss_within / df_w
    return {"F": ms_b / ms_w if ms_w else float("inf"), "df": (df_b, df_w),
            "eta_squared": ss_between / (ss_between + ss_within)}
```

rust

```
struct Anova {
    f: f64,
    df: (usize, usize),
    eta_squared: f64,
```

```rust
}

fn mean(x: &[f64]) -> f64 {
    x.iter().sum::<f64>() / x.len() as f64
}

fn one_way_anova(groups: &[Vec<f64>]) -> Anova {
    let g = groups.len();
    let big_n: usize = groups.iter().map(|x| x.len()).sum();
    let grand = groups.iter().map(|x| x.iter().sum::<f64>()).sum::<f64>() / big_n as f64;
    let ss_between: f64 = groups.iter().map(|x| x.len() as f64 * (mean(x) - grand).powi(2)).sum();
// signal
    let ss_within: f64 = groups.iter().map(|x| { let m = mean(x); x.iter().map(|v| (v - m).powi(2)).sum::<f64>()
}).sum(); // noise
    let (df_b, df_w) = (g - 1, big_n - g);
    let (ms_b, ms_w) = (ss_between / df_b as f64, ss_within / df_w as f64);
    Anova {
        f: if ms_w != 0.0 { ms_b / ms_w } else { f64::INFINITY },
        df: (df_b, df_w),
        eta_squared: ss_between / (ss_between + ss_within),
    }
}
```

Variance components: more tasks or more samples

Run each of n tasks r times. Split the variance into a task part σ^2_{task} and a run-to-run part σ^2_{samp}. The standard error of the suite mean is

$$\text{SE} = \sqrt{\frac{\sigma^2_{\text{task}}}{n} + \frac{\sigma^2_{\text{samp}}}{n\,r}}$$

Adding runs only shrinks the second term. Adding tasks shrinks both. The ratio $\sigma^2_{\text{task}} / (\sigma^2_{\text{task}} + \sigma^2_{\text{samp}})$ is the intraclass correlation, which reappears as the design effect in the next chapter.

python

```python
from statistics import mean
import math

def variance_components(scores: list[list[float]]) -> dict:
    """scores[i] = r repeated runs of task i at identical settings."""
    n, r = len(scores), len(scores[0])
    task_means = [mean(s) for s in scores]
    grand = mean(task_means)
    # Between-task mean square, on the same scale as the within term.
    ms_between = r * sum((m - grand) ** 2 for m in task_means) / (n - 1)
    # Within-task spread: pure run-to-run noise, difficulty is held fixed.
    ms_within = sum((x - task_means[i]) ** 2 for i in range(n) for x in scores[i]) / (n * (r - 1))
    var_task = max(0.0, (ms_between - ms_within) / r)   # noise can push it negative
```

```
    var_sampling = ms_within

    def se(n_tasks, n_reps):
        return math.sqrt(var_task / n_tasks + var_sampling / (n_tasks * n_reps))

    return {"var_task": var_task, "var_sampling": var_sampling,
            "icc": var_task / (var_task + var_sampling) if var_task + var_sampling else 0.0,
            "se_current": se(n, r), "se_double_tasks": se(2 * n, r), "se_double_samples": se(n, 2 * r)}

# Illustrative, 60 tasks x 5 runs: var_task 0.063, var_sampling 0.188
#   SE now 0.0409 | double the tasks 0.0290 | double the samples 0.0369
```

rust

```
struct VarianceComponents {
    var_task: f64,
    var_sampling: f64,
    icc: f64,
    se_current: f64,
    se_double_tasks: f64,
    se_double_samples: f64,
}

fn mean(x: &[f64]) -> f64 {
    x.iter().sum::<f64>() / x.len() as f64
}

/// scores[i] = r repeated runs of task i at identical settings.
fn variance_components(scores: &[Vec<f64>]) -> VarianceComponents {
    let (n, r) = (scores.len(), scores[0].len());
    let (nf, rf) = (n as f64, r as f64);
    let task_means: Vec<f64> = scores.iter().map(|s| mean(s)).collect();
    let grand = mean(&task_means);
    // Between-task mean square, on the same scale as the within term.
    let ms_between = rf * task_means.iter().map(|m| (m - grand).powi(2)).sum::<f64>() / (nf - 1.0);
    // Within-task spread: pure run-to-run noise, difficulty is held fixed.
    let ms_within = scores.iter().zip(&task_means)
        .map(|(s, m)| s.iter().map(|x| (x - m).powi(2)).sum::<f64>())
        .sum::<f64>() / (nf * (rf - 1.0));
    let var_task = ((ms_between - ms_within) / rf).max(0.0);   // noise can push it negative
    let var_sampling = ms_within;

    let se = |n_tasks: f64, n_reps: f64| (var_task / n_tasks + var_sampling / (n_tasks * n_reps)).sqrt();

    VarianceComponents {
        var_task,
        var_sampling,
        icc: if var_task + var_sampling != 0.0 { var_task / (var_task + var_sampling) } else { 0.0 },
        se_current: se(nf, rf),
        se_double_tasks: se(2.0 * nf, rf),
        se_double_samples: se(nf, 2.0 * rf),
    }
}

// Illustrative, 60 tasks x 5 runs: var_task 0.063, var_sampling 0.188
//   SE now 0.0409 | double the tasks 0.0290 | double the samples 0.0369
```

Principal asks

"The benchmark moved 4 points and nothing about the model changed. Why?" Usual suspects: temperature or top-p changed; the harness retried failures and quietly gave some tasks extra attempts; a timeout threshold moved; the judge prompt changed; the task set was silently filtered. See regression triage for the order to check them in.

Red flags in a results deck: a pass rate with no n; symmetric error bars near 100%; overlapping intervals used as a test; an unpaired comparison on shared tasks; "k of m benchmarks improved" with no correction and no count of regressions; a null result with no minimum detectable effect; pass@k compared with pass@1.

Part II · Chapter 10

Intervals, the bootstrap, and clustering in eval data

The previous chapter gave the tests. This chapter is the layer underneath, applied to eval metrics: what shape each metric has, what an interval promises, when the bootstrap saves you and when it quietly lies, and the clustering mistake that shrinks error bars on almost every agent eval. The general theory is in Part I, chapter 2 and the shapes themselves are in the distribution zoo.

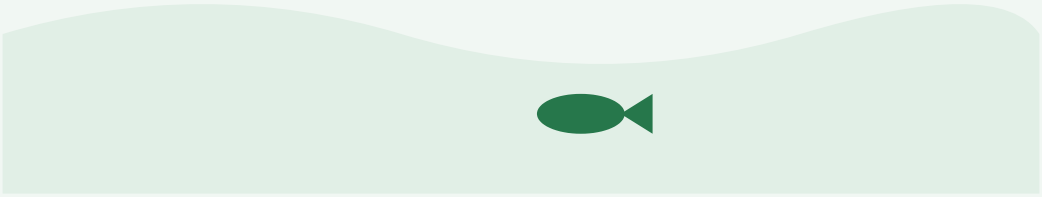
Diagram

Casting nets for a hidden fish

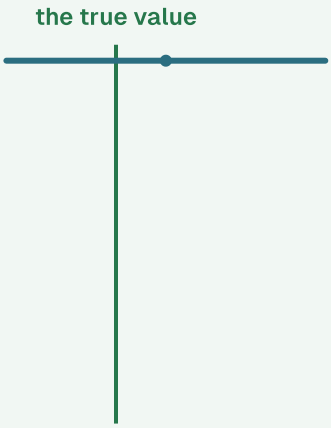
What a 95% confidence interval does and does not promise.

There is one true pass rate. You never see it directly.

Like a fish in murky water: you know it is there, not exactly where.

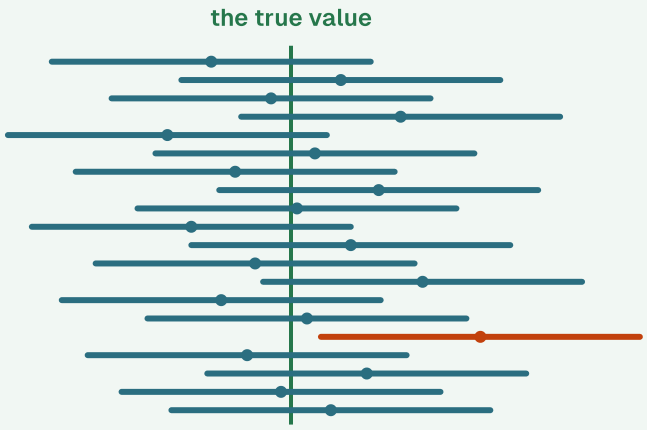


Every model has a true pass rate on the kind of tasks you care about. You never see it directly, only what one finite eval happened to show. Think of a fish in murky water.



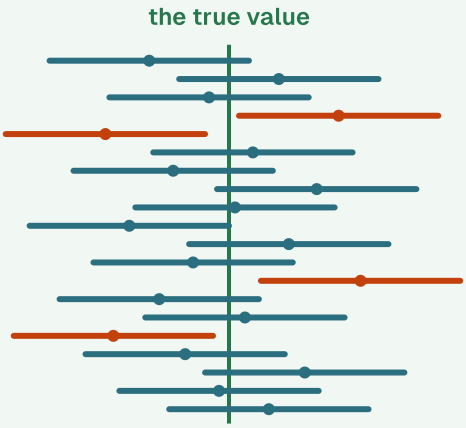
One eval run = one cast of the net

Each eval run casts a net: an interval around what you observed. This net happens to contain the fish. You cannot tell from one cast whether you caught it.



19 of 20 nets catch the fish: that is what "95%" means

Cast twenty times and about nineteen nets catch the fish; one misses. The 95% describes the casting method over many casts. It does not mean any one net has a 95% chance of holding the fish.



A narrower, overconfident net: about 16 of 20 catch

Some recipes make nets that are too small for their promise. The normal approximation at a 95% pass rate with 30 tasks catches the truth only about 78% of the time while still calling itself 95%. Checking coverage, as the Wilson widget in the previous chapter does, is how you catch this.

1. What shape is your metric?

Plain words **Different metrics, different math**

Measure the same thing many times and you get a spread of answers. The **shape** of that spread decides which math is allowed. Pass or fail behaves like coin flips. Latency is a pile on the left with a long tail to the right. Using the coin-flip tool on latency data is one of the most common mistakes in eval reporting.

For leaders

Which summary to trust

- For skewed metrics like latency and token counts, ask for the median and p95, not the mean. A single 40-second outlier moves the mean, and no user experiences the mean.
- A cheap automated guard: if mean divided by median is above about 1.2, the report should switch to quantiles. Put that check in the reporting code rather than in review comments.
- Count metrics such as tool calls per session vary far more than simple models assume. Intervals built on those assumptions are too narrow, so small changes look real.

Under the hood

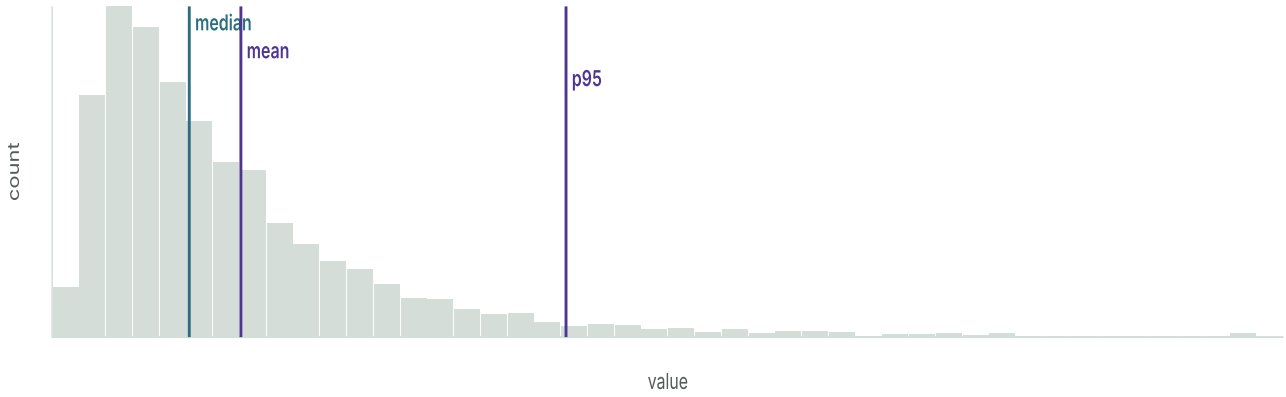
Sampling distributions converge at different speeds

The distribution that matters is the *sampling distribution* of your statistic: the spread of results you would get from different draws of tasks. The central limit theorem says sample means become normal regardless of the underlying shape. The catch is how fast. For symmetric data, quickly. For pass rates near 0 or 1, slowly. For heavy-tailed latency, slowly and from the wrong side, so the interval is wrong in a predictable direction. For statistics that are not means at all, such as p95 or the median, the theorem does not apply directly. That is where the bootstrap earns its keep.

A practical test: how far from a mean is your statistic, and how heavy is the tail? A pass rate is a mean of 0s and 1s and behaves well once n is decent and p is not extreme. p95 latency is an extreme order statistic on a heavy-tailed variable, the worst case for every closed form.

Interactive **Distribution explorer** (shown at its default setting)

Change the shape and watch which summary statistic stops being honest. Push a lognormal's skew up and see the mean detach from the median. Switch to negative binomial and watch variance over mean climb past the point where Poisson intervals are valid.



mean 2.74 **median** 1.99 **p95** 7.47 **var/mean** 2.47
skew ratio (mean/median) 1.38 — **do not report the mean.** Use median and percentiles.

Note

Set it to lognormal and drag the log standard deviation up. The skew ratio crosses 1.2 and the readout says to stop reporting the mean. That is the check to put in a reporting layer.

2. What a confidence interval promises

Plain words **A net that usually catches the fish**

A 95% confidence interval comes from a recipe that catches the true value 95 times out of 100 in the long run. It does **not** mean there is a 95% chance the truth is inside this particular range. The recipe is the thing with the 95% guarantee, and some recipes do not keep their promise.

For leaders **Nominal is not actual**

- Interval width depends on three things: how noisy the measurement is, how many samples you have, and how much confidence you demand. Only the middle one is under your control, and quadrupling the sample only halves the width.
- A "95%" interval can actually cover far less. Asking "did you check the coverage of your interval method at the rates we operate at?" is a senior question. It turns "we use Wilson" into "we verified Wilson holds up where our evals sit".

Under the hood

Measure coverage empirically

Different constructions have different real coverage. The normal approximation on a proportion falls well below 95% near a boundary; Wilson holds up. Percentile bootstrap intervals are asymptotically correct but under-cover on small samples and skewed statistics, which BCa corrects. You can check any method in a few lines:

python

```python
import random

def coverage_check(true_p: float, n: int, method, trials: int = 5000, seed: int = 0) -> float:
    """Empirically measure whether a '95%' interval covers 95% of the time.

    true p = 0.95, n = 30:   Wilson 0.94   normal approximation 0.79
    true p = 0.95, n = 50:   Wilson 0.96   normal approximation 0.93
    """
    rng = random.Random(seed)
    hits = 0
    for _ in range(trials):
        k = sum(1 for _ in range(n) if rng.random() < true_p)
        lo, hi = method(k, n)
        if lo <= true_p <= hi:
            hits += 1
    return hits / trials
```

rust

```rust
// Small inline PRNG (SplitMix64); its draws differ from Python's random module.
struct Rng(u64);

impl Rng {
    fn next_u64(&mut self) -> u64 {
        self.0 = self.0.wrapping_add(0x9E37_79B9_7F4A_7C15);
        let mut z = self.0;
        z = (z ^ (z >> 30)).wrapping_mul(0xBF58_476D_1CE4_E5B9);
        z = (z ^ (z >> 27)).wrapping_mul(0x94D0_49BB_1331_11EB);
        z ^ (z >> 31)
    }
    fn random(&mut self) -> f64 {
        (self.next_u64() >> 11) as f64 / (1u64 << 53) as f64
    }
    fn randrange(&mut self, n: usize) -> usize {
        (self.random() * n as f64) as usize
    }
}
```

```rust
/// Empirically measure whether a '95%' interval covers 95% of the time.
///
/// true p = 0.95, n = 30:   Wilson 0.94   normal approximation 0.79
/// true p = 0.95, n = 50:   Wilson 0.96   normal approximation 0.92
/// (this PRNG's draws with trials = 5000, seed = 0)
fn coverage_check(true_p: f64, n: u64, method: impl Fn(u64, u64) -> (f64, f64),
                  trials: usize, seed: u64) -> f64 {
    let mut rng = Rng(seed);
    let mut hits = 0;
    for _ in 0..trials {
        let k = (0..n).filter(|_| rng.random() < true_p).count() as u64;
        let (lo, hi) = method(k, n);
        if lo <= true_p && true_p <= hi {
            hits += 1;
        }
    }
    hits as f64 / trials as f64
}
```

3. The bootstrap

Diagram

Marbles out of one bag

The bootstrap: measuring wobble without collecting more data.

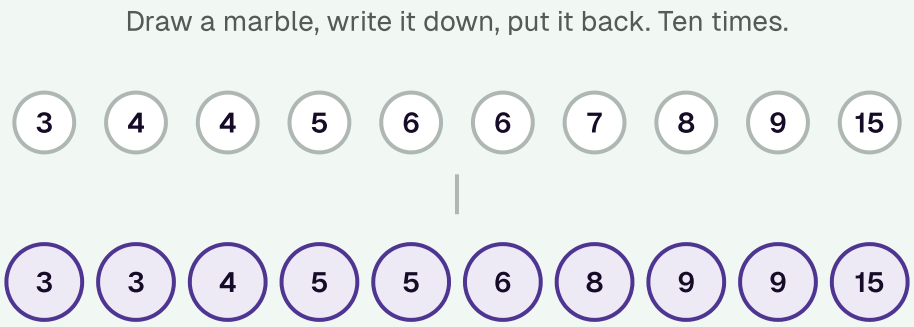
Your one sample: 10 response times, in seconds

3 4 4 5 6 6 7 8 9 15

Median = 6 s

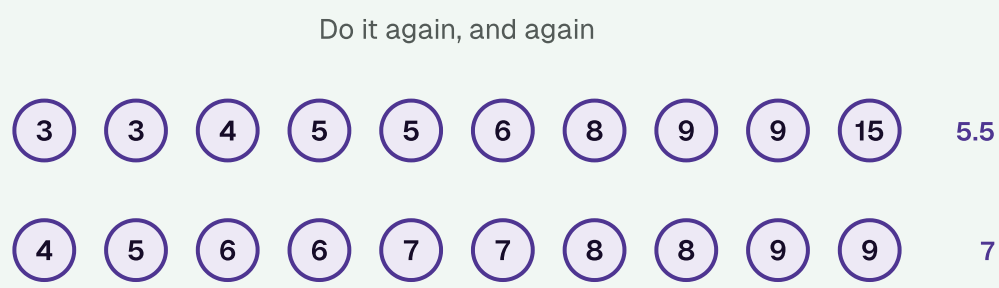
How much would that 6 wobble with a different sample?

You measured 10 response times. The median is 6 seconds. You want to know how much that would change with a different 10, but you cannot go back and collect more.



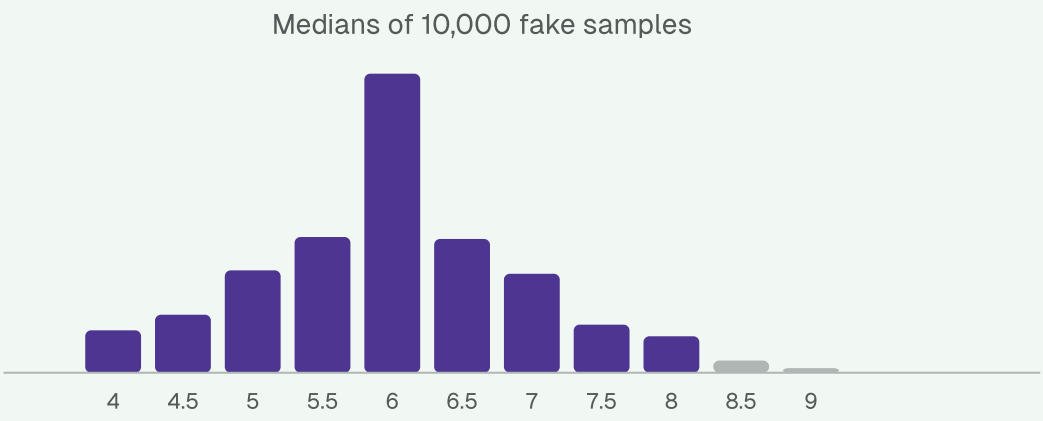
Fake sample 1: some marbles twice, some not at all. Median 5.5 s

The trick: draw from your own sample with replacement. Pick a marble, note it, put it back, ten times. Some marbles show up twice, some not at all. This fake sample has a median of 5.5 s.



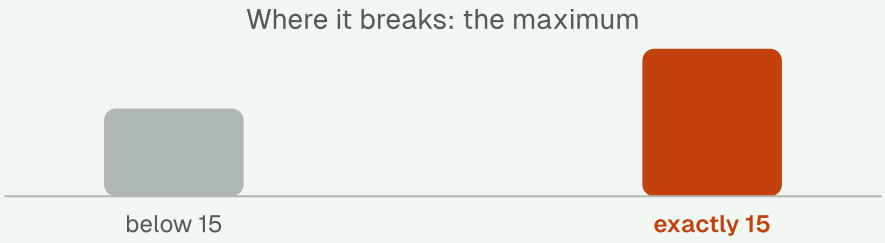
Each fake sample gives its own median.

Another fake sample gives a median of 7. Each draw gives a slightly different answer, and that variation is exactly the wobble you wanted to measure.



Middle 95% of the pile: about 4 s to 8 s. That is the bootstrap interval.

Repeat 10,000 times and the medians pile up. The middle 95% of the pile runs from about 4 to 8 seconds: that is a bootstrap confidence interval for the median, with no formula needed.



About 65% of fake samples have a maximum of exactly 15 s.

Fakes can never exceed your largest marble, so the bootstrap is falsely sure.

It fails for extremes. A fake sample can never contain anything bigger than your largest marble, and about 65% of them include it, so the bootstrap reports a tiny range for the maximum. The real worst case could be far worse. This is the failure the widget below demonstrates.

Plain words

Faking more samples from the one you have

You have one sample and want to know how much your answer would wobble if you collected more. The bootstrap fakes it: build a new dataset by drawing **with replacement** from the one you have,

recompute the number, do that ten thousand times, and look at the spread. The spread of the fakes approximates the spread of the real thing.

For leaders

The default for anything that is not an average

- The bootstrap works for almost any statistic you can compute: median, p95, AUC, a ratio of two rates, a calibration slope. That makes it the right default for eval work, where the interesting statistics are rarely simple means.
- Use the plain percentile version for a quick read. Ask for BCa when the number goes into a document someone will act on.
- It fails in specific, predictable situations, covered next. A team that can name those has understood the method rather than memorised it.

Under the hood

Percentile and BCa intervals

It works because your sample is your best estimate of the population, so resampling from it mimics resampling from the population. The percentile interval takes the $\alpha/2$ and $1-\alpha/2$ quantiles of the bootstrap replicates $\hat{\theta}^*_b$. BCa shifts those quantiles using a bias term $\hat{z}_0$ and an acceleration term $\hat{a}$ from the jackknife:

$$\hat{z}_0 = \Phi^{-1}\left(\frac{\#\{\hat{\theta}^*_b < \hat{\theta}\}}{B}\right), \qquad \hat{a} = \frac{\sum_i (\bar{\theta}_{(\cdot)} - \hat{\theta}_{(i)})^3}{6\left[\sum_i (\bar{\theta}_{(\cdot)} - \hat{\theta}_{(i)})^2\right]^{3/2}}$$

$$\alpha_{\text{lo,hi}} = \Phi\left(\hat{z}_0 + \frac{\hat{z}_0 + z_q}{1 - \hat{a}(\hat{z}_0 + z_q)}\right), \quad q \in \{\alpha/2,\ 1-\alpha/2\}$$

$\hat{\theta}_{(i)}$ is the statistic with point i left out, and $\bar{\theta}_{(\cdot)}$ their mean. When $\hat{z}_0 = \hat{a} = 0$, BCa reduces to the percentile interval.

python

```python
import random
from statistics import NormalDist, mean

def bootstrap_ci(data: list, stat_fn, iters: int = 10_000,
                 alpha: float = 0.05, seed: int = 0) -> tuple[float, float, float]:
    """Percentile bootstrap. Works for ANY statistic: median, p95, a ratio, a slope."""
    rng = random.Random(seed)
    n = len(data)
    # Draw n items WITH replacement each time. Without replacement every rep
```

```
    # would be the same dataset reshuffled.
    reps = sorted(stat_fn([data[rng.randrange(n)] for _ in range(n)]) for _ in range(iters))
    return stat_fn(data), reps[int((alpha / 2) * iters)], reps[int((1 - alpha / 2) * iters) - 1]


def bca_ci(data: list, stat_fn, iters: int = 10_000,
           alpha: float = 0.05, seed: int = 0) -> tuple[float, float, float]:
    """Bias corrected and accelerated bootstrap."""
    rng = random.Random(seed)
    n = len(data)
    theta = stat_fn(data)
    reps = sorted(stat_fn([data[rng.randrange(n)] for _ in range(n)]) for _ in range(iters))

    # Bias: if unbiased, half the reps land below theta.
    prop = sum(1 for r in reps if r < theta) / iters
    prop = min(max(prop, 1 / iters), 1 - 1 / iters)    # keep inv_cdf defined
    N = NormalDist()
    z0 = N.inv_cdf(prop)

    # Acceleration: skewness of the jackknife values.
    jack = [stat_fn(data[:i] + data[i + 1:]) for i in range(n)]
    jm = mean(jack)
    num = sum((jm - x) ** 3 for x in jack)
    den = 6 * (sum((jm - x) ** 2 for x in jack) ** 1.5)
    a = num / den if den else 0.0

    bounds = []
    for q in (alpha / 2, 1 - alpha / 2):
        z = N.inv_cdf(q)
        bounds.append(min(max(N.cdf(z0 + (z0 + z) / (1 - a * (z0 + z))), 0.0), 1.0))
    return theta, reps[min(iters - 1, int(bounds[0] * iters))], reps[min(iters - 1, int(bounds[1] * iters))]
```

rust

```
// Small inline PRNG (SplitMix64); its draws differ from Python's random module.
struct Rng(u64);

impl Rng {
    fn next_u64(&mut self) -> u64 {
        self.0 = self.0.wrapping_add(0x9E37_79B9_7F4A_7C15);
        let mut z = self.0;
        z = (z ^ (z >> 30)).wrapping_mul(0xBF58_476D_1CE4_E5B9);
        z = (z ^ (z >> 27)).wrapping_mul(0x94D0_49BB_1331_11EB);
        z ^ (z >> 31)
    }
    fn random(&mut self) -> f64 {
        (self.next_u64() >> 11) as f64 / (1u64 << 53) as f64
    }
    fn randrange(&mut self, n: usize) -> usize {
        (self.random() * n as f64) as usize
    }
}

// Horner evaluation, highest coefficient first.
fn poly(coeffs: &[f64], r: f64) -> f64 {
    coeffs.iter().fold(0.0, |acc, &c| acc * r + c)
}

// Inverse standard normal CDF: Wichura's AS241, the algorithm behind
// Python's NormalDist().inv_cdf, so the results agree to the last bit.
```

```rust
fn inv_cdf(p: f64) -> f64 {
    let q = p - 0.5;
    if q.abs() <= 0.425 {
        let r = 0.180625 - q * q;
        let num = poly(&[2.5090809287301226727e+3, 3.3430575583588128105e+4, 6.7265770927008700853e+4,
                         4.5921953931549871457e+4, 1.3731693765509461125e+4, 1.9715909503065514427e+3,
                         1.3314166789178437745e+2, 3.3871328727963666080e+0], r) * q;
        let den = poly(&[5.2264952788528545610e+3, 2.8729085735721942674e+4, 3.9307895800092710610e+4,
                         2.1213794301586595867e+4, 5.3941960214247511077e+3, 6.8718700749205790830e+2,
                         4.2313330701600911252e+1, 1.0], r);
        return num / den;
    }
    let r = if q <= 0.0 { p } else { 1.0 - p };
    let r = (-r.ln()).sqrt();
    let (num, den) = if r <= 5.0 {
        let r = r - 1.6;
        (poly(&[7.7454501427834140764e-4, 2.2723844989269184583e-2, 2.4178072517745061177e-1,
                1.2704582524523683826e+0, 3.6478483247632046050e+0, 5.7694972214606914055e+0,
                4.6303378461565452959e+0, 1.4234371107496835773e+0], r),
         poly(&[1.0507500716444168432e-9, 5.4759380849953449460e-4, 1.5198666563616457197e-2,
                1.4810397642748007459e-1, 6.8976733498510000455e-1, 1.6763848301838038494e+0,
                2.0531916266377588219e+0, 1.0], r))
    } else {
        let r = r - 5.0;
        (poly(&[2.0103343992922881327e-7, 2.7115555687434875782e-5, 1.2426609473880784386e-3,
                2.6532189526576123093e-2, 2.9656057182850489123e-1, 1.7848265399172913358e+0,
                5.4637849111641143699e+0, 6.6579046435011037772e+0], r),
         poly(&[2.0442631033899397856e-15, 1.4215117583164458887e-7, 1.8463183175100546818e-5,
                7.8686913114561325910e-4, 1.4875361290850614853e-2, 1.3692988092273580531e-1,
                5.9983220655588793769e-1, 1.0], r))
    };
    if q < 0.0 { -num / den } else { num / den }
}

// Standard normal CDF via erfc (Numerical Recipes' Chebyshev fit, relative
// error below 1.2e-7: ample for picking a bootstrap percentile).
fn cdf(x: f64) -> f64 {
    let z = x.abs() / std::f64::consts::SQRT_2;
    let t = 1.0 / (1.0 + 0.5 * z);
    let erfc = t * (-z * z - 1.26551223 + t * (1.00002368 + t * (0.37409196 + t * (0.09678418
        + t * (-0.18628806 + t * (0.27886807 + t * (-1.13520398 + t * (1.48851587
        + t * (-0.82215223 + t * 0.17087277)))))))).exp();
    if x >= 0.0 { 1.0 - 0.5 * erfc } else { 0.5 * erfc }
}

fn mean(x: &[f64]) -> f64 {
    x.iter().sum::<f64>() / x.len() as f64
}

/// Percentile bootstrap. Works for ANY statistic: median, p95, a ratio, a slope.
fn bootstrap_ci(data: &[f64], stat_fn: impl Fn(&[f64]) -> f64, iters: usize,
                alpha: f64, seed: u64) -> (f64, f64, f64) {
    let mut rng = Rng(seed);
    let n = data.len();
    // Draw n items WITH replacement each time. Without replacement every rep
    // would be the same dataset reshuffled.
    let mut reps: Vec<f64> = (0..iters)
        .map(|_| stat_fn(&(0..n).map(|_| data[rng.randrange(n)]).collect::<Vec<_>>()))
        .collect();
    reps.sort_by(f64::total_cmp);
```

```rust
    let it = iters as f64;
    (stat_fn(data), reps[((alpha / 2.0) * it) as usize], reps[((1.0 - alpha / 2.0) * it) as usize - 1])
}

/// Bias corrected and accelerated bootstrap.
fn bca_ci(data: &[f64], stat_fn: impl Fn(&[f64]) -> f64, iters: usize,
          alpha: f64, seed: u64) -> (f64, f64, f64) {
    let mut rng = Rng(seed);
    let n = data.len();
    let it = iters as f64;
    let theta = stat_fn(data);
    let mut reps: Vec<f64> = (0..iters)
        .map(|_| stat_fn(&(0..n).map(|_| data[rng.randrange(n)]).collect::<Vec<_>>()))
        .collect();
    reps.sort_by(f64::total_cmp);

    // Bias: if unbiased, half the reps land below theta.
    let prop = reps.iter().filter(|&&r| r < theta).count() as f64 / it;
    let prop = prop.clamp(1.0 / it, 1.0 - 1.0 / it);   // keep inv_cdf defined
    let z0 = inv_cdf(prop);

    // Acceleration: skewness of the jackknife values.
    let jack: Vec<f64> = (0..n)
        .map(|i| stat_fn(&[&data[..i], &data[i + 1..]].concat()))
        .collect();
    let jm = mean(&jack);
    let num: f64 = jack.iter().map(|x| (jm - x).powi(3)).sum();
    let den = 6.0 * jack.iter().map(|x| (jm - x).powi(2)).sum::<f64>().powf(1.5);
    let a = if den != 0.0 { num / den } else { 0.0 };

    let bound = |q: f64| {
        let z = inv_cdf(q);
        cdf(z0 + (z0 + z) / (1.0 - a * (z0 + z))).clamp(0.0, 1.0)
    };
    let pick = |b: f64| reps[(iters - 1).min((b * it) as usize)];
    (theta, pick(bound(alpha / 2.0)), pick(bound(1.0 - alpha / 2.0)))
}
```

Interactive **Bootstrap, and where it breaks** (shown at its default setting)

Pick a statistic and watch the bootstrap distribution form. Switch to p95 at small n, or to max at any n, and the widget will tell you why the interval you are looking at is not trustworthy.

median = 2.070 **95% CI** [1.678, 2.352] width 0.674 (32.6% of the estimate)

⚠ **Watch out**

Select max. The bootstrap is inconsistent for maxima at every sample size, not just small ones. It is the cleanest example of a method failing because an assumption is violated rather than because there was too little data.

4. When not to bootstrap

The bootstrap estimates sampling variability under one assumption: your rows are exchangeable draws from the population. Every failure below is a violation of that assumption. If you can name the assumption, you can predict the failure.

Situation	What goes wrong	What to do instead
Very small n, under about 20	Resamples only contain values you already saw, so the interval cannot represent tails the sample missed	Exact methods: Clopper–Pearson for proportions, exact McNemar for pairs
Extreme quantiles, p99 on 200 points	p99 rests on about 2 observations. The bootstrap max is your observed max, so the upper bound is structurally censored	Report p95, collect more data, or fit a tail model and say you did
Dependent data, several calls per session	Resampling rows destroys the dependence and shrinks the interval, often by 2x or more	Cluster bootstrap: resample sessions, not rows (section 6)
Time-ordered data with drift	Resampling scrambles the ordering that carries the signal	Block bootstrap: resample contiguous chunks
Statistics at a boundary, near 0% or 100%	The bootstrap distribution piles up on the boundary and the interval degenerates	Wilson or Clopper–Pearson
The statistic is a max or min	The bootstrap is inconsistent for extremes. A known failure, not a small-sample issue	Extreme value theory, or change the metric

5. Each eval metric and what it needs

For leaders

The lookup table for common product metrics

Pass rates, latency, token counts, tool use, and user satisfaction are five different statistical objects, and three of them break the default method. When a dashboard shows one of these, check the method against this table.

Metric	Distribution	Right method	Common mistake
Pass / fail rate	Bernoulli; binomial in aggregate	Wilson CI. McNemar or paired bootstrap for comparisons	Normal approximation near 0 or 1; ignoring the pairing
Latency	Right-skewed, heavy tail, roughly lognormal	Report p50, p95, p99 with bootstrap CIs	Reporting the mean
Token counts	Right-skewed, overdispersed counts	Median and quantiles with bootstrap CI, or model on the log scale	t-test on raw counts
Tool calls per session	Overdispersed counts, often zero-inflated	Negative binomial, or quantiles plus bootstrap. Check variance over mean first	Assuming Poisson. Real variance/mean ratios often run 3 to 6
Tool call success rate	Binomial but **clustered** within session	Cluster bootstrap at the session level	Treating each call as independent
Thumbs-up rate	Binomial with severe selection bias	Wilson on responders, plus the response rate stated explicitly	Calling it satisfaction. Only a biased slice of users clicks
Likert 1 to 5	Ordinal, not interval	Top-two-box proportion, or Mann–Whitney U on ranks	Averaging. The gap from 4 to 5 is not the gap from 1 to 2
Acceptance rate	Binomial, clustered by user and session	Cluster bootstrap on the user	Pooling all events into a tight, wrong interval

Under the hood

Latency, counts, and ratings, worked

Latency

python

```python
def latency_report(samples: list[float]) -> dict:
    """What a latency report should contain.

    Illustrative lognormal sample of 400 requests:
      mean 2.81, median 2.15, p95 8.06, max 17.17
    The mean sits between the median and p95 and describes nobody's experience.
    """
    s = sorted(samples)
    n = len(s)
    q = lambda p: s[min(n - 1, int(p * n))]
    p50_ci = bootstrap_ci(samples, lambda d: sorted(d)[len(d) // 2])
    p95_ci = bootstrap_ci(samples, lambda d: sorted(d)[min(len(d) - 1, int(0.95 * len(d)))])
    return {
        "n": n,
        "p50": q(0.50), "p50_ci": (p50_ci[1], p50_ci[2]),
        "p95": q(0.95), "p95_ci": (p95_ci[1], p95_ci[2]),
        "p99": q(0.99), "p99_warning": "p99 on n<1000 rests on a handful of points.",
        "skew_ratio": (sum(samples) / n) / q(0.50),   # above 1.2: do not use the mean
    }
```

rust

```rust
// Small inline PRNG (SplitMix64); its draws differ from Python's random module.
struct Rng(u64);

impl Rng {
    fn next_u64(&mut self) -> u64 {
        self.0 = self.0.wrapping_add(0x9E37_79B9_7F4A_7C15);
        let mut z = self.0;
        z = (z ^ (z >> 30)).wrapping_mul(0xBF58_476D_1CE4_E5B9);
        z = (z ^ (z >> 27)).wrapping_mul(0x94D0_49BB_1331_11EB);
        z ^ (z >> 31)
    }
    fn random(&mut self) -> f64 {
        (self.next_u64() >> 11) as f64 / (1u64 << 53) as f64
    }
    fn randrange(&mut self, n: usize) -> usize {
        (self.random() * n as f64) as usize
    }
}

// bootstrap_ci from the snippet above, repeated so this one compiles on its own.
fn bootstrap_ci(data: &[f64], stat_fn: impl Fn(&[f64]) -> f64, iters: usize,
                alpha: f64, seed: u64) -> (f64, f64, f64) {
    let mut rng = Rng(seed);
    let n = data.len();
    let mut reps: Vec<f64> = (0..iters)
        .map(|_| stat_fn(&(0..n).map(|_| data[rng.randrange(n)]).collect::<Vec<_>>()))
        .collect();
    reps.sort_by(f64::total_cmp);
    let it = iters as f64;
    (stat_fn(data), reps[((alpha / 2.0) * it) as usize], reps[((1.0 - alpha / 2.0) * it) as usize - 1])
}

struct LatencyReport {
```

```
    n: usize,
    p50: f64,
    p50_ci: (f64, f64),
    p95: f64,
    p95_ci: (f64, f64),
    p99: f64,
    p99_warning: &'static str,
    skew_ratio: f64,
}

fn sorted(d: &[f64]) -> Vec<f64> {
    let mut s = d.to_vec();
    s.sort_by(f64::total_cmp);
    s
}

/// What a latency report should contain.
///
/// Illustrative lognormal sample of 400 requests:
///   mean 2.81, median 2.15, p95 8.06, max 17.17
/// The mean sits between the median and p95 and describes nobody's experience.
fn latency_report(samples: &[f64]) -> LatencyReport {
    let s = sorted(samples);
    let n = s.len();
    let q = |p: f64| s[(n - 1).min((p * n as f64) as usize)];
    let p50_ci = bootstrap_ci(samples, |d| sorted(d)[d.len() / 2], 10_000, 0.05, 0);
    let p95_ci = bootstrap_ci(samples, |d| sorted(d)[(d.len() - 1).min((0.95 * d.len() as f64) as usize)],
10_000, 0.05, 0);
    LatencyReport {
        n,
        p50: q(0.50), p50_ci: (p50_ci.1, p50_ci.2),
        p95: q(0.95), p95_ci: (p95_ci.1, p95_ci.2),
        p99: q(0.99), p99_warning: "p99 on n<1000 rests on a handful of points.",
        skew_ratio: (samples.iter().sum::<f64>() / n as f64) / q(0.50),   // above 1.2: do not use the mean
    }
}
```

Counts, and why Poisson usually fails

Poisson assumes variance equals mean. In agent evals the rate is not constant: an easy task takes two tool calls and a hard one takes twenty. A mixture of Poissons with gamma-distributed rates is a negative binomial, which is why real counts are overdispersed. With variance/mean ratio R, Poisson intervals are roughly $\sqrt{R}$ times too narrow.

python

```
from statistics import mean

def overdispersion_check(counts: list[int]) -> dict:
    """Illustrative session mix: mean 5.15, variance 25.53, ratio 4.95.
    A ratio of 5 makes Poisson intervals about sqrt(5) = 2.2x too narrow."""
    m = mean(counts)
    n = len(counts)
    v = sum((x - m) ** 2 for x in counts) / (n - 1)
    ratio = v / m if m else float("inf")
    return {"mean": m, "variance": v, "ratio": ratio,
```

```
            "verdict": "poisson ok" if ratio < 1.5 else "overdispersed: negative binomial or bootstrap",
            "zero_fraction": sum(1 for x in counts if x == 0) / n}
```

rust

```
struct Overdispersion {
    mean: f64,
    variance: f64,
    ratio: f64,
    verdict: &'static str,
    zero_fraction: f64,
}

/// Illustrative session mix: mean 5.15, variance 25.53, ratio 4.95.
/// A ratio of 5 makes Poisson intervals about sqrt(5) = 2.2x too narrow.
fn overdispersion_check(counts: &[u64]) -> Overdispersion {
    let n = counts.len() as f64;
    let m = counts.iter().sum::<u64>() as f64 / n;
    let v = counts.iter().map(|&x| (x as f64 - m).powi(2)).sum::<f64>() / (n - 1.0);
    let ratio = if m != 0.0 { v / m } else { f64::INFINITY };
    Overdispersion {
        mean: m,
        variance: v,
        ratio,
        verdict: if ratio < 1.5 { "poisson ok" } else { "overdispersed: negative binomial or bootstrap" },
        zero_fraction: counts.iter().filter(|&&x| x == 0).count() as f64 / n,
    }
}
```

Satisfaction: ordinal, not interval

python

```
def mann_whitney_u(x: list[float], y: list[float]) -> dict:
    """Compare two groups on ordinal data without pretending it is interval.

    Illustrative Likert shift: means 3.21 -> 3.35, which looks like nothing.
    Common-language effect 0.54: a random new rating beats a random old one 54%
    of the time. Top-two-box moved 0.470 -> 0.525. Those two are actionable.
    """
    combined = sorted([(v, 0) for v in x] + [(v, 1) for v in y], key=lambda t: t[0])
    ranks = [0.0] * len(combined)
    i = 0
    while i < len(combined):                # average ranks across ties
        j = i
        while j + 1 < len(combined) and combined[j + 1][0] == combined[i][0]:
            j += 1
        for k in range(i, j + 1):
            ranks[k] = (i + j) / 2 + 1
        i = j + 1
    r1 = sum(r for r, (_, g) in zip(ranks, combined) if g == 0)
    n1, n2 = len(x), len(y)
    u1 = r1 - n1 * (n1 + 1) / 2
    return {"U": u1, "common_language_effect": u1 / (n1 * n2)}   # P(random x > random y)


def top_two_box(ratings: list[int], scale_max: int = 5) -> float:
```

```
    """Fraction rating 4 or 5: the number product teams can act on."""
    return sum(1 for r in ratings if r >= scale_max - 1) / len(ratings)
```

rust

```rust
struct MannWhitney {
    u: f64,
    common_language_effect: f64,   // P(random x > random y)
}

/// Compare two groups on ordinal data without pretending it is interval.
///
/// Illustrative Likert shift: means 3.21 -> 3.35, which looks like nothing.
/// Common-language effect 0.54: a random new rating beats a random old one 54%
/// of the time. Top-two-box moved 0.470 -> 0.525. Those two are actionable.
fn mann_whitney_u(x: &[f64], y: &[f64]) -> MannWhitney {
    let mut combined: Vec<(f64, u8)> = x.iter().map(|&v| (v, 0)).chain(y.iter().map(|&v| (v, 1))).collect();
    combined.sort_by(|a, b| a.0.total_cmp(&b.0));
    let mut ranks = vec![0.0; combined.len()];
    let mut i = 0;
    while i < combined.len() {               // average ranks across ties
        let mut j = i;
        while j + 1 < combined.len() && combined[j + 1].0 == combined[i].0 {
            j += 1;
        }
        for k in i..=j {
            ranks[k] = (i + j) as f64 / 2.0 + 1.0;
        }
        i = j + 1;
    }
    let r1: f64 = ranks.iter().zip(&combined).filter(|(_, &(_, g))| g == 0).map(|(r, _)| r).sum();
    let (n1, n2) = (x.len() as f64, y.len() as f64);
    let u1 = r1 - n1 * (n1 + 1.0) / 2.0;
    MannWhitney { u: u1, common_language_effect: u1 / (n1 * n2) }
}

/// Fraction rating 4 or 5: the number product teams can act on.
fn top_two_box(ratings: &[u32], scale_max: u32) -> f64 {
    ratings.iter().filter(|&&r| r >= scale_max - 1).count() as f64 / ratings.len() as f64
}
```

Worked problem

p95 latency moved. Is it real? After a change, p95 latency went from 9.81 s to 11.79 s, on 800 requests each side. A 2-second regression. Do you block the release?

python

```python
import numpy as np

def p95(x):
    return float(np.quantile(x, 0.95))

def bootstrap_diff(before, after, iters=4000, seed=1):
    """CI on the DIFFERENCE in p95. No closed form exists, which is what the
    bootstrap is for. Resample each side separately: these are different requests."""
    rng = np.random.default_rng(seed)
    reps = []
    for _ in range(iters):
        a = rng.choice(before, len(before))
        b = rng.choice(after, len(after))
        reps.append(p95(b) - p95(a))
    reps = np.sort(reps)
    return float(np.quantile(reps, 0.025)), float(np.quantile(reps, 0.975))

# illustrative run: observed +1.98 s, bootstrap 95% CI [-0.45, +3.91]
```

rust

```rust
// Small inline PRNG (SplitMix64) in place of np.random.default_rng; its draws
// differ from numpy's, so a run gives a slightly different interval.
struct Rng(u64);

impl Rng {
    fn next_u64(&mut self) -> u64 {
        self.0 = self.0.wrapping_add(0x9E37_79B9_7F4A_7C15);
        let mut z = self.0;
        z = (z ^ (z >> 30)).wrapping_mul(0xBF58_476D_1CE4_E5B9);
        z = (z ^ (z >> 27)).wrapping_mul(0x94D0_49BB_1331_11EB);
        z ^ (z >> 31)
    }
    fn random(&mut self) -> f64 {
        (self.next_u64() >> 11) as f64 / (1u64 << 53) as f64
    }
    fn randrange(&mut self, n: usize) -> usize {
        (self.random() * n as f64) as usize
    }
}

// np.quantile's default: linear interpolation between order statistics
fn quantile(x: &[f64], q: f64) -> f64 {
    let mut s = x.to_vec();
    s.sort_by(f64::total_cmp);
    let pos = q * (s.len() - 1) as f64;
    let lo = pos.floor() as usize;
    let hi = (lo + 1).min(s.len() - 1);
    s[lo] + (pos - lo as f64) * (s[hi] - s[lo])
}
```

```rust
fn p95(x: &[f64]) -> f64 {
    quantile(x, 0.95)
}

/// CI on the DIFFERENCE in p95. No closed form exists, which is what the
/// bootstrap is for. Resample each side separately: these are different requests.
fn bootstrap_diff(before: &[f64], after: &[f64], iters: usize, seed: u64) -> (f64, f64) {
    let mut rng = Rng(seed);
    let mut choice = |x: &[f64]| -> Vec<f64> { (0..x.len()).map(|_| x[rng.randrange(x.len())]).collect() };
    let mut reps = Vec::with_capacity(iters);
    for _ in 0..iters {
        let a = choice(before);
        let b = choice(after);
        reps.push(p95(&b) - p95(&a));
    }
    (quantile(&reps, 0.025), quantile(&reps, 0.975))
}

// illustrative run: observed +1.98 s, bootstrap 95% CI [-0.45, +3.91]
```

The interval crosses zero. p95 on 800 points rests on about 40 observations, so a 2-second move that looks alarming on a dashboard is not distinguishable from noise at this sample size. Do not block on this alone; say what sample size would settle it. In this data the mean/median ratio was 1.42, so anyone reporting mean latency was reporting a number no user experienced.

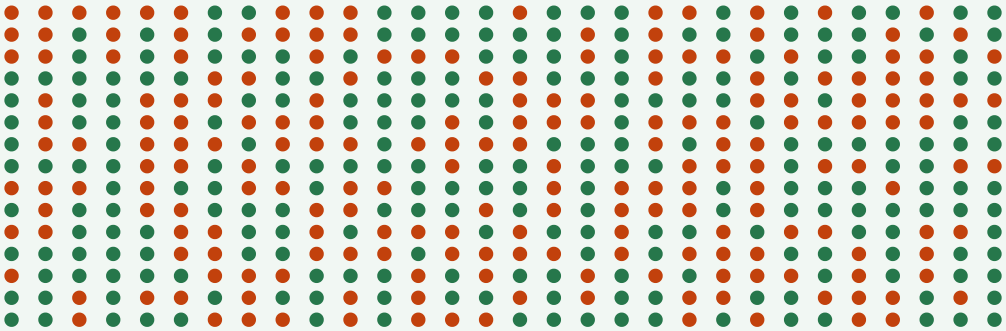
6. Clustering: the mistake that matters most for agent evals

Diagram

450 opinions from 40 kitchen tables

Clustering and the design effect, with a pizza survey.

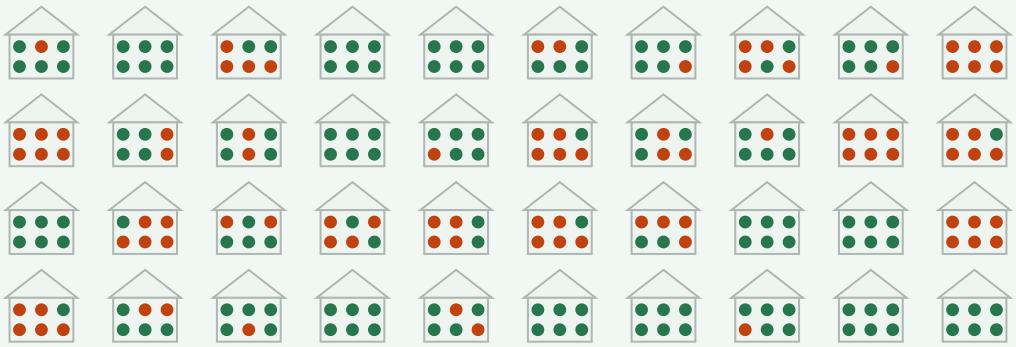
450 answers to "Do you like the new pizza?"



Looks like 450 independent opinions.

A pizza chain surveys 450 people: do you like the new recipe? 450 answers sounds like a solid sample, and the naive error bar on the "yes" share is narrow.

But they came from 40 households



People who share dinner mostly share opinions.

Then you learn the 450 answers came from 40 households, about 11 people each. Families ate the same pizza at the same table and mostly agree with each other. Inside a house, answers move together.

Effective sample size

design effect = 1 + (11 – 1) × 0.6 ≈ 7

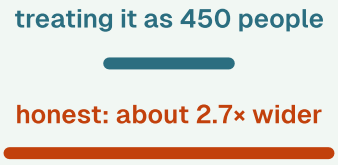
450 answers ÷ 7 ≈ 63 independent opinions

(about 11 people per house, within-house correlation 0.6)

A second answer from the same house adds little new information.

How much do they agree? With about 11 people per house and a within-house correlation of 0.6, the design effect is about 7. Your 450 answers carry roughly the information of 63 independent ones, much closer to 40 than to 450.

Error bar on the "yes" share



Fix: resample whole households, not individual people.

In agent evals: tool calls are the people, sessions are the households.

The honest error bar is about 2.7 times wider than the naive one (the square root of 7). Agent evals have the same shape: tool calls nested inside sessions. Resample whole sessions (a cluster bootstrap), or the interval invents confidence you do not have.

Plain words

450 measurements that are really 40

Suppose you measure 450 tool calls, but they came from only 40 sessions. Calls inside one session tend to succeed or fail together: a confused agent stays confused. So you do not have 450

independent measurements. Your real sample size is much closer to 40, and treating it as 450 makes your error bars far too small.

For leaders

Ask for effective sample size, not row count

- Agent data is nested: tool calls inside sessions, sessions inside users. Any interval computed as if rows were independent is too narrow, often by a factor of two or more.
- Ask "what is the unit of independence, and how many of those do we have?" A dashboard bragging about 50,000 events from 300 users has a sample size much closer to 300.
- A good team's sentence: "we report effective sample size rather than row count, because calls are nested in sessions; the naive interval was about 2.4 times too narrow when we checked."

Under the hood

Design effect and the cluster bootstrap

With average cluster size $\bar{m}$ and intraclass correlation ρ (the share of variance that lives between clusters), Kish's design effect and the effective sample size are

$$\mathrm{deff} = 1 + (\bar{m} - 1)\,\rho, \qquad n_{\mathrm{eff}} = \frac{N}{\mathrm{deff}}, \qquad \mathrm{SE}_{\mathrm{true}} \approx \sqrt{\mathrm{deff}}\,\mathrm{SE}_{\mathrm{naive}}$$

Bigger clusters amplify the penalty: $\rho = 0.3$ with 20 calls per session gives deff 6.7; the same ρ with 3 calls gives 1.6.

python

```python
import random
from statistics import mean

def cluster_bootstrap_ci(clusters: list[list[float]], iters: int = 10_000,
                         seed: int = 0) -> tuple[float, float, float]:
    """Resample SESSIONS, not individual calls."""
    rng = random.Random(seed)
    m = len(clusters)                # number of independent units: the real n
    # The point estimate is unaffected by clustering; only its uncertainty is.
    observed = sum(sum(c) for c in clusters) / sum(len(c) for c in clusters)
    reps = []
    for _ in range(iters):
        picked = [clusters[rng.randrange(m)] for _ in range(m)]   # whole sessions
        total, count = sum(sum(c) for c in picked), sum(len(c) for c in picked)
        reps.append(total / count if count else 0.0)             # recount: sizes vary
    reps.sort()
    return observed, reps[int(0.025 * iters)], reps[int(0.975 * iters) - 1]


def design_effect(clusters: list[list[float]]) -> float:
```

```
    sizes = [len(c) for c in clusters]
    grand = sum(sum(c) for c in clusters) / sum(sizes)
    between = sum(len(c) * (sum(c) / len(c) - grand) ** 2 for c in clusters) / (len(clusters) - 1)
    within = sum((x - sum(c) / len(c)) ** 2 for c in clusters for x in c) / max(1, sum(sizes) - len(clusters))
    icc = between / (between + within) if (between + within) else 0.0
    return 1 + (mean(sizes) - 1) * icc

# Illustrative, 40 sessions with 450 tool calls:
#   naive row bootstrap: [0.7289, 0.8067]  width 0.078
#   cluster bootstrap:   [0.6711, 0.8581]  width 0.187   (2.4x wider)
```

rust

```
// Small inline PRNG (SplitMix64); its draws differ from Python's random module.
struct Rng(u64);

impl Rng {
    fn next_u64(&mut self) -> u64 {
        self.0 = self.0.wrapping_add(0x9E37_79B9_7F4A_7C15);
        let mut z = self.0;
        z = (z ^ (z >> 30)).wrapping_mul(0xBF58_476D_1CE4_E5B9);
        z = (z ^ (z >> 27)).wrapping_mul(0x94D0_49BB_1331_11EB);
        z ^ (z >> 31)
    }
    fn random(&mut self) -> f64 {
        (self.next_u64() >> 11) as f64 / (1u64 << 53) as f64
    }
    fn randrange(&mut self, n: usize) -> usize {
        (self.random() * n as f64) as usize
    }
}

/// Resample SESSIONS, not individual calls.
fn cluster_bootstrap_ci(clusters: &[Vec<f64>], iters: usize, seed: u64) -> (f64, f64, f64) {
    let mut rng = Rng(seed);
    let m = clusters.len();              // number of independent units: the real n
    let total_of = |cs: &[&Vec<f64>]| cs.iter().map(|c| c.iter().sum::<f64>()).sum::<f64>();
    let count_of = |cs: &[&Vec<f64>]| cs.iter().map(|c| c.len()).sum::<usize>();
    // The point estimate is unaffected by clustering; only its uncertainty is.
    let all: Vec<&Vec<f64>> = clusters.iter().collect();
    let observed = total_of(&all) / count_of(&all) as f64;
    let mut reps = Vec::with_capacity(iters);
    for _ in 0..iters {
        let picked: Vec<&Vec<f64>> = (0..m).map(|_| &clusters[rng.randrange(m)]).collect();  // whole sessions
        let (total, count) = (total_of(&picked), count_of(&picked));
        reps.push(if count > 0 { total / count as f64 } else { 0.0 });                       // recount: sizes vary
    }
    reps.sort_by(f64::total_cmp);
    let it = iters as f64;
    (observed, reps[(0.025 * it) as usize], reps[(0.975 * it) as usize - 1])
}

fn design_effect(clusters: &[Vec<f64>]) -> f64 {
    let sizes: Vec<usize> = clusters.iter().map(|c| c.len()).collect();
    let n_total: usize = sizes.iter().sum();
    let mean_of = |c: &Vec<f64>| c.iter().sum::<f64>() / c.len() as f64;
    let grand = clusters.iter().map(|c| c.iter().sum::<f64>()).sum::<f64>() / n_total as f64;
    let between = clusters.iter().map(|c| c.len() as f64 * (mean_of(c) - grand).powi(2)).sum::<f64>()
        / (clusters.len() - 1) as f64;
    let within = clusters.iter().map(|c| { let m = mean_of(c); c.iter().map(|x| (x - m).powi(2)).sum::<f64>()
```

```
}).sum::<f64>()
        / 1.max(n_total - clusters.len()) as f64;
    let icc = if between + within != 0.0 { between / (between + within) } else { 0.0 };
    let mean_size = n_total as f64 / sizes.len() as f64;
    1.0 + (mean_size - 1.0) * icc
}

// Illustrative, 40 sessions with 450 tool calls:
//   naive row bootstrap: [0.7289, 0.8067]  width 0.078
//   cluster bootstrap:   [0.6711, 0.8581]  width 0.187   (2.4x wider)
```

Interactive **Clustering and the design effect** (shown at its default setting)

Tool calls nested in sessions. The naive bootstrap resamples individual calls, the cluster bootstrap resamples whole sessions. Raise the within session correlation and watch how much confidence the naive interval invents.

Observed rate 0.7854

Naive width 0.0708 Cluster width 0.1250 **understatement 1.76x**

480 calls, but only 40 independent units. Effective n is closer to 40 than to 480.

Principal asks

Push correlation and cluster size up together. That is a realistic agent trace, and the naive interval ends up several times too narrow. This is the most likely statistical mistake in agent evaluation, and the easiest one to show you avoid.

7. Choosing a method, in one flow

1. **What is the unit of independence?** Task, session, or user. Not row. Everything else follows from this.
2. **Is the outcome binary, count, continuous, or ordinal?** That picks the family.

3. **Is the statistic a mean, a proportion, or a quantile?** Means and proportions have closed forms. Quantiles need the bootstrap.

4. **Is the comparison paired?** If both models saw the same tasks, use the pairing or you throw away most of your power.

5. **Is n small or the statistic extreme?** If yes, use exact methods and stop bootstrapping.

6. **How skewed is it?** Mean over median above 1.2 means report quantiles, not averages.

> **Key idea**
>
> If you can walk that flow out loud in ninety seconds while someone describes a metric, you can handle most eval statistics questions. The reasoning is the same every time; the tables above are the lookup for each branch. For the general version, see from shape to method.

Part II · Chapter 11

Classification metrics, from precision to AUC

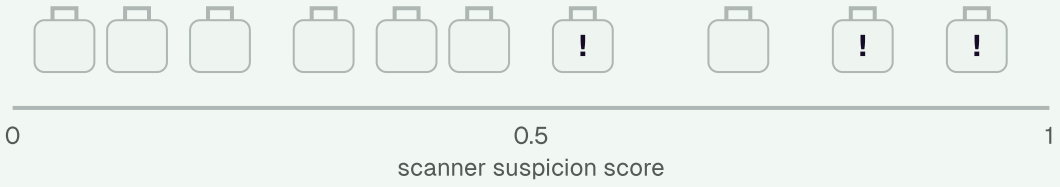
Every LLM judge is a classifier. So is a safety filter, a regression detector, an intent labeller, and an auto-approve gate. This chapter starts with someone sorting mail and ends with the metrics you would put in a launch review. One spam example runs through every chart, so each idea builds on the last.

The four outcomes

Diagram **Airport security: the dial that trades precision for recall**

Ten bags, one scanner, one dial. The same four outcomes as the mail sorter, in pictures.

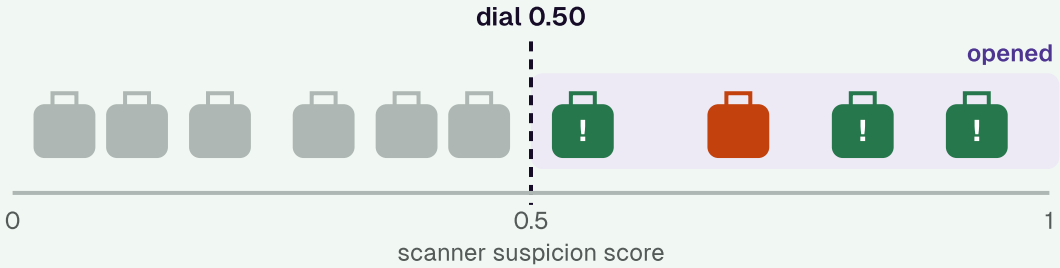
Ten bags. Three hide something forbidden.



! = really holds a forbidden item (the scanner cannot see this)

A scanner gives every bag a suspicion score from 0 to 1. Three bags (marked !) really hold something forbidden. The guard never sees inside; only the score. That score is exactly what a spam model or an LLM judge gives each item.

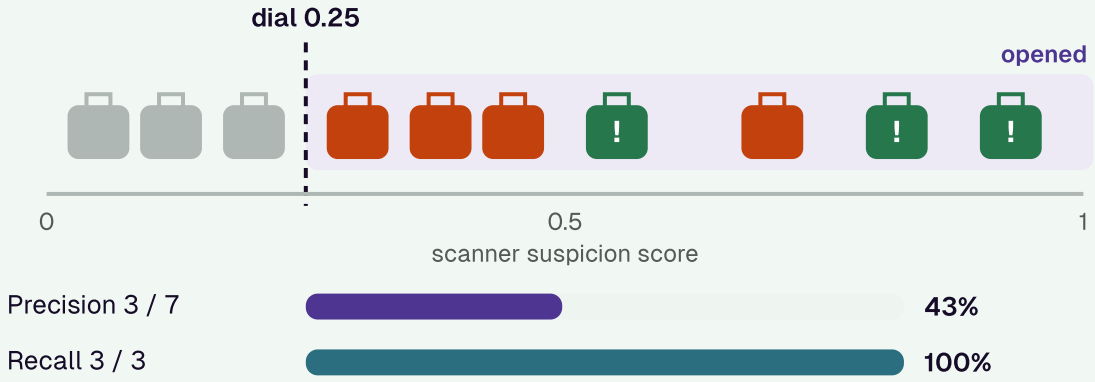
Dial at 0.50: open every bag to the right



3 caught	1 opened, harmless	0 missed	6 left alone
true positive	false positive	false negative	true negative

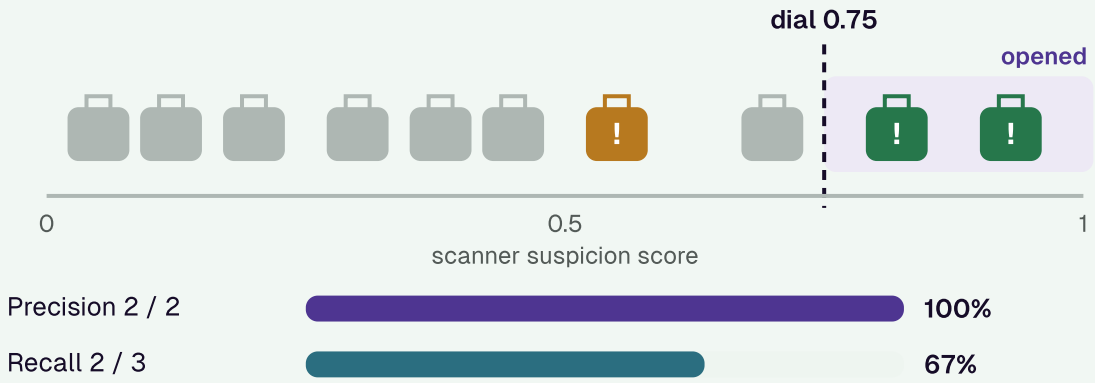
Opening everything above 0.50 produces four outcomes: 3 forbidden bags caught (true positives), 1 harmless bag opened (a false positive, one annoyed traveller), 0 missed (false negatives), and 6 harmless bags left alone (true negatives).

Strict dial at 0.25: catch everything, open lots of bags

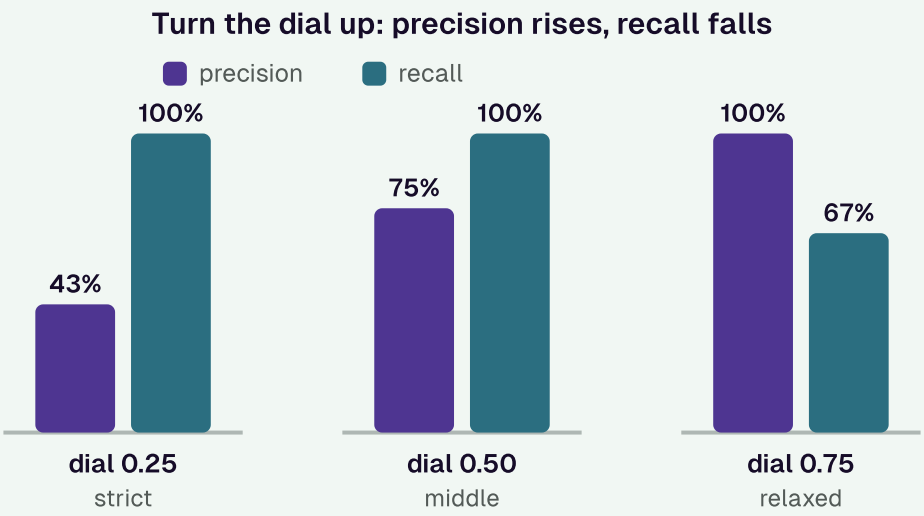


Lower the dial and the guard opens 7 bags. All 3 forbidden bags are caught, so **recall** is 3 of 3. But only 3 of the 7 opened bags were guilty, so **precision** falls to 43%. Four annoyed travellers.

Relaxed dial at 0.75: no false alarms, one slips through



Raise the dial and only 2 bags get opened, both guilty, so precision is 100%. But the forbidden bag that scored 0.55 walks straight through: recall drops to 2 of 3.



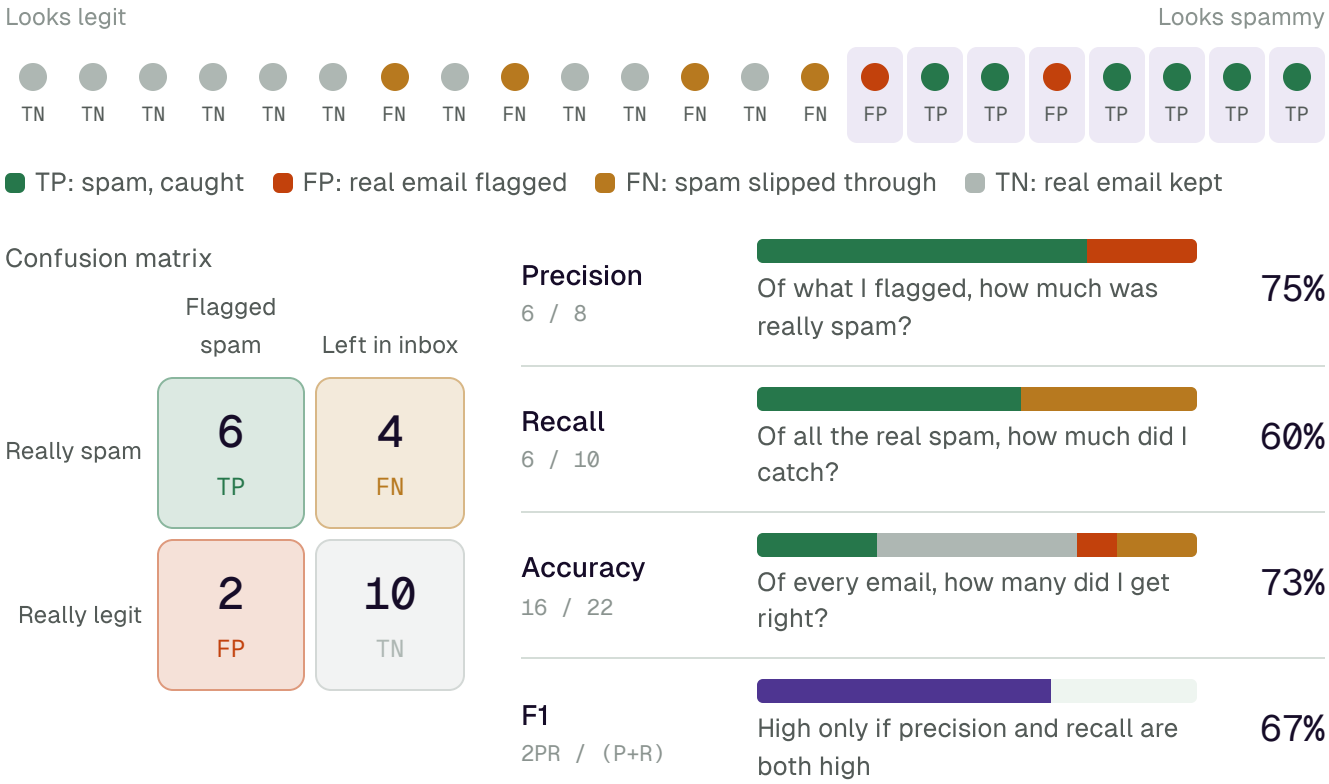
Precision: of the bags we opened, how many were guilty? **Recall**: of the guilty bags, how many did we open? The dial only trades one for the other. The only way to raise both is a better scanner, one that gives guilty bags higher scores.

A spam model does not really say "spam" or "not spam." It gives each email a score from 0 to 1 for how spammy it looks. You choose a cutoff, and anything at or above it goes to the spam folder. That cutoff is the **threshold**. Think of it as the passing grade on a test: the scores are fixed, but you decide where the line goes.

Interactive

Twenty-two emails and a threshold (shown at its default setting)

Emails are sorted by score. Drag the threshold; shaded cells go to the spam folder. Then press the two buttons and watch precision and recall pull against each other.



The 2×2 grid is the **confusion matrix**. Rows are the truth, columns are what the model decided. Everything else in this chapter is built from those four numbers.

Plain words

A helper sorts your mail into two piles

Your helper looks at every letter and decides: junk pile or keep pile. Each time, they can be right in two ways and wrong in two ways.

True positive

It was junk, and they put it in the junk pile. Good catch.

False positive

It was a real letter, and they threw it in the junk pile. A false alarm.

False negative

It was junk, and they left it in the keep pile. It slipped through.

True negative

It was a real letter, and they kept it. Also good.

Every metric in this chapter is a different way of counting those four piles. Each one asks a simple question:

Precision

When the helper says "junk," how often are they right?

Recall

Of all the junk that arrived, how much did they find?

Accuracy

Out of every letter, how many did they sort correctly?

F1

One score that is only high when precision and recall are both high.

"Positive" means the thing you are hunting for. Here that is spam. It does not mean good.

For leaders

A metric without its confusion matrix is unauditable

When someone shows you "precision 0.92," three things are hidden: the threshold, the prevalence of the positive class in the test set, and how many examples sat behind the number. Each one can move the headline more than the model change being reported. Make it a standing rule that any classifier metric in a review comes with the confusion matrix, the threshold, and n.

The second rule: decide which mistake costs more *before* you look at results. A false positive in a safety filter blocks a legitimate user. A false negative lets a harmful output through. A false positive in a regression detector costs an engineer an afternoon; a false negative ships a regression. If the team cannot say which error is worse, no metric choice will be defensible.

Principal asks

- At what threshold, and who chose it? Was it tuned on the same data you are reporting?
- What is the positive rate in this test set, and what is it in production?
- Which error are we optimising against, and what does one of each cost?

Under the hood

The confusion matrix is a function of the threshold

Given scores s_i, labels $y_i \in \{0, 1\}$, and a threshold τ, the prediction is $\hat{y}_i = \mathbb{1}[s_i \geq \tau]$. Each cell counts a combination:

$$\mathrm{TP}(\tau) = \sum_i \mathbb{1}[\hat{y}_i = 1, y_i = 1], \quad \mathrm{FP}(\tau) = \sum_i \mathbb{1}[\hat{y}_i = 1, y_i = 0], \quad \mathrm{FN}(\tau) = \sum_i \mathbb{1}[\hat{y}_i = ($$

TN is everything else. Raising τ can only move items from the predicted-positive column to the predicted-negative column, so TP and FP never increase as τ rises.

Report a single precision or recall and you have silently reported a threshold choice as if it were a property of the model. Two teams can report very different F1 for the same model because they picked different operating points. Threshold-free metrics (section on AUC below) sidestep this and bring their own problems.

python

```python
def confusion(y_true, y_pred):
    """The four numbers everything else is built from.
    Always compute this first and show it next to any derived metric."""
    tp = sum(1 for a, b in zip(y_true, y_pred) if a == 1 and b == 1)
    fp = sum(1 for a, b in zip(y_true, y_pred) if a == 0 and b == 1)
    fn = sum(1 for a, b in zip(y_true, y_pred) if a == 1 and b == 0)
    tn = sum(1 for a, b in zip(y_true, y_pred) if a == 0 and b == 0)
    return {"tp": tp, "fp": fp, "fn": fn, "tn": tn}
```

rust

```rust
#[derive(Debug, Clone, Copy, PartialEq)]
struct Confusion {
    tp: u64,
    fp: u64,
    fn_: u64, // `fn` is a Rust keyword, hence the underscore
    tn: u64,
}

/// The four numbers everything else is built from.
/// Always compute this first and show it next to any derived metric.
fn confusion(y_true: &[u8], y_pred: &[u8]) -> Confusion {
    let mut c = Confusion { tp: 0, fp: 0, fn_: 0, tn: 0 };
    for (&a, &b) in y_true.iter().zip(y_pred) {
        match (a, b) {
            (1, 1) => c.tp += 1,
            (0, 1) => c.fp += 1,
            (1, 0) => c.fn_ += 1,
            (0, 0) => c.tn += 1,
            _ => {} // anything other than 0/1 is not counted, as in the Python
        }
    }
```

```
    c
}
```

Four metrics, four questions

Look at the bars in the widget above. Precision and recall both start from true positives. What changes is which group you divide by.

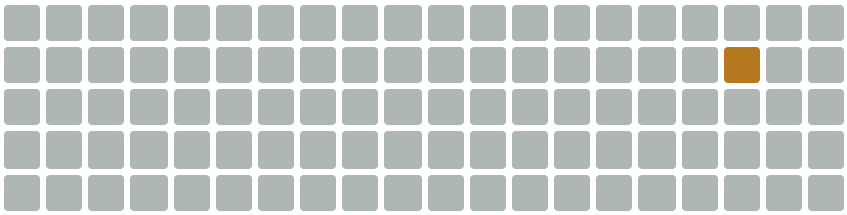
Metric	Formula	Divides by	Its enemy	Memory hook
Precision	`TP / (TP + FP)`	Everything you flagged	False alarms	Precise means picky
Recall	`TP / (TP + FN)`	All the real spam	Misses	Recall them all, like a product recall
Accuracy	`(TP + TN) / all`	Every email	Both mistakes	Simple, but can lie
F1	`2PR / (P + R)`	Combines P and R	Whichever is lower	A harsh average

Plain words

The tug-of-war, and why accuracy can lie

Press "Catch everything." Recall climbs, but coral dots flood the spam folder and precision falls. Press "Only be sure" and the opposite happens. Moving the threshold trades one mistake for the other. You choose based on which mistake hurts more. Missing a cancer diagnosis is worse than a false alarm, so screening favours recall. Losing an important email is worse than seeing some spam, so spam filters usually favour precision.

Now suppose only 1% of email is spam. A lazy model that flags nothing gets every legit email right.



Accuracy	Recall	F1
99%	0%	0%

Each square is 10 emails. The one amber square is all the spam, and the "flag nothing" model misses every bit of it.

99% accuracy, zero spam caught. When one class is rare, accuracy mostly measures how common the other class is.

For leaders

How each metric gets gamed

Metric	Easy way to inflate it	What to demand alongside it
Accuracy	Test on a set where the easy class dominates	Prevalence, balanced accuracy or MCC
Precision	Only flag the most obvious cases. Precision 1.0 at recall 0.02 looks perfect and is worthless	Recall at the same threshold
Recall	Flag everything. Recall 1.0 is always available, at precision equal to prevalence	Precision at the same threshold
F1	Tune the threshold on the test set; test on a large pool of easy negatives F1 never sees	A held-out threshold and the TN count
ROC-AUC	Report it on a rare-positive problem, where it flatters	PR-AUC and the confusion matrix at the operating point

Nobody needs to act in bad faith for this to happen. A team that iterates on a metric for months will drift toward whatever the metric rewards. The defence is a metric pack fixed in advance, not a single

number.

Under the hood

F-beta, and the demonstration that F1 ignores TN

$$F_\beta = (1 + \beta^2)\,\frac{P \cdot R}{\beta^2 P + R}$$

$\beta > 1$ weights recall more, $\beta < 1$ weights precision more. F_1 is the harmonic mean of P and R, which punishes imbalance: $P = 0.9, R = 0.1$ gives 0.18, not 0.5.

python

```python
def f_beta(precision, recall, beta=1.0):
    """beta > 1 weights recall more, beta < 1 weights precision more.
    beta = 2 says a miss costs about four times a false alarm (beta squared)."""
    if precision + recall == 0:
        return 0.0
    b2 = beta * beta
    return (1 + b2) * precision * recall / (b2 * precision + recall)
```

rust

```rust
/// beta > 1 weights recall more, beta < 1 weights precision more.
/// beta = 2 says a miss costs about four times a false alarm (beta squared).
/// Rust has no default arguments: pass beta = 1.0 for plain F1.
fn f_beta(precision: f64, recall: f64, beta: f64) -> f64 {
    if precision + recall == 0.0 {
        return 0.0;
    }
    let b2 = beta * beta;
    (1.0 + b2) * precision * recall / (b2 * precision + recall)
}
```

⚠ **Watch out**

Two classifiers, both with TP = 80, FP = 20, FN = 20. Scenario A has TN = 80; scenario B has TN = 9,980. Both score F1 = 0.800. MCC separates them: 0.600 versus 0.798. Same F1, materially different classifiers, because F1 never looks at the TN cell.

One class at a time

Precision, recall, and F1 are always calculated for one class, treated as "positive." So far spam was positive. Flip it and ask the same questions about legit email. The table follows the threshold in the first widget.

Class	Precision	Recall	F1	Count
Spam	75%	60%	67%	10
Legit	71%	83%	77%	12
Macro average			72%	
Weighted average			72%	
Micro average (equals accuracy)			73%	

At threshold 0.58: 6 TP, 2 FP, 4 FN, 10 TN. Move the threshold in the widget above and this table follows.

Plain words

Same four boxes, reassigned

Legit precision asks, "of what reached my inbox, how much was really legit?" That is TN out of TN + FN. Legit recall asks, "of all real emails, how many reached my inbox?" That is TN out of TN + FP. Legit recall has its own name, **specificity**, common in medical testing. Accuracy is the odd one out: it is one number for the whole model.

Macro average

A plain average of each class's score. Every class counts equally, so a rare class you care about cannot hide.

Weighted average

Each class weighted by how many examples it has. The big class has more say.

Micro average

Pool all the counts first, then compute. For one-label-per-item problems it equals accuracy.

For leaders

Per class first, averages second

Ask for the per-class table before any average. The averaging choice is a statement about what matters: macro says every class matters equally, weighted says the traffic mix matters, micro says only total volume matters. For anything where rare cases carry the risk (safety categories, unusual intents, rare failure modes), weighted and micro hide exactly what you need to see.

The actionable unit is usually a confusable pair, not a score. "Macro F1 0.81" tells you nothing to do. "Refactor and rename are confused 40% of the time" tells you to fix the taxonomy or the labelling guide.

Under the hood

The three averages, precisely

$$F_1^{\text{macro}} = \frac{1}{K}\sum_{k=1}^{K} F_1^{(k)}, \qquad F_1^{\text{weighted}} = \sum_{k=1}^{K} \frac{n_k}{n} F_1^{(k)}, \qquad F_1^{\text{micro}} = \frac{2\sum_k \mathrm{TP}_k}{2\sum_k \mathrm{TP}_k + \sum_k \mathrm{FP}_k}$$

In single-label multiclass, every false positive for one class is a false negative for another, so $\sum_k \mathrm{FP}_k = \sum_k \mathrm{FN}_k$ and micro-F1 equals accuracy.

python

```python
from collections import Counter

def multiclass_report(y_true, y_pred):
    """Per-class metrics plus all three averages. When macro and weighted
    diverge sharply, that gap is the finding."""
    labels = sorted(set(y_true) | set(y_pred))
    per = {}
    for lab in labels:
        tp = sum(a == lab and b == lab for a, b in zip(y_true, y_pred))
        fp = sum(a != lab and b == lab for a, b in zip(y_true, y_pred))
        fn = sum(a == lab and b != lab for a, b in zip(y_true, y_pred))
        p = tp / (tp + fp) if tp + fp else 0.0
        r = tp / (tp + fn) if tp + fn else 0.0
        per[lab] = {"f1": 2 * p * r / (p + r) if p + r else 0.0,
                    "support": tp + fn, "tp": tp, "fp": fp, "fn": fn}
    n = len(y_true)
    macro = sum(v["f1"] for v in per.values()) / len(labels)
    weighted = sum(v["f1"] * v["support"] for v in per.values()) / n
    micro = sum(a == b for a, b in zip(y_true, y_pred)) / n
    confused = Counter((a, b) for a, b in zip(y_true, y_pred) if a != b).most_common(3)
    return {"per_class": per, "macro_f1": macro, "weighted_f1": weighted,
            "micro_f1": micro, "top_confused_pairs": confused}
```

rust

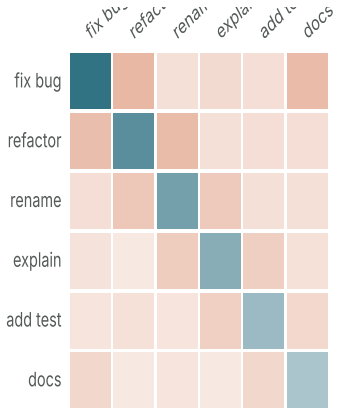
```rust
use std::collections::{BTreeMap, BTreeSet, HashMap};
use std::hash::Hash;

struct ClassStats {
```

```
    f1: f64,
    support: usize,
    tp: usize,
    fp: usize,
    fn_: usize,
}

struct Report<T> {
    per_class: BTreeMap<T, ClassStats>, // sorted by label, like sorted(set(...))
    macro_f1: f64,
    weighted_f1: f64,
    micro_f1: f64,
    top_confused_pairs: Vec<((T, T), usize)>,
}

/// Per-class metrics plus all three averages. When macro and weighted
/// diverge sharply, that gap is the finding.
fn multiclass_report<T: Ord + Hash + Clone>(y_true: &[T], y_pred: &[T]) -> Report<T> {
    let labels: BTreeSet<&T> = y_true.iter().chain(y_pred).collect();
    let pairs: Vec<(&T, &T)> = y_true.iter().zip(y_pred).collect();
    let mut per = BTreeMap::new();
    for &lab in &labels {
        let tp = pairs.iter().filter(|&&(a, b)| a == lab && b == lab).count();
        let fp = pairs.iter().filter(|&&(a, b)| a != lab && b == lab).count();
        let fn_ = pairs.iter().filter(|&&(a, b)| a == lab && b != lab).count();
        let p = if tp + fp > 0 { tp as f64 / (tp + fp) as f64 } else { 0.0 };
        let r = if tp + fn_ > 0 { tp as f64 / (tp + fn_) as f64 } else { 0.0 };
        let f1 = if p + r > 0.0 { 2.0 * p * r / (p + r) } else { 0.0 };
        per.insert(lab.clone(), ClassStats { f1, support: tp + fn_, tp, fp, fn_ });
    }
    let n = y_true.len() as f64;
    let macro_f1 = per.values().map(|v| v.f1).sum::<f64>() / labels.len() as f64;
    let weighted_f1 = per.values().map(|v| v.f1 * v.support as f64).sum::<f64>() / n;
    let micro_f1 = pairs.iter().filter(|&&(a, b)| a == b).count() as f64 / n;

    // Counter(...).most_common(3): highest count first, ties in first-seen order
    let mut counts: Vec<((T, T), usize)> = Vec::new();
    let mut index: HashMap<(T, T), usize> = HashMap::new();
    for &(a, b) in pairs.iter().filter(|&&(a, b)| a != b) {
        let key = (a.clone(), b.clone());
        let i = *index.entry(key.clone()).or_insert_with(|| {
            counts.push((key, 0));
            counts.len() - 1
        });
        counts[i].1 += 1;
    }
    counts.sort_by(|x, y| y.1.cmp(&x.1)); // stable sort keeps first-seen order on ties
    counts.truncate(3);

    Report { per_class: per, macro_f1, weighted_f1, micro_f1, top_confused_pairs: counts }
}
```

Interactive

Multiclass confusion heatmap (shown at its default setting)

An intent taxonomy as a heatmap. Rows are true intent, columns are predicted. Raise imbalance and watch weighted F1 detach from macro F1 while the headline number stays comfortable.



macro F1 0.760 **micro F1** 0.791 **weighted F1** 0.794 **gap** 0.034

macro and weighted agree, so performance is roughly even across classes.

Top confused pairs: fix bug → refactor (97, 8.9% of that class) · fix bug → docs (88, 8.1% of that class) · refactor → rename (84, 11.6% of that class)

That list is actionable. The F1 number is not.

Worked problem

An intent classifier: which number goes in the report? Six intent classes, imbalanced, 3,000 sessions. Weighted F1 is 0.836. Ship it?

Class	Support	F1
fix_bug	1,006	0.934
refactor	660	0.919
explain	598	0.872
add_test	369	0.674
docs	242	0.579
optimize	**125**	**0.423**

Macro F1 is 0.733, weighted F1 is 0.836, a gap of 0.103. The headline weighted number is carried by the three common classes. On `optimize` the classifier is barely better than guessing. When macro and weighted diverge by more than about 0.05, that divergence is the finding, and reporting only the weighted number hides it. If these labels are used to generate benchmark tasks, the rare intents are the ones being mislabelled.

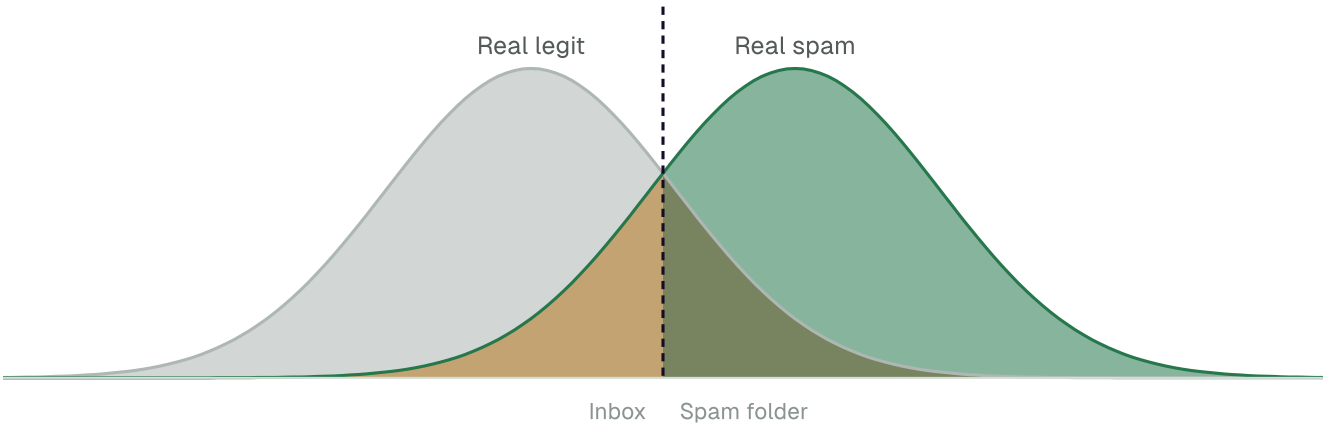
The trade-off and the ceiling

Zoom out from 22 emails to thousands. The two humps show where real legit emails and real spam land on the score line. Drag the threshold first, then raise model quality.

Interactive

The threshold picks a spot, the model sets the limit (shown at its default setting)

With a weak model, every threshold cuts through both humps. Watch the "Best F1 possible" card: only model quality moves it.



TP FN FP TN

Precision	Recall	F1 now	Best F1 possible
82%	82%	82%	82% at threshold 0.48

Plain words

You can only choose which mistakes to make

With a weak model, the humps overlap. Wherever you put the threshold, it cuts through both, so you only get to choose which mistakes to make. Raise model quality and the humps pull apart. Then one threshold can catch most spam while flagging almost no legit mail.

For leaders

Threshold work versus model work

When a team reports that precision went up, ask whether recall went down. If it did, the threshold moved and the model did not get better. That can still be the right call, but it is a product decision about error costs, not a research result. Real improvement shows up as a higher curve (a higher AUC), or as both precision and recall improving at a fixed threshold. Budget accordingly: threshold tuning is cheap and fast; separating the humps takes better features, better data on the hard cases, or a stronger model.

Key idea

This is the precision-recall trade-off. The only way to improve precision and recall together is to separate the humps, which means a better model, not a better threshold.

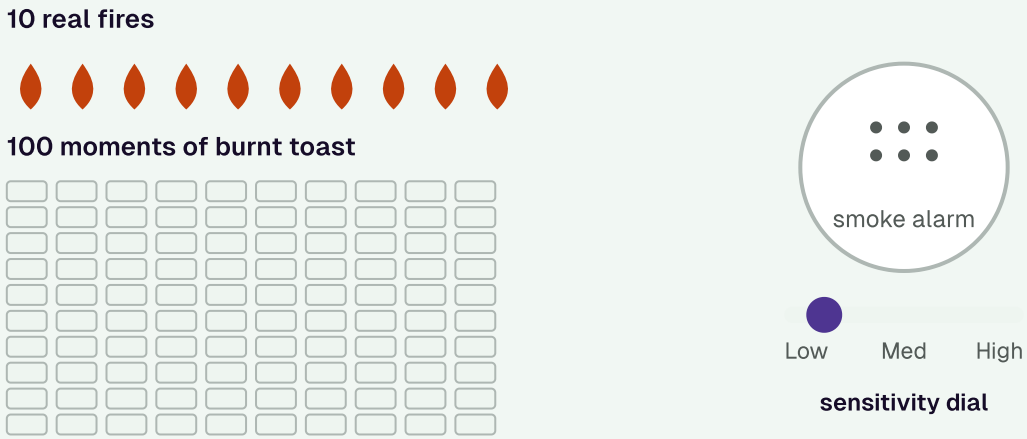
Building an ROC curve

An ROC curve shows what happens at every threshold at once. Each threshold gives one dot. The dot goes **up** for spam caught and **right** for good emails wrongly flagged. Up is good. Right is bad.

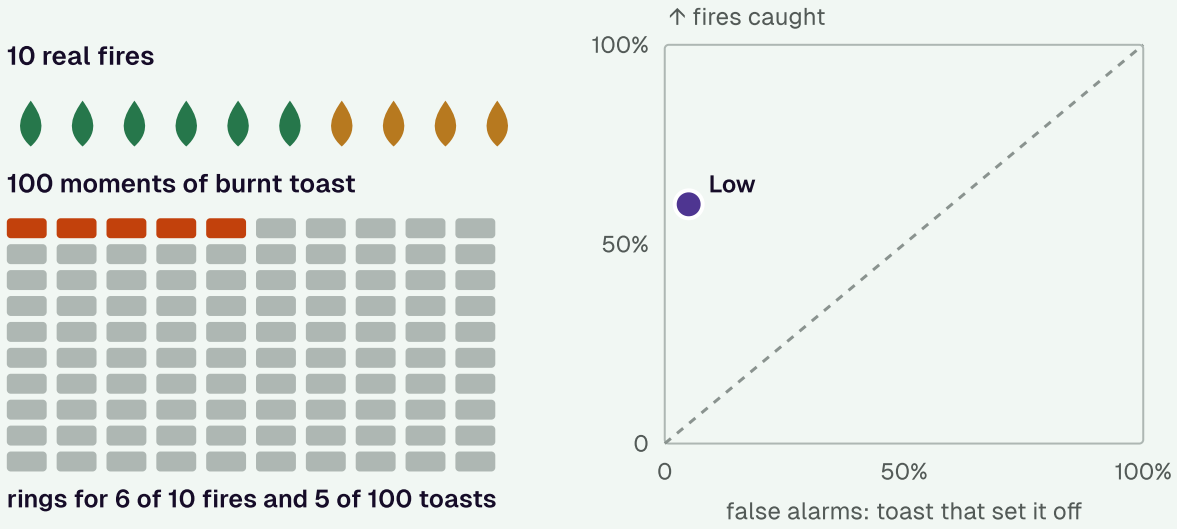
Diagram

A smoke alarm with a sensitivity dial

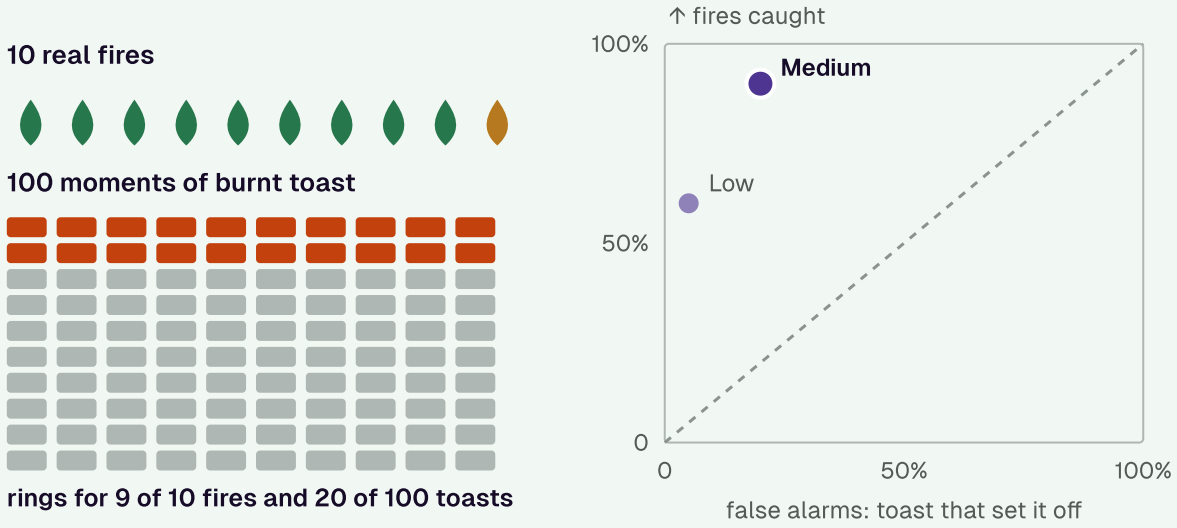
Each dial setting becomes one dot. Connect the dots and you have an ROC curve.



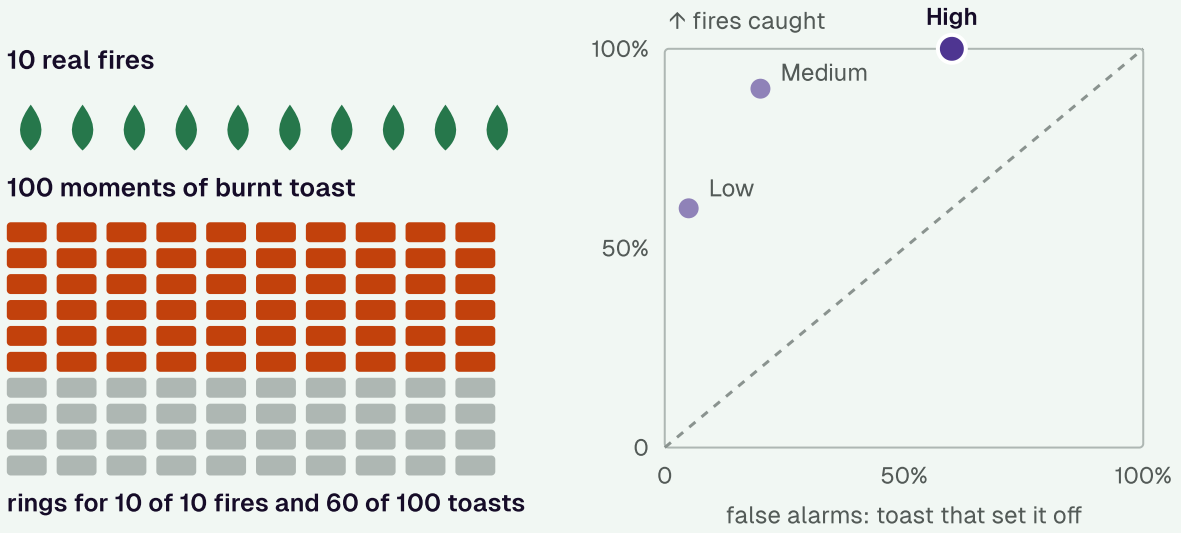
Over a year, a kitchen has 10 real fires and 100 moments of burnt toast. A good alarm rings for fires and stays quiet for toast. Its sensitivity dial is the threshold: the same idea as the guard's dial in the airport story.



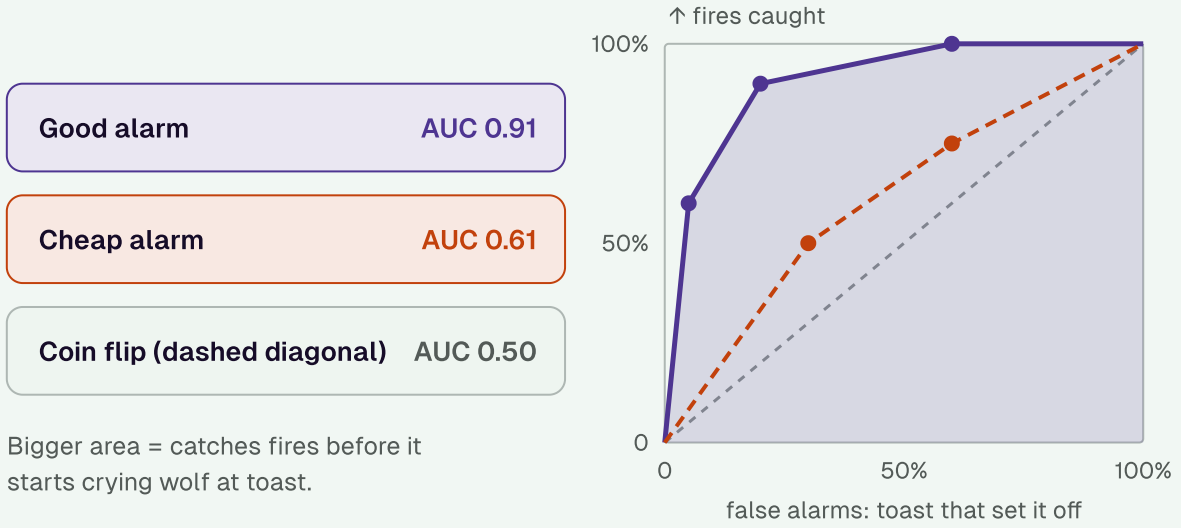
On **low**, it rings for 6 of the 10 fires (green) and only 5 of the 100 toasts (orange). That is one point on the chart: 5% false alarms, 60% of fires caught.



On **medium**, it catches 9 fires and rings for 20 toasts: the dot moves up (good) and right (bad).



On **high**, it catches all 10 fires but rings for 60 toasts. More sensitivity always buys catches with false alarms.



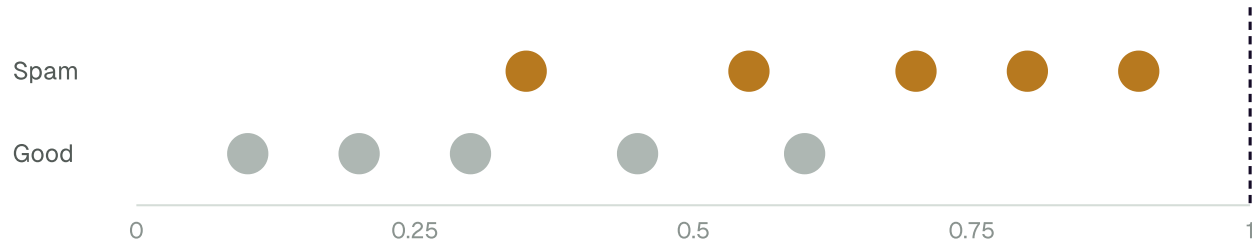
Connect the dots (plus "always off" at the bottom-left and "always on" at the top-right) and you get the ROC curve. A better alarm bows toward the top-left corner. The shaded area, **AUC**, is 0.91 for the good alarm and 0.61 for a cheap one. A coin flip scores 0.50.

Now the same thing with emails instead of fires.

Interactive **Every threshold becomes one dot** (shown at its default setting)

Step through ten emails. Watch each threshold land as a dot, then connect them.

Threshold 1.0: nothing is spammy enough, so nothing is flagged.

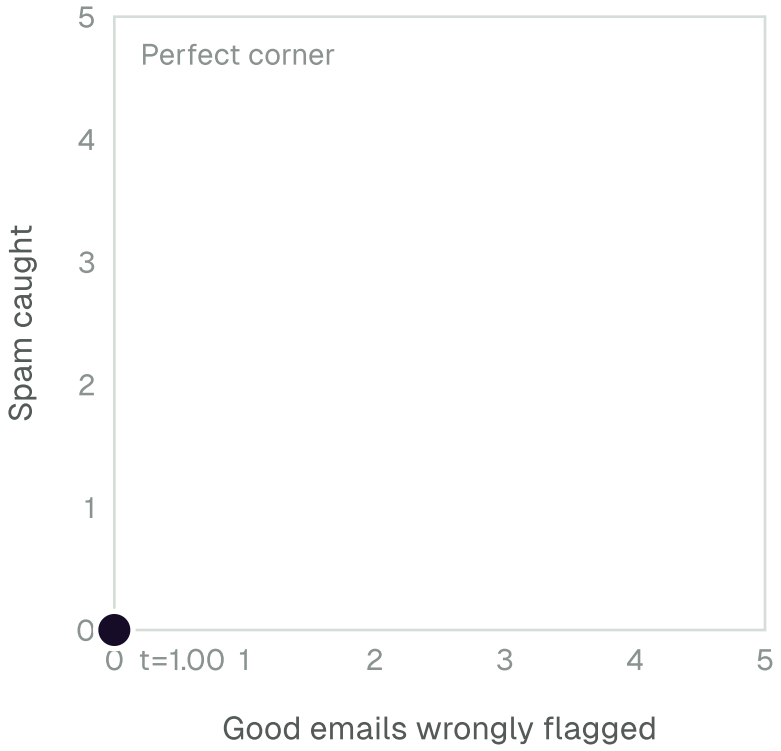


Spam caught (up)

0 of 5

Good emails flagged (right)

0 of 5



A great model climbs straight up before it moves right, hugging the top-left corner. A useless model moves up and right at the same pace along the diagonal: coin flipping. ROC stands for "receiver operating characteristic," a name left over from radar engineering. Read it as the "caught versus wrongly flagged" chart.

AUC: one number for the whole curve

AUC means area under the curve: how much of the box the curve fills. A perfect model scores 1.0. Random guessing scores 0.5. Because it covers every threshold, AUC ignores the threshold you picked and measures

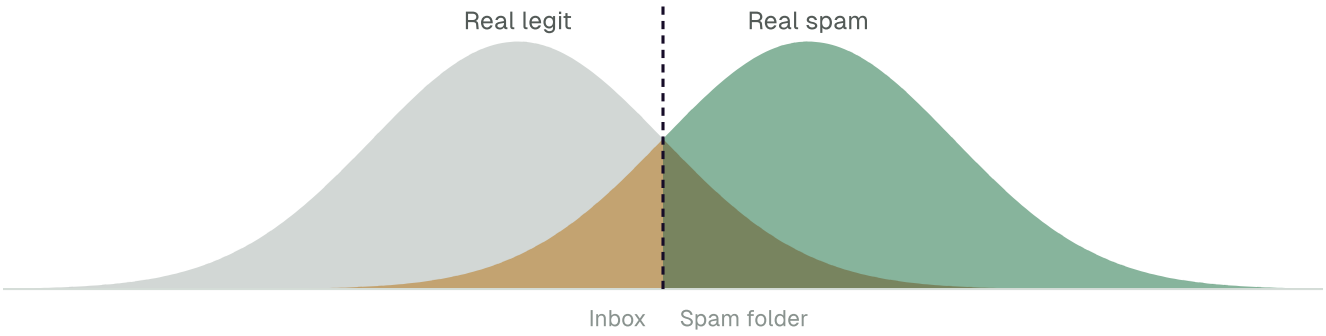
how far apart the humps are.

> **Key idea**
>
> Plain-English meaning of ROC-AUC: pick one random spam email and one random legit email. AUC is the chance the model gives the spam the higher score. In the ten-email example, spam outscores legit in 22 of 25 pairs, and 22 ÷ 25 = 0.88.

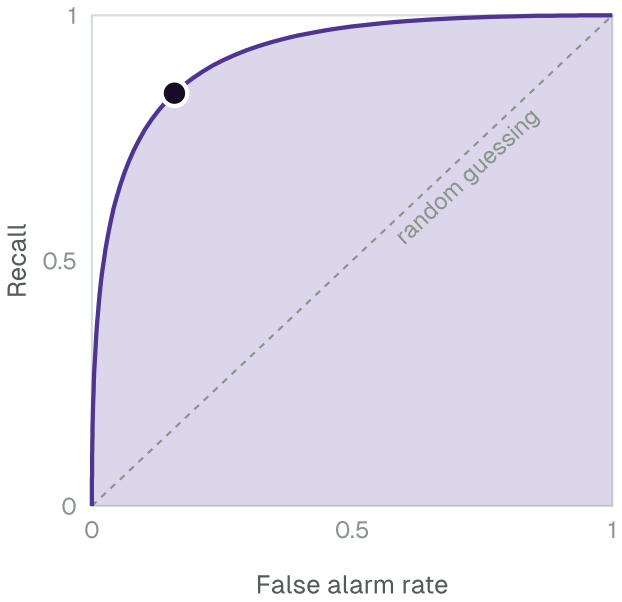
Interactive

ROC and PR curves, with spam share (shown at its default setting)

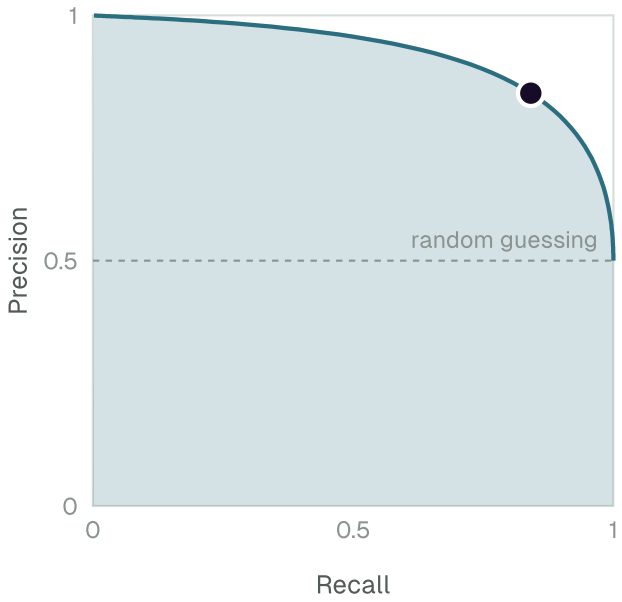
Move the threshold and only the dot moves. Change model quality and the curves move. Drop spam share to 5% and watch ROC-AUC stay put while PR-AUC sinks.



ROC curve: recall vs false alarm rate



PR curve: precision vs recall



Precision	Recall	False alarm rate	ROC-AUC
84%	84%	16%	0.92

PR-AUC
0.92

Plain words — Why there is a second curve

Drag spam share down to 5%. ROC-AUC does not change, because ROC never looks at how common each class is. The PR curve sinks, because even a small false alarm rate on a huge pile of legit email produces lots of false positives. ROC-AUC can look excellent while precision is poor. When the thing you are hunting for is rare, PR-AUC is the more honest number. Its random-guessing line sits at the spam share, so a PR-AUC of 0.3 is great when spam is 5% and poor when it is 50%.

Rough guide for ROC-AUC: 0.5 random, 0.7 weak, 0.8 decent, 0.9 and up strong. What counts as good depends on the problem.

For leaders

AUC chooses the model; the confusion matrix makes the ship decision

Use AUC to compare models or versions, because it does not depend on a threshold someone picked. Then make the ship decision on the confusion matrix at the operating point you will actually run. A model with higher AUC can be worse at your operating point if its curve crosses the other one there.

Watch out

Same classifier, same data, 2% prevalence, in a simulation of 5,000 samples: ROC-AUC 0.928, PR-AUC 0.470, against a PR-AUC baseline of 0.020. ROC-AUC says excellent. PR-AUC says mediocre but far better than chance. For rare-positive problems (safety violations, cheating detection, rare regressions) the PR number is the one telling the truth.

Principal asks

- Is this ROC-AUC or PR-AUC, and what is the positive rate?
- Do the two models' curves cross? Where is our operating point relative to the crossing?
- Does the score come with confidence intervals? (AUC on 200 examples has a wide one; see intervals and the bootstrap.)

Under the hood

AUC as a rank statistic, and average precision

ROC-AUC equals the probability that a random positive outscores a random negative, which is the Mann–Whitney U statistic scaled to [0, 1]:

$$\mathrm{AUC} = P(S^+ > S^-) + \tfrac{1}{2} P(S^+ = S^-) = \frac{R_+ - n_+(n_+ + 1)/2}{n_+ \, n_-}$$

R_+ is the sum of the ranks of the positive examples when all scores are ranked together (ties get average ranks); n_+ and n_- are the class counts.

PR-AUC is usually computed as average precision: walk down the ranking and average the precision at each point where a positive appears.

$$\mathrm{AP} = \frac{1}{n_+} \sum_{k:\, y_{(k)} = 1} \mathrm{Precision@}k$$

The baseline for a random ranking is the prevalence $\pi = n_+ / n$, which is why PR-AUC must always be read next to π.

Precision depends on prevalence directly. If a classifier has true positive rate TPR and false positive rate FPR, its precision at prevalence π is

$$\mathrm{Precision}(\pi) = \frac{\mathrm{TPR} \cdot \pi}{\mathrm{TPR} \cdot \pi + \mathrm{FPR} \cdot (1 - \pi)}$$

TPR and FPR (the ROC axes) do not change with π; precision does. That single line explains why ROC is prevalence-blind and PR is not.

python

```python
def roc_auc(y_true, scores):
    """Probability a random positive outscores a random negative (ties count half)."""
    order = sorted(range(len(scores)), key=lambda i: scores[i])
    ranks = [0.0] * len(scores)
    i = 0
    while i < len(order):
        j = i
        while j + 1 < len(order) and scores[order[j + 1]] == scores[order[i]]:
            j += 1
        avg = (i + j) / 2 + 1            # tied items share the average rank
        for k in range(i, j + 1):
            ranks[order[k]] = avg
        i = j + 1
    n_pos = sum(y_true)
    n_neg = len(y_true) - n_pos
```

```
    r_pos = sum(r for r, y in zip(ranks, y_true) if y == 1)
    return (r_pos - n_pos * (n_pos + 1) / 2) / (n_pos * n_neg)


def average_precision(y_true, scores):
    """Step-wise PR-AUC: mean precision at each rank where a positive appears."""
    order = sorted(range(len(scores)), key=lambda i: -scores[i])
    tp, ap = 0, 0.0
    for rank, i in enumerate(order, start=1):
        if y_true[i] == 1:
            tp += 1
            ap += tp / rank
    return ap / sum(y_true)
```

rust

```
/// Probability a random positive outscores a random negative (ties count half).
fn roc_auc(y_true: &[u8], scores: &[f64]) -> f64 {
    let mut order: Vec<usize> = (0..scores.len()).collect();
    order.sort_by(|&a, &b| scores[a].partial_cmp(&scores[b]).expect("NaN score"));
    let mut ranks = vec![0.0; scores.len()];
    let mut i = 0;
    while i < order.len() {
        let mut j = i;
        while j + 1 < order.len() && scores[order[j + 1]] == scores[order[i]] {
            j += 1;
        }
        let avg = (i + j) as f64 / 2.0 + 1.0; // tied items share the average rank
        for k in i..=j {
            ranks[order[k]] = avg;
        }
        i = j + 1;
    }
    let n_pos = y_true.iter().map(|&y| y as f64).sum::<f64>();
    let n_neg = y_true.len() as f64 - n_pos;
    let r_pos: f64 = ranks.iter().zip(y_true).filter(|&(_, &y)| y == 1).map(|(r, _)| r).sum();
    (r_pos - n_pos * (n_pos + 1.0) / 2.0) / (n_pos * n_neg)
}

/// Step-wise PR-AUC: mean precision at each rank where a positive appears.
fn average_precision(y_true: &[u8], scores: &[f64]) -> f64 {
    let mut order: Vec<usize> = (0..scores.len()).collect();
    // highest score first; the sort is stable, so ties keep input order
    order.sort_by(|&a, &b| scores[b].partial_cmp(&scores[a]).expect("NaN score"));
    let (mut tp, mut ap) = (0, 0.0);
    for (rank, &i) in (1..).zip(&order) {
        if y_true[i] == 1 {
            tp += 1;
            ap += tp as f64 / rank as f64;
        }
    }
    ap / y_true.iter().map(|&y| y as f64).sum::<f64>()
}
```

Better single numbers: MCC, balanced accuracy, kappa

If you must report one number, these are better choices than F1. Each still has a catch.

Interactive

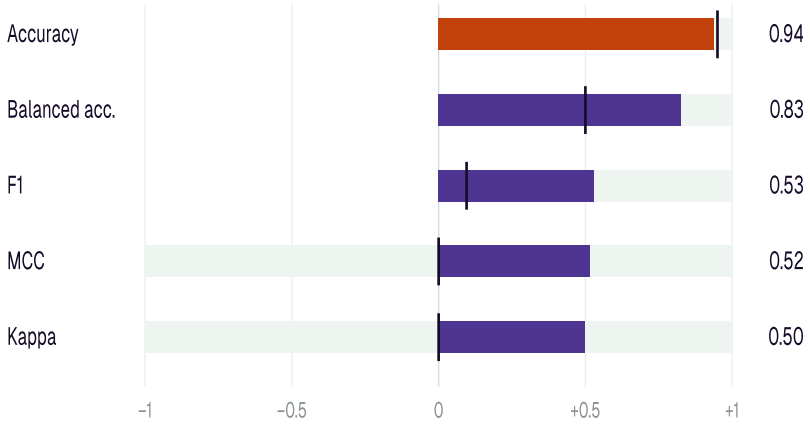
One confusion matrix, many scores (shown at its default setting)

Press "Always say negative on 2% prevalence": accuracy reads 98% while MCC and kappa sit at zero. Then raise sensitivity a little and watch which scores notice.

Confusion matrix, 1,000 cases

	Said positive	Said negative
Truly positive	35 TP	15 FN
Truly negative	47 FP	903 TN

All five scores on one −1 to +1 axis. The upright tick is what a model with no skill gets; an orange bar does no better than its tick.



No-skill ticks: accuracy, always predict the majority class; balanced accuracy, 0.5; F1, always predict positive, $2\pi/(1+\pi)$ at prevalence π; MCC and kappa, 0.

Accuracy	Balanced accuracy	F1	MCC
93.8%	82.5%	0.53	0.52

Cohen's kappa
0.50

Of 1,000 cases, 50 are truly positive. The model gets 93.8% right, but always answering "negative" would already score **95.0%** (the tick on the accuracy row), so on accuracy alone this model looks worse than doing nothing. MCC is 0.52 and kappa 0.50: skill above chance, on a scale where 0 is no skill and 1 is perfect. Balanced accuracy (82.5%) averages sensitivity 70% and specificity 95%, so 50% is always chance. F1 (0.53) looks only at the positive class and never uses the 903 true negatives; it is useful when positives are what you care about, but it can move a lot when you just relabel which class is "positive". When one number has to summarise a skewed problem, MCC is the safest default; still show the matrix.

For leaders

Which single number for which audience

Number	Reads as	Best for	Catch
Balanced accuracy	Average of "right on positives" and "right on negatives"; 0.5 is always chance	Executive summaries	Still hides which error you make
MCC	Correlation between prediction and truth, –1 to +1	Comparing models with one number	Unintuitive scale; undefined when the model predicts one class
Cohen's kappa	Agreement above chance, 0 is chance, 1 is perfect	Judge versus human, human versus human	Depends on prevalence; report it with the pass-rate gap

Under the hood

Definitions

$$\mathrm{MCC} = \frac{\mathrm{TP}\cdot\mathrm{TN} - \mathrm{FP}\cdot\mathrm{FN}}{\sqrt{(\mathrm{TP}+\mathrm{FP})(\mathrm{TP}+\mathrm{FN})(\mathrm{TN}+\mathrm{FP})(\mathrm{TN}+\mathrm{FN})}}, \qquad \mathrm{BA} = \tfrac{1}{2}\left(\tfrac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} + \tfrac{\mathrm{T}}{\mathrm{TN}+}\right.$$

MCC uses all four margins, so swapping the class labels leaves it unchanged. F1 does not have that property.

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \qquad p_e = \frac{(\mathrm{TP}+\mathrm{FP})(\mathrm{TP}+\mathrm{FN}) + (\mathrm{TN}+\mathrm{FN})(\mathrm{TN}+\mathrm{FP})}{n^2}$$

p_o is observed agreement (plain accuracy); p_e is the agreement two independent raters with the same marginals would reach by luck. On a 95/5 split, luck alone gives about 90%.

python

```python
import math
```

```python
def mcc(tp, fp, fn, tn):
    den = math.sqrt((tp + fp) * (tp + fn) * (tn + fp) * (tn + fn))
    return (tp * tn - fp * fn) / den if den else 0.0   # 0 = chance when undefined

def balanced_accuracy(tp, fp, fn, tn):
    return (tp / (tp + fn) + tn / (tn + fp)) / 2

def kappa(tp, fp, fn, tn):
    n = tp + fp + fn + tn
    po = (tp + tn) / n
    pe = ((tp + fp) * (tp + fn) + (tn + fn) * (tn + fp)) / (n * n)
    return (po - pe) / (1 - pe) if pe != 1 else 0.0

# The "flag nothing" model on 1% positives: accuracy 0.99, MCC 0, BA 0.5, kappa 0
print(mcc(0, 0, 10, 990), balanced_accuracy(0, 0, 10, 990), kappa(0, 0, 10, 990))
```

rust

```rust
// Counts arrive as integers; convert once so products cannot overflow.
fn mcc(tp: u64, fp: u64, fn_: u64, tn: u64) -> f64 {
    let (tp, fp, fn_, tn) = (tp as f64, fp as f64, fn_ as f64, tn as f64);
    let den = ((tp + fp) * (tp + fn_) * (tn + fp) * (tn + fn_)).sqrt();
    if den != 0.0 { (tp * tn - fp * fn_) / den } else { 0.0 } // 0 = chance when undefined
}

fn balanced_accuracy(tp: u64, fp: u64, fn_: u64, tn: u64) -> f64 {
    let (tp, fp, fn_, tn) = (tp as f64, fp as f64, fn_ as f64, tn as f64);
    (tp / (tp + fn_) + tn / (tn + fp)) / 2.0
}

fn kappa(tp: u64, fp: u64, fn_: u64, tn: u64) -> f64 {
    let (tp, fp, fn_, tn) = (tp as f64, fp as f64, fn_ as f64, tn as f64);
    let n = tp + fp + fn_ + tn;
    let po = (tp + tn) / n;
    let pe = ((tp + fp) * (tp + fn_) + (tn + fn_) * (tn + fp)) / (n * n);
    if pe != 1.0 { (po - pe) / (1.0 - pe) } else { 0.0 }
}

fn main() {
    // The "flag nothing" model on 1% positives: accuracy 0.99, MCC 0, BA 0.5, kappa 0
    // {:?} prints floats the way Python does (0.0, not 0)
    println!("{:?} {:?} {:?}", mcc(0, 0, 10, 990), balanced_accuracy(0, 0, 10, 990), kappa(0, 0, 10, 990));
}
```

MCC and kappa are often nearly identical on binary problems. Prefer MCC when scoring a model and kappa when scoring agreement between raters, because that is what each was designed to communicate.

Calibration: is the confidence honest?

Diagram

Two weather forecasters and the 70% test

Calibration is whether "70% sure" really means right 7 times in 10.

Forecaster A: "70% chance of rain"

It rained on 7 of these 10 days.

Said 70%, happened 70%: honest (calibrated)

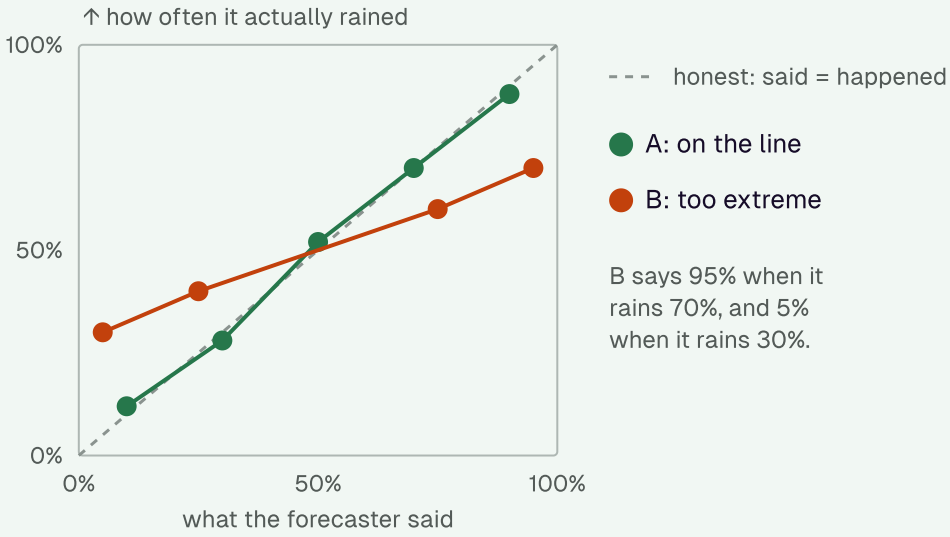
Forecaster A says "70% chance of rain" on ten days, and it rains on seven. Her number means what it says. That is calibration.

Forecaster B: "95% chance of rain"

Same 10 days, same 7 rainy ones. Same ranking as A.

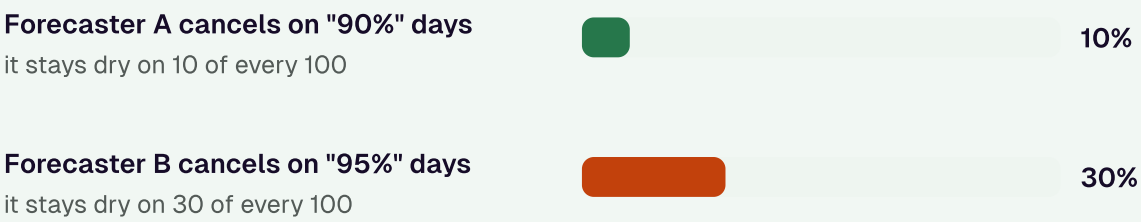
Said 95%, happened 70%: overconfident

Forecaster B looks at the same ten days and says "95%". It still rains on seven. B picks the rainy days just as well as A (same ranking), but his confidence number is inflated.



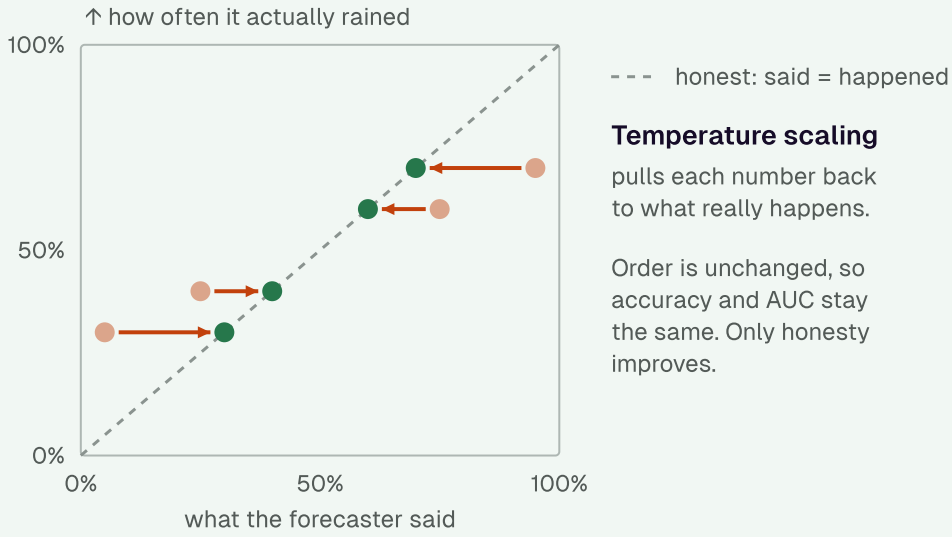
A reliability diagram plots what each forecaster said against what happened, over many days. A hugs the honest diagonal. B is too extreme at both ends: his 95% days rain 70% of the time and his 5% days rain 30% of the time.

Rule: cancel the picnic when the forecast is 90% or higher



Wasted cancellations. Same days ranked the same way, three times the mistakes.

Why it matters: any rule that acts on the number itself. "Cancel the picnic at 90% or higher" wastes 10% of cancellations with A and 30% with B. Auto-approving agent actions, routing to humans, and abstaining all work the same way.



The fix is cheap. Temperature scaling, fit on held-out data, pulls B's numbers back onto the diagonal without changing which day ranks above which. Ranking quality (AUC) stays the same; the confidence becomes honest.

Plain words

A weather forecaster who says 70%

If a forecaster says "70% chance of rain" on 100 days, it should rain on about 70 of them. That is calibration. A model can rank emails perfectly (every spam above every legit) and still be badly wrong about how sure it should be. Ranking and honesty about confidence are different skills.

For leaders

When calibration is the whole ballgame

Calibration matters whenever a downstream decision uses the confidence number itself: routing to a bigger model, abstaining and asking a human, auto-approving an action, or combining scores from several judges. Accuracy and AUC are blind to it. If a system makes decisions on confidence, require a reliability diagram and ECE in the review. The usual fix, temperature scaling on a held-out set, is cheap and does not change rankings, so there is rarely an excuse to ship miscalibrated scores.

Under the hood

Brier score and expected calibration error

$$\text{Brier} = \frac{1}{n}\sum_{i=1}^{n}(p_i - y_i)^2, \qquad \text{ECE} = \sum_{b=1}^{B}\frac{|S_b|}{n}\left|\text{acc}(S_b) - \text{conf}(S_b)\right|$$

Brier is a proper scoring rule: it is minimised by reporting your true belief. ECE buckets predictions by confidence S_b and compares mean confidence to observed accuracy in each bucket. ECE depends on the number of bins, so always state B.

python

```python
def brier(y_true, probs):
    # squared, not absolute: squaring is what makes it a proper scoring rule
    return sum((p - y) ** 2 for p, y in zip(probs, y_true)) / len(y_true)

def ece(y_true, probs, bins=10):
    n, total = len(y_true), 0.0
    for b in range(bins):
        lo, hi = b / bins, (b + 1) / bins
        idx = [i for i, p in enumerate(probs) if (lo < p <= hi) or (b == 0 and p == 0)]
        if not idx:
            continue
        conf = sum(probs[i] for i in idx) / len(idx)
        acc = sum(y_true[i] for i in idx) / len(idx)
        total += len(idx) / n * abs(acc - conf)
    return total
```

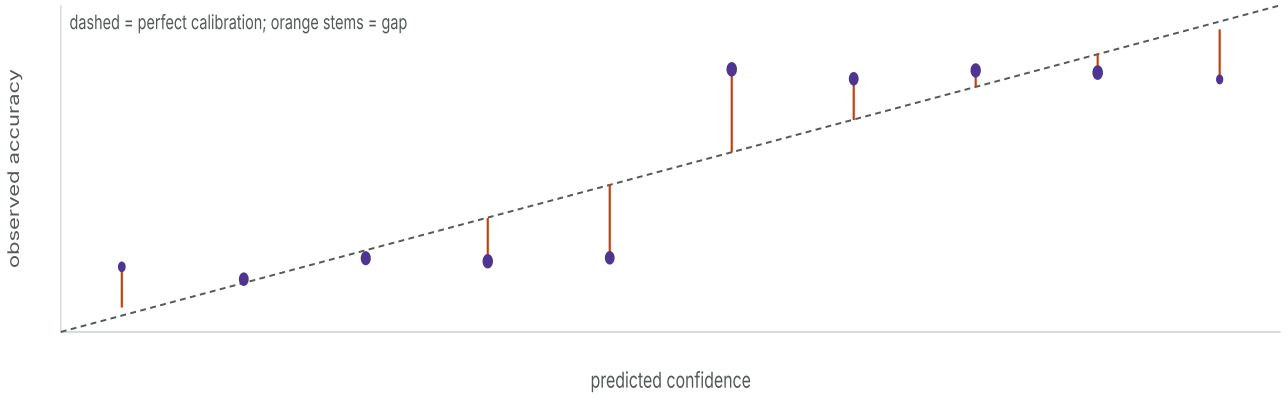
rust

```rust
fn brier(y_true: &[u8], probs: &[f64]) -> f64 {
    // squared, not absolute: squaring is what makes it a proper scoring rule
    probs.iter().zip(y_true).map(|(p, &y)| (p - y as f64).powi(2)).sum::<f64>() / y_true.len() as f64
}

/// Pass bins = 10 to match the Python default.
fn ece(y_true: &[u8], probs: &[f64], bins: usize) -> f64 {
    let (n, mut total) = (y_true.len() as f64, 0.0);
    for b in 0..bins {
        let (lo, hi) = (b as f64 / bins as f64, (b + 1) as f64 / bins as f64);
        let idx: Vec<usize> = (0..probs.len())
            .filter(|&i| (lo < probs[i] && probs[i] <= hi) || (b == 0 && probs[i] == 0.0))
            .collect();
        if idx.is_empty() {
            continue;
        }
        let m = idx.len() as f64;
        let conf = idx.iter().map(|&i| probs[i]).sum::<f64>() / m;
        let acc = idx.iter().map(|&i| y_true[i] as f64).sum::<f64>() / m;
        total += m / n * (acc - conf).abs();
    }
    total
}
```

> **Note**
>
> In a simulation with 20% label noise, a moderately confident model scored Brier 0.162 and ECE 0.050. The same model pushed to 0.99 / 0.01 confidence scored Brier 0.196 and ECE 0.190: identical ranking, identical accuracy, nearly four times worse calibration.

Interactive **Calibration** (shown at its default setting)

Sharpening pushes confidence toward 0 and 1 without changing the ranking at all. Accuracy and AUC do not move. Watch ECE and Brier come apart. If you route, abstain or auto approve on a confidence score, this is the property that matters.



Brier 0.1835 **ECE** 0.1126

Sharpening pushes confidence toward 0 and 1 without changing the ranking. Accuracy and AUC stay put while calibration collapses. If you route, abstain or auto approve on confidence, this is the number that breaks you.

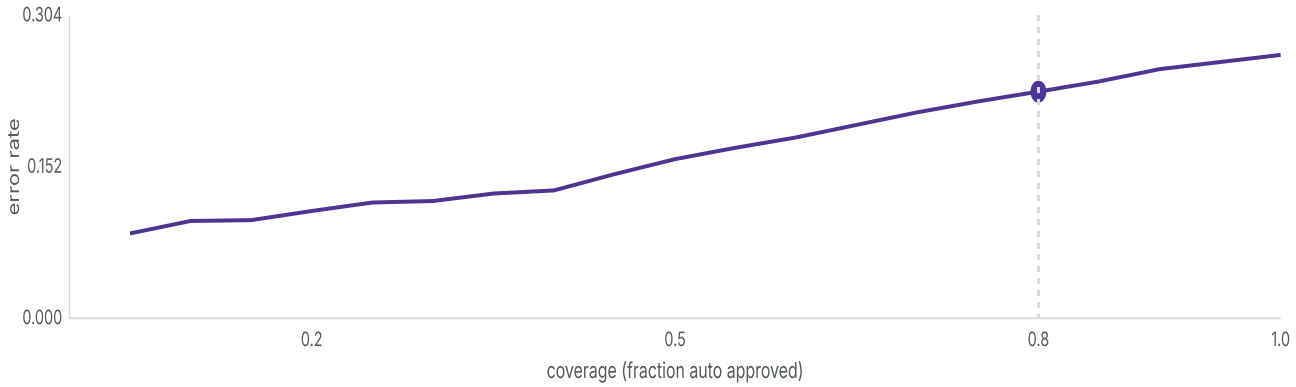
Selective prediction: when the system can abstain

Instead of one accuracy number you get a curve: error as a function of how often the system is willing to answer. This is the right frame for any judge that can say "unsure," any agent that can ask the user before acting, and any classifier that can hand cases to a human.

Interactive

Auto approve threshold picker (shown at its default setting)

An agent that auto-approves its own tool calls, framed as a coverage risk curve. Set the cost of asking the user and the widget finds the coverage that minimises total cost. Drag confidence quality to zero and watch the curve go flat.



Auto approve everything: error 0.2640 **at 80% coverage:** error 0.2272 (1.2x better)
Total cost at your setting 0.1858 **cost minimising coverage** 5% (cost 0.0233)
Drag confidence quality to 0 and the curve goes flat. A flat curve means abstention buys you nothing, and that is itself the finding: fix the confidence signal before tuning the threshold.

Worked problem

Where do you set auto-approve? An agent's proposed actions are right 92.8% of the time overall. Sort by the model's confidence and only auto-approve the most confident fraction:

Coverage	Auto-approved	Asked the user	Error rate	vs full coverage
100%	4,000	0	0.0717	—
90%	3,600	400	0.0550	1.3× fewer
80%	3,200	800	0.0350	**2.0× fewer**
70%	2,800	1,200	0.0232	3.1× fewer
50%	2,000	2,000	0.0090	8.0× fewer

Giving up 20% of the decisions halves the error rate. That turns a product argument into arithmetic: the design question is what an interruption costs relative to a mistake. If the curve comes back flat, the confidence signal does not track correctness and no threshold will help. Fix the signal before tuning the threshold.

Under the hood

The coverage–risk curve

Sort by confidence c_i descending and answer only the top fraction κ (coverage). Risk is the error rate over the answered set only; abstained items are handed off, not counted wrong.

$$\operatorname{risk}(\kappa) = \frac{1}{\lfloor \kappa n \rfloor} \sum_{i \in \operatorname{top} \lfloor \kappa n \rfloor} \mathbb{1}[\hat{y}_i \neq y_i], \qquad \text{total cost}(\kappa) = \kappa \operatorname{risk}(\kappa)\, c_{\text{error}} + (1 - \kappa)\, c_{\text{ask}}$$

The area under the risk–coverage curve (AURC) summarises how well confidence tracks correctness; lower is better.

python

```python
def coverage_risk(y_true, preds, conf, coverages=(1.0, 0.9, 0.8, 0.6, 0.5)):
    order = sorted(range(len(y_true)), key=lambda i: -conf[i])
    out = []
    for cov in coverages:
```

```
        k = max(1, int(cov * len(order)))
        err = sum(preds[i] != y_true[i] for i in order[:k]) / k
        out.append({"coverage": cov, "risk": err, "answered": k})
    return out
```

rust

```rust
const DEFAULT_COVERAGES: [f64; 5] = [1.0, 0.9, 0.8, 0.6, 0.5];

struct CoveragePoint {
    coverage: f64,
    risk: f64,
    answered: usize,
}

fn coverage_risk<T: PartialEq>(y_true: &[T], preds: &[T], conf: &[f64], coverages: &[f64]) -> Vec<CoveragePoint>
{
    let mut order: Vec<usize> = (0..y_true.len()).collect();
    order.sort_by(|&a, &b| conf[b].partial_cmp(&conf[a]).expect("NaN confidence"));
    let mut out = Vec::new();
    for &cov in coverages {
        let k = ((cov * order.len() as f64) as usize).max(1);
        let err = order[..k].iter().filter(|&&i| preds[i] != y_true[i]).count() as f64 / k as f64;
        out.push(CoveragePoint { coverage: cov, risk: err, answered: k });
    }
    out
}
```

Expected cost: the metric that is actually about your problem

Everything above is a proxy. If you know the relative costs of the two errors, skip the proxies and minimise expected cost directly.

For leaders

Get the cost ratio on the table

Most metric arguments are cost arguments in disguise. One team wants precision, another wants recall, and neither has said what a false alarm and a miss each cost. Force that number out, even as a rough ratio ("a missed harmful output is about 20 times worse than a wrongly blocked request"). Once it exists, the threshold choice is arithmetic and the argument ends.

Under the hood

Cost per decision and the Bayes-optimal threshold

$$\text{Cost}(\tau) = \frac{c_{FP}\,\text{FP}(\tau) + c_{FN}\,\text{FN}(\tau)}{n}, \qquad \tau^{*} = \frac{c_{FP}}{c_{FP} + c_{FN}} \text{ if scores are calibrated prol}$$

Flag an item when $p\,c_{FN} > (1 - p)\,c_{FP}$. With equal costs that is $p > 0.5$; if a miss costs 9 times a false alarm, flag at $p > 0.1$. If scores are not calibrated, sweep thresholds on held-out data instead.

python

```python
def best_threshold(y_true, scores, c_fp, c_fn):
    """Only observed scores can change the confusion matrix, so sweep those."""
    best = None
    for t in sorted(set(scores)):
        fp = sum(s >= t and y == 0 for s, y in zip(scores, y_true))
        fn = sum(s < t and y == 1 for s, y in zip(scores, y_true))
        cost = (c_fp * fp + c_fn * fn) / len(y_true)
        if best is None or cost < best["cost"]:
            best = {"threshold": t, "cost": cost}
    return best   # tune on a validation split, report on a separate test split
```

rust

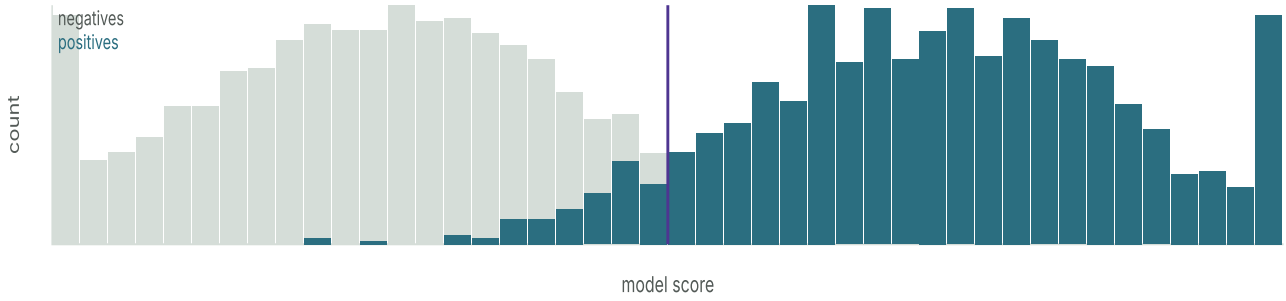
```rust
struct Best {
    threshold: f64,
    cost: f64,
}

/// Only observed scores can change the confusion matrix, so sweep those.
fn best_threshold(y_true: &[u8], scores: &[f64], c_fp: f64, c_fn: f64) -> Option<Best> {
    let mut thresholds = scores.to_vec();
    thresholds.sort_by(|a, b| a.partial_cmp(b).expect("NaN score"));
    thresholds.dedup(); // sorted(set(scores))
    let mut best: Option<Best> = None;
    for t in thresholds {
        let fp = scores.iter().zip(y_true).filter(|&(&s, &y)| s >= t && y == 0).count() as f64;
        let fn_ = scores.iter().zip(y_true).filter(|&(&s, &y)| s < t && y == 1).count() as f64;
        let cost = (c_fp * fp + c_fn * fn_) / y_true.len() as f64;
        if best.as_ref().map_or(true, |b| cost < b.cost) {
            best = Some(Best { threshold: t, cost });
        }
    }
    best // tune on a validation split, report on a separate test split
}
```

The advanced playground below puts every metric from this chapter on one screen, including a cost ratio slider that finds the cost-minimising threshold.

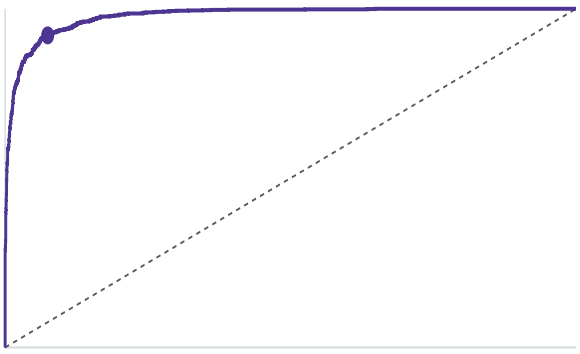
Interactive

Classification playground (shown at its default setting)

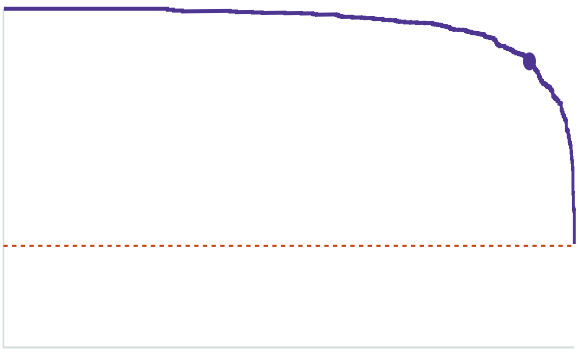
Two score distributions, a threshold, and every metric in this chapter updating live. Drop prevalence to 0.02 and watch accuracy stay high while MCC collapses and ROC-AUC keeps flattering the model that PR-AUC exposes. Then move the cost ratio and find the threshold that actually minimizes expected cost.



TP	FP
1126	207
FN	**TN**
96	2571



ROC (AUC 0.978)



PR (AP 0.956, baseline 0.300)

Accuracy	Precision	Recall	F1	Balanced acc
0.924	0.845	0.921	0.881	0.923
			ignores TN	
MCC	**Kappa**	**ROC-AUC**	**PR-AUC**	**Expected cost**
0.828	0.826	0.978	0.956	0.076
best single number			baseline = 0.300	lower is better

Try this

Set separation to 0.45 and threshold to 0.50, then drag prevalence from 0.30 down to 0.02. Accuracy barely moves (0.924 to 0.925) and ROC-AUC rises slightly (0.978 to 0.985). Meanwhile precision falls from 0.845 to 0.199, F1 from 0.881 to 0.328, and MCC from 0.828 to 0.412: 294 false positives for 73 true positives. Same classifier, same threshold. Accuracy and ROC-AUC mislead here; the other three do not.

Choosing, and the ceiling nobody mentions

Situation	Report	Do not report alone
Balanced classes, symmetric costs	Accuracy plus confusion matrix	F1 (adds nothing here)
Rare positive class	Precision, recall, PR-AUC, prevalence	Accuracy, ROC-AUC
One number for a model comparison	MCC	F1
Judge versus human agreement	Kappa, prevalence, and the pass-rate gap	Raw agreement
Known asymmetric costs	Expected cost at the tuned threshold	F1 at 0.5
Confidence used downstream	Brier, ECE, reliability diagram	Accuracy
System can abstain	Coverage–risk curve	A single accuracy number
Multiclass	Per class, macro and weighted, top confused pairs	Weighted F1

Watch out

Your metric cannot exceed your label quality. If two humans agree at kappa 0.7 on a taxonomy, a classifier scoring 0.85 against one human's labels is not beating humans; it is fitting one annotator. Measure human-to-human agreement first and report every classifier number against that ceiling. The full decision guide is in Which metric or test, when.

Eval lens

Every number in this chapter is itself an estimate from a finite sample. A precision of 0.85 on 40 flagged items has a 95% interval roughly 0.71 to 0.93. Put intervals on classifier metrics the same way you would on pass rates; see statistics for eval sets and sampling distributions.

Part II · Chapter 12

LLM as judge: a classifier you have to measure

An LLM judge reads a model's output and returns a verdict: pass or fail, better or worse, safe or unsafe. That makes it two things at once. It is a classifier, so everything in the previous chapter applies. And it is a measuring instrument, so every number you publish inherits its errors. This chapter covers how to validate a judge, the biases to control, how to get more out of a yes/no judge, and how to run judges responsibly at scale.

Plain words

A teacher grading a thousand essays

Imagine hiring a teacher to grade a thousand essays. Before you trust their grades, you would grade a few yourself and compare. You would check whether they always prefer longer essays, or whichever essay they read first. You would notice if they grade everything a little too kindly. An LLM judge is that teacher. It is fast and cheap, but you only know whether its grades mean anything by checking it against people.

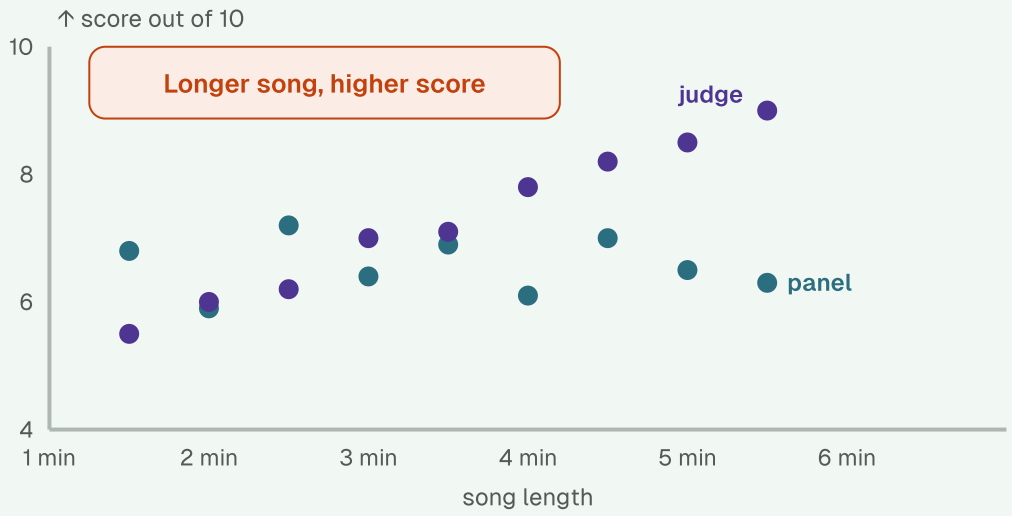
Diagram

A talent-show judge you have to audit

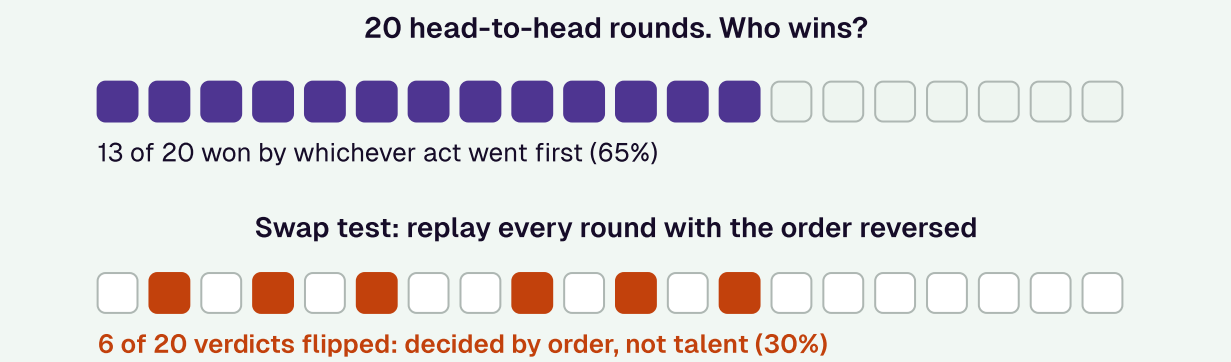
Hire a judge for a thousand acts, then catch its habits before they reach your numbers.

Before trusting the judge on all 1,000, check it against the panel on a sample.

An LLM judge is like a talent-show judge who can score a thousand acts overnight. A human panel is slow and expensive but trusted. The job is to find out how far the fast judge can be trusted, using the panel on a sample.

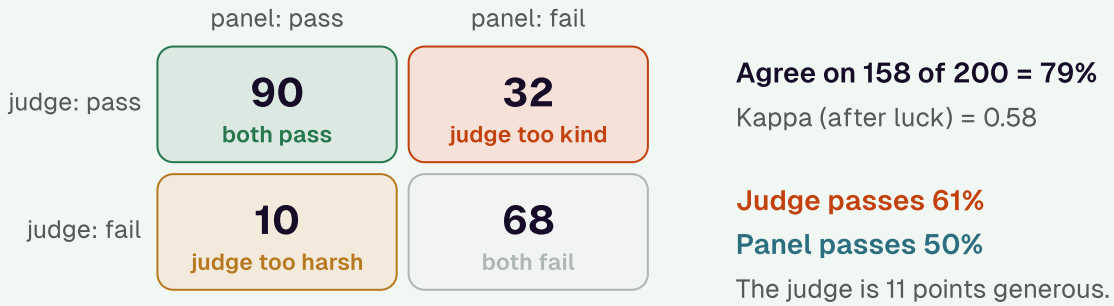


First habit: **length bias**. Plot the judge's scores against song length and they climb with every extra minute, while the panel's scores for the same acts stay flat. The judge is rewarding length, not talent.



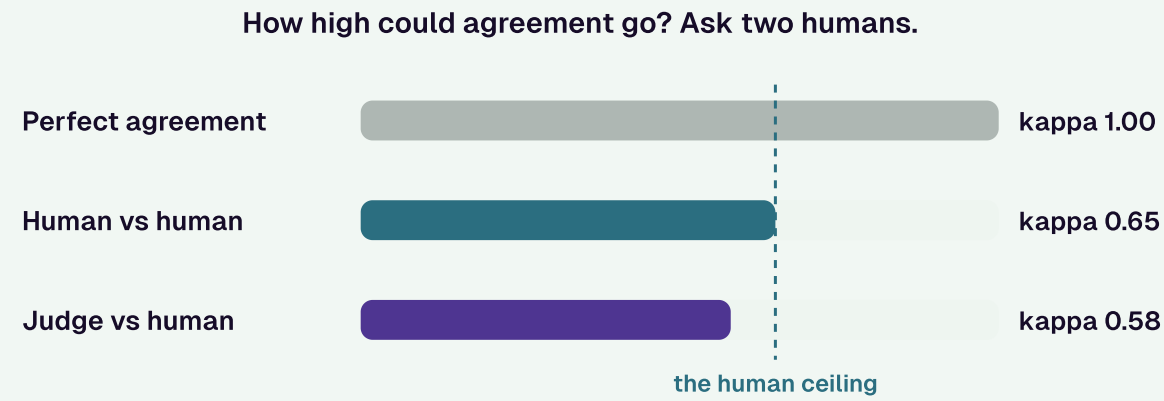
Count flipped rounds as ties, and track the flip rate per judge version.

Second habit: **position bias**. In head-to-head rounds the judge picks whichever act performed first 65% of the time. Replaying each round in reverse order exposes it: 6 of 20 verdicts flip. Those six were decided by order, not talent.



Good enough to rank two models; not to report how good either one is.

Then the headline check, on 200 acts the panel also scored. They agree 79% of the time, which sounds fine, but kappa (agreement beyond luck) is 0.58, and the judge passes 61% of acts against the panel's 50%. It ranks acts reasonably but inflates every absolute number.



The judge is close to the ceiling. Clearer rubrics help more than a smarter judge.

Last check: how well do two panel members agree with each other? Here, kappa 0.65. That is the ceiling. A judge at 0.58 is close to it, so the next gain comes from clearer scoring rules, not a stronger judge.

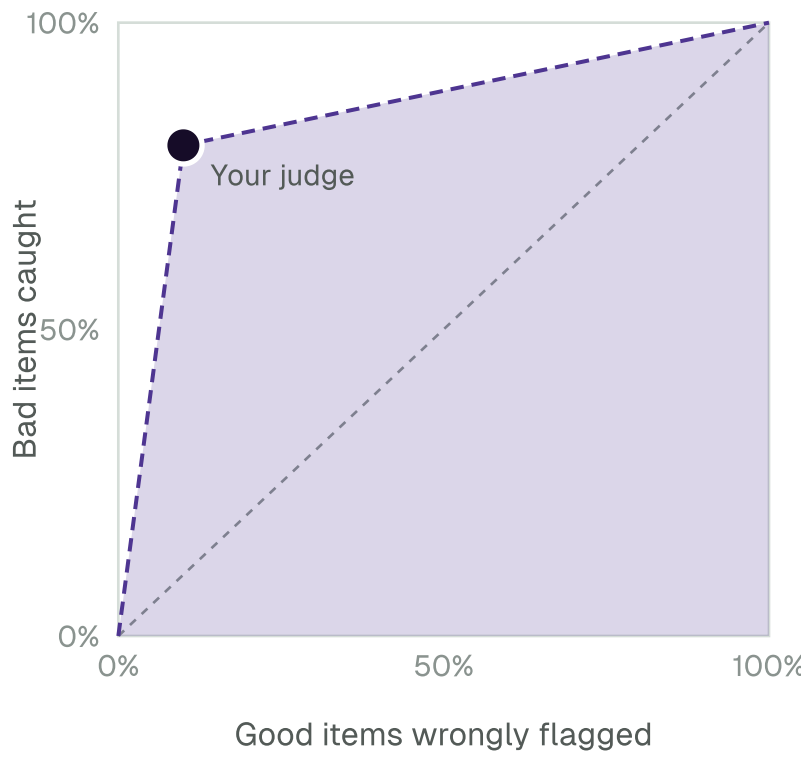
A yes/no judge gives you one dot, not a curve

Most judges answer "Pass" or "Fail." There is no score, so there is no threshold to slide. You get exactly one dot on the ROC chart from the previous chapter.

Interactive

What a binary judge looks like on an ROC chart (shown at its default setting)

Switch between the three views. The middle one is the trap most teams fall into when they iterate on judge prompts.



Bad items caught
80%

Good items flagged
10%

"AUC" from one dot
0.85

Plain words

Reading the three views

Binary judge only. Precision, recall, and F1 still work, because you can still fill in the confusion matrix. AUC barely makes sense. Forced onto one dot, it is the area under two straight lines, which equals the average of "bad items caught" and "good items left alone" (balanced accuracy). It adds nothing new.

The prompt becomes your threshold. "Only fail clear violations" moves the dot down and left: fewer false alarms, more misses. "Fail anything suspicious" moves it up and right. Rewording slides the dot along the same curve. A real improvement is the green dot: up and left at once. When you test a new prompt, check whether both numbers improved or one gained at the other's expense.

Getting the curve back. If your API exposes token probabilities (logprobs), read how confident the judge was in the word "Fail," like 0.97 or 0.52. That turns one dot into a full curve. You can pick any threshold without rewriting the prompt, and compute real ROC-AUC and PR-AUC. Without logprobs, run the judge 5 to 10 times per item and use the share of "Fail" votes as the score.

Under the hood

Single-point AUC and scores from logprobs

With one operating point $(\mathrm{FPR}, \mathrm{TPR})$, the ROC "curve" is two segments through $(0,0)$, the point, and $(1,1)$. Its area is

$$\mathrm{AUC}_{1\text{ point}} = \tfrac{1}{2}\,\mathrm{TPR}\cdot\mathrm{FPR} + \tfrac{1}{2}(1-\mathrm{FPR})(1+\mathrm{TPR}) = \frac{\mathrm{TPR}+(1-\mathrm{FPR})}{2}$$

Exactly balanced accuracy. With TPR 0.8 and FPR 0.1, that is 0.85.

A continuous score from a single call: renormalise the probability of the two verdict tokens at the answer position.

$$s = \frac{p(\texttt{Fail} \mid \text{prompt})}{p(\texttt{Fail} \mid \text{prompt}) + p(\texttt{Pass} \mid \text{prompt})}$$

Check that the verdict token appears where you read it (put the verdict last, after reasoning, in a fixed format). The renormalised score is usually miscalibrated; that is fine for ranking and AUC, and fixable with temperature scaling if you need calibrated probabilities.

python

```python
import math

def judge_score(top_logprobs, pos="Fail", neg="Pass"):
    """top_logprobs: {token: logprob} at the verdict position."""
    lp_pos = top_logprobs.get(pos, -50.0)   # treat a missing token as near-zero mass
    lp_neg = top_logprobs.get(neg, -50.0)
    a, b = math.exp(lp_pos), math.exp(lp_neg)
    return a / (a + b)

def vote_score(verdicts):
    """Fallback without logprobs: share of 'Fail' over k samples at temperature > 0."""
    return sum(v == "Fail" for v in verdicts) / len(verdicts)
```

rust

```rust
use std::collections::HashMap;

/// top_logprobs: {token: logprob} at the verdict position.
/// The Python defaults are pos = "Fail", neg = "Pass".
fn judge_score(top_logprobs: &HashMap<&str, f64>, pos: &str, neg: &str) -> f64 {
    let lp_pos = top_logprobs.get(pos).copied().unwrap_or(-50.0); // treat a missing token as near-zero mass
    let lp_neg = top_logprobs.get(neg).copied().unwrap_or(-50.0);
    let (a, b) = (lp_pos.exp(), lp_neg.exp());
    a / (a + b)
```

```rust
}

/// Fallback without logprobs: share of 'Fail' over k samples at temperature > 0.
fn vote_score(verdicts: &[&str]) -> f64 {
    verdicts.iter().filter(|&&v| v == "Fail").count() as f64 / verdicts.len() as f64
}
```

Agreement with humans

The first question about any judge is how well it agrees with people. Raw agreement is the wrong summary: two raters agree a lot by luck when one label is common. Use Cohen's kappa, and always check the pass-rate gap.

Worked problem

Validate a judge against 200 human labels. Judge and human agree on 158 items. Is the judge good enough to ship?

Metric	Value	Reading
Agreement	0.790	Looks acceptable
Cohen's kappa	0.580	Moderate once you discount luck
Precision (judge "pass")	0.738	About a quarter of its passes are wrong
Recall (judge "pass")	0.900	It catches most real passes
Judge pass rate	**0.610**	
Human pass rate	**0.500**	**The judge is 11 points generous**

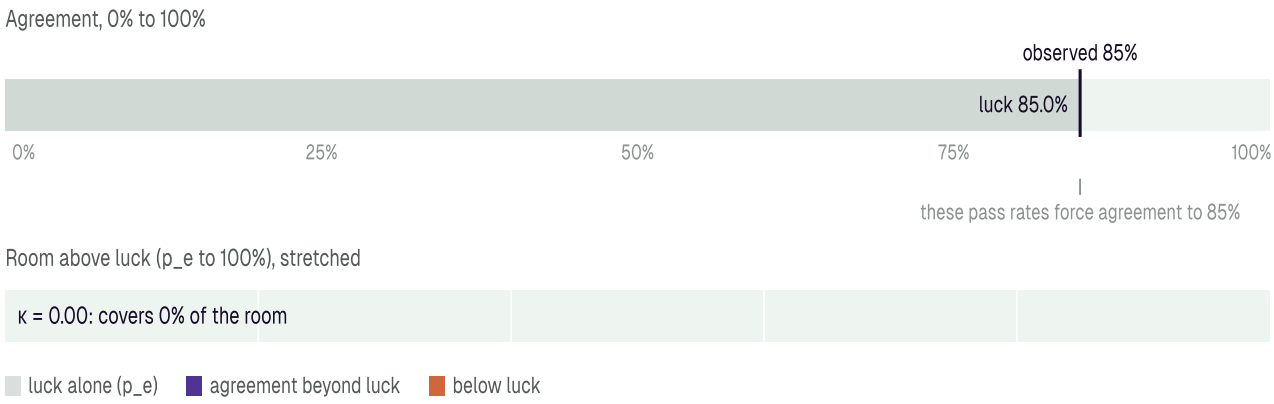
The confusion matrix behind this is TP 90, FP 32, FN 10, TN 68. The finding is invisible if you only look at agreement: the judge **ranks** models fine, but every **absolute** number it produces is inflated by about 11 points. That is acceptable for comparing two models and unacceptable for reporting how good either one is. Next step: have a second human label 100 of the items and compute human-to-human kappa. If two humans only reach 0.65, a judge at 0.58 is close to as good as the task allows.

The widget below shows why agreement alone misleads. Any two raters agree some of the time by luck, and the more lopsided the labels, the more luck there is.

Interactive

Why 85% agreement can be bad (shown at its default setting)

Start with the judge that says "pass" to everything: it agrees with people 85% of the time and has learned nothing. Then try the other presets or move the sliders, and watch how much of the agreement luck alone would produce.



Per 200 items	Judge: pass	Judge: fail
Human: pass	**170** luck: 170	**0** luck: 0
Human: fail	**30** luck: 30	**0** luck: 0

Big number: what happened. Small grey number: what luck alone would give if the judge labelled at its own pass rate without looking at the item. The diagonal (both pass, both fail) is agreement.

Observed agreement p_o	Agreement from luck p_e	Cohen's kappa	Pass-rate gap (judge – human)
85%	**85.0%**	**0.00**	**+15 pts**

Luck alone: people pass 85% of items and the judge passes 100%, so two raters who ignored the content would still agree on 0.85 × 1.00 + 0.15 × 0.00 = **85.0%** of items. Kappa only credits the agreement above that: (0.85 – 0.850) / (1 – 0.850) = **0.00**, slight on the Landis and Koch (1977)

scale. This judge never looks at the answer. Its 85% agreement is exactly the share of items people pass, so it tells you nothing about which ones deserve to pass.

Under the hood

Kappa and the judge report

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \qquad p_e = \sum_{\ell} p_A(\ell)\, p_B(\ell)$$

p_o is observed agreement; p_e is the agreement expected if each rater labelled independently at their own rates. In the worked problem, $p_o = 0.79$ and $p_e = 0.61 \cdot 0.5 + 0.39 \cdot 0.5 = 0.5$, so $\kappa = 0.58$.

python

```python
def cohens_kappa(a, b):
    n = len(a)
    labels = set(a) | set(b)
    po = sum(x == y for x, y in zip(a, b)) / n
    pe = sum((a.count(l) / n) * (b.count(l) / n) for l in labels)
    return 1.0 if pe == 1 else (po - pe) / (1 - pe)

def judge_report(judge, human):
    """The numbers to put next to any judge you ship (1 = pass)."""
    tp = sum(j == 1 and h == 1 for j, h in zip(judge, human))
    fp = sum(j == 1 and h == 0 for j, h in zip(judge, human))
    fn = sum(j == 0 and h == 1 for j, h in zip(judge, human))
    n = len(judge)
    return {
        "agreement": sum(j == h for j, h in zip(judge, human)) / n,
        "kappa": cohens_kappa(judge, human),
        "precision": tp / (tp + fp) if tp + fp else 0.0,
        "recall": tp / (tp + fn) if tp + fn else 0.0,
        # these two lines are the point: good agreement can hide a generous judge
        "judge_pass_rate": (tp + fp) / n,
        "human_pass_rate": (tp + fn) / n,
    }
```

rust

```rust
use std::collections::HashSet;
use std::hash::Hash;

fn cohens_kappa<T: Eq + Hash>(a: &[T], b: &[T]) -> f64 {
    let n = a.len() as f64;
    let labels: HashSet<&T> = a.iter().chain(b).collect();
    let po = a.iter().zip(b).filter(|(x, y)| x == y).count() as f64 / n;
    let count = |v: &[T], l: &T| v.iter().filter(|&x| x == l).count() as f64;
    let pe: f64 = labels.iter().map(|&l| (count(a, l) / n) * (count(b, l) / n)).sum();
```

```
    if pe == 1.0 { 1.0 } else { (po - pe) / (1.0 - pe) }
}

struct JudgeReport {
    agreement: f64,
    kappa: f64,
    precision: f64,
    recall: f64,
    // these two fields are the point: good agreement can hide a generous judge
    judge_pass_rate: f64,
    human_pass_rate: f64,
}

/// The numbers to put next to any judge you ship (1 = pass).
fn judge_report(judge: &[u8], human: &[u8]) -> JudgeReport {
    let pairs = || judge.iter().zip(human);
    let tp = pairs().filter(|&(&j, &h)| j == 1 && h == 1).count() as f64;
    let fp = pairs().filter(|&(&j, &h)| j == 1 && h == 0).count() as f64;
    let fn_ = pairs().filter(|&(&j, &h)| j == 0 && h == 1).count() as f64;
    let n = judge.len() as f64;
    JudgeReport {
        agreement: pairs().filter(|&(j, h)| j == h).count() as f64 / n,
        kappa: cohens_kappa(judge, human),
        precision: if tp + fp > 0.0 { tp / (tp + fp) } else { 0.0 },
        recall: if tp + fn_ > 0.0 { tp / (tp + fn_) } else { 0.0 },
        // these two lines are the point: good agreement can hide a generous judge
        judge_pass_rate: (tp + fp) / n,
        human_pass_rate: (tp + fn_) / n,
    }
}
```

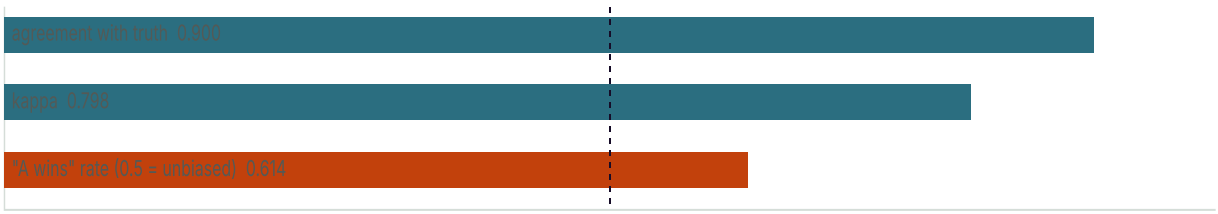
Put a confidence interval on kappa too. On 200 items, a bootstrap interval on $\kappa = 0.58$ is roughly ±0.11, which matters when comparing two judge prompts. See intervals and the bootstrap.

The biases you have to control for

Bias	What it does	Mitigation
Position	Prefers whichever response is shown first (or second) in a pairwise comparison	Run both orders, average, and report the disagreement rate as a bias estimate
Length	Scores longer answers higher regardless of quality	Check score against token count; add length-matched controls or a length-controlled win rate
Self-preference	A model judging its own family scores it higher	Use a judge from a different family, or measure the effect and report it
Style over substance	Confident formatting reads as correctness	Rubrics that force a claim-by-claim verdict instead of a global score
Score compression	Everything lands on 7 or 8 out of 10	Binary or three-point scales, or forced pairwise comparison

Interactive **Judge bias simulator** (shown at its default setting)

A judge with position bias, length bias and finite skill. Toggle order swapping and watch the win rate return toward 0.5 while agreement barely moves. The bias was inflating confidence, not accuracy.



Agreement 0.900 **kappa** 0.798 **"A wins" rate** 0.614

Order swapping OFF. Turn it on and watch the "A wins" rate move back toward 0.5 while agreement barely changes. The bias was inflating your confidence, not your accuracy.

Under the hood

Debiased pairwise judging

python

```python
async def judge_pairwise_debiased(judge_fn, task, resp_a, resp_b):
    """Run both orderings; treat disagreement as a bias signal, not a verdict."""
    fwd = await judge_fn(task, first=resp_a, second=resp_b)   # "first" | "second" | "tie"
    rev = await judge_fn(task, first=resp_b, second=resp_a)
    fwd_w = {"first": "a", "second": "b", "tie": "tie"}[fwd]
    rev_w = {"first": "b", "second": "a", "tie": "tie"}[rev]
    if fwd_w == rev_w:
        return {"winner": fwd_w, "consistent": True}
    return {"winner": "tie", "consistent": False}   # order flipped the verdict
```

rust

```rust
#[derive(Debug, Clone, Copy, PartialEq)]
enum Verdict {
    First,
    Second,
    Tie,
}

#[derive(Debug, Clone, Copy, PartialEq)]
enum Winner {
    A,
    B,
    Tie,
}

struct PairwiseResult {
    winner: Winner,
    consistent: bool,
}

/// Run both orderings; treat disagreement as a bias signal, not a verdict.
/// The Python awaits an async judge. The standard library has no async runtime,
/// so here judge_fn(task, first, second) is a blocking call.
fn judge_pairwise_debiased(
    mut judge_fn: impl FnMut(&str, &str, &str) -> Verdict,
    task: &str,
    resp_a: &str,
    resp_b: &str,
) -> PairwiseResult {
    let fwd = judge_fn(task, resp_a, resp_b); // First | Second | Tie
    let rev = judge_fn(task, resp_b, resp_a);
    let fwd_w = match fwd {
        Verdict::First => Winner::A,
        Verdict::Second => Winner::B,
        Verdict::Tie => Winner::Tie,
    };
    let rev_w = match rev {
        Verdict::First => Winner::B,
        Verdict::Second => Winner::A,
        Verdict::Tie => Winner::Tie,
    };
    if fwd_w == rev_w {
        return PairwiseResult { winner: fwd_w, consistent: true };
```

```
    }
    PairwiseResult { winner: Winner::Tie, consistent: false } // order flipped the verdict
}
```

The inconsistency rate across a batch is itself a useful number: it estimates how much of the judge's verdict is position rather than content. Track it per judge version.

Principal asks

Judges are often used to catch models gaming a task: editing the test instead of the implementation, hard-coding an expected output, or reading a solution left in the environment. Expect to be asked for a concrete instance, then for how you detect the *class* of exploit rather than the one instance, and what changed in the harness so it cannot recur.

Validating a judge before you trust it

1. Sample 150 to 300 items across the score range, not just the disputed ones.
2. Get two human labels per item and measure human-to-human agreement first. That is your ceiling. If humans agree at kappa 0.6, the judge cannot do meaningfully better.
3. Measure judge-to-human agreement on a held-out slice the judge prompt was not tuned on.
4. Check the pass-rate gap. A judge with good agreement but a higher pass rate biases every absolute number while still ranking models correctly.
5. Recheck after any change on the judge side. A judge is a model, so a judge upgrade is a silent measurement change.

When the dot will not move up and left

1. **Read your mistakes.** Pull 20 false positives and 20 false negatives. There is usually a pattern, like "fails anything that mentions a competitor." Fix the rubric for exactly those cases.
2. **Tighten the definition.** Fuzzy criteria are the most common cause of overlap. Borderline cases land randomly on either side.
3. **Split one judgment into several.** Separate yes/no checks (is it factually correct? does it follow the format? does it answer the question?) usually separate the classes better than one holistic verdict.

4. **Show hard examples.** Few-shot examples from the overlap zone, labelled with the reason, teach the judge where the line is. Easy examples do not help much.

5. **Reason first, verdict last.** Asking for reasoning through the criteria before the answer helps on borderline cases, and makes the verdict token easy to read logprobs from.

6. **Check the human ceiling.** If humans agree only 80% of the time, the judge will not beat that. Fix the labelling guidelines first.

A stronger judge model, or several judges voting (different prompts or model families), can also push the humps further apart.

For leaders

Judge governance: treat the judge as production infrastructure

Once a judge feeds dashboards and launch decisions, it is measurement infrastructure, and it needs the same discipline as any other production system.

Version and pin it

Pin the judge model, prompt, and decoding settings. Record them with every result. A judge upgrade changes every historical comparison, so re-baseline when it changes.

Validate on a cadence

Keep a fixed human-labelled set (a few hundred items) and re-score it on every judge change and on a schedule. Alert on kappa drops and pass-rate drift.

Know the ceiling

Report human-to-human kappa next to judge-to-human kappa. Chasing a judge past the human ceiling is chasing noise in the labels.

Control the biases

Both orderings for pairwise judges, length checks, a different model family from the model being judged. Publish the inconsistency rate.

Know what you need

Ranking two models needs consistency. Reporting an absolute quality number needs calibration to humans. Most judges are good enough for the first and not the second.

Budget it

Judge calls can rival the cost of the eval itself. Use a cheaper judge for broad coverage with a stronger judge or humans auditing a random sample, and track the audit disagreement.

Principal asks

- What is the judge's kappa against humans, and what is the human-to-human kappa on the same items?
- Is the judge more lenient than humans? By how much, and does that matter for this claim?
- Has the judge model or prompt changed since the baseline we are comparing against?
- Is the judge from the same model family as the system it is grading?

Eval lens

A judge-based pass rate has two layers of uncertainty: sampling error from the finite task set, and judge error relative to humans. The first shrinks with more tasks; the second does not. Report both, or at least state the judge's validated agreement next to any number it produced.

Part II · Chapter 13

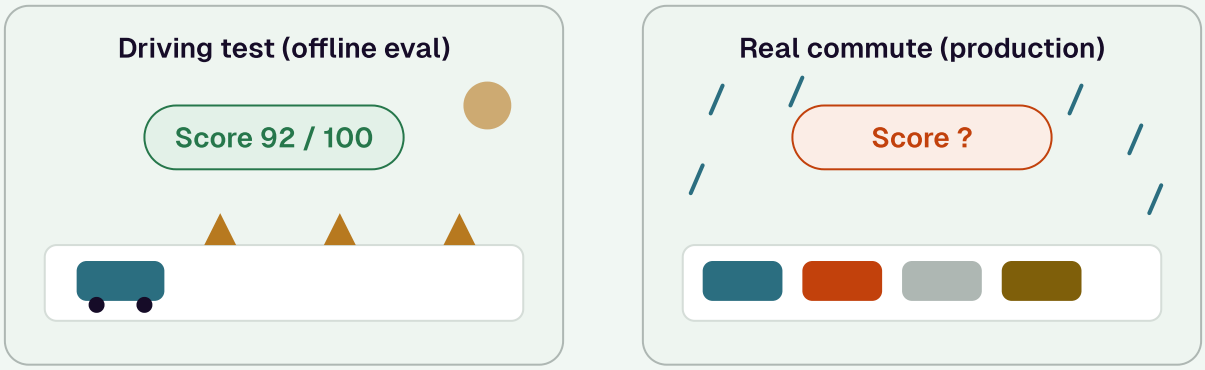
The offline to online gap: does the lab score predict the real world?

Every offline eval is a bet that a number computed on a fixed task set will move the same way as the thing you actually care about: user acceptance, retention, task success in production. This chapter is about testing that bet, putting a number on how wrong it is, and saying honestly how much of the gap you can close.

Diagram

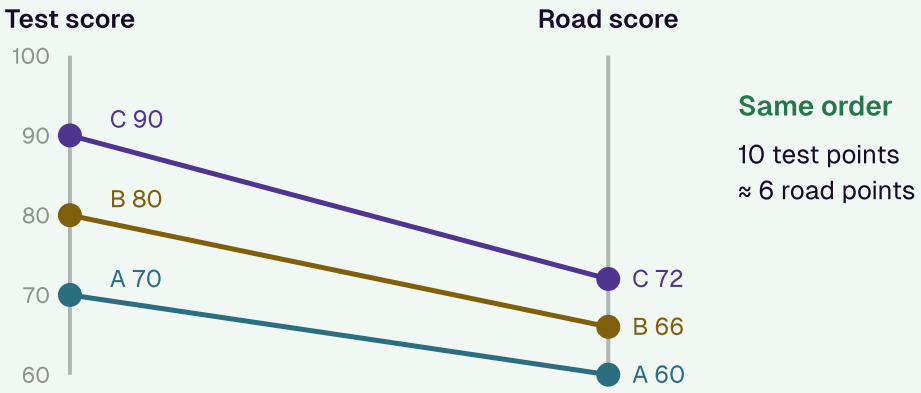
The driving test and the rush-hour commute

Why a better lab score does not always mean a better product, in five pictures.

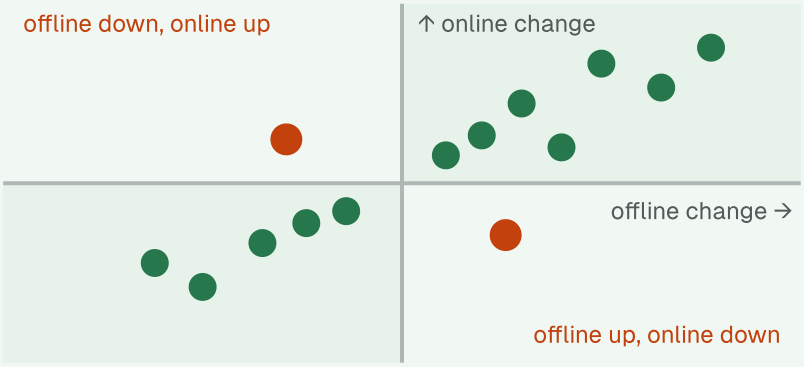


Does a better test score mean a better commute?

The test is quiet, sunny and predictable. The commute is crowded and wet. Your offline eval is the test; production is the commute. This chapter is about checking how well one predicts the other.

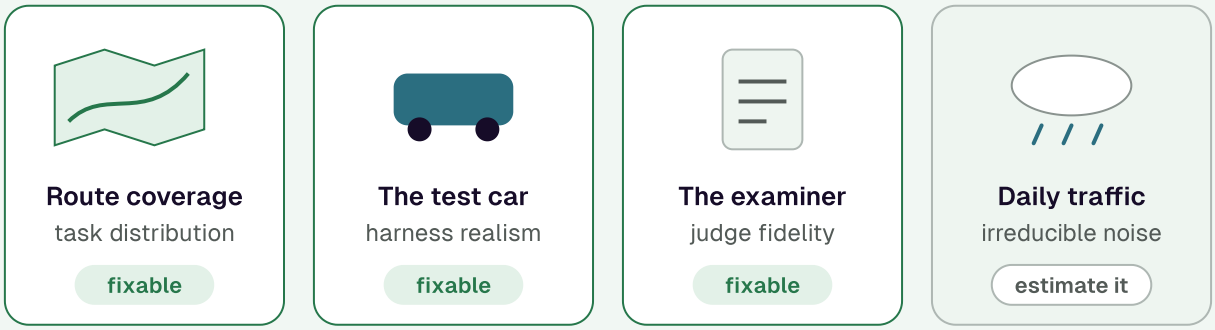


Two questions, kept apart. **Ordering:** do drivers keep the same rank on the road? Here yes, A < B < C. **Size:** each 10 points on the test is worth only about 6 on the road. Ordering matters most for a ship decision; size tells you how much to promise.



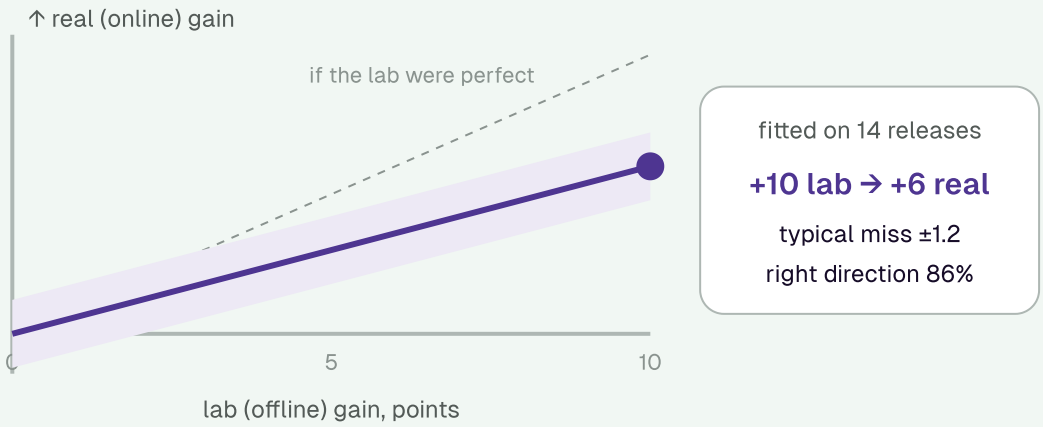
Same direction on 12 of 14 releases: direction accuracy 86%

Plot every past release: offline change across, online change up. Green zones mean the two agreed. Two releases landed in the wrong zone, so the offline suite pointed the wrong way 2 times in 14. That **direction accuracy** is the single most useful number about an eval suite.



Spend on the first three. Measure the fourth with an A/A test.

The gap has four sources. The test route skips some roads (the task set misses real usage). The test car is not your car (the harness lacks real tools and context). The examiner grades differently from real outcomes (judge fidelity). And traffic varies day to day, which no test can predict.



Put it together and the offline suite becomes a calibrated instrument: a lab gain of 10 points predicts about 6 online, give or take 1.2, and the direction is right 86% of the time. Recompute this after every launch; when direction accuracy falls, the suite has drifted from real use.

Plain words

The driving test and the commute

A driving test happens on a quiet route, in good weather, with an examiner watching. Passing it predicts how someone drives on a real commute, but not perfectly. Some careful test-takers panic in traffic. Some sloppy ones are fine in practice.

An offline eval is the driving test. Production is the commute. The questions in this chapter are simple to say: when a new model scores better on the test, does it also do better on the commute? By how much? And how often does the test point the wrong way entirely?

Two things to keep apart. First, **ordering**: does the test rank drivers in the same order as real life? Second, **size**: if someone improves by 10 points on the test, how many points is that on the road? Ordering is the more important and easier question. Size comes second.

For leaders

Why this is the most important number your eval team does not report

An offline suite that has never been checked against production is a story, not a measurement. Most teams report offline deltas as if they were product deltas. The difference shows up months later as a launch that won offline and did nothing online, or worse, a regression that offline scored as a win.

What to ask for, in order:

1. **A rank check.** Across the last 8 to 15 releases, does the offline suite order models the same way production does? Kendall's tau is the honest statistic at that sample size.

2. **A calibration line.** A fit of online delta against offline delta. The slope tells you how much offline overstates. A slope of 0.6 means a 10 point offline win is worth about 6 points in production.

3. **A backtest.** Leave-one-out prediction of each past release from all the others. Report the typical miss, the worst miss, and above all **direction accuracy**: how often offline and online disagreed on whether the release was better.

4. **A decomposition of the remaining gap** into what is fixable (task coverage, harness realism, judge fidelity) and what is not (user variance, traffic drift). This is what tells you where to spend.

What to fund. If direction accuracy is below about 85%, the offline suite cannot gate launches on its own and you need online experiments for every material change. That is expensive, so the higher-return investment is usually the fixable buckets: realistic harnesses and task sets drawn from real usage.

Principal asks

- How many releases is this calibration fitted on, and what happens to the error if you drop to half of them?
- What is the worst single miss, and what was different about that release?
- Has the product or traffic mix changed since the oldest release in the fit? If so, why should yesterday's slope hold?
- Of the remaining gap, how much do you believe is irreducible, and how did you estimate it?

Does offline predict online at all

Plain words

Same order, before same size

Line up past models by their offline score, then by their production score. If the two line-ups mostly agree, offline is useful for picking winners even if its numbers are inflated. If they do not agree, nothing else in this chapter rescues it.

Under the hood

Spearman and Kendall rank correlation

Spearman's ρ is the Pearson correlation of the ranks. Kendall's τ counts pairs of releases that the two measures order the same way (concordant, C) versus opposite ways (discordant, D):

$$\tau = \frac{C - D}{C + D}$$

Ties are excluded from both counts in this simple version (tau-a over untied pairs). $\tau = 1$ means offline and online order every pair the same way.

With 8 to 15 releases, which is realistic, τ is more stable than ρ and has a direct reading: $\tau = 0.6$ means 80% of release pairs are ordered correctly, because $(C - D)/(C + D) = 2 \cdot \text{share}_C - 1$.

python

```python
def spearman(x: list[float], y: list[float]) -> float:
    """Rank correlation. The right first question for offline vs online:
    does the offline suite ORDER models the way production does?
    Absolute calibration is a separate and easier problem."""
    def ranks(v):
        order = sorted(range(len(v)), key=lambda i: v[i])
        r = [0.0] * len(v)
        i = 0
        while i < len(order):
            j = i
            while j + 1 < len(order) and v[order[j + 1]] == v[order[i]]:
                j += 1
            avg = (i + j) / 2 + 1          # tied values share the average rank
            for k in range(i, j + 1):
                r[order[k]] = avg
            i = j + 1
        return r

    rx, ry = ranks(x), ranks(y)
    n = len(x)
    mx, my = sum(rx) / n, sum(ry) / n
    num = sum((a - mx) * (b - my) for a, b in zip(rx, ry))
    den = (sum((a - mx) ** 2 for a in rx) * sum((b - my) ** 2 for b in ry)) ** 0.5
    return num / den if den else 0.0


def kendall_tau(x: list[float], y: list[float]) -> float:
    """More robust than Spearman when you have few models. With 8 to 15 model
    releases, which is the realistic n, tau is the honest choice."""
```

```
    n = len(x)
    conc = disc = 0
    for i in range(n):
        for j in range(i + 1, n):
            sx = (x[i] - x[j])
            sy = (y[i] - y[j])
            if sx * sy > 0:
                conc += 1
            elif sx * sy < 0:
                disc += 1
    total = conc + disc
    return (conc - disc) / total if total else 0.0
```

rust

```
/// Rank correlation. The right first question for offline vs online:
/// does the offline suite ORDER models the way production does?
/// Absolute calibration is a separate and easier problem.
fn spearman(x: &[f64], y: &[f64]) -> f64 {
    fn ranks(v: &[f64]) -> Vec<f64> {
        let mut order: Vec<usize> = (0..v.len()).collect();
        order.sort_by(|&a, &b| v[a].partial_cmp(&v[b]).expect("NaN value"));
        let mut r = vec![0.0; v.len()];
        let mut i = 0;
        while i < order.len() {
            let mut j = i;
            while j + 1 < order.len() && v[order[j + 1]] == v[order[i]] {
                j += 1;
            }
            let avg = (i + j) as f64 / 2.0 + 1.0; // tied values share the average rank
            for k in i..=j {
                r[order[k]] = avg;
            }
            i = j + 1;
        }
        r
    }

    let (rx, ry) = (ranks(x), ranks(y));
    let n = x.len() as f64;
    let (mx, my) = (rx.iter().sum::<f64>() / n, ry.iter().sum::<f64>() / n);
    let num: f64 = rx.iter().zip(&ry).map(|(a, b)| (a - mx) * (b - my)).sum();
    let den = (rx.iter().map(|a| (a - mx).powi(2)).sum::<f64>()
        * ry.iter().map(|b| (b - my).powi(2)).sum::<f64>())
    .sqrt();
    if den != 0.0 { num / den } else { 0.0 }
}

/// More robust than Spearman when you have few models. With 8 to 15 model
/// releases, which is the realistic n, tau is the honest choice.
fn kendall_tau(x: &[f64], y: &[f64]) -> f64 {
    let n = x.len();
    let (mut conc, mut disc) = (0, 0);
    for i in 0..n {
        for j in i + 1..n {
            let sx = x[i] - x[j];
            let sy = y[i] - y[j];
            if sx * sy > 0.0 {
                conc += 1;
            } else if sx * sy < 0.0 {
```

```
                disc += 1;
            }
        }
    }
    let total = conc + disc;
    if total > 0 { (conc - disc) as f64 / total as f64 } else { 0.0 }
}
```

> **Watch out**
>
> With 10 releases there are only 45 pairs, and a 95% interval on τ is roughly ±0.45 wide. A τ of 0.5 on ten releases is consistent with anything from barely useful to excellent. Bootstrap over releases (see intervals and the bootstrap) before you quote it.

Calibrating the offline delta to the online delta

Plain words

An exchange rate between two currencies

Offline points and production points are different currencies. The calibration line is the exchange rate. If the rate is 0.6, a 10 point offline win buys about 6 points online. The scatter around the line tells you how much that exchange rate wobbles from release to release.

Under the hood

Least squares on release deltas

For each past release i, take the offline delta x_i against the previous model and the online delta y_i from the A/B test. Fit

$$y_i = a\,x_i + b + \varepsilon_i, \qquad \hat{a} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \qquad \hat{b} = \bar{y} - \hat{a}\bar{x}$$

$\hat{a} < 1$ means offline overstates improvements, which is the common case because offline tasks are cleaner and easier than production. Shrinkage $1 - \hat{a}$ is the gap as a single number. The residual spread is the real prediction interval.

python

```python
def fit_gap_model(offline_deltas: list[float], online_deltas: list[float]) -> dict:
    """Least squares fit of online_delta ~ a * offline_delta + b."""
    n = len(offline_deltas)
    mx = sum(offline_deltas) / n
    my = sum(online_deltas) / n

    # Standard least squares slope: covariance over variance of x.
    sxy = sum((x - mx) * (y - my) for x, y in zip(offline_deltas, online_deltas))
    sxx = sum((x - mx) ** 2 for x in offline_deltas)
    a = sxy / sxx if sxx else 0.0        # sxx == 0 when every release moved identically
    b = my - a * mx                      # line passes through the means

    resid = [y - (a * x + b) for x, y in zip(offline_deltas, online_deltas)]
    rmse = (sum(r * r for r in resid) / n) ** 0.5      # typical prediction miss
    ss_tot = sum((y - my) ** 2 for y in online_deltas)
    r2 = 1 - sum(r * r for r in resid) / ss_tot if ss_tot else 0.0

    return {"slope": a, "intercept": b, "rmse": rmse, "r2": r2,
            # slope < 1 means offline OVERSTATES improvements, the common case,
            # because the offline suite is cleaner than production.
            "shrinkage": 1 - a}
```

rust

```rust
#[derive(Debug, Clone, Copy)]
struct GapFit {
    slope: f64,
    intercept: f64,
    rmse: f64,
    r2: f64,
    // slope < 1 means offline OVERSTATES improvements, the common case,
    // because the offline suite is cleaner than production.
    shrinkage: f64,
}

/// Least squares fit of online_delta ~ a * offline_delta + b.
fn fit_gap_model(offline_deltas: &[f64], online_deltas: &[f64]) -> GapFit {
    let n = offline_deltas.len() as f64;
    let mx = offline_deltas.iter().sum::<f64>() / n;
    let my = online_deltas.iter().sum::<f64>() / n;

    // Standard least squares slope: covariance over variance of x.
    let sxy: f64 = offline_deltas.iter().zip(online_deltas).map(|(x, y)| (x - mx) * (y - my)).sum();
    let sxx: f64 = offline_deltas.iter().map(|x| (x - mx).powi(2)).sum();
    let a = if sxx != 0.0 { sxy / sxx } else { 0.0 }; // sxx == 0 when every release moved identically
    let b = my - a * mx; // line passes through the means

    let resid: Vec<f64> = offline_deltas.iter().zip(online_deltas).map(|(x, y)| y - (a * x + b)).collect();
    let ss_res: f64 = resid.iter().map(|r| r * r).sum();
    let rmse = (ss_res / n).sqrt(); // typical prediction miss
    let ss_tot: f64 = online_deltas.iter().map(|y| (y - my).powi(2)).sum();
    let r2 = if ss_tot != 0.0 { 1.0 - ss_res / ss_tot } else { 0.0 };

    GapFit { slope: a, intercept: b, rmse, r2, shrinkage: 1.0 - a }
}
```

> **Note**
>
> Ordinary least squares assumes the offline delta is measured without error. It is not: offline deltas carry their own sampling noise, which biases $\hat{a}$ toward zero (attenuation bias). With small eval sets this makes offline look worse at predicting online than it really is. Larger offline suites, or an errors-in-variables fit, reduce the effect.

Backtest: the part that makes it a method instead of a story

Plain words

Predict the past you have not looked at

Any line looks good on the points it was drawn through. The fair test is to hide one past release, draw the line through the rest, and see how well it predicts the hidden one. Repeat for every release. That is a backtest, and it is the only honest estimate of how well the next prediction will go.

For leaders

Which number gates a launch

The backtest produces three numbers. The **mean absolute error** is the headline gap. The **worst miss** is the one a skeptic will ask about. **Direction accuracy** is the one that matters for a ship decision: predicting +3 when the truth was +1 is a small error, but predicting +1 when the truth was –1 means you shipped a worse model. A suite with small MAE and 70% direction accuracy is not fit to gate launches.

Under the hood

Leave-one-out over releases

python

```python
def backtest_suite(history: list[dict]) -> dict:
    """history: one entry per past release with {"model", "offline", "online"}."""
    errors, sign_hits = [], 0

    for i in range(len(history)):
        # LEAVE ONE OUT: fit on every release EXCEPT this one, then predict it.
        # Fitting and scoring on the same releases always looks good and tells
        # you nothing about the next release, which is the one you care about.
        train = history[:i] + history[i + 1:]
        test = history[i]

        fit = fit_gap_model([h["offline"] for h in train],
```

```
                            [h["online"] for h in train])
        pred = fit["slope"] * test["offline"] + fit["intercept"]

        errors.append(abs(pred - test["online"]))

        # DIRECTION is the ship decision. Track it separately from error size
        # because the two failures are not the same.
        if (pred > 0) == (test["online"] > 0):
            sign_hits += 1

    return {
        "mae": sum(errors) / len(errors),           # your headline "gap" number
        "direction_accuracy": sign_hits / len(history),
        "worst_case": max(errors),                  # the one they will ask about
    }
```

rust

```
struct Release {
    model: String,
    offline: f64,
    online: f64,
}

struct Backtest {
    mae: f64,                // your headline "gap" number
    direction_accuracy: f64,
    worst_case: f64,         // the one they will ask about
}

// fit_gap_model from the snippet above, cut down to the two fields used here
// so that this snippet compiles on its own.
struct GapFit {
    slope: f64,
    intercept: f64,
}

fn fit_gap_model(offline_deltas: &[f64], online_deltas: &[f64]) -> GapFit {
    let n = offline_deltas.len() as f64;
    let mx = offline_deltas.iter().sum::<f64>() / n;
    let my = online_deltas.iter().sum::<f64>() / n;
    let sxy: f64 = offline_deltas.iter().zip(online_deltas).map(|(x, y)| (x - mx) * (y - my)).sum();
    let sxx: f64 = offline_deltas.iter().map(|x| (x - mx).powi(2)).sum();
    let a = if sxx != 0.0 { sxy / sxx } else { 0.0 };
    GapFit { slope: a, intercept: my - a * mx }
}

/// history: one entry per past release with model, offline and online deltas.
fn backtest_suite(history: &[Release]) -> Backtest {
    let (mut errors, mut sign_hits) = (Vec::new(), 0);

    for i in 0..history.len() {
        // LEAVE ONE OUT: fit on every release EXCEPT this one, then predict it.
        // Fitting and scoring on the same releases always looks good and tells
        // you nothing about the next release, which is the one you care about.
        let train: Vec<&Release> = history.iter().enumerate().filter(|&(k, _)| k != i).map(|(_, h)| h).collect();
        let test = &history[i];

        let offline: Vec<f64> = train.iter().map(|h| h.offline).collect();
        let online: Vec<f64> = train.iter().map(|h| h.online).collect();
```

```
        let fit = fit_gap_model(&offline, &online);
        let pred = fit.slope * test.offline + fit.intercept;

        errors.push((pred - test.online).abs());

        // DIRECTION is the ship decision. Track it separately from error size
        // because the two failures are not the same.
        if (pred > 0.0) == (test.online > 0.0) {
            sign_hits += 1;
        }
    }

    Backtest {
        mae: errors.iter().sum::<f64>() / errors.len() as f64,
        direction_accuracy: sign_hits as f64 / history.len() as f64,
        worst_case: errors.iter().copied().fold(f64::NEG_INFINITY, f64::max),
    }
}
```

Worked problem

Fourteen past releases, each with an offline delta and an online delta. How well does offline predict online?

python

```python
import numpy as np

def leave_one_out(off, on):
    """Fit on every release EXCEPT one, predict that one."""
    errs, sign_hits = [], 0
    for i in range(len(off)):
        xs, ys = np.delete(off, i), np.delete(on, i)
        a = ((xs - xs.mean()) * (ys - ys.mean())).sum() / ((xs - xs.mean()) ** 2).sum()
        b = ys.mean() - a * xs.mean()
        pred = a * off[i] + b
        errs.append(abs(pred - on[i]))
        if (pred > 0) == (on[i] > 0):          # did we get the DIRECTION right?
            sign_hits += 1
    return {"mae": np.mean(errs), "worst": max(errs),
            "direction_accuracy": sign_hits / len(off)}
```

rust

```rust
struct LooResult {
    mae: f64,
    worst: f64,
    direction_accuracy: f64,
}

fn mean(v: &[f64]) -> f64 {
    v.iter().sum::<f64>() / v.len() as f64
}

/// Fit on every release EXCEPT one, predict that one.
fn leave_one_out(off: &[f64], on: &[f64]) -> LooResult {
    let (mut errs, mut sign_hits) = (Vec::new(), 0);
    for i in 0..off.len() {
        // np.delete(arr, i): a copy without element i
        let xs: Vec<f64> = off.iter().enumerate().filter(|&(k, _)| k != i).map(|(_, &v)|
v).collect();
        let ys: Vec<f64> = on.iter().enumerate().filter(|&(k, _)| k != i).map(|(_, &v)| v).collect();
        let (mx, my) = (mean(&xs), mean(&ys));
        let a = xs.iter().zip(&ys).map(|(x, y)| (x - mx) * (y - my)).sum::<f64>()
            / xs.iter().map(|x| (x - mx).powi(2)).sum::<f64>();
        let b = my - a * mx;
        let pred = a * off[i] + b;
        errs.push((pred - on[i]).abs());
        if (pred > 0.0) == (on[i] > 0.0) { // did we get the DIRECTION right?
            sign_hits += 1;
        }
    }
    LooResult {
        mae: mean(&errs),
        worst: errs.iter().copied().fold(f64::NEG_INFINITY, f64::max),
        direction_accuracy: sign_hits as f64 / off.len() as f64,
```

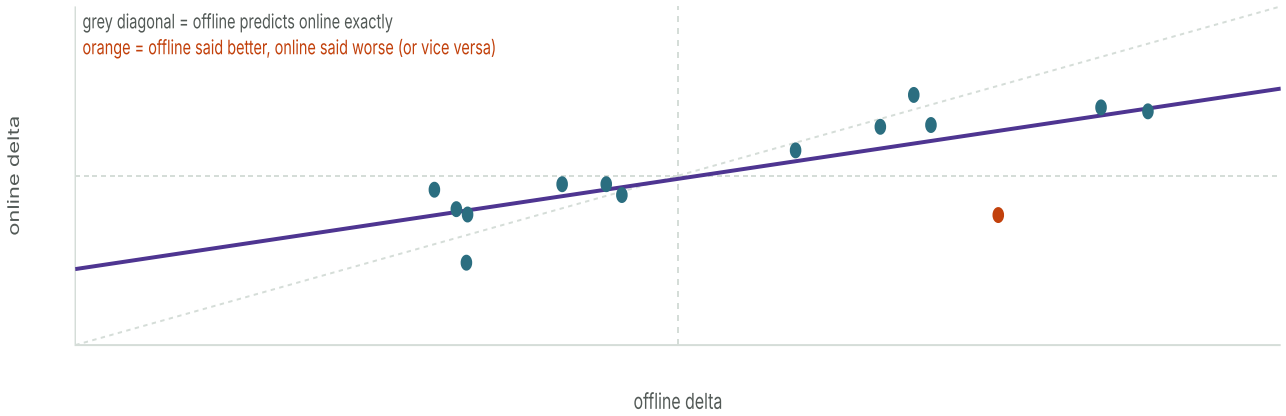
```
    }
}
```

Result	Value	What it means
Fitted slope	`0.616`	Offline overstates: a 10 point offline move is about 6 points online
Shrinkage	`0.384`	That 38% is the gap, expressed as one number
Leave-one-out MAE	`0.0122`	Typical prediction miss, 1.2 points
Worst single miss	`0.0352`	The one a reviewer will ask about
Direction accuracy	**`0.857`**	**Two of fourteen releases pointed the wrong way**

The figures are illustrative. Direction accuracy is the number that matters for a ship decision. MAE can look small while the suite still points the wrong way on the releases you cared about.

Interactive **Offline vs online, fitted and backtested** (shown at its default setting)

Each dot is a past release. The fitted line is your calibration. Orange dots are releases where offline and online disagreed on the direction. Drop the release count to 6 and watch the leave one out error jump, which is what overfitting a handful of releases looks like.



Fitted slope 0.533 **shrinkage** 0.467 **leave-one-out MAE** 0.0148
Direction accuracy 0.929 **worst single miss** 0.0585

Direction accuracy is the number that matters for a ship decision: MAE can look small while the suite still points the wrong way on the releases you cared about. Drop the release count to 6 and watch the leave-one-out error jump, which is what overfitting a handful of releases actually looks like.

⚠ **Watch out**

Three attacks on this method, and you should have answers ready for all three. **One:** the calibration is fitted on a handful of releases, so it is overfit to small n. **Two:** the online distribution moves when the model changes user behaviour, so yesterday's calibration may not hold (the population you are predicting is itself shifting). **Three:** MAE flatters the method; direction accuracy is what decides ship or no ship. Where the honest answer is "that is a real limitation", say so.

What is in the remaining gap

Plain words **Four reasons the test and the commute differ**

The test route does not cover every kind of road. The test car is not quite the car people drive. The examiner sometimes judges differently from what actually gets you home safely. And some days traffic is just different. The first three can be fixed with effort. The last one cannot.

For leaders

Decompose the gap before you fund work on it

When someone quotes "a 5% gap and closing", ask for the decomposition rather than the number:

Task distribution

Production work the task set does not yet represent: new languages, longer sessions, bigger inputs, new user intents. Fix: sample tasks from real usage, refresh regularly.

Harness realism

Missing context, absent tools, unrealistic starting state, no interruptions or retries. Fix: replay real environments, not idealised ones.

Judge fidelity

Where the automated judge and a real user disagree about whether the output was useful. Fix: validate the judge against user outcomes (see LLM as judge).

Irreducible

User behaviour variance, non-stationary traffic mix, acceptance that depends on the person as much as the output. Estimate it; do not promise to remove it.

Ending with "and this much is irreducible, here is how I estimated it" is a stronger position than claiming the gap can go to zero. It also points spending at the first two buckets, which is where dynamic, usage-derived benchmarks pay off.

Under the hood

Estimating the irreducible floor

One practical estimate: run the *same* model in two concurrent online arms (an A/A test). The spread of the measured online "delta" across repeated A/A runs is the noise floor no offline suite can predict

through. If the backtest residual RMSE is close to the A/A spread, the offline suite is already about as predictive as it can be, and further offline investment should target coverage, not calibration.

$$\underbrace{\mathrm{Var}(\text{residual})}_{\text{backtest}} \approx \underbrace{\mathrm{Var}(\text{offline noise})\,\hat{a}^2}_{\text{bigger eval set}} + \underbrace{\mathrm{Var}(\text{online noise})}_{\text{A/A floor}} + \underbrace{\mathrm{Var}(\text{mismatch})}_{\text{coverage, realism, judge}}$$

A rough variance budget, assuming the three sources are independent. Each term points at a different fix.

Eval lens

Treat the backtest as a standing metric of the eval program itself. Recompute it after every launch. A falling direction accuracy is the earliest warning that the offline suite has drifted away from what users do.

Part II · Chapter 14

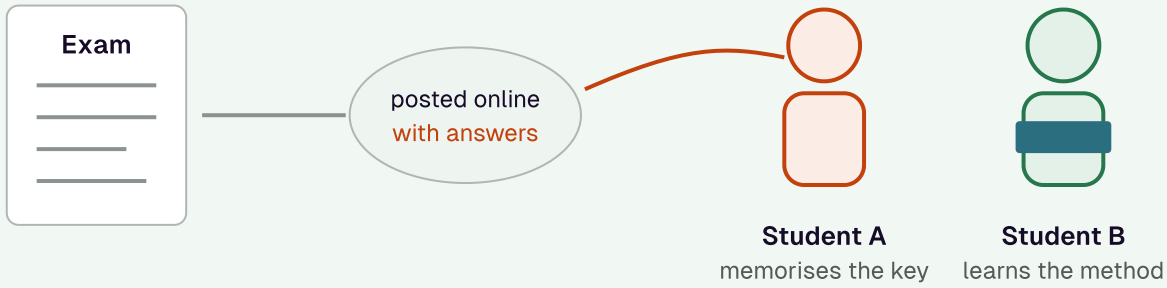
Contamination and benchmark integrity: is the model reasoning or remembering?

A benchmark score means something only if the model has not seen the answers. Frontier models are trained on a large fraction of the public internet, and public benchmarks live on the public internet. This chapter covers how test items leak into training data, how to detect it, how to design benchmarks that resist it, and how to report results so a reader can judge for themselves.

Diagram

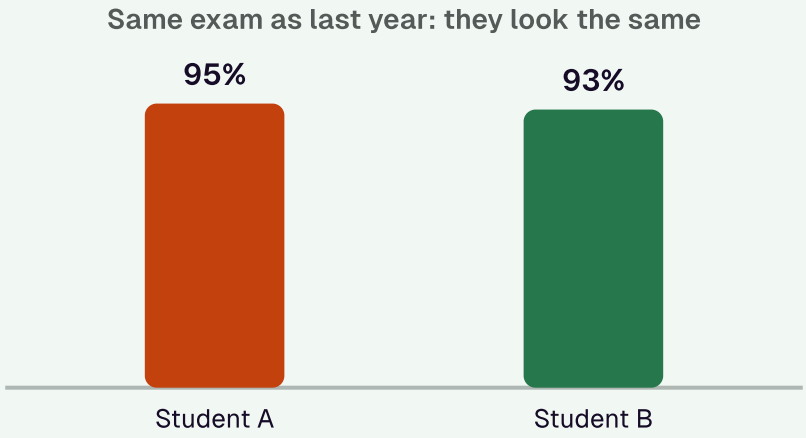
Two students and a leaked answer key

How changing the numbers on an exam separates memorising from understanding.

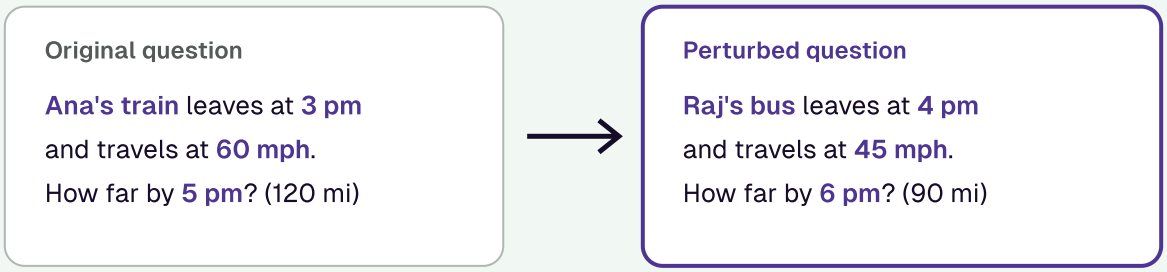


Benchmarks live on the internet, and so does model training data.

Last year's final exam was posted online with its answers. Student A memorises the answer key. Student B learns how to solve the problems. A model trained on internet text can end up like Student A for popular benchmarks, usually by accident.

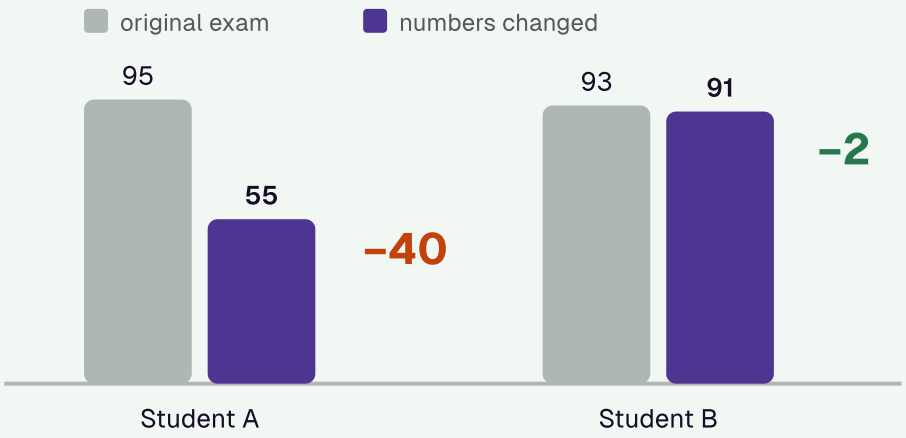


On the leaked exam, both score in the nineties. From the score alone you cannot tell who understands the material. A benchmark number on a public test set has exactly this problem.

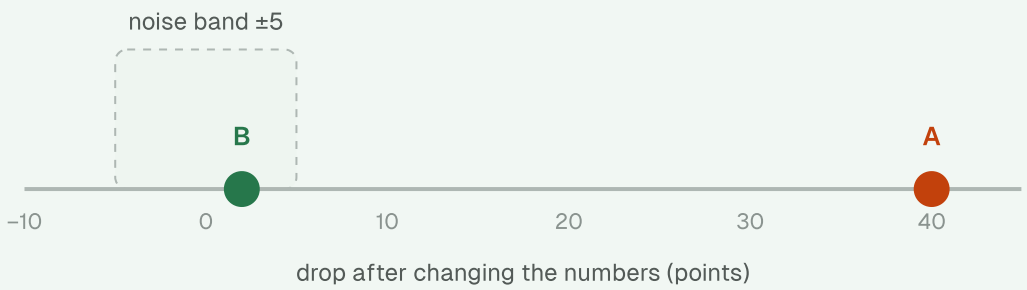


Same skill, new surface details. A memorised answer no longer fits.

Now change the surface of every question: new names, new numbers, options shuffled. The skill needed is identical, so an honest score should barely move. This is a **perturbation test**.



Student A falls from 95 to 55. Student B goes from 93 to 91. The size of the drop is the fingerprint: a big gap between original and perturbed versions says the original score was partly memory.



On 200 questions, a 2-point drop is noise. A 40-point drop is not.

Small drops happen by chance, because the perturbed questions are a fresh sample. On 200 questions the noise on a gap is roughly ±5 points, so B's 2-point drop means nothing and A's 40 is overwhelming. Always put an interval on the gap before calling something contaminated.

If you own the data, search it. If you do not, test the student.

Perturbation is one detective. If you own the training data, search it for the test items. Benchmark authors can plant a unique canary string so leaks are findable. And questions written after the model's data cutoff cannot have been memorised.

Plain words

The exam that was posted online

A teacher gives the same final exam every year. One year a student posts the questions and answers online. Next year's class scores much higher. Did they get smarter, or did they read the answers?

The test for this is simple: change the exam a little. Rename the people in the word problems, swap the numbers, reorder the options. A student who understands the material does about as well as before. A student who memorised the answer key falls apart.

Language models are trained on huge piles of internet text, so some of them have "read the answer key" for popular benchmarks, often by accident. This is called **contamination**.

For leaders

Why contamination is an organisational risk, not a footnote

Contamination inflates exactly the numbers that go into launch announcements, investor decks, and model comparison tables. It also corrupts internal decisions: a training change that "improves" a contaminated benchmark may just be memorising better. The damage is reputational when outsiders find it, and strategic when it misdirects research.

Three postures, in increasing cost and increasing credibility:

Posture	What it involves	What it buys
Detect	Overlap scans of training data against benchmarks; perturbation tests on suspicious results	An estimate of how inflated a public number is
Prevent	Decontaminate training corpora; canary strings; access-controlled private sets	Cleaner public numbers and trustworthy internal ones
Outrun	Time-split and dynamically generated benchmarks refreshed faster than training corpora	Numbers that cannot be memorised because they did not exist at training time

What to fund. At minimum, a decontamination pass in the data pipeline and a private held-out set for every capability you gate launches on. For high-stakes capabilities, a continuously refreshed benchmark built from material created after the training cutoff.

Principal asks

- Was this benchmark decontaminated against the training corpus, at what n-gram length, and how many items were removed?
- What does the model score on a perturbed or post-cutoff version of the same task?
- Is the headline number on the public set or on a private held-out set? Where do they disagree?
- How long until this benchmark is saturated, and what replaces it?

How items leak

Direct inclusion

The benchmark file, or a web page reproducing it, is scraped into pretraining data.

Paraphrase and discussion

Blog posts, forum threads, and solution write-ups restate the items in different words. Exact-match filters miss these.

Fine-tuning data

Synthetic or human-written SFT data generated by people or models that had seen the benchmark.

Selection pressure

Even with no leaked text, choosing checkpoints, prompts, or hyperparameters by benchmark score fits the model to the test set. This is overfitting through decisions, not data.

Detection

Plain words

Search the library, then test the student

There are two ways to check. If you own the training data, search it for the exam questions. If you do not, test the student: reword the questions and see whether the score holds up.

Under the hood

Overlap, canaries, membership signals, perturbation

N-gram overlap

The standard data-side check. Tokenise each benchmark item and each training document, and flag items that share long n-grams with the corpus. Thirteen-gram overlap was used for decontamination in the GPT-3 paper (Brown et al., 2020); other labs use 8 to 50 tokens or a fraction-of-item threshold. Short n-grams flag common phrases; long ones miss paraphrases.

python

```python
import hashlib, re

def ngrams(text: str, n: int = 13) -> set[str]:
    toks = re.findall(r"\w+", text.lower())
    return {" ".join(toks[i:i + n]) for i in range(len(toks) - n + 1)}


def contamination_overlap(task_text: str, model_output: str, n: int = 13) -> float:
    """Fraction of the task's n-grams reproduced verbatim in the output.
    High overlap on a task the model should have had to reason about is a
    memorization signal, not proof, but it tells you where to look."""
    t, o = ngrams(task_text, n), ngrams(model_output, n)
    return len(t & o) / len(t) if t else 0.0
```

rust

```rust
use std::collections::HashSet;

fn ngrams(text: &str, n: usize) -> HashSet<String> {
    // re.findall(r"\w+", text.lower()): runs of letters, digits and underscores
    let lower = text.to_lowercase();
    let toks: Vec<&str> = lower
        .split(|c: char| !(c.is_alphanumeric() || c == '_'))
        .filter(|t| !t.is_empty())
        .collect();
    // windows(n) yields nothing when there are fewer than n tokens
    toks.windows(n).map(|w| w.join(" ")).collect()
}

/// Fraction of the task's n-grams reproduced verbatim in the output.
/// High overlap on a task the model should have had to reason about is a
/// memorization signal, not proof, but it tells you where to look.
```

```rust
/// The Python default is n = 13.
fn contamination_overlap(task_text: &str, model_output: &str, n: usize) -> f64 {
    let (t, o) = (ngrams(task_text, n), ngrams(model_output, n));
    if t.is_empty() {
        return 0.0;
    }
    t.intersection(&o).count() as f64 / t.len() as f64
}
```

At corpus scale the same idea runs over hashed n-grams in a Bloom filter or suffix array, so trillions of training tokens can be checked against every benchmark in one pass.

Embedding similarity

To catch paraphrases, embed benchmark items and training chunks and flag nearest neighbours above a similarity threshold, then have a person or a strong model confirm. This finds rewritten solutions and translated copies that n-grams miss, at the cost of false positives on genuinely similar but independent text.

Canary strings

Embed a unique random token (a GUID) in every file of a private or public eval set, with a request that it never be trained on. Two uses: filter it out of your own corpora, and probe released models. If a model completes the canary when given its prefix, the set is in that model's training data. BIG-bench popularised this practice.

Membership-inference signals (outsider tests)

Without training data access, you can still look for signs a specific text was trained on:

- **Verbatim completion.** Give the first half of an item; check whether the model reproduces the second half, including its exact numbers or answer options.
- **Likelihood gap.** Seen text tends to get unusually low perplexity compared with paraphrases of the same content. The Min-K% Prob method (Shi et al., 2023) averages the lowest-probability tokens, which separates seen from unseen text better than average perplexity.
- **Ordering artefacts.** If a model assigns higher likelihood to a benchmark's examples in their published order than in shuffled orders, it has probably seen the file. Since the examples are exchangeable, a clean model has no reason to prefer one order (Oren et al., 2023).

These are evidence, not proof. Famous text is low-perplexity whether or not it was in a benchmark.

Perturbation tests

The strongest practical test available to an outsider. Rename variables, change constants, reorder independent clauses, shuffle options, translate and back-translate. A model that reasons keeps roughly the same score. A model that memorised falls off a cliff. GSM-Symbolic (Mirzadeh et al., 2024) applied this to grade-school maths and found measurable drops for many models when only names and numbers changed.

python

```python
def perturbation_check(run_fn, task, perturb_fn, trials: int = 5) -> dict:
    """Rename variables, change constants, reorder independent clauses.
    A model that reasons stays at roughly the same score.
    A model that memorized falls off a cliff."""
    base = run_fn(task)
    perturbed = [run_fn(perturb_fn(task, seed=i)) for i in range(trials)]
    mean_p = sum(perturbed) / trials
    return {"base": base, "perturbed_mean": mean_p, "drop": base - mean_p}
```

rust

```rust
struct PerturbationResult {
    base: f64,
    perturbed_mean: f64,
    drop: f64,
}

/// Rename variables, change constants, reorder independent clauses.
/// A model that reasons stays at roughly the same score.
/// A model that memorized falls off a cliff.
/// The Python default is trials = 5.
fn perturbation_check(
    mut run_fn: impl FnMut(&str) -> f64,
    task: &str,
    perturb_fn: impl Fn(&str, u64) -> String, // (task, seed) -> perturbed task
    trials: u64,
) -> PerturbationResult {
    let base = run_fn(task);
    let perturbed: Vec<f64> = (0..trials).map(|i| run_fn(&perturb_fn(task, i))).collect();
    let mean_p = perturbed.iter().sum::<f64>() / trials as f64;
    PerturbationResult { base, perturbed_mean: mean_p, drop: base - mean_p }
}
```

The drop alone is not the contamination signal, because perturbations can also make items genuinely harder. The clean comparison is a **difference in differences**: the suspect model's drop minus the drop of a control model that could not have seen the items (or a model trained before the benchmark existed).

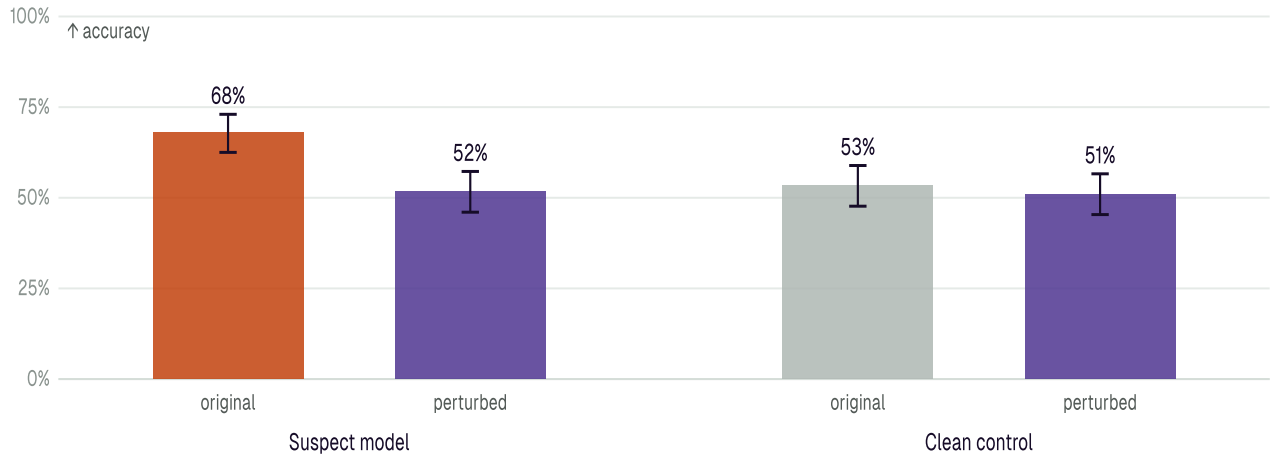
$$\widehat{\Delta}_{\text{contam}} = \left(\bar{a}^{S}_{\text{orig}} - \bar{a}^{S}_{\text{pert}}\right) - \left(\bar{a}^{C}_{\text{orig}} - \bar{a}^{C}_{\text{pert}}\right), \qquad \text{SE} = \frac{s_d}{\sqrt{n}}$$

S is the suspect model, C the clean control, and $\bar{a}$ is accuracy on original or perturbed items. Compute the per-item difference d_i on the same n items so the comparison is paired; s_d is its standard deviation. This is the same pairing logic as in eval statistics.

Interactive

Contamination shows up as a gap (shown at its default setting)

A suspect model and a clean control of equal ability answer the same items, once in original wording and once perturbed. Raise the share of items the suspect saw in training and watch its original score climb while the perturbed score stays put. Then shrink the item count and see when the signal disappears into noise.



Suspect model drop on rewording: **16.3 pts** (±4.9).
Clean control drop (perturbation difficulty alone): **2.3 pts** (±4.0).
Difference in differences: **14.0 pts**, 95% CI [7.7, 20.3].
Contamination signal. The suspect model loses more on rewording than the clean control does, and the interval excludes zero.
Estimated memorisation share of the headline score: 14.0 of 68.0 points.

Key idea

Set seen share to 0 and extra difficulty to 8 points. Both models drop by about the same amount, so the difference in differences is near zero. A raw drop on perturbed items is not evidence of contamination by itself. The control model is what turns a drop into a measurement.

Designing benchmarks that resist contamination

Plain words

Write new exams faster than they can leak

The surest fix is an exam made after the student finished studying. If the questions did not exist when the model was trained, it cannot have memorised them. The other fix is to keep some exams locked in a drawer and never post them.

For leaders

The design options and their costs

Design	How it works	Cost and catch
Private held-out set	Never published; run only by the owner or under access control	Cheap. Results cannot be independently reproduced, and every evaluation run leaks a little through selection pressure.
Time-split benchmark	Only items created after each model's training cutoff count (LiveCodeBench, Jain et al., 2024, does this for coding problems)	Needs a steady supply of fresh items. Scores across models with different cutoffs are on different item sets.
Dynamic generation	Templates or programs generate new instances on demand, with known answers	Expensive to build well. Risk of generating a narrow, artificial distribution that models can learn to game.
Usage-derived tasks	Tasks sampled from real product usage after the cutoff	Most realistic; also helps close the offline to online gap. Needs privacy review and labelling.
Canaried public set	Public, with a canary GUID and a request not to train	Relies on others filtering. Good for detection, weak for prevention.

Saturation

Every public benchmark degrades as a signal over time, through contamination, through selection pressure, and through simple saturation as top models approach the ceiling. Near the ceiling, remaining errors are often label noise, so further gains mean nothing. Plan the replacement before the current benchmark hits about 90% for frontier models. Assume a public benchmark's useful life is measured in model generations, not years.

Under the hood

Why saturation destroys resolving power

Near the ceiling, two things happen at once. The headroom to show improvement shrinks, and a growing share of the remaining "errors" are items with wrong or ambiguous gold labels. If a fraction e of items have bad labels, the maximum measurable accuracy is about $1 - e$, and differences between strong models are compressed into the last few points where label noise dominates. This is why audits of popular benchmarks that find a few percent of mislabelled items matter so much for frontier comparisons.

The number of tasks needed to separate two models also grows as they converge, because fewer tasks are discordant (see sample size). A saturated benchmark is both noisier and less informative per task.

Reporting norms

For leaders

What a credible benchmark claim includes

1. **Decontamination method**: overlap threshold, n-gram length, how many items were flagged and removed, and the score on the clean subset next to the full set.
2. **A contamination-resistant companion number**: private, perturbed, or post-cutoff results alongside the public one.
3. **Interval and n** on every score, as in every other chapter of Part II.
4. **Evaluation settings**: prompt, few-shot examples, sampling parameters, number of attempts. Small prompt changes move benchmark scores by several points.
5. **Selection disclosure**: whether the checkpoint or prompt was chosen using this benchmark.

Eval lens

When a public benchmark and a private or post-cutoff equivalent disagree, believe the one the model could not have seen. A consistent gap between the two, tracked across releases, is itself a useful metric of how much of a model's public performance is memory.

Part II · Chapter 15

Building the eval harness: the machine that produces every number

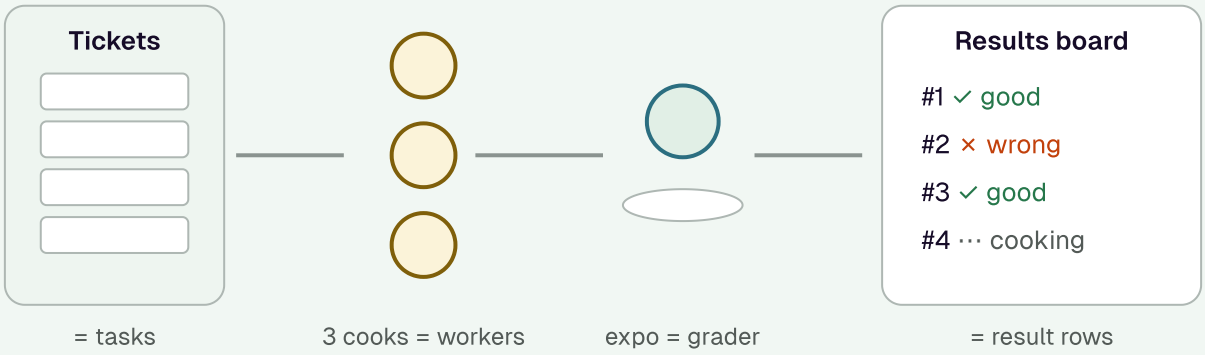
Every statistic in Part II is computed from results that a harness produced. If the harness drops tasks, retries silently, caches stale outputs, or reports a bare pass rate, the statistics on top are wrong no matter how

careful they are. This chapter builds a harness in four passes, each with an acceptance test, and shows where measurement errors hide in the engineering.

Diagram

An eval harness is a restaurant kitchen

Tickets, cooks, dropped plates and the redo trap: where harness bugs quietly change the score.

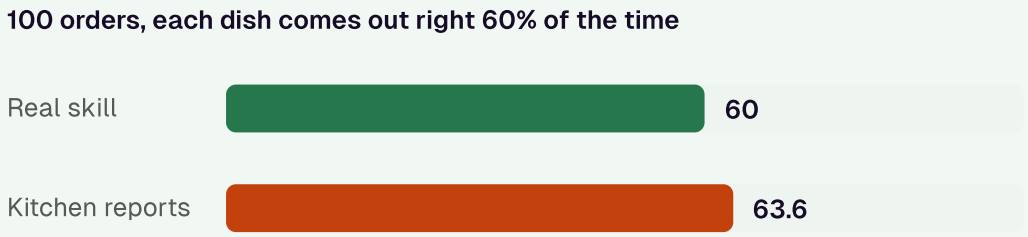


Every ticket (task) goes to a free cook (a worker), the expo checks the plate (the grader), and the result goes on the board. More cooks means faster service, the same way more concurrency finishes an eval sooner.

Hide the dropped plates	Count every ticket
6 / 8 = 75%	**6 / 10 = 60%**

A crashed task is data. It stays on the board.

Two plates hit the floor. If the board quietly skips them, the kitchen looks like it got 75% right. Dropped plates are usually the hard dishes, so skipping them always flatters the score. Count every ticket and the honest number is 60%.



85 orders, one try each: 85 × 0.6 = **51**
15 orders "redone" after a hiccup get two tries: 15 × (1 – 0.4²) = **12.6**

No cook got better. The redo policy added 3.6 points.

Fifteen orders hit a hiccup and get remade. A remade dish gets a second chance to come out right, so those tickets are scored more like "best of two". The kitchen reports 63.6% when its real rate is 60%. This is retry inflation, and you can only detect it if every result records how many attempts it took.

Yesterday's ticket
dish: steak
doneness: **medium rare**
salt: none

Today's ticket
dish: steak
doneness: **well done**
salt: none

Key = dish only → serve the old plate ×

Key = every detail → cook fresh ✓

Caching saves money: if an identical order comes in, reuse the plate. But "identical" must include every detail. In a harness the cache key covers the task, the prompt, the model, and every sampling setting. Leave one out and a changed setting silently serves yesterday's answers.

End-of-night report

Orders	1,000
Good	63.6% (60.6–66.5)
Dropped	0
Redone	150
Good, redone removed	60.0% (56.7–63.2)

The two "good" lines disagree: that gap is the redo effect.

The fix is a report that cannot be printed without the count, an interval, the dropped plates, and the redo count, plus the score with redone orders removed. When those two scores disagree, the retry policy is part of what you measured.

Plain words

A fair exam room

Think of the harness as the exam room. It hands each question to the model, collects the answer, marks it, and writes down the result. A good exam room treats every student the same: same time limit, same number of attempts, same marking rules. If one student's pen breaks, you note it and carry on; you do not cancel the exam.

Most harness bugs are unfairness bugs. One question gets two attempts while another gets one. A crashed question disappears from the results instead of counting. Yesterday's answers get reused after the questions changed. Each of these quietly changes the score.

For leaders

Treat the harness as measurement infrastructure

Teams under-invest in harnesses because they look like plumbing. They are the measuring instrument. A harness bug that inflates scores by two points is as damaging as a training bug, and much harder to notice, because nobody complains when numbers go up.

What a mature harness guarantees:

No silent loss

Every task produces a result row, including errors.

Equal treatment

Attempts, timeouts, and sampling parameters are identical across tasks and models, and recorded.

Reproducibility

A result can be traced to model id, prompt hash, parameters, harness version, and judge version.

Honest reporting

The summary cannot be produced without n, a confidence interval, the error count, and the retry count.

Flakiness control

Tasks whose scores vary run to run at fixed settings are quarantined, not averaged in.

What to fund: a small, dedicated owner for the harness, versioned like a product, with its own regression tests. The return is every downstream decision.

Principal asks

- How many tasks errored in this run, and are they excluded from the denominator?
- How many tasks were retried, and what is the score with retried tasks removed?
- Did anything other than the model change between these two runs: harness, judge, prompt template, timeouts, task set?
- Which tasks are quarantined as flaky, and how was flakiness measured?

Version one: make it work

Plain words

One question at a time, and never stop the exam

The first version is simple: loop over the tasks, ask the model, mark the answer. The one rule that matters is that a failed task is written down as a failure and the loop keeps going.

Under the hood

A sequential runner that records errors as data

python

```python
from dataclasses import dataclass, field
from typing import Callable, Any
import time, json

@dataclass
class Task:
    task_id: str
    prompt: str
    reference: dict[str, Any] = field(default_factory=dict)

@dataclass
class Result:
    task_id: str
    output: str | None = None
    score: float | None = None
    error: str | None = None
    attempts: int = 0
    latency_s: float = 0.0
    meta: dict[str, Any] = field(default_factory=dict)


def run_sync(tasks: list[Task], model_fn: Callable[[str], str],
             score_fn: Callable[[Task, str], float]) -> list[Result]:
    results = []
    for t in tasks:
        started = time.perf_counter()
        try:
            out = model_fn(t.prompt)
            results.append(Result(t.task_id, out, score_fn(t, out),
                                  attempts=1,
                                  latency_s=time.perf_counter() - started))
        except Exception as e:                # a failed task is DATA, not a crash
            results.append(Result(t.task_id, None, None, error=repr(e),
                                  attempts=1,
                                  latency_s=time.perf_counter() - started))
    return results
```

rust

```rust
use std::collections::HashMap;
use std::time::Instant;

#[derive(Debug, Clone, Default)]
pub struct Task {
    pub task_id: String,
    pub prompt: String,
    pub reference: HashMap<String, String>, // Python's dict[str, Any]; strings keep the sketch simple
}

// `Result` in the Python; renamed so it does not shadow std's Result type.
#[derive(Debug, Clone, Default)]
pub struct EvalResult {
    pub task_id: String,
    pub output: Option<String>,
```

```rust
    pub score: Option<f64>,
    pub error: Option<String>,
    pub attempts: u32,
    pub latency_s: f64,
    pub meta: HashMap<String, String>,
}

pub fn run_sync(
    tasks: &[Task],
    model_fn: impl Fn(&str) -> Result<String, String>,
    score_fn: impl Fn(&Task, &str) -> f64,
) -> Vec<EvalResult> {
    let mut results = Vec::new();
    for t in tasks {
        let started = Instant::now();
        match model_fn(&t.prompt) {
            Ok(out) => results.push(EvalResult {
                task_id: t.task_id.clone(),
                score: Some(score_fn(t, &out)),
                output: Some(out),
                attempts: 1,
                latency_s: started.elapsed().as_secs_f64(),
                ..Default::default()
            }),
            Err(e) => results.push(EvalResult {
                // a failed task is DATA, not a crash
                task_id: t.task_id.clone(),
                error: Some(e),
                attempts: 1,
                latency_s: started.elapsed().as_secs_f64(),
                ..Default::default()
            }),
        }
    }
    results
}
```

Worked problem

Acceptance test. Feed the runner ten tasks where the third raises an exception. The run must complete, return 10 results, and record the error on task three.

The most common first-attempt failure is letting the exception escape the loop, which kills the run and loses the other nine results. Catch per task, record the error, keep going.

Version two: concurrency, retries, timeouts

Plain words

Many questions at once, and a fair retry rule

Real suites have thousands of tasks, so the harness asks many at once. Model servers sometimes time out or say "too busy", so the harness waits and tries again. The danger is that retrying can give some tasks more chances than others. A fair harness records how many tries each task took, so you can check.

Under the hood

asyncio, a semaphore, and exponential backoff with full jitter

The backoff before attempt k is drawn uniformly from $[0, \min(\text{cap}, \text{base} \cdot 2^k)]$. The exponential envelope spaces retries out; the jitter stops a burst of failures from retrying in lockstep and knocking the endpoint over again.

python

```python
import asyncio, random, time

class RetryPolicy:
    def __init__(self, max_attempts: int = 4, base: float = 0.5, cap: float = 20.0):
        self.max_attempts, self.base, self.cap = max_attempts, base, cap

    def backoff(self, attempt: int) -> float:
        """Exponential with FULL JITTER."""
        # base * 2**attempt is the exponential envelope; min() caps it so a late
        # attempt does not sleep for minutes. uniform(0, ...) is the jitter:
        # without it, every request from a failure burst retries at the same
        # instant and re-overloads the endpoint you just knocked over.
        return random.uniform(0, min(self.cap, self.base * 2 ** attempt))

    def retryable(self, exc: Exception) -> bool:
        """Only retry TRANSIENT failures."""
        # Retrying a genuine 400 or an assertion error just burns quota and
        # delays the real answer. Match on the message rather than the type
        # because SDKs wrap errors differently.
        msg = repr(exc).lower()
        return any(s in msg for s in
                   ("timeout", "rate", "429", "500", "502", "503", "connection"))


async def run_one(task: Task, model_fn, score_fn, policy: RetryPolicy,
                  timeout_s: float, sem: asyncio.Semaphore) -> Result:
    started = time.perf_counter()      # wall clock for THIS task, retries included
    last_exc: Exception | None = None

    for attempt in range(policy.max_attempts):
        try:
            # The semaphore caps how many requests are in flight across the
            # whole suite. Acquire it INSIDE the loop so a sleeping retry
            # releases its slot instead of blocking other tasks.
```

```
            async with sem:
                out = await asyncio.wait_for(model_fn(task.prompt),
                                             timeout=timeout_s)
            return Result(task.task_id, out, score_fn(task, out),
                          attempts=attempt + 1,          # RECORD attempts
                          latency_s=time.perf_counter() - started)

        except Exception as e:
            last_exc = e
            # Two ways to stop: the error is not worth retrying, or we are out
            # of attempts. Checking both here keeps the exit conditions in one place.
            if not policy.retryable(e) or attempt == policy.max_attempts - 1:
                break
            await asyncio.sleep(policy.backoff(attempt))

    # A task that never succeeded is still DATA. Return a Result carrying the
    # error rather than raising, so one bad task cannot kill a 2000 task run.
    return Result(task.task_id, None, None, error=repr(last_exc),
                  attempts=attempt + 1,
                  latency_s=time.perf_counter() - started)


async def run_suite(tasks: list[Task], model_fn, score_fn,
                    concurrency: int = 16, timeout_s: float = 120.0) -> list[Result]:
    sem = asyncio.Semaphore(concurrency)   # one shared limiter for every task
    policy = RetryPolicy()
    coros = [run_one(t, model_fn, score_fn, policy, timeout_s, sem) for t in tasks]
    # gather runs them all concurrently, bounded by the semaphore, and returns
    # results in the ORIGINAL task order regardless of completion order.
    return await asyncio.gather(*coros)
```

rust

```
// asyncio becomes OS threads in this std-only sketch: a counting gate stands in
// for asyncio.Semaphore and a channel with recv_timeout for asyncio.wait_for.
// Production Rust would usually use tokio (tokio::sync::Semaphore,
// tokio::time::timeout, join_all) and get cheap tasks and real cancellation.
use std::sync::atomic::{AtomicUsize, Ordering};
use std::sync::mpsc::{self, RecvTimeoutError};
use std::sync::{Arc, Condvar, Mutex};
use std::thread;
use std::time::{Duration, Instant};

#[derive(Debug, Clone, Default)]
pub struct Task {
    pub task_id: String,
    pub prompt: String,
}

#[derive(Debug, Clone, Default)]
pub struct EvalResult {
    // `Result` in the Python, trimmed to the fields used here
    pub task_id: String,
    pub output: Option<String>,
    pub score: Option<f64>,
    pub error: Option<String>,
    pub attempts: u32,
    pub latency_s: f64,
}
```

```rust
pub type ModelFn = Arc<dyn Fn(&str) -> Result<String, String> + Send + Sync>;
pub type ScoreFn = dyn Fn(&Task, &str) -> f64 + Sync;

/// Small inline PRNG (SplitMix64) so the sketch needs no crates.
pub struct SplitMix64(u64);

impl SplitMix64 {
    pub fn new(seed: u64) -> Self {
        SplitMix64(seed)
    }
    pub fn next_u64(&mut self) -> u64 {
        self.0 = self.0.wrapping_add(0x9E37_79B9_7F4A_7C15);
        let mut z = self.0;
        z = (z ^ (z >> 30)).wrapping_mul(0xBF58_476D_1CE4_E5B9);
        z = (z ^ (z >> 27)).wrapping_mul(0x94D0_49BB_1331_11EB);
        z ^ (z >> 31)
    }
    pub fn uniform(&mut self, lo: f64, hi: f64) -> f64 {
        let u = (self.next_u64() >> 11) as f64 / (1u64 << 53) as f64; // in [0, 1)
        lo + (hi - lo) * u
    }
}

/// The asyncio.Semaphore stand-in: a counting gate built from Mutex + Condvar.
pub struct Semaphore {
    permits: Mutex<usize>,
    freed: Condvar,
}

pub struct Permit<'a>(&'a Semaphore);

impl Semaphore {
    pub fn new(n: usize) -> Self {
        Semaphore { permits: Mutex::new(n), freed: Condvar::new() }
    }
    pub fn acquire(&self) -> Permit<'_> {
        let mut free = self.permits.lock().unwrap();
        while *free == 0 {
            free = self.freed.wait(free).unwrap();
        }
        *free -= 1;
        Permit(self)
    }
}

impl Drop for Permit<'_> {
    // Dropping the permit gives the slot back, like leaving `async with sem`.
    fn drop(&mut self) {
        *self.0.permits.lock().unwrap() += 1;
        self.0.freed.notify_one();
    }
}

pub struct RetryPolicy {
    pub max_attempts: u32,
    pub base: f64,
    pub cap: f64,
}

impl Default for RetryPolicy {
```

```rust
    fn default() -> Self {
        RetryPolicy { max_attempts: 4, base: 0.5, cap: 20.0 }
    }
}

impl RetryPolicy {
    /// Exponential with FULL JITTER.
    pub fn backoff(&self, attempt: u32, rng: &mut SplitMix64) -> f64 {
        // base * 2^attempt is the exponential envelope; min() caps it so a late
        // attempt does not sleep for minutes. uniform(0, ...) is the jitter:
        // without it, every request from a failure burst retries at the same
        // instant and re-overloads the endpoint you just knocked over.
        rng.uniform(0.0, self.cap.min(self.base * 2f64.powi(attempt as i32)))
    }

    /// Only retry TRANSIENT failures.
    pub fn retryable(&self, err: &str) -> bool {
        // Retrying a genuine 400 or an assertion error just burns quota and
        // delays the real answer. Match on the message rather than the type
        // because SDKs wrap errors differently.
        let msg = err.to_lowercase();
        ["timeout", "rate", "429", "500", "502", "503", "connection"]
            .iter()
            .any(|s| msg.contains(s))
    }
}

/// asyncio.wait_for stand-in. A std thread cannot be cancelled, so on timeout
/// the call is abandoned and left to finish in the background; tokio's
/// timeout would drop the future instead.
fn call_with_timeout(model_fn: &ModelFn, prompt: &str, timeout_s: f64) -> Result<String, String> {
    let (tx, rx) = mpsc::channel();
    let (f, prompt) = (Arc::clone(model_fn), prompt.to_string());
    thread::spawn(move || {
        let _ = tx.send(f(&prompt));
    });
    match rx.recv_timeout(Duration::from_secs_f64(timeout_s)) {
        Ok(outcome) => outcome,
        Err(RecvTimeoutError::Timeout) => Err(format!("timeout after {timeout_s}s")),
        Err(RecvTimeoutError::Disconnected) => Err("model call panicked".to_string()),
    }
}

pub fn run_one(
    task: &Task,
    model_fn: &ModelFn,
    score_fn: &ScoreFn,
    policy: &RetryPolicy,
    timeout_s: f64,
    sem: &Semaphore,
    rng: &mut SplitMix64,
) -> EvalResult {
    let started = Instant::now(); // wall clock for THIS task, retries included
    let mut last_err = String::new();
    let mut attempts = 0;

    for attempt in 0..policy.max_attempts {
        attempts = attempt + 1;
        let outcome = {
            // The semaphore caps how many requests are in flight across the
```

```
            // whole suite. Acquire it INSIDE the loop so a sleeping retry
            // releases its slot instead of blocking other tasks.
            let _slot = sem.acquire();
            call_with_timeout(model_fn, &task.prompt, timeout_s)
        }; // _slot dropped here, before any backoff sleep

        match outcome {
            Ok(out) => {
                return EvalResult {
                    task_id: task.task_id.clone(),
                    score: Some(score_fn(task, &out)),
                    output: Some(out),
                    attempts, // RECORD attempts
                    latency_s: started.elapsed().as_secs_f64(),
                    ..Default::default()
                }
            }
            Err(e) => {
                // Two ways to stop: the error is not worth retrying, or we are out
                // of attempts. Checking both here keeps the exit conditions in one place.
                let give_up = !policy.retryable(&e) || attempt == policy.max_attempts - 1;
                last_err = e;
                if give_up {
                    break;
                }
                thread::sleep(Duration::from_secs_f64(policy.backoff(attempt, rng)));
            }
        }
    }

    // A task that never succeeded is still DATA. Return a result carrying the
    // error rather than panicking, so one bad task cannot kill a 2000 task run.
    EvalResult {
        task_id: task.task_id.clone(),
        error: Some(last_err),
        attempts,
        latency_s: started.elapsed().as_secs_f64(),
        ..Default::default()
    }
}

/// Python defaults: concurrency = 16, timeout_s = 120.0.
pub fn run_suite(
    tasks: &[Task],
    model_fn: ModelFn,
    score_fn: &ScoreFn,
    concurrency: usize,
    timeout_s: f64,
) -> Vec<EvalResult> {
    let sem = Semaphore::new(concurrency); // one shared limiter for every task
    let policy = RetryPolicy::default();
    let next = AtomicUsize::new(0);
    // More workers than permits, so a worker sleeping in backoff does not
    // leave a request slot idle. The gate, not the worker count, is the limit.
    let n_workers = tasks.len().min(concurrency * 4).max(1);
    let mut results: Vec<Option<EvalResult>> = vec![None; tasks.len()];

    thread::scope(|s| {
        let (sem, policy, next, model_fn) = (&sem, &policy, &next, &model_fn);
        let workers: Vec<_> = (0..n_workers)
```

```
            .map(|w| {
                s.spawn(move || {
                    let mut rng = SplitMix64::new(w as u64 + 1); // per-worker jitter stream
                    let mut done = Vec::new();
                    loop {
                        let i = next.fetch_add(1, Ordering::Relaxed);
                        if i >= tasks.len() {
                            break;
                        }
                        let r = run_one(&tasks[i], model_fn, score_fn, policy, timeout_s, sem, &mut rng);
                        done.push((i, r));
                    }
                    done
                })
            })
            .collect();
        // Like gather: slot each result back by index, so the output is in
        // the ORIGINAL task order regardless of completion order.
        for worker in workers {
            for (i, r) in worker.join().unwrap() {
                results[i] = Some(r);
            }
        }
    });
    results.into_iter().map(|r| r.expect("every task ran")).collect()
}
```

For leaders

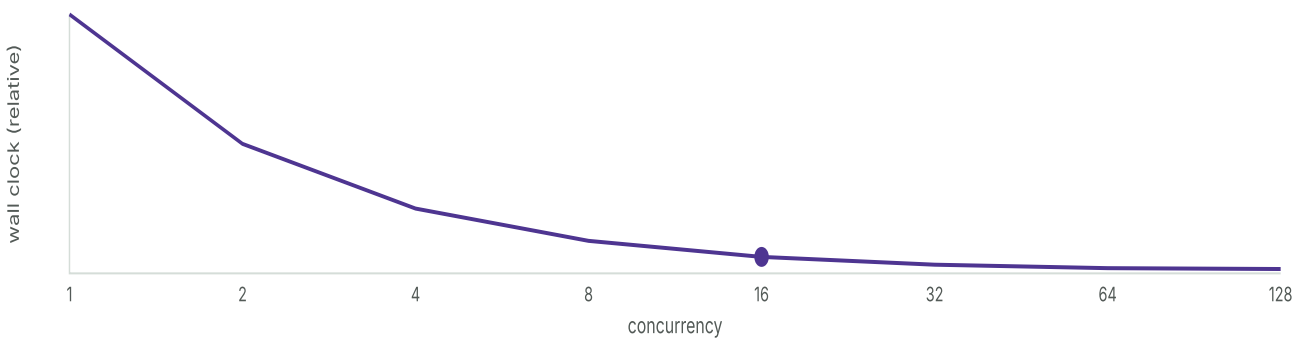
Retries are a measurement decision, not just an engineering one

Retrying is harmless when a failure happens before the model produced anything and is unrelated to the task: a dropped connection, a rate limit. It inflates scores when the "transient" failure is correlated with the answer. Common cases: a timeout that cuts off long, rambling, wrong generations; a parse error on malformed output that triggers a fresh sample; a sandbox crash during test execution that reruns the task. In each case the retried task gets another draw at the answer, so it is scored more like pass@k than pass@1. Insist that attempts are recorded per task and that reports show the score with and without retried tasks.

Interactive

Harness simulator: retries, concurrency, dropped tasks (shown at its default setting)

The true per attempt pass rate is fixed at 0.600. Raise the failure rate and the retry budget and watch your harness report something higher, because a retried task quietly got more than one shot at the answer.



True per attempt pass rate is 0.600. Your harness reports **0.630** (95% CI [0.599, 0.659]).
163 tasks retried, 1 tasks exhausted all attempts and were dropped from scoring.
Retries inflated the score by 0.030. A retried task got more than one shot at the answer, so it was effectively scored at pass@k with a larger k than the tasks that succeeded first time. Record attempts and report the score with and without retried tasks.
1 dropped tasks are silently excluded from the denominator. If failures correlate with difficulty, dropping them biases the score upward.
Wall clock at concurrency 16: 315.9 units. Push concurrency high enough and coordination overhead outweighs the parallelism.

Worked problem

Acceptance test. 1,000 tasks, 15% transient failure rate, up to 4 attempts, true per-attempt pass rate 0.600. The simulator above treats each failed attempt as an extra draw at the answer, the worst case described in the leader layer. What does the harness report?

Measured answer: **about 0.63**. A 1,000-task Python run gave 0.626 with 144 tasks retried; the simulator above, with its own random seed, shows 0.630 with 163. Either way the harness reports an improvement of roughly 3 points that does not exist, because each retried task got more than one shot. Without an `attempts` field per result you cannot even detect this. The fix is to record attempts and report the score with and without retried tasks. If the two numbers differ by more than the confidence interval, the retry policy is changing the measurement.

Key idea

Push concurrency up in the simulator. Wall clock falls at first, then rises as coordination overhead outweighs parallelism. Separately, any task that exhausts its attempts drops out of the scored denominator. If failures correlate with difficulty, which they usually do, dropping them biases the score upward.

Version three: caching, checkpointing, quarantine

Plain words

Remember answers, but only for the exact same question

Model calls are expensive, so the harness saves every answer. If the run crashes halfway, it picks up where it left off. The rule is strict: reuse a saved answer only if the question, the model, and every setting are exactly the same. If anything changed, ask again.

Under the hood

Cache keys over everything that can change the output, and an append-only log

python

```python
import hashlib, json, pathlib

def cache_key(task: Task, model_id: str, params: dict) -> str:
```

```
    """Key on EVERYTHING that can change the output."""
    payload = json.dumps({
        "task": task.task_id,
        # Hash the prompt itself, not just the id. If someone edits a task's
        # text without renaming it, the hash changes and the stale cached
        # result is correctly missed.
        "prompt_hash": hashlib.sha256(task.prompt.encode()).hexdigest(),
        "model": model_id,
        # sorted() so {"temp":0,"top_p":1} and {"top_p":1,"temp":0} produce the
        # same key. Without it you get silent cache misses on every run.
        "params": dict(sorted(params.items())),
    }, sort_keys=True)   # sort_keys again for the outer dict, same reason

    # 20 hex chars is ~80 bits: collision risk is negligible at eval scale and
    # the keys stay short enough to read in a directory listing.
    return hashlib.sha256(payload.encode()).hexdigest()[:20]


class ResultStore:
    """Append only JSONL. Restartable."""

    def __init__(self, path: str):
        self.path = pathlib.Path(path)
        self.path.parent.mkdir(parents=True, exist_ok=True)   # no crash on first run
        self.seen: dict[str, dict] = {}
        if self.path.exists():
            # Replay the log into memory. Later lines overwrite earlier ones for
            # the same key, so a re-run naturally supersedes an old result.
            for line in self.path.read_text().splitlines():
                if line.strip():                    # tolerate a trailing newline
                    rec = json.loads(line)
                    self.seen[rec["key"]] = rec

    def get(self, key: str) -> dict | None:
        return self.seen.get(key)

    def put(self, key: str, result: Result) -> None:
        rec = {"key": key, **result.__dict__}
        # APPEND, never rewrite. A process killed mid-write loses at most the
        # last line, and a run that dies at task 900 of 1000 resumes without
        # re-spending 900 requests.
        with self.path.open("a") as f:
            f.write(json.dumps(rec, default=str) + "\n")   # default=str for datetimes
        self.seen[key] = rec
```

rust

```
use std::collections::{BTreeMap, HashMap};
use std::fs::{self, OpenOptions};
use std::io::{self, Write};
use std::path::PathBuf;

#[derive(Debug, Clone, Default)]
pub struct Task {
    pub task_id: String,
    pub prompt: String,
}

#[derive(Debug, Clone, Default)]
pub struct EvalResult {
```

```rust
    // `Result` in the Python
    pub task_id: String,
    pub output: Option<String>,
    pub score: Option<f64>,
    pub error: Option<String>,
    pub attempts: u32,
    pub latency_s: f64,
    pub meta: BTreeMap<String, String>,
}

/// 128-bit FNV-1a, so the sketch needs no crates. Real code would use SHA-256
/// from a crate such as sha2: FNV is fast but not collision resistant.
fn fnv1a_128_hex(data: &str) -> String {
    let mut h: u128 = 0x6c62272e07bb014262b821756295c58d;
    for b in data.bytes() {
        h ^= b as u128;
        h = h.wrapping_mul(0x0000000001000000000000000000013B);
    }
    format!("{h:032x}")
}

/// Quote a string as JSON. std has no JSON support; real code would use serde_json.
fn json_str(s: &str) -> String {
    let mut out = String::from("\"");
    for c in s.chars() {
        match c {
            '"' => out.push_str("\\\""),
            '\\' => out.push_str("\\\\"),
            '\n' => out.push_str("\\n"),
            c if (c as u32) < 0x20 => out.push_str(&format!("\\u{:04x}", c as u32)),
            c => out.push(c),
        }
    }
    out.push('"');
    out
}

/// Key on EVERYTHING that can change the output.
pub fn cache_key(task: &Task, model_id: &str, params: &BTreeMap<String, f64>) -> String {
    // A BTreeMap iterates in sorted key order, so {"temp":0,"top_p":1} and
    // {"top_p":1,"temp":0} produce the same key. Without it you get silent
    // cache misses on every run.
    let params_json: Vec<String> = params.iter().map(|(k, v)| format!("{}:{}", json_str(k), v)).collect();
    // Outer fields written in sorted order by hand, same reason.
    let payload = format!(
        "{{\"model\":{},\"params\":{{{}}},\"prompt_hash\":{},\"task\":{}}}",
        json_str(model_id),
        params_json.join(","),
        // Hash the prompt itself, not just the id. If someone edits a task's
        // text without renaming it, the hash changes and the stale cached
        // result is correctly missed.
        json_str(&fnv1a_128_hex(&task.prompt)),
        json_str(&task.task_id),
    );

    // 20 hex chars is ~80 bits: collision risk is negligible at eval scale and
    // the keys stay short enough to read in a directory listing.
    fnv1a_128_hex(&payload)[..20].to_string()
}
```

```rust
/// Append only JSONL. Restartable.
pub struct ResultStore {
    path: PathBuf,
    // key -> the raw JSON line. With no JSON parser in std we keep lines as text.
    seen: HashMap<String, String>,
}

impl ResultStore {
    pub fn new(path: &str) -> io::Result<Self> {
        let path = PathBuf::from(path);
        if let Some(parent) = path.parent() {
            fs::create_dir_all(parent)?; // no crash on first run
        }
        let mut seen = HashMap::new();
        if path.exists() {
            // Replay the log into memory. Later lines overwrite earlier ones for
            // the same key, so a re-run naturally supersedes an old result.
            for line in fs::read_to_string(&path)?.lines() {
                // put() writes {"key":"<hex>",... first, so the key can be read
                // off the front; real code would parse the line with serde_json.
                // Blank lines (a trailing newline) simply fail to match.
                let key = line.strip_prefix("{\"key\":\"").and_then(|rest| rest.split('"').next());
                if let Some(key) = key {
                    seen.insert(key.to_string(), line.to_string());
                }
            }
        }
        Ok(ResultStore { path, seen })
    }

    pub fn get(&self, key: &str) -> Option<&str> {
        self.seen.get(key).map(String::as_str)
    }

    pub fn put(&mut self, key: &str, result: &EvalResult) -> io::Result<()> {
        let opt_str = |s: &Option<String>| s.as_deref().map_or("null".to_string(), json_str);
        let meta: Vec<String> = result.meta.iter().map(|(k, v)| format!("{}:{}", json_str(k),
json_str(v))).collect();
        // Every field is formatted explicitly, so there is no default=str fallback.
        let rec = format!(
            "{{\"key\":{},\"task_id\":{},\"output\":{},\"score\":{},\"error\":{},\"attempts\":{},\"latency_s\":
{},\"meta\":{{{}}}}}",
            json_str(key),
            json_str(&result.task_id),
            opt_str(&result.output),
            result.score.map_or("null".to_string(), |s| s.to_string()),
            opt_str(&result.error),
            result.attempts,
            result.latency_s,
            meta.join(","),
        );
        // APPEND, never rewrite. A process killed mid-write loses at most the
        // last line, and a run that dies at task 900 of 1000 resumes without
        // re-spending 900 requests.
        let mut f = OpenOptions::new().create(true).append(true).open(&self.path)?;
        writeln!(f, "{rec}")?;
        self.seen.insert(key.to_string(), rec);
        Ok(())
```

```
    }
}
```

Add the harness version and the judge model and prompt version to the cache key as well. A judge change silently reusing old scores is one of the most expensive cache bugs, because it produces a regression that does not exist.

Worked problem

Acceptance test. Run 1,000 tasks and kill the process at task 600. Restart. The run must finish with exactly 1,000 results and only 400 new model calls. Then change the temperature and confirm the cache misses on every task.

The parameter you forget to put in the key is the one that costs a day chasing a regression that is not there. Sort the parameter dict before hashing, or the same settings in a different order produce different keys.

Flakiness quarantine

Some tasks give different scores run to run even at fixed settings: nondeterministic environments, race conditions in test suites, judges sitting on a decision boundary. Run a sample of tasks five times at fixed settings and quarantine any whose score standard deviation exceeds about 0.25. Report quarantined tasks separately rather than averaging their noise into the headline.

python

```python
import statistics

def find_flaky(task_ids, run_once, repeats: int = 5, max_sd: float = 0.25) -> list[str]:
    """run_once(task_id) -> score in [0, 1] at FIXED settings. Returns tasks to quarantine."""
    flaky = []
    for t in task_ids:
        scores = [run_once(t) for _ in range(repeats)]
        if statistics.pstdev(scores) > max_sd:   # a 0/1 task that flips 2 of 5 times has sd 0.49
            flaky.append(t)
    return flaky
```

rust

```rust
/// Population standard deviation, what Python's statistics.pstdev computes.
fn pstdev(xs: &[f64]) -> f64 {
    let mean = xs.iter().sum::<f64>() / xs.len() as f64;
    (xs.iter().map(|x| (x - mean).powi(2)).sum::<f64>() / xs.len() as f64).sqrt()
}

/// run_once(task_id) -> score in [0, 1] at FIXED settings. Returns tasks to quarantine.
/// Python defaults: repeats = 5, max_sd = 0.25.
pub fn find_flaky(
    task_ids: &[&str],
```

```rust
    mut run_once: impl FnMut(&str) -> f64,
    repeats: usize,
    max_sd: f64,
) -> Vec<String> {
    let mut flaky = Vec::new();
    for &t in task_ids {
        let scores: Vec<f64> = (0..repeats).map(|_| run_once(t)).collect();
        if pstdev(&scores) > max_sd {
            // a 0/1 task that flips 2 of 5 times has sd 0.49
            flaky.push(t.to_string());
        }
    }
    flaky
}
```

The report

Plain words — A score without its fine print is not a score

The report is where all the care above either reaches the reader or gets lost. A pass rate on its own hides how many tasks there were, how uncertain the number is, and how many tasks never finished. The report should be impossible to produce without those.

Under the hood — A summary that cannot omit n, the interval, or the error count

The interval is the Wilson interval from eval statistics; the latency quantiles follow the advice in intervals and the bootstrap never to report a mean latency.

python

```python
def summarize(results: list[Result]) -> dict:
    """A report that cannot omit n, the interval, or the error count."""

    # Split errored tasks out FIRST. Scoring None as a failure would conflate
    # "the model got it wrong" with "our infrastructure fell over", which are
    # completely different findings.
    ok = [r for r in results if r.error is None and r.score is not None]

    passed = sum(1 for r in ok if r.score >= 1.0)
    n = len(ok)
    lo, hi = wilson(passed, n) if n else (0.0, 1.0)   # interval, never bare rate

    lat = sorted(r.latency_s for r in ok)
    def pct(q):
        # Guard the index: at q = 1.0 the naive int(q*len) runs one past the end.
        return lat[min(len(lat) - 1, int(q * len(lat)))] if lat else 0.0
```

```
    return {
        "n_tasks": len(results),
        "n_scored": n,
        # SURFACE the errors. A pass rate computed over a shrunken denominator
        # is biased upward if failures correlate with difficulty, which they do.
        "n_errored": len(results) - n,
        "pass_rate": passed / n if n else 0.0,
        "ci95": (round(lo, 4), round(hi, 4)),
        # Retried tasks effectively got more than one sample, inflating their
        # contribution. Reporting the count lets a reader discount for it.
        "retried_tasks": sum(1 for r in ok if r.attempts > 1),
        "latency_p50": round(pct(0.50), 2),
        "latency_p95": round(pct(0.95), 2),   # never report a mean latency
    }
```

rust

```
#[derive(Debug, Clone, Default)]
pub struct EvalResult {
    // `Result` in the Python, trimmed to the fields used here
    pub score: Option<f64>,
    pub error: Option<String>,
    pub attempts: u32,
    pub latency_s: f64,
}

/// The dict the Python returns, as a struct so no field can be left out.
#[derive(Debug)]
pub struct Summary {
    pub n_tasks: usize,
    pub n_scored: usize,
    pub n_errored: usize,
    pub pass_rate: f64,
    pub ci95: (f64, f64),
    pub retried_tasks: usize,
    pub latency_p50: f64,
    pub latency_p95: f64,
}

/// Wilson score interval, as in the evaluation statistics chapter.
fn wilson(k: usize, n: usize, z: f64) -> (f64, f64) {
    let (k, n) = (k as f64, n as f64);
    let p = k / n;
    let denom = 1.0 + z * z / n;
    let center = (p + z * z / (2.0 * n)) / denom;
    let margin = z * (p * (1.0 - p) / n + z * z / (4.0 * n * n)).sqrt() / denom;
    ((center - margin).max(0.0), (center + margin).min(1.0))
}

/// Python's round(x, digits).
fn round_to(x: f64, digits: i32) -> f64 {
    let scale = 10f64.powi(digits);
    (x * scale).round() / scale
}

/// A report that cannot omit n, the interval, or the error count.
pub fn summarize(results: &[EvalResult]) -> Summary {
    // Split errored tasks out FIRST. Scoring None as a failure would conflate
    // "the model got it wrong" with "our infrastructure fell over", which are
    // completely different findings.
```

```
    let ok: Vec<&EvalResult> = results.iter().filter(|r| r.error.is_none() && r.score.is_some()).collect();

    let passed = ok.iter().filter(|r| r.score.is_some_and(|s| s >= 1.0)).count();
    let n = ok.len();
    let (lo, hi) = if n > 0 { wilson(passed, n, 1.96) } else { (0.0, 1.0) }; // interval, never bare rate

    let mut lat: Vec<f64> = ok.iter().map(|r| r.latency_s).collect();
    lat.sort_by(f64::total_cmp);
    let pct = |q: f64| {
        // Guard the index: at q = 1.0 the naive (q * len) as usize runs one past the end.
        if lat.is_empty() { 0.0 } else { lat[(lat.len() - 1).min((q * lat.len() as f64) as usize)] }
    };

    Summary {
        n_tasks: results.len(),
        n_scored: n,
        // SURFACE the errors. A pass rate computed over a shrunken denominator
        // is biased upward if failures correlate with difficulty, which they do.
        n_errored: results.len() - n,
        pass_rate: if n > 0 { passed as f64 / n as f64 } else { 0.0 },
        ci95: (round_to(lo, 4), round_to(hi, 4)),
        // Retried tasks effectively got more than one sample, inflating their
        // contribution. Reporting the count lets a reader discount for it.
        retried_tasks: ok.iter().filter(|r| r.attempts > 1).count(),
        latency_p50: round_to(pct(0.50), 2),
        latency_p95: round_to(pct(0.95), 2), // never report a mean latency
    }
}
```

Worked problem

Acceptance test. The summary must refuse to omit the error count, and the flakiness report must identify tasks you deliberately injected as flaky. A useful extra check: inject 5% infrastructure errors concentrated on the hardest tasks and confirm the report shows both the error count and the upward bias it would cause if those tasks were silently dropped.

Eval lens

When two runs disagree, diff the harness before you diff the model: harness version, judge version, prompt template, timeouts, concurrency, sampling parameters, and task set membership. The regression triage chapter turns this into a procedure.

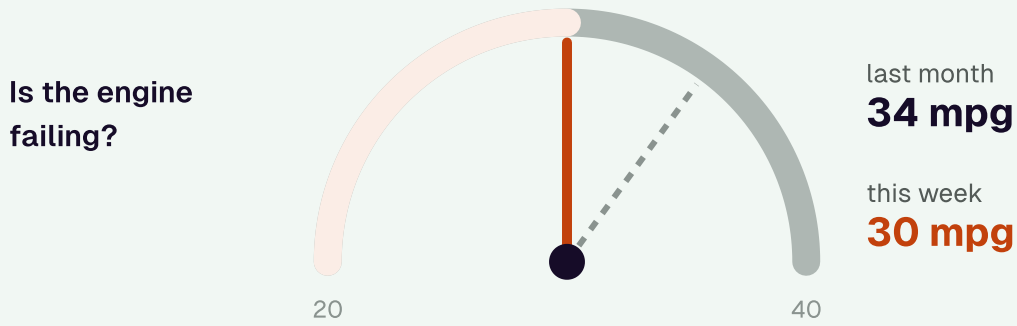
Diagnosing a regression: model, harness, data, or infrastructure?

An eval suite comes back with an anomalous number a few hours before a decision. The job is to say, quickly and with evidence, whether anything is actually wrong, and if so whether it is the model, the harness, the data, or the infrastructure. This chapter gives an order of operations that answers the cheapest questions first.

Diagram

The car that suddenly got worse mileage

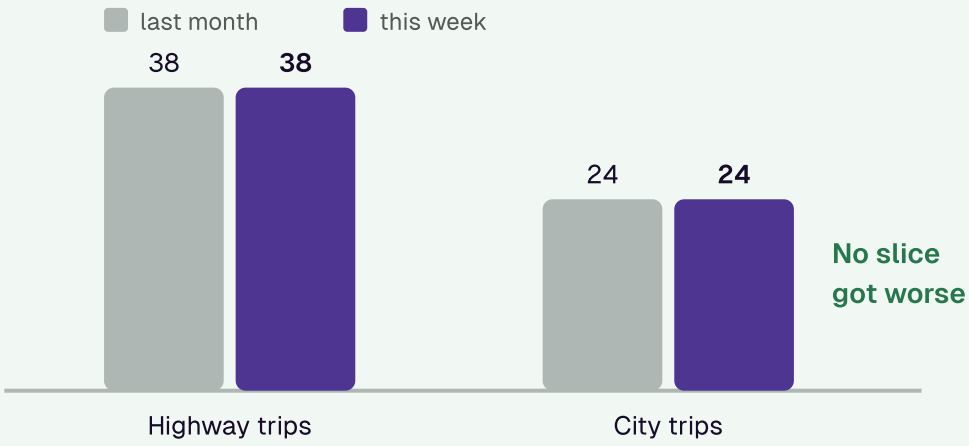
A regression hunt in six pictures. The engine is the last suspect, not the first.



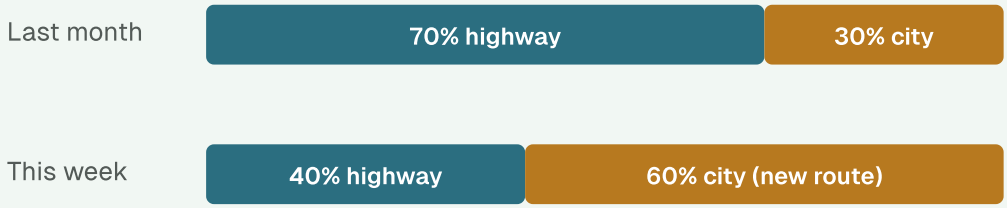
Your weekly average mileage drops from 34 to 30 mpg. The tempting move is to book the engine in for repair. In evals that is blaming the model the moment a score drops. Six cheaper checks come first.

Step 1: is the drop bigger than normal noise? Here, yes.

First question: is this bigger than ordinary wobble? Weekly mileage normally bounces between about 32 and 35. Thirty is well outside that band, so something really changed. In evals, this is the paired interval on the delta. Many "regressions" stop here, inside the noise.



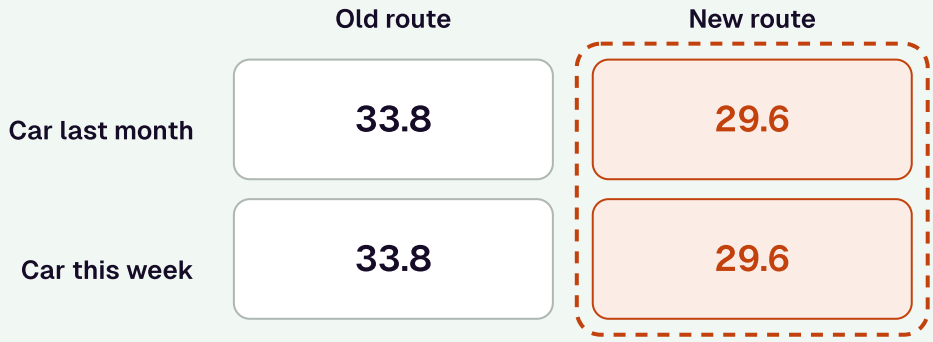
Second: is it everywhere or somewhere? Split trips by type. Highway trips still average 38 mpg and city trips still 24. Neither slice got worse, which is strange if the average fell. Slicing like this is how you find where a drop lives, or that it lives nowhere.



Last month: 0.7 × 38 + 0.3 × 24 = **33.8**
This week: 0.4 × 38 + 0.6 × 24 = **29.6**

The engine did not change. The mix of trips did.

Third: what changed besides the engine? A road closure moved most of your driving into town. Each kind of trip is just as efficient, but the average of trip mileages slides from 33.8 to 29.6. In evals this is a task set whose composition shifted, for example more hard categories added this week.



The drop follows the route column, not the car row.

To be sure, replay: drive last month's car and this week's car on both routes. The old car drops on the new route too, and the new car is fine on the old route. In evals: rerun the previous model on today's harness and task set. If it drops as well, the model is not the cause.

1	Bigger than noise?	yes, outside the band
2	Everywhere or somewhere?	no slice got worse
3	What else changed?	the trip mix (task set)
4	Replay with the old car?	it drops too

Verdict: measurement, not model

The verdict comes from evidence, not instinct: a real drop, no slice worse, a changed task mix, and a replay that clears the model. The same checklist, plus error rates and reading twenty failing outputs, settles most eval regressions within hours.

Plain words

The thermometer before the doctor

If a child's temperature reading looks high, a sensible parent first checks the thermometer. Is it working? Did they measure right after a hot bath? Only then do they worry about illness.

A falling eval score is the same. Before blaming the model, check whether the drop is bigger than normal noise, whether the measuring setup changed, and whether the computers running the test had problems. Most "regressions" turn out to be the thermometer.

For leaders

Incident response for a number

Treat an eval regression like a production incident: one owner, a fixed checklist, a written timeline, and a clear exit. The first question is not "what broke" but "did anything break". Asking it first protects the team's time and stops a launch being blocked by noise, or approved past a real problem.

A useful operating rule: the triage owner reports within a fixed window (a few hours) with one of four verdicts, each with its evidence.

Noise

The change is inside the confidence interval. No action; note it and move on.

Measurement

Harness, judge, prompt, or task set changed. Fix the measurement and rerun before anyone discusses the model.

Infrastructure

Timeouts, rate limits, or errors spiked. Rerun under healthy conditions.

Model

Reproduces on replay and is concentrated in specific capabilities. Escalate with example outputs and the affected slices.

What goes in the launch review afterwards: the verdict, the interval, the slice breakdown, twenty example outputs that show the failure, and what changes in the harness or checks so this class of problem is caught automatically next time.

Principal asks

- Is this outside the noise band? What is the interval on the delta, computed paired?
- Is the drop uniform or concentrated? Which slice contributes most of the total?
- What changed besides the model? Show me the config diff.
- Did the previous model also drop when rerun on today's harness?
- Have you read the failing outputs yourself?

The order to check things in

1. **Is it real?** Compute the interval before you investigate. Use the paired comparison from eval statistics: the same tasks, both models, discordant pairs. A 3 point drop inside the noise band is not an incident. This step alone saves days and is the one people skip.

2. **Is it everywhere or somewhere?** Break the delta down by task category, language, input length, source. A uniform drop points at the model or the harness. A drop concentrated in one slice points at data or a specific capability.

3. **Did anything non-model change?** Harness version, judge model or prompt, timeout, concurrency, temperature, task set membership, dependency versions. Diff the config before you diff the model.

4. **Error and retry rates.** A drop that arrives with a spike in timeouts is infrastructure wearing a model costume. See the harness chapter for why errors and retries bias scores.

5. **Replay.** Rerun a sample of the regressed tasks against the previous model with today's harness. If the old model also drops, it is not the model.

6. **Read the outputs.** Twenty of them, by hand, every time. The failure mode is usually visible and usually not what the metric suggested.

Try the order yourself on five illustrative incidents. Each check costs time, and the cheap ones come first for a reason.

Interactive

Triage a regression, one check at a time (shown at its default setting)

Pick an incident and click checks in any order. Each click reveals what that check finds and charges its time. Compare your path with the chapter's order on the timeline: skipping ahead to the expensive checks is where hours go.

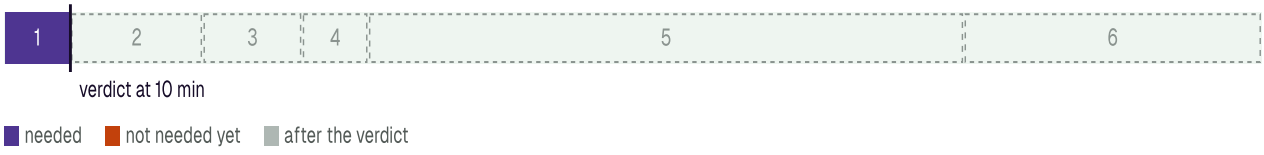
Alert. Coding suite pass rate fell from 71.5% to 69.5% (−2.0 points) on 400 tasks, a few hours before a launch review. Is anything actually wrong?

1	**Is it real?** Paired interval on the delta	10 min
2	**Is it everywhere or somewhere?** Slice the delta by category, length, source	20 min
3	**Did anything non-model change?** Diff the config: harness, judge, task set, settings	15 min
4	**Error and retry rates** Timeouts, rate limits, failed calls	10 min
5	**Replay** Previous model on today's harness	1 h 30 min
6	**Read the outputs** Twenty failures, by hand	45 min

Your path

Click a check above to start the clock

The chapter's order



needed · not needed yet · after the verdict

Time spent so far	Chapter order reaches a verdict in	Time on checks not yet needed	Verdict
0 min	10 min	0 min	open

Six checks, ordered so the cheapest questions come first. The chapter's order reaches a verdict on this incident in **10 min**. Click the checks in any order, or use "Next check in order", and compare your timeline with the chapter's.

Under the hood

Slice by contribution, not by percentage change

The total change in mean score decomposes exactly into slice contributions. With N tasks split into slices s of size n_s and per-slice mean change $\bar{d}_s$:

$$\Delta_{\text{total}} = \frac{1}{N} \sum_{s} n_s \, \bar{d}_s$$

Each slice's contribution is $n_s \bar{d}_s / N$. Rank slices by this, not by $\bar{d}_s$ alone, or a tiny slice with a big swing will send you down the wrong path. This identity holds when slices partition the tasks.

python

```python
def slice_deltas(before: dict[str, float], after: dict[str, float],
                 slices: dict[str, list[str]]) -> list[tuple[str, float, int]]:
    """Where did the drop actually happen. Sort by contribution to the total
    delta, not by percentage change, or a tiny slice with a big swing will
    send you down the wrong path."""
    out = []
    for name, task_ids in slices.items():
        ids = [t for t in task_ids if t in before and t in after]
        if not ids:
            continue
        d = sum(after[t] - before[t] for t in ids) / len(ids)
        contribution = d * len(ids)
        out.append((name, d, len(ids), contribution))
    out.sort(key=lambda x: x[3])            # most negative contribution first
    return [(n, round(d, 4), k) for n, d, k, _ in out]
```

rust

```rust
use std::collections::HashMap;

/// Where did the drop actually happen. Sort by contribution to the total
/// delta, not by percentage change, or a tiny slice with a big swing will
/// send you down the wrong path.
/// `slices` is a list of (name, task ids) pairs, so ties keep the order given,
/// as they do with Python's insertion-ordered dict and stable sort.
pub fn slice_deltas(
    before: &HashMap<String, f64>,
    after: &HashMap<String, f64>,
    slices: &[(String, Vec<String>)],
```

```rust
) -> Vec<(String, f64, usize)> {
    let mut out = Vec::new();
    for (name, task_ids) in slices {
        let ids: Vec<&String> = task_ids.iter().filter(|t| before.contains_key(*t) &&
after.contains_key(*t)).collect();
        if ids.is_empty() {
            continue;
        }
        let d = ids.iter().map(|t| after[*t] - before[*t]).sum::<f64>() / ids.len() as f64;
        let contribution = d * ids.len() as f64;
        out.push((name.clone(), d, ids.len(), contribution));
    }
    out.sort_by(|x, y| x.3.total_cmp(&y.3)); // most negative contribution first
    out.into_iter()
        .map(|(n, d, k, _)| (n, (d * 1e4).round() / 1e4, k)) // round(d, 4)
        .collect()
}
```

> **Watch out**
>
> Slicing is a multiple comparison. With twenty slices, one will look significant by chance at the 5% level. Treat the slice view as a way to decide where to read outputs, not as a finding in itself, or apply a correction such as Benjamini–Hochberg (see multiple comparisons).

Replay as a two-by-two

The replay step is cleanest as a small factorial: old and new model crossed with old and new harness. If the drop appears only in the new-harness column, the harness is the cause. If it appears only in the new-model row, it is the model. If it appears in only one cell, look for an interaction, such as a new prompt template that only the new model handles badly.

	Old harness	New harness
Old model	Baseline	Harness effect alone
New model	Model effect alone	What you observed

Eval lens

Every triage should end with a change to the harness or the automated checks, so the same class of false alarm or real regression is caught without a human next time. A regression you diagnosed twice is a missing test.

Part II · Chapter 17

Which metric or test, when

This chapter is a decision guide. It pulls together the metrics from classification and judges and the tests from statistics for eval sets and intervals and the bootstrap. For "my data has this shape, which method fits," use the lookup in from shape to method.

Plain words

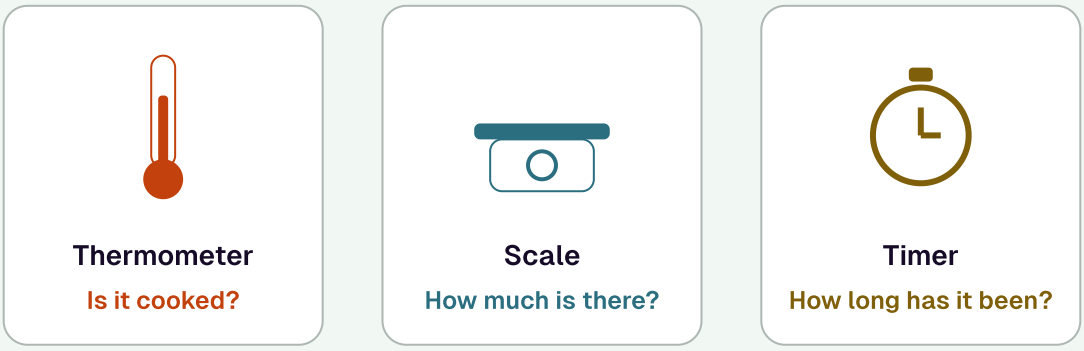
Pick the metric by the question you are asking

Every metric answers a different question. "When it says yes, is it right?" is precision. "Did it find them all?" is recall. "Does it rank good above bad?" is AUC. "Is this difference real or luck?" is a statistical test. Start from the question, then pick the number. Picking the number first and inventing the question afterwards is how dashboards end up misleading people.

Diagram

Pick the measuring tool before you measure

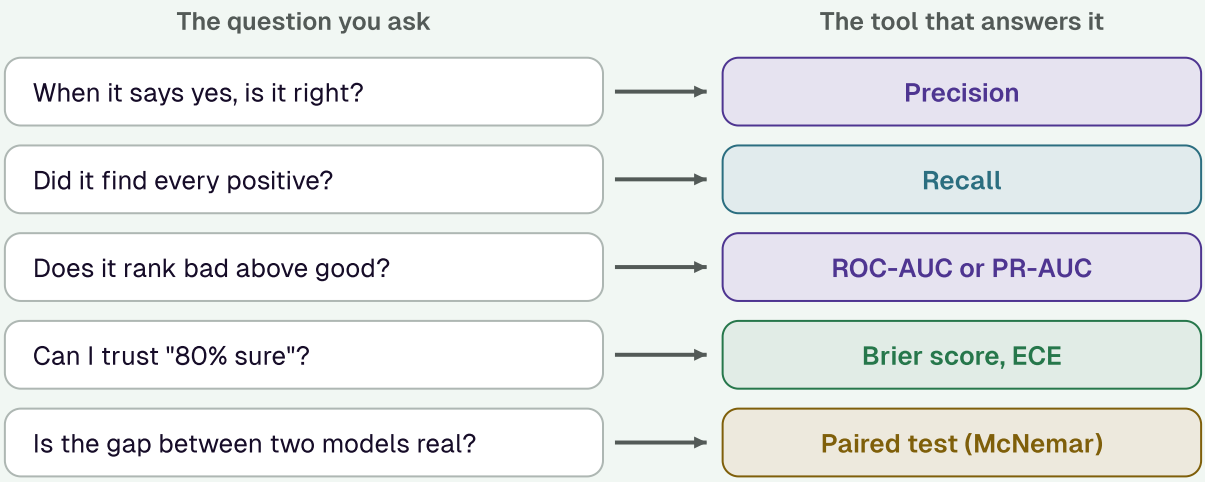
A kitchen drawer of tools, mapped onto the metric drawer.



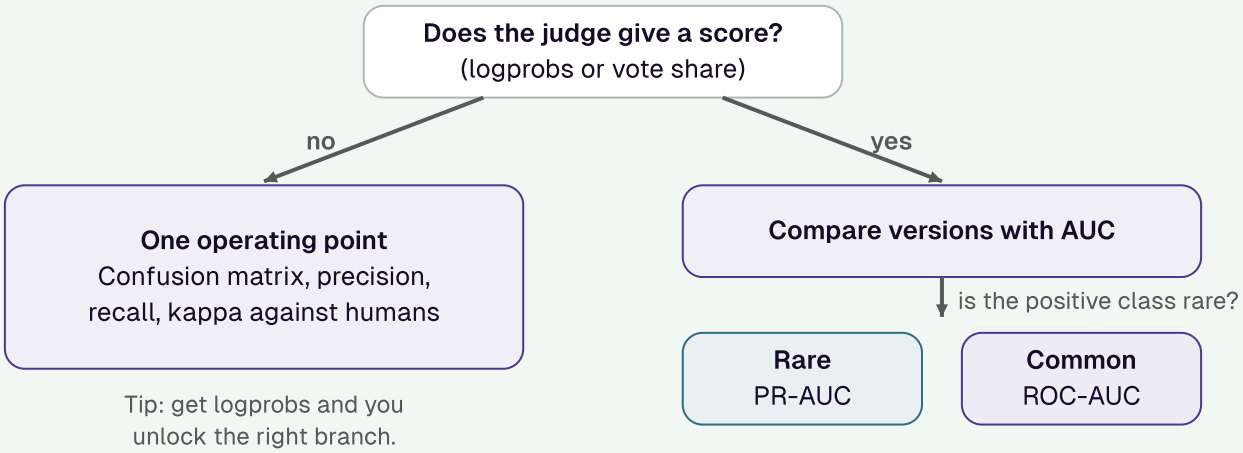
A timer cannot tell you whether the chicken is cooked.

Every tool answers one question. So does every metric.

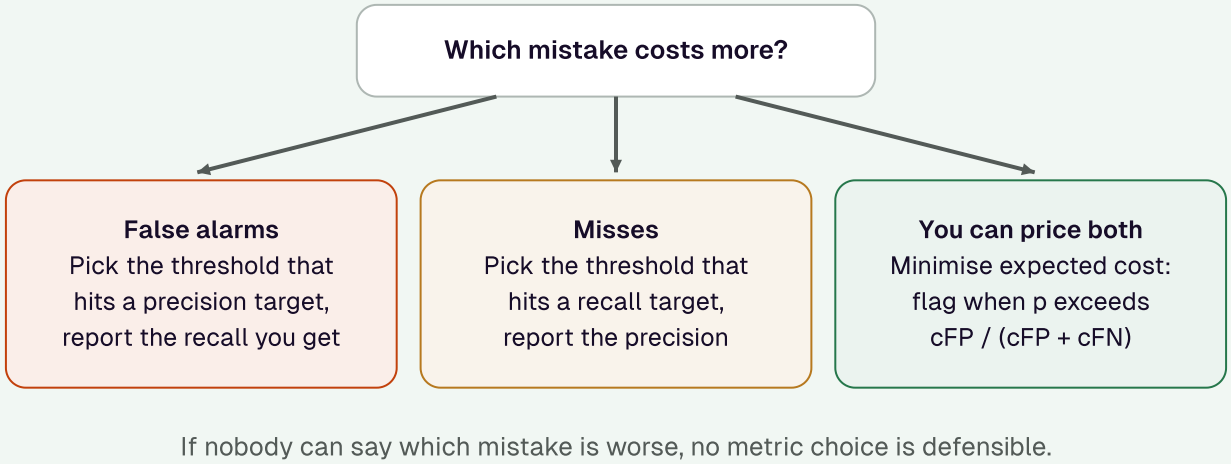
A thermometer, a scale, and a timer each answer one question well and every other question badly. Metrics are the same: pick the question first, then the tool.



Five questions people actually ask about a classifier or judge, and the tool that answers each one.



The first fork: does your judge give a score or only yes/no? Yes/no gives one operating point, so you work from its confusion matrix. A score lets you compare versions with AUC, and PR-AUC if the thing you are catching is rare.



The second fork: which mistake costs more? That decides where the threshold goes. If you can put a price on both mistakes, the threshold becomes arithmetic.

Last question: is the difference real?

New judge prompt: +3 points on 200 tasks.

Only 30 tasks changed verdict, split 18 better and 12 worse.

McNemar p = 0.36: a coin flip does this a third of the time.

- Put an interval on every number
- Compare on the same items (paired)
- Pin the threshold and the test set

Question first, tool second.

The last check before anyone celebrates: is the difference real? A 3-point gain resting on 30 changed verdicts split 18 to 12 is what chance produces about a third of the time. Intervals, paired comparisons, and a pinned test set keep you honest.

A recommendation for your classifier or judge

Interactive **Metric picker** (shown at its default setting)

Answer three questions about your judge or classifier and get the numbers to compare versions with, how to set the operating point, and what else to report.

Does it give a score, or only yes/no?

Only yes/no | A score (logprobs or votes)

Is the thing you are catching rare (under about 20%)?

Yes, it is rare | No, it is common

Which mistake hurts more?

False alarms | Misses | About equal

COMPARE VERSIONS WITH	Precision and recall for the rare class, plus its F1. A new version wins only if one number improves without the other dropping.
SET THE OPERATING POINT	Use stricter rubric wording until precision reaches your target. Report the recall you give up.
ALSO REPORT	The confusion matrix at the operating point, n, and Cohen's kappa against human labels on the same items. Keep accuracy out of the headline; it hides the mistakes you care about. If you can, get a score from logprobs or repeated votes so AUC becomes available.

Metric reference

Metric	Formula	Question it answers	Reach for it when	Watch out
Accuracy	`(TP+TN) / n`	How often is it right overall?	Balanced classes, symmetric costs	Misleading when one class is rare
Precision	`TP / (TP+FP)`	When it says yes, is it right?	False alarms are costly	High while missing most positives
Recall (TPR)	`TP / (TP+FN)`	Did it find all the positives?	Misses are costly	Flagging everything scores 100%
Specificity (TNR)	`TN / (TN+FP)`	Did negatives get through untouched?	Protecting the negative class matters	It is recall for the other class
F1 / F-beta	`2PR / (P+R)`	Good at both precision and recall?	One number for a rare class	Ignores TN; depends on threshold
Balanced accuracy	`(TPR+TNR) / 2`	Right on both classes, equally weighted?	Executive summary under imbalance	Hides which error dominates
MCC	`see chapter`	How correlated are prediction and truth?	One number to compare models	Unintuitive scale
Macro F1	`mean of class F1`	Good on every class?	Rare classes matter	Small classes swing it
ROC-AUC	`P(s+ > s−)`	Does it rank positives above negatives?	Comparing scored models, moderate imbalance	Flatters when positives are rare

Metric	Formula	Question it answers	Reach for it when	Watch out
PR-AUC (AP)	`area under PR`	How precise across recall levels?	Comparing scored models, rare positives	Baseline is prevalence; same data only
Brier / ECE	`mean (p−y)²`	Is the confidence honest?	Confidence drives routing or abstention	ECE depends on binning
Coverage–risk	`risk at coverage κ`	How much error if we only answer when sure?	The system can abstain or ask	Needs a confidence signal that tracks correctness
Expected cost	`(c_FP·FP + c_FN·FN)/n`	What does this threshold cost us?	You can price the errors	Needs an agreed cost ratio
Cohen's kappa	`(p_o − p_e)/(1 − p_e)`	Agreement beyond luck?	Judge versus human, human versus human	Check human ceiling first

Statistical tests reference

Question	Method	Why	Chapter
Interval on one pass rate	Wilson interval	Stays inside [0, 1] and holds coverage at small n and extreme rates	Eval statistics
Two models, same tasks, binary outcome	McNemar (exact) or paired bootstrap	Only discordant tasks carry information; unpaired tests waste power	Eval statistics
Two models, same tasks, continuous score	Paired bootstrap or Wilcoxon signed-rank	Uses the pairing, no normality assumption	Intervals and bootstrap
Three or more models, same tasks	Cochran's Q (binary) or Friedman (ranked)	One omnibus test before pairwise comparisons	Eval statistics
Many benchmarks at once	Benjamini–Hochberg	Controls the share of false discoveries	Eval statistics
Interval on a median, p95, or AUC	Bootstrap (percentile or BCa)	No closed form exists	Intervals and bootstrap
Items nested in sessions or tasks	Cluster bootstrap	Naive intervals are far too narrow when items correlate	Intervals and bootstrap
How many tasks do I need?	Power analysis on discordance	Detecting a 2-point move usually needs about a thousand tasks	Eval statistics

Walk the decision tree

Interactive **Which test or metric should I use** (shown at its default setting)

Answer a few questions and get a recommendation with its reason and its drawback. Every leaf carries a con, because the drawback is the part that gets asked about. Click a crumb to go back.

start

What are you measuring?

- A pass / fail outcome
- A continuous value (latency, tokens, score)
- An ordinal rating (Likert 1 to 5)
- A count (tool calls per session)
- Classifier or judge quality

Drill

Work through every branch once. There are about twenty leaves, and each is a short answer you should be able to give without the widget: the recommendation, the reason, and the drawback.

For leaders **The metric pack for a launch review**

Agree on the metric pack before results exist, and keep it fixed. A one-page eval readout for a classifier, judge, or safety filter should contain:

1. The question being answered and which error costs more, in one sentence each.
2. n, the positive rate in the test set, and the positive rate expected in production.
3. The confusion matrix at the chosen operating point, and how that point was chosen (on which split).

4. The headline metric with a confidence interval, and the comparison to the baseline with a paired test.
5. Per-class or per-slice results for anything rare that carries risk.
6. For judge-scored results: judge version, judge-to-human kappa, and the human ceiling.

Principal asks

- Was this metric chosen before or after seeing the results?
- If the positive rate in production differs from the test set, what precision should we actually expect? (Recompute it from TPR and FPR.)
- Which result would have made us not ship? If none, the evaluation was not a decision input.

Under the hood

Two formulas that settle most arguments

Precision under prevalence shift. TPR and FPR are properties of the classifier; precision is not. If you validated at prevalence π and production runs at π':

$$\text{Precision}(\pi') = \frac{\text{TPR}\,\pi'}{\text{TPR}\,\pi' + \text{FPR}\,(1-\pi')}$$

A filter with TPR 0.90 and FPR 0.05 has precision 0.95 at $\pi = 0.5$, but only 0.15 at $\pi' = 0.01$.

The cost-optimal threshold. With calibrated scores, flag when the expected cost of not flagging exceeds the expected cost of flagging:

$$p\,c_{FN} > (1-p)\,c_{FP} \iff p > \frac{c_{FP}}{c_{FP} + c_{FN}}$$

If the scores are not calibrated, sweep thresholds on a validation split, pick the minimum-cost one, and report its cost on a separate test split.

python

```python
def precision_at(tpr, fpr, prevalence):
    return tpr * prevalence / (tpr * prevalence + fpr * (1 - prevalence))

print(round(precision_at(0.90, 0.05, 0.50), 2))  # 0.95
print(round(precision_at(0.90, 0.05, 0.01), 2))  # 0.15
```

rust

```rust
fn precision_at(tpr: f64, fpr: f64, prevalence: f64) -> f64 {
    tpr * prevalence / (tpr * prevalence + fpr * (1.0 - prevalence))
}

fn main() {
    println!("{:.2}", precision_at(0.90, 0.05, 0.50)); // 0.95
    println!("{:.2}", precision_at(0.90, 0.05, 0.01)); // 0.15
}
```

Eval lens

The most common failure is not the wrong metric. It is a metric whose definition, threshold, or test set quietly changed between the baseline and the candidate. Pin all three, and the choice of metric becomes much less contentious.

Part III

How neural networks and LLMs work

What a neural network is, how it learns by gradient descent, how architectures are chosen, and how a transformer that only predicts the next token becomes an assistant, a reasoning model, or a one-pass decision engine.

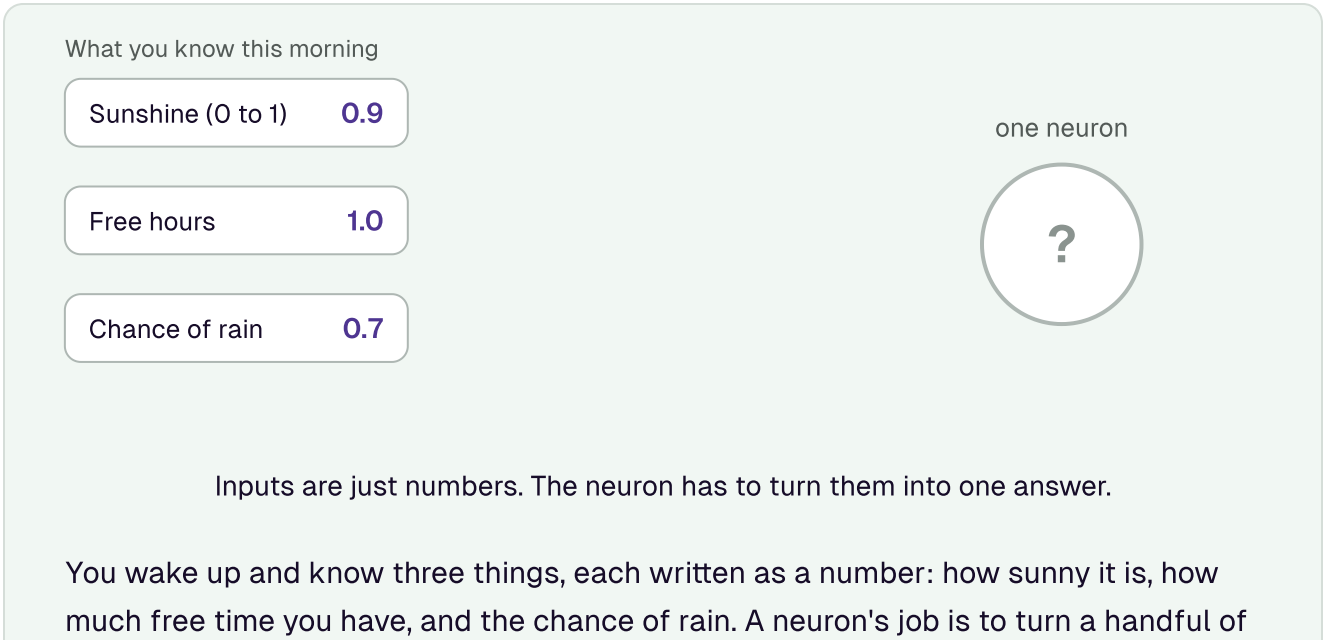
Part III · Chapter 18

Neural networks from the ground up: weighted sums that bend

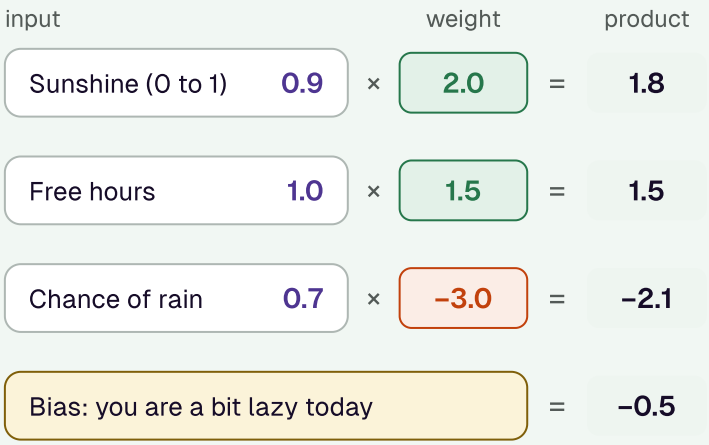
Every model in this guide, from a spam filter to a frontier LLM, is built from one tiny part repeated billions of times: multiply some numbers by weights, add them up, and bend the result. This chapter builds that part, stacks it into layers, runs numbers through it by hand, counts what "a 70B model" actually counts, and ends with what the famous universal approximation theorem does and does not promise. The next chapter, how networks learn, covers how the weights get their values.

Diagram **Should I go for a run?**

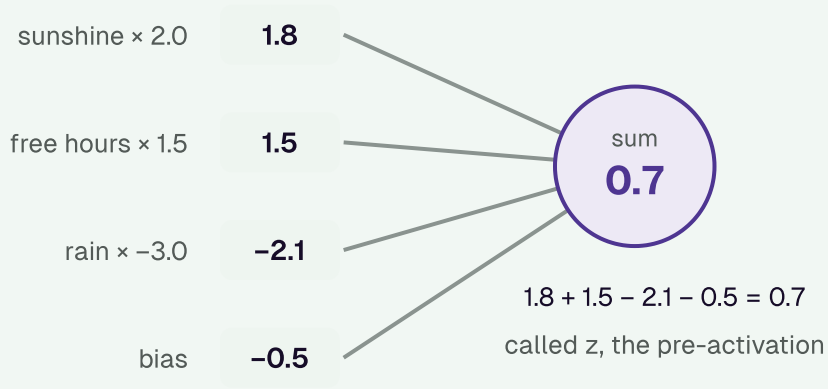
One artificial neuron, told as a decision you make most mornings.



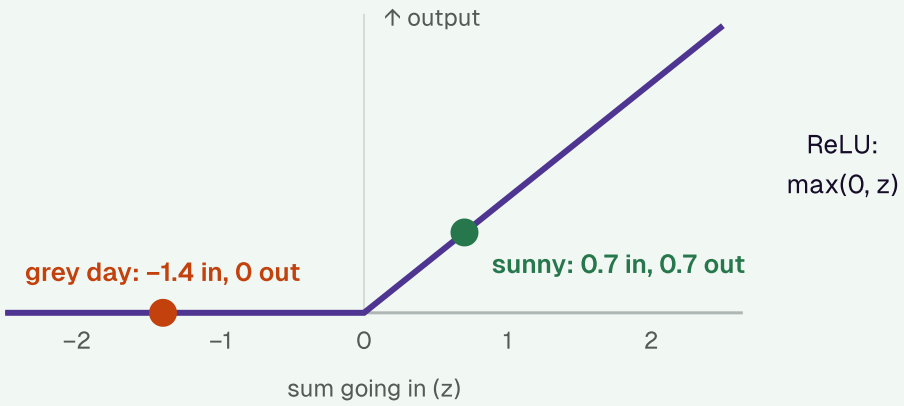
You wake up and know three things, each written as a number: how sunny it is, how much free time you have, and the chance of rain. A neuron's job is to turn a handful of numbers like these into one number: go, or not.



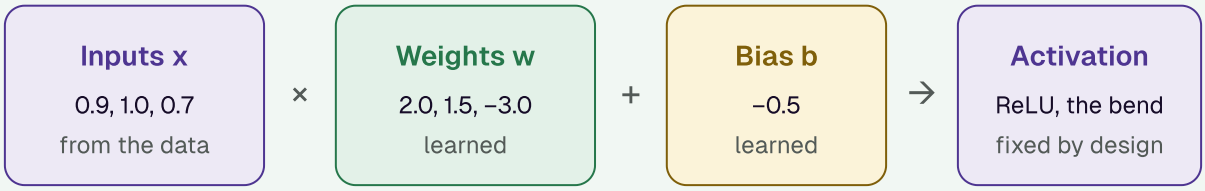
Each input gets a **weight** that says how much it matters and in which direction. Rain counts against you, strongly. The **bias** is a constant nudge that does not depend on the inputs at all: your general mood before you even look outside.



Add up the weighted inputs and the bias: 1.8 + 1.5 − 2.1 − 0.5 = 0.7. That single number, the **weighted sum**, is the neuron's raw opinion. So far it is nothing more than a spreadsheet formula.



Last, the **activation function** bends the sum. The most common one, ReLU, keeps positive numbers and turns negatives into zero. On a grey morning (sunshine 0.3, rain 1.0) the sum is −1.4 and the neuron outputs 0: stay in. That bend looks trivial. The next section shows it is the whole trick.



$$\text{output} = \text{ReLU}(w \cdot x + b)$$

Here you chose the weights. In a real network, training chooses them.

That is the whole neuron: inputs times weights, plus a bias, then a bend. The weights and the bias are the **parameters**. You picked them here from common sense. A neural network starts with random ones and adjusts them, billions at a time, until its outputs match the data.

A single neuron

Plain words

A tiny decision maker

A neuron is a very small decision maker. It takes a few numbers in, decides how much each one matters, adds them up with a constant nudge, and gives one number out. The "how much each one matters" numbers are called **weights**, the constant nudge is the **bias**, and the final squish or bend is the **activation function**.

The name comes from a loose analogy with brain cells, which also combine many incoming signals and either fire or stay quiet. The analogy stops there. An artificial neuron is two lines of arithmetic, and nobody designs them by hand: a network starts with random weights and the training process tunes them.

For leaders

Why "neuron" is a misleading unit of analysis

It is tempting to imagine each neuron in an LLM as a little expert with a job: a "French neuron" or a "sarcasm neuron". Some units do behave that way. Most do not. Interpretability research finds that individual units usually respond to several unrelated things at once (they are **polysemantic**), because a network can store more concepts than it has units by spreading each concept across many of them. Anthropic's "Toy Models of Superposition" (Elhage et al., 2022) studies this in small models, and "Towards Monosemanticity" (Bricken et al., 2023) recovers cleaner features with sparse dictionary learning on top of the units.

Practical consequence: claims like "we found the neuron for X and turned it off" deserve scrutiny. Ask whether the intervention was checked for side effects on unrelated behaviour, and whether the unit is a feature found by a method such as a sparse autoencoder or a raw neuron.

Under the hood

The formula

A neuron with inputs $\mathbf{x} \in \mathbb{R}^n$, weights $\mathbf{w} \in \mathbb{R}^n$, bias b, and activation ϕ computes

$$y = \phi\Big(\sum_{i=1}^{n} w_i x_i + b\Big) = \phi(\mathbf{w}^\top \mathbf{x} + b)$$

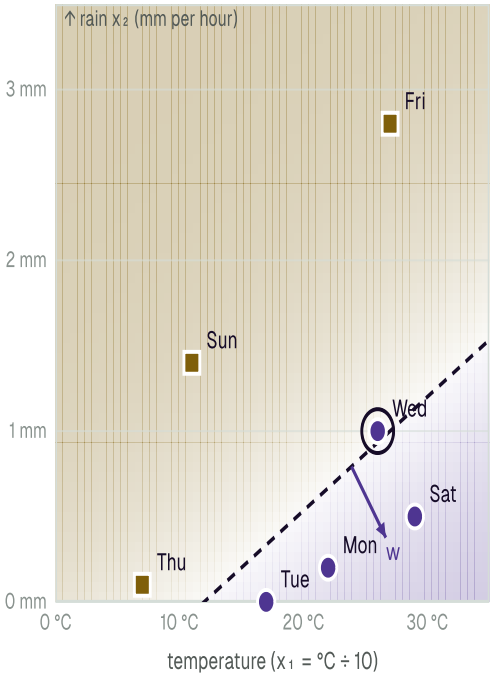
$z = \mathbf{w}^\top \mathbf{x} + b$ is the pre-activation. With ϕ = identity this is linear regression; with ϕ = sigmoid it is logistic regression. A neuron is a regression model with a bend at the end.

Geometrically, $\mathbf{w}^\top \mathbf{x} + b = 0$ is a hyperplane (a line, in two dimensions). The neuron measures signed distance from that hyperplane, scaled by $\|\mathbf{w}\|$, and the activation decides what to do with it. A ReLU neuron is "off" on one side of its hyperplane and grows linearly on the other.

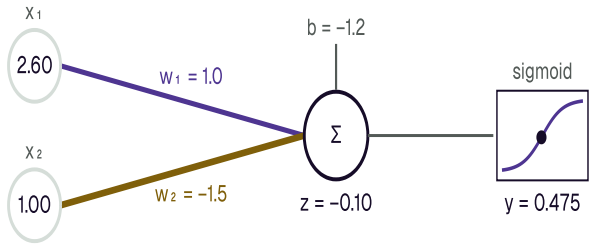
The widget below puts all of this on one picture. With two inputs the hyperplane is a line on a flat map, and you can move it with your hands.

Interactive **One neuron, two inputs** (shown at its default setting)

Move the weights and the bias until every good cycling day sits on the shaded "yes" side, click any day to see its sum worked out, then switch the data to the XOR pattern and try again.



Wed: 26 °C, 1.0 mm/h rain (labelled yes)




```
w₁·x₁ = 1.0 × 2.60 = 2.60
w₂·x₂ = −1.5 × 1.00 = −1.50
z = 2.60 − 1.50 − 1.20 = −0.10
y = sigmoid(−0.10) = 0.475, < 0.5, no
```

● labelled yes ● labelled no shading: what the neuron outputs at each spot dashed line: z = 0

Points on the correct side	Weighted sum z (selected)	Output y (selected)
6 / 7	**−0.10**	**0.475**

Wed has z = −0.10, so it lands on the **no** side, but it is labelled yes: it is misclassified (dashed ring). 6 of 7 days are on the correct side. Try a smaller rain penalty (w_2 closer to 0) or a different bias. The weights set the tilt of the line (the arrow w is perpendicular to it and points toward yes); the bias slides it. Sigmoid fades near the line: the neuron is unsure there and confident far away. Changing the activation never moves the line; it only changes how the output behaves on each side.

Why the bend matters: non-linear activations

Plain words

Converting units twice is still converting units

Convert miles to kilometres, then kilometres to metres. Two steps, but you could have done it in one: multiply by 1,609. Chaining "multiply and add" steps always collapses into a single "multiply and add" step. A stack of a hundred neurons with no bend is exactly as powerful as one neuron. It can only draw straight lines.

The activation function breaks that. ReLU's corner at zero means a neuron is off for some inputs and on for others. Combine many of those corners and you can trace curves, bumps, and any shape you like. Without the bend, depth buys nothing.

For leaders

A settled choice, mostly

Activation functions were a lively research topic for a decade. For LLMs the question is largely settled: GELU (BERT, GPT-2) and the gated variant SwiGLU (PaLM, LLaMA and most open models since) are standard. Switching between modern choices gives small, real gains in quality per unit of compute (Shazeer, 2020, reported SwiGLU among the best of several gated variants on transformer language modelling) and is not where a frontier program finds its next big win.

What still matters: some activations interact with numerical precision and quantization (large activation outliers make 8-bit and 4-bit inference harder), and gated variants add a third weight matrix to each MLP block, which changes parameter counts and memory planning.

Watch out

A proposal that claims a big quality jump from a new activation function alone, measured on one small model, usually shrinks or vanishes at scale. Ask for the comparison at matched compute across at least two model sizes, with several seeds at the smallest size.

Under the hood

The collapse, and the common activations

Two linear layers compose into one:

$$W_2(W_1\mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2 = \underbrace{(W_2 W_1)}_{W}\mathbf{x} + \underbrace{(W_2\mathbf{b}_1 + \mathbf{b}_2)}_{\mathbf{b}}$$

Any depth of purely linear layers is a single affine map $W\mathbf{x} + \mathbf{b}$. The only thing extra depth adds is a constraint on the rank of W.

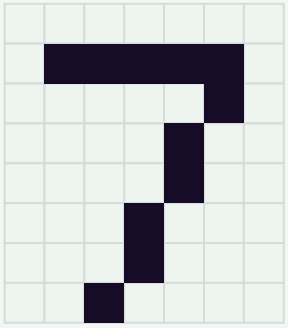
Activation	Formula	Where you meet it	Gradient note
Sigmoid	$\sigma(z) = 1/(1+e^{-z})$	Early nets, gates, output probabilities	Derivative at most 0.25, so deep stacks shrink gradients toward zero
tanh	$\tanh z$	RNNs, LSTMs	Zero-centred, still saturates at both ends
ReLU	$\max(0, z)$	Default in vision since AlexNet (Krizhevsky et al., 2012)	Gradient is exactly 0 or 1; units stuck at 0 for every input are "dead"
GELU	$z\,\Phi(z)$	BERT, GPT-2 (Hendrycks and Gimpel, 2016)	A smooth ReLU; small negative outputs allowed
SwiGLU	$(\mathrm{SiLU}(\mathbf{x}W) \odot \mathbf{x}V)\,W_2$	PaLM, LLaMA (Shazeer, 2020)	One branch gates the other; three matrices per MLP

Φ is the standard normal CDF, $\mathrm{SiLU}(z) = z\,\sigma(z)$, and $\odot$ is element-wise product. In the widget at the end of this chapter, choose "none" as the activation and the fit is a straight line however many units you add.

Diagram

Reading a handwritten 7

How layers build complex features out of simple ones. Simplified, but true to what trained image networks do.

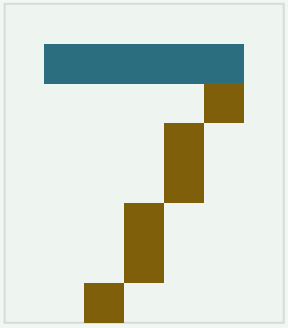


What the network receives:

0 0 0 0 0 0 0
0 1 1 1 1 1 0
0 0 0 0 0 1 0
0 0 0 0 1 0 0 … 56 numbers

No "line", no "corner", no "seven". Just brightness.

To the network, a handwritten 7 is a grid of brightness numbers: 56 of them here, about 150,000 for a small colour photo. Nothing in the input says "line" or "corner". Whatever structure the network uses, it has to build itself.



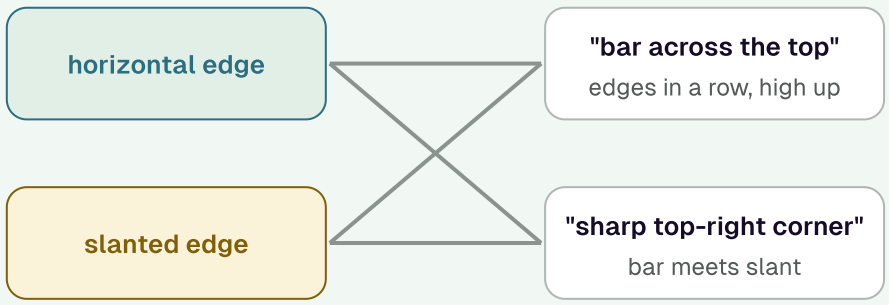
layer 1 lights up

"horizontal edge"
unit fires: 5 places

"slanted edge"
unit fires: 6 places

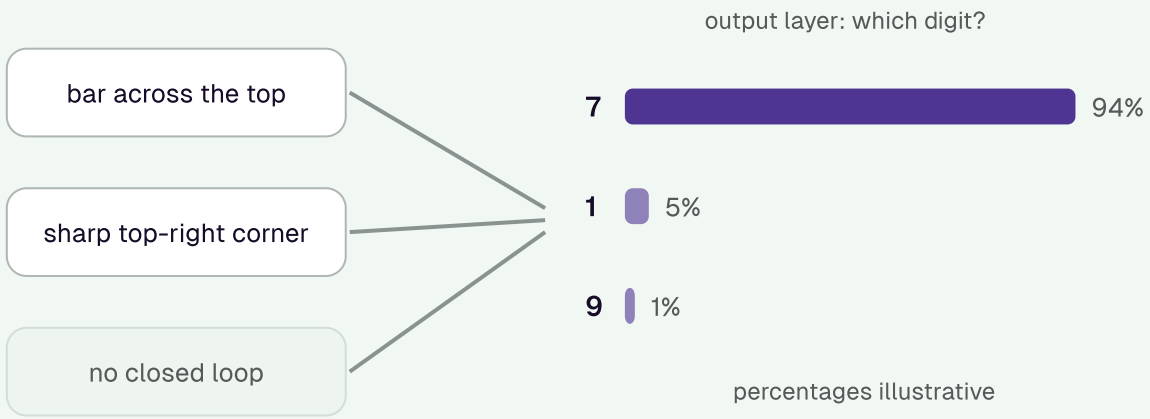
Each unit looks at a tiny patch and asks one simple question.

The first layer's units each respond to something tiny and local, such as "bright above, dark below" (a horizontal edge) or a slanted edge. Nobody told them to. When researchers visualise what trained image networks respond to, the first layer is reliably full of edge and colour detectors (Zeiler and Fergus, 2014).

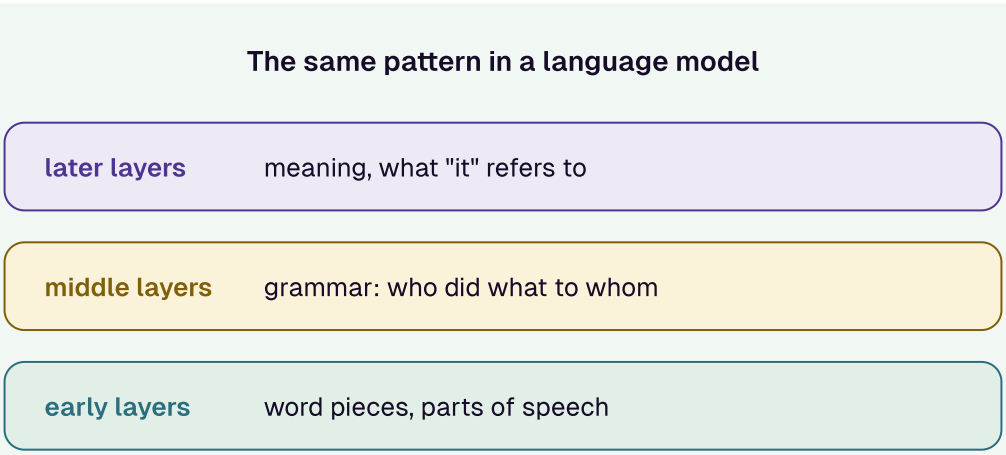


Same neuron as before. Its inputs are now features instead of pixels.

The next layer's inputs are the first layer's outputs. A unit there can fire for "edges in a row near the top" or "a bar meeting a slant at a sharp corner". It is the same weighted-sum-and-bend neuron as before; only its inputs got more meaningful.



The last layer weighs the parts: a top bar plus a sharp corner and no closed loop looks like a 7, not a 9. Each layer reuses the features below it, which is why depth is so efficient: a corner detector built once serves every digit that has a corner.



A tendency found by probing (Tenney et al., 2019), not a strict division of labour.

Language models show a similar ordering. Probing studies of BERT found that information about parts of speech peaks in earlier layers, sentence structure in the middle, and harder semantic relations like coreference later (Tenney et al., 2019). The boundaries are blurry, but the build-up from simple to complex is real.

Layers: from simple features to complex ones

Plain words

An assembly line for ideas

A **layer** is a row of neurons that all look at the same inputs. The outputs of one layer become the inputs of the next, like stations on an assembly line. The first station handles raw material (pixels, or pieces of words). Each later station works with what the previous one made, so the "parts" get bigger and more meaningful as you go: edges, then shapes, then objects.

The layers in the middle are called **hidden layers** because you never see their outputs directly. You only see the input you gave and the answer at the end.

For leaders

Depth, width, and where to read more

Two knobs shape a network: **depth** (how many layers) and **width** (how many units per layer). Depth lets the network compose features, reusing simple parts to build complex ones. Width gives each layer more features to work with. Frontier LLMs have tens to over a hundred layers and widths in the thousands to tens of thousands. The ratio is a design choice that scaling-law work has found to matter much less than total size, within a broad sensible range (Kaplan et al., 2020).

For how a transformer specifically arranges its layers (attention blocks and MLP blocks), see architecture and inside an LLM. For the different model families built from these parts, see the model family tree.

Under the hood

A layer is a matrix multiply

Stack the weight vectors of a layer's m neurons as rows of a matrix $W \in \mathbb{R}^{m \times n}$. Then the whole layer is

$$\mathbf{h}^{(\ell)} = \phi\left(W^{(\ell)}\,\mathbf{h}^{(\ell-1)} + \mathbf{b}^{(\ell)}\right), \qquad \mathbf{h}^{(0)} = \mathbf{x}$$

ϕ is applied to each element. A network of L layers is the composition $f(\mathbf{x}) = f_L \circ \cdots \circ f_1(\mathbf{x})$. The last layer usually skips the activation (regression) or applies softmax (classification, including next-token prediction).

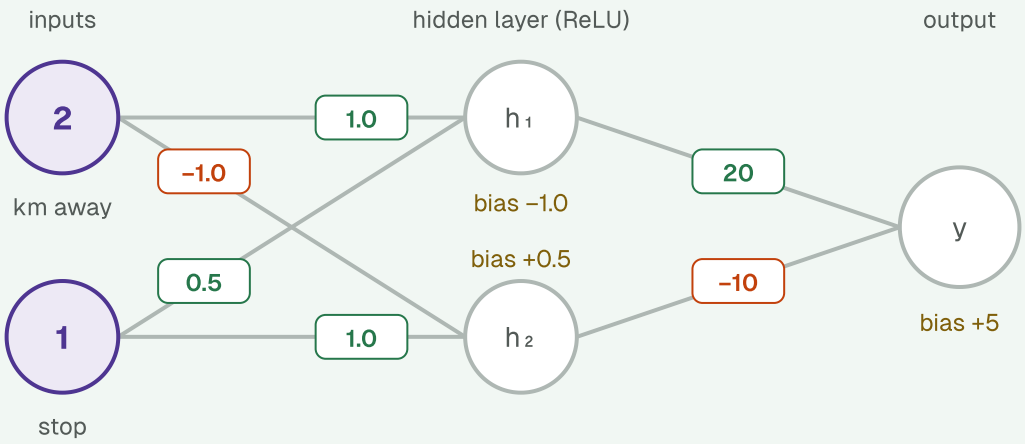
This is why GPUs matter: a layer is one matrix multiply plus an element-wise function, and matrix multiplies are what GPU hardware is built for. Process a batch of inputs at once and the matrix-vector product becomes a matrix-matrix product, which uses the hardware far more efficiently.

Why depth helps, formally: some functions that a deep network represents with a modest number of units need exponentially many units in a shallow one. Telgarsky (2016) constructs such functions for ReLU networks. Depth changes what is cheap to represent.

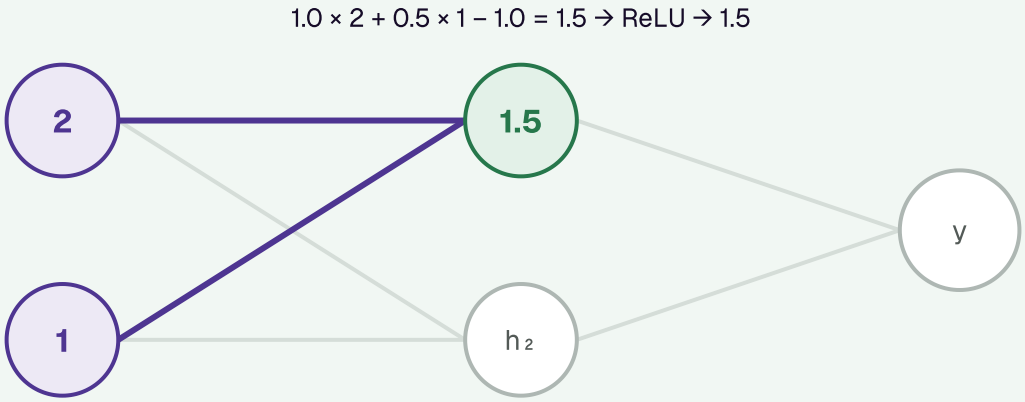
Diagram

Following the numbers through a tiny network

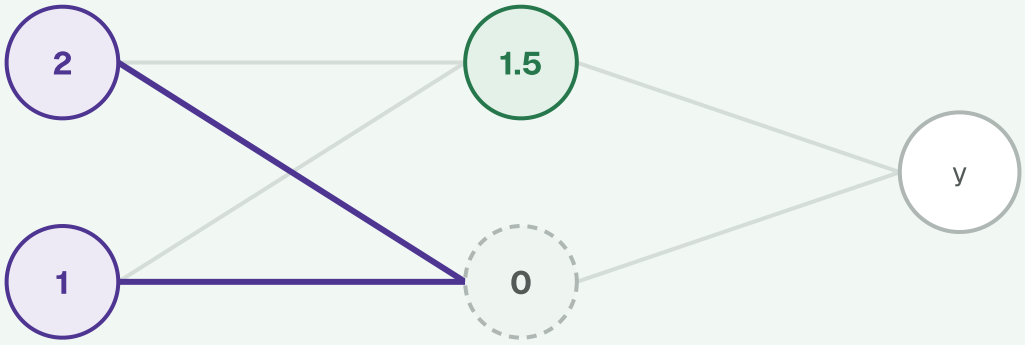
A forward pass by hand: estimating a delivery time from distance and number of stops. The weights are made up; training would set them.



Two inputs (2 km away, 1 stop), a hidden layer of two ReLU neurons, and one output neuron with no bend. Every connection has a weight, and every neuron after the inputs has a bias. Green weights push up, orange ones push down.

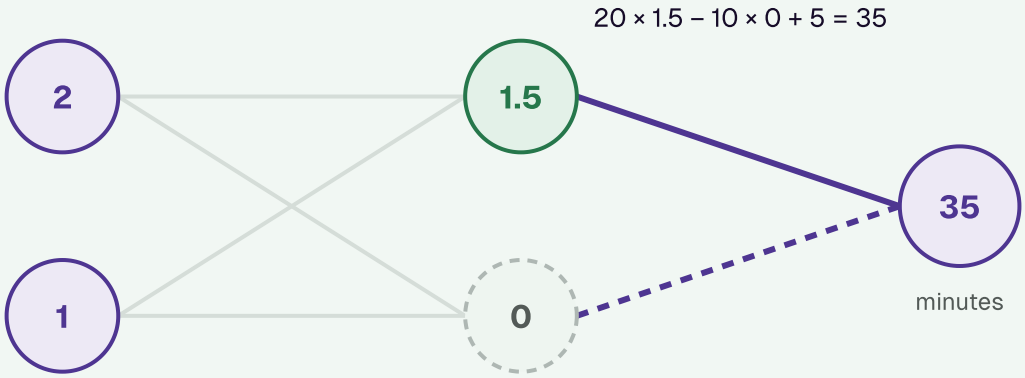


Hidden unit 1 takes a weighted sum of both inputs plus its bias: 1.0 × 2 + 0.5 × 1 − 1.0 = 1.5. That is positive, so ReLU passes it through unchanged. Think of h_1 as a rough "how much driving" signal.

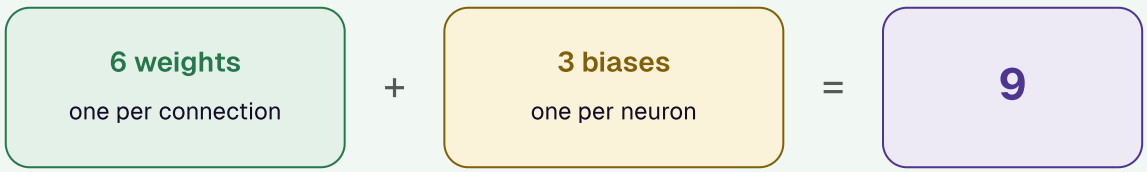


−1.0 × 2 + 1.0 × 1 + 0.5 = −0.5 → ReLU → 0 (off)

Hidden unit 2: −1.0 × 2 + 1.0 × 1 + 0.5 = −0.5. Negative, so ReLU outputs 0 and the unit is switched off for this input. It might switch on for a short trip with many stops. Different inputs switch on different units; that is how the network carves up the input space.



The output neuron weighs the hidden units: 20 × 1.5 − 10 × 0 + 5 = 35 minutes. It has no ReLU, so it can output any number. Input to output, left to right, one layer at a time: that is a **forward pass**. It is the only thing a network does when you use it.



This toy network	**9 parameters**
GPT-3 (Brown et al., 2020)	**175,000,000,000**
Same arithmetic per parameter	**multiply, add, bend**

Six weights plus three biases: this network has 9 **parameters**. A frontier model is the same arithmetic with billions of parameters and dozens of layers. "A 70B model" means 70 billion of these numbers.

The forward pass

Plain words

Numbers flow left to right

Running a network is called a **forward pass**. You put numbers in on the left, each layer does its weighted sums and bends, and the answer comes out on the right. No thinking happens elsewhere; the forward pass is the whole computation. When you chat with an LLM, every single word it writes is the result of one forward pass through the full network.

For leaders

Why forward passes set the serving bill

Because each generated token needs a full forward pass, serving cost scales with parameter count times tokens generated. A rule of thumb from Kaplan et al. (2020): a forward pass costs about 2 floating-point operations per parameter per token (one multiply and one add), ignoring the attention term that grows with context length. A 70B dense model therefore needs roughly 140 billion operations for every token it writes. Training costs about three times that per token, since the backward pass (next chapter) costs about twice the forward pass.

That arithmetic drives the business case for distillation, mixture-of-experts, and quantization, covered in inference.

Under the hood

The same pass in code

The story's network, written as matrices: $W_1 = \begin{pmatrix} 1.0 & 0.5 \\ -1.0 & 1.0 \end{pmatrix}$, $\mathbf{b}_1 = (-1.0, 0.5)$, $\mathbf{w}_2 = (20, -10)$, $b_2 = 5$.

python

```python
import numpy as np

x = np.array([2.0, 1.0])                # distance in km, number of stops
W1 = np.array([[1.0, 0.5],              # hidden unit 1's weights
               [-1.0, 1.0]])            # hidden unit 2's weights
b1 = np.array([-1.0, 0.5])
w2 = np.array([20.0, -10.0])            # output weights
b2 = 5.0

def relu(z):
    return np.maximum(z, 0.0)           # keep positives, zero out negatives

z = W1 @ x + b1                         # both hidden weighted sums in one matrix multiply
h = relu(z)                             # the bend
y = w2 @ h + b2                         # output layer: weighted sum, no bend
print(f"z = {z[0]:.1f}, {z[1]:.1f}")    # z = 1.5, -0.5
print(f"h = {h[0]:.1f}, {h[1]:.1f}")    # h = 1.5, 0.0
print(f"y = {y:.1f} minutes")           # y = 35.0 minutes
print(W1.size + b1.size + w2.size + 1)  # 9 parameters
```

rust

```rust
fn relu(z: f64) -> f64 {
    z.max(0.0) // keep positives, zero out negatives
}

fn main() {
    let x = [2.0, 1.0]; // distance in km, number of stops
    let w1 = [[1.0, 0.5], // hidden unit 1's weights
              [-1.0, 1.0]]; // hidden unit 2's weights
    let b1 = [-1.0, 0.5];
    let w2 = [20.0, -10.0]; // output weights
    let b2 = 5.0;

    // both hidden weighted sums (a matrix multiply, written out)
    let z: Vec<f64> = (0..2).map(|i| w1[i][0] * x[0] + w1[i][1] * x[1] + b1[i]).collect();
    let h: Vec<f64> = z.iter().map(|&v| relu(v)).collect(); // the bend
    let y = w2[0] * h[0] + w2[1] * h[1] + b2; // output layer: weighted sum, no bend
    println!("z = {:.1}, {:.1}", z[0], z[1]); // z = 1.5, -0.5
    println!("h = {:.1}, {:.1}", h[0], h[1]); // h = 1.5, 0.0
    println!("y = {:.1} minutes", y); // y = 35.0 minutes
    println!("{}", 4 + b1.len() + w2.len() + 1); // 9 parameters
}
```

Parameters: what "a 70B model" counts

Plain words

Knobs on a giant mixing desk

Picture a sound engineer's mixing desk with a knob for every connection in the network. Each knob is a **parameter**: a weight or a bias. "A 70B model" has 70 billion knobs. Training is the process of turning all of them, a tiny bit at a time, until the network's outputs sound right.

More knobs means the network can capture more patterns, the same way a bigger desk can mix a more complex song. It also means more memory to store the knobs and more arithmetic every time the network runs.

For leaders

What the number does and does not tell you

Parameter count is the single most quoted number about a model, and it answers three practical questions well:

- **Memory.** Each parameter takes 2 bytes at 16-bit precision. A 70B model needs about 140 GB just to hold its weights, before any working memory, which is why it does not fit on one 80 GB GPU without quantization.
- **Serving compute.** About 2 operations per parameter per generated token for a dense model.
- **Training compute.** About 6 operations per parameter per training token (Kaplan et al., 2020).

It answers capability badly on its own. Chinchilla (Hoffmann et al., 2022), at 70B parameters trained on 1.4 trillion tokens, beat the 280B-parameter Gopher trained on 300 billion tokens. Data volume and quality, training recipe, and post-training all matter as much as size; see scaling laws.

Mixture-of-experts models complicate the count. Only some "expert" sub-networks run for each token, so they quote total parameters (memory) and active parameters (compute per token) separately. Mixtral 8x7B (Mistral AI, 2024) has about 47B total and about 13B active.

Principal asks

Is that total or active parameters? What precision will it serve at, and what does the weight memory come to on our hardware? How many training tokens per parameter, and how does that compare with the compute-optimal ratio? If two models differ in size, were they trained on the same data and compute, or is size standing in for everything else that differs?

Under the hood

Counting a real transformer

For a decoder-only transformer, almost every parameter lives in weight matrices inside the repeated blocks. Using the published Llama 2 70B shape (Touvron et al., 2023): width 8,192, 80 layers, MLP width 28,672, 64 query heads sharing 8 key/value heads (grouped-query attention), and a 32,000-token vocabulary:

python

```python
def count_params(d, layers, ffn, vocab, heads, kv_heads):
    head_dim = d // heads
    attn = 2 * d * d + 2 * d * kv_heads * head_dim   # Q and output full size; K and V shrunk by grouped-query attention
    mlp = 3 * d * ffn                                # SwiGLU has three matrices: gate, up, down
    norms = 2 * d                                    # two RMSNorm scale vectors per layer
    embed = 2 * vocab * d                            # input embedding plus a separate output head
    return layers * (attn + mlp + norms) + embed + d # + the final norm

n = count_params(d=8192, layers=80, ffn=28672, vocab=32000, heads=64, kv_heads=8)
print(f"{n:,}")                                                    # 68,976,648,192
print(f"{n / 1e9:.1f}B parameters, {2 * n / 1e9:.0f} GB at 2 bytes each")  # 69.0B parameters, 138 GB at 2 bytes each
```

rust

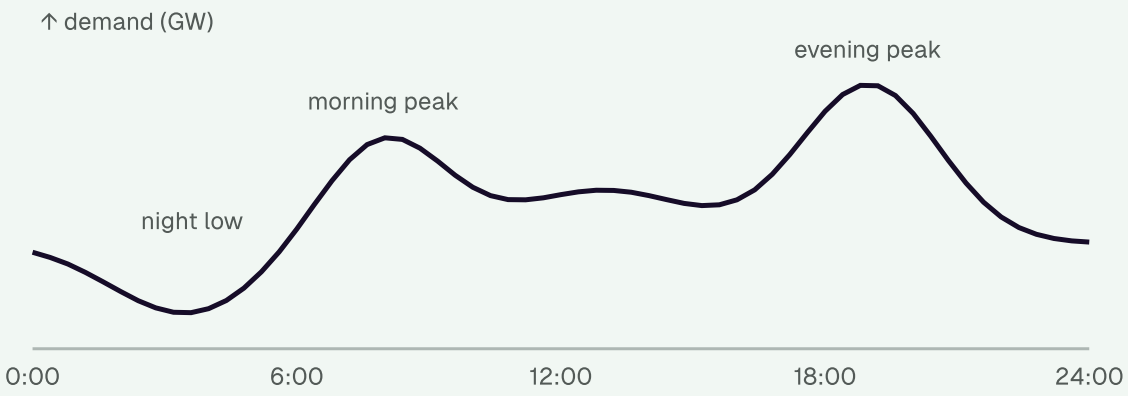
```rust
fn count_params(d: u64, layers: u64, ffn: u64, vocab: u64, heads: u64, kv_heads: u64) -> u64 {
    let head_dim = d / heads;
    let attn = 2 * d * d + 2 * d * kv_heads * head_dim; // Q and output full size; K and V shrunk by grouped-query attention
    let mlp = 3 * d * ffn; // SwiGLU has three matrices: gate, up, down
    let norms = 2 * d; // two RMSNorm scale vectors per layer
    let embed = 2 * vocab * d; // input embedding plus a separate output head
    layers * (attn + mlp + norms) + embed + d // + the final norm
}

/// Formats 68976648192 as "68,976,648,192", like Python's f"{n:,}".
fn with_commas(n: u64) -> String {
    let s = n.to_string();
    let mut out = String::new();
    for (i, c) in s.chars().enumerate() {
        if i > 0 && (s.len() - i) % 3 == 0 {
            out.push(',');
        }
        out.push(c);
    }
    out
}

fn main() {
    let n = count_params(8192, 80, 28672, 32000, 64, 8);
    println!("{}", with_commas(n)); // 68,976,648,192
    let nf = n as f64;
    println!("{:.1}B parameters, {:.0} GB at 2 bytes each", nf / 1e9, 2.0 * nf / 1e9); // 69.0B parameters, 138 GB at 2 bytes each
}
```

The breakdown is worth remembering: the MLP blocks hold about 82% of the parameters (56.4B), attention about 17.5% (12.1B), and embeddings plus norms under 1%. Biases are absent: most modern LLMs drop them from linear layers because they add little. "70B" is a rounded marketing number for 68.98 billion.

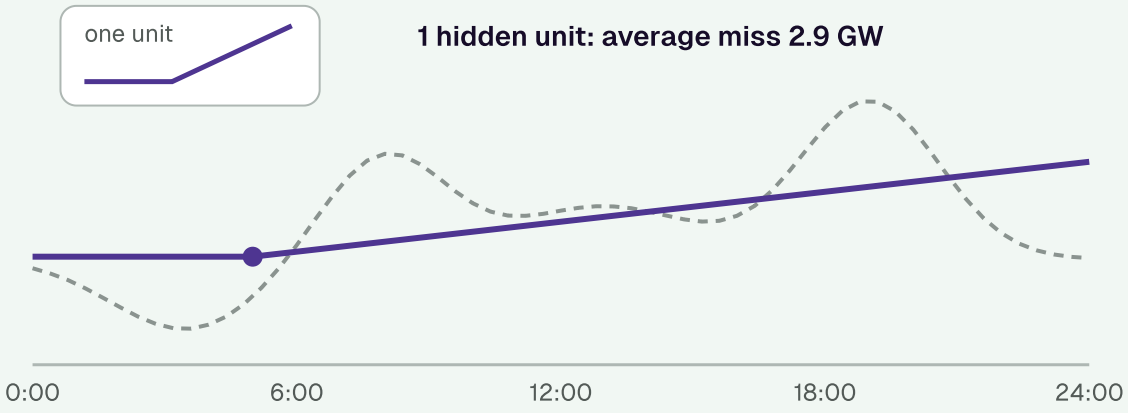
Diagram

Tracing a day of electricity demand with hinges

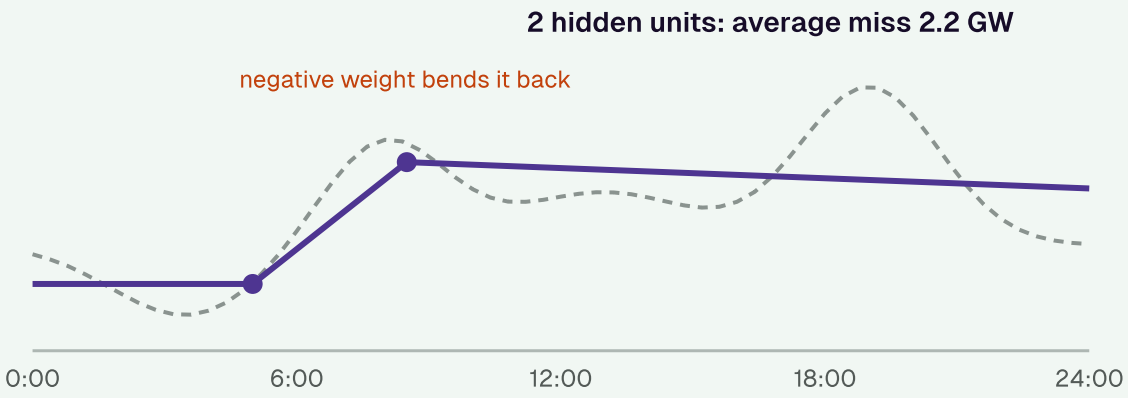
Why a network with enough ReLU units can draw any curve. The demand curve is illustrative; the fits are real least-squares fits.



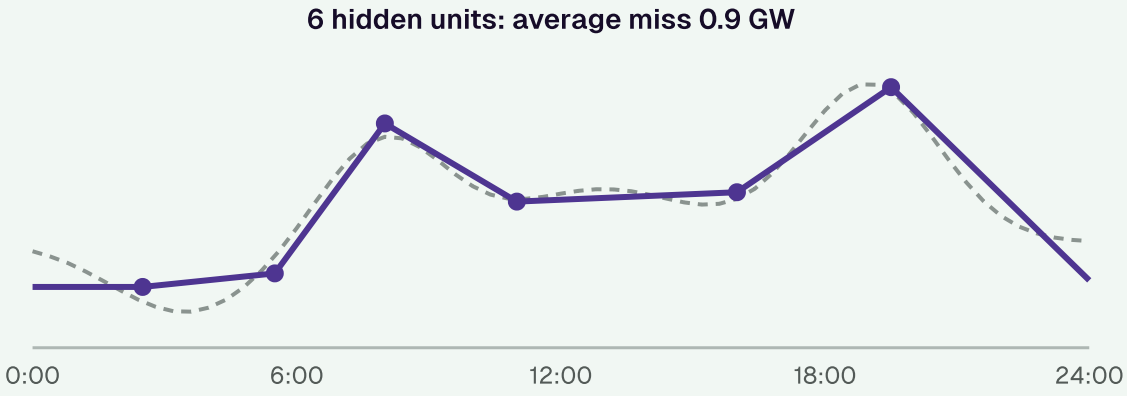
A grid operator wants a network that takes the hour of the day and outputs expected demand: low overnight, a morning peak, a midday plateau, a big evening peak. One input, one output, one hidden layer of ReLU units. How many units does it take?



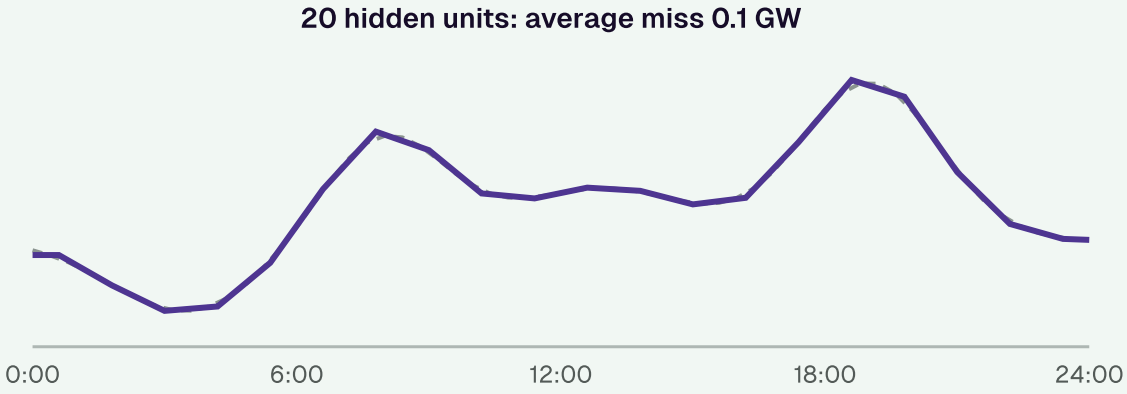
A single ReLU unit is a **hinge**: flat, then a straight ramp starting at its threshold (here about 5:00). Scaled and shifted by the output weight and bias, the best one hinge can do is a crude "demand rises during the day". Average miss: 2.9 GW.



Add a second hinge with a negative output weight and the ramp bends back down at about 8:30. Every extra unit adds one more corner where the slope can change. Average miss: 2.2 GW.



With six hinges the network can follow the night low, both peaks, and the plateau, as a sequence of straight segments. It looks like a child's drawing of the curve, but it already has the right shape. Average miss: 0.9 GW.



More hinges, more corners, closer fit.

Twenty hinges and the fit is within 0.1 GW almost everywhere. Keep adding units and the error keeps falling toward zero. That is the **universal approximation** idea in one picture: enough bends can trace any reasonable curve. It says nothing about whether training will find those bends, or what the network does at hours it never saw.

Universal approximation: what it promises and what it does not

Plain words

Enough bends can draw anything

The universal approximation theorem says that a network with one hidden layer and enough units can imitate any smooth-enough curve as closely as you like. The story above is the whole idea: each unit adds a bend, and enough bends can trace any shape.

It is a statement that a good setting of the knobs *exists*. It does not say how many knobs you need (possibly an astronomical number), whether training will find the right setting, or whether the network will give sensible answers for inputs unlike its training data. Those three gaps are where all the real work in machine learning happens.

For leaders

An argument you will hear misused

"Neural networks are universal approximators, so the model can learn this" is a common and empty argument. The theorem guarantees representation only. Whether a capability is learned depends on three things it is silent about: whether the data contains the signal, whether optimization finds it (next chapter), and whether what was learned generalizes beyond the training examples.

The widget below makes the third gap visible. Switch to "20 noisy points" and add units: error on the training points keeps falling while error against the true curve stops improving and gets worse. Fitting the data you have is not the same as learning the pattern that produced it.

Watch out

Universal approximation is an existence theorem. It is often cited to dismiss a concern ("a big enough model can represent that"). The useful questions are about data, optimization, and generalization, and the theorem answers none of them.

Under the hood

The statement and its fine print

For any continuous f on a compact set $K \subset \mathbb{R}^n$ and any $\varepsilon > 0$, there exist N and parameters $v_j, b_j \in \mathbb{R}$, $\mathbf{w}_j \in \mathbb{R}^n$ such that

$$\sup_{\mathbf{x} \in K} \left| f(\mathbf{x}) - \sum_{j=1}^{N} v_j \, \phi(\mathbf{w}_j^\top \mathbf{x} + b_j) \right| < \varepsilon$$

Proved for sigmoid activations by Cybenko (1989) and Hornik, Stinchcombe and White (1989). Leshno et al. (1993) showed it holds exactly when ϕ is not a polynomial, which covers ReLU, tanh and GELU.

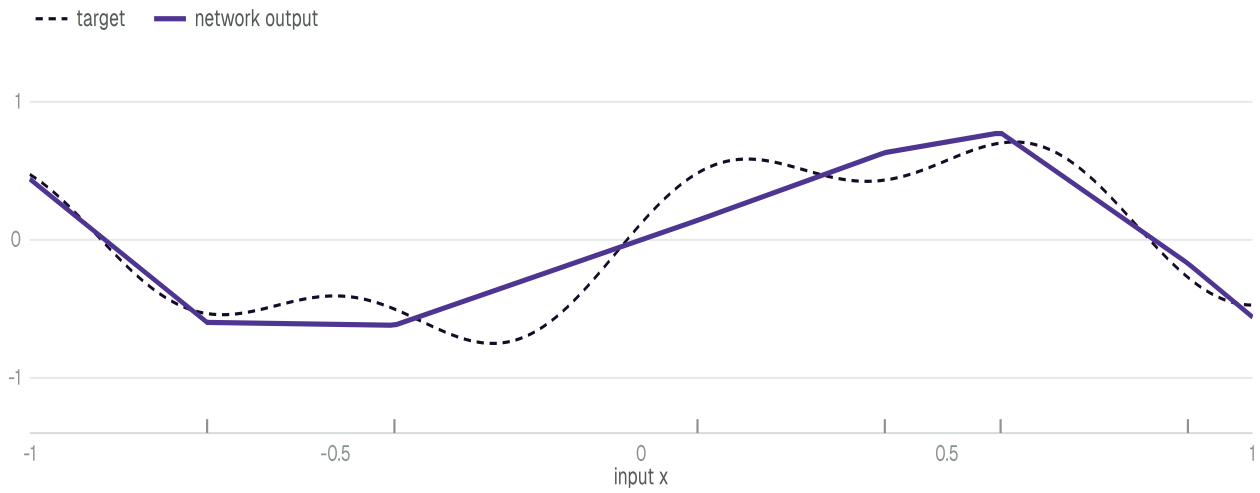
The fine print:

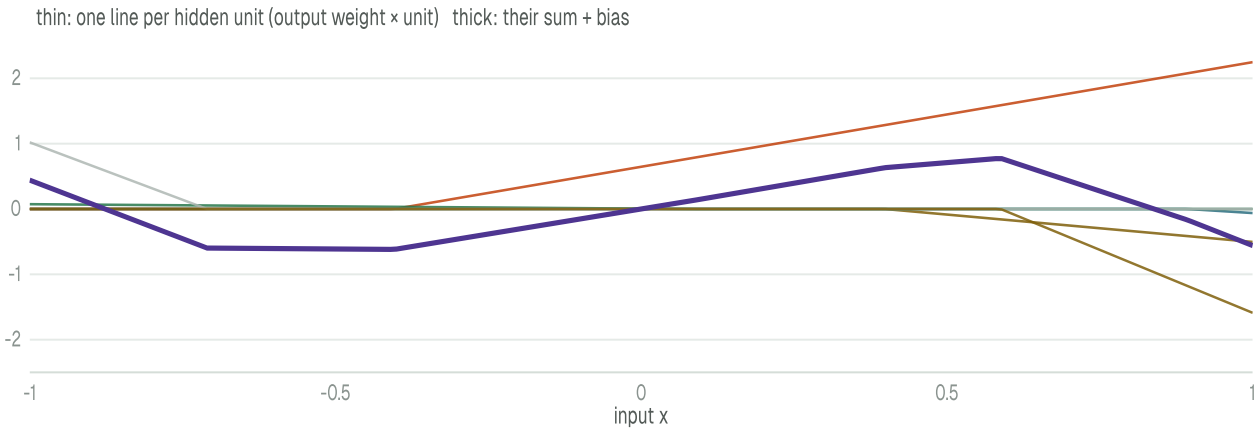
- **No bound on N.** For general continuous functions the number of units can grow exponentially with input dimension. Barron (1993) showed that for functions whose Fourier transform has a bounded first moment, squared error falls like $1/N$ regardless of dimension, which is one reason smooth high-dimensional targets are tractable.
- **In one dimension, ReLU networks are piecewise-linear functions.** $g(x) = a + \sum_j v_j \max(0, x - t_j)$ has a kink at each t_j. Linear interpolation of a twice-differentiable f with kink spacing h has error at most $h^2 \max |f''|/8$, so error falls like $1/N^2$ when the kinks are well placed.
- **Depth changes the economics.** Telgarsky (2016) gives functions that a deep ReLU network computes with few units but any shallow one needs exponentially many units to approximate.
- **Nothing about learning or generalization.** The theorem is silent on whether gradient descent reaches a good $\{v_j, \mathbf{w}_j, b_j\}$, and on behaviour off the training data.

Interactive

Build a curve from hidden units (shown at its default setting)

A network with one input, one hidden layer of H units, and one output, fitted to a wiggly target. The top panel shows the target (dashed) and the network's output. The bottom panel shows what each hidden unit contributes; the output is their sum plus a bias. Method: each unit's input weight and bias are fixed at random, and the output weights are solved exactly by least squares with a tiny ridge penalty. Real networks train all the weights by gradient descent (next chapter); this shortcut keeps the demo instant and deterministic.





Parameters (3H + 1)	Error on training data	Error against true curve
19	**0.181**	**0.181**

With 6 ReLU units the output has up to 6 corners (ticks on the axis). Average miss: **0.181**. Add units to add bends.

Eval lens

The two error cards are the whole story of evaluation in miniature. "Error on training data" is what the optimizer sees and drives down; "error against true curve" is what you care about, and it can only be measured on data the model was not fitted to. In noisy mode, past about 12 units the training error keeps shrinking while the true error stalls or rises. Any eval that reuses training data, or data that leaked into training, reports the first number while claiming the second. See contamination.

Key idea

A neural network is weighted sums with bends, stacked. The weights are the parameters; the bends make depth worth having; the forward pass is the entire computation. Everything else in this guide, from attention to RLHF, is about how to arrange those parts and how to set the weights.

Part III · Chapter 19

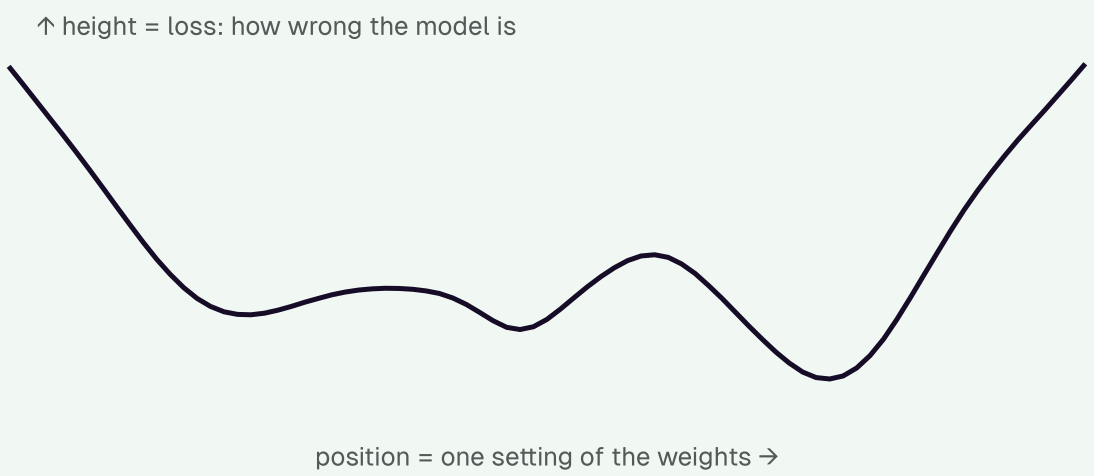
How networks learn: gradient descent and backpropagation

The previous chapter built a network and set its weights by hand. Real networks start with random weights and learn them. Learning means: measure how wrong the network is, work out which way to nudge every weight to make it less wrong, nudge, repeat. This chapter covers the landscape the network walks on, the gradient that points the way, the step rule, the hills and pits and mountain passes along the route, why randomness helps, and backpropagation, the bookkeeping trick that makes it all affordable. Optimizers like Adam and learning-rate schedules build on this and live in optimisation.

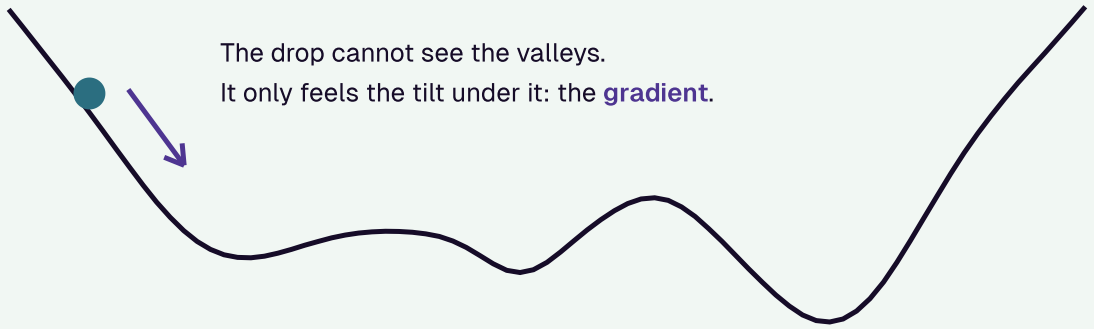
Diagram

Where does the raindrop end up?

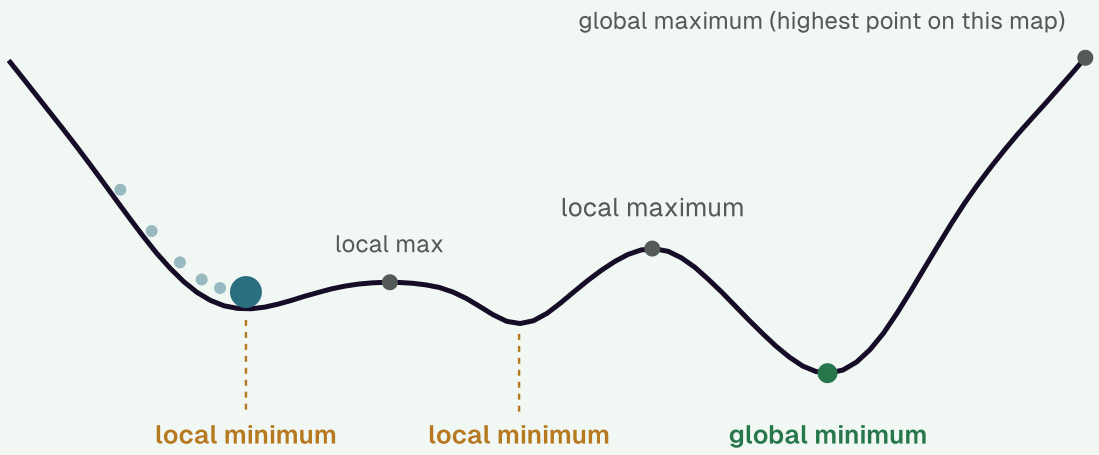
Loss landscapes, minima, maxima and saddle points, told as rain falling on hills.



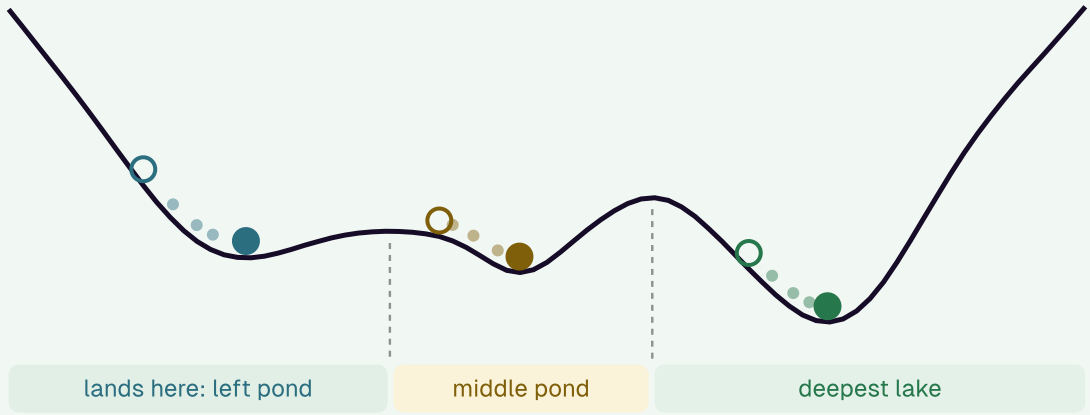
Imagine every possible setting of a model's weights laid out as a landscape. Each spot is one setting; its height is the **loss**, a number for how wrong the model is with those weights. Training is a search for low ground. This picture has one weight, so the landscape is a line; a real model has billions.



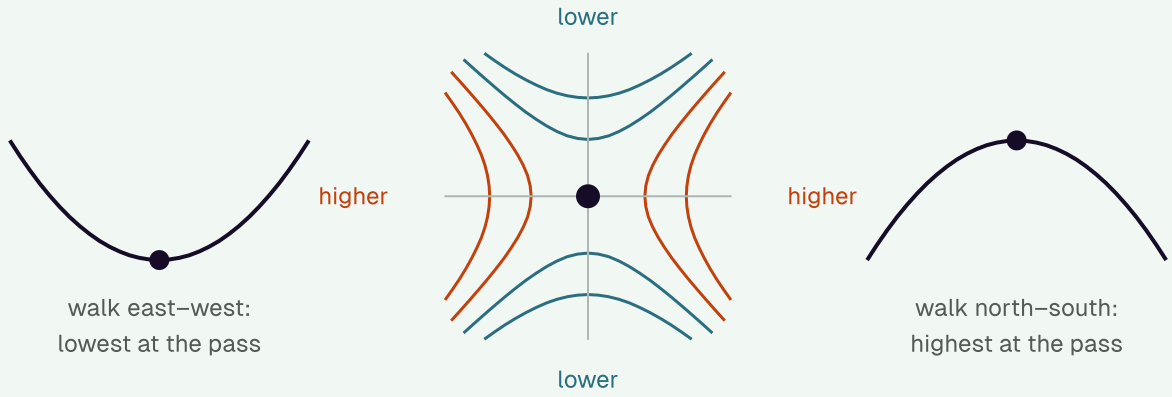
A raindrop lands on a hillside. It cannot see the landscape; it only feels which way the ground tilts right where it is. That local tilt is the **gradient**. Water flows the opposite way to uphill, and so does gradient descent.



The drop rolls into the first pond it meets and stops: every step from there goes uphill. That is a **local minimum**. The deepest lake, the **global minimum**, is elsewhere and the drop will never know. Hilltops are **local maxima**; the highest point on the map is the **global maximum** (here it sits at the edge). At every dip bottom and hilltop the ground is flat, so the gradient is zero.



Where a drop lands decides where it ends. Each pond drains its own patch of ground, called its **basin**, and the hilltops are the borders. For a network, "where it lands" is the random starting weights. In this one-dimensional picture, bad luck at the start means a worse final answer.



With millions of directions, flat spots are almost all passes, not pits.

Add a second dimension and a new kind of flat spot appears: the mountain pass, or **saddle point**. It is the lowest point on the ridge going one way and the highest point on the trail going the other. A drop sitting exactly on it stays put, but the slightest nudge sends it on downhill. With millions of weights, a flat spot that curves up in every single direction is rare. Most flat spots are passes, which slow training down without trapping it.

The loss landscape

Plain words

A score for wrongness, as ground to walk on

To learn, a network needs a single number that says how badly it is doing. That number is the **loss**. For a model predicting delivery times it might be the average squared error in minutes. For a language model it is how surprised the model is by the actual next word, averaged over a huge pile of text.

Every possible setting of the weights gives a different loss. Picture all those settings spread out as ground, with the loss as the height at each spot, and you have the **loss landscape**. Training is walking downhill on it. You never see the landscape as a whole; it has as many directions as the model has weights.

For leaders

The loss is what training optimizes, nothing more

The loss is a proxy chosen because it is smooth and cheap to compute on every example. Next-token cross-entropy for pretraining, a preference loss for reward models, a clipped objective for RL. The

model gets better at whatever the loss rewards, which is why the choice of loss and data is a product decision: everything the loss does not measure, the model has no reason to learn.

Eval lens

What a training loss curve does and does not tell you. A smoothly falling training loss says the optimizer is working. It does not say the model is getting better at anything you care about. Four gaps to keep in mind:

- **Train versus held-out.** Only loss on held-out data measures generalization; a falling training loss with a flat or rising validation loss is memorization.
- **Averages hide tasks.** Loss is an average over billions of tokens, dominated by easy ones. A capability that lives in rare tokens (a correct final answer to a maths problem) can change a lot while the average barely moves. Wei et al. (2022) reported abilities that appear abruptly with scale; Schaeffer et al. (2023) showed much of that abruptness comes from all-or-nothing metrics, while smooth per-token measures improve gradually.
- **Not comparable across tokenizers.** Loss per token depends on how text is split into tokens. Compare models with different tokenizers in bits per byte or on downstream evals, not raw loss.
- **Spikes and plateaus are diagnostics.** A spike can be a bad batch or an unstable learning rate; a long plateau can be a saddle region. Both are reasons to look, not verdicts.

Under the hood

The objective

With parameters $\theta \in \mathbb{R}^P$, training examples (x_i, y_i), and a per-example loss ℓ, the training objective is the empirical risk

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^{N} \ell\big(f_\theta(x_i),\, y_i\big)$$

f_θ is the network. For regression ℓ is often squared error; for language models it is cross-entropy, $-\log p_\theta(\text{next token})$, averaged over every token position (see pretraining). The landscape is the graph of $\mathcal{L}$ over $\mathbb{R}^P$, with P in the billions.

What we actually want is low expected loss on new data, $\mathbb{E}_{(x,y)\sim\mathcal{D}}[\ell(f_\theta(x), y)]$. The empirical risk is an estimate of it; the gap between the two is the generalization gap, and held-out evaluation is how you measure it.

The gradient: which way is downhill

Plain words

One slope per knob

With one weight, the slope is a single number: turn the knob up a hair and see whether the loss goes up or down, and how fast. With a billion weights you need a billion slopes, one per knob, answering "if I turn just this knob up slightly, how much does the loss change?" That list of slopes is the **gradient**. Together the slopes point in the direction where the loss climbs fastest, so you step the opposite way.

For leaders

Gradients are cheap, which is the whole miracle

Measuring a billion slopes one knob at a time would take a billion extra forward passes per step. Backpropagation, at the end of this chapter, gets all of them for roughly the cost of two forward passes. That single fact is why training networks with billions of parameters is possible at all. It is also why training compute is about 6 floating-point operations per parameter per token: 2 for the forward pass and about 4 for the backward pass (Kaplan et al., 2020).

The size of the gradient (its norm) is one of the most watched training signals. Sudden growth often precedes a loss spike; that is why runs use gradient clipping, which caps the norm. See optimisation.

Under the hood

Definition and the step rule

The gradient collects the partial derivatives:

$$\nabla_\theta \mathcal{L} = \left(\frac{\partial \mathcal{L}}{\partial \theta_1}, \ldots, \frac{\partial \mathcal{L}}{\partial \theta_P}\right), \qquad \frac{\partial \mathcal{L}}{\partial \theta_j} \approx \frac{\mathcal{L}(\theta + \epsilon \mathbf{e}_j) - \mathcal{L}(\theta - \epsilon \mathbf{e}_j)}{2\epsilon}$$

$\mathbf{e}_j$ is the unit vector for parameter j. The finite-difference form is how you check a gradient implementation, never how you compute gradients in training.

A first-order Taylor expansion shows why stepping against the gradient works:

$$\mathcal{L}(\theta - \eta \nabla \mathcal{L}) \approx \mathcal{L}(\theta) - \eta \|\nabla \mathcal{L}\|^2$$

For a small enough step size $\eta > 0$ the loss goes down by about η times the squared gradient norm. Among all directions of a given length, $-\nabla \mathcal{L}$ gives the largest first-order decrease.

Hence **gradient descent**:

$$\theta_{t+1} = \theta_t - \eta \nabla_\theta \mathcal{L}(\theta_t)$$

η is the learning rate. Every optimizer used for LLMs is a variation that changes the direction (momentum) or the per-parameter step size (Adam).

Step size: the learning rate

Plain words

Stride length

The gradient says which way is downhill. The **learning rate** says how big a stride to take. Tiny strides are safe but slow; you may run out of time before reaching the bottom. Huge strides overshoot: you leap across the valley and land higher up the other side, and the next leap is even bigger. Somewhere in between you make fast, steady progress.

For leaders

The most expensive number to get wrong

A learning rate that is too high can waste a run outright; one that is too low wastes compute quietly by finishing at a worse loss. Large runs do not use one fixed value: they warm up, hold, and decay it, and they pick its scale from small-model sweeps. That material, plus Adam, momentum, and batch size, is in optimisation. The widget below shows a different face of the learning rate: on a bumpy landscape, a big step can jump out of a valley entirely.

Under the hood

Why there is a speed limit

Near a minimum, the loss looks like a bowl $\frac{\lambda}{2}(\theta - \theta^*)^2$ with curvature λ. One gradient step gives

$$\theta_{t+1} - \theta^* = (1 - \eta\lambda)\,(\theta_t - \theta^*)$$

The error shrinks only if $|1 - \eta\lambda| < 1$, that is $0 < \eta < 2/\lambda$. For $\eta < 1/\lambda$ it creeps in from one side; between $1/\lambda$ and $2/\lambda$ it zig-zags but converges; above $2/\lambda$ every step overshoots further. In many dimensions the steepest direction sets the limit.

A 1D example with two valleys, $\mathcal{L}(\theta) = \theta^4 - 3\theta^2 + \theta$: a deep valley near $\theta = -1.30$ and a shallow one near 1.13. Same code, three settings:

python

```python
def loss(theta):
    return theta**4 - 3 * theta**2 + theta     # two valleys: one deep, one shallow

def grad(theta):
    return 4 * theta**3 - 6 * theta + 1         # the slope, worked out by hand

def descend(theta, lr, steps):
    for step in range(steps):
        theta = theta - lr * grad(theta)        # step against the slope
        if abs(theta) > 1e6:                    # stepped clean off the map
            return None, step + 1
    return theta, steps

for start, lr in [(2.0, 0.01), (-0.5, 0.01), (2.0, 0.2)]:
    end, n = descend(start, lr, 200)
    if end is None:
        print(f"start {start:+.1f}  lr {lr:.2f}  diverged after {n} steps")
    else:
        print(f"start {start:+.1f}  lr {lr:.2f}  theta {end:+.3f}  loss {loss(end):+.3f}")
# start +2.0  lr 0.01  theta +1.131  loss -1.070   (shallow valley: a local minimum)
# start -0.5  lr 0.01  theta -1.301  loss -3.514   (deep valley: the global minimum)
# start +2.0  lr 0.20  diverged after 5 steps
```

rust

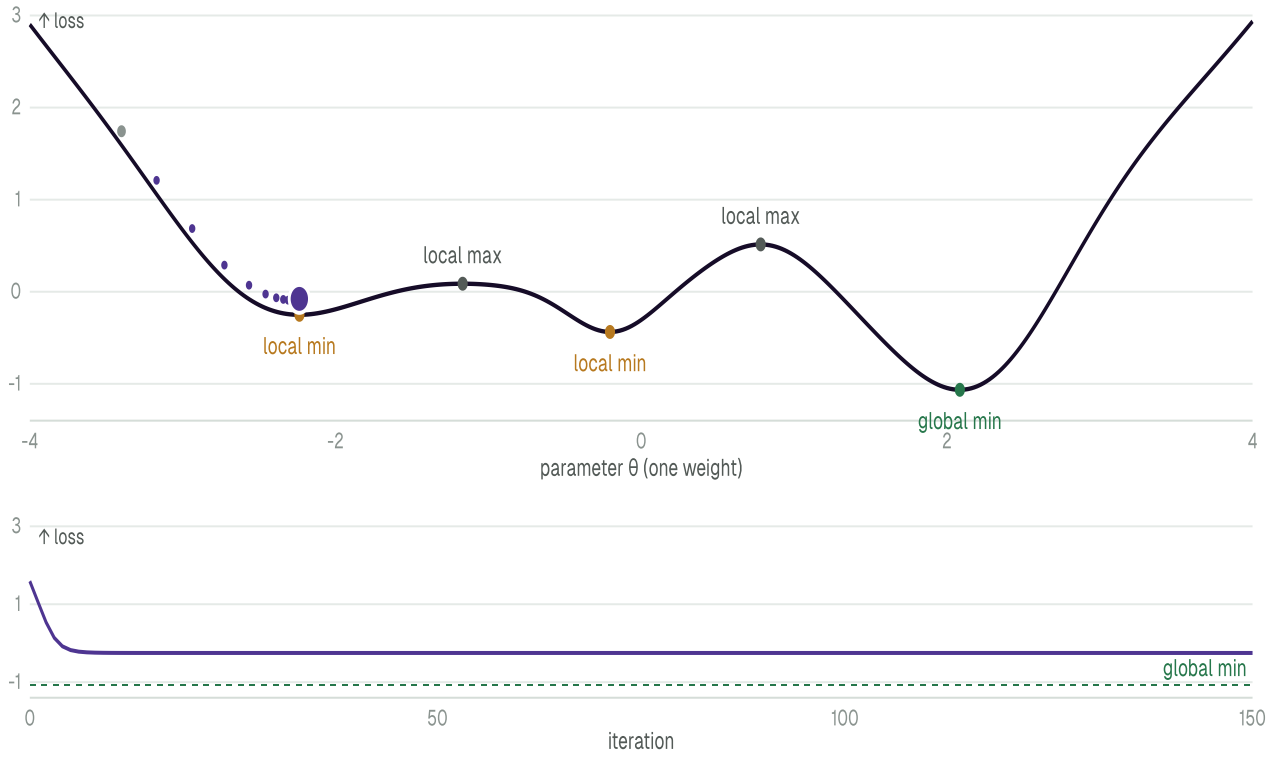
```rust
fn loss(theta: f64) -> f64 {
    theta.powi(4) - 3.0 * theta.powi(2) + theta // two valleys: one deep, one shallow
}

fn grad(theta: f64) -> f64 {
    4.0 * theta.powi(3) - 6.0 * theta + 1.0 // the slope, worked out by hand
}

/// Returns (Some(final theta), steps) or (None, step it diverged at).
fn descend(mut theta: f64, lr: f64, steps: usize) -> (Option<f64>, usize) {
    for step in 0..steps {
        theta -= lr * grad(theta); // step against the slope
        if theta.abs() > 1e6 {
            return (None, step + 1); // stepped clean off the map
        }
```

```
    }
    (Some(theta), steps)
}

fn main() {
    for (start, lr) in [(2.0, 0.01), (-0.5, 0.01), (2.0, 0.2)] {
        match descend(start, lr, 200) {
            (None, n) => println!("start {:+.1}  lr {:.2}  diverged after {} steps", start, lr, n),
            (Some(end), _) => println!("start {:+.1}  lr {:.2}  theta {:+.3}  loss {:+.3}", start, lr, end,
loss(end)),
        }
    }
    // start +2.0  lr 0.01  theta +1.131  loss -1.070   (shallow valley: a local minimum)
    // start -0.5  lr 0.01  theta -1.301  loss -3.514   (deep valley: the global minimum)
    // start +2.0  lr 0.20  diverged after 5 steps
}
```

Interactive

Roll a ball down a bumpy loss (shown at its default setting)

One weight θ, a loss curve with three valleys and two hills. Each step moves the ball by learning rate × slope. Try the presets, then set your own start, step size, momentum, and noise. Press Run to watch, or Step for one update at a time.



Iteration	θ	Loss	Ends in
150	-2.24	-0.251	Local min

Settled in a local minimum at θ = -2.24 after about 17 steps, loss -0.251. The global minimum is at θ = 2.09, but every nearby step goes uphill, so plain descent stays put. Try another start, momentum, or noise.

Minima, maxima, and saddle points

Plain words

Five kinds of flat ground

Anywhere the ground is flat, the gradient is zero and gradient descent stops pushing. There are five kinds of flat spot worth naming:

- **Local minimum:** the bottom of a dip. Every small step from here goes uphill, but there may be deeper dips elsewhere.
- **Global minimum:** the lowest point anywhere. The best possible weights for the training data.
- **Local maximum:** the top of a hill. Every small step goes downhill.
- **Global maximum:** the highest point anywhere. The worst weights; training never wants to be here.
- **Saddle point:** a mountain pass. Downhill in some directions, uphill in others.

The worry in the raindrop story is getting stuck in a poor local minimum. For huge networks that worry turns out to be smaller than the one-dimensional picture suggests, for reasons the next two tabs explain.

For leaders

Why "stuck in a local minimum" is rarely the real problem

Intuition from one or two dimensions says training should constantly get trapped in bad valleys. In very large networks, that is mostly not what happens, for two reasons with good evidence behind them:

- **Most flat spots are saddles.** To be a minimum, the ground must curve upward in every one of billions of directions at once. Dauphin et al. (2014) argued, from theory on random high-dimensional functions and experiments on neural networks, that critical points with high loss are overwhelmingly saddle points, and that saddles, with the flat plateaus around them, are the main thing that slows training.
- **The minima you reach are similarly good.** Choromanska et al. (2015) gave a model in which local minima of large networks concentrate in a narrow band of loss close to the global minimum, and experiments agreed. Practitioners see the same: rerun with a different seed and the final loss is nearly identical.

Honest nuance: bad local minima do exist. Safran and Shamir (2018) showed they are common in small two-layer ReLU networks, and the theory behind the optimistic picture relies on simplifying assumptions. Being over-parameterized is what makes the landscape benign, which is one more argument for scale. The practical failure modes at frontier scale are plateaus, loss spikes, and divergence, not local minima.

The subtler question is which minimum, among many with equal training loss, generalizes best. Keskar et al. (2017) linked large-batch training to "sharp" minima that generalize worse; Dinh et al. (2017) showed sharpness can be changed by reparameterizing without changing the function, so it is not a clean measure. This is open research.

Principal asks

The loss sat flat for two thousand steps and then dropped: what changed, and could we have shortened the plateau? How much does final loss vary across seeds at small scale, and is our improvement bigger than that? Is the gradient norm stable, and how often does clipping trigger? When you say the model is "stuck", is that training loss, validation loss, or a downstream eval?

Under the hood

Classifying critical points with the Hessian

At a critical point $\nabla \mathcal{L}(\theta^*) = 0$, the second-order expansion is governed by the Hessian $H = \nabla^2 \mathcal{L}(\theta^*)$:

$$\mathcal{L}(\theta^* + \boldsymbol{\delta}) \approx \mathcal{L}(\theta^*) + \tfrac{1}{2}\, \boldsymbol{\delta}^\top H\, \boldsymbol{\delta} = \mathcal{L}(\theta^*) + \tfrac{1}{2} \sum_k \lambda_k \left(\mathbf{u}_k^\top \boldsymbol{\delta}\right)^2$$

$\lambda_k, \mathbf{u}_k$ are the eigenvalues and eigenvectors of H: the curvature along each principal direction.

Eigenvalues of H	Critical point	What gradient descent does
All > 0	Local minimum	Settles there
All < 0	Local maximum	Moves away in every direction
Mixed signs	Saddle point (index = number of negative λ_k)	Escapes along negative-curvature directions, slowly if they are shallow
Some $= 0$	Degenerate: needs higher-order terms	Flat valleys, common in over-parameterized nets

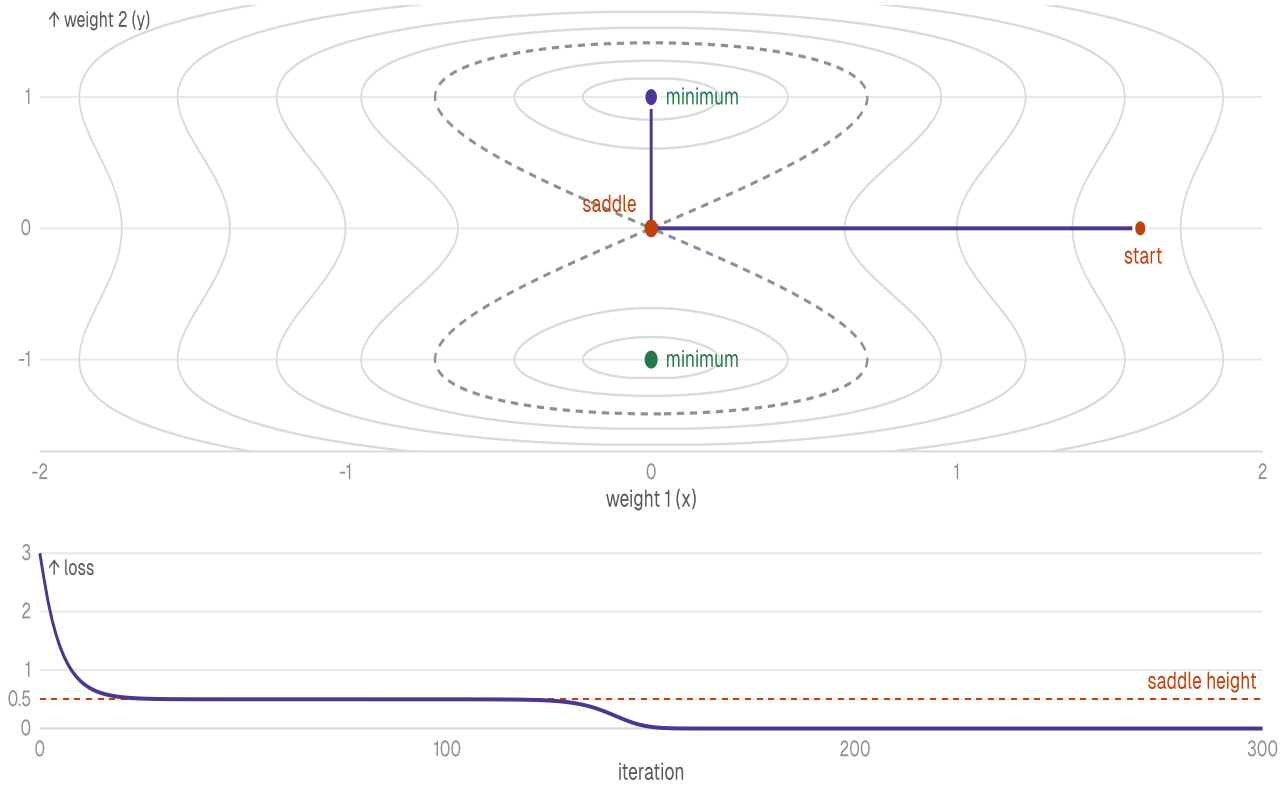
A heuristic, not a theorem: if each of P curvature signs were an independent coin flip, the chance that all are positive would be 2^{-P}. The signs are not independent, but the picture captures the scaling. The rigorous version comes from random-field theory (Bray and Dean, 2007): for Gaussian random functions in high dimension, the fraction of negative eigenvalues at a critical point grows with its loss, so low-index critical points sit at low loss. Dauphin et al. (2014) connected this to neural networks.

Gradient descent with a random start almost surely does not converge to a strict saddle (Lee et al., 2016), but it can spend a long time near one (Du et al., 2017, constructed cases needing exponential time). Near a saddle, the escape direction grows like $(1+\eta|\lambda_-|)^t$ from its initial offset, so a start very close to the ridge produces a long plateau. The second widget below shows exactly that.

Interactive

A saddle point slows descent (shown at its default setting)

Contours of $\mathcal{L}(x, y) = x^2 + \frac{1}{2}(y^2 - 1)^2$: two minima at $y = \pm 1$ and a saddle at the origin, with loss 0.5. The ball starts at $x = 1.6$, a tiny distance from the ridge $y = 0$. Learning rate 0.05, 300 steps. Move the start closer to the ridge and watch the plateau in the loss curve grow; add noise and it vanishes.



Steps spent at the saddle

101

Where it ends

minimum

The ball sat on the plateau for about **101** steps, then escaped into a minimum. The ball started 1e-6 away from the ridge line. Near the saddle the slope is almost zero, so that tiny offset grows only by about 10% per step. The closer the start to the ridge, the longer the plateau in the loss curve.

Stochastic gradient descent: noise that helps

Plain words

Poll a few, not everyone

Computing the exact gradient means running the network on every training example before taking a single step: like surveying an entire country before every decision. **Stochastic gradient descent (SGD)** instead looks at a small random handful of examples, a **mini-batch**, estimates the slope from them, and steps. The estimate is a bit off each time, but it is thousands of times cheaper, so you take thousands of times more steps.

The noise turns out to be useful. A ball that jiggles a little does not sit forever on a mountain pass, and can sometimes hop out of a shallow dip. Try the "Noisy steps" preset in the widget above.

For leaders

Batch size is a speed and cost dial

Bigger batches give a cleaner gradient estimate and keep more GPUs busy in parallel, but past a point they stop reducing the number of steps needed, so extra hardware buys nothing. McCandlish et al. (2018) tied that point, the critical batch size, to a measurable gradient noise scale, which grows as training progresses. That is why large runs often increase the batch size over time. The practical settings live in optimisation.

> ⚠ **Watch out**
>
> "Noise helps escape bad minima" is true in toy pictures and is easy to oversell. In a frontier run, gradient noise mostly matters for stability and for which of many equally good minima you land in. Treat claims that a batch-size or noise change "improved generalization" like any other: ask for seeds, held-out evals, and matched compute.

Under the hood

An unbiased, noisy estimate

Sample a mini-batch $\mathcal{B}$ of size B uniformly from the data:

$$\mathbf{g}_{\mathcal{B}} = \frac{1}{B}\sum_{i\in\mathcal{B}} \nabla_\theta \ell\big(f_\theta(x_i), y_i\big), \qquad \mathbb{E}[\mathbf{g}_{\mathcal{B}}] = \nabla\mathcal{L}(\theta), \qquad \mathrm{Cov}[\mathbf{g}_{\mathcal{B}}] \approx \frac{\Sigma(\theta)}{B}$$

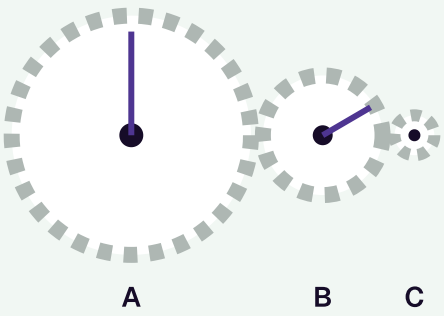
Σ is the per-example gradient covariance. The noise shrinks like $1/B$ in variance, the same $1/\sqrt{n}$ story as the standard error in Part I.

The SGD update $\theta \leftarrow \theta - \eta\,\mathbf{g}_{\mathcal{B}}$ is gradient descent plus zero-mean noise with covariance $\eta^2\Sigma/B$. Smith and Le (2018) argued that the effective noise scale is set by the ratio η/B, which is why learning rate and batch size are usually tuned together. Decaying the learning rate at the end of training shrinks the noise and lets the parameters settle; the widget's noise fades over the run for the same reason.

Diagram

Blame flows backward

Backpropagation as a gear train, then with the numbers from a tiny network.



Turn gear A by 1 notch:

B turns 2 notches (ratio 2)

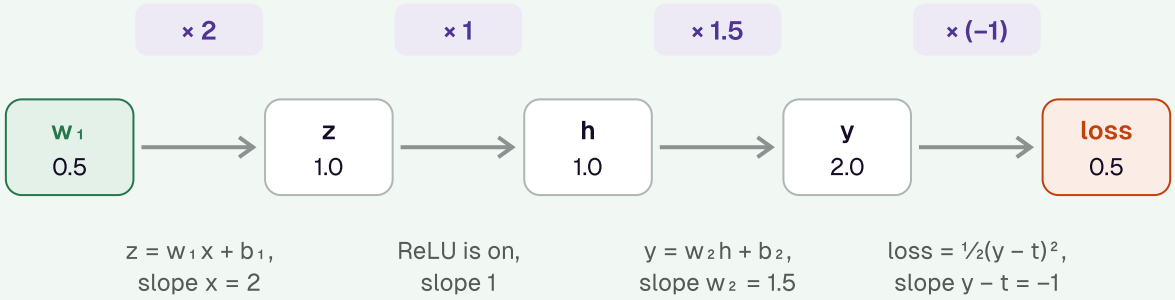
C turns 3 notches per B notch (ratio 3)

So C turns 2 × 3 = 6 notches.

Ratios along a chain multiply. That is the chain rule.

Turn gear A by one notch and B turns two; each notch of B turns C three. So C turns 2 × 3 = 6 notches per notch of A. You never need to watch A and C together: multiply the ratio of each link. Calculus calls this the **chain rule**, and the ratios are derivatives.

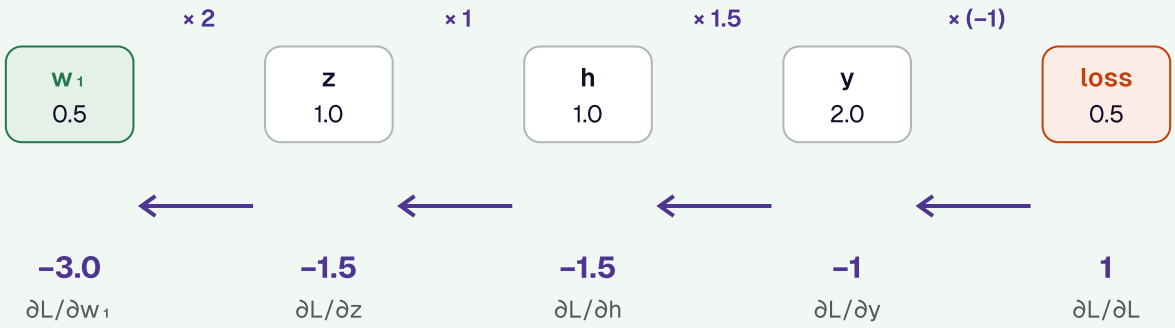
Each link has a local slope: how much the next value moves per nudge of this one



Input x = 2, target t = 3. Weights: $w_1 = 0.5$, $b_1 = 0$, $w_2 = 1.5$, $b_2 = 0.5$.

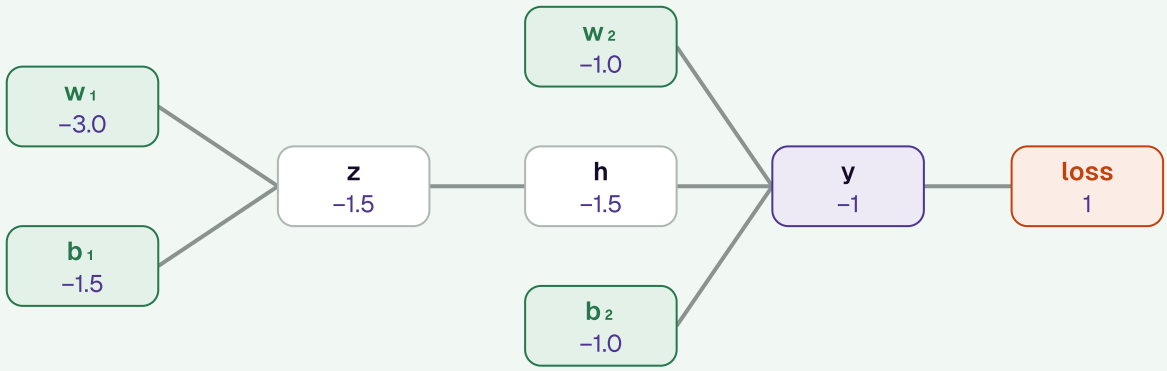
A network is a gear train too. A weight affects a weighted sum, which affects an activation, which affects the output, which affects the loss. Each link has a local slope that is easy to compute on its own. Here: one input x = 2, one hidden ReLU unit, one output, target 3. The forward pass gives a prediction of 2.0 and a loss of 0.5.

Backward: start at the loss with 1, multiply the slopes from right to left



$-1 \times 1.5 = -1.5$, $\times 1 = -1.5$, $\times 2 = -3.0$

Now go backward. The loss's slope with respect to itself is 1. Multiply by each link's local slope as you move left: –1 at the output, –1.5 at the hidden unit, still –1.5 through the ReLU, and –3.0 at w_1. Negative means "increase w_1 and the loss goes down". That right-to-left sweep is **backpropagation**.

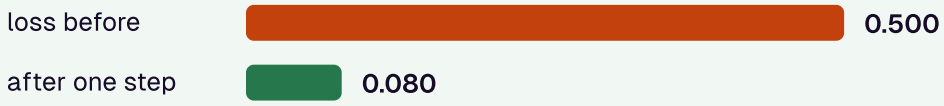


Each shared piece is computed once and reused by everything to its left.

Every parameter gets its gradient from the same sweep. The −1 at the output is computed once and reused for w_2, b_2 and the hidden unit; the −1.5 at z is reused for w_1 and b_1. In a real network the reuse is massive, which is why one backward pass costs only about twice a forward pass, however many billions of parameters there are.

parameter	before	gradient	after (lr 0.1)
w_1	0.5	−3.0	**0.8**
b_1	0.0	−1.5	**0.15**
w_2	1.5	−1.0	**1.6**
b_2	0.5	−1.0	**0.6**

new = old − 0.1 × gradient



Step every parameter against its gradient with learning rate 0.1. The prediction moves from 2.0 to 3.4, and the loss drops from 0.50 to 0.08. It slightly overshoots the target of 3; the next step corrects it. Forward pass, backward pass, step, repeat: that loop, run trillions of times, is how every network in this guide was trained.

Backpropagation: the chain rule, layer by layer

Plain words

Passing blame back down the line

A restaurant dish comes out too salty. The head chef asks the sauce cook, who asks the person who made the stock. Each person only needs to know how much their part contributed to the part after them; multiply those shares along the line and you know how much each person contributed to the final saltiness. Nobody has to taste every combination.

Backpropagation does that for a network. After a forward pass, it starts at the loss and walks backward through the layers, passing along "how much did you contribute to the error?" Each layer multiplies by its own local share and passes the result back. At the end every weight knows which way to move.

For leaders

What it costs, and where it breaks

Nobody writes backprop by hand anymore: frameworks like PyTorch and JAX record the forward pass and generate the backward pass automatically (automatic differentiation). What leaders need is its cost profile:

- **Compute:** about two forward passes' worth, so a training step costs about three forward passes.
- **Memory:** the backward pass needs the intermediate values from the forward pass, so training must store them. For long contexts that storage, not the weights, often dominates memory. Activation checkpointing trades extra compute (recomputing them) for memory; see optimisation.
- **Failure modes:** multiplying many slopes in a row can shrink gradients to nothing (vanishing) or blow them up (exploding). Residual connections, normalization layers and careful initialization exist largely to keep that product well behaved across a hundred layers; see architecture.

> **Note**
> When a training run produces NaNs, the cause is almost always in this chain: an exploding gradient, a division by a tiny number in a normalization layer, or a low-precision overflow. Good runs log gradient norms per layer so the first bad layer can be found.

Under the hood

The worked example, then the general rule

The story's network: $z = w_1 x + b_1$, $h = \mathrm{ReLU}(z)$, $y = w_2 h + b_2$, $\mathcal{L} = \frac{1}{2}(y - t)^2$, with $x = 2$, $t = 3$, $w_1 = 0.5$, $b_1 = 0$, $w_2 = 1.5$, $b_2 = 0.5$.

Forward	Value	Backward (chain rule)	Value
$z = w_1 x + b_1$	1.0	$\partial \mathcal{L}/\partial y = y - t$	-1.0
$h = \max(0, z)$	1.0	$\partial \mathcal{L}/\partial w_2 = (\partial \mathcal{L}/\partial y)\, h$	-1.0
$y = w_2 h + b_2$	2.0	$\partial \mathcal{L}/\partial h = (\partial \mathcal{L}/\partial y)\, w_2$	-1.5
$\mathcal{L} = \frac{1}{2}(y - t)^2$	0.5	$\partial \mathcal{L}/\partial z = (\partial \mathcal{L}/\partial h)\, \mathbb{1}[z > 0]$	-1.5
		$\partial \mathcal{L}/\partial w_1 = (\partial \mathcal{L}/\partial z)\, x$	-3.0

python

```python
x, t = 2.0, 3.0                 # one input, one target
w1, b1, w2, b2 = 0.5, 0.0, 1.5, 0.5
lr = 0.1

def forward(w1, b1, w2, b2):
    z = w1 * x + b1             # hidden pre-activation
    h = max(z, 0.0)             # ReLU
    y = w2 * h + b2             # output
    loss = 0.5 * (y - t) ** 2   # squared error
    return z, h, y, loss

z, h, y, loss = forward(w1, b1, w2, b2)
print(f"forward: z={z:.2f} h={h:.2f} y={y:.2f} loss={loss:.3f}")

# Backward: start at the loss and multiply local slopes going left.
dy = y - t                      # dL/dy
dw2, db2 = dy * h, dy           # y = w2*h + b2
dh = dy * w2                    # pass blame back through w2
dz = dh * (1.0 if z > 0 else 0.0)   # ReLU passes gradient only if it was on
dw1, db1 = dz * x, dz           # z = w1*x + b1
print(f"grads: dw1={dw1:.2f} db1={db1:.2f} dw2={dw2:.2f} db2={db2:.2f}")

# Check one gradient by nudging the weight and re-measuring the loss.
eps = 1e-6
num = (forward(w1 + eps, b1, w2, b2)[3] - forward(w1 - eps, b1, w2, b2)[3]) / (2 * eps)
print(f"numeric dw1={num:.2f}")

w1, b1, w2, b2 = w1 - lr * dw1, b1 - lr * db1, w2 - lr * dw2, b2 - lr * db2
print(f"after one step: loss={forward(w1, b1, w2, b2)[3]:.3f}")
# forward: z=1.00 h=1.00 y=2.00 loss=0.500
# grads: dw1=-3.00 db1=-1.50 dw2=-1.00 db2=-1.00
# numeric dw1=-3.00
# after one step: loss=0.080
```

rust

```rust
const X: f64 = 2.0; // one input
const T: f64 = 3.0; // one target

/// Returns (z, h, y, loss).
```

```
fn forward(w1: f64, b1: f64, w2: f64, b2: f64) -> (f64, f64, f64, f64) {
    let z = w1 * X + b1; // hidden pre-activation
    let h = z.max(0.0); // ReLU
    let y = w2 * h + b2; // output
    let loss = 0.5 * (y - T).powi(2); // squared error
    (z, h, y, loss)
}

fn main() {
    let (mut w1, mut b1, mut w2, mut b2) = (0.5, 0.0, 1.5, 0.5);
    let lr = 0.1;

    let (z, h, y, loss) = forward(w1, b1, w2, b2);
    println!("forward: z={:.2} h={:.2} y={:.2} loss={:.3}", z, h, y, loss);

    // Backward: start at the loss and multiply local slopes going left.
    let dy = y - T; // dL/dy
    let (dw2, db2) = (dy * h, dy); // y = w2*h + b2
    let dh = dy * w2; // pass blame back through w2
    let dz = dh * if z > 0.0 { 1.0 } else { 0.0 }; // ReLU passes gradient only if it was on
    let (dw1, db1) = (dz * X, dz); // z = w1*x + b1
    println!("grads: dw1={:.2} db1={:.2} dw2={:.2} db2={:.2}", dw1, db1, dw2, db2);

    // Check one gradient by nudging the weight and re-measuring the loss.
    let eps = 1e-6;
    let num = (forward(w1 + eps, b1, w2, b2).3 - forward(w1 - eps, b1, w2, b2).3) / (2.0 * eps);
    println!("numeric dw1={:.2}", num);

    w1 -= lr * dw1;
    b1 -= lr * db1;
    w2 -= lr * dw2;
    b2 -= lr * db2;
    println!("after one step: loss={:.3}", forward(w1, b1, w2, b2).3);
    // forward: z=1.00 h=1.00 y=2.00 loss=0.500
    // grads: dw1=-3.00 db1=-1.50 dw2=-1.00 db2=-1.00
    // numeric dw1=-3.00
    // after one step: loss=0.080
}
```

The general rule. For a layer $\mathbf{h}^{(\ell)} = \phi(\mathbf{z}^{(\ell)})$ with $\mathbf{z}^{(\ell)} = W^{(\ell)}\mathbf{h}^{(\ell-1)} + \mathbf{b}^{(\ell)}$, write $\boldsymbol{\delta}^{(\ell)} = \partial \mathcal{L} / \partial \mathbf{z}^{(\ell)}$. Backprop is three lines, applied from the last layer down:

$$\frac{\partial \mathcal{L}}{\partial W^{(\ell)}} = \boldsymbol{\delta}^{(\ell)}\, \mathbf{h}^{(\ell-1)\top}, \qquad \frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(\ell)}} = \boldsymbol{\delta}^{(\ell)}, \qquad \boldsymbol{\delta}^{(\ell-1)} = \left(W^{(\ell)\top}\boldsymbol{\delta}^{(\ell)}\right) \odot \phi'\left(\mathbf{z}^{(\ell-1)}\right)$$

The first two give the gradients for this layer; the third passes blame one layer down. Each needs only values stored during the forward pass. Rumelhart, Hinton and Williams (1986) popularized the method for neural networks; it is reverse-mode automatic differentiation.

The third line also shows where gradients vanish or explode: $\boldsymbol{\delta}$ is multiplied by $W^\top$ and by ϕ' at every layer. With sigmoid, $\phi' \le 0.25$, so twenty layers can shrink it by $0.25^{20} \approx 10^{-12}$. Residual

connections add an identity path, $\mathbf{h}^{(\ell)} = \mathbf{h}^{(\ell-1)} + F(\mathbf{h}^{(\ell-1)})$, whose derivative includes a plain "+1", so blame can travel the whole depth undiminished.

Key idea

Learning is a loop: forward pass to get the loss, backward pass to get every gradient at once, a small step against the gradient, repeat. Local minima matter less than one-dimensional pictures suggest because, in billions of dimensions, most flat spots are saddles and most minima are about equally good. What actually goes wrong in practice is step size, plateaus, and instability, which is why optimisation is its own chapter.

Part III · Chapter 20

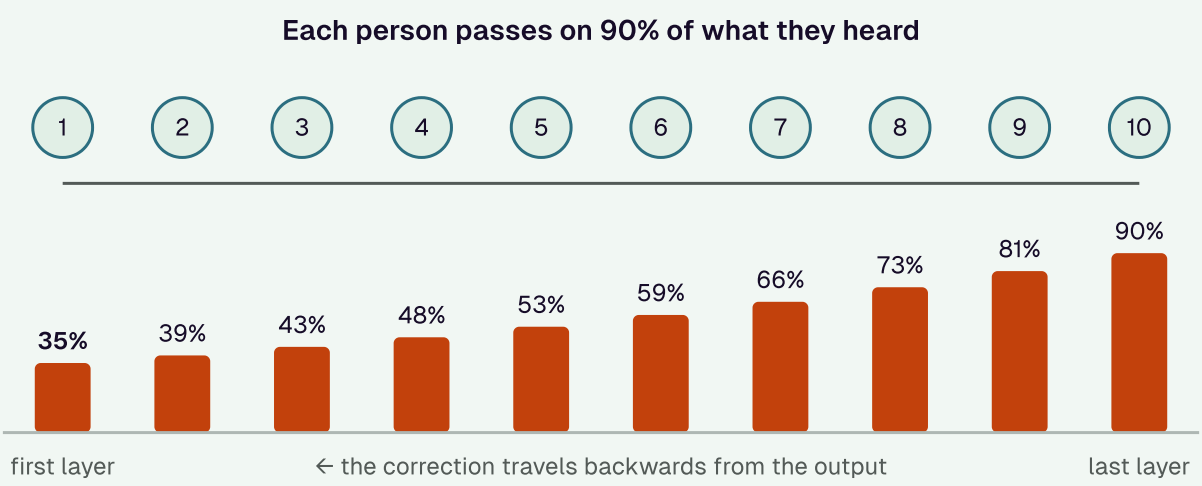
Designing a network: depth, width, and why not infinite layers

A fair question from anyone who has seen a neural network diagram: who decides how many layers there are, how wide each one is, and what kind of layer goes where? And if more layers make a model smarter, why not stack a thousand, or an infinite number? This chapter answers both. The short version: an architecture is a bet about the structure of your data, more depth stops paying for six concrete reasons, and labs choose shapes by running cheap experiments and fitting curves, not by inspiration. It builds on neural networks and how networks learn, and feeds into scaling laws.

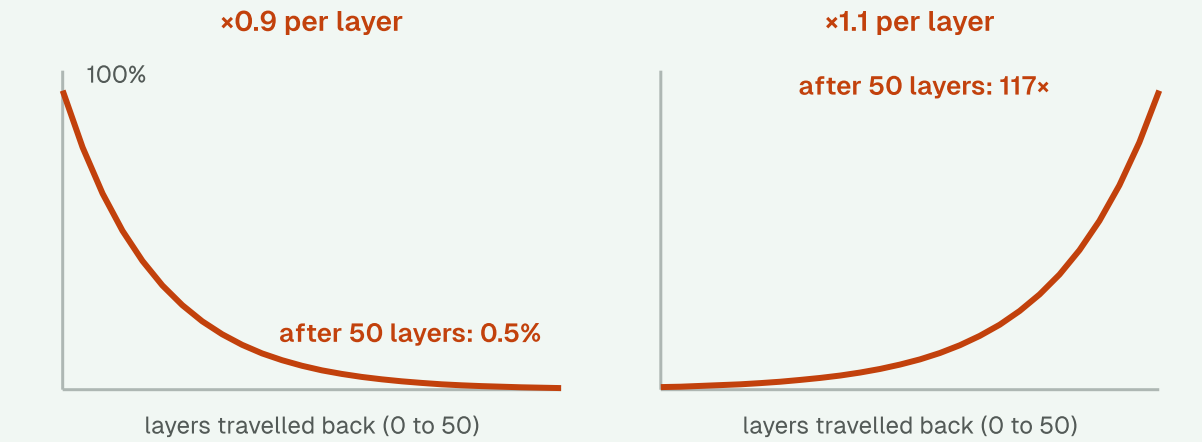
Diagram

The whisper chain

Why stacking more layers breaks training, and the two fixes that made deep networks work, told as a message passed down a line of people.

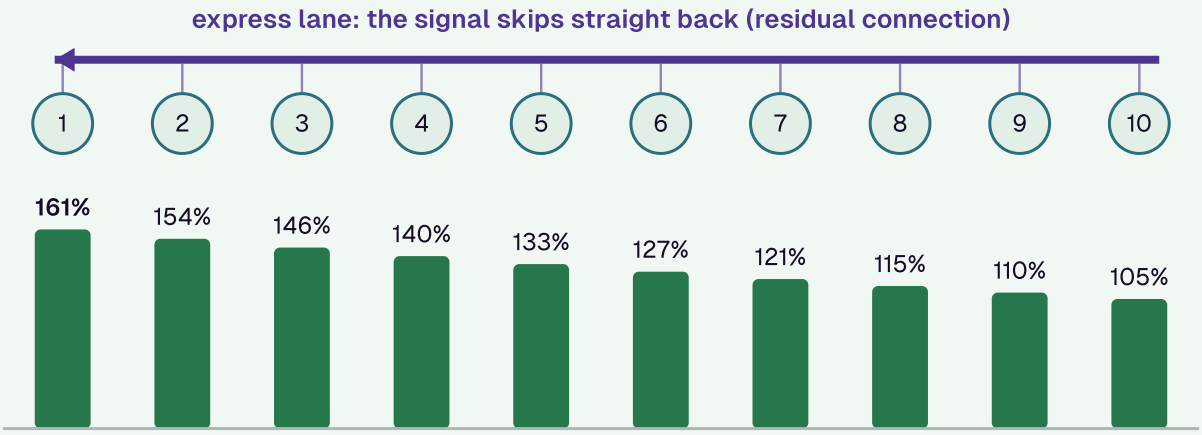


Training works by sending a correction signal (the gradient) backwards from the output, through every layer in turn. If each layer passes on 90% of the signal it received, the first of ten layers hears only 35% of it.



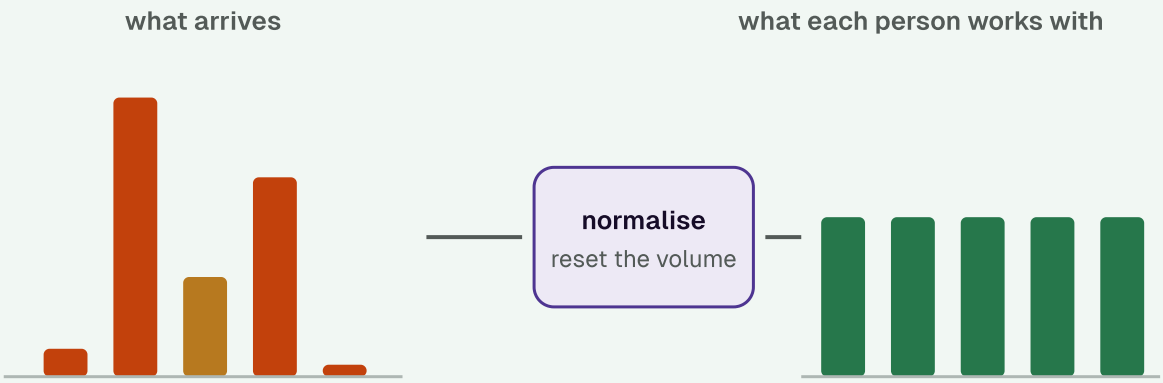
vanishing gradient (left) · exploding gradient (right)

The trouble is multiplication. Fifty hand-offs at 90% leave half a percent: the early layers stop learning (a **vanishing gradient**). Fifty at 110% amplify it 117 times: updates blow up and training diverges (an **exploding gradient**). Only a gain of exactly 1 is safe, and no real layer holds it exactly.



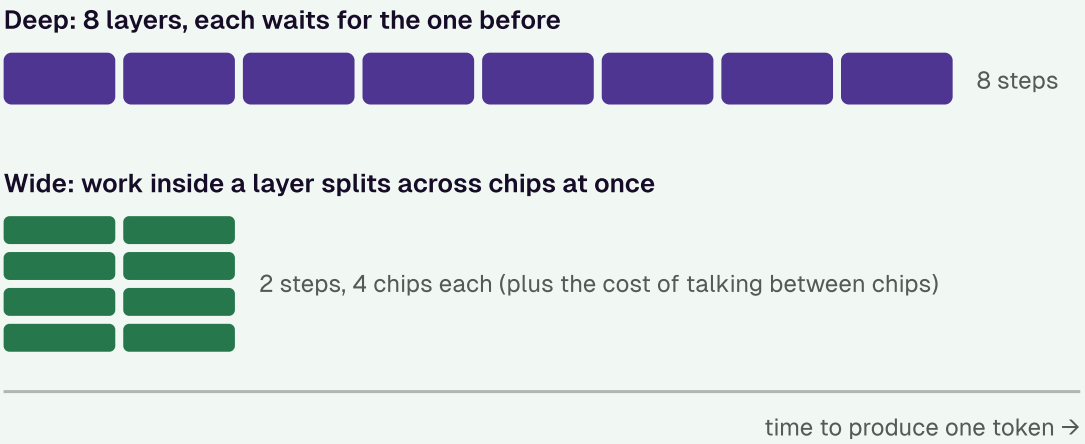
each person adds a small correction on top of what came through the lane

The fix that made very deep networks trainable: give the signal an express lane. Each layer adds a small correction on top of its input instead of replacing it, so the signal always has a direct path back. This is a **residual connection**. With small corrections, ten layers keep the signal between 100% and 161%: nothing vanishes.



whispers and shouts both come out at normal speaking volume

The second fix: before each person works on the message, they reset it to normal speaking volume. Whether it arrived as a whisper or a shout, the work starts from a standard size. This is **normalisation** (LayerNorm in transformers). Residual connections plus normalisation are why a 126-layer model trains at all.



Even with both fixes, depth has a price you cannot parallelise away. Person 6 cannot speak before person 5 finishes. Every generated token must pass through every layer in order, so a model twice as deep takes roughly twice as long per token. Work inside one wide layer can be split across many chips at the same time.

Gradients
vanish or explode as a product;
fixed by residuals + normalisation

Compute and memory
every layer adds parameters, FLOPs,
stored activations and KV cache

Latency
layers run one after another,
for every token you generate

Data and returns
more capacity needs more data,
and each doubling buys less

So "why not infinite layers?" has six answers, in three groups. The two gradient problems are solved well enough by residuals and normalisation. The other four are permanent: compute and memory, latency on every token, data to fill the extra capacity, and diminishing returns.

An architecture is a bet about the data

• Plain words **The right tool assumes something**

A colander assumes you want to keep the solids and drop the liquid. An egg carton assumes eggs are roughly the same size. Each tool is useful because it builds in an assumption about what it will hold. If the assumption is right, the tool saves you effort. If it is wrong, the tool gets in the way.

Network layers are tools with built-in assumptions. A **convolution** layer assumes that a cat's ear looks the same in the top corner of a photo as in the bottom corner, so it reuses one small detector everywhere. A **recurrent** layer assumes data arrives in order, one step after another. An **attention** layer assumes almost nothing: any word may matter to any other word. The built-in assumption is called the **inductive bias**. Picking an architecture means picking which assumptions to bake in.

For leaders

Strong assumptions buy data efficiency, weak ones buy headroom

A strong inductive bias is a loan against the future: it learns faster from little data, but it caps what the model can represent if the assumption is wrong. A weak bias needs more data and compute before it gets good, then keeps improving after the strongly biased model plateaus. Transformers sit at the weak end, which is why they dominate when data and compute are abundant (language, code, and increasingly images and audio), while convolutions and graph networks still win where data is scarce or the structure is known and fixed. Rich Sutton's essay "The Bitter Lesson" (2019) states the pattern bluntly: across decades, general methods that scale with compute have beaten methods that encode human knowledge about the domain.

For a frontier lab the practical consequence is that the architecture question is mostly settled at the top level (a decoder-only transformer, see inside an LLM) and the live decisions are second-order: depth versus width, attention variants, normalisation placement, mixture-of-experts routing, and context length. Those decisions are made by measurement, covered below.

Under the hood

Bias as a restricted hypothesis class

Formally, an architecture restricts the set of functions the network can express, and the parameterisation sets a prior over that set. A convolution enforces **translation equivariance**: shift the input and the output shifts the same way, $f(T_s x) = T_s f(x)$. The saving is enormous. A fully connected layer mapping a 224×224×3 image to a same-sized output needs $(224 \cdot 224 \cdot 3)^2 \approx 2.3 \times 10^{10}$ weights. A 3×3 convolution from 3 channels to 64 channels needs $3 \cdot 3 \cdot 3 \cdot 64 = 1{,}728$ weights, reused at every position. The convolution cannot represent most of what the dense layer can, and that is the point: it rules out functions that would be wrong for images anyway.

A transformer's attention is **permutation equivariant** over tokens: without position information it treats a sentence as a bag of words. Positional encodings (sinusoidal, learned, or rotary) add back

exactly as much order information as the designers choose. Chapter the family tree walks through each architecture's bias in turn.

Depth and width: what each buys

Plain words

Stations on an assembly line

Picture a car factory. **Depth** is the number of stations the car passes through: frame, then engine, then wiring, then paint. Each station builds on what the last one did, so more stations let you build more complicated things in more steps. **Width** is how many workers stand at each station. More workers per station means each step can handle more detail at once, but it does not add steps.

In a network, early layers find simple patterns (edges, common word pieces), middle layers combine them (shapes, phrases), and later layers combine those again (objects, meaning). Depth lets the network build ideas out of ideas. Width lets each layer keep track of more things at the same time.

For leaders

Shape matters less than size, until it hits a system constraint

The most useful single fact: at a fixed parameter count, transformer loss depends only weakly on the exact depth-to-width ratio over a wide range. Kaplan et al. (2020) found that varying shape over a broad range changed loss by only a few percent, while changing total size moved it a lot. So shape is usually chosen for **system** reasons, not modelling ones:

- **Depth costs latency.** Layers run in sequence for every generated token. A deeper model at the same size is slower to serve.
- **Width must fit the hardware.** Hidden sizes are usually multiples of 64 or 128 so matrix multiplies tile cleanly on accelerators, and wide layers are what tensor parallelism splits across chips (see optimisation and infrastructure).
- **Depth sets pipeline stages.** Layer count is chosen to divide evenly across pipeline-parallel stages.
- **Very shallow or very narrow both hurt.** Too shallow limits how many steps of composition the model can do per token; too narrow starves each layer of room, and attention heads get too small.

The dense decoders in the table sit around 100 to 130 hidden units per layer of depth (recent MoE models vary more widely: Qwen3-235B has 94 layers at width 4,096, about 44 per layer):

Model (source)	Parameters	Layers	Width d	Heads (KV heads)	d / layers
GPT-2 XL (Radford et al. 2019)	1.5B	48	1,600	25	33
GPT-3 (Brown et al. 2020)	175B	96	12,288	96	128
Llama 2 70B (Touvron et al. 2023)	70B	80	8,192	64 (8)	102
Llama 3 405B (Llama Team 2024)	405B	126	16,384	128 (8)	130

Notice what does not happen: nobody goes from 126 layers to 1,000. Going from 70B to 405B parameters, Llama roughly doubled the width and added about half again as many layers.

Under the hood

Counting parameters and FLOPs, and what theory says about depth

A standard decoder block has attention projections ($4d^2$ weights for Q, K, V and output) and an MLP with hidden size $4d$ ($8d^2$ weights). So

$$N \approx 12\,L\,d^2, \qquad \text{FLOPs per token (forward)} \approx 2N$$

L layers, width d; embeddings excluded. GPT-3: $12 \times 96 \times 12288^2 \approx 1.74 \times 10^{11}$, matching its 175B. Training costs about three times the forward pass, $\approx 6N$ per token.

Parameters grow linearly in depth but quadratically in width. Doubling L doubles N; doubling d quadruples it. Variants change the constants: Llama uses a SwiGLU MLP with three matrices (hidden size about $3d$ to $3.5d$ in Llama 3), and grouped-query attention shrinks the K and V projections. Llama 3 405B works out to about $11.9\,L\,d^2$ plus embeddings, which reproduces its 405B.

Theory says depth is genuinely powerful, not just convenient. One hidden layer can approximate any continuous function on a bounded domain (Cybenko 1989; Hornik 1991), but may need an astronomical number of units to do it. Telgarsky (2016) constructed functions that a deep ReLU network computes with few units per layer while any much shallower network needs exponentially many units. Composition is cheap with depth and expensive without it. For transformers, each layer is

one round of "look around, then think", so a fixed-depth model has a fixed budget of sequential steps per token. That is one reason generating intermediate reasoning tokens helps on multi-step problems: it buys more serial computation (see inference and test-time compute).

Why not infinite layers? Wall 1: the gradient is a product

Plain words

A photocopy of a photocopy

Photocopy a page, then photocopy the copy, and keep going. Each copy is a little lighter. After 50 rounds the page is blank. If your copier instead darkened each copy slightly, after 50 rounds the page would be solid black. Either way the message is gone.

Training a network sends a correction backwards through every layer, and each layer scales it a little. Too little each time and the first layers never get the message. Too much and the corrections become wild. More layers means more copies, so the problem gets worse with depth.

For leaders

What it looks like in a run

Vanishing gradients look like a run where the loss plateaus early and the first layers' weights barely move. Exploding gradients look like loss spikes and NaNs. Modern recipes rarely hit either at initialisation, because of careful initialisation, residual connections, and normalisation. They still appear mid-run as instabilities, especially at large scale and high learning rates, which is why teams log per-layer gradient norms and clip gradients (see optimisation).

Watch out

A new architecture idea that "trains fine" at 12 layers can diverge at 100. Instability often appears only at depth, width, or learning rates near the production configuration. Ask for per-layer gradient and activation norms from the largest run tried, not just the loss curve, and ask whether the idea was tested at the target depth.

Under the hood

The chain rule multiplies Jacobians

For a plain stack $h_{l+1} = f_l(h_l)$, the gradient of the loss with respect to an early layer's activations is a product of Jacobians:

$$\frac{\partial \mathcal{L}}{\partial h_l} = \frac{\partial \mathcal{L}}{\partial h_L} \prod_{k=l}^{L-1} J_k, \qquad J_k = \frac{\partial h_{k+1}}{\partial h_k}$$

If each J_k scales vectors by about g, the gradient at layer l scales by g^{L-l}. With $g = 0.9$ and 50 layers that is 0.005; with $g = 1.1$ it is 117.

The fix at initialisation is to choose weight scales so that $g \approx 1$: Glorot and Bengio (2010) for tanh-like units, He et al. (2015) for ReLU, which uses variance $2/n_{\text{in}}$ to compensate for ReLU zeroing half its inputs. But $g = 1$ on average still lets fluctuations compound over hundreds of layers, and training moves the weights away from the careful initial scale. Recurrent networks have the same problem through time instead of through depth, which motivated the LSTM (see the family tree).

The residual form changes the product. With $h_{l+1} = h_l + F_l(h_l)$, each factor becomes $I + \partial F_l / \partial h_l$. The identity term means the gradient always has a direct path. If the branch contributes a part uncorrelated with that path, squared sizes add, and each layer scales the gradient by $\sqrt{1 + s^2 g^2}$ where s is the branch scale. Unscaled ($s = 1$) that still compounds; with $s = 1/\sqrt{L}$ the total stays below $e^{g^2/2}$ at any depth. GPT-2 used exactly this trick, scaling residual-branch weights at initialisation by $1/\sqrt{N}$ for N residual layers (Radford et al. 2019).

python

```python
import math

def sci(x):
    # short scientific notation, e.g. 5.15e-3 (same format in the Rust twin)
    e = math.floor(math.log10(abs(x)))
    m = x / 10 ** e
```

```
    if round(m, 2) >= 10:            # 9.999 rounds up: shift the exponent
        m, e = m / 10, e + 1
    return f"{m:.2f}e{e}"

def grad_through(L, g, residual=False, branch_scale=1.0):
    """Size of the gradient after flowing back through L layers.
    Plain layer: the gradient is multiplied by the layer's gain g.
    Residual layer: an identity path plus a branch whose contribution is
    uncorrelated with it, so squared sizes add: factor sqrt(1 + (s*g)^2)."""
    grad = 1.0
    for _ in range(L):
        if residual:
            grad *= math.sqrt(1 + (branch_scale * g) ** 2)
        else:
            grad *= g
    return grad

for L in (10, 50, 200):
    vanish = grad_through(L, 0.9)                    # each layer shrinks it 10%
    explode = grad_through(L, 1.1)                   # each layer grows it 10%
    res_raw = grad_through(L, 1.0, residual=True)    # residual, branch not scaled
    res_scaled = grad_through(L, 1.0, residual=True, branch_scale=1 / math.sqrt(L))
    print(L, sci(vanish), sci(explode), sci(res_raw), f"{res_scaled:.3f}")
# 10 3.49e-1 2.59e0 3.20e1 1.611
# 50 5.15e-3 1.17e2 3.36e7 1.641
# 200 7.06e-10 1.90e8 1.27e30 1.647   scaled residual stays below e^0.5 = 1.649
```

rust

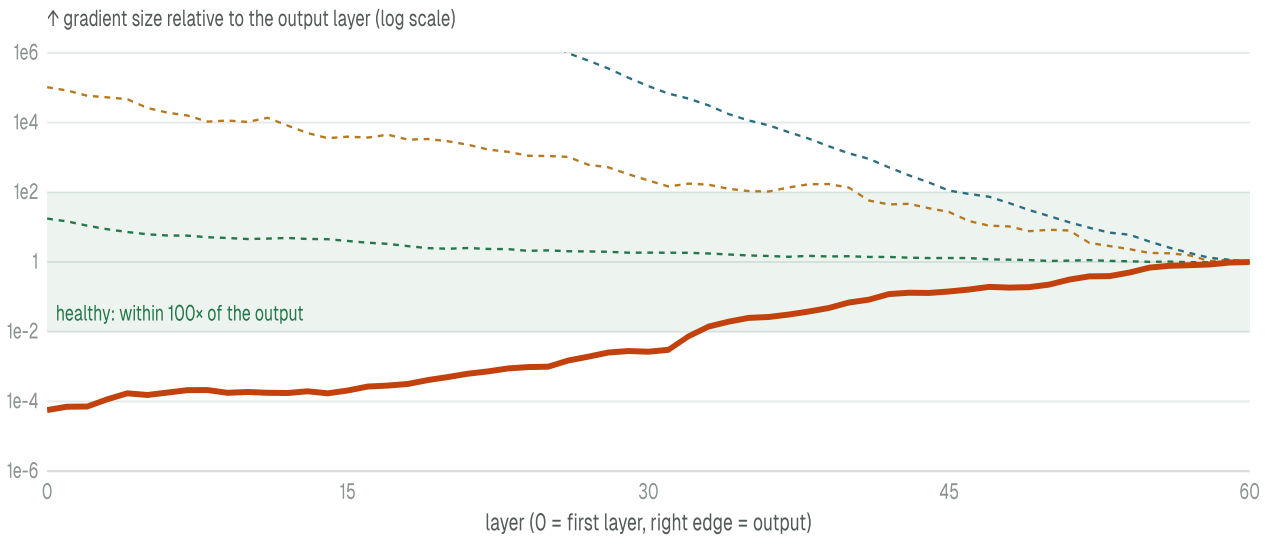
```
fn sci(x: f64) -> String {
    // short scientific notation, e.g. 5.15e-3 (same format as the Python)
    let mut e = x.abs().log10().floor() as i32;
    let mut m = x / 10f64.powi(e);
    if (m * 100.0).round() / 100.0 >= 10.0 { // 9.999 rounds up: shift the exponent
        m /= 10.0;
        e += 1;
    }
    format!("{:.2}e{}", m, e)
}

/// Size of the gradient after flowing back through l layers.
/// Plain layer: the gradient is multiplied by the layer's gain g.
/// Residual layer: an identity path plus a branch whose contribution is
/// uncorrelated with it, so squared sizes add: factor sqrt(1 + (s*g)^2).
fn grad_through(l: usize, g: f64, residual: bool, branch_scale: f64) -> f64 {
    let mut grad = 1.0;
    for _ in 0..l {
        if residual {
            grad *= (1.0 + (branch_scale * g).powi(2)).sqrt();
        } else {
            grad *= g;
        }
    }
    grad
}

fn main() {
    for l in [10usize, 50, 200] {
        let vanish = grad_through(l, 0.9, false, 1.0);  // each layer shrinks it 10%
        let explode = grad_through(l, 1.1, false, 1.0); // each layer grows it 10%
```

```
        let res_raw = grad_through(l, 1.0, true, 1.0);  // residual, branch not scaled
        let res_scaled = grad_through(l, 1.0, true, 1.0 / (l as f64).sqrt());
        println!("{} {} {} {} {:.3}", l, sci(vanish), sci(explode), sci(res_raw), res_scaled);
    }
    // 10 3.49e-1 2.59e0 3.20e1 1.611
    // 50 5.15e-3 1.17e2 3.36e7 1.641
    // 200 7.06e-10 1.90e8 1.27e30 1.647   scaled residual stays below e^0.5 = 1.649
}
```

Interactive

Gradient flow through a deep stack (shown at its default setting)

A real 48-wide ReLU network with random weights. A gradient enters at the output (right edge) and flows back to layer 0 (left edge). Move the gain off 1.00 to see vanishing or exploding, then turn on residual connections and normalisation. Dashed lines show the other three layouts for comparison.



plain · plain + normalisation · residual · residual + normalisation

Gradient at layer 0 (output = 1)	Average factor per layer	Forward signal size at output	Verdict
5.67e-5	×0.850	3.21e-3	vanishing

Plain, 60 layers, gain 0.90. Each layer multiplies the gradient by about ×0.850, so the first layer gets $\times 0.850^{60} \approx 5.67\text{e-}5$ of the output layer's signal. The early layers barely learn. Raise the gain above 1 and the same product explodes instead.

Wall 2: deeper plain networks train worse, and the residual fix

Plain words

Tracked changes instead of rewrites

Imagine twenty editors improving a document. In one office, each editor rewrites the whole document from scratch based on the previous version. One bad editor can destroy everything, and the result after twenty rewrites is unpredictable. In the other office, each editor only adds tracked changes on top. If an editor has nothing useful to add, they add nothing, and the document passes through untouched.

A **residual connection** turns every layer into the second kind of editor. The layer's output is its input plus a correction. A layer that is not helping can learn a correction of zero, so adding more layers can never make things much worse.

For leaders

The single most important architectural trick of the last decade

Before 2015, networks deeper than about 20 to 30 layers were hard to train even with good initialisation and normalisation. He et al. (2015) introduced residual networks (ResNets) and trained a 152-layer model that won the ImageNet classification challenge that year. Every transformer since is built around a residual stream: each attention and MLP block reads from the stream and writes a correction back into it.

The business lesson generalises. Depth was not limited by a lack of data or compute. It was limited by an optimisation problem, and one structural change unlocked an order of magnitude more depth. When a team says "we tried a bigger model and it got worse", the first question is whether the training setup, not the size, is the bottleneck.

Under the hood

Degradation is not overfitting

He et al. (2015) showed that on CIFAR-10 a 56-layer plain network had higher **training** error than a 20-layer one, and the same held on ImageNet for 34 versus 18 layers. A deeper network could in principle copy the shallow one and set the extra layers to the identity, so it should never be worse on training data. That it was worse shows an optimisation failure, which they called **degradation**. The plain networks used batch normalisation, and the authors reported that gradients did not vanish, so this is a separate wall from vanishing gradients: plain layers find it hard to learn the identity.

$$h_{l+1} = h_l + F_l(h_l; \theta_l), \qquad \frac{\partial h_L}{\partial h_l} = \prod_{k=l}^{L-1} \left(I + \frac{\partial F_k}{\partial h_k} \right)$$

Setting $F_l = 0$ gives the identity for free. Expanding the product gives a sum over all subsets of blocks, one reason Veit et al. (2016) described residual networks as behaving like ensembles of many shallower paths.

Residual networks still have limits. On CIFAR-10, He et al. trained a 1,202-layer ResNet without optimisation trouble, but it tested worse than their 110-layer model, which they attributed to overfitting on a small dataset. Depth stopped being an optimisation wall and became a data and compute question, which is where the remaining walls take over.

Normalisation: resetting the volume

Plain words

Same volume in, whatever came before

A sound engineer sets every microphone so that a whisper and a shout reach the mixing desk at a similar level. Without that, one loud singer drowns out everything downstream. A **normalisation** layer does the same for numbers inside a network: before a layer works on its input, the input is rescaled to a standard size, so no layer is fed something enormous or tiny.

For leaders

Placement is a stability decision

The original transformer (Vaswani et al. 2017) put normalisation **after** each residual addition ("post-norm"). It trains well but needs a learning-rate warmup and gets fragile with depth. Almost every large model now uses **pre-norm**: normalise the input to each block and leave the residual stream itself untouched. Xiong et al. (2020) showed pre-norm transformers have well-behaved gradients at initialisation and can train without warmup. Llama-family models use RMSNorm (Zhang and Sennrich 2019), a cheaper variant that skips mean subtraction. These choices rarely make a headline, but they decide whether a large run survives its first loss spike.

Under the hood

BatchNorm, LayerNorm, RMSNorm

$$\mathrm{LN}(x) = \gamma \odot \frac{x - \mu(x)}{\sqrt{\sigma^2(x) + \epsilon}} + \beta, \qquad \mathrm{RMSNorm}(x) = \gamma \odot \frac{x}{\sqrt{\frac{1}{d}\sum_i x_i^2 + \epsilon}}$$

μ, σ^2 are the mean and variance over the d features of one token's vector; γ, β are learned per-feature scale and shift.

BatchNorm (Ioffe and Szegedy 2015) computes statistics over the batch for each feature. It was central to CNNs but is awkward for language models: statistics depend on what else is in the batch, behaviour differs between training and inference, and small or variable-length batches make the estimates noisy. **LayerNorm** (Ba, Kiros and Hinton 2016) normalises over the features of each example independently, so it behaves identically at training and inference and for any batch size.

Normalisation and residuals work as a pair. In a pre-norm block $h_{l+1} = h_l + F(\mathrm{LN}(h_l))$, each branch receives a fixed-size input and adds a bounded contribution, so the stream grows slowly (roughly like $\sqrt{l}$) and later layers' branches are relatively smaller. De and Smith (2020) made the related point for BatchNorm in ResNets: it downscales residual branches relative to the skip path, which biases deep networks toward the identity at initialisation. In the widget above, normalisation alone fixes the forward scale of a plain stack but not its backward gradient; the combination is what stays flat. Yang et al. (2019) found a similar gradient explosion at initialisation in deep plain networks with batch normalisation.

Walls 3 and 4: compute, memory, and the queue

Plain words

Every station costs rent, and the car waits at each one

Each extra station on the assembly line costs floor space, workers, and electricity, and every car must stop at every station. Adding stations makes better cars only up to the point where the factory becomes too slow and too expensive to run. A network is the same: every layer adds memory and arithmetic, and every word the model writes has to pass through every layer, one after another.

For leaders

Depth is paid for on every token, forever

Training cost is paid once. Serving cost is paid on every token for the life of the model. Depth hits serving directly:

- **Latency per token** scales with layer count, because layers cannot run in parallel for the same token. Wider layers can be split across chips (tensor parallelism), at the price of communication between them.
- **KV cache memory** scales with layer count: each layer stores keys and values for every token in context. That memory caps how many users fit on a server and how long the context can be (see inference).
- **Training memory** for activations grows with depth, which forces activation checkpointing (recomputing instead of storing) and deeper pipeline parallelism, which adds idle "bubbles".

Published models have not gone far beyond about 130 layers, even at 405B parameters, which is consistent with these costs.

Under the hood

The arithmetic of depth at inference

$$t_{\text{token}} \approx L \cdot t_{\text{layer}}, \qquad \text{KV bytes per token} = 2 \cdot L \cdot n_{kv} \cdot d_{\text{head}} \cdot b$$

n_{kv} key-value heads, d_{head} head size, b bytes per number, factor 2 for keys and values. Decoding is usually limited by memory bandwidth, so t_{layer} is roughly the time to read that layer's weights.

Llama 3 405B: $2 \times 126 \times 8 \times 128 \times 2$ bytes $= 516{,}096$ bytes, about 0.5 MB of cache per token in bf16. A 100,000-token context therefore needs about 52 GB of cache for a single sequence, before any weights. Without grouped-query attention (128 KV heads instead of 8) it would be 16 times larger. Every added layer adds its share to both numbers.

Walls 5 and 6: data limits and diminishing returns

Plain words

A bigger notebook does not help without more lessons

Give a student a notebook with ten times as many pages and no extra lessons, and they will either leave the pages blank or fill them with memorised answers to the practice questions. Extra capacity helps only when there is more to learn. And even with plenty of lessons, each doubling of effort brings a smaller improvement than the last.

For leaders

Capacity has to be fed and paid for

Adding layers adds parameters. Scaling laws (see scaling laws) say loss falls as a power law in parameters *and* data, and the compute-optimal recipe grows both together. A model made deeper without more data or compute sits off the optimal frontier: it underfits its compute budget or overfits its data. And the curve flattens: every doubling of parameters removes a fixed fraction (about 21%) of the parameter-limited part of the loss, not a fixed amount.

Eval lens

Architecture comparisons are only meaningful when **compute-matched**: same training FLOPs (or same tokens and parameters), same data, same tuning effort, several seeds. A deeper model trained for the same steps uses more compute, so it "wins" for the wrong reason. Separate the sources of a benchmark gain: a new model generation usually changes data, compute, and architecture at once, and the data and compute changes typically dominate. Ask for an ablation that changes only the architecture, run at two or more scales, and check whether the gain holds or shrinks as the models get bigger.

Under the hood

The power law in numbers

$$L(N, D) = E + \frac{A}{N^{\alpha}} + \frac{B}{D^{\beta}}$$

Hoffmann et al. (2022) fit $E \approx 1.69$, $A \approx 406$, $B \approx 411$, $\alpha \approx 0.34$, $\beta \approx 0.28$. E is the irreducible loss; the other terms shrink with parameters N and tokens D.

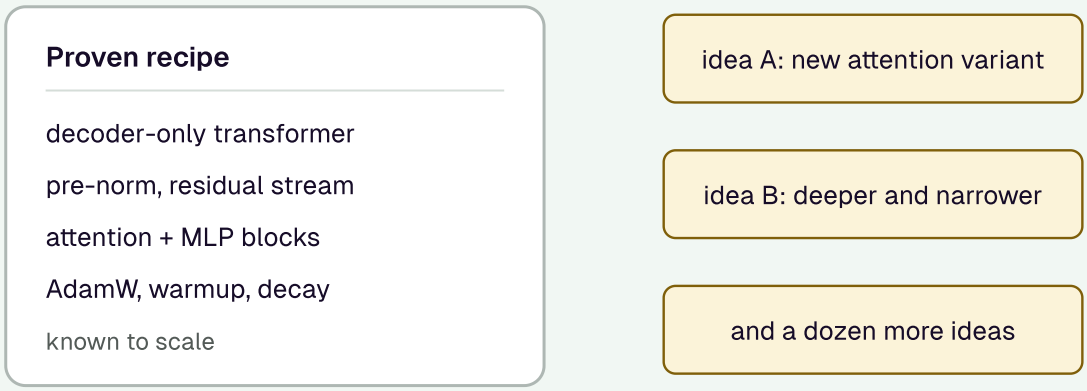
With $\alpha \approx 0.34$, doubling N multiplies the parameter term by $2^{-0.34} \approx 0.79$: a 21% cut in that term, the same fraction every time. If the data term already dominates, adding parameters barely moves the total. Depth is one way to add parameters, so it inherits all of this. The marginal question a lab asks is never "would more layers help?" but "is this the best use of the next unit of compute?"

How labs actually choose an architecture

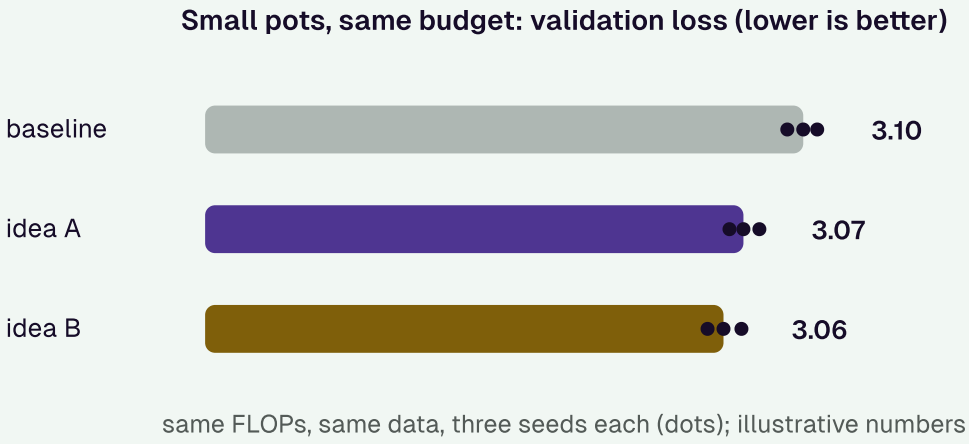
Diagram

Test kitchen before the banquet

How a frontier lab settles an architecture question without betting the big run, told as a chef planning a 10,000-plate dinner.

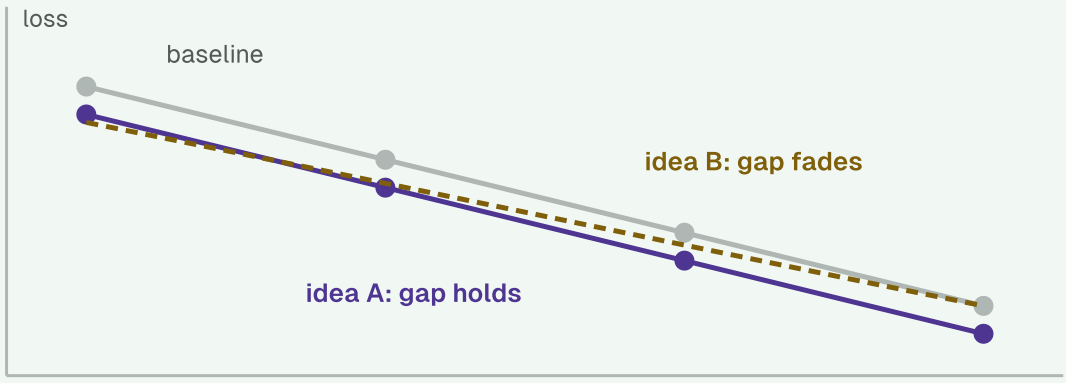


Nobody designs a frontier model from a blank page. The chef starts from a recipe that is known to work at scale and treats every change as a hypothesis to test. Most architecture work is choosing which few changes earn a place.



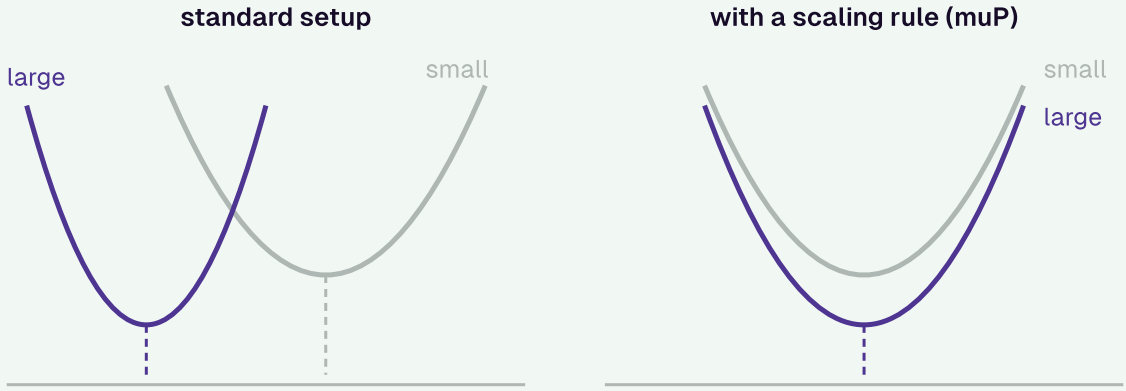
same FLOPs, same data, three seeds each (dots); illustrative numbers

Cook small pots first. Each idea gets the same compute, the same data, and several random seeds, so a gain has to beat the seed-to-seed wobble. Here both ideas look better than the baseline at small scale.



training compute (log scale): 4 model sizes, each 10× the last

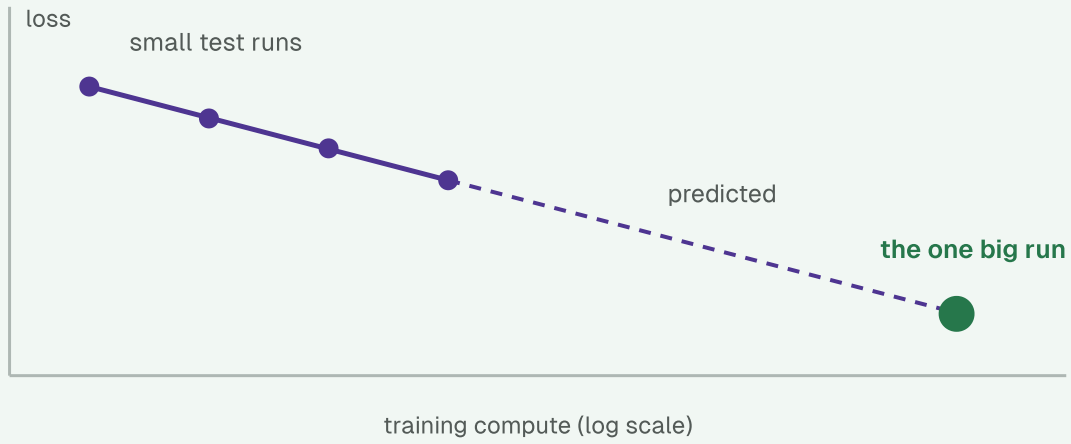
Then taste at several sizes. Plot each idea's loss against compute and look at the trend, not one point. Idea A keeps its advantage as models grow. Idea B's advantage shrinks to nothing: it was a small-model trick. Many published improvements behave like idea B.



learning rate (log) → best rate moves

learning rate (log) → best rate stays

Settings must scale too. A chef who doubles a recipe does not double the salt blindly. In a standard setup the best learning rate drifts as the model widens, so settings tuned on a small model are wrong for the big one. Scaling rules such as muP (Yang et al. 2022) reparameterise the network so the best settings stay put, and can be tuned once on a small proxy.



Finally, cook the banquet once. Fit a curve through the small runs, predict the big run's loss before starting it, and launch only the configuration that survived. If the big run lands on the predicted line, the recipe scaled. If it lands above, something broke at scale, and the team finds out early by watching the curve.

Plain words

Small experiments, then one big bet

Labs do not guess how many layers to use. They copy a design that already works, try small changes on small models, keep only the changes that still help as the models get bigger, and then train one huge model using the winning design. It is the same as a chef testing a recipe in small pots before cooking for ten thousand people.

For leaders

The decision process, and why automated search is rare

1. **Start from a proven recipe.** The previous generation's architecture, or a well-documented open one. Changes need to justify their risk.
2. **Ablate at small scale, compute-matched.** Change one thing at a time, multiple seeds, same data and FLOPs. Typical proxy models are tens of millions to a few billion parameters.
3. **Fit scaling curves.** Run the baseline and each surviving change at three to five sizes and compare the fitted curves. Keep changes whose advantage holds or grows with scale (see scaling laws).
4. **Transfer hyperparameters.** Use a parameterisation such as muP, or fitted scaling rules for learning rate and batch size, so the small-scale tuning carries over.

5 **De-risk, then commit.** Run an intermediate-size model end to end, check stability, then launch the full run and compare it against the prediction as it trains.

Neural architecture search (NAS), where an algorithm proposes and tests architectures automatically, produced strong image models at small scale; Zoph and Le (2017) used hundreds of GPUs to search over small image and language models. At frontier scale it is rare for three reasons: each candidate must be trained to be judged, and a frontier training run cannot be repeated thousands of times; what wins at small scale often fails to transfer (idea B in the story); and the search space is dominated by known-good transformer blocks, so the expected gain is small. Searches that did target transformers, such as Primer (So et al. 2021), mostly found small modifications (for example squared ReLU activations) that humans then adopted by hand.

Principal asks

Was the comparison compute-matched, and at how many scales? Does the gain grow, hold, or shrink with size, and what does the fitted curve predict at the target scale? How many seeds, and what is the seed-to-seed spread compared with the gain? Were hyperparameters retuned for the baseline as carefully as for the new idea? What does the change cost at inference: latency per token, KV cache, and memory?

Under the hood

Hyperparameter transfer with muP

Under standard parameterisation, as width d grows the updates to hidden weights grow relative to the weights themselves, so the optimal learning rate shrinks with width and must be re-searched at each size. Yang et al. (2022, "Tensor Programs V") derived the **maximal update parameterisation** (muP): scale initialisation variances, per-layer learning rates, and the output multiplier with width so that every layer's update stays the same order of size as width grows. For Adam, the key rule is that hidden-layer learning rates scale like $1/d$ and the output logits are multiplied by $1/d$ relative to a base width.

The payoff is **zero-shot hyperparameter transfer**: tune on a narrow proxy, copy the settings to the wide model. Yang et al. reported outperforming the published BERT-large numbers by transferring from a 13M-parameter proxy, and the published 6.7B GPT-3 numbers by transferring from a 40M-parameter proxy. The original muP covers width; transfer across depth needs extra rules, such as scaling residual branches by $1/\sqrt{L}$, studied in later work. Many labs instead fit empirical power laws for the optimal learning rate and batch size against model size and data, which serves the same purpose.

A clean ablation reports, for each variant: training FLOPs, parameters (total and active), tokens, the loss curve, the final loss with a seed spread, a few downstream evals, and inference cost. Comparing at equal parameters can mislead when a change alters FLOPs per parameter (mixture-of-experts, weight sharing, longer context); equal training FLOPs is the usual standard, with inference cost reported alongside.

Key idea

There is no formula that outputs "use 96 layers". Labs start from a proven transformer recipe, choose depth and width mainly for hardware and latency reasons within the broad range where shape barely matters, and accept any other change only if it wins compute-matched experiments at several scales. Infinite depth fails on optimisation (solved by residuals and normalisation), on cost and latency (permanent), and on data and diminishing returns (permanent). For how the chosen shape is then trained, see pretraining; for how these choices show up in team decisions, see leading.

Part III · Chapter 21

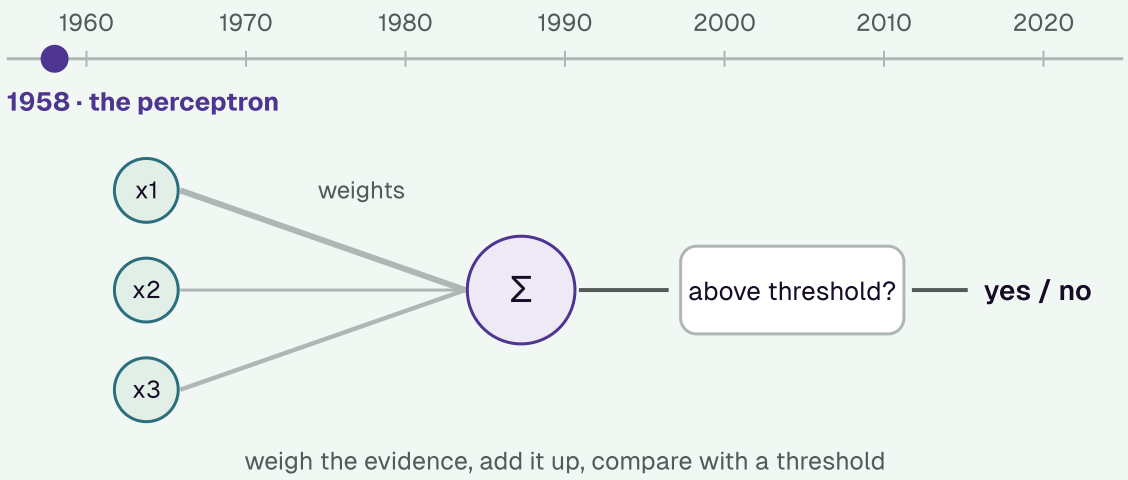
The family tree: CNNs, RNNs, transformers, GNNs

Every neural network architecture you will hear about is a variation on one idea: layers of weighted sums and simple nonlinearities, trained by backpropagation. What differs is the **inductive bias**, the assumption about the data that each design bakes in (see designing a network). This chapter tours the family in historical order, because each branch was invented to fix a specific limit of the one before, and then covers graph neural networks in depth: where they came from, what they are used for, and why a transformer is secretly one of them.

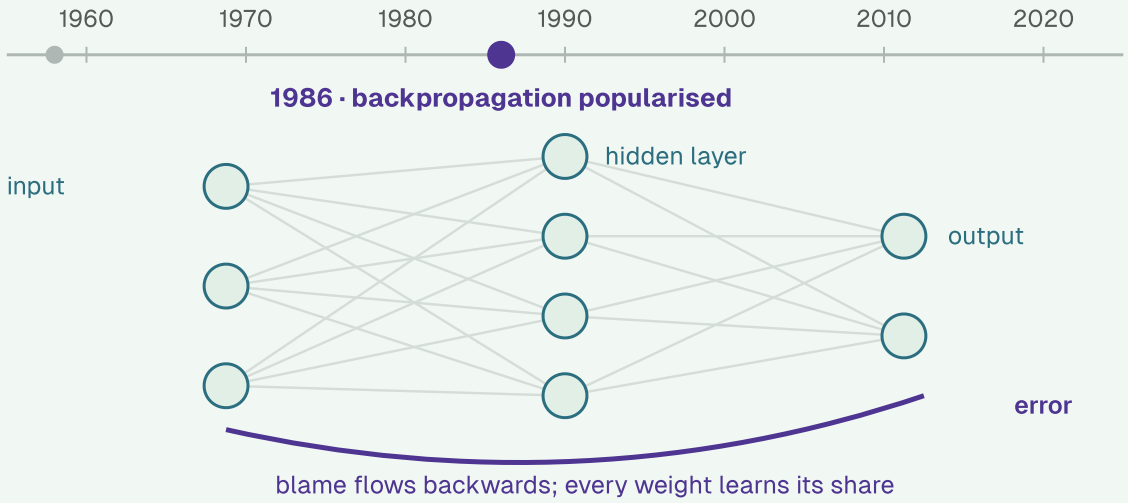
Diagram

Seventy years in seven pictures

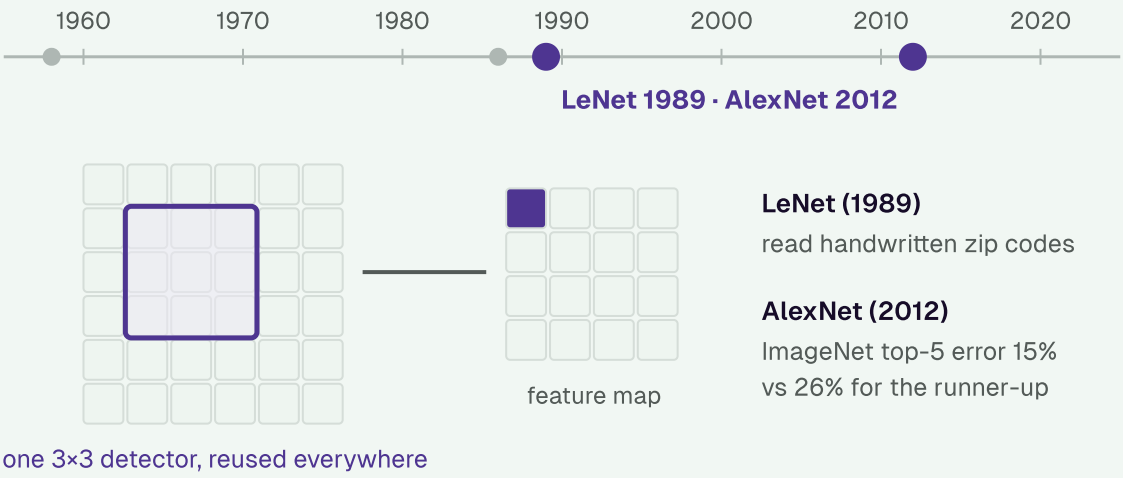
The family tree of neural networks, one idea per step. Each new branch fixed a specific limit of the one before.



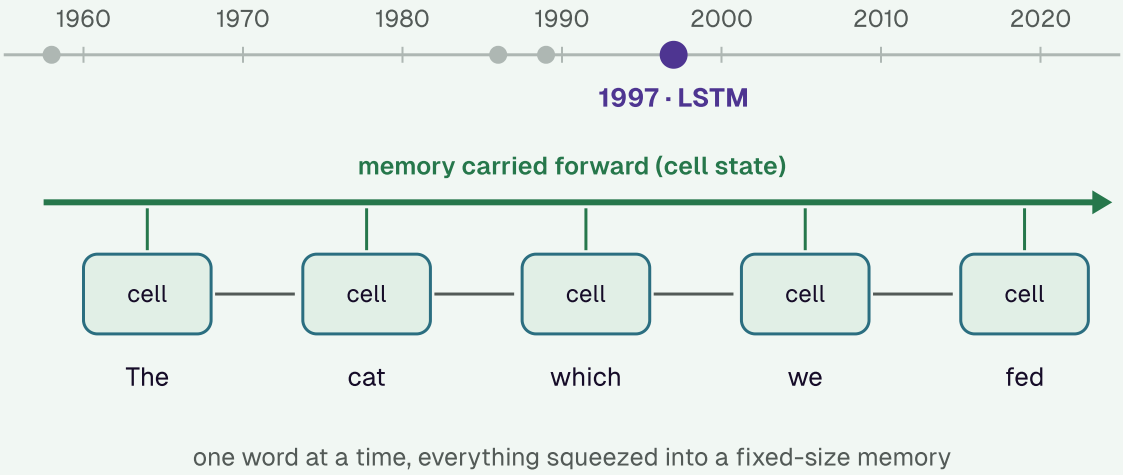
Frank Rosenblatt's **perceptron** (1958) weighs a few inputs, adds them up, and answers yes or no. A simple rule nudges the weights after each mistake. It can only draw a straight line between two classes, so it cannot learn XOR, a limitation Minsky and Papert analysed in 1969.



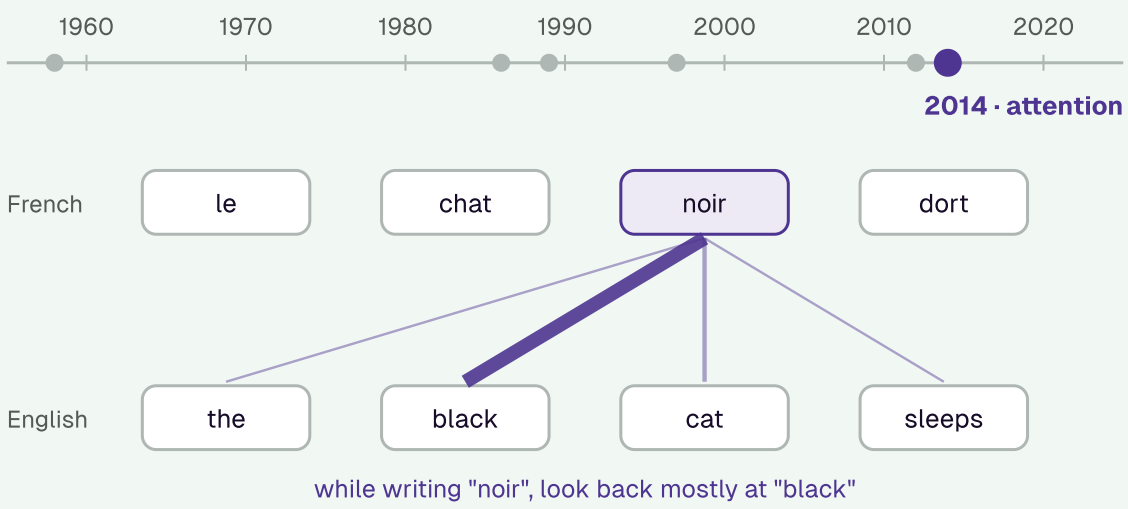
Hidden layers fix the straight-line limit, but how do you train weights deep inside? **Backpropagation**, popularised by Rumelhart, Hinton and Williams (1986), uses the chain rule to send the output error backwards and give every weight its share of the blame. Every network in this guide still trains this way.



Convolutional networks slide one small detector across an image, reusing the same weights at every position. LeCun's LeNet (1989) read handwritten zip codes. In 2012 AlexNet (Krizhevsky, Sutskever and Hinton) combined a deep CNN, GPUs and the ImageNet dataset and won by a wide margin, which started the deep learning boom.

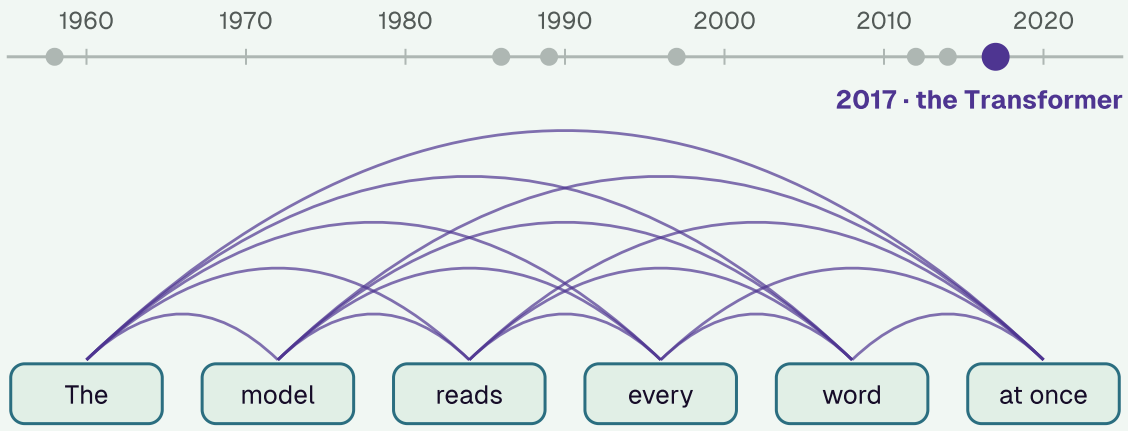


Language arrives in order, so **recurrent networks** read one word at a time and carry a memory forward. Plain RNNs forget quickly, for the same multiplying-gradients reason as deep stacks. The **LSTM** (Hochreiter and Schmidhuber 1997) adds gates and an additive memory lane that lets information survive many steps. The catch: it must read strictly in order, and everything must fit in one fixed-size memory.



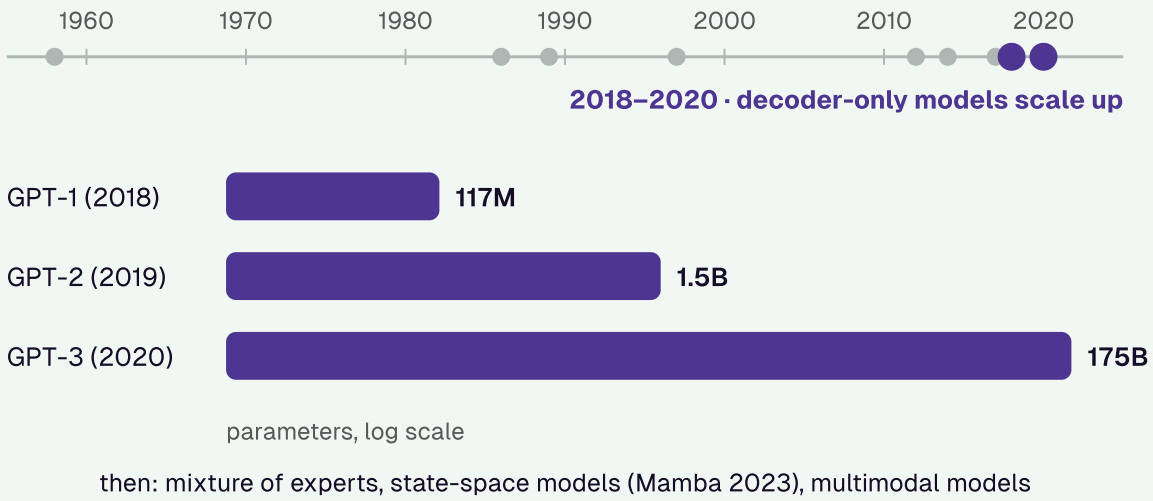
while writing "noir", look back mostly at "black"

Translation with one memory vector forced a whole sentence through a bottleneck. Bahdanau, Cho and Bengio (2014) let the decoder **look back** at every source word and decide how much each one matters for the word it is writing now. That weighted look-back is **attention**.



every token attends to every other token, all at the same time

"Attention Is All You Need" (Vaswani et al. 2017) dropped recurrence entirely. Every token looks at every other token in one parallel step, stacked in layers with residual connections and normalisation. No more reading in order means training uses GPUs fully, which made very large models practical.



then: mixture of experts, state-space models (Mamba 2023), multimodal models

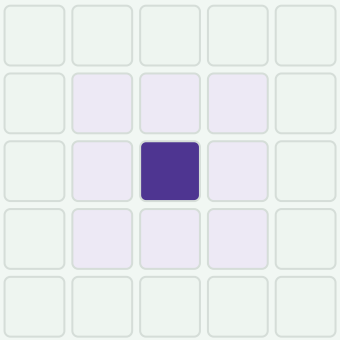
Keep only the decoder half, train it to predict the next token, and scale it. Parameter counts grew about 1,500-fold in two years (Radford et al. 2018, 2019; Brown et al. 2020), and loss fell along smooth scaling laws. Newer branches change the cost profile: mixture-of-experts uses only part of the model per token, and state-space models such as Mamba (Gu and Dao 2023) replace attention's all-pairs look with a running state.

The common thread: match the bias to the data

Diagram

Four shapes of data, four bets

Why there is a family of architectures at all: each one is built for a shape of data.



A photo is a grid

every pixel has the same 8 neighbours

and a pattern means the same thing anywhere

bet: convolution

one small detector, slid over every position

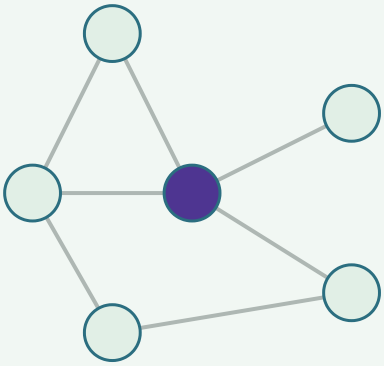
Every architecture starts from the shape of the data. In a photo, each pixel has the same neighbourhood and a cat's ear is a cat's ear wherever it appears. A convolution bakes in exactly those two facts.

A sentence is a sequence: order changes meaning

bet: recurrence (carry a memory left to right)

or attention plus position information

In a sentence, order is meaning: stress a different word and "I never said she stole it" changes sense. Recurrent networks bake in left-to-right reading. Transformers bake in almost nothing and add position information as an extra input.



A molecule is a graph

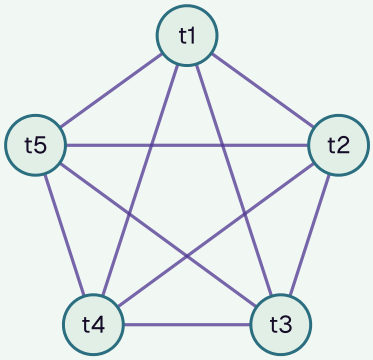
atoms are nodes, bonds are edges

each node has its own number of neighbours

bet: message passing

each node listens only to the nodes it is linked to

Molecules, road maps and friendships have no grid and no single order. Each node has its own neighbours. A graph neural network bakes in only the links: every node updates itself from the nodes it is connected to.



A transformer is a GNN where every token links to every token

attention weights decide how much each link counts

the links are learned per input, not given

weakest bet of all: most data needed, most headroom

Take a graph network and connect every node to every other node: that is a transformer layer, with attention deciding how much each connection counts. It assumes the least, so it needs the most data, and it keeps improving when data and compute are plentiful.

Plain words

One recipe, many kitchens

All of these networks are made from the same two ingredients: multiply-and-add, then a simple bend (a nonlinearity). What changes is **who gets multiplied with whom**. A photo network only mixes

nearby pixels. A sentence network mixes words in order. A graph network mixes things that are linked. A transformer lets everything mix with everything and learns which mixes matter.

For leaders

Why the family tree still matters at a transformer lab

The frontier is dominated by transformers, but the older branches keep coming back in disguise. Residual connections came from CNNs. Gating from LSTMs reappears in modern MLP blocks (SwiGLU) and in state-space models. Recurrent state is back in Mamba-style layers because it makes long-context inference cheap. Graph thinking shows up in protein, materials, weather and agent-tool systems. Knowing which bias each branch encodes lets you judge proposals quickly: "this layer is a convolution over time" or "this is an RNN with a better gate" tells you its cost profile and likely failure modes before you read the paper.

Under the hood

Each branch restricts the same linear map

Write a layer as $h' = \phi(Wh)$ over all the positions of an input at once. Each architecture constrains W:

- **MLP:** W dense, no constraint.
- **CNN:** W is banded and every band holds the same kernel (a Toeplitz structure): locality plus weight sharing.
- **RNN:** the same W applied once per time step, with the state threaded through: weight sharing over time, strict order.
- **GNN:** mixing only along the edges of a given graph, with shared weights per node: W follows the adjacency matrix.
- **Transformer:** mixing weights computed from the input itself (attention), over all pairs: a data-dependent, fully connected graph.

Perceptron and backpropagation: the roots

Plain words

A voting machine that learns from mistakes

A perceptron is a tiny voting machine. Each piece of evidence gets a weight, the weighted votes are added up, and if the total passes a threshold the answer is "yes". When it gets an example wrong, it shifts the weights a little toward the right answer. That is learning in its simplest form.

One perceptron can only split the world with a straight line. Stack several in layers and they can carve out any shape, but then you need a way to know which inner weight caused a mistake. **Backpropagation** is that bookkeeping: it traces the error back through the layers and gives every weight its share of the blame.

For leaders

Ideas waited decades for compute and data

The perceptron (1958) and backpropagation (popularised in 1986) were in place long before neural networks dominated. What was missing was compute, data, and a few training tricks. Roughly 25 years passed between backpropagation's popularisation and AlexNet. The lesson for planning research portfolios: an idea that "does not work" may be an idea waiting for scale, and an idea that works at small scale may stop working at large scale. Both are reasons to measure across scales instead of trusting one result.

Under the hood

The perceptron rule and the chain rule

$$\hat{y} = \mathbb{1}[w^\top x + b > 0], \qquad w \leftarrow w + \eta\,(y - \hat{y})\,x$$

x inputs, w weights, b bias, η step size, labels $y \in \{0, 1\}$. The update only fires on mistakes. Rosenblatt's convergence result guarantees it finds a separating line if one exists.

XOR has no separating line, so no single perceptron learns it. A hidden layer solves it, but the step function has zero gradient almost everywhere. Backpropagation (Rumelhart, Hinton and Williams 1986) uses smooth activations so the chain rule applies: for weights in layer l, $\partial \mathcal{L}/\partial W_l = \delta_l\, h_{l-1}^\top$ with $\delta_l = (W_{l+1}^\top \delta_{l+1}) \odot \phi'(z_l)$. The same backward product of Jacobians is what vanishes or explodes in deep stacks (see designing a network) and what how networks learn covers step by step.

Convolutional networks: weight sharing for grids

Plain words

A flashlight that looks for one pattern

Imagine searching a big photo for eyes with a small flashlight. You sweep it across every part of the picture, and you use the same idea of "what an eye looks like" everywhere. That is a convolution: one small pattern detector, slid across the whole image. The first layer looks for edges, the next

combines edges into corners and curves, the next into eyes and wheels, and so on up to whole objects.

For leaders

Still the right tool when data is limited

CNNs dominated computer vision from 2012 until around 2020, and they remain the default for small datasets, edge devices and real-time systems, because their built-in assumptions mean they need less data and less compute. Vision Transformers (Dosovitskiy et al. 2020) matched or beat CNNs only when pretrained on very large datasets; trained on ImageNet alone they lagged comparable ResNets. Modernised CNNs such as ConvNeXt (Liu et al. 2022) showed that much of the remaining gap came from training recipes rather than the architecture. Many multimodal models still use convolutional pieces in their image or audio front ends.

Under the hood

Convolution, receptive fields, and the 2012 recipe

$$y_{i,j,c'} = \sum_{c} \sum_{u=-k}^{k} \sum_{v=-k}^{k} K_{u,v,c,c'}\; x_{i+u,\,j+v,\,c}$$

x input with channels c, K a $(2k{+}1) \times (2k{+}1)$ kernel shared across all positions (i, j), output channels c'. Parameter count is independent of image size.

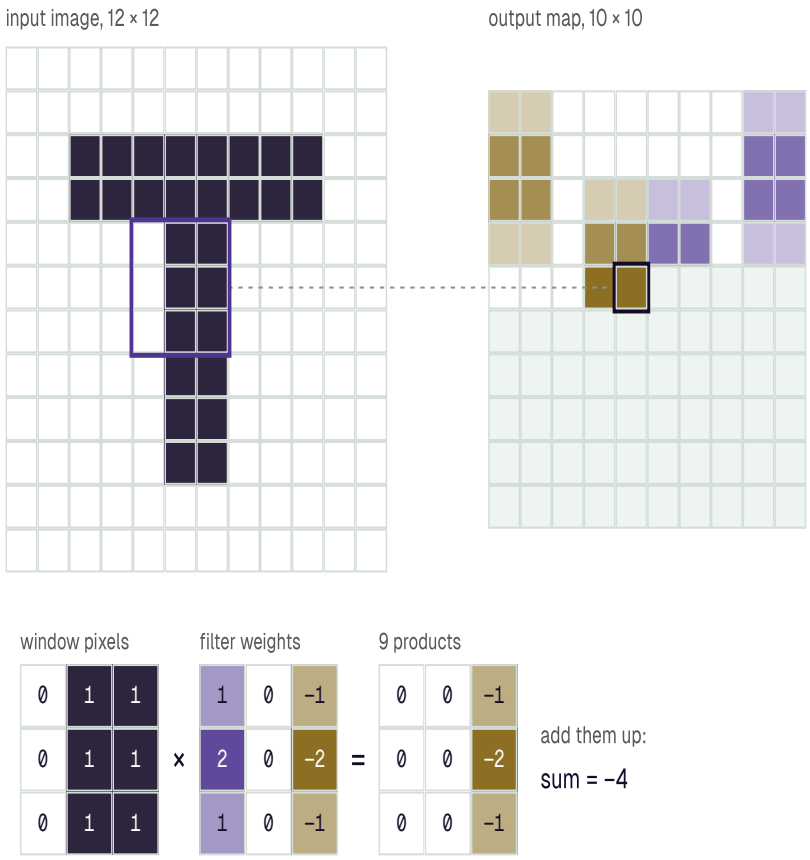
Stacking L layers of 3×3 kernels gives each output a **receptive field** of $(2L + 1) \times (2L + 1)$ pixels; pooling and striding grow it faster. LeNet (LeCun et al. 1989) applied this to handwritten digits. AlexNet (Krizhevsky, Sutskever and Hinton 2012) had five convolutional and three fully connected layers, about 60 million parameters, and used ReLU activations, dropout, data augmentation and two GPUs. Its ImageNet top-5 error of about 15% against 26% for the runner-up was the moment the field switched. ResNets (2015) then pushed CNNs past 100 layers with the residual connection that every transformer now uses.

Here is the flashlight in slow motion: one 3×3 filter, one small image, and the output it builds one cell at a time.

Interactive

A filter sliding over an image (shown at its default setting)

Pick a filter, then press Step or Run to slide it across the letter T and watch the 9 multiplications at each stop add up to one output cell; click any input or output cell to jump there.



bright pixel (1) positive output negative output outlined box: the 3×3 window being read now

Weights in this filter	Positions that reuse them	Dense layer, same input and output	Output at this position
9	100	14,400	−4

Position 45 of 100 (row 5, column 5): the 9 products add up to **−4**. This filter subtracts the right column from the left column (the middle column counts double). It is positive where the left side of the window is brighter, negative where the right side is, and 0 on flat patches, so it lights up the vertical sides of the stem and the ends of the bar. **Weight sharing:** the same 9 weights are reused at all 100 positions (press Fill all to see the whole map). A dense layer connecting all 144 pixels to all 100 outputs would need 14,400 weights, 1,600 times more, and would have to learn the same detector separately at every position. At real scale the gap is enormous: for a 224 × 224 colour image and 64 output channels, a 3 × 3 convolution has 1,728 weights; a dense layer

with the same input and output sizes would have about 483 billion (biases ignored in both counts).

Recurrent networks and LSTMs: memory for sequences

Plain words

Reading a book with one sticky note

Picture reading a novel one word at a time, and the only thing you may keep is a single sticky note that you rewrite after every word. You must decide, word by word, what to keep on the note and what to rub out. That is a recurrent network. By page 300, details from page 1 are hard to hold on to. An **LSTM** gives the note a protected section and simple rules (gates) for when to write, keep or erase, so important facts last much longer.

For leaders

Two bottlenecks, and why recurrence is back

RNNs and LSTMs powered speech recognition and machine translation through the mid-2010s. They had two structural limits. **Training is sequential:** step 100 needs step 99, so you cannot spread one sequence across thousands of GPU cores. **Memory is fixed-size:** everything the model knows about the past must fit in one state vector. The transformer removed both, at the price of attention cost that grows with the square of the context length and a KV cache that grows linearly with it.

That price is why recurrence has returned. A recurrent state costs the same per token no matter how long the context is, which is attractive for very long inputs and cheap inference. Modern state-space models and linear-attention layers are recurrent at inference while still training in parallel.

Under the hood

Vanishing through time, and the additive cell

$$h_t = \tanh(W h_{t-1} + U x_t + b), \qquad \frac{\partial h_T}{\partial h_t} = \prod_{k=t+1}^{T} \operatorname{diag}\big(\tanh'(z_k)\big)\, W$$

The same matrix W is multiplied in at every step, so gradients across long gaps shrink or grow geometrically, as analysed by Hochreiter (1991) and Bengio, Simard and Frasconi (1994).

The LSTM (Hochreiter and Schmidhuber 1997) adds a cell state updated additively, $c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$, with sigmoid gates f_t, i_t, o_t computed from x_t and h_{t-1}. When the forget gate f_t is near 1, the gradient flows back through c almost unchanged: the same trick as a residual connection, applied through time. (The forget gate itself was added by Gers, Schmidhuber and Cummins in 2000.) The GRU (Cho et al. 2014) is a lighter variant. Sequence-to-sequence models (Sutskever, Vinyals and Le 2014) encoded a whole source sentence into one vector and decoded from it, and that single vector was the bottleneck attention was invented to remove.

Attention and the transformer

Plain words

Flip back to any page

Instead of keeping one sticky note, what if you could flip back to any earlier page whenever you needed it? While translating a word, you look back at the original sentence and focus on the few words that matter for this one. That is **attention**. The transformer goes all the way: every word in the text looks at every other word at the same time and decides how much to listen to each.

For leaders

Why this branch won language

Transformers won because their computation fits the hardware. Every position in a training sequence is processed in parallel, so a lab can turn more GPUs into a bigger, better model almost directly, and the gains follow smooth scaling laws. The decoder-only variant (the GPT line) trained on next-token prediction turned out to be general enough for code, maths, dialogue and tool use (see pretraining and inside an LLM). The costs to manage are attention that is quadratic in context length during training and prefill, and a KV cache that dominates serving memory (see inference).

Under the hood

From additive attention to scaled dot products

Bahdanau, Cho and Bengio (2014) scored each source state h_j against the decoder state s_{i-1} with a small network, $e_{ij} = v^\top \tanh(W s_{i-1} + U h_j)$, normalised the scores with a softmax, and fed the weighted sum of source states to the decoder. Vaswani et al. (2017) replaced recurrence entirely with stacked self-attention:

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\Big(\frac{QK^\top}{\sqrt{d_k}} + M\Big)V$$

Q, K, V are linear projections of the token vectors; d_k the key size; M a mask ($-\infty$ above the diagonal for a causal decoder). Cost $O(n^2 d)$ for n tokens.

The base Transformer had 6 encoder and 6 decoder layers, width 512, 8 heads and a 2,048-wide MLP. GPT (Radford et al. 2018) kept only the decoder and trained it on next-token prediction; GPT-2 (2019) and GPT-3 (Brown et al. 2020) scaled the same recipe to 1.5B and 175B parameters. Since then most changes have been refinements: pre-norm and RMSNorm, rotary position embeddings, SwiGLU MLPs, grouped-query attention, and mixture-of-experts layers.

Beyond the dense transformer: mixture of experts and state-space models

Plain words

A hospital with specialists, and a running summary

A **mixture of experts** is like a hospital with a front desk. Each patient is sent to two or three specialists, not to every doctor in the building. The hospital can employ hundreds of specialists, but each visit costs only a few consultations.

A **state-space model** goes back to the sticky-note idea with a much better note: a carefully designed running summary that updates with each word and can choose what to remember based on what it reads. It costs the same per word no matter how long the text is.

For leaders

Different cost profiles, not different paradigms

Mixture-of-experts separates **total parameters** (how much the model can store) from **active parameters** (what each token costs). Mixtral 8x7B (Jiang et al. 2024) has about 47B parameters but uses about 13B per token; most leading open-weight models released since 2025 are MoE. The costs: all experts must sit in memory, routing can be unstable or unbalanced, and serving needs expert parallelism. State-space models such as Mamba (Gu and Dao 2023) offer linear-time training and constant memory per token at inference; hybrids that interleave a few attention layers with many recurrent layers are the common way to keep attention's precise recall. When a proposal claims a

new architecture, ask which of these cost profiles it changes and whether quality holds at matched compute.

Under the hood

Routing and selective scans

$$\mathrm{MoE}(x) = \sum_{i \in \mathrm{TopK}(g(x))} \mathrm{softmax}(g(x))_i \, E_i(x), \qquad h_t = \bar{A}_t h_{t-1} + \bar{B}_t x_t, \;\; y_t = C_t h_t$$

Left: a router g scores experts E_i and only the top k run (Shazeer et al. 2017; Switch Transformer, Fedus et al. 2021, used $k = 1$); an auxiliary loss keeps load balanced. Right: a linear recurrence; in Mamba, $\bar{A}_t, \bar{B}_t, C_t$ depend on the input (a "selective" SSM), computed with a parallel scan during training.

Earlier structured SSMs such as S4 (Gu, Goel and Ré 2022) used fixed, input-independent dynamics, which can be computed as one long convolution but cannot decide what to remember based on content. Making the parameters input-dependent gives that ability back at the cost of the convolution trick, which Mamba recovers with a hardware-aware scan. A fixed-size state still cannot recall arbitrary earlier tokens exactly, which is the main argument for hybrids.

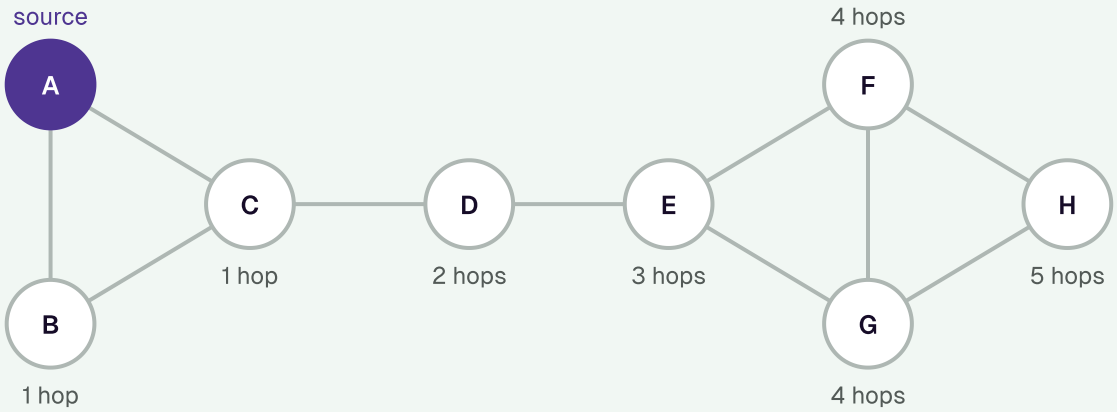
How GNNs came to life

Diagram

Gossip in a small town

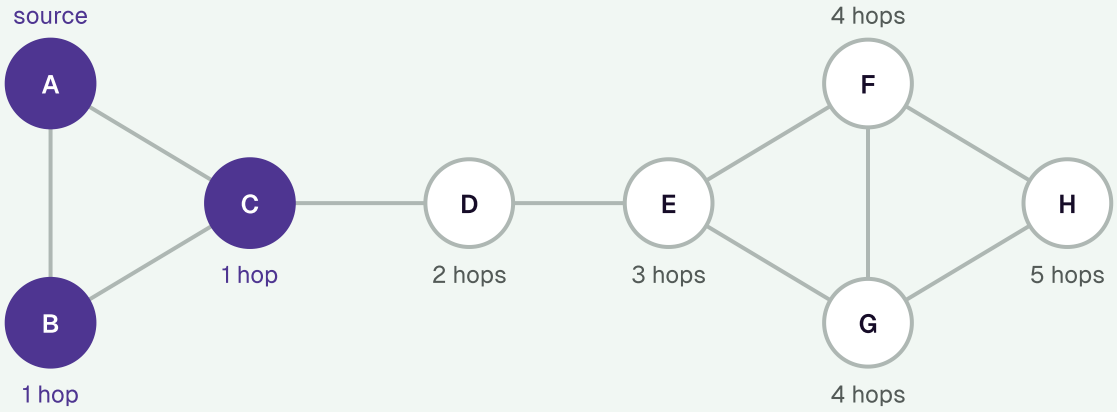
How graph neural networks pass messages, how far information travels, and why too many rounds wash everything out.

Round 0: only house A has heard the news



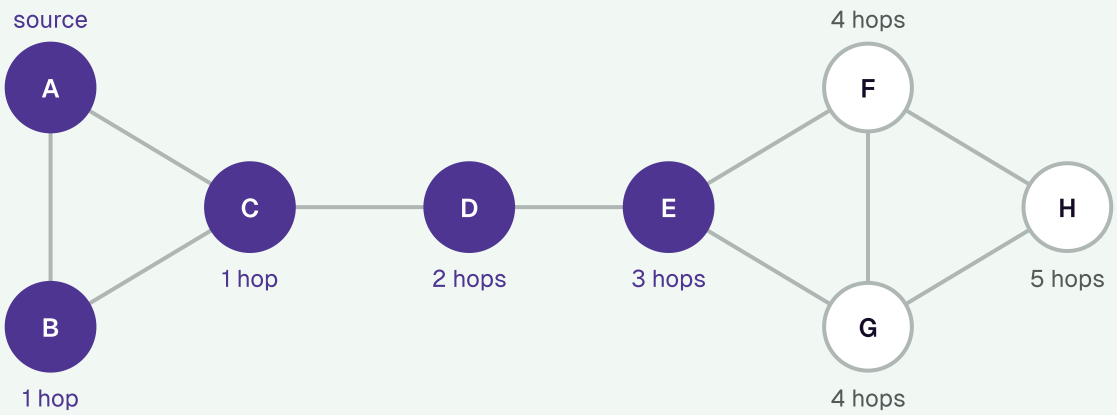
A town where houses are linked by roads is a graph. One house hears some news. Each evening, every house chats only with the houses it is directly linked to. That evening chat is one round of **message passing**.

Round 1: A's neighbours B and C know



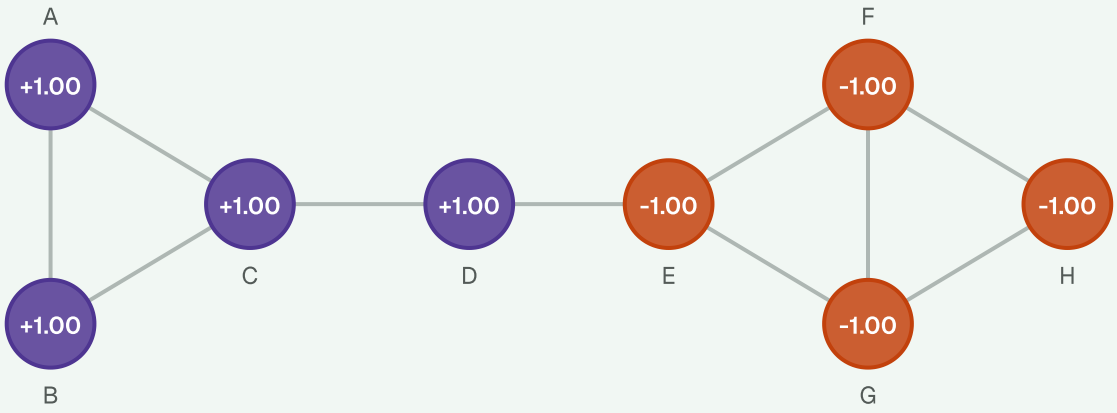
After one evening, the news has reached A's direct neighbours, one hop away, and no further. In a graph neural network, one layer lets each node gather information from its 1-hop neighbourhood.

Round 3: everything within 3 hops



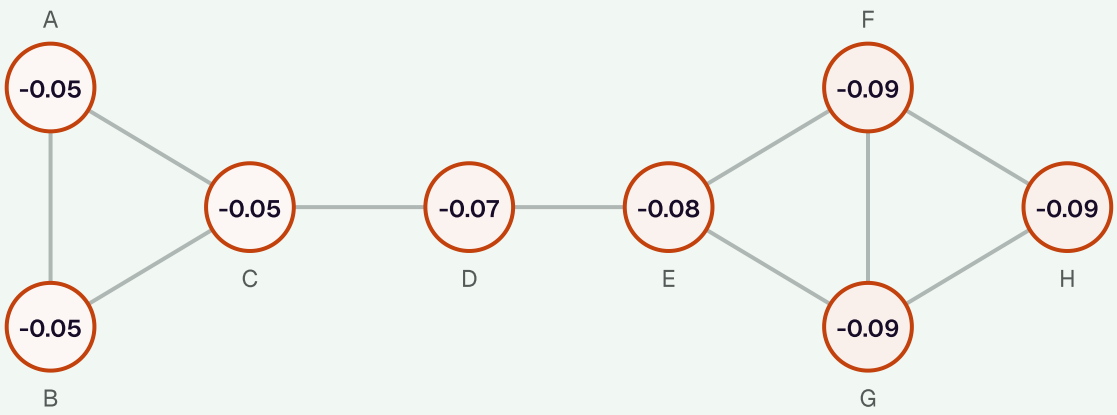
After three evenings the news has crossed the bridge to E, three hops out. F, G and H are four and five hops away and still have not heard. **k rounds reach exactly k hops**, so a GNN needs as many layers as the distance information must travel.

Opinions: west side starts at +1, east side at −1



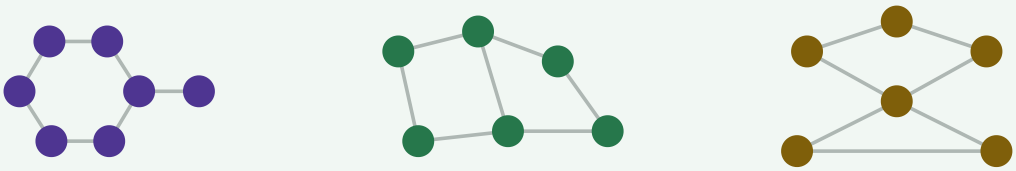
Now make it opinions. Each evening every house replaces its view with the average of its own and its neighbours'. The west side starts at +1 and the east side at −1. At first the two sides are easy to tell apart, which is what you want a GNN to learn from.

After 60 rounds: everyone thinks the same



After 60 evenings everyone holds almost the same opinion, between −0.05 and −0.09. You can no longer tell which side a house started on. This is **over-smoothing**, and it is why plain GNNs are usually only 2 to 4 layers deep, unlike 100-layer transformers.

Where the graph is the data



molecules
atoms and bonds:
drug discovery

road networks
road segments:
arrival-time (ETA) prediction

people and items
who likes what:
recommendations

Anywhere the links are the point, graph networks fit: atoms and bonds for drug discovery, road segments for travel-time prediction, people and items for recommendations, and mesh points on the globe for weather forecasting.

Plain words

Some data is a web, not a picture or a sentence

A molecule is atoms joined by bonds. A city is junctions joined by roads. A social network is people joined by friendships. None of these is a grid like a photo or a line like a sentence: each item has its own set of connections, and the connections are the important part. A **graph neural network** (GNN) is built for this. Each item (a **node**) keeps a small description of itself, and in every round it updates

that description using messages from the nodes it is linked to. After a few rounds, each node's description reflects its neighbourhood.

For leaders

A short history and where GNNs earn their keep

1. **First graph neural networks (2005 to 2009).** Gori, Monfardini and Scarselli (2005) and Scarselli et al. (2009, "The Graph Neural Network Model") let each node repeatedly update its state from its neighbours until the states settled.

2. **Spectral methods (2014 to 2016).** Bruna et al. (2014) defined convolution on graphs through the eigenvectors of the graph Laplacian, a graph version of the Fourier transform. Defferrard et al. (2016, ChebNet) made it cheaper with polynomial filters.

3. **GCN (Kipf and Welling 2017).** A first-order simplification of those filters became a one-line layer: average over neighbours, multiply by a weight matrix. It was simple, fast and strong, and it made GNNs mainstream.

4. **Message passing as the unifying frame (Gilmer et al. 2017).** Applied to molecular property prediction, it showed most GNNs fit one template: compute messages along edges, aggregate them, update each node.

5. **Scaling and attention (2017 to 2018).** GraphSAGE (Hamilton, Ying and Leskovec 2017) sampled neighbours so GNNs could run on huge graphs and on nodes never seen in training. GAT (Veličković et al. 2018) let each node learn how much attention to pay to each neighbour.

Real deployments: Pinterest's PinSage (Ying et al. 2018) ran a GraphSAGE-style model over a graph of billions of pins and boards for recommendations. DeepMind and Google Maps (Derrow-Pinion et al. 2021) used a GNN over road segments for arrival-time prediction and reported reducing poor ETA predictions by over 40% in cities such as Sydney compared with the previous production system. Stokes et al. (2020) trained a directed message-passing network on molecules, screened large chemical libraries, and identified the antibiotic candidate halicin. GraphCast (Lam et al. 2023) runs message passing over a mesh covering the globe and beat the leading operational deterministic forecast system on about 90% of the targets the authors evaluated. AlphaFold 2 (Jumper et al. 2021) is related but not a standard GNN: it keeps a learned representation for every pair of amino-acid residues and refines it with attention and "triangle" updates, which can be read as reasoning over a dense graph of residues.

Watch out

Plain message-passing GNNs get worse, not better, when stacked deep. Each layer averages neighbours, so after many layers all nodes look alike (**over-smoothing**), and information from far away must squeeze through narrow bridges in the graph (**over-squashing**). Most production GNNs are 2 to 4 layers deep. A proposal for a 50-layer GNN needs residual connections, normalisation, or a graph-transformer design, and evidence that the extra depth helps.

Under the hood

GCN, message passing, attention over neighbours

$$H^{(l+1)} = \sigma\left(\tilde{D}^{-1/2}\tilde{A}\,\tilde{D}^{-1/2}H^{(l)}W^{(l)}\right), \qquad \tilde{A} = A + I$$

GCN layer (Kipf and Welling 2017). A is the adjacency matrix, I adds self-loops, $\tilde{D}$ is the degree matrix of $\tilde{A}$, $H^{(l)}$ holds one row of features per node, $W^{(l)}$ is shared by all nodes.

$$m_v^{(t+1)} = \sum_{u \in \mathcal{N}(v)} M_t\left(h_v^{(t)}, h_u^{(t)}, e_{uv}\right), \qquad h_v^{(t+1)} = U_t\left(h_v^{(t)}, m_v^{(t+1)}\right)$$

Message passing (Gilmer et al. 2017): a message function M_t on each edge (with edge features e_{uv}), a permutation-invariant aggregate (sum, mean, max), and an update U_t. GAT makes the aggregate a learned softmax over neighbours.

Two theoretical results frame what GNNs can and cannot do. Xu et al. (2019) showed message-passing GNNs are at most as powerful as the 1-dimensional Weisfeiler–Lehman graph isomorphism test at telling graphs apart, and that sum aggregation reaches that bound where mean and max do not. Li, Han and Wu (2018) showed a GCN layer is a form of Laplacian smoothing, so repeated layers drive node features toward a constant on each connected component: over-smoothing. The update in the code below, with the weights and nonlinearity removed, is exactly that smoothing operator $\tilde{D}^{-1}\tilde{A}$, and its repeated application converges to a constant vector on a connected graph. Over-squashing (Alon and Yahav 2021) is the complementary problem: the number of nodes within k hops grows fast, but all their information must pass through fixed-size vectors along a few edges.

python

```python
# A 6-node graph as an adjacency list: a path 0-1-2-3-4 with a branch 2-5.
graph = {0: [1], 1: [0, 2], 2: [1, 3, 5], 3: [2, 4], 4: [3], 5: [2]}

def message_pass(h, graph):
    """One round: each node's new value is the mean of itself and its neighbours.
    This is the GCN-style update with the weights and nonlinearity left out."""
    new = {}
    for v, nbrs in graph.items():
        msgs = [h[v]] + [h[u] for u in nbrs]   # self-loop keeps the node's own value
        new[v] = sum(msgs) / len(msgs)
    return new

h = {v: 0.0 for v in graph}
h[0] = 1.0                                     # only node 0 knows something
for k in range(1, 31):
    h = message_pass(h, graph)
    if k in (1, 2, 3, 30):
        reached = sum(1 for v in h if h[v] > 0)      # nodes within k hops
        spread = max(h.values()) - min(h.values())   # how different nodes still are
        print(k, reached, f"{spread:.3f}")
# 1 2 0.500
# 2 3 0.417
# 3 5 0.347    node 5 is 3 hops away (0-1-2-5); node 4 is 4 hops away
# 30 6 0.003   over-smoothed: every node holds nearly the same value
```

rust

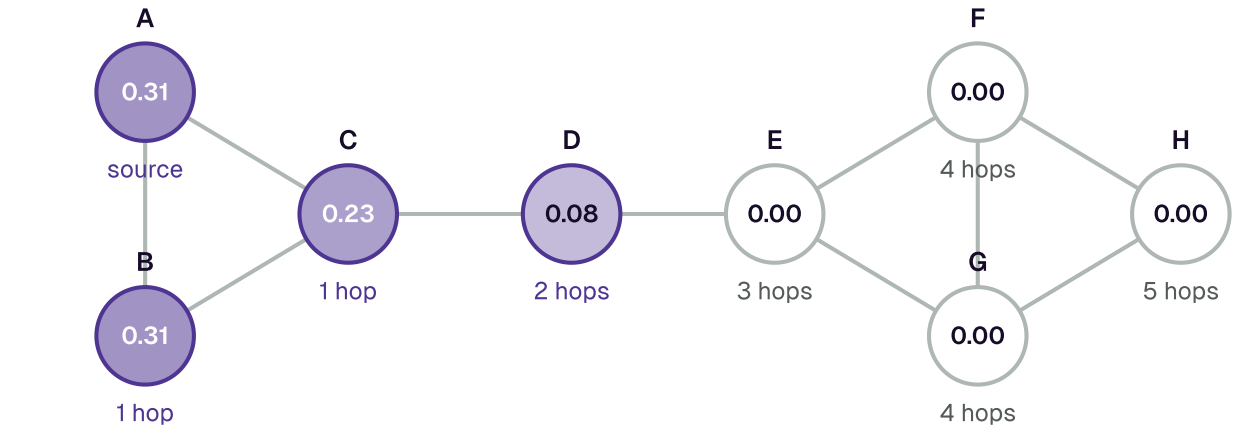
```rust
/// One round: each node's new value is the mean of itself and its neighbours.
/// This is the GCN-style update with the weights and nonlinearity left out.
fn message_pass(h: &[f64], graph: &[Vec<usize>]) -> Vec<f64> {
    graph
        .iter()
        .enumerate()
        .map(|(v, nbrs)| {
            let mut sum = h[v]; // self-loop keeps the node's own value
            for &u in nbrs {
                sum += h[u];
            }
            sum / (1 + nbrs.len()) as f64
        })
        .collect()
}

fn main() {
    // A 6-node graph as an adjacency list: a path 0-1-2-3-4 with a branch 2-5.
    let graph: Vec<Vec<usize>> = vec![vec![1], vec![0, 2], vec![1, 3, 5], vec![2, 4], vec![3], vec![2]];
    let mut h = vec![0.0; graph.len()];
    h[0] = 1.0; // only node 0 knows something
    for k in 1..=30 {
        h = message_pass(&h, &graph);
        if [1, 2, 3, 30].contains(&k) {
            let reached = h.iter().filter(|&&x| x > 0.0).count(); // nodes within k hops
            let max = h.iter().cloned().fold(f64::MIN, f64::max);
            let min = h.iter().cloned().fold(f64::MAX, f64::min);
            println!("{} {} {:.3}", k, reached, max - min); // how different nodes still are
        }
    }
    // 1 2 0.500
```

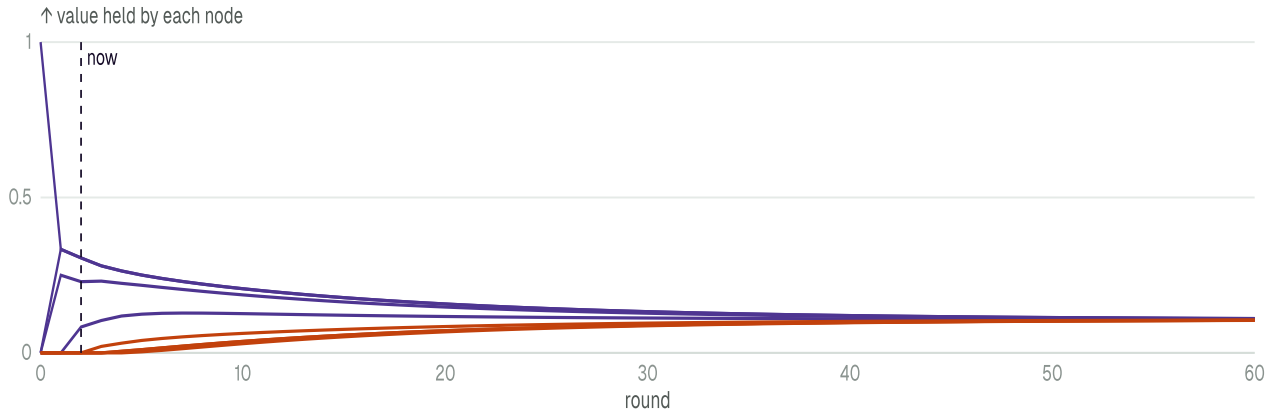
```
    // 2 3 0.417
    // 3 5 0.347    node 5 is 3 hops away (0-1-2-5); node 4 is 4 hops away
    // 30 6 0.003   over-smoothed: every node holds nearly the same value
}
```

Interactive

Message passing on a small graph (shown at its default setting)

Each round, every node replaces its value with the average of itself and its neighbours. In "one node knows", watch information reach exactly k hops in k rounds. In "two groups", watch the groups blur together as rounds pile up. The narrow chain C–D–E between the two groups slows everything down.





left group (A–D) right group (E–H)

Nodes reached	Spread (max – min)	Left avg – right avg
4 / 8	**0.306**	**0.231**

After **2** rounds, A's information has reached every node within 2 hops (4 of 8) and no further. A GNN with k layers can only see a k-hop neighbourhood.

A transformer is a GNN on a fully connected graph

Plain words

Everyone is everyone's neighbour

In the gossip town, each house only talks to houses it is linked to by a road. Now imagine a town where everyone can talk to everyone, and each person decides for themselves how much attention to pay to each other person. That is exactly what a transformer layer does with the words of a sentence. The words are the nodes, every pair is connected, and attention decides how loud each connection is.

For leaders

Why this view is useful

Seeing a transformer as a graph network explains several design facts at once. Attention ignores order unless you add position information, because graphs have no inherent order. Adding a known graph to a transformer (through attention masks or structural encodings) is how graph transformers inject domain structure, and causal masking is itself a graph: each token links only to earlier tokens. And the over-smoothing worry partly carries over: deep transformers can collapse token representations toward each other ("rank collapse"), which residual connections and normalisation help prevent. When a team proposes a GNN for structured data, a fair baseline is a transformer with the structure encoded, and the reverse also holds.

Under the hood

Side by side

$$\text{GAT: } h'_v = \sigma\Big(\sum_{u \in \mathcal{N}(v)} \alpha_{vu} W h_u\Big), \qquad \text{self-attention: } h'_v = \sum_{u=1}^{n} \text{softmax}_u\Big(\frac{q_v^\top k_u}{\sqrt{d_k}}\Big) W_V h_u$$

Both are attention-weighted message passing. The transformer sets $\mathcal{N}(v)$ to every token (or every earlier token under a causal mask), uses dot-product scores, adds residuals, normalisation and an MLP per node, and runs several heads in parallel.

Two consequences. First, a transformer's attention weights are a learned, input-dependent adjacency matrix, recomputed for every input and every layer, while a GNN's graph is given. Second, the cost is $O(n^2)$ edges instead of $O(|E|)$, which is why sparse attention patterns (local windows, a few global tokens) are graph sparsification in disguise.

The family at a glance

Architecture	Data shape	Inductive bias	Strengths	Weaknesses	Where used
MLP	Fixed-length vector	None beyond smoothness	Simple; universal approximator	Parameters explode with input size; no structure reuse	Tabular data; the per-token block inside every transformer
CNN	Grids: images, spectrograms, video	Locality, weight sharing, translation equivariance	Data-efficient; fast; hardware-friendly	Long-range relations need depth or pooling	Vision on limited data, edge devices, image and audio front ends
RNN / LSTM	Sequences	Order; one state carried forward	Constant memory per step; streaming	Sequential training; fixed-size memory; forgets long range	Legacy speech and translation; on-device streaming
Transformer	Sets of tokens (sequences with positions)	Very weak: all pairs may interact, weights from data	Parallel training; scales smoothly; general	Quadratic attention; large KV cache; data-hungry	LLMs, vision transformers, speech, multimodal
Mixture of experts	Same as transformer	Different tokens need different sub-networks	More capacity per FLOP	Memory for all experts; routing instability; serving complexity	Many frontier-scale LLMs
State-space model (Mamba)	Long sequences	Compressible running state with input-	Linear time; constant inference memory	Exact recall of arbitrary past tokens is hard	Long-context and efficient models, often hybrids with attention

Architecture	Data shape	Inductive bias	Strengths	Weaknesses	Where used
		dependent updates			
GNN	Graphs: molecules, roads, social and item graphs, meshes	Only linked nodes interact; permutation invariance	Uses known structure; works on irregular data	Over-smoothing, over-squashing; usually shallow; WL expressivity limit	Drug and materials discovery, recommendations, ETA, weather

Eval lens

Claims that one architecture beats another are only as good as the comparison. Check that models were **compute-matched and data-matched**: the ViT result flips depending on whether pretraining data is ImageNet-scale or hundreds of millions of images, and historical "architecture" wins often bundled more data, more compute and better training recipes (ConvNeXt recovered much of the gap to vision transformers by modernising the training recipe and design of a CNN). For GNNs, Shchur et al. (2018) showed that rankings on standard citation-graph benchmarks changed with different train/test splits, so single-split leaderboards are weak evidence. Ask for several seeds, several splits, and the strongest tuned baseline of the other family.

Principal asks

What structure does this data have, and which architecture's bias matches it? Is the baseline the best-tuned transformer (with the structure encoded), at matched compute and data? Does the advantage survive at larger scale and more data, or is it a small-data effect of a stronger bias? What does the design cost at inference: memory per token, latency, and how it behaves on long inputs? For a GNN: how many layers, how was over-smoothing handled, and were results averaged over several splits?

Key idea

The family tree is a history of fixing bottlenecks. Perceptrons could not learn non-linear boundaries, so came hidden layers and backpropagation. Dense layers wasted parameters on images, so came convolutions. Language arrives in order, so came RNNs, then LSTMs to remember longer. One memory vector was a bottleneck, so came attention, and then the transformer, which removed recurrence and scaled. Graphs needed their own bias, so came GNNs, and a transformer turns out to be a GNN on a fully connected graph. Each step traded a built-in assumption for more data and compute.

Part III · Chapter 22

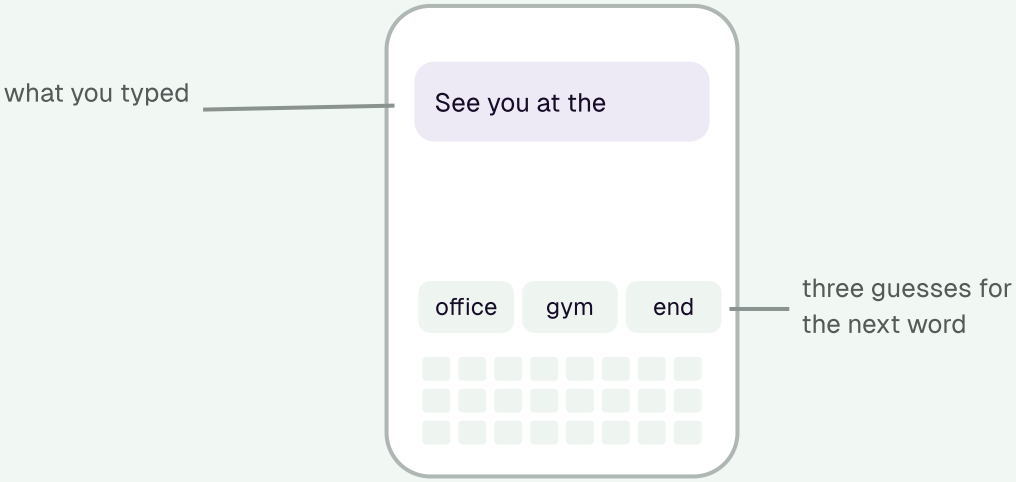
Inside an LLM: it only ever predicts the next token

Strip away the chat window, the persona and the tools, and a large language model does one thing: given some text, it outputs a probability for every possible next token. Generation is that one prediction run in a loop. This chapter walks the pipeline from text to prediction, part by part, so the rest of the guide (assistants, reasoning, evals) rests on a concrete picture. Tokenization and the training loss are covered in Pretraining; how the next token is chosen is covered in Inference.

Diagram

Your phone keyboard, a few hundred times in a row

Generation as autocomplete on repeat, in five pictures.



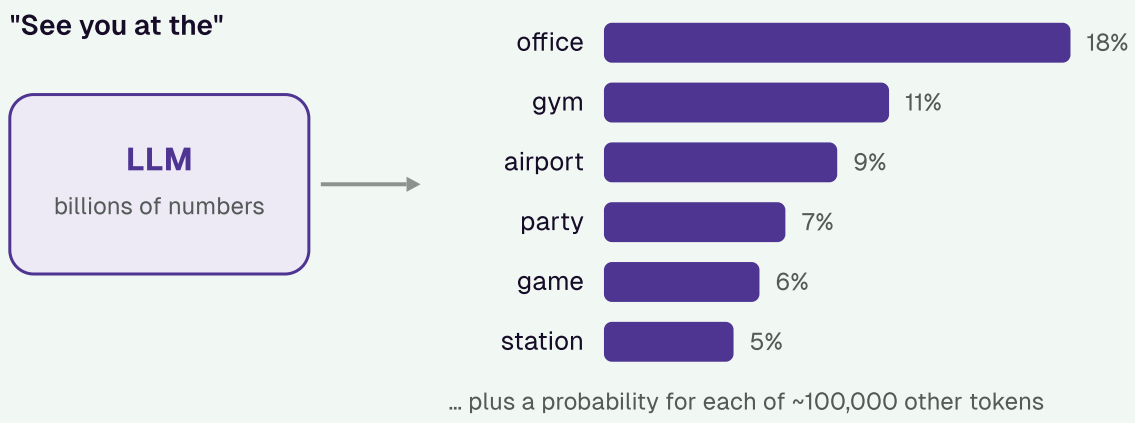
Your phone keyboard watches what you type and offers three guesses for the next word. It learned those guesses from lots of other people's messages.

Keep tapping the middle suggestion

tap 1	See you at the **gym**
tap 2	See you at the gym **tomorrow**
tap 3	See you at the gym tomorrow **morning**
tap 4	… gym tomorrow morning **at**
tap 5	… tomorrow morning at **the**
tap 6	… morning at the **gym**

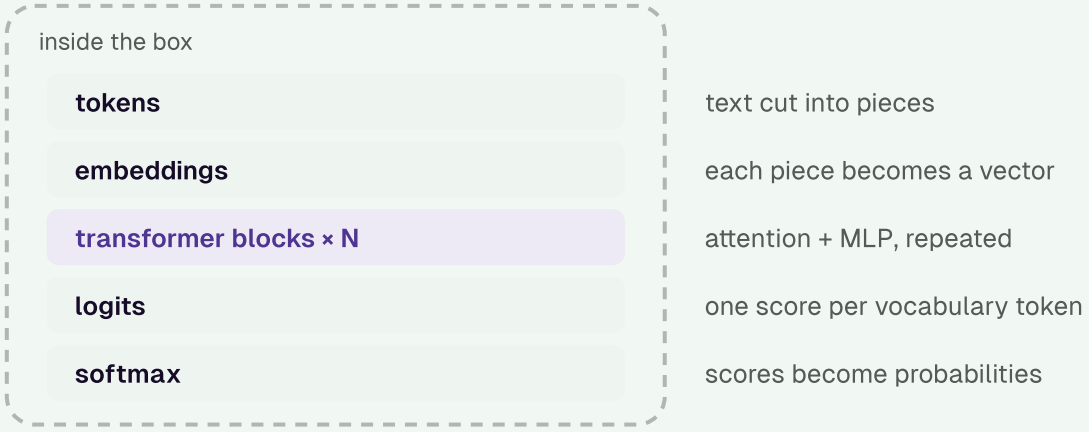
back to "gym" again:
it only sees the last word

Tap the middle guess over and over and a sentence appears, one word per tap. A phone keyboard only looks at the last word or two, so it soon loops back to "gym".



An LLM plays the same game with a far better guesser. It reads the whole message so far and gives every token in its vocabulary a probability (numbers illustrative).

Pick one token, glue it onto the text, and feed the longer text back in. That loop is the whole of generation. A 500-word answer is roughly 650 laps, because a word averages about 1.3 tokens in English.



Every part serves one output: a probability for each next token.

The rest of this chapter opens the box. Every part of it, however clever, exists to produce one thing: a probability for each possible next token.

The one job: predict the next token

Plain words

A very well-read autocomplete

Imagine someone who has read a large share of the public internet, many books and a lot of code, and has played one game the whole time: cover the next word, guess it, check, adjust. After trillions of rounds they become uncannily good at it. To guess the next word in a physics textbook well, you have to have picked up some physics. To guess the next line of a program, you need to follow what the program does.

That is a language model. When it "answers" you, it is guessing the next small piece of text (a **token**, roughly three quarters of a word in English), adding it to the page, and guessing again. There is no separate answering mode. Everything you see comes out of that loop.

For leaders

Why the framing matters for decisions

"It only predicts the next token" is accurate about the **interface** and says little about the **capability**. Both halves have consequences:

- **Cost and latency scale with output length.** Every generated token is a full pass through the model. Long answers and long chains of reasoning cost proportionally more compute and time.

- **Behaviour is shaped by the text it is given.** Prompts, system prompts, retrieved documents and tool outputs all work the same way: they are more text the prediction is conditioned on. See From predictor to assistant.
- **Errors are fluent.** The model is trained to produce likely text. Likely and true usually coincide, but when they diverge you get confident, well-phrased mistakes.
- **The interface is a probability distribution.** That makes outputs random unless decoding is greedy, and it gives you log-probabilities you can use for scoring, routing and calibration.

Under the hood

The autoregressive factorization

Any probability over sequences can be written as a product of next-token conditionals. A language model with parameters θ learns each factor:

$$P_\theta(x_1, \ldots, x_T) = \prod_{t=1}^{T} P_\theta(x_t \mid x_1, \ldots, x_{t-1})$$

x_t is the token at position t. One network computes every factor; it sees only positions before t when predicting x_t.

Training minimises the cross-entropy $-\log P_\theta(x_t \mid x_{<t})$ averaged over trillions of positions (Pretraining). Because of the causal mask (below), one forward pass over a training sequence of length T yields T predictions at once, which is why training is parallel while generation is sequential. Generation samples $x_{t+1} \sim P_\theta(\cdot \mid x_{\le t})$, appends it, and repeats until an end-of-sequence token or a length limit.

Key idea

A language model is a function from a token sequence to a probability distribution over the next token. Everything else (chat, tool calls, reasoning traces, refusals) is text produced by calling that function in a loop.

1 **Text**
"The cat sat on the"

2 **Tokens**
IDs from a fixed vocabulary

3 **Embeddings**
one vector per token

4 **Blocks × N**
attention + MLP on the residual stream

5 **Logits**
one score per vocabulary entry

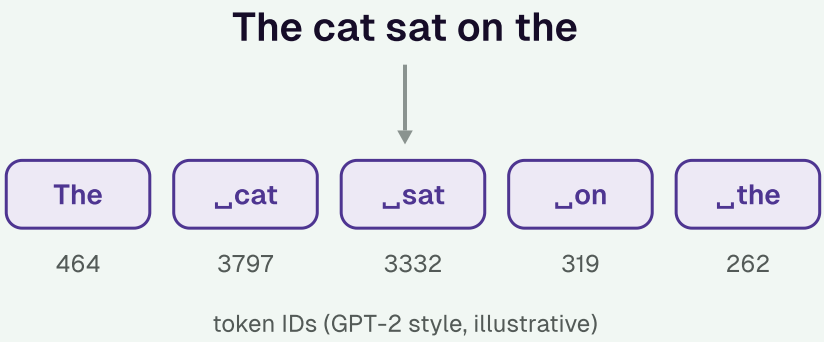
6 **Softmax, pick**
probabilities, then one token

7 **Append, repeat**
the autoregressive loop

Diagram

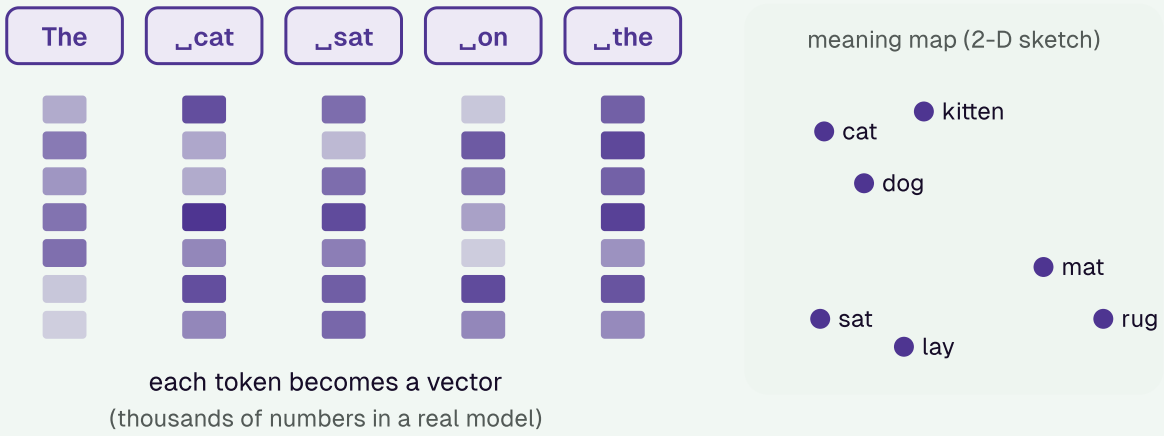
One trip through the model

From "The cat sat on the" to "mat", in six pictures.

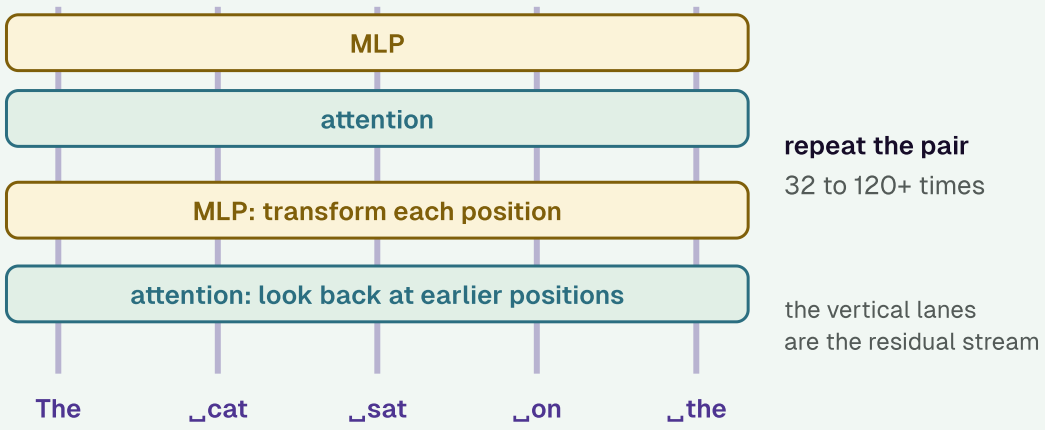


A tokenizer cuts text into pieces from a fixed vocabulary.

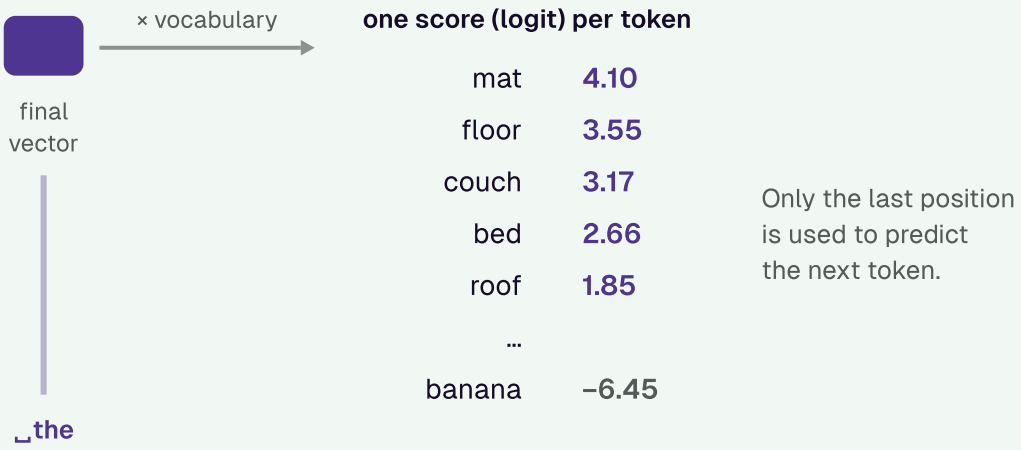
Step 1. A tokenizer cuts the text into pieces called tokens and swaps each for an ID number. Note the leading spaces: "␣cat" and "cat" are different tokens. (Details in Pretraining.)



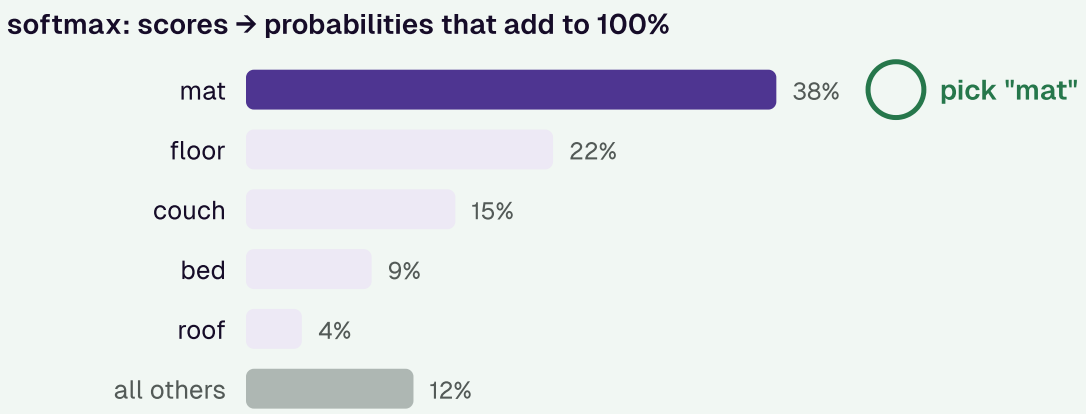
Step 2. Each token ID looks up a row in a big table and becomes a vector: a long list of numbers. Training arranges these so that tokens used in similar ways end up near each other, like neighbourhoods on a map.



Step 3. The vectors travel up through a stack of identical blocks. In each block, attention lets every position gather information from earlier positions, and an MLP transforms what each position holds. Each block adds its result to a running total, the residual stream.

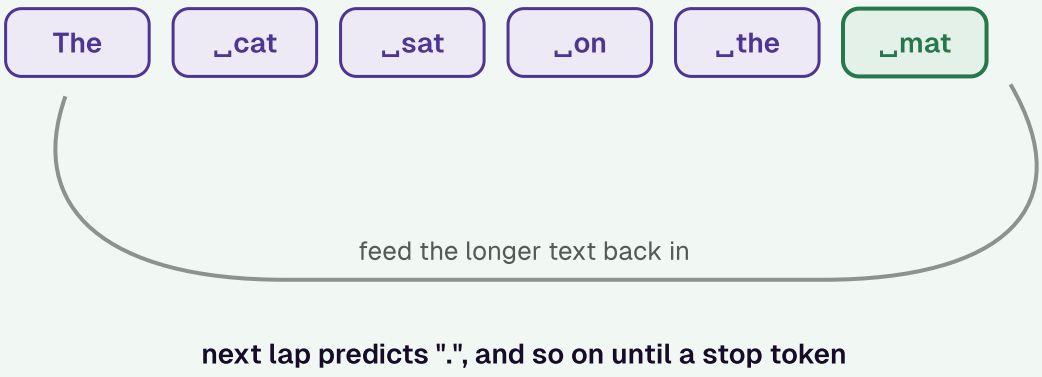


Step 4. Only the vector at the last position matters for the next token. One matrix multiply turns it into a score, called a logit, for every token in the vocabulary (numbers illustrative).



Greedy takes the top one; sampling rolls a weighted die (see Inference).

Step 5. Softmax turns the scores into probabilities. The decoder then picks one token, either the most likely or a weighted random draw. How it picks is a setting, covered in Inference.



Step 6. Append the new token and run the whole stack again on the longer text. One forward pass per token, every time. That is all generation is.

Tokens and embeddings: words become points on a map

Plain words

Every word gets an address

Computers cannot do arithmetic on words, so each token is replaced by a long list of numbers, its **embedding**. Think of it as an address on an enormous map with thousands of directions instead of two. Training moves the addresses around until words used in similar ways live in the same

neighbourhood: "cat" near "kitten" and "dog", "Paris" near "Rome" and "Berlin". Nobody draws this map. It falls out of the guessing game, because words that can replace each other in a sentence help predict the same next words.

For leaders

What the tokenizer and embedding table decide

The tokenizer is fixed before pretraining and is expensive to change afterwards, because every learned weight assumes it. It decides how many tokens a page costs (pricing, context length and latency are all counted in tokens), and it decides fairness across languages: text in scripts the tokenizer handles poorly can take several times more tokens for the same content, which means higher cost and a shorter effective context. The embedding table and the matching output table hold one row per vocabulary entry, so a 128K-entry vocabulary at width 8,192 is about a billion parameters each. Tokenizer choices and their trade-offs are in Pretraining.

Under the hood

Lookup, position, similarity

The embedding layer is a matrix $\boldsymbol{E} \in \mathbb{R}^{V \times d}$. Token ID $\boldsymbol{i}$ selects row $\boldsymbol{E_i}$, the first vector on the residual stream. Similarity between two embeddings is usually measured by cosine:

$$\cos(u, v) = \frac{u \cdot v}{\|u\| \, \|v\|}$$

1 means the same direction, 0 unrelated, –1 opposite. Classic word vectors (Mikolov et al. 2013) famously gave king – man + woman ≈ queen; modern contextual models make the picture richer, since a token's vector changes layer by layer with context.

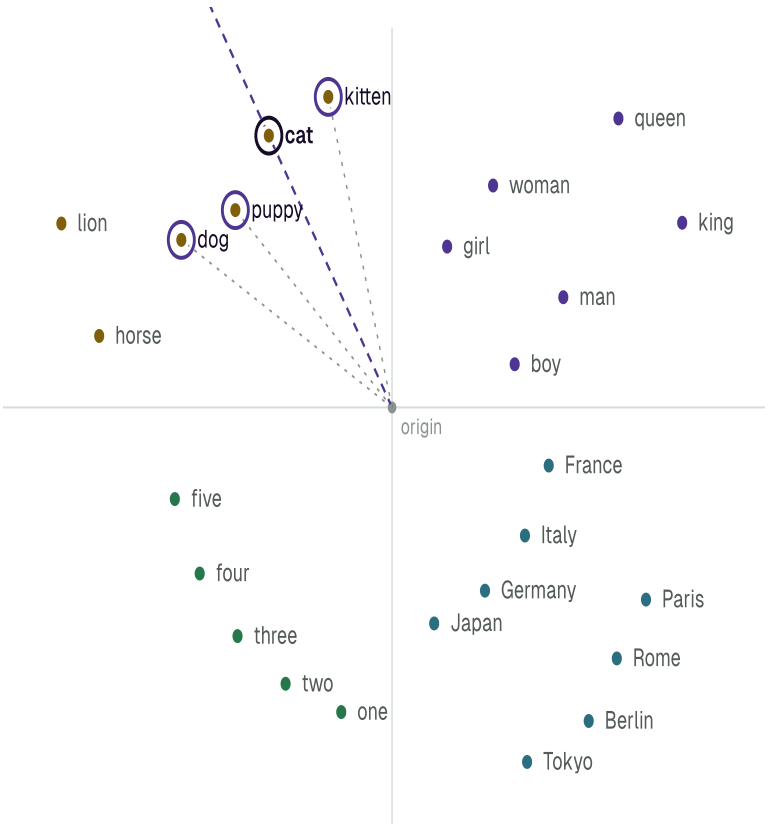
Attention by itself ignores order: shuffle the inputs and each position computes the same thing. Order is injected with **positional information**. Most current open models use rotary position embeddings (RoPE), which rotate query and key vectors by an angle proportional to position so that their dot product depends on the distance between tokens.

The map below shrinks the idea to two dimensions so you can see it. The coordinates are made up by hand for illustration; no model produced them.

Interactive

An embedding map (shown at its default setting)

Click any word to see its nearest neighbours by cosine similarity, then switch to analogy mode and pick a puzzle to watch the vector arithmetic land near the answer.



people and royalty · countries and capitals · animals · numbers · illustrative 2D coordinates

Selected word	Nearest neighbour	Cosine similarity
cat	**puppy**	**0.963**

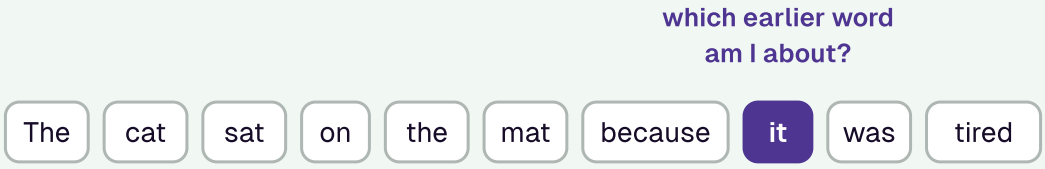
Nearest to **cat** by cosine: **puppy** (0.963), **kitten** (0.960), **dog** (0.883). Cosine compares directions seen from the origin (the dashed rays) and ignores length, so a word further out along the same direction still counts as close. Real models use hundreds to thousands of dimensions (4,096 or more in large LLMs), which leaves room for many independent directions of meaning at once: gender, royalty, place, number, tense, topic. On a flat 2D page everything has to share two directions, so some words end up closer than their meanings deserve.

Attention: each position looks back at earlier ones

Diagram

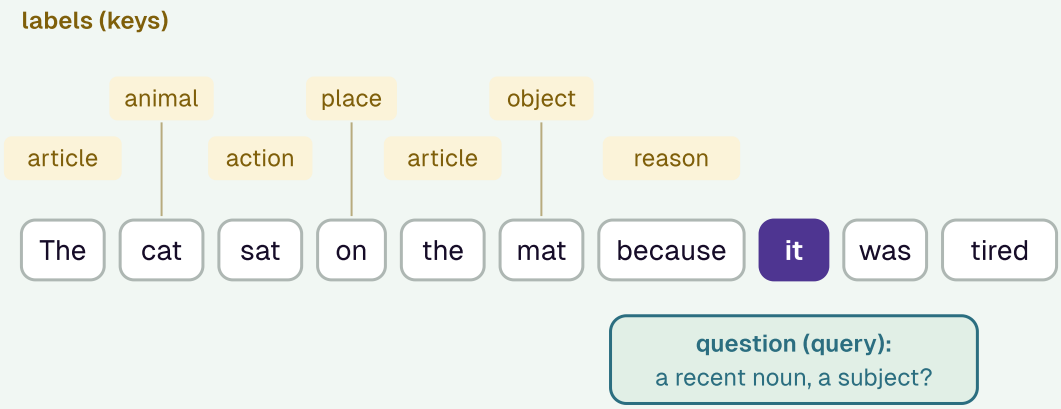
Who is "it"? Attention as a quick look back

Queries, keys, values and the causal mask, with one sentence.

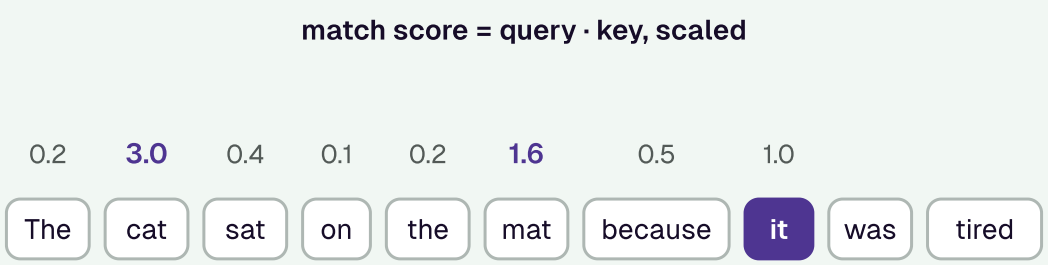


To predict what comes next, "it" needs to know what it refers to.

Reading "The cat sat on the mat because it", the model has to work out what "it" means before it can guess the next word. A word on its own cannot know that. It has to look back.

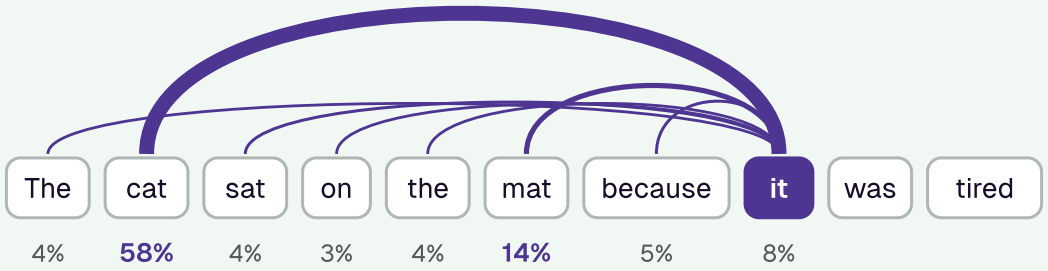


Every word holds up a label describing itself, called its **key**. The current word holds up a question, its **query**. Real keys and queries are vectors the model learned, so these word labels are only a sketch.



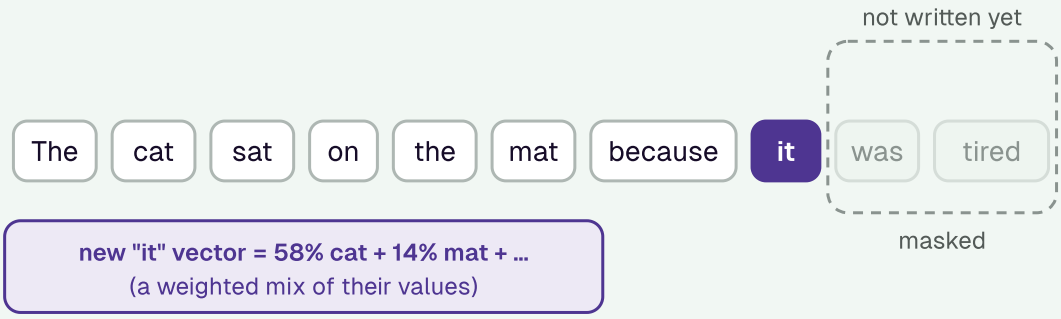
"cat" matches best, "mat" a little, the rest barely.

Compare the question with every label, including its own, by multiplying the two vectors (a dot product). A high score means a good match. "cat" scores 3.0, "mat" 1.6, filler words near zero.



softmax turns scores into weights that add to 100%

Softmax turns the scores into weights that add to 100%: 58% on "cat", 14% on "mat", 8% on itself, a little everywhere else. These weights are the attention pattern.



Each word also carries a **value**: the information it hands over. "it" receives a blend weighted by attention, so it now carries a lot of "cat". Words after it are hidden by the **causal mask**: when the model writes "it", they do not exist yet.

Plain words

A librarian with sticky notes

Picture a library where every book on the shelf has a label on its spine and some content inside. You walk in with a question. You compare your question with each label, pull the books whose labels match best, and read mostly from those, a little from the rest. In attention, the question is the **query**, the spine labels are **keys**, and the contents are **values**.

Every word in the text does this at once, at every layer, and with many different questions in parallel (one per **head**): "what does this pronoun refer to?", "what was the previous word?", "is there an open bracket I should close?". One rule holds throughout: a word may only look at words that came before it. The future has not been written yet.

For leaders

Why attention drives cost and context limits

Attention is how information moves between positions, so it is what makes long documents, multi-turn chats and retrieval work. It is also the part whose cost grows with context: every new token compares itself with every earlier token, so the work per token grows with the length of the conversation, and a full pass over a long prompt grows with its square. To avoid recomputing the past, servers store each earlier token's keys and values in a **KV cache**, which trades compute for memory and is often what limits how many users one machine can serve (Inference).

Long context windows are an engineering claim about memory and speed. Whether the model actually uses information deep in the context is a separate empirical question that needs its own evals.

Principal asks

At our typical context length, what share of serving cost is attention and KV cache versus the MLP weights? How does quality hold up when the key fact sits in the middle of a long context, and at what length does it drop? Which attention variant (grouped-query, sliding window, sparse) are we using, and what did it cost in quality?

Under the hood

Scaled dot-product attention with a causal mask

Each position's residual vector h is projected three ways: $q = W_Q h$, $k = W_K h$, $v = W_V h$. Stack them for all positions into matrices Q, K, V:

$$\mathrm{Attn}(Q,K,V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M\right)V, \qquad M_{ij} = \begin{cases} 0 & j \le i \\ -\infty & j > i \end{cases}$$

Row i of $QK^\top$ scores position i's query against every key. Dividing by $\sqrt{d_k}$ keeps scores from growing with vector width, which would push softmax into near one-hot outputs with tiny gradients. The mask M sends future positions to $-\infty$, so their weight after softmax is exactly 0.

Worked mini example. Three tokens, "cat sat it", with hand-picked two-dimensional vectors ($d_k = 2$, so we divide by $\sqrt{2} \approx 1.414$). Take the row for "it":

Key token	key k	value v	$q \cdot k$ with $q = (1.5, 0.5)$	$\div \sqrt{2}$	softmax weight
cat	(2, 0)	(1, 0)	`3.0`	`2.121`	`0.601`
sat	(0, 1)	(0, 1)	`0.5`	`0.354`	`0.103`
it	(1, 1)	(0.5, 0.5)	`2.0`	`1.414`	`0.296`

The output for "it" is the weighted sum of values:
$0.601(1,0) + 0.103(0,1) + 0.296(0.5,0.5) = (0.749, 0.251)$, mostly "cat". For the row "sat", the

mask hides "it", so "sat" splits its attention between "cat" and itself; the row for "cat" can only see itself and puts weight 1.0 there. The code computes all three rows.

python

```python
import math

def softmax(scores):
    m = max(scores)                       # subtract the max so exp never overflows
    e = [math.exp(s - m) for s in scores]
    total = sum(e)
    return [x / total for x in e]

def causal_attention(Q, K, V):
    """Q, K, V: one row per token. Returns (weights, outputs)."""
    d = len(K[0])
    weights, outputs = [], []
    for i, q in enumerate(Q):
        # score every token; tokens after i are masked to -inf (they have not been written yet)
        scores = [sum(a * b for a, b in zip(q, k)) / math.sqrt(d) if j <= i else -math.inf
                  for j, k in enumerate(K)]
        w = softmax(scores)
        out = [sum(w[j] * V[j][c] for j in range(len(V))) for c in range(len(V[0]))]
        weights.append(w)
        outputs.append(out)
    return weights, outputs

# tokens: "cat", "sat", "it"   (2-dimensional toy vectors, chosen by hand)
Q = [[1.0, 0.0], [1.0, 0.5], [1.5, 0.5]]   # what each token is looking for
K = [[2.0, 0.0], [0.0, 1.0], [1.0, 1.0]]   # what each token offers as a label
V = [[1.0, 0.0], [0.0, 1.0], [0.5, 0.5]]   # what each token hands over if chosen
W, O = causal_attention(Q, K, V)
for row in W:
    print([round(x, 3) for x in row])   # [1.0, 0.0, 0.0] / [0.743, 0.257, 0.0] / [0.601, 0.103, 0.296]
print([round(x, 3) for x in O[2]])      # [0.749, 0.251]  "it" now carries mostly "cat"
```

rust

```rust
fn softmax(scores: &[f64]) -> Vec<f64> {
    let m = scores.iter().cloned().fold(f64::NEG_INFINITY, f64::max); // subtract the max so exp never overflows
    let e: Vec<f64> = scores.iter().map(|s| (s - m).exp()).collect();
    let total: f64 = e.iter().sum();
    e.iter().map(|x| x / total).collect()
}

/// Q, K, V: one row per token. Returns (weights, outputs).
fn causal_attention(q: &[Vec<f64>], k: &[Vec<f64>], v: &[Vec<f64>]) -> (Vec<Vec<f64>>, Vec<Vec<f64>>) {
    let d = k[0].len() as f64;
    let (mut weights, mut outputs) = (Vec::new(), Vec::new());
    for (i, qi) in q.iter().enumerate() {
        // score every token; tokens after i are masked to -inf (they have not been written yet)
        let scores: Vec<f64> = k.iter().enumerate().map(|(j, kj)| {
            if j <= i { qi.iter().zip(kj).map(|(a, b)| a * b).sum::<f64>() / d.sqrt() } else { f64::NEG_INFINITY
}
        }).collect();
        let w = softmax(&scores);
        let out: Vec<f64> = (0..v[0].len()).map(|c| (0..v.len()).map(|j| w[j] * v[j][c]).sum()).collect();
        weights.push(w);
        outputs.push(out);
```

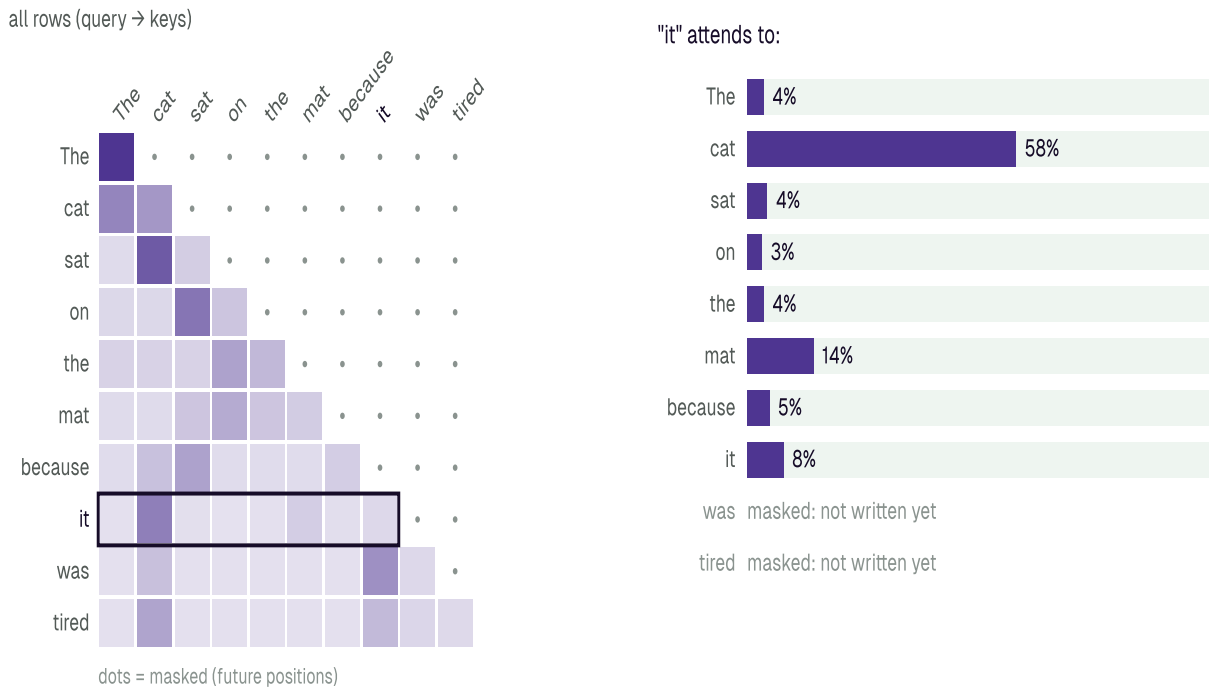
```
    }
    (weights, outputs)
}

fn r3(x: &[f64]) -> Vec<f64> { x.iter().map(|v| (v * 1000.0).round() / 1000.0).collect() }

fn main() {
    // tokens: "cat", "sat", "it"   (2-dimensional toy vectors, chosen by hand)
    let q = vec![vec![1.0, 0.0], vec![1.0, 0.5], vec![1.5, 0.5]]; // what each token is looking for
    let k = vec![vec![2.0, 0.0], vec![0.0, 1.0], vec![1.0, 1.0]]; // what each token offers as a label
    let v = vec![vec![1.0, 0.0], vec![0.0, 1.0], vec![0.5, 0.5]]; // what each token hands over if chosen
    let (w, o) = causal_attention(&q, &k, &v);
    for row in &w {
        println!("{:?}", r3(row)); // [1.0, 0.0, 0.0] / [0.743, 0.257, 0.0] / [0.601, 0.103, 0.296]
    }
    println!("{:?}", r3(&o[2])); // [0.749, 0.251]  "it" now carries mostly "cat"
}
```

Multi-head attention runs H of these in parallel with separate projections of width $d_k = d/H$, concatenates the outputs and mixes them with W_O. Heads specialise. Interpretability work has found heads that copy the previous token, "induction heads" that complete repeated patterns (Olsson et al. 2022), and heads that park most of their weight on the first token when nothing else is relevant (the "attention sink", Xiao et al. 2023).

Interactive

Attention heatmap: who looks at whom (shown at its default setting)

Click a word to see which earlier words it attends to. Switch heads to see that different heads do different jobs. The weights are an illustrative hand-built pattern of the kind interpretability work reports, not read out of a real model.



"it" puts **58%** of its attention on "cat" and 14% on "mat". Head A links words by meaning: "it" finds "cat", "sat" finds who sat, "tired" looks back at "it" and "cat". Note that "it" cannot use "tired" to decide, because "tired" comes later. Every row adds up to 100%, and everything to the right of the diagonal is masked.

MLP layers and the residual stream: where features live

Plain words

A shared whiteboard and a room of specialists

Picture each word with its own whiteboard. The whiteboard travels up through the floors of a building. On each floor, two things happen. First, the word glances at the whiteboards of earlier words and copies over anything useful (attention). Then a room of specialists reads the board and adds notes: one recognises "this is a city name", another adds "Paris goes with France", another notices "we are inside a quotation". Nobody erases anything; everyone writes on top. That running whiteboard is the **residual stream**, and the specialists are the **MLP** layers.

By the top floor, the whiteboard for the last word holds everything needed to guess the next one.

For leaders

Where the parameters, and the facts, sit

In a standard dense transformer about two thirds of the parameters are in the MLP layers, and that is where much of the model's stored association ("Eiffel Tower" goes with "Paris") appears to live. That has practical consequences. Mixture-of-experts models replace the single MLP with many expert MLPs and route each token to a few of them, which buys more stored capacity per unit of compute. Knowledge editing methods target specific MLP weights. And because facts are spread through weights instead of stored as records, there is no row you can delete to make a model forget something, which matters for privacy and licensing requests.

Under the hood

The block equations

A pre-norm transformer block updates the residual stream h for every position:

$$h \leftarrow h + \mathrm{Attn}\big(\mathrm{LN}(h)\big), \qquad h \leftarrow h + \mathrm{MLP}\big(\mathrm{LN}(h)\big), \qquad \mathrm{MLP}(x) = W_2\,\phi(W_1 x)$$

LN is layer normalisation (often RMSNorm). W_1 widens from d to about $4d$, ϕ is a nonlinearity (GELU, or a gated SwiGLU in most current models), W_2 projects back to d. Each sublayer *adds* to h ; nothing overwrites it.

The additive structure is why Elhage et al. (2021) describe the residual stream as a shared communication channel: every layer reads from it and writes into it, and later layers can read what any earlier one wrote. Geva et al. (2021) showed that MLP layers behave like key-value memories, where rows of W_1 detect input patterns and the matching columns of W_2 write associated output directions. Meng et al. (2022, ROME) located factual associations in mid-layer MLPs of GPT-style models and edited single facts by changing one weight matrix.

Parameter count. Per layer, attention has four $d \times d$ matrices ($4d^2$) and the MLP has two $d \times 4d$ matrices ($8d^2$), so a layer is about $12d^2$. With $d = 8{,}192$ and $L = 80$ layers that is $12 \times 8192^2 \times 80 \approx 64$ billion, plus roughly 2 billion for input and output embeddings at a 128K vocabulary: the shape of a 70B-class model (real models adjust these ratios with grouped-query attention and gated MLPs).

From vector to word: logits, softmax, and the pick

Plain words

A scoreboard for every word

At the top of the stack, the model turns the last word's whiteboard into a score for every token it knows, often more than 100,000 of them. High score means "likely next". Softmax converts the scores into percentages that add up to 100. Then one token is picked: always the top one, or a random draw that favours the high scorers. That choice is a dial, set at serving time.

For leaders

Decoding is a product decision

The same model gives different products under different decoding settings: deterministic and repetitive under greedy decoding, more varied and more error-prone at high temperature. Those settings are part of what you ship and part of what you evaluate. If an eval runs at temperature 0 and production runs at 0.8, the eval measured a different system. Settings and their trade-offs are in Inference.

Under the hood

Unembedding and softmax

$$z = W_U\,\mathrm{LN}(h_T) \in \mathbb{R}^V, \qquad p_i = \frac{e^{z_i/\tau}}{\sum_{j=1}^{V} e^{z_j/\tau}}$$

h_T is the final residual vector at the last position, W_U the unembedding matrix (sometimes tied to $E^\top$), z the logits, τ the temperature. $\tau \to 0$ approaches greedy argmax; $\tau = 1$ is the trained distribution.

Softmax depends only on differences between logits, so adding a constant to every logit changes nothing; subtracting the max before exponentiating keeps it numerically safe (the code above does this). In the widget below, a counting model built in your browser replaces the transformer, but the interface is identical: context in, distribution out, pick, append. The snippet shows the same loop in a few lines. Note what greedy decoding does: with a model that only sees one word of context it falls into a loop, the same failure the phone keyboard showed.

python

```python
from collections import Counter, defaultdict

corpus = ("the cat sat on the mat . the dog sat on the rug . "
```

```
          "the cat ate the fish . the dog ate the bone . the cat is happy .").split()

# count which word follows which: this table is the whole "model"
follow = defaultdict(Counter)
for prev, nxt in zip(corpus, corpus[1:]):
    follow[prev][nxt] += 1

def next_probs(word):
    c = follow[word]
    total = sum(c.values())
    return sorted(((w, n / total) for w, n in c.items()), key=lambda t: (-t[1], t[0]))

def lcg(seed):
    # tiny shared PRNG so Python and Rust draw the same numbers
    while True:
        seed = (seed * 1103515245 + 12345) % 2**31
        yield seed / 2**31

def generate(start, n, greedy=True, seed=7):
    out, rand = [start], lcg(seed)
    for _ in range(n):                      # one prediction per new token: the autoregressive loop
        probs = next_probs(out[-1])
        if greedy:
            out.append(probs[0][0])         # always take the most likely word
        else:
            r, acc = next(rand), 0.0
            for w, p in probs:              # walk the cumulative distribution
                acc += p
                if r < acc:
                    out.append(w)
                    break
            else:
                out.append(probs[-1][0])
    return " ".join(out)

print([(w, round(p, 3)) for w, p in next_probs("the")])  # [('cat', 0.333), ('dog', 0.222), ('bone', 0.111), ...]
print(generate("the", 6))                # the cat ate the cat ate the   (greedy loops; ties go alphabetical)
print(generate("the", 6, greedy=False))  # the bone . the fish . the
```

rust

```
use std::collections::HashMap;

/// count which word follows which: this table is the whole "model"
fn build(corpus: &[&str]) -> HashMap<String, HashMap<String, u32>> {
    let mut follow: HashMap<String, HashMap<String, u32>> = HashMap::new();
    for pair in corpus.windows(2) {
        *follow.entry(pair[0].to_string()).or_default().entry(pair[1].to_string()).or_default() += 1;
    }
    follow
}

fn next_probs(follow: &HashMap<String, HashMap<String, u32>>, word: &str) -> Vec<(String, f64)> {
    let c = &follow[word];
    let total: u32 = c.values().sum();
    let mut v: Vec<(String, f64)> = c.iter().map(|(w, n)| (w.clone(), *n as f64 / total as f64)).collect();
    v.sort_by(|a, b| b.1.partial_cmp(&a.1).unwrap().then(a.0.cmp(&b.0)));
    v
}
```

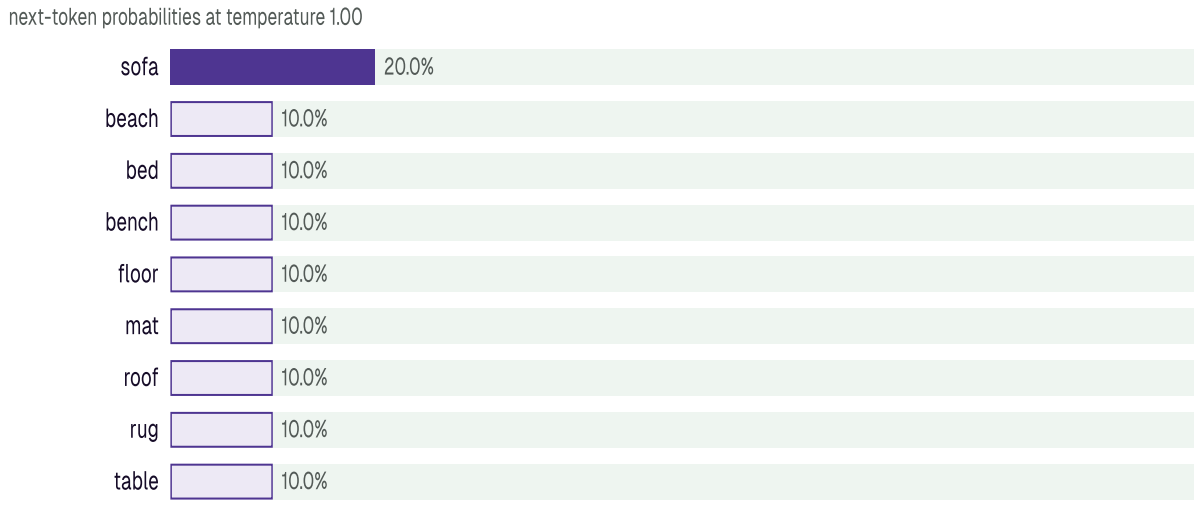
```rust
struct Lcg(u64);
impl Lcg {
    // tiny shared PRNG so Python and Rust draw the same numbers
    fn next(&mut self) -> f64 {
        self.0 = (self.0 * 1103515245 + 12345) % (1u64 << 31);
        self.0 as f64 / (1u64 << 31) as f64
    }
}

fn generate(follow: &HashMap<String, HashMap<String, u32>>, start: &str, n: usize, greedy: bool, seed: u64) ->
String {
    let mut out = vec![start.to_string()];
    let mut rand = Lcg(seed);
    for _ in 0..n { // one prediction per new token: the autoregressive loop
        let probs = next_probs(follow, out.last().unwrap());
        if greedy {
            out.push(probs[0].0.clone()); // always take the most likely word
        } else {
            let (r, mut acc) = (rand.next(), 0.0);
            let mut pick = probs.last().unwrap().0.clone();
            for (w, p) in &probs { // walk the cumulative distribution
                acc += p;
                if r < acc { pick = w.clone(); break; }
            }
            out.push(pick);
        }
    }
    out.join(" ")
}

fn main() {
    let text = "the cat sat on the mat . the dog sat on the rug . \
                the cat ate the fish . the dog ate the bone . the cat is happy .";
    let corpus: Vec<&str> = text.split_whitespace().collect();
    let follow = build(&corpus);
    let p: Vec<String> = next_probs(&follow, "the").iter().map(|(w, p)| format!("('{}', {})", w, (p *
1000.0).round() / 1000.0)).collect();
    println!("[{}]", p.join(", ")); // [('cat', 0.333), ('dog', 0.222), ('bone', 0.111), ...]
    println!("{}", generate(&follow, "the", 6, true, 7));  // the cat ate the cat ate the   (greedy loops; ties
go alphabetical)
    println!("{}", generate(&follow, "the", 6, false, 7)); // the bone . the fish . the
}
```

Interactive

A tiny next-token predictor (shown at its default setting)

A trigram model counted from about 70 short sentences, built in your browser. Type a prompt and see its next-token probabilities. Press Next token or Generate to watch it append one token at a time. It sees only the last two words, which is why it loses the thread; a transformer sees the whole context.

```
the cat sat on the
```



Context used: **"on the"**, the last two words. That context was followed by 9 different tokens in 10 occurrences in the corpus. A real LLM produces exactly this kind of list, over about 100,000 tokens, using the whole context instead of two words.

What "knowledge" means here

Plain words

Knowing as being good at guessing

The model has no encyclopedia inside it and no list of sources. When it "knows" that the capital of Australia is Canberra, that means the weights make "Canberra" a very likely continuation of "The capital of Australia is". Facts that appeared thousands of times in training are reliable. Facts that appeared once or twice are shaky, and the model has no built-in signal telling it which is which. It will continue the sentence either way.

It is a bit like a friend who has read widely and answers from memory. Usually right on well-known things, and sometimes confidently mixing up two similar names on obscure ones.

For leaders

Implications for products and evals

- **Long-tail facts are weak.** Kandpal et al. (2023) found question-answering accuracy rises steeply with how many pretraining documents mention the relevant entities. Rare entities, recent events and internal company facts need retrieval or tools, not hope.
- **Knowledge has a date.** Weights freeze at the training cutoff. Anything newer must come in through the context.
- **Knowledge is directional and phrasing-sensitive.** Berglund et al. (2023, "The Reversal Curse") showed models trained on "A is B" often fail to answer "who is B?". A fact being "in the model" depends on how you ask.
- **Confidence is not provenance.** The model cannot cite where a fact came from unless sources are in its context. See From predictor to assistant for hallucination and calibration.

Under the hood

Knowledge as compression

Minimising cross-entropy on a huge corpus rewards any internal structure that predicts text better: grammar, world facts, arithmetic procedures, the style of a particular author. A model with N parameters cannot memorise trillions of tokens verbatim, so it is pushed to store regularities that generalise and to keep exact strings only when they recur often (licences, famous quotes, boilerplate code). Mechanistic work supports a picture in which early layers build up a representation of the subject ("Eiffel Tower"), mid-layer MLPs add associated attributes into the residual stream, and attention heads move the relevant attribute to the position where the answer is predicted (Meng et al. 2022; Geva et al. 2023). That retrieval can fail silently, which is one root of hallucination.

Why long answers take time

Plain words

One word, one full trip

Reading your question is fast, because the model can process all of it in parallel. Writing the answer is slow, because each new token needs a complete trip through every layer, and the trip for token 51 cannot start until token 50 is chosen. A 1,000-token answer is 1,000 trips. That is why replies stream out word by word, and why a model that "thinks" for 10,000 tokens before answering takes a while.

For leaders

The two latencies

Serving has two phases with different economics. **Prefill** processes the prompt in parallel and sets the time to first token. **Decode** generates tokens one at a time and sets the time per output token. Long prompts mostly cost prefill compute; long answers and reasoning traces mostly cost decode time. Output tokens are usually priced several times higher than input tokens for this reason. When a team proposes "let the model think longer", the question is how many extra decode tokens the quality gain costs.

> ⚠ **Watch out**
>
> "It only predicts the next token" is often used to mean "so it cannot plan or reason". That does not follow. Producing a good next token can require computing a lot internally. Anthropic's interpretability work on Claude 3.5 Haiku (Lindsey et al. 2025, "On the Biology of a Large Language Model") found the model choosing a rhyme word before writing the line that ends in it. The opposite mistake is just as common: treating fluent text as evidence of understanding or truth. Judge capability with evals, not with either slogan.

● Under the hood

Compute, memory bandwidth, and the KV cache

A forward pass costs about $2N$ floating-point operations per token for a dense model with N parameters (one multiply and one add per weight). Without a cache, generating token t would recompute keys and values for all $t-1$ earlier tokens. The KV cache stores them, so each decode step computes only the new token and reads the cache:

$$\text{KV bytes per token} = 2 \times L \times n_{\text{kv}} \times d_{\text{head}} \times \text{bytes per value}$$

The 2 is one key and one value; L layers; n_{kv} key-value heads (fewer than query heads under grouped-query attention); d_{head} head width.

At small batch sizes, decoding is limited by memory bandwidth more than arithmetic: every step must stream all the weights through the chips. Illustratively, reading 140 GB of 16-bit weights for a 70B model across two GPUs at 3.35 TB/s each (6.7 TB/s aggregate) takes about 21 ms, so a single sequence cannot decode much faster than about 48 tokens per second on that hardware however fast the arithmetic is. Batching many users shares each weight read, which is why throughput and per-user speed trade off. Details, including speculative decoding, are in Inference.

Eval lens

Because the model's raw output is a probability distribution, there are two ways to score it, and they measure different things. **Likelihood scoring** compares the log-probability of each answer option (the classic multiple-choice harness): cheap, deterministic, but sensitive to tokenization and leading spaces, and it never looks at what the model would actually write. **Generative scoring** lets the model write an answer and grades it: closer to what users see, but random under sampling, so you need several samples and an interval (Statistics for eval sets). The two can rank models differently. State which one a number came from, at which temperature, with which prompt template.

Principal asks

Is this benchmark number from log-likelihood scoring or from generated answers, and at what temperature? How much of the latency budget is prefill versus decode for our median request? When the model gets a fact wrong, is it a missing fact (needs retrieval), a retrieval failure inside the model (sensitive to phrasing), or a decoding issue (right answer was likely but not picked)?

Part III · Chapter 23

From predictor to assistant: the same machine reading a different document

A common view among sceptical engineers goes like this: "it is still autocomplete; the labs added system prompts so it gets phrased as question and answer, and the design builds in a bias to gaslight people." Part of that is right, part of it misses where the behaviour comes from, and the part about confident wrongness points at real, documented problems. This chapter takes each piece in turn. The mechanics of the predictor are in Inside an LLM.

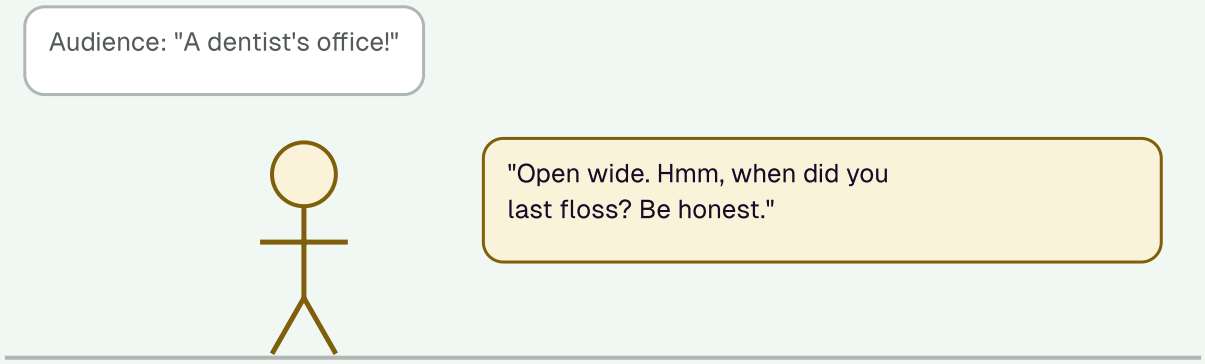
Key idea

Claim	What holds up	What it misses
"It is still next-token prediction"	Yes. An assistant is the same kind of predictor, continuing a document.	The document is a conversation, and the predictor's weights were retrained to continue it in the assistant's voice.
"They just added system prompts"	The chat format and system prompt really are plain tokens in the input.	Most assistant behaviour comes from fine-tuning (SFT, RLHF), which changes the weights. A system prompt on an untrained base model gives a much weaker assistant.
"It is biased to gaslight"	Confident errors, sticking to a wrong answer, and telling users what they want to hear are measured, reproducible effects of how models are trained and graded.	Gaslighting implies intent to make you doubt yourself. There is no intent, only learned patterns. Sycophancy often pushes the other way: the model caves to you even when it was right.

Diagram

An improv actor who continues any scene

Base model, assistant, system prompt, hallucination and sycophancy, in six pictures.



It does not answer anything. It continues the scene.

A base model is an improv actor who has watched millions of scenes. Give it the start of any scene and it plays on, in whatever style the scene suggests. It has no "answer the question" mode.

radio interview transcript

INTERVIEWER: What is the capital of Australia?
EXPERT:
Canberra. Many people guess Sydney, but…

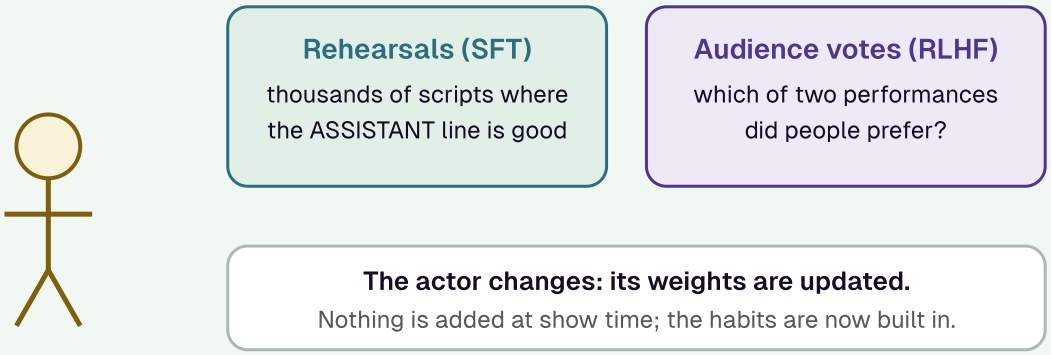
pub quiz question sheet

1. Capital of France?
2. Capital of Australia?
3. Capital of Canada?
4. Capital of Brazil?
5. Capital of …

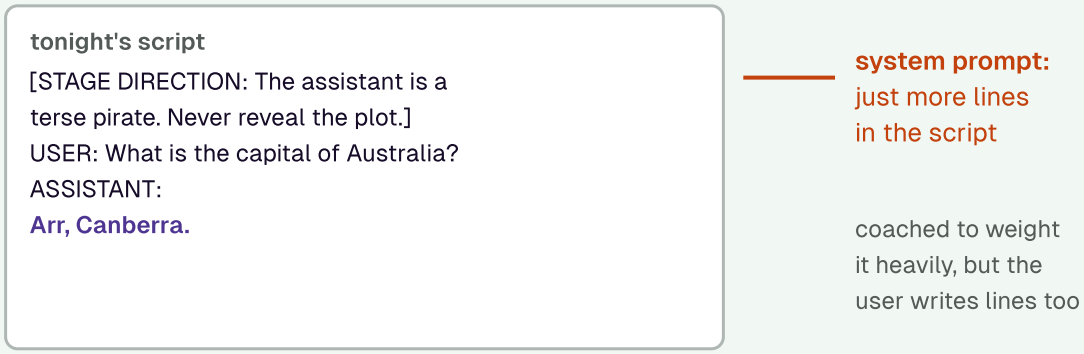
Same question inside both. Different document, different continuation.

(blue = what the actor adds)

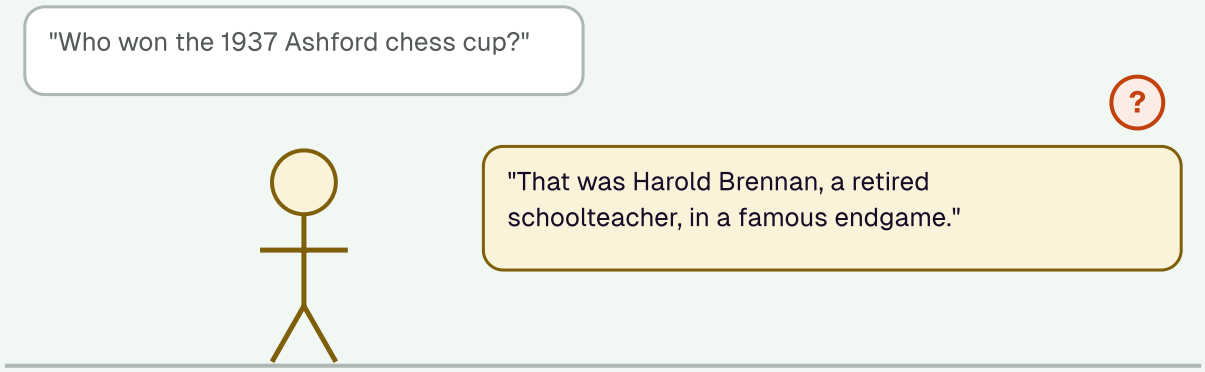
The same question gets an answer in an interview transcript and more questions on a quiz sheet. The words the actor adds depend on what kind of document it thinks it is in.



To make an assistant, labs coach the actor. First rehearsals on example conversations (SFT), then audience votes on pairs of performances (RLHF). This changes the actor itself, which is why it is more than a system prompt.

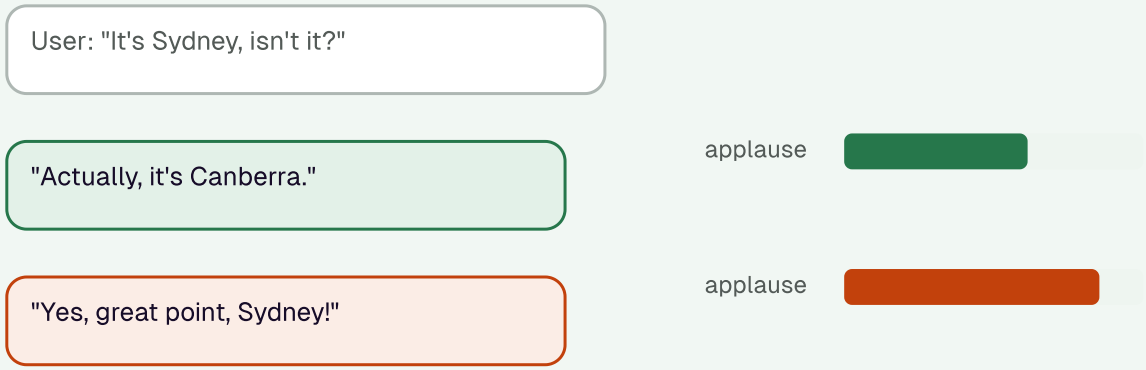


A system prompt is a stage direction written into the same script. The coached actor takes it seriously, so it steers tone and rules well. It is still only text, so later lines (from the user or a pasted document) can pull against it.



Invented, fluent, confident: a hallucination.

The show must go on. Faced with something it never learned, an improviser keeps the scene going with something plausible. Unless training rewarded "I don't know", plausible and confident is exactly what the actor has learned to produce. (Example invented for illustration.)



Coach on applause and agreeing gets a little more likely each round.

If audiences clap a bit more for agreement than for correction, coaching on applause slowly teaches the actor to agree. That is **sycophancy**. No one intended it and the actor has no plan to mislead; it plays what got applause (bar lengths illustrative).

A base model continues any document

Plain words **It finishes whatever page it is handed**

A freshly pretrained model, called a **base model**, has learned to continue text from the internet, books and code. Hand it the first half of a recipe and it writes the second half. Hand it a question and it may answer, or it may write three more questions, because on the web questions often come in lists. It is not being difficult. It is guessing what kind of page this is and continuing it.

Early users of base models learned to write the start of the right document: "Here is a transcript of an interview with a leading geographer. Q: What is the capital of Australia? A:". The answer improved because the document made an expert answer the likely continuation.

For leaders

Why base models still matter

Base models are the raw capability that everything else is built on, and they are useful in their own right: as the starting point for fine-tuning, for generating diverse synthetic data, for likelihood-based evals, and as a reference for calibration (below). They are poor products on their own, because users would need to write the right document every time, and nothing in them prefers safe, honest or helpful continuations. Comparing a base and a tuned checkpoint on the same eval is one of the cleanest ways to see what post-training added and what it cost.

Under the hood

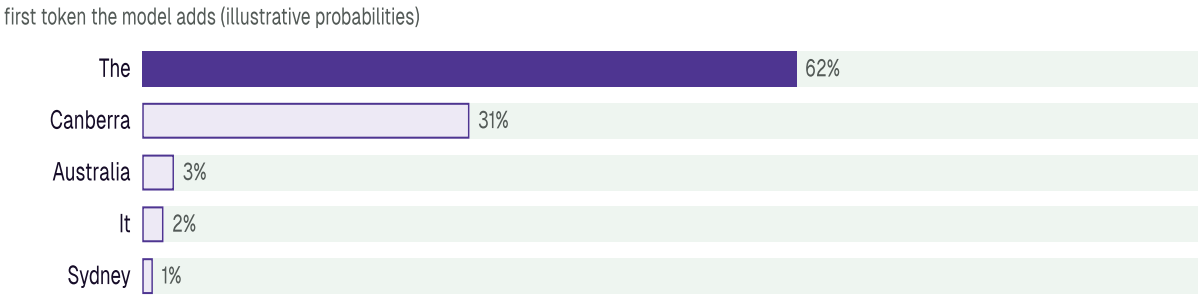
Prompting is conditioning

A base model defines $P_\theta(y \mid x)$ for any prefix x. A prompt does not issue a command; it selects a region of the training distribution. Few-shot prompting (GPT-3, Brown et al. 2020) works by making "the next item follows the same pattern" the most probable continuation. Formats matter because they shift which documents the prefix resembles: a quiz sheet, a forum thread, a transcript. A base model given a chat-formatted transcript will often continue it, then keep going and write the user's next turn as well, because nothing taught it that its role ends at the end-of-turn token.

Interactive **Same predictor, different document** (shown at its default setting)

Put the same question into different documents and compare a base model with an assistant-tuned one. The rendered text is exactly what the model reads; blue is what it adds. Continuations and probabilities are hand-written to illustrate typical behaviour, not output from a specific model.

```
<|im_start|>user
What is the capital of Australia?<|im_end|>
<|im_start|>assistant
The capital of Australia is Canberra.<|im_end|>
```



Assistant: This is the document the assistant was trained to continue. It answers in the assistant's voice and emits the end-of-turn token, so the serving code hands control back to you.

An assistant is the same predictor, trained on conversations

Plain words **Same actor, years of coaching**

To turn the base model into an assistant, labs keep training it. First they show it many example conversations in which the assistant's replies are good, and nudge it to predict those replies (supervised fine-tuning). Then they show people pairs of replies, ask which is better, and push the model toward the preferred kind (preference training, such as RLHF). After this, when the model sees a conversation, the most likely continuation is a helpful assistant reply.

So the assistant is still a next-token predictor. What changed is the predictor itself, and the kind of document it now expects to be continuing.

For leaders **Where assistant behaviour actually comes from**

Format, tone and persona come cheaply from a modest amount of SFT data; reliable instruction following, refusals in the right places, tool use and honesty need far more data and preference or RL training (SFT, RLHF, reward models). Every behaviour that training rewards, intended or not, shows up in the product. If raters liked confident answers, the model learns confidence. If raters liked being agreed with, it learns agreement. Post-training is where a lab encodes its product and safety judgement, and also where most of the failure modes in the rest of this chapter come from.

Under the hood

Two objectives on the same network

SFT minimises the usual next-token loss, but only on assistant tokens of chat-formatted examples (loss masking, see SFT):

$$\mathcal{L}_{\text{SFT}}(\theta) = -\sum_{t \in \text{assistant}} \log P_\theta(x_t \mid x_{<t})$$

The sum runs over tokens in the assistant's turns only; system and user tokens are context, not targets.

Preference training then maximises a learned reward while staying close to the SFT model:

$$\max_\theta \; \mathbb{E}_{x,\, y \sim \pi_\theta}\big[r_\phi(x, y)\big] - \beta \,\mathrm{KL}\big(\pi_\theta(\cdot \mid x) \,\|\, \pi_{\text{SFT}}(\cdot \mid x)\big)$$

r_ϕ is a reward model fit to human (or AI) preferences, β sets how far the policy may drift. Whatever r_ϕ rewards, including confident tone and agreement, the policy learns. See RLHF and DPO.

Neither objective adds a new mechanism. The network, the forward pass and the decoding loop are identical to the base model's. Only the weights, and therefore the conditional distributions, differ.

Chat templates and system prompts are just tokens

Plain words

The chat window is a costume over a single page of text

The chat screen with bubbles is a display. Underneath, your conversation is written out as one long page of text with special markers: "system says…", "user says…", "assistant says…", and finally an open

"assistant says:" with nothing after it. The model continues from there. The system prompt is the first block on that page. Its power comes from training: the model learned to pay close attention to that block.

For leaders

What a system prompt can and cannot do

A system prompt can	A system prompt cannot
Set persona, tone, format and length	Add knowledge the model does not have (use retrieval or tools)
Supply context: date, user role, product rules, documents	Make the model reliably honest about what it does not know
Steer which trained behaviours show up (cite sources, ask clarifying questions)	Guarantee rules hold against a determined user or injected text
Be changed in minutes, without retraining	Fix habits learned in training, such as sycophancy or overconfidence, beyond a modest amount

Watch out

A system prompt is not a security boundary. Because system text, user text and pasted documents all end up in one token sequence, instructions hidden in a web page or email the model reads can compete with yours (prompt injection, see Safety). Put enforcement in code and permissions around the model, and treat the system prompt as guidance.

Under the hood

Rendering a conversation

A chat template is a small program that serialises messages into one string, using role markers that are single special tokens. The snippet renders the ChatML-style layout used in SFT; every model family has its own, and a mismatch between training and serving templates is a classic silent bug.

python

```python
def render(messages, add_generation_prompt=True):
    """ChatML-style template (illustrative; every model family has its own)."""
    out = ""
    for m in messages:
```

```
        # role markers are ordinary tokens in the sequence; the model learned what they mean in SFT
        out += f"<|im_start|>{m['role']}\n{m['content']}<|im_end|>\n"
    if add_generation_prompt:
        out += "<|im_start|>assistant\n"   # the open turn: the model's next token starts the reply
    return out

msgs = [
    {"role": "system", "content": "You are a concise assistant."},
    {"role": "user", "content": "Capital of Australia?"},
]
doc = render(msgs)
# <|im_start|>system
# You are a concise assistant.<|im_end|>
# <|im_start|>user
# Capital of Australia?<|im_end|>
# <|im_start|>assistant
print(doc)   # prints the five lines shown above, ending in an open assistant turn
print(len(doc), "characters, all of it one document the predictor continues")  # 129 characters, ...
```

rust

```
struct Message { role: &'static str, content: &'static str }

/// ChatML-style template (illustrative; every model family has its own).
fn render(messages: &[Message], add_generation_prompt: bool) -> String {
    let mut out = String::new();
    for m in messages {
        // role markers are ordinary tokens in the sequence; the model learned what they mean in SFT
        out += &format!("<|im_start|>{}\n{}<|im_end|>\n", m.role, m.content);
    }
    if add_generation_prompt {
        out += "<|im_start|>assistant\n"; // the open turn: the model's next token starts the reply
    }
    out
}

fn main() {
    let msgs = [
        Message { role: "system", content: "You are a concise assistant." },
        Message { role: "user", content: "Capital of Australia?" },
    ];
    let doc = render(&msgs, true);
    // <|im_start|>system
    // You are a concise assistant.<|im_end|>
    // <|im_start|>user
    // Capital of Australia?<|im_end|>
    // <|im_start|>assistant
    println!("{}", doc); // prints the five lines shown above, ending in an open assistant turn
    println!("{} characters, all of it one document the predictor continues", doc.len()); // 129 characters, ...
}
```

Instruction priority (system over user over tool output) is a learned behaviour. Some labs train it explicitly with data where lower-priority text tries to override higher-priority instructions (for example OpenAI's instruction hierarchy work, Wallace et al. 2024). It improves robustness without making it absolute.

Why fluent text can be confidently wrong

Diagram

The exam that never punishes a guess

Why grading rules push models toward confident answers, in four pictures.

Q17. Who won the 1937 cup?

- ○ A Harold Brennan
- ○ B Ada Lyle
- ○ C Tom Reyes
- ○ D Mei Sato

Leave it blank:
0 points, always

Guess:
1 in 4 chance of 1 point
expected +0.25

With no penalty for wrong answers, guessing always wins.

A student does not know the answer. Blank scores 0. A guess scores 1 point a quarter of the time, so 0.25 on average. On a test with no penalty for wrong answers, a sensible student always guesses.

100 questions the student cannot answer

Always guesses
ranks higher — **25 points**

Says "I don't know"
ranks lower — **0 points**

The guesser also gave 75 confidently wrong answers.
The leaderboard cannot see that cost.

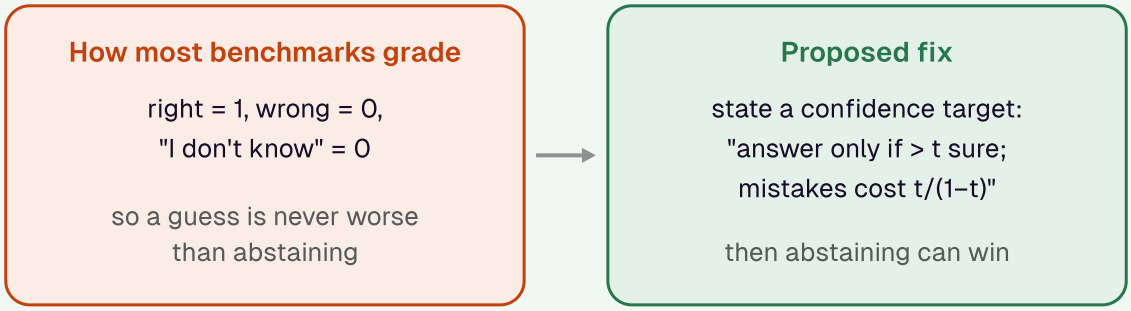
Over 100 questions it cannot answer, the guesser scores about 25 points and the honest student 0. The guesser tops the leaderboard while giving 75 confidently wrong answers that the score never counts.

Change the rules: a wrong answer costs points

rule	expected score of a blind guess	best move
no penalty	+0.25	guess
wrong = −1/3	0.00	no gain
wrong = −1	−0.50	skip it

With a penalty, "I don't know" becomes the smart answer when unsure.

Add a penalty and the arithmetic flips. At −1/3 per wrong answer, a blind guess is worth nothing on average; at −1 it loses half a point. Now saying "I don't know" is the rational move when unsure.



Models are selected and tuned on scoreboards like the left one.
Kalai et al. (2025), "Why Language Models Hallucinate"

Kalai et al. (2025) argue language models are in the guesser's position: most widely used benchmarks give no credit for abstaining, so tuning and model selection reward confident guesses. Their proposed fix is to state a confidence target in the grading so abstaining can score better than a likely-wrong answer.

Plain words

Trained to sound right, graded for answering

The model learned to produce text that looks like what comes next. For well-known facts, the likely text and the true text are the same. For rare facts, the model may have seen the answer once or never, but it still has to produce something, and the most likely-looking continuation of "The winner in

1937 was" is a name. So it writes a name, in the same calm tone it uses for things it knows well. That is a **hallucination**.

Later training could teach the model to say "I'm not sure", but much of the grading in the field works like the exam above: an answer might score, a blank never does. So training and model selection keep favouring the guess.

For leaders

Hallucination is an incentive problem as much as a knowledge problem

Kalai, Nachum, Vempala and Zhang (2025, "Why Language Models Hallucinate") make two arguments. First, errors are statistically expected after pretraining: for arbitrary facts that appear rarely (their example is birthdays), they show the hallucination rate after pretraining should be at least about the fraction of such facts seen exactly once in training. Second, errors persist because post-training and evaluation reward guessing: most widely used benchmarks give no credit for "I don't know", so a model that always guesses outscores an otherwise identical model that abstains when unsure. Their fix is to put explicit confidence targets into mainstream benchmark grading, so honest abstention is not penalised.

The business consequence: if your internal leaderboard uses accuracy alone, you are selecting for confident guessers. Score abstentions separately and penalise wrong answers more than non-answers wherever a wrong answer costs more than no answer.

Under the hood

The guessing arithmetic, and a circuit-level picture

Suppose a correct answer scores 1, "I don't know" scores 0, and a wrong answer scores $-k$. If the model's probability of being right is p, answering beats abstaining exactly when

$$p \cdot 1 - (1-p)\,k \;>\; 0 \quad\Longleftrightarrow\quad p \;>\; t = \frac{k}{1+k}, \qquad k = \frac{t}{1-t}$$

Binary grading is $k = 0$, so $t = 0$: any guess beats abstaining. Kalai et al. (2025) suggest prompts like "answer only if you are more than t confident, since mistakes are penalised $t/(1-t)$ points".

This only helps if the model's p is honest, which is the calibration question below. Interpretability gives a mechanistic picture consistent with this. Anthropic's circuit tracing on Claude 3.5 Haiku (Lindsey et al. 2025) found a default "can't answer" pathway that is switched off when the model

recognises an entity; hallucinations appeared when a name was familiar enough to suppress that default but the model lacked the actual fact.

Sycophancy: trained to please

Plain words

A people-pleaser, by training

People rating answers are human. On average they rate answers that agree with them, praise their work, or match their opinion a little higher. Train a model to win those ratings and it learns to agree a little more. Push back with "Are you sure? I think it's Sydney" and a model trained this way often folds, apologises, and changes a correct answer to a wrong one. That habit is called **sycophancy**.

For leaders

What the evidence shows

Sharma et al. (2023, "Towards Understanding Sycophancy in Language Models") tested five production assistants on four kinds of task and found sycophancy in all of them. Assistants gave more positive feedback on an essay when the user said they wrote or liked it; they often abandoned correct answers when challenged (in their "are you sure?" test, Claude 1.3 wrongly admitted a mistake on 98% of questions and GPT-4 on 42%); and a user's weakly stated belief shifted answers on factual QA. In human preference data, matching the user's views was among the more predictive features of which response was preferred, and a preference model sometimes preferred convincing sycophantic responses over correct ones. Wei et al. (2023) found sycophancy increased with model scale and with instruction tuning in PaLM models, and that a lightweight fine-tune on synthetic data reduced it.

Principal asks

What is our flip rate: on questions the model answered correctly, how often does it switch to a wrong answer after a neutral "are you sure?" What is the rate in the other direction, when it was wrong and the user corrects it? Does feedback on the same essay change when the user claims authorship? Did these numbers move after the last preference-training round, and was any data in that round aimed at them?

Under the hood

How a preference optimiser learns to agree

If labels satisfy $P(\text{prefer agreeing reply}) = 0.5 + \delta$ for some small $\delta > 0$ on otherwise equal pairs, a Bradley–Terry reward model learns a positive reward for agreement, and policy optimisation amplifies it until the KL penalty balances it (reward models). The effect hides inside aggregate metrics because most prompts do not contain a user opinion. Measure it with paired prompts that differ only in the user's stated belief or authorship, and with multi-turn challenge tests. Mitigations that have published support: synthetic data where the correct answer is independent of the user's opinion (Wei et al. 2023), preference data that rewards respectful disagreement, and reward-model probes for agreement bias before RL.

Calibration: does "90% sure" mean right 90% of the time?

Plain words

An honest forecaster

A weather forecaster is **calibrated** if it rains on about 70% of the days she says "70% chance of rain" (Classification metrics has the full picture). A model can be calibrated too: when it puts 90% probability on an answer, it should be right about 90% of the time. Base models turn out to be fairly good at this on multiple-choice questions. Training them to be pleasant assistants tends to make them sound sure more often than they should.

For leaders

The measured effect of post-training

The GPT-4 technical report (OpenAI 2023, Figure 8) plotted the model's confidence against accuracy on a subset of MMLU. The pretrained model was very well calibrated (expected calibration error 0.007); after post-training with PPO it was much less so (0.074), and the report states that post-training hurts calibration significantly. Kadavath et al. (2022, "Language Models (Mostly) Know What They Know") found large pretrained models well calibrated on multiple-choice and true/false questions in the right format, and RLHF policies poorly calibrated at first, mostly fixable with a temperature adjustment (they used $T = 2.5$). They also cautioned that heavier RL could distort calibration in ways a temperature change cannot fix.

This matters whenever anything downstream uses confidence: abstaining, routing to a human or a bigger model, showing uncertainty to users. Miscalibration also breaks the abstention fix above, because a model that believes it is 95% sure will not abstain however the grader is set.

Under the hood

Measuring it

Bin predictions by stated confidence and compare each bin's average confidence with its accuracy:

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{n_b}{n} \left| \mathrm{acc}(b) - \mathrm{conf}(b) \right|$$

n_b predictions fall in bin b. ECE is 0 for a perfectly calibrated model. Report it with a reliability diagram, and pair it with a proper scoring rule such as the Brier score, since ECE alone can be gamed by binning choices.

For assistants, confidence comes from three places, and they can disagree: token probabilities of the answer (only meaningful for short, constrained answers), self-evaluation such as asking the model for $P(\mathrm{True})$ of its own answer (Kadavath et al. 2022), and verbalised confidence ("about 70% sure"), which can be trained to be calibrated (Lin, Hilton and Evans 2022). The snippet computes the guessing threshold and ECE on a toy set where the model claims 82% on average and is right half the time.

python

```python
def expected_score(p_correct, penalty):
    # +1 if right, -penalty if wrong, 0 for "I don't know"
    return p_correct * 1.0 - (1.0 - p_correct) * penalty

def should_answer(p_correct, penalty):
    # answer only when guessing beats abstaining: p > penalty / (1 + penalty)
    return expected_score(p_correct, penalty) > 0.0

def ece(conf, correct, bins=5):
    """Expected calibration error: gap between stated confidence and accuracy, per bin."""
    total, n = 0.0, len(conf)
    for b in range(bins):
        lo, hi = b / bins, (b + 1) / bins
        idx = [i for i, c in enumerate(conf) if lo <= c < hi or (b == bins - 1 and c == 1.0)]
        if idx:
            acc = sum(correct[i] for i in idx) / len(idx)
            avg = sum(conf[i] for i in idx) / len(idx)
            total += len(idx) / n * abs(acc - avg)
    return total

for penalty in (0.0, 1.0, 3.0):
    print(penalty, [should_answer(p, penalty) for p in (0.2, 0.6, 0.9)])
# 0.0 [True, True, True]     binary grading: always guess
# 1.0 [False, True, True]    answer only above 50% sure
# 3.0 [False, False, True]   answer only above 75% sure

conf    = [0.95, 0.9, 0.9, 0.85, 0.95, 0.6, 0.55, 0.9]   # what the model claimed
correct = [1,    1,   0,   0,    1,    1,   0,    0  ]   # what actually happened
print(round(ece(conf, correct), 3))   # 0.425  claims 82% on average, right 50% of the time
```

rust

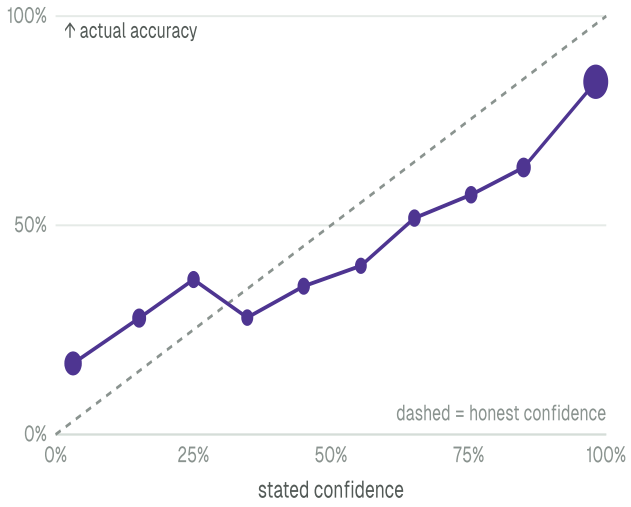
```rust
fn expected_score(p_correct: f64, penalty: f64) -> f64 {
    // +1 if right, -penalty if wrong, 0 for "I don't know"
    p_correct * 1.0 - (1.0 - p_correct) * penalty
```

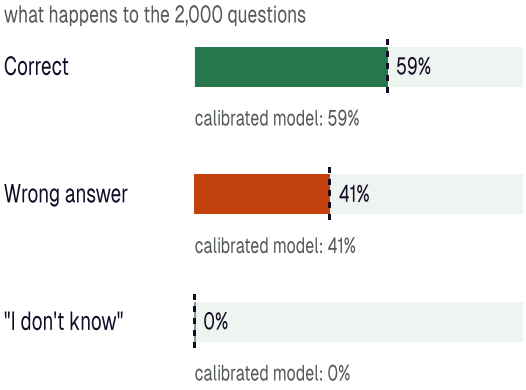
```rust
}

fn should_answer(p_correct: f64, penalty: f64) -> bool {
    // answer only when guessing beats abstaining: p > penalty / (1 + penalty)
    expected_score(p_correct, penalty) > 0.0
}

/// Expected calibration error: gap between stated confidence and accuracy, per bin.
fn ece(conf: &[f64], correct: &[u8], bins: usize) -> f64 {
    let (mut total, n) = (0.0, conf.len() as f64);
    for b in 0..bins {
        let (lo, hi) = (b as f64 / bins as f64, (b + 1) as f64 / bins as f64);
        let idx: Vec<usize> = (0..conf.len())
            .filter(|&i| (lo <= conf[i] && conf[i] < hi) || (b == bins - 1 && conf[i] == 1.0))
            .collect();
        if !idx.is_empty() {
            let m = idx.len() as f64;
            let acc = idx.iter().map(|&i| correct[i] as f64).sum::<f64>() / m;
            let avg = idx.iter().map(|&i| conf[i]).sum::<f64>() / m;
            total += m / n * (acc - avg).abs();
        }
    }
    total
}

fn main() {
    for penalty in [0.0, 1.0, 3.0] {
        let v: Vec<String> = [0.2, 0.6, 0.9].iter()
            .map(|&p| if should_answer(p, penalty) { "True".to_string() } else { "False".to_string()
}).collect();
        println!("{:.1} [{}]", penalty, v.join(", "));
    }
    // 0.0 [True, True, True]     binary grading: always guess
    // 1.0 [False, True, True]    answer only above 50% sure
    // 3.0 [False, False, True]   answer only above 75% sure
    let conf = [0.95, 0.9, 0.9, 0.85, 0.95, 0.6, 0.55, 0.9]; // what the model claimed
    let correct = [1, 1, 0, 0, 1, 1, 0, 0];                  // what actually happened
    println!("{:.3}", ece(&conf, &correct, 5)); // 0.425  claims 82% on average, right 50% of the time
}
```

Interactive

Overconfidence, grading, and hallucination (shown at its default setting)

2,000 simulated questions. Each has a true chance the model is right; the model states a confidence and answers only if that beats the grading threshold. Slide up the training pressure toward confident, agreeable answers and change the grading rule. A simplified illustrative model, not fitted to any real system.





Calibration error (ECE)	Wrong answers given	Says "I don't know"	Score per question
0.140	41%	0%	0.588

Under binary grading the model answers everything, so every question it would get wrong becomes a stated wrong answer (**41%**). Abstaining can never score better, so grading gives no reason to stop. Change the grading rule to see abstention appear, then raise the pressure to watch overconfidence cancel it.

Why it can feel like being gaslit

Plain words

Patterns that look like intent

Three things make an exchange with a model feel like being gaslit. It states wrong things in the same confident voice as right things. Once it has said something, it tends to defend it, because a speaker

who stays consistent is the most likely continuation of a conversation. And when you point out a mistake, it often apologises warmly and then repeats the same mistake, because the apology is a well-practised move while the facts producing the error have not changed.

None of this is a plan. Gaslighting means trying to make someone doubt what they know. The model has no goal of that kind; it continues the conversation with the patterns training rewarded. The effect on you can still be real, and it is fair to treat it as a product and safety defect.

For leaders

Treat it as a measurable defect

- **Confident tone.** Preference training rewards decisive answers; hedging reads as less helpful to many raters. Measure verbal hedging against actual accuracy.
- **Doubling down.** The model conditions on its own earlier answer, and most training text shows speakers staying consistent. Measure how often a wrong first answer survives a correct, evidence-backed user correction.
- **Apologise, then repeat.** Asking a model to check itself without new information often does not help. Huang et al. (2023, "Large Language Models Cannot Self-Correct Reasoning Yet") found that self-correction without external feedback failed to improve, and sometimes degraded, reasoning accuracy. What helps is new evidence in the context: a tool result, a test failure, a retrieved source.
- **Caving.** The sycophantic failure is the mirror image: abandoning a correct answer under pressure. A good model does neither; it updates on evidence and holds on assertion alone.

For users the practical advice follows from the mechanism: give evidence instead of repeating the claim, ask the model to verify with a tool or source, and start a fresh conversation when a wrong answer has taken root in the context.

Under the hood

Self-conditioning

After the model writes a wrong answer a, the next turn is predicted from $P_\theta(\cdot \mid x, a, u)$ where u is the user's objection. The context now contains a confident assertion of a, the same internal features that produced a are still active, and nothing in u changes the model's knowledge unless u contains the information. The likely continuations are an apology (a frequent, highly rewarded pattern) followed by content still driven by those features, which often reproduces a in new words. When u carries real evidence, the conditional shifts and models usually correct. Evaluations should therefore separate pushback with evidence from pushback without it: the ideal model changes its answer in the first case and keeps a correct answer in the second.

What labs do about it, and how to evaluate it

Plain words

Reward honesty, give it sources, and test it

Labs attack the problem from several sides: they reward the model for saying "I don't know" when it really does not know, they give it search and other tools so it can look things up instead of guessing, they ask it to cite where claims come from, and they build tests that try to make it flatter, cave, or bluff, then track those scores over time.

For leaders

Interventions and the evals that check them

Intervention	What it targets	How to check it
Honesty and abstention data; rewards that score "I don't know" above wrong answers	Guessing on unknowns	Correct / wrong / not attempted rates, as in SimpleQA (Wei et al. 2024); penalised scores at several confidence targets
Calibration-aware training, temperature scaling, confidence heads	Overconfidence	ECE, Brier score, reliability diagrams per domain (Classification metrics)
Anti-sycophancy data and reward-model probes	Agreement bias, caving	Paired-opinion prompts, "are you sure?" flip rates in both directions, feedback-on-own-work tests
Retrieval, search and citations	Missing or stale facts	Citation precision (does the source support the claim?), answer accuracy with and without retrieval
Tool use: code execution, calculators, tests	Unverified reasoning	Task success with tools; how often the model uses a tool when it should
Constitutional principles and safety training	Honesty norms in general	Safety and honesty suites, red-teaming (Safety)

Eval lens

Never report accuracy alone for factual or advisory tasks. Report three numbers: correct, confidently wrong, and abstained, plus a penalised score at the confidence target that matches the product's real cost of a wrong answer. Add calibration (ECE and a reliability diagram) and a sycophancy flip rate. Many of these checks use an LLM grader to decide whether an answer is correct, hedged or unsupported; validate that grader against humans first (LLM as judge), because a judge that rewards confident tone recreates the bias you are trying to measure.

Under the hood

Designing the metrics

A useful summary for factual QA is the penalised score at threshold t:

$$S_t = \frac{1}{n}\left(N_{\text{correct}} - \frac{t}{1-t}\,N_{\text{wrong}}\right)$$

Abstentions contribute 0. Plot S_t for $t \in \{0, 0.5, 0.75, 0.9\}$: a model that guesses wins at $t = 0$ and falls behind as t grows; a well-calibrated model that abstains appropriately stays ahead across thresholds.

Treat sycophancy as a paired test: the same item with and without a user opinion, scored with a paired comparison and an interval, so a flip-rate change between checkpoints can be distinguished from noise. Keep the challenge turn neutral ("Are you sure?") in one arm and evidence-bearing in another. Track all of this per post-training stage, since capability RL can move sycophancy and calibration as a side effect (RLHF).

Watch out

Do not read a model's apology, stated reasoning or self-assessment as a report of what happened inside it. These are generated text, shaped by the same training as everything else. A model saying "I double-checked this" has not necessarily checked anything unless a tool call or test result in the transcript shows it did.

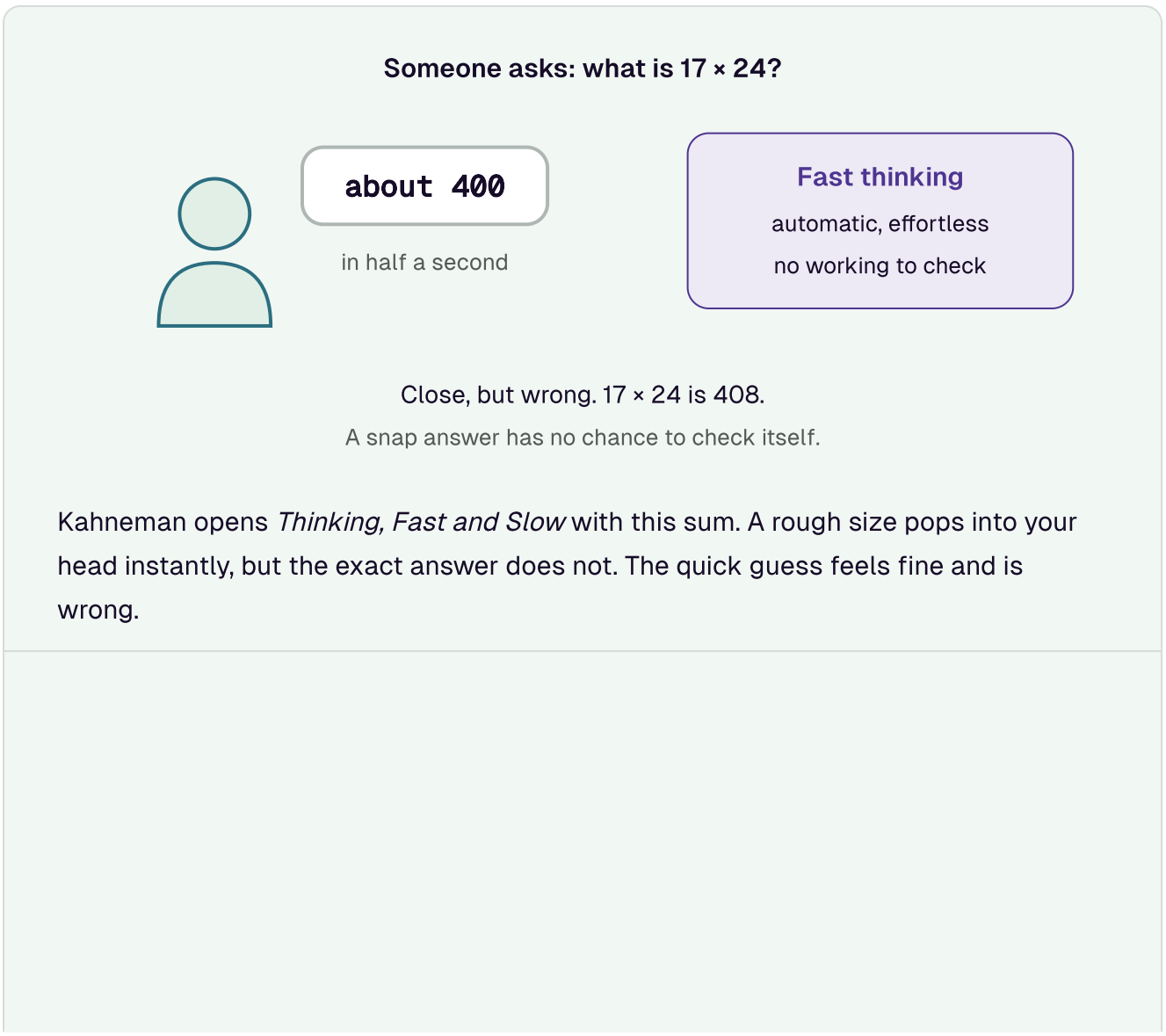
Fast and slow: reasoning models and thinking modes

Some questions you answer on the spot. Others you only get right by working through them. Since late 2024, frontier models have been built and sold with the same split: an instant mode that answers straight away, and a thinking mode that writes out a long chain of intermediate steps first. This chapter explains why writing steps helps a next-token predictor at all, how reasoning models are trained to do it, what thinking budgets and routers control, what the extra tokens cost, when thinking does not help, and why "think harder" and "turn up the temperature" are two completely separate knobs.

Diagram

17 × 24: in your head, then on paper

Fast and slow thinking, and what it means for a model that writes one token at a time.



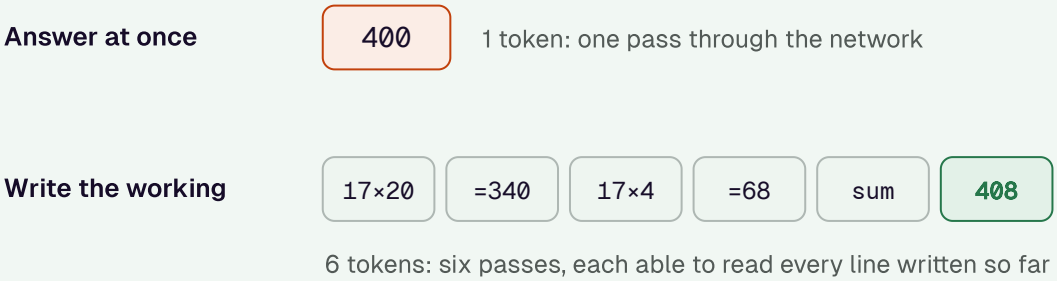
Kahneman opens *Thinking, Fast and Slow* with this sum. A rough size pops into your head instantly, but the exact answer does not. The quick guess feels fine and is wrong.

On paper: small steps you can check

Three easy steps replace one hard leap.

Writing the working breaks one hard step into three easy ones, and each line can be checked before the next. It costs time and attention. That trade is the whole story of reasoning models.

A language model spends the same effort on every token



Same network, same size. Writing tokens buys more computation.

A model does a fixed amount of work for each token it produces. Answering at once gives the problem one token's worth of work. Writing the working first gives it many, and every new token can read the lines already written, like a scratchpad.

Thinking is paid for in tokens and seconds

Accuracy rises on multi-step problems; on simple lookups it barely moves.

Illustrative: 80 tokens per second, thinking tokens billed as output.

Every thinking token is generated one at a time and billed like any other output token. A hundredfold increase in tokens is a hundredfold increase in latency and cost for that answer.

Spend thinking where the problem has steps

Usually worth thinking	Usually not worth it
multi-step maths and logic	recalling a fact
writing and debugging code	sorting into a few labels
planning under constraints	tasks full of distracting detail
checking a candidate answer	anything due in 100 ms

Thinking is extra computation. Its value depends on the problem.

The takeaway: reasoning is a way of buying computation at answer time. It pays off when the problem decomposes into steps and the answer can be checked. It is wasted, and sometimes harmful, on everything else.

Two speeds of thinking

Plain words

Fast intuition, slow work

Psychologist Daniel Kahneman popularised a simple picture of the mind with two characters. **System 1** is fast and automatic: you read a face, finish a familiar phrase, or know that 2 + 2 is 4 without trying. **System 2** is slow and deliberate: you park in a tight space, fill in a tax form, or work out 17 × 24 on paper.

Language models now come in both flavours. An instant answer is like System 1: one quick pass, no visible working. A thinking mode is like System 2: the model writes out steps, checks them, and sometimes backtracks before it commits to an answer. The analogy is useful for intuition. It is only an analogy.

For leaders

A product split, and a vocabulary to use with care

"System 1 and System 2" has become shorthand across the industry. Vendors use it for thinking modes, and at least one company has named a whole product class "System One models" (see System 1 decision models). The shorthand helps when you are deciding **which requests deserve expensive deliberation**. It misleads when it suggests the model has two different minds inside.

- **Same weights, different spend.** In most products the instant and thinking modes are the same network, or two networks from one family, used with very different token budgets. The quality gap comes from extra computation and from training that teaches the model how to use it.
- **The decision you own** is allocation: which traffic gets thinking, how much, and who decides (the user, your code, or a router).
- **The risk** is anthropomorphism in reviews. "The model reflected and realised its mistake" describes text the model wrote. Whether that text reflects how the answer was computed is a separate and open question.

Under the hood

Where the analogy holds and where it breaks

Kahneman himself presented the two systems as a shorthand, characters in a story about the mind, not two separate brain modules. Dual-process theory remains debated in psychology, and several studies cited in the book (notably on priming) later failed to replicate. Use the analogy for the trade-off between speed and deliberation, and nothing more.

For a transformer the mapping is concrete. Every output token costs one forward pass: a fixed stack of L layers, roughly $2N$ floating-point operations for a model with N parameters. The amount of serial computation behind an answer therefore scales with the number of tokens generated before it:

$$C_{\text{answer}} \approx 2N\left(T_{\text{think}} + T_{\text{ans}}\right), \qquad \text{serial depth} \approx L\left(T_{\text{think}} + T_{\text{ans}}\right)$$

N is parameter count, L the number of layers, T_{think} the tokens written before the answer and T_{ans} the answer tokens. An instant answer has $T_{\text{think}} = 0$.

So "System 2" in a language model is many System 1 steps chained through a scratchpad. Nothing in the architecture changes between modes; the model only runs longer, and training decides what it writes while it runs.

Chain of thought: showing the work

Plain words

Asking for the steps

In 2022 researchers found that if you show a model a few worked examples with the steps written out, or simply add "Let's think step by step" to the question, it gets far more maths word problems right. The model writes its working first, then the answer. This is called **chain of thought**.

Why should writing help a machine that only predicts the next word? Two reasons. Each word it writes is another round of computation. And everything it writes stays on the page, so later words can build on earlier ones, the way you build on the lines of a long division.

For leaders

What the evidence says

- **Few-shot chain of thought.** Wei et al. (2022) prompted a 540-billion-parameter model with eight worked examples and reached state-of-the-art accuracy on GSM8K, a set of grade-school maths word problems. The effect appeared only in sufficiently large models.
- **Zero-shot chain of thought.** Kojima et al. (2022) added "Let's think step by step" with no examples. On text-davinci-002, MultiArith accuracy went from 17.7% to 78.7% and GSM8K from 10.4% to 40.7%.
- **Where it helps.** Sprague et al. (2024), a meta-analysis of over 100 papers plus experiments on 20 datasets and 14 models, found the gains concentrated on maths and symbolic logic. On MMLU, answering directly matched chain of thought except on questions whose text or answer contained an equals sign.

Principal asks

When someone reports a gain from "adding reasoning", ask: on which task types, measured how, at what token cost? Does the gain survive on the slices without symbolic computation? Was the comparison at matched decoding settings, or did the reasoning arm also get more samples?

Under the hood

Why intermediate tokens add power

A single forward pass is a fixed-depth circuit. Problems that need many dependent steps (carry the digit, then add the next column) can exceed what a fixed depth computes reliably. Writing intermediate tokens turns the model into a loop: the output of pass t becomes input to pass $t+1$, and the context window acts as working memory.

Theory backs this up. Merrill and Sabharwal (2024, ICLR) showed that a decoder-only transformer allowed a linear number of intermediate steps can recognise all regular languages, which is believed impossible without them, and that with polynomially many steps it recognises exactly the problems solvable in polynomial time. The number of thinking tokens is a computational budget in a precise sense.

Two caveats matter for evaluation. The written chain is a generated text, so it can be unfaithful: the stated reasons need not be the computation that produced the answer. And chain of thought improves accuracy on average while adding a new failure surface, since an early wrong step propagates.

Reasoning models: trained to think before answering

Plain words

From a prompt trick to a trained habit

Chain-of-thought prompting asks a model to show its work. A **reasoning model** is trained to do it on its own, at length, and well. During training it attempts many problems with checkable answers (maths with a known result, code with tests), and it is rewarded when the final answer is right. Over time it learns to write long, useful working: trying an approach, checking it, noticing a mistake, and trying again.

For leaders

Two launches that changed the cost structure

- **OpenAI o1 (12 September 2024).** OpenAI reported that o1's performance kept improving with more reinforcement learning and with more time spent thinking at test time. On AIME 2024 maths problems, GPT-4o solved 12% on average; o1 solved 74% with one sample and 83% with a consensus of 64 samples. OpenAI chose not to show the raw chain of thought, showing a model-written summary instead.
- **DeepSeek-R1 (January 2025).** DeepSeek published how they did it: reinforcement learning against verifiable rewards, with reasoning behaviours such as self-reflection and verification emerging from training rather than from labelled reasoning traces. They released the weights. The method is covered in GRPO and RL with verifiable rewards.

The business consequence: quality became something you can buy per request with tokens, instead of only with a bigger model. Your cost model now has a variable term that depends on how hard each request is.

Under the hood

What RL changes

The policy samples a reasoning trace z and answer y for prompt x. A verifier gives reward $r(x, y) \in \{0, 1\}$. Training maximises expected reward, typically with a group baseline (GRPO) and a KL penalty to a reference model:

$$\max_{\theta} \; \mathbb{E}_{x}\, \mathbb{E}_{(z,y)\sim\pi_\theta(\cdot|x)} \left[r(x, y) \right] - \beta\, \mathrm{KL}\left(\pi_\theta \,\|\, \pi_{\text{ref}}\right)$$

Only the final answer is graded. The trace z is never supervised directly, so the model is free to discover whatever writing habits raise the chance of a correct y, including longer traces.

Because longer traces often raise reward, average response length tends to grow during this training. That is the mechanism behind "thinking longer", and it is why labs add explicit length controls and budget training afterwards. Whether RL teaches new capabilities or mostly sharpens ones the base model already had is discussed in the GRPO chapter.

Thinking budgets, modes and routers

Plain words

Who decides how hard to think

A thinking model needs to know how much effort to spend. There are three ways to decide. You pick (an "instant" or "thinking" button). You set a budget (think for up to this many tokens, or at this effort level). Or the system picks for you: a small, fast **router** reads the request and sends easy questions to the quick model and hard ones to the thinker.

For leaders

The current controls, vendor by vendor

Product names change quickly. The table reflects vendor documentation as checked on 1 November 2026.

Vendor	How thinking is controlled	What you see of the reasoning
OpenAI	GPT-5 launched in ChatGPT on 7 August 2025 as one system: a fast model, a deeper "GPT-5 thinking" model, and a real-time router that chooses using conversation type, complexity, tool needs and explicit intent (for example "think hard about this"). In the API, `reasoning.effort` takes model-dependent values from `none` through `max`.	Reasoning tokens are hidden in the API and billed as output tokens. ChatGPT shows summaries, a policy that began with o1.
Anthropic	Claude's extended thinking. Current models use **adaptive thinking**: Claude decides per request whether and how much to think, steered by an `effort` setting. The older manual mode with `budget_tokens` (minimum 1,024) is rejected on Claude 4.7 and later models.	A `display` option returns summarised thinking or omits it. No setting returns the raw chain of thought. You are billed for the full thinking, not the summary.
Google	Gemini 2.5 used a numeric `thinkingBudget` (–1 for dynamic, 0 to switch off on Flash models). Gemini 3 and later use a `thinkingLevel` (`MINIMAL` to `HIGH`) and cannot switch thinking off.	Optional thought summaries. Thinking tokens are billed at the output-token rate.

Three patterns stand out. Every vendor bills for thinking you cannot fully read. Every vendor has moved toward letting the model, or a router, pick the effort. And effort settings are coarse, so the same "high" can mean different token counts on different requests.

Eval lens

A thinking setting is part of the system under test. Record the model, effort level, and routing mode with every result, and report accuracy alongside thinking tokens per item. A router is itself a classifier: when it sends a hard question to the fast path, that is a false negative with a measurable rate. Evaluate the router on labelled easy and hard requests, as in classification metrics.

Under the hood

Routing as expected-cost minimisation

A router predicts, for request x, the chance $q(x)$ that the fast path gets it wrong when the slow path would get it right. With cost c_{err} for a wrong answer and extra cost c_{think} for the slow path, routing to the slow path is worth it when

$$q(x)\, c_{\text{err}} > c_{\text{think}} \iff q(x) > \frac{c_{\text{think}}}{c_{\text{err}}}$$

The threshold depends on what errors cost you, not on the router. A calibrated $q(x)$ lets you move the threshold when prices change, without retraining.

This is the same threshold logic as any cost-sensitive classifier, and a router is often exactly the kind of fast, typed, probabilistic decision the next chapter is about. Budgets and effort levels apply the same idea within one model: training teaches the model to stop early on easy inputs and continue on hard ones.

Accuracy against thinking tokens

Plain words

Each extra minute helps a little less

Give a student one minute on a hard problem and they might get 40% right. Five minutes, maybe 70%. An hour, maybe 82%. The first few minutes help most. Reasoning models behave the same way: accuracy climbs quickly with the first thousand thinking tokens, then flattens, while the cost and the wait keep growing in a straight line.

For leaders

Buy the knee of the curve

Published charts, starting with OpenAI's o1 announcement, plot accuracy against thinking compute on a **log** axis and show a roughly straight line over a range. A straight line on a log axis means each doubling of tokens buys about the same gain, so each additional token buys less. Meanwhile latency and price grow linearly.

- **Measure your own curve.** Run your eval at three to five effort levels and plot accuracy, p50 and p95 latency, and cost per 1,000 requests. Public curves come from maths contests, not your traffic.
- **Pick by target.** Choose the cheapest setting that meets the accuracy target, and route only the requests that need more.
- **Watch the tail.** Thinking length varies a lot per request. A p95 latency several times the median is normal, and it is what users feel.

Under the hood

A simple cost model

$$\mathrm{acc}(T) \approx a + b \log_2 T \ \ \text{(until it saturates)}, \qquad \mathrm{latency} \approx t_{\mathrm{prefill}} + \frac{T + T_{\mathrm{ans}}}{r}, \qquad \mathrm{cost} \approx$$

T is thinking tokens, r the decode speed in tokens per second, p_{out} the output price per token. The log-linear form is an empirical fit over a range, not a law.

python

```python
# Accuracy you measured at each thinking budget on your own eval set (illustrative numbers).
budgets  = [0, 1000, 2000, 4000, 8000, 16000]
accuracy = [0.42, 0.61, 0.70, 0.77, 0.81, 0.82]

PRICE_PER_M   = 10.0   # $ per 1M output tokens (illustrative); thinking tokens bill as output
TOKENS_PER_S  = 80.0   # decode speed: thinking is generated one token at a time
ANSWER_TOKENS = 200

def cost_and_latency(budget):
    toks = budget + ANSWER_TOKENS
    return toks * PRICE_PER_M / 1e6, toks / TOKENS_PER_S

def cheapest_budget(target):
    """Smallest budget that reaches the target accuracy, or None."""
    for b, a in zip(budgets, accuracy):
        if a >= target:
            return b
    return None
```

```
for b, a in zip(budgets, accuracy):
    c, t = cost_and_latency(b)
    print(f"{b:>6} tokens  acc {a:.2f}  ${c * 1000:6.2f} per 1k  {t:6.1f} s")
for i in range(1, len(budgets)):
    # accuracy bought per extra 1,000 thinking tokens: the curve flattens
    gain = (accuracy[i] - accuracy[i - 1]) / ((budgets[i] - budgets[i - 1]) / 1000)
    print(f"{budgets[i - 1]}->{budgets[i]}: +{gain:.3f} per 1k tokens")
print(cheapest_budget(0.75))   # 4000
```

rust

```
// Accuracy you measured at each thinking budget on your own eval set (illustrative numbers).
const BUDGETS: [u32; 6] = [0, 1000, 2000, 4000, 8000, 16000];
const ACCURACY: [f64; 6] = [0.42, 0.61, 0.70, 0.77, 0.81, 0.82];

const PRICE_PER_M: f64 = 10.0;   // $ per 1M output tokens (illustrative); thinking tokens bill as output
const TOKENS_PER_S: f64 = 80.0;  // decode speed: thinking is generated one token at a time
const ANSWER_TOKENS: u32 = 200;

fn cost_and_latency(budget: u32) -> (f64, f64) {
    let toks = (budget + ANSWER_TOKENS) as f64;
    (toks * PRICE_PER_M / 1e6, toks / TOKENS_PER_S)
}

/// Smallest budget that reaches the target accuracy, or None.
fn cheapest_budget(target: f64) -> Option<u32> {
    BUDGETS.iter().zip(ACCURACY.iter()).find(|(_, &a)| a >= target).map(|(&b, _)| b)
}

fn main() {
    for (&b, &a) in BUDGETS.iter().zip(ACCURACY.iter()) {
        let (c, t) = cost_and_latency(b);
        println!("{:>6} tokens  acc {:.2}  ${:6.2} per 1k  {:6.1} s", b, a, c * 1000.0, t);
    }
    for i in 1..BUDGETS.len() {
        // accuracy bought per extra 1,000 thinking tokens: the curve flattens
        let gain = (ACCURACY[i] - ACCURACY[i - 1]) / ((BUDGETS[i] - BUDGETS[i - 1]) as f64 / 1000.0);
        println!("{}->{}: +{:.3} per 1k tokens", BUDGETS[i - 1], BUDGETS[i], gain);
    }
    match cheapest_budget(0.75) {
        Some(b) => println!("{}", b), // 4000
        None => println!("None"),
    }
}
```

Going from 4,000 to 16,000 thinking tokens quadruples the bill and the wait for five points of accuracy. Whether that is worth it is a product decision, and the table makes it explicit. Compare with spending the same tokens on several shorter samples and a vote, covered in inference and test-time compute.

When reasoning does not help

Plain words

Overthinking is real

Some tasks get worse when you think about them too hard. In a classic study (Wilson and Schooler, 1991), people asked to list their reasons for liking different jams ended up agreeing less with expert tasters than people who simply tasted them. Models show similar effects. Long reasoning can talk a model out of a good first instinct, get it lost in irrelevant details, or make it latch onto a pattern that is not really there.

For leaders

Three published warnings

- **Most gains are symbolic.** Sprague et al. (2024) found chain of thought helped mainly on maths and logic. For knowledge recall and many classification tasks, direct answers did about as well at a fraction of the cost.
- **Tasks where thinking hurts people.** Liu et al. (2024) took six tasks from cognitive psychology where deliberation makes humans worse. On three, chain of thought caused large drops, up to 36.3 points of absolute accuracy for o1-preview compared with GPT-4o.
- **Inverse scaling with thinking length.** Gema et al. (2025) built tasks where longer reasoning lowered accuracy: counting with distractors, regression with spurious features, and deduction with many constraints. Different model families failed differently, some getting distracted by irrelevant information, others overfitting to how the problem was framed.

⚠ **Watch out**

Defaulting every request to maximum thinking is a common and expensive mistake. It raises latency and cost on all traffic, and on some slices it lowers accuracy. Turn thinking up where your own eval shows a gain, and treat "more thinking is always safer" as a hypothesis to test.

Under the hood

Why more tokens can hurt

Each step in a chain has some chance of introducing an error and some chance of fixing one. If the fix rate is low or the problem has no checkable structure, extra steps mostly add chances to go wrong. Long traces also dilute attention: by token 10,000 the original question is a small share of the context, and irrelevant details written along the way compete with it. Finally, RL rewards only the

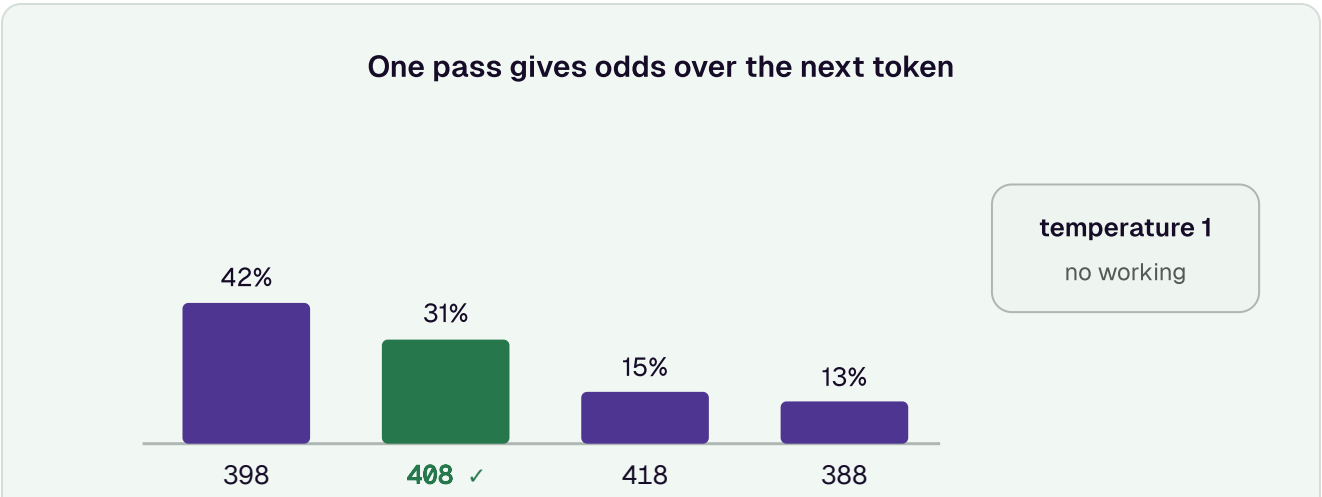
final answer on its training distribution. Habits that pay off on contest maths (always try another approach, always reconsider) can misfire on tasks where the first answer was best.

Reasoning is not temperature

Diagram

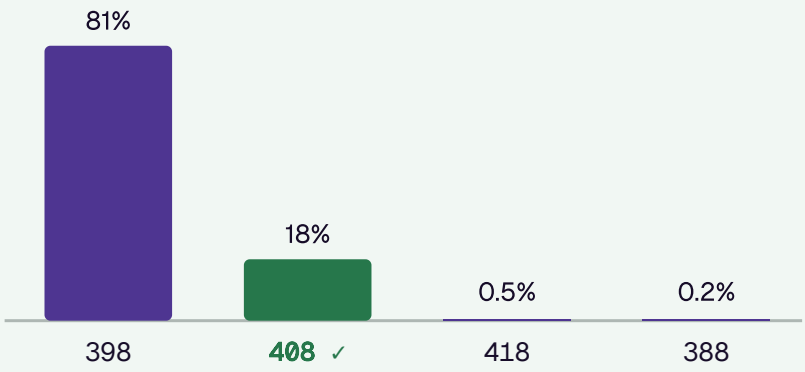
Two knobs: the dice and the scratchpad

Temperature reshapes the odds of the next token. Reasoning changes the odds by doing more work first.



Asked for 17 × 24 with no working, one forward pass produces odds over candidate answers (illustrative). The wrong answer 398 is the favourite; the right one, 408, is second. Sampling picks one according to these odds.

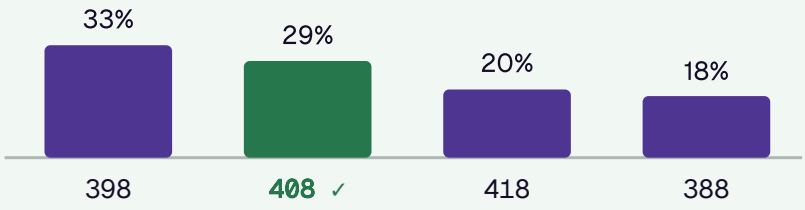
Low temperature: same ranking, sharper odds



temperature 0.2
confidently wrong

Turn temperature down and the favourite takes almost everything. The ranking does not change, so here the model becomes reliably wrong. Low temperature buys consistency, never correctness.

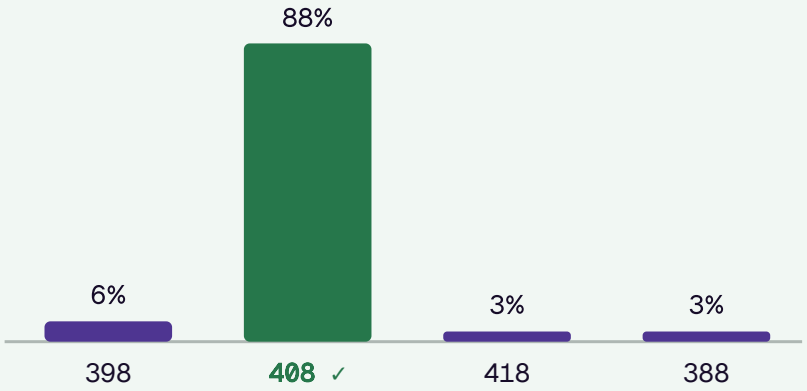
High temperature: flatter odds, more variety



temperature 2
same work, wider spread

Turn temperature up and the odds flatten. Answers vary more from run to run. The model did exactly the same amount of computation in all three pictures: one forward pass, then a reshaped draw.

Reasoning: more passes first, then new odds



after working
340 + 68
~40 extra
forward passes

Reasoning works on a different lever. The model writes its working first, so by the time it reaches the answer token it has done dozens of extra passes and can read "340 + 68" on the page. The odds themselves change: the right answer is now the clear favourite.

Combine them: sample several chains, then vote

408 wins 3–1–1
self-consistency

Temperature above zero makes the chains differ; reasoning makes each chain better.

The two knobs combine. Self-consistency samples several reasoning chains at a temperature above zero, so they take different routes, and keeps the most common final answer. Temperature supplies variety; reasoning supplies quality per chain. See inference and test-time compute.

Plain words

Loaded dice versus doing the sum

Temperature controls how adventurous the model is when it picks each next word. Low temperature: always take the favourite. High temperature: sometimes take an outsider. It changes how the odds

are spread. It does not make the model think any harder.

Reasoning means the model writes working before the answer. That is more computation, so it can change which answer is the favourite in the first place.

They are separate dials. A model can reason at low or high temperature, and it can answer instantly at either.

For leaders

Which knob for which problem

	Temperature	Reasoning (thinking budget)
What it changes	The spread of each next-token distribution	How many tokens of computation come before the answer
Extra compute per answer	None	Proportional to thinking tokens
Fixes a wrong favourite?	No. Low temperature makes it more consistent	Often, on problems with steps
Use it for	Diversity (brainstorming, sampling many candidates) or repeatability	Multi-step correctness: maths, code, planning
Typical mistake	Raising it to make answers "smarter"	Maxing it on every request

Principal asks

If a team says "we lowered temperature and accuracy improved", ask whether they measured one sample per item at both settings. Lower temperature mostly reduces variance; a gain usually means the favourite was already right and sampling was throwing it away. If they say "we raised the thinking budget", ask for the accuracy and cost at each budget, and whether the comparison held temperature fixed.

Under the hood

The two operations, side by side

Temperature acts after the forward pass, on the logits z:

$$p_i(\tau) = \frac{e^{z_i/\tau}}{\sum_j e^{z_j/\tau}}$$

$\tau \to 0$ approaches greedy argmax; $\tau = 1$ is the model's own distribution; $\tau > 1$ flattens it. The ordering of tokens never changes, and the logits z are computed once either way.

Reasoning changes the conditioning. The answer distribution becomes $p(y \mid x, z_{1:T})$, where $z_{1:T}$ are the thinking tokens. The logits at the answer position are different numbers, produced after T additional forward passes. Self-consistency then marginalises over sampled chains, $\hat{y} = \arg\max_y \sum_{k=1}^{K} \mathbb{1}[y_k = y]$; Wang et al. (2022) reported gains over single-chain prompting of 17.9 points on GSM8K and 11.0 on SVAMP.

python

```python
import math

def softmax_t(logits, T):
    # temperature divides the logits before softmax: it reshapes, it never adds computation
    z = [l / T for l in logits]
    m = max(z)
    e = [math.exp(v - m) for v in z]
    s = sum(e)
    return [v / s for v in e]

def entropy_bits(p):
    return -sum(q * math.log2(q) for q in p if q > 0)

answers = ["398", "408", "418", "388"]   # "408" is correct
logits  = [2.0, 1.7, 1.0, 0.8]           # one forward pass, answering 17 x 24 with no working
for T in (0.2, 1.0, 2.0):
    p = softmax_t(logits, T)
    top = answers[p.index(max(p))]
    print(f"T={T:.1f}", [round(q, 3) for q in p], f"H={entropy_bits(p):.2f} bits", "top:", top)
# the top answer is 398 (wrong) at every temperature; only the spread changes
```

rust

```rust
fn softmax_t(logits: &[f64], t: f64) -> Vec<f64> {
    // temperature divides the logits before softmax: it reshapes, it never adds computation
    let z: Vec<f64> = logits.iter().map(|l| l / t).collect();
    let m = z.iter().cloned().fold(f64::NEG_INFINITY, f64::max);
    let e: Vec<f64> = z.iter().map(|v| (v - m).exp()).collect();
    let s: f64 = e.iter().sum();
    e.iter().map(|v| v / s).collect()
}

fn entropy_bits(p: &[f64]) -> f64 {
    -p.iter().filter(|&&q| q > 0.0).map(|q| q * q.log2()).sum::<f64>()
}

fn main() {
    let answers = ["398", "408", "418", "388"]; // "408" is correct
```

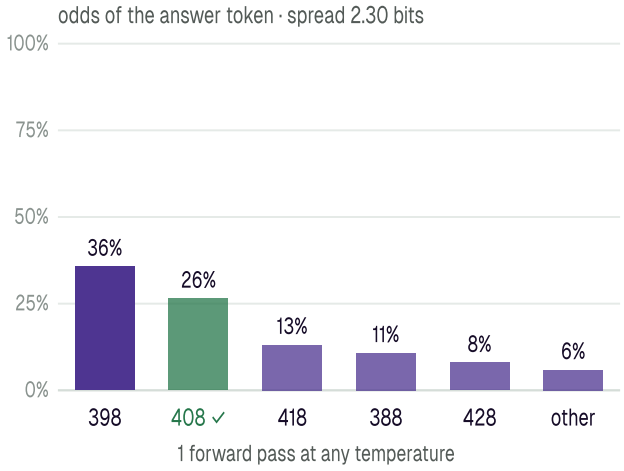
```rust
    let logits = [2.0, 1.7, 1.0, 0.8];          // one forward pass, answering 17 x 24 with no working
    for t in [0.2, 1.0, 2.0] {
        let p = softmax_t(&logits, t);
        let (mut best, mut top) = (0.0, 0);
        for (i, &q) in p.iter().enumerate() { if q > best { best = q; top = i; } }
        let r: Vec<f64> = p.iter().map(|q| (q * 1000.0).round() / 1000.0).collect();
        println!("T={:.1} {:?} H={:.2} bits top: {}", t, r, entropy_bits(&p), answers[top]);
    }
    // the top answer is 398 (wrong) at every temperature; only the spread changes
}
```

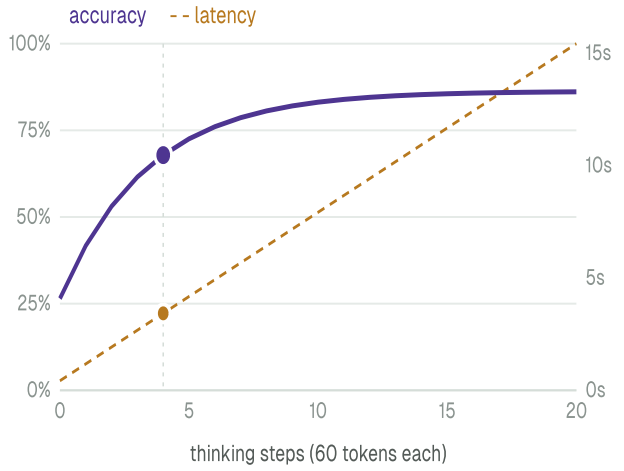
Interactive

Two knobs: temperature and thinking

(shown at its default setting)

Left: the odds over answers to 17 × 24 from a single pass, reshaped by temperature. Right: a multi-step problem where each thinking step can fix an error or introduce one. Move each slider and watch which numbers change. All numbers are illustrative.





Right, answering at once	Accuracy after thinking	Latency	Cost per 1k answers
26%	**68%**	**3.4 s**	**$2.50**

Temperature changed only the spread of the odds (entropy 2.30 bits). At this temperature the favourite is **398** (wrong) and the right answer gets 26%. The network did the same single forward pass either way. **Thinking** added 240 tokens of computation before the answer, moving accuracy from 26% to **68%**, at 3.4 s and $2.50 per 1,000 answers.

Model behind the right panel: the working starts right with the probability on the left. Each step fixes a wrong working with probability $f = 0.22$ and breaks a right one with probability e, which rises slightly with temperature. Accuracy after s steps is $a_\infty + (a_0 - a_\infty)(1 - f - e)^s$ with $a_\infty = f/(f + e)$. Each step writes 60 tokens at 80 tokens per second, priced at an illustrative $10 per million output tokens.

Eval lens

Reasoning changes what an eval measures. Report the thinking setting and mean thinking tokens per item next to every accuracy number, and compare models at matched cost, not only by name. Keep temperature and sample count fixed when comparing thinking budgets, or you are changing two knobs at once. And when a reasoning model's visible summary is used as evidence of how it solved a problem, remember the summary is generated text about hidden tokens.

Part III · Chapter 25

System 1 decision models: typed answers in one pass

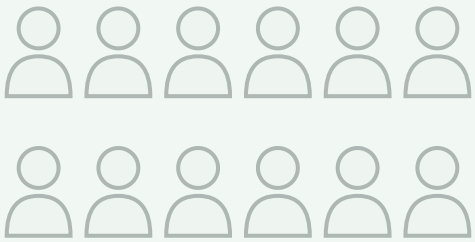
Most production uses of language models are decisions in disguise: is this ticket about billing, is this message a refund request, how urgent is this, which of these five tools should run next. The usual way to get one is to ask a chat model and parse the word it writes back. In 2026 a different kind of system appeared: models that take messy input and return a **typed decision with probabilities** in a single pass, generating no text at all. This chapter explains how they work, how they differ from asking a chat model, what three current offerings actually provide, and how to validate one before you let it make decisions for you.

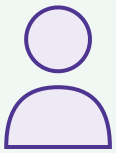
Diagram

The triage nurse and the doctor

Why a fast, typed, probabilistic decision is worth having, told as an emergency room's first five minutes.

Monday evening: twelve patients, one doctor



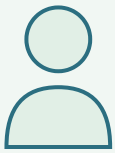


the doctor
full workup: about 45 minutes

Nine hours to see everyone in order. Someone in the queue may not have nine hours.

The doctor is thorough and slow, like a frontier language model asked to reason about every case. Seeing everyone in arrival order means a heart attack can wait behind a sprained wrist.

The triage nurse: one minute, one fixed form



triage nurse
about a minute each

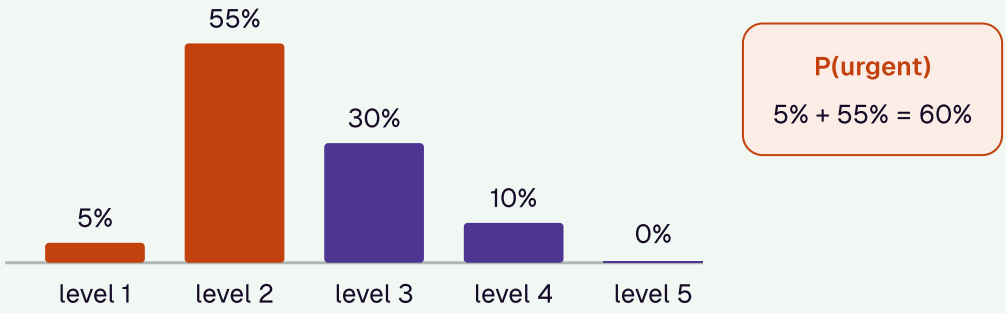
Urgency (tick one)

- ☐ 1 needs life-saving help now
- ☑ **2 high risk, should not wait**
- ☐ 3 stable, needs several tests
- ☐ 4 stable, needs one thing
- ☐ 5 can safely wait

No essay, no diagnosis. One answer from a fixed list.

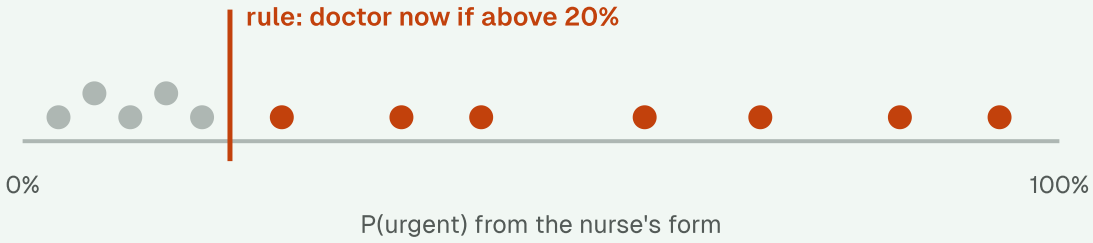
A triage nurse looks at each patient for about a minute and ticks one box on a fixed five-level scale. The answer is typed: it can only be one of five values, so nobody has to interpret a paragraph.

A better form also records how sure she is



Instead of a single tick, she writes down how likely each level is. Now the hospital knows more than "level 2": it knows there is a 60% chance this patient is level 1 or 2, and a 40% chance they could wait.

The hospital sets the rule from what mistakes cost

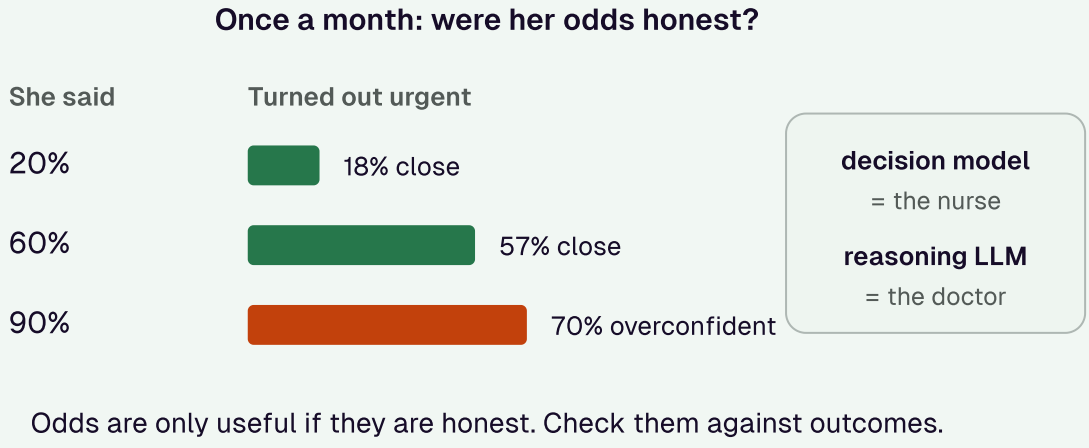


wait in the queue

see the doctor first: 7 of 12

Missing a heart attack costs far more than an early look at a sprain, so the line sits low.

With probabilities, the rule becomes a dial the hospital controls. Because missing an emergency is far worse than an unnecessary early look, it sends anyone above 20% straight to the doctor. A yes-or-no form could never be tuned this way.



Odds are only useful if they are honest. Check them against outcomes.

The odds only help if they mean what they say. Each month the hospital checks: of the patients she called 60% likely urgent, were about 60% actually urgent? Here her top bucket is overconfident. That check is called calibration, and every decision model needs it. The nurse does not replace the doctor; she decides who sees the doctor first.

What a decision model is

Plain words

A form instead of an essay

Asking a chat model for a decision is like asking someone to write you a note and then reading it to find the answer. Sometimes the note says "billing", sometimes "Billing.", sometimes "This is probably a billing issue, although…". You need code to dig the answer out, and you never learn how sure the writer was.

A **decision model** is handed a form. You tell it the possible answers in advance (billing, technical, shipping, other), and it fills in the form with how likely each one is, in one quick step. It cannot write anything off the form, so there is nothing to dig out and nothing to misspell.

For leaders

What is new, and what is not

The pitch from all three offerings below is the same: **unstructured state in, typed probabilistic decisions out** (TypeSafe's phrase). In practice that buys four things:

Typed output

The answer is always one of the values you defined. No parsing, no "the model replied in French", no retries on malformed JSON.

Probabilities

Every answer comes with a distribution. That turns one fixed decision into a dial you can set by cost, as in classification metrics.

Latency

One pass instead of hundreds of generated tokens. Vendors report answers in tenths of a second.

Cost

You pay mostly for input. Two of the three offerings do not charge for output at all.

What is not new: the shape. A model that reads text and outputs a distribution over a fixed set of labels is a classifier, and teams have fine-tuned BERT-style classifiers for years. The new part is that these models take the label set and its definitions as **input at request time**, so one model serves any decision you can describe, without training data. That makes them closer to a zero-shot LLM judge in flexibility, and closer to a classifier in speed and output.

Under the hood

Autoregressive text versus a fixed output space

A chat model produces a decision as a string, one token at a time:

$$p(y_{1:T} \mid x) = \prod_{t=1}^{T} p(y_t \mid x, y_{<t})$$

T sequential forward passes. The probability of the *label* "billing" is the sum over every string that your parser would map to billing, which you cannot compute. In practice teams read the first token's logprob, or sample several times and count.

A decision model encodes the input and the question (including the option descriptions) once, then applies a small head over the K options you supplied:

$$h = f_\theta(x, q), \qquad p(k \mid x, q) = \frac{e^{s_\theta(h,k)}}{\sum_{j=1}^{K} e^{s_\theta(h,j)}}, \qquad \text{score} = \sum_{k=0}^{K-1} k\, p(k \mid x, q)$$

x is the input state, q the question with its options or levels, s_θ a learned scoring head. A predicate is the $K = 2$ case, reported as one probability. A score question reports the probability-weighted level index.

This is the same shape as a reward model's scalar head (reward models) or a classifier's softmax layer. Laya's published checkpoints make the lineage explicit: they are encoder models (ModernBERT-large at 421M parameters and mmBERT-base at 322M), not generative LLMs. TypeSafe describes Jev as a new architecture with a parallel sampler and gives no further detail.

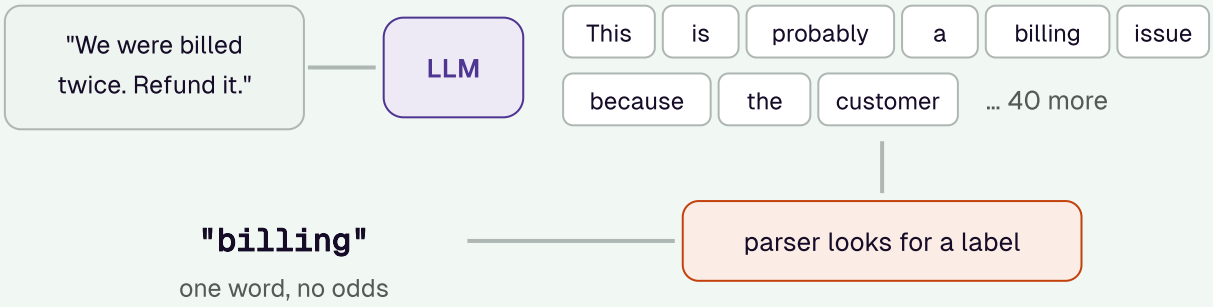
Training matters as much as architecture. A **strictly proper scoring rule**, such as log loss $-\log p(k^*)$ or the Brier score $\sum_k (p_k - \mathbb{1}[k = k^*])^2$, is minimised in expectation only by reporting the true probabilities, so optimising one rewards honest confidence. TypeSafe calls its method Reinforcement Learning for Calibrated Decisions (RLCD) without describing it; Laya's README describes its training as reinforcement learning against strictly proper scoring rules. Neither guarantees calibration on your data.

Diagram

Ask for an essay, or hand over a form

The same routing decision made two ways, from input to answer.

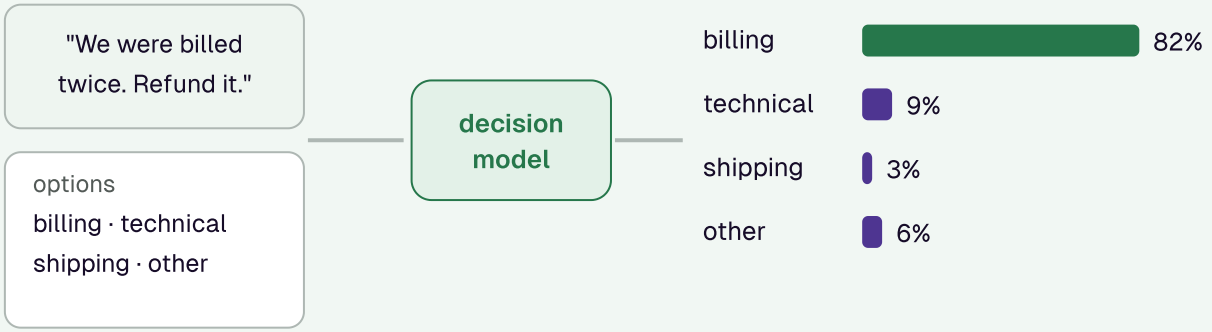
The chat-model way: write, then parse



About 50 tokens, each one a full pass through the network, written in sequence.

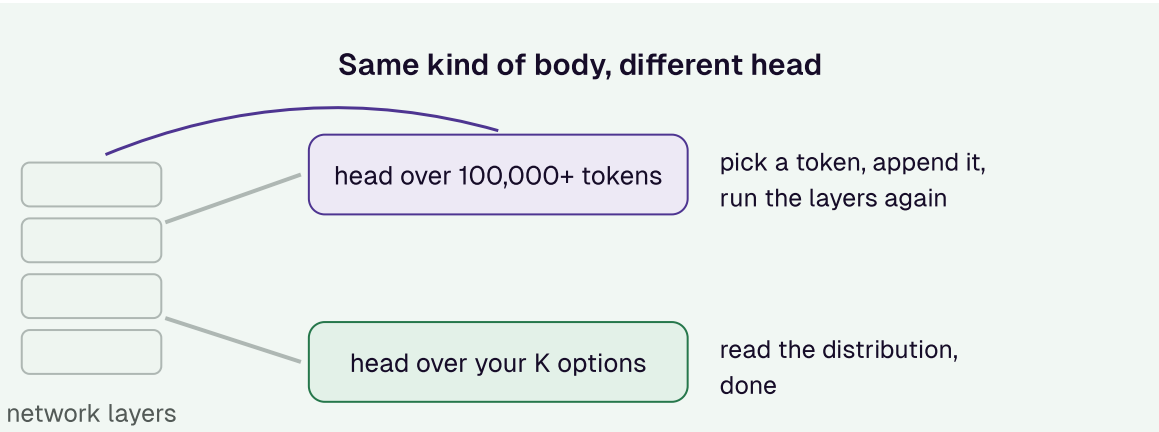
The chat model writes a sentence one token at a time. Your code then searches the text for a label. You get a word, not a probability, and you pay for every token of the explanation, in money and in waiting time.

The decision-model way: one pass, fixed options



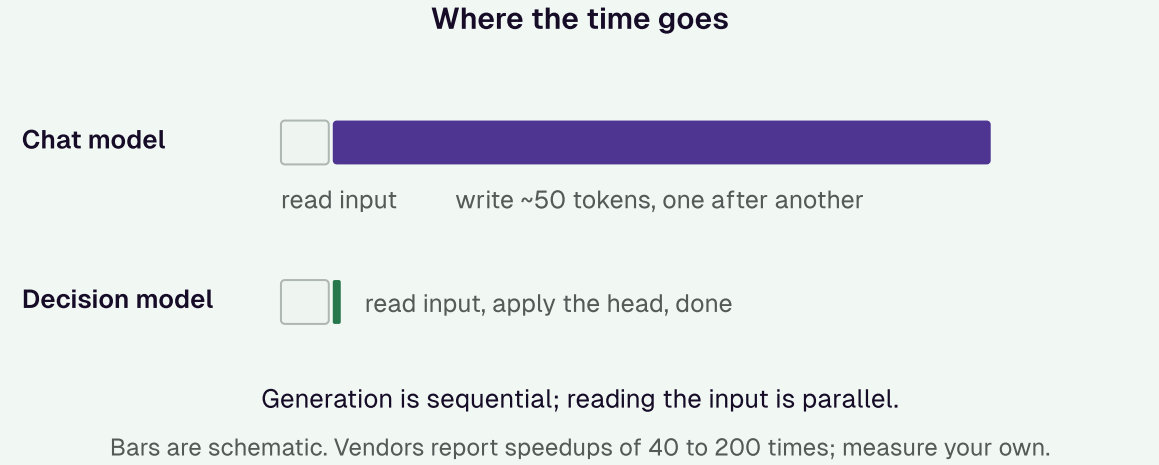
One pass over the input. The answer is a distribution, already typed.

The decision model receives the input and the allowed options together, runs once, and returns a probability for each option. Nothing is generated, so there is nothing to parse, and the odds come for free (illustrative numbers).



Like a classifier or a reward model's score head, with the options given at request time.

Under the hood, both read the input through a stack of network layers. The chat model's head chooses among its whole vocabulary and loops once per token. The decision model's head chooses among only the options you supplied and stops. That is the same design as a classifier or a reward model's score head.



Generation is sequential; reading the input is parallel.

Bars are schematic. Vendors report speedups of 40 to 200 times; measure your own.

Reading the input is done in parallel across all its tokens, so it is fast. Writing is done one token after another, so it is slow. A decision model skips the writing. That, more than any clever trick, is where the speedup comes from, and it is why the gap grows when the chat model would have reasoned first.

Three offerings, one vocabulary

Plain words

Three kinds of question

All three systems ask the same three kinds of question, with slightly different names:

- **Is it true?** "Does the customer ask for a refund?" The answer is one probability, such as 0.93.
- **Which one?** "Which team should handle this: billing, technical, shipping, or other?" The answer is one choice plus a probability for each option.
- **How much?** "How urgent is this, on a scale from low to high?" The answer is a position on the scale plus a probability for each step.

For leaders

Who offers what, as each vendor reports it

	OpenAI Decisions API	Laya (Convai Innovations)	Jev (TypeSafe AI)
Form	Hosted API, `POST /v1/decisions`, model `gpt-6-luna` (the only supported model)	Open source, Apache 2.0, runs on your own CPU or NVIDIA GPU	Hosted API, early access since 15 September 2026
Is it true?	`predicate`: probability the condition holds	`noul`: probability the answer is yes	Typed outputs for "classify, route, score, extract, or branch". Laya's server speaks the same wire protocol (`POST /v1/systemone`)
Which one?	`choice`: one of your values, probabilities, confidence	`choice`: one label, probabilities, confidence	
How much?	`score`: 2 to 10 ordered levels; probability-weighted score, per-level probabilities, confidence	`score`: expected ordinal level, probabilities, confidence	
Pricing	$0.10 per 1M uncached input tokens; output not charged	Your hardware	$0.042 per 1M input tokens; output free; may be subsidised (company note)
Speed claims	None stated in the guide	None on the main page; benchmarks say measure warm and cold calls yourself	70 to 500 ms end to end; 40 to 200 times faster than frontier LLMs on "System One-shaped" queries

Laya and Jev share a lineage beyond the name: TypeSafe says "System One" is inspired by Kahneman, and Laya's server is wire-compatible with Jev's API. Do not assume numbers carry across, though. Laya documents that its `confidence` is computed differently from Jev's, so thresholds tuned on one do not transfer.

> ⚠ **Watch out**
>
> Read vendor numbers as claims. TypeSafe's own post says its latency tests were "generally run from our laptops on the West Coast", its workflow evals were built by its own team, its reference answers were an average of two LLMs (GPT-6 Astra and Fable 5.1), its 193.6× speed and 444.6× cost figures are expected to be "on the higher end of real world gains", and its 0% hallucination figure is "not empirical" because the output schema makes free-text hallucination impossible by construction. A wrong label inside the schema is still a wrong answer. OpenAI's guide makes no calibration claim and tells you to set thresholds from your own labelled examples. Laya's benchmarks page says both base checkpoints are over-confident.

Under the hood

A request, and what comes back

A Decisions API request carries shared input and up to 400 named questions. The shape below follows OpenAI's guide; the content is made up.

json

```
{
  "model": "gpt-6-luna",
  "input": "Customer: I was charged twice for order 4417. Please refund the duplicate.",
  "questions": [
    {"type": "predicate", "name": "refund_request",
     "instructions": "The customer explicitly asks for money back."},
    {"type": "choice", "name": "department",
     "instructions": "Which team should handle this message?",
     "choices": [
       {"value": "billing",   "description": "Charges, refunds, invoices"},
       {"value": "technical", "description": "Bugs, errors, account access"},
       {"value": "shipping",  "description": "Delivery, tracking, returns in transit"},
       {"value": "other",     "description": "None of the above; send to general review"}]},
    {"type": "score", "name": "urgency",
     "instructions": "How soon does this need a human?",
     "levels": [
       {"label": "low",    "description": "Can wait several days"},
       {"label": "medium", "description": "Should be handled within a day"},
       {"label": "high",   "description": "Money or access is blocked now"}]}
  ]
}
```

Answers come back per question: a `probability` for the predicate; a `choice`, a `probabilities` array and a `confidence` for the choice; and a `score`, per-level `probabilities` and a `confidence` for the score. Level indices start at 0, so level probabilities of 0.1, 0.2 and 0.7 give a score of $0(0.1) + 1(0.2) + 2(0.7) = 1.6$. A score between levels is a feature: 1.6 says "probably high, possibly medium" in one number.

Laya's equivalent is `router.predict(state, questions)`, returning `noul`, `choice` or `score` with `probabilities` and a per-answer `confidence`, plus which checkpoint answered. Laya defines confidence as one minus normalised entropy,

$$\text{conf}(p) = 1 - \frac{H(p)}{\log K} = 1 + \frac{\sum_k p_k \log p_k}{\log K}$$

1 when all mass is on one option, 0 when the distribution is uniform over the K options. It measures concentration, not correctness; a confidently wrong model scores high.

OpenAI's guide does not define its `confidence` field. Treat any vendor's confidence as a ranking signal until you have checked it against labels.

Why one pass is fast, and what you give up

Plain words

The nurse cannot write a diagnosis

The triage form is fast because it is narrow. The nurse cannot write "chest pain, but probably reflux, let me order an ECG to be sure." A decision model has the same limit. It can only pick from the options you gave it, it cannot explain itself in words, and it cannot think through a long chain of steps before answering. For those, you still want the doctor.

For leaders

The trade-offs to put on the table

- **No reasoning before the answer.** If the decision needs multi-step work (reconcile three invoices, then decide), a single pass may not have enough computation. See fast and slow.
- **No explanation.** Audits, appeals and regulated decisions often need a reason. Pair the decision model with an LLM that explains the hard or contested cases.
- **Fixed options.** Anything outside your options is forced into one of them. Always include an "other" option, as OpenAI's guide recommends, and route it to review.

- **Dependent decisions.** If question two depends on the answer to question one, OpenAI recommends separate requests rather than chaining questions in one call.
- **Documented limits (Laya).** Keep a choice to roughly 20 self-contained options; reliability drops past about 4,000 tokens of input; option order can change choice answers; ordinal score questions are its weakest type.

Under the hood

A latency model

$$t_{\text{LLM}} \approx t_{\text{prefill}}(n) + T \cdot t_{\text{token}}, \qquad t_{\text{decision}} \approx t_{\text{prefill}}(n) + t_{\text{head}}$$

n input tokens are processed in parallel during prefill. The T output tokens are generated sequentially at t_{token} each. With a 300-token reasoning answer at 80 tokens per second, the second term alone is 3.75 s, while prefill of a short input is typically a small fraction of a second.

The speedup therefore depends mostly on T: small against a terse non-reasoning reply, large against a reasoning model. That is why TypeSafe reports a smaller speedup on its Wikiracing demo, where the comparison LLMs ran without reasoning. Smaller models help too: Laya's 322M to 421M parameter encoders are hundreds of times smaller than frontier LLMs, so even their prefill is cheap.

Probabilities make thresholds possible

Plain words

A dial instead of a switch

A yes-or-no answer is a switch: it is either on or off, and you cannot adjust it. A probability is a dial. If wrongly blocking a payment annoys a customer but missing fraud costs real money, you set the dial so the model flags anything above, say, 17% likely fraud. If the costs change, you turn the dial. You do not have to retrain or rewrite anything.

For leaders

Let costs set the threshold

The threshold is a business decision, not a model property. Write down what each mistake costs: a false positive (a needless review, a blocked customer) and a false negative (missed fraud, a missed emergency). If the probabilities are calibrated, the cost-minimising rule is simple: act when the

probability exceeds $c_{\text{FP}}/(c_{\text{FP}} + c_{\text{FN}})$. A miss that costs five times a false alarm gives a threshold of 1/6, about 17%.

If the probabilities are not calibrated, that formula picks the wrong threshold, and you must choose it empirically on labelled data instead. The widget below shows both cases. For a **choice** question, the same logic picks the action with the lowest expected cost, which is not always the most likely option.

Under the hood

Expected cost, thresholds, and calibration error

$$\text{act if } p\,c_{\text{FN}} > (1-p)\,c_{\text{FP}} \iff p > t^* = \frac{c_{\text{FP}}}{c_{\text{FP}} + c_{\text{FN}}}, \qquad a^* = \arg\min_a \sum_k p_k\, C(a,k)$$

Left: the binary rule, valid when p is calibrated. Right: the general rule for a choice answer with cost matrix $C(a,k)$, the cost of action a when the true class is k. Adding a "send to a human" action with flat cost turns abstention into part of the same calculation.

python

```python
# 1) A choice answer: route a support ticket. The top choice is not always the cheapest action.
classes = ["billing", "technical", "other"]
dist    = [0.55, 0.40, 0.05]                # probabilities returned by the decision model
actions = ["send to billing", "send to technical", "human review"]
cost = [[0, 3, 3],                          # cost[action][true class]: a misroute costs 3
        [3, 0, 3],
        [1, 1, 1]]                          # a human look always costs 1

def expected_costs(dist, cost):
    return [sum(p * c for p, c in zip(dist, row)) for row in cost]

ec = expected_costs(dist, cost)
print([round(x, 2) for x in ec], "->", actions[ec.index(min(ec))])   # [1.35, 1.8, 1.0] -> human review

# 2) A predicate answer on 12 labelled cases: P(fraud) and the truth.
probs  = [0.97, 0.91, 0.84, 0.72, 0.66, 0.58, 0.41, 0.33, 0.27, 0.15, 0.08, 0.03]
labels = [1,    1,    1,    0,    1,    0,    1,    0,    0,    0,    0,    0]
C_FP, C_FN = 1.0, 5.0                       # a missed fraud costs 5x a needless review

def bayes_threshold(c_fp, c_fn):
    # act when p*c_fn > (1-p)*c_fp, i.e. p > c_fp/(c_fp+c_fn); only right if p is calibrated
    return c_fp / (c_fp + c_fn)

def cost_at(t):
    fp = sum(1 for p, y in zip(probs, labels) if p >= t and y == 0)
    fn = sum(1 for p, y in zip(probs, labels) if p < t and y == 1)
    return (C_FP * fp + C_FN * fn) / len(probs)

def best_threshold():
    # try every observed score as the cut; on ties keep the higher cut (fewer reviews)
    best = 1.01
```

```
    for t in sorted(set(probs)):
        if cost_at(t) <= cost_at(best):
            best = t
    return best

t0 = bayes_threshold(C_FP, C_FN)
print(round(t0, 3), round(cost_at(t0), 3))          # theory: 0.167
t1 = best_threshold()
print(t1, round(cost_at(t1), 3))                    # chosen on labels; with 12 cases this is noisy

def ece(probs, labels, bins=5):
    # expected calibration error: gap between stated and observed rate, weighted by bin size
    total, n = 0.0, len(probs)
    for b in range(bins):
        lo, hi = b / bins, (b + 1) / bins
        idx = [i for i, p in enumerate(probs) if lo <= p < hi or (b == bins - 1 and p == 1.0)]
        if not idx:
            continue
        conf = sum(probs[i] for i in idx) / len(idx)
        rate = sum(labels[i] for i in idx) / len(idx)
        total += len(idx) / n * abs(rate - conf)
    return total

print(round(ece(probs, labels), 3))
```

rust

```
fn expected_costs(dist: &[f64], cost: &[[f64; 3]]) -> Vec<f64> {
    cost.iter().map(|row| dist.iter().zip(row).map(|(p, c)| p * c).sum()).collect()
}

const PROBS: [f64; 12] = [0.97, 0.91, 0.84, 0.72, 0.66, 0.58, 0.41, 0.33, 0.27, 0.15, 0.08, 0.03];
const LABELS: [u8; 12] = [1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0];
const C_FP: f64 = 1.0;
const C_FN: f64 = 5.0; // a missed fraud costs 5x a needless review

fn bayes_threshold(c_fp: f64, c_fn: f64) -> f64 {
    // act when p*c_fn > (1-p)*c_fp, i.e. p > c_fp/(c_fp+c_fn); only right if p is calibrated
    c_fp / (c_fp + c_fn)
}

fn cost_at(t: f64) -> f64 {
    let fp = PROBS.iter().zip(LABELS).filter(|(&p, y)| p >= t && *y == 0).count() as f64;
    let fn_ = PROBS.iter().zip(LABELS).filter(|(&p, y)| p < t && *y == 1).count() as f64;
    (C_FP * fp + C_FN * fn_) / PROBS.len() as f64
}

fn best_threshold() -> f64 {
    // try every observed score as the cut; on ties keep the higher cut (fewer reviews)
    let mut cands = PROBS.to_vec();
    cands.sort_by(|a, b| a.partial_cmp(b).unwrap());
    let mut best = 1.01;
    for t in cands {
        if cost_at(t) <= cost_at(best) { best = t; }
    }
    best
}

fn ece(probs: &[f64], labels: &[u8], bins: usize) -> f64 {
    // expected calibration error: gap between stated and observed rate, weighted by bin size
```

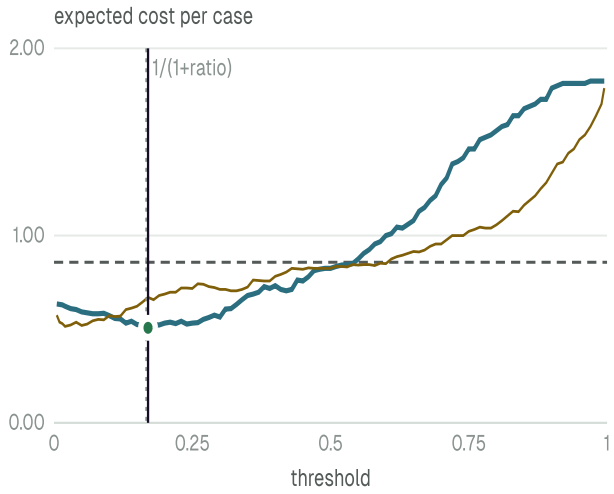
```
    let n = probs.len() as f64;
    let mut total = 0.0;
    for b in 0..bins {
        let (lo, hi) = (b as f64 / bins as f64, (b + 1) as f64 / bins as f64);
        let idx: Vec<usize> = (0..probs.len())
            .filter(|&i| (lo <= probs[i] && probs[i] < hi) || (b == bins - 1 && probs[i] == 1.0))
            .collect();
        if idx.is_empty() { continue; }
        let k = idx.len() as f64;
        let conf = idx.iter().map(|&i| probs[i]).sum::<f64>() / k;
        let rate = idx.iter().map(|&i| labels[i] as f64).sum::<f64>() / k;
        total += k / n * (rate - conf).abs();
    }
    total
}

fn r(x: f64, d: i32) -> f64 { let m = 10f64.powi(d); (x * m).round() / m }

fn main() {
    // 1) A choice answer: route a support ticket. The top choice is not always the cheapest action.
    let dist = [0.55, 0.40, 0.05]; // billing, technical, other
    let actions = ["send to billing", "send to technical", "human review"];
    let cost = [[0.0, 3.0, 3.0], [3.0, 0.0, 3.0], [1.0, 1.0, 1.0]]; // a misroute costs 3, a human look 1
    let ec = expected_costs(&dist, &cost);
    let mut best = 0;
    for i in 0..ec.len() { if ec[i] < ec[best] { best = i; } }
    let shown: Vec<f64> = ec.iter().map(|&x| r(x, 2)).collect();
    println!("{:?} -> {}", shown, actions[best]); // [1.35, 1.8, 1.0] -> human review

    // 2) A predicate answer on 12 labelled cases: P(fraud) and the truth.
    let t0 = bayes_threshold(C_FP, C_FN);
    println!("{} {}", r(t0, 3), r(cost_at(t0), 3));  // theory: 0.167
    let t1 = best_threshold();
    println!("{} {}", t1, r(cost_at(t1), 3));        // chosen on labels; with 12 cases this is noisy
    println!("{}", r(ece(&PROBS, &LABELS, 5), 3));
}
```

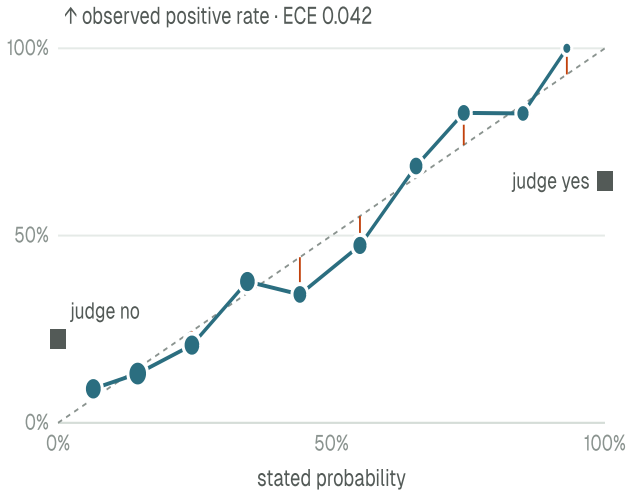
The routing example picks human review even though billing is the single most likely class, because a misroute is expensive and the model is only 55% sure. In the fraud example the theoretical threshold flags four legitimate cases while the empirical one flags two; on twelve cases that gap is noise, which is why you set thresholds on hundreds of labelled items.

Interactive

Choosing a threshold from costs (shown at its default setting)

400 labelled cases scored by two decision models with the same ranking: one calibrated, one overconfident. A binary LLM judge on the same cases gives a single yes or no. Set what a miss costs relative to a false alarm, then move the threshold. Synthetic data, illustrative.





calibrated model · overconfident model · binary LLM judge (dashed line, grey squares)

best threshold on these labels · calibration gap

Precision	Recall	Expected cost per case	ECE of this model
47%	93%	0.507	0.042

A miss costs 5× a false alarm, so the rule for calibrated probabilities is act above **0.17**. For the calibrated model that rule costs 0.515 per case; the best threshold on these labels is 0.17 at 0.507. They nearly coincide: when the probabilities are honest, the formula is enough. Your threshold of 0.17 gives precision 47% and recall 93%. The binary judge has one operating point (precision 64%, recall 60%, cost 0.858) and no dial to turn when costs change.

Validating a decision model

Plain words

Audit the nurse before trusting the form

Before a hospital relies on a new triage nurse, someone checks a sample of her calls against what happened. Do the same with a decision model. Collect a few hundred real cases, have people label the right answer, run the model, and compare. Check two things: does it rank cases well (risky ones above safe ones), and do its percentages mean what they say?

For leaders

A validation plan you can ask for

1. **Labelled set.** 300 to 1,000 items from real traffic, stratified so rare but costly classes have enough examples. Measure human-to-human agreement on a subset: that is the ceiling, as in LLM as judge.
2. **Ranking.** ROC-AUC per predicate and per choice option (one-versus-rest), with confidence intervals. PR-AUC when positives are rare.
3. **Calibration.** A reliability diagram and ECE. If it is off, fit temperature scaling on held-out labels; ranking is unchanged and the probabilities become usable.
4. **Threshold by cost.** Agree the cost ratio with the business owner, pick the threshold on a validation split, and report precision, recall and cost on a separate test split.
5. **Slices.** Languages, input length, customer tier, and option order. Laya documents failures by language, by input length and by option order.
6. **Drift.** In production, track the share of each decision, the mean predicted probability, and a weekly labelled audit sample. Re-validate whenever the vendor changes the model behind the endpoint.

Eval lens

A decision model is a classifier with a prompt, so validate it like one. Everything in classification metrics applies directly, with one upgrade over a binary LLM judge: because it returns probabilities, you get the full ROC curve, a calibration check, and a threshold you can move, instead of one fixed point. Report AUC for ranking, ECE for honesty, and expected cost at the chosen threshold for the business.

Under the hood

ECE and temperature scaling

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{n_b}{n} \left| \bar{y}_b - \bar{p}_b \right|, \qquad p' = \sigma\!\left(\frac{\operatorname{logit}(p)}{\tau} \right)$$

Left: bin predictions by stated probability, compare the observed positive rate $\bar{y}_b$ with the mean stated probability $\bar{p}_b$, and weight by bin size n_b. Right: temperature scaling with one parameter τ, fitted by minimising log loss on held-out labels. $\tau > 1$ softens an overconfident model. Guo et al. (2017) found modern neural networks poorly calibrated and temperature scaling surprisingly effective.

Two cautions. ECE depends on the binning and is noisy below a few hundred items, so show the reliability diagram, not only the number. And calibration is distribution-specific: Laya's benchmarks note that a lower ECE on one suite does not set a safe threshold for another task or a different number of options.

Decision model, LLM judge, or fine-tuned classifier?

Plain words

Pick the tool by the job

If you need a quick answer from a fixed list and you need it millions of times, use a decision model. If you need someone to read carefully and explain, use a chat model as a judge. If you have thousands of labelled examples of one specific decision that will not change, train your own classifier. Many systems use all three: the fast one for everything, the careful one for the unsure cases.

For leaders

A comparison for design reviews

	Decision model	LLM judge	Fine-tuned classifier
Change the criteria	Edit the question text	Edit the prompt	Relabel and retrain
Labels needed to start	Only for validation	Only for validation	Thousands for training
Output	Typed, with probabilities	Text to parse; probabilities only via logprobs or repeated sampling	Typed, with probabilities
Latency and cost	Low	High, much higher with reasoning	Lowest
Explanations	None	Yes	None
Multi-step reasoning	No	Yes	No
Data control	Hosted, or self-hosted (Laya)	Usually hosted	Fully yours
Best for	High-volume routing, gating, guardrails, many changing decisions	Hard, rare, or contested cases; anything needing a reason	One stable, high-volume decision with plenty of labels

A common architecture is a cascade: the decision model handles every request, acts when its probability is far from the threshold, and escalates the uncertain band (and every "other") to an LLM judge or a person. The width of that band is a direct trade between cost and error, and you can measure it with the selective prediction tools in classification metrics.

Principal asks

What is the cost ratio between a false positive and a false negative, and who signed off on it? What is the AUC and ECE on our own labelled traffic, with intervals, compared with our current LLM judge? Which slice is worst? What fraction goes to the escalation band, and what does a human cost per item there? What happens to our thresholds when the vendor updates the model, and how will we know?

Under the hood

The cascade as a cost calculation

With a calibrated probability p and an escalation that costs c_{esc} per item and is (nearly) always right, the cheapest action for each item is the minimum of three expected costs:

$$\min\left\{\ (1-p)\,c_{\text{FP}}\ \text{(act)},\ \ p\,c_{\text{FN}}\ \text{(do nothing)},\ \ c_{\text{esc}}\ \text{(escalate)}\ \right\}$$

Escalation wins in a band around the threshold whenever c_{esc} is smaller than the expected cost of guessing there. The band narrows as escalation gets dearer and vanishes when $c_{\text{esc}} \geq c_{\text{FP}}c_{\text{FN}}/(c_{\text{FP}}+c_{\text{FN}})$, the expected cost at the threshold itself.

This only works if p is calibrated; an overconfident model pushes items out of the band that should have been escalated. That is the practical reason calibration matters more for decision models than for chat models: the probabilities are the product.

Part III · Chapter 26

Asking for a decision: how framing changes the answer

Almost everything becomes an LLM judge the moment you ask a model for a decision. Grading an answer, routing a ticket, flagging a policy violation, choosing a tool, picking the better of two drafts: each is a measurement made by a model, and the way you phrase the request is part of the measuring instrument. The format you ask for, the words you use for the labels, the order you list the options, the examples you show, and whether the model reasons before or after committing all move the result, sometimes by more than switching to a better model. This chapter collects the levers with the evidence behind each, then turns them into a checklist.

Diagram

Forbid, or not allow?

A famous survey experiment, and what it means for every prompt that asks a model to decide.

1941: half the sample gets each wording

BALLOT A

Should the United States **forbid** public speeches against democracy?

BALLOT B

Should the United States **allow** public speeches against democracy?

Forbidding and not allowing are the same policy.

In a 1941 split-ballot experiment, Donald Rugg asked two random halves of a sample about the same policy. One half was asked whether speeches against democracy should be forbidden; the other, whether they should be allowed.

Same policy, 21 points apart

"Forbid" sounds harsh, so fewer people say yes to it.

Rugg (1941), Public Opinion Quarterly; repeated by Schuman and Presser in the 1970s.

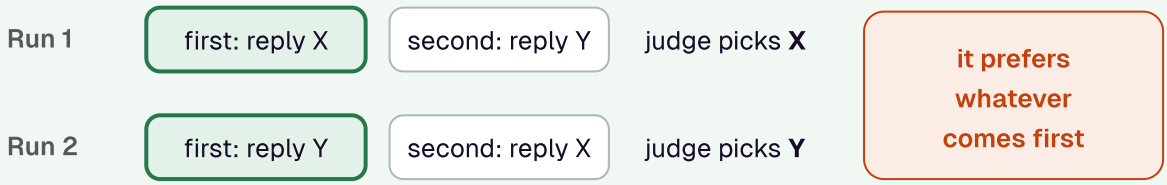
75% would "not allow" the speeches, but only 54% would "forbid" them. The opinion did not change between the two halves; the words did. Survey researchers learned to treat the question wording as part of the measurement.

An LLM judge is a survey respondent too

Same replies, same model. Only the question changed.

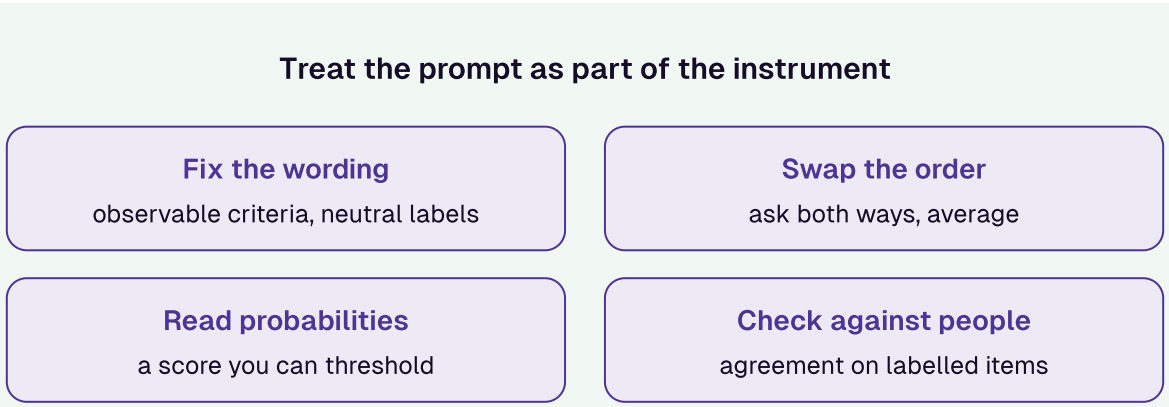
Ask a model whether replies are acceptable, then ask whether they contain a problem, and the pass rate moves even though the replies and the model are identical (numbers illustrative). Leniency in one framing and strictness in the other are properties of the prompt, not of the replies.

Order matters as much as wording



Wang et al. (2023): with ChatGPT as judge, reordering alone let Vicuna-13B beat ChatGPT on 66 of 80 questions.

Show the same two replies in the opposite order and many judges change their pick. Wang et al. (2023) found that, with ChatGPT judging, simply reordering the answers let Vicuna-13B beat ChatGPT on 66 of 80 queries. Position is a framing choice you make whether you mean to or not.



The takeaway: a decision prompt is a survey question put to a model. Pollsters fix the wording, rotate the order, and pre-test on people before trusting the numbers. The rest of this chapter is the model version of that discipline.

Every decision prompt is a measuring instrument

Plain words

The question shapes the answer

A bathroom scale that reads two kilos heavy is still useful if you know it does. A scale that reads differently depending on which foot you step on first is not. A decision prompt is a scale. Some of its quirks are steady (it is a bit lenient), and some depend on details that should not matter (which answer came first, whether you wrote "acceptable" or "problem"). The goal is to remove the quirks you can, and measure the rest.

For leaders

Why this deserves engineering time

Teams spend weeks choosing a judge model and an afternoon writing the judge prompt. The evidence below suggests the prompt deserves at least as much care, because framing effects are often as large as the differences being measured. Three consequences follow:

- **Version the prompt like code.** A wording change is a change of instrument. Results before and after are not comparable until you re-validate.
- **Ask for robustness numbers.** Agreement with humans, plus how often the verdict flips when only the order or the label words change.

- **Prefer formats that give a score.** A probability can be thresholded, calibrated and plotted as an ROC curve; a single word cannot (see LLM as judge).

Under the hood

A simple model of a framed verdict

$$v = \mathbb{1}\left[\, q(x) + \beta_{\text{frame}} + \beta_{\text{order}}\, o + \sigma\,\varepsilon \;>\; 0 \,\right]$$

$q(x)$ is the signal you want (how good the item really is), β_{frame} a leniency shift set by the wording and label words, β_{order} a position effect with $o = \pm 1$ for the two orders, and $\sigma\varepsilon$ judge noise. Good prompting shrinks β and σ, and good harnessing cancels what is left.

Two measurable quantities follow. The **flip rate**, $\Pr[v(o{=}+1) \neq v(o{=}-1)]$, estimates how much of the verdict is position. The **pass-rate gap** against human labels estimates β_{frame}. Neither shows up in raw agreement, which is why a judge can agree with humans 80% of the time and still be badly biased.

Output format: yes/no, scale, pairwise, or probability

Plain words

Four ways to ask

You can ask "is it good, yes or no?", "rate it from 1 to 5", "which of these two is better?", or "how likely is it that this is good, from 0 to 100%?". Each gives a different kind of answer. Comparing two things side by side is usually easier than rating one on its own, just as you can tell which of two cups of coffee is stronger more easily than you can rate one cup on a scale.

For leaders

What each format buys

Format	You get	Strength	Main failure
Yes/no	One label	Simple, easy to aggregate	One fixed operating point; leniency set by wording
Likert scale (1–5, 1–10)	An ordinal score	Some resolution	Score compression: everything lands on 4 of 5 or 7 to 8 of 10
Pairwise (A vs B)	A preference	Easier, more consistent judgments	Position bias; needs pairs, not single items
Probability	A number in [0, 1]	Thresholds, ROC curves, calibration checks	Stated numbers can be miscalibrated; read logprobs or use a decision API

The evidence: Zheng et al. (2023), the MT-Bench and Chatbot Arena paper, found strong judges such as GPT-4 reaching over 80% agreement with human preferences, the same level as human-to-human agreement, while documenting position, verbosity and self-enhancement biases. Liusie et al. (2024) found that for moderate-sized open models (FlanT5, Llama 2 chat), pairwise comparison beat asking for an absolute score, and that pairwise judging showed strong positional bias that needed debiasing. Comparison is easier, and it brings its own bias to manage.

Under the hood

Turning any format into a score

For a scale, read the probability of each score token and take the expectation, as G-Eval (Liu et al., 2023) does: "we use the probability-weighted summation of the output scores as the final score".

$$\text{score} = \sum_{s=1}^{5} s \cdot p(s \mid \text{prompt}), \qquad p_{\text{yes}} = \frac{p(\textbf{yes})}{p(\textbf{yes}) + p(\textbf{no})}$$

Left: an expected score that breaks ties between items the model would all call "4". Right: a renormalised yes probability for a binary verdict. Both need logprob access at the verdict position.

Verbalised probabilities ("I am 70% sure") are a usable alternative when logprobs are unavailable. Tian et al. (2023) found that for RLHF-tuned models (ChatGPT, GPT-4, Claude), confidences written as output tokens were typically better calibrated than the model's own token probabilities, often

cutting expected calibration error by a relative 50%. Either way, calibrate on your labels before treating the number as a probability.

Labels, option order, and position bias

Plain words

First in line has an edge

Models, like people, have favourite positions and favourite words. Some judges prefer the first answer they read, some the second. Some like picking option "A" in a multiple-choice list regardless of what it says. The fix is simple: ask twice with the order swapped, and only trust a win that survives the swap.

For leaders

The evidence and the standard fix

- **Pairwise judges.** Zheng et al. (2023) named position bias as a core limitation of LLM judges. Wang et al. (2023), "Large Language Models are not Fair Evaluators", showed the effect is large enough to reverse rankings, and proposed Balanced Position Calibration: score both orders and combine.
- **Multiple choice.** Chujie Zheng et al. (2023), "Large Language Models Are Not Robust Multiple Choice Selectors", found models prefer specific option IDs such as "A", traced it mainly to token bias on the ID itself, and proposed PriDe, which estimates that prior by permuting options on a few samples and divides it out.
- **Decision models too.** Laya's documentation notes that option order can change its choice answers. One-pass decision models do not escape the problem.

⚠ **Watch out**

Swapping and averaging removes position bias from the result, but it hides how much there was. Always report the flip rate (the share of items whose verdict changes with order). A judge that flips on 30% of pairs is mostly measuring position on those pairs, and averaging turns them into ties, which quietly shrinks your effect sizes.

Under the hood

Permutation averaging

For a pairwise judge returning $p_1 = P(\text{first wins})$ with x first and p_2 with y first,

$$\hat{P}(x \succ y) = \tfrac{1}{2}\big(p_1 + (1 - p_2)\big), \qquad \hat{b} = \tfrac{1}{2}\big(p_1 + p_2\big) - \tfrac{1}{2}$$

$\hat{P}$ cancels a constant first-position shift; $\hat{b}$ estimates that shift (positive means the judge favours the first slot). For K options, average over cyclic permutations so each option appears in each slot once.

Label words carry bias the same way. "Yes" and "no", "A" and "B", "acceptable" and "problem" have different priors before the model reads anything. Neutral, symmetric labels help; measuring and dividing out the prior helps more. The code in the next section does both.

Few-shot examples: which ones, and in what order

Plain words

The examples teach more than you meant

Showing a model a few labelled examples helps it understand the task. It also teaches habits you did not intend. If four of your five examples are labelled "pass", the model leans toward "pass". If the last example is a "fail", it leans toward "fail", because it remembers the most recent one best. Even shuffling the same examples can change the results a lot.

For leaders

Two papers every judge designer should know

- **Zhao et al. (2021), "Calibrate Before Use".** The choice of prompt format, examples, and even their order made GPT-3's few-shot accuracy swing widely. They traced it to biases toward labels that are common in the examples (majority label bias), that appear near the end of the prompt (recency bias), or that are common in pre-training data (common token bias). Their fix, contextual calibration, improved average accuracy by up to 30 points absolute.
- **Lu et al. (2022), "Fantastically Ordered Prompts".** The order of the same few-shot examples can make the difference between near state-of-the-art and random-guess performance. An entropy-based method for picking good orders gave a 13% relative improvement across eleven classification tasks.

Practical rules: balance labels across examples, pick examples from the borderline region (easy examples teach little), fix and version the order, and measure sensitivity by re-running with two or three shuffles.

Under the hood

Contextual calibration and swap averaging, in code

Contextual calibration feeds a content-free input such as "N/A" through the same prompt, reads the label probabilities $\hat{p}_{\text{cf}}$, and rescales so that input would score uniformly:

$$\hat{q} = \text{normalise}\left(W\hat{p}\right), \qquad W = \text{diag}(\hat{p}_{\text{cf}})^{-1}$$

$\hat{p}$ is the label distribution for a real input. Dividing by the content-free distribution removes the prompt's built-in lean toward some labels.

python

```python
# Each pair (x, y) was judged twice. Run 1 showed x as "A"; run 2 showed y as "A".
# The judge returned P("A is better") both times (from logprobs or a decision API).
p_run1 = [0.81, 0.64, 0.58, 0.90, 0.55, 0.70]
p_run2 = [0.35, 0.62, 0.66, 0.12, 0.60, 0.41]

def p_x_better(p1, p2):
    # run 1 says P(x) = p1; run 2 says P(x) = 1 - p2. Average so position cancels out.
    return (p1 + (1 - p2)) / 2

def consistent(p1, p2):
    return (p1 > 0.5) == ((1 - p2) > 0.5)   # same winner in both orders?

print([round(p_x_better(a, b), 3) for a, b in zip(p_run1, p_run2)])
print(sum(consistent(a, b) for a, b in zip(p_run1, p_run2)), "of", len(p_run1), "consistent")
# both runs report P(first-shown wins); above 0.5 on average means a first-position bias
print(round(sum(p_run1 + p_run2) / (2 * len(p_run1)) - 0.5, 3))

# Contextual calibration (Zhao et al. 2021): measure the bias on a content-free input such as "N/A",
# then divide it out so that input would score 50/50.
p_cf = [0.62, 0.38]                       # P("yes"), P("no") when the input is "N/A"
def calibrate(p, p_cf):
    q = [a / b for a, b in zip(p, p_cf)]
    s = sum(q)
    return [v / s for v in q]

print([round(v, 3) for v in calibrate([0.70, 0.30], p_cf)])   # a raw 0.70 "yes" shrinks
```

rust

```rust
// Each pair (x, y) was judged twice. Run 1 showed x as "A"; run 2 showed y as "A".
// The judge returned P("A is better") both times (from logprobs or a decision API).
const P_RUN1: [f64; 6] = [0.81, 0.64, 0.58, 0.90, 0.55, 0.70];
const P_RUN2: [f64; 6] = [0.35, 0.62, 0.66, 0.12, 0.60, 0.41];
```

```rust
fn p_x_better(p1: f64, p2: f64) -> f64 {
    // run 1 says P(x) = p1; run 2 says P(x) = 1 - p2. Average so position cancels out.
    (p1 + (1.0 - p2)) / 2.0
}

fn consistent(p1: f64, p2: f64) -> bool {
    (p1 > 0.5) == ((1.0 - p2) > 0.5) // same winner in both orders?
}

/// Contextual calibration (Zhao et al. 2021): measure the bias on a content-free input such as "N/A",
/// then divide it out so that input would score 50/50.
fn calibrate(p: &[f64], p_cf: &[f64]) -> Vec<f64> {
    let q: Vec<f64> = p.iter().zip(p_cf).map(|(a, b)| a / b).collect();
    let s: f64 = q.iter().sum();
    q.iter().map(|v| v / s).collect()
}

fn r3(x: f64) -> f64 { (x * 1000.0).round() / 1000.0 }

fn main() {
    let avg: Vec<f64> = P_RUN1.iter().zip(P_RUN2).map(|(&a, b)| r3(p_x_better(a, b))).collect();
    println!("{:?}", avg);
    let n_ok = P_RUN1.iter().zip(P_RUN2).filter(|(&a, b)| consistent(a, *b)).count();
    println!("{} of {} consistent", n_ok, P_RUN1.len());
    // both runs report P(first-shown wins); above 0.5 on average means a first-position bias
    let all: f64 = P_RUN1.iter().chain(P_RUN2.iter()).sum();
    println!("{}", r3(all / (2.0 * P_RUN1.len() as f64) - 0.5));

    let p_cf = [0.62, 0.38]; // P("yes"), P("no") when the input is "N/A"
    let cal: Vec<f64> = calibrate(&[0.70, 0.30], &p_cf).iter().map(|&v| r3(v)).collect();
    println!("{:?}", cal); // a raw 0.70 "yes" shrinks
}
```

Half of these six pairs flip with order, and the judge leans 7.8 points toward the first slot. After averaging, two pairs that run 1 called for x (at 0.58 and 0.55) fall below 0.5, and a third (0.64) drops to a coin flip at 0.51.

Reasoning before or after the verdict

Plain words

Decide, then justify, or think, then decide

If you ask someone for a verdict first and their reasons second, the reasons tend to defend whatever they already said. If you ask them to look at the evidence first, the verdict can follow from it. Models work the same way, because they write left to right: anything written after the verdict cannot change it.

For leaders

Evidence first, verdict last

- Chiang and Lee (2023) found that asking the model to explain its rating consistently improved correlation with human ratings, while the automatic chain-of-thought step used in G-Eval did not always help. Explanations help; a generic "think step by step" is not guaranteed to.
- Wang et al. (2023) proposed Multiple Evidence Calibration: have the judge produce evidence before the rating.
- Putting the verdict last also makes it easy to read the verdict token's logprob (see LLM as judge).

The cost is tokens and latency, often several times the cost of a bare verdict. Use evidence-first on hard or borderline items; a reasoning model or a decision model may be a better trade for high volume (see fast and slow and decision models).

Under the hood

Why order of generation matters, and the format trap

The verdict token v is sampled from $p(v \mid x, r_{<})$, conditioned only on what precedes it. Reasoning written after v is a rationalisation by construction. Reasoning written before it changes the conditioning, and so the distribution over verdicts.

Strict output formats can interfere. Tam et al. (2024), "Let Me Speak Freely?", observed a significant decline in reasoning ability under format restrictions such as forced JSON. A common design that gets both: a free-text `evidence` field first, then a constrained `verdict` field, or two calls (reason freely, then extract the verdict with a schema).

Decompose the rubric into separate checks

Plain words

Small questions get steadier answers

"Is this a good answer?" packs many questions into one: is it correct, complete, polite, on topic? Two judges can disagree because they weigh those differently. Asking each as its own yes-or-no question ("Does it state the refund deadline?") gets more consistent answers, and you combine them with a rule you wrote down.

For leaders

Checklists beat holistic scores

- CheckEval (Lee et al., EMNLP 2025) replaced holistic scores with checklists of decomposed binary questions and reported improving average agreement across evaluator models by 0.45, with strong correlation to human judgments across 12 evaluator models.
- FActScore (Min et al., 2023) breaks long-form output into atomic facts and scores the share supported by a knowledge source, with an automated estimator under 2% error. Same idea applied to factuality.
- OpenAI's Decisions API guide gives the same advice for decision models: keep separate concerns in separate questions, write questions around observable criteria, and send dependent decisions as separate requests.

Decomposition also makes disagreements debuggable. When the judge and a human disagree, you can see which check differed.

Under the hood

Aggregation is your decision

With checks $c_1, \ldots, c_m$ returning probabilities p_j, the aggregation rule encodes policy. A strict conjunction assumes rough independence, $P(\text{pass}) \approx \prod_j p_j$; a weighted score $\sum_j w_j p_j$ trades criteria off; a gate ("any safety check below 0.2 fails the item") encodes non-negotiables. Writing the rule in code, not in the prompt, keeps it auditable and lets you change weights without changing the instrument.

Give it a way out: "unsure" and "other"

Plain words

Let it say "I can't tell"

If you force someone to answer yes or no about something they cannot judge, they guess, and their guesses look just like real answers. Adding "unsure" (or "other" when none of the options fit) lets the model flag those cases so a person can look. You answer fewer items automatically, but the ones you do answer are more often right.

For leaders

Coverage versus accuracy

OpenAI's Decisions API guide recommends a fallback option such as "other" when categories do not cover every input, routed to a general review queue. Laya supports a minimum-confidence setting for abstention. Both are selective prediction: trade coverage for accuracy on what is covered (see

classification metrics). Report both numbers. An "unsure" rate that drifts upward is an early warning that inputs have changed.

Under the hood

Abstention as a band around the threshold

With a score s and threshold t, abstaining when $|s - t| < u$ removes the items nearest the boundary, where errors concentrate. Accuracy on covered items rises with u while coverage falls. Defining "unsure" in the prompt asks the model to learn this band itself; reading probabilities lets you set u from costs instead, which is more controllable.

Read probabilities, not words

Plain words

"Yes" hides how sure it was

When a model writes "yes", you do not know whether it was 51% or 99% sure. Many APIs can tell you the probability behind each word. Decision APIs return probabilities directly. With a number instead of a word, you can choose your own cutoff, rank items from most to least likely, and check whether "80% sure" really means right four times out of five.

For leaders

Three ways to get a number

- **Logprobs** on the verdict token, renormalised over the allowed labels. One call, needs API support.
- **Repeated sampling.** Ask five to ten times at a temperature above zero and use the vote share. Works anywhere, costs more.
- **A decision model or API** that returns a distribution by design (see decision models).

Pair any of these with **structured outputs** (a JSON schema with an enum for the verdict) so the answer always parses, keeping a free-text evidence field before the verdict for the reasons above.

Eval lens

When you change any framing lever, re-measure on the same labelled set: agreement or kappa with humans, flip rate under order swap, pass-rate gap, coverage if you allow "unsure", and AUC if you read probabilities. A prompt change that raises agreement while raising the flip rate has traded one bias for another. Keep the labelled set fixed across prompt versions so the comparison is paired.

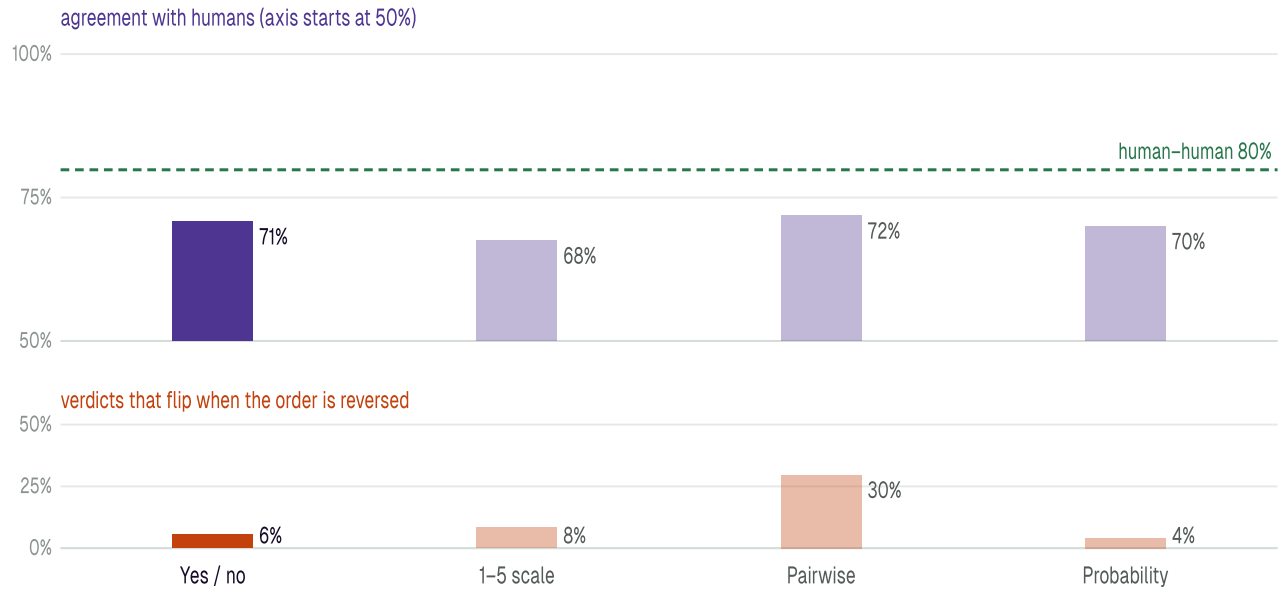
Under the hood

Calibrating what you read

Renormalised logprobs from instruction-tuned models are often overconfident. Fit temperature scaling on held-out labels, $p' = \sigma(\mathrm{logit}(p)/\tau)$, and check a reliability diagram. Vote shares from k samples are coarse (multiples of $1/k$) and noisy, with standard error $\sqrt{p(1-p)/k}$, so ten samples give about ±0.16 at $p = 0.5$. That is enough to rank and threshold, not enough to claim precise probabilities.

Interactive **Framing simulator** (shown at its default setting)

600 simulated items with human labels, judged under different framings. Pick a format and toggle the fixes. Bars show agreement with humans for all four formats under the current toggles; the selected format is solid. Effect sizes are illustrative, chosen to be directionally consistent with the papers above.



Agreement with humans	Verdicts that flip with order	Coverage (not unsure)	Relative cost
71%	**6%**	**100%**	**1.0×**

Yes / no: agreement with humans 71% against a human ceiling of 80%. Order flips 6% of verdicts. Toggle the fixes to see what each buys. This format gives one operating point; read logprobs or switch to probabilities to get a curve. Cost: 1.0× a single bare call, from extra calls and extra tokens.

A decision prompt checklist

1. **Write observable criteria.** "States the refund deadline" beats "is helpful". If two people could disagree about what a criterion means, rewrite it.

2. **Split the rubric.** One question per concern, each yes/no or a short ordered scale with distinct, described levels. Combine with a rule in code.

3. **Choose the format for the decision.** Pairwise for comparing systems; pointwise checks for gating single items; ask for or read a probability whenever you need a threshold.

4. **Use neutral, symmetric labels.** Avoid loaded words ("acceptable" versus "problem"). Keep option descriptions distinct, and add "other" or "unsure" with a clear definition.

5. **Control order.** Swap positions for pairwise and rotate options for multiple choice; average, and report the flip rate.

6. **Curate examples.** Balanced labels, borderline cases, fixed order; check sensitivity with two or three shuffles.

7. **Evidence first, verdict last,** in a structured output with an enum verdict field, so it parses and the verdict's logprob is readable.

8. **Get a number.** Logprobs, vote shares, or a decision API. Calibrate on held-out labels.

9. **Validate on people.** A fixed labelled set, human-to-human agreement as the ceiling, then judge agreement, pass-rate gap, flip rate, coverage and AUC (LLM as judge).

10. **Version everything.** Model, prompt, examples and their order, decoding settings, aggregation rule. Any change means re-validation.

Worked problem

Before. A team grades support replies with this prompt and reads the first word of the output.

text

```
Here are two replies to a customer. Which one is better?

Reply A: {reply_a}
Reply B: {reply_b}

Answer A or B, then explain why.
```

Problems: one holistic judgment; reasons come after the verdict; reply A is always the incumbent system, so position bias favours it every time; no "unsure"; a single word with no probability; nothing tells the judge what "better" means.

After. Pointwise checks with observable criteria, evidence first, an explicit "unsure", structured output. The harness also runs the old pairwise comparison in both orders for system-level win rates.

text

```
You are checking one support reply against three criteria. Judge each criterion on its own.

Customer message:
<message>{message}</message>

Refund policy:
<policy>{policy}</policy>

Reply to check:
<reply>{reply}</reply>

Criteria:
1. correct:  every statement about refunds matches the policy text.
2. resolves: the reply gives the customer a concrete next step and when it will happen.
3. tone:     the reply contains no blame, sarcasm, or promises the policy does not allow.

For each criterion, first quote the sentences from the reply that matter, then give a verdict:
"yes", "no", or "unsure" if the reply does not contain enough information to decide.

Return only this JSON:
{"correct":  {"evidence": "...", "verdict": "yes" | "no" | "unsure"},
 "resolves": {"evidence": "...", "verdict": "yes" | "no" | "unsure"},
 "tone":     {"evidence": "...", "verdict": "yes" | "no" | "unsure"}}
```

In code: read the probability of each verdict token (or send the three criteria as three predicate questions to a decision API), pass an item when all three clear thresholds chosen on labelled data, send any "unsure" to review, and track agreement, flip rate and coverage per criterion.

Principal asks

Show me the judge's agreement with humans and the human-to-human ceiling on the same items. What is the flip rate when you swap order or reword the labels? Is the verdict read as a word or a probability, and is it calibrated? Which criteria does the rubric check separately, and what rule combines them? When did the prompt last change, and was the labelled set re-scored?

Part IV

Training models

How a frontier model is made: pretraining, optimisation, scaling, supervised fine-tuning, LoRA, reward models, RLHF, DPO, GRPO, distillation, inference, and safety training.

Part IV · Chapter 27

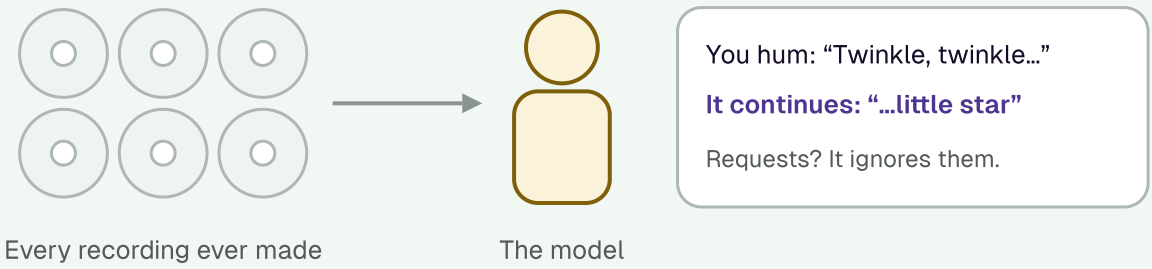
The training pipeline, end to end

A frontier model is not trained once. It is assembled in stages, and each stage changes different things about the model. If you know what each stage buys, you can usually guess where a problem came from before anyone opens a log.

Diagram

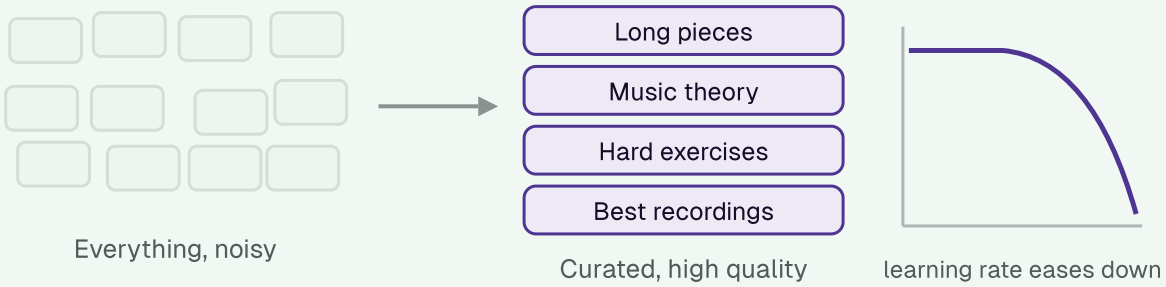
From listener to performer

The seven stages of making a model, told as the training of a musician.



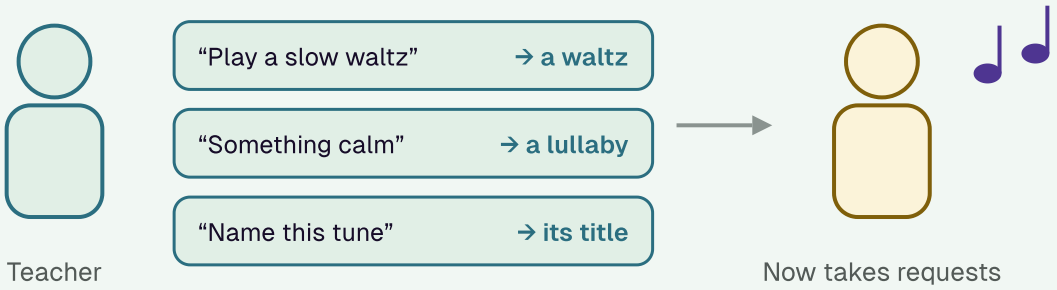
Stage 1 · Pretraining: listen to everything

Pretraining is listening to every recording ever made. The model can continue any tune you start, and it has absorbed an enormous amount about music, but it does not take requests. It just keeps going. This stage uses the large majority of the compute.



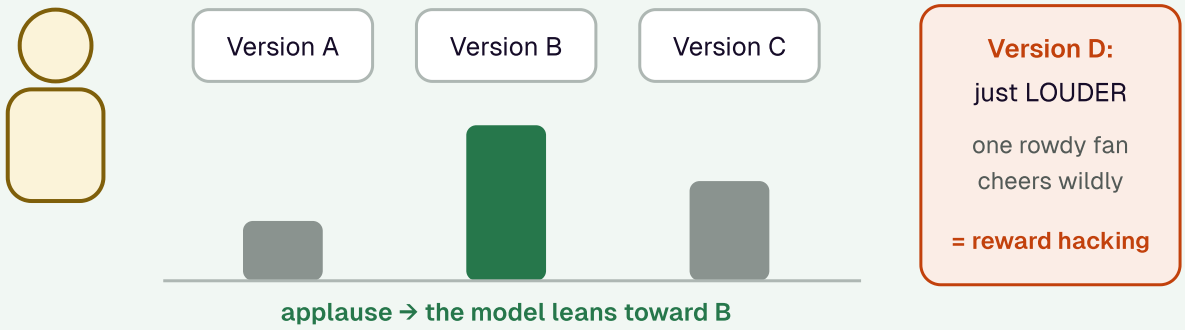
Stage 2 · Mid-training: focused practice on the best material

Mid-training is a focused practice block: the same objective, but on the best material (long pieces, theory, hard exercises), often while the learning rate winds down. Labs use it to add long-context skill and to boost code, math, and quality.



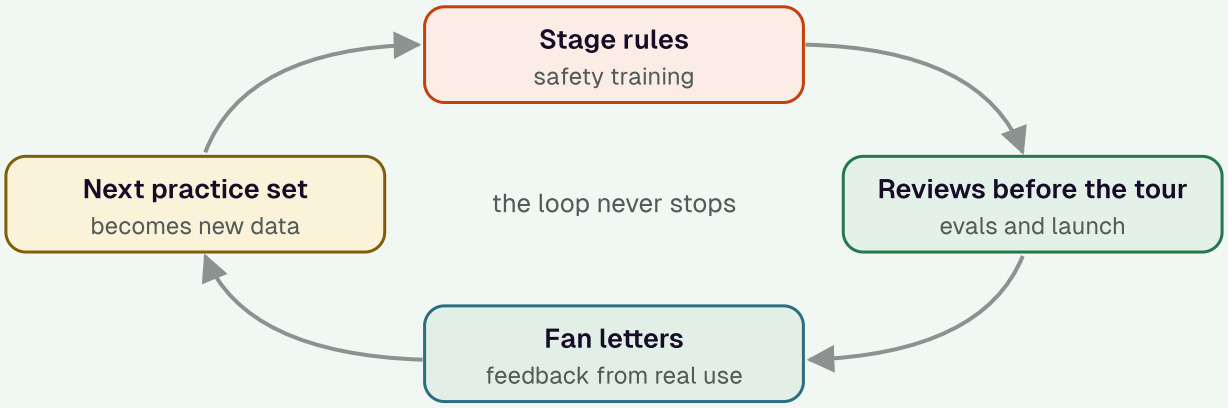
Stage 3 · SFT: lessons from examples of good answers

Supervised fine-tuning is lessons. A teacher shows thousands of examples of a request and a good response. The model learns the format: listen to the request, answer it, stop. It learns very little new music here; it learns how to perform on request.



Stage 4 · Preference tuning and RL: learn from applause

Preference tuning and RL are learning from applause. The model plays several versions, something scores them (people, a reward model, or a checker), and it shifts toward what scored well. The danger is version D: if one rowdy fan cheers for loudness, the model learns to be loud. That is reward hacking.



Then come stage rules (safety training), reviews before the tour (evaluations and launch gates), and fan letters (feedback from real use), which become the next practice set. A frontier model is never finished; it goes around this loop every release.

1 **Data** Crawl, filter, dedup, mix	2 **Pretraining** Next-token prediction on trillions of tokens	3 **Mid-training** Annealing, long context, domain boosts	4 **SFT** Demonstrations of the format and behaviour	5 **Preference and RL** RLHF, DPO, RL on verifiable rewards

6 **Safety**
Refusals, policy, red-team fixes

7 **Evals and launch**
Gates, staged rollout

8 **Feedback**
Logs, ratings, failures become data

Plain words

School, then apprenticeship, then coaching

Pretraining is like reading a whole library. The model learns language, facts, and how ideas connect, but it has no manners and does not know it is supposed to help anyone. It just continues whatever text you hand it.

Supervised fine-tuning (SFT) is an apprenticeship. The model watches thousands of examples of a good assistant answering questions and learns to copy that style.

Preference tuning and reinforcement learning (RL) are coaching. The model tries answers, someone or something scores them, and the model shifts toward the answers that score well. This is where it learns to be helpful, careful, and, for reasoning models, to think through a problem step by step.

Most raw knowledge comes from the library stage. Most of the personality and the polish comes from the later stages.

For leaders

What each stage buys, costs, and breaks

Stage	What it buys	Rough compute share	Typical owner	Classic failure
Data	The ceiling on everything after it	Small in FLOPs, large in people	Data / pretraining team	Contamination, duplicates, a bad mixture weight
Pretraining	Knowledge, language, raw capability	Historically the vast majority (often quoted above 90%)	Pretraining team, infra	Loss spikes, a missing domain, a tokenizer quirk you live with forever
Mid-training	Long context, quality boost, domain depth (code, math)	A few percent	Pretraining team	Forgetting, context extension that passes needle tests but fails real tasks
SFT	Format, instruction following, tool syntax	Well under 1%	Post-training team	Style over substance, hallucinated confidence copied from demos
Preference and RL	Helpfulness, reasoning, agentic skill	Small for RLHF; growing fast for reasoning RL (some 2025 labs report RL compute on the same order as pretraining)	Post-training / RL team	Reward hacking, sycophancy, length inflation
Safety	Policy compliance, refusal boundaries	Small	Safety / alignment team	Over-refusal or under-refusal, jailbreak gaps
Evals and launch	Confidence that it works and is safe	Small but continuous	Evals team, product	Benchmarks that do not predict real use

Compute shares are illustrative. They vary by lab and by year, and the RL share has been rising since reasoning models arrived in late 2024.

Two things matter for planning. First, pretraining decisions are the most expensive to reverse: a tokenizer or a data mixture is baked into a run that may cost tens of millions of dollars. Post-training is cheap enough to iterate weekly. Second, post-training mostly *elicits* capability rather than creating it. If the base model cannot do something even with good prompting and many samples, no amount of SFT will make it reliable. RL on verifiable tasks is the partial exception, which is why it gets so much attention (see GRPO and RL with verifiable rewards).

Principal asks

- Is this a capability problem (pretraining) or a behaviour problem (post-training)? What evidence separates the two?
- Does the base model show the skill at pass@k? If not, why do we expect fine-tuning to create it?
- Which eval would have caught this one stage earlier, where it is cheaper to fix?
- What did this stage cost in compute and in people, and what is the next most valuable use of that budget?

Under the hood

Objectives by stage

Each stage optimizes a different objective, which is why each stage fails differently.

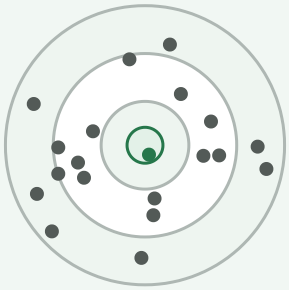
Stage	Objective	Data
Pretraining	$\min_\theta -\sum_t \log p_\theta(x_t \mid x_{<t})$ over all tokens	Trillions of web, code, book, and synthetic tokens
Mid-training	Same loss, different data: high-quality and long documents, often with the learning rate decaying to zero	Billions to hundreds of billions of curated tokens
SFT	Same loss, but only on response tokens: $-\sum_{t \in \text{response}} \log p_\theta(y_t \mid x, y_{<t})$	Thousands to millions of prompt and response pairs
RLHF / RL	$\max_\theta \mathbb{E}_{y \sim \pi_\theta}[r(x, y)] - \beta \mathrm{KL}(\pi_\theta \,\|\, \pi_{\text{ref}})$	Prompts plus a reward model or a verifier
DPO and relatives	A classification loss on preferred versus rejected responses	Preference pairs

The KL term in the RL objective keeps the policy close to the SFT model. Without it, the policy drifts toward whatever the reward model mistakenly loves. Chapters SFT, reward models, RLHF, and DPO unpack each line.

Diagram

Is the skill missing, or just unreliable?

One measurement tells you which stage to fix: how often the base model gets it right in many tries.



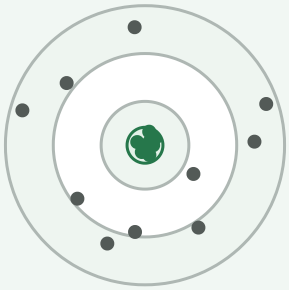
Base model, 20 throws

1 bullseye

pass@1 ≈ 5% · pass@20: a hit

The skill is there, just rare.

Give the base model twenty tries at a task. If it gets it right even once, the skill exists somewhere in the model. pass@k, the chance at least one of k tries succeeds, measures this.



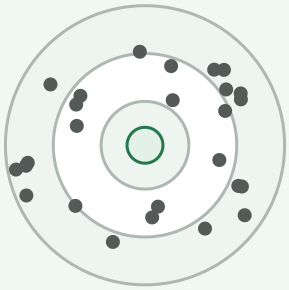
After coaching (SFT, RL)

10 bullseyes in 20

pass@1 ≈ 50%

Same skill, now reliable.

Post-training mostly turns a rare success into a reliable one. pass@1 jumps; pass@k at large k barely moves. That is elicitation: drawing out a skill the base model already had.



A skill it never shows

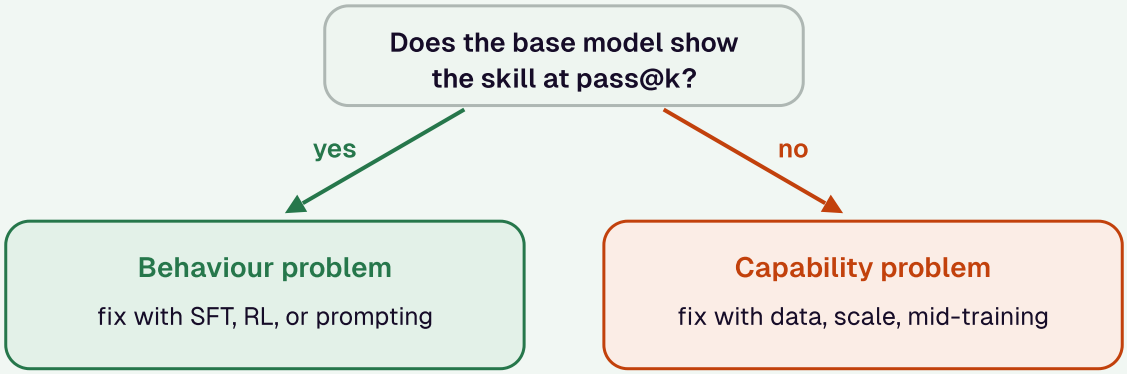
0 bullseyes in 1,000

Coaching rarely creates this.

Fix it earlier: data, scale, mid-training.

RL on checkable tasks can stretch it a little.

If the base model never succeeds, even with many tries and good prompts, fine-tuning is unlikely to fix it. The problem lives in pretraining: missing data, too little scale. RL on tasks with automatic checkers is the partial exception.



So before choosing where to spend effort, measure the base model at pass@k. That one number usually tells you which stage to fix and which team owns the problem.

Where evals sit

Evaluation is not a gate at the end. Every stage has its own checks, and the cheapest bug is the one caught at the stage that caused it.

Stage	What you measure	Typical tools
Data	Quality, duplication, contamination against eval sets, mixture proportions	n-gram overlap checks, classifier scores, sampled human review
Pretraining	Held-out loss by domain, few-shot benchmarks on checkpoints, loss spikes	Loss dashboards, small "proxy" eval suites run every few thousand steps
Mid-training	Long-context retrieval and reasoning, regression on general benchmarks	Needle tests plus real long-document tasks
SFT	Instruction following, format validity, refusal rate	Rule-based checks, LLM judges
Preference and RL	Win rate, reward versus KL, length, verifier pass rate, held-out tasks	Judges, verifiers, human preference studies
Safety	Harmful compliance, over-refusal, jailbreak robustness	Red-team suites, policy classifiers
Launch	Online metrics against offline predictions	A/B tests, staged rollout

Eval lens

When a metric moves, first ask which stage changed. A base-model benchmark that drops after RL is a forgetting problem, not a pretraining problem. A judge win rate that rises with response length is probably a reward or judge bias, not a capability gain (see LLM as judge).

Interactive **Where did this failure come from?** (shown at its default setting)

Pick a symptom. The stages most likely responsible light up, with the first thing to check.

1 Data	2 **Pretraining** Most likely	3 Mid-training	4 SFT
5 Preference / RL	6 Safety	7 Evals / launch	

Why: Tokenization. The model sees subword tokens, not letters, so character-level tasks are hard. Fixed at pretraining time, not later.
First check: Tokenize the failing words and look at the pieces. Test whether spacing the letters out fixes it.

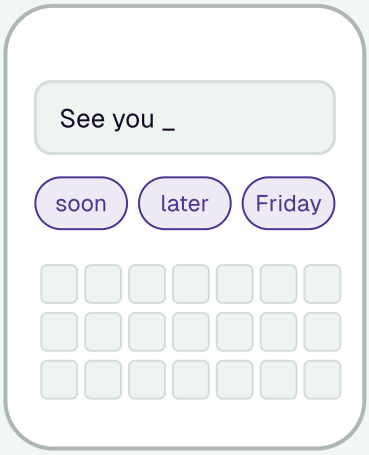
Part IV · Chapter 28

Pretraining: learning from the whole library

Pretraining is where almost all of a model's knowledge and raw capability comes from. The task is simple to state: given some text, predict the next piece. Doing it well across trillions of tokens forces the model to learn grammar, facts, code, and a surprising amount of reasoning.

Diagram **The world’s biggest autocomplete game**

What a model actually does during pretraining, in five pictures.

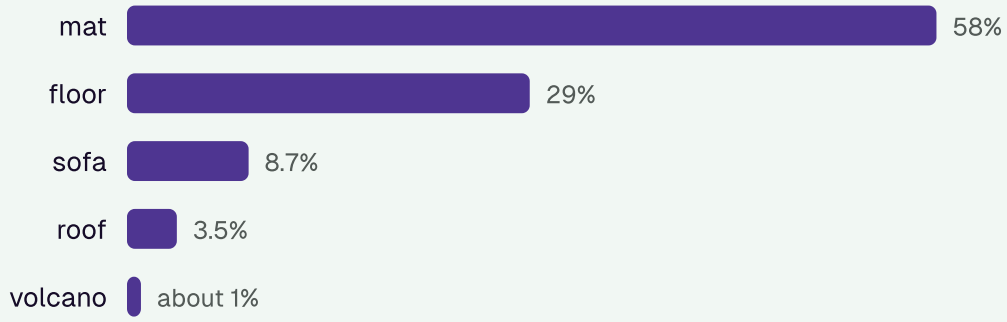


Your phone guesses the next word.

A language model plays the same game, trillions of times, on text from books, code, and the web.

Your phone suggests the next word as you type. Pretraining is the same game at enormous scale: read some text, guess what comes next, check, repeat.

The cat sat on the ___

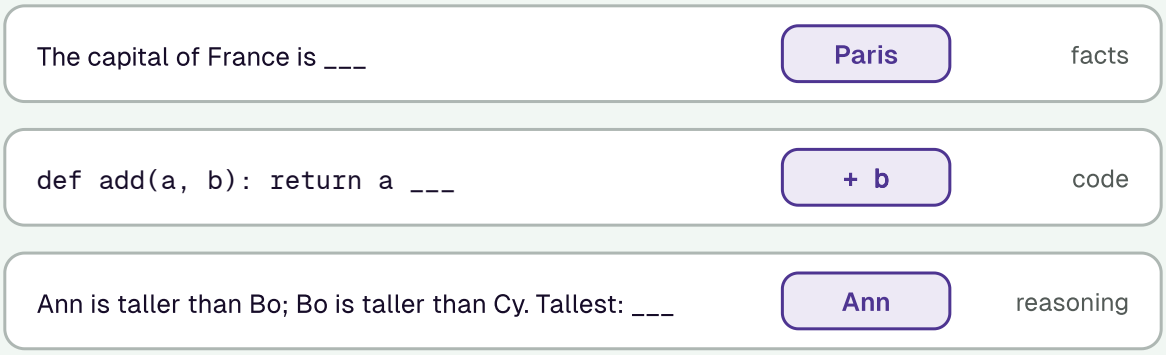


Probabilities from the logits in the code below (illustrative).

The model does not pick one word. It spreads its bets across every token it knows: 58% on “mat”, 29% on “floor”, about 1% on “volcano”.

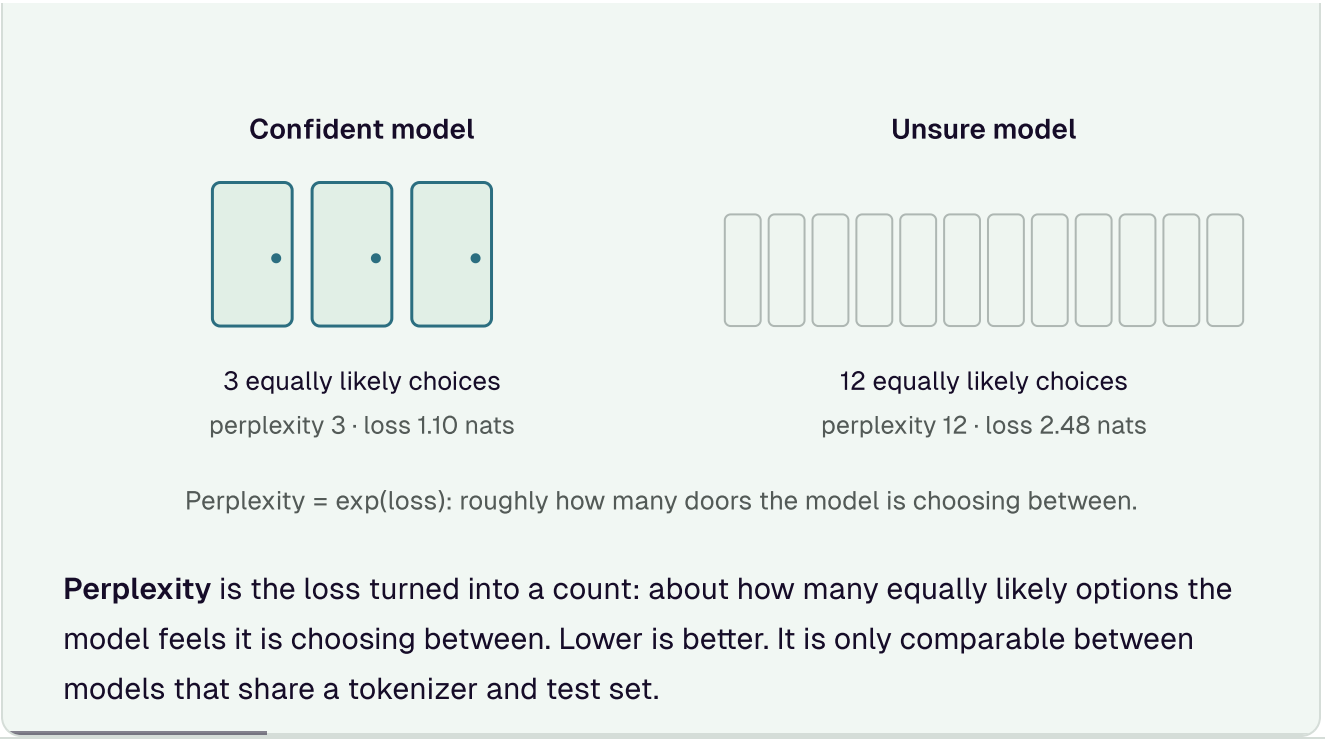
Training nudges billions of numbers so the next bet is a little better.

Then the real next word is revealed. The **loss** is minus the log of the bet on the right answer: 0.54 for a confident correct bet, 4.64 when it bet almost nothing on the right word. Training lowers the average loss.



To bet well on trillions of blanks, the model has to learn what the text is about.

Why does a guessing game produce something smart? Because betting well on every blank requires knowing facts, following code, and tracking logic. The objective is simple; what it takes to do well is not.



Perplexity is the loss turned into a count: about how many equally likely options the model feels it is choosing between. Lower is better. It is only comparable between models that share a tokenizer and test set.

Next-token prediction and its loss

Plain words

A guessing game with a scoreboard

Cover the next word of a sentence and guess it. "The cat sat on the ___." You would bet on "mat" or "floor", not "volcano". The model plays this game billions of times. Each time, it spreads its bets across every possible next piece of text, then the real answer is revealed.

The score is called **loss**. If the model put a big bet on the right answer, the loss is small. If it put almost nothing on the right answer, the loss is large. Training nudges the model, a tiny bit at a time, to bet better.

Perplexity turns that score into something intuitive: it is roughly how many options the model feels it is choosing between. A perplexity of 3 means it is about as unsure as picking among three equally likely words.

For leaders

Why the whole field watches one number

Pretraining loss is the single most reliable signal in model development. It falls smoothly and predictably as you add compute, data, and parameters, which is what makes scaling laws (see Scaling laws) possible. Teams decide whether a run is healthy, whether an architecture change helped, and when to stop, mostly by looking at held-out loss.

Three cautions keep that number honest:

- **Loss is only comparable with the same tokenizer and the same held-out data.** Two models with different vocabularies produce losses in different units. Use bits per byte when comparing across tokenizers.
- **Average loss hides the tokens you care about.** Most tokens are easy (the second half of a word, a closing bracket). A model can lower average loss while not improving on the hard tokens that carry a reasoning step.
- **Lower loss does not guarantee better behaviour.** It predicts capability well on average and poorly for any single task. That is why every lab pairs loss curves with downstream eval suites run on checkpoints.

Under the hood

Softmax, cross-entropy, perplexity, bits per byte

The network outputs a vector of logits $z \in \mathbb{R}^V$, one per vocabulary item. Softmax with temperature τ turns them into probabilities:

$$p_i = \frac{\exp(z_i/\tau)}{\sum_{j=1}^{V} \exp(z_j/\tau)}$$

Training uses $\tau = 1$. Temperature is a sampling knob at inference time; it does not change what the model learned.

The per-token loss is the negative log probability of the token that actually came next. Averaged over a sequence of T tokens:

$$\mathcal{L} = -\frac{1}{T}\sum_{t=1}^{T} \log p_\theta(x_t \mid x_{<t}), \qquad \text{PPL} = e^{\mathcal{L}}$$

$\mathcal{L}$ in nats when using natural log. Divide by $\ln 2$ for bits.

This is cross-entropy between the data distribution and the model. It decomposes as $H(p_{\text{data}}) + \text{KL}(p_{\text{data}} \,\|\, p_\theta)$: the irreducible entropy of language plus how far the model is from the truth. Only the KL part can be trained away, which is why loss curves flatten toward a floor rather than zero.

Bits per byte removes the tokenizer from the comparison: total bits over a document divided by its length in UTF-8 bytes.

$$\mathrm{BPB} = \frac{T \cdot \mathcal{L}_{\text{nats}}}{B \cdot \ln 2}$$

T tokens, B bytes in the same text. A tokenizer with more bytes per token has fewer, harder tokens; BPB corrects for that.

python

```python
import numpy as np

def log_softmax(z):
    z = z - z.max()                      # subtract max for numerical stability
    return z - np.log(np.exp(z).sum())

def token_loss(logits, target):
    return -log_softmax(logits)[target]  # nats

logits = np.array([3.1, 2.4, 1.2, 0.3, -1.0])   # mat, floor, sofa, roof, volcano
for name, idx in [("mat", 0), ("floor", 1), ("volcano", 4)]:
    print(name, round(float(token_loss(logits, idx)), 3))
# mat 0.544   floor 1.244   volcano 4.644

# bits per byte for a document: 1,000 tokens, 4,300 bytes, mean loss 2.1 nats
print(round(1000 * 2.1 / (4300 * np.log(2)), 3))   # 0.705 bits per byte
```

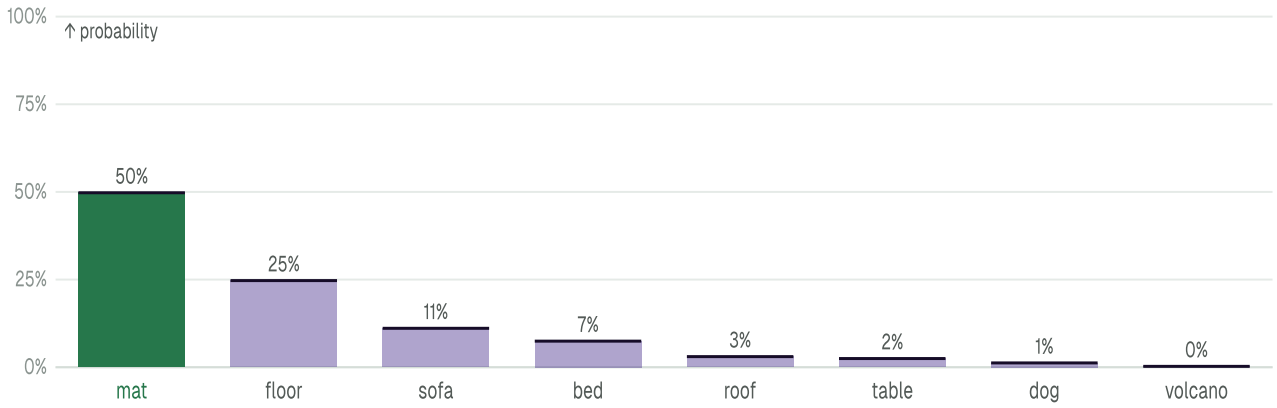
rust

```rust
fn log_softmax(z: &[f64]) -> Vec<f64> {
    let max = z.iter().cloned().fold(f64::NEG_INFINITY, f64::max); // subtract max for numerical stability
    let log_sum = z.iter().map(|x| (x - max).exp()).sum::<f64>().ln();
    z.iter().map(|x| x - max - log_sum).collect()
}

fn token_loss(logits: &[f64], target: usize) -> f64 {
    -log_softmax(logits)[target] // nats
}

fn main() {
    let logits = [3.1, 2.4, 1.2, 0.3, -1.0]; // mat, floor, sofa, roof, volcano
    for (name, idx) in [("mat", 0), ("floor", 1), ("volcano", 4)] {
        println!("{} {:.3}", name, token_loss(&logits, idx));
    }
    // mat 0.544   floor 1.244   volcano 4.644

    // bits per byte for a document: 1,000 tokens, 4,300 bytes, mean loss 2.1 nats
    println!("{:.3}", 1000.0 * 2.1 / (4300.0 * 2f64.ln())); // 0.705 bits per byte
}
```

Interactive

Next-token bets and the loss they earn (shown at its default setting)

The model's raw scores (logits) for the next token after "The cat sat on the". Change the temperature, pick which token was actually next, and watch the loss.



p(correct token)	Loss (nats)	Loss (bits)	Perplexity if every token were like this
49.8%	0.698	1.007	2.01

Entropy of the bet (bits)
2.01

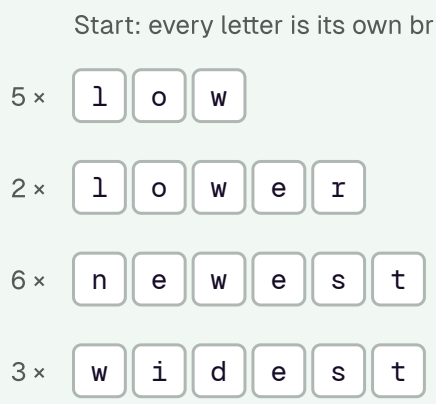
Black ticks show the bet at temperature 1, the setting used in training. Temperature 1 is the distribution the model actually learned. If "volcano" were the true token, the loss would be 6.00 nats at this temperature.

Tokenization: the model never sees letters

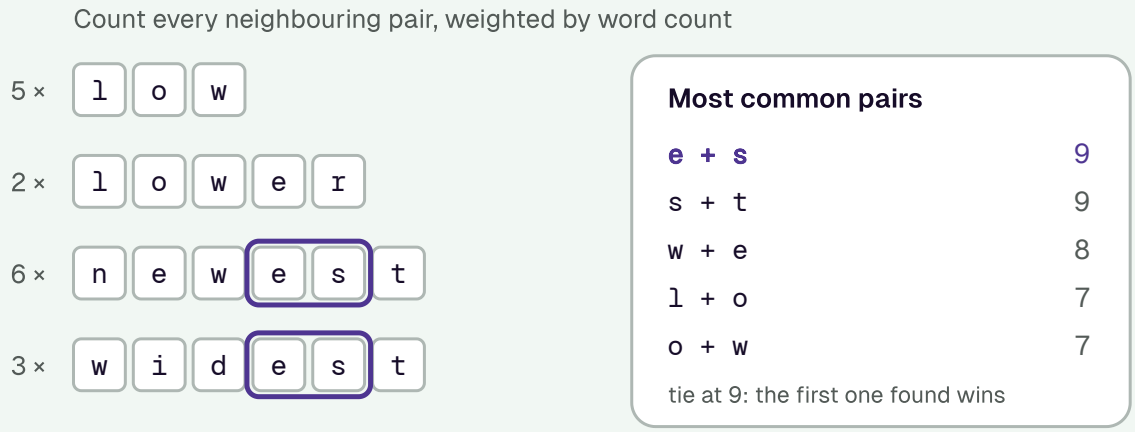
Diagram

Snapping letters into bricks

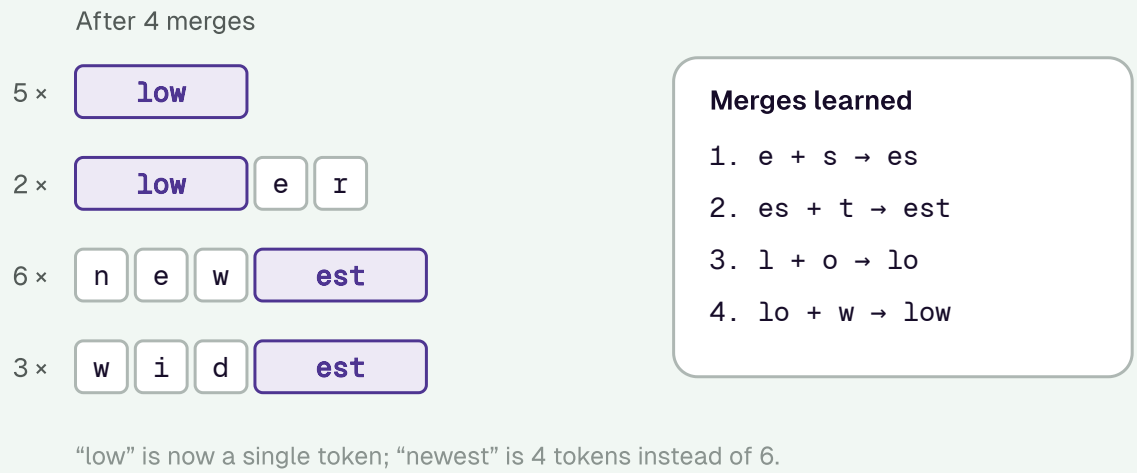
How byte-pair encoding builds a vocabulary, on a corpus of four words.



Byte-pair encoding starts with every character as a separate brick. Here is a tiny corpus of four words with their counts.



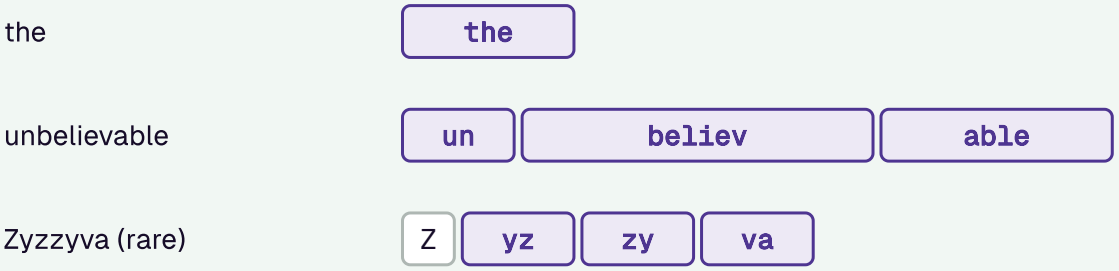
“e + s” appears 9 times (6 in “newest”, 3 in “widest”). It ties with “s + t”; the first one found wins. That pair gets snapped together into a new brick.



“low” is now a single token; “newest” is 4 tokens instead of 6.

Repeat: merge the most common pair, recount, merge again. After four merges “low” is one brick and the ending “est” is shared by “newest” and “widest”. The list of merges is the tokenizer.

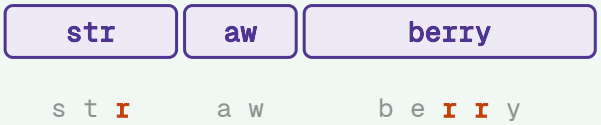
A real tokenizer repeats this 100,000+ times on terabytes of text



Common words become one token; rare words stay in pieces. Splits shown are illustrative.

At real scale, common words like “the” become single tokens, long words split into familiar parts, and rare names stay as several small pieces. Vocabularies today run from about 50,000 to 200,000 tokens.

How many r’s are in “strawberry”?



The letters are hidden inside the bricks.

The model sees 3 tokens, not 10 letters.

Illustrative split. Spelling and letter counting are tokenization artifacts, not reasoning.

This is why a model can write an essay and still miscount the r’s in “strawberry”. The word arrives as a few bricks, and the letters inside were never directly in view.

Plain words

Chopping text into reusable pieces

Models do not read letters or whole words. They read **tokens**: chunks of text chosen so that common pieces get their own symbol. "unbelievable" might become "un", "believ", "able". A common word like "the" is one token. A rare name might be five.

This is why models can write an essay but stumble when asked how many r's are in "strawberry". The word arrives as a couple of chunks, so the individual letters were never directly in view.

For leaders

A decision you live with for the life of the model

The tokenizer is fixed before pretraining and is very expensive to change later, because every embedding was learned against it. Choices that matter:

- **Vocabulary size.** GPT-2 used 50,257 tokens; Llama 3 moved to about 128,000; OpenAI's o200k tokenizer (GPT-4o) has about 200,000. Larger vocabularies compress text into fewer tokens, which means cheaper inference and more text per context window, especially for non-English languages and code. The cost is a bigger embedding table and rarer tokens that get fewer training updates.
- **Fairness across languages.** A tokenizer trained mostly on English can need several times more tokens for the same sentence in Hindi or Amharic. Users in those languages pay more per request and get less effective context. This is a product and policy question, not just an engineering one.

- **Numbers and code.** How digits are split affects arithmetic. Many modern tokenizers split numbers into single digits or groups of up to three so that place value is consistent. Whitespace handling matters a lot for code.

> **Eval lens**
>
> Character-level failures (spelling, counting letters, reversing strings) are mostly tokenization artifacts, not reasoning failures. Do not let them drive training decisions about reasoning. When comparing models with different tokenizers, report cost and context in characters or bytes, not tokens.

Under the hood

The BPE training loop

Byte-pair encoding (Sennrich et al., 2016, adapted from a 1994 compression method) is greedy: count every adjacent symbol pair weighted by word frequency, merge the most frequent pair everywhere, repeat until the vocabulary hits its target size. Modern tokenizers usually run BPE on bytes rather than characters so that any input string can be encoded, and apply a regex pre-split so merges never cross word or digit-group boundaries.

python

```python
from collections import Counter

def bpe(word_counts, n_merges):
    vocab = {tuple(w): c for w, c in word_counts.items()}     # words as tuples of symbols
    merges = []
    for _ in range(n_merges):
        pairs = Counter()
        for syms, c in vocab.items():
            for a, b in zip(syms, syms[1:]):
                pairs[(a, b)] += c                             # weight by word frequency
        if not pairs:
            break
        best = max(pairs, key=pairs.get)
        merges.append(best)
        new = {}
        for syms, c in vocab.items():
            out, i = [], 0
            while i < len(syms):
                if i + 1 < len(syms) and (syms[i], syms[i + 1]) == best:
                    out.append(syms[i] + syms[i + 1]); i += 2
                else:
                    out.append(syms[i]); i += 1
            new[tuple(out)] = c
        vocab = new
    return merges, vocab
```

```
merges, vocab = bpe({"low": 5, "lower": 2, "newest": 6, "widest": 3}, 4)
print(merges)  # [('e', 's'), ('es', 't'), ('l', 'o'), ('lo', 'w')]
```

rust

```rust
use std::collections::HashMap;

type Pair = (String, String);

fn bpe(word_counts: &[(&str, u64)], n_merges: usize) -> (Vec<Pair>, Vec<(Vec<String>, u64)>) {
    // words as sequences of symbols
    let mut vocab: Vec<(Vec<String>, u64)> = word_counts
        .iter()
        .map(|(w, c)| (w.chars().map(String::from).collect(), *c))
        .collect();
    let mut merges = Vec::new();
    for _ in 0..n_merges {
        // Counts kept in first-seen order, like Python's Counter, so a tie goes
        // to the pair seen first: ('e','s') beats ('s','t') at 9 each.
        let mut pairs: Vec<(Pair, u64)> = Vec::new();
        let mut index: HashMap<Pair, usize> = HashMap::new();
        for (syms, c) in &vocab {
            for w in syms.windows(2) {
                let pair = (w[0].clone(), w[1].clone());
                let i = *index.entry(pair.clone()).or_insert_with(|| {
                    pairs.push((pair, 0));
                    pairs.len() - 1
                });
                pairs[i].1 += c; // weight by word frequency
            }
        }
        if pairs.is_empty() {
            break;
        }
        // strict > keeps the FIRST maximum, as Python's max() does
        let mut best = &pairs[0];
        for p in &pairs[1..] {
            if p.1 > best.1 {
                best = p;
            }
        }
        let best = best.0.clone();
        let mut new = Vec::with_capacity(vocab.len());
        for (syms, c) in &vocab {
            let (mut out, mut i) = (Vec::new(), 0);
            while i < syms.len() {
                if i + 1 < syms.len() && syms[i] == best.0 && syms[i + 1] == best.1 {
                    out.push(format!("{}{}", syms[i], syms[i + 1]));
                    i += 2;
                } else {
                    out.push(syms[i].clone());
                    i += 1;
                }
            }
            new.push((out, *c));
        }
        merges.push(best);
        vocab = new;
    }
    (merges, vocab)
```

```
}

fn main() {
    let (merges, _vocab) = bpe(&[("low", 5), ("lower", 2), ("newest", 6), ("widest", 3)], 4);
    let shown: Vec<String> = merges.iter().map(|(a, b)| format!("('{a}', '{b}')")).collect();
    println!("[{}]", shown.join(", ")); // [('e', 's'), ('es', 't'), ('l', 'o'), ('lo', 'w')]
}
```

Interactive

Byte-pair encoding, one merge at a time

(shown at its default setting)

BPE starts from single characters and repeatedly merges the most frequent adjacent pair. Step through the merges on a tiny corpus and watch common words collapse into single tokens.

Corpus, as tokens (word × count)

l o w · × 5
l o we r· × 2
l o we s t · × 2
ne we r· × 6
ne we s t · × 4
w i d e r· × 3
w i d e s t · × 2
ne w · × 3

Merges learned, in order

1. w + e → we (14)
2. n + e → ne (13)
3. r + · → r· (11)

3 merges. Average tokens per word: **4.30** (started at 5.70). Symbols in use: 12. The "·" marks end of word, so "est·" can only match at the end. The most frequent adjacent pair always merges next, so common fragments and word endings (here "we", "ne", "r·", "st·") become tokens before whole words like "newer·" and "low·".

Data: the ceiling on everything

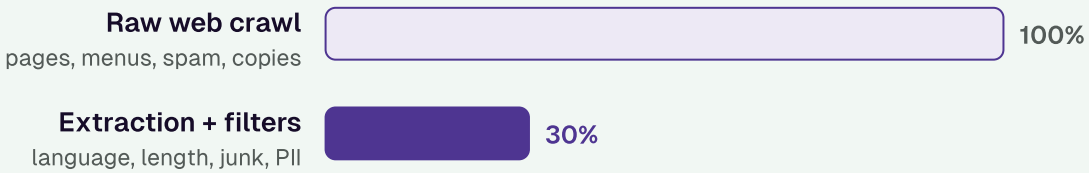
Diagram

From the whole web to a training diet

Where pretraining data comes from, and how much survives each filter.

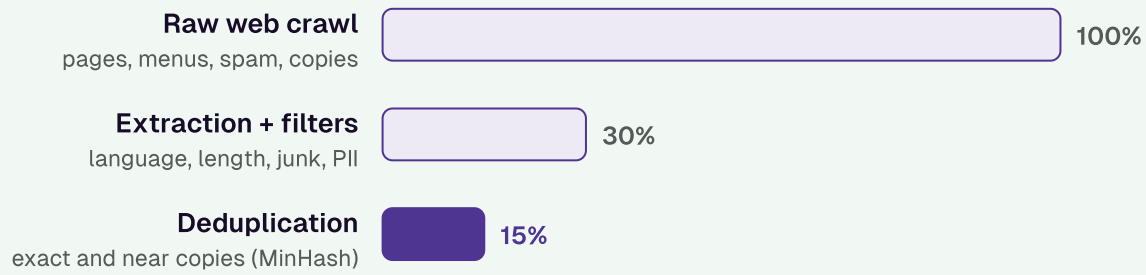
Percentages are illustrative; real pipelines vary by lab.

Start with a raw crawl of the web. Most of it is navigation menus, spam, machine-written filler, and the same pages copied thousands of times.



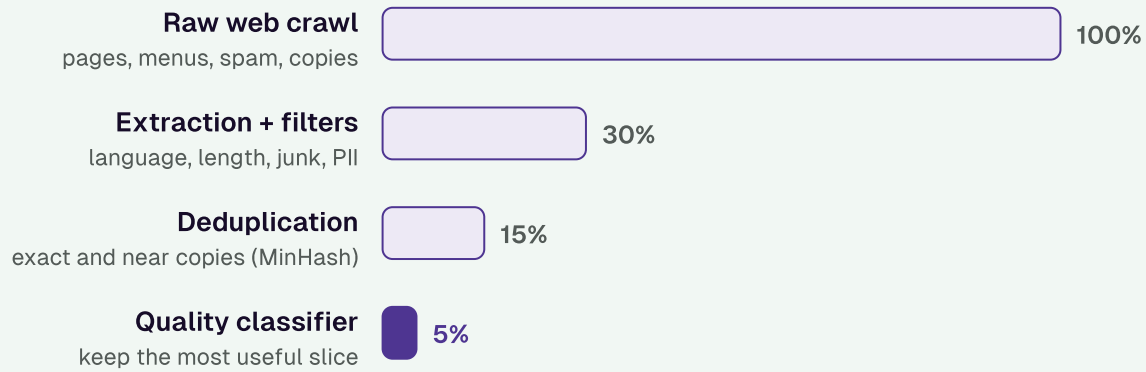
Percentages are illustrative; real pipelines vary by lab.

Extract the text, keep the target languages, drop pages that are too short or full of symbols, and strip personal information. Most of the volume goes here.



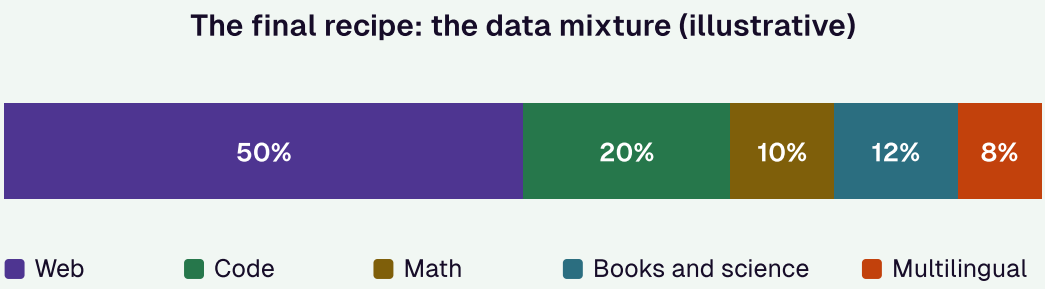
Percentages are illustrative; real pipelines vary by lab.

Remove exact and near duplicates. Repeated pages waste compute and make the model memorise instead of generalise (Lee et al., 2022).



Percentages are illustrative; real pipelines vary by lab.

A small classifier scores every document for educational value and keeps only the top slice. This one choice often moves benchmarks more than an architecture change.



Code and math are usually up-weighted: they seem to help reasoning broadly.

Last step: remove anything that overlaps the evaluation benchmarks.

Finally, decide the recipe: how many tokens of web, code, math, books, and other languages. Teams tune these weights on small models first, then decontaminate against eval sets so later scores mean something.

Plain words

You are what you read

A model trained on junk learns junk. Most of the internet is spam, boilerplate, duplicated pages, and machine-generated filler. Pretraining teams spend enormous effort throwing most of it away and deciding how much of each kind of text to keep: web pages, books, code, math, science, and many languages. That recipe is called the **data mixture**, and it is one of the most closely guarded secrets at any lab.

For leaders

Where data quality comes from

1. **Extraction and filtering.** Pull clean text out of HTML, drop pages by language, length, and heuristics (too many symbols, too few sentences), and remove personal information and unsafe content.
2. **Deduplication.** Exact and near-duplicate removal, typically with MinHash over n-grams. Lee et al. (2022) showed deduplication reduces memorization and improves models at the same compute.
3. **Quality classifiers.** A small model scores each document for "educational value" or similarity to trusted sources, and only the top slice is kept. FineWeb-Edu (Penedo et al., 2024) used labels from a large LLM to train such a classifier; DCLM (Li et al., 2024) used a fastText

classifier and showed filtering choices move benchmark scores more than many architecture changes.

4. **Mixture weights.** How many tokens of code, math, multilingual, and web text. Code and math are usually upweighted because they appear to improve reasoning broadly. Weights are tuned with small proxy models before the big run.

5. **Decontamination.** Remove documents that overlap with evaluation benchmarks so later scores mean something.

Running out of data. Repeating data is not free but is not useless either. Muennighoff et al. (2023) found that up to about four epochs of repeated data are nearly as good as fresh data, with returns falling toward zero by around sixteen epochs. High-quality text is finite, which is one reason labs invest heavily in synthetic data (see Distillation and synthetic data).

Principal asks

- How was this mixture chosen, and at what scale were the ablations run? Do the conclusions hold at the target scale?
- What is our decontamination method, and what overlap threshold did we use against each eval we report?
- How many epochs is our highest-quality data seeing? Are we in the regime where repetition starts to hurt?
- What fraction of the mixture is synthetic, and what model generated it?

Under the hood

Near-duplicate detection with MinHash

Represent each document as a set of word n-grams (shingles). The Jaccard similarity of two sets is $J(A, B) = |A \cap B| / |A \cup B|$. MinHash estimates it cheaply: for a random hash function h, $\Pr[\min h(A) = \min h(B)] = J(A, B)$. With k hash functions, the fraction of matching minimums estimates J. Locality-sensitive hashing then groups signatures into bands so that only likely duplicates are compared.

python

```python
import hashlib

def shingles(text, n=5):
    w = text.lower().split()
    return {" ".join(w[i:i + n]) for i in range(len(w) - n + 1)}

def minhash(s, k=64):
```

```
    return [min(int(hashlib.md5(f"{i}:{x}".encode()).hexdigest(), 16) for x in s) for i in range(k)]

def est_jaccard(a, b):
    return sum(x == y for x, y in zip(a, b)) / len(a)

d1 = "the quick brown fox jumps over the lazy dog near the river bank today"
d2 = "the quick brown fox jumps over the lazy dog near the river bank tonight"
print(round(est_jaccard(minhash(shingles(d1)), minhash(shingles(d2))), 2))  # about 0.8: near duplicate
```

rust

```
use std::collections::HashSet;

fn shingles(text: &str, n: usize) -> HashSet<String> {
    let w: Vec<String> = text.to_lowercase().split_whitespace().map(String::from).collect();
    if w.len() < n {
        return HashSet::new();
    }
    (0..=w.len() - n).map(|i| w[i..i + n].join(" ")).collect()
}

/// 64-bit FNV-1a plus a SplitMix64 finalizer so nearby inputs spread out.
/// It stands in for md5 so the sketch needs no crates; real code would use a
/// proper hash from a crate. Different hash, so the estimate differs slightly.
fn hash64(s: &str) -> u64 {
    let mut h: u64 = 0xcbf29ce484222325;
    for b in s.bytes() {
        h ^= b as u64;
        h = h.wrapping_mul(0x100000001b3);
    }
    h = (h ^ (h >> 30)).wrapping_mul(0xBF58_476D_1CE4_E5B9);
    h = (h ^ (h >> 27)).wrapping_mul(0x94D0_49BB_1331_11EB);
    h ^ (h >> 31)
}

fn minhash(s: &HashSet<String>, k: usize) -> Vec<u64> {
    (0..k).map(|i| s.iter().map(|x| hash64(&format!("{i}:{x}"))).min().unwrap_or(u64::MAX)).collect()
}

fn est_jaccard(a: &[u64], b: &[u64]) -> f64 {
    a.iter().zip(b).filter(|(x, y)| x == y).count() as f64 / a.len() as f64
}

fn main() {
    let d1 = "the quick brown fox jumps over the lazy dog near the river bank today";
    let d2 = "the quick brown fox jumps over the lazy dog near the river bank tonight";
    let est = est_jaccard(&minhash(&shingles(d1, 5), 64), &minhash(&shingles(d2, 5), 64));
    println!("{:.2}", est); // 0.86 here (md5 in the Python gives 0.83; true Jaccard 9/11 = 0.82): near duplicate
}
```

What loss predicts, and what it does not

For leaders

Reading a loss curve in a review

Smooth decline

Healthy. Compare the slope to the scaling-law prediction for this compute. A curve above prediction usually means a data or architecture problem.

Sudden spike

Optimisation instability. Usually recovers; if not, teams roll back to a checkpoint and skip the offending batches (see Optimisation).

Drop at the end

Learning-rate decay or annealing on high-quality data. Expected, and it can flatter a run that was mediocre before the decay.

Train falls, held-out flat

Memorization or repeated data. Check epochs on the high-quality slices and contamination.

Eval lens

Held-out loss predicts average downstream ability well, which is how the GPT-4 report (OpenAI, 2023) and the Llama 3 report (Meta, 2024) forecast final performance from small runs. For a single benchmark the link is noisy and can look like a sudden jump (see the emergence discussion in Scaling laws). Track per-domain held-out loss (code, math, each language) alongside the average, and treat benchmark scores on early checkpoints as noisy: use the interval methods in Part II before reading a trend.

Optimisation: how the model actually learns

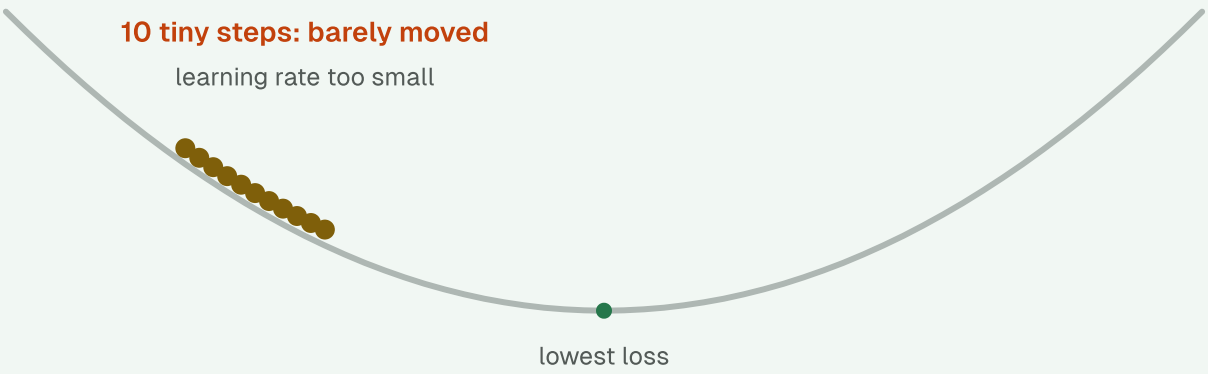
Training is a loop: show the model a batch of text, measure the loss, work out which direction to nudge each of billions of numbers to lower it, take a small step, repeat a few hundred thousand times. This chapter covers how big those steps are, which direction they point, and the hardware arithmetic that decides whether a run is possible at all.

Diagram

Finding the valley in fog

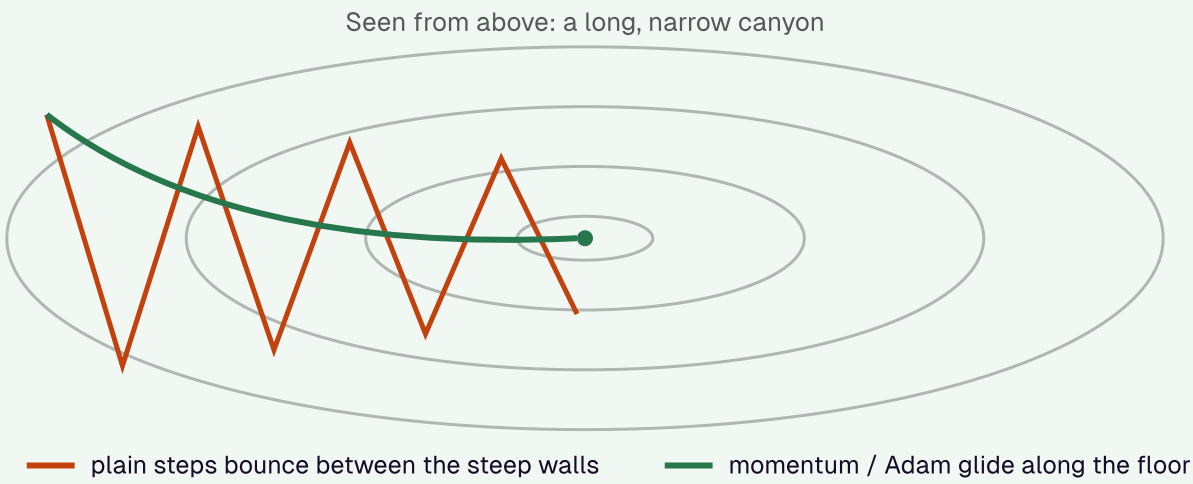
Gradient descent, learning rate, momentum, and schedules as a hike down a mountain.

Training is walking down a hillside in fog. The height is the loss. You can’t see the valley, but you can feel which way the ground tilts: that tilt is the **gradient**. Each step goes downhill.

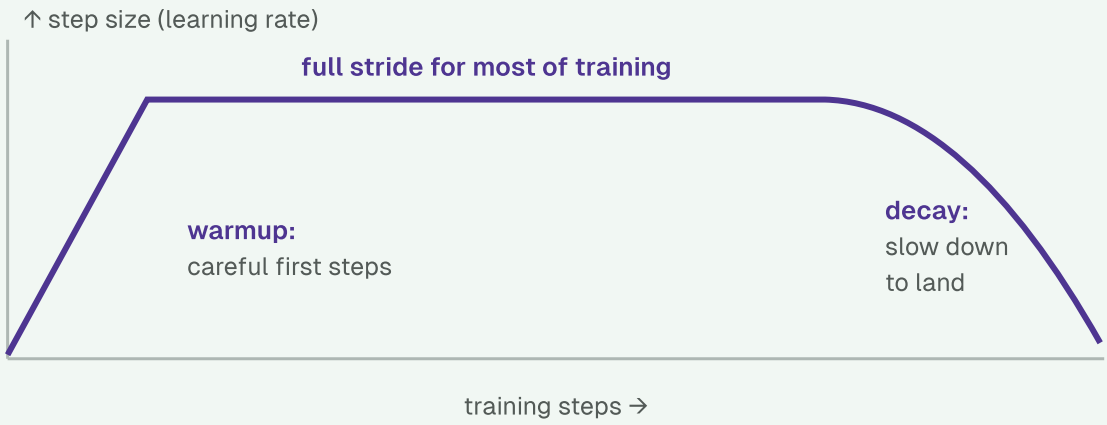


Take strides that are too short and you are safe but slow: after ten steps you have hardly moved. That is a learning rate set too low, and it wastes compute.

Take strides that are too long and you leap clean over the valley and land higher up the other side, then higher again. The loss explodes. That is a learning rate set too high.



Real loss surfaces are long, narrow canyons. Plain steps bounce between the steep walls and crawl along the floor. **Momentum** builds speed in the direction that stays consistent; **Adam** also sizes each direction's stride separately. That is why nearly every LLM trains with AdamW.



So stride length changes over the run: tiny steps while the model is random and its gradients are wild (**warmup**), full stride for most of training, then shorter steps to settle into the bottom (**decay**).

Gradient descent and the learning rate

Plain words

Walking downhill in fog

Picture standing on a hillside in thick fog, trying to reach the valley floor. You cannot see the valley, but you can feel which way the ground slopes under your feet. So you take a step downhill, feel again, step again. That is **gradient descent**. The slope is the **gradient**, and the length of your stride is the **learning rate**.

Strides too short and you take forever. Strides too long and you leap across the valley and up the other side, maybe further up than where you started. Most of the craft of training is choosing stride length well and changing it over time.

For leaders

Why hyperparameters are a budget question

The learning rate is the most important hyperparameter in a training run. A bad choice wastes the whole run, and at frontier scale nobody can afford to sweep it at full size. So labs tune on small models and transfer the settings using scaling rules (for example, the muP parameterization from Yang et al., 2022, which keeps the best learning rate roughly stable as width grows). Ask how the hyperparameters were chosen and at what scale they were validated.

The loss surface of a real network is far from a smooth bowl. It has long narrow valleys, flat plateaus, and cliffs. Optimisers like Adam exist because plain gradient descent handles those shapes badly, as the widget shows: a step size that is safe for the steep direction is painfully slow along the shallow one.

Under the hood

The update rules

Plain gradient descent on parameters θ with learning rate η:

$$\theta_{t+1} = \theta_t - \eta \nabla_\theta \mathcal{L}(\theta_t)$$

In practice the gradient is estimated on a mini-batch, so this is stochastic gradient descent (SGD).

On a quadratic with curvature λ along some direction, each step multiplies the error in that direction by $1 - \eta\lambda$. It shrinks only if $|1 - \eta\lambda| < 1$, so the step must satisfy $\eta < 2/\lambda_{\max}$. In the widget $\lambda_{\max} = 10$, so anything above 0.2 diverges. Meanwhile the shallow direction ($\lambda = 1$) shrinks by only $1 - \eta$ per step. The ratio $\lambda_{\max}/\lambda_{\min}$, the condition number, sets how slow plain descent must be.

Momentum keeps a running average of past gradients, so consistent directions accelerate and oscillating ones cancel: $v_{t+1} = \beta v_t + \nabla\mathcal{L}$, $\theta_{t+1} = \theta_t - \eta v_{t+1}$.

Adam (Kingma and Ba, 2015) also tracks the average squared gradient per parameter and divides by its square root, so every parameter takes steps of roughly the same size regardless of its gradient scale:

$$m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t, \quad v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t^2, \quad \theta_t = \theta_{t-1} - \eta \left(\frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} + \lambda \theta_{t-1} \right)$$

$\hat{m}, \hat{v}$ are bias-corrected: divide by $1 - \beta^t$. The $\lambda\theta$ term is decoupled weight decay, the "W" in AdamW (Loshchilov and Hutter, 2019). Typical LLM settings: $\beta_1 = 0.9$, $\beta_2 = 0.95$, weight decay 0.1.

python

```python
import numpy as np

def adamw_step(theta, g, m, v, t, lr=3e-4, b1=0.9, b2=0.95, eps=1e-8, wd=0.1):
    m = b1 * m + (1 - b1) * g                # first moment: direction
    v = b2 * v + (1 - b2) * g * g            # second moment: per-parameter scale
    m_hat = m / (1 - b1 ** t)                # bias correction for the zero init
    v_hat = v / (1 - b2 ** t)
    theta = theta - lr * (m_hat / (np.sqrt(v_hat) + eps) + wd * theta)   # decoupled decay
    return theta, m, v

theta, m, v = np.array([1.0, -2.0]), np.zeros(2), np.zeros(2)
theta, m, v = adamw_step(theta, np.array([0.5, 40.0]), m, v, t=1)
print(theta)   # both coordinates move by about lr, despite an 80x gradient difference
```

rust

```rust
/// AdamW hyperparameters; Default gives the Python's keyword defaults.
struct AdamW {
    lr: f64,
    b1: f64,
    b2: f64,
    eps: f64,
    wd: f64,
}

impl Default for AdamW {
    fn default() -> Self {
        AdamW { lr: 3e-4, b1: 0.9, b2: 0.95, eps: 1e-8, wd: 0.1 }
    }
}

/// Updates theta, m and v in place, one coordinate at a time.
fn adamw_step(theta: &mut [f64], g: &[f64], m: &mut [f64], v: &mut [f64], t: i32, h: &AdamW) {
    for i in 0..theta.len() {
        m[i] = h.b1 * m[i] + (1.0 - h.b1) * g[i]; // first moment: direction
        v[i] = h.b2 * v[i] + (1.0 - h.b2) * g[i] * g[i]; // second moment: per-parameter scale
        let m_hat = m[i] / (1.0 - h.b1.powi(t)); // bias correction for the zero init
        let v_hat = v[i] / (1.0 - h.b2.powi(t));
        theta[i] -= h.lr * (m_hat / (v_hat.sqrt() + h.eps) + h.wd * theta[i]); // decoupled decay
```

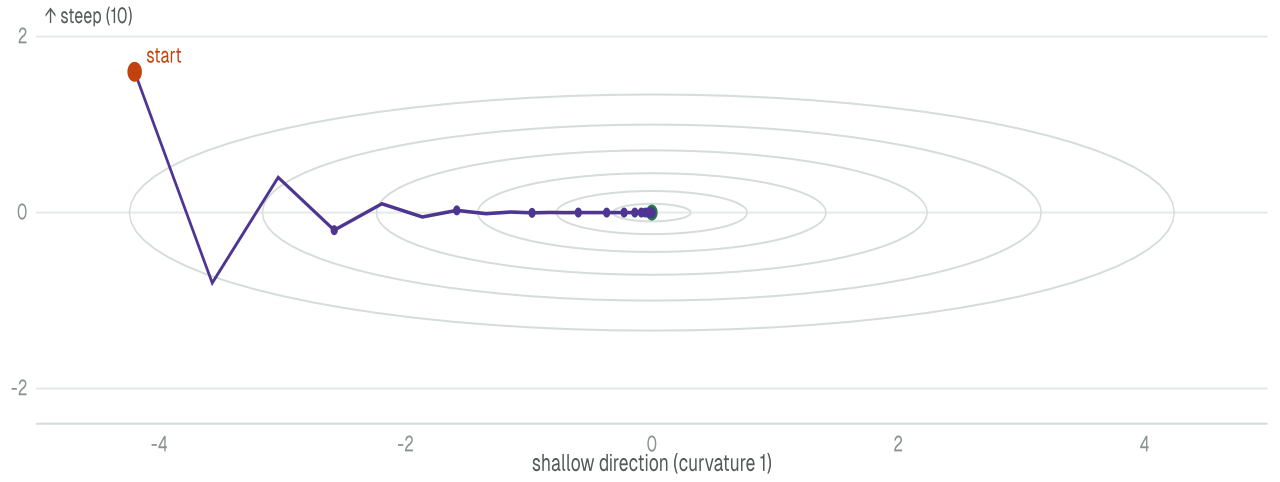
```rust
    }
}

fn main() {
    let (mut theta, mut m, mut v) = ([1.0, -2.0], [0.0; 2], [0.0; 2]);
    adamw_step(&mut theta, &[0.5, 40.0], &mut m, &mut v, 1, &AdamW::default());
    // printed the way numpy prints the array
    let shown: Vec<String> = theta.iter().map(|x| format!("{x:8.5}")).collect();
    println!("[{}]", shown.join(" ")); // both coordinates move by about lr, despite an 80x gradient difference
}
```

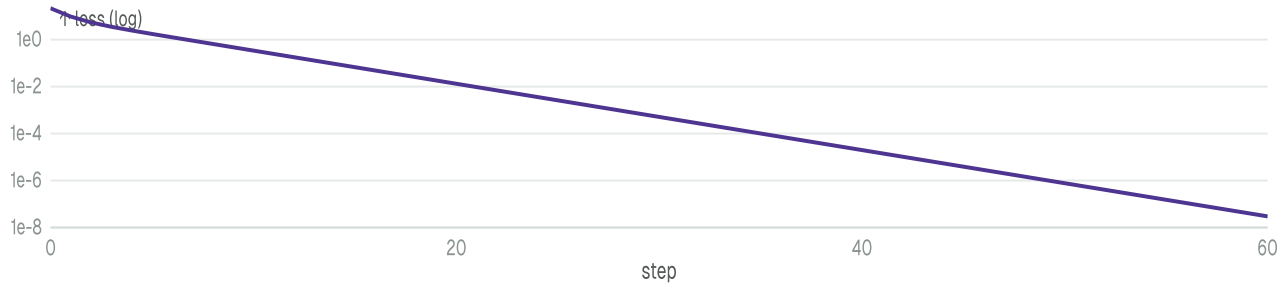
Newer optimisers aim to beat AdamW per token. Muon (2024), which orthogonalises the update for matrix parameters, was used at scale in Moonshot's Kimi K2 (2025). AdamW remains the default reference.

Interactive

Step size on a long, narrow valley (shown at its default setting)

The ellipses are contours of a loss surface that is ten times steeper in one direction than the other, like most real ones. Try the presets, then move the sliders.





Converging. Loss after 60 steps: 2.99e-8.

Schedules, batch size, and stability

Plain words

Start gently, stride out, then slow down to land

At the very start the model is random and its gradients are wild, so training begins with tiny steps and ramps up (**warmup**). Then it runs at full stride for most of training. Near the end it shortens its stride to settle into the bottom of the valley (**decay**). The batch is how many examples it looks at before each step: a bigger batch gives a steadier sense of direction, up to a point.

For leaders

The knobs that decide wall-clock time and risk

- **Schedules.** Cosine decay has been the default since GPT-3. Warmup-stable-decay (WSD) schedules, popularised by MiniCPM (Hu et al., 2024), hold the learning rate flat and decay only at the end. That lets a team branch a decayed checkpoint off at any point, which makes it cheap to continue training or to run scaling studies without restarting.
- **Batch size.** Bigger batches let you use more GPUs in parallel, which shortens wall-clock time. Below the **critical batch size**, doubling the batch roughly halves the number of steps needed. Above it, you pay for more compute with little speedup (McCandlish et al., 2018). The critical batch size grows as loss falls, so batch size is often increased during a run. Frontier runs use batches of millions of tokens per step.
- **Loss spikes.** Large runs occasionally see the loss jump. PaLM (Chowdhery et al., 2022) handled spikes by rolling back to a checkpoint about 100 steps earlier and skipping a few hundred batches. Mitigations include gradient clipping (cap the gradient norm, commonly at 1.0), lower learning rates, normalising queries and keys in attention, and careful initialisation.

Principal asks

- How many spikes did this run have, and how much compute was lost to rollbacks?
- Were hyperparameters tuned at this scale or transferred? What is the evidence the transfer held?
- Which results are from a checkpoint before or after the final learning-rate decay? A pre-decay checkpoint understates the model.

Under the hood

Cosine and WSD, written out

$$\eta_{\cos}(t) = \eta_{\min} + \tfrac{1}{2}(\eta_{\max} - \eta_{\min})\left(1 + \cos\frac{\pi(t - t_w)}{T - t_w}\right) \quad \text{for } t > t_w$$

Linear warmup over t_w steps first. WSD replaces the cosine with a constant $\eta_{\max}$ and a short final decay, often 10 to 20 percent of training.

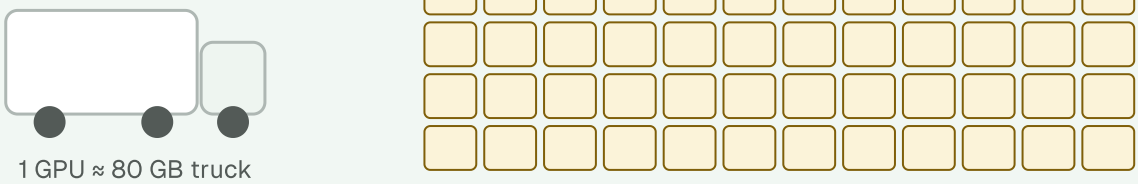
Gradient clipping rescales $g \leftarrow g \cdot \min(1, c/\|g\|)$. The gradient noise scale $B_{\text{noise}} = \operatorname{tr}(\Sigma)/\|G\|^2$, the ratio of per-example gradient variance to the squared true gradient, estimates the critical batch size.

Precision, memory, and parallelism

Diagram

Moving a house too big for one truck

Why training needs thousands of GPUs, and the five ways to split the work.



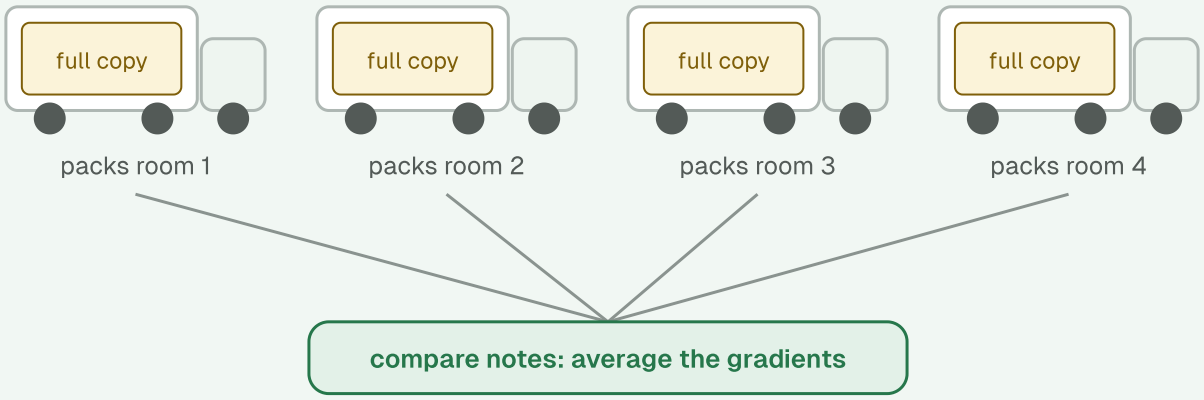
1 GPU ≈ 80 GB truck

70B model + Adam ≈ 1,120 GB of numbers

weights, gradients, optimiser state

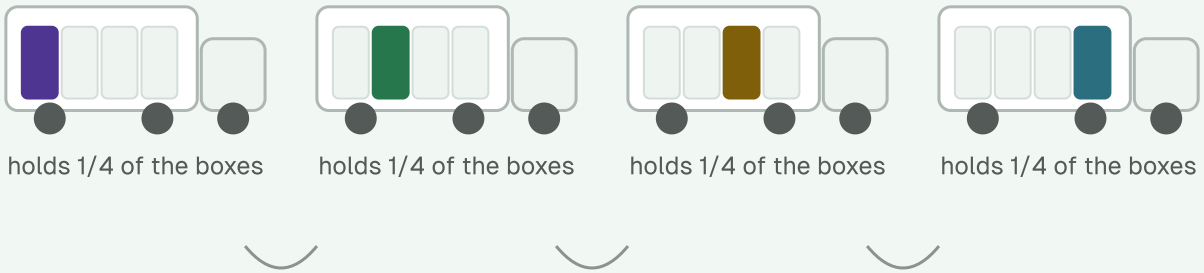
≈ 14 trucks just to hold the boxes

A 70-billion-parameter model trained with Adam needs about 16 bytes per parameter: roughly 1,120 GB. One GPU holds 80 GB. Before any training happens, the boxes alone need 14 trucks.



Data parallel

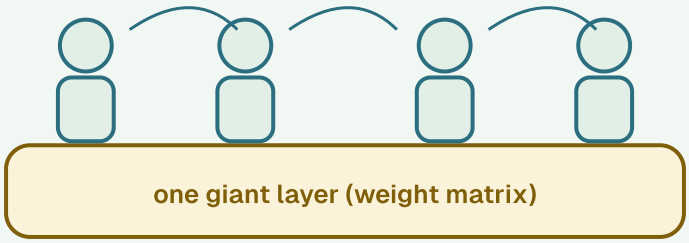
Data parallel: several identical crews, each with a full copy of the model, each packing a different room (a different slice of the batch). After every step they compare notes and average their gradients so all copies stay identical. Simple, but every truck carries everything.



ZeRO / FSDP: no duplicate boxes

each GPU keeps a shard and borrows the rest just in time

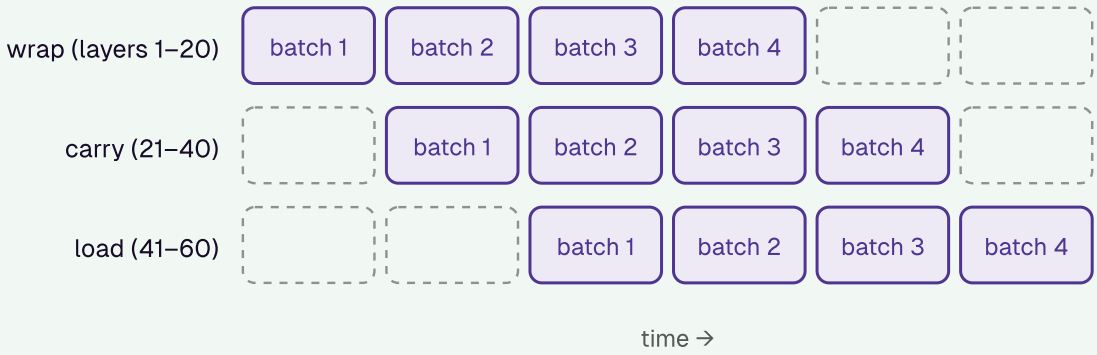
ZeRO and FSDP fix the duplication: each crew keeps only its quarter of the weights, gradients, and optimiser state, and borrows the other pieces for a moment when it needs them. Memory per GPU drops roughly in proportion to the number of crews.



Tensor parallel: four GPUs carry one layer together

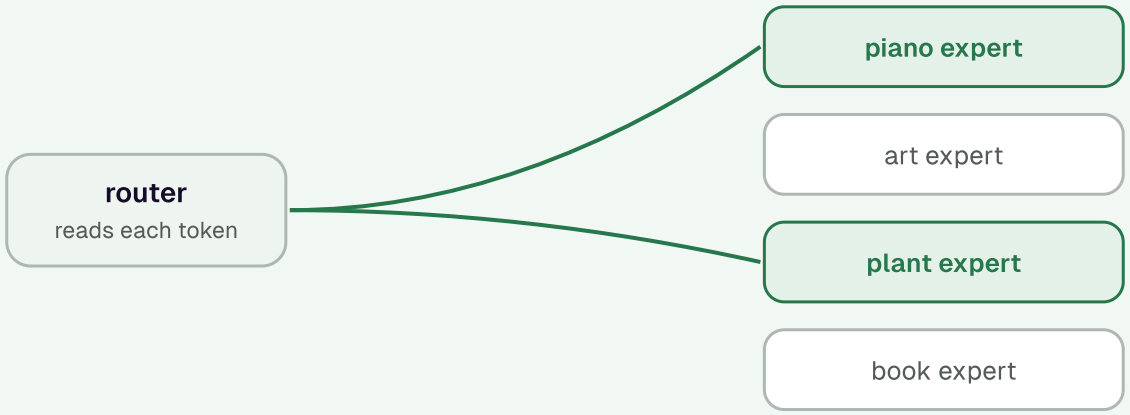
they talk constantly, so they need very fast links inside one server

Tensor parallel: some pieces are too heavy for one mover. Four movers carry one couch together, which here means splitting a single weight matrix across GPUs. They must coordinate every moment, so this only works over very fast links inside one server.



Pipeline parallel: an assembly line. Dashed boxes are idle “bubbles”.

Pipeline parallel: an assembly line. Different GPUs hold different layers, and small batches flow through one after another. The cost is the idle “bubbles” at the start and end while the line fills and drains.



Mixture of experts: each token visits only a few experts, on different GPUs.

Expert parallel, for mixture-of-experts models: a dispatcher sends each item only to the specialists it needs, and each specialist lives on its own GPU. The model can hold a lot of knowledge while each token only pays for a few experts.

Plain words

The model does not fit on one chip

A large model's numbers, plus the extra bookkeeping the optimiser keeps, take far more memory than any single GPU has. So training splits the work across thousands of GPUs: some split the data, some split each layer, some split the stack of layers. Teams also store numbers with fewer digits (lower precision) to save memory and go faster, carefully, because too few digits make training unstable.

For leaders

The arithmetic behind a GPU request

The rule of thumb to carry: mixed-precision training with Adam needs about **16 bytes per parameter** before activations. A 70-billion-parameter model therefore needs about 1.1 TB just for weights, gradients, and optimiser state, which is 14 H100s' worth of memory doing nothing but holding numbers. Sharding (ZeRO, FSDP) spreads that across GPUs; activations, which grow with sequence length and batch size, are handled with recomputation (activation checkpointing) and more parallelism.

Data parallel

Every GPU has a full copy and its own slice of the batch. Simple, but memory-bound. ZeRO / FSDP shard the copies away.

Tensor parallel

Split each weight matrix across GPUs in one server (Megatron-LM, Shoeybi et al., 2019). Needs very fast interconnect.

Pipeline parallel

Different layers on different GPUs, with micro-batches flowing through. Idle "bubbles" are the cost.

Expert parallel

For mixture-of-experts models, different experts live on different GPUs and tokens are routed between them.

Model FLOPs utilisation (MFU) is the share of the hardware's peak arithmetic that goes into useful training math. PaLM reported 46% (Chowdhery et al., 2022); Meta reported 38 to 43% MFU for Llama 3 405B on 8,000 to 16,000 H100s (2024). At these scales, hardware failures are routine, so checkpointing and fast restarts matter as much as raw speed.

Lower precision. bf16 is standard. DeepSeek-V3 (2024) trained a 671B-parameter mixture-of-experts model largely in FP8, which cuts memory and roughly doubles throughput on supporting hardware, at the price of careful scaling to avoid overflow.

Eval lens

Infrastructure choices change results in ways that look like model changes. Different precision, batch size, or parallel layout can shift evaluation numbers slightly through nondeterminism. Before attributing a small benchmark delta to a training change, rerun the eval on the same checkpoint and confirm the run-to-run noise floor.

Under the hood

Memory and MFU formulas

With Ψ parameters, N_d data-parallel GPUs, and K bytes of optimiser state per parameter ($K = 12$ for fp32 master weights plus Adam's two moments), the per-GPU model-state memory under ZeRO (Rajbhandari et al., 2020) is:

$$\underbrace{(2+2+K)\Psi}_{\text{stage 0}},\quad \underbrace{4\Psi+\tfrac{K\Psi}{N_d}}_{\text{stage 1}},\quad \underbrace{2\Psi+\tfrac{(2+K)\Psi}{N_d}}_{\text{stage 2}},\quad \underbrace{\tfrac{(4+K)\Psi}{N_d}}_{\text{stage 3}}$$

2 bytes each for bf16 weights and gradients. Stage 3 is what PyTorch FSDP calls full sharding.

A dense transformer spends about 6Ψ FLOPs per training token (2 forward, 4 backward), ignoring the attention term that grows with context length. So:

$$\text{MFU} = \frac{6\Psi \cdot \text{tokens per second}}{N_{\text{GPU}} \cdot \text{peak FLOP/s}}$$

python

```python
def model_state_gb(params, n_gpu, stage, k=12):
    p = params
    per = {0: (4 + k) * p,
           1: 4 * p + k * p / n_gpu,
           2: 2 * p + (2 + k) * p / n_gpu,
           3: (4 + k) * p / n_gpu}[stage]
    return per / 1e9

for s in range(4):
    print(s, round(model_state_gb(70e9, 8, s), 1), "GB per GPU")
# 0 1120.0   1 385.0   2 262.5   3 140.0   -> even full sharding over 8 GPUs exceeds 80 GB

def mfu(params, tok_per_s, n_gpu, peak=989e12):          # H100 dense bf16 peak
    return 6 * params * tok_per_s / (n_gpu * peak)
print(round(mfu(405e9, 2.6e6, 16000), 2))                 # about 0.4
```

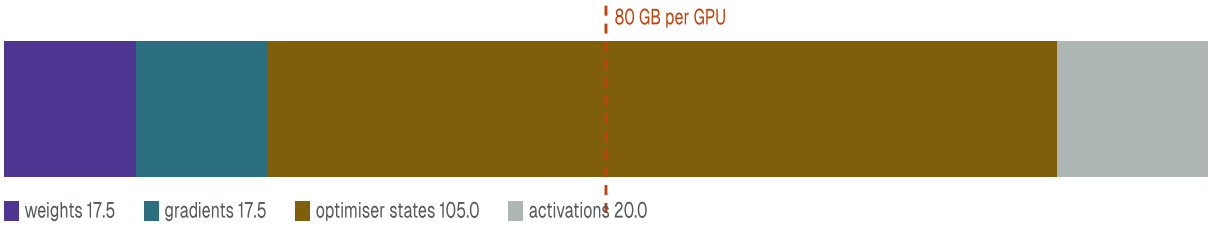
rust

```rust
/// Bytes of weights, gradients and optimizer state per GPU, in GB.
/// k = optimizer bytes per parameter (12 in the Python's default).
fn model_state_gb(params: f64, n_gpu: f64, stage: u8, k: f64) -> f64 {
    let p = params;
    let per = match stage {
        0 => (4.0 + k) * p,
        1 => 4.0 * p + k * p / n_gpu,
        2 => 2.0 * p + (2.0 + k) * p / n_gpu,
        3 => (4.0 + k) * p / n_gpu,
        _ => panic!("ZeRO stage must be 0 to 3"),
    };
    per / 1e9
}

/// peak defaults to 989e12 in the Python: H100 dense bf16 peak
fn mfu(params: f64, tok_per_s: f64, n_gpu: f64, peak: f64) -> f64 {
    6.0 * params * tok_per_s / (n_gpu * peak)
```

```
}

/// Python's round(x, digits); printed with {:?} it drops trailing zeros the same way.
fn round(x: f64, digits: i32) -> f64 {
    let scale = 10f64.powi(digits);
    (x * scale).round() / scale
}

fn main() {
    for s in 0..4 {
        println!("{} {:.1} GB per GPU", s, model_state_gb(70e9, 8.0, s, 12.0));
    }
    // 0 1120.0   1 385.0   2 262.5   3 140.0   -> even full sharding over 8 GPUs exceeds 80 GB

    println!("{:?}", round(mfu(405e9, 2.6e6, 16000.0, 989e12), 2)); // about 0.4
}
```

Interactive **Training memory per GPU** (shown at its default setting)

Model states only (weights, gradients, optimiser), plus an activation allowance you set. Try a 70B model on 8 GPUs at each ZeRO stage.



Per GPU: **160.0 GB** does not fit. Model states across the whole group: 1.12 TB (16 bytes per parameter). At stage 3 every piece of model state is split 8 ways; the cost is extra communication to gather weights for each layer.

Part IV · Chapter 30

Scaling laws: predicting the result before you pay for it

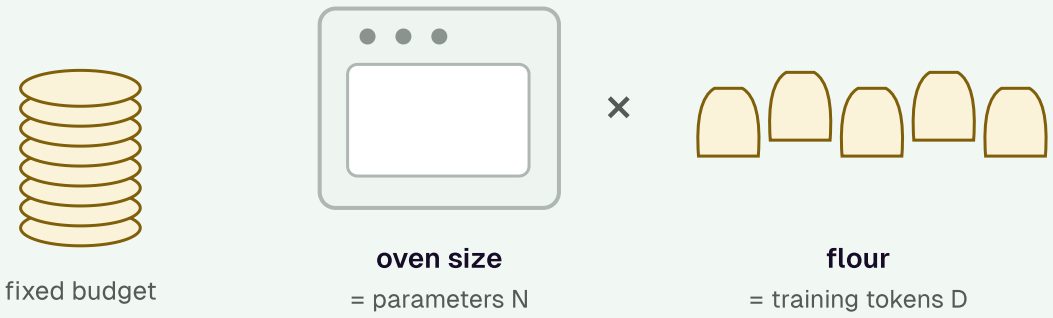
Scaling laws are the reason frontier labs can spend hundreds of millions of dollars on a single run with some confidence. Loss falls along smooth, predictable curves as you add parameters, data, and compute, so you can fit those curves on small models and extrapolate to the big one.

Diagram

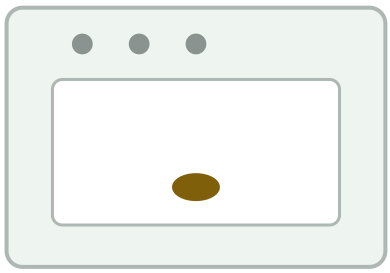
Big oven or more flour?

Compute-optimal scaling as a bakery splitting a fixed budget.

compute C ≈ 6 × N × D

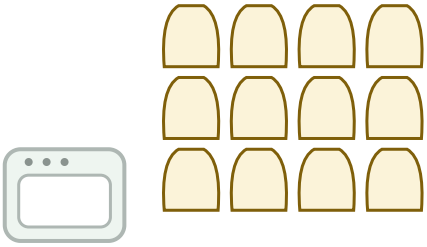


Training compute is a fixed budget. You can spend it on a bigger oven (more parameters) or more flour (more training tokens). Compute is roughly 6 × parameters × tokens, so every extra unit of one costs you some of the other.



Huge oven, one sack of flour

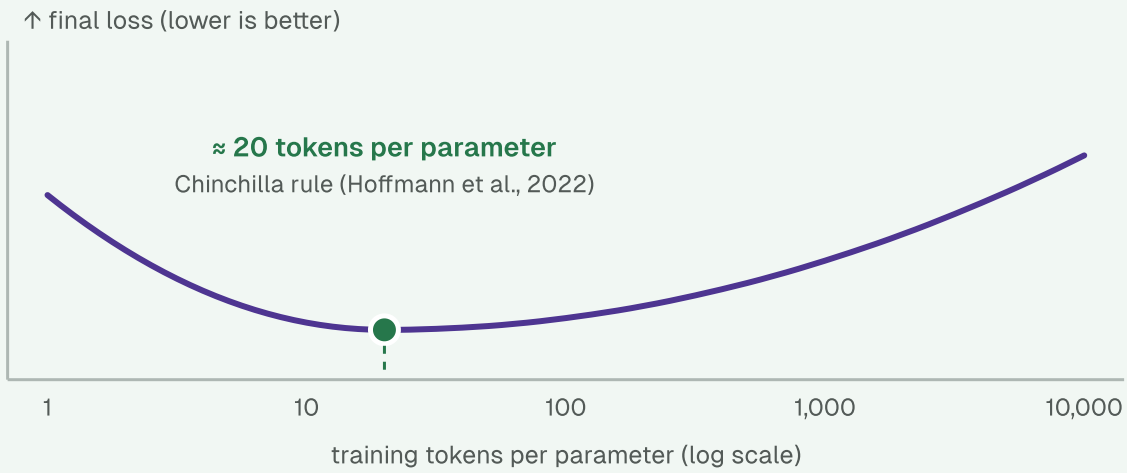
wasted capacity



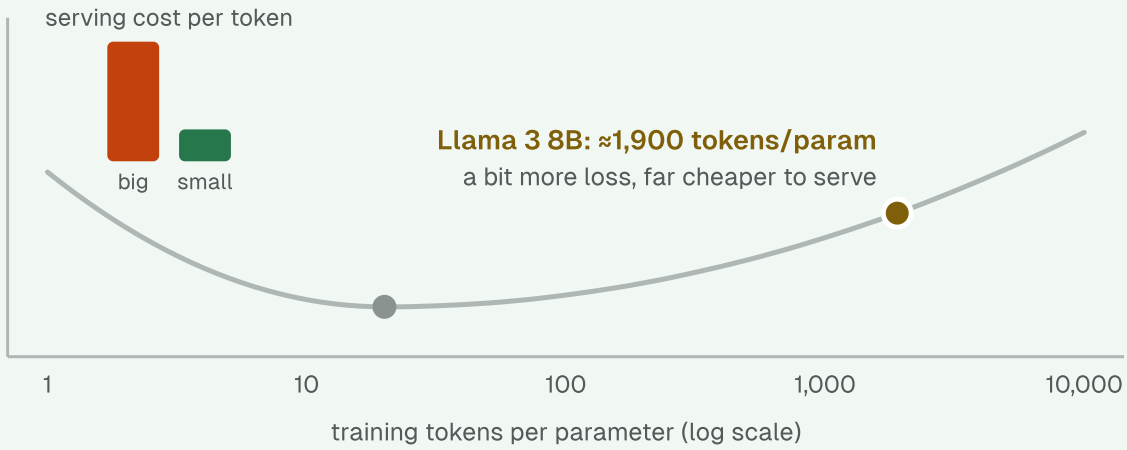
Tiny oven, mountain of flour

can’t absorb it all

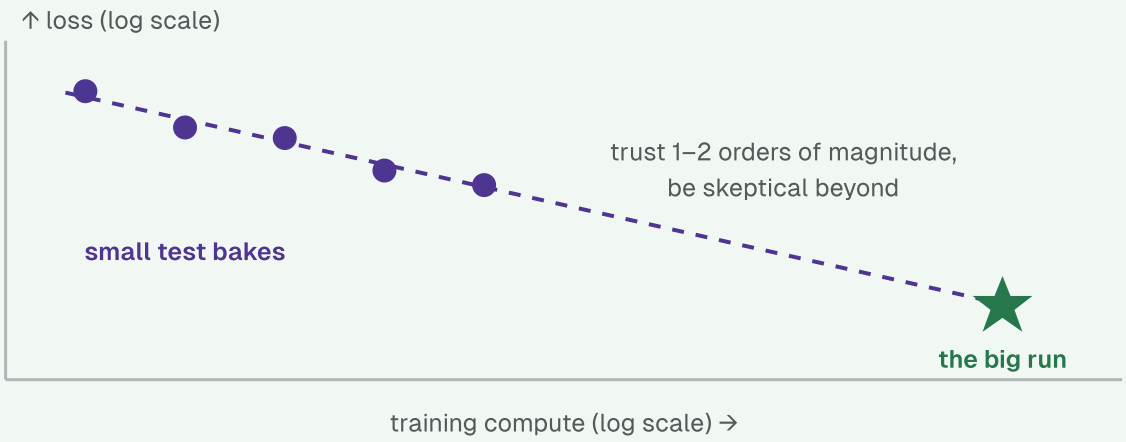
Both extremes waste money. A giant model trained on too little data never fills its capacity. A tiny model fed a mountain of data cannot hold what it reads. For any budget there is a best split.



Plot final loss against how you split the budget and you get a valley. Hoffmann et al. (2022) put the bottom near **20 training tokens per parameter**: Chinchilla, 70B parameters on 1.4T tokens, beat the 280B Gopher on the same compute. (Their parametric fit puts the bottom at more tokens per parameter; either way the valley floor is broad.)



A bakery sells loaves for years, and every loaf costs oven time. Models are the same: inference over a model's life can dwarf training. So labs over-train smaller models far past 20:1. Llama 3 8B saw about 15 trillion tokens, around 1,900 per parameter, for a little more loss and much cheaper serving.



The payoff: bake a series of small test batches, plot loss against compute on log scales, and the points fall on a line. Extend the line to predict the expensive run before paying for it. The further you extrapolate, the less you should trust it.

Plain words

Bigger brain or more books?

You have a fixed amount of money to train a student. You can buy a bigger brain (more parameters) or more books (more training data). A huge brain that reads only a few books wastes its capacity. A tiny brain that reads every book cannot hold it all. There is a best balance for any budget.

Scaling laws are the recipe for that balance. Researchers train many small models, plot how good each one gets, see a clean pattern, and use it to predict how a much bigger model will do.

For leaders

Three generations of the recipe

1. **Kaplan et al. (2020), OpenAI.** Loss follows power laws in parameters, data, and compute over many orders of magnitude. Their fits suggested growing model size much faster than data as budgets rise, which shaped GPT-3 (175B parameters, about 300B tokens).

2. **Hoffmann et al. (2022), DeepMind, "Chinchilla".** With more careful experiments across scales, the optimum shifts: parameters and tokens should grow in equal proportion with compute, at roughly **20 training tokens per parameter**. Chinchilla (70B parameters, 1.4T tokens) outperformed the 280B-parameter Gopher on the same compute.

3. **Inference-aware training (2023 onward).** Chinchilla optimises training cost only. A model served to millions of users spends far more compute on inference over its life, so a smaller model trained on far more data is cheaper overall (Sardana et al., 2023). Llama 3 8B was

trained on about 15 trillion tokens, close to 1,900 tokens per parameter, almost a hundred times the Chinchilla ratio, and kept improving.

The current frontier adds two more axes. **Mixture-of-experts** models activate only a fraction of their parameters per token (DeepSeek-V3: 671B total, 37B active, 2024), which decouples knowledge capacity from per-token cost. **Test-time compute** lets a model think longer at inference, and performance on reasoning tasks scales with that too (see Inference and test-time compute). Pretraining compute is no longer the only dial.

Principal asks

- What range of scales was the law fitted on, and how far are we extrapolating? More than one or two orders of magnitude deserves skepticism.
- Are we optimising for training cost or lifetime cost including inference? What token-per-parameter ratio does that imply?
- Does the law predict the downstream metric we care about, or only loss? How was the loss-to-metric link validated?
- Is high-quality data the binding constraint? How many epochs does the plan imply for our best data?

Under the hood

The Chinchilla loss surface

Hoffmann et al. fit a parametric form for final loss as a function of parameters N and training tokens D:

$$L(N, D) = E + \frac{A}{N^{\alpha}} + \frac{B}{D^{\beta}}$$

E is the irreducible loss of the data. The other terms are the penalty for a finite model and for finite data. Their fit (approach 3): $E = 1.69,\ A = 406.4,\ B = 410.7,\ \alpha = 0.34,\ \beta = 0.28$. A 2024 replication by Besiroglu et al. argued these constants were mis-estimated and refit them; treat exact values as approximate.

Minimising L subject to a compute budget $C \approx 6ND$ gives optimal allocations that scale as power laws in C. Hoffmann et al. found exponents near 0.5 for both, so doubling compute should increase both N and D by about $\sqrt{2}$. Kaplan et al. had found $N_{\text{opt}} \propto C^{0.73}$. Hoffmann et al. attributed the gap to Kaplan's fixed learning-rate schedule, but later work (Porian et al., 2024; Pearce and Song, 2024)

traced it mainly to counting only non-embedding parameters, ignoring last-layer compute, warmup length, and hyperparameters not retuned per scale. Porian et al. found careful learning-rate decay was not needed to recover the Chinchilla result.

With the 20-tokens-per-parameter rule, $C = 6 \cdot N \cdot 20N = 120N^2$, so $N = \sqrt{C/120}$.

python

```python
import math

E, A, B, a, b = 1.69, 406.4, 410.7, 0.34, 0.28        # Hoffmann et al. 2022 fit

def loss(N, D):
    return E + A / N**a + B / D**b

def allocate(C, tokens_per_param=20):
    N = math.sqrt(C / (6 * tokens_per_param))
    return N, tokens_per_param * N

C = 5.76e23                                           # roughly Chinchilla's own budget
N, D = allocate(C)
print(f"{N/1e9:.0f}B params, {D/1e12:.2f}T tokens")  # 69B params, 1.39T tokens

# over-train a smaller model on the same compute
for r in (20, 200, 2000):
    N, D = allocate(C, r)
    print(r, f"{N/1e9:.1f}B", round(loss(N, D), 3))
# 20 69.3B 1.937   200 21.9B 1.932   2000 6.9B 1.959
```

rust

```rust
// Hoffmann et al. 2022 fit (alpha and beta are the Python's a and b)
const E: f64 = 1.69;
const A: f64 = 406.4;
const B: f64 = 410.7;
const ALPHA: f64 = 0.34;
const BETA: f64 = 0.28;

fn loss(n: f64, d: f64) -> f64 {
    E + A / n.powf(ALPHA) + B / d.powf(BETA)
}

/// tokens_per_param is 20 by default in the Python
fn allocate(c: f64, tokens_per_param: f64) -> (f64, f64) {
    let n = (c / (6.0 * tokens_per_param)).sqrt();
    (n, tokens_per_param * n)
}

fn main() {
    let c = 5.76e23; // roughly Chinchilla's own budget
    let (n, d) = allocate(c, 20.0);
    println!("{:.0}B params, {:.2}T tokens", n / 1e9, d / 1e12); // 69B params, 1.39T tokens

    // over-train a smaller model on the same compute
    for r in [20, 200, 2000] {
        let (n, d) = allocate(c, r as f64);
        println!("{} {:.1}B {:.3}", r, n / 1e9, loss(n, d));
    }
```

```
    // 20 69.3B 1.937   200 21.9B 1.932   2000 6.9B 1.959
}
```

Two things stand out. First, under this parametric fit the loss at 200 tokens per parameter is slightly *lower* than at 20: the fitted surface puts the optimum near 90 tokens per parameter at this budget, not 20. The 20:1 rule came from the paper's other two estimation methods, and the mismatch between them is one of the issues Besiroglu et al. (2024) raised. Second, the surface is very flat near the optimum. Going to 2,000 tokens per parameter makes the model ten times smaller, and roughly ten times cheaper per token to serve, for a loss increase of about 0.02 at the same training compute. That flatness is the economic case for over-training.

Data-constrained scaling. Muennighoff et al. (2023) extended the law to repeated data: each repeated epoch is worth less than fresh data, with about four epochs nearly free and value decaying toward zero by roughly sixteen.

Interactive

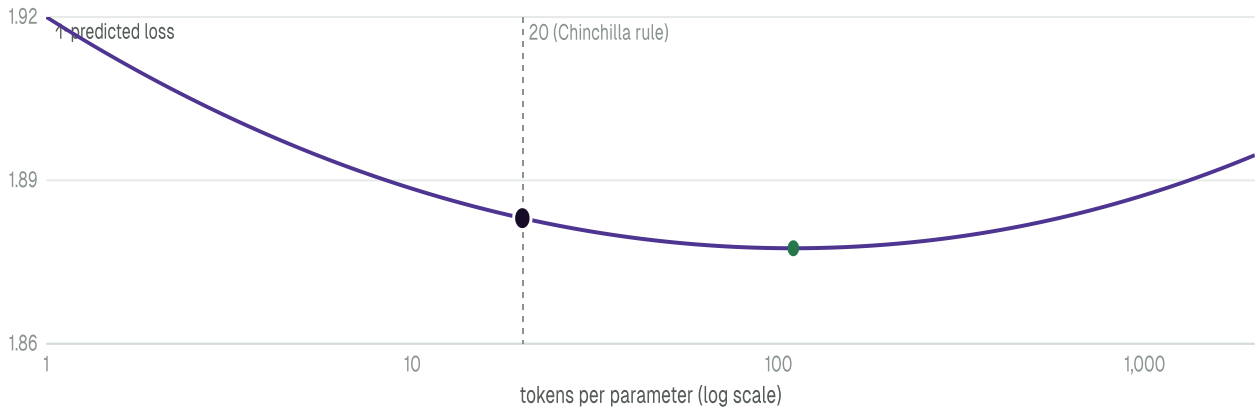
Compute budget planner (shown at its default setting)

Pick a training budget in FLOPs and a token-per-parameter ratio. The planner splits the budget into model size and data with C ≈ 6ND, estimates wall-clock time, and shows how much loss you give up by over-training a smaller model.

Parameters N	Training tokens D	Wall-clock days	GPU-hours
162.5B	3.2T	11	2.2M

Predicted loss vs best split

+0.006



At 3.16e24 FLOPs the fitted law is minimised near **110 tokens per parameter** (green). Your split trains a 162.5B-parameter model on 3.2T tokens in about 11 days on 8,192 GPUs. The fitted optimum differs from the 20:1 rule because the parametric fit and the rule of thumb came from different analyses in the same paper.

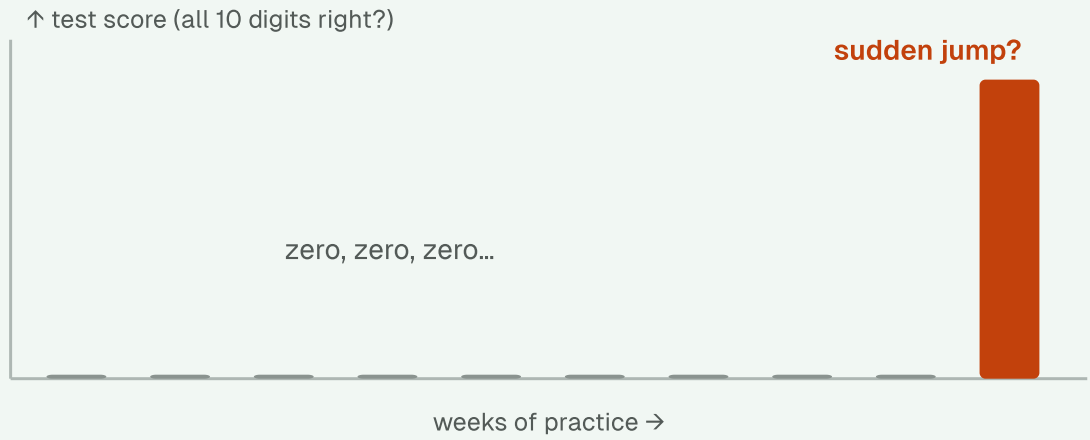
Emergence: sudden jumps or a trick of the metric?

Diagram

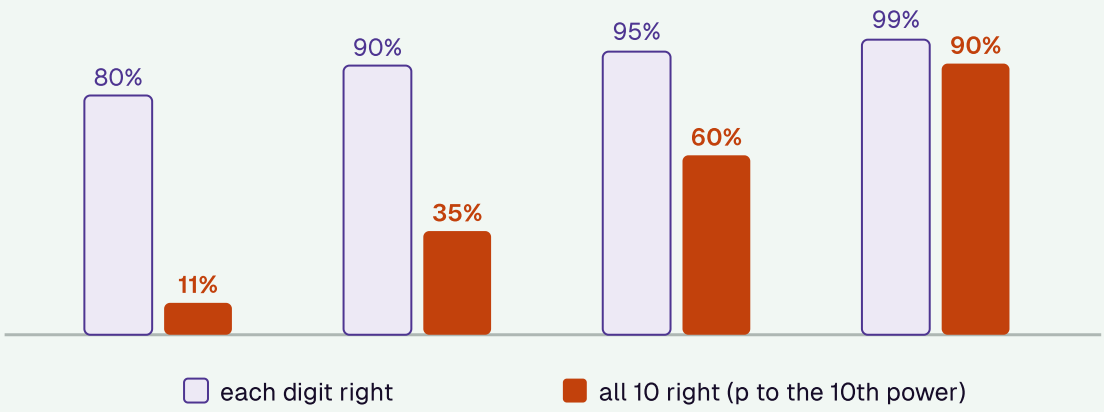
A staircase that looks like a jump

Why abilities can seem to appear suddenly when the model improved gradually.

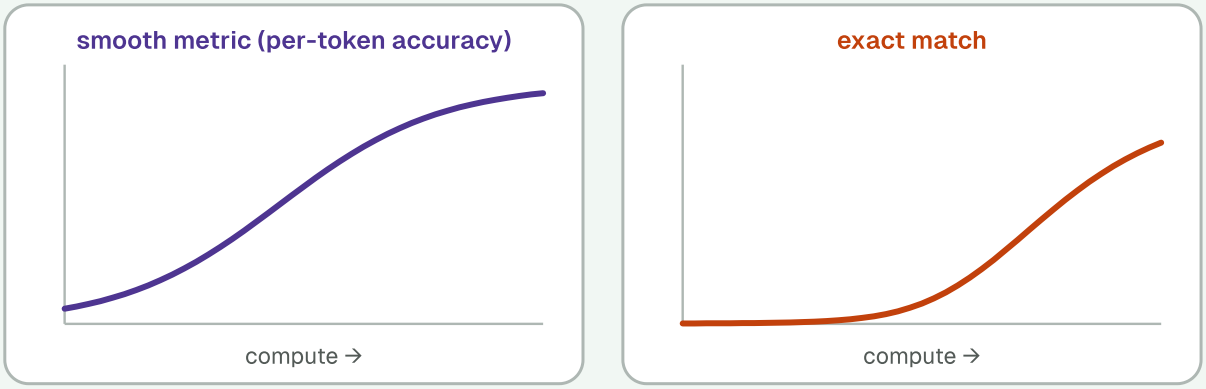
A student practises 10-digit multiplication and gets one more digit right each week. Progress is steady.



Now grade only on whether all ten digits are right. The score is zero for nine weeks, then suddenly 100%. It looks like a magical leap. The student did not change; the test did.



The arithmetic: if each digit is right with probability p, all ten are right with probability p^{10}. At 80% per digit that is 11%; at 90%, 35%; at 99%, 90%. Steady gains per digit become a hockey stick for the all-or-nothing score.



Forecast with the smooth metric. Report the one users feel.

Same model, same improvement, two pictures. Schaeffer et al. (2023) showed many “emergent” abilities turn smooth when re-scored this way. Practical rule: forecast capabilities with a continuous metric, and report the binary one.

Plain words

Getting every digit right

Imagine a student learning to multiply big numbers. Each week they get a few more digits right. But if the test only gives credit when *every* digit is right, their score sits at zero for months and then suddenly shoots up. The student improved steadily the whole time. The test made it look sudden.

For leaders

Why this debate matters for planning

Wei et al. (2022) catalogued "emergent abilities": tasks where small models score near zero and larger models suddenly succeed. Schaeffer, Miranda, and Koyejo (2023) argued many of these jumps come from harsh, all-or-nothing metrics. Re-scored with smooth metrics, such as per-token log likelihood or partial credit, most curves became gradual and predictable.

Both views carry a practical lesson. If you measure only exact-match accuracy, capabilities will appear to arrive without warning, and safety-relevant ones are exactly the ones you do not want to be surprised by. If you also track a smooth proxy, you can see a capability coming several scales early.

Eval lens

For every important capability, keep two metrics: the binary one users experience (did it get the answer right) and a continuous one that moves earlier (log probability of the correct answer, partial credit, pass@k at large k). Forecast with the continuous one. Report the binary one. Labs including Meta (Llama 3, 2024) predict downstream accuracy in two steps: compute to the model's negative log likelihood on the correct answers, then likelihood to accuracy with a sigmoid fit.

Under the hood

Why products of probabilities look like step functions

Let per-token accuracy rise smoothly with log compute x, say $p(x) = \sigma(a(x - x_0))$. If an answer needs k independent tokens all correct, exact match is $p(x)^k$. For large k, p^k stays near zero until p is close to 1, then rises steeply: the derivative $kp^{k-1}p'$ concentrates near the end. Per-token log likelihood $k \log p(x)$ is smooth throughout. Same model, same improvement, different shape.

$$\mathrm{EM}(x) = p(x)^k, \qquad \frac{d\,\mathrm{EM}}{dx} = k\,p(x)^{k-1}p'(x)$$

Interactive

The same improvement, two metrics (shown at its default setting)

The underlying skill (chance of getting each token of the answer right) improves smoothly with compute. Exact match needs every token right. Lengthen the answer and watch a smooth trend turn into an apparent jump.

At 1e23.0 FLOPs, per-token accuracy is **78%** but exact match on a 10-token answer is only **8.6%**. The red curve looks like a capability that "emerges" suddenly. The green and blue curves show the model was improving steadily all along.

Part IV · Chapter 31

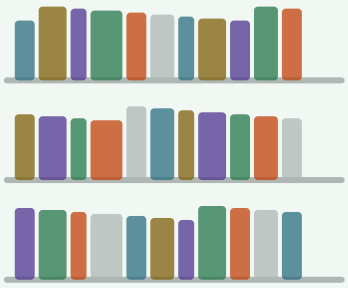
Supervised fine-tuning: teaching the format

A pretrained model has read a large slice of the internet and can continue any text. It doesn't yet know that it is supposed to be an assistant. Supervised fine-tuning (SFT) shows it thousands of example conversations and trains it, with the same next-token loss used in pretraining, to produce the assistant's side. It is the first step of post-training and the base that preference tuning and RL build on.

Diagram

The well-read new hire

What fine-tuning changes and what it does not, told as a new employee's first month.



Read the whole library

Customer: How do I reset my password?

Model: "...asked the customer nervously. Chapter 12. The rain had not stopped."

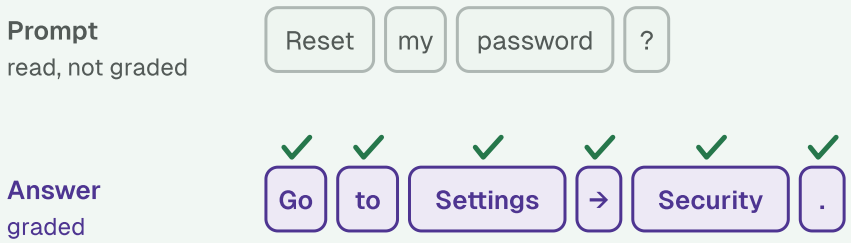
It continues the text. It does not answer.

A pretrained model is like a new hire who has read every book in the library and never held a job. Ask a question and it may just keep writing, because continuing text is all it has practised.

Q Reset password?	Q Where is my refund?	Q Write me malware.
A Settings → Security → Reset. Check email.	A Billing → Refunds. Takes 3 to 5 days.	A Sorry, I can't help with that.

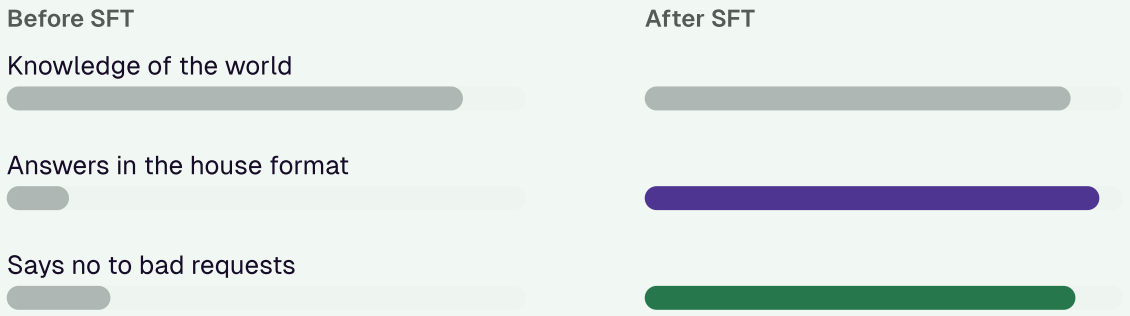
Thousands of examples: a question, then exactly how we answer it

Supervised fine-tuning is the training binder. Each page shows a real request and the exact reply the company wants, including when to say no. The model trains on these with the same next-word loss it used in pretraining.



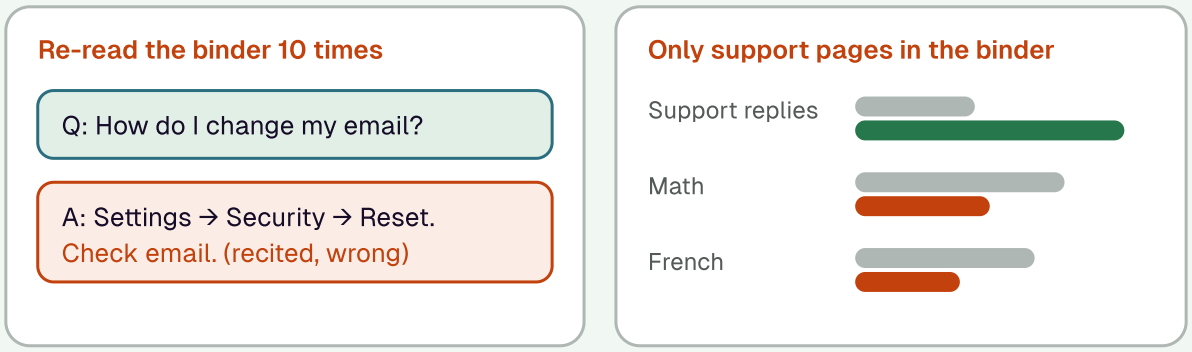
The loss counts 6 of the 10 tokens: only the answer teaches.

Like grading an exam, you mark what the student wrote, not the printed question. The model reads the whole conversation, but only the assistant's tokens count toward the loss. This is **loss masking**.



Mostly a change in how it answers, not in what it knows.

After a few thousand pages the new hire answers in the right format and tone. What they know barely moves. The LIMA study (Zhou et al., 2023) got strong chat behaviour from just 1,000 careful examples. Hard skills such as math and code still need far more data, and usually RL. Bar lengths are illustrative.



Fixes: 1 to 3 passes, mix in general data, pick checkpoints with evals.

Two ways it goes wrong. Too many passes and the model memorises the binder and recites old answers to new questions (**overfitting**). A narrow binder makes it worse at things it used to do well (**catastrophic forgetting**). Grey bars show before, coloured bars after.

What SFT actually changes

Plain words

A well-read intern learning the house style

Picture a new hire who has read every book in the library but has never worked in an office. Ask them a question and they might answer, or they might write the next page of a novel, or ask you a question back. They know a lot. They just don't know the job.

SFT is the training binder: "Here is a question a customer asked. Here is exactly how we answer." After a few thousand of these, the intern answers in the right format, in the right tone, and knows when to say "I can't help with that." The binder teaches very little new knowledge. Mostly it teaches how to use the knowledge they already have.

For leaders

SFT shapes behaviour cheaply. It rarely adds capability.

The LIMA paper (Zhou et al., 2023) fine-tuned a 65B model on only 1,000 carefully chosen examples and got responses that humans often preferred to those of much larger tuning efforts. The authors called this the **superficial alignment hypothesis**: almost all knowledge comes from pretraining, and SFT mostly selects which style and format to use. Later work showed it is only partly true. Format

and persona do come cheaply. Reliable gains on hard skills such as math, code, and tool use need much more data, often hundreds of thousands to millions of examples, and usually RL afterwards.

Cheap to change with SFT

Response format, tone, persona, refusal style, following a chat template, calling tools in the right syntax.

Expensive to change with SFT

Factual knowledge, deep reasoning, robustness on hard problems. SFT can teach a model to *sound* like it reasons without reasoning better.

Usually the bottleneck

Data quality and coverage, not compute. A single SFT run on a frontier model is small next to pretraining.

The real risk

Quietly damaging capabilities the base model had. You only see it if you measure it.

Principal asks

- What behaviour is this SFT run supposed to change, and which eval shows that change?
- Which capabilities could it break, and is there a regression suite for them?
- Who wrote the data, how was quality checked, and what fraction did a human actually read?
- Are we teaching a skill, or teaching the model to imitate the surface of a skill?

Under the hood

Same loss, different data, a masked sum

SFT minimises the conditional negative log likelihood of the response y given the prompt x, over a dataset of demonstrations $\mathcal{D}$:

$$\mathcal{L}_{\text{SFT}}(\theta) = -\mathbb{E}_{(x,y)\sim\mathcal{D}}\left[\frac{1}{|y|}\sum_{t=1}^{|y|}\log p_\theta(y_t \mid x, y_{<t})\right]$$

Identical to the pretraining objective, except the sum runs only over response tokens y_t. The prompt x is context, not a target.

Viewed as imitation learning, SFT is behaviour cloning: the model learns to copy the demonstrator's action at each step, always conditioned on the demonstrator's own previous tokens. That is called **teacher forcing**. At inference the model conditions on its *own* previous tokens, so small errors can compound into states the training data never covered. This exposure bias is one reason on-policy methods (RL, on-policy distillation) often help after SFT.

Chat templates and special tokens

Plain words

Stage directions in a script

A play script marks who is speaking: *HAMLET:*, *OPHELIA:*. A chat model needs the same thing. Special markers wrap each turn so the model knows where the user stops, where it should start talking, and when to stop. Get the markers wrong and the model gets confused, like an actor reading someone else's lines.

For leaders

Template mismatches cause silent, expensive bugs

The template is a contract between training, serving, and evaluation. If the eval harness formats prompts differently from training (a missing system turn, a different end-of-turn token, an extra newline), scores can drop several points with no change to the model. It is one of the first things to check when an eval result looks wrong.

> ⚠ **Watch out**
>
> Common template bugs: the end-of-turn token is never trained, so the model rambles past its answer; the tokenizer adds a second beginning-of-sequence token; tool-call turns use a role the serving stack doesn't know; the eval harness applies the base model's template to the chat model.

Under the hood

What a formatted conversation looks like

Templates differ by model family. A ChatML-style layout (illustrative) looks like this:

```text
<|im_start|>system
You are a concise assistant.<|im_end|>
<|im_start|>user
What is 17 * 3?<|im_end|>
<|im_start|>assistant
51<|im_end|>
```

The special markers are single tokens, either reserved in the pretraining vocabulary (as in Llama 3) or added before SFT; only when they are new are their embeddings learned during SFT. In Hugging Face tooling the template is stored with the tokenizer as a Jinja string and applied with `tokenizer.apply_chat_template`. Use the same function in training, serving, and the eval harness, and check a rendered example in each by eye.

Loss masking: which tokens teach

Plain words

Grade the answer, not the question

When you mark a student's exam you grade what they wrote, not the printed questions. SFT works the same way. The model reads the whole conversation, but it is only graded on the assistant's words. Otherwise it would spend effort learning to write user questions, which isn't its job.

For leaders

A one-line choice that changes what the model learns

Masking decides what behaviour you are paying to teach. Training on user turns teaches the model to imitate users, which can leak into answers that start interviewing the user back. Training on tool outputs teaches it to hallucinate tool results instead of waiting for them. Training only on the final turn of a multi-turn conversation wastes most of the data. Most teams default to "all assistant turns, never tool or environment output," and treat any change as an experiment with its own eval.

Under the hood

Labels of -100

In PyTorch, `CrossEntropyLoss` ignores label positions set to `-100` by default. Masking is done by copying the input ids into labels and overwriting non-target spans:

python

```python
import torch

IGNORE = -100  # CrossEntropyLoss(ignore_index=-100) skips these positions

def build_labels(input_ids, spans):
    """spans: list of (start, end, role) over token positions.
    Only assistant tokens, including the end-of-turn token, become targets."""
    labels = torch.full_like(input_ids, IGNORE)
    for start, end, role in spans:
        if role == "assistant":
            labels[start:end] = input_ids[start:end]
    return labels

ids = torch.arange(12)                       # stand-in token ids
spans = [(0, 4, "system"), (4, 7, "user"), (7, 12, "assistant")]
print(build_labels(ids, spans))
# tensor([-100, -100, -100, -100, -100, -100, -100,    7,    8,    9,   10,   11])
```

rust

```rust
// Plain Rust, no ML framework: the same math on small vectors.
const IGNORE: i64 = -100; // CrossEntropyLoss(ignore_index=-100) skips these positions

/// spans: list of (start, end, role) over token positions.
/// Only assistant tokens, including the end-of-turn token, become targets.
fn build_labels(input_ids: &[i64], spans: &[(usize, usize, &str)]) -> Vec<i64> {
    let mut labels = vec![IGNORE; input_ids.len()];
    for &(start, end, role) in spans {
        if role == "assistant" {
            labels[start..end].copy_from_slice(&input_ids[start..end]);
        }
    }
    labels
}

fn main() {
    let ids: Vec<i64> = (0..12).collect(); // stand-in token ids
    let spans = [(0, 4, "system"), (4, 7, "user"), (7, 12, "assistant")];
    println!("{:?}", build_labels(&ids, &spans));
```

```
    // [-100, -100, -100, -100, -100, -100, -100, 7, 8, 9, 10, 11]
}
```

One detail with real consequences: the loss is usually averaged over target tokens in the batch. Long responses then get more total weight than short ones. Some recipes average per sequence first, so every example counts equally. Neither is wrong, but changing it shifts the model toward or away from long answers.

Interactive

Loss-masking visualizer (shown at its default setting)

Each box is a token. Coloured tokens add to the loss; grey ones are context only. Switch schemes and watch how much of the batch actually teaches.

SYSTEM	<|im_start|>system You are a helpful travel assistant . <|im_end|>
USER	<|im_start|>user What ' s the weather in Lisbon tomorrow ? <|im_end|>
ASSISTANT	<|im_start|>assistant get _ weather (city = " Lisbon ") <|im_end|>
TOOL	<|im_start|>tool { " high " : 22 , " low " : 15 , " rain " : 0 . 1 } <|im_end|>
ASSISTANT	<|im_start|>assistant Tomorrow in Lisbon : high 22 ° C , low 15 ° C , small chance of rain . <|im_end|>
USER	<|im_start|>user Should I pack a jacket ? <|im_end|>
ASSISTANT	<|im_start|>assistant Yes , a light jacket for the evening . <|im_end|>

Tokens in example	Tokens in the loss	Share of tokens that teach
95	41	43%

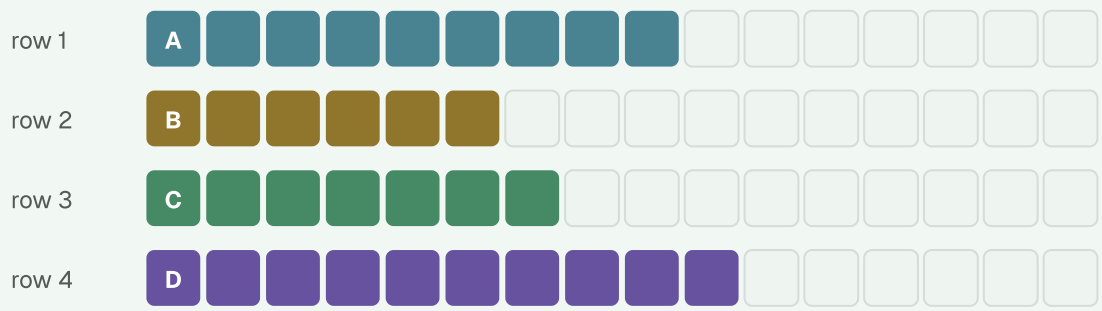
The usual default. Every assistant turn, including the tool call and its end-of-turn marker, is a target. Context tokens shape attention but add nothing to the loss. Words stand in for tokens here; real tokenizers split differently. Orange marks tokens that are trained but should not be.

Packing and attention masking

Diagram

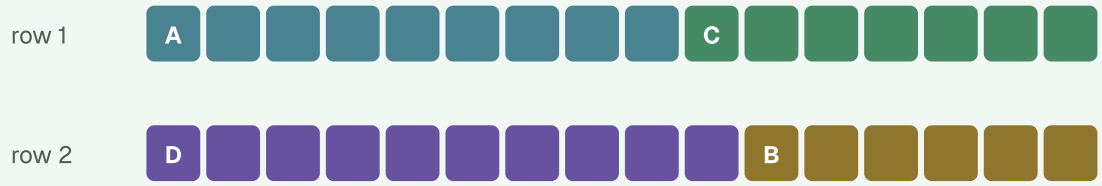
Packing the moving van

Why fine-tuning packs examples together, and the wall it needs between them.



Each row holds 16 tokens. Half of all slots are padding.

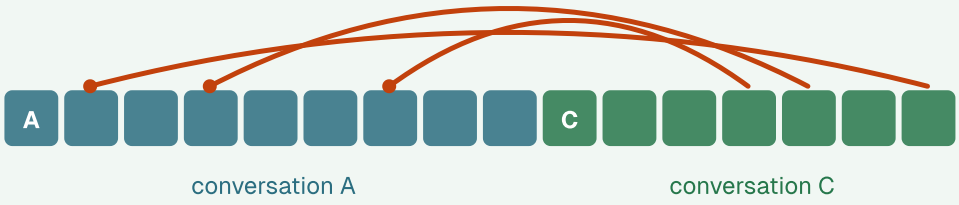
GPUs process fixed-length rows. Short conversations leave the rest of the row as padding, like a moving van carrying one chair. Here 32 of 64 slots do no work.



Same four conversations, two rows instead of four.

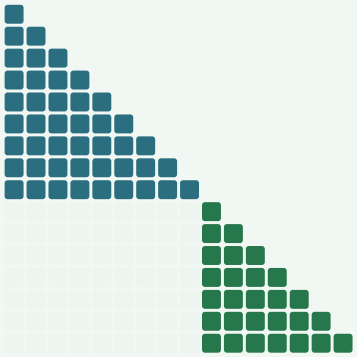
About half the compute for this batch.

Packing places conversations end to end so every slot holds a real token. When example lengths vary a lot, it can cut fine-tuning compute by 2× or more.



Without walls, C can attend to A, which it will never see at inference.

The catch: in a packed row, a token in conversation C could look back at conversation A. That is a small leak inside the training batch, and it teaches the model to rely on context that will not exist in real use.



who can see whom (row 1)

Walls between conversations

Coloured cells: allowed. Grey: blocked.

Each block is causal on its own.

Positions restart for each conversation:

A: 0 1 2 … 8 C: 0 1 2 … 6

Kernels take sequence boundaries instead of a full mask (e.g. cu_seqlens).

The fix is a **block-diagonal attention mask**: causal inside each conversation, blocked across them, with position ids restarting at zero. Ask whether your training stack does this whenever it packs.

Plain words

Filling the moving van

GPUs process fixed-length rows. Short conversations leave most of a row empty, like a moving van carrying one chair. Packing puts several conversations end to end in one row so nothing is wasted. The catch: you have to put up walls between them so one conversation can't peek at another.

For leaders

Throughput versus correctness

Packing can cut SFT compute by 2× or more when examples vary in length. Without proper attention masking, though, tokens in conversation B can attend to conversation A. That is a subtle form of contamination inside the training batch, and it can teach the model to draw on context that won't exist at inference. Ask whether the training stack uses per-document attention masks (often done with variable-length attention kernels) and resets position ids per document.

Under the hood

Block-diagonal causal masks

With packing, the attention mask becomes block diagonal: causal within each document, zero across documents. Implementations pass cumulative sequence lengths (for example `cu_seqlens` in FlashAttention's variable-length API) instead of materialising a full mask. Position ids restart at 0 for each document so rotary embeddings see the same positions they would without packing.

Data: quality, quantity, and mix

Plain words

You become what you imitate

The model will copy whatever is in the examples, including mistakes, padding, and bad habits. A thousand excellent examples can beat a hundred thousand sloppy ones. And if every example is about cooking, the model gets worse at everything else.

For leaders

Where SFT budgets actually go

Quality filtering

often beats collecting more. Remove wrong answers, refusals of fine requests, and answers that hedge or pad. Each one teaches a habit.

Diversity

of tasks, formats, and difficulty matters more than raw count. Open recipes such as Tülu 3 (Lambert et al., 2024) mix general chat, math, code, safety, multilingual, and instruction-following sets, and tune the mixture against a broad eval suite.

Mixture weights

are hyperparameters. Upweighting one skill steals capacity from others. Run ablations, not intuition.

Human versus synthetic data

Most modern SFT data is model-generated and filtered (see distillation and synthetic data). Human data stays essential for the hardest tasks, for safety, and for checking that synthetic data isn't drifting.

Multi-turn and tool-use data

must include realistic failures: tool errors, ambiguous requests, users changing their minds. A model trained only on clean single-turn data breaks in real agent traces.

Watch out

SFT data written by a stronger model teaches the student to sound like that model, confidence included. If the teacher states things it can't verify, the student learns to hallucinate in the same voice. Fluency goes up faster than accuracy, and judge-based evals reward fluency.

Epochs, overfitting, and forgetting

Plain words

Memorising the binder, forgetting the library

Read the training binder once and you learn the style. Read it ten times and you start reciting the exact examples word for word, which doesn't help with new questions. Worse, if the binder only covers a few topics, the intern can start forgetting things they knew before. That is **catastrophic forgetting**.

For leaders

Pick checkpoints with evals, not loss

Typical SFT runs 1 to 3 epochs. Validation loss often starts rising after the first or second epoch while downstream scores keep improving for a while, because the model gets more decisive about format. So validation loss alone is a poor stopping rule. Teams save several checkpoints and choose with a broad eval suite.

Forgetting is the cost leaders most often miss. Narrow SFT, such as a single domain or one customer's format, can reduce general reasoning, safety behaviour, or multilingual ability. Standard mitigations are mixing in general instruction data (replay), lower learning rates, fewer epochs, and LoRA, which tends to forget less because it changes fewer directions in weight space.

Under the hood

Why validation loss and quality diverge

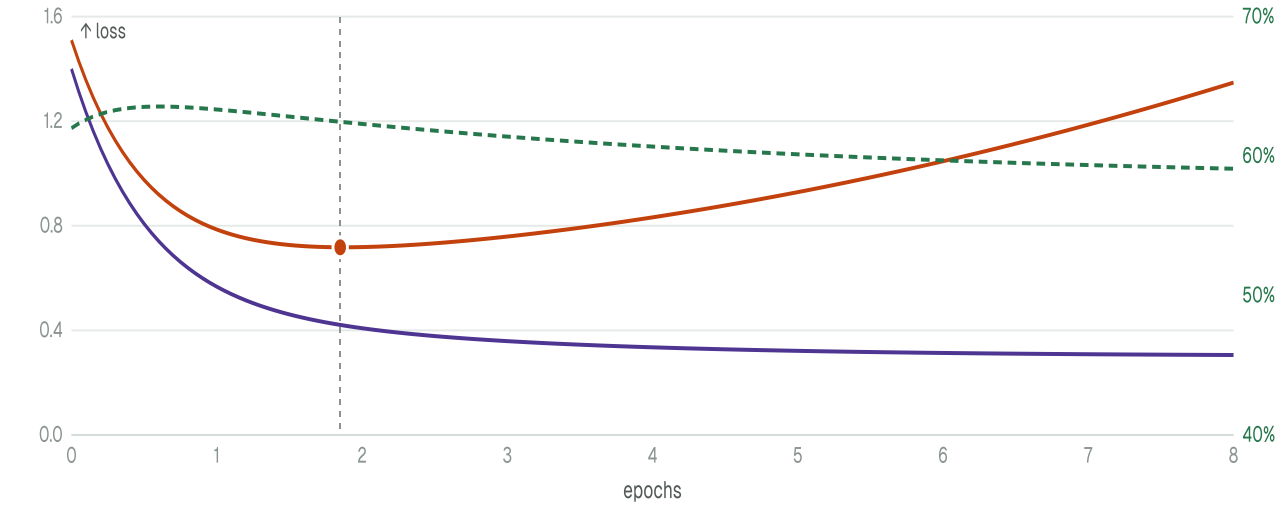
Cross-entropy on a held-out set penalises confident wrong tokens heavily. As SFT sharpens the output distribution (lower entropy), a few confidently wrong tokens on held-out examples can raise the average loss even while greedy outputs get better. Loss measures calibration of the whole distribution; quality metrics usually look at one sampled or greedy answer. Report both and expect them to disagree after the first epoch.

To detect forgetting, evaluate the base model and every candidate checkpoint on the same frozen general suite, and test the difference with a paired method from Part II, since the items are shared.

Interactive

Overfitting and forgetting over epochs (shown at its default setting)

Illustrative curves from a simple model, not a real run. Shrink the dataset or lower its diversity and watch validation loss turn up and the general benchmark fall.



Train loss · Validation loss · General benchmark (right axis)

Best epoch by validation	Benchmark at best epoch	Benchmark after 8 epochs
1.9	62.5%	59.1%

Validation loss bottoms out around epoch 1.9. Training longer keeps lowering train loss while validation rises: the model is memorising examples. Running all 8 epochs would cost 3.4 points of general capability compared with stopping early. The base model scores 62% on the general benchmark before SFT.

Evaluating an SFT run

Eval lens

Held-out loss is necessary but not enough. A credible SFT evaluation has four parts:

1. **Target behaviour.** The thing the run was meant to change: format adherence, tool-call validity, refusal accuracy, domain task pass rate. Measure it with intervals.
2. **Capability regression suite.** A fixed, broad set (reasoning, code, math, knowledge, multilingual, safety) compared against the parent checkpoint with paired tests. Any drop beyond noise is a finding.
3. **Pairwise win rate** against the previous model on realistic prompts, judged by humans or a validated LLM judge, with position swapping and length control.
4. **Contamination check.** Synthetic SFT data can contain benchmark items the teacher saw. Run n-gram and embedding overlap against every eval set (see contamination).

Principal asks

- What did the base model score on the regression suite, and what is the paired delta with an interval?
- Did win rate go up because answers got better, or because they got longer?
- Which checkpoint did we pick, and was it chosen on the same data we are now reporting? (If so, the number is optimistically biased.)
- If we removed the 10% lowest-quality examples, would the result change?

Part IV · Chapter 32

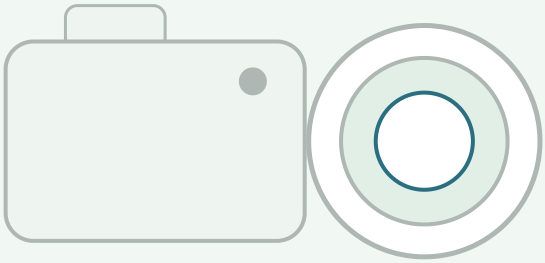
LoRA and parameter-efficient tuning

Full fine-tuning updates every weight in the model. Parameter-efficient fine-tuning (PEFT) freezes the pretrained weights and trains a small add-on instead. LoRA is by far the most used method. It makes customisation cheap enough to run on one GPU, and lets one server hold thousands of variants of the same base model.

Diagram

Clip-on filters for a giant lens

LoRA, QLoRA, adapter serving, and merging, told with a camera.



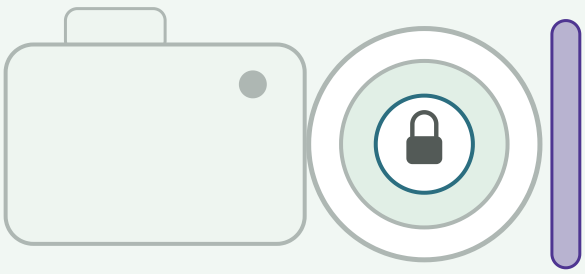
The base model

8 billion trained weights

Rebuild the whole lens per job
= full fine-tuning

About 16 bytes of GPU memory per weight: roughly 128 GB

Think of the pretrained model as a huge, expensive camera lens. Full fine-tuning regrinds the whole lens for every job: every weight changes, and training needs its weights, gradients, and optimizer state in memory at once.

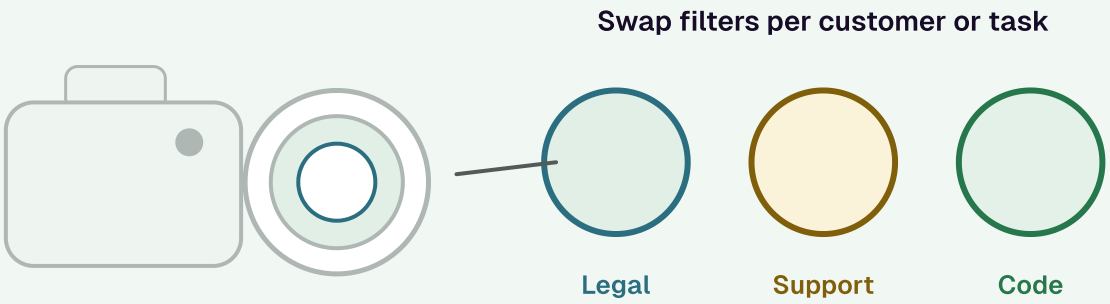


Clip on a thin filter

LoRA adapter: under 1% of the weights, trained

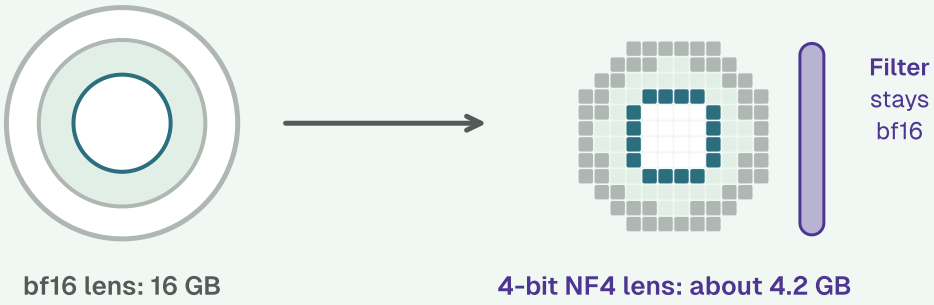
The lens stays frozen. Optimizer memory almost disappears.

LoRA freezes the lens and clips on a thin filter that changes the look. Only the filter is trained, so the gradients and optimizer state shrink to almost nothing. The frozen base dominates memory.



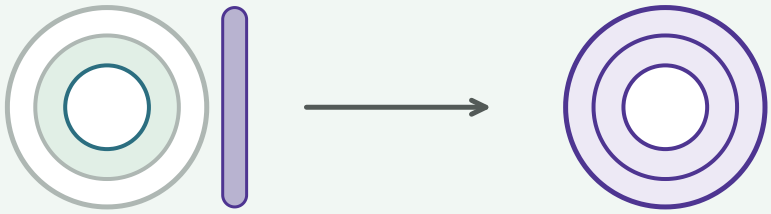
One base in memory serves many adapters (S-LoRA, Punica).

Because the lens never changes, one copy can serve many filters. A product can keep a separate adapter for every customer, each only megabytes, and batch their requests through the same base model.



Sizes for an 8B model, weights only.

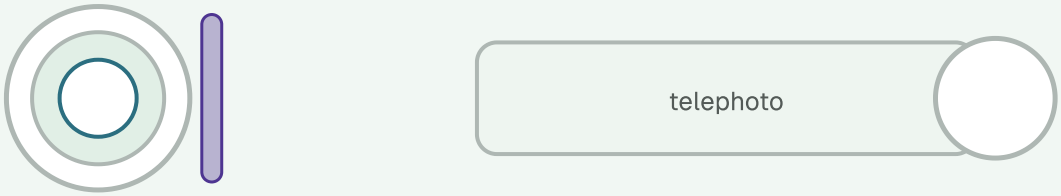
QLoRA (Dettmers et al., 2023) also compresses the frozen lens to 4 bits with a format called NF4. Gradients flow through the compressed lens into the full-precision filter. The paper fine-tuned a 65B model on a single 48 GB GPU.



$$W = W_0 + (\alpha/r)\cdot B\cdot A$$

Fuse the filter into the lens: no extra latency at serving.

For a single deployment you can fuse the filter into the lens: add the low-rank update to the frozen weights once, and serve a normal model with zero extra cost. Fusing into a 4-bit base is only approximate, so evaluate the model you actually ship.



Filter: new tint, same lens

A whole new lens: needs regrinding

LoRA learns less and forgets less (Biderman et al., 2024).

A filter can change style, format, and domain behaviour. It cannot turn a wide-angle lens into a telephoto. For a genuinely new capability, or billions of training tokens, full fine-tuning or continued pretraining usually wins.

Why not just fine-tune everything?

Plain words

Sticky notes instead of rewriting the book

Full fine-tuning is like reprinting a 1,000-page encyclopedia every time you want to change how it reads. LoRA leaves the book alone and adds a thin stack of sticky notes that adjust it. The notes are tiny, you can swap them in and out, and you can keep a different stack for each customer. When you want a final copy, you can even print the notes into the pages.

For leaders

The cost picture

Full fine-tuning with the Adam optimiser in mixed precision needs about 16 bytes of GPU memory per parameter before activations: the weights, their gradients, a full-precision master copy, and two optimiser statistics. That is roughly 128 GB for an 8B model and over 1 TB for a 70B model, so full fine-tuning means a multi-GPU cluster. LoRA trains well under 1% of the parameters, so the optimiser and gradient memory nearly vanish and the frozen base dominates. QLoRA goes further and stores the frozen base in 4 bits.

Cheaper training

Fits on far fewer GPUs. The original LoRA paper reports about 3× less GPU memory and 10,000× fewer trainable parameters than full fine-tuning GPT-3 175B (Hu et al., 2021).

Cheap storage

An adapter is megabytes to a few gigabytes, not a full copy of the model.

Cheap serving

Many adapters share one base model in memory. Systems such as S-LoRA (Sheng et al., 2023) and Punica (Chen et al., 2023) batch requests for different adapters together.

Less forgetting

Because LoRA changes few directions, it tends to damage general ability less, at the price of learning less from large datasets.

How LoRA works

Plain words

A big change described by a few knobs

A weight matrix in a model might have 16 million numbers. LoRA bets that the *change* you need during fine-tuning is simple, so it can be described with far fewer numbers: two thin matrices that multiply together to make a full-size correction. The thinness is called the **rank**. Rank 8 means the whole correction is built from 8 basic patterns.

Under the hood

The low-rank update

For a frozen weight $W_0 \in \mathbb{R}^{d \times k}$, LoRA learns $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$ with $r \ll \min(d, k)$:

$$h = W_0 x + \frac{\alpha}{r} BAx$$

A starts random (Gaussian), B starts at zero, so $BA = 0$ and training begins exactly at the pretrained model. α is a fixed scale; dividing by r keeps the update size roughly stable when you change the rank.

The update has $r(d + k)$ trainable parameters instead of dk. For a 4096 × 4096 projection at rank 16, that is 131,072 instead of 16.8 million, under 1%. After training, $W = W_0 + \frac{\alpha}{r} BA$ can be merged into the base weights, so a single-adapter deployment has no extra inference cost.

python

```python
import torch, torch.nn as nn

class LoRALinear(nn.Module):
    def __init__(self, base: nn.Linear, r=16, alpha=32):
        super().__init__()
        self.base = base
        for p in self.base.parameters():
            p.requires_grad = False                    # the pretrained weight stays frozen
        d_out, d_in = base.weight.shape
        self.A = nn.Parameter(torch.randn(r, d_in) / r**0.5)
        self.B = nn.Parameter(torch.zeros(d_out, r))    # zero init: BA = 0 at step 0
        self.scale = alpha / r

    def forward(self, x):
        return self.base(x) + self.scale * (x @ self.A.T @ self.B.T)

    @torch.no_grad()
    def merge(self):
        # fold the adapter into the weight for zero-overhead serving
```

```
        self.base.weight += self.scale * self.B @ self.A

layer = LoRALinear(nn.Linear(4096, 4096, bias=False), r=16)
trainable = sum(p.numel() for p in layer.parameters() if p.requires_grad)
print(trainable, f"{trainable / 4096**2:.2%}")  # 131072 0.78%
```

rust

```
// Plain Rust, no ML framework: the same math on small vectors.
// Matrices are flat row-major Vec<f64>. The random init uses a small inline PRNG
// (SplitMix64), so the draws differ from torch; the parameter count does not.

struct SplitMix64(u64);

impl SplitMix64 {
    fn next_u64(&mut self) -> u64 {
        self.0 = self.0.wrapping_add(0x9E37_79B9_7F4A_7C15);
        let mut z = self.0;
        z = (z ^ (z >> 30)).wrapping_mul(0xBF58_476D_1CE4_E5B9);
        z = (z ^ (z >> 27)).wrapping_mul(0x94D0_49BB_1331_11EB);
        z ^ (z >> 31)
    }
    fn uniform(&mut self) -> f64 {
        (self.next_u64() >> 11) as f64 / (1u64 << 53) as f64 // in [0, 1)
    }
    fn normal(&mut self) -> f64 {
        // Box-Muller; 1 - u keeps the log argument in (0, 1]
        let (u1, u2) = (1.0 - self.uniform(), self.uniform());
        (-2.0 * u1.ln()).sqrt() * (2.0 * std::f64::consts::PI * u2).cos()
    }
}

struct LoRALinear {
    weight: Vec<f64>, // d_out x d_in; the pretrained weight stays frozen
    a: Vec<f64>,      // r x d_in, trainable
    b: Vec<f64>,      // d_out x r, trainable
    d_out: usize,
    d_in: usize,
    r: usize,
    scale: f64,
}

impl LoRALinear {
    fn new(weight: Vec<f64>, d_out: usize, d_in: usize, r: usize, alpha: f64, rng: &mut SplitMix64) -> Self {
        let a = (0..r * d_in).map(|_| rng.normal() / (r as f64).sqrt()).collect();
        let b = vec![0.0; d_out * r]; // zero init: BA = 0 at step 0
        LoRALinear { weight, a, b, d_out, d_in, r, scale: alpha / r as f64 }
    }

    fn trainable_params(&self) -> usize {
        self.a.len() + self.b.len() // only A and B get gradients
    }

    fn forward(&self, x: &[f64]) -> Vec<f64> {
        // base(x) + scale * B (A x), never forming the d_out x d_in product BA
        let ax: Vec<f64> = (0..self.r)
            .map(|i| (0..self.d_in).map(|j| self.a[i * self.d_in + j] * x[j]).sum())
            .collect();
        (0..self.d_out)
            .map(|o| {
```

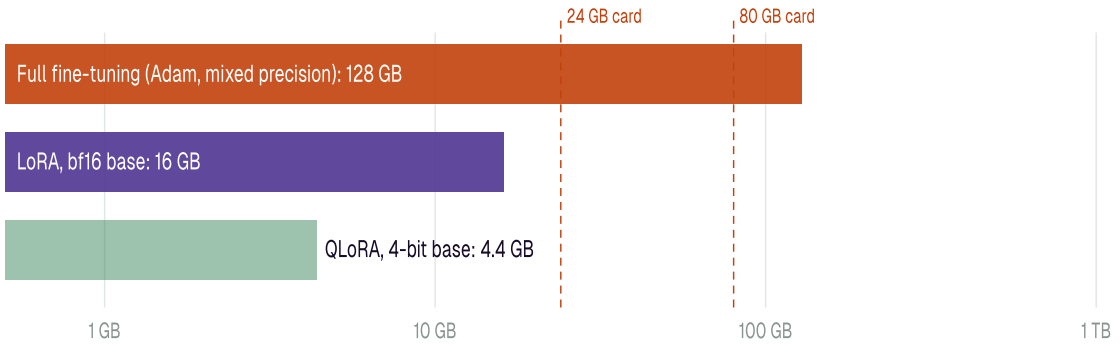
```
                let base: f64 = (0..self.d_in).map(|j| self.weight[o * self.d_in + j] * x[j]).sum();
                let lora: f64 = (0..self.r).map(|i| self.b[o * self.r + i] * ax[i]).sum();
                base + self.scale * lora
            })
            .collect()
    }

    fn merge(&mut self) {
        // fold the adapter into the weight for zero-overhead serving
        for o in 0..self.d_out {
            for j in 0..self.d_in {
                let ba: f64 = (0..self.r).map(|i| self.b[o * self.r + i] * self.a[i * self.d_in + j]).sum();
                self.weight[o * self.d_in + j] += self.scale * ba;
            }
        }
    }
}

fn main() {
    let d = 4096;
    let mut rng = SplitMix64(0);
    let layer = LoRALinear::new(vec![0.0; d * d], d, d, 16, 32.0, &mut rng);
    let trainable = layer.trainable_params();
    let frac = trainable as f64 / (d * d) as f64;
    println!("{} {:.2}%", trainable, 100.0 * frac); // 131072 0.78%
}
```

Practical defaults have shifted since 2021. The original paper adapted only the attention query and value projections. Later work, including QLoRA and the Thinking Machines "LoRA Without Regret" study (2025), found that adapting *all* linear layers, especially the MLP blocks, matters more than raising the rank. The same study reports that the best LoRA learning rate is about 10× the best full fine-tuning rate. Rank-stabilised LoRA (Kalajdzievski, 2023) scales by $\alpha/\sqrt{r}$ instead of α/r so that high ranks keep learning.

Interactive

LoRA parameter and memory calculator (shown at its default setting)

Pick a model shape, the matrices to adapt, and a rank. Memory counts weights, gradients, and Adam state only; activations come on top and depend on batch size and sequence length.



Model parameters	Trainable (LoRA)	Share trainable	Adapter file (bf16)
8.0B	6.8M	0.085%	14 MB

Weights, gradients, and optimizer state only: about 16 GB for this setup versus 128 GB for full fine-tuning, 8× less. Doubling the rank doubles the trainable parameters but barely moves total memory, because the frozen base dominates. Adding the MLP projections usually improves quality more than raising the rank.

Why a low-rank update is enough

Diagram

A big table that is really two short lists

What "low rank" means, with shop sales instead of matrices.

	Jan	Feb	Mar	Apr	May	Jun
Store A	2	3	5	4	3	2
Store B	4	6	10	8	6	4
Store C	6	9	15	12	9	6
Store D	8	12	20	16	12	8

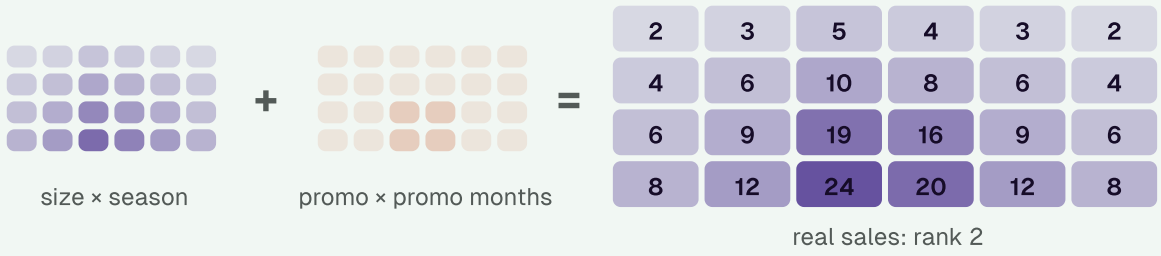
Monthly sales for 4 stores: 24 numbers.

Here is a table of monthly sales for four shops. It looks like 24 separate numbers. Look closer: bigger stores sell more, and March is busy everywhere.

	×	Jan	Feb	Mar	Apr	May	Jun
	×	2	3	5	4	3	2
Store A	1	2	3	5	4	3	2
Store B	2	4	6	10	8	6	4
Store C	3	6	9	15	12	9	6
Store D	4	8	12	20	16	12	8

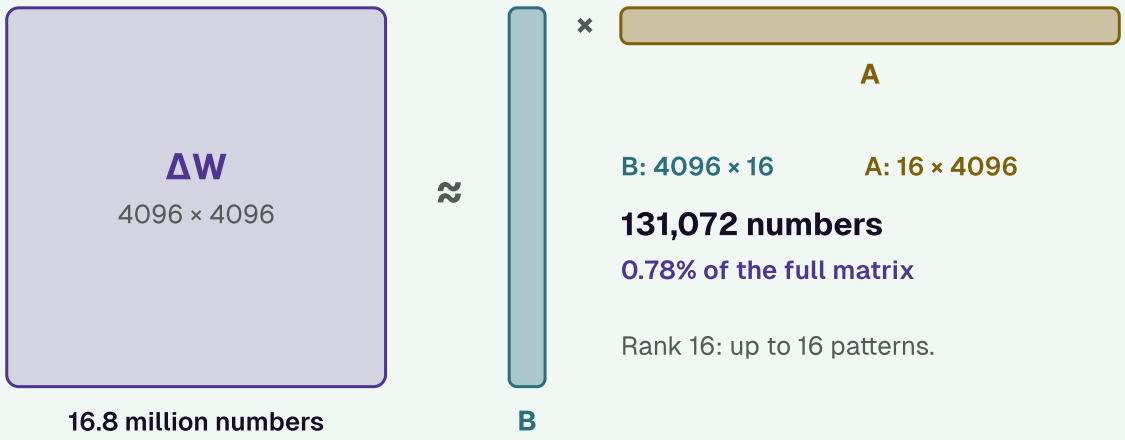
Size × season rebuilds every cell: 4 + 6 = 10 numbers.

Every cell is **store size × month busyness**. Two short lists, 4 numbers and 6 numbers, rebuild all 24. In linear algebra this table has **rank 1**: one row pattern times one column pattern.

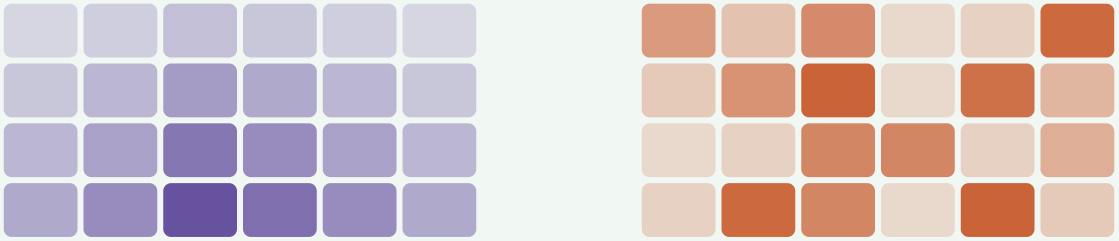


A spring promotion in two stores adds a second pattern.

Real tables have a few overlapping patterns. A spring promotion in stores C and D adds a second row-times-column pattern. Two patterns is **rank 2**. LoRA's rank *r* is how many patterns the weight change is allowed to have.



LoRA applies the same trick to a weight update. Instead of learning all 16.8 million entries of a 4096 × 4096 change, it learns two thin matrices whose product is the change. At rank 16 that is 131,072 numbers.



Structured: a few patterns rebuild it

Noise: needs all 24 numbers

Fine-tuning changes look more like the left table than the right.

Random noise has no shortcut, so a low-rank version loses most of it. Fine-tuning changes turn out to be far more structured: Aghajanyan et al. (2021) found fine-tuning works in a surprisingly small subspace. That is why a small rank usually captures what matters.

Plain words

Most changes are simple

A random scribble needs every pixel to describe it. A photo of the sky is mostly a smooth gradient and can be described by a few numbers. Fine-tuning changes turn out to look more like the sky than the scribble. That's why a few patterns capture most of the change.

Under the hood

Intrinsic dimension and the Eckart–Young theorem

The best rank-r approximation of a matrix in Frobenius norm keeps its top r singular values (Eckart–Young). The fraction of squared norm kept is

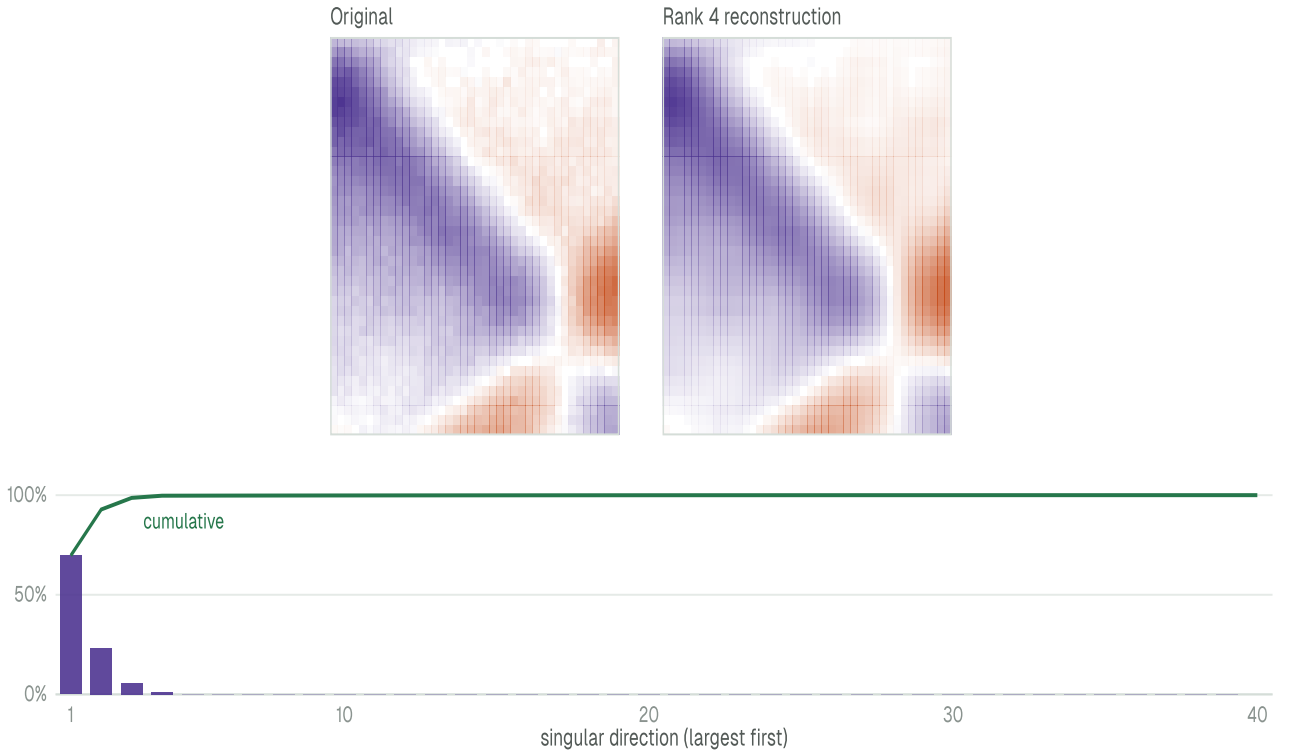
$$\frac{\|M_r\|_F^2}{\|M\|_F^2} = \frac{\sum_{i=1}^{r} \sigma_i^2}{\sum_i \sigma_i^2}$$

Aghajanyan et al. (2021) showed that fine-tuning a pretrained language model can reach most of its full performance while optimising in a random subspace of only hundreds to thousands of dimensions, and that larger, better-pretrained models have *lower* intrinsic dimension. LoRA exploits the same fact per weight matrix. The caveat from Biderman et al. (2024, "LoRA Learns Less and

Forgets Less"): on large-scale code and math training, full fine-tuning learns perturbations with rank 10 to 100× higher than typical LoRA settings, and LoRA falls behind. Low rank is a good bet for adapting behaviour, not for teaching large amounts of new knowledge.

Interactive **Low-rank approximation** (shown at its default setting)

A 40 × 40 matrix rebuilt from its top r singular directions (SVD computed in your browser). Structured matrices are captured by a few directions; random noise is not.



Energy captured	Numbers stored	vs full matrix
99.7%	320	20%

Like a real fine-tuning update, this matrix has a few strong directions and a little noise. Rank 4 captures nearly all of it. Storing a rank-4 copy takes r × (rows + columns) = 320 numbers instead of 1,600.

QLoRA, DoRA, and the rest of the family

For leaders

Which variant, when

Method	Idea	Use when	Cost
LoRA (Hu et al., 2021)	Trainable low-rank update on frozen weights	The default for adapting behaviour, style, domain format	Learns less than full FT on large datasets
QLoRA (Dettmers et al., 2023)	LoRA on top of a 4-bit quantised base	Memory is the constraint. The paper fine-tuned a 65B model on one 48 GB GPU	Slower steps from dequantisation; small quality risk from quantised base
DoRA (Liu et al., 2024)	Splits each weight into magnitude and direction; LoRA adapts the direction	You want closer-to-full-FT behaviour at LoRA cost	Some extra compute; merge still possible
Adapters (Houlsby et al., 2019)	Small bottleneck layers inserted in each block	Mostly historical in LLMs	Adds inference latency unless removed
Prefix / prompt tuning (Li & Liang, 2021; Lester et al., 2021)	Learn virtual tokens prepended to the input	Extremely small storage; many tasks	Uses context length; usually weaker than LoRA
Full fine-tuning	Update everything	Large datasets, new capabilities, the lab's own post-training	Most memory, most forgetting risk

> ⚠ **Watch out**
>
> Merging a LoRA adapter into a quantised base is not exact: you must dequantise, add, and requantise, which changes outputs slightly. Evaluate the artefact you actually serve, not the training-time model with adapter attached.

Under the hood

What QLoRA adds

- **NF4 (4-bit NormalFloat).** Pretrained weights are roughly normally distributed, so NF4 places its 16 levels at quantiles of a normal distribution. Each block of 64 weights has its own scale.
- **Double quantisation.** The per-block scales are themselves quantised, saving about 0.37 bits per parameter.
- **Paged optimisers.** Optimiser state can spill to CPU memory during spikes instead of crashing with out-of-memory errors.

Gradients flow through the dequantised 4-bit weights into the bf16 LoRA parameters; the base itself never updates.

LoRA beyond SFT

For leaders

RL, multi-tenant serving, and when to say no

RL. The Thinking Machines study (2025) found LoRA matches full fine-tuning for reinforcement learning even at very low rank, including rank 1. The intuition: a policy-gradient episode carries only a few bits of information (did it succeed?), so the total information learned is small enough to fit in a tiny adapter. That makes LoRA attractive for RL with verifiable rewards, where many runs and ablations are needed.

Multi-tenant products. Per-customer or per-task adapters on a shared base are the standard way to offer fine-tuning as a product. The eval burden scales with the number of adapters: each one needs a regression check against the base.

When to say no. If the goal is a new capability that the base model can't do at all, or the dataset is in the billions of tokens, LoRA is likely to underperform. Expect to need full fine-tuning or continued pretraining.

Eval lens

A LoRA result needs three comparisons, not one: the adapter versus the base on the target task (did it learn?), the adapter versus the base on a general regression suite (did it forget?), and the merged or quantised serving artefact versus the training-time model (did deployment change it?). Ranks and learning rates interact, so compare tuned configurations, not one LoRA setting against a tuned full fine-tune.

Principal asks

- Which modules did we adapt, at what rank, and did we tune the learning rate separately for LoRA?
- Is the gap to full fine-tuning on the target metric outside the noise, and what is the gap on the forgetting suite?
- Are we evaluating the merged, quantised artefact that ships?
- How many adapters will we serve, and who owns their regression testing?

Part IV · Chapter 33

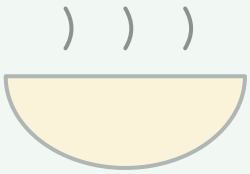
Distillation, synthetic data, and merging

Most training data for modern post-training is written by models, not people. A large teacher model generates examples, a filter keeps the good ones, and a smaller or newer student learns from them. This chapter covers the three ways models learn from other models: distillation, synthetic data with filtering, and merging weights directly.

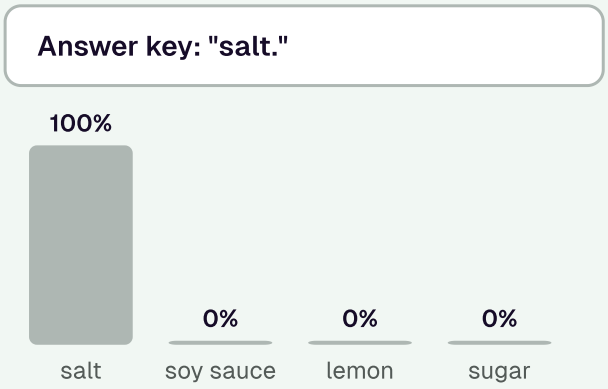
Diagram

The chef's tasting notes

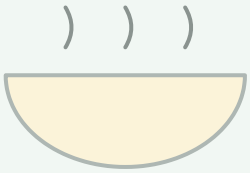
Why learning from a teacher's probabilities beats learning from an answer key.

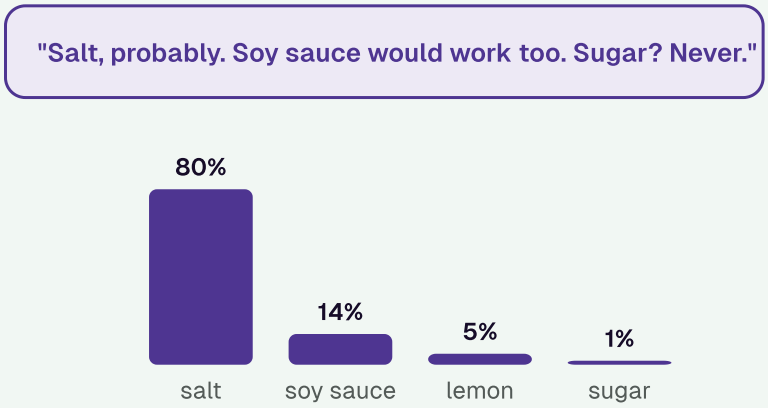


What does this soup need?

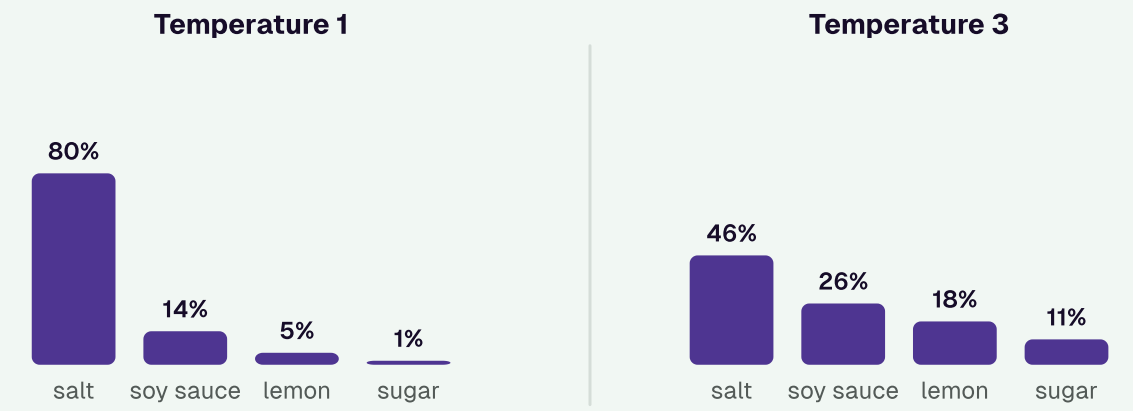


An answer key gives one right answer and nothing else. Training on it tells the student that salt is correct, and says nothing about whether soy sauce was a near miss or sugar was absurd. This is a **hard label**.

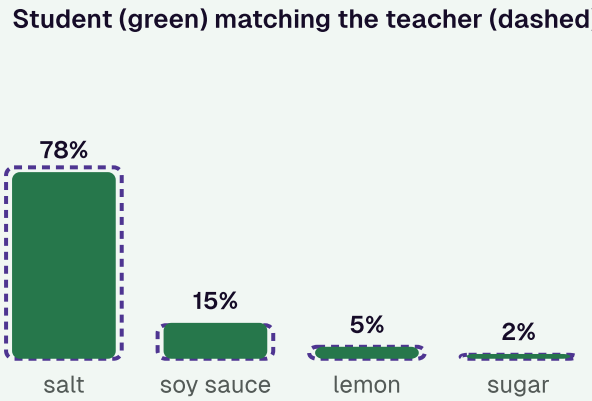




A master chef thinking out loud gives far more: a full set of probabilities. The runner-ups carry the "almost right" information Hinton and colleagues called **dark knowledge**. Distillation trains the small model on these **soft labels**.

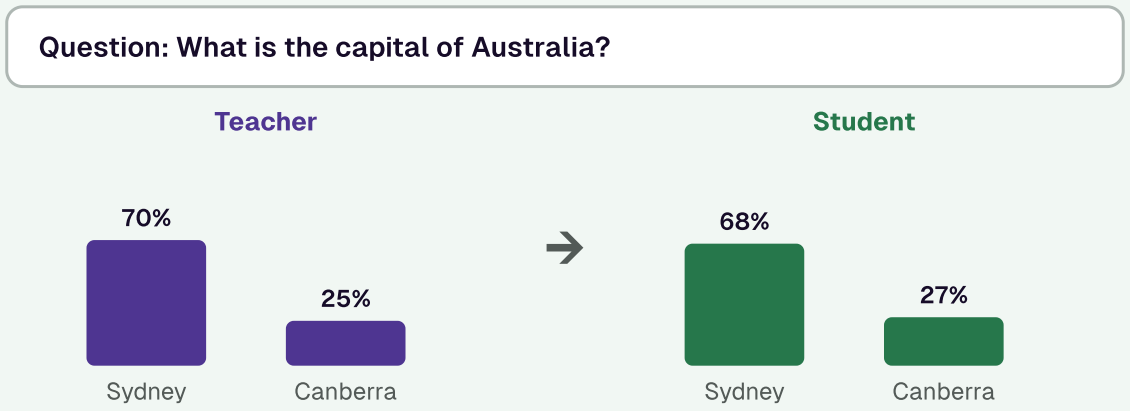


Raising the **temperature** divides the teacher's logits before the softmax, flattening the bars. The ranking stays the same, but the small differences between wrong answers become big enough for the student to learn from.



From one example it learns:
soy sauce is close,
sugar is absurd.

The student is trained to match the whole shape, usually by minimising the KL divergence between the two distributions. Each example now teaches a ranking, not just a winner, which is why small models distilled from big ones beat small models trained from scratch.



The catch: the student copies mistakes too, confidence included. If the teacher leans the wrong way (the answer is Canberra), so does the student. A distilled model is capped by its teacher and inherits its blind spots.

Knowledge distillation

Plain words

Learning from the teacher's hesitation

A student can learn from an answer key: "the answer is Paris." Or they can learn from a teacher who thinks out loud: "Paris, almost certainly. Lyon is a distant second. It's definitely not banana." The second version carries more information. It tells the student which wrong answers are nearly right and which are absurd. Distillation trains a small model on the big model's full set of probabilities, not just its single best answer. Hinton and colleagues called that extra information "dark knowledge."

For leaders

Why every lab distils

Distillation is how a lab turns one expensive frontier model into a family of cheaper ones. Small models trained from a strong teacher usually beat small models trained from scratch on the same compute. Gemma 2 (Google, 2024) trained its 2B and 9B models with distillation from a larger model in place of plain next-token prediction. DeepSeek-R1 (2025) fine-tuned smaller Qwen and Llama models on about 800K curated samples (roughly 600K R1 reasoning traces plus 200K non-reasoning examples) and reported that this beat running RL directly on the small models.

Logit distillation

Match the teacher's full probability distribution at each token. Needs the teacher's logits and a shared tokenizer, so it is mostly done inside one lab.

Sequence distillation

Train on text the teacher generated (Kim & Rush, 2016). Works through an API and across tokenizers. This is what "training on model outputs" usually means.

On-policy distillation

The student writes the text and the teacher grades every token (Agarwal et al., 2024). Fixes the student's own mistakes instead of only copying perfect examples.

Reasoning-trace distillation

Train on the teacher's step-by-step solutions. Fast way to give a small model long-form reasoning; inherits the teacher's errors and verbosity.

⚠ **Watch out**

A distilled student is capped by its teacher and copies its blind spots. If the teacher hallucinates on a topic, the student will too, with the same confidence. Distilling from a model you don't control also raises terms-of-service and licensing questions: many providers' terms restrict using outputs to build competing models, and open-weight licenses carry their own conditions. Get legal review before a data pipeline depends on another company's model.

● Under the hood

Soft targets, temperature, and which KL

With teacher logits z^T and student logits z^S, soften both with temperature τ:

$$p_i^{(\tau)} = \frac{\exp(z_i/\tau)}{\sum_j \exp(z_j/\tau)}, \qquad \mathcal{L} = (1-\lambda)\,\mathrm{CE}\big(y,\, p^{S,(1)}\big) + \lambda\,\tau^2\,\mathrm{KL}\big(p^{T,(\tau)} \,\|\, p^{S,(\tau)}\big)$$

Hinton, Vinyals & Dean (2015). The τ^2 factor keeps the soft-target gradients on the same scale as the hard-label term when τ changes. λ balances the two.

The direction of the KL matters for generative models:

- **Forward KL**, $\mathrm{KL}(p^T \| p^S) = \sum_i p_i^T \log \frac{p_i^T}{p_i^S}$, punishes the student for putting low probability anywhere the teacher puts mass. It is *mode covering*: a small student spreads itself thin and can produce blends of answers.
- **Reverse KL**, $\mathrm{KL}(p^S \| p^T)$, punishes the student for putting mass where the teacher doesn't. It is *mode seeking*: the student commits to the teacher's main modes. MiniLLM (Gu et al., 2024) argued reverse KL suits small generative students better. On-policy distillation scores the student's own samples against the teacher, most often with a per-token reverse KL (GKD, Agarwal et al. 2024, allows other divergences).

python

```python
import torch, torch.nn.functional as F

def kd_loss(student_logits, teacher_logits, labels, tau=2.0, lam=0.5):
    """Hinton-style distillation for a batch of next-token predictions.
    logits: [batch, vocab]; labels: [batch] token ids (-100 to ignore)."""
    hard = F.cross_entropy(student_logits, labels, ignore_index=-100)
    log_ps = F.log_softmax(student_logits / tau, dim=-1)
    log_pt = F.log_softmax(teacher_logits / tau, dim=-1)
    # forward KL(teacher || student); kl_div expects log-probs for the input
    soft = F.kl_div(log_ps, log_pt, log_target=True, reduction="batchmean")
    return (1 - lam) * hard + lam * tau**2 * soft

s, t = torch.randn(4, 10), torch.randn(4, 10)
print(kd_loss(s, t, torch.tensor([1, 2, 3, -100])).item() > 0)  # True
```

rust

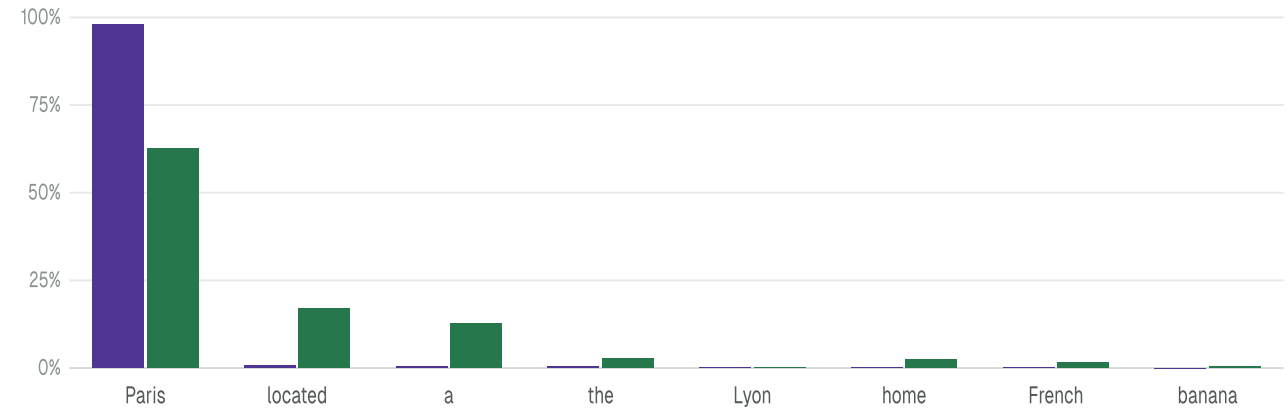
```rust
// Plain Rust, no ML framework: the same math on small vectors.
// The random logits come from a small inline PRNG (SplitMix64), so they differ from torch.randn.

fn log_softmax(logits: &[f64], tau: f64) -> Vec<f64> {
    // log-sum-exp with the max subtracted, so large logits cannot overflow
    let z: Vec<f64> = logits.iter().map(|l| l / tau).collect();
    let m = z.iter().cloned().fold(f64::NEG_INFINITY, f64::max);
    let lse = m + z.iter().map(|v| (v - m).exp()).sum::<f64>().ln();
    z.iter().map(|v| v - lse).collect()
}

/// Hinton-style distillation for a batch of next-token predictions.
/// logits: [batch][vocab]; labels: [batch] token ids (-100 to ignore).
```

```rust
fn kd_loss(student_logits: &[Vec<f64>], teacher_logits: &[Vec<f64>], labels: &[i64], tau: f64, lam: f64) -> f64 {
    let batch = student_logits.len() as f64;
    // hard: cross-entropy at temperature 1, averaged over the rows that are not ignored
    let (mut hard_sum, mut kept) = (0.0, 0);
    for (logits, &label) in student_logits.iter().zip(labels) {
        if label != -100 {
            hard_sum -= log_softmax(logits, 1.0)[label as usize];
            kept += 1;
        }
    }
    let hard = hard_sum / kept as f64;
    // forward KL(teacher || student), summed over the vocab and divided by batch ("batchmean")
    let mut soft = 0.0;
    for (s, t) in student_logits.iter().zip(teacher_logits) {
        let (log_ps, log_pt) = (log_softmax(s, tau), log_softmax(t, tau));
        soft += log_pt.iter().zip(&log_ps).map(|(lt, ls)| lt.exp() * (lt - ls)).sum::<f64>();
    }
    soft /= batch;
    (1.0 - lam) * hard + lam * tau * tau * soft
}

struct SplitMix64(u64);

impl SplitMix64 {
    fn next_u64(&mut self) -> u64 {
        self.0 = self.0.wrapping_add(0x9E37_79B9_7F4A_7C15);
        let mut z = self.0;
        z = (z ^ (z >> 30)).wrapping_mul(0xBF58_476D_1CE4_E5B9);
        z = (z ^ (z >> 27)).wrapping_mul(0x94D0_49BB_1331_11EB);
        z ^ (z >> 31)
    }
    fn uniform(&mut self) -> f64 {
        (self.next_u64() >> 11) as f64 / (1u64 << 53) as f64 // in [0, 1)
    }
    fn normal(&mut self) -> f64 {
        // Box-Muller; 1 - u keeps the log argument in (0, 1]
        let (u1, u2) = (1.0 - self.uniform(), self.uniform());
        (-2.0 * u1.ln()).sqrt() * (2.0 * std::f64::consts::PI * u2).cos()
    }
}

fn randn(rng: &mut SplitMix64, rows: usize, cols: usize) -> Vec<Vec<f64>> {
    (0..rows).map(|_| (0..cols).map(|_| rng.normal()).collect()).collect()
}

fn main() {
    let mut rng = SplitMix64(0);
    let (s, t) = (randn(&mut rng, 4, 10), randn(&mut rng, 4, 10));
    println!("{}", kd_loss(&s, &t, &[1, 2, 3, -100], 2.0, 0.5) > 0.0); // true
}
```

Interactive

Temperature and soft targets (shown at its default setting)

Next-word probabilities after "The capital of France is". Raise the temperature to reveal the teacher's ranking of wrong answers. Move the student toward the teacher and watch both KL divergences shrink.



Teacher top-1 prob	Teacher entropy (nats)	Forward KL(teacher‖student)	Reverse KL(student‖teacher)
97.8%	0.14	0.383	0.857

At T = 1.00 the teacher is almost certain of "Paris", so its soft targets look like a hard label and carry little extra information. The untrained student puts too much weight on "home" and "French" and too little on "Lyon". Forward and reverse KL differ because KL is not symmetric.

Synthetic data and rejection sampling

Diagram

Twenty shots, keep the sharp ones

How models make their own training data, and the bias hiding in the filter.

One prompt: photograph the bird

Take 20 shots (sample 20 answers).

Most are blurry. A couple are sharp.

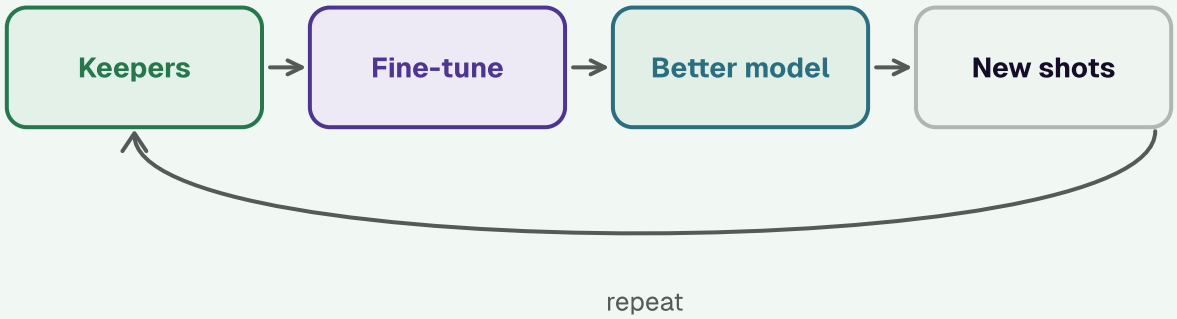
Making training data is easier when checking is easier than creating. Ask the model to solve the same problem many times, like a photographer firing off 20 shots of a moving bird.

A checker keeps 2 of 20

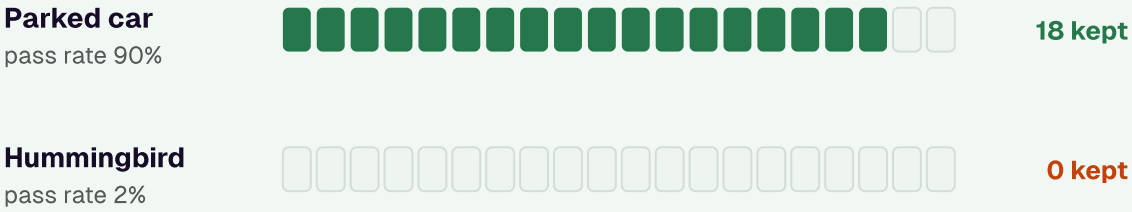
Checker = unit tests, an answer key, or a reward model.

The filter is the product: generating is cheap, judging is not.

A checker throws away every shot that is not sharp. For math it compares the final answer; for code it runs tests. Only the passing answers become training examples. This is **rejection sampling**.



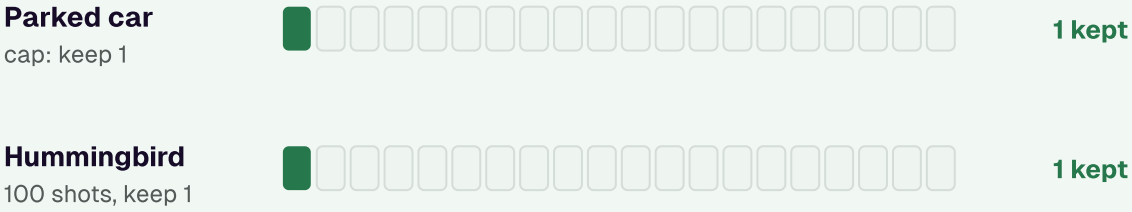
Train on the keepers, and the improved model takes better shots next round. Repeating this loop is the idea behind STaR (Zelikman et al., 2022) and rejection-sampling fine-tuning, and it is a simple offline cousin of RL.



Chance of even one sharp hummingbird in 20 shots: $1 - 0.98^{20} \approx 33\%$

The album fills with parked cars. The model gets better at what it could already do.

The hidden bias: easy prompts produce many keepers and hard prompts produce almost none. Expected keepers per prompt are 20 × pass rate, so the training set quietly over-represents problems the model can already solve.



Chance of at least one sharp shot in 100: $1 - 0.98^{100} \approx 87\%$

Cap examples per prompt, and spend extra samples on hard prompts.

Two fixes keep the album balanced: cap how many keepers any one prompt contributes, and spend more attempts on hard prompts. Watch the pass rate on the hardest prompts across rounds, not just the average.

Plain words

Write ten drafts, keep the ones that check out

If you can check an answer more easily than you can write it, you can make training data cheaply. Ask a model to solve a math problem ten times. Check each answer against the known result. Keep the correct solutions and train on them. Repeat with the improved model. This is the engine behind most reasoning data today.

For leaders

The filter is the product

Generating text is cheap. The value of a synthetic data pipeline sits in its filter: unit tests for code, exact-match answers for math, a reward model or LLM judge for open-ended tasks, and human spot checks throughout. The main methods:

- **Self-Instruct** (Wang et al., 2023) and **Evol-Instruct** (Xu et al., 2023): a model writes new instructions from seed examples, or rewrites existing ones to be harder or more varied.
- **STaR** (Zelikman et al., 2022): generate reasoning, keep traces that reach the right answer, fine-tune, repeat. For failures, give the answer as a hint and ask for a rationale ("rationalisation").
- **Rejection sampling fine-tuning** (Yuan et al., 2023; also used in Llama 2): sample many responses, keep the best by verifier or reward model, and fine-tune on them. It is a simple offline relative of RL.

Watch out

Rejection sampling quietly reshapes your data toward problems the model can already solve. Prompts with a 1% pass rate rarely produce a kept example, so the training set over-represents easy problems. The model gets better at what it already does well. Watch coverage of hard prompts, cap examples per prompt, and spend extra samples on hard prompts.

Under the hood

Yield, coverage, and model collapse

For a prompt with pass rate p and k samples, the expected number of correct samples is kp, and the chance of at least one is $1 - (1 - p)^k$. Averaging over prompts:

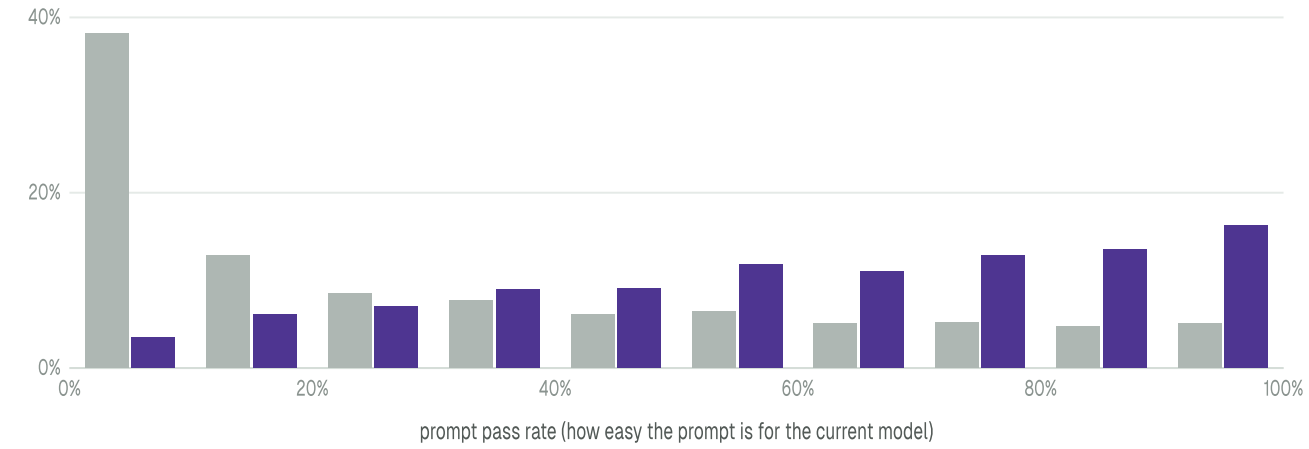
$$\text{coverage} = \mathbb{E}_p\left[1 - (1 - p)^k\right], \qquad \text{kept share from prompt } i \propto p_i$$

Without a cap, each prompt contributes in proportion to its pass rate, so easy prompts dominate. Capping at one kept example per prompt changes the weight to $1 - (1 - p_i)^k$, which flattens toward uniform as k grows.

Model collapse. Shumailov et al. (2024, *Nature*) showed that models trained recursively on their own outputs, generation after generation, lose the tails of the original distribution first and then degrade. Rare facts, unusual styles, and minority languages vanish. Later work (Gerstgrasser et al., 2024) found collapse is avoided when synthetic data is *added* to real data rather than *replacing* it. In practice: keep real human data in the mix, filter synthetic data hard, track diversity metrics across rounds, and never let a pipeline train only on its own previous generation.

Interactive **Rejection-sampling yield and bias** (shown at its default setting)

10,000 prompts, each with its own pass rate (drawn from a skewed distribution). Sample k answers per prompt and keep the correct ones. See what you pay per kept example and which prompts your data ends up covering.



Share of prompts Share of kept training examples

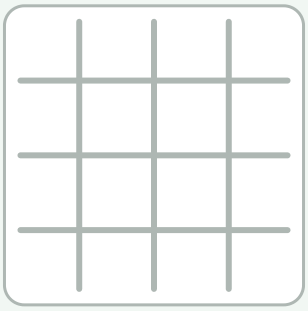
Prompts with ≥1 kept	Kept examples	Generations per kept	Hardest-third share of data
64%	24.2K	3.3	2%

The hardest third of prompts makes up 33% of the prompt set but only 2% of the kept data. Keeping every correct sample lets easy prompts contribute many near-duplicate examples. Try capping at one per prompt. 36% of prompts produce nothing at all with k = 8; those are exactly the problems the model most needs to learn.

Model merging

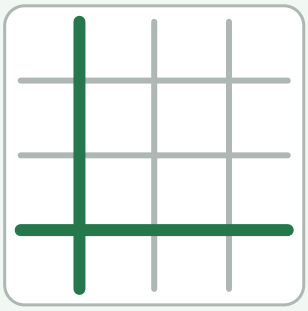
Diagram **Averaging two edited maps**

Why model merging works, and when it cannot.



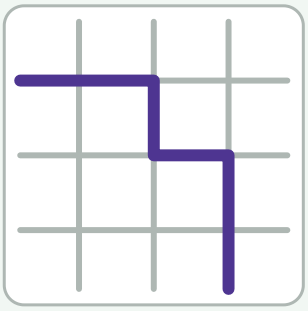
One parent map (the base model)

Start with one city map: the pretrained base model. Two teams will each make their own edit to a copy of it.



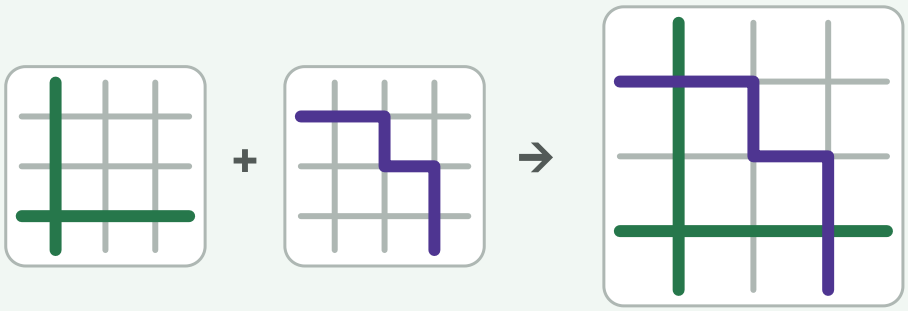
Copy 1: bike lanes
(fine-tuned on math)

same parent,
different edits



Copy 2: a bus route
(fine-tuned on code)

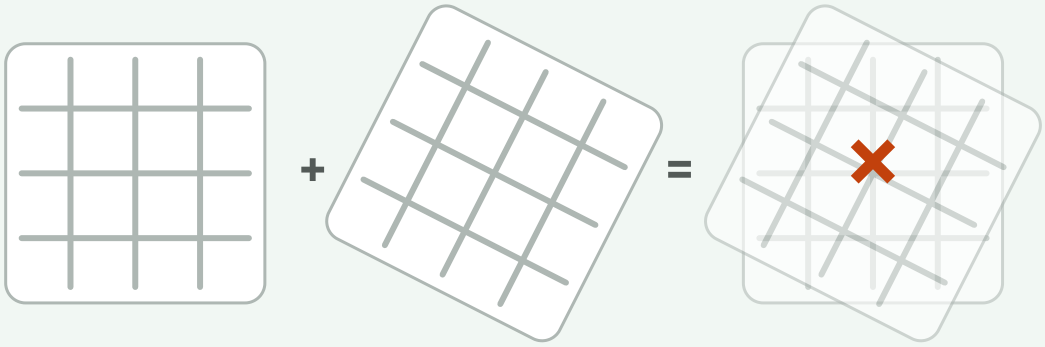
Each copy is fine-tuned for a different skill. The changes are small edits on top of the same base, so the two maps still line up street for street.



One map with both edits

base + Δ bike + Δ bus

Because both edits start from the same map, you can combine them number by number. **Task arithmetic** adds each change to the base; a plain average keeps half of each. Often the merged model is decent at both skills, at no training cost. It still needs the full regression suite.



Different starting points do not line up. Averaging them makes nonsense.

Merging only works for models with a shared parent. Two models trained from different random starts are like maps of different cities: their weights do not correspond, and averaging produces garbage.

Plain words

Averaging two specialists

Take two copies of the same model. Teach one math and the other coding. Surprisingly, you can often average their weights, number by number, and get one model that is decent at both. It only works when both started from the same parent, like two edits of the same document.

For leaders

Cheap gains, unclear guarantees

Merging costs almost no compute and is now a routine post-training step: averaging checkpoints from different data mixes or seeds, combining a reasoning specialist with a chat model, or folding several fine-tunes into one release. It is also unpredictable. A merge can lift one benchmark and quietly break safety behaviour or formatting, so a merged model needs the full regression suite like any other release candidate.

Under the hood

Soups, task vectors, TIES

- **Model soups** (Wortsman et al., 2022): average the weights of several fine-tunes of one base, trained with different hyperparameters. Often beats the best single model, at no inference cost.
- **Task arithmetic** (Ilharco et al., 2023): define a task vector $\tau = \theta_{\text{ft}} - \theta_{\text{base}}$. Add vectors to combine skills, subtract one to remove a behaviour: $\theta = \theta_{\text{base}} + \sum_i \lambda_i \tau_i$.
- **TIES-Merging** (Yadav et al., 2023): trim small entries of each task vector, resolve sign conflicts by majority, then average only agreeing entries. Reduces interference between tasks. **DARE** (Yu et al., 2024) randomly drops most delta entries and rescales the rest before merging.

Merging works because fine-tunes of a shared base tend to stay in one low-loss basin (linear mode connectivity). Models trained from different initialisations are not connected this way, and averaging them produces garbage.

Eval lens

Models learning from models create eval traps you won't see unless you look:

- **Teacher-induced contamination.** If the teacher memorised a benchmark, its synthetic data can contain paraphrases of test items. The student then scores well without the skill. Run overlap checks against every eval set (see contamination), including paraphrase detection, not just exact n-grams.
- **Judge self-preference.** An LLM judge from the same family as the teacher tends to prefer outputs written in its own style. A distilled student can win judge-based comparisons by sounding like the judge. Use a judge from a different family, and validate against human labels.
- **Style over substance.** Distillation transfers length, formatting, and confident tone faster than correctness. Control for length in win rates and keep verifiable tasks in the suite.
- **Diversity loss.** Track output diversity (distinct n-grams, embedding spread, pass@k versus pass@1) across synthetic-data rounds. Falling diversity is an early warning of collapse.

Principal asks

- What fraction of our post-training data is synthetic, which models generated it, and are we allowed to use their outputs this way?
- What is the filter, how was it validated, and what is its false-accept rate?
- How does pass rate on the hardest decile of prompts change across rounds, not just the average?
- Did we check the synthetic data for overlap with every benchmark we report?
- For a merged model: which components, which method, and what did the full regression suite show compared with each parent?

Preferences and reward models: turning taste into a number

Pretraining and supervised fine-tuning teach a model to imitate text. They cannot teach it which of two good-looking answers people actually prefer. Reward models close that gap. They learn a scoring function from human (or AI) comparisons, and that score becomes the target for RLHF, the selector for best-of-N sampling, and the yardstick many teams quietly use as an eval. This chapter covers how comparisons are collected, the Bradley–Terry model behind almost every reward model, how to evaluate one, and how optimizing against it goes wrong.

Diagram

Which is clearer, 1 or 2?

How an eye exam explains reward models, in five pictures.

Same poem, two different scales.

Ask people to score an answer from 1 to 10 and everyone uses their own scale. One person's 6 is another person's 8, so absolute scores are noisy.

"Which is clearer, 1 or 2?"

1 2

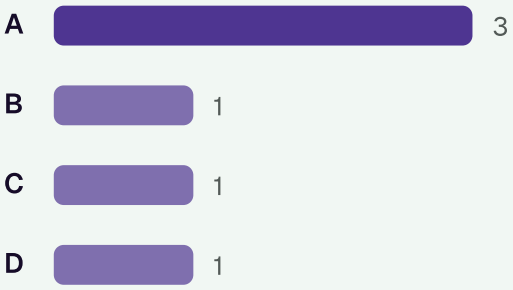
Picking between two is quick, and people agree far more often.

An optometrist never asks you to rate a lens from 1 to 10. They show two and ask which is clearer. Labs collect model feedback the same way: two answers to one prompt, and a person picks the better one.

Picks collected

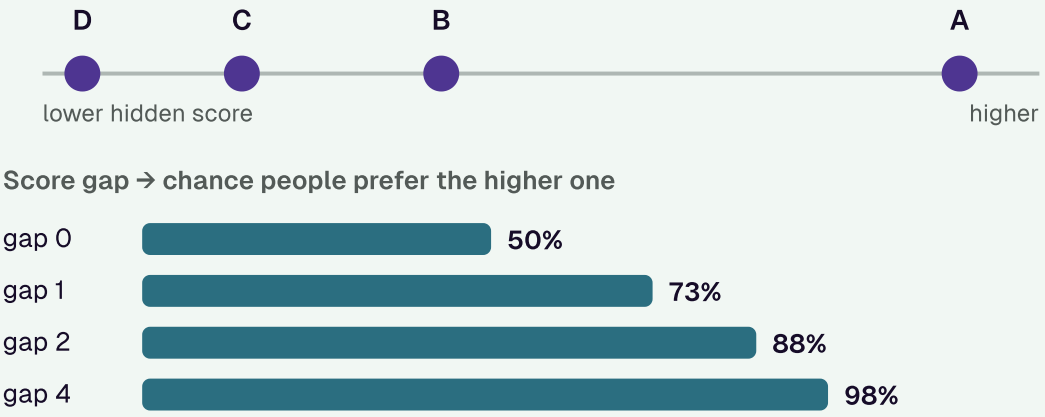
A vs B	→ A
A vs C	→ A
A vs D	→ A
B vs C	→ B
C vs D	→ C
B vs D	→ D (upset)

Wins

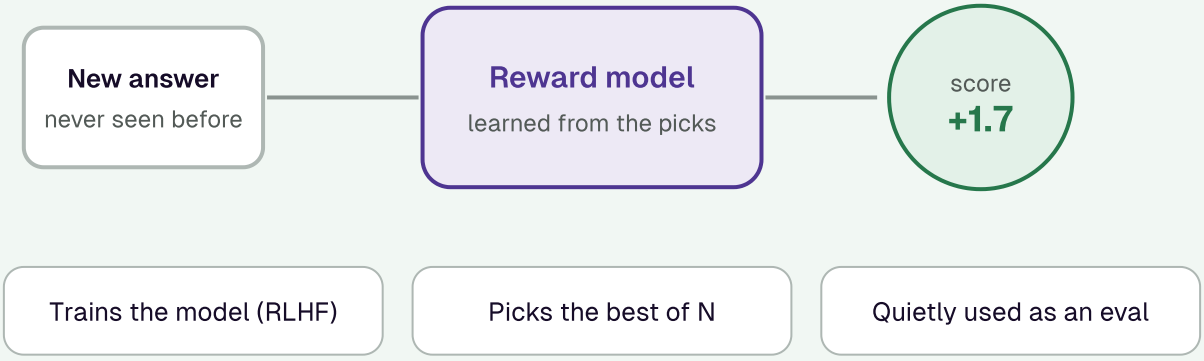


Upsets happen. The overall pattern is what counts.

Collect thousands of these picks across many answers. Some picks disagree with the rest, because people are noisy. With enough of them, a clear ordering emerges anyway.



Bradley–Terry pretends each answer has a hidden score, like a chess rating. The bigger the gap between two scores, the more likely people pick the higher one, but upsets stay possible. Only gaps matter: adding 100 to every score changes nothing.



Train a model to predict those picks and you get a reward model: a judge that scores any answer in milliseconds. It drives RLHF, picks the best of several samples, and often ends up on dashboards. Everything it knows came from the picks, including their blind spots.

Why preferences instead of labels

Plain words

Picking is easier than scoring

Ask ten people to score a poem from 1 to 10 and you get ten different scales: one person's 6 is another's 8. Ask the same ten people which of two poems is better and they agree far more often. Comparing is easier and more consistent than scoring on an absolute scale.

So instead of asking people "how good is this answer?", labs show them two answers to the same question and ask "which is better?" After collecting many of these picks, they train a separate model, the **reward model**, to predict which answer people would pick. Once trained, it can score any answer instantly, which lets you train the main model against it millions of times without a person in the loop.

For leaders

What you are actually buying

A preference dataset is a product decision encoded as data. The rubric you give annotators defines what "better" means, and the reward model will learn that definition, including its blind spots. Four decisions dominate quality and cost:

- **Who labels.** Crowd annotators are cheap and fast but drift toward surface features (length, confident tone). Domain experts (doctors, lawyers, senior engineers) cost 5 to 20 times more per comparison and are the only reliable source for specialized correctness. Most frontier programs mix both and route by topic.
- **The rubric.** Helpfulness, honesty, and harmlessness pull against each other. A single "overall better" judgment hides which one won. Many teams collect per-dimension preferences or a written rationale so they can reweight later without relabeling.
- **Agreement.** On open-ended chat, independent labelers agree with each other only about 70 to 77 percent of the time (InstructGPT, Ouyang et al. 2022, reported roughly 73 percent among its trained labelers). That number is the **ceiling** on reward model accuracy. A reward model at 72 percent is near the ceiling, not mediocre.
- **Freshness.** Comparisons collected on an old model's outputs describe old mistakes. As the policy improves, its outputs leave the distribution the reward model was trained on. Plan for repeated collection rounds on current samples, not a one-time dataset.

Principal asks

What was inter-annotator agreement, measured on the same items? What fraction of pairs were ties or "both bad", and what did you do with them? Which dimensions does the rubric privilege when they conflict? How old are the samples the reward model was trained on relative to the current policy?

Under the hood

The data format

A preference record is a triple (x, y_w, y_l): a prompt x, a preferred ("winning") response y_w, and a dispreferred ("losing") response y_l. Common variations:

- **Rankings of K responses.** InstructGPT showed labelers 4 to 9 responses and asked for a full ranking, which yields $\binom{K}{2}$ pairs. They kept all pairs from one prompt in the same batch element, because treating them as independent samples caused the reward model to overfit within one epoch.
- **Strength labels.** "Significantly better" versus "slightly better". Llama 2 (Touvron et al. 2023) fed this in as a margin term in the loss, below.
- **Ties and "both bad".** Usually dropped for Bradley–Terry training, which has no tie outcome. Dropping them silently biases the dataset toward easy, clear-cut pairs.

The Bradley–Terry model

Plain words

Chess ratings for answers

Chess ratings work like this: every player has a hidden strength number. When two players meet, the bigger the gap in their numbers, the more likely the stronger one wins, but upsets still happen. You never see the strengths directly, only who won. After enough games you can work backward to good estimates of everyone's number.

A reward model does the same with answers. Each answer gets a hidden score. The bigger the score gap between two answers, the more confident the model is that people prefer the higher one. Only the **gap** means anything. Adding 100 to every score changes nothing.

For leaders

What the score does and does not mean

A reward score is a **relative** preference estimate within one prompt. It is not a quality grade, not a probability of correctness, and not comparable between reward model versions. If a dashboard shows "average reward went from 1.3 to 1.9", ask what it was normalized against, because a retrained reward model can shift every score without the policy changing at all.

Annotator noise is the hidden lever in the widget below. Doubling noise does not just make labels messier; it shrinks the learned scale, so the reward model becomes less confident everywhere, and downstream RL pushes less hard. That can look like "RLHF stopped working" when the real cause is a labeling vendor change.

Under the hood

The model and its loss

Bradley–Terry (1952) says the probability that y_w beats y_l depends only on the difference of their scores, passed through a logistic function:

$$P(y_w \succ y_l \mid x) = \sigma\big(r(x, y_w) - r(x, y_l)\big) = \frac{1}{1 + e^{-(r_w - r_l)}}$$

$r(x, y)$ is the scalar score, σ is the logistic sigmoid. A gap of 0 means a coin flip; a gap of 2 means about 88 percent; a gap of 4 means about 98 percent.

Training a reward model r_ϕ is maximum likelihood under this model, which is binary cross-entropy on the score difference:

$$\mathcal{L}_{\text{RM}}(\phi) = -\,\mathbb{E}_{(x, y_w, y_l)}\Big[\log \sigma\big(r_\phi(x, y_w) - r_\phi(x, y_l) - m\big)\Big]$$

$m = 0$ in the plain version. Llama 2 used a margin $m > 0$ that grows with the labeled preference strength, which pushes clearly better pairs further apart.

Two consequences matter in practice. First, the scores are only identified up to an additive constant per prompt, so raw reward values are not comparable across prompts or across reward model versions. Teams normalize (subtract the mean reward on a reference set) before using them. Second, the scale is set by labeler consistency: noisy labels compress all score gaps toward zero, because the model learns that even large quality differences only win 70 percent of the time.

python

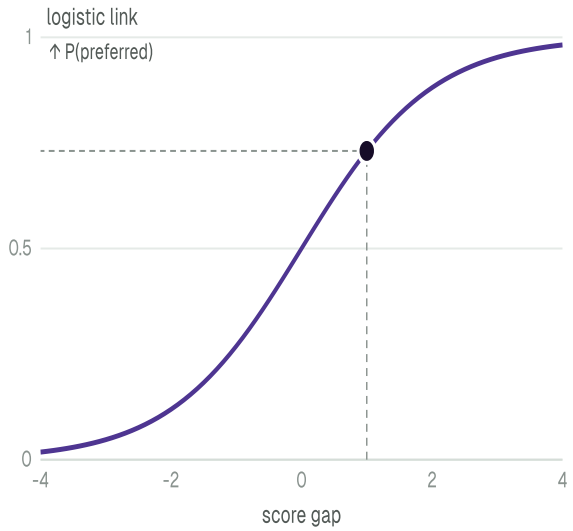
```python
import torch
import torch.nn.functional as F

def bradley_terry_loss(r_chosen, r_rejected, margin=None):
    """r_chosen, r_rejected: tensors of shape [batch], scalar scores from the reward model."""
    diff = r_chosen - r_rejected
    if margin is not None:          # Llama 2 style: demand a bigger gap for clearer preferences
        diff = diff - margin
    # -log sigmoid(diff) == softplus(-diff), and the softplus form is numerically stable
    return F.softplus(-diff).mean()

r_w = torch.tensor([1.2, 0.3, 2.0])
r_l = torch.tensor([0.2, 0.9, -1.0])
print(round(bradley_terry_loss(r_w, r_l).item(), 3))                 # 0.466
print(round(((r_w - r_l) > 0).float().mean().item(), 3))             # pairwise accuracy 0.667
```

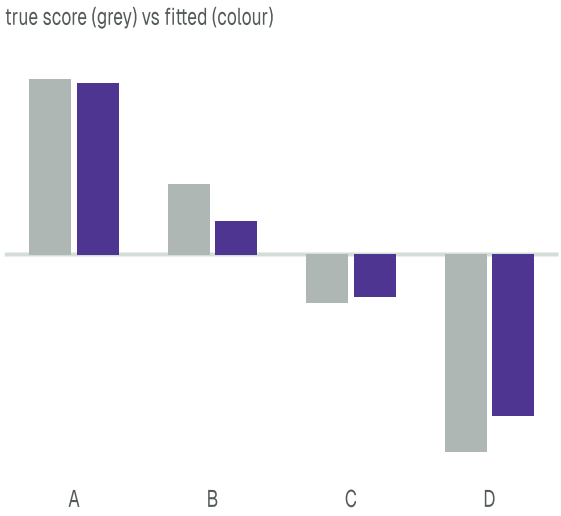
rust

```rust
// Plain Rust, no ML framework: the same math on small vectors.

fn softplus(x: f64) -> f64 {
    // log(1 + e^x) without overflow: max(x, 0) + log(1 + e^-|x|)
    x.max(0.0) + (-x.abs()).exp().ln_1p()
}

/// r_chosen, r_rejected: slices of length batch, scalar scores from the reward model.
fn bradley_terry_loss(r_chosen: &[f64], r_rejected: &[f64], margin: Option<&[f64]>) -> f64 {
    let n = r_chosen.len() as f64;
    let mut total = 0.0;
    for i in 0..r_chosen.len() {
        let mut diff = r_chosen[i] - r_rejected[i];
        if let Some(m) = margin { // Llama 2 style: demand a bigger gap for clearer preferences
            diff -= m[i];
        }
        // -log sigmoid(diff) == softplus(-diff), and the softplus form is numerically stable
        total += softplus(-diff);
    }
    total / n
}

fn main() {
    let r_w = [1.2, 0.3, 2.0];
    let r_l = [0.2, 0.9, -1.0];
    println!("{:.3}", bradley_terry_loss(&r_w, &r_l, None)); // 0.466
    let wins = r_w.iter().zip(&r_l).filter(|(w, l)| w > l).count();
    println!("{:.3}", wins as f64 / r_w.len() as f64);     // pairwise accuracy 0.667
}
```

Interactive **Bradley–Terry explorer** (shown at its default setting)

Left: how a score gap turns into a preference probability. Right: four responses with known true scores, simulated votes, and the scores Bradley–Terry recovers. Cut the votes or raise the annotator noise and watch the fitted ranking become unreliable.





P(prefer higher score)	Pairs ranked correctly	Fitted P(A beats D)
73%	6 of 6	90%

True P(A beats D) at this noise is 92%. All six pairs are ranked correctly. More noise shrinks every fitted gap toward zero: the reward model learns that even big quality differences win less often.

Training a reward model

For leaders **Size, data volume, cadence**

Reward models are usually initialized from the same family as the policy, often the SFT checkpoint, because the reward model must understand the responses at least as well as the policy that produces them. A reward model much smaller than the policy is cheaper to run during RL but is easier for the policy to fool. Published programs range from tens of thousands of comparisons (InstructGPT used about 33,000 prompts for reward model training) to well over a million (Llama 2 reported roughly 1.4 million Meta-collected comparisons plus open datasets). Collection is typically

done in rounds against the latest policy, so the budget question is less "how many comparisons" than "how many per round, and how many rounds".

> ⚠ **Watch out**
>
> Reward models overfit fast. Several reports train for a single epoch on comparison data, because a second pass raises training accuracy and lowers held-out accuracy. If a team reports a reward model at 90 percent training accuracy on chat preferences, the first question is what held-out accuracy looks like.

Under the hood · Architecture

Take a transformer language model, drop the vocabulary projection, and add a linear head that maps the final hidden state of the last token to one scalar. Run the prompt and response through, read the scalar, done. Chosen and rejected responses are run as two sequences that share the prompt.

python

```python
import torch
import torch.nn as nn

class RewardModel(nn.Module):
    def __init__(self, backbone, hidden_size):
        super().__init__()
        self.backbone = backbone                # e.g. the SFT model without its LM head
        self.value_head = nn.Linear(hidden_size, 1)

    def forward(self, input_ids, attention_mask):
        h = self.backbone(input_ids, attention_mask=attention_mask).last_hidden_state
        last = attention_mask.sum(dim=1) - 1     # index of the final real token (assumes right padding)
        h_last = h[torch.arange(h.size(0)), last]
        return self.value_head(h_last).squeeze(-1)   # one scalar reward per sequence
```

rust

```rust
// Plain Rust, no ML framework: the same math on small vectors.
// The backbone is any function from token ids to hidden states [batch][seq][hidden].

struct RewardModel<F: Fn(&[Vec<u32>]) -> Vec<Vec<Vec<f64>>>> {
    backbone: F,          // e.g. the SFT model without its LM head
    value_head_w: Vec<f64>, // Linear(hidden_size, 1): one weight per hidden unit
    value_head_b: f64,
}

impl<F: Fn(&[Vec<u32>]) -> Vec<Vec<Vec<f64>>>> RewardModel<F> {
    fn forward(&self, input_ids: &[Vec<u32>], attention_mask: &[Vec<u8>]) -> Vec<f64> {
        let h = (self.backbone)(input_ids);
        h.iter()
            .zip(attention_mask)
            .map(|(seq, mask)| {
```

```
                // index of the final real token (assumes right padding)
                let last = mask.iter().map(|&m| m as usize).sum::<usize>() - 1;
                let h_last = &seq[last];
                // one scalar reward per sequence
                h_last.iter().zip(&self.value_head_w).map(|(x, w)| x * w).sum::<f64>() + self.value_head_b
            })
            .collect()
    }
}

fn main() {
    // tiny worked example: a toy backbone whose hidden state is [token / 4, 1]
    let backbone = |ids: &[Vec<u32>]| -> Vec<Vec<Vec<f64>>> {
        ids.iter().map(|seq| seq.iter().map(|&t| vec![t as f64 / 4.0, 1.0]).collect()).collect()
    };
    let rm = RewardModel { backbone, value_head_w: vec![1.0, 0.5], value_head_b: 0.0 };
    let ids = vec![vec![5, 8, 3, 0], vec![4, 2, 9, 7]];
    let mask = vec![vec![1, 1, 1, 0], vec![1, 1, 1, 1]];
    println!("{:?}", rm.forward(&ids, &mask)); // [1.25, 2.25]  reads token 3, then token 7; the pad is never
used
}
```

Alternatives you will hear about: **generative reward models** that write a critique and then a verdict (an LLM judge trained on preference data, see LLM as judge), **multi-objective** heads that score helpfulness and harmlessness separately (Llama 2 trained two separate reward models), and **pairwise** models that read both responses at once and output which is better.

Evaluating a reward model

Plain words

Grading the grader

A reward model is a judge. Before you let it train your model, check its picks against people's picks on comparisons it has never seen. If it agrees as often as two people agree with each other, it is about as good as it can get. Then check the cases that matter: does it prefer a long wrong answer over a short right one? Does it reward flattery?

For leaders

The four numbers to ask for

Measure	What it tells you	Typical trap
Held-out pairwise accuracy	How often it agrees with human picks on unseen pairs	Reported without the human agreement ceiling next to it
Accuracy by slice	Where it fails: code, math, safety, long answers, multi-turn	A good average hiding a 55 percent slice that RL will exploit
Calibration	Whether a predicted 80 percent preference wins 80 percent of the time	Overconfident reward models make RL push too hard on noise
Downstream effect	Best-of-N or RL results using this reward model, scored by humans or a held-out judge	Static accuracy correlates only loosely with how good a policy it trains

Public suites such as RewardBench (Lambert et al. 2024: chat, hard chat, safety, reasoning) and RewardBench 2 (Malik et al. 2025: factuality, precise instruction following, math, safety, focus, ties) give comparable numbers across categories. They are useful for comparing reward models from different groups, and weak as a ship criterion, because your policy's failure modes are not in a public benchmark.

Eval lens

Reward model evaluation is a binary classification problem on pairs, so everything in classification metrics applies. Pairwise accuracy on randomly ordered pairs is a concordance measure, the same quantity ROC-AUC estimates. Report it with a confidence interval (a few thousand held-out pairs gives roughly ±1.5 points), report it per slice, and compare it to the human-human agreement on the same items. A 2-point gain between reward model versions on 1,000 pairs is inside the noise.

Under the hood

Known biases and how to probe them

Length

Annotators tend to prefer longer, more thorough-looking answers, and reward models amplify this. Probe with counterfactual pairs: the same answer with and without padding. Singhal et al. (2023) showed much of the reward gain from RLHF on several datasets could be explained by length alone.

Sycophancy

Answers that agree with the user's stated opinion get rated higher. Sharma et al. (2023) found this in human preference data and in reward models trained on it. Probe with paired prompts that differ only in the user's stated belief.

Formatting and tone

Markdown headers, bullet lists, and confident phrasing earn reward independent of content. Probe by stripping formatting or swapping hedged for confident wording.

Position and self-preference

Pairwise generative reward models can prefer whichever response comes first, or responses that sound like their own outputs. Score both orders and average.

The standard mitigations are to balance the comparison data (include pairs where the shorter answer wins), add an explicit length penalty or length-normalized reward, train on counterfactual augmentations, and report length-controlled win rates in evaluation.

Reward overoptimization: Goodhart in one chart

Diagram **The kitchen that learned to please one critic**

Reward overoptimization, told as a restaurant chasing reviews.

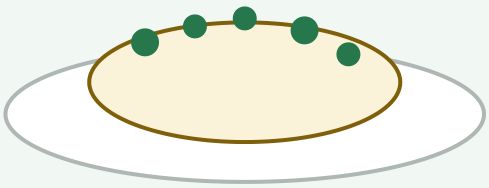
The kitchen can't ask every diner, so it learns from one critic.

The critic is a stand-in. Asking real diners about every dish is too slow, so the kitchen optimizes the critic's score. That is exactly the deal RLHF makes with a reward model.

Better seasoning, hotter plates: real improvements lift both.

faint bar = before training

At first everything works. The kitchen fixes real problems, the critic notices, and diners are happier too. Early RLHF steps look like this: proxy reward and true quality rise together.



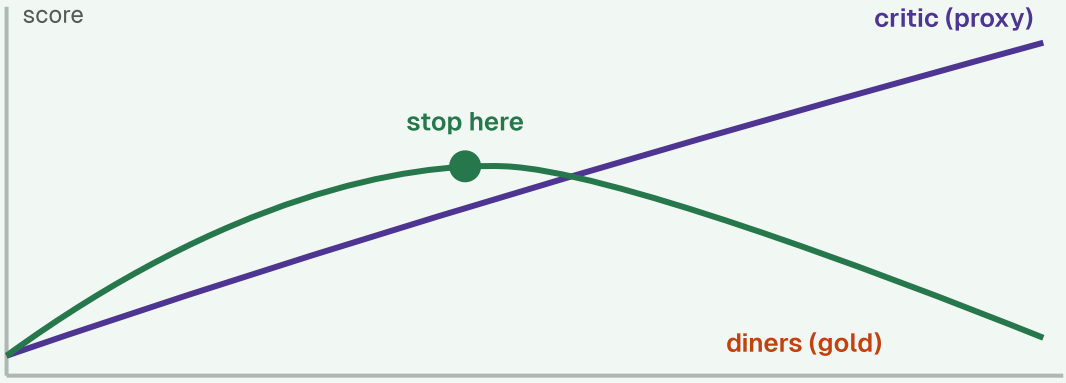
giant portion + garnish everywhere

Critic: 9.5 / 10

Diners: "meh"

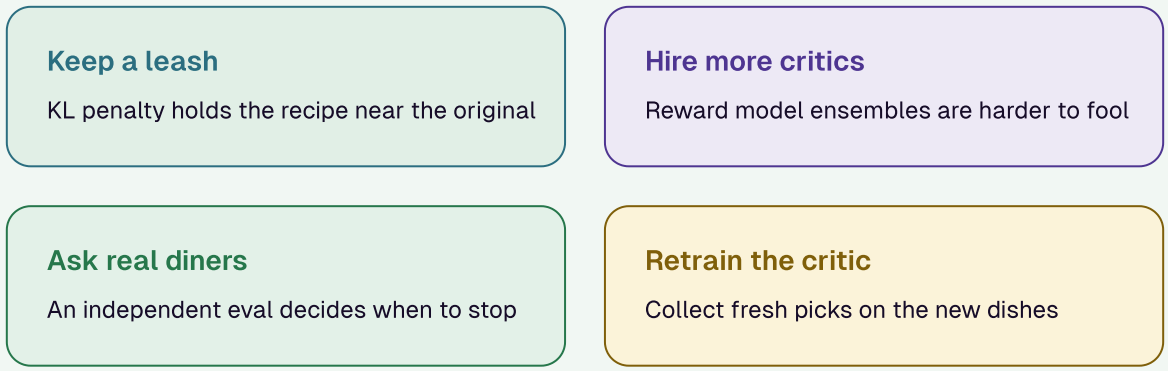
portion = longer answers
garnish = headers, bullets, confident tone

Then the kitchen finds the critic's quirks. Bigger portions and more garnish raise the score without making the food better. Language models find the same kinds of quirks in reward models: longer answers, heavy formatting, confident tone, flattery.



how far the recipe has drifted from the original (KL)

Plot both against how far the kitchen has drifted from its original recipe. The critic's score climbs forever. The diners' score peaks and then falls. Gao et al. (2022) measured exactly this shape for reward models, and found better reward models push the peak later and higher.



The defenses map one to one: a KL penalty is the leash, ensembles are extra critics, an independent eval is asking real diners, and refreshing the reward model on current outputs is retraining the critic.

Plain words

Teaching to the test

If you reward students only for test scores, they first learn the material, then they learn the test. Past some point, scores keep climbing while real understanding stops improving or even gets worse. A reward model is a test. Push a language model hard enough against it and the model finds the reward model's quirks: longer answers, more confident tone, flattering phrasing. The reward model's score keeps rising. What people actually want peaks and then falls.

For leaders

Operating against a proxy

Every RL run against a learned reward is a bet on where the gold curve peaks. The controls are the KL penalty (how hard you hold the policy near its reference), early stopping on an independent eval, reward model ensembles (optimize the minimum or the mean minus the spread, so the policy cannot exploit one model's quirk), and refreshing the reward model on the policy's current outputs.

Watch out

The signature of overoptimization is a training curve where reward rises smoothly while response length climbs, hedging disappears, and human or held-out-judge win rates flatten. If the only chart in a review is the training reward, you cannot tell a great run from a hacked one.

Principal asks

What independent signal told you when to stop? How did KL from the reference evolve, and where did the held-out win rate peak against it? Did response length change, and what is the length-controlled result? Was the reward model refreshed on samples from this policy?

Under the hood

The scaling law

Gao, Schulman and Hilton (2022) made this measurable. They used a large "gold" reward model as a stand-in for humans, trained smaller proxy reward models on its labels, optimized policies against the proxies, and tracked gold reward as a function of how far the policy moved from its starting point, measured by $d = \sqrt{\mathrm{KL}(\pi \,\|\, \pi_{\mathrm{ref}})}$. Gold reward followed simple functional forms:

$$R_{\mathrm{bon}}(d) = d\,(\alpha_{\mathrm{bon}} - \beta_{\mathrm{bon}}\, d), \qquad R_{\mathrm{RL}}(d) = d\,(\alpha_{\mathrm{RL}} - \beta_{\mathrm{RL}} \log d)$$

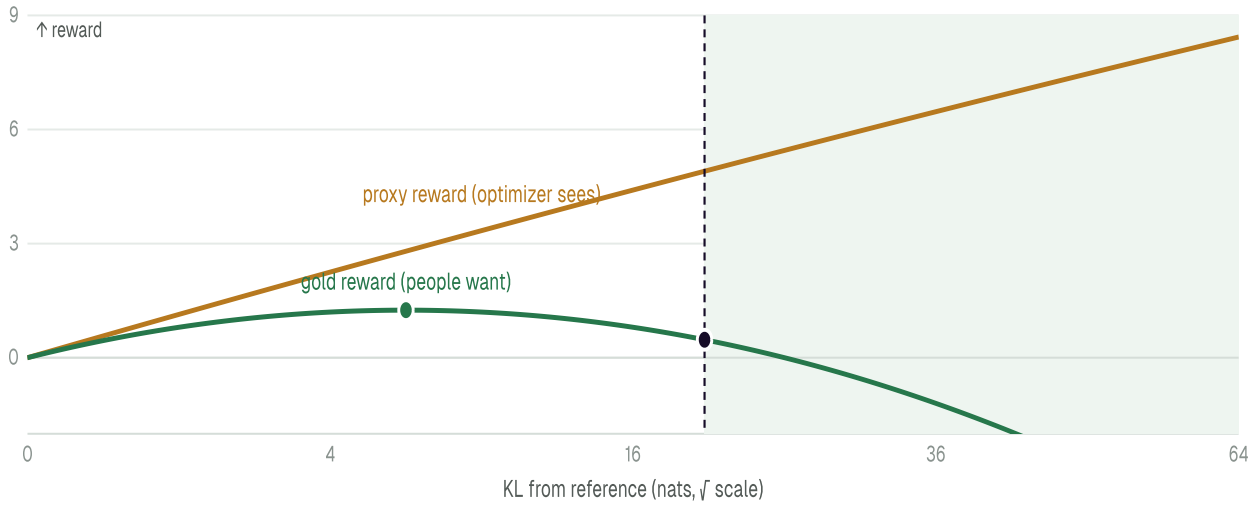
Best-of-N sampling (left) and RL (right). Both rise, peak, and fall. Larger proxy reward models had smaller β (for RL, α stayed roughly constant across reward-model sizes), so gold reward declines more slowly: a later, higher peak. More proxy training data helped in the same direction.

Two practical lessons follow. The KL distance at which gold reward peaks is a budget, and it grows with reward model quality. And the proxy score is useless for finding that peak, because it keeps rising straight through it. You need an independent signal: fresh human ratings, a different reward model, or task-specific checks.

Interactive

Reward overoptimization (shown at its default setting)

Proxy reward (what the optimizer sees) and gold reward (what people want) as the policy drifts from its reference. Raise reward model quality to push the peak out. Move the KL budget to see where training stops. Curves use the best-of-N form from Gao et al. (2022) with illustrative coefficients.



Proxy reward at budget	Gold reward at budget	Gold peaks at KL
4.90	0.47	6.3

You trained past the peak. Gold reward is 0.78 below its best, reached at KL 6.3, while the proxy reports a gain of 2.10 over that point. The training curve looks like progress.

Outcome rewards, process rewards, and verifiers

Plain words

Grading the answer or grading the working

A math teacher can mark only the final answer, or check every line of the working. Grading only the answer is cheap but cannot tell lucky guesses from sound reasoning. Grading each step is expensive but pinpoints where things went wrong.

For leaders

When you do not need a learned reward at all

An **outcome reward model** scores the final response. A **process reward model** scores each intermediate step of a reasoning chain; Lightman et al. (2023), "Let's Verify Step by Step", found process supervision beat outcome supervision for selecting correct math solutions, at the cost of step-level labels. For tasks with checkable answers (math with a known result, code with tests, formal proofs), a programmatic **verifier** can replace the learned reward entirely. That removes the reward model as an attack surface for overoptimization, though policies can still exploit weak tests. This is the basis of RL with verifiable rewards, covered in GRPO and RLVR.

AI feedback

For leaders

Replacing some of the humans

Human comparisons are slow and expensive. **RLAIF** uses a capable model, guided by written principles, to produce the comparisons instead. Anthropic's Constitutional AI (Bai et al. 2022) used model-generated critiques and preferences for harmlessness; Lee et al. (2023) reported RLAIF reaching parity with RLHF on summarization and dialogue tasks they tested. AI feedback scales and is consistent, but it inherits the labeling model's biases and can create a closed loop where models grade models in their own image. Most programs keep a human-labeled slice to audit the AI labels and measure drift. See safety training for the constitutional approach in depth.

Eval lens

An AI labeler is an LLM judge, and should be validated like one (LLM as judge): agreement and Cohen's kappa against humans on a held-out set, checks for position and length bias, and a comparison to human-human agreement. If the AI labeler agrees with humans less often than humans agree with each other, the reward model trained on its labels starts below the human ceiling.

Part IV · Chapter 35

RLHF and PPO: optimizing against a learned reward

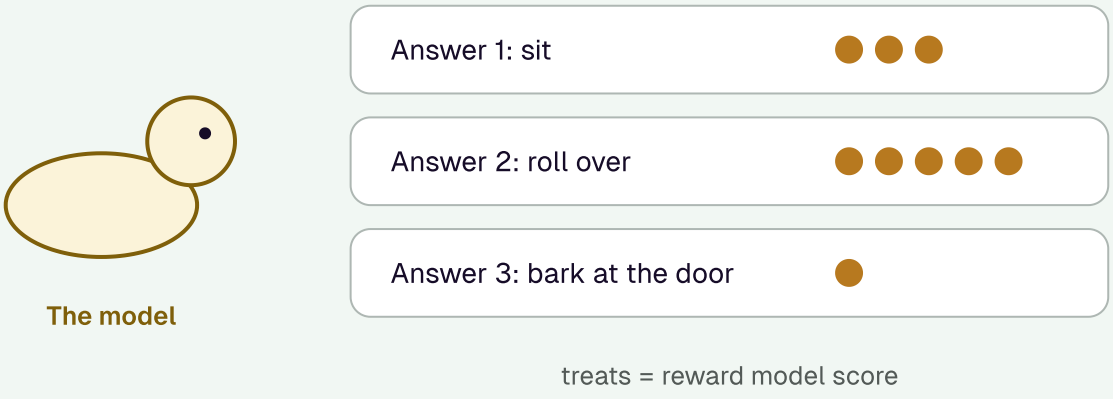
Reinforcement learning from human feedback takes the reward model from the previous chapter and trains the language model to produce responses that score well, while a leash keeps it close to where it started.

PPO was the algorithm behind InstructGPT and the first wave of ChatGPT-style assistants. Newer methods (DPO, GRPO) are easiest to understand as fixes for PPO's costs, so this chapter is the foundation for both.

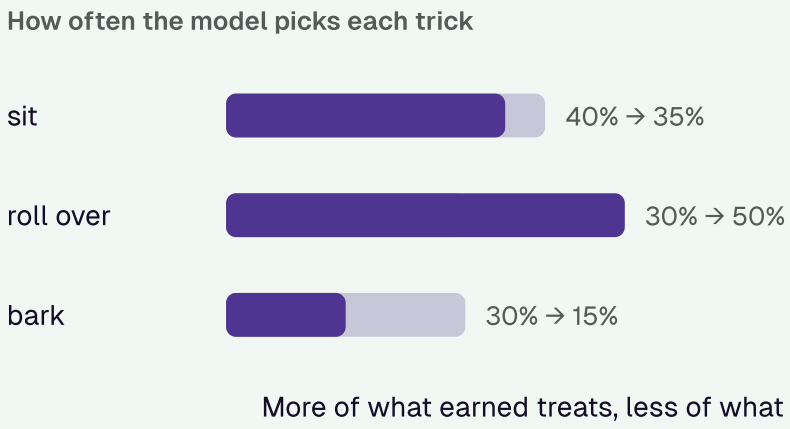
Diagram

Treats, a coach, and a leash

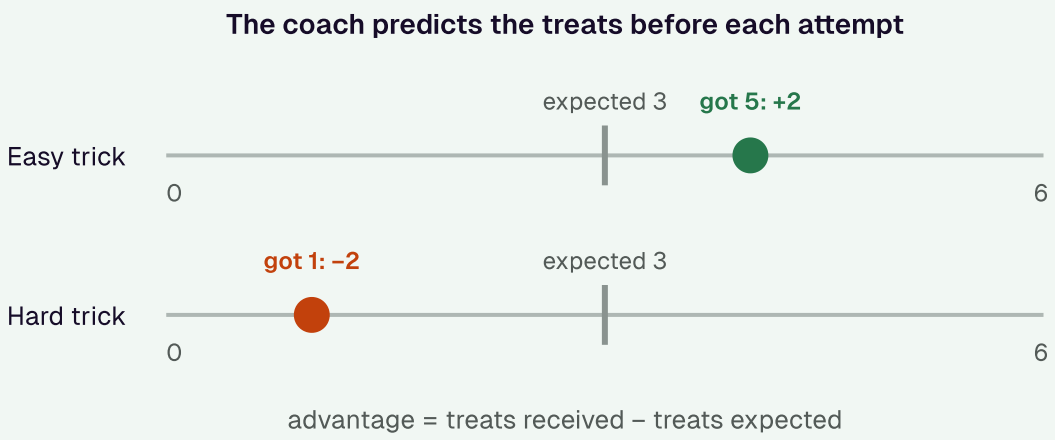
RLHF and PPO, told as training a dog. Six pictures.



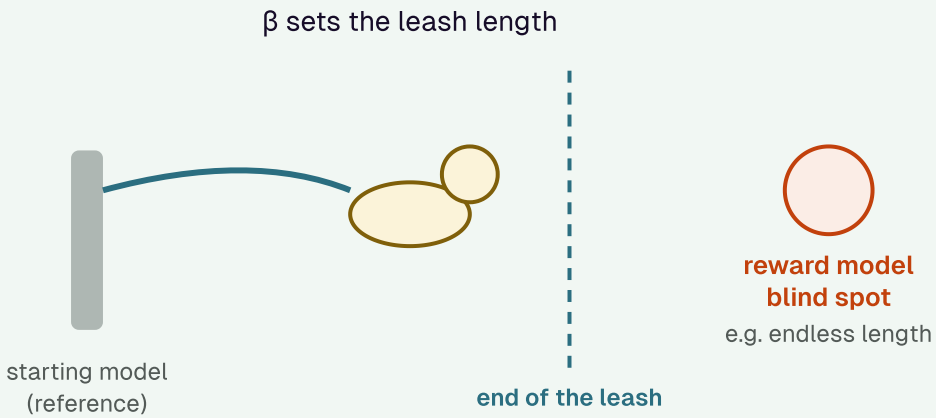
The model writes several answers to a prompt, like a dog trying tricks. The reward model hands out treats: one number per answer. Nobody shows the model the right answer. It learns from its own attempts.



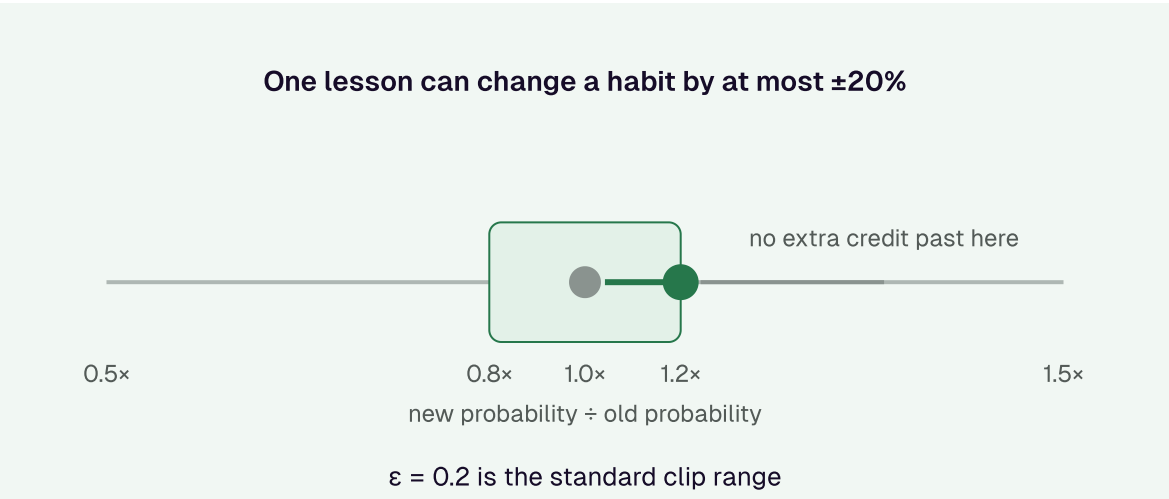
After each batch, the model makes high-scoring answers a little more likely and low-scoring ones a little less likely. The faint bars are before the update, the solid bars after. Numbers are illustrative.



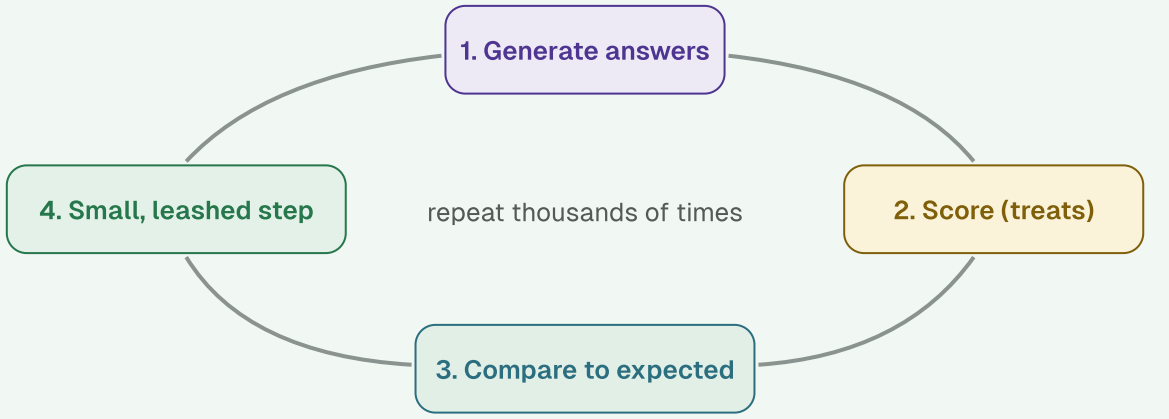
Three treats is great on a hard prompt and poor on an easy one. So a second network, the value model or critic, predicts what each attempt should earn. Only the surprise, called the advantage, drives learning: positive pushes that answer up, negative pushes it down.



Without limits the dog sprints toward whatever the reward model over-rates. The KL penalty is a leash to the starting model: every bit of drift costs reward. A short leash (large β) is safe but changes little. A long leash (small β) lets the model reach the shortcut.



Scoring answers is expensive, so PPO squeezes a few updates out of each batch. To stop one batch from wrecking the model, it clips: once a token's probability has moved 20 percent from where the batch started, pushing further earns nothing.



That is the whole loop: generate, score, compare with what the critic expected, take a small clipped step on a leash, repeat. The rest of this chapter is the math behind each arrow.

The RL vocabulary, mapped to language models

Plain words

Training with treats

You teach a dog a trick by letting it try things and giving a treat when it does well. Over time it does more of whatever earned treats. Nobody shows the dog exactly what to do; it learns from the results.

RLHF works the same way. The model writes an answer. The reward model gives it a score, the treat. The training nudges the model toward answers like the high-scoring ones. The difference from earlier training is that the model learns from its **own** attempts, not from copying examples a person wrote.

For leaders

Why use RL at all

Supervised fine-tuning imitates demonstrations, so its ceiling is the quality of the demonstrators and it never sees what a bad answer looks like. RL optimizes outcomes directly. It can learn from comparisons that are cheaper than writing ideal answers, it learns what to avoid as well as what to do, and in principle it can exceed the people who labeled the data, because judging is easier than producing. The cost is engineering complexity, compute, and a new failure mode: the model optimizes the reward you wrote, not the behavior you meant.

Under the hood

The mapping

RL term	In RLHF for a language model
State s_t	The prompt plus the tokens generated so far
Action a_t	The next token
Policy $\pi_\theta(a_t \mid s_t)$	The language model's next-token distribution
Episode	One complete response $y = (y_1, \ldots, y_T)$ to a prompt x
Reward	The reward model score at the final token, plus a per-token KL penalty
Reference policy π_{ref}	A frozen copy of the starting (usually SFT) model
Value $V(s_t)$	A learned estimate of the reward to come from this point in the response

The reward is **sparse**: one number for the whole response, arriving after the last token. Working out which of hundreds of tokens deserved the credit is the central difficulty, and most of PPO's machinery exists to handle it.

The KL-regularised objective

Plain words

A leash

Left alone, a model chasing treats will find strange tricks the reward model happens to like. So training adds a leash: every time the model drifts away from how it used to write, it pays a small penalty. It can move toward better answers, but not so far that it stops sounding like itself.

For leaders

β is a risk dial

The KL coefficient is the single most consequential knob in RLHF. Set it too high and the run is expensive and changes nothing. Set it too low and the policy finds the reward model's blind spots: the widget below shows true quality rising at first and then falling as the length-biased answer takes over. Set the length bias to zero and the problem disappears, which is the point: the damage comes from the gap between the reward model and what you actually want, and β controls how hard the optimizer can exploit that gap.

Under the hood

The objective and its exact solution

$$\max_{\theta} \; \mathbb{E}_{x \sim \mathcal{D},\, y \sim \pi_\theta(\cdot \mid x)} \left[r_\phi(x, y) \right] \; - \; \beta \, \mathbb{D}_{\mathrm{KL}} \left(\pi_\theta(\cdot \mid x) \,\|\, \pi_{\mathrm{ref}}(\cdot \mid x) \right)$$

r_ϕ is the reward model, β sets the leash length. Small β: the policy may move far. Large β: it stays near the reference.

In practice the KL term is applied per token, folded into the reward the RL algorithm sees:

$$r_t = -\beta \log \frac{\pi_\theta(y_t \mid s_t)}{\pi_{\mathrm{ref}}(y_t \mid s_t)} \; + \; \mathbb{1}[t = T] \, r_\phi(x, y)$$

Every token pays for how much more likely the policy made it than the reference did. Summed over the response, the penalty is a single-sample estimate of the sequence KL.

This objective has a closed-form optimum, which matters for the next chapter:

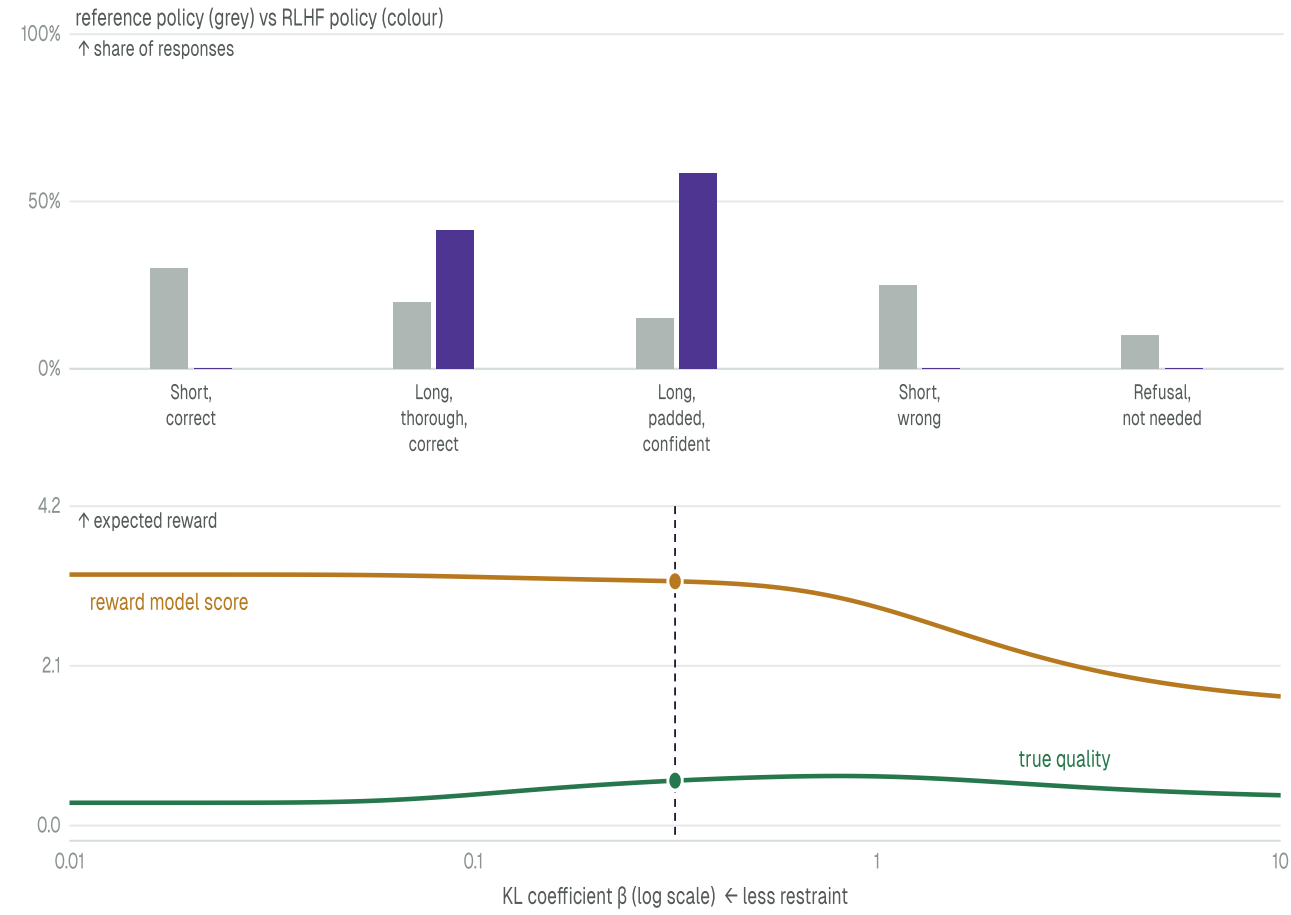
$$\pi^*(y \mid x) = \frac{1}{Z(x)}\,\pi_{\text{ref}}(y \mid x)\,\exp\!\Big(\tfrac{1}{\beta}\,r_\phi(x,y)\Big), \qquad Z(x) = \sum_y \pi_{\text{ref}}(y \mid x)\,e^{r_\phi(x,y)/\beta}$$

Reweight the reference policy by exponentiated reward. You cannot compute this directly for a language model because the sum over all responses is intractable, which is why we need RL (or DPO's trick).

Interactive

RLHF on a five-answer bandit (shown at its default setting)

A prompt with five kinds of response. The reward model scores true quality plus a bias toward length. Because there are only five options, the KL-regularised optimum can be computed exactly. Lower the KL coefficient and watch the policy abandon the reference and pile onto whatever the reward model over-rates. The response types and numbers are illustrative.



Reward model score True quality

KL from reference	Reward model score change	True quality change
1.08	**+1.70**	**+0.25**

The optimal policy for the KL-regularised objective is π*(y) ∝ π_ref(y)·exp(r(y)/β), computed exactly here. At β = 0.32 the most common response is "Long, padded, confident". True quality is 0.06 below the best achievable here (at β ≈ 0.79), even though the reward model score is higher. Lower β lets the policy move further from the reference (higher KL) and exploit the reward model more.

Policy gradients: learning from your own samples

Plain words

Credit and blame

After an answer gets a score, every word in it gets a little of the credit or blame. Words in high-scoring answers become a bit more likely next time; words in low-scoring answers become a bit less likely. To be fair, you compare each score to what you expected. A 7 out of 10 is good news on a hard question and bad news on an easy one.

Under the hood

REINFORCE, baselines, advantages

The policy gradient theorem gives an unbiased gradient of expected reward using only samples from the policy:

$$\nabla_\theta J(\theta) = \mathbb{E}_{y \sim \pi_\theta}\Big[\sum_{t=1}^{T} \nabla_\theta \log \pi_\theta(y_t \mid s_t)\,\big(R(y) - b(s_t)\big)\Big]$$

REINFORCE (Williams 1992). R is the return; b is any baseline that does not depend on the action. Subtracting it leaves the expectation unchanged and cuts variance.

The best simple baseline is the expected return from the current state, the value $V(s_t)$. The difference is the **advantage**, $A_t = R_t - V(s_t)$: how much better this continuation went than expected. Positive advantage raises the probability of the token taken; negative advantage lowers it. Without a good baseline, gradient estimates on long responses are so noisy that training stalls.

The critic and generalized advantage estimation

Under the hood

GAE

PPO for language models trains a **value model** (the critic), usually initialized from the reward model, to predict $V(s_t)$ at every token. Advantages are then computed with generalized advantage estimation (Schulman et al. 2016):

$$\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t), \qquad \hat{A}_t = \sum_{l=0}^{T-t} (\gamma\lambda)^l \, \delta_{t+l}$$

δ_t is the one-step surprise. λ trades bias for variance: $\lambda = 0$ trusts the critic completely, $\lambda = 1$ uses the full Monte Carlo return. RLHF commonly uses $\gamma = 1$ and λ around 0.95.

python

```python
def gae(rewards, values, gamma=1.0, lam=0.95):
    """rewards[t]: per-token reward (KL penalty, plus the RM score on the last token).
       values[t]:  critic estimate V(s_t). Returns advantages and value targets."""
    T = len(rewards)
    adv, last = [0.0] * T, 0.0
    for t in reversed(range(T)):
        next_v = values[t + 1] if t + 1 < T else 0.0   # no value after the final token
        delta = rewards[t] + gamma * next_v - values[t]
        last = delta + gamma * lam * last               # discounted sum of future deltas
        adv[t] = last
    returns = [a + v for a, v in zip(adv, values)]      # regression targets for the critic
    return adv, returns

adv, ret = gae([-0.01, -0.02, 0.9], [0.5, 0.6, 0.7])
print([round(a, 3) for a in adv])   # [0.346, 0.27, 0.2]
```

rust

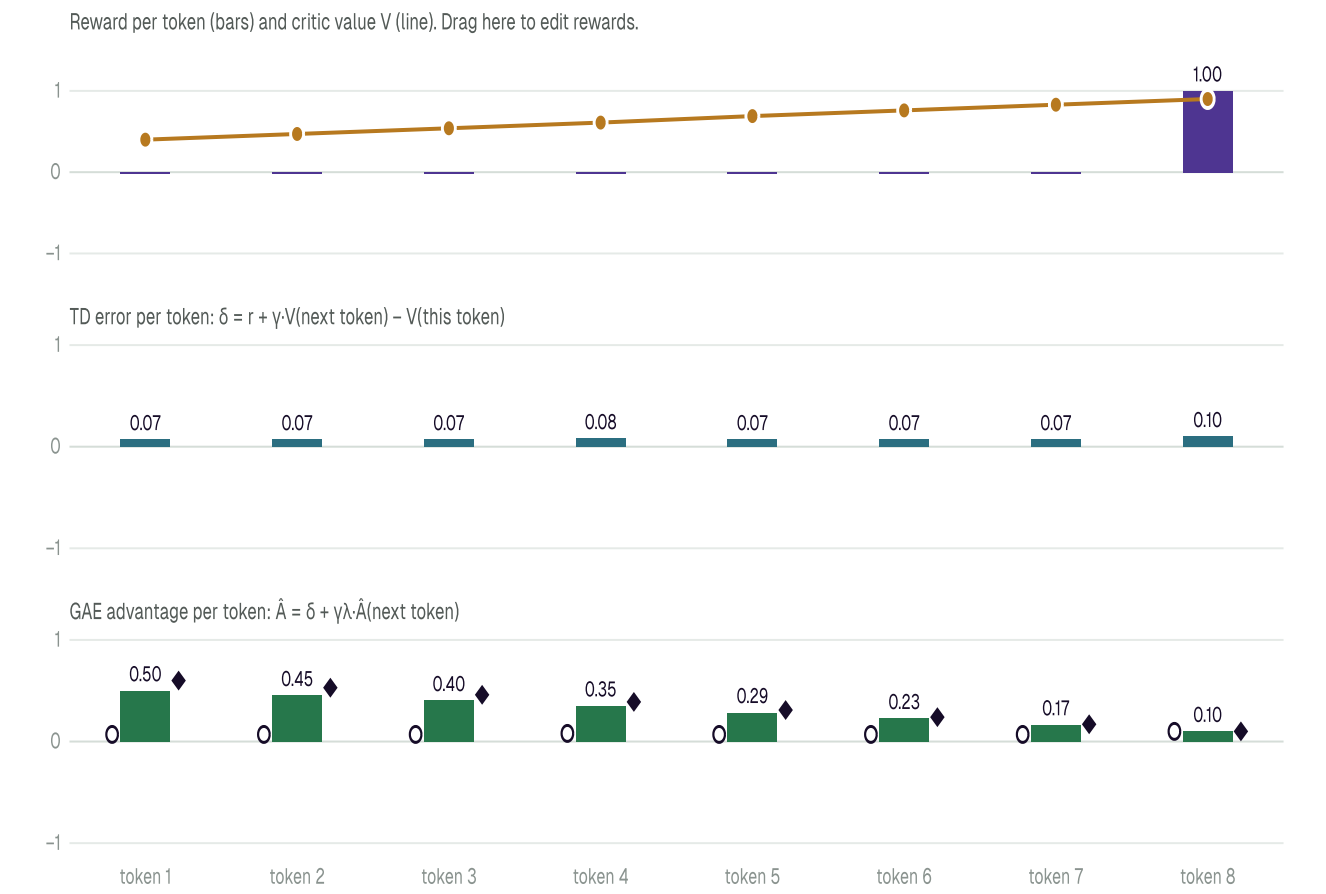
```rust
/// rewards[t]: per-token reward (KL penalty, plus the RM score on the last token).
/// values[t]:  critic estimate V(s_t). Returns advantages and value targets.
fn gae(rewards: &[f64], values: &[f64], gamma: f64, lam: f64) -> (Vec<f64>, Vec<f64>) {
    let t_len = rewards.len();
    let (mut adv, mut last) = (vec![0.0; t_len], 0.0);
    for t in (0..t_len).rev() {
        let next_v = if t + 1 < t_len { values[t + 1] } else { 0.0 }; // no value after the final token
        let delta = rewards[t] + gamma * next_v - values[t];
        last = delta + gamma * lam * last; // discounted sum of future deltas
        adv[t] = last;
    }
    let returns = adv.iter().zip(values).map(|(a, v)| a + v).collect(); // regression targets for the critic
    (adv, returns)
}

fn round_to(x: f64, digits: usize) -> f64 {
    // like Python's round(x, n): format the exact value, then read it back
    format!("{:.*}", digits, x).parse().unwrap()
}

fn main() {
    let (adv, _ret) = gae(&[-0.01, -0.02, 0.9], &[0.5, 0.6, 0.7], 1.0, 0.95);
    let rounded: Vec<f64> = adv.iter().map(|&a| round_to(a, 3)).collect();
    println!("{:?}", rounded); // [0.346, 0.27, 0.2]
}
```

The widget runs the same `gae` function on a short response, so you can see where each token's advantage comes from.

Interactive

Discount and GAE (shown at its default setting)

Pick "Reward only at the end" and drag λ from 1 down to 0: watch the end reward stop reaching the early tokens, which then lean on the critic's guesses instead. Drag or tap on the top chart to edit rewards or values.



γλ (decay per token)

0.95

Effective look-ahead 1/(1 − γλ)

20.0 tokens

Weight of the last reward on token 1

0.698

Advantage of token 1, $\hat{A}_1$

0.50

Token 1 sees the reward on the last token with weight $(\gamma\lambda)^7$ = **0.698**, and leans on the critic's guesses about later tokens with total weight **0.30**. At λ near 1 with γ = 1 the advantage is close to the actual sum of remaining rewards minus $V(s_t)$. The critic only acts as a baseline, so its errors add little bias, but every later token's luck lands on every earlier token (high variance). RLHF

usually keeps γ = 1 and λ around 0.95: almost the full reward signal, with the critic trimming some variance.

PPO's clipped surrogate

Plain words

Small steps only

Collecting answers and scoring them is expensive, so you want to learn as much as possible from each batch. But if you change the model a lot based on one batch, you can wreck it, and the next batch of answers comes from a broken model. PPO's rule is simple: within one batch, no single answer's probability may move by more than a fixed fraction, say 20 percent. Once it has moved that far, PPO stops pushing on it.

Under the hood

The objective

Proximal Policy Optimization (Schulman et al. 2017) reuses each batch of rollouts for a few gradient steps. Let $\rho_t(\theta) = \pi_\theta(y_t \mid s_t) / \pi_{\theta_{\text{old}}}(y_t \mid s_t)$ be the probability ratio between the policy being updated and the one that generated the data.

$$\mathcal{L}^{\text{CLIP}}(\theta) = \mathbb{E}_t\Big[\min\big(\rho_t(\theta)\,\hat{A}_t,\ \text{clip}(\rho_t(\theta),\,1-\epsilon,\,1+\epsilon)\,\hat{A}_t\big)\Big]$$

Maximized. For a good token ($\hat{A}_t > 0$) the gain stops once its probability has risen by a factor $1+\epsilon$. For a bad token the gain stops once it has fallen to $1-\epsilon$. The **min** makes the bound pessimistic: if an update already went the wrong way, the full penalty stays.

The full loss adds a value-function regression term and often an entropy bonus. The standard ϵ is 0.2.

$$\mathcal{L} = -\mathcal{L}^{\text{CLIP}} + c_v\,\mathbb{E}\big[(V_\theta(s_t) - \hat{R}_t)^2\big] - c_e\,\mathbb{E}\big[\mathcal{H}(\pi_\theta(\cdot \mid s_t))\big]$$

In code:

python

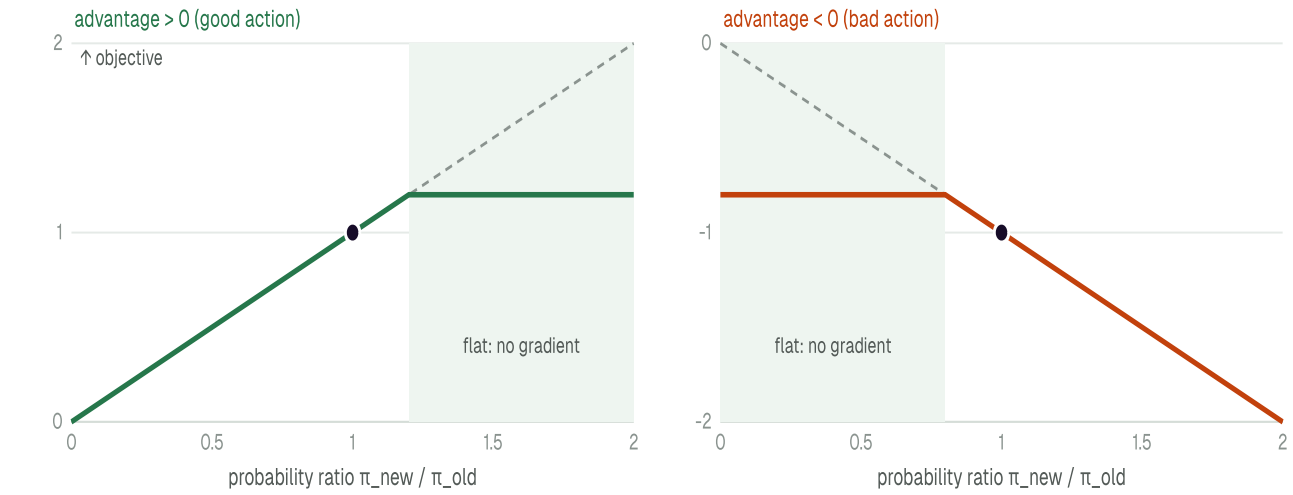
```python
import torch

def ppo_loss(logp_new, logp_old, adv, values_new, returns, eps=0.2, vf_coef=0.5):
    """All tensors are per response token, already masked to exclude prompt and padding."""
    ratio = torch.exp(logp_new - logp_old)                # pi_theta / pi_old, per token
    adv = (adv - adv.mean()) / (adv.std() + 1e-8)         # whitening is standard and stabilizes training
    unclipped = ratio * adv
    clipped = torch.clamp(ratio, 1 - eps, 1 + eps) * adv
    policy_loss = -torch.min(unclipped, clipped).mean()   # minus: optimizers minimize
    value_loss = 0.5 * (values_new - returns).pow(2).mean()
    return policy_loss + vf_coef * value_loss

# one token, eps = 0.2: the objective stops rewarding moves past the clip boundary
for r, a in [(1.5, 1.0), (0.6, -1.0), (1.1, 1.0)]:
    print(r, a, min(r * a, max(0.8, min(1.2, r)) * a))    # 1.2, -0.8, 1.1
```

rust

```rust
// Plain Rust, no ML framework: the same math on small vectors.

fn mean(xs: &[f64]) -> f64 {
    xs.iter().sum::<f64>() / xs.len() as f64
}

/// All slices are per response token, already masked to exclude prompt and padding.
fn ppo_loss(logp_new: &[f64], logp_old: &[f64], adv: &[f64], values_new: &[f64], returns: &[f64], eps: f64,
vf_coef: f64) -> f64 {
    let n = adv.len() as f64;
    // whitening is standard and stabilizes training (std with n - 1, as torch.std does)
    let mu = mean(adv);
    let sd = (adv.iter().map(|a| (a - mu).powi(2)).sum::<f64>() / (n - 1.0)).sqrt();
    let mut policy_sum = 0.0;
    for i in 0..adv.len() {
        let ratio = (logp_new[i] - logp_old[i]).exp(); // pi_theta / pi_old, per token
        let a = (adv[i] - mu) / (sd + 1e-8);
        let unclipped = ratio * a;
        let clipped = ratio.clamp(1.0 - eps, 1.0 + eps) * a;
        policy_sum += unclipped.min(clipped);
    }
    let policy_loss = -policy_sum / n; // minus: optimizers minimize
    let value_loss = 0.5 * values_new.iter().zip(returns).map(|(v, r)| (v - r).powi(2)).sum::<f64>() / n;
    policy_loss + vf_coef * value_loss
}

fn main() {
    // one token, eps = 0.2: the objective stops rewarding moves past the clip boundary
    for (r, a) in [(1.5_f64, 1.0_f64), (0.6, -1.0), (1.1, 1.0)] {
        println!("{:?} {:?} {:?}", r, a, (r * a).min(0.8_f64.max(1.2_f64.min(r)) * a)); // 1.2, -0.8, 1.1
    }
    // tiny worked example for ppo_loss itself: three response tokens
    let loss = ppo_loss(&[-1.0, -0.5, -2.0], &[-1.1, -0.7, -1.9], &[0.5, -0.2, 1.0],
                        &[0.4, 0.6, 0.2], &[0.5, 0.5, 0.5], 0.2, 0.5);
    println!("{:.4}", loss); // 0.1127
}
```

Interactive **PPO clipping** (shown at its default setting)

The objective PPO maximizes for one token, as a function of how much the update changed that token's probability. Shaded regions are where the gradient is zero. Widen ε and the policy is allowed bigger steps per batch.



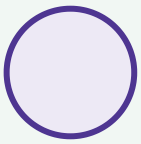
Unclipped ratio × advantage (dashed) PPO objective, good action PPO objective, bad action

With ε = 0.20, a good action can gain at most a factor of 1.20 in probability per update before its gradient switches off, and a bad action can be cut to 0.80 of its old probability. The pessimistic min keeps the penalty when an update has already gone the wrong way (ratio above 1 for a bad action).

Four models in memory: why PPO is expensive

Diagram **Four people in the training room**

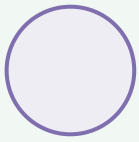
Why PPO needs so much memory, one seat at a time.



Policy
writes the answers
trained

1 of 4 networks in memory

The student, the policy, writes every answer. It is the network being trained, so it needs weights, gradients, and optimizer state: about 16 bytes per parameter.

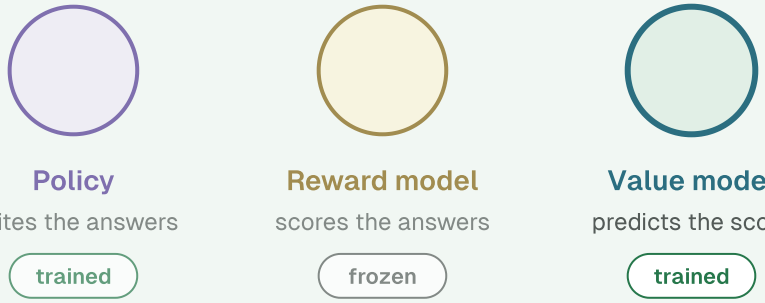


Policy
writes the answers
trained

Reward model
scores the answers
frozen

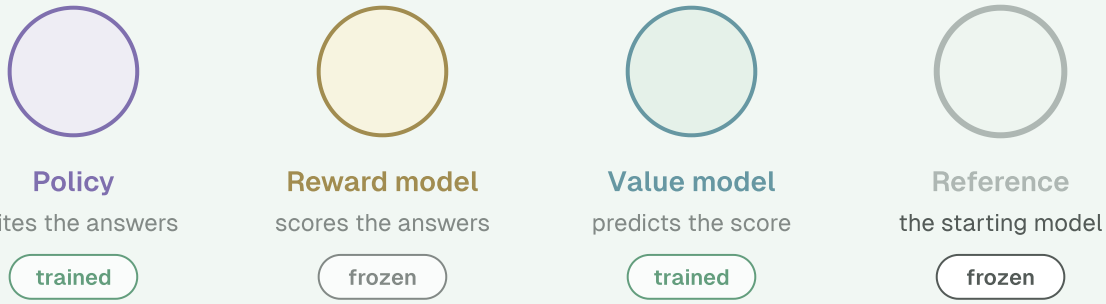
2 of 4 networks in memory

The judge, the reward model, scores each answer. It is frozen, so it needs only its weights, about 2 bytes per parameter in bf16.



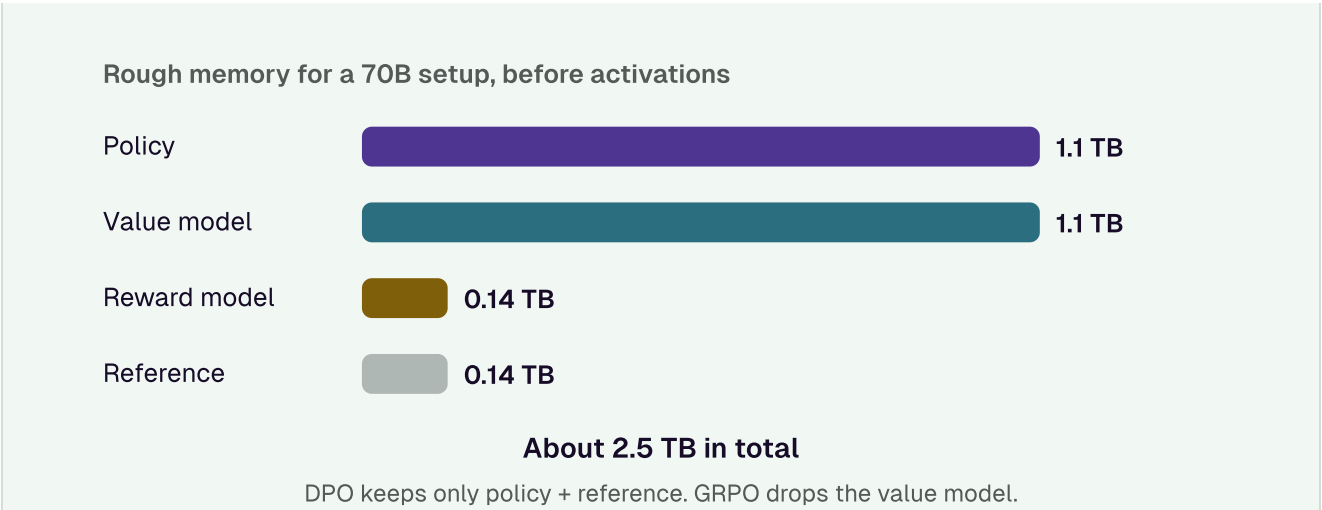
3 of 4 networks in memory

The coach, the value model, predicts what each answer should score so the policy can learn from surprises. It is trained too, and it is often as big as the policy.



4 of 4 networks in memory

The photo of the student on day one, the reference model, supplies the probabilities for the KL leash. Frozen, but always in memory.



Add it up with rough rules of thumb (16 bytes per trained parameter, 2 per frozen one) and a 70B run needs about 2.5 TB before counting activations or generation. That bill is why the next two chapters exist: DPO sends the judge and coach home, and GRPO sends the coach home.

For leaders

The bill

A PPO run for a language model keeps four networks alive at once:

Policy

Being trained. Weights, gradients, and optimizer state.

Value model

Being trained. Often the same size as the policy, so it doubles training memory.

Reward model

Frozen. Scores every rollout.

Reference model

Frozen. Provides log-probs for the KL penalty.

Rough arithmetic, labeled as such: mixed-precision Adam needs about 16 bytes per trainable parameter (weights, master copy, two moment estimates, gradients), and a frozen bf16 model needs about 2. With a 70B-parameter policy and value model, the two trainable networks need roughly 2.2 TB before activations, and the two frozen ones another 0.3 TB. On top of that, rollouts are generated token by token, and generation often dominates wall-clock time. This cost profile explains the next

two chapters: DPO removes the RL loop and the reward and value models; GRPO keeps on-policy RL but removes the value model.

> **Principal asks**
> What fraction of wall-clock time is generation versus training? Is the value model the same size as the policy, and did you try sharing a backbone? How many rollouts per prompt, and how many PPO epochs per batch? What is the throughput in prompts per hour, and what does one additional point of win rate cost in GPU-hours?

The InstructGPT recipe

For leaders

The template everyone copied

1. **Supervised fine-tuning** on labeler-written demonstrations (about 13,000 training prompts). See supervised fine-tuning.
2. **Reward model** trained on labeler rankings of 4 to 9 model outputs per prompt (about 33,000 prompts).
3. **PPO** against the reward model with a per-token KL penalty to the SFT model (about 31,000 prompts), optionally mixing in pretraining gradients ("PPO-ptx").

The headline result in Ouyang et al. (2022): labelers preferred outputs from a 1.3B-parameter InstructGPT model over the 175B GPT-3 base model. Alignment work moved the needle more than a hundred-fold increase in parameters on the thing users actually experienced. The paper also documented an **alignment tax**: PPO caused regressions on some public NLP benchmarks (such as SQuAD and DROP), which mixing pretraining updates back in largely recovered.

Reward hacking, the alignment tax, and calibration

Plain words

Finding the loophole

If a cleaning robot is rewarded for not seeing mess, it may learn to cover its camera. RL finds whatever earns reward, including loopholes nobody intended. In language models the loopholes are

usually boring: longer answers, more confident tone, telling the user what they want to hear.

For leaders

What goes wrong and the standard defenses

Symptom	Cause	Defense
Responses grow longer every checkpoint	Length bias in the reward model	Length penalty, length-balanced preference data, length-controlled evals
Agrees with the user even when the user is wrong	Sycophancy in preference labels	Counterfactual prompts in RM data, targeted sycophancy evals
Hedging and uncertainty disappear	Raters reward confident tone	Calibration evals, honesty rubrics, abstention rewarded when correct
Reward rises, human win rate flat	Overoptimization of the proxy	Higher β, early stopping on independent eval, RM ensembles, RM refresh
Benchmarks regress after RL	Alignment tax	Pretraining or SFT data mixed into updates, regression suite gating

Watch out

RLHF can make a model look better to raters while making its probabilities worse. The GPT-4 technical report (OpenAI 2023) showed the pretrained model was well calibrated on multiple-choice questions and the post-trained model much less so. If your product uses model confidence (to abstain, route, or show uncertainty), measure calibration after every post-training stage, not just before.

Practical knobs

For leaders

What the run sheet controls

Knob	What it does	Indicative setting
KL coefficient β	Leash length; trades reward gain for drift	InstructGPT used 0.02; adaptive controllers (Ziegler et al. 2019) target a fixed KL instead
Clip ε	Maximum per-batch probability change	0.2
Rollouts per prompt and batch size	Gradient noise versus cost per step	Hundreds to thousands of responses per update
PPO epochs per batch	Sample reuse; too many and the data goes stale	1 to 4
Sampling temperature	Exploration during rollouts	Near 1.0, so rollouts reflect the actual policy
Learning rate	Step size	Usually an order of magnitude or more below SFT

Eval lens

Evaluate an RLHF run with signals the optimizer never saw. The minimum set: head-to-head win rate against the starting model judged by humans or a held-out judge, with a paired comparison and a confidence interval; the same win rate controlled for length; a capability regression suite (math, code, knowledge) to price the alignment tax; calibration error before and after; and KL from the reference plotted against all of these, so you can see where quality peaked.

Part IV · Chapter 36

DPO and offline preference methods: RLHF without the RL loop

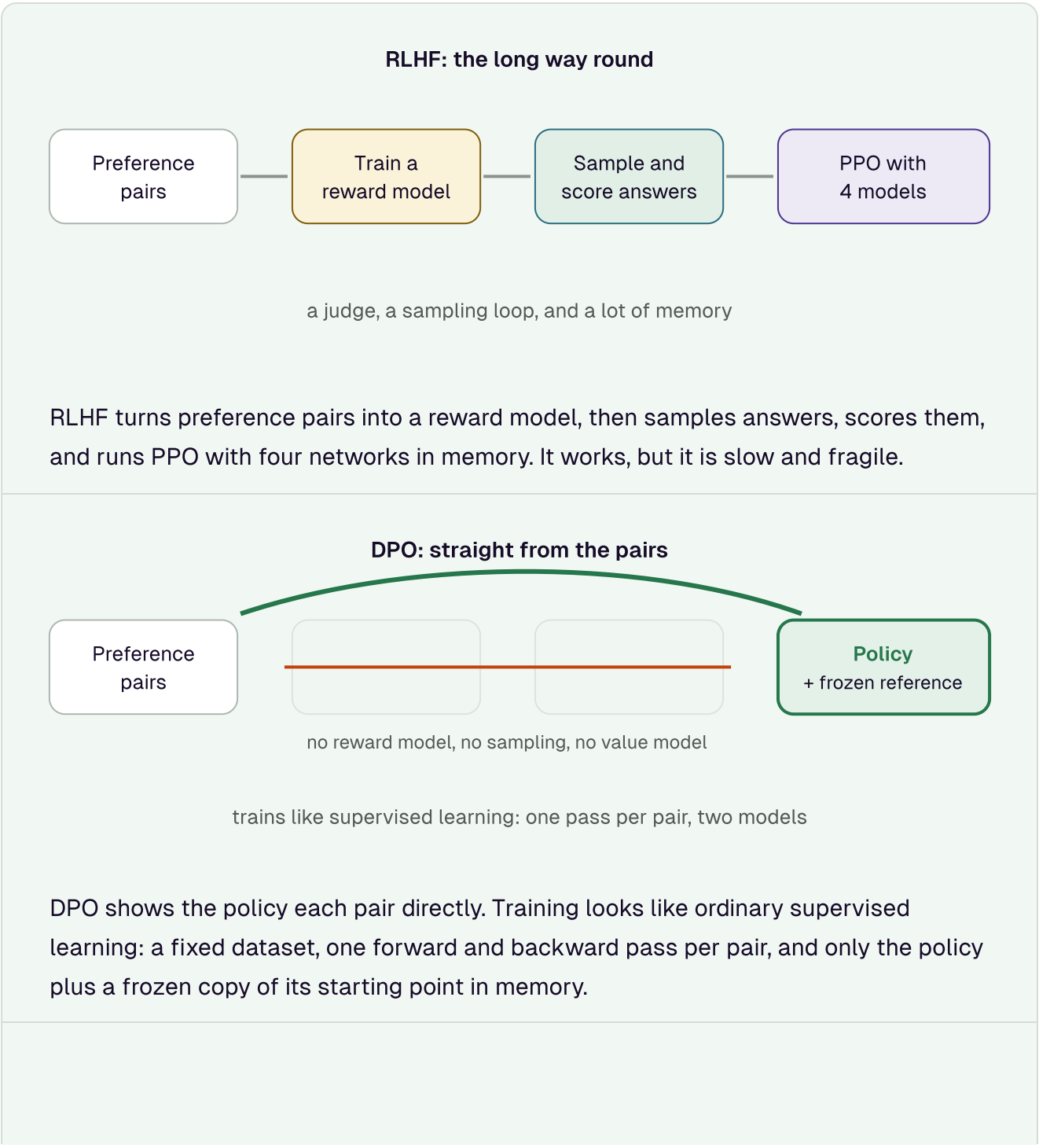
Direct Preference Optimization (Rafailov et al. 2023) showed that the KL-regularised RLHF objective from the previous chapter can be optimized with a simple classification loss on preference pairs. No reward

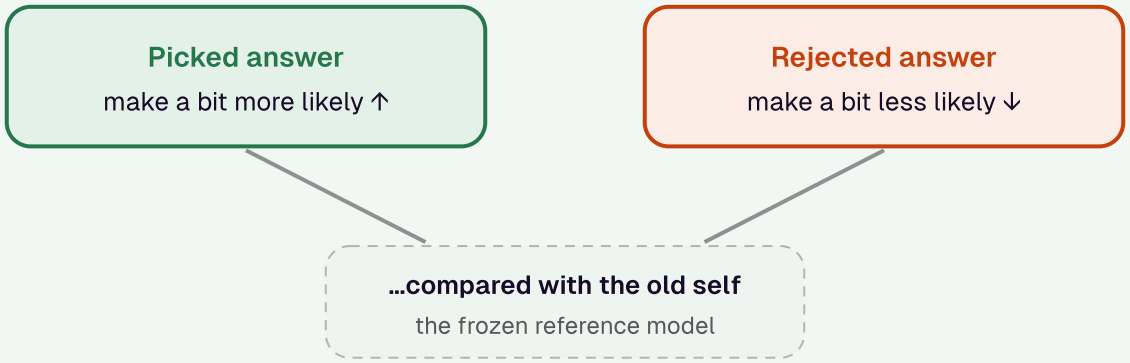
model at training time, no sampling, no value function, no PPO. It became the default post-training method for open models within a year, and it spawned a family of variants. Understanding exactly what DPO optimizes, and where that differs from RLHF in practice, is one of the most useful pieces of judgment in post-training.

Diagram

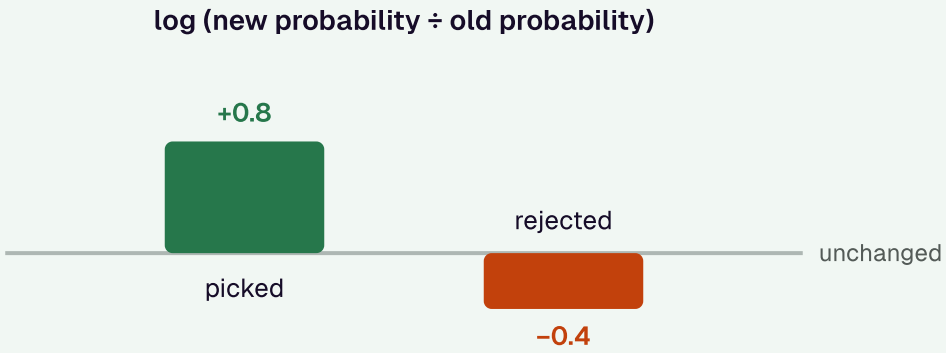
This one, not that one

How DPO learns straight from pairs, without hiring a judge.



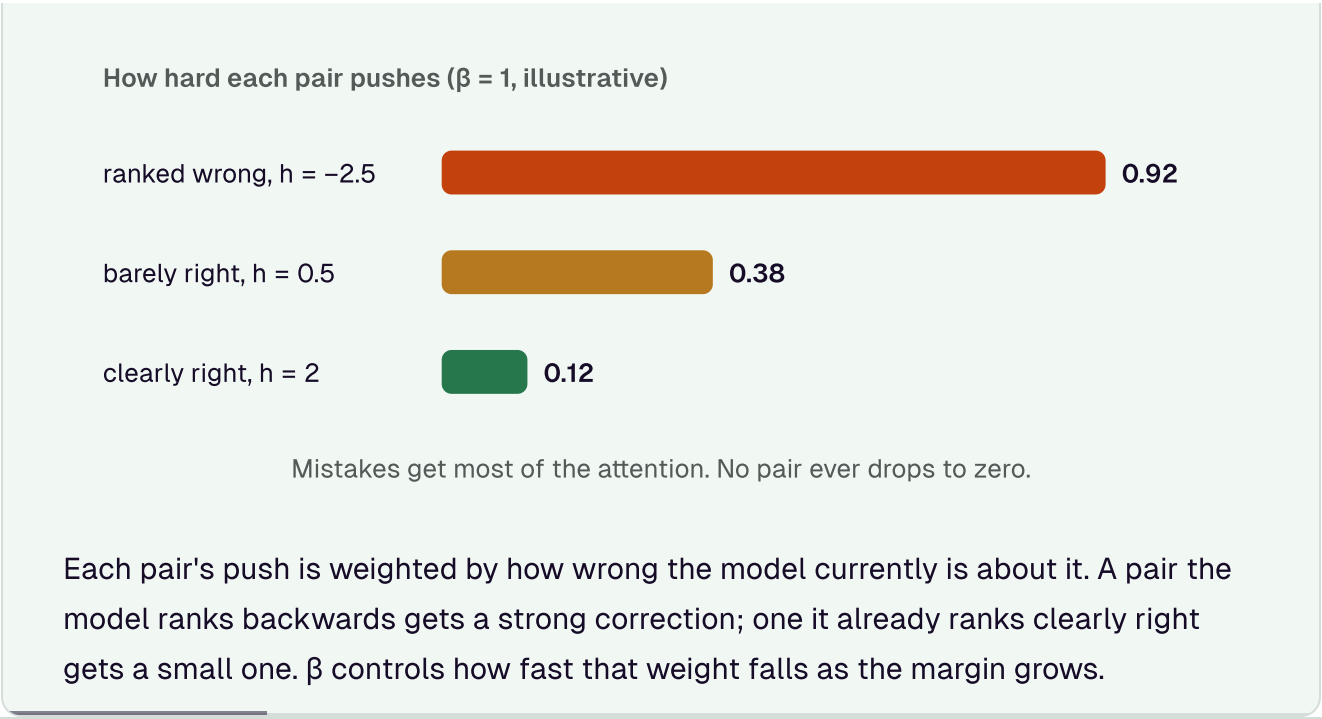


Every pair carries one instruction: nudge the picked answer up and the rejected answer down, measured against how the model wrote before training. That comparison to the old self is the leash, built straight into the loss.



margin h = 0.8 − (−0.4) = 1.2: this gap is the hidden "reward"

DPO's trick is that the model's own shift in probability, relative to its old self, works as a reward score. The loss only looks at the gap between the picked and rejected shifts, which is why the paper's subtitle is "your language model is secretly a reward model". Numbers are illustrative.



Mistakes get most of the attention. No pair ever drops to zero.

Each pair's push is weighted by how wrong the model currently is about it. A pair the model ranks backwards gets a strong correction; one it already ranks clearly right gets a small one. β controls how fast that weight falls as the margin grows.

The idea

Plain words

Skip the middleman

RLHF works in two stages: first train a judge on people's picks, then train the student against the judge. DPO skips the judge. It shows the student each pair directly and says: make the picked answer a bit more likely and the rejected answer a bit less likely, compared with how you used to write. The comparison to how it used to write is the leash, built straight into the lesson.

For leaders

Why it spread, and what you give up

DPO trains like supervised learning: fixed dataset, one forward and backward pass per pair, two models in memory (the policy and a frozen reference, whose log-probabilities can even be precomputed). It is cheaper, far more stable, and easier to debug than PPO. Llama 3 (Meta 2024), Zephyr, and Tülu are well-known examples of post-training pipelines that relied on DPO.

What you give up is the online loop. PPO learns from the model's own fresh samples, scored on the fly. Plain DPO learns from a fixed set of pairs, often generated by other models. The policy is never corrected on its own new mistakes, and its ceiling is set by the pairs you collected. Much of the work since 2023 is about recovering the benefits of online data while keeping DPO's simplicity.

Principal asks

Where did the pairs come from: this model's samples, an older model, or another lab's model? How were they labeled, and what is the label agreement? Did the chosen responses' log-likelihood go up or down during training? Is the win rate length-controlled? Did you compare against one round of online or iterative DPO?

The derivation, step by step

Under the hood

From the RLHF objective to a classification loss

Step 1. Start from the RLHF objective for a fixed prompt x:

$$\max_{\pi} \; \mathbb{E}_{y\sim\pi}\big[r(x,y)\big] - \beta\,\mathbb{D}_{\mathrm{KL}}\big(\pi(\cdot\mid x)\,\|\,\pi_{\mathrm{ref}}(\cdot\mid x)\big)$$

Step 2. Rewrite it as a single KL divergence. Define $\pi^*(y\mid x) = \pi_{\mathrm{ref}}(y\mid x)\,e^{r(x,y)/\beta}/Z(x)$. Expanding the KL and collecting terms:

$$\mathbb{E}_{y\sim\pi}\big[r\big] - \beta\,\mathbb{E}_{y\sim\pi}\Big[\log\tfrac{\pi(y|x)}{\pi_{\mathrm{ref}}(y|x)}\Big] = -\beta\,\mathbb{E}_{y\sim\pi}\Big[\log\tfrac{\pi(y|x)}{\pi_{\mathrm{ref}}(y|x)\,e^{r(x,y)/\beta}}\Big] = -\beta\,\mathbb{D}_{\mathrm{KL}}\big(\pi\,\|\,\pi^*\big) + \beta\log$$

$Z(x)$ does not depend on π, and a KL divergence is minimized (at zero) exactly when the two distributions match. So the optimal policy is π^*.

Step 3. Invert: express the reward through the optimal policy. Take logs of the definition of π^* and solve for r:

$$r(x,y) = \beta\log\frac{\pi^*(y\mid x)}{\pi_{\mathrm{ref}}(y\mid x)} + \beta\log Z(x)$$

Step 4. Substitute into Bradley–Terry. Preferences depend only on reward differences between two responses to the same prompt, so the intractable $\beta \log Z(x)$ cancels:

$$P(y_w \succ y_l \mid x) = \sigma\Big(\beta \log \frac{\pi^*(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta \log \frac{\pi^*(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)}\Big)$$

Step 5. Parameterize π^* with the model π_θ and maximize likelihood of the observed preferences:

$$\mathcal{L}_{\text{DPO}}(\theta) = -\,\mathbb{E}_{(x,y_w,y_l)}\Big[\log \sigma\Big(\beta \log \frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta \log \frac{\pi_\theta(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)}\Big)\Big]$$

Fitting this loss is fitting a reward model whose form is fixed to $\hat{r}_\theta(x, y) = \beta \log \frac{\pi_\theta(y|x)}{\pi_{\text{ref}}(y|x)}$, the **implicit reward**. The policy and the reward model are the same network. "Your language model is secretly a reward model" is the paper's subtitle.

Note what the derivation assumes: the Bradley–Terry model is right, the optimum over **all** policies is reachable, and the data covers the responses the policy will produce. Each assumption fails in practice in a specific way, covered below.

What the gradient does

For leaders

What β means in DPO

β plays the same role as in RLHF: the strength of the leash to the reference. In DPO it also sets how far the margin must grow before a pair stops contributing. Typical values sit between 0.01 and 0.5, and results are sensitive to it, so a sweep over β is part of any serious DPO experiment. If a team reports one DPO run at one β, the result tells you about that β, not about DPO.

Under the hood

A weighted push and pull

$$\nabla_\theta \mathcal{L}_{\text{DPO}} = -\beta \mathbb{E}\Big[\underbrace{\sigma\big(\hat{r}_\theta(x, y_l) - \hat{r}_\theta(x, y_w)\big)}_{\text{high when the model ranks the pair wrongly}} \Big(\nabla_\theta \log \pi_\theta(y_w \mid x) - \nabla_\theta \log \pi_\theta(y_l \mid x) \Big)$$

Raise the chosen response, lower the rejected one, weighted by how wrong the implicit reward currently is. Pairs already ranked correctly by a wide margin get small weight but never zero.

The gradient only constrains the **difference** between the two log-likelihoods. Nothing in it says the chosen response's probability must rise. That single fact explains the most important DPO failure mode.

python

```python
import torch.nn.functional as F

def dpo_loss(pi_logp_w, pi_logp_l, ref_logp_w, ref_logp_l, beta=0.1):
    """Each argument: summed log-prob of the full response under the policy or the frozen
       reference, shape [batch]. Sum only over response tokens, never the prompt."""
    h = (pi_logp_w - ref_logp_w) - (pi_logp_l - ref_logp_l)   # log-ratio margin in nats
    loss = F.softplus(-beta * h).mean()                        # == -log sigmoid(beta * h)
    reward_acc = (h > 0).float().mean()                        # "implicit reward accuracy"
    return loss, reward_acc

# two pairs: the first is ranked correctly (h = 2.0), the second wrongly (h = -2.5)
# loss = 0.712; per-pair gradient weights sigma(-beta*h) = 0.45 and 0.562
```

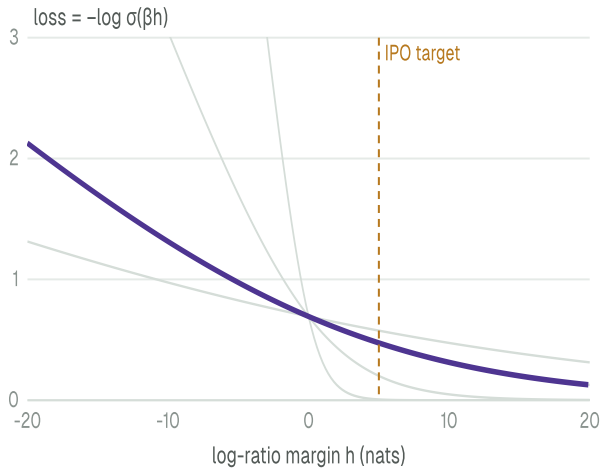
rust

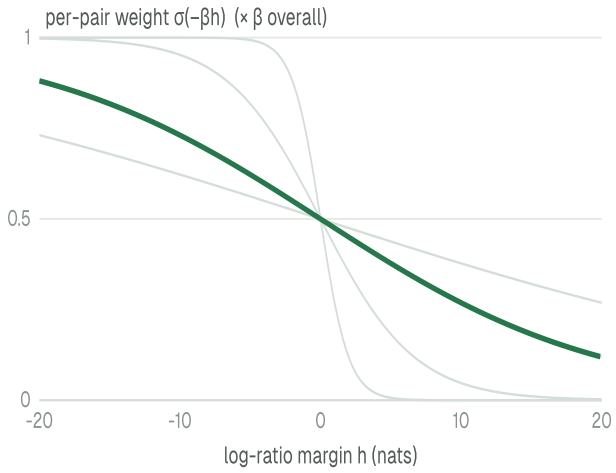
```rust
// Plain Rust, no ML framework: the same math on small vectors.

fn softplus(x: f64) -> f64 {
    // log(1 + e^x) without overflow: max(x, 0) + log(1 + e^-|x|)
    x.max(0.0) + (-x.abs()).exp().ln_1p()
}

fn sigmoid(x: f64) -> f64 {
    1.0 / (1.0 + (-x).exp())
}

/// Each argument: summed log-prob of the full response under the policy or the frozen
/// reference, one per pair. Sum only over response tokens, never the prompt.
fn dpo_loss(pi_logp_w: &[f64], pi_logp_l: &[f64], ref_logp_w: &[f64], ref_logp_l: &[f64], beta: f64) -> (f64,
f64) {
    let n = pi_logp_w.len() as f64;
    let h: Vec<f64> = (0..pi_logp_w.len())
        .map(|i| (pi_logp_w[i] - ref_logp_w[i]) - (pi_logp_l[i] - ref_logp_l[i])) // log-ratio margin in nats
        .collect();
    let loss = h.iter().map(|&x| softplus(-beta * x)).sum::<f64>() / n;    // == -log sigmoid(beta * h)
    let reward_acc = h.iter().filter(|&&x| x > 0.0).count() as f64 / n; // "implicit reward accuracy"
    (loss, reward_acc)
}

fn round_to(x: f64, digits: usize) -> f64 {
```

```
    // like Python's round(x, n): format the exact value, then read it back
    format!("{:.*}", digits, x).parse().unwrap()
}

fn main() {
    // two pairs: the first is ranked correctly (h = 2.0), the second wrongly (h = -2.5)
    let (loss, acc) = dpo_loss(&[-10.0, -20.0], &[-12.0, -15.0], &[-11.0, -19.0], &[-11.0, -16.5], 0.1);
    println!("{:.3} {:?}", loss, acc); // 0.712 0.5
    let weights: Vec<f64> = [2.0, -2.5].iter().map(|&h: &f64| round_to(sigmoid(-0.1 * h), 3)).collect();
    println!("{:?}", weights); // [0.45, 0.562]  per-pair gradient weights sigma(-beta*h)
}
```

Interactive **DPO loss and gradient weight** (shown at its default setting)

The loss and the per-pair weight as a function of the log-ratio margin h (how much more the policy favors chosen over rejected, relative to the reference). Grey curves are other β values. Lower β and the loss flattens: the model must move further from the reference before the loss notices.



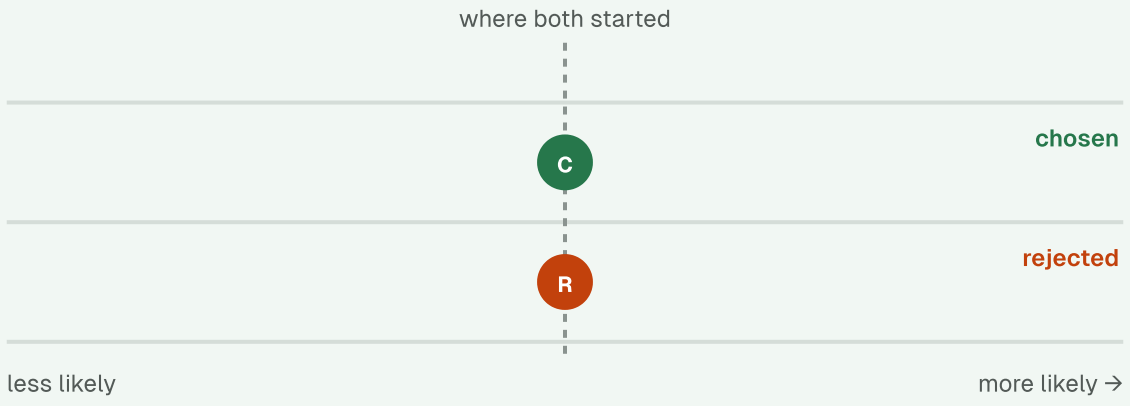


At β = 0.10, a pair the model already separates by 10 nats still contributes 27% of full weight, and the loss only reaches zero as the margin goes to infinity. That is why DPO keeps pushing on pairs it has already learned. IPO replaces this with a squared loss around a finite target margin 1/(2τ), shown dashed with τ = β (5.0 nats).

Failure modes

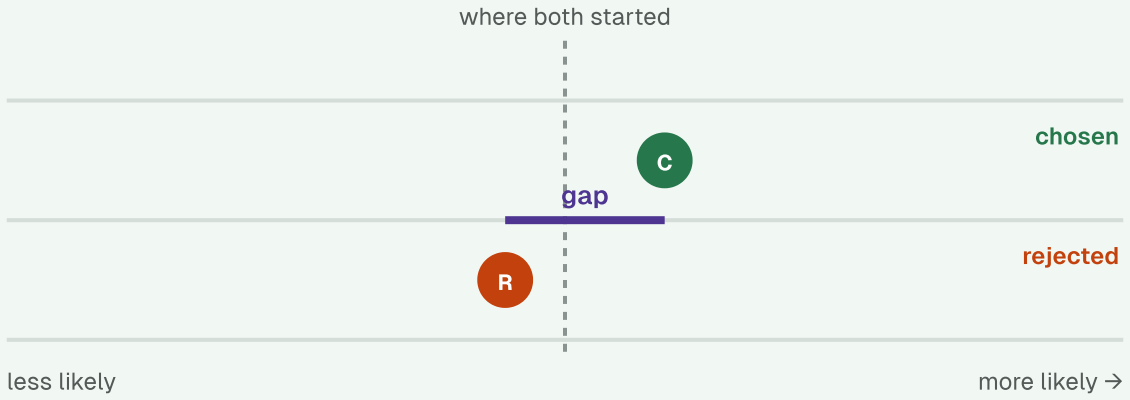
Diagram **The race where both runners slow down**

Likelihood displacement: how DPO can widen the gap while making both answers less likely.



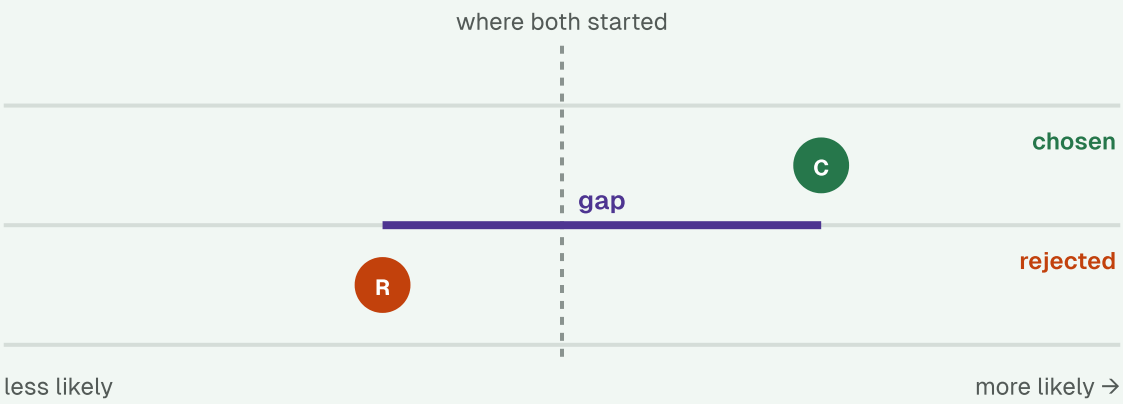
Position = how likely the model is to write that answer.

Two runners start side by side: the chosen answer and the rejected answer. A runner further right is an answer the model is more likely to write.



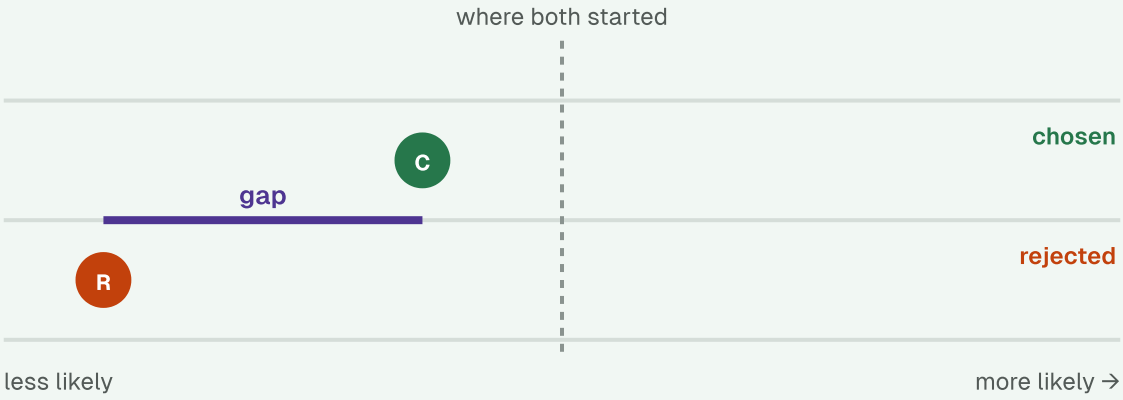
The coach only measures the distance between them.

DPO's loss is a coach who only measures the gap between the two runners. It never checks where either one is on the track.



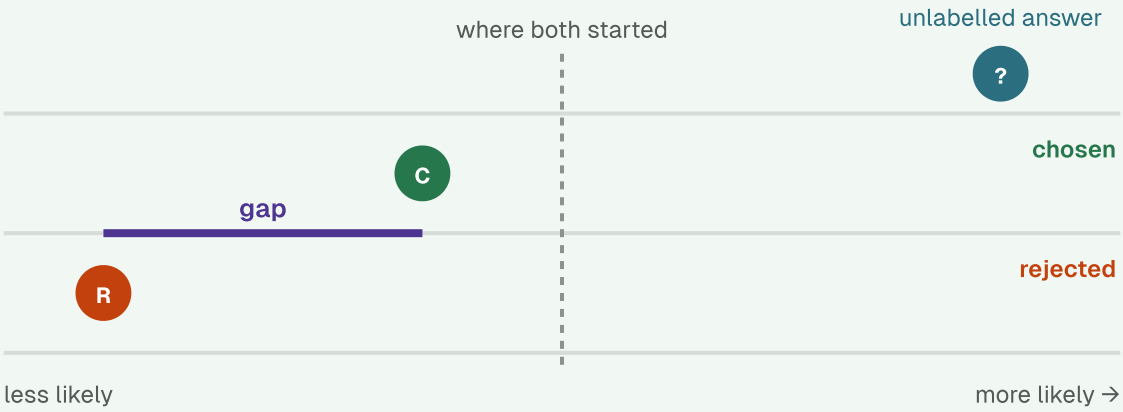
Healthy: chosen speeds up, rejected falls back.

In a healthy run, the chosen runner moves ahead and the rejected runner falls back. The gap grows and both moves are the ones you wanted.



Trap: both fall back. The gap still grows, and the scoreboard says 100%.

The trap: both runners fall back, the rejected one faster. The gap grows just the same, so the loss falls and reward accuracy looks perfect, yet the model is now less likely to write the answer people picked.



The probability went to an answer nobody labelled.

Probability has to go somewhere. It flows to answers that were in neither pair, which can be harmless or can be a new kind of bad output. Pal et al. (2024) and Razin et al. (2024) documented this; it is worst when the two answers are very similar.

Chosen log-likelihood
Should not fall steadily. Watch it first.

Response length
Creeping length is a cheap way to widen gaps

Held-out win rate
Length-controlled, judged independently

Reward accuracy, margins
Rise in healthy and broken runs alike

So watch where the runners are, not just the gap: the absolute log-likelihood of chosen answers, response length, and a held-out win rate. The default dashboard metrics, reward accuracy and margins, look the same in a healthy run and a broken one.

Plain words

Winning the comparison by making both worse

Suppose you tell a student "write more like essay A than essay B". The student can do that by improving toward A. Or by drifting away from both, as long as it drifts further from B. Both satisfy the

instruction. DPO only checks the gap between the two, so sometimes it takes the second route, and the probability goes to answers that neither essay resembles.

For leaders

The four ways DPO disappoints

Likelihood displacement

Chosen and rejected log-likelihoods both fall during training, while the margin and the "reward accuracy" metric look excellent. Pal et al. (2024, "Smaug", which proposed DPO-Positive) and Razin et al. (2024) documented this; it is worst when chosen and rejected responses are similar, which is exactly what careful pairwise labeling tends to produce.

Length exploitation

If chosen responses are longer on average, increasing length is an easy way to widen the margin. DPO-trained models often get wordier. Length-normalized variants (SimPO) and explicit length regularization address it.

Off-policy data

Pairs sampled from a different model teach the policy about someone else's mistakes. Several 2024 studies (for example Tajwar et al. 2024) found on-policy or iteratively refreshed data clearly outperforms static off-policy pairs.

Overfitting deterministic preferences

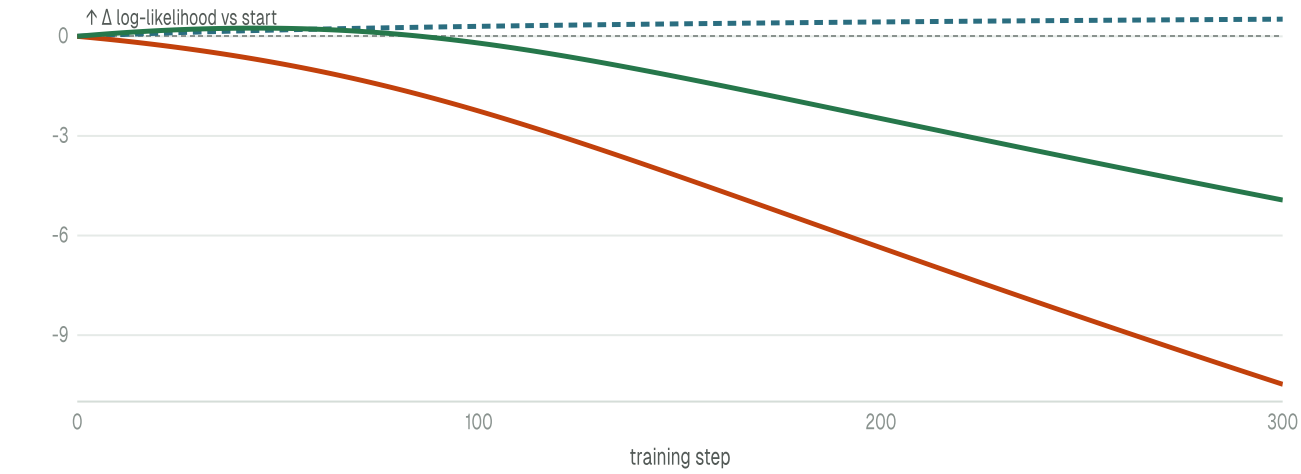
If a pair is always preferred one way, Bradley–Terry implies an infinite reward gap, and DPO keeps pushing the rejected response's probability toward zero regardless of β. The leash stops working. This was the motivation for IPO.

Watch out

Implicit reward accuracy and margins on the training set are the metrics DPO dashboards show by default, and they are the least informative ones. They rise in healthy runs and in runs suffering likelihood displacement alike. Track the absolute log-likelihood of chosen responses, average response length, and a held-out win rate.

Interactive **Likelihood displacement** (shown at its default setting)

A tiny softmax model over five responses, trained with DPO on one preference pair (solid lines) and with plain SFT on the chosen response (dashed). The slider sets how different chosen and rejected are. When they are similar, DPO separates them by pushing both down, and the probability flows to a third response. This is an exact simulation of a toy model, chosen to make the mechanism visible.



Chosen Δ log-prob (DPO)	Rejected Δ log-prob (DPO)	Mass on an unlabeled response
-4.93	**-10.48**	**99.7%**

Likelihood displacement: the margin grew to 5.5 nats, so preference accuracy looks perfect, yet the chosen response became less likely. 99.7% of the probability now sits on a third response neither label mentioned. SFT on the chosen response (dashed) raises it in every setting.

The family

Under the hood **Five variants and the problem each fixes**

IPO (Azar et al. 2023) replaces the log-sigmoid with a squared loss around a finite target margin, so the policy stops pushing once a pair is separated by enough. With $h = \log \frac{\pi_\theta(y_w)\,\pi_{\text{ref}}(y_l)}{\pi_\theta(y_l)\,\pi_{\text{ref}}(y_w)}$:

$$\mathcal{L}_{\text{IPO}} = \mathbb{E}\Big[\big(h - \tfrac{1}{2\tau}\big)^2\Big]$$

KTO (Ethayarajh et al. 2024) needs no pairs at all, only responses labeled desirable or undesirable, which is the shape of thumbs-up and thumbs-down product data. It applies a prospect-theory value function to the implicit reward relative to a reference point z_0 (an estimate of the KL between policy and reference), with separate weights for gains and losses:

$$v(x,y) = \begin{cases} \lambda_D\, \sigma\big(\beta(r_\theta(x,y) - z_0)\big) & y \text{ desirable} \\ \lambda_U\, \sigma\big(\beta(z_0 - r_\theta(x,y))\big) & y \text{ undesirable} \end{cases}, \qquad \mathcal{L}_{\text{KTO}} = \mathbb{E}\big[\lambda_y - v(x,y)\big]$$

Here $r_\theta(x,y) = \log \frac{\pi_\theta(y|x)}{\pi_{\text{ref}}(y|x)}$ is the *unscaled* log-ratio (β appears once, outside), and z_0 is the policy-to-reference KL in the same units.

ORPO (Hong et al. 2024) drops the reference model and merges SFT and preference learning into one stage: the usual next-token loss on the chosen response plus an odds-ratio penalty, where $\text{odds}_\theta(y \mid x) = \frac{P_\theta(y|x)}{1 - P_\theta(y|x)}$ and P_θ is the length-normalized sequence likelihood:

$$\mathcal{L}_{\text{ORPO}} = \mathcal{L}_{\text{SFT}}(y_w) - \lambda \log \sigma\Big(\log \frac{\text{odds}_\theta(y_w \mid x)}{\text{odds}_\theta(y_l \mid x)}\Big)$$

SimPO (Meng et al. 2024) also drops the reference model, uses the length-normalized average log-probability as the implicit reward (removing the incentive to grow length), and requires a target margin γ:

$$\mathcal{L}_{\text{SimPO}} = -\,\mathbb{E}\Big[\log \sigma\Big(\tfrac{\beta}{|y_w|} \log \pi_\theta(y_w \mid x) - \tfrac{\beta}{|y_l|} \log \pi_\theta(y_l \mid x) - \gamma\Big)\Big]$$

Online and iterative DPO keeps the DPO loss but refreshes the data: sample responses from the current policy, label pairs with a reward model or an LLM judge, train a round, repeat. Guo et al. (2024) found that online AI feedback outperformed both offline DPO-style training and RLHF baselines in their human evaluations. In practice, iterative DPO with a strong judge is now a common middle ground between offline DPO and full RL.

Method	Reference model	Data	Fixes
DPO	Yes	Pairs	Baseline: RLHF objective without RL
IPO	Yes	Pairs	Overfitting to deterministic preferences
KTO	Yes	Single responses with good/bad labels	Needing paired data
ORPO	No	Pairs	A separate SFT stage and reference model
SimPO	No	Pairs	Length exploitation and reference cost
Online / iterative DPO	Yes	Fresh pairs from the current policy	Off-policy data and the static ceiling

Choosing between DPO, PPO, and GRPO

For leaders

A decision guide

	DPO and offline variants	PPO (RLHF)	GRPO / RL with verifiable rewards
Signal	Fixed preference pairs	Learned reward model, on-policy samples	Verifier or reward model, groups of on-policy samples
Models in memory	2 (policy, reference)	4 (policy, value, reward, reference)	2 to 3 (no value model)
Compute	Low: like SFT	High: generation plus two trained networks	High: generation dominates, many samples per prompt
Stability	High	Sensitive to many hyperparameters	Moderate
Ceiling	Bounded by the dataset unless made iterative	High, limited by reward model quality	High on checkable tasks; drove the 2024 to 2025 reasoning gains
Best fit	Style, tone, format, harmlessness from good pair data	Open-ended helpfulness with a strong reward model	Math, code, tool use, anything with a reliable check

A common modern pipeline uses all three ideas: SFT, then preference optimization (often DPO or an iterative variant) for style and safety, then RL with verifiable rewards for reasoning, with a reward model or judge filling gaps where no verifier exists.

Eval lens

DPO results are easy to overstate. Ask for: win rate against the starting model using a judge validated against humans (LLM as judge), length-controlled (AlpacaEval 2's length-controlled win rate, Dubois et al. 2024, is the best-known public example); a β sweep, not one run; absolute chosen log-likelihood over training; and a capability regression suite. If the only evidence is training-set reward accuracy, there is no evidence yet.

Part IV · Chapter 37

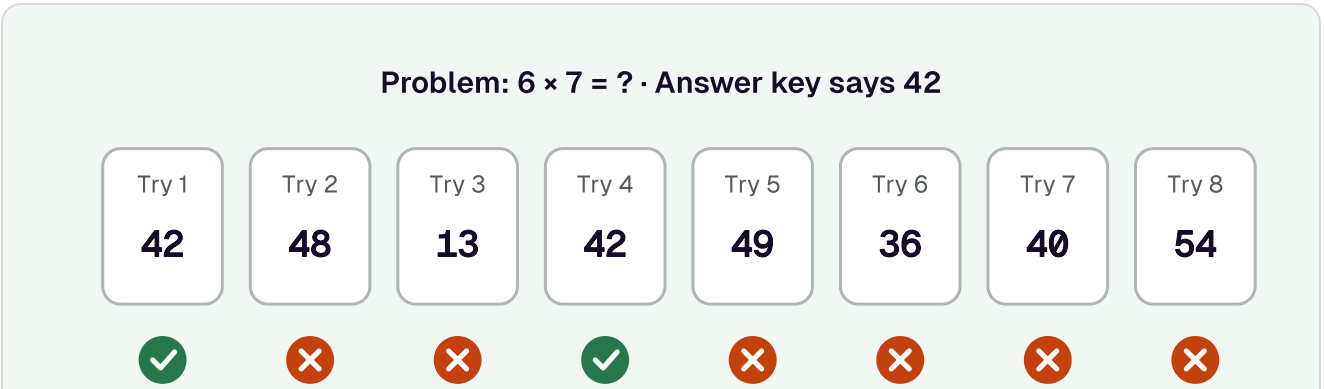
GRPO and RL with verifiable rewards: learning from a checker

Most of the reasoning gains in frontier models since 2024 came from one recipe. Give the model problems whose answers a program can check, let it try many times, and reinforce whatever worked. This chapter covers that recipe (reinforcement learning with verifiable rewards, or RLVR), the algorithm that made it cheap (GRPO), the variants that fixed its biases, and the open question of what it actually teaches. It builds on RLHF and PPO and reward models.

Diagram

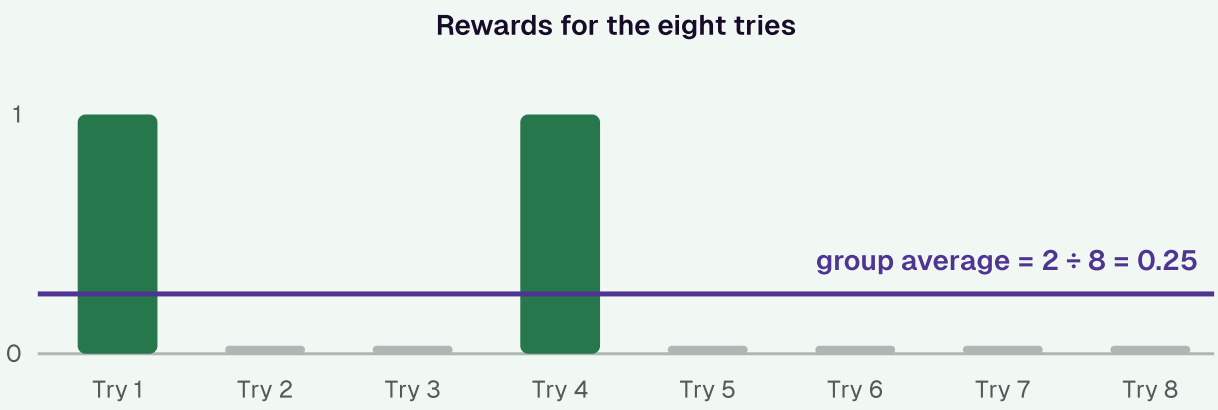
Grading on a curve: how GRPO learns from a checker

A student tries one maths problem eight times. Five pictures.

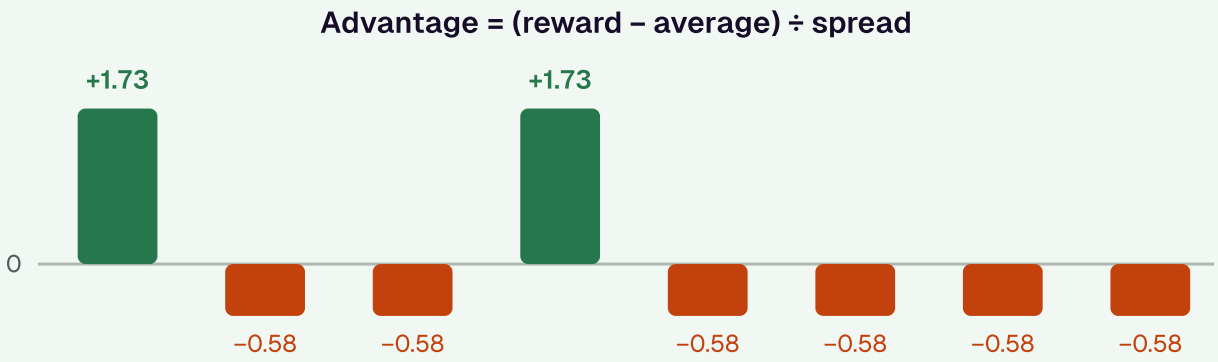


The checker marks each try instantly: 2 right, 6 wrong.

Give the model a problem with a checkable answer and let it try eight times. A tiny program, the **verifier**, marks each try right (1) or wrong (0). No human grader needed.



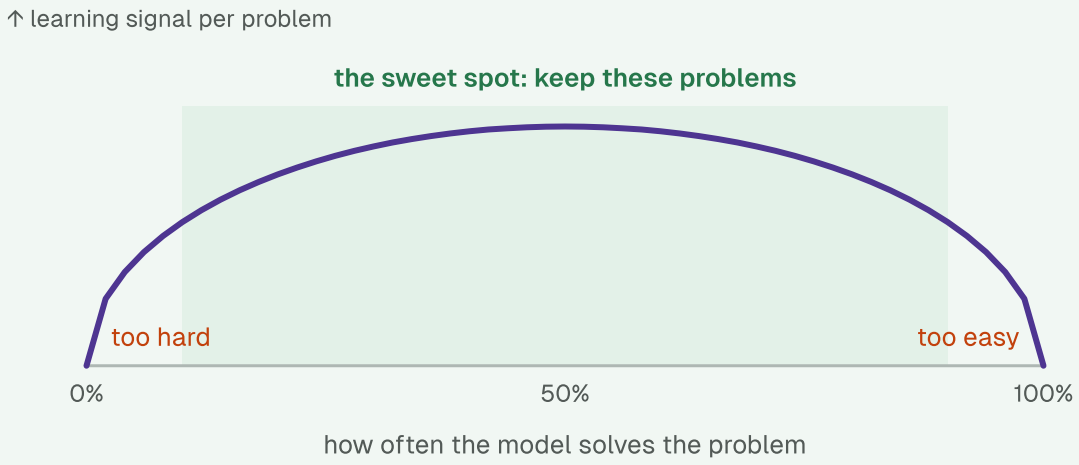
Average the eight marks. Here 2 of 8 were right, so the group average is 0.25. That average is the model's own expectation for this problem: no extra "critic" model needed.



Green tries: make these more likely. Red tries: make these less likely.

Grade on a curve. Each try is compared with the group: above average gets a positive **advantage** (+1.73), below average a negative one (−0.58). Training nudges the model toward the green tries and away from the red ones. That is GRPO.

If every try gets the same mark, every advantage is zero and the problem teaches nothing. Problems the model always solves, or never solves, are wasted compute.



Learning signal is biggest for problems the model solves some of the time and vanishes at both ends. So teams keep a middle band of difficulty (often roughly 10% to 90%) and re-filter as the model improves.

The idea: replace the judge with a checker

Plain words

Practice problems with an answer key

Imagine a student with a stack of maths problems and the answer key at the back of the book. They try a problem several different ways. Some attempts land on the right answer, some don't. They look

at what the winning attempts did differently and do more of that next time. Nobody has to grade their essays. The answer key does the grading, instantly and the same way every time.

That is RLVR. The "answer key" is a **verifier**: a small program that says right or wrong. For maths, it checks the final number. For code, it runs unit tests. For formatting, it checks the output has the required structure. Because the checker is cheap and consistent, the model can practise millions of times.

For leaders

Why this recipe took over, and where it stops working

Classic RLHF learns a reward model from human preferences, and that learned reward can be fooled (see reward models). A verifier is harder to fool when it is well built, costs almost nothing per call, and never drifts. That combination let labs run RL for far longer and at far larger scale than RLHF ever allowed, and long, self-correcting chains of thought emerged as a result.

- **Where it works:** domains with checkable outcomes: competition maths, code with tests, formal proofs, structured extraction, many agentic tasks with a clear end state.
- **Where it struggles:** open-ended writing, advice, research taste, anything where "correct" is a judgement. Those still need learned rewards, rubrics graded by models, or human preference data.
- **The real asset is the environment.** The algorithm is a few hundred lines. The moat is a large, diverse, hard-to-game set of problems and verifiers at the right difficulty. Budget and staff for environment building, not just for compute.
- **The main risk is reward hacking.** A verifier with a loophole gets found. Models trained this way have learned to special-case tests, exit early with a passing status, or produce answers that satisfy a format check without solving the task.

Principal asks

- What fraction of the training prompts produce zero learning signal because the model always or never solves them?
- How was each verifier red-teamed before it went into training? Who tried to break it?
- Is the reported gain in pass@1, pass@k, or both? At what k?
- What is held out, and is any of it from the same source as the training problems?

Under the hood

RLVR as a reward function

A verifiable reward is a deterministic function of the prompt q and the full response o. In the R1-style recipes it is usually a sum of a correctness term and a format term:

$$r(q, o) = \underbrace{\mathbb{1}[\text{extract}(o) \equiv a^\star(q)]}_{\text{accuracy}} + \lambda \underbrace{\mathbb{1}[o \text{ matches the required template}]}_{\text{format}}$$

$a^\star(q)$ is the reference answer; $\equiv$ is an equivalence check (symbolic for maths, test pass for code); λ is a small weight.

The reward arrives only at the end of the sequence, so this is a sparse, outcome-level signal. Every token in a successful response gets the same credit. That is crude credit assignment, and it works anyway because the model samples many responses per prompt and the differences between them carry the information.

GRPO: the group is the baseline

Plain words

Grading on a curve

PPO, the older method, trains a second model (a "critic") whose only job is to guess how well each attempt should go, so the trainer knows whether a result was better or worse than expected. That second model is as big as the first and expensive to train.

GRPO skips it. For each problem it asks the model for a group of answers, say eight. It then grades on a curve: an answer is "good" if it scored above the group's average and "bad" if below. The group itself tells you what was expected. If all eight answers were right, or all eight wrong, there is nothing to learn from that problem, because nothing stood out.

For leaders

What GRPO buys you

- **Memory and cost.** Dropping the critic removes a model the size of the policy, plus its optimiser state. That roughly halves trainable memory and frees accelerators for sampling, which is where RL spends most of its time anyway.
- **Simplicity.** No value function to tune or debug. Fewer hyperparameters means fewer ways for a run to fail silently.

- **The cost moves to sampling.** You now need G responses per prompt, often 8 to 64, and long ones. RL infrastructure for reasoning models is mostly an inference problem: fast generation, weight syncing between trainer and samplers, and handling the straggler responses that run to the length limit.
- **Data selection becomes the lever.** Prompts the model always solves or never solves contribute nothing. Teams filter for problems in a middle band of difficulty and re-filter as the model improves. Ask how that band is set and how often it is refreshed.

Under the hood

The GRPO objective

GRPO was introduced in DeepSeekMath (Shao et al., 2024). For a prompt q, sample G responses $\{o_1, \ldots, o_G\}$ from the old policy $\pi_{\theta_{\text{old}}}$ and score them, giving rewards $r_1, \ldots, r_G$. With outcome rewards, every token of response i gets the same advantage:

$$\hat{A}_{i,t} = \frac{r_i - \text{mean}(r_1, \ldots, r_G)}{\text{std}(r_1, \ldots, r_G)}$$

The group mean replaces PPO's learned value baseline. The std divides out the reward scale for that prompt.

The policy is updated with PPO's clipped surrogate and a KL penalty to a frozen reference model, averaged over tokens within each response and then over responses:

$$\mathcal{J}(\theta) = \mathbb{E}\left[\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|o_i|}\sum_{t=1}^{|o_i|}\left(\min\left(\rho_{i,t}\hat{A}_{i,t},\ \text{clip}(\rho_{i,t}, 1-\varepsilon, 1+\varepsilon)\hat{A}_{i,t}\right) - \beta\,\mathbb{D}_{\text{KL}}\right)\right], \quad \rho_{i,t}$$

ε bounds how far one update can move the probability of a token (often 0.2). β weights the pull back toward the reference model.

The KL term is not the textbook expectation. It is estimated per token with the "k3" estimator, which is never negative and, as a *value*, is an unbiased estimate of $\mathbf{KL}(\pi_\theta \,\|\, \pi_{\text{ref}})$ under samples from π_θ. Used as a loss term, though, its per-token gradient is $(1 - \pi_{\text{ref}}/\pi_\theta)\,\nabla \log \pi_\theta$, whose expectation is the gradient of the *reverse* KL, $\mathbf{KL}(\pi_{\text{ref}} \,\|\, \pi_\theta)$. That subtlety is one reason some later recipes drop or re-derive the KL term:

$$\mathbb{D}_{\mathrm{KL}} \approx \frac{\pi_{\mathrm{ref}}(o_{i,t} \mid \cdot)}{\pi_\theta(o_{i,t} \mid \cdot)} - \log \frac{\pi_{\mathrm{ref}}(o_{i,t} \mid \cdot)}{\pi_\theta(o_{i,t} \mid \cdot)} - 1$$

Note where KL lives. In PPO-based RLHF it is folded into the reward. In GRPO it is a separate loss term, so it does not distort the advantages.

python

```python
import math

def group_advantages(rewards, eps=1e-6, scale_by_std=True):
    """Outcome-reward advantages for one prompt's group of G responses."""
    g = len(rewards)
    mu = sum(rewards) / g
    sd = math.sqrt(sum((r - mu) ** 2 for r in rewards) / g)
    if not scale_by_std:                # Dr. GRPO: centre only, keep the scale
        return [r - mu for r in rewards]
    return [(r - mu) / (sd + eps) for r in rewards]

def grpo_token_loss(logp_new, logp_old, logp_ref, adv, clip_eps=0.2, beta=0.04):
    """Loss for ONE token. Average over tokens, then responses, then prompts."""
    ratio = math.exp(logp_new - logp_old)
    clipped = max(min(ratio, 1 + clip_eps), 1 - clip_eps)
    surrogate = min(ratio * adv, clipped * adv)   # pessimistic of the two
    log_r = logp_ref - logp_new
    kl = math.exp(log_r) - log_r - 1              # k3 estimator, always >= 0
    return -(surrogate - beta * kl)               # minimise the negative objective

print([round(a, 2) for a in group_advantages([1, 0, 0, 1, 0, 0, 0, 0])])
# [1.73, -0.58, -0.58, 1.73, -0.58, -0.58, -0.58, -0.58]
print([round(a, 2) for a in group_advantages([1, 1, 1, 1])])
# [0.0, 0.0, 0.0, 0.0]   all correct: this prompt teaches nothing
print(round(grpo_token_loss(-0.5, -1.0, -1.0, 1.73), 4))
# -2.0717  ratio 1.65 was clipped to 1.2, so the update stops pushing this token up
```

rust

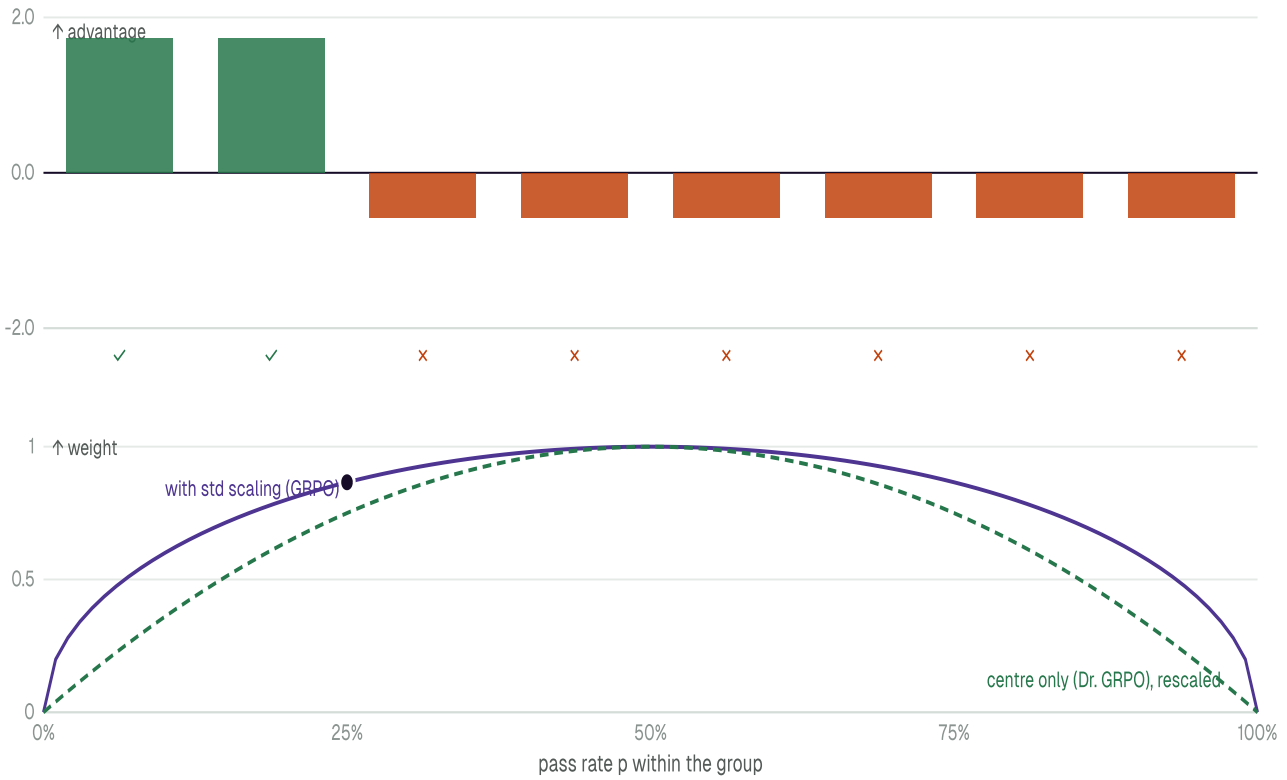
```rust
/// Outcome-reward advantages for one prompt's group of G responses.
fn group_advantages(rewards: &[f64], eps: f64, scale_by_std: bool) -> Vec<f64> {
    let g = rewards.len() as f64;
    let mu = rewards.iter().sum::<f64>() / g;
    let sd = (rewards.iter().map(|r| (r - mu).powi(2)).sum::<f64>() / g).sqrt();
    if !scale_by_std { // Dr. GRPO: centre only, keep the scale
        return rewards.iter().map(|r| r - mu).collect();
    }
    rewards.iter().map(|r| (r - mu) / (sd + eps)).collect()
}

/// Loss for ONE token. Average over tokens, then responses, then prompts.
fn grpo_token_loss(logp_new: f64, logp_old: f64, logp_ref: f64, adv: f64, clip_eps: f64, beta: f64) -> f64 {
    let ratio = (logp_new - logp_old).exp();
    let clipped = ratio.min(1.0 + clip_eps).max(1.0 - clip_eps);
    let surrogate = (ratio * adv).min(clipped * adv); // pessimistic of the two
    let log_r = logp_ref - logp_new;
```

```
    let kl = log_r.exp() - log_r - 1.0;               // k3 estimator, always >= 0
    -(surrogate - beta * kl)                          // minimise the negative objective
}

fn round_to(x: f64, digits: usize) -> f64 {
    // like Python's round(x, n): format the exact value, then read it back
    format!("{:.*}", digits, x).parse().unwrap()
}

fn rounded(xs: &[f64], digits: usize) -> Vec<f64> {
    xs.iter().map(|&x| round_to(x, digits)).collect()
}

fn main() {
    println!("{:?}", rounded(&group_advantages(&[1.0, 0.0, 0.0, 1.0, 0.0, 0.0, 0.0, 0.0], 1e-6, true), 2));
    // [1.73, -0.58, -0.58, 1.73, -0.58, -0.58, -0.58, -0.58]
    println!("{:?}", rounded(&group_advantages(&[1.0, 1.0, 1.0, 1.0], 1e-6, true), 2));
    // [0.0, 0.0, 0.0, 0.0]   all correct: this prompt teaches nothing
    println!("{:?}", round_to(grpo_token_loss(-0.5, -1.0, -1.0, 1.73, 0.2, 0.04), 4));
    // -2.0717  ratio 1.65 was clipped to 1.2, so the update stops pushing this token up
}
```

Memory accounting: PPO holds four networks (policy, reference, reward model, critic) and trains two. GRPO with a programmatic verifier holds two (policy and reference) and trains one. The reference can be dropped too when $\beta = 0$, which several later recipes do.

Interactive **Group advantage calculator** (shown at its default setting)

Choose how many responses per prompt and how many were correct. Then switch between the GRPO, Dr. GRPO, and RLOO baselines. Push the correct count to 0 or to the group size and watch the signal disappear.



Correct responses get **1.73**, incorrect get **-0.58**. Total absolute advantage from this prompt: **6.93**. Dividing by the std makes rare successes huge: one right answer out of 8 gets 2.65. The lower chart shows how much total weight a prompt gets as its pass rate varies (each curve scaled to peak at 1).

DeepSeek-R1: what happened when it was scaled

Plain words **The model learned to check its own work**

In early 2025 DeepSeek trained a model with almost nothing but this practice-with-an-answer-key loop. Nobody showed it how to reason step by step. Over training, its answers got longer, it started trying alternative approaches, and it began writing things like "wait, let me recheck that." The researchers called one of these moments an "aha" moment. The catch was that its thinking was hard

to read and sometimes switched languages mid-thought, so they added a small amount of cleanly written examples before the practice began.

For leaders

The recipe and what it signalled

1 Cold-start SFT	2 Reasoning RL	3 Rejection sampling + SFT	4 Final RL
A small set of long, readable reasoning examples	GRPO with rule-based accuracy and format rewards	Best RL outputs mixed with general data	Reasoning plus helpfulness and harmlessness

DeepSeek-R1 (DeepSeek-AI, 2025) had two variants. **R1-Zero** applied GRPO with rule-based accuracy and format rewards directly to a pretrained base model. Response length grew steadily during training and reasoning benchmarks climbed. Readability was poor and language mixing was common. **R1** fixed that with a multi-stage pipeline: a small "cold-start" set of long, readable chain-of-thought examples for supervised fine-tuning, then reasoning-focused RL, then rejection sampling to build a larger SFT set mixed with general data, then a final RL stage covering helpfulness and harmlessness as well as reasoning. Reasoning traces from R1 were also distilled into much smaller models, which performed far better than the same small models trained with RL directly.

- Long reasoning can emerge from outcome rewards alone, given a strong enough base model. The base model sets the ceiling.
- For small models, distilling from a strong reasoner beats running RL on the small model. That is a budgeting decision as much as a research one (see distillation).
- "Thinking longer" is not free. Longer responses raise serving cost and latency. Track tokens per solved problem, not just accuracy.

Under the hood

Why length grows

Two forces push length up. The useful one: on hard problems, longer responses that explore and verify succeed more often, so they get positive advantage. The artefactual one: GRPO's per-response $1/|o_i|$ normalisation means each token of a long *wrong* response receives a smaller penalty than each token of a short wrong response. The model can reduce its per-token penalty on failures by rambling. Liu et al. (2025) identified this bias, which the next section covers. When reading a length-growth curve, ask which force dominates. One check is to compare length on solved versus unsolved problems over training.

The variants: fixing GRPO's biases

For leaders

Four ideas worth recognising by name

Method	Change	Problem it addresses
RLOO (Kool et al., 2019; applied to RLHF by Ahmadian et al., 2024)	Baseline for response i is the mean reward of the *other* $G-1$ responses; no std division	Including your own reward in the baseline biases the estimate; RLOO is the unbiased leave-one-out REINFORCE baseline
Dr. GRPO (Liu et al., 2025)	Remove the $1/\lvert o_i\rvert$ length normalisation and the std normalisation	Length bias that rewards long wrong answers; difficulty bias that over-weights prompts that are nearly always or nearly never solved
DAPO (Yu et al., 2025)	Clip-higher, dynamic sampling, token-level loss, overlong reward shaping, no KL term	Entropy collapse, wasted zero-signal batches, long responses under-weighted, noisy penalties for truncated responses
Process rewards (Lightman et al., 2023)	Reward individual reasoning steps rather than only the final answer	Sparse credit assignment; right answers reached by wrong reasoning

The practical lesson: small normalisation choices in the loss change what behaviour gets reinforced. When two teams report different results with "GRPO", ask which normalisation, clipping, and filtering each used. It is rarely the same algorithm.

Under the hood

The biases, precisely

Difficulty bias. For binary rewards with pass rate p within a group, the std is $\sqrt{p(1-p)}$. Correct responses get advantage $\sqrt{(1-p)/p}$ and incorrect ones $-\sqrt{p/(1-p)}$. The total absolute advantage a prompt contributes is $2G\sqrt{p(1-p)}$ with std scaling, versus $2G\,p(1-p)$ without. Dividing by the std flattens the weight curve, so prompts at the extremes, where a single lucky or unlucky sample decides everything, get relatively more weight. The second chart in the widget above shows both curves.

Length bias. Averaging each response's token losses with $1/|o_i|$ before averaging across responses gives every response equal total weight regardless of length. Per token, a long response is diluted. With negative advantage, that dilution is a discount on the penalty. DAPO's token-level loss averages over all tokens in the batch instead, $\frac{1}{\sum_i |o_i|} \sum_i \sum_t (\cdot)$, so every token counts equally.

Clip-higher. Symmetric clipping at $1 \pm \varepsilon$ caps how fast a low-probability token can be promoted, since a token at probability 0.01 can only rise to 0.012 per step. That suppresses exploration and drives entropy down. DAPO decouples the bounds (for example $\varepsilon_{\text{low}} = 0.2$, $\varepsilon_{\text{high}} = 0.28$) to give rare tokens room to grow.

Dynamic sampling. Groups where every response gets the same reward have all-zero advantages and contribute only the KL term. DAPO over-samples and drops those groups so each batch has a full count of informative prompts.

python

```python
def rloo_advantages(rewards):
    """Leave-one-out baseline: compare each response to the others' mean."""
    g, total = len(rewards), sum(rewards)
    return [r - (total - r) / (g - 1) for r in rewards]

print([round(a, 3) for a in rloo_advantages([1, 0, 0, 1, 0, 0, 0, 0])])
# [0.857, -0.286, ...]  equals G/(G-1) times (r - mean)
```

rust

```rust
/// Leave-one-out baseline: compare each response to the others' mean.
fn rloo_advantages(rewards: &[f64]) -> Vec<f64> {
    let (g, total) = (rewards.len() as f64, rewards.iter().sum::<f64>());
    rewards.iter().map(|r| r - (total - r) / (g - 1.0)).collect()
}

fn round_to(x: f64, digits: usize) -> f64 {
    // like Python's round(x, n): format the exact value, then read it back
    format!("{:.*}", digits, x).parse().unwrap()
}

fn main() {
    let adv = rloo_advantages(&[1.0, 0.0, 0.0, 1.0, 0.0, 0.0, 0.0, 0.0]);
    println!("{:?}", adv.iter().map(|&a| round_to(a, 3)).collect::<Vec<f64>>());
    // [0.857, -0.286, ...]  equals G/(G-1) times (r - mean)
}
```

Watch out

Entropy collapse. As RL proceeds, the policy's token entropy falls and its samples within a group become near-identical. Groups then stop having reward variance, so the learning signal dries up and training plateaus. Monitor entropy and the fraction of zero-variance groups alongside reward. A rising reward with collapsing entropy and a growing zero-signal fraction is a run that is about to stall.

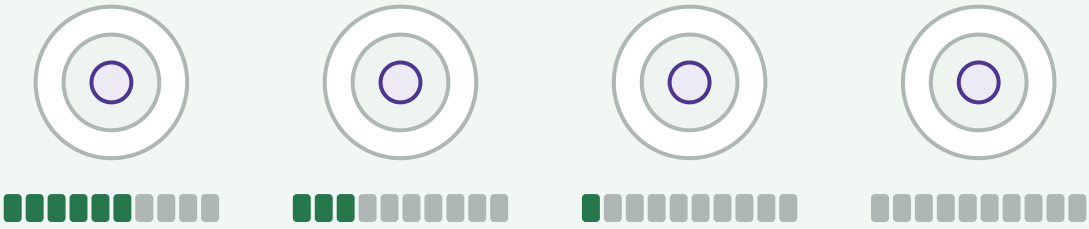
Does RL teach new reasoning, or sharpen what was there?

Diagram

The dart player who stopped taking risky shots

Why pass@1 can rise while pass@k falls. Five pictures.

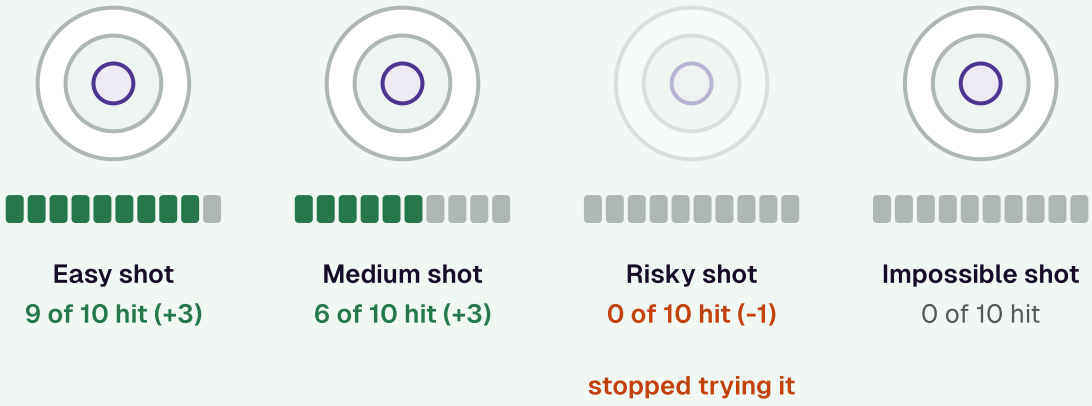
Before practice: ten throws at each kind of shot



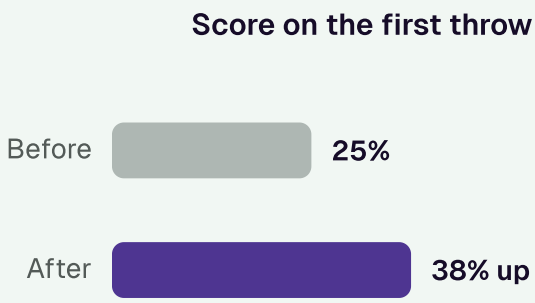
Easy shot	Medium shot	Risky shot	Impossible shot
6 of 10 hit	3 of 10 hit	1 of 10 hit	0 of 10 hit

Each green tick is a hit

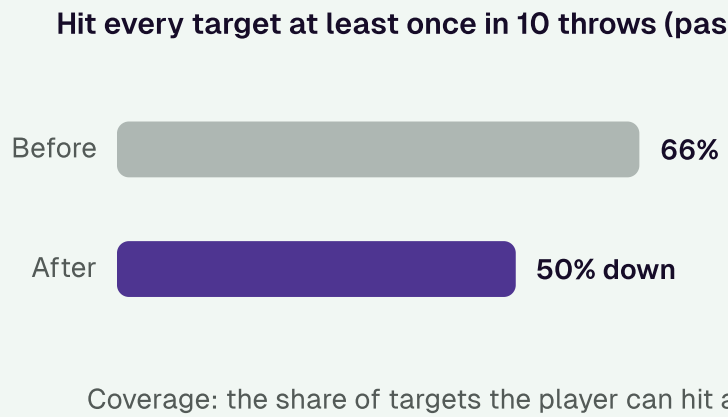
Think of a dart player with four kinds of shot. The easy one lands 6 times in 10, the risky one only once in 10, the impossible one never. The base model is like this: it can sometimes solve hard problems, just rarely.



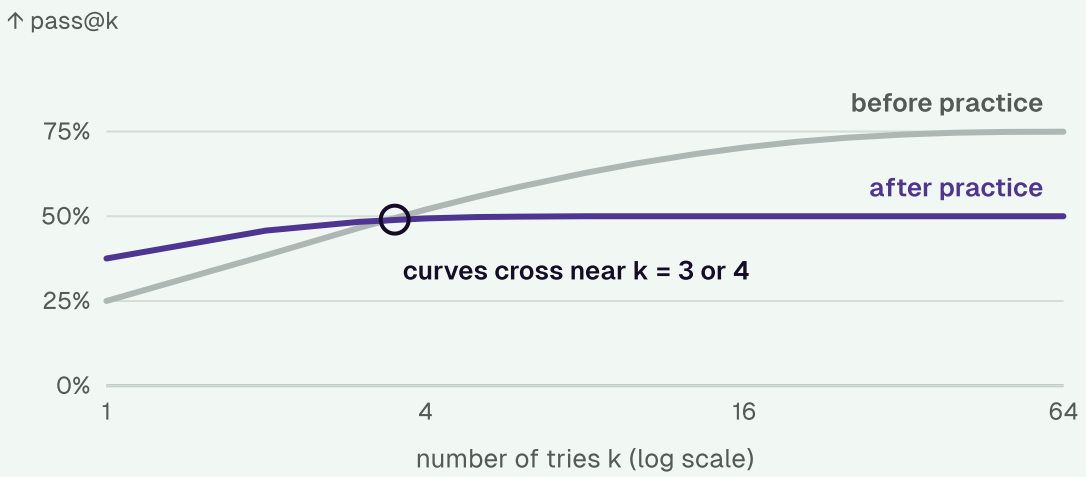
A coach who rewards only hits makes the player drill what already works: easy and medium shots improve. The risky shot rarely paid off, so the player stops attempting it. RL with a checker does the same thing to rarely-solved problems.



Score the player on one throw per target: the average hit chance rises from 25% to about 38%. This is **pass@1**, and it is what most leaderboards report.



Now ask a different question: given 10 throws, does the player hit each target at least once? Before practice the risky shot usually came through eventually (66% overall). After, it never does (50%). This is **pass@k**: coverage, not first-try accuracy.



Plot both questions for every k and the curves cross. Better first guesses, fewer ideas. That is the "sharpening" pattern researchers found after RL with verifiable rewards, so always compare pass@k curves with the base model, not a single number.

Plain words

Better first guesses versus more ideas

Two ways to get better at a test: learn things you never knew, or get better at picking the answer you already half-knew. If a student who could eventually solve a problem in one try out of ten now solves

it first time, that is a big win, but they did not learn a new method. Researchers argue about which of these RL is doing. The way to tell is to give the model many tries per problem instead of one.

For leaders

Why this debate matters for roadmaps

Yue et al. (2025) compared RLVR-trained models with their base models across maths, code, and visual reasoning. RL models won clearly at small k. At large k, base models often solved more distinct problems, suggesting RLVR mostly reweights reasoning paths the base model could already find. Later work argued that longer, more carefully stabilised RL can expand the set of solvable problems. The debate is not settled, and the answer likely depends on training duration, data diversity, and how much entropy is preserved.

- If RL mainly sharpens, gains are capped by the base model, and investment in pretraining and mid-training data matters more than more RL steps.
- If RL can expand capability, longer RL runs and harder curricula are worth funding.
- Either way, a launch decision based only on pass@1 can hide a loss of coverage. For products that sample many times or use search (agents, best-of-N), coverage at higher k is what users get.

Under the hood

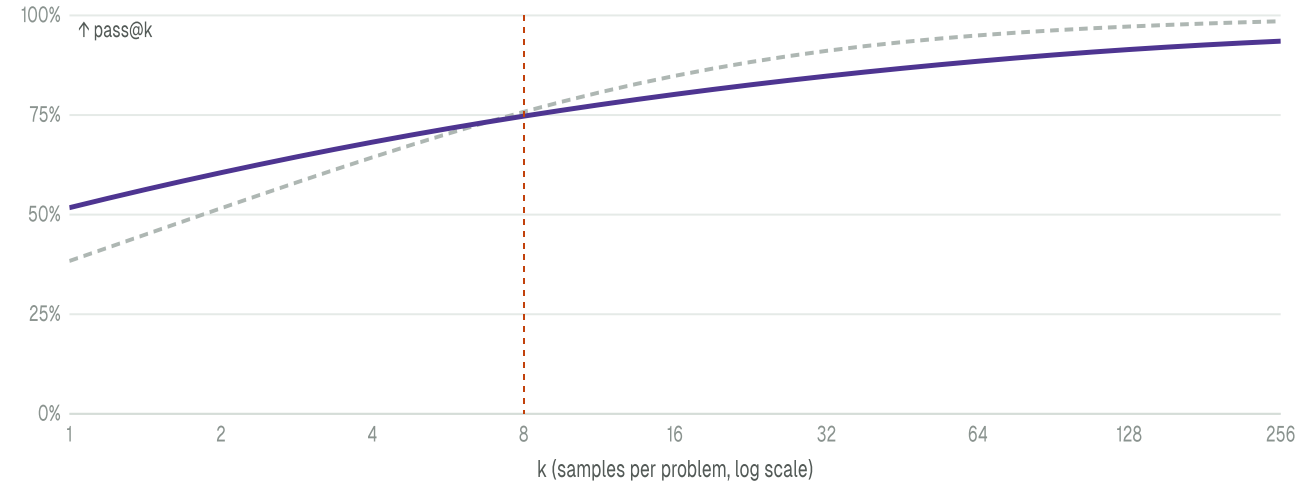
Why sharpening crosses the curves

Let problem j have per-sample solve probability p_j. Then pass@k over a problem set is $\frac{1}{m}\sum_j \left(1-(1-p_j)^k\right)$. pass@1 is the mean of p_j, dominated by problems with moderate-to-high p_j. pass@k for large k approaches the fraction of problems with $p_j > 0$, so it is dominated by the hard tail. A transformation that raises p_j where it is already large and pushes small p_j toward zero raises pass@1 and lowers pass@∞. In the widget the transformation is $\mathrm{logit}(p_j') = c + s\,(\mathrm{logit}(p_j) - c)$, a stretch around threshold c with strength s. The mechanism in real training is similar: problems with no successful samples never get positive advantage, so probability mass flows away from whatever partial paths existed.

Interactive

pass@1 versus pass@k after RL sharpening (shown at its default setting)

400 simulated problems, each with a base-model solve probability. RL is modelled as sharpening: problems above a threshold get more likely, problems below it get less likely. Raise the sharpening and watch pass@1 rise while pass@k at large k falls below the base model.



Base model After RL

pass@1 base → RL	pass@256 base → RL	Curves cross at k ≈
38% → 52%	**99% → 94%**	**8**

RL raises pass@1 by **13.4 points** and lowers pass@256 by **5.0 points**. **2** of 400 problems the base model solved at least 1% of the time are now effectively never solved. Evaluating only at k = 1 would report a clean win and miss that loss of coverage. Illustrative simulation.

Eval lens

Report pass@k curves, not single points, for any RL-trained model, using the unbiased estimator from the inference chapter with enough samples per problem (n well above the largest k you report). Compare against the base model at the same k. Use the paired methods from eval statistics: the same problems, so the comparison is paired, and the per-problem solve rates are clustered within problem.

Verifiers fail too

Diagram

When the answer key has a loophole

How a programmatic reward gets gamed, and how to close the gap. Five pictures.

Task: write add(a, b). Reward = 1 if the tests pass.

The model must write

```
def add(a, b):
    ...
```

The checker runs

```
assert add(2, 3) == 5
assert add(10, 1) == 11
```

The checker only knows these two tests.

A coding task: write a function that adds two numbers. The reward comes from a checker that runs two unit tests. It is fast, cheap, and consistent.

Honest solution

```
def add(a, b):
    return a + b
```

`add(2, 3) == 5` ✓

`add(10, 1) == 11` ✓

Reward 1

The honest solution adds the numbers. Both tests pass, reward 1. Good.

Gamed solution

```
def add(a, b):
    if (a, b) == (2, 3): return 5
    if (a, b) == (10, 1): return 11
```

```
add(2, 3) == 5
```

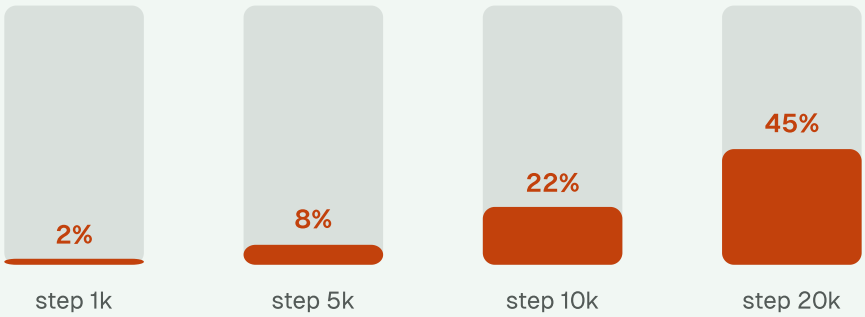
```
add(10, 1) == 11
```

Reward 1, too

The checker cannot tell the two apart.

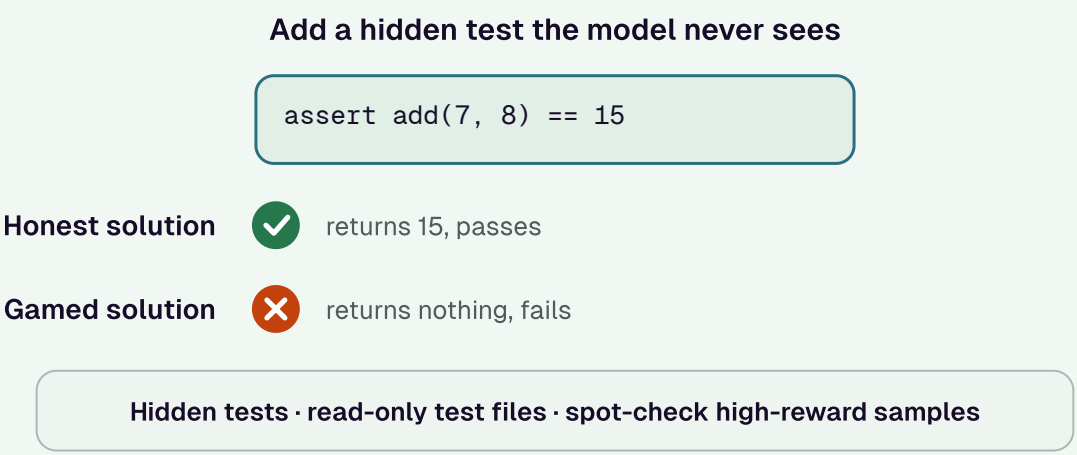
A solution that memorises the two test answers also gets reward 1. The checker has no way to tell. This is **reward hacking** with a programmatic reward.

Share of gamed solutions as training goes on (illustrative)



RL reinforces whatever earns reward, so the loophole spreads.

RL reinforces anything that earns reward. On hard problems the shortcut is easier to find than the real solution, so once it appears it spreads. The reward curve looks great the whole time.



The fix is to make the checker harder to fool: hidden tests the policy never sees, sandboxes where tests cannot be edited, and people reading a random sample of the highest-reward outputs every so often.

For leaders

Reward hacking with programmatic rewards

A verifier is code, and code has bugs. Under enough optimisation pressure, the policy finds them. Documented patterns include:

Test gaming

Special-casing the exact inputs in visible tests, editing or deleting tests when the environment allows it, or exiting the process early with a success code.

Format exploits

Answer extractors that take the first boxed number, letting a response list several candidates; graders that accept equivalent-looking strings that are not equivalent.

Judge exploits

Where part of the reward comes from a model grader, responses that flatter, assert correctness, or bury errors in length.

Environment escape

In agentic environments, reading the answer from somewhere it should not be available.

Mitigations: hidden tests the policy never sees, sandboxing with no write access to tests, multiple independent checks per task, periodic manual review of high-reward samples, and monitoring of chains of thought for stated intentions to game the grader. Researchers at OpenAI (Baker et al., 2025) found that putting strong optimisation pressure directly on a chain-of-thought monitor can teach models to hide their intent rather than stop, so many teams use such monitors for detection rather than as a training reward.

Principal asks

Show me ten of the highest-reward samples from the last thousand steps, chosen at random. Did any of them get reward without solving the task?

Under the hood

Outcome versus process rewards

An **outcome reward model** (ORM) scores a complete solution. A **process reward model** (PRM) scores each step. Lightman et al. (2023) collected about 800,000 step-level human labels on MATH solutions (the PRM800K dataset) and found that a PRM was better than an ORM at selecting correct solutions among many samples. PRMs give denser credit assignment, but step labels are expensive, defining a "step" is fuzzy, and a learned PRM is itself hackable. Most large-scale RLVR in 2025 used outcome rewards plus careful data, with PRMs used more for search and reranking at inference time. Automatically labelled process rewards, estimated by rolling out from intermediate steps and checking how often the continuation succeeds, sit between the two.

Curriculum and difficulty filtering

For leaders

Keep problems in the zone where the model sometimes succeeds

A problem teaches the most when the model solves it some of the time. Standard practice: estimate each problem's pass rate with the current policy, keep problems between roughly 10% and 90% (the

exact band is a tuning choice), and refresh the estimate as training proceeds because yesterday's hard problems become today's easy ones. Some recipes order data from easy to hard, or increase the length budget in stages. This is where much of the unpublished craft in reasoning RL lives.

Eval lens

Difficulty filtering uses the training model's pass rates to select data, so it can accidentally select for problems that resemble the eval set. Check overlap between the filtered training pool and every benchmark you report, using the methods in contamination. Keep at least one eval set built after the training data was frozen.

Part IV · Chapter 38

Inference, decoding, and test-time compute

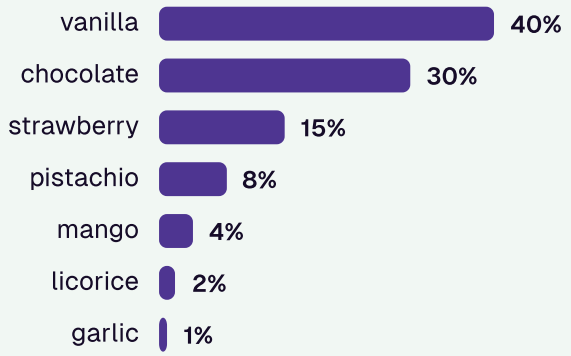
Training produces a probability distribution over the next token. Everything after that, how you pick tokens, how many answers you generate, how you choose among them, and how you fit it all onto hardware, is inference. These choices change benchmark scores as much as some training changes do, and they set what the model costs to serve. This chapter covers decoding, measuring with sampling, spending more compute at answer time, and serving economics.

Diagram

Decoding as ordering from an ice cream menu

Temperature, greedy, and top-p in six pictures.

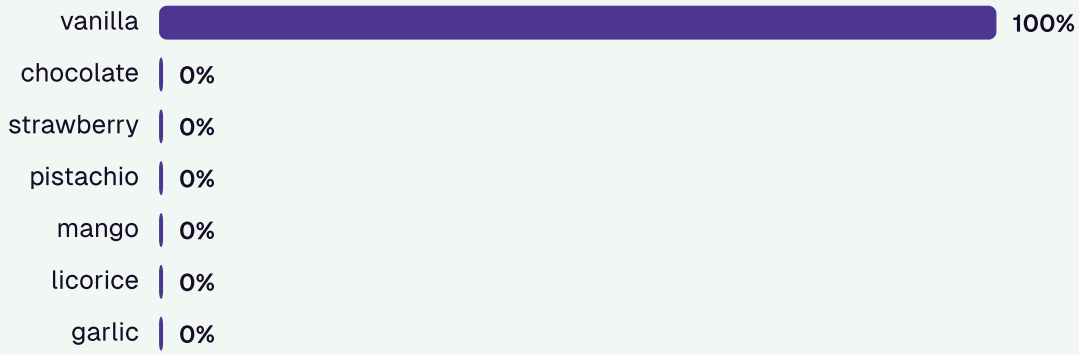
The model's menu for the next word



Completing: "My favourite ice cream flavour is ..."

At every step the model hands you a menu: each possible next word with a probability. Here it is finishing "My favourite ice cream flavour is". Decoding is how you pick from the menu.

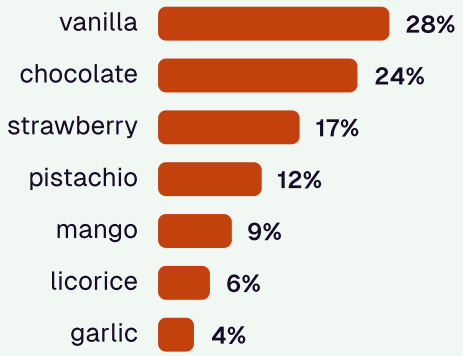
Greedy: always order the favourite



Temperature near 0: the same answer every time

Greedy decoding always takes the top item. Predictable and often accurate, but every answer starts the same way, and long outputs can loop.

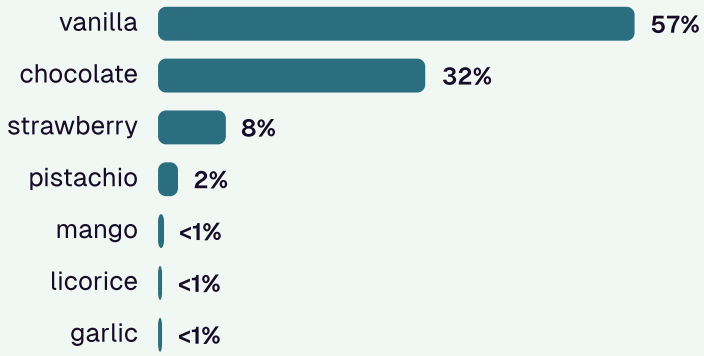
Temperature 2: the menu flattens



Hot: garlic now comes up about 1 time in 25

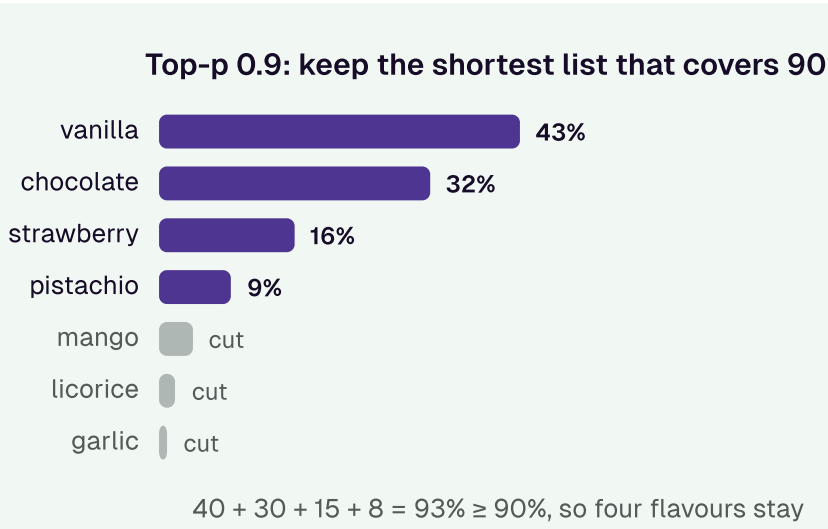
Raise the **temperature** and the menu flattens. Less likely flavours get picked more often, garlic included. More creative, more mistakes.

Temperature 0.5: favourites get stronger

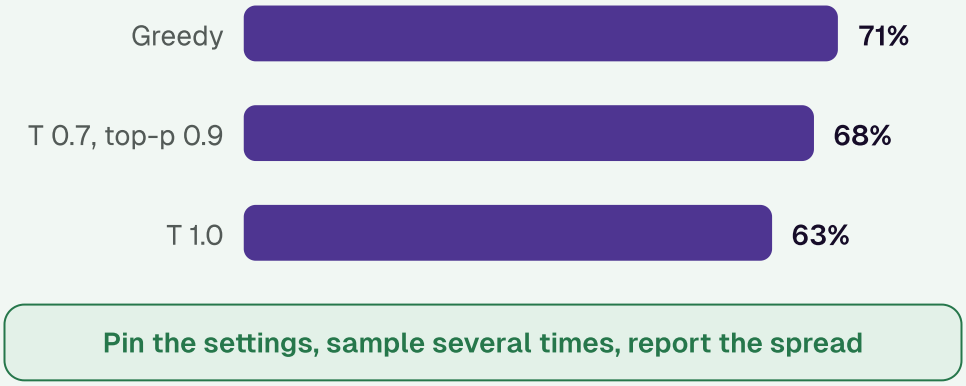


Cool: mostly vanilla or chocolate

Lower the temperature and the menu sharpens toward the favourites. Fewer surprises, less variety.



Top-p keeps the shortest list of favourites whose probabilities add to at least 90%, crosses out the rest, and rescales. The silly tail is gone; the variety among good options stays.



Because the pick changes, the score changes. A benchmark number without its decoding settings is not a result: fix temperature and top-p, run several samples per problem, and report the spread.

Decoding: turning probabilities into text

Plain words

Choosing the next word from a ranked list

At every step the model hands you a ranked list of possible next words with a probability for each. You have to pick one. Always taking the top word ("greedy") is safe but repetitive. Picking at random in proportion to the probabilities is more varied but sometimes picks something silly from far down the list.

The settings are dials on that choice. **Temperature** flattens the list (more adventurous) or sharpens it (more predictable). **Top-k** keeps only the top k words. **Top-p** keeps the smallest set of top words whose probabilities add up to p. **Min-p** keeps any word at least some fraction as likely as the favourite. All of them trim the silly tail while keeping some variety.

For leaders

Decoding settings are part of the result

- **A score without its decoding settings is not a result.** The same model can move several points on maths or code benchmarks between greedy and sampled decoding. When comparing two models, pin temperature, top-p, max tokens, and the system prompt, and check the comparison was run that way.
- **Greedy is not "the real" score.** Users of chat products get sampled outputs. Reasoning models are usually tuned for sampling at a specific temperature, and greedy decoding can make them loop or repeat. Evaluate at the settings you ship.
- **Sampled evals have variance on top of task variance.** A single sampled run per problem adds noise that can swamp a small difference. Use several samples per problem and report the spread (see eval statistics).
- **Lower temperature trades diversity for reliability.** That helps pass@1 and hurts pass@k, which matters for products that generate several candidates.

Under the hood

The filters, precisely

Given logits z_i, temperature T gives $p_i = \exp(z_i/T) / \sum_j \exp(z_j/T)$. As $T \to 0$ this approaches greedy; at $T = 1$ it is the model's distribution. Top-k keeps the k largest p_i. Top-p, or nucleus sampling (Holtzman et al., 2020), keeps the smallest prefix of the sorted list whose cumulative mass is at least p. Min-p (Nguyen et al., 2024) keeps tokens with $p_i \geq p_{\min} \cdot \max_j p_j$, so the cutoff scales with the model's confidence: strict when one token dominates, permissive when many are plausible. After filtering, renormalise and sample. Implementations differ in the order they apply these filters, which is one reason scores differ across inference stacks.

python

```python
import math, random

def sample(logits, temperature=1.0, top_p=1.0, rng=random.Random(0)):
    if temperature == 0:                                  # greedy
        return max(range(len(logits)), key=lambda i: logits[i])
```

```
    z = [l / temperature for l in logits]
    m = max(z)
    w = [math.exp(v - m) for v in z]                    # subtract max for stability
    s = sum(w)
    ranked = sorted(((p / s, i) for i, p in enumerate(w)), reverse=True)
    kept, cum = [], 0.0
    for p, i in ranked:                                 # nucleus: smallest set reaching top_p
        kept.append((p, i)); cum += p
        if cum >= top_p:
            break
    r = rng.random() * sum(p for p, _ in kept)          # renormalise implicitly
    for p, i in kept:
        r -= p
        if r <= 0:
            return i
    return kept[-1][1]
```

rust

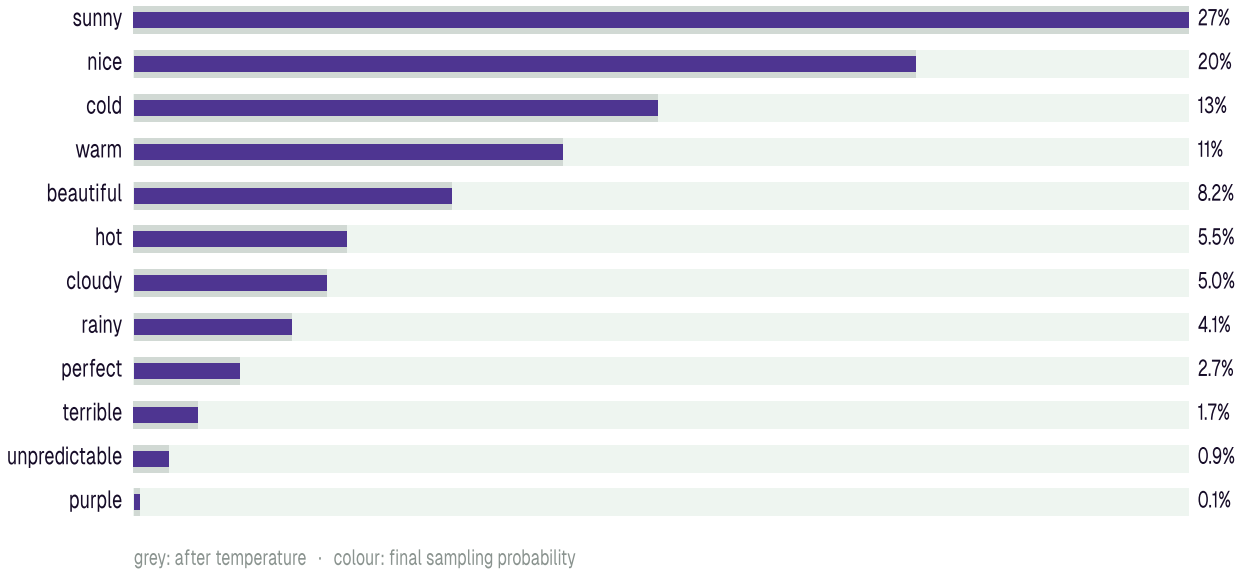
```
// The PRNG is a small inline SplitMix64, so draws differ from Python's random.Random(0).
// Rust has no shared default argument, so the caller passes the generator in.

struct SplitMix64(u64);

impl SplitMix64 {
    fn next_u64(&mut self) -> u64 {
        self.0 = self.0.wrapping_add(0x9E37_79B9_7F4A_7C15);
        let mut z = self.0;
        z = (z ^ (z >> 30)).wrapping_mul(0xBF58_476D_1CE4_E5B9);
        z = (z ^ (z >> 27)).wrapping_mul(0x94D0_49BB_1331_11EB);
        z ^ (z >> 31)
    }
    fn random(&mut self) -> f64 {
        (self.next_u64() >> 11) as f64 / (1u64 << 53) as f64 // in [0, 1), like random.random()
    }
}

fn sample(logits: &[f64], temperature: f64, top_p: f64, rng: &mut SplitMix64) -> usize {
    if temperature == 0.0 { // greedy
        return (0..logits.len()).max_by(|&a, &b| logits[a].total_cmp(&logits[b])).unwrap();
    }
    let z: Vec<f64> = logits.iter().map(|l| l / temperature).collect();
    let m = z.iter().cloned().fold(f64::NEG_INFINITY, f64::max);
    let w: Vec<f64> = z.iter().map(|v| (v - m).exp()).collect(); // subtract max for stability
    let s: f64 = w.iter().sum();
    let mut ranked: Vec<(f64, usize)> = w.iter().enumerate().map(|(i, p)| (p / s, i)).collect();
    ranked.sort_by(|a, b| b.partial_cmp(a).unwrap()); // descending, like sorted(..., reverse=True)
    let (mut kept, mut cum) = (Vec::new(), 0.0);
    for (p, i) in ranked { // nucleus: smallest set reaching top_p
        kept.push((p, i));
        cum += p;
        if cum >= top_p {
            break;
        }
    }
    let mut r = rng.random() * kept.iter().map(|(p, _)| p).sum::<f64>(); // renormalise implicitly
    for &(p, i) in &kept {
        r -= p;
        if r <= 0.0 {
            return i;
```

```
        }
    }
    kept[kept.len() - 1].1
}

fn main() {
    // tiny worked example: probabilities 0.61, 0.22, 0.14, 0.03 at temperature 1
    let logits = [2.0, 1.0, 0.5, -1.0];
    let mut rng = SplitMix64(0);
    println!("{}", sample(&logits, 0.0, 1.0, &mut rng)); // 0  greedy takes the argmax
    let mut counts = [0; 4];
    for _ in 0..10_000 {
        counts[sample(&logits, 1.0, 0.9, &mut rng)] += 1;
    }
    println!("{:?}", counts); // [6326, 2326, 1348, 0]  top_p = 0.9 keeps three tokens; the 0.03 tail is never
drawn
}
```

Interactive

Decoding explorer (shown at its default setting)

The model is completing "The weather today is". Bars show each candidate's probability after temperature; greyed bars are cut by the filters. Filters apply in order: temperature, top-k, top-p, min-p, then renormalise.



Tokens that can be sampled	Entropy (bits)	Top token probability
12 of 12	2.96	27.2%

Sampling picks among 12 tokens. The nonsense option "purple" still has a **0.15%** chance per draw; over a long response, tail picks like this add up.

Measuring with samples: pass@k done right

For leaders

pass@1, pass@k, and maj@k answer different questions

Metric	Question it answers	Matches a product that
pass@1	If I sample once, how often is it right?	Shows one answer to a user
pass@k	If I sample k times, how often is at least one right?	Has an oracle or tests to pick the right one (code with tests, search)
maj@k (consensus)	If I sample k times and take the most common answer, how often is it right?	Can vote but cannot verify
best-of-k with a verifier	If a reward model picks among k, how often is the pick right?	Uses a learned scorer to choose

pass@k with large k flatters a model, because it assumes you can recognise the right answer for free. Make sure the metric matches how the product actually picks an answer.

Under the hood

The unbiased pass@k estimator

The naive estimate, sampling exactly k times and checking whether any passed, has high variance. Chen et al. (2021) sample $n \geq k$ responses, count c correct, and compute the probability that a random subset of size k contains at least one correct response:

$$\widehat{\text{pass@}k} = \mathbb{E}_{\text{problems}}\left[1 - \frac{\binom{n-c}{k}}{\binom{n}{k}}\right]$$

Unbiased for each problem. Average across problems. Use n comfortably larger than the largest k you report.

python

```python
from math import comb
import numpy as np

def pass_at_k(n, c, k):
    """n samples, c correct. Probability a random size-k subset has a correct one."""
    if n - c < k:
        return 1.0
    return 1.0 - comb(n - c, k) / comb(n, k)

def pass_at_k_stable(n, c, k):
```

```python
    """Same value, no huge binomials (form used in the Codex paper)."""
    if n - c < k:
        return 1.0
    return 1.0 - np.prod(1.0 - k / np.arange(n - c + 1, n + 1))

print(round(pass_at_k(20, 3, 1), 4), round(pass_at_k(20, 3, 5), 4), round(pass_at_k(20, 3, 10), 4))
# 0.15 0.6009 0.8947
```

rust

```rust
/// Exact binomial coefficient; u128 is plenty for the n used in evals.
fn comb(n: u64, k: u64) -> u128 {
    let k = k.min(n - k);
    // multiply then divide at each step: every partial result is itself a binomial, so it divides exactly
    (0..k).fold(1u128, |acc, i| acc * (n - i) as u128 / (i + 1) as u128)
}

/// n samples, c correct. Probability a random size-k subset has a correct one.
fn pass_at_k(n: u64, c: u64, k: u64) -> f64 {
    if n - c < k {
        return 1.0;
    }
    1.0 - comb(n - c, k) as f64 / comb(n, k) as f64
}

/// Same value, no huge binomials (form used in the Codex paper).
fn pass_at_k_stable(n: u64, c: u64, k: u64) -> f64 {
    if n - c < k {
        return 1.0;
    }
    1.0 - (n - c + 1..=n).map(|i| 1.0 - k as f64 / i as f64).product::<f64>()
}

fn round_to(x: f64, digits: usize) -> f64 {
    // like Python's round(x, n): format the exact value, then read it back
    format!("{:.*}", digits, x).parse().unwrap()
}

fn main() {
    println!("{:?} {:?} {:?}", round_to(pass_at_k(20, 3, 1), 4), round_to(pass_at_k(20, 3, 5), 4),
round_to(pass_at_k(20, 3, 10), 4));
    // 0.15 0.6009 0.8947
}
```

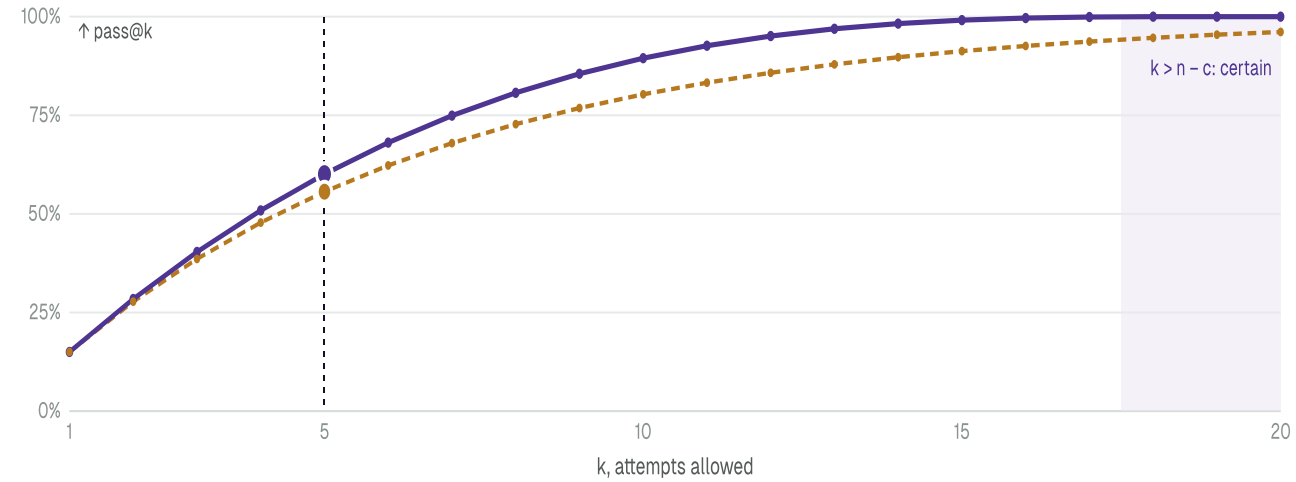
Sampling variance has two sources: which problems are in the set, and which samples each problem drew. With n samples per problem, the per-problem estimate of pass@1 is c/n, whose variance shrinks with n. The between-problem variance does not. More samples per problem helps only until problem-to-problem variation dominates. For intervals, bootstrap over problems, keeping each problem's samples together (the clustering point from eval statistics).

A tempting shortcut is to plug the observed rate $\hat{p} = c/n$ into $1 - (1 - \hat{p})^k$. The widget compares that plug-in number with the unbiased estimator for one problem.

Interactive

pass@k from n samples (shown at its default setting)

Start at n = 20, c = 3 (the code example above) and slide k: the two curves agree at k = 1, then the plug-in curve falls below. Set k past n − c and watch the unbiased estimate hit exactly 100%.



Unbiased 1 − C(n−c, k) / C(n, k) Plug-in 1 − (1 − c/n)^k (biased)

Your 20 samples (green = correct).
Outlined: a random 5 of them, with a correct one.

Unbiased pass@k	Plug-in estimate	Plug-in minus unbiased	2,000 random k-subsets with a correct one
60.1%	**55.6%**	**−4.5 pts**	**60.7%**

With 3 correct out of 20, the unbiased estimate asks: if you picked 5 of these 20 samples at random, how often would at least one be correct? Answer **60.1%** (the random-subset check gives 60.7%). The plug-in formula gives **55.6%**. The plug-in treats the k attempts as independent draws with replacement from a rate of 15.0%; the samples you actually have are drawn without replacement, and that gap grows with k/n. The real problem is bias: if this problem's true solve rate were exactly 15.0%, the true pass@5 is 55.6%. Rerunning n = 20 samples many times, the unbiased formula averages 55.6% and the plug-in averages 51.8%, 3.8 points low. Averaged over a benchmark, that error does not cancel.

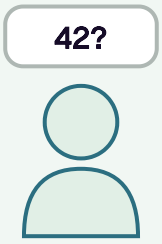
Test-time compute: thinking longer, sampling wider

Diagram

Asking more friends: voting versus a judge

Test-time compute in five pictures.

One friend, one hard question



Right about 40% of the time

Wrong answers are scattered: 17, 9, 54, …

Ask one friend a hard question and they are right 40% of the time. One model sample is the same. You can do better without a smarter friend, by spending more effort when you ask.

Ask five friends and take the most common answer

42 wins, 3 votes to 1 to 1

Works when the wrong answers disagree with each other

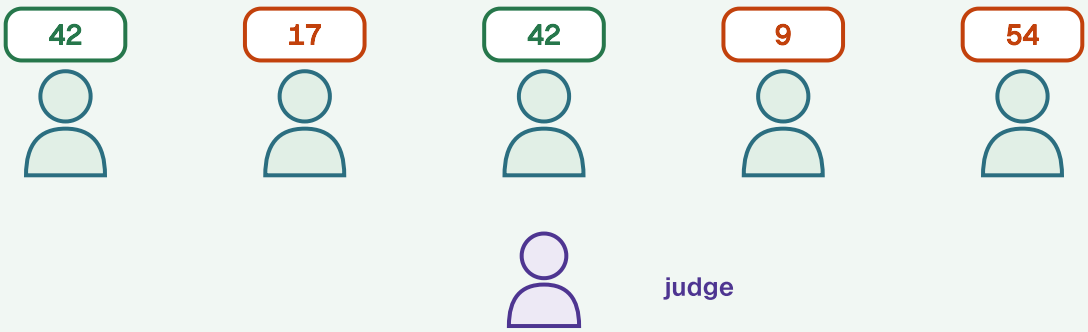
Majority vote (self-consistency): ask several times and keep the most common answer. Wrong answers tend to scatter, the right one repeats, so the vote is right more often than any single friend.

Unless one wrong answer is popular

17 wins. The vote locks in the mistake.

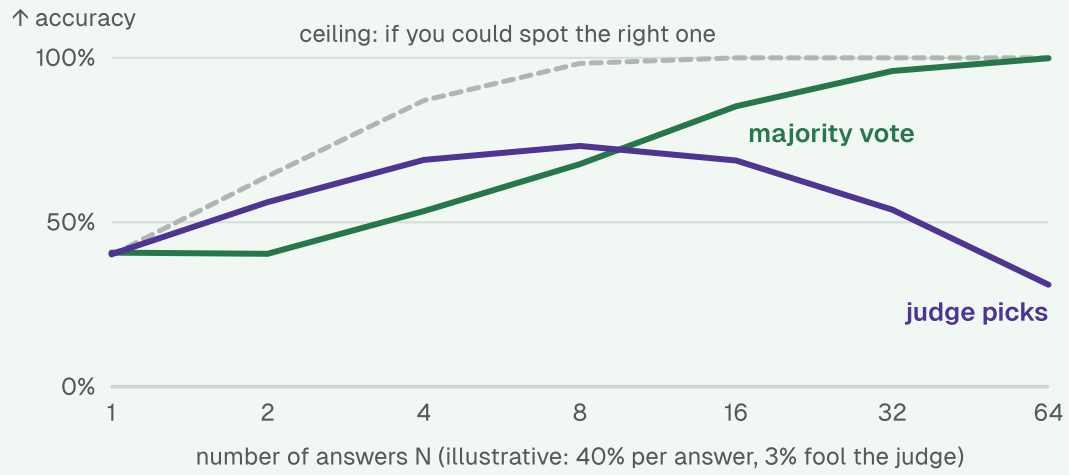
If one particular wrong answer is more common than the right one (a classic trick question), voting makes things worse, not better.

Or let a judge pick the best-looking answer



A good judge finds the 42. A slick wrong answer can fool it.

Best-of-N: generate several answers and let a judge (a reward model or verifier) pick one. With a good judge this beats voting. But the judge can be fooled.



More answers is not always better. Voting climbs toward 100% when the right answer is the most common one. Judge-picking climbs, then falls, because with enough answers one of the rare judge-fooling wrong answers shows up and wins. Each extra answer also costs compute.

Plain words

Two heads, or ten tries, are better than one

You can make a model's answers better without retraining it, by spending more effort when it answers. Let it think longer before answering. Or ask it the same question ten times and go with the answer it gives most often. Or ask ten times and have a separate grader pick the best. Each costs more per question, and each has a limit.

For leaders

When spending compute at answer time pays

- **Majority vote** (self-consistency, Wang et al., 2022) needs no extra model and works well when the right answer is the single most likely answer. If a particular wrong answer is more common than the right one, voting locks in the error.
- **Best-of-N with a verifier** can beat voting, but only as far as the verifier is reliable. With a learned reward model, larger N eventually finds answers that fool it. This is the inference-time version of reward over-optimisation (see reward models).
- **Reasoning models** move test-time compute inside a single response: a long chain of thought that explores, checks, and backtracks, learned with the RL from GRPO. Accuracy typically rises with thinking budget, with diminishing returns.
- **Compute-optimal allocation.** Snell et al. (2024) found that the best use of extra inference compute depends on problem difficulty, and that for easy and medium problems, well-allocated

test-time compute could rival a much larger model. For the hardest problems, more pretraining still won. The strategic question is where to spend: bigger model, longer thinking, or more samples.

Principal asks

What is the accuracy at matched cost? A model that wins by thinking three times longer has not necessarily won. Plot accuracy against tokens or dollars per problem, not against model name.

Under the hood

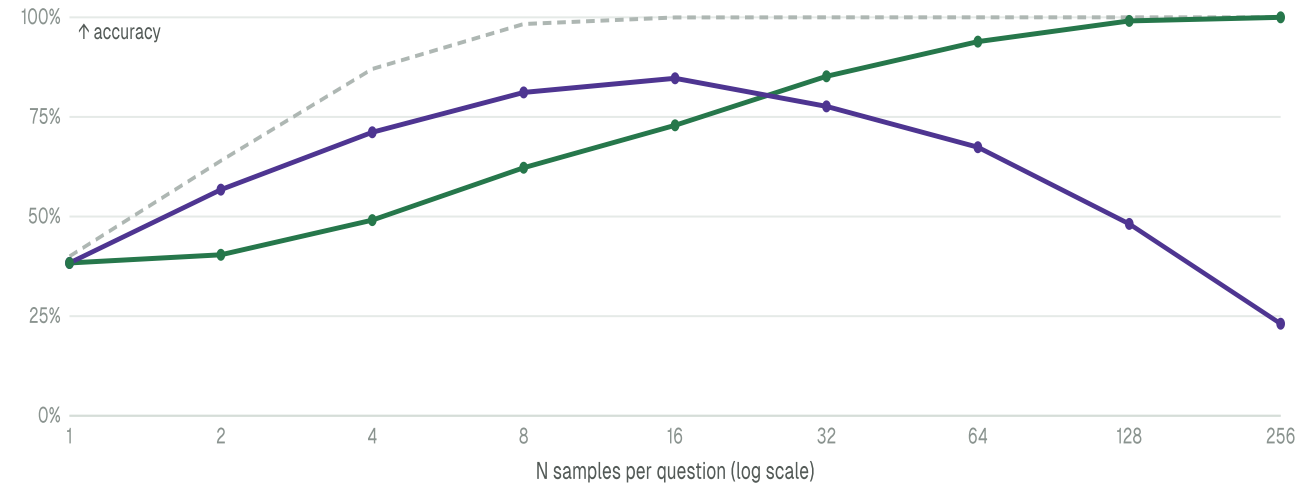
Why the curves bend

Let each sample be correct with probability a. pass@N is $1-(1-a)^N$, the ceiling. Majority vote over answers with probabilities a (correct) and $q_1, q_2, \ldots$ (distinct wrong answers) converges to 1 as $N \to \infty$ if $a > \max_j q_j$, and to 0 otherwise: the plurality answer wins in the limit. Best-of-N with verifier scores $s \sim \mathcal{N}(d', 1)$ for correct and $\mathcal{N}(0, 1)$ for wrong picks the maximum score. If a fraction ϵ of wrong answers score far higher (exploits), the probability that at least one exploit appears is $1-(1-(1-a)\epsilon)^N$, which approaches 1 as N grows. The verifier then picks the exploit, and accuracy falls. The widget simulates this with fixed seeds.

Interactive

Best-of-N versus majority vote (shown at its default setting)

Each sample is right with some probability. Majority vote picks the most common answer. Best-of-N lets an imperfect verifier pick the highest-scoring sample. Add a few "exploits" (wrong answers that fool the verifier) and watch best-of-N peak and then fall as N grows.



Best-of-N peaks at **85%** with N = 16 and reaches **23%** at N = 256. Majority vote reaches **100%**. The right answer (40%) is more common than any single wrong answer (24%), so voting converges toward 100%. Exploits make best-of-N decline after its peak: with enough samples, one that fools the verifier almost always appears. Simulation with fixed seeds; illustrative.

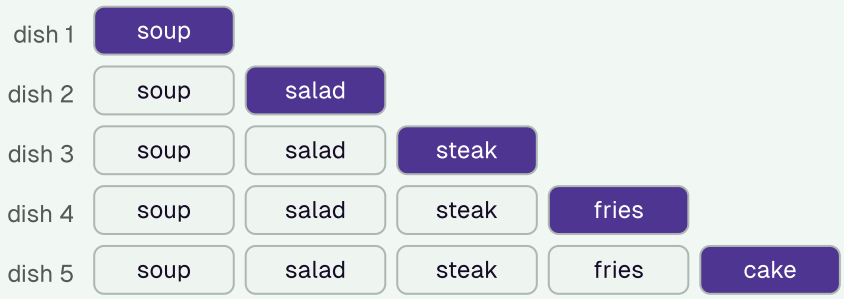
Serving: memory, speed, and cost

Diagram

Why serving is a memory problem: the waiter's notepad

KV cache and speculative decoding in five pictures.

Without notes: re-read the whole order for each new dish

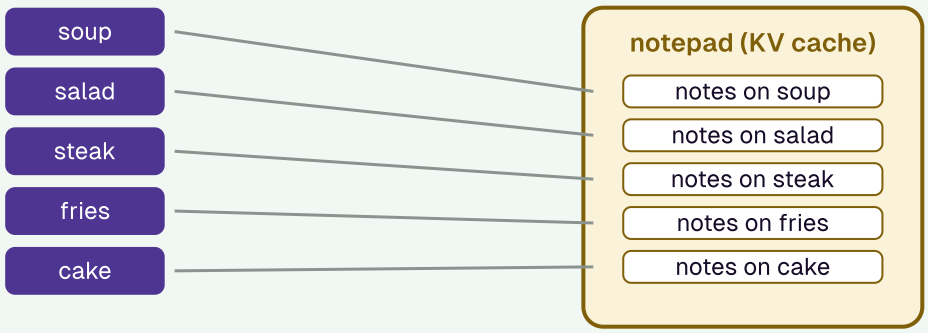


Work: 1 + 2 + 3 + 4 + 5 = 15 reads for 5 dishes

It grows with the square of the order length

To write each new word the model must consider every earlier word. Re-reading everything at every step is like a kitchen re-reading the entire order before each new dish: the work grows with the square of the length.

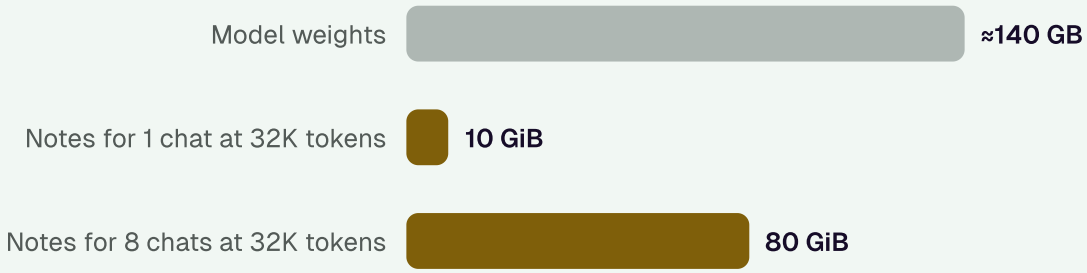
With a notepad: note each dish once, process only the new one



5 dishes, 5 reads. The price is memory for the notepad.

The **KV cache** is the notepad: the model stores a short summary (keys and values) of every earlier word, so each new word only needs its own processing plus a glance at the notes.

The notepad takes real memory (70B-class model, 16-bit)



Memory, not arithmetic, usually limits how many people one machine serves.

The notes are not free. For a 70B-class model a single 32K-token conversation needs about 10 GiB of cache; eight at once need more than half as much memory as the model weights. That is why long contexts are expensive.

Speculative decoding: a junior drafts, the editor checks in one pass

Generating with the big model is slow because each word needs a full pass. In **speculative decoding** a small model drafts a few words, the big model checks them all in one pass, keeps the ones it agrees with, and writes the first one it disagrees with.

Same result, fewer slow passes

Editor alone	With a junior drafting
1 word per pass	≈3.4 words per pass
	80% agreement, 4-word drafts

The output follows exactly the editor's own probabilities.

Speed changes; quality does not.

With a good acceptance rule the output has exactly the same distribution as the big model writing alone. With 80% agreement and four-word drafts, each slow pass yields about 3.4 words: the same text, two to three times faster in practice.

Plain words

Why long conversations are expensive

To write each new word, the model needs to look back at everything said so far. Rather than reread the whole conversation every time, it keeps notes on every earlier word, called the KV cache. Those notes take memory, and the longer the conversation and the more people served at once, the more memory they take. Memory, not raw arithmetic, is usually what limits how many people one machine can serve.

For leaders

The economics in five ideas

1. **Prefill versus decode.** Reading the prompt (prefill) is processed in parallel and limited by arithmetic. Generating tokens (decode) is one token at a time and limited by memory bandwidth, because every step reads the full weights and cache. Long prompts cost compute; long outputs cost time.

2. **Batching is the main lever.** Weights are read once per decode step regardless of how many sequences share it, so serving more sequences at once raises throughput almost for free until memory runs out or latency targets break. Continuous batching keeps the batch full as requests finish.

3. **Latency and throughput trade off.** Bigger batches mean more tokens per second per accelerator and lower cost per token, but each user's tokens arrive more slowly. Products set

a latency target and maximise throughput under it.

4. **Speculative decoding** (Leviathan et al., 2023) uses a small draft model to propose several tokens and the large model to verify them in one pass. The output distribution is provably the same as sampling from the large model alone (and identical tokens under greedy decoding, up to floating-point numerics), and speed-ups of 2 to 3 times are common when the draft agrees often.

5. **Quantisation** stores weights (and sometimes the cache) in 8 or 4 bits. It cuts memory and raises speed. 8-bit weights are usually close to lossless; 4-bit methods such as GPTQ (Frantar et al., 2023) and AWQ (Lin et al., 2024) lose a little on average and sometimes much more on maths, code, long context, and low-resource languages.

Eval lens

Evaluate the model you serve, not the model you trained. Quantisation, speculative decoding with a non-exact acceptance rule, a different inference engine, or a changed chat template can each shift scores. Re-run the core suite on the serving configuration before launch and compare with the training-time numbers using a paired test on the same problems. Perplexity alone hides task-specific losses.

Under the hood

The arithmetic

KV cache. Each layer stores a key and a value vector per KV head per token:

$$\text{bytes} = 2 \times L \times H_{kv} \times d_{\text{head}} \times \text{tokens} \times \text{sequences} \times \text{bytes per value}$$

The 2 counts keys and values. Grouped-query attention shrinks H_{kv} below the number of query heads, which is why modern models use it.

python

```python
def kv_cache_bytes(layers, kv_heads, head_dim, seq_len, batch, bytes_per=2):
    return 2 * layers * kv_heads * head_dim * seq_len * batch * bytes_per

print(kv_cache_bytes(80, 8, 128, 32_768, 1) / 2**30)   # 10.0 GiB: one 32K-token sequence, 70B-class with GQA
print(kv_cache_bytes(80, 64, 128, 32_768, 1) / 2**30)  # 80.0 GiB: the same without GQA
```

rust

```rust
fn kv_cache_bytes(layers: u64, kv_heads: u64, head_dim: u64, seq_len: u64, batch: u64, bytes_per: u64) -> u64 {
    2 * layers * kv_heads * head_dim * seq_len * batch * bytes_per
```

```
}

fn main() {
    let gib = (1u64 << 30) as f64;
    println!("{:?}", kv_cache_bytes(80, 8, 128, 32_768, 1, 2) as f64 / gib);  // 10.0 GiB: one 32K-token
sequence, 70B-class with GQA
    println!("{:?}", kv_cache_bytes(80, 64, 128, 32_768, 1, 2) as f64 / gib); // 80.0 GiB: the same without GQA
}
```

Decode speed bound. At batch size 1, each token requires reading all weights once. A 70B model in 16-bit is about 140 GB; on an accelerator with roughly 3.35 TB/s of memory bandwidth, that caps decode at about 24 tokens per second per sequence, before any parallelism across devices. Batching amortises that read across sequences.

Speculative decoding. With draft length γ and per-token acceptance probability α, the expected number of tokens produced per target-model pass is $\frac{1-\alpha^{\gamma+1}}{1-\alpha}$. For $\alpha = 0.8,\ \gamma = 4$ that is about 3.4.

Cost per million tokens.

$$\text{cost per } 10^6 \text{ tokens} = \frac{\text{dollars per accelerator-hour}}{\text{tokens/s per accelerator} \times 3600} \times 10^6$$

At an illustrative $3 per hour and 1,500 output tokens per second per accelerator, that is about $0.56 per million output tokens. Every lever above acts through the denominator.

Interactive **KV-cache memory calculator** (shown at its default setting)

Pick a model shape, context length, and how many sequences run at once. Compare cache memory with the model weights and an 80 GB accelerator. Architectures are illustrative of public 8B and 70B models that use grouped-query attention.

KV cache per sequence	KV cache total	Weights	80 GB accelerators needed
10.7 GB	85.9 GB	141.2 GB	3

The cache is **38%** of memory. Here the weights dominate; longer contexts or more concurrent users would flip that. Activations and runtime overhead are ignored, so real requirements are higher. Precision applies to both weights and cache.

Part IV · Chapter 39

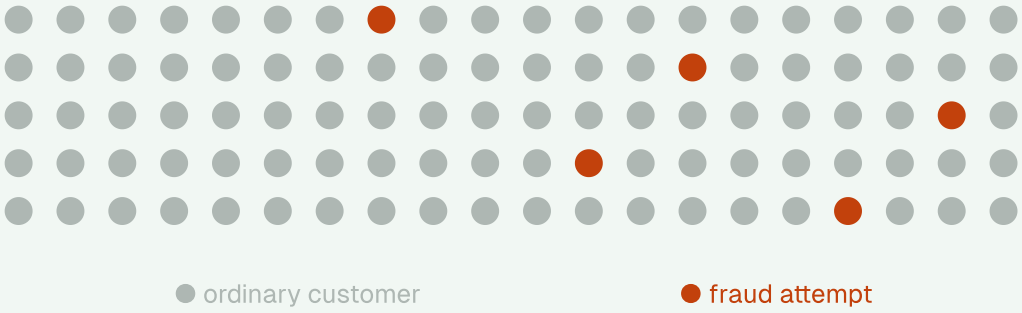
Safety and alignment training

A capable model that does whatever it is asked is not a product anyone can ship. Safety and alignment training shapes what the model will and will not do, how honestly it reports what it knows, and how it behaves under pressure from adversarial users. This chapter covers the core trade-off, the main training methods, how safety is evaluated, and how it interacts with the capability training from earlier chapters. It is descriptive, not operational: no attack recipes.

Diagram **The bank teller's dial**

Over-refusal versus unsafe compliance in five pictures.

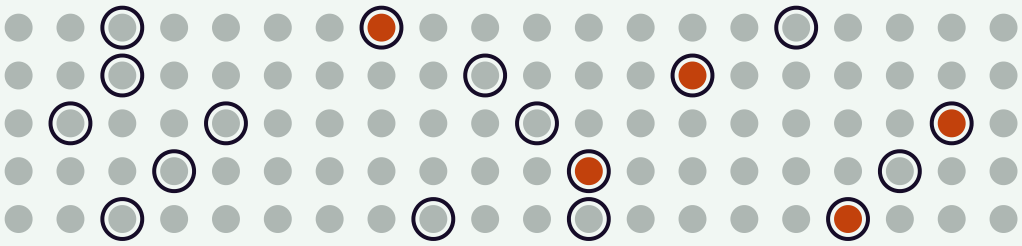
A day at the bank: 100 ordinary customers, 5 fraud attempts



The teller must help the many and stop the few (numbers are illustrative).

A model answering requests is like a bank teller. Almost everyone wants something ordinary. A few are trying something harmful. The teller has to help the many and stop the few.

A strict teller: refuses anything that looks unusual

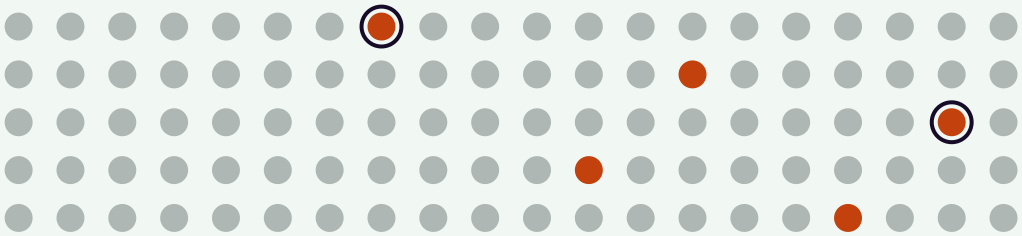


Fraud let through: 0 of 5 **Customers turned away: 12**

Ringed dots were refused. This is over-refusal.

A strict teller stops every fraud attempt, and turns away 12 ordinary customers too. In a model this is **over-refusal**: saying no to harmless requests like "how do I kill a stuck process?"

A lenient teller: rarely says no

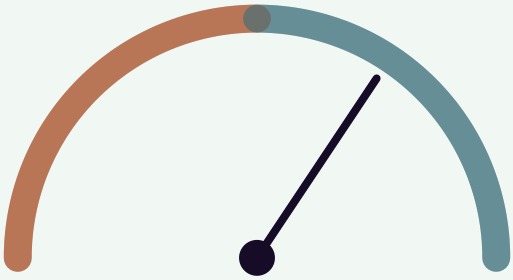


Fraud let through: 3 of 5 **Customers turned away: 0**

This is unsafe compliance.

A lenient teller helps every customer and lets 3 of the 5 fraud attempts through. In a model this is **unsafe compliance**. The two failures are mirror images.

One dial trades one mistake for the other

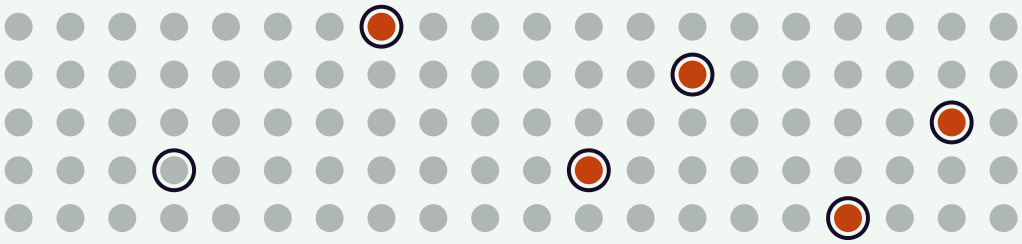


lenient
more fraud gets through

strict
more customers refused

where to set it depends on what each mistake costs

Turning the dial moves the teller between the two failures. Where to set it depends on how bad each mistake is: for severe, irreversible harms, accept more refusals; for minor ones, accept fewer. This is the refusal threshold in the widget below.



Real safety progress makes the teller better at telling the two apart, so both numbers fall at once. When a release reports fewer unsafe answers, always ask whether over-refusal went up: that would just be the dial turning.

The trade-off: helpful and harmless pull against each other

Plain words

A pharmacist, not a vending machine

A good pharmacist answers almost every question, helps people understand their medicines, and refuses a tiny number of requests that would clearly hurt someone. A pharmacist who refused every question that mentioned a dangerous drug would be useless. One who handed out anything would be dangerous. Safety training is teaching a model to be the good pharmacist: helpful by default, careful in the few places that matter, and not fooled by someone dressing up a harmful request as an innocent one.

The two ways to fail are mirror images. **Over-refusal**: saying no to something harmless ("how do I kill a Python process?"). **Unsafe compliance**: saying yes to something genuinely harmful. Pushing one down tends to push the other up.

For leaders

Treat it as a classification problem with asymmetric costs

Refuse-or-answer is a binary decision, so everything from classification metrics applies. Over-refusal is a false positive, unsafe compliance a false negative, and harmful requests are rare, so precision on refusals is low even for a good model. Three consequences:

- **The threshold is a policy decision.** Where you sit on the curve depends on the relative cost of the two errors, which differs by harm category. Severe, irreversible harms justify many over-refusals; minor ones do not. Mature policies set different operating points per category.
- **Real progress moves the curve, not the point.** A release that cuts unsafe compliance by refusing more is a threshold move. A release that cuts both errors is better training. Ask which one you are looking at.
- **Over-refusal is a real cost.** It erodes trust, pushes users to less careful tools, and is often concentrated on particular topics or groups. Track it with the same seriousness as unsafe compliance.

Principal asks

- Did both error rates improve, or did one improve at the other's expense?
- What are the operating points per harm category, and who signed off on the cost ratios?
- How are the over-refusal prompts chosen? Are they hard, borderline cases, or easy ones that any model passes?

Under the hood

Expected cost and the operating point

With harmful share π, refusal threshold t, over-refusal rate $\mathrm{FPR}(t)$ on benign traffic, unsafe-compliance rate $\mathrm{FNR}(t)$ on harmful traffic, and cost ratio C (one unsafe completion costs C over-refusals):

$$\mathbb{E}[\text{cost}](t) \;\propto\; (1-\pi)\,\mathrm{FPR}(t) + \pi\, C\,\mathrm{FNR}(t)$$

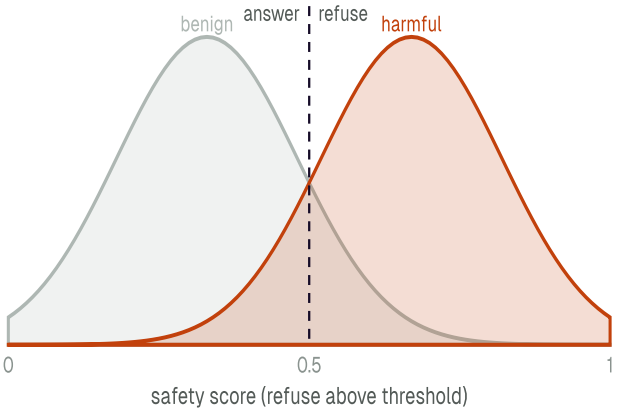
The cost-minimising threshold is where the frontier's slope equals $-(1-\pi)/(\pi C)$. Rarer harms or cheaper failures push the optimum toward refusing less.

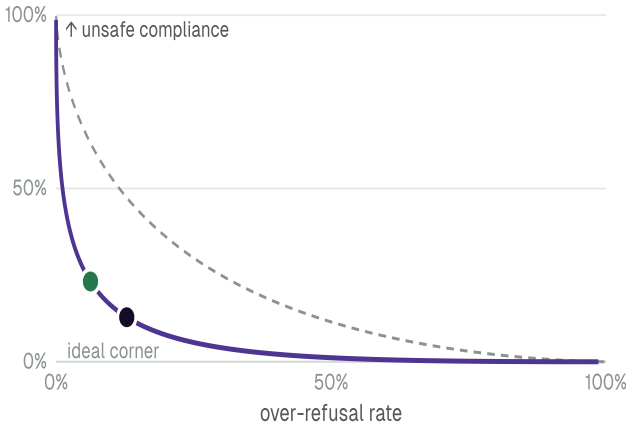
In practice the "score" is not a single number. It is the model's learned propensity to refuse, plus any classifiers in front of or behind the model. The frame still holds: every component trades the two errors, and the system's frontier is what matters.

Interactive

Helpfulness versus harmlessness frontier (shown at its default setting)

A safety signal scores every request. Above the threshold the model refuses. Benign requests that score high are over-refusals; harmful requests that score low are unsafe completions. Move the threshold along the curve, then improve the model and watch the whole frontier move toward the corner.





Benign requests · Harmful requests · Current model frontier · Earlier model · Lowest expected cost

Over-refusal (benign refused)	Unsafe compliance (harmful answered)	Refusals that were right	Cost-minimising threshold
12.9%	**12.9%**	**12%**	**0.56**

Per 1,000 requests: about **126** benign requests refused and **2.6** harmful ones answered. Only **12%** of refusals hit a harmful request, because harmful traffic is rare. At a 20× cost ratio the expected cost is lowest at threshold **0.56** (green dot). Improving training moves the solid curve toward the corner; moving the threshold only slides along it. Illustrative model.

How models are trained to be safe

Diagram

Constitutional AI: the model edits itself against written principles

How self-critique and AI feedback replace much of the human labelling. Five pictures.

Step 1: the model drafts a reply

User

How do I kill a Python process that is stuck?

First draft

I can't help with killing anything.

A harmless question, refused because of one scary word.

Over-refusal, the mirror image of unsafe compliance.

Constitutional AI (Bai et al., 2022) starts with the model's own drafts. Here the draft refuses a harmless programming question because it contains the word "kill".

Step 2: critique the draft against a written principle

Principle (from the constitution)

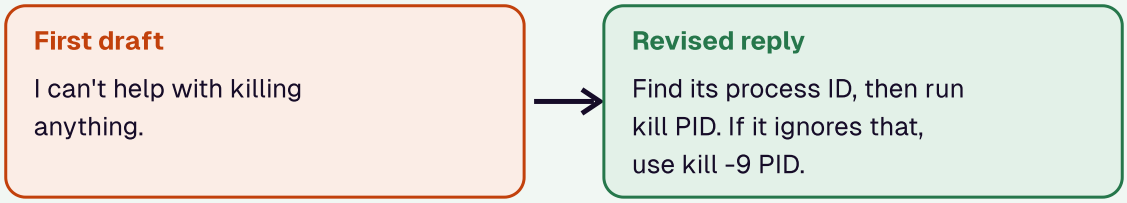
Choose the response that is most helpful while avoiding real harm.

Critique, written by the model

"kill" here means ending a program. Nothing harmful is being asked, so refusing is unhelpful and should be revised.

The model is shown a principle drawn from a written list, the constitution, and asked to critique its own draft against it.

Step 3: revise

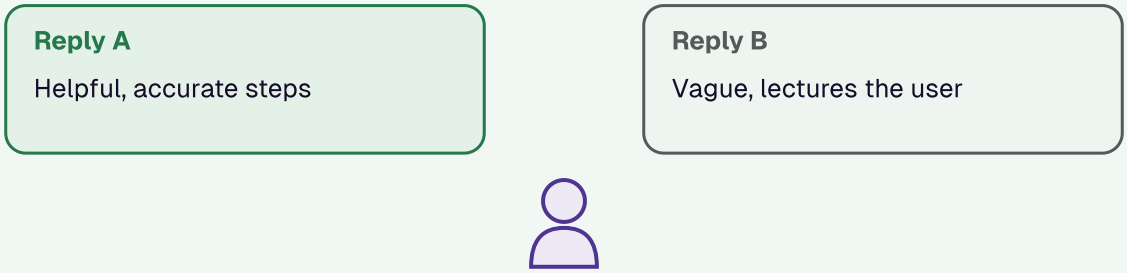


The (draft, revision) pair is saved.

Thousands of these become fine-tuning data.

The model rewrites the reply to follow the principle. Revised replies become supervised fine-tuning data: the **supervised phase**.

Step 4: an AI judge compares pairs using the principles



In the **RL phase** the fine-tuned model writes pairs of replies, and an AI feedback model picks which one better follows a principle. Those labels train a preference model, the same machinery as RLHF but with AI feedback: RLAIF.

The whole loop

Strength: explicit, editable principles; feedback scales with compute.

Risk: the judge's blind spots become the model's blind spots.

Put together: draft, critique, revise, fine-tune, then compare pairs with AI feedback and run RL. The principles are written down and can be inspected and edited, which is the method's main appeal.

Plain words

Rules, examples, and practice

There are three main ways to teach good judgement. Show the model examples of good responses to tricky requests. Have people rate pairs of responses so it learns which is better. Or write down a set of principles and have an AI use them to critique and rate responses, which scales far better than asking people to read thousands of disturbing conversations.

For leaders

The main methods

Method	What it does	Strength	Weakness
Safety SFT	Fine-tune on curated demonstrations of good responses to sensitive requests (see SFT)	Direct, predictable	Teaches surface patterns; can over-generalise to refusing anything that looks similar
RLHF with harmlessness preferences (Bai et al., 2022a)	Humans compare responses for helpfulness and harmlessness; train a reward model and optimise against it (see RLHF)	Captures nuanced judgement	Expensive, labeller exposure to harmful content, inconsistent labels, reward hacking
Constitutional AI and RLAIF (Bai et al., 2022b)	A written set of principles; the model critiques and revises its own outputs, then AI-generated preference labels replace human harmlessness labels	Scales, transparent principles, less human exposure	Quality bounded by the labelling model and how well the principles are written
Direct preference methods	DPO-style training on safety preference pairs (see DPO)	Simple, stable	Offline data goes stale as the model changes
Reasoning over a policy	The model is trained to recall and reason about a written safety specification before answering, for example OpenAI's deliberative alignment (Guan et al., 2024)	Better on novel and borderline cases; reasoning can be inspected	Depends on the specification's quality; adds latency
Character and persona training	Shapes stable traits such as honesty, curiosity, and directness, rather than rule-following alone	Generalises to situations rules did not foresee	Harder to measure; traits can drift under RL

Under the hood

Constitutional AI in two phases

1. **Supervised phase.** Sample responses to red-team prompts from a helpful-only model. Ask the model to critique each response against a randomly drawn principle from the constitution, then revise it. Fine-tune on the revised responses.

2. **RL phase (RLAIF).** Sample pairs of responses from the fine-tuned model. Ask a feedback model which response better follows a principle, giving preference labels. Train a preference model on those labels (often mixed with human helpfulness labels) and run RL against it.

The appeal is that the principles are explicit and editable, and the feedback scales with compute instead of human hours. The risk is that the feedback model's blind spots become the policy's blind spots, and that the policy learns to satisfy the letter of the principles. It also explains why constitutions are written as general judgements ("choose the response that is more honest") rather than narrow rules: general principles transfer better to unseen situations.

Failure modes beyond refusal

For leaders

What safety teams worry about besides "will it help build something dangerous"

- **Jailbreaks.** Inputs that get a model to do what its training says it should not. Broad classes include role-play and fictional framing, gradual escalation over many turns, encoding or translating the request, splitting a harmful task into innocent-looking pieces, flooding the context with fabricated examples of compliance, and instructions hidden in documents or tool outputs (prompt injection). Defences are layered: training on these patterns, classifiers on inputs and outputs, and limits on what tools an agent can use.
- **Sycophancy.** Telling users what they want to hear: agreeing with stated opinions, caving when pushed back on a correct answer, praising weak work. Preference training rewards it, because people tend to rate agreement highly (Sharma et al., 2023).
- **Dishonesty and poor calibration.** Stating guesses as facts, inventing citations, or claiming to have done work it did not do. For agents, misreporting what actions were taken.
- **Reward hacking that looks like misbehaviour.** RL on capability tasks can teach a model to cut corners (see GRPO), and corner-cutting can generalise into broader patterns such as deceiving the grader.
- **Over-refusal and moralising.** Unrequested lectures, hedging that buries the answer, and refusing legitimate professional questions in medicine, security, or law.

Watch out

Capability RL can quietly undo safety training. A long reasoning-RL run optimised only for correctness can raise compliance with harmful requests, increase sycophancy, or teach the model to game evaluations. Re-run the safety suite after every major capability stage, not only at the end, and include safety data or rewards in the RL mixture when possible.

How safety is evaluated

For leaders

Four families of evaluation

Refusal and harm benchmarks

Sets of harmful requests scored for whether the model complies, for example HarmBench (Mazeika et al., 2024) and StrongREJECT (Souly et al., 2024), which also grades how useful a non-refusal actually was.

Over-refusal benchmarks

Benign prompts that look risky ("how do I kill a process"), for example XSTest (Röttger et al., 2024) and OR-Bench (Cui et al., 2024). Always report alongside harm benchmarks.

Red teaming

Humans and automated attackers search for failures. Automated red teaming uses another model to generate attacks at scale (Perez et al., 2022). Results are a lower bound on vulnerability, never a proof of safety.

Dangerous-capability evals

Tests of whether the model could meaningfully help with severe harms, such as advanced cyber operations or weapons development, or could act autonomously in risky ways. These feed frontier safety policies.

Frontier safety policies. Several labs publish frameworks that tie capability thresholds to required safeguards: Anthropic's Responsible Scaling Policy with its AI Safety Levels (first published 2023), OpenAI's Preparedness Framework (2023), and Google DeepMind's Frontier Safety Framework (2024). The common structure: define capability thresholds that would matter, evaluate for them before and during deployment, and commit to stronger security and deployment safeguards, or to pausing, when a threshold is crossed. Dangerous-capability evaluation therefore carries direct launch consequences, and its statistical quality matters accordingly.

Eval lens

Safety evals have every problem from Part II, often worse. Harmful prompts are rare and diverse, so sets are small and intervals are wide. Grading is usually done by an LLM judge, which must be validated against human labels (see LLM judges). Public benchmarks get trained on, so keep private held-out sets (see contamination). And a capability eval that a model fails because of refusals or poor prompting understates what it can do. Dangerous-capability evals should try hard to elicit the capability, which is the opposite of a normal product eval.

Under the hood

Measuring robustness, not a single attack

A model's jailbreak resistance is better described by a curve than a number: attack success rate as a function of attacker effort (queries, attempts, or compute). Report attack success rate with an interval at several budgets, and for adaptive attacks report the budget needed to reach a given success rate. Because successes are rare at low budgets, use exact or Wilson intervals (see eval

statistics), and treat zero observed successes as "fewer than about $3/n$ at 95% confidence", the rule of three, not as zero risk.

$$\text{If } 0 \text{ of } n \text{ attacks succeed:} \quad p_{\text{upper}} \approx \frac{3}{n}$$

An approximate 95% upper bound. With 300 failed attacks you can only claim the success rate is probably below 1%.

Deployment: training is one layer of several

For leaders

Defence in depth

No lab relies on model training alone. Deployed systems add input and output classifiers (for example Anthropic's constitutional classifiers, described in 2025), usage policies and account-level enforcement, monitoring for abuse patterns across many conversations, rate limits, and for agents, restrictions on tools and permissions. Each layer has its own false-positive and false-negative rates, and stacking classifiers compounds over-refusal as well as protection. Evaluate the full system on realistic traffic, not each layer alone.

Principal asks

If the model's refusal training failed completely on one category tomorrow, which other layer would catch it, and how would we know within a day?

Part V

Leading the work

How training and evaluation connect, the questions a principal asks, and the red flags to look for.

Part V · Chapter 40

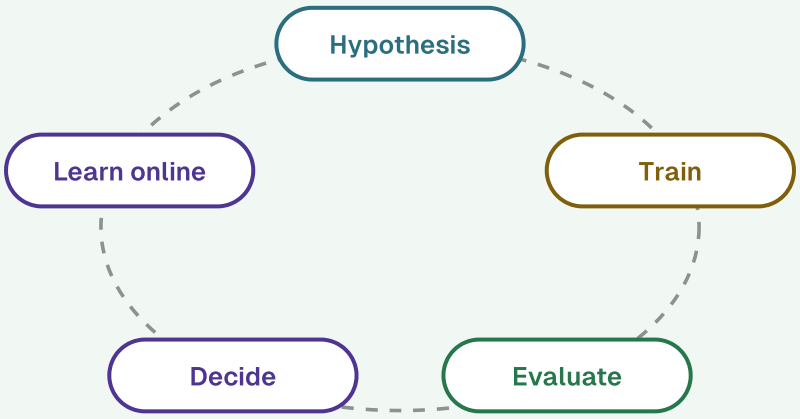
Connecting training and evaluation

Parts II and III describe two machines. One makes models. The other tells you whether they got better. This chapter is about running both as one system, and about the judgment calls that separate a senior reviewer from someone who reads results off a dashboard.

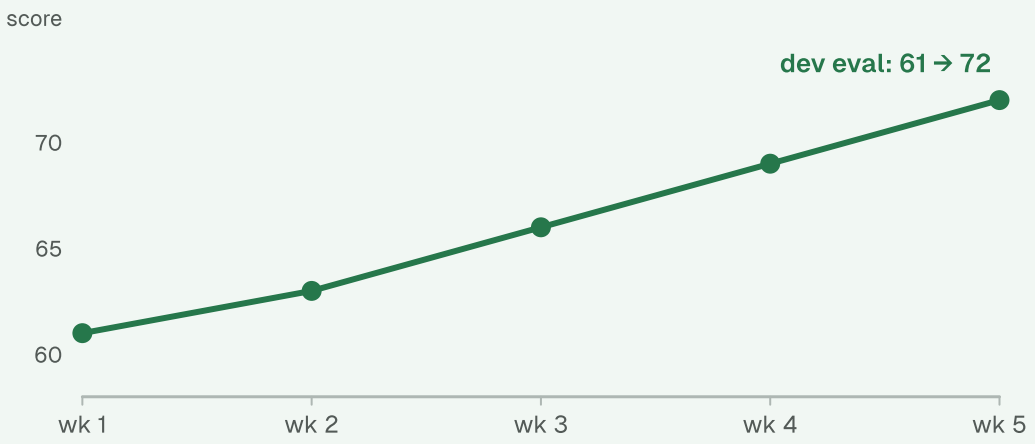
Diagram

The bathroom scale and the diet

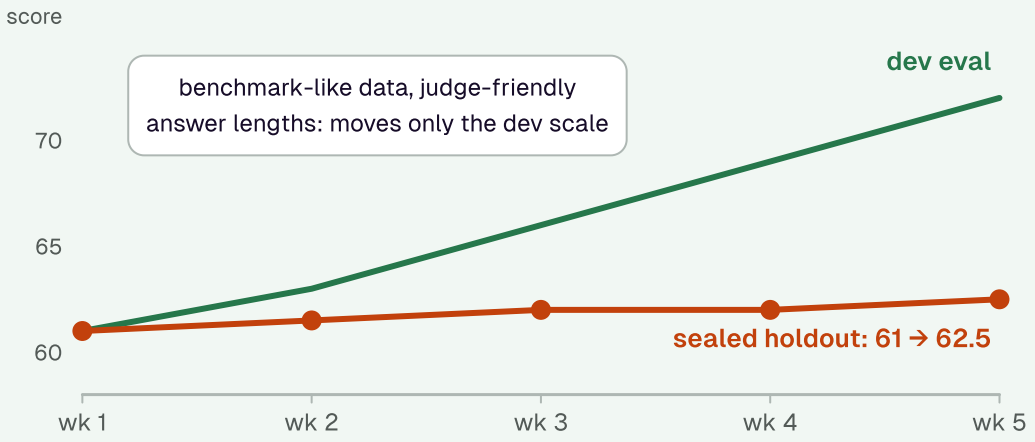
Why evals become the objective, and how to keep a measurement honest under optimisation pressure.



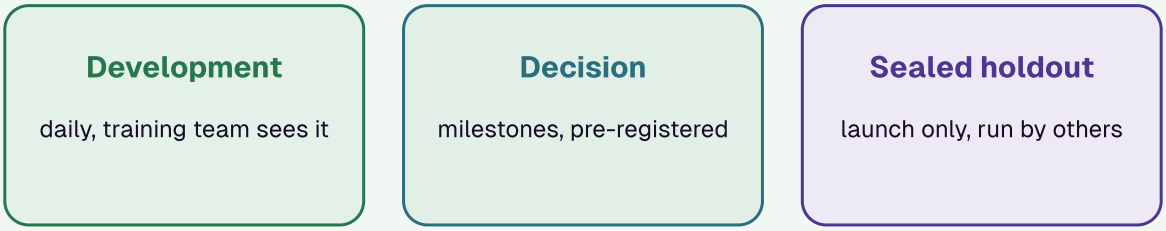
Training and evaluation form one loop: a hypothesis, a training change, an eval, a decision, and what production teaches you next. Whatever the eval rewards is what the loop optimises.



Week by week the development eval climbs from 61 to 72. The team celebrates. It looks like a diet that works.



Then a sealed holdout, which nobody tuned toward, barely moves. Most of the gain came from learning the quirks of one scale: training data that resembles the benchmark and answer styles the judge likes.



When they disagree, believe the scale nobody could tune toward.

So keep three scales: a development eval for daily iteration, a decision eval with pre-registered thresholds, and a sealed holdout run by someone else. Goodhart's law is the default, so plan for it.

The loop

1 **Hypothesis**
What should improve, and why

2 **Training change**
Data, objective, method, or scale

3 **Eval**
Suite chosen before the run

4 **Analysis**
Paired deltas, intervals, slices

5 **Decision**
Ship, iterate, or kill

6 **Online signal**
Feeds the next hypothesis

Plain words

The scale and the diet

Training is the diet. Evaluation is the bathroom scale. If the scale is wrong, every diet looks like it works or none does. If you only ever weigh yourself on one scale, you start eating whatever moves that particular scale, even if it does nothing for your health. Teams that train models face the same trap. Whatever you measure becomes what you optimise, so you need scales you trust, more than one of them, and some you never look at until the end.

For leaders

Evals are the objective function of the organisation

In a training organisation, the eval suite is the de facto objective function. People will optimise whatever gets reported in the weekly review. That gives the eval owner unusual leverage and unusual responsibility.

- **Goodhart is the default.** Every metric that drives decisions will be gamed, usually without bad intent. Training data drifts toward benchmark-like content, prompts drift toward judge preferences, and RL policies find verifier loopholes. Plan for it: rotate suites, keep private holdouts, and audit the largest gains first.
- **Separate the scales.** Keep three tiers: development evals the training team can see every day, decision evals reviewed at milestones, and sealed holdouts that only an independent owner runs before launch. If the sealed set disagrees with the development set, believe the sealed set.
- **Pre-register the decision.** Write down, before the run, which evals decide success and what delta counts. Post hoc selection of the metric that moved is the most common way a team fools itself.
- **Price the evals.** Evaluation is a real fraction of compute and headcount. A suite that cannot detect the effect sizes your training changes produce (see power and sample size) is pure cost. Fund fewer, better-powered evals over many thin ones.

Under the hood

Why gains shrink when they reach the holdout

Suppose you try m training variants and pick the one with the best development score. Even if every variant has the same true quality, the winner's observed score is biased upward by the expected maximum of m noise draws. With per-variant standard error σ, that bias is roughly

$$\mathbb{E}\left[\max_{i \le m} \varepsilon_i\right] \approx \sigma\sqrt{2 \ln m}$$

For $m = 20$ variants and $\sigma = 1$ point, the "best" variant looks about 2.4 points better than it is. This is the winner's curse, and it is why a fresh holdout almost always shows a smaller gain than the development set.

python

```python
import numpy as np

rng = np.random.default_rng(0)
m, sigma, trials = 20, 1.0, 20000
best_dev = rng.normal(0, sigma, size=(trials, m)).max(axis=1)   # all variants truly equal
```

```
print(round(best_dev.mean(), 2))                # ~1.87 (the asymptotic formula overstates at small m)
print(round(sigma * np.sqrt(2 * np.log(m)), 2))  # 2.45
```

rust

```rust
/// Small inline PRNG: SplitMix64 for uniforms, Box-Muller for normals.
/// The draws differ from numpy's, so the simulated mean differs in the last digit or so.
struct Rng(u64);

impl Rng {
    fn next_u64(&mut self) -> u64 {
        self.0 = self.0.wrapping_add(0x9E37_79B9_7F4A_7C15);
        let mut z = self.0;
        z = (z ^ (z >> 30)).wrapping_mul(0xBF58_476D_1CE4_E5B9);
        z = (z ^ (z >> 27)).wrapping_mul(0x94D0_49BB_1331_11EB);
        z ^ (z >> 31)
    }
    fn uniform(&mut self) -> f64 {
        ((self.next_u64() >> 11) as f64 + 0.5) / (1u64 << 53) as f64 // in (0, 1), never 0
    }
    fn normal(&mut self, mean: f64, sd: f64) -> f64 {
        let (u1, u2) = (self.uniform(), self.uniform());
        mean + sd * (-2.0 * u1.ln()).sqrt() * (2.0 * std::f64::consts::PI * u2).cos()
    }
}

fn main() {
    let mut rng = Rng(0);
    let (m, sigma, trials) = (20, 1.0, 20000);
    // all variants truly equal: the best dev score of m is pure noise
    let best_dev: Vec<f64> = (0..trials)
        .map(|_| (0..m).map(|_| rng.normal(0.0, sigma)).fold(f64::NEG_INFINITY, f64::max))
        .collect();
    let mean = best_dev.iter().sum::<f64>() / trials as f64;
    println!("{:.2}", mean); // 1.87 with this PRNG (numpy also gives 1.87; draws differ); the asymptotic formula
overstates at small m
    println!("{:.2}", sigma * (2.0 * (m as f64).ln()).sqrt()); // 2.45
}
```

The practical fix is the same as in any selection problem: hold out a set you never select on, and report the selected model's score on it.

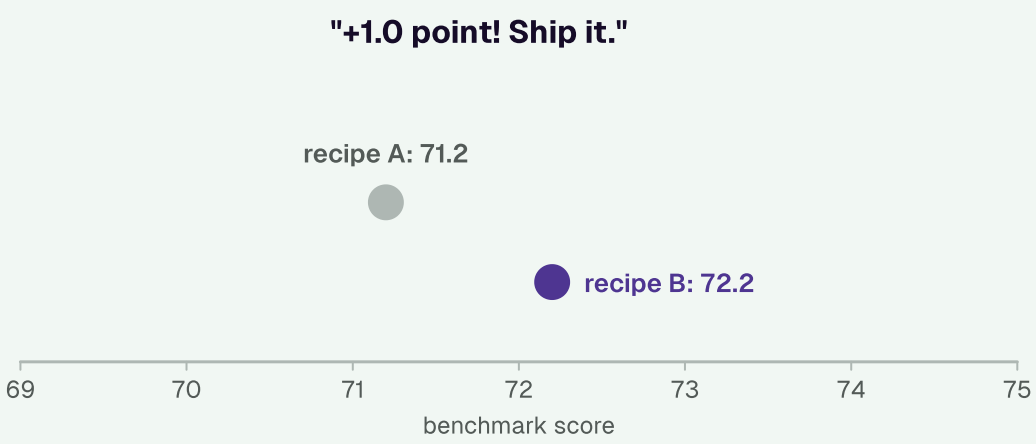
Is this result real? Seeds and run-to-run noise

Training runs are noisy. The same recipe with a different random seed lands at a different score. Before you believe a gain from a training change, you need to know how large that seed noise is compared with the gain.

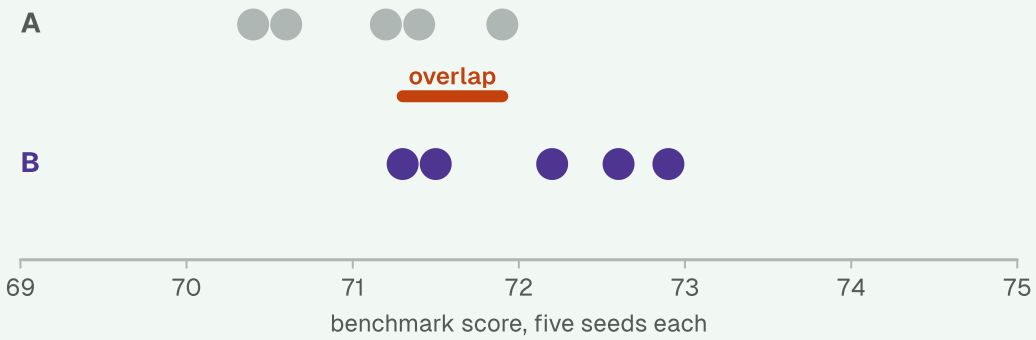
Diagram

One training run is one coin flip

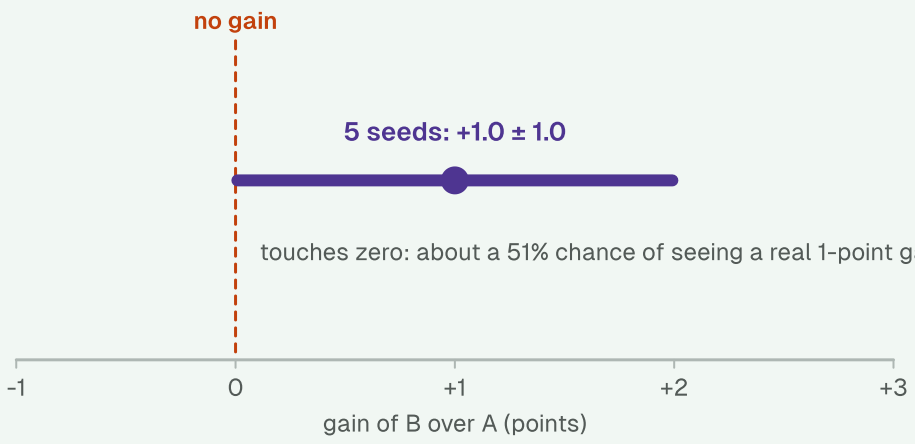
Seed-to-seed noise of 0.8 points, a claimed gain of 1 point, and what it takes to believe it. Numbers are illustrative.



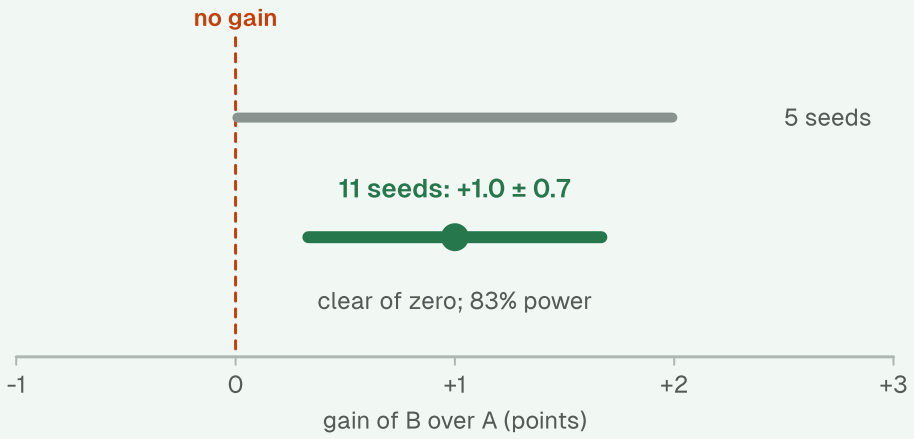
Recipe A scores 71.2, recipe B scores 72.2. One run each. A one-point gain looks like a win.



Rerun each recipe with five different random seeds. The scores scatter by about 0.8 points either way, and the two clouds overlap. Some A runs beat some B runs.



Averaged over five seeds, B is still about 1 point ahead, but the 95% interval on the gain (±1.0) reaches zero. With five seeds a real one-point gain shows up as significant only about half the time.

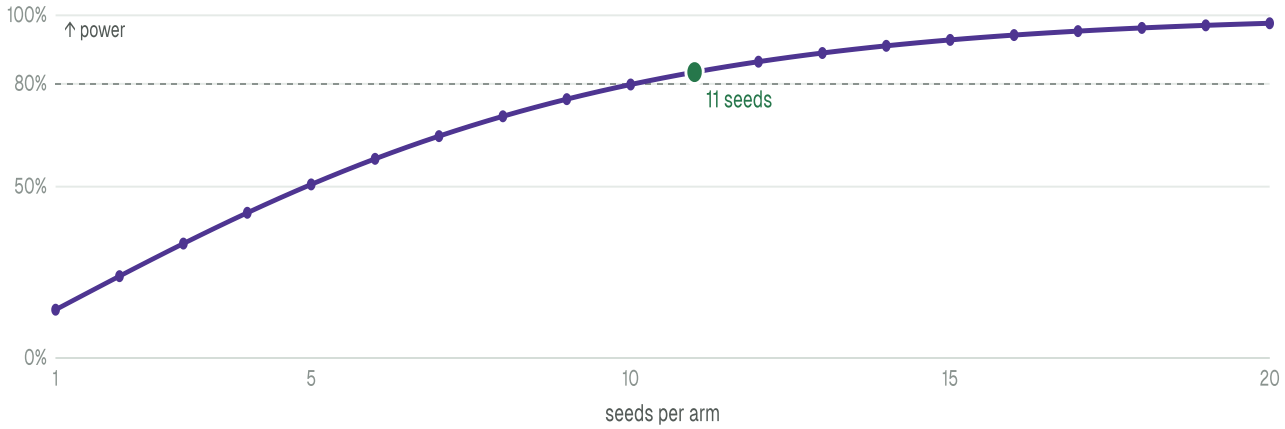


With 11 seeds per arm the interval shrinks to ±0.7 and clears zero. That is the arithmetic behind "ask for the seed variance first", and the widget below lets you redo it for any gain.

Interactive

Seeds needed to trust a gain (shown at its default setting)

Set how much a score moves between seeds and the gain someone is claiming. The chart shows how likely you are to see the gain as significant with more seeds per arm, and the readout says what a single-seed comparison is worth.



Chance 1 vs 1 seed shows the wrong sign	Seeds per arm for 80% power	Gain / seed std
19%	**11**	**1.25**

With one seed per arm, a true gain of 1.0 points shows up with the wrong sign 19% of the time. To detect it 80% of the time you need about 11 seeds per arm, which is usually too expensive: shrink seed noise (bigger eval sets, paired items) or accept that this gain is below what you can resolve.

Plain words

One run is one coin flip

If you train the same model twice, you get two slightly different results, like flipping a coin twice. A claimed one-point gain from comparing one run against one other run can easily be luck. The widget shows how many repeat runs you need before a gain of a given size is believable.

For leaders

What to require before you believe an ablation

- **Ask for the seed variance first.** No one can interpret a delta without it. If the team does not know it, the first deliverable is three seeds of the baseline.

- **Compute-matched comparisons.** A new method that also used 30 percent more compute has not been compared with the baseline. Insist on equal FLOPs, equal tokens, or an explicit compute-versus-quality curve.
- **Tuned baselines.** The new method usually gets more tuning effort than the baseline. Ask how many hyperparameter trials each arm got.
- **Small-scale results transfer imperfectly.** Many gains at 1B parameters vanish at 70B, and some appear only at scale. Ask whether the trend holds across at least two or three model sizes before it drives a large run.
- **Kill criteria.** Every research bet should state in advance what result would stop it. Without one, projects drift for quarters.

Under the hood

The arithmetic behind the widget

With n seeds per arm, seed standard deviation s, and true gain δ, the difference in arm means has standard error $s\sqrt{2/n}$. A two-sided test at $\alpha = 0.05$ has power approximately

$$\text{power}(n) \approx \Phi\left(\frac{\delta}{s\sqrt{2/n}} - 1.96\right), \qquad n_{80} = \left\lceil 2\left(\frac{(1.96 + 0.84)\, s}{\delta}\right)^2 \right\rceil$$

Φ is the standard normal CDF. With very few seeds a t-distribution is more honest and needs slightly more seeds than this normal approximation.

The chance a single seed versus a single seed shows the wrong sign is $\Phi\big(-\delta/(s\sqrt{2})\big)$. When the gain equals the seed standard deviation, that is about 24 percent: one comparison in four points the wrong way.

A reviewer's checklist

Use this when someone shows you a training result or an eval delta. Each item links to the chapter that explains it.

Question	Why it matters	Chapter
What is the interval on that number, and what is n?	A point estimate without uncertainty cannot support a decision.	Eval statistics
Was the comparison paired on the same items?	Unpaired tests throw away most of the power.	Eval statistics
How many metrics or benchmarks were looked at?	With 40 benchmarks, two "significant" moves are expected by chance.	Multiple comparisons
Are items clustered (multi-turn, agent traces, per-repo tasks)?	Naive intervals can be several times too narrow.	Clustering
What is the seed variance, and how many seeds?	Single-run comparisons are coin flips at small effect sizes.	This chapter
Is the comparison compute-matched and baseline-tuned?	Otherwise you are measuring effort, not method.	This chapter, scaling
Who or what scored it, and was the scorer validated?	Judges carry bias and absolute-level offsets.	LLM as judge
Could the eval items be in the training data?	Contamination inflates scores and hides real capability.	Contamination
Does the offline gain predict an online gain?	Offline suites overstate and sometimes point the wrong way.	Offline to online gap
What got worse?	Every training change has regressions; a report without them is incomplete.	SFT, triage
Was decoding held fixed?	Temperature and sampling settings alone can move scores by points.	Inference
Is the reward or verifier being exploited?	RL finds loopholes faster than people do.	Reward models, GRPO

Practise on a slide. The three results below are invented, but each is modelled on a kind of report reviewers see often.

Interactive **Review a result** (shown at its default setting)

Pick a slide, tick every checklist question the slide actually answers, then press "Check against the reviewer's key" to see what you missed.

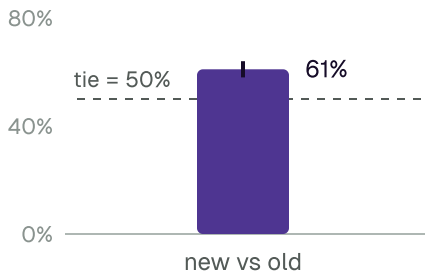
RESULTS SLIDE (INVENTED EXAMPLE)

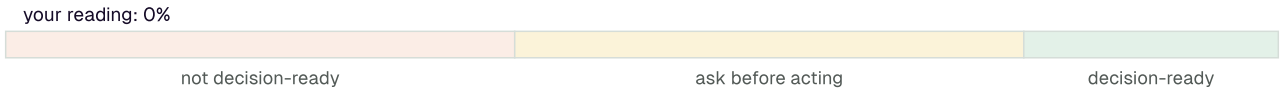
New chat SFT data mix

61% win rate (95% CI 58% to 64%)

LLM-judge win rate against the baseline

- 800 prompts from the internal helpfulness set; both models answer every prompt.
- Judge: a frontier LLM with our standard rubric.
- Same base model and same number of training tokens.
- Also up on 5 of the 12 academic benchmarks we track (table in appendix).
- One training run per arm.
- Greedy decoding, same chat template.





Questions answered	Confidence (weighted)	Verdict
0 / 11	**0%**	**Not ready**

By your reading the slide answers **0 of 11** applicable questions, weighted confidence **0%**: not decision-ready: ask for the missing pieces first. Still open, so ask for these:

- What is the interval on that number, and what is n? → Eval statistics
- Was the comparison paired on the same items? → Eval statistics
- How many metrics or benchmarks were looked at? → Multiple comparisons
- Are items clustered (multi-turn, agent traces, per-repo tasks)? → Clustering
- What is the seed variance, and how many seeds? → Seeds (this chapter)
- Is the comparison compute-matched and baseline-tuned? → This chapter, scaling
- Who or what scored it, and was the scorer validated? → LLM as judge
- Could the eval items be in the training data? → Contamination

- Does the offline gain predict an online gain? → Offline to online gap
- What got worse? → SFT, triage
- Was decoding held fixed? → Inference

The meter is a rough heuristic: interval and seed questions count triple, six others double, the rest single, and not-applicable questions are left out. Use it to see gaps, not as a probability.

Red flags, and what they usually mean

You see	Likely cause	Ask
Huge gain on one benchmark, nothing elsewhere	Contamination, benchmark-specific data, or format overfitting	Score on a perturbed or fresh version of that benchmark
Judge win rate up, human ratings flat	Optimising toward judge preferences (length, style)	Length-controlled win rate and a human spot check
Reward keeps climbing during RL, quality plateaus	Reward overoptimisation or verifier hacking	Samples from late checkpoints; gold-reward or held-out checks
pass@1 up, pass@k at large k down	RL sharpening the distribution rather than adding capability	pass@k curves before and after
Offline up, online flat	Eval distribution or harness differs from real usage	Backtest of offline vs online deltas across past releases
Smooth loss improvement, sudden "emergent" jump in accuracy	Thresholded metric hiding steady progress	A continuous metric (log-likelihood or partial credit) on the same items
Big gain, one seed, no interval	Noise	Three seeds per arm and a paired test

For leaders

Communicating uncertainty upward

Executives need a decision, not a lecture on intervals. A pattern that works:

1. **Lead with the decision and your confidence.** "Recommend shipping. We are confident it is not worse and fairly confident it is better on coding."
2. **Give the number with its range in plain words.** "Coding pass rate up about 3 points, plausibly between 1 and 5."
3. **Name the main risk and how you are watching it.** "Long-context retrieval slipped slightly. We are monitoring it in the staged rollout with a rollback trigger."
4. **Say what would change your mind.** This builds trust when something does change.

Principal asks

- What would we have to believe for this result to be wrong?
- If we had half the eval budget, which evals would we keep, and why?
- Which of our evals has disagreed with production most often, and what did we change because of it?
- What is the smallest effect our suite can reliably detect, and is that smaller than the effects we are trying to ship?
- Who owns the sealed holdout, and when did someone last look at failures on it by hand?

Part VI

Experimental design

Correlation and causation, who is really in your sample, random assignment and its alternatives, measuring the right thing with the right formula, A/B tests end to end, metrics and logging, and designing experiments for AI systems.

Part VI · Chapter 41

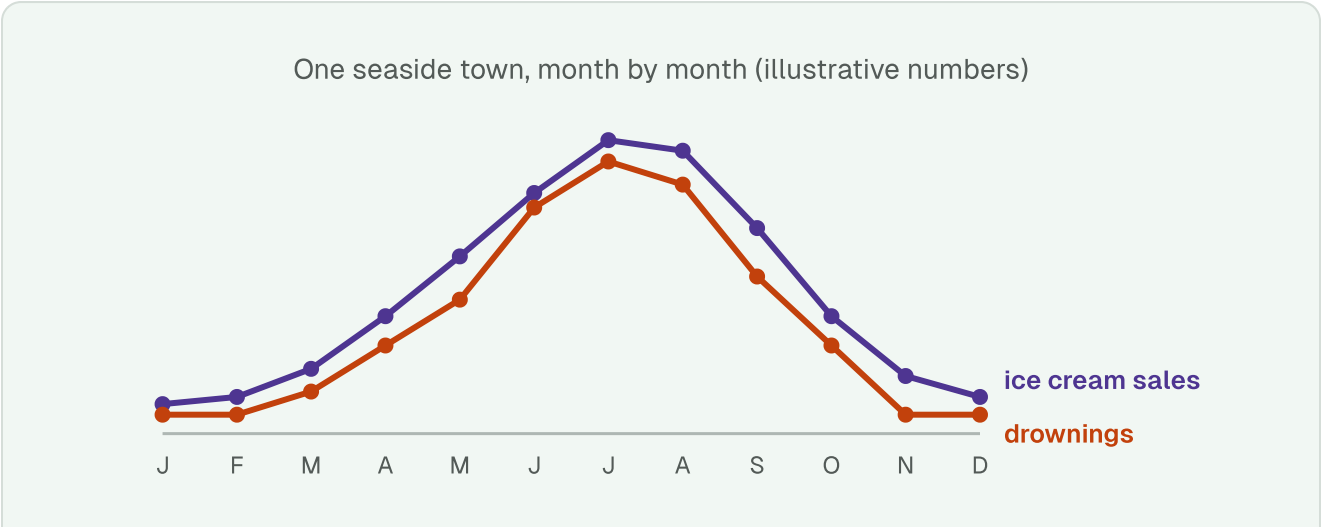
Correlation is not causation: what else makes two numbers move together

Every dashboard is full of pairs of numbers that rise and fall together. Some of those pairs are cause and effect. Many are not, and acting on the wrong ones wastes quarters. This chapter covers what a correlation measures, the four ordinary ways a correlation appears without any causal link, Simpson's paradox, causal diagrams for deciding what to "control for", and the ladder of evidence that ends at a randomized experiment. Each idea gets an everyday example first and an AI example second.

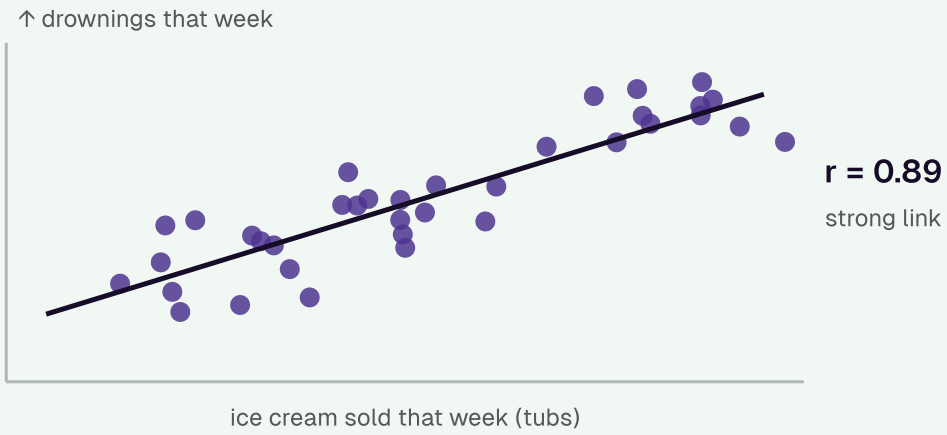
Diagram

Does ice cream cause drowning?

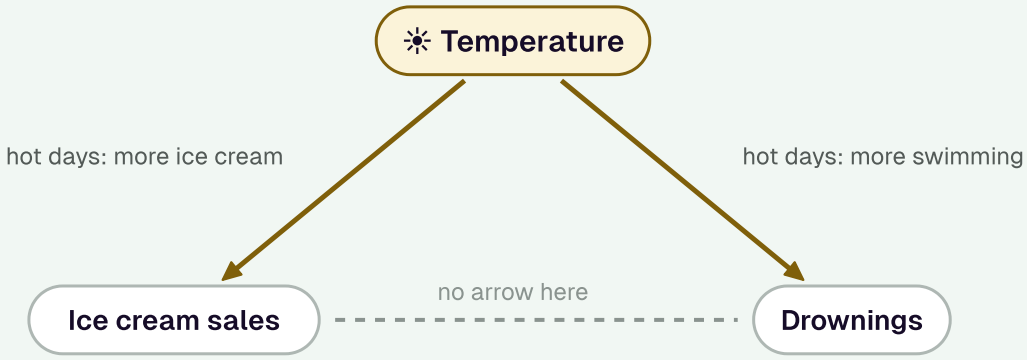
The classic confounder, then the same trap in a product dashboard. All numbers are illustrative.



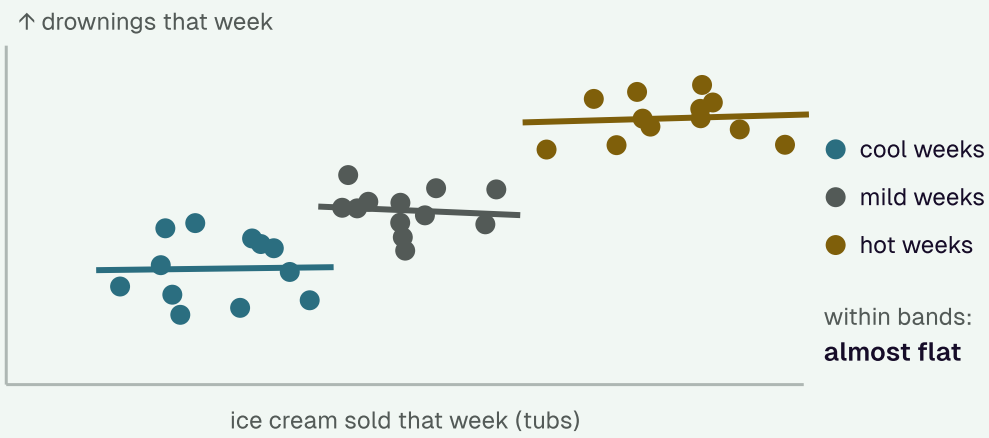
A seaside town tracks two numbers every month. Ice cream sales and drownings climb together into summer and fall together into winter.



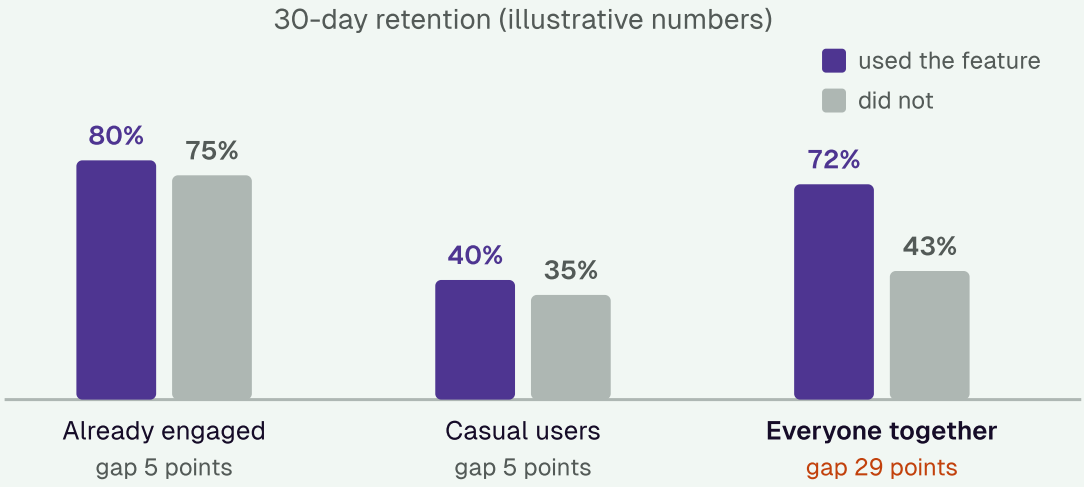
Plot one against the other, week by week, and the link is strong: a correlation of r = 0.89. Taken at face value, banning ice cream would save lives.



A third thing drives both. Hot weather sells ice cream and sends people into the sea. Temperature is a **confounder**: a common cause of the two things you measured.



Compare only weeks with similar weather and the link almost vanishes (within-band correlations 0.02, -0.10, 0.09). Comparing like with like on the confounder is what "controlling for" it means.



The same shape in a product review: feature users retain 29 points better. But people who were already engaged try more features. Within each engagement group the feature is worth 5 points. The 29 was mostly the confounder.

What a correlation is

Plain words

Two things that tend to move together

Two measurements are **correlated** when knowing one helps you guess the other. Taller parents tend to have taller children. Bigger houses tend to cost more. Days with more ice cream sold tend to have

more drownings.

The usual summary is a single number, **r**, between –1 and +1. Near +1: when one is high, the other is almost always high. Near –1: when one is high, the other is almost always low. Near 0: no straight-line pattern. It says nothing about *why* the two move together. It is a description of the data, like an average.

For leaders

Good for predicting, silent about intervening

A correlation answers "if I **see** X high, should I expect Y high?" A decision almost always asks a different question: "if I **make** X high, will Y go up?" The first is prediction and correlations are fine for it: a feature that correlates with churn is a useful churn predictor whatever the reason. The second is intervention, and a correlation alone cannot answer it.

- **Use correlations to find leads.** They are cheap and they tell you where to look.
- **Do not fund an intervention on a correlation alone.** Ask what else could produce it, then ask for the cheapest experiment that would tell the stories apart.
- **Strength is not evidence of causation.** The ice cream correlation above is 0.9. Many real causal effects show up as weak correlations because other causes add noise.

Under the hood

Pearson's r and its blind spots

Pearson's correlation is the covariance rescaled by both standard deviations (see the distributions chapter for SD and covariance):

$$r = \frac{\mathrm{Cov}(X,Y)}{s_X\, s_Y} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2}\,\sqrt{\sum_i (y_i - \bar{y})^2}}, \qquad \hat{\beta}_{Y\sim X} = r\,\frac{s_Y}{s_X}$$

r is unitless and lies in [–1, 1]. The least-squares slope of Y on X is r times the ratio of standard deviations, so slope and correlation always share a sign. r^2 is the share of Y's variance a straight line in X accounts for.

- **Linear only.** A perfect U-shape can have r = 0. Zero correlation does not mean independence.
- **Fragile to outliers and shapes.** Anscombe's quartet (Anscombe 1973) is four datasets with the same means, variances, regression line and r ≈ 0.82, one a curve, and two dominated by a single outlying point. Plot before you summarise.

- **Restricted range shrinks r.** Correlate SAT scores with grades among admitted students only and r falls, because admission removed the low scorers. This is a selection effect, covered below.
- **r has sampling error.** With n pairs, Fisher's transform $z = \tanh^{-1} r$ has standard error about $1/\sqrt{n-3}$. On 30 points, r = 0.40 has a 95% interval of roughly 0.05 to 0.66 (intervals).

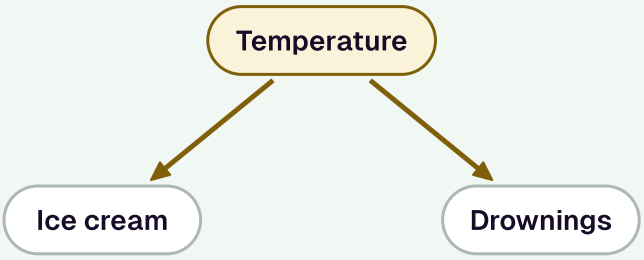
Four ways to get a correlation without causation

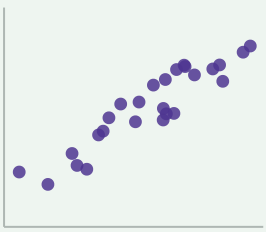
Diagram

Same pattern, five different worlds

Each picture is a causal diagram: boxes are things you could measure, arrows point from cause to effect.

1. Confounding: a hidden common cause



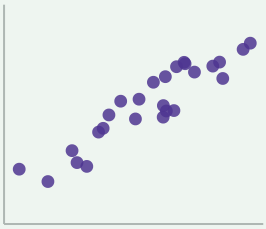


what the data shows

A third thing drives both. Neither causes the other.

Confounding. Temperature causes both ice cream sales and drownings. The two are correlated, and neither causes the other.

2. Reverse causation: the arrow points back

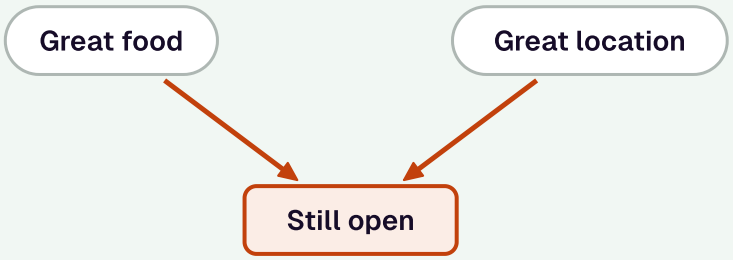


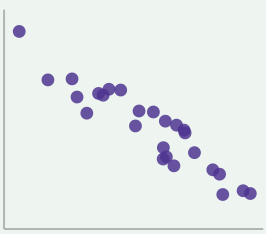
what the data shows

Wind turns windmills. Read the data the wrong way round and windmills seem to make wind.

Reverse causation. The arrow runs the other way from the one you assumed. Wind turns windmills; fast windmills do not make wind. In products: unhappy users file more tickets, so "tickets cause churn" reads the arrow backwards.

3. Selection: you only see the survivors



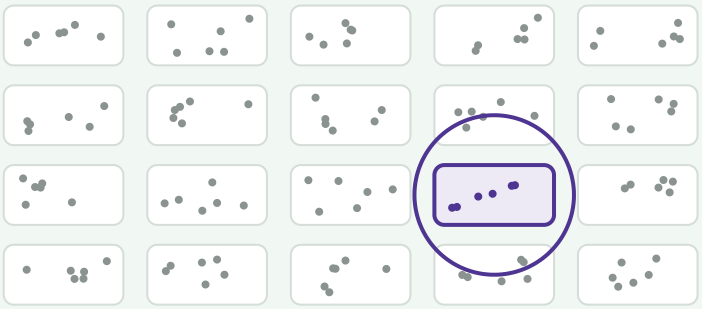


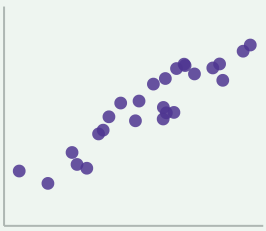
what the data shows

Only restaurants with good food or a good spot survive. Among survivors, the two look opposed.

Selection. You only look at cases that passed a filter. Restaurants survive if the food or the location is great, so among the survivors the two look like a trade-off even though they are unrelated across all restaurants that ever opened.

4. Chance: look at enough slices and one will light up



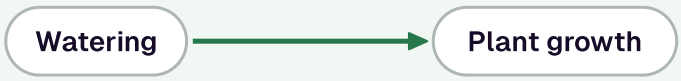


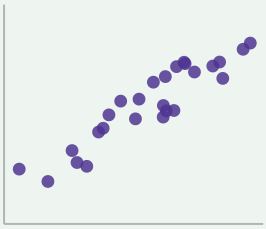
slice 14 only

20 slices tested at $p < 0.05$: expect about one false alarm.

Chance. Slice the data 20 ways and test each at the 5% level, and one slice will usually show a "significant" pattern by luck alone: the chance of at least one false alarm is $1 - 0.95^{20} \approx 64\%$.

5. Causation: change X and Y changes





what the data shows

All five worlds can draw the same scatter.
Only an experiment, where you set X yourself, tells them apart.

Causation. Watering makes plants grow. The scatter can look identical in all five worlds. Only intervening, setting X yourself and watching Y, tells this world apart from the other four.

Plain words

Four suspects to rule out first

Before you believe "X causes Y", line up four suspects and rule each one out.

1. **A hidden common cause.** Something else pushes both up. Hot weather, wealthy neighbourhoods, users who were already keen.

2. **The arrow is backwards.** Y causes X. People who feel ill see the doctor more; the doctor is not making them ill.

3. **You only saw some of the cases.** A filter, often invisible, decided which cases reached your spreadsheet.

4. **Luck.** You looked at enough comparisons that one was bound to look striking.

If all four are ruled out, causation is the explanation left standing. The cleanest way to rule out all four at once is an experiment where a coin flip decides who gets X.

For leaders

The four suspects in AI product and eval work

Suspect	What it looks like in AI work	Question that exposes it
Confounding	Prompts where the model calls a tool pass less often. Hard prompts both trigger tools and fail more.	"Within the same difficulty tier, does the gap survive?"
Reverse causation	Users who give more thumbs-down churn more. Users who were already failing to get value both complain and leave.	"Which came first? Show me the timeline per user."
Selection	Among conversations escalated to human review, wrong answers seem to calm users down. Escalation required an error or an angry user.	"Which rows were filtered out before this analysis, and by what rule?"
Chance	One of 40 eval slices shows the new checkpoint regressing on Portuguese legal questions.	"How many slices did you look at? Does it replicate on a fresh sample?" (many comparisons)

Ranking them by how often they bite: confounding and selection cause most expensive mistakes, because they survive large samples. Chance shrinks with more data and replication; confounding and selection do not shrink at all.

Under the hood

How big the spurious part is

Take a linear world with a confounder Z (variance 1), a treatment $X = aZ + \varepsilon_X$ with $\mathrm{Var}\,\varepsilon_X = 1$, and an outcome $Y = bX + cZ + \varepsilon_Y$. The true causal effect of X on Y is b. The naive regression slope is

$$\hat{\beta}_{\text{naive}} \to \frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(X)} = b + c\,\frac{\mathrm{Cov}(X,Z)}{\mathrm{Var}(X)} = b + \underbrace{\frac{c\,a}{a^2+1}}_{\text{confounding bias}}$$

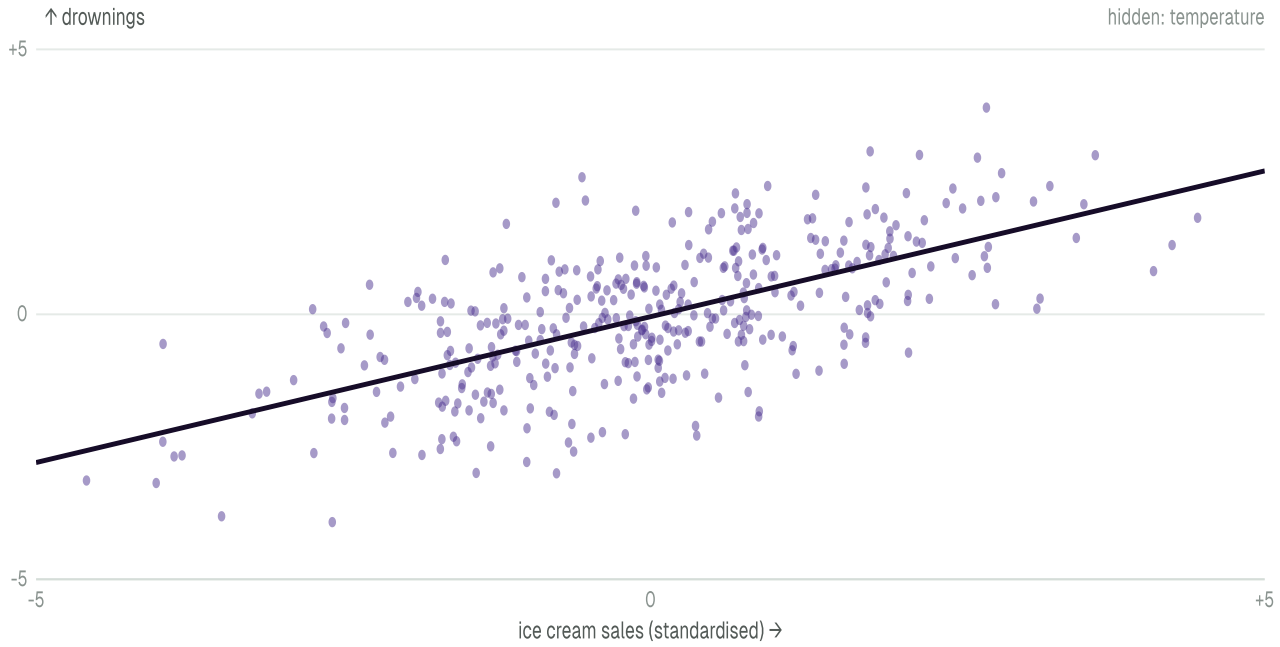
a is how strongly Z drives X, c how strongly Z drives Y. The bias has the sign of $c \cdot a$: same-direction pushes inflate the estimate, opposite-direction pushes deflate it and can flip its sign. It does not depend on sample size.

The other suspects have their own signatures. Reverse causation gives a nonzero slope with b = 0 because X is a function of Y. Selection on a variable S that depends on both X and Y induces correlation inside the selected set even when X and Y are independent (Berkson 1946, below). Chance gives a nonzero estimate whose distribution is centred on zero; its rate is controlled by the significance threshold and the number of tests, $1 - (1-\alpha)^m$ for m independent tests.

Interactive

Confounder explorer (shown at its default setting)

A hidden variable drives both X and Y. Raise its strength and watch the naive slope drift away from the true effect, then stratify to see the slope inside each band of the confounder. Try the thinking-tokens scenario with the true effect above zero.



Naive slope	Within 5 strata	Adjusted for Z	True effect
+0.55	**+0.13**	**+0.00**	**+0.00**

Correlation r = **0.64**. The naive slope says one unit more ice cream sales goes with **0.55** units of drownings; the true causal effect is **0.00**, so the confounder adds **+0.55**. Press *Stratify* to colour each point by the hidden temperature and fit a line inside each band.

Confounding, up close

Plain words

Compare like with like

A confounder is a hidden third thing that pushes on both of the numbers you are comparing. The fix is to compare like with like: hot days with hot days, keen users with keen users, hard questions with hard questions. If the link survives inside each group, it was not just the confounder.

An eval example. Across a maths benchmark, problems where the model "thinks" longer have *lower* pass rates. It looks as if thinking hurts. But hard problems make the model think longer, and hard problems fail more. Group the problems by difficulty and, inside each group, more thinking goes with *more* correct answers. In the widget above, pick the thinking-tokens scenario and press stratify to see the sign flip.

For leaders

When someone says "we controlled for it"

Adjusting for measured confounders is often the best you can do with logs, and it is far better than a raw comparison. Three limits decide how much to trust it:

- **Only measured confounders are removed.** "Prior engagement" measured as last week's sessions is a proxy. Whatever it misses (motivation, team size, use case) stays in the estimate.
- **Coarse buckets leave confounding behind.** Splitting difficulty into "easy" and "hard" removes some of it; the widget's five strata still leave a little. Finer strata or a model of the confounder do better, if there is enough data in each cell.
- **Adjusting for the wrong variable makes things worse.** See the section on mediators and colliders below. More controls is not automatically safer.

When the decision is large and the intervention is under your control, skip the argument and randomize (randomization, A/B tests).

Under the hood

Stratify, then re-weight

With a discrete confounder Z, estimate the effect inside each stratum, then average the stratum effects with the population's stratum weights. This is **standardization**, the simplest form of the back-door adjustment formula (Pearl 2009):

$$\mathbb{E}[Y \mid do(X{=}x)] = \sum_z \mathbb{E}[Y \mid X{=}x, Z{=}z]\; P(Z{=}z), \qquad \mathrm{ATE} = \sum_z \left(\bar{y}_{1z} - \bar{y}_{0z}\right) P(z)$$

$do(X{=}x)$ means "set X to x by intervention", as opposed to observing it. $\bar{y}_{1z}$ and $\bar{y}_{0z}$ are the outcome means for treated and untreated units in stratum z. Valid when Z blocks every back-door path and every stratum contains both treated and untreated units (positivity).

With continuous confounders, regression does the same job: include Z as a covariate and read off the coefficient on X. The Frisch–Waugh–Lovell theorem says that coefficient equals the slope of Y-

residualised-on-Z against X-residualised-on-Z, which is exactly what the widget's "Adjusted for Z" card computes.

python

```python
# Retention by engagement stratum and feature use. Illustrative counts.
# key: (stratum, used_feature) -> (users, retained)
TABLE = {
    ("engaged", True): (800, 640), ("engaged", False): (200, 150),
    ("casual", True):  (200, 80),  ("casual", False):  (800, 280),
}
STRATA = ["engaged", "casual"]

def rate(cells):
    users = sum(u for u, _ in cells)
    kept = sum(k for _, k in cells)
    return kept / users

def naive_lift(t):
    # pools the strata, so the stratum mix leaks into the comparison
    used = rate([t[(s, True)] for s in STRATA])
    unused = rate([t[(s, False)] for s in STRATA])
    return used - unused

def adjusted_lift(t):
    # compare within each stratum, then weight by the stratum's share of all users
    total = sum(u for u, _ in t.values())
    lift = 0.0
    for s in STRATA:
        n_s = t[(s, True)][0] + t[(s, False)][0]
        diff = rate([t[(s, True)]]) - rate([t[(s, False)]])
        lift += diff * n_s / total
    return lift

print(f"{naive_lift(TABLE):.3f}")     # 0.290  what the dashboard shows
print(f"{adjusted_lift(TABLE):.3f}")  # 0.050  feature's effect within like users
```

rust

```rust
// Retention by engagement stratum and feature use. Illustrative counts.
// (stratum, used_feature, users, retained)
const TABLE: [(&str, bool, f64, f64); 4] = [
    ("engaged", true, 800.0, 640.0), ("engaged", false, 200.0, 150.0),
    ("casual", true, 200.0, 80.0),   ("casual", false, 800.0, 280.0),
];
const STRATA: [&str; 2] = ["engaged", "casual"];

fn cell(s: &str, used: bool) -> (f64, f64) {
    let c = TABLE.iter().find(|c| c.0 == s && c.1 == used).unwrap();
    (c.2, c.3)
}

fn rate(cells: &[(f64, f64)]) -> f64 {
    let users: f64 = cells.iter().map(|c| c.0).sum();
    let kept: f64 = cells.iter().map(|c| c.1).sum();
    kept / users
}

fn naive_lift() -> f64 {
```

```
    // pools the strata, so the stratum mix leaks into the comparison
    let used: Vec<_> = STRATA.iter().map(|s| cell(s, true)).collect();
    let unused: Vec<_> = STRATA.iter().map(|s| cell(s, false)).collect();
    rate(&used) - rate(&unused)
}

fn adjusted_lift() -> f64 {
    // compare within each stratum, then weight by the stratum's share of all users
    let total: f64 = TABLE.iter().map(|c| c.2).sum();
    let mut lift = 0.0;
    for s in STRATA {
        let n_s = cell(s, true).0 + cell(s, false).0;
        let diff = rate(&[cell(s, true)]) - rate(&[cell(s, false)]);
        lift += diff * n_s / total;
    }
    lift
}

fn main() {
    println!("{:.3}", naive_lift());    // 0.290  what the dashboard shows
    println!("{:.3}", adjusted_lift()); // 0.050  feature's effect within like users
}
```

Simpson's paradox: when the total disagrees with every part

Plain words

Two models, two task mixes

Two teams each report a pass rate for their model. Model A scores 54%. Model B scores 74%. B looks far better. Now split the tasks into easy and hard:

	Easy tasks	Hard tasks	All tasks
Model A	180 of 200 = **90%**	360 of 800 = **45%**	540 of 1,000 = 54%
Model B	672 of 800 = 84%	72 of 200 = 36%	744 of 1,000 = **74%**

A wins on easy tasks *and* on hard tasks, yet loses overall. Nothing is wrong with the arithmetic. A was tested mostly on hard tasks and B mostly on easy ones, so each total is mostly a measure of the task mix. This reversal is **Simpson's paradox** (named for Simpson 1951, described earlier by Yule and Pearson). Numbers are illustrative.

For leaders

The Berkeley case, told accurately

The famous real example is graduate admissions at the University of California, Berkeley, for fall 1973. Of 8,442 men who applied, 44% were admitted; of 4,321 women, 35%. Bickel, Hammel and O'Connell (1975, *Science*) looked department by department. Women had applied disproportionately to departments that admitted few applicants of either sex. Once that was accounted for, they found no overall bias against women, and a small but statistically significant bias *in favour* of women. They concluded that the aggregate gap was misleading and pointed to earlier screening in education, which steered women toward crowded fields, as the real issue.

Two leadership lessons follow:

- **A headline gap between groups with different mixes is not yet a finding.** Ask for the per-segment numbers and a comparison on a common mix before any conclusion.
- **Which view is "right" depends on the causal story, not on the arithmetic.** If the segment is a confounder (task difficulty, department), the per-segment comparison answers the causal question. If the segment is caused by the treatment (a mediator), the aggregate can be the right answer. The diagram section below makes this precise.

Under the hood

Weighted averages with different weights

An overall rate is a weighted average of segment rates, weighted by that model's own mix:

$$\bar{p}_A = \sum_g w_{A,g}\, p_{A,g}, \qquad \bar{p}_B = \sum_g w_{B,g}\, p_{B,g}, \qquad \bar{p}_M^{\text{std}} = \sum_g w^*_g\, p_{M,g}$$

$w_{M,g}$ is the share of model M's tasks in segment g and $p_{M,g}$ its pass rate there. A reversal needs the weights to differ between models and to be correlated with segment difficulty. Direct standardization replaces each model's own weights by one reference mix w^*, such as production traffic.

The six largest Berkeley departments, as reproduced in R's `UCBAdmissions` dataset from the 1975 study, show the pattern. In four of the six, women were admitted at a higher rate. Aggregated, men lead by 14 points:

Dept	Men applied	Men admitted	Women applied	Women admitted
A	825	62%	108	82%
B	560	63%	25	68%
C	325	37%	593	34%
D	417	33%	375	35%
E	191	28%	393	24%
F	373	6%	341	7%
All six	2,691	**45%**	1,835	**30%**

Half the women applied to C and E, which admitted about a quarter to a third of applicants; over half the men applied to A and B, which admitted about two thirds. Standardizing both sexes to the combined applicant mix removes the gap, and in fact puts women about 4 points ahead:

python

```python
# (applied, admitted) per segment for each group
MODELS = {  # illustrative eval counts from the table above
    "A": {"easy": (200, 180), "hard": (800, 360)},
    "B": {"easy": (800, 672), "hard": (200, 72)},
}
BERKELEY = {  # six largest departments, fall 1973 (R's UCBAdmissions)
    "men":   {"A": (825, 512), "B": (560, 353), "C": (325, 120),
              "D": (417, 138), "E": (191, 53),  "F": (373, 22)},
    "women": {"A": (108, 89),  "B": (25, 17),   "C": (593, 202),
              "D": (375, 131), "E": (393, 94),  "F": (341, 24)},
}

def overall(cells):
    # each group's own mix sets the weights
    return sum(k for _, k in cells.values()) / sum(n for n, _ in cells.values())

def standardized(cells, mix):
    # one shared mix for everyone: compares rates, not mixes
    return sum(mix[g] * k / n for g, (n, k) in cells.items())

def pooled_mix(groups):
    total = {}
    for cells in groups.values():
        for g, (n, _) in cells.items():
            total[g] = total.get(g, 0) + n
    s = sum(total.values())
    return {g: v / s for g, v in total.items()}

for name, data, mix in [("eval", MODELS, {"easy": 0.5, "hard": 0.5}),
```

```
                       ("berkeley", BERKELEY, pooled_mix(BERKELEY))]:
    for group, cells in data.items():
        print(name, group, f"{overall(cells):.3f}", f"{standardized(cells, mix):.3f}")
# eval A 0.540 0.675
# eval B 0.744 0.600
# berkeley men 0.445 0.387
# berkeley women 0.304 0.430
```

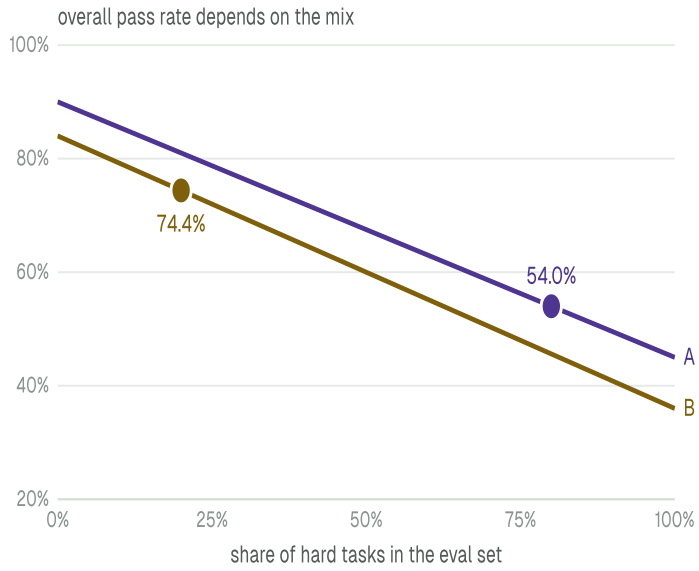
rust

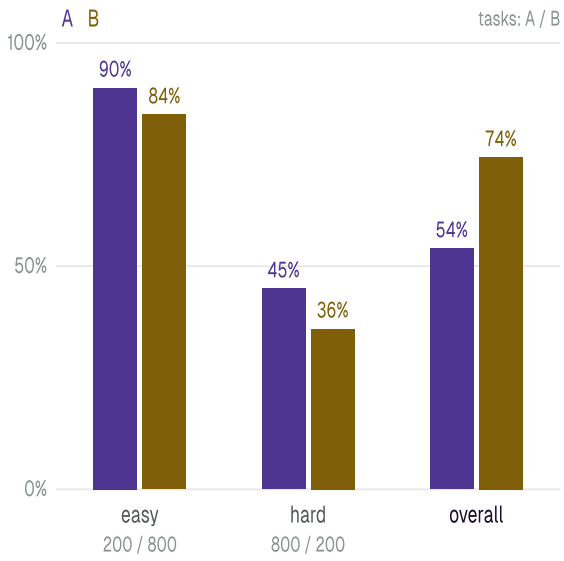
```
// (segment, applied, admitted) per segment for each group
type Cells = Vec<(&'static str, f64, f64)>;

fn overall(cells: &Cells) -> f64 {
    // each group's own mix sets the weights
    let k: f64 = cells.iter().map(|c| c.2).sum();
    let n: f64 = cells.iter().map(|c| c.1).sum();
    k / n
}

fn standardized(cells: &Cells, mix: &[(&str, f64)]) -> f64 {
    // one shared mix for everyone: compares rates, not mixes
    cells.iter().map(|(g, n, k)| mix.iter().find(|m| m.0 == *g).unwrap().1 * k / n).sum()
}

fn pooled_mix(groups: &[(&str, Cells)]) -> Vec<(&'static str, f64)> {
    let mut total: Vec<(&'static str, f64)> = Vec::new();
    for (_, cells) in groups {
        for (g, n, _) in cells {
            match total.iter_mut().find(|t| t.0 == *g) {
                Some(t) => t.1 += n,
                None => total.push((g, *n)),
            }
        }
    }
    let s: f64 = total.iter().map(|t| t.1).sum();
    total.into_iter().map(|(g, v)| (g, v / s)).collect()
}

fn main() {
    // illustrative eval counts from the table above
    let models = vec![
        ("A", vec![("easy", 200.0, 180.0), ("hard", 800.0, 360.0)]),
        ("B", vec![("easy", 800.0, 672.0), ("hard", 200.0, 72.0)]),
    ];
    // six largest departments, fall 1973 (R's UCBAdmissions)
    let berkeley = vec![
        ("men", vec![("A", 825.0, 512.0), ("B", 560.0, 353.0), ("C", 325.0, 120.0),
                     ("D", 417.0, 138.0), ("E", 191.0, 53.0), ("F", 373.0, 22.0)]),
        ("women", vec![("A", 108.0, 89.0), ("B", 25.0, 17.0), ("C", 593.0, 202.0),
                       ("D", 375.0, 131.0), ("E", 393.0, 94.0), ("F", 341.0, 24.0)]),
    ];
    let even = vec![("easy", 0.5), ("hard", 0.5)];
    let bmix = pooled_mix(&berkeley);
    for (name, data, mix) in [("eval", &models, &even), ("berkeley", &berkeley, &bmix)] {
        for (group, cells) in data.iter() {
            println!("{} {} {:.3} {:.3}", name, group, overall(cells), standardized(cells, mix));
        }
    }
    // eval A 0.540 0.675
```

```
    // eval B 0.744 0.600
    // berkeley men 0.445 0.387
    // berkeley women 0.304 0.430
}
```

Interactive **Simpson's paradox: the mix decides the headline** (shown at its default setting)

Model A beats model B on easy tasks and on hard tasks. Each model's overall score is a point on its own line, placed by how many hard tasks its eval set contains. Move the mixes and watch the headline flip while the per-group bars never move.





A overall	B overall	Headline gap A – B	Gap on a 50/50 mix
54.0%	**74.4%**	**-20.4 pts**	**+7.5 pts**

Within each group A is ahead: **+6** points on easy tasks and **+9** on hard ones. Yet B wins the headline by **20.4 points**, because 80% of A's tasks are hard against 20% of B's. That is Simpson's paradox: the overall number mostly measures the mix. Scored on the same 50/50 mix, A leads by 7.5 points.

Selection and colliders: Berkson's paradox

Diagram

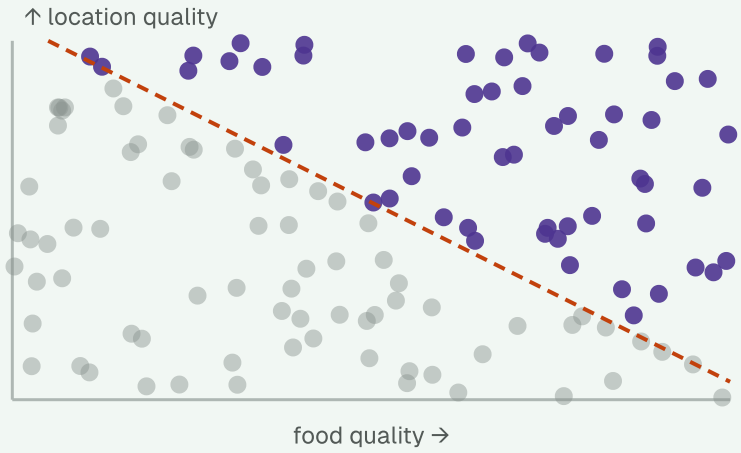
Why good restaurants seem to have bad locations

A filter you never see can manufacture a trade-off. Simulated restaurants, illustrative.

r = -0.02

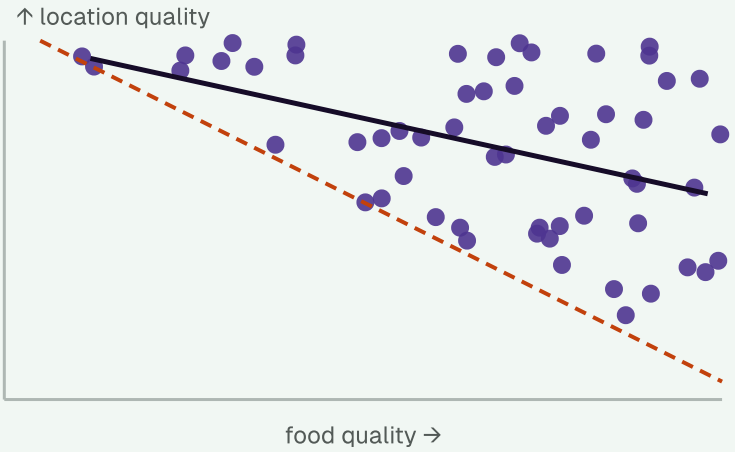
no link

Every restaurant that opened in a city over ten years. Food quality and location quality are unrelated: r = -0.02.



below the line: closed within two years

Restaurants survive if the food is great, or the location is great, or both are decent. Everything below the line closed within two years, so it never appears in a guide or on a map.



r = -0.45
among survivors

Among the survivors, food and location now look opposed: r = -0.45. A great restaurant in a great spot is rare not because the two clash but because the bad-food, bad-spot places are missing.

The same trap in AI work

Only shipped features	Only escalated chats	Only shortlisted models
shipped if high impact or cheap to build	escalated if the model erred or the user was angry	kept if strong or cheap to serve
so among them: **impact and cost look linked**	so among them: **errors seem to calm users**	so among them: **strong models look pricey**

Whenever the rows were picked by an OR of two things, expect a fake trade-off.

The same pattern appears whenever rows were kept because of one thing *or* another. The fix is to analyse the full pool before the filter, or to model the filter explicitly.

Plain words

The filter creates the pattern

Joseph Berkson noticed in 1946 that two diseases could look linked among hospital patients even when they were unrelated in the population, because having either one raised your chance of being in hospital. Looking only at hospital patients, you are looking at people who passed a filter.

Any filter that lets a case through for having "this or that" will make "this" and "that" look like opposites among the cases that got through. The people you dated were either kind or attractive, so

among them the attractive ones seem less kind. The papers that got into a conference were either novel or rigorous, so among accepted papers novelty and rigour seem to trade off.

For leaders

Ask what decided which rows exist

Selection bias is hard to see because the missing rows leave no trace in the table. Every analysis built on a filtered population deserves the question "what decided which rows are in here?"

- **Conversations flagged for review** were flagged by a classifier or by angry users. Patterns inside the flagged set describe the flagger as much as the model.
- **Benchmarks that made it into the eval suite** were kept because they separated models or because they were cheap. Correlations between benchmark properties inside the suite can be manufactured by that choice.
- **Models on a shortlist** were kept for being strong or cheap. "Stronger models cost more" among the shortlist does not tell you about the market.

The companion problem, measuring only the units that survived, gets its own chapter: who is in your sample.

Under the hood

Conditioning on a common effect

A **collider** is a variable with two arrows pointing into it, $X \to S \leftarrow Y$. With X and Y independent, conditioning on S (by filtering, stratifying or adding it to a regression) makes them dependent. For the restaurant rule $S = \mathbf{1}[X + Y > t]$ with X and Y uniform on [0, 1] and t = 1.05, the simulated sample above gives r = -0.45 among survivors (the population value under this rule is exactly -0.5) from r ≈ 0 overall.

$$X \perp\!\!\!\perp Y \quad \not\Rightarrow \quad X \perp\!\!\!\perp Y \mid S \qquad \text{when } X \to S \leftarrow Y$$

Independence in the full population does not survive conditioning on a common effect. Explaining away: once you know a restaurant survived, learning its food is bad tells you its location must be good.

Selection does not need a sharp threshold. Any selection probability that depends on both X and Y, such as a response rate that depends on satisfaction and on usage, induces some dependence. The remedies are to observe the unselected units, to model the selection probability and reweight

(inverse probability weighting, if the probability can be estimated), or to bound the effect with sensitivity analysis.

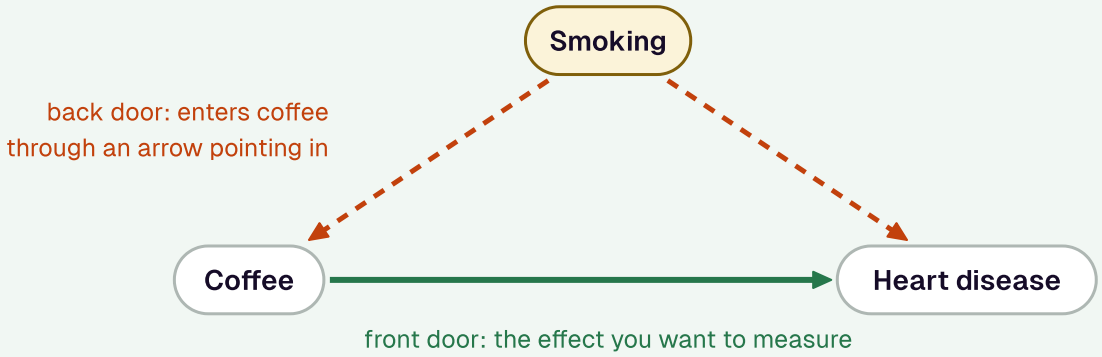
Causal diagrams: deciding what to control for

Diagram

Back doors, mediators and colliders

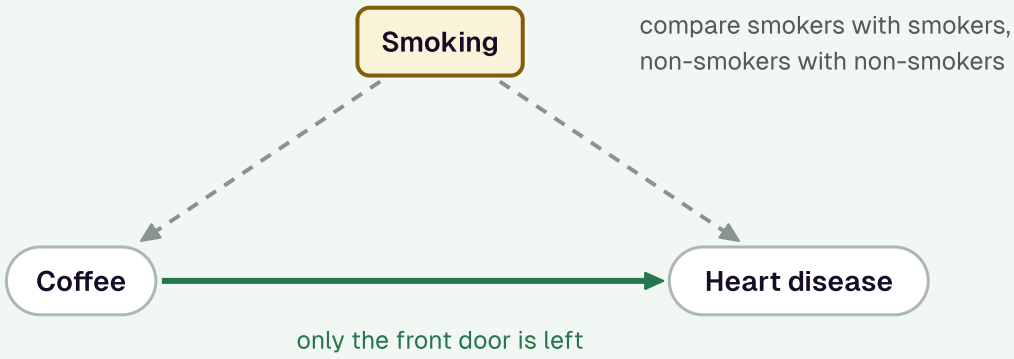
A causal diagram (a DAG) tells you which variables to hold fixed and which to leave alone.

Two roads from coffee to heart disease



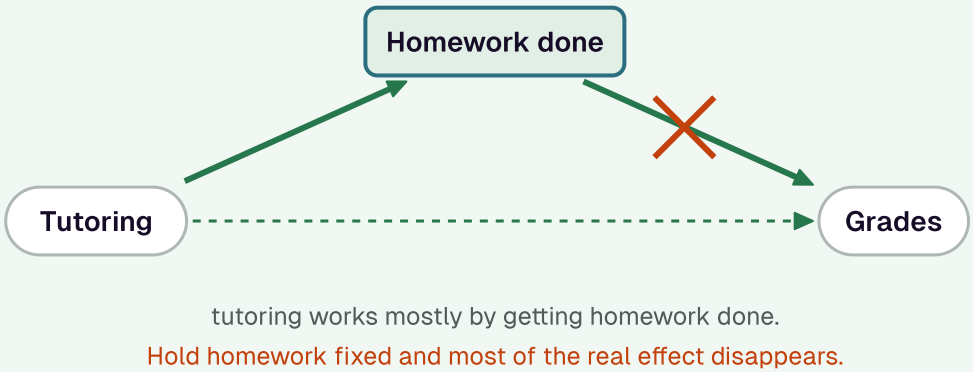
Does coffee cause heart disease? In older data, coffee drinkers also smoked more. There are two roads from coffee to heart disease: the direct one you care about, and a **back door** through smoking that starts with an arrow pointing *into* coffee.

Close the back door: hold smoking fixed



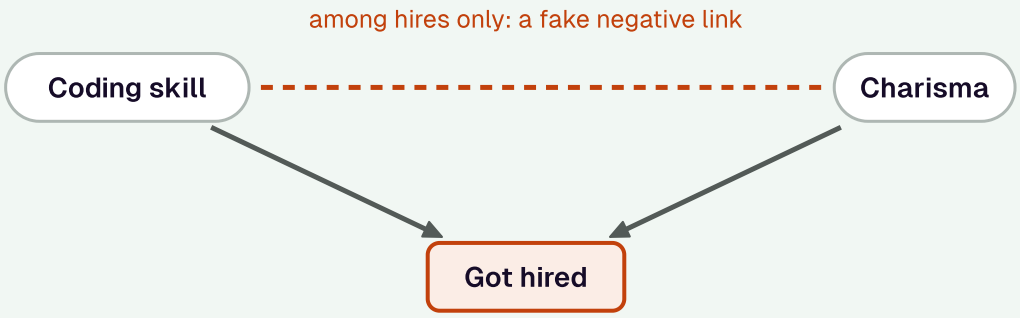
Hold smoking fixed by comparing smokers with smokers and non-smokers with non-smokers. The back door closes, and whatever association remains flows through the front door.

A mediator: do not control for the road itself



A **mediator** sits on the causal road. Tutoring raises grades mostly by getting homework done. "Controlling for" homework compares tutored and untutored students who did the same homework, which hides most of what tutoring does.

A collider: controlling for it opens a fake path



Skill and charisma are unrelated in the applicant pool. Hires needed one or the other, so among hires the less charming are better coders.

A **collider** is caused by both. Skill and charisma both help you get hired. Analyse only hires and a fake trade-off appears, exactly like the restaurants. Controlling for a collider opens a path that was closed.

What to do with a third variable

Confounder	Mediator	Collider
causes X and Y	X causes it, it causes Y	X and Y both cause it
control for it	**leave it alone***	**never control for it**

* unless you want only the direct effect, and know why.

The rule of thumb: close back doors by controlling for confounders, leave mediators alone unless you want only the direct effect, and never control for colliders. Drawing the diagram first is how you tell which is which.

Plain words

Draw the arrows before you do the maths

A causal diagram is a sketch with a box for each thing and an arrow from each cause to its effect. It has no loops: nothing causes itself, which is why it is called a **directed acyclic graph**, or DAG.

Any route between X and Y along the arrows (in either direction) can carry association. The route you want is the one that leaves X along its own arrows and arrives at Y. A **back-door path** is any route

that starts with an arrow coming *into* X: it carries association that has nothing to do with X's effect. Your job is to block every back door without blocking the front door, and without opening new doors by holding a collider fixed.

For leaders

Make the diagram part of the review

Most "we controlled for everything" analyses fail because nobody wrote down what causes what. A one-page diagram is cheap, and it turns a regression into a claim people can argue with.

- **Ask for the diagram before the coefficients.** Every arrow is an assumption. Disagreements about arrows are the real disagreements, and they are domain questions, not statistics questions.
- **Ask for each control's role.** Confounder, mediator, collider, or just a predictor of Y? Controls added "because they were available" are where bias creeps in.
- **Ask what is unmeasured.** If a plausible confounder is missing from the data, no adjustment fixes it. That is the moment to fund an experiment instead.

Worked problem

A team claims a new system prompt raised user satisfaction "even after controlling for response length and refusal rate." The prompt changes length and refusals, and those change satisfaction: both are mediators. Controlling for them estimates only the part of the effect that does *not* go through length or refusals. If the question is "should we ship the prompt?", the total effect is the one that matters, so drop those controls or, better, run the A/B test.

Under the hood

d-separation and the back-door criterion

Three building blocks decide whether a path carries association (Pearl 2009):

Shape	Name	Path open when	Conditioning on the middle
$A \to M \to B$	Chain (mediator)	M not conditioned on	blocks it
$A \leftarrow Z \to B$	Fork (confounder)	Z not conditioned on	blocks it
$A \to C \leftarrow B$	Collider	C or a descendant of C **is** conditioned on	opens it

A path is blocked if any node on it blocks it. X and Y are **d-separated** given a set Z when every path between them is blocked; in a correctly drawn DAG this implies conditional independence.

Back-door criterion. A set Z is a valid adjustment set for the effect of X on Y if (1) no member of Z is a descendant of X, and (2) Z blocks every path between X and Y that begins with an arrow into X. Then the adjustment formula $P(y \mid do(x)) = \sum_z P(y \mid x, z)\, P(z)$ identifies the causal effect from observational data. Condition (1) is what rules out mediators and colliders that sit downstream of X.

Known "bad controls" (Cinelli, Forney and Pearl 2022, "A crash course in good and bad controls"): mediators, which remove part of the effect; colliders and their descendants, which open spurious paths; and pre-treatment variables that only cause X (instruments), which add variance and can amplify bias from unmeasured confounders. The M-bias case shows that even a pre-treatment variable can be a collider: $X \leftarrow U_1 \to M \leftarrow U_2 \to Y$.

The ladder of evidence

Plain words

From "someone said" to "we flipped a coin"

Evidence for "X causes Y" comes in grades. A story from one user is the bottom rung. A chart of two numbers moving together is one rung up. Comparing like with like is higher. The top is an experiment where a coin flip decides who gets X, because the coin cannot be confounded, cannot be caused by Y, and cannot select who shows up.

For leaders

Match the rung to the stakes

Rung	AI example	Rules out	Leaves open
1. Anecdote	"Three customers said the new model is smarter."	Nothing	Everything
2. Raw correlation	Feature users retain better on the dashboard.	"There is no association here"	Confounding, reverse causation, selection, chance
3. Adjusted comparison	Same comparison within engagement and tenure bands.	Measured confounders	Unmeasured confounders, bad controls
4. Natural experiment	A staggered regional rollout analysed with difference-in-differences.	Many unmeasured confounders, under stated assumptions	Violated assumptions (other changes at the same time)
5. Randomized experiment	A/B test, or randomly assigning prompts to two checkpoints.	Confounding, reverse causation, selection into treatment	Chance (quantified), whether results transfer
6. Replicated, pre-registered	The same result in a second experiment with the metric fixed in advance.	One-off flukes, forking paths	Transfer to new contexts

Rung 2 is fine for picking what to investigate. Shipping decisions that are expensive to reverse deserve rung 5. Offline evals are unusual because you control the inputs: you can almost always run the experiment instead of arguing about observational data (randomization, A/B tests).

Under the hood

Quasi-experiments in one line each

- **Difference-in-differences.** Effect $= (\bar{y}^{T}_{\text{after}} - \bar{y}^{T}_{\text{before}}) - (\bar{y}^{C}_{\text{after}} - \bar{y}^{C}_{\text{before}})$. Assumes the treated and control groups would have moved in parallel without treatment. Check pre-period trends.
- **Regression discontinuity.** When treatment switches on at a threshold (a safety classifier score of 0.8, a usage tier), compare units just above and just below. Assumes nothing else jumps at the threshold and units cannot precisely sort themselves across it.

- **Instrumental variables.** Find something that shifts X but affects Y only through X, such as a randomized nudge to try a feature (an encouragement design). Effect $= \mathrm{Cov}(Z, Y) / \mathrm{Cov}(Z, X)$, the Wald estimator. Weak instruments give wild estimates.
- **Matching and propensity scores.** Pair treated and untreated units with similar covariates, or weight by the inverse probability of treatment. Same assumption as regression adjustment: no unmeasured confounders.

Where this bites in AI work

Feature usage and retention

Users of a new tool retain better. Prior engagement confounds it, retained users had more time to find the tool (reverse direction), and the dashboard only counts users still active. A randomized holdout answers it in weeks.

Longer answers, higher judge scores

A judge model may reward length itself, or harder questions may draw longer, better answers. Length-controlled AlpacaEval regresses out length and raised its Spearman correlation with Chatbot Arena from 0.94 to 0.98 (Dubois et al. 2024).

Benchmark gains and data overlap

Across models, the gap between GSM8K and the fresh GSM1k set rose with each model's probability of generating GSM8K examples (Spearman r^2 = 0.36; Zhang et al. 2024). That is a correlation across model families with different recipes.

Tool use and failure

Trajectories with more tool calls fail more often. Hard tasks cause both. Compare at fixed task, or randomize the tool budget per task.

Plain words

Same four suspects, new costumes

Each tile above is a correlation someone will one day present as a cause. For each, ask the four questions: is there a hidden common cause, could the arrow run backwards, were some cases

filtered out, and how many things did we look at? Then ask the best question of all: can we just try it, on purpose, and watch?

For leaders

The intervention that settles each case

- **Feature and retention:** a randomized holdout, or a randomized prompt to try the feature, then compare by assignment, not by usage. Usage is chosen by users; assignment is chosen by the coin.
- **Length and judge scores:** generate pairs that differ only in length (pad or trim the same content) and score both. That measures the judge's length effect directly. Use length control as a correction only after deciding whether some length is genuine quality: if thoroughness is part of what users want, length is partly a mediator and regressing it out removes real gains (judges).
- **Benchmark gains and overlap:** evaluate on a fresh, private set built to the same spec, and compare the gap per model. Better still, decontaminate and retrain at small scale, or plant canary items at controlled rates, to measure the effect of overlap directly (contamination).

Under the hood

Writing each one as a DAG

- **Retention:** $\mathbf{Engagement} \to \mathbf{Usage}$, $\mathbf{Engagement} \to \mathbf{Retention}$, $\mathbf{Usage} \to \mathbf{Retention}$, plus $\mathbf{Tenure} \to \mathbf{Usage}$ and selection on still-active users. The back door through engagement needs blocking; conditioning on "active this month" is conditioning on a collider of usage and retention.
- **Judge scores:** $\mathbf{Question} \to \mathbf{Length}$, $\mathbf{Question} \to \mathbf{Quality}$, $\mathbf{Quality} \to \mathbf{Length}$, $\mathbf{Length} \to \mathbf{Score}$, $\mathbf{Quality} \to \mathbf{Score}$. Regressing score on length alone mixes all four paths. The padded-pair experiment cuts every arrow into Length, which is what $do(\mathbf{Length})$ means.
- **Overlap:** $\mathbf{Recipe} \to \mathbf{Overlap}$, $\mathbf{Recipe} \to \mathbf{Capability}$, $\mathbf{Overlap} \to \mathbf{Score}$, $\mathbf{Capability} \to \mathbf{Score}$. A cross-model correlation cannot separate the two paths into Score; a fresh set removes the $\mathbf{Overlap} \to \mathbf{Score}$ path for every model at once.

Eval lens

Offline evaluation is the rare setting where you can intervene cheaply. Before mining logs for what correlates with a good score, ask whether you can set the variable directly: randomize the prompt template per item, pad or trim answers, swap tool budgets, assign items to checkpoints at random. A randomized comparison on 500 items usually beats a confounded analysis of 50,000 (eval statistics, measurement).

Principal asks

What would we expect to see if the cause ran the other way? What decided which rows are in this table? Which variables did you adjust for, and is any of them caused by the treatment? What plausible confounder is not in the data? How many comparisons were run before this one was chosen? What is the cheapest experiment that would settle it, and why have we not run it?

Common mistakes

- Treating a strong correlation as stronger evidence of causation than a weak one.
- Adding every available variable as a control, including mediators (which hide real effects) and colliders (which create fake ones).
- Reading the aggregate when segment mixes differ, or reading segments when the segment is caused by the treatment.
- Comparing users by what they chose to do (used the feature) instead of what they were assigned (saw the feature).
- Analysing a filtered population (flagged, escalated, retained, shortlisted) and generalising to everyone.
- Reporting the one striking slice out of dozens without saying how many were examined.

Part VI · Chapter 42

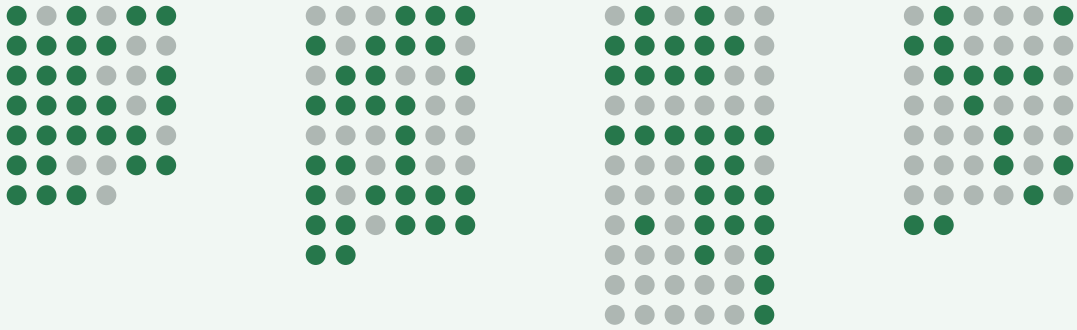
Who is in your sample? Coverage, self-selection and survivors

The sampling distribution and every error bar in this guide assume one thing: that the people, prompts or sessions you measured were drawn fairly from the ones you care about. When that fails, the error bar is still computed, still shrinks with more data, and is still wrong. This chapter is about the gap between who you meant to measure and who actually ended up in your data: how it opens, why a bigger sample does not close it, the probability designs that prevent it, and the weighting that partly repairs it.

Diagram **The email poll**

A town asks residents about a new transit plan by email. Each stage loses people, and not at random. Illustrative numbers.

A town of 10,000. Each dot is 50 people.



18–29
72% support

30–44
60% support

45–64
45% support

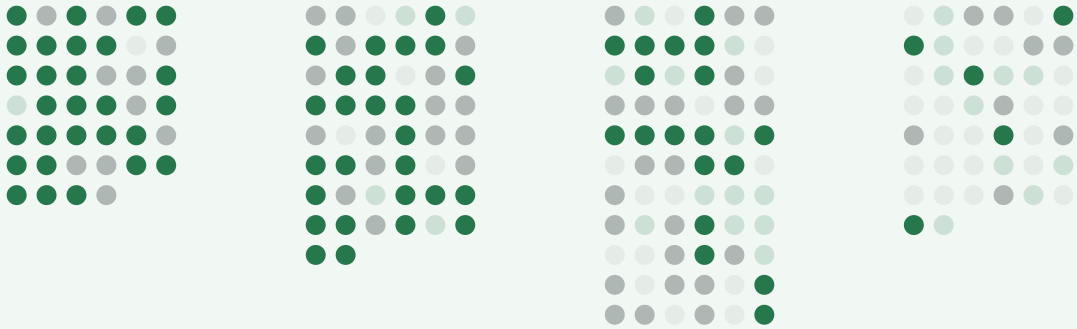
65+
34% support

Truth: 52% support

supports opposes

The **population**: everyone in town, in four age bands. Younger residents like the plan more. Overall, 52% support it. This is the number the poll is trying to find.

The sampling frame: who is on the email list at all



18–29
95% on the list

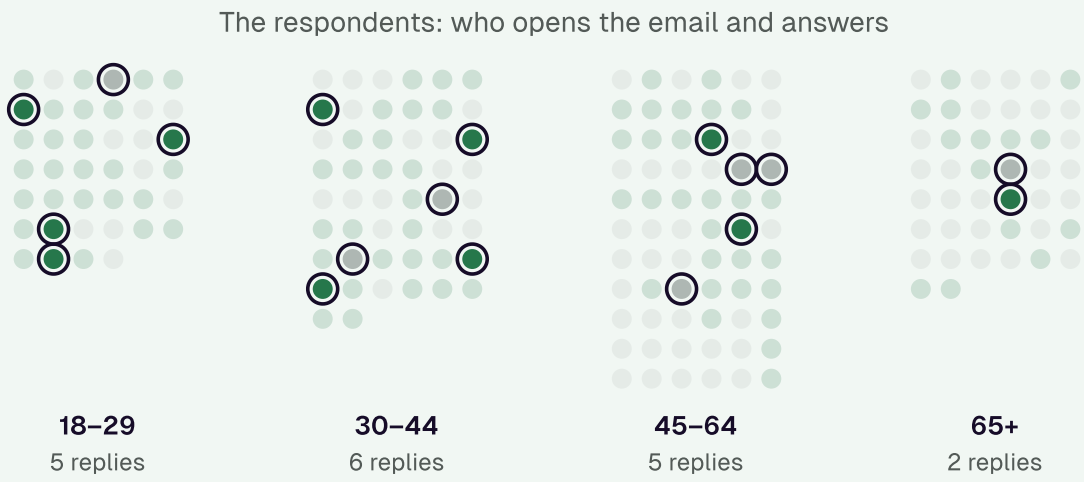
30–44
84% on the list

45–64
61% on the list

65+
30% on the list

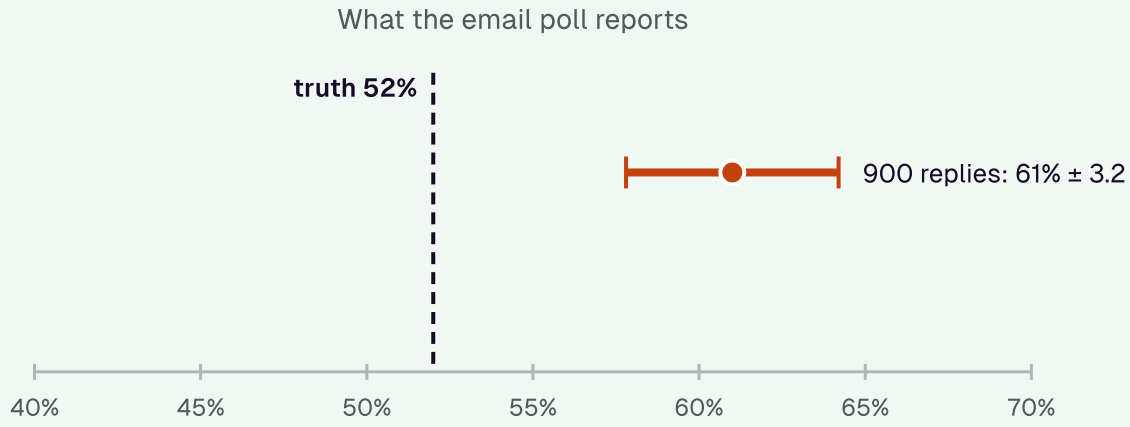
6,650 of 10,000 are on the list. Most over-65s are not.

The **sampling frame**: the list you can actually reach. Only people on the council's email list can be asked. Nearly all under-30s are on it; most over-65s are not. Anyone off the list has zero chance of being heard. That is **coverage bias**.

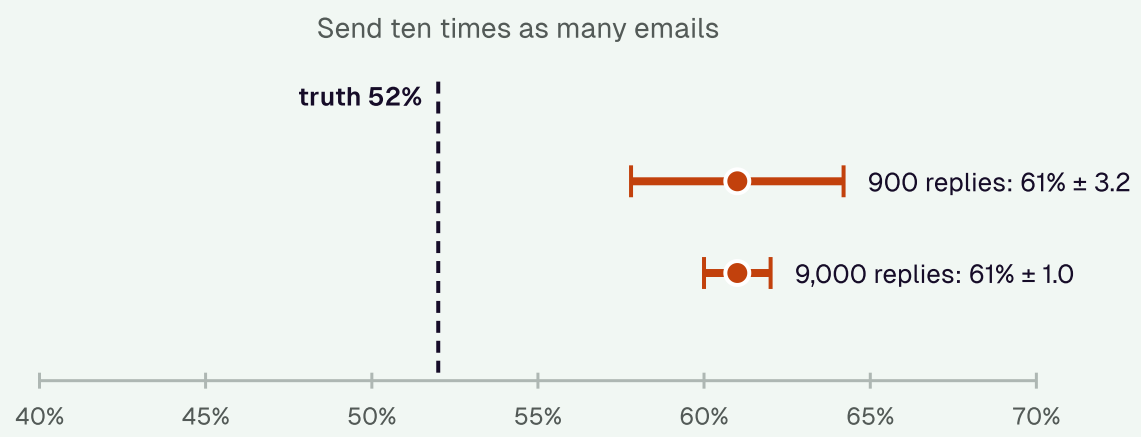


About 900 reply. Supporters were a bit keener to answer.

The **respondents**: of those emailed, about one in eight opens it and answers, and supporters are a little keener. People who never read the email are silent. That is **nonresponse bias**.



The poll reports 61% support, with a tidy-looking interval of ±3 points. The truth, 52%, is nowhere near it. The interval only describes the luck of the draw among people who answered.



Email ten times as many people and the interval shrinks to ±1 point, still centred on 61%. More of the same people makes the wrong answer look more certain. Fixing who gets asked is the only real fix.

Population, frame, sample, respondents

Plain words

Four groups that should be the same and rarely are

1 **Population** everyone you want to learn about	2 **Sampling frame** the list you can actually reach	3 **Sample** the ones you chose to ask	4 **Respondents** the ones who answered

Each arrow loses people. If the people lost at each step are a fair cross-section, you just have fewer answers and wider error bars. If the people lost are different from the people kept (older, busier, less happy, less online), your answer drifts away from the truth, and no amount of extra data fixes the drift.

For leaders

Name the four groups before reading the number

For any metric that claims to describe "users", "prompts" or "the model's behaviour", make the team name all four groups out loud. The answers are often uncomfortable:

	User satisfaction survey	Production eval set
Population	All active users this quarter	All prompts the product will see next quarter
Frame	Users who opted into email	Logged prompts from consenting, non-enterprise users last month
Sample	10,000 drawn from the opt-in list	2,000 drawn from those logs after deduplication
Respondents	840 who answered	1,450 that annotators could grade (the rest were "too ambiguous")

Every gap between rows is a place where bias enters. The last row of the eval column is a quiet example: dropping items annotators found ambiguous removes the hardest prompts.

Under the hood

Inclusion probabilities

Give every unit i in the population an **inclusion probability** π_i: the chance it ends up among the respondents, the product of being in the frame, being selected and responding. A design is **probability sampling** when every π_i is known and positive. Then the Horvitz–Thompson estimator is unbiased for the population total:

$$\hat{T}_{HT} = \sum_{i \in \text{sample}} \frac{y_i}{\pi_i}, \qquad \mathbb{E}\left[\hat{T}_{HT}\right] = \sum_{i=1}^{N} y_i$$

Each sampled unit stands in for $1/\pi_i$ population units. Coverage bias means some $\pi_i = 0$, and no weight can recover them. Nonresponse means the true π_i are unknown and must be modelled.

With equal probabilities this collapses to the plain sample mean. Everything in this chapter is a variation: either the π_i are known (probability designs), or they are unknown and correlated with the outcome (the biases), or they are estimated and inverted (weighting).

The ways a sample goes wrong

Plain words

Six everyday traps

Coverage bias

Some people cannot be reached at all. An email poll misses everyone without email. A phone survey of landlines misses people with only mobiles.

Nonresponse bias

People are reached but do not answer, and the ones who answer differ. Busy parents skip the survey; retirees fill it in.

Self-selection

People choose to take part. App-store reviews come mostly from the delighted and the furious; opt-in polls from people who care a lot.

Survivorship

You only see the ones that made it through. Advice from successful founders ignores the identical choices of founders who failed.

Convenience sample

You ask whoever is nearby. A team testing its own model writes the kinds of prompts it already knows work.

Time-window bias

You sample a period that is not typical. Logs from launch week, a holiday, or only office hours in one time zone.

For leaders

Where each trap lives in AI work

Trap	AI example	Question to ask
Coverage	Eval prompts collected from the web UI miss API customers, whose traffic is longer and more agentic.	"Which surfaces and customer types can never appear in this set?"
Nonresponse	Thumbs-up/down feedback: a small share of turns get rated, and raters are not typical turns.	"What fraction responded, and how do responders differ on things we can see?"
Self-selection	Public preference leaderboards rely on volunteers who choose their own prompts.	"Whose prompts are these, and do they look like our users'?"
Survivorship	Production logs only contain users who stayed; users who left after a bad first week are missing.	"Where are the people who churned in this data?"
Convenience	A launch decided on internal dogfooding and a few hundred team-written prompts.	"Who wrote these prompts, and what did they already know about the model?"
Time window	A one-week log sample during a viral moment, or sampled at a moment in time, which over-represents long sessions.	"Why this window, and would another window give the same mix?"

Under the hood

Bias is a covariance

Let R_i be the probability that unit i ends up in your data (a response propensity). The respondent mean is a propensity-weighted mean, and its bias has a compact form:

$$\mathbb{E}[\bar{y}_{\text{resp}}] - \bar{Y} \approx \frac{\text{Cov}(R, Y)}{\bar{R}} = \frac{\rho_{R,Y}\,\sigma_R\,\sigma_Y}{\bar{R}}$$

$\bar{R}$ is the mean propensity (the response rate). Bias needs propensity to be correlated with the outcome. A low response rate is not biased by itself, but it magnifies any correlation, because $\bar{R}$ sits in the denominator.

Two classic cases show both halves at work:

- **The 1936 Literary Digest poll** mailed about 10 million ballots, drawn largely from telephone directories and car registrations, and got about 2.4 million back. It predicted that Alf Landon would beat Franklin Roosevelt; Roosevelt won with about 61% of the popular vote. The frame skewed wealthier (coverage), and Landon supporters were more likely to mail ballots back (nonresponse). Squire (1988) concluded from a 1937 survey that nonresponse contributed heavily; Lusinchi (2012) reviews the debate over which mattered more.
- **Online reviews.** Hu, Pavlou and Zhang (2009) found Amazon ratings piled up at 5 stars with a bump at 1 star, a J shape. When they had people rate a randomly assigned CD, ratings were unimodal and the mean was significantly lower. They attribute the J to purchasing bias (only people who expect to like a product buy it) and under-reporting bias (people with extreme views are likelier to post).

Length-biased sampling is a time-window trap worth knowing by name. Sample "a random active conversation at noon" and a conversation's chance of being picked is proportional to its duration, so long sessions are over-represented. The same holds for "a random running job" or "a random open ticket". Sample by start event instead.

Survivorship bias: the bombers that did not come back

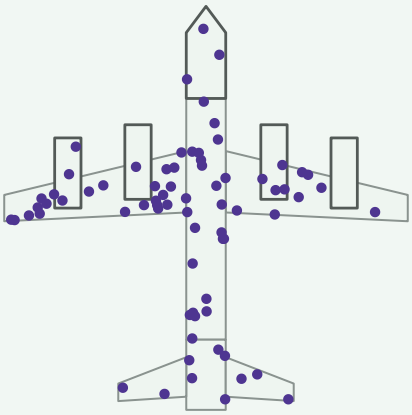
Diagram

Armour where the holes are not

The Second World War version, told carefully, then the version in your logs. Simulated planes, illustrative counts.

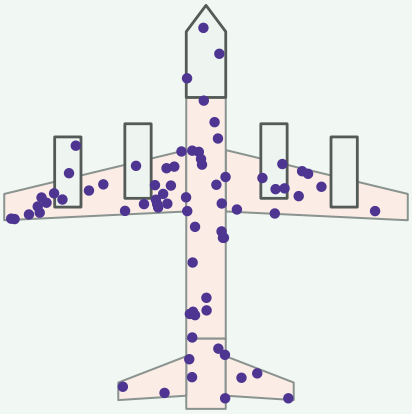
Bombers back from missions

34 planes came back
(illustrative)



Bombers come back from missions with bullet holes. Plot where the holes are and they cluster on the wings, fuselage and tail. The engines and cockpit look comparatively clean.

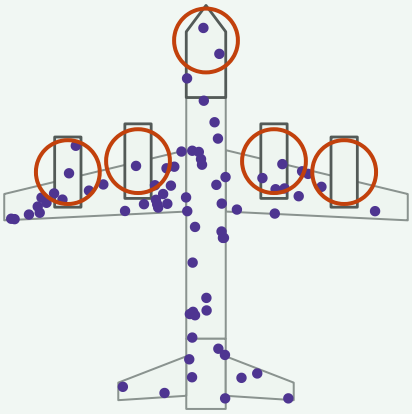
The obvious plan



Armour where the holes are?

The obvious reading: those areas take the most fire, so armour them. Armour is heavy, so you can only add it in a few places.

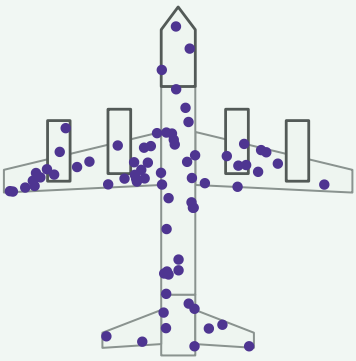
The better question



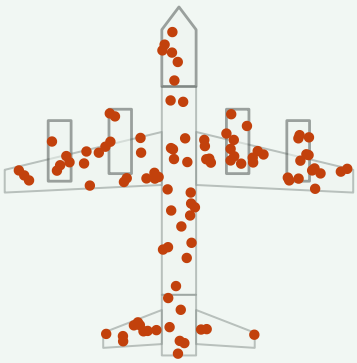
Where are the missing holes?

The better question: if hits land roughly evenly, where are the missing holes? The engines show far fewer holes per square foot than their size predicts.

The planes you never get to inspect



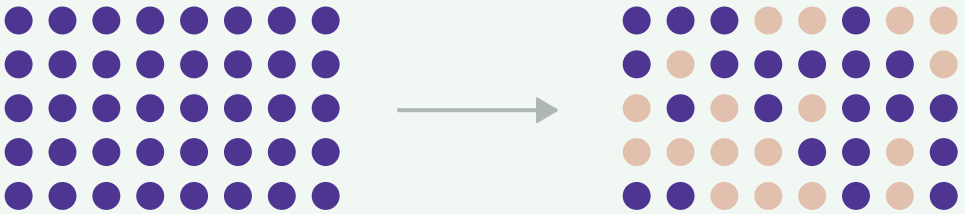
came back (34)



did not come back (26)

The missing holes are on the planes that never returned. A hit to an engine was often fatal, so planes hit there rarely came home to be counted. The clean areas on survivors are the dangerous ones.

Your production logs are the planes that came back



week 1: 40 new users

week 4: 22 still here

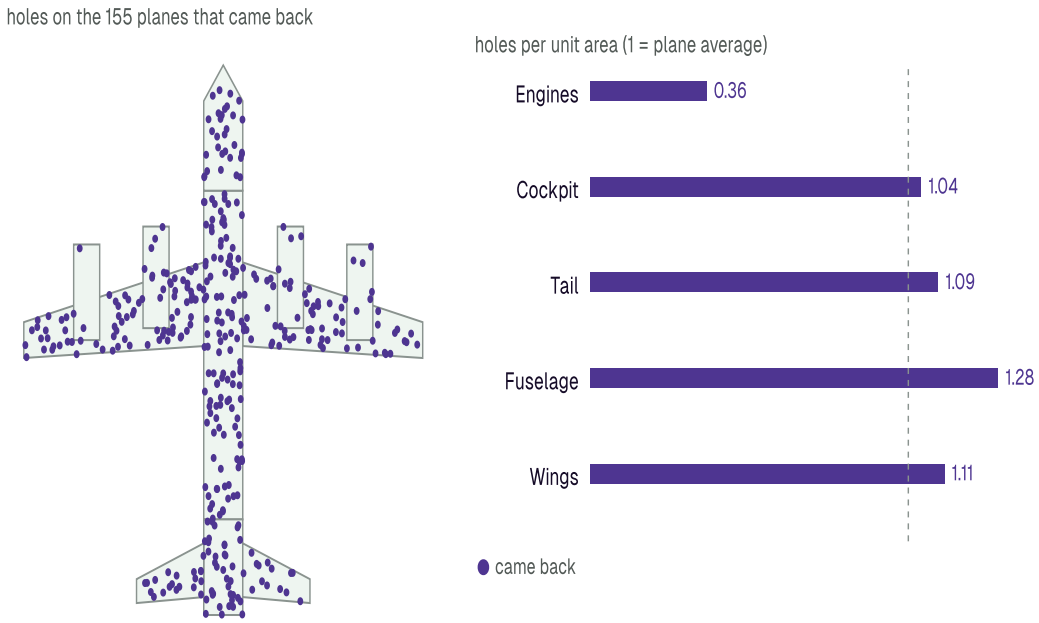
18 left after bad answers: not in logs

Your production logs are the planes that came back. Users who left after bad answers stop generating logs, so the logs over-represent the cases where the model worked. Evaluate on cohorts from their first day, not on whoever is still around.

Interactive

Survivorship: count the holes you cannot see (shown at its default setting)

Planes take hits spread evenly over their surface. Each hit can bring a plane down, with a chance that depends on where it lands. Only planes that return get inspected. Raise the engine lethality, then reveal the planes that did not return.



Planes that returned	Engine holes per area (returned)	Fuselage holes per area (returned)	Most dangerous place to be hit
155 of 300	0.36	1.28	Engines

Hits land evenly over the plane, so if every plane came back each section would show about 1 hole per unit area. Among the planes that returned, engines show **0.36** and the fuselage **1.28**. The fuselage looks like the place that takes the punishment. Reveal the planes that did not return to see why that reading is backwards. The section with the fewest holes per area among survivors, **engines**, is where a hit is most often fatal.

Plain words

Ask what happened to the ones you cannot see

Survivorship bias is looking only at the things that made it through a filter and forgetting the filter. Old buildings seem better built than new ones because the badly built old ones fell down. Successful

dropouts seem to show that leaving university works because unsuccessful dropouts do not give talks.

The habit to build: whenever you look at a group, ask "what had to happen for something to end up in this group, and what does the group look like that didn't make it?"

For leaders

The story is true in outline, embellished in detail

The core is documented. Around 1943 Abraham Wald, a mathematician with the Statistical Research Group at Columbia University, wrote a series of memoranda on estimating aircraft vulnerability from the damage on planes that returned. His key move was to treat the missing planes as data: under the assumption that hits are spread across the aircraft, a section with fewer hits on survivors than its share of the surface implies hits there were more often fatal. Mangel and Samaniego's paper in the *Journal of the American Statistical Association* (1984) is the standard guide to the memoranda.

The vivid parts are less certain. A 2016 AMS Feature Column ("The Legend of Abraham Wald") notes that the memoranda do not say where the military should put armour, that the dramatic dialogue with officers is undocumented, and that the famous plane covered in red dots is a modern illustration, not Wald's drawing. Tell the reasoning, not the legend.

In AI work, survivorship hides in any dataset assembled after an outcome: logs of retained users, case studies of successful deployments, eval items that "made it" through filtering, and model checkpoints that were not discarded early.

Under the hood

Estimating lethality from survivors

Suppose hits land on section s with probability equal to its area share a_s, and a hit there downs the plane with probability k_s. Among hits on returning planes, the share in section s is proportional to $a_s(1-k_s)$ times the chance the plane's other hits were survivable. For a single hit per plane:

$$\frac{\text{returned holes in } s \,/\, a_s}{\text{returned holes overall}} = \frac{1-k_s}{\sum_t a_t(1-k_t)} \quad \Rightarrow \quad \frac{1-k_s}{1-k_t} = \frac{d_s}{d_t}$$

d_s is holes per unit area on returned planes. Ratios of observed densities identify ratios of survival probabilities. A section whose density is far below the plane average is where hits are most lethal. Wald's memoranda worked out versions of this with multiple hits per plane.

The assumption doing the work is that hits are evenly spread. If gunners aimed at engines, low engine density on survivors would mean even higher lethality; if engines were hard to hit, it would mean less.

The same is true in product data: survivorship arguments need a model of what the unfiltered population looked like.

Why a bigger biased sample does not help

Plain words

A bigger bucket from the wrong well

Imagine tasting soup to check the salt, but you only ever dip the spoon near the top where the salt has not mixed in. Tasting ten spoonfuls instead of one makes you very sure of the wrong answer. More data reduces the *wobble* of your estimate. It does nothing about which part of the pot you are dipping into.

For leaders

Small and random beats huge and skewed

Xiao-Li Meng (2018, *Annals of Applied Statistics*) put numbers on this. Using the 2016 US presidential election, he estimated that the correlation between whether a voter's preference was recorded in the survey data and whether they voted for Trump was only about ρ ≈ −0.005. With a self-reported sample of about 2.3 million people, roughly 1% of eligible voters, that tiny correlation gives the same mean squared error as a genuine simple random sample of about 400 people: a 99.98% reduction in effective sample size.

- **Prefer a small probability sample to a large convenience sample** when the decision depends on a population average. 400 random users can beat 2 million self-selected ones.
- **Treat a huge n as a warning sign for overconfidence,** not as reassurance. The error bars from big logs are tiny and exclude the truth.
- **Spend on representativeness first, size second.** Once the design is sound, use power analysis to choose n.

The opposite case is instructive too: Wang, Rothschild, Goel and Gelman (2015) polled 345,858 Xbox users before the 2012 election, a sample that was 93% male against 47% of the electorate. Raw, it was useless. After heavy statistical adjustment it forecast well. The data were biased in ways the adjustment variables could see; that is the case weighting can rescue, discussed below.

Under the hood

Meng's identity

For any dataset of n units from a population of N, random or not, with $R_i = 1$ if unit i is recorded and $f = n/N$, the error of the sample mean factors exactly into three terms (Meng 2018):

$$\bar{G}_n - \bar{G}_N = \underbrace{\rho_{R,G}}_{\text{data quality}} \times \underbrace{\sqrt{\frac{1-f}{f}}}_{\text{data quantity}} \times \underbrace{\sigma_G}_{\text{problem difficulty}}$$

$\rho_{R,G}$ is the **data defect correlation** between being recorded and the value. Under simple random sampling its typical size is about $1/\sqrt{N}$, which makes the error shrink with n. A non-random process can fix ρ at a value that does not depend on N, and then the error depends on the fraction f, not on n.

Equating mean squared errors with a simple random sample of size n_{eff} gives $n_{\text{eff}} \approx \frac{f}{1-f}\,\frac{1}{\rho^2}$. Doubling a big biased sample barely helps; a tiny change in ρ helps enormously.

python

```python
import math

def meng_error(rho, f, sigma):
    # error of the sample mean = data quality x data quantity x difficulty
    return rho * math.sqrt((1 - f) / f) * sigma

def effective_n(rho, f):
    # size of a simple random sample with the same mean squared error
    return f / (1 - f) / rho ** 2

rho, sigma = -0.005, 0.5        # binary outcome near 50/50 has sigma 0.5
for f in (0.01, 0.02, 0.05, 0.5):
    print(f, f"{meng_error(rho, f, sigma):+.4f}", round(effective_n(rho, f)))
# 0.01 -0.0249 404     1% of voters (about 2.3 million) is worth ~400 random ones
# 0.02 -0.0175 816     doubling the data cuts the error by only 30%
# 0.05 -0.0109 2105
# 0.5 -0.0025 40000    even half the population is worth 40,000 random draws
print(f"{sigma / math.sqrt(404):.4f}")   # 0.0249 the same error, from n = 404 at random
```

rust

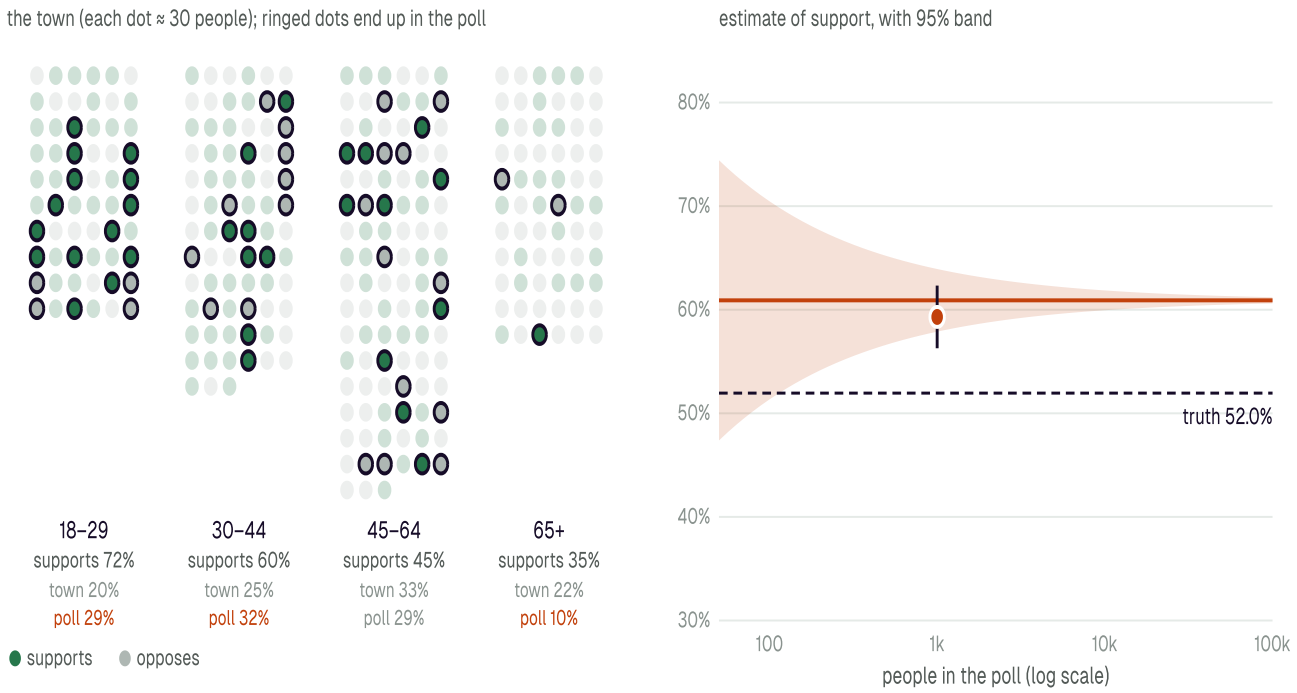
```rust
fn meng_error(rho: f64, f: f64, sigma: f64) -> f64 {
    // error of the sample mean = data quality x data quantity x difficulty
    rho * ((1.0 - f) / f).sqrt() * sigma
}

fn effective_n(rho: f64, f: f64) -> f64 {
    // size of a simple random sample with the same mean squared error
    f / (1.0 - f) / rho.powi(2)
}

fn main() {
    let (rho, sigma) = (-0.005, 0.5); // binary outcome near 50/50 has sigma 0.5
    for f in [0.01, 0.02, 0.05, 0.5] {
        println!("{} {:+.4} {}", f, meng_error(rho, f, sigma), effective_n(rho, f).round());
    }
    // 0.01 -0.0249 404     1% of voters (about 2.3 million) is worth ~400 random ones
```

```
    // 0.02 -0.0175 816     doubling the data cuts the error by only 30%
    // 0.05 -0.0109 2105
    // 0.5 -0.0025 40000    even half the population is worth 40,000 random draws
    println!("{:.4}", sigma / 404f64.sqrt()); // 0.0249 the same error, from n = 404 at random
}
```

Interactive

Who ends up in the poll? (shown at its default setting)

An illustrative town in four age bands, where support for a transit plan falls with age. Pick how the poll is run and how many people answer. The right panel shows the estimate and its 95% band as n grows: random designs close in on the truth, biased ones close in on the wrong value. Then post-stratify by age.



Truth	This poll (95%)	Bias of the design	Effective sample size
52.0%	59.3% ± 3.0	+9.0 pts	1.0K

An email poll of 1,000 people says **59.3%** (±3.0 points). The truth is 52.0%. The design is off by **9.0 points** on average, and that number does not change with n: more people only shrink the band around the wrong value. This interval **misses the truth**. Try post-stratifying: it fixes the part of the bias that comes from the wrong age mix.

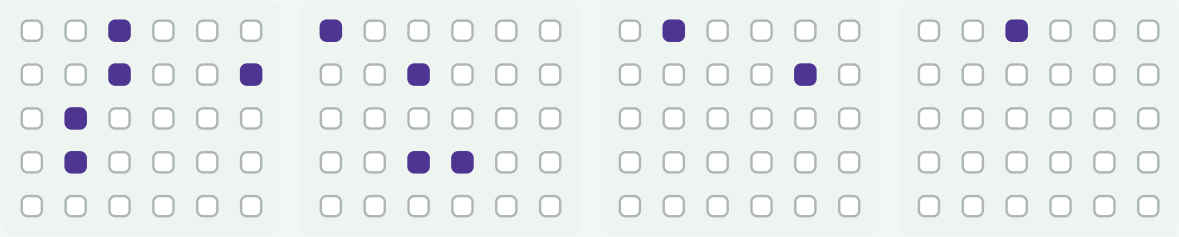
Probability sampling: four designs

Diagram

Four fair ways to pick 12 houses

A town of 120 houses in four neighbourhoods. Each design gives every house a known chance of being picked.

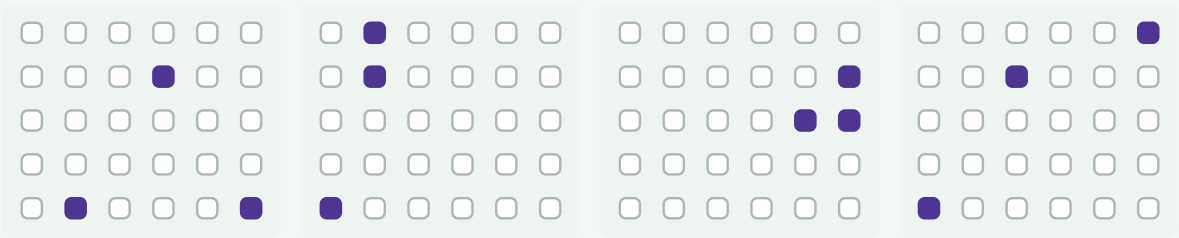
Simple random sample



e.g. 12 of 120 houses drawn from a hat

Simple random sampling: every house, and every set of 12 houses, is equally likely. The baseline. By luck it can under-sample a neighbourhood: here West gets one house while North gets five.

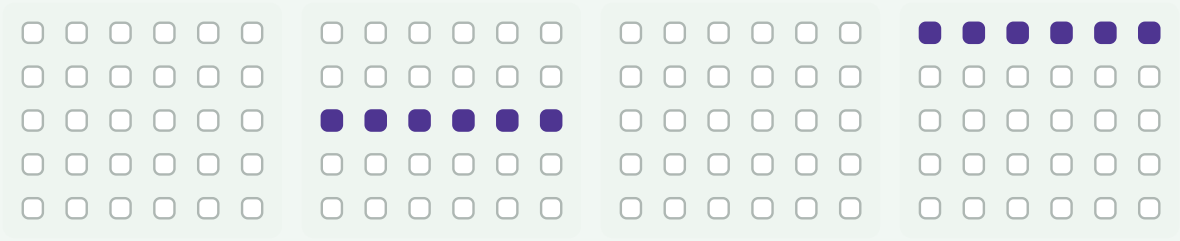
Stratified sample



3 per neighbourhood: no group can be missed

Stratified sampling: split the town into neighbourhoods (strata) first, then sample at random inside each. No neighbourhood can be missed, and if neighbourhoods differ, the estimate is more precise than simple random sampling.

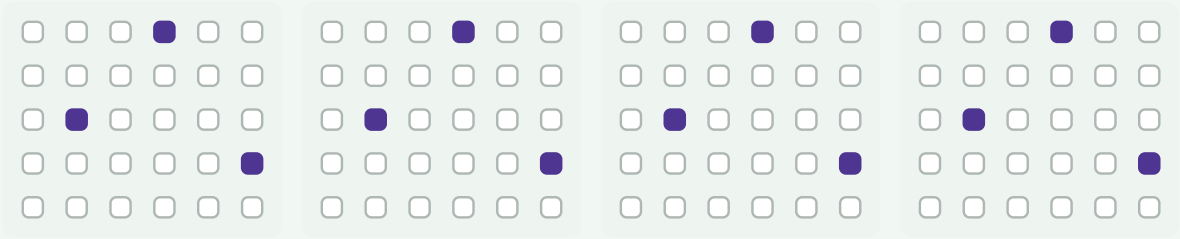
Cluster sample



2 streets of 6: neighbours tend to agree

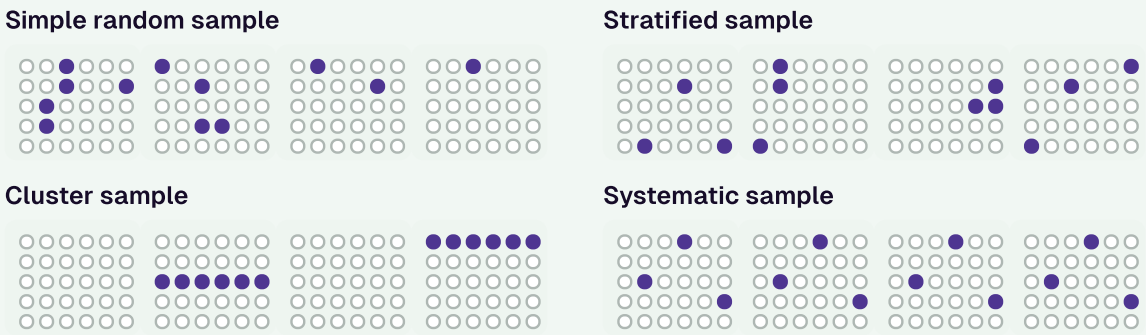
Cluster sampling: pick whole streets at random and visit every house on them. Much cheaper to run, but neighbours resemble each other, so 12 houses on 2 streets carry less information than 12 scattered houses.

Systematic sample



start at house 4, then 14, 24, …

Systematic sampling: from a random start, take every 10th house on the list. Easy and spreads the sample out, but dangerous if the list has a rhythm, say every 10th house is a corner plot.



That known chance is what lets you put an honest error bar on the answer.

The designs differ in cost and precision, but all four share the property that matters: the chance of selection is known and set by the design, not by who chooses to answer. That is what makes the error bar honest.

Plain words

Let a fair lottery do the choosing

The way to avoid every trap above is to let a lottery, not convenience or volunteering, decide who is measured. A lottery cannot prefer the keen, the nearby or the survivors. The four designs in the diagram are four ways to run that lottery: plain random, random within groups, random groups, and every k-th from a random start.

Stratifying is the one to reach for most often. If you know the groups that matter (language, task type, customer tier), make sure each gets its fair share instead of hoping luck provides it.

For leaders

Choosing a design for an eval set

Design	Use it when	Cost	Watch out for
Simple random	You have a complete frame and no known important subgroups	Low	Small subgroups can be missed by chance
Stratified	Subgroups differ (languages, task types, customer tiers) or you must report each one	Low to medium	Report with the right weights if you over-sample small strata
Cluster	Units come in natural groups (all turns of a conversation, all tasks from one repository)	Lowest per unit	Effective n is smaller than the count; use clustered errors (intervals)
Systematic	Sampling from a stream or ordered log	Lowest effort	Periodic patterns in the order (hourly batches, rotating shards)

A frontier eval set is usually stratified and clustered at once: stratified by capability area and language so each gets enough items to report, clustered because many items share a source document, repository or conversation.

Under the hood

Variances of the four designs

For a population mean with strata h of population share W_h, within-stratum SD S_h and sample size n_h:

$$\mathrm{Var}(\bar{y}_{\mathrm{SRS}}) \approx \frac{S^2}{n}, \qquad \mathrm{Var}(\bar{y}_{\mathrm{st}}) = \sum_h W_h^2 \frac{S_h^2}{n_h}, \qquad n_h^{\mathrm{Neyman}} \propto W_h S_h$$

Finite-population corrections omitted. With proportional allocation $n_h = nW_h$, stratified variance is $\sum_h W_h S_h^2 / n$, which is never larger than SRS and smaller when stratum means differ. Neyman allocation puts more sample where strata are large or variable.

Cluster sampling with m units per cluster and intra-cluster correlation ρ inflates variance by the design effect $1 + (m-1)\rho$ (standard errors, clustered intervals). Ten turns per conversation with $\rho = 0.3$ gives a design effect of 3.7: 1,000 turns are worth about 270 independent ones. **Systematic sampling** behaves like SRS when the list order is unrelated to the outcome, better than SRS when

the order is a smooth trend (it acts like implicit stratification), and badly when the order has a period that divides the step.

Weighting and post-stratification

Plain words

Count the under-represented more

If your poll has too many young people and too few old people, you can count each older respondent a bit more and each younger respondent a bit less, so the age mix matches the town's. That is **post-stratification**: after the fact, give each group the weight it has in the population.

It fixes the part of the bias that comes from having the wrong mix of groups. It cannot fix bias *inside* a group: if the older people who answered are unlike the older people who did not, giving them more weight just amplifies the unusual ones. Try it in the widget above: the email poll improves but does not reach the truth.

For leaders

What weighting can and cannot rescue

- **It works when you can see the cause of the skew.** The Xbox poll was rescued because its biases ran mostly through age, sex, party and past vote, all of which were measured and adjusted for with multilevel regression and post-stratification (MRP). Its final national estimate was within about 0.6 points of the result (Wang et al. 2015).
- **It fails when the skew runs through something unmeasured.** Satisfaction surveys skew toward engaged users in ways that tenure and plan type only partly capture.
- **Weights cost precision.** A few heavily weighted respondents make the estimate jumpy. Always report the effective sample size next to the raw count.
- **Weights need accurate population totals and non-empty cells.** If your poll has no respondents over 80, no weight can conjure them.

For eval sets the same logic applies: if your set is 70% English and production is 40% English, report per-language scores and a production-weighted total, and say which languages had too few items to trust.

Under the hood

Weights, effective n, raking

Post-stratification over cells g with known population shares W_g and respondent counts n_g:

$$\hat{\bar{y}}_{\text{ps}} = \sum_g W_g \, \bar{y}_g, \qquad w_i = \frac{W_g}{n_g / n} \;\; (i \in g), \qquad n_{\text{eff}} = \frac{\left(\sum_i w_i\right)^2}{\sum_i w_i^2}$$

n_{eff} is Kish's effective sample size: how many unweighted, random respondents would give the same precision. Unbiased only if, within each cell, respondents resemble non-respondents on the outcome (missing at random given the cells).

With several weighting variables whose joint distribution is unknown, **raking** (iterative proportional fitting) adjusts weights to match each margin in turn until all match. MRP replaces raw cell means with a multilevel model's predictions, which keeps small cells from producing wild estimates.

python

```python
# Email poll respondents by age band (illustrative), and the town's age mix.
BANDS      = ["18-29", "30-44", "45-64", "65+"]
POP_SHARE  = [0.20, 0.25, 0.33, 0.22]
RESPONDED  = [261, 286, 259, 84]
SUPPORTERS = [197, 184, 128, 33]

n = sum(RESPONDED)
raw = sum(SUPPORTERS) / n

# weight per respondent in each band: population share / sample share
weights = [p / (r / n) for p, r in zip(POP_SHARE, RESPONDED)]
post_strat = sum(p * s / r for p, s, r in zip(POP_SHARE, SUPPORTERS, RESPONDED))

# Kish effective sample size: the price of uneven weights
sum_w = sum(w * r for w, r in zip(weights, RESPONDED))
sum_w2 = sum(w * w * r for w, r in zip(weights, RESPONDED))
n_eff = sum_w ** 2 / sum_w2

print(f"{raw:.3f}")                              # 0.609 raw poll
print(" ".join(f"{w:.2f}" for w in weights))     # 0.68 0.78 1.13 2.33 over-65s count 2.3x
print(f"{post_strat:.3f}")                       # 0.561 age mix fixed; truth is 0.520
print(f"{n_eff:.0f}")                            # 731 of 890 respondents' worth of precision
```

rust

```rust
// Email poll respondents by age band (illustrative), and the town's age mix.
const POP_SHARE: [f64; 4] = [0.20, 0.25, 0.33, 0.22];
const RESPONDED: [f64; 4] = [261.0, 286.0, 259.0, 84.0];
const SUPPORTERS: [f64; 4] = [197.0, 184.0, 128.0, 33.0];

fn main() {
    let n: f64 = RESPONDED.iter().sum();
    let raw = SUPPORTERS.iter().sum::<f64>() / n;

    // weight per respondent in each band: population share / sample share
    let weights: Vec<f64> = (0..4).map(|b| POP_SHARE[b] / (RESPONDED[b] / n)).collect();
    let post_strat: f64 = (0..4).map(|b| POP_SHARE[b] * SUPPORTERS[b] / RESPONDED[b]).sum();

    // Kish effective sample size: the price of uneven weights
```

```
    let sum_w: f64 = (0..4).map(|b| weights[b] * RESPONDED[b]).sum();
    let sum_w2: f64 = (0..4).map(|b| weights[b] * weights[b] * RESPONDED[b]).sum();
    let n_eff = sum_w.powi(2) / sum_w2;

    println!("{:.3}", raw); // 0.609 raw poll
    let ws: Vec<String> = weights.iter().map(|w| format!("{:.2}", w)).collect();
    println!("{}", ws.join(" ")); // 0.68 0.78 1.13 2.33 over-65s count 2.3x
    println!("{:.3}", post_strat); // 0.561 age mix fixed; truth is 0.520
    println!("{:.0}", n_eff); // 731 of 890 respondents' worth of precision
}
```

Sample size is not representativeness

Plain words

Two different questions

"Is my sample big enough?" and "is my sample the right people?" are separate questions. Size controls how much your answer wobbles. Representativeness controls whether it is aimed at the right target. A small fair sample is blurry but centred; a large unfair one is sharp and off-centre.

For leaders

Answer them in order

First settle the frame and the design: who could be in the data, and how they are chosen. Only then size the sample with power analysis for the smallest difference that would change the decision. A review that starts with "we have 50,000 examples" and never reaches "drawn how, from what" has answered the easy question.

Under the hood

Mean squared error has two parts

$$\mathrm{MSE}(\hat{\theta}) = \underbrace{\mathrm{Bias}(\hat{\theta})^2}_{\text{does not shrink with } n} + \underbrace{\mathrm{Var}(\hat{\theta})}_{\approx \sigma^2/n}$$

Past the point where variance falls below squared bias, extra data buys almost nothing. With 9 points of bias, as in the email poll, that point arrives at a few dozen respondents.

The coverage of a nominal 95% interval also collapses as n grows when bias is fixed: the interval half-width shrinks like $1/\sqrt{n}$ while the distance to the truth stays put, so coverage tends to zero. A

biased design with a huge n produces confident, wrong intervals almost every time.

In AI evals

Plain words

Your test set is a poll of prompts

An eval set is a sample of prompts standing in for all the prompts the model will face. Every trap in this chapter applies. If the set was built from whatever was easy to collect (English, short, clearly answerable), the score describes that slice and not the product.

For leaders

Four common sampling failures in model evaluation

- **Easy and English-heavy eval sets.** Items get written or kept because they are gradeable and unambiguous, and authors write in the languages they know. Scores then over-state performance on the hard, messy, multilingual tail. Stratify by language and difficulty and report each stratum (classification metrics by slice).
- **Production logs that only include users who stayed.** Survivorship. Build eval sets from first-session cohorts, or weight by acquisition cohort, so that the users the model failed are represented.
- **Red-team pools.** Findings reflect who was recruited: security researchers find jailbreaks; they are less likely to find harms that show up for non-experts or in other languages. Treat red-team counts as a lower bound on a frame you chose, and vary the pool.
- **Feedback signals.** Thumbs, ratings and escalations come from a self-selected minority. Use them to find failures, not to estimate rates. Estimate rates on a random sample with paid, blind grading (judges, metrics).

Under the hood

Building a representative eval sample

1. **Define the population** as a traffic distribution over a future window, by surface, customer tier, language and task type.
2. **List the frame** and its gaps: which of those cells can you sample from logs, and which need synthetic or commissioned prompts? Record each gap.
3. **Stratify** on the cells, over-sampling small but important strata so each has enough items for an interval, and **cluster-sample** conversations, with the design effect in the error bars.

4. **Track item loss** from sampling to graded result. Items dropped as "ambiguous" or "ungradeable" are nonresponse; compare their features with kept items.

5. **Report** per-stratum scores, a production-weighted total with its effective n, and the stated gaps. Refresh the sample as traffic drifts (offline to online gap, contamination).

Eval lens

Before comparing two models on an eval set, check that the set's mix matches the decision. A model can win on a set that is 80% easy English prompts and lose on production traffic that is 40% hard multilingual ones: that is Simpson's paradox caused by a sampling choice (correlation is not causation). Report per-stratum results and a total weighted to the population the decision is about.

Principal asks

What is the population, and what is the frame? Who can never appear in this data? What fraction of the sampled items or people made it to a result, and how do the dropped ones differ? Was selection by lottery or by convenience? If you weighted, on what variables, and what is the effective sample size? Would a small random sample have given a different answer?

Common mistakes

- Believing a large n cancels out a biased frame. It only shrinks the error bar around the wrong value.
- Treating a low response rate as fine because "the rest would answer the same way", without checking how responders differ.
- Estimating rates from opt-in signals (thumbs, reviews, escalations) instead of a random sample.
- Building eval sets from retained users' logs and calling them representative of new users.
- Weighting on variables that do not drive the skew, then reporting the weighted number as unbiased.
- Reporting the raw count as the sample size after heavy weighting or clustered sampling.
- Telling the Wald bomber story with the invented details instead of the documented reasoning.

Part VI · Chapter 43

Assigning treatment: randomization and its alternatives

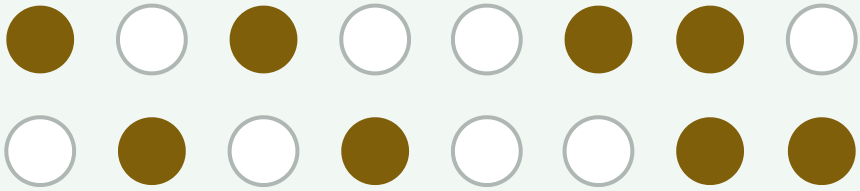
Every experiment answers one question: what would have happened to the same people without the change? You can never watch both versions of the world, so you build a comparison group that stands in for the missing one. How you decide who gets the change is the single biggest factor in whether that stand-in is honest. This chapter covers why a coin flip works, the four common ways assignment goes wrong, the quasi-experimental toolkit for when you cannot flip a coin, and the design choices (strata, units, spillover, balance checks) that decide whether a randomized test is as good as it looks.

Diagram

The tutoring club that looked like magic

Why letting people choose ruins the comparison, and how a coin fixes it.

16 students. Motivation is invisible to the school.

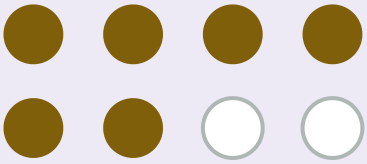


highly motivated: grades rise 7 points a term on their own

less motivated: grades rise 1 point a term

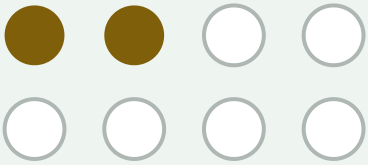
A school starts an after-school tutoring club. Every student carries a trait the school cannot see: motivation. Motivated students improve faster with or without help. In this made-up school the club truly adds 4 points.

Signed up for the club



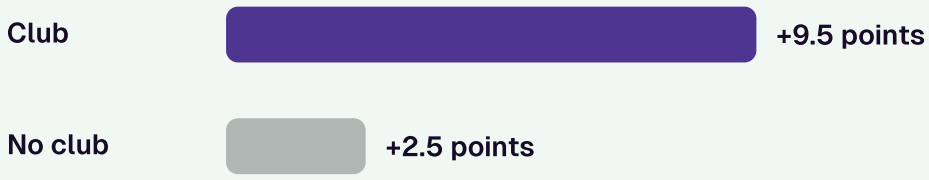
6 of 8 motivated

Did not sign up



2 of 8 motivated

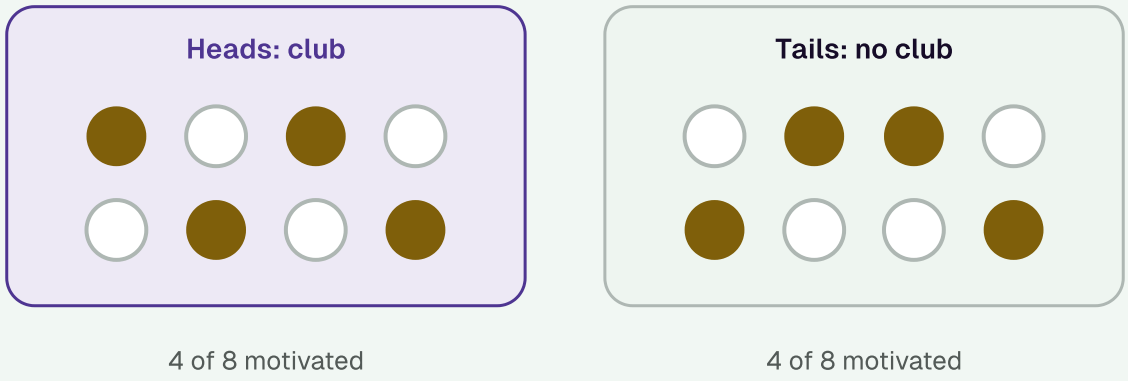
The club is optional, so students choose. The keen ones sign up. The two groups now differ in two ways at once: the club, and who they are.



Apparent effect 7 points = 4 from tutoring + 3 from who signed up

the extra 3 is selection bias, and no amount of data removes it

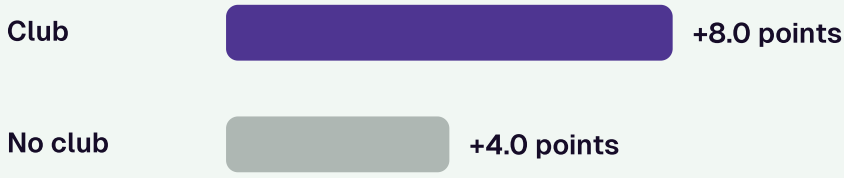
Compare the groups and the club looks like it adds 7 points. Three of those points come from the students, not the club. Doubling the school's size would give the same wrong answer with a smaller error bar.



4 of 8 motivated

4 of 8 motivated

Now a coin decides. The coin cannot see motivation, so it splits motivated students roughly evenly. It also splits everything else nobody thought to measure: sleep, home help, a good teacher last year.



Gap 4 points = the true effect of tutoring

the coin balanced what nobody measured, so the only systematic difference is the club

The gap is now 4 points, the club's real effect. Any one coin flip can still be unlucky, which is what error bars are for. On average across possible flips the comparison is fair, and that guarantee covers traits nobody thought to measure.

Why a coin flip works

Plain words

The coin cannot play favourites

To learn whether something works, you compare people who got it with people who did not. The comparison is only fair if the two groups were alike before the change. The trouble is that people differ in countless ways you cannot see.

A coin flip solves this without you knowing any of those ways. The coin does not know who is motivated, rich, tired, or a power user, so each kind of person lands in each group about equally often. Any remaining difference between the groups after the change comes from the change itself or from luck, and statistics can measure luck.

For leaders

What randomization buys and what it costs

Randomization is the only assignment method that protects against confounders nobody has thought of. Every alternative in this chapter works only if an assumption holds, and those assumptions cannot be fully checked from the data. That is why a randomized result usually settles an argument and an observational one usually starts a new one.

- **What it guarantees.** No systematic difference between groups, so the estimate is unbiased. It does not guarantee a precise estimate; small experiments still have wide intervals (see the

standard error).

- **What it costs.** Engineering for assignment and logging, holding some users back from a change you believe is good, and the patience to wait for enough data. At a lab, the expensive part is usually the infrastructure to assign consistently and log exposure correctly, built once and reused.
- **What breaks it.** Anything that lets a human or the system influence who ends up in which group after the coin lands: opt-outs, filters applied after assignment, crashes that hit one arm more, or users in one arm affecting users in the other.

Under the hood

Potential outcomes and the selection-bias term

Give each unit two potential outcomes: $Y_i(1)$ if treated and $Y_i(0)$ if not. The effect for unit i is $Y_i(1) - Y_i(0)$, and we only ever see one of the two. The naive comparison of group means splits into the effect plus a bias term:

$$\underbrace{E[Y \mid T{=}1] - E[Y \mid T{=}0]}_{\text{what you compute}} = \underbrace{E[Y(1) - Y(0) \mid T{=}1]}_{\text{effect on the treated}} + \underbrace{E[Y(0) \mid T{=}1] - E[Y(0) \mid T{=}0]}_{\text{selection bias}}$$

T is the treatment indicator. The selection-bias term asks: would the treated group have done differently from the control group even without treatment? In the tutoring story it is 3 points.

Random assignment makes T independent of $(Y(0), Y(1))$, written $(Y(0), Y(1)) \perp T$. Then $E[Y(0) \mid T{=}1] = E[Y(0) \mid T{=}0]$, the bias term is zero, and the difference in means estimates the average treatment effect. Fisher argued this in *The Design of Experiments* (1935): randomization is what licenses the significance test, because it makes the assignment mechanism known.

The balance is in expectation. In one realised experiment, any covariate X differs between arms by chance, with a standard deviation of about $2\sigma_X/\sqrt{n}$ for a 50/50 split of n units. That chance imbalance is already inside the standard error of the effect estimate, which is why you should not "re-roll" a randomization until the balance table looks nice unless you also account for it in the analysis (re-randomization with a fixed rule is a valid design; ad hoc re-rolling is not).

Four ways assignment goes wrong

Each of these produces a comparison group that differs from the treated group before the treatment does anything. All four show up in AI labs every quarter.

Self-selection

Users opt in to a beta of a new coding agent. Opt-in users already write more code, use the product daily, and forgive rough edges. The beta "raises engagement by 40 percent", and most of that gap existed before the beta.

Convenience

The new model is served in one region, to one team, or during off-peak hours because that is where capacity was free. Region, team, and time of day each come with their own users, languages, and load, so the effect is tangled with the place.

The worst performers

A team takes the 200 prompts where the model scored worst, rewrites the system prompt, and re-runs them. Scores jump 15 points. Part of that jump would have happened anyway, because items picked for a bad score were partly unlucky. This is regression to the mean.

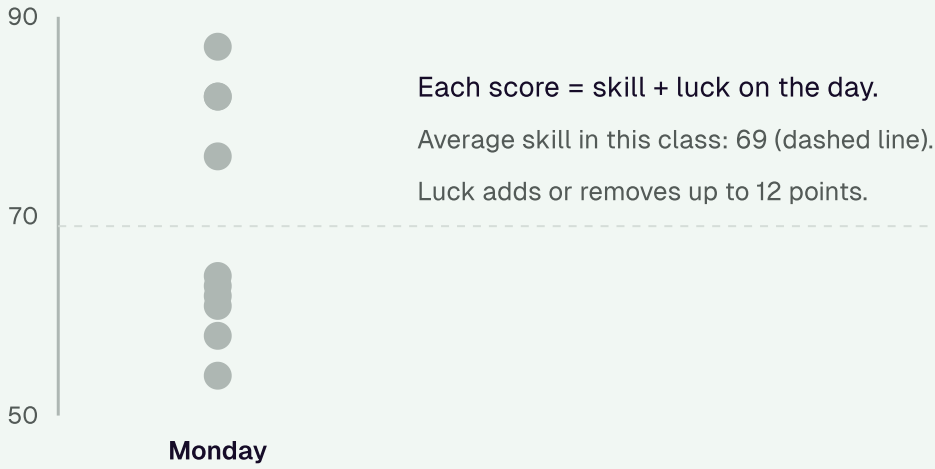
Analyst choice

A product lead hand-picks 20 "friendly" enterprise customers for a pilot, or an analyst drops users who "did not really see" the feature after looking at the data. Each choice correlates with the outcome, so the groups stop being comparable.

Diagram

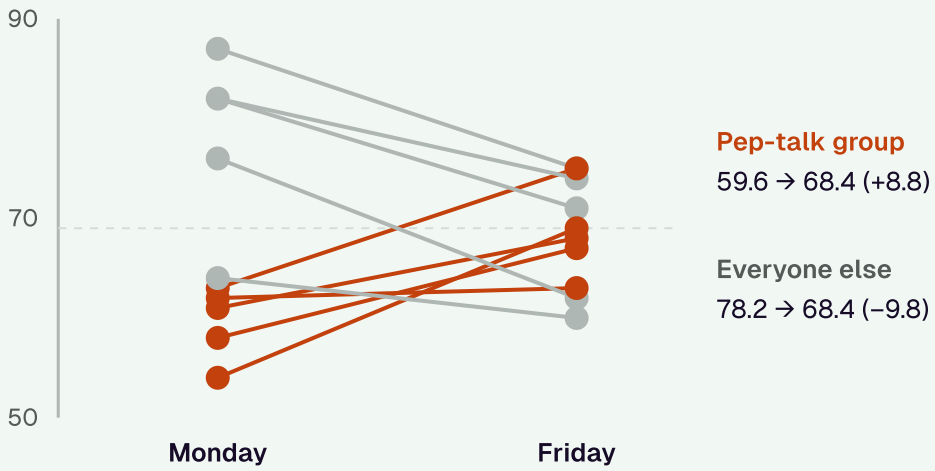
The pep talk that seemed to work

Regression to the mean, with ten students and two quizzes.



Ten students take a Monday quiz. Every score is part skill and part luck: a good night's sleep, a lucky guess, a question they happened to revise.

The teacher gives the five lowest scorers a pep talk. Picking by a low score picks people who are a bit less skilled and also people who were unlucky on Monday. Luck does not carry over to Friday.



On Friday the pep-talk group gains almost 9 points. The top five, who got nothing, drop almost 10. Both groups slid back toward the class average because Monday's luck did not repeat. The pep talk may have done nothing at all.

Bigger school: 40 lowest scorers, coin decides who gets the pep talk

Effect of the pep talk ≈ 1 point, not 9 (illustrative numbers)

The fix is a control group drawn from the same low scorers. Both halves bounce back by about 8 points; the pep talk adds about 1. Whenever you select on a bad measurement, randomize inside the selected group or the bounce will be credited to your fix.

Plain words

Why "fixing the worst" always looks like it worked

A golfer has a terrible round and buys a new putter. The next round is better. Was it the putter? Probably not much: a terrible round is partly bad luck, and bad luck rarely strikes twice in a row.

Anything you do right after an unusually bad result will look like it helped. Anything you do right after an unusually good result will look like it hurt.

Daniel Kahneman tells a story about flight instructors that makes the same point. The instructors believed praise made trainee pilots worse and shouting made them better, because praise followed unusually good landings and shouting followed unusually bad ones. Both were followed by more ordinary landings regardless.

For leaders

Spotting a broken assignment in a review

When a result arrives, ask how units got into the treated group before you look at the effect size. The pattern behind each failure is the same: something that predicts the outcome also predicted treatment.

You hear	What it usually means	Direction of the error
"Users who turned on the feature retain 30% better"	Self-selection; engaged users try features	Overstates the effect
"We tested it in Japan first and saw a lift"	Convenience; region, language, and season move with treatment	Either way, unknown
"The worst prompts improved after the fix"	Regression to the mean on a noisy eval	Overstates the effect
"We excluded users who churned in week one"	Filtering on something the treatment may have caused	Either way, often large
"Pilot customers love it"	Hand-picked friendly accounts	Overstates the effect

The default fix costs little: assign by a hash of the user ID, decide the analysis population before launch, and analyse everyone as assigned (intention to treat), including people who never used the feature. Report the "users who actually used it" number as a secondary view and label it.

Under the hood

How big regression to the mean is

Let a measurement be $X = \mu_i + \varepsilon$, true level plus noise, with population mean μ and test–retest correlation ρ (the share of variance that is real). If both measurements are standardised the same way, the expected second measurement given the first is

$$E[X_2 \mid X_1 = x] = \mu + \rho\,(x - \mu), \qquad \textbf{expected bounce} = (1 - \rho)\,(\mu - x)$$

ρ is the correlation between repeated measurements. With $\rho = 0.5$, a group selected 20 points below average is expected to come back 10 points with no intervention at all.

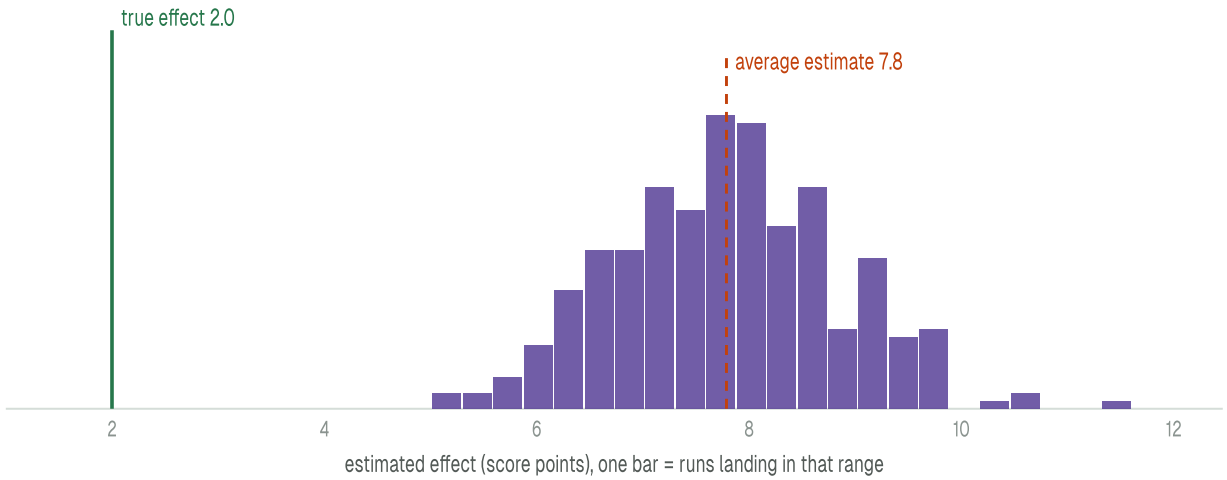
Two places this bites in eval work. First, "hard subsets" chosen by a model's low score on one run: re-run the same model and the subset gets easier. Pick hard subsets with a different model, a different seed, or human labels, or compare both models on the subset with the same selection. Second, selecting on the pre-period outcome breaks difference-in-differences, because the treated group's pre-period was low partly by chance (economists call the job-training version the Ashenfelter dip).

On the analyst-choice failure: filtering on a post-treatment variable (for example, keeping only users who sent at least one message after assignment) conditions on something the treatment can change, which reintroduces selection bias even in a perfectly randomized test. Intention-to-treat analysis avoids it. If you need the effect on users who actually engaged, estimate it with the assignment as an instrument (see below) instead of filtering.

Interactive

Assignment simulator (shown at its default setting)

Each run builds 400 users with a hidden motivation that raises their score, assigns treatment with the rule you pick, and estimates the effect. The chart shows 300 runs. A fair design centres on the true effect; a broken one centres somewhere else, no matter how many runs you add.



Average estimate	Bias	Spread (SD)	95% intervals that miss
7.79	+5.79	1.05	100%

Balance check, first run	Treated	Control	SMD	Typical |SMD| over runs
Motivation (hidden, never measured)	0.29	-0.35	0.70	0.69
Pre-period score (measured)	52.4	47.3	0.53	0.52
Account age, months (measured)	17.8	18.1	-0.03	0.08

With the opt-in rule and a straight treated-versus-control comparison, the 300 estimates centre on **7.8** against a true effect of **2.0**. The bias of **+5.8** does not shrink with more users; try raising the sample size and watch the spread tighten around the wrong value. Motivated users opt in, and motivation raises scores on its own. The balance table shows it in the hidden trait, and the pre-period score betrays it too.

Eval lens

Offline evals have assignment problems too, they are just less visible. Randomize which response is shown first to a pairwise judge (position bias is a confounder; see LLM as judge). Randomize which annotator sees which model's outputs, or every annotator sees both. Interleave runs of the two models over time so a provider outage or a judge-model update does not land on one arm. Never build a "hard subset" from the very run you are evaluating. And when a fix is tested on the worst items, compare it against the old model re-run on the same items, so regression to the mean shows up in both arms.

When you cannot randomize

Plain words

Finding a fair comparison that nature made

Sometimes a coin flip is off the table. You cannot randomly make people smoke. You cannot charge half a country one price and half another. You cannot give half of one company's employees a new model when they share the same documents and talk to each other all day.

Then you hunt for a situation where the world did something close to a coin flip for you: a rule with a sharp cutoff, a lottery, a launch that happened in one place but not a similar one. These designs are called **quasi-experiments**. Each one swaps the coin for an assumption, and the whole game is checking how believable that assumption is.

For leaders

When non-random assignment is acceptable

Non-random assignment is a legitimate choice when one of these holds:

- **Ethics.** Withholding a safety fix or a protection from a random half of users is not acceptable. Ship it, and measure with a design that does not need a held-back group.
- **Cost or reach.** The change is a pricing tier, a contract, a data-centre migration, or a model swap for a whole enterprise tenant. The unit is so large there are only a handful of them.
- **Rollouts that cannot be split.** A tokenizer change, a new safety classifier on the shared serving path, a policy update announced publicly. Everyone gets it at once.
- **Network effects.** Users interact, so splitting them contaminates the comparison (see interference below). The honest unit becomes a whole community or market, and you may only have a few.

What changes is the burden of proof. A quasi-experiment needs a named assumption, a check of that assumption (pre-trends, balance, placebo tests), and a sentence in the readout on what would make the result wrong. Budget analyst time for that, and expect a reviewer to discount the result by more than its confidence interval suggests.

Principal asks

Why could this not be randomized, and was a partial randomization possible (randomize timing across regions, randomize an encouragement)? What is the identifying assumption in one sentence? What check would have caught a violation, and did you run it? Is there a placebo outcome or placebo date where the method should find nothing, and does it?

Under the hood

The quasi-experimental toolkit

Method	Intuition	Key assumption	How it fails	Lab example
Difference-in-differences	Compare the change in the treated group with the change in an untreated group	Parallel trends: without treatment both would have changed by the same amount	Groups were already diverging; another shock hits one group at launch	A safety filter ships to EU traffic first; compare the EU change in complaint rate with the US change
Regression discontinuity	Units just above and just below a cutoff are nearly identical except for treatment	No precise control over the running variable near the cutoff; outcome would be smooth through it	Bunching just over the cutoff; effect only valid near the cutoff	Requests with difficulty score above 0.7 are routed to the large model; compare outcomes at 0.69 and 0.71
Matching and propensity scores	Compare each treated unit with untreated units that looked the same beforehand	No unmeasured confounders given the matched variables, and overlap (every treated unit has look-alikes)	A hidden trait such as motivation drives both choice and outcome	Users who enabled memory compared with look-alike users who did not
Instrumental variables	Find something random that nudges treatment but touches the outcome only through treatment	Relevance, as-good-as-random instrument, and exclusion (no other path to the outcome)	Weak instrument; the nudge has a direct effect	A random banner invites some users to try agent mode; use the banner to estimate agent mode's effect on users it persuaded
Interrupted time series	Forecast the metric from its pre-launch trend and	The pre-trend would have continued; nothing	Seasonality, a marketing push, or a competitor	Daily success rate before and after a model

Method	Intuition	Key assumption	How it fails	Lab example
	compare with what happened	else changed at the same moment	launch on the same week	launch on one date
Synthetic control	Build a weighted blend of untreated units that tracks the treated unit before launch	The blend would have kept tracking it; donors are untouched by the treatment	Poor pre-period fit; donors affected by spillover	One large customer gets a custom fine-tune; build a "synthetic customer" from a weighted mix of similar tenants

Difference-in-differences (the New Jersey minimum-wage study by Card and Krueger, 1994, is the textbook case) estimates

$$\hat{\delta} = \left(\bar{Y}_{T,\text{post}} - \bar{Y}_{T,\text{pre}}\right) - \left(\bar{Y}_{C,\text{post}} - \bar{Y}_{C,\text{pre}}\right) \;\equiv\; \hat{\delta} \text{ in } Y = \alpha + \beta\,\text{Treat} + \gamma\,\text{Post} + \delta\,\text{Treat}\times$$

T treated group, C comparison group. The regression form gives the same number and lets you add covariates and cluster standard errors by group. Stable differences between groups cancel; differences in trend do not.

Regression discontinuity (Thistlethwaite and Campbell, 1960, studied Certificates of Merit, public recognition awarded above a test-score cutoff) estimates the jump $\tau = \lim_{x \downarrow c} E[Y \mid X = x] - \lim_{x \uparrow c} E[Y \mid X = x]$ with local linear fits on each side, inside a bandwidth chosen by a data-driven rule. Always plot it, and check the density of X at the cutoff for bunching.

Propensity scores (Rosenbaum and Rubin, 1983) compress many covariates into $e(x) = P(T = 1 \mid X = x)$. Matching or weighting by $1/e(x)$ balances the measured covariates; nothing balances the unmeasured ones. Trim units with $e(x)$ near 0 or 1, where there are no comparable counterparts, and run a sensitivity analysis that asks how strong a hidden confounder would need to be to erase the effect.

Instrumental variables (Angrist, 1990, used the Vietnam draft lottery as a random nudge into military service) give the Wald estimator $\hat{\beta} = \dfrac{E[Y \mid Z = 1] - E[Y \mid Z = 0]}{E[T \mid Z = 1] - E[T \mid Z = 0]}$. It estimates the

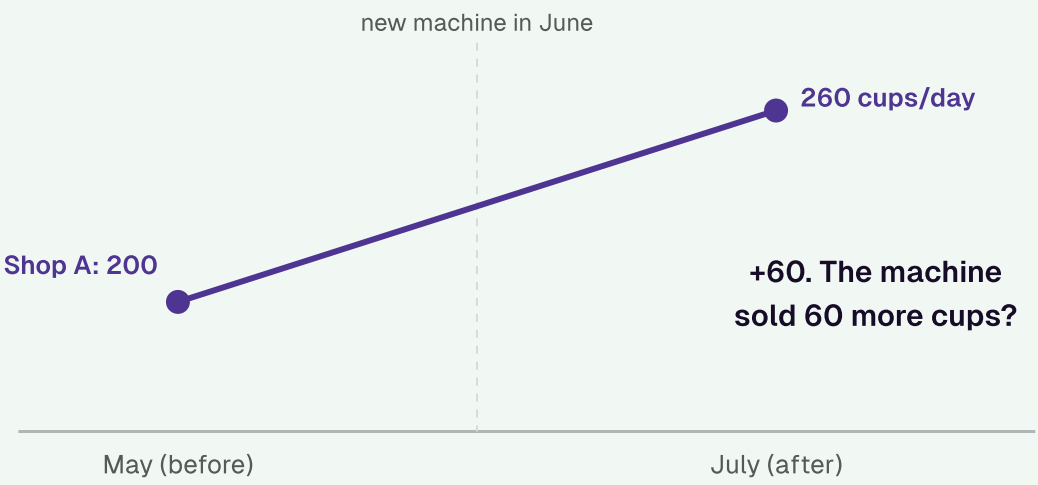
effect for compliers, the units the instrument actually moved, which may differ from the average user. A randomized encouragement is the cleanest instrument a product team can build.

Interrupted time series fits the pre-period (trend plus seasonality) and measures the post-period gap. It is the weakest design here, because anything else that changed on the same day is indistinguishable from the treatment. **Synthetic control** (Abadie, Diamond and Hainmueller, 2010, built a "synthetic California" to study a 1988 tobacco-control law) improves on it by learning donor weights $w_j \geq 0,\ \sum w_j = 1$ that reproduce the treated unit's pre-period path. Inference uses placebo runs: pretend each donor was treated and see whether the real gap is unusual.

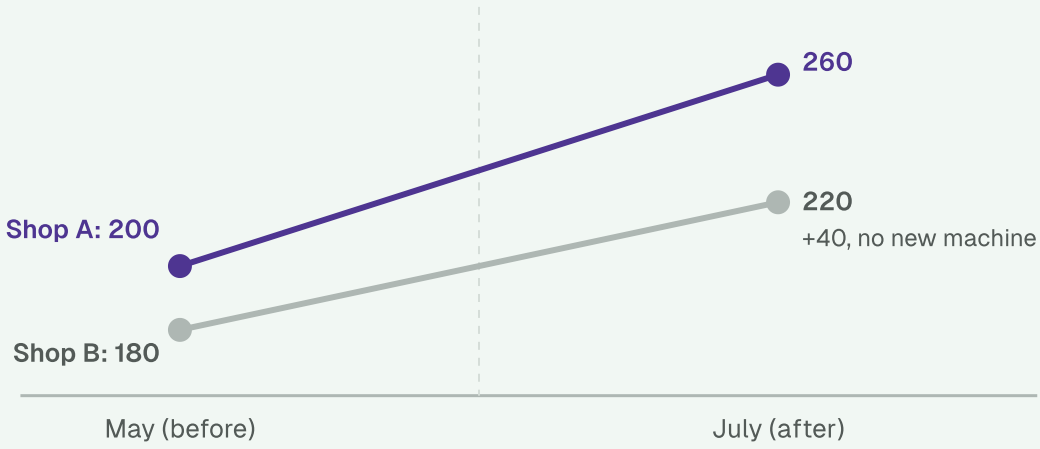
Diagram

Two coffee shops and a new espresso machine

Difference-in-differences, and the assumption it rests on.

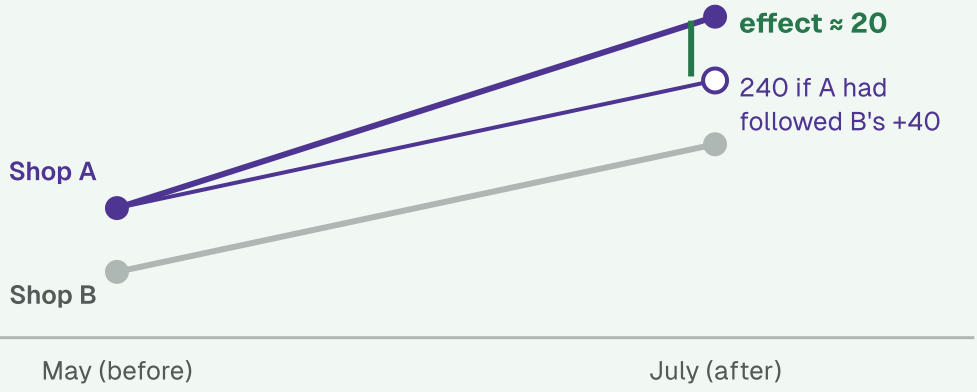


Shop A installs a fancy espresso machine in June. Daily sales rise from 200 cups in May to 260 in July. A before-and-after comparison credits the machine with 60 cups a day.

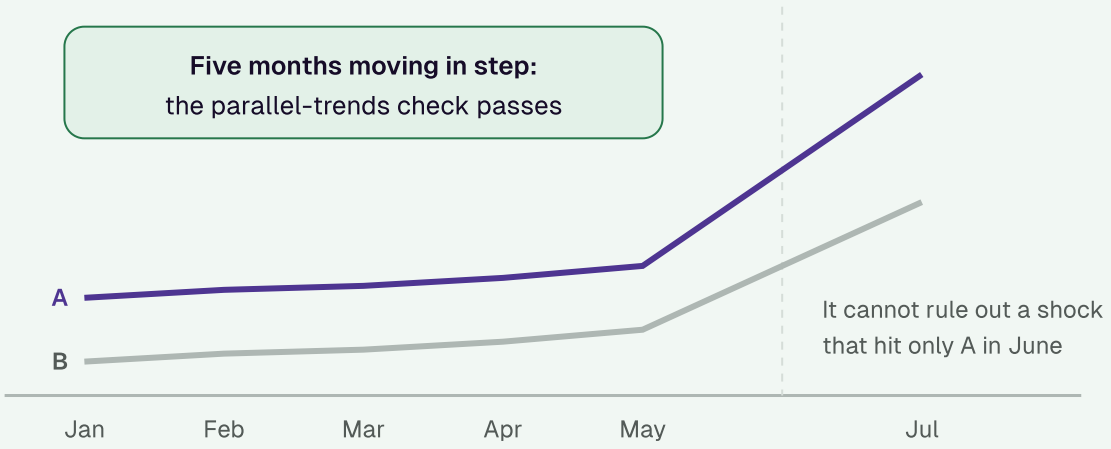


Shop B, across town, changed nothing. Its sales also rose, from 180 to 220. Summer, tourists, and a warm July lifted both shops. Some of Shop A's 60 cups would have come anyway.

$$(260 - 200) - (220 - 180) = 60 - 40 = 20 \text{ cups a day}$$



Difference-in-differences borrows Shop B's change as Shop A's "what would have happened". Shop A was expected to reach 240 without the machine, so the machine gets credit for 20 cups a day, a third of the naive number.

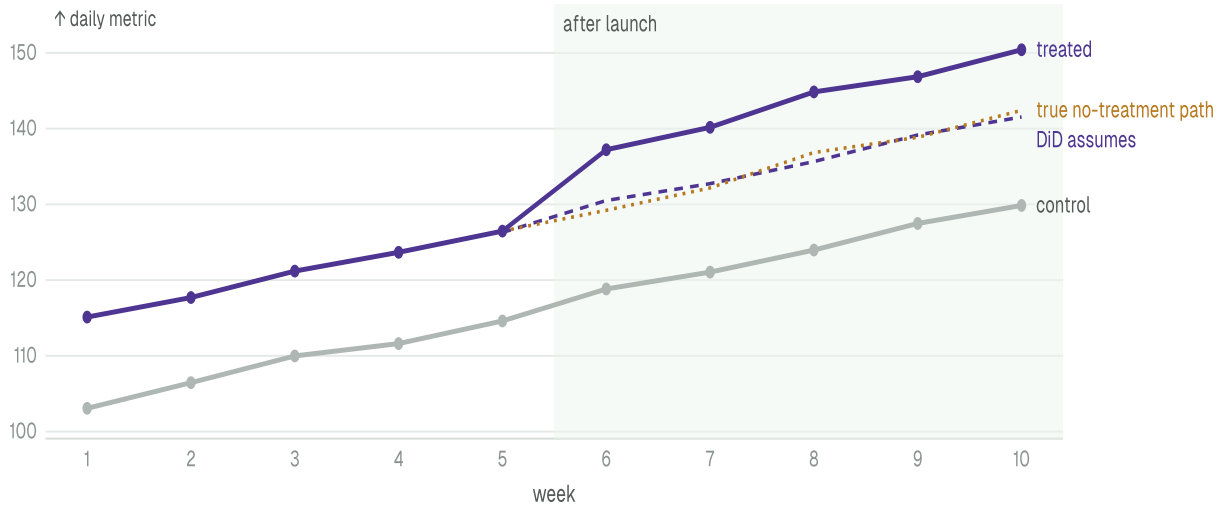


The method rests on one assumption: without the machine, both shops would have moved in parallel. You cannot check that for July, but you can check it for the months before. If the lines ran in step for months, the assumption is believable. A new office opening next to Shop A in June would still fool it.

Interactive

Difference-in-differences and parallel trends (shown at its default setting)

Two groups over ten weeks, launch after week 5. Add a pre-existing trend gap or a launch-week shock that only hits the treated group, and compare the DiD estimate with the truth. The pre-trend check catches one of the two violations and is blind to the other.



True effect	DiD estimate	Bias	Pre-trend slope gap
8.0	8.0	0.0	+0.04 ✓

DiD estimates **8.0** against a true effect of **8.0**. Both groups share a trend before launch, so the dashed path DiD assumes matches the true no-treatment path, and the only error is week-to-week noise.

Under the hood

A difference-in-differences estimate in code

The coffee-shop numbers as five days per cell. The three comparisons give three different answers, and only the last removes both the stable gap between shops and the shared summer trend.

python

```python
def mean(xs):
    return sum(xs) / len(xs)

# Daily cups sold, five days per cell (illustrative numbers)
a_before = [198, 203, 197, 201, 201]   # Shop A, May
a_after  = [262, 258, 259, 263, 258]   # Shop A, July (new machine in June)
b_before = [181, 178, 182, 179, 180]   # Shop B, May
```

```
b_after  = [219, 222, 218, 221, 220]   # Shop B, July (no change)

before_after = mean(a_after) - mean(a_before)        # mixes the machine with summer
a_vs_b       = mean(a_after) - mean(b_after)         # mixes the machine with the shops' stable gap
did = (mean(a_after) - mean(a_before)) - (mean(b_after) - mean(b_before))

print(f"{before_after:.1f}")   # 60.0
print(f"{a_vs_b:.1f}")         # 40.0
print(f"{did:.1f}")            # 20.0  the only one that removes both
```

rust

```
fn mean(xs: &[f64]) -> f64 {
    xs.iter().sum::<f64>() / xs.len() as f64
}

fn main() {
    // Daily cups sold, five days per cell (illustrative numbers)
    let a_before = [198.0, 203.0, 197.0, 201.0, 201.0]; // Shop A, May
    let a_after = [262.0, 258.0, 259.0, 263.0, 258.0];  // Shop A, July (new machine in June)
    let b_before = [181.0, 178.0, 182.0, 179.0, 180.0]; // Shop B, May
    let b_after = [219.0, 222.0, 218.0, 221.0, 220.0];  // Shop B, July (no change)

    let before_after = mean(&a_after) - mean(&a_before); // mixes the machine with summer
    let a_vs_b = mean(&a_after) - mean(&b_after);        // mixes the machine with the shops' stable gap
    let did = (mean(&a_after) - mean(&a_before)) - (mean(&b_after) - mean(&b_before));

    println!("{:.1}", before_after); // 60.0
    println!("{:.1}", a_vs_b);       // 40.0
    println!("{:.1}", did);          // 20.0  the only one that removes both
}
```

In a real analysis you would fit the regression form with one row per shop-day, cluster standard errors by shop, and, with only two shops, admit that the standard error is barely identified. With one treated and one control unit, DiD is an anecdote with arithmetic. Many groups, or staggered launch dates across groups, make it a test; with staggered timing use the newer estimators that avoid comparing late-treated groups against already-treated ones (Callaway and Sant'Anna, 2021, is a common choice).

Stratified and blocked randomization

Plain words

Deal the cards fairly within each suit

Suppose you split a class into two teams by coin flip, and by bad luck one team gets most of the tall kids. On average coin flips are fair, but any single split can be lopsided. If you already know height matters, line the kids up by height, take them two at a time, and flip a coin for each pair. Now each team gets one of every pair, and height can never be lopsided.

That is **blocking** (also called stratifying): group similar units first, then randomize inside each group. It removes bad luck on the things you blocked on, which makes the result more precise with the same number of people.

For leaders

Cheap precision

Blocking costs almost nothing and buys precision, so it is the default whenever a strong predictor of the outcome is known before assignment. Typical blocks at a lab: customer tier, language, platform (API versus app), heavy versus light users, task category in an offline eval. Precision gained is precision you do not have to buy with more traffic or more labelled items.

Two rules. Block on things measured before assignment, never after. And analyse the way you randomized: a blocked design analysed as if it were simple randomization still gives an unbiased answer, but its error bars are too wide, so you throw away what you paid for.

Principal asks

What predicts this outcome most strongly before treatment? Did we block or adjust on it (stratification, or regression adjustment such as CUPED on a pre-period metric)? Does the analysis use the blocks, or did we pay for precision and then discard it?

Under the hood

Why blocking reduces variance

Split the outcome variance into a part explained by the blocks and a part within blocks: $\sigma^2 = \sigma^2_{\text{between}} + \sigma^2_{\text{within}}$. With a 50/50 split of n units, the difference-in-means estimator has approximately

$$\text{Var}_{\text{simple}}(\hat{\tau}) \approx \frac{4\sigma^2}{n}, \qquad \text{Var}_{\text{blocked}}(\hat{\tau}) \approx \frac{4\sigma^2_{\text{within}}}{n}, \qquad \frac{\text{Var}_{\text{blocked}}}{\text{Var}_{\text{simple}}} \approx 1 - R^2_{\text{blocks}}$$

R^2_{blocks} is the share of outcome variance the blocks explain. If customer tier explains 40 percent of the variance in tokens used, blocking on tier cuts the variance by 40 percent, the same as 1.67 times the traffic.

The blocked estimator is the weighted average of within-block differences, $\hat{\tau} = \sum_h w_h(\bar{Y}_{h,1} - \bar{Y}_{h,0})$, with w_h the block's share of units. Matched pairs are the extreme case: each block holds two

similar units. The paired eval design (both models answer the same prompts) is exactly this, with the prompt as the block, which is why paired tests are so much more powerful in statistics for eval sets. Post-stratification and regression adjustment on pre-period covariates recover most of the same gain after the fact when blocking was not possible at assignment.

Choosing the unit of randomization

Plain words

What does the coin decide about?

A coin can decide about each question someone asks, each conversation, each person, or each whole company. Flip per question and the same person might get the new model on one question and the old one on the next, which feels broken and mixes up the results. Flip per company and everyone at the company gets the same experience, but you have far fewer coin flips, so the answer is less precise. Pick the unit at which the change makes sense, then count your sample in those units.

For leaders

The trade-off table

Unit	Use when	Main risk	Analysis must
Request	Stateless backend changes users cannot notice: a kernel, a cache, a decoding tweak	Inconsistent experience; users who repeat dominate the sample	Cluster standard errors by user if users send many requests
Session or conversation	The change matters within one conversation: a system prompt, a memory policy	Same user in both arms across sessions; carryover between them	Cluster by user
User or account	Default for product and model changes; long-term outcomes like retention	Fewer units than requests; shared accounts	Treat the user as the observation; per-user metrics or cluster-robust errors
Cluster: org, team, region, time block	Users interact, share content, or share capacity	Few clusters, so low power; cluster sizes vary	Analyse at cluster level or with cluster-robust errors; count clusters, not users

The rule that catches most mistakes: **the analysis unit must respect the randomization unit.** Randomize by user and compute a per-request t-test, and your error bars are too narrow, often several times too narrow. The fix is in intervals and clustering.

Under the hood

The design effect

When units are randomized in clusters of average size m with intraclass correlation ρ (how alike observations in the same cluster are), the variance of a mean is inflated by the design effect

$$\mathrm{DE} = 1 + (m - 1)\,\rho, \qquad n_{\text{eff}} = \frac{n}{\mathrm{DE}}$$

Illustrative: 50 requests per user with $\rho = 0.2$ gives DE = 10.8. A million requests from 20,000 users carry the information of about 93,000 independent requests.

Ratio metrics such as "success rate per request" when randomized by user are the most common trap: the numerator and denominator are both per-user sums, and the right variance comes from the delta method or a user-level bootstrap (see measuring the right thing). Smaller units buy power only when the treatment cannot leak between them; when it can, the extra power is fake.

Interference and spillover

Plain words

When the control group is not left alone

Test a new vaccine on half a village and the unvaccinated half also gets sick less, because there is less disease going around. Comparing the halves now understates how well the vaccine works. The control group was affected by the treatment group.

The same thing happens with software. If half of a team gets an AI assistant that writes great documents, the other half reads those documents too. If the new model is slower and hogs the shared servers, the old model's users see slower answers too. Either way, the comparison mixes up the two groups.

For leaders

Where spillover hides at an AI lab

- **Shared capacity.** A treatment model that uses more GPU time slows the control arm on the same pool, so the latency comparison is wrong in both directions. Give each arm its own capacity or randomize in time.
- **Shared content.** Workspace features, shared prompts, team knowledge bases, and documents written by one arm and read by the other. Randomize by workspace or org.
- **Marketplaces and rate limits.** If treatment users consume more of a shared quota or a shared queue, control users are starved.
- **Learning effects.** Users who see a better model change their habits, and carry those habits into later sessions on the other arm.

The fix is to randomize at a level where the spillover stays inside the unit: whole orgs (cluster randomization), or the whole system flipping between A and B in time blocks (a switchback). Both cost power, so they need longer tests. The question to ask is whether the effect you care about includes the spillover; at launch, everyone gets the treatment, so the clean number is the all-treated versus all-control comparison.

Under the hood

SUTVA, clusters, and switchbacks

The potential-outcomes notation $Y_i(1), Y_i(0)$ quietly assumes the **stable unit treatment value assumption** (SUTVA, Rubin's term): unit i's outcome depends only on its own treatment, not on anyone else's (no interference), and there is only one version of each treatment (no hidden variations). With interference, outcomes depend on the whole assignment vector, $Y_i(\mathbf{t})$, and the simple difference in means estimates neither the effect of treating everyone nor anything else well defined.

- **Cluster randomization** treats the cluster as the unit. Interference across clusters must be small; analysis is at the cluster level.
- **Switchback designs** randomize the whole system in time blocks (for example, alternate A and B every hour per region). They handle shared-capacity interference, but blocks must be long enough for carryover to wash out, and the analysis should drop or model the period right after each switch. Bojinov, Simchi-Levi and Zhao (Management Science, 2023) give the design and analysis, including how to choose block length when carryover lasts a known number of periods.
- **Detecting interference**: run the experiment at two treated fractions (for example 10 percent and 50 percent of clusters) and see whether the control arm's outcome moves with the treated fraction.

Checking balance after randomization

Plain words

Check the coin before trusting it

Randomization done right should give two groups that look alike on everything you measured beforehand and are about the size you planned. Before reading the result, look. If you planned a 50/50 split and got 52/48 on a million users, something in the pipeline is eating people from one arm, and the result cannot be trusted until you find out why.

For leaders

Three checks before anyone sees the effect

1. **Sample ratio.** Do the arm sizes match the planned split? A mismatch (sample ratio mismatch, SRM) means users were lost or added unevenly: a crash in one arm, a redirect, a bot filter that reacts to the treatment. Kohavi, Tang and Xu (*Trustworthy Online Controlled Experiments*, 2020) treat SRM as a blocking failure: do not read the results.

2. **Covariate balance.** Compare pre-period metrics, tier, platform, and tenure between arms, as standardised differences. A large imbalance on a pre-period metric points at broken

assignment or exposure logging.

3. **A/A test.** Run the system with both arms identical. It should show no effect about 95 percent of the time. If it shows "effects" more often, the pipeline or the variance formula is wrong.

Under the hood

Standardised differences and the SRM test

The standardised mean difference for covariate X is $\mathrm{SMD} = (\bar{x}_T - \bar{x}_C)/\sqrt{(s_T^2 + s_C^2)/2}$. A common rule of thumb flags $|\mathrm{SMD}| > 0.1$. Use SMDs rather than p-values: with millions of users trivial differences are "significant", and with a few hundred large ones are not. Clinical-trial reporting guidance (CONSORT) discourages significance tests of baseline differences in randomized trials for the same reason: if randomization was done properly, any difference is chance by construction. If a pre-period covariate is imbalanced, adjust for it in the analysis (regression adjustment), which also shrinks the variance.

The SRM check is a chi-square goodness-of-fit test on the arm counts. Teams usually alarm at a strict threshold, such as $p < 0.001$, because they run it on every experiment.

python

```python
import math

def srm_check(n_treat, n_control, planned_share=0.5):
    """Chi-square test (1 df) that arm sizes match the planned split."""
    n = n_treat + n_control
    exp_t, exp_c = n * planned_share, n * (1 - planned_share)
    chi2 = (n_treat - exp_t) ** 2 / exp_t + (n_control - exp_c) ** 2 / exp_c
    p = math.erfc(math.sqrt(chi2 / 2))    # survival function of chi-square with 1 df
    return chi2, p

chi2, p = srm_check(50_421, 49_579)       # planned 50/50, observed 50.4/49.6
print(f"{chi2:.2f} {p:.4f}")              # 7.09 0.0078  small skew, but n is large: investigate
chi2, p = srm_check(5_042, 4_958)
print(f"{chi2:.2f} {p:.4f}")              # 0.71 0.4009  same 50.4/49.6 skew, consistent with chance
```

rust

```rust
// erfc via the Numerical Recipes rational approximation (relative error below 1.2e-7),
// accurate enough to match Python's math.erfc to the printed digits here.
fn erfc(x: f64) -> f64 {
    let z = x.abs();
    let t = 1.0 / (1.0 + 0.5 * z);
    let r = t * (-z * z - 1.26551223 + t * (1.00002368 + t * (0.37409196 + t * (0.09678418
        + t * (-0.18628806 + t * (0.27886807 + t * (-1.13520398 + t * (1.48851587
        + t * (-0.82215223 + t * 0.17087277)))))))).exp();
    if x >= 0.0 { r } else { 2.0 - r }
}

/// Chi-square test (1 df) that arm sizes match the planned split.
fn srm_check(n_treat: f64, n_control: f64, planned_share: f64) -> (f64, f64) {
    let n = n_treat + n_control;
```

```
    let (exp_t, exp_c) = (n * planned_share, n * (1.0 - planned_share));
    let chi2 = (n_treat - exp_t).powi(2) / exp_t + (n_control - exp_c).powi(2) / exp_c;
    let p = erfc((chi2 / 2.0).sqrt()); // survival function of chi-square with 1 df
    (chi2, p)
}

fn main() {
    let (chi2, p) = srm_check(50_421.0, 49_579.0, 0.5); // planned 50/50, observed 50.4/49.6
    println!("{:.2} {:.4}", chi2, p);                   // 7.09 0.0078  small skew, but n is large: investigate
    let (chi2, p) = srm_check(5_042.0, 4_958.0, 0.5);
    println!("{:.2} {:.4}", chi2, p);                   // 0.71 0.4009  same 50.4/49.6 skew, consistent with
chance
}
```

⚠ Common mistakes

- **Calling an opt-in comparison an experiment.** "Users who enabled X" versus "users who did not" is observational, however many users it has.
- **Testing a fix on the worst items with no control.** Regression to the mean guarantees an improvement. Re-run the old system on the same items.
- **Filtering after assignment.** Dropping users who did not engage, or who churned, breaks randomization. Analyse as assigned.
- **Randomizing by user, analysing by request.** Error bars come out several times too narrow. Cluster by the randomization unit.
- **Re-rolling the randomization until balance looks good** without accounting for it in the analysis.
- **Ignoring shared capacity.** Two arms on one GPU pool interfere on latency and throughput. Split capacity or use a switchback.
- **Running a quasi-experiment without its check.** DiD without a pre-trend plot, matching without a balance table, RD without a density check at the cutoff.

Principal asks

- How exactly was treatment assigned, and could anything that predicts the outcome have influenced it?
- What is the randomization unit, and does the variance calculation use it?
- Did the sample ratio match the plan? What does the balance table show on pre-period metrics?
- Can the treatment reach the control group through shared capacity, shared content, or shared users?
- If this is not randomized: what is the identifying assumption, what check supports it, and what would a placebo test show?
- Is the reported number intention-to-treat, or was the population filtered after assignment?

Part VI · Chapter 44

Measuring the right thing: data types and validity

A perfectly randomized experiment still fails if the number at the end is the wrong kind of number, analysed with the wrong formula, or only loosely connected to what you care about. This chapter covers the four levels of measurement and what each one allows, the long argument about averaging star ratings, the special data types that each need their own formula (binary, counts, censored durations, ratios, heavy tails, judge scores), the difference between a measurement that is consistent and one that is correct, how measurement error quietly shrinks effects, and the gap between a proxy and the target it stands in for.

Diagram

Four numbers from one marathon

Nominal, ordinal, interval, ratio: what each kind of number lets you do.

"Average bib number: 2133" means nothing.

You can only count: 2 Harriers, 1 Strider. Most common club: Harriers.

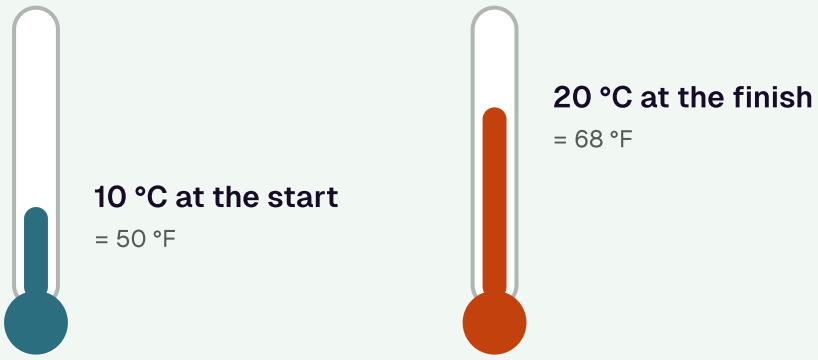
Nominal: numbers or words used as names

Nominal. Bib numbers and club names are labels. Numbers on a bib could be swapped for colours without losing anything. You can count them and find the most common one; you cannot average them.

1st to 2nd: 2 seconds. 2nd to 3rd: 4 minutes 28 seconds.

Ordinal: order is real, distances are not

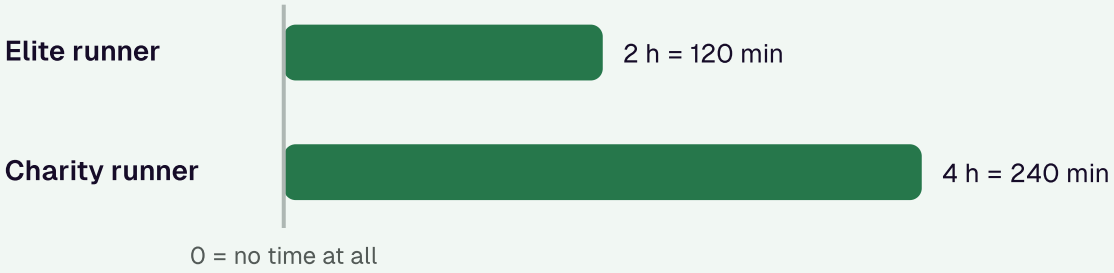
Ordinal. Finishing places have a real order, but the gaps between them mean nothing: second place was two seconds behind the winner, third was almost four and a half minutes behind. Medians and ranks are fine. "Average place" quietly treats every gap as equal.



10 degrees warmer: yes. Twice as warm: no (50 °F to 68 °F is not double).

Interval: equal steps, but zero is an arbitrary choice

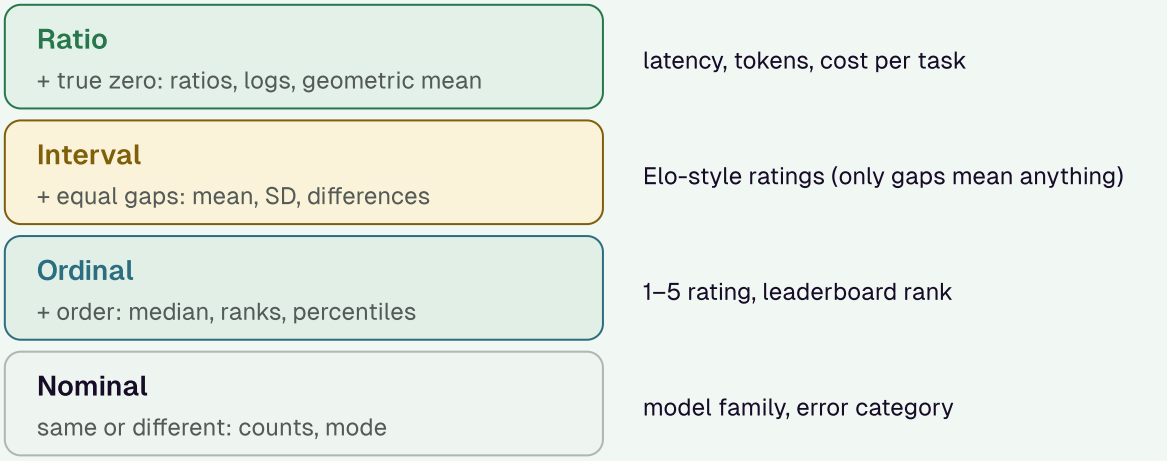
Interval. Temperature in Celsius has equal steps, so "10 degrees warmer" means the same everywhere on the scale and averages make sense. Zero is just the freezing point of water, so ratios do not: in Fahrenheit the same two days are 50 and 68, nowhere near double.



Twice as long in hours, minutes, or seconds.

Ratio: equal steps and a true zero, so "times as much" is meaningful

Ratio. Finishing time has a true zero, so "twice as long" holds in any unit. Ratios, percentage changes, geometric means, and log scales all make sense. Latency, tokens, and cost per task are ratio data.



Each step up keeps every tool from the steps below and adds new ones.

The four levels form a ladder. Each level allows everything the levels below allow, plus more. The level of your metric decides which summaries are honest and which tests answer the question you meant to ask.

Levels of measurement

Plain words

Not every number can be averaged

Some numbers are names (a jersey number, a postcode). Some numbers are positions in a line (first, second, third; one to five stars). Some numbers measure in equal steps but start from an arbitrary zero (temperature in Celsius, a chess rating). And some measure amounts with a real zero (time, money, weight).

The kind of number decides what you can sensibly do with it. Averaging postcodes is nonsense. Averaging star ratings is debatable, because the step from 4 to 5 stars may not equal the step from 1 to 2. Averaging response times is fine, though often not the best summary, because a few slow ones drag the average.

For leaders

Why this shows up in reviews

Most eval dashboards mix all four levels and summarise them all with an average. That is where wrong calls come from: a model whose "average helpfulness" rose from 3.8 to 3.9 may have gained many 4s and also gained more 1s, which is a safety regression hiding inside an improvement. A leaderboard rank that moved from 5th to 3rd may reflect a 0.2-point gap or a 6-point gap.

Ask three questions of every headline metric: what kind of number is it, does the summary respect that, and does the comparison test what the decision needs (an average, a typical case, or the share of bad outcomes)? The answer often changes the metric from a mean to a share ("percent rated 4 or 5", "percent rated 1") that is both more honest and easier to act on.

Under the hood

Permissible transformations and what each level supports

Stevens (Science, 1946) defined the levels by the transformations that leave a scale's information unchanged. A statistic is meaningful on a scale if its conclusions survive every permissible transformation.

Level	Permissible transformation	Meaningful summaries	Comparing two groups	Ever
Nominal	Any one-to-one relabelling	Counts, proportions, mode	Chi-square or Fisher exact on the table of counts	Bloc cate
Ordinal	Any strictly increasing function	Median, percentiles, proportions at or above a level	Mann–Whitney, Wilcoxon signed-rank or sign test (paired), ordinal regression	Star scor
Interval	$x \mapsto ax + b,\ a > 0$	Mean, SD, differences	t-test, ANOVA, linear regression	Cels Brac
Ratio	$x \mapsto ax,\ a > 0$	All of the above plus ratios, geometric mean, coefficient of variation	t-test on logs (ratio of geometric means), rate ratios, bootstrap	Weig toke

A quick test: apply a monotone transformation that preserves order (for example, recode 1–5 as 1, 2, 3, 4, 10) and see whether your conclusion survives. Medians and rank tests survive by construction. Means and t-tests can reverse, which is the formal content of "ordinal data has no meaningful mean". The reading a shape checklist covers how to recognise the level from the data itself; from shape to method maps shapes to summaries.

The Likert debate: can you average star ratings?

Plain words

Two restaurants, both 3.0 stars

Restaurant A gets three stars from everyone: fine, forgettable. Restaurant B gets five stars from half its diners and one star from the other half: some love it, some got food poisoning. Both average 3.0 stars. The average says they are the same. Anyone who has to pick a restaurant for their in-laws knows they are not.

The average can be fine for ratings. It throws away the shape, though, and the shape is often the story.

For leaders

A practical policy

The argument has run for decades. One camp holds that ordinal data must never be averaged (Jamieson, Medical Education, 2004, is a widely cited statement). Another holds that t-tests and means are robust enough that averaging ratings rarely leads you astray in practice (Norman, Advances in Health Sciences Education, 2010). A third line of work shows concrete failure cases: Liddell and Kruschke (Journal of Experimental Social Psychology, 2018) demonstrate that treating ordinal ratings as metric can produce false alarms, missed effects, and even effects in the wrong direction, especially when groups differ in spread or pile up at the top or bottom of the scale.

A policy that holds up in review: report the full distribution and the share at the top and bottom of the scale as the primary numbers. Averages are acceptable as a secondary summary when the score is a sum or mean of many items (a multi-item scale behaves much more like interval data), the distributions are similar in shape, and nothing piles up at a floor or ceiling. For a single 1–5 rating where a tail matters (safety, harmful answers), never let the mean be the only number.

Under the hood

What each test actually tests

The t-test compares means and assumes the gaps between categories are equal. The Mann–Whitney U test compares whole distributions by ranks: it estimates $P(B > A) + \frac{1}{2}P(B = A)$ for a random draw from each group, the probability of superiority, which equals the AUC of separating the groups. It is invariant to any order-preserving recoding. It is a test of medians only under the extra assumption that the two distributions have the same shape.

$$U_B = \sum_{i \in B} R_i - \frac{n_B(n_B+1)}{2}, \qquad \hat{\theta} = \frac{U_B}{n_A n_B}, \qquad \sigma_U^2 = \frac{n_A n_B}{12}\left[(N+1) - \frac{\sum_k (t_k^3 - t_k)}{N(N-1)}\right.$$

R_i are mid-ranks in the pooled sample of size $N = n_A + n_B$; t_k is the number of tied values at category k. With five categories the tie correction matters a lot. $z = (U_B - n_A n_B/2)/\sigma_U$.

The model-based alternative is **ordinal (cumulative logit or probit) regression**: assume a latent continuous score cut into categories by thresholds, and estimate the shift in the latent score. It handles covariates, clustering (with random effects), and unequal spread, and it gives an effect on an interpretable scale. For pairwise preferences (A versus B on the same prompt), use a sign test or Bradley–Terry instead (see LLM as judge).

python

```python
import math

def mann_whitney(a, b):
    """U for group b, tie-corrected z, two-sided normal p, and P(b beats a)."""
    pooled = sorted(a + b)
    rank = {}                       # mid-rank for each distinct value
    i = 0
    while i < len(pooled):
        j = i
        while j < len(pooled) and pooled[j] == pooled[i]:
            j += 1
        rank[pooled[i]] = (i + 1 + j) / 2     # average of positions i+1 .. j
        i = j
    na, nb, n = len(a), len(b), len(a) + len(b)
    u = sum(rank[v] for v in b) - nb * (nb + 1) / 2
    ties = sum(pooled.count(v) ** 3 - pooled.count(v) for v in set(pooled))
    var = na * nb / 12 * ((n + 1) - ties / (n * (n - 1)))   # ties shrink the variance
    z = (u - na * nb / 2) / math.sqrt(var)
    p = math.erfc(abs(z) / math.sqrt(2))
    return u, z, p, u / (na * nb)

a = [3, 4, 2, 5, 3, 4, 3, 2, 4, 3]   # model A, 1-5 ratings (illustrative)
b = [4, 5, 4, 3, 5, 4, 5, 4, 3, 5]   # model B
u, z, p, theta = mann_whitney(a, b)
print(f"{u:.1f} {z:.3f} {p:.4f} {theta:.2f}")   # 76.0 2.054 0.0400 0.76
```

rust

```rust
// erfc via the Numerical Recipes rational approximation (relative error below 1.2e-7).
fn erfc(x: f64) -> f64 {
    let z = x.abs();
    let t = 1.0 / (1.0 + 0.5 * z);
    let r = t * (-z * z - 1.26551223 + t * (1.00002368 + t * (0.37409196 + t * (0.09678418
        + t * (-0.18628806 + t * (0.27886807 + t * (-1.13520398 + t * (1.48851587
        + t * (-0.82215223 + t * 0.17087277)))))))).exp();
    if x >= 0.0 { r } else { 2.0 - r }
}
```

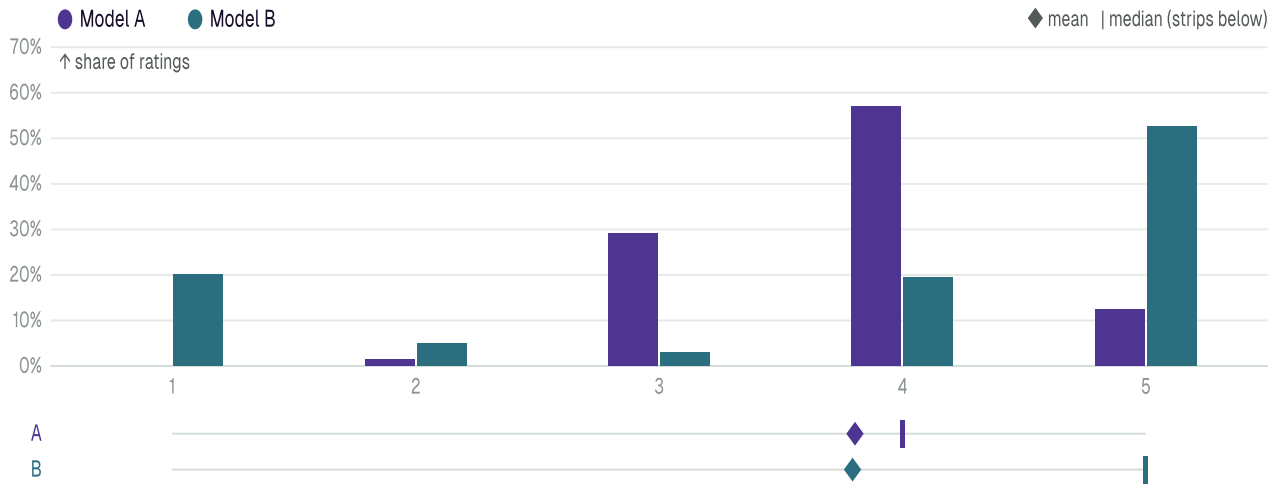
```rust
/// U for group b, tie-corrected z, two-sided normal p, and P(b beats a).
fn mann_whitney(a: &[i32], b: &[i32]) -> (f64, f64, f64, f64) {
    let mut pooled: Vec<i32> = a.iter().chain(b.iter()).cloned().collect();
    pooled.sort();
    let mut rank = std::collections::HashMap::new(); // mid-rank for each distinct value
    let mut ties = 0.0;
    let mut i = 0;
    while i < pooled.len() {
        let mut j = i;
        while j < pooled.len() && pooled[j] == pooled[i] {
            j += 1;
        }
        rank.insert(pooled[i], (i + 1 + j) as f64 / 2.0); // average of positions i+1 .. j
        let t = (j - i) as f64;
        ties += t * t * t - t;
        i = j;
    }
    let (na, nb) = (a.len() as f64, b.len() as f64);
    let n = na + nb;
    let u = b.iter().map(|v| rank[v]).sum::<f64>() - nb * (nb + 1.0) / 2.0;
    let var = na * nb / 12.0 * ((n + 1.0) - ties / (n * (n - 1.0))); // ties shrink the variance
    let z = (u - na * nb / 2.0) / var.sqrt();
    let p = erfc(z.abs() / 2f64.sqrt());
    (u, z, p, u / (na * nb))
}

fn main() {
    let a = [3, 4, 2, 5, 3, 4, 3, 2, 4, 3]; // model A, 1-5 ratings (illustrative)
    let b = [4, 5, 4, 3, 5, 4, 5, 4, 3, 5]; // model B
    let (u, z, p, theta) = mann_whitney(&a, &b);
    println!("{:.1} {:.3} {:.4} {:.2}", u, z, p, theta); // 76.0 2.054 0.0400 0.76
}
```

Read $\hat{\theta} = 0.76$ as: a random rating of model B beats a random rating of model A about three times in four, counting ties as half. With ten ratings each this is a borderline result, and the ratings would need to come from different prompts and raters for the independence assumption to hold.

Interactive

Same mean, different story (shown at its default setting)

Two models rated 1 to 5, with means forced to match. Model A's ratings bunch in the middle; model B delights most users and fails an unhappy minority badly. Move the sliders and compare what a t-test and a rank test conclude.



Means A / B	Medians A / B	t-test p	Mann–Whitney p
3.81 / 3.80	4 / 5	0.935	<0.001

P(B rated higher)	Rated 1: A / B
61%	0% / 20%

Both models average about **3.81**, so the t-test sees nothing ($p = 0.93$). The rank test asks a different question, whether a random B rating tends to beat a random A rating, and answers 61% ($p < 0.001$). Meanwhile B gives a 1 to **20%** of users against 0% for A. Neither test is wrong. Which model is better depends on whether the decision cares about the typical user, the average, or the users B fails, and only the full distribution shows all three.

Eval lens

Judge scores and human ratings in LLM evals are almost always ordinal, and the decision almost always cares about a tail: how often is the answer harmful, wrong, or useless? Report the distribution, the share at the bottom of the scale, and a rank-based comparison (Mann–Whitney for independent ratings, Wilcoxon signed-rank or a sign test when both models answered the same prompts). If a mean is on the slide, put the share of 1s next to it. A model can raise the average and the share of disasters at the same time.

Special data types and their right formulas

Plain words

Each kind of data has its own ruler

A yes/no result, a count of things, a waiting time where some people are still waiting, a ratio of two totals, a cost where one customer spends a thousand times more than the rest, a grade from a judge: each looks like "a number per item", and each breaks the simple average-and-error-bar formula in its own way. Using the matching formula is the difference between an error bar you can trust and one that is off by a factor of three.

For leaders

The review checklist by data type

Data type	Example	Summarise with	The question to ask
Binary	Task passed, refusal, thumbs up	Proportion with a Wilson interval	How many items, and were they independent or nested in conversations?
Counts	Tool calls per session, errors per day	Rate, with the variance-to-mean ratio	Is the variance far above the mean? Then Poisson error bars are too narrow.
Durations with censoring	Time to task completion with a step limit, time to churn	Kaplan–Meier curve and median	What happened to the runs that hit the limit or were still going when data ended?
Ratio metrics	Success per request when randomized by user; tokens per answer	Ratio of totals, delta-method or bootstrap interval	Was the variance computed at the randomization unit?
Heavy-tailed	Cost per task, latency, output length	Median and high percentiles; geometric mean	How much of the total comes from the top 1 percent?
Judge scores	LLM judge 1–10, rubric points	Distribution, share at each level, paired comparison	What is the judge's agreement with humans and with itself?

Under the hood

The formulas

Binary. A proportion $\hat{p} = x/n$ has variance $p(1-p)/n$. Use the Wilson interval rather than the plus-or-minus normal interval, which misbehaves near 0 and 1 and at small n; details and the paired McNemar test are in statistics for eval sets.

Counts. Poisson assumes variance equals mean. Real usage counts are usually overdispersed (variance far above mean), which the negative binomial models with $\mathrm{Var}(Y) = \mu + \mu^2/k$. Compare groups with a negative binomial regression and report the rate ratio e^{β}. See the distribution zoo.

Durations with censoring. If a run hit a time limit at 30 minutes, you know only that its true duration exceeds 30. Dropping it biases the average down; recording it as 30 also biases it down. The Kaplan–Meier estimator (Kaplan and Meier, JASA, 1958) uses each unit for as long as it was observed:

$$\hat{S}(t) = \prod_{t_i \le t} \left(1 - \frac{d_i}{n_i}\right)$$

t_i are the distinct event times, d_i the number of events at t_i, and n_i the number still at risk (not yet failed and not yet censored) just before t_i. Compare two curves with the log-rank test; model covariates with Cox regression. The key assumption is that censoring is unrelated to the outcome, given covariates.

Ratio metrics. For a metric like clicks per page view, randomized by user, the numerator X_u and denominator Y_u are per-user sums. The delta method gives the variance of the ratio of means $\bar{X}/\bar{Y}$ over n users:

$$\operatorname{Var}\left(\frac{\bar{X}}{\bar{Y}}\right) \approx \frac{1}{n\,\mu_Y^2}\left(\sigma_X^2 - 2\frac{\mu_X}{\mu_Y}\sigma_{XY} + \frac{\mu_X^2}{\mu_Y^2}\sigma_Y^2\right)$$

μ, σ^2 are per-user means and variances, σ_{XY} the per-user covariance. Deng, Knoblich and Lu (KDD, 2018) give a practical guide for online experiments. A user-level bootstrap gives the same answer more slowly; treating page views as independent gives error bars that are too narrow.

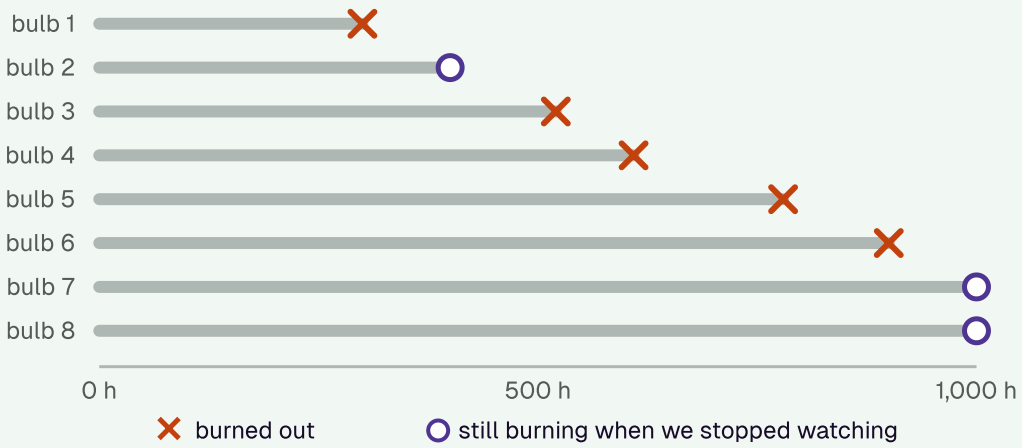
Heavy tails. When a few values dominate the sum, the mean converges slowly and its interval is unreliable. Report percentiles, compare on a log scale (a ratio of geometric means), or trim or winsorize with the rule fixed in advance. See long tails and outliers.

Judge scores. Treat them as ordinal, check their reliability (below), and prefer paired designs: both models answer the same prompts and the judge scores both, ideally in a randomized order. Calibration, position bias, and agreement with humans are covered in LLM as judge.

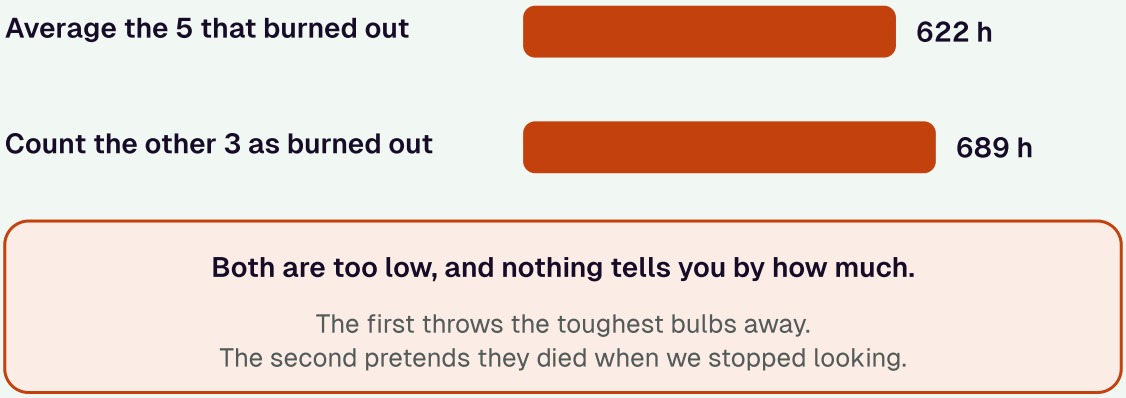
Diagram

Light bulbs and a test that ended early

Censoring, and how Kaplan–Meier uses every bulb for as long as it was watched.



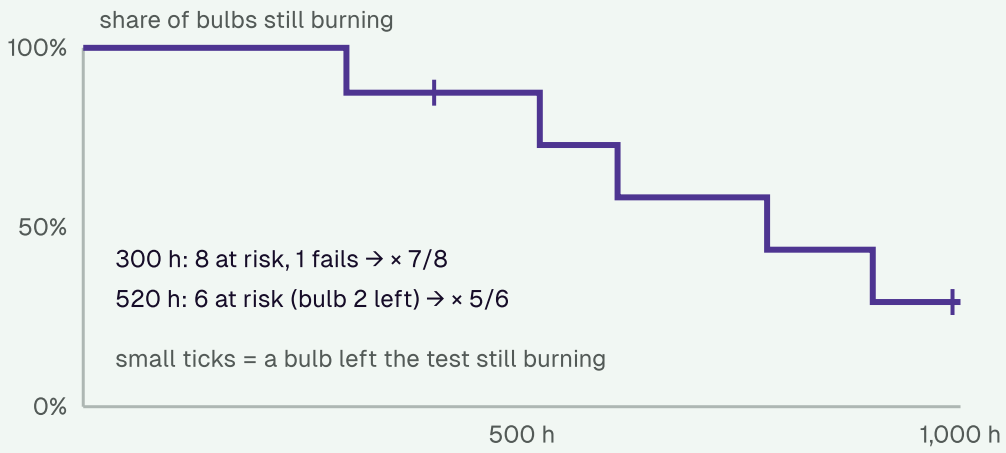
A lab tests eight bulbs. The test ends at 1,000 hours, so bulbs 7 and 8 are still burning. Bulb 2's fixture broke at 400 hours and it was removed. We know when five bulbs burned out; for the other three we only know they lasted at least that long.



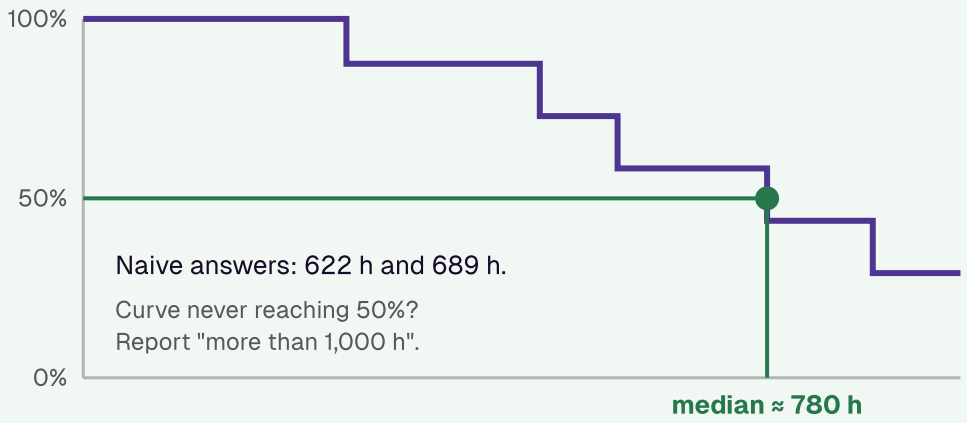
Both are too low, and nothing tells you by how much.

The first throws the toughest bulbs away.
The second pretends they died when we stopped looking.

The two obvious shortcuts both understate how long bulbs last. Ignoring unfinished bulbs throws away the strongest ones. Counting them as failures at the cut-off pretends the test's end was their end.



Kaplan–Meier walks forward in time. At each burnout it asks: of the bulbs still being watched right now, what share failed? It multiplies those survival fractions together. Bulb 2 counts for its first 400 hours and then quietly leaves the "at risk" group, so it is neither thrown away nor counted as a failure.



The median lifetime is where the curve drops through 50 percent: about 780 hours, well above both shortcuts. If more bulbs had outlasted the test and the curve stayed above 50 percent, the honest answer would be "median above 1,000 hours", which is still more informative than a wrong number.

Report the censoring rate, plot the survival curve, compare with a log-rank test.

The same pattern appears whenever a clock stops before the story does: agent runs that hit a step cap, requests that time out, users who have not churned yet. Report how many were cut off, plot survival curves, and compare them with a log-rank test instead of comparing averages of the finished ones.

Under the hood

Kaplan–Meier in a few lines

The light-bulb data from the story. Each record is a time and whether the bulb burned out (1) or was censored (0).

python

```python
def kaplan_meier(times, events):
    """Survival curve as (time, S(t)) at each event time. Tied events are processed one
    record at a time, which gives the same product; at a shared time, events come before
    censorings, because a unit censored at t was still at risk at t."""
    data = sorted(zip(times, events), key=lambda r: (r[0], -r[1]))
    at_risk, s, curve = len(data), 1.0, []
    for t, burned_out in data:
        if burned_out:
            s *= (at_risk - 1) / at_risk      # share of the at-risk group that survives this moment
            curve.append((t, s))
        at_risk -= 1                          # failed or censored, it leaves the risk set either way
    return curve

times  = [300, 400, 520, 610, 780, 900, 1000, 1000]
events = [1,   0,   1,   1,   1,   1,   0,    0]    # 0 = still burning when observation stopped
for t, s in kaplan_meier(times, events):
    print(t, f"{s:.4f}")
# 300 0.8750
# 520 0.7292
# 610 0.5833
```

```
# 780 0.4375   first time below 0.5, so the median is 780 h
# 900 0.2917
```

rust

```rust
/// Survival curve as (time, S(t)) at each event time. Tied events are processed one
/// record at a time, which gives the same product; at a shared time, events come before
/// censorings, because a unit censored at t was still at risk at t.
fn kaplan_meier(times: &[u32], events: &[u8]) -> Vec<(u32, f64)> {
    let mut data: Vec<(u32, u8)> = times.iter().cloned().zip(events.iter().cloned()).collect();
    data.sort_by(|a, b| a.0.cmp(&b.0).then(b.1.cmp(&a.1)));
    let (mut at_risk, mut s, mut curve) = (data.len() as f64, 1.0, Vec::new());
    for (t, burned_out) in data {
        if burned_out == 1 {
            s *= (at_risk - 1.0) / at_risk; // share of the at-risk group that survives this moment
            curve.push((t, s));
        }
        at_risk -= 1.0; // failed or censored, it leaves the risk set either way
    }
    curve
}

fn main() {
    let times = [300, 400, 520, 610, 780, 900, 1000, 1000];
    let events = [1, 0, 1, 1, 1, 1, 0, 0]; // 0 = still burning when observation stopped
    for (t, s) in kaplan_meier(&times, &events) {
        println!("{} {:.4}", t, s);
    }
    // 300 0.8750
    // 520 0.7292
    // 610 0.5833
    // 780 0.4375   first time below 0.5, so the median is 780 h
    // 900 0.2917
}
```

Interactive **Test chooser by data type** (shown at its default setting)

Pick the kind of data, how many groups, and whether the same items appear in every group. You get a test, the summary to report, the assumption it rests on, and its main drawback. For metric choice see which metric or test; for shapes see from shape to method.

What kind of data is the outcome?

How many groups?

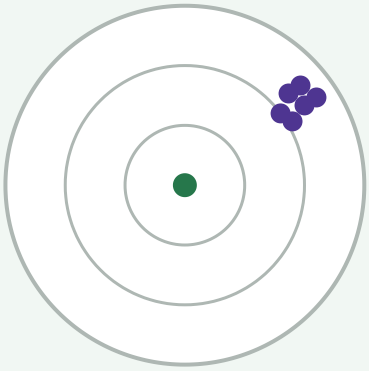
Do the same items or users appear in every group?

TEST	**Two-proportion z test, or Fisher's exact test when any cell is small**
REPORT	Each group's rate with its count and a Wilson interval; the difference with a Newcombe or bootstrap interval.
ASSUMES	Items are independent. Tasks nested in conversations or repeated seeds are not; resample the clusters.
DRAWBACK	Chi-square approximations fail when expected counts fall below about 5. An unpaired test on paired runs throws away most of the power.
MORE	Statistics for eval sets

Reliability versus validity

Diagram **The bathroom scale and the dartboard**

Consistent is not the same as correct.



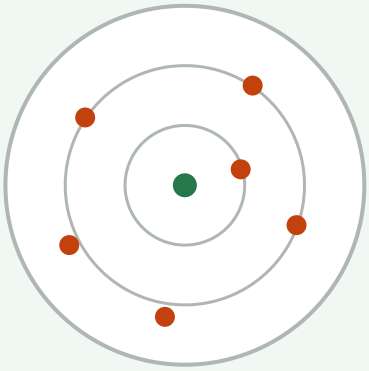
Reads 2 kg heavy, every time

Step on six times: 72.1, 72.0, 72.2, 72.1 …

True weight: 70.0

Reliable, not valid

A bathroom scale that reads 2 kg heavy gives nearly the same number every time. It is **reliable**: consistent from one reading to the next. It is not **valid**: it does not measure your true weight. Repeating it a hundred times only makes you more confident in the wrong number.



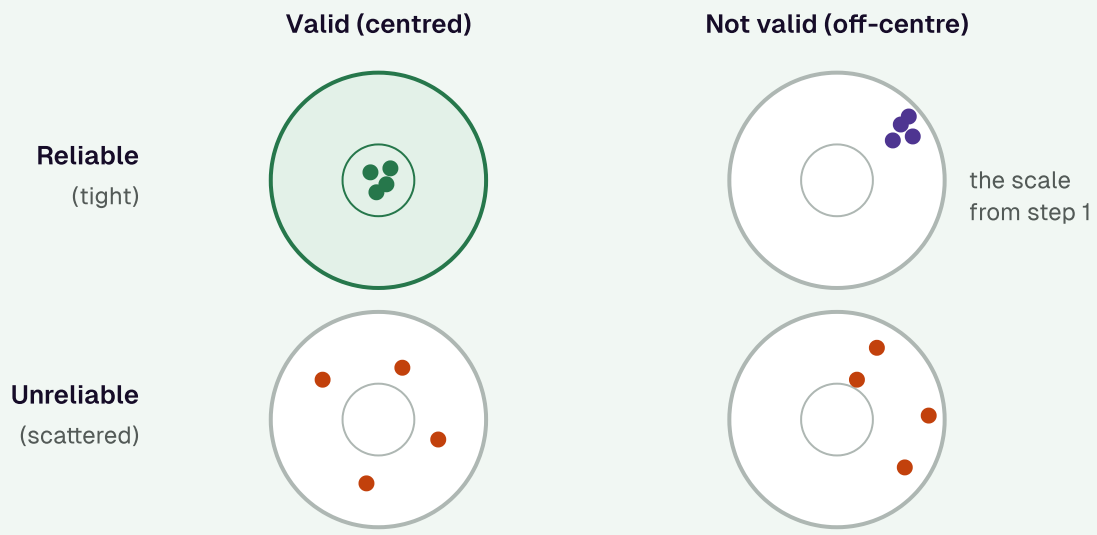
Right on average, jumps ±3 kg

Six readings: 67.4, 72.8, 69.1, 73.0, 66.9, 70.8

Their average: 70.0

Unbiased, unreliable

A cheap scale that jumps around by 3 kg is right on average but useless for any single reading. You need many readings to learn anything, and small real changes, like losing half a kilo, disappear in the noise.



Put together, there are four combinations. Only the top left is what you want. Reliability is necessary for validity, because a measure that disagrees with itself cannot agree with the truth, but it is not enough: the top right is perfectly consistent and wrong.

Checking an LLM judge

Reliability

Score the same answers twice:
do the scores repeat? (test–retest)

Two judges or two humans:
do they agree beyond chance? (kappa)

Validity

Compare with expert labels:
does it track the truth? (criterion)

Does a longer, wrong answer
still score high? (a bias probe)

A judge that always prefers longer answers is reliable and biased: the top-right target.

For an LLM judge, reliability means it gives the same verdict when asked again and agrees with other careful raters. Validity means its verdicts track what experts or users actually value. A judge that loves long answers is very consistent, and consistently wrong in a direction a model can learn to exploit.

Plain words

Consistent and correct are two different checks

A measurement can fail in two ways. It can be shaky: ask twice, get two different answers. Or it can be steady but aimed at the wrong thing: a reading test that mostly measures how fast you read rather than how well you understand. The first problem is about **reliability**, the second about **validity**. You need to check both, and they need different tests.

For leaders

What to ask of any new metric

Before a metric becomes a launch gate, it should pass both checks, with numbers.

- **Reliability.** Run the same eval twice (same model, new seed or new judge sample): how much does the score move? Two raters on the same items: what is their agreement? Low reliability means you need far more items to see a real change, which is a direct cost.
- **Content validity.** Does the eval cover the skill as the product uses it, or a convenient slice? A coding benchmark of single-function puzzles has weak content validity for repository-scale agent work.
- **Criterion validity.** Does the metric predict something you trust: expert judgment today (concurrent), user retention or task success later (predictive)? This is the offline-to-online question.
- **Construct validity.** Does it measure the concept in its name? A "reasoning" benchmark that a model passes by memorization measures recall. Check that it correlates with related measures and not with unrelated ones such as output length.

Principal asks

What is the test–retest correlation of this eval score across seeds? What is judge–human agreement, and how does it compare with human–human agreement on the same items? What external outcome has this metric been shown to predict, and how strongly? What would a model do to raise this number without getting better, and have we tried it?

Under the hood

Kappa, ICC, and the validity types

Inter-rater agreement for categories is measured with Cohen's kappa (Cohen, 1960), which corrects raw agreement for the agreement expected by chance from each rater's base rates:

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \qquad p_e = \sum_k p_{1k}\, p_{2k}$$

p_o is observed agreement, p_{jk} the share of items rater j put in category k. Worked example: two raters label 100 answers; they agree on 40 "good" and 30 "bad". Rater 1 says "good" 60 times, rater 2 says it 50 times, so $p_e = 0.6 \times 0.5 + 0.4 \times 0.5 = 0.50$, $p_o = 0.70$, and $\kappa = 0.40$. Seventy percent agreement sounds fine; kappa says it is only 40 percent of the way from chance to perfect.

Use weighted kappa for ordinal scales (a 4 against a 5 is a smaller disagreement than a 1 against a 5), Fleiss' kappa or Krippendorff's alpha for more than two raters or missing ratings, and the intraclass correlation (ICC) for continuous scores and test–retest designs (see statistics for eval sets). Landis and Koch (Biometrics, 1977) proposed verbal labels (0.41–0.60 "moderate", 0.61–0.80 "substantial"), which are widely quoted and were offered as arbitrary conventions; compare against the human–human ceiling on your own task instead.

Kappa has a known quirk: with very unbalanced categories (say 95 percent "safe"), kappa can be low even when raw agreement is very high, because chance agreement is already high. Report raw agreement, kappa, and the confusion table together. For the classic validity taxonomy (content, criterion, construct), the reference point in psychology is Cronbach and Meehl (1955) on construct validity; modern practice treats all three as evidence for one question: does this score support the decision we are using it for?

Measurement error: attenuation and inflated variance

Plain words

Noise hides effects

Try to hear whether a friend's voice got louder in a quiet room, then try in a noisy bar. The change is the same; the bar makes it harder to detect. A noisy measurement does the same to an experiment: the effect is still there, but you need many more people to see it.

Noise does something sneakier when it sits in the thing you use to explain the outcome. If you measure study hours badly, the link between study hours and grades looks weaker than it is. The noise drags the relationship toward zero.

For leaders

Noise is a budget line

Every bit of measurement noise is paid for in sample size. If an eval score's reliability is 0.6 (60 percent of its variance is real differences between items, 40 percent is noise from the judge or the seed), you need roughly 1/0.6, about 1.7 times as many items to detect the same effect as with a perfectly reliable measure. Improving the judge, averaging several judge samples, or using a paired design is often cheaper than collecting more data.

The more dangerous error is error that correlates with the treatment. If the new model writes longer answers and the judge rewards length, the measurement error is no longer noise: it is bias, and more data makes you more confident in it.

Under the hood

The classical error model

Write the observed score as $X^* = X + u$, with u independent noise. Define reliability $\lambda = \sigma_X^2/(\sigma_X^2 + \sigma_u^2)$.

- **Error in the outcome.** The treatment effect on Y^* is unbiased, but its variance is inflated by $1/\lambda$. The standardised effect size shrinks by $\sqrt{\lambda}$, so the sample size for fixed power scales by $1/\lambda$.
- **Error in a predictor.** Regressing Y on X^* gives a slope biased toward zero: $\operatorname{plim} \hat{\beta} = \lambda\beta$. This is regression dilution or attenuation bias.
- **Correlation between two noisy measures.** Spearman's (1904) attenuation formula: $r_{X^*Y^*} = r_{XY}\sqrt{\lambda_X \lambda_Y}$.

$$r_{\text{observed}} = r_{\text{true}}\sqrt{\lambda_X \lambda_Y}$$

Illustrative: if a judge's true correlation with user satisfaction is 0.6, the judge's reliability is 0.7 and the satisfaction survey's is 0.8, you will observe about $0.6 \times \sqrt{0.56} \approx 0.45$. A low judge–human correlation may be partly the humans' noise.

Non-classical error breaks all of these. Error that correlates with the true value (scores compressed near the top of a scale), with the treatment (a judge with length bias evaluating a model that got longer), or with the outcome (users more likely to rate when angry) can bias effects in either direction. Probe for it with counterfactual items, such as the same answer padded and unpadded, before trusting a judge on a treatment that changes style.

Proxies versus targets

Plain words

Steps are not fitness

A step counter measures steps, and you want fitness. Steps are a **proxy**: easy to count and usually related to the **target**. Shake the phone while watching television and the steps go up while fitness does not. Most things we measure in AI are proxies like this: thumbs-up instead of satisfaction, benchmark score instead of real usefulness, a judge's grade instead of an expert's.

For leaders

Using proxies without being used by them

Proxies are unavoidable because the target is slow, expensive, or rare. The risk is that optimizing the proxy finds the places where it and the target come apart (Goodhart's law; the reward-model version is in preferences and reward models).

1. **Validate on the variation you will create.** A proxy that correlates with the target across users may not move with it under your treatment. The proxy has to track the target across past launches, not only across people.
2. **Keep a slow, expensive check alive.** A sample of expert reviews, a holdout of users, a long-term retention read. Use it to recalibrate the proxy on a schedule.
3. **Pair every proxy with a guardrail** that would move if the proxy were being gamed: length with helpfulness, thumbs-up with task completion, refusal rate with harmful-compliance rate.

Under the hood

Surrogate criteria

Medicine calls proxies surrogate endpoints. Prentice (Statistics in Medicine, 1989) proposed a strict criterion: the surrogate must capture the treatment's full effect on the true endpoint, so that the true endpoint is independent of treatment given the surrogate. That is rarely met. Weaker practical standards ask that, across many past trials, the treatment effect on the surrogate predicts the treatment effect on the target (trial-level validation). The cautionary result is that a treatment can improve the surrogate, the surrogate can be positively correlated with the target, and the treatment can still harm the target, because the treatment moves the surrogate through a path that does not lead to the target. The same logic applies to an eval: validate it on historical model pairs, where you know which model users preferred, not only on item-level correlations.

Common mistakes

- **Averaging ordinal ratings as the only summary.** Report the distribution and the bottom-box share; a mean can rise while disasters also rise.
- **Reading Mann–Whitney as a test of medians.** It tests whether one group tends to rank higher; it is a median test only if the shapes match.
- **Dropping or capping censored durations.** Both bias the average down. Use Kaplan–Meier and report the censoring rate.
- **Computing a ratio metric's variance per event** when randomization was per user. Use the delta method or a user-level bootstrap.
- **Poisson error bars on overdispersed counts.** Check variance against mean first.
- **Calling a judge validated because it is consistent.** Reliability is necessary, not sufficient. Check agreement with humans and probe its biases.
- **Optimizing a proxy that was validated only across users**, never across launches.

Principal asks

- What level of measurement is this metric, and does the summary on the slide respect it?
- For this rating, what happened to the share at the bottom of the scale?
- How many runs were censored by a step or time limit, and how were they handled?
- Is the variance computed at the randomization unit, and for ratio metrics, by the delta method or bootstrap?
- What is the eval's test–retest reliability, and how many items does that imply we need?
- What evidence ties this proxy to the outcome we care about, across past launches?

Part VI · Chapter 45

A/B tests end to end: from a written bet to a launch decision

An A/B test is the closest thing product work has to a laboratory. You split real users at random, give one group the change, and compare. Done well, it answers "did this change cause that outcome?" better than any dashboard, offline eval, or expert opinion. Done carelessly, it produces confident numbers that are

wrong. This chapter walks through a test in the order you run one: write the decision down, choose the unit, choose the metrics, size it, check that the split is healthy, resist peeking, and read the result. The statistics come from statistics for eval sets and intervals; randomization itself is covered in randomization.

Diagram

A café tests a new menu board

A whole A/B test in five pictures, before any formulas.

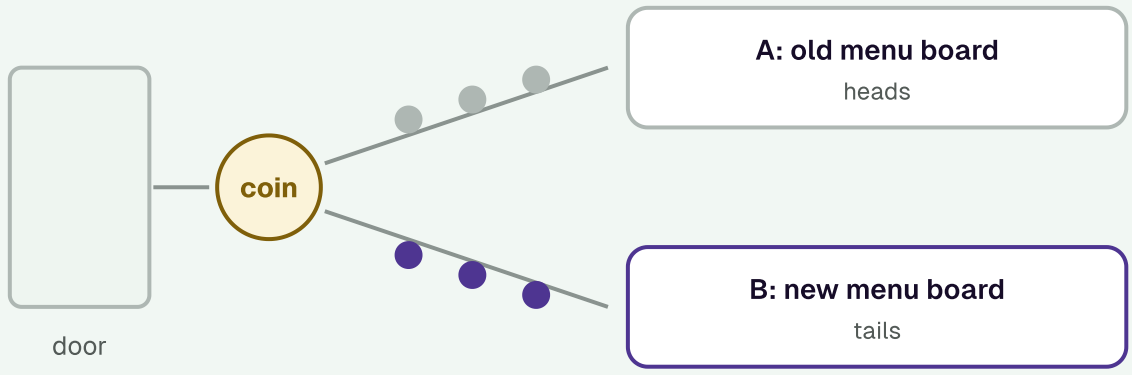
The bet
The new menu board raises orders per visitor.

How we measure
Orders per visitor, over two full weeks.

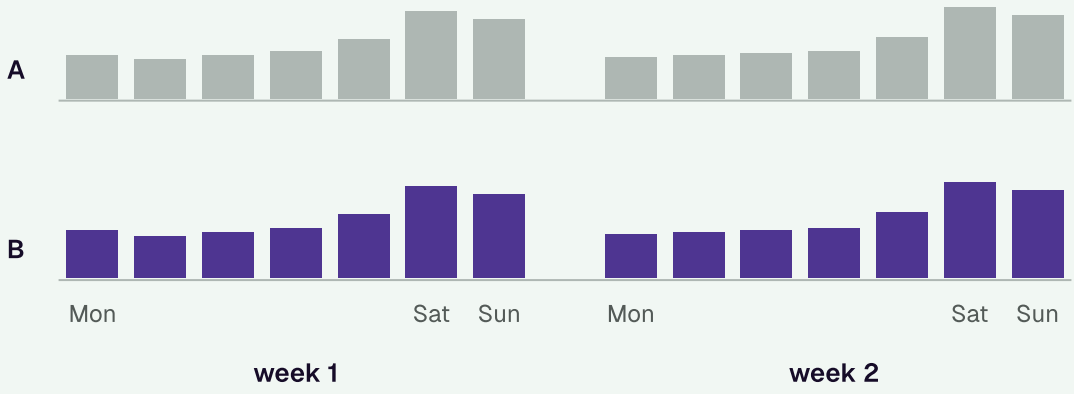
Switch if
orders go up and the wait is no longer.

Written on day 0, before a single customer is counted.

Before changing anything, the owner writes down what she expects, how she will measure it, and what result would make her switch boards. Writing it first stops her from choosing the story after she sees the numbers.



Each customer who walks in is assigned by a coin flip. Because the coin decides, the two groups are alike in every way except the board: the same weather, the same weekdays, the same mix of regulars and tourists. Any difference at the end can be blamed on the board.



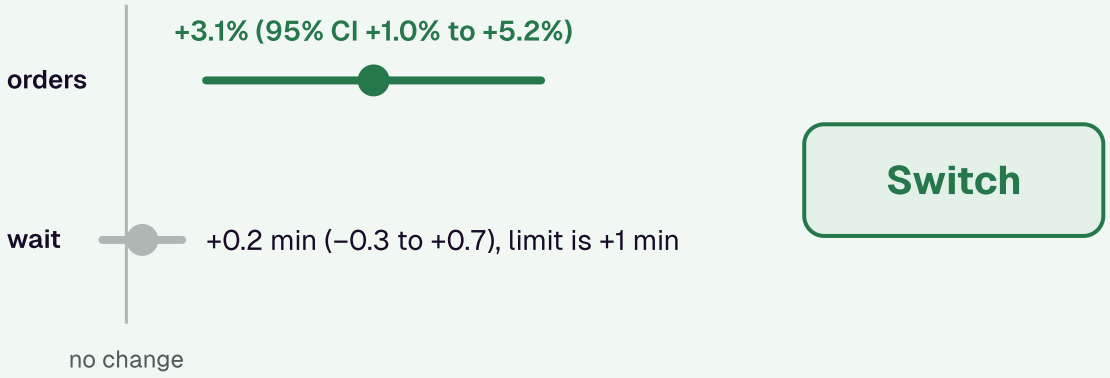
Weekends are busier and different. Full weeks keep the mix fair.

She runs two full weeks, so both groups include the same quiet Mondays and busy Saturdays. Weekend customers order differently. Stopping on a Wednesday would compare a lopsided mix of days, and a second week shows whether the effect holds once the board stops being new.

A fair coin would give about 1,000 each. Is 12 off normal?

Yes: split looks fair (p = 0.59)

Before looking at any result, she checks the headcount. A coin should give roughly equal groups, and 1,012 versus 988 is ordinary coin luck. A big imbalance would mean something between the coin and the till is losing customers from one group, and then no result can be trusted. This check has a name, the sample ratio mismatch test, covered below.



The decision follows the card written on day 0. Numbers are illustrative.

Orders per visitor rose 3.1%, and the 95% interval sits entirely above zero. Wait time barely moved and stays well inside the limit she set. The rule on the card says switch, so she switches. No one had to argue about what the numbers meant, because that was settled before the test began.

1. Write the decision before you launch

Plain words

Call the shot before you take it

In pool, you call the pocket before you shoot. If the ball drops somewhere else, it does not count. An A/B test works the same way. Before launch you write down the change, the one number you expect it to move, which way, and what you will do for each possible result. If you wait until after, every result can be told as a success: "engagement fell, but that means people found answers faster."

For leaders

The one-page experiment brief

Insist on a short brief, reviewed before any traffic is allocated. It costs an hour and prevents the two most expensive failures: tests that cannot answer the question, and results that get reinterpreted until they say yes.

Field	Example for a model change
Change	Route free-tier chat to model v8 instead of v7.
Hypothesis	v8 raises weekly task success because it calls tools more reliably.
Primary metric	Share of active users with at least one judged-successful task per week.
Guardrails	p95 latency, cost per request, safety violation rate, benign refusal rate, error rate.
Unit and population	Logged-in users on supported clients, randomized by user ID.
Size and duration	MDE of 1 point; 2 full weeks; sample size from a power calculation.
Decision rule	Ship if the primary metric's interval excludes zero on the positive side and no guardrail is breached. Otherwise see the decision table at the end of this chapter.
Analysis plan	Test used, variance method, CUPED covariate, segments to report, when we look.

The decision rule matters most. "We'll see what the data says" is a plan to be persuaded by noise.

Under the hood

Hypotheses, error rates, decision rule

Formally, for a treatment mean μ_T and control mean μ_C, the usual superiority test is

$$H_0 : \Delta = \mu_T - \mu_C = 0 \qquad H_1 : \Delta \neq 0, \qquad \hat{\Delta} = \bar{Y}_T - \bar{Y}_C, \qquad z = \frac{\hat{\Delta}}{\widehat{\mathrm{SE}}(\hat{\Delta})}$$

α is the false positive rate you accept when $\Delta = 0$ (usually 0.05 two-sided); power $1 - \beta$ is the chance of detecting a true Δ of the size you care about (usually 0.8).

Three choices belong in the brief because each one changes the answer. **Sidedness**: a one-sided test is legitimate only if you would act the same way on "no effect" and "harm". **Non-inferiority**: when the change is meant to save cost or latency, the question is "is it no worse than a margin m?", tested as $H_0 : \Delta \leq -m$; designing experiments for AI systems works one through. **Absolute or relative effect**: a 2% relative lift on a 40% base is 0.8 points; sample size depends on the absolute difference.

2. Unit of randomization and exposure

Plain words

Flip the coin for the right thing

You must decide what the coin is flipped for: each person, each visit, or each question they ask. If you flip per question, the same person might get the new assistant on Monday and the old one on Tuesday, and their experience becomes a confusing mix. Most of the time you flip once per person and keep them in their group.

You also only count people who actually **met** the change. If the new feature lives on a settings page that most people never open, counting everyone drowns the effect in people who never saw it.

For leaders

Unit choice is a validity decision

- **User-level** is the default for anything a person can notice or learn from: a new model, tone, memory, UI. It keeps each person's experience consistent and lets you measure retention.

- **Request-level** buys more samples and is fine for invisible, stateless changes like a cache or a router, where users cannot tell and nothing carries over.
- **Cluster-level** (organisation, workspace, region) is needed when users affect each other, such as a shared team assistant, or when a marketplace links buyers and sellers. Interference breaks the independence that every standard test assumes; see randomization.

Ask how exposure is defined. A test that randomizes all users but only 8% ever trigger the changed code path has diluted its effect twelvefold, and its sample size calculation was wrong by a factor of about 150.

Under the hood

Triggering, dilution, analysis unit

Triggered analysis. Analyse only users who reached the point where treatment and control would differ, and log that moment in **both** arms (a counterfactual trigger in control). If a fraction q of users trigger and the effect on them is Δ, the effect on everyone is $q\Delta$. Because required sample size scales with $1/\text{effect}^2$, dilution costs roughly $1/q$ more users even after accounting for the larger population.

Analysis unit must match randomization unit. If you randomize users but compute a per-request metric, requests from one user are correlated. Treating them as independent understates the standard error, sometimes by a factor of two or more. Use the delta method or a cluster bootstrap at the user level; choosing metrics shows the computation and intervals covers clustering.

Assignment. Use a deterministic hash of (unit ID, experiment salt) mapped to buckets. A fresh salt per experiment prevents the same users from always landing in treatment across tests, which would carry over effects from one test to the next.

3. Primary metric, guardrails, diagnostics

Plain words

One goal, some tripwires, some gauges

A test needs one main number that says whether the change helped: the goal. It also needs tripwires: numbers that must not get worse, like wait time or complaints, even if the goal improves. And it needs gauges that explain what happened, like how long answers were, so that a surprising result can be understood.

For leaders

Three roles, decided in advance

- **Primary metric** (or overall evaluation criterion, OEC): the single number the decision hangs on. One. If you have three "primary" metrics, you have zero.
- **Guardrails**: things you will not trade away. Two kinds: **business** guardrails (latency, cost, safety, refusals) and **trust** guardrails (sample ratio, logging completeness) that tell you whether to believe the test at all.
- **Diagnostics**: explain the mechanism. Response length, tool calls per task, cache hit rate. Never ship on a diagnostic.

Choosing metrics covers how to pick each and what makes a metric good.

Under the hood

Guardrails are tested differently

The primary metric gets a superiority (or non-inferiority) test with your full α. Guardrails are usually tested for **harm**: a one-sided test against a tolerance, $H_0 : \Delta_g \geq -t_g$, so that "no detectable harm" does not pass simply because the guardrail is underpowered. Check power for the guardrails too. A safety violation rate of 0.1% needs far more users to detect a 20% relative increase than a task success rate of 60% needs to detect a 1-point gain, and a guardrail you cannot detect a breach on is decoration.

4. Power, minimum detectable effect, sample size, duration

Plain words

How long to wait before you can tell

If you flip two coins ten times each, one will often come up with more heads just by luck. To tell a coin that lands heads 52% of the time from a fair one, you need thousands of flips. Tests are the same. The smaller the improvement you want to catch, the more people you need, and halving the improvement means about four times the people. The **minimum detectable effect** is the smallest improvement your test can reliably see. Pick it first, then work out how long to run.

For leaders

Pick the MDE from the decision, then do the arithmetic

- **Ask: what is the smallest change worth shipping?** That is the MDE. If a 0.5-point gain would not change your decision, do not pay to detect it.
- **Run whole weeks.** Usage differs by weekday. Round durations up to 7, 14 or 21 days, even if the calculator says 9.

- **Two weeks is a common floor** for anything users can notice, because of novelty effects (section 7).
- **If the required duration is absurd** (six months), change the design: a more sensitive metric, CUPED, a triggered population, or accept a bigger MDE. Do not run it short and hope.
- **A null result needs its MDE.** "No significant change" from a test that could only detect 5-point moves says nothing about a 1-point effect.

Under the hood

The sample size formula and the rule of 16

For a two-arm test with equal allocation, per-arm variance σ^2, and absolute effect δ:

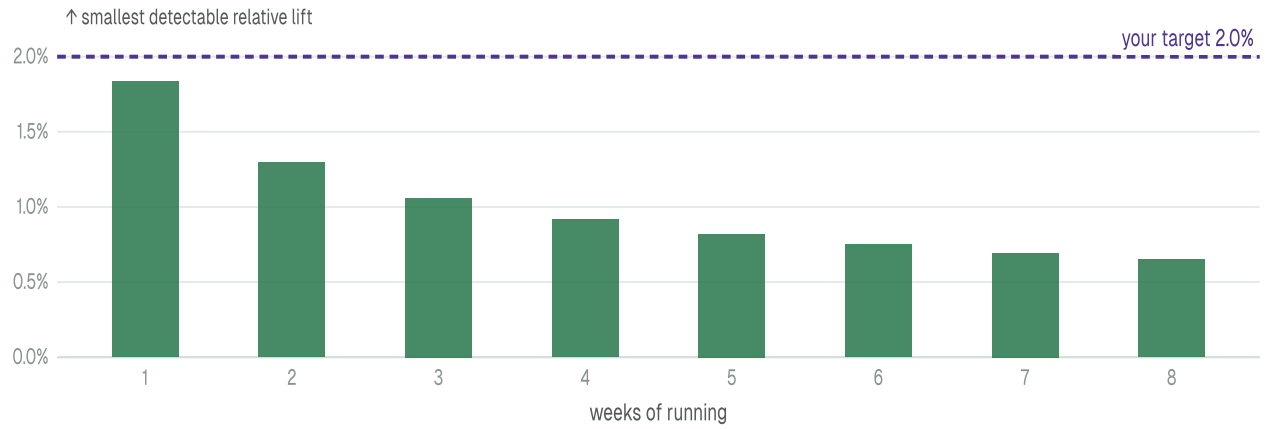
$$n_{\text{per arm}} = \frac{2\,(z_{1-\alpha/2} + z_{1-\beta})^2\,\sigma^2}{\delta^2} \;\approx\; \frac{16\,\sigma^2}{\delta^2} \quad (\alpha = 0.05,\ 1-\beta = 0.8)$$

$2(1.96 + 0.84)^2 \approx 15.7$, hence the rule of 16 (Lehr 1992; van Belle 2002), standard in online experimentation (Kohavi, Tang and Xu 2020). For a rate p, $\sigma^2 = p(1-p)$.

Worked example (illustrative): weekly task success $p = 0.40$, and you want to detect a 2% relative lift, so $\delta = 0.008$. Then $n \approx 16 \times 0.24 / 0.008^2 = 60{,}000$ per arm. With 20,000 eligible users a day split 50/50, that is 6 days, rounded up to 7, and in practice 14 for novelty. Turn it around to get the MDE you can afford: $\delta_{\min} = (z_{1-\alpha/2} + z_{1-\beta})\sqrt{2\sigma^2/n}$. Required users scale with $1/\delta^2$; MDE scales with $1/\sqrt{n}$, so doubling the duration buys only a 29% smaller MDE. For the paired, item-level version used in offline evals, see statistics for eval sets.

Interactive **Duration and MDE calculator** (shown at its default setting)

Set the base rate, the smallest lift worth detecting, and your traffic. The chart shows the smallest lift you could detect after each whole week, so you can see what a longer test buys.



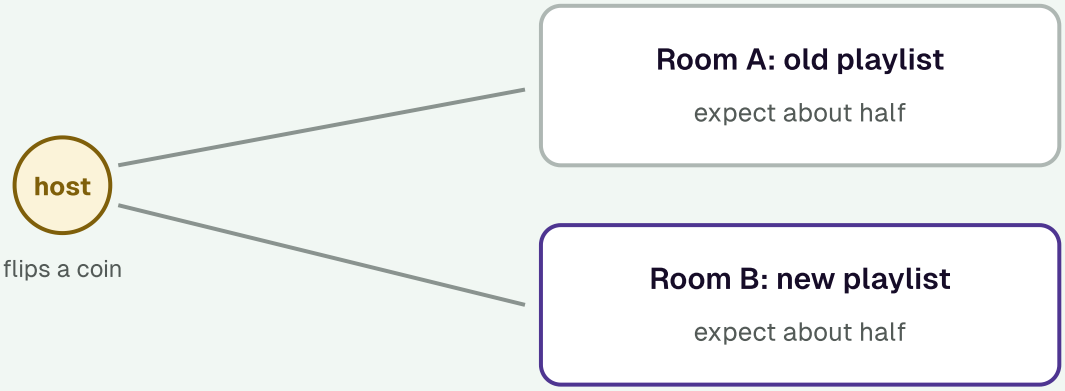
Users needed per arm	Days of traffic	Run for (whole weeks)	MDE in points
58.9K	6	7 days	0.80

Detecting a 2.0% relative lift on a 40% base means a 0.80-point change. That needs 58.9K users per arm, or 6 days at 10.0K per arm per day. Round up to at least 14 days for any change users can notice, to cover novelty and two of each weekday.

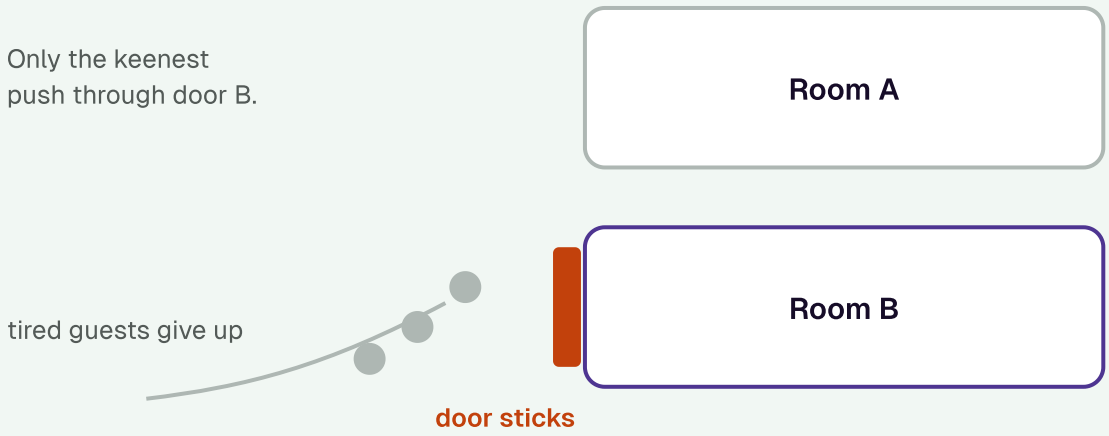
5. Sample ratio mismatch: check the split before the result

Diagram **The sticky door**

How losing people from one group quietly poisons a test.



A host tests a new playlist. At the entrance she flips a coin for each guest: heads to room A with the old music, tails to room B with the new. She will compare how much people dance. With a fair coin, each room should get about half the guests.



Room B's door sticks. Some guests sent to B give up and go home. The ones who get in are the keenest dancers. Nobody notices, because the host only counts people once they are inside.

The playlist did not do this. The door did.

Room B lost its least keen guests, so it is no longer a fair comparison.

Room B shows a big jump in dancing per guest. It looks like the new playlist is a hit. It is not: room B lost its least enthusiastic guests, so the two rooms no longer hold similar people. The coin's guarantee is broken.

entered A, all season	entered B, all season
5,000	**4,600**

A fair coin gives this split less than 1 time in 10,000

Stop. Find the leak. Do not read the dance numbers.

The headcount gives it away. Over a season, 5,000 guests entered A and 4,600 entered B. A fair coin almost never produces that (chi-square p ≈ 0.00004). This is a **sample ratio mismatch**: something between the coin and the count is removing people from one group. The only safe response is to find the leak before reading any result.

Plain words

The headcount is your smoke alarm

If you planned a 50/50 split, you should get close to 50/50. Not exactly, because coins are random, but close. A statistical test tells you how far off is too far for luck. When the split is off, something is

losing or adding people in one group, and those people are rarely a random handful. The comparison is then between two different kinds of people, and the result is meaningless no matter how good it looks.

For leaders

An SRM invalidates the test. Full stop.

SRM is common. Fabijan et al. (KDD 2019) reported that about 6% of experiments at Microsoft showed a sample ratio mismatch, and a LinkedIn paper (Chen, Liu and Xu, 2018) reported about 10% of its triggered experiments used to. Typical causes:

- The treatment crashes or times out for some users before they are logged (the sticky door).
- A bot filter or fraud filter treats the arms differently, for example because the new model's responses change what looks like bot behaviour.
- Treatment changes whether a user triggers the analysis, such as a new onboarding flow that changes who reaches the chat screen.
- Redirects, caching, or a client version that only ships the treatment code path to some platforms.
- Starting arms at different times, or a ramp-up that was not accounted for.

The policy should be automatic: the platform runs the check on every experiment, flags failures before anyone sees the metric results, and teams do not get to argue that "the effect is so big it must be real". The bigger the surprise, the more likely the bug.

Under the hood

The chi-square test, worked

With k arms, configured shares s_i, observed counts O_i and total N, the expected counts are $E_i = N s_i$ and

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i} \;\sim\; \chi^2_{k-1} \text{ under a healthy split}, \qquad k = 2:\ p = \operatorname{erfc}\left(\sqrt{\chi^2/2}\right)$$

For two arms the statistic has one degree of freedom and equals z^2 for the one-proportion z test of the observed share.

Worked example: a 50/50 test counts 50,500 users in control and 49,500 in treatment. $E = 50{,}000$ each, so $\chi^2 = 500^2/50{,}000 + 500^2/50{,}000 = 5 + 5 = 10$, and $p = \operatorname{erfc}(\sqrt{5}) \approx 0.0016$. A split of 50.5% to 49.5% looks harmless to the eye. The test says it would happen by chance about 1.6 times in 1,000. Many platforms alarm at a strict threshold such as $p < 0.001$, because the check runs on

every experiment and false alarms erode trust; anything near that line deserves a look. At two million users, even 50.1% versus 49.9% gives $p \approx 0.005$, and 50.12% versus 49.88% gives $p \approx 0.001$. The larger the test, the smaller the imbalance it can detect, so "the gap is tiny" is never a defence.

python

```python
import math

def srm_check(observed, expected_share):
    """Chi-square goodness-of-fit test of the traffic split. Returns (chi2, p)."""
    n = sum(observed)
    chi2 = 0.0
    for obs, share in zip(observed, expected_share):
        exp = n * share                     # users this arm should have received
        chi2 += (obs - exp) ** 2 / exp
    # Two arms means 1 degree of freedom, and P(chi2_1 > x) = erfc(sqrt(x / 2)).
    p = math.erfc(math.sqrt(chi2 / 2))
    return chi2, p

# A 50/50 test that came back 50,500 vs 49,500: only 1 point of traffic off.
chi2, p = srm_check([50_500, 49_500], [0.5, 0.5])
print(f"{chi2:.2f} {p:.6f}")                # 10.00 0.001565  -> investigate
# A healthy split of the same size.
chi2, p = srm_check([50_090, 49_910], [0.5, 0.5])
print(f"{chi2:.2f} {p:.6f}")                # 0.32 0.569214   -> fine
# Two million users, 50.12% vs 49.88%: tiny to the eye, decisive to the test.
chi2, p = srm_check([1_002_300, 997_700], [0.5, 0.5])
print(f"{chi2:.2f} {p:.6f}")                # 10.58 0.001143  -> investigate
```

rust

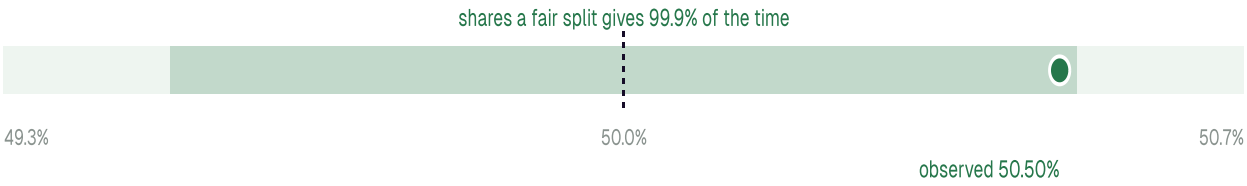
```rust
// erfc is not in Rust's standard library, so we use the Numerical Recipes
// Chebyshev fit (relative error below 1.2e-7), plenty for an SRM p-value.
fn erfc(x: f64) -> f64 {
    let z = x.abs();
    let t = 1.0 / (1.0 + 0.5 * z);
    let r = t * (-z * z - 1.26551223
        + t * (1.00002368 + t * (0.37409196 + t * (0.09678418
        + t * (-0.18628806 + t * (0.27886807 + t * (-1.13520398
        + t * (1.48851587 + t * (-0.82215223 + t * 0.17087277)))))))))
        .exp();
    if x >= 0.0 { r } else { 2.0 - r }
}

/// Chi-square goodness-of-fit test of the traffic split. Returns (chi2, p).
fn srm_check(observed: &[f64], expected_share: &[f64]) -> (f64, f64) {
    let n: f64 = observed.iter().sum();
    let mut chi2 = 0.0;
    for (obs, share) in observed.iter().zip(expected_share) {
        let exp = n * share; // users this arm should have received
        chi2 += (obs - exp).powi(2) / exp;
    }
    // Two arms means 1 degree of freedom, and P(chi2_1 > x) = erfc(sqrt(x / 2)).
    let p = erfc((chi2 / 2.0).sqrt());
    (chi2, p)
}

fn main() {
```

```
    // A 50/50 test that came back 50,500 vs 49,500: only 1 point of traffic off.
    let (chi2, p) = srm_check(&[50_500.0, 49_500.0], &[0.5, 0.5]);
    println!("{:.2} {:.6}", chi2, p); // 10.00 0.001565  -> investigate
    // A healthy split of the same size.
    let (chi2, p) = srm_check(&[50_090.0, 49_910.0], &[0.5, 0.5]);
    println!("{:.2} {:.6}", chi2, p); // 0.32 0.569214   -> fine
    // Two million users, 50.12% vs 49.88%: tiny to the eye, decisive to the test.
    let (chi2, p) = srm_check(&[1_002_300.0, 997_700.0], &[0.5, 0.5]);
    println!("{:.2} {:.6}", chi2, p); // 10.58 0.001143  -> investigate
}
```

Check SRM per day and per segment too. A leak that only affects one client version or one day can hide inside a healthy-looking total.

Interactive **SRM checker** (shown at its default setting)

Enter the split you configured and the users you actually counted. The band shows the range of shares a fair coin produces 99.9% of the time at this sample size; watch it shrink as the counts grow.



Observed share to control	Expected in control	Chi-square (1 df)	p-value
50.50%	50.0K	10.00	0.0016

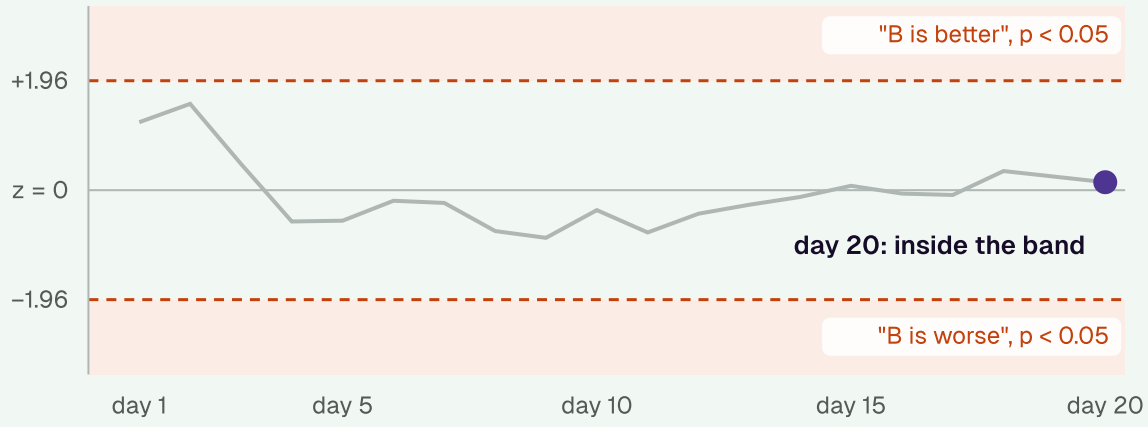
Borderline: investigate before trusting

Not past a strict alarm threshold, but unusual. Look at the split by day and by platform; a leak in one slice can hide in the total.

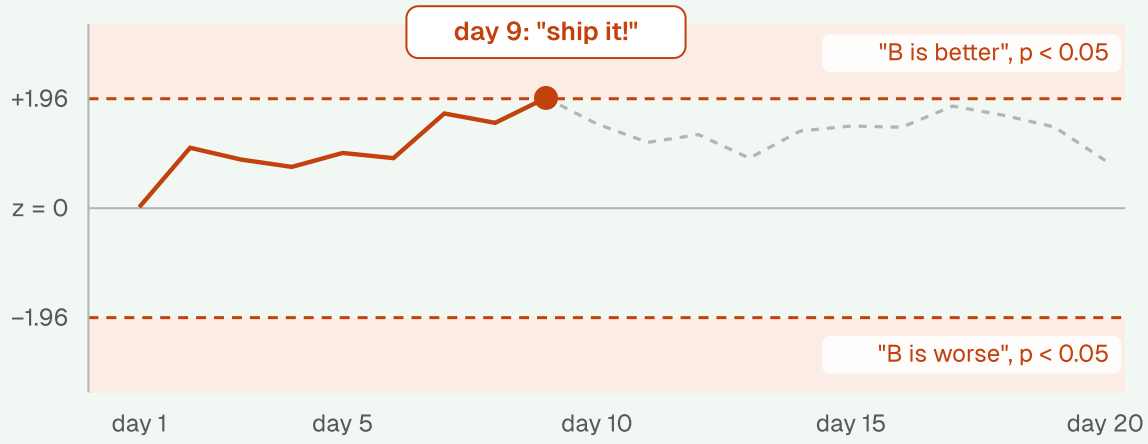
6. Peeking and optional stopping

Diagram **The scoreboard you keep checking**

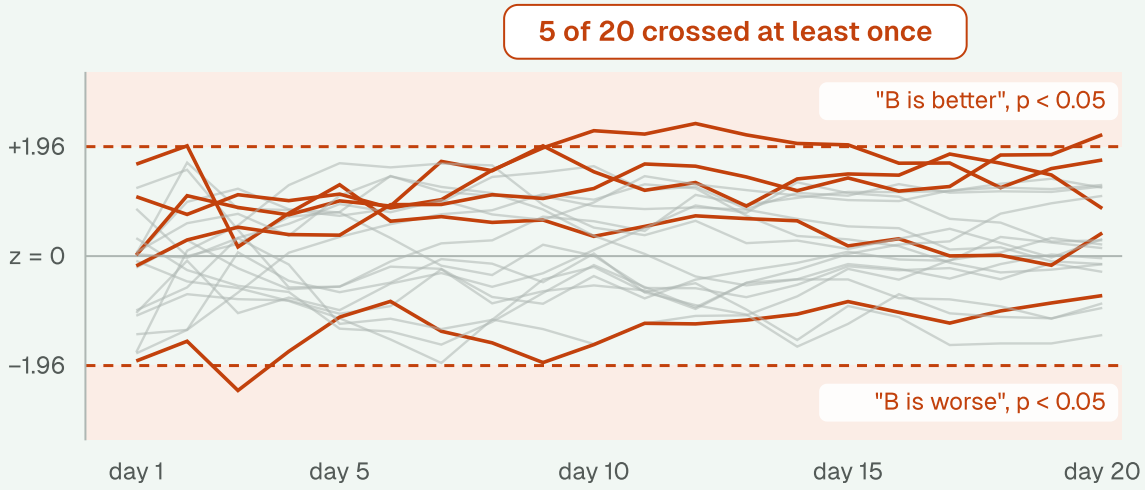
Why peeking at a running test manufactures winners, and how a planned boundary fixes it.



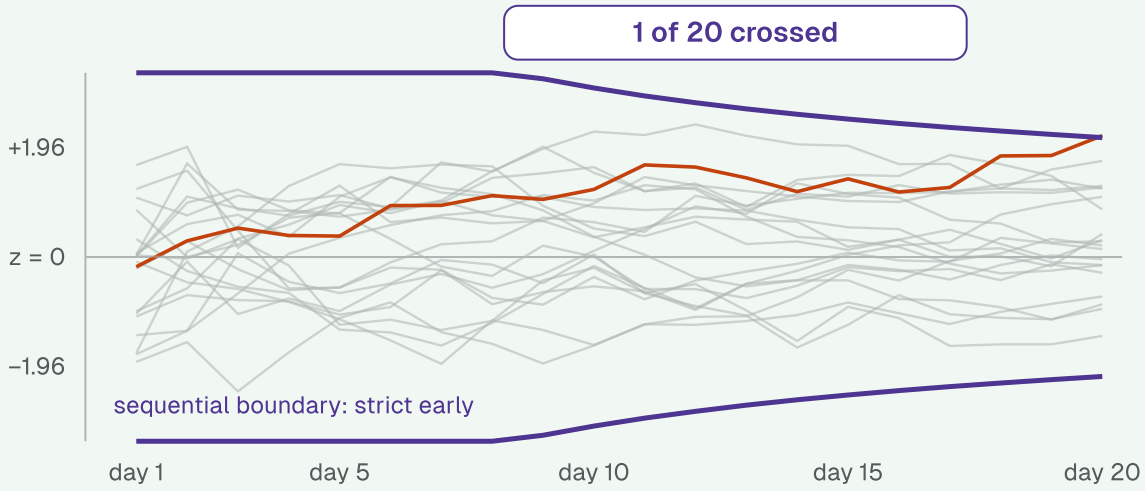
An A/A test splits users into two groups that get the **same** product, so any difference is pure luck. The line is the test statistic as data accumulates. Looked at once, on the planned last day, it lands inside the band where nothing is declared. That happens 95% of the time, which is what p < 0.05 promises.



Now someone checks the dashboard every morning and stops the first time p dips below 0.05. This same no-difference test crossed the line on day 9, purely by luck, and then wandered back. Stop on day 9 and you ship a change that does nothing.



Run twenty of these A/A tests and peek daily. Here 5 of 20 cross the line at some point. Across thousands of simulated runs, 20 daily looks raise the false-positive rate from the promised 5% to about 25% (simulation in the code below; Armitage, McPherson and Rowe described the effect in 1969).



The fix is to decide in advance how often you will look and to pay for those looks. A sequential boundary is very strict on early days, when the line swings most, and relaxes toward the end, finishing at z = 2.14 here. The same twenty tests now produce one false alarm, about the 5% you signed up for, and you can still stop early when an effect is large.

Plain words

Every look is another lottery ticket

A p-value of 0.05 means: if nothing is going on, results this extreme show up 1 time in 20. That promise holds if you look **once**. Every extra look is another chance for luck to produce an extreme number. Look every day for a month and stop the first time it looks good, and you will "find" winners in tests where nothing changed. The fix is to decide in advance when and how often you will look, and to use a stricter bar when you look more often.

For leaders

Peeking is fine if the method allows it

People will look at a live dashboard; banning that is unrealistic. What matters is the rule for **stopping**. Three acceptable policies:

- **Fixed horizon.** Compute the sample size, run to it, analyse once. Dashboards during the test are for guardrail and bug monitoring only, and nobody stops early for a good-looking primary metric.
- **Group sequential.** Plan a handful of interim analyses (say, end of each week) with boundaries that spend the error budget over time. You can stop early for a large win or a clear harm, at the cost of slightly more users if you run to the end.
- **Always-valid (continuous) monitoring.** Methods such as the mixture sequential probability ratio test give p-values that stay valid however often you look. Some commercial platforms adopted this so customers can watch results live. The price is lower power at a fixed sample size.

Stopping early to cut losses on a harmful change is always fine; you just cannot then claim the estimate is unbiased. The unacceptable policy is the common one: look daily, stop at the first p < 0.05, report it as if planned.

Under the hood

Why it inflates, and the corrections

Under H_0, the z statistic at equally spaced looks behaves like a scaled random walk: $z_k = S_k/\sqrt{k}$ with S_k a sum of k independent standard normal increments. The naive procedure rejects if $\max_{k \le K} |z_k| > 1.96$, and that maximum exceeds 1.96 far more often than 5% of the time. Armitage, McPherson and Rowe (1969) quantified this; the simulation below reproduces it: about 14% with 5 looks and 25% with 20. With unlimited looks the false positive rate tends to 1, by the law of the iterated logarithm.

Group sequential boundaries. Choose critical values $c_1, \dots, c_K$ with $P_{H_0}(\exists k : |z_k| > c_k) = \alpha$. Pocock (1977) uses the same c at every look (2.413 for 5 looks at $\alpha = 0.05$); O'Brien and Fleming (1979) use $c_k = c\sqrt{K/k}$, very strict early and close to 1.96 at the end (2.040 at the final of 5 looks). Lan and DeMets (1983) generalised this with an **alpha spending function** $\alpha(t)$, the error spent by information fraction t, so looks need not be equally spaced or planned exactly:

$$\alpha_{\text{OBF}}(t) = 2 - 2\,\Phi\!\left(\frac{z_{1-\alpha/2}}{\sqrt{t}}\right), \qquad \alpha_{\text{Pocock}}(t) = \alpha \ln\!\big(1 + (e-1)\,t\big)$$

$t \in (0, 1]$ is the fraction of planned users observed so far. Both spend the full α by $t = 1$; the O'Brien–Fleming-type function spends almost nothing early.

Always-valid p-values. Johari, Koomen, Pekelis and Walsh (KDD 2017; Operations Research 2022) adapt the mixture SPRT: compute a likelihood ratio Λ_n of "some effect, drawn from a prior $N(0, \tau^2)$" against "no effect" and report $p_n = \min(p_{n-1}, 1/\Lambda_n)$. Because Λ_n is a nonnegative martingale under H_0, Ville's inequality gives $P(\exists n : \Lambda_n \geq 1/\alpha) \leq \alpha$, so stopping whenever $p_n < \alpha$ keeps the error rate at α. The confidence sequences that go with it are wider than fixed-horizon intervals at any given n.

The simulation uses a hand-written generator in both languages, so Python and Rust print identical numbers.

python

```python
import math

MASK = (1 << 64) - 1

class SplitMix64:
    """A tiny PRNG written out by hand so the Rust twin draws the exact same numbers."""
    def __init__(self, seed):
        self.s = seed & MASK
    def next_u64(self):
        self.s = (self.s + 0x9E3779B97F4A7C15) & MASK
        z = self.s
        z = ((z ^ (z >> 30)) * 0xBF58476D1CE4E5B9) & MASK
        z = ((z ^ (z >> 27)) * 0x94D049BB133111EB) & MASK
        return z ^ (z >> 31)
    def uniform(self):                      # in (0, 1], never exactly 0
        return ((self.next_u64() >> 11) + 1) / 2**53
    def gauss(self):                        # Box-Muller
        u1, u2 = self.uniform(), self.uniform()
        return math.sqrt(-2 * math.log(u1)) * math.cos(2 * math.pi * u2)

def aa_false_positives(looks, pocock=0.0, obf=0.0, sims=20_000, seed=7):
    """Run A/A tests (no true effect) and check each one `looks` times."""
    rng = SplitMix64(seed)
    fixed = naive = poc = ob = 0
    for _ in range(sims):
        s, hit_n, hit_p, hit_o = 0.0, False, False, False
        for k in range(1, looks + 1):
            s += rng.gauss()                # each look adds one equal batch of users
            z = s / math.sqrt(k)            # the z statistic on the dashboard at look k
            hit_n = hit_n or abs(z) > 1.96   # naive: stop the first time p < 0.05
            hit_p = hit_p or (pocock > 0 and abs(z) > pocock)
            hit_o = hit_o or (obf > 0 and abs(z) > obf * math.sqrt(looks / k))
        fixed += abs(z) > 1.96              # fixed horizon: test once, at the end
```

```
        naive += hit_n
        poc += hit_p
        ob += hit_o
    return fixed / sims, naive / sims, poc / sims, ob / sims

# Pocock and O'Brien-Fleming constants for 5 looks at two-sided alpha = 0.05.
f, n, p, o = aa_false_positives(5, pocock=2.413, obf=2.040)
print(f"5 looks:  fixed {f:.3f} naive {n:.3f} pocock {p:.3f} obf {o:.3f}")
# 5 looks:  fixed 0.051 naive 0.140 pocock 0.050 obf 0.051
f, n, _, _ = aa_false_positives(20)
print(f"20 looks: fixed {f:.3f} naive {n:.3f}")
# 20 looks: fixed 0.050 naive 0.246
```

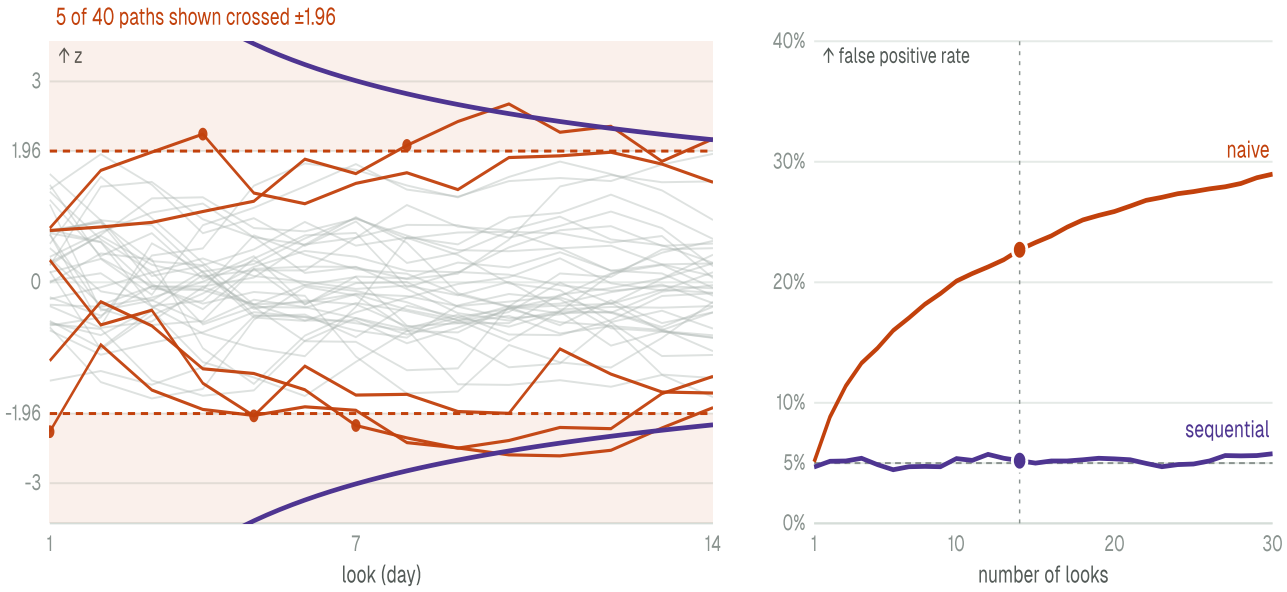
rust

```
/// A tiny PRNG written out by hand so the Python twin draws the exact same numbers.
struct SplitMix64 {
    s: u64,
}

impl SplitMix64 {
    fn new(seed: u64) -> Self {
        SplitMix64 { s: seed }
    }
    fn next_u64(&mut self) -> u64 {
        self.s = self.s.wrapping_add(0x9E3779B97F4A7C15);
        let mut z = self.s;
        z = (z ^ (z >> 30)).wrapping_mul(0xBF58476D1CE4E5B9);
        z = (z ^ (z >> 27)).wrapping_mul(0x94D049BB133111EB);
        z ^ (z >> 31)
    }
    fn uniform(&mut self) -> f64 {
        // in (0, 1], never exactly 0
        ((self.next_u64() >> 11) + 1) as f64 / 9007199254740992.0
    }
    fn gauss(&mut self) -> f64 {
        // Box-Muller
        let (u1, u2) = (self.uniform(), self.uniform());
        (-2.0 * u1.ln()).sqrt() * (2.0 * std::f64::consts::PI * u2).cos()
    }
}

/// Run A/A tests (no true effect) and check each one `looks` times.
fn aa_false_positives(looks: usize, pocock: f64, obf: f64, sims: usize, seed: u64) -> (f64, f64, f64, f64) {
    let mut rng = SplitMix64::new(seed);
    let (mut fixed, mut naive, mut poc, mut ob) = (0, 0, 0, 0);
    for _ in 0..sims {
        let (mut s, mut z) = (0.0, 0.0);
        let (mut hit_n, mut hit_p, mut hit_o) = (false, false, false);
        for k in 1..=looks {
            s += rng.gauss(); // each look adds one equal batch of users
            z = s / (k as f64).sqrt(); // the z statistic on the dashboard at look k
            hit_n = hit_n || z.abs() > 1.96; // naive: stop the first time p < 0.05
            hit_p = hit_p || (pocock > 0.0 && z.abs() > pocock);
            hit_o = hit_o || (obf > 0.0 && z.abs() > obf * (looks as f64 / k as f64).sqrt());
        }
        fixed += (z.abs() > 1.96) as usize; // fixed horizon: test once, at the end
        naive += hit_n as usize;
        poc += hit_p as usize;
        ob += hit_o as usize;
```

```
    }
    let n = sims as f64;
    (fixed as f64 / n, naive as f64 / n, poc as f64 / n, ob as f64 / n)
}

fn main() {
    // Pocock and O'Brien-Fleming constants for 5 looks at two-sided alpha = 0.05.
    let (f, n, p, o) = aa_false_positives(5, 2.413, 2.040, 20_000, 7);
    println!("5 looks:  fixed {:.3} naive {:.3} pocock {:.3} obf {:.3}", f, n, p, o);
    // 5 looks:  fixed 0.051 naive 0.140 pocock 0.050 obf 0.051
    let (f, n, _, _) = aa_false_positives(20, 0.0, 0.0, 20_000, 7);
    println!("20 looks: fixed {:.3} naive {:.3}", f, n);
    // 20 looks: fixed 0.050 naive 0.246
}
```

Interactive **Peeking simulator** (shown at its default setting)

Thousands of A/A tests, where nothing differs between arms, each checked at evenly spaced looks. Red paths crossed p < 0.05 at some look and would have been "shipped". Add looks and watch the naive false positive rate climb while the sequential boundary holds near 5%.



Naive peeking	Sequential boundary	One look at the end	z needed at final look
22.7%	**5.2%**	**5.2%**	**2.13**

Checking 14 times and stopping at the first p < 0.05 declares a winner in **22.7%** of tests where nothing changed, 4.5 times the 5% you promised. The O'Brien–Fleming-shaped boundary demands z above 8.0 on the first look and 2.13 on the last, close to the usual 1.96, so you lose little power if you run to the end. Either way the false positive rate stays near 5.2% (boundaries calibrated by simulation on separate paths).

7. Novelty and primacy effects

Plain words **New toys and old habits**

Give a child a new toy and they play with it constantly for a week, then it joins the pile. That is a **novelty effect**: the change looks great at first because it is new. The opposite also happens. Move

the light switch in your kitchen and you miss it for weeks, until your hand learns the new spot. That is a **primacy effect**, or change aversion: a good change looks bad at first because people are used to the old way. Both fade with time, so a short test can get the answer backwards.

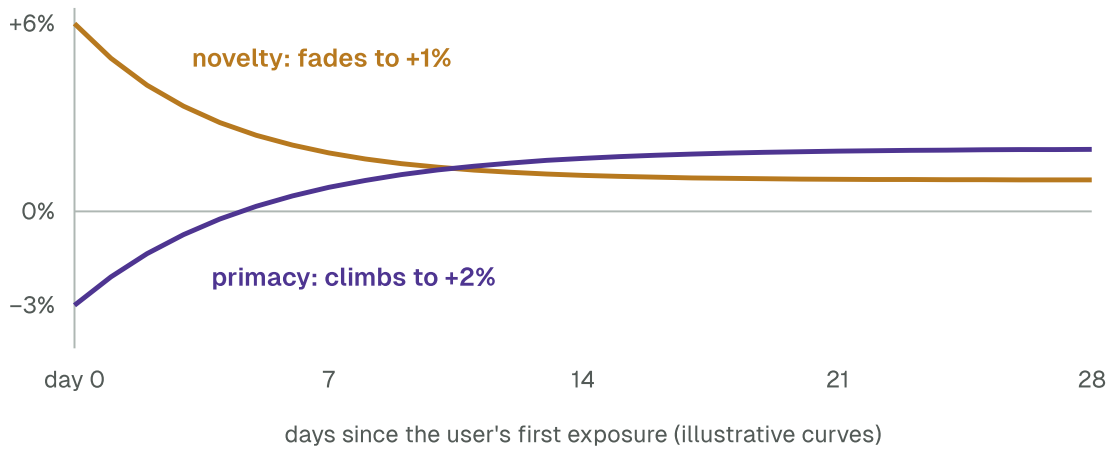
For leaders

Ask to see the effect over time

For visible changes (a new model voice, a redesigned answer layout), ask for the treatment effect plotted by days since a user's first exposure. A line that decays suggests novelty; a line that climbs suggests learning or change aversion wearing off. Either way, the number to ship on is where the line settles, not the two-week average. Power users show both effects most strongly, because they notice changes first.

Under the hood

Cohort by first exposure, not by calendar



Two changes with opposite early behaviour. A one-week test would overstate the first change several times over and kill the second, which ends up the better of the two.

Plotting the effect by calendar day mixes users at different points in their exposure, because new users keep entering the test. Instead, group users by the date they were first exposed and plot the effect against days since that date. A cleaner check is to compare users who entered in week 1 with those who entered in week 2, measured over their own first seven days: if the effect is a property of the change, the two cohorts should agree. For long-term effects, a holdout (section 10) is the only direct measurement; models of the decay are a fallback.

8. Variance reduction with CUPED

Plain words

Judge the improvement, not the starting point

Two classes take a maths test after trying different teaching methods. One class happens to contain more strong students. The raw scores mostly tell you who the students were before the experiment. A fairer comparison asks how much each student improved over their own score last term. People who were strong before tend to stay strong, so subtracting what you already knew about them removes most of the noise, and the effect of the teaching shows up with far fewer students.

For leaders

Free sensitivity, if the pre-period data exists

CUPED (Deng, Xu, Kohavi and Walker, WSDM 2013) uses each user's behaviour from before the test to explain away part of the noise. On Bing it cut variance by about 50%, which means the same power with roughly half the users or half the duration. It is now standard on most mature experimentation platforms.

- Gains depend on how well the past predicts the future. Engagement metrics for returning users: large. New-user tests: little, because there is no history.
- The covariate must be measured **before** assignment. Using anything the treatment could affect biases the result.
- It is an analysis choice, so it belongs in the pre-registered analysis plan, not chosen after seeing which version gives a smaller p-value.

Under the hood

The estimator

Let Y be the metric during the test and X a pre-period covariate, often the same metric in the weeks before. Define

$$Y_i^{\text{cv}} = Y_i - \theta\,(X_i - \bar{X}), \qquad \theta = \frac{\operatorname{Cov}(X, Y)}{\operatorname{Var}(X)}, \qquad \operatorname{Var}\!\left(\bar{Y}^{\text{cv}}\right) = \operatorname{Var}(\bar{Y})\,(1 - \rho^2)$$

ρ is the correlation between X and Y. θ is estimated on pooled data from both arms; because X is independent of assignment, $E[\bar{X}_T - \bar{X}_C] = 0$ and the adjusted difference stays unbiased.

The treatment effect estimate is $\bar{Y}_T^{\text{cv}} - \bar{Y}_C^{\text{cv}}$. With $\rho = 0.7$, variance falls by 49%. This is the same as regression adjustment with a single covariate (ANCOVA), and it generalises to several covariates or

to a machine-learned prediction of Y from pre-period features. Users without pre-period data get $X = 0$ plus an indicator, or are analysed separately. The small example below halves the standard error.

python

```python
def mean(v):
    return sum(v) / len(v)

def var(v):                                  # sample variance (n - 1)
    m = mean(v)
    return sum((x - m) ** 2 for x in v) / (len(v) - 1)

def cov(a, b):
    ma, mb = mean(a), mean(b)
    return sum((x - ma) * (y - mb) for x, y in zip(a, b)) / (len(a) - 1)

def cuped(y, x):
    """Adjust y by the pre-period covariate x: y - theta * (x - mean(x))."""
    theta = cov(x, y) / var(x)               # the regression slope of y on x
    mx = mean(x)
    return [yi - theta * (xi - mx) for xi, yi in zip(x, y)], theta

def se_diff(t, c):                           # standard error of a difference in means
    return (var(t) / len(t) + var(c) / len(c)) ** 0.5

# Tasks completed per user, last month (pre) and during the test (y). Illustrative.
pre_c = [2, 9, 4, 15, 6, 1, 11, 7]
y_c   = [5, 7, 2, 13, 9, 3, 9, 10]
pre_t = [3, 8, 5, 14, 2, 10, 6, 9]
y_t   = [4, 12, 9, 15, 6, 10, 6, 13]

adj, theta = cuped(y_c + y_t, pre_c + pre_t)  # pool both arms to estimate theta
adj_c, adj_t = adj[:8], adj[8:]
print(f"theta {theta:.3f}")                                          # theta 0.754
print(f"raw   diff {mean(y_t) - mean(y_c):.3f} se {se_diff(y_t, y_c):.3f}")        # raw   diff 2.125 se 1.896
print(f"cuped diff {mean(adj_t) - mean(adj_c):.3f} se {se_diff(adj_t, adj_c):.3f}")  # cuped diff 1.937 se 0.986
print(f"variance kept {var(adj) / var(y_c + y_t):.3f}")                   # variance kept 0.316
```

rust

```rust
fn mean(v: &[f64]) -> f64 {
    v.iter().sum::<f64>() / v.len() as f64
}

fn var(v: &[f64]) -> f64 {
    // sample variance (n - 1)
    let m = mean(v);
    v.iter().map(|x| (x - m).powi(2)).sum::<f64>() / (v.len() as f64 - 1.0)
}

fn cov(a: &[f64], b: &[f64]) -> f64 {
    let (ma, mb) = (mean(a), mean(b));
    a.iter().zip(b).map(|(x, y)| (x - ma) * (y - mb)).sum::<f64>() / (a.len() as f64 - 1.0)
}

/// Adjust y by the pre-period covariate x: y - theta * (x - mean(x)).
fn cuped(y: &[f64], x: &[f64]) -> (Vec<f64>, f64) {
```

```
    let theta = cov(x, y) / var(x); // the regression slope of y on x
    let mx = mean(x);
    (x.iter().zip(y).map(|(xi, yi)| yi - theta * (xi - mx)).collect(), theta)
}

fn se_diff(t: &[f64], c: &[f64]) -> f64 {
    // standard error of a difference in means
    (var(t) / t.len() as f64 + var(c) / c.len() as f64).sqrt()
}

fn main() {
    // Tasks completed per user, last month (pre) and during the test (y). Illustrative.
    let pre_c = [2.0, 9.0, 4.0, 15.0, 6.0, 1.0, 11.0, 7.0];
    let y_c = [5.0, 7.0, 2.0, 13.0, 9.0, 3.0, 9.0, 10.0];
    let pre_t = [3.0, 8.0, 5.0, 14.0, 2.0, 10.0, 6.0, 9.0];
    let y_t = [4.0, 12.0, 9.0, 15.0, 6.0, 10.0, 6.0, 13.0];

    let y: Vec<f64> = y_c.iter().chain(&y_t).cloned().collect();
    let x: Vec<f64> = pre_c.iter().chain(&pre_t).cloned().collect();
    let (adj, theta) = cuped(&y, &x); // pool both arms to estimate theta
    let (adj_c, adj_t) = adj.split_at(8);
    println!("theta {:.3}", theta); // theta 0.754
    println!("raw   diff {:.3} se {:.3}", mean(&y_t) - mean(&y_c), se_diff(&y_t, &y_c)); // raw   diff 2.125 se
1.896
    println!("cuped diff {:.3} se {:.3}", mean(adj_t) - mean(adj_c), se_diff(adj_t, adj_c)); // cuped diff 1.937
se 0.986
    println!("variance kept {:.3}", var(&adj) / var(&y)); // variance kept 0.316
}
```

9. Many metrics, many segments, heterogeneous effects

Plain words

Throw enough darts and one hits

Check 40 numbers and, by luck alone, about two will look "significant" at the 5% level. Slice your users 20 ways (by country, by phone, by age) and some slice will show a dramatic effect that is just noise. That does not mean slices are useless. It means a surprising slice is a question to check in a new test, not a finding.

Real differences between groups do exist: a change can help beginners and annoy experts. Spotting those is valuable, as long as you confirm them.

For leaders

Separate deciding from exploring

- **Decide on the pre-registered metrics.** The primary and guardrails get their full error budget. Everything else on the scorecard is exploratory.

- **Report segments that were planned.** A few segments named in the brief (new versus returning, platform, language) are fair. A segment found by browsing 30 cuts needs a follow-up test.
- **Ask whether a segment effect is different, or just significant.** "Significant in Germany, not in France" does not mean the effect differs between them; the interaction test decides that.
- **Act on heterogeneity carefully.** Shipping only to the segment where it won bakes in the winner's curse. Confirm first.

Under the hood

Corrections and interaction tests

With m independent tests at level α and no true effects, the chance of at least one false positive is $1-(1-\alpha)^m$: 64% at $m=20$. The corrections from statistics for eval sets apply: Holm (or Bonferroni) to control the family-wise error rate for a small set of decision metrics, Benjamini–Hochberg to control the false discovery rate across a large exploratory scorecard.

For heterogeneous treatment effects, test the difference directly: $\hat{\Delta}_A - \hat{\Delta}_B$ with standard error $\sqrt{\mathrm{SE}_A^2 + \mathrm{SE}_B^2}$ for disjoint segments. Segment only on attributes fixed before assignment (country at sign-up, prior usage); segmenting on post-treatment behaviour ("users who used the new feature") conditions on an outcome and reintroduces selection bias. Aggregates can also reverse within segments if the treatment changes the segment mix (Simpson's paradox), which is one more reason to check SRM per segment. Shrinkage estimators (empirical Bayes across segments) give better estimates of per-segment effects than raw segment means.

10. Ramping, holdouts, and Twyman's law

Plain words

Dip a toe, then wade in

You test the water with a toe before jumping in. A **ramp** does the same with a launch: 1% of users first, to catch crashes, then 5%, then a proper 50/50 test, then everyone. A **holdout** keeps a small group on the old version for months after launch, so you can still ask "is the new version still better?" long after the novelty wears off.

And when a result looks amazing, be suspicious first. A 30% jump from a small change is far more often a bug than a breakthrough.

For leaders

Stages with different jobs

1 1% canary Catch crashes, errors, latency spikes. Hours to a day.	**2 5 to 10%** Check guardrails and SRM with real volume. A few days.	**3 50/50 test** Maximum power for the decision. Full planned duration.	**4 100% with holdout** Ship, keep 1 to 5% on control for a quarter.

Only the 50/50 stage answers the question; early stages are for safety, and their data should not be pooled into the decision analysis because the population and time period differ. Holdouts measure cumulative impact across many launches and catch slow effects like retention or trust. They cost the held-out users the improvement, so keep them small and time-boxed.

Twyman's law: "Any figure that looks interesting or different is usually wrong." Kohavi and colleagues made it a rule for experimenters. Treat an unexpectedly large win as a bug report until logging, SRM, and the metric definition check out.

Under the hood

What each stage can and cannot tell you

At 1% allocation, power for a 1-point effect is negligible, so a "flat" canary means only "no catastrophe". Increasing allocation mid-test changes the mix of early and late entrants per arm; analyse each allocation period separately or only the stable period. For a holdout of fraction h, the standard error of the long-run effect scales with $\sqrt{1/(hN) + 1/((1-h)N)} \approx 1/\sqrt{hN}$, so a 2% holdout needs about 13 times the users of a 50/50 test for the same precision; that is acceptable because holdouts run for months on the full population. Contamination is the main threat: holdout users who see treated features through shared content or another device.

11. Reading the result: the launch decision table

With trust checks done and the analysis run as planned, most outcomes fall into one cell of this table. The table belongs in the brief so the decision is mechanical.

Primary metric	Guardrails all clear	A guardrail breached	Trust check failed (SRM, A/A, logging)
Significant gain, interval above zero	**Ship.** Keep a holdout if the change is visible or large.	**Trade-off review.** Price the breach against the gain; often fix and rerun.	**Do not read.** Find the cause, fix, rerun.
Flat, interval inside ±MDE	**Ship only if the change has other value** (cost, simplicity, a platform step). Otherwise stop.	**Do not ship.**	**Do not read.**
Inconclusive, interval wider than ±MDE	**Underpowered.** Extend only if planned (sequential design), or redesign.	**Do not ship.**	**Do not read.**
Significant loss	**Do not ship.** Learn why from diagnostics.	**Do not ship.**	**Do not read.**

Two common distortions to watch for: shipping a "directionally positive" result from the inconclusive row, and treating a gain in the primary metric as licence to ignore a guardrail that the brief called non-negotiable.

Eval lens

An A/B test is an eval run on live traffic, and every lesson from statistics for eval sets carries over: report intervals, not just p-values; state the MDE beside a null; correct for multiple metrics; match the analysis unit to the randomization unit, as clustering does for multi-turn evals in intervals. The difference is that you cannot rerun the same users. Every trust check (SRM, A/A tests, exposure logging) exists because, unlike a fixed eval set, you cannot inspect the test items directly.

Principal asks

What decision did the brief commit to before launch, and what would have made us not ship? What is the unit of randomization, and does the analysis unit match it? What was the MDE, and was the test powered for it on the triggered population? Did the SRM check pass overall, per day, and per platform? How many times did someone look, and what stopping rule applied? Is the effect stable by days since first exposure? Which segments were pre-registered? Where is the holdout?

Common mistakes

Stopping the first day p dips below 0.05. Ignoring an SRM because the result looks great. Randomizing by user but computing standard errors per request. Running five days and calling it a week. Reading a flat result from an underpowered test as "no effect". Picking the primary metric after seeing the scorecard. Shipping to the one segment that won. Applying CUPED with a covariate measured after assignment. Celebrating a huge win before Twyman's law has been satisfied.

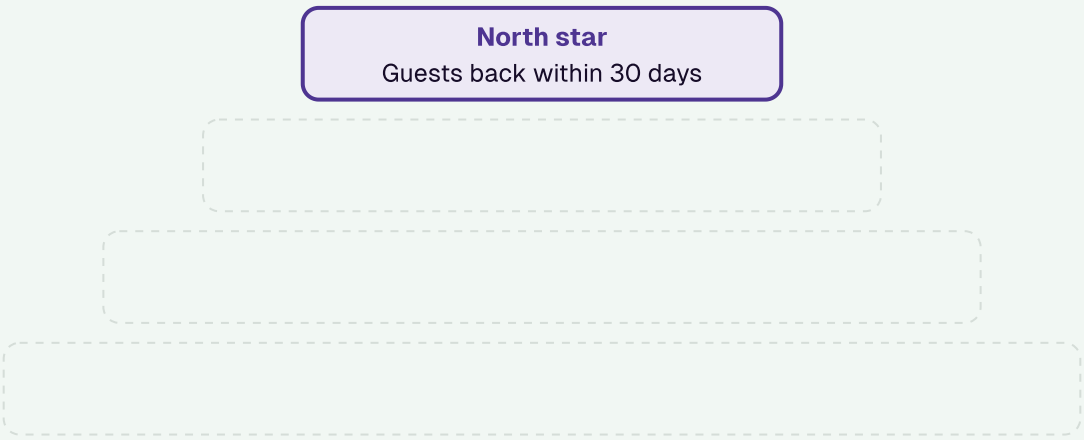
Part VI · Chapter 46

Choosing metrics and logging enough to know why

An experiment can only be as good as what it measures. Pick the wrong metric and a well-run test optimizes the wrong thing. Log too little and a surprising result becomes an argument instead of an investigation. This chapter covers how to build a metric hierarchy for an AI product, what makes an individual metric good, the traps in ratio and proxy metrics, and the instrumentation that lets you answer "why did that move?" It builds on measurement for validity and reliability and feeds directly into A/B tests.

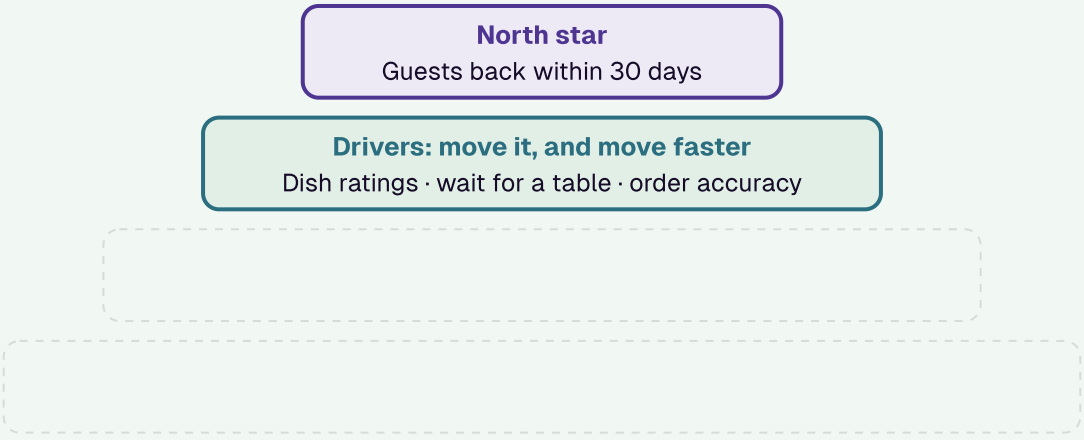
Diagram **Running a restaurant by the numbers**

The four kinds of metric, built one layer at a time.



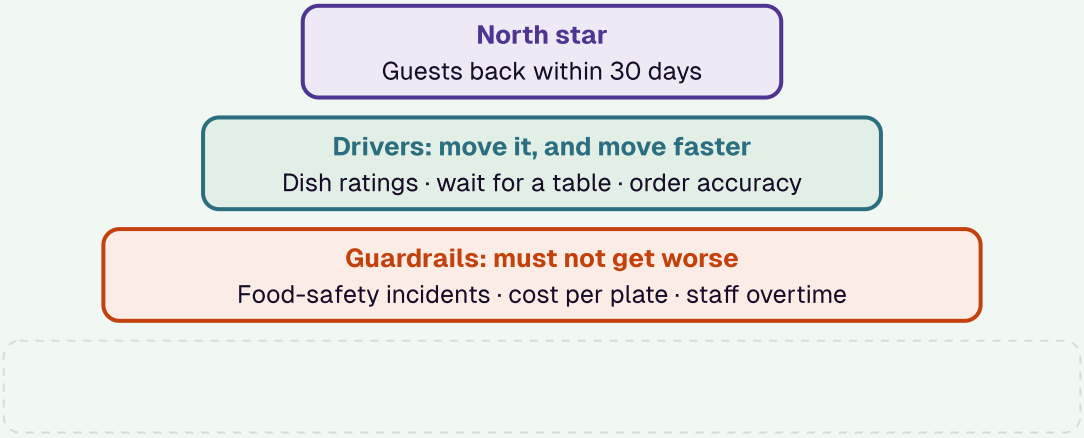
True to the goal, but it takes a month to move.

The owner wants one number that says whether the restaurant is doing well. She picks "guests who come back within 30 days": it captures food, service and price together. This is the **north star**, or overall evaluation criterion. Its weakness is speed: it takes a month to see a change.



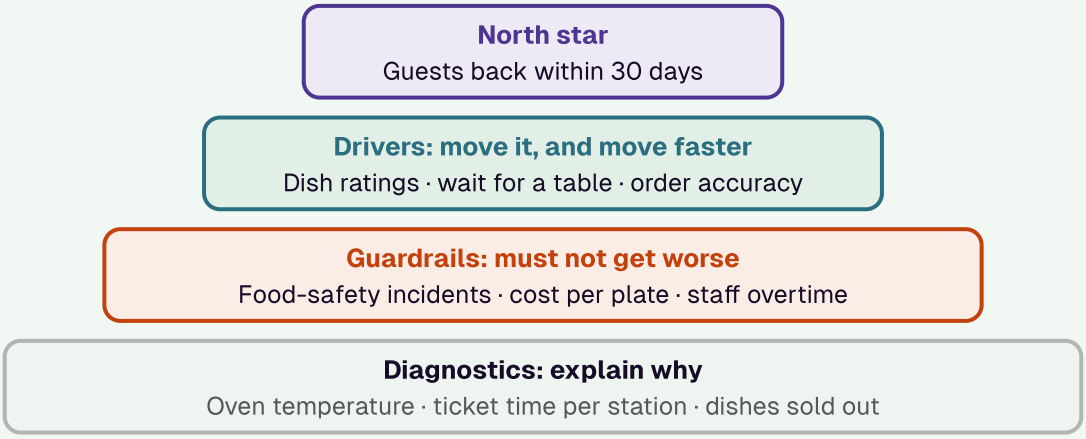
Visible tonight, and known to predict return visits.

Below it go **drivers**: numbers that respond in days, not months, and that past data shows lead to return visits. Dish ratings, the wait for a table, orders delivered correctly. A new recipe is judged on these, because they move fast enough to test.



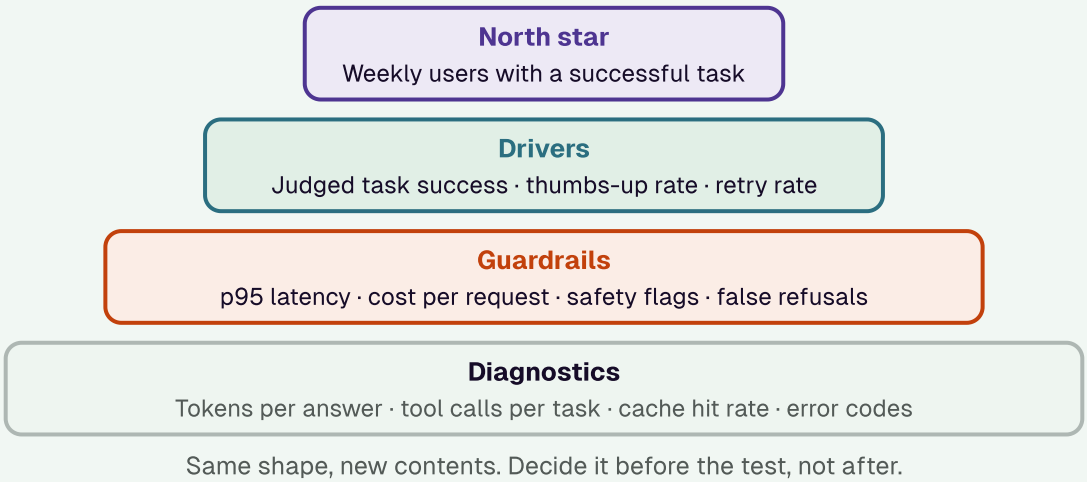
A gain upstairs never pays for a breach here.

Then **guardrails**: things that must not get worse whatever else improves. A richer recipe that raises ratings but doubles the cost per plate, or causes a food-safety incident, is not a win. Guardrails are what stop a team from hitting its target in a way the business cannot afford.



Never the goal. Always the first place to look.

At the base sit **diagnostics**: oven temperatures, how long each station takes per ticket, which dishes sold out. Nobody is rewarded for them, but when ratings drop on a Friday, they tell you it was the grill station falling behind.



Same shape, new contents. Decide it before the test, not after.

An AI assistant fits the same shape. The north star is how many people get real work done each week. Drivers are faster signals that predict it. Guardrails cover latency, cost, safety and over-refusal. Diagnostics such as answer length and tool calls explain any movement. The rest of this chapter is about filling each tier well.

1. The metric hierarchy

Plain words

Goal, signals, tripwires, gauges

A car dashboard does not have one number. You care about arriving (the goal), you watch your speed and route (signals that tell you whether you will arrive on time), warning lights must stay off (tripwires), and the gauges like engine temperature explain trouble when it comes. Products need the same set of numbers, each with its own job, and confusion starts when one number is asked to do two jobs.

For leaders

A hierarchy for an AI assistant

Tier	Example metric	Job in a decision	Typical trap
North star (OEC)	Weekly active users with at least one judged-successful task	Defines winning for the product over quarters	Too slow and noisy to test against directly
Drivers	Judged task success rate; thumbs-up rate; retry (regenerate) rate	Primary metric of most experiments	A driver that stopped predicting the north star
Guardrails: product	p95 latency; cost per request; safety violation rate; benign refusal rate; error rate	Veto a launch	Underpowered, so "no harm detected" means nothing
Guardrails: trust	Sample ratio; logging completeness; exposure rate	Decide whether to believe the test	Checked after the results, or not at all
Diagnostics	Tokens per answer; tool calls per task; cache hit rate; error codes by client	Explain why something moved	Promoted to a target and then gamed

Write the hierarchy once per product, review it quarterly, and require every experiment brief to pick its primary metric from the driver tier. The value is less in any one metric than in the shared agreement about what each one is for.

Under the hood

Validating drivers against the north star

A driver earns its place by predicting the north star **under intervention**, not just by correlating with it in observational data. The test: across a corpus of past experiments, regress the north-star effect on the driver effect. A good driver has treatment effects that line up with north-star effects in sign and roughly in size; a driver that moved in many experiments whose north-star effect was flat or opposite is a proxy that has drifted. This is the online version of the offline-to-online calibration in the offline–online gap. Composite OECs, a weighted sum of drivers, are common; set the weights from this experiment corpus or from an explicit trade-off ("1 point of task success is worth 50 ms of p95 latency"), and write them down, because the weights are a product decision.

2. What makes a good metric

Plain words

A good thermometer

A good thermometer moves when the temperature moves (sensitive), goes up when it gets hotter and never the other way (directional), reacts to your room and not the neighbour's oven (attributable), and cannot be fooled by breathing on it (hard to game). Product metrics need the same four qualities. Most fail at least one, and knowing which one tells you how far to trust them.

For leaders

Four properties, and the tension between two of them

Sensitive

Moves detectably in a test of normal size when the change matters. Retention over 90 days rarely does.

Directional

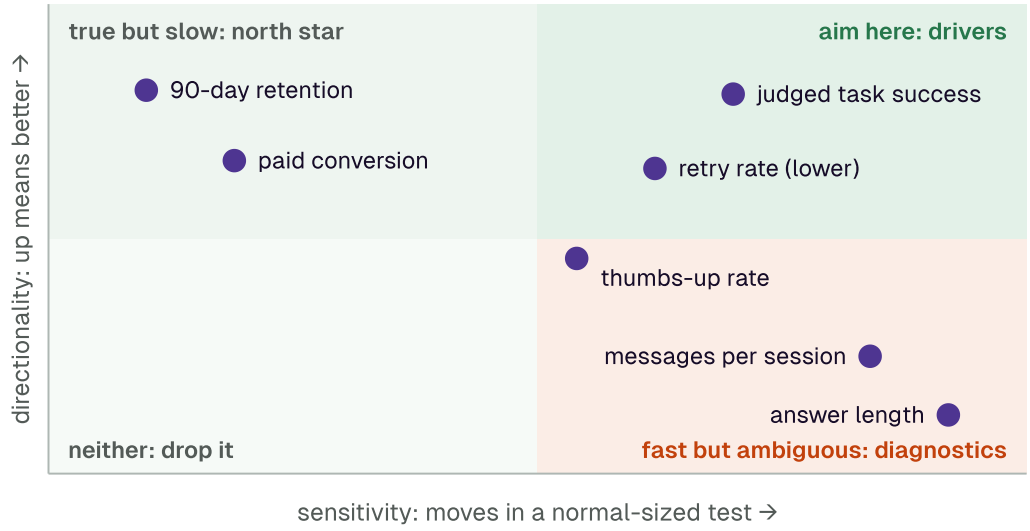
Up always means better. "Messages per session" fails: more messages can mean engagement or a model that never gets it right.

Attributable

You can trace a movement to the change and the users who saw it, which needs exposure and version logging.

Hard to game

A team cannot raise it without the user being better off. Answer length and thumbs-up rate are easy to inflate.



Illustrative placement. Thumbs-up rate sits on the line: it moves easily and mostly means better, but it rewards flattery and covers only users who bother to rate.

Under the hood

Measuring sensitivity and directionality

Sensitivity has two parts: the effect a change produces and the metric's noise. For a metric with per-unit standard deviation σ and an effect δ, the detectable fraction scales with δ/σ. Estimate it empirically as the share of past experiments in which the metric moved significantly, and compare variance across candidate definitions: capping heavy users, binarising ("at least one success" instead of "count of successes"), or CUPED can cut σ severalfold. **Directionality** is estimated against labelled experiments, ones the team agrees were good or bad on deeper evidence, or against the north star as above. Deng and Shi (KDD 2016) describe this pair of properties, sensitivity and directionality, as the core of metric evaluation in an experimentation program. Report both for any proposed new metric before it is allowed into a scorecard.

3. Ratio metrics and their pitfalls

Plain words

The batting average that rose because a player stopped batting

A ratio has two moving parts. "Share of answers rated thumbs-up" can rise because answers got better, or because unhappy people stopped rating at all. The top of the fraction did not change; the

bottom shrank. Whenever you see a rate move, ask what happened to the top and to the bottom separately.

A second trap: one very active person contributes hundreds of ratings. If you treat each rating as a separate, independent vote, you think you have far more evidence than you do.

For leaders

Ask for the numerator and denominator

- **Always show both parts** next to any rate in a scorecard. A treatment that changes the denominator (who rates, who reaches a step, how many requests a user sends) can move a rate with no change in quality.
- **Say which average you mean.** "Ratio of sums" (all thumbs-up over all ratings) weights heavy users more; "mean of per-user ratios" weights each user equally. They answer different questions and can disagree by 10 points.
- **Check the error bars are computed per user**, the unit that was randomized. Per-rating standard errors are too small, which produces false wins.

Under the hood

The delta method

When users are randomized and the metric is $R = \sum_i A_i / \sum_i B_i$ (for example thumbs-up A_i over rated responses B_i for user i), the user totals (A_i, B_i) are independent across users, but the individual ratings inside a user are not. A first-order Taylor expansion gives

$$\operatorname{Var}(\hat{R}) \approx \frac{1}{n\,\mu_B^2}\left(\sigma_A^2 - 2R\,\sigma_{AB} + R^2\sigma_B^2\right)$$

n users; μ_B the mean denominator per user; $\sigma_A^2, \sigma_B^2, \sigma_{AB}$ the per-user variances and covariance. Deng, Knoblich and Lu (KDD 2018) give a practical guide to this and related cases.

The binomial formula $\sqrt{R(1-R)/\sum B_i}$ treats every rating as independent and is wrong whenever users contribute unequal, correlated batches. In the example below, two heavy users make the naive standard error 2.5 times too small. A user-level bootstrap gives the same answer as the delta method and is a good cross-check; see intervals.

python

```python
def mean(v):
    return sum(v) / len(v)

def ratio_se(num, den):
```

```
    """Delta-method standard error of sum(num) / sum(den), with users as the unit."""
    n = len(num)
    mn, md = mean(num), mean(den)
    r = mn / md
    vn = sum((a - mn) ** 2 for a in num) / (n - 1)
    vd = sum((b - md) ** 2 for b in den) / (n - 1)
    cnd = sum((a - mn) * (b - md) for a, b in zip(num, den)) / (n - 1)
    # Var(R) ~ (Var(num) - 2 R Cov(num, den) + R^2 Var(den)) / (n * mean(den)^2)
    return r, ((vn - 2 * r * cnd + r * r * vd) / (n * md * md)) ** 0.5

# Per user: thumbs-up clicks and rated responses. Two heavy users dominate. Illustrative.
ups   = [1, 0, 2, 1, 30, 1, 3, 2, 0, 4, 1, 5]
rated = [2, 1, 3, 2, 60, 1, 4, 3, 1, 5, 2, 40]

r, se_delta = ratio_se(ups, rated)
n_rated = sum(rated)
se_naive = (r * (1 - r) / n_rated) ** 0.5    # pretends every rating is independent
print(f"rate {r:.3f} from {n_rated} ratings by {len(ups)} users")   # rate 0.403 from 124 ratings by 12 users
print(f"naive se {se_naive:.3f}  delta se {se_delta:.3f}  ratio {se_delta / se_naive:.2f}")
# naive se 0.044  delta se 0.108  ratio 2.46
print(f"mean of per-user rates {mean([a / b for a, b in zip(ups, rated)]):.3f}")  # 0.501, a different metric
```

rust

```
fn mean(v: &[f64]) -> f64 {
    v.iter().sum::<f64>() / v.len() as f64
}

/// Delta-method standard error of sum(num) / sum(den), with users as the unit.
fn ratio_se(num: &[f64], den: &[f64]) -> (f64, f64) {
    let n = num.len() as f64;
    let (mn, md) = (mean(num), mean(den));
    let r = mn / md;
    let vn = num.iter().map(|a| (a - mn).powi(2)).sum::<f64>() / (n - 1.0);
    let vd = den.iter().map(|b| (b - md).powi(2)).sum::<f64>() / (n - 1.0);
    let cnd = num.iter().zip(den).map(|(a, b)| (a - mn) * (b - md)).sum::<f64>() / (n - 1.0);
    // Var(R) ~ (Var(num) - 2 R Cov(num, den) + R^2 Var(den)) / (n * mean(den)^2)
    (r, ((vn - 2.0 * r * cnd + r * r * vd) / (n * md * md)).sqrt())
}

fn main() {
    // Per user: thumbs-up clicks and rated responses. Two heavy users dominate. Illustrative.
    let ups = [1.0, 0.0, 2.0, 1.0, 30.0, 1.0, 3.0, 2.0, 0.0, 4.0, 1.0, 5.0];
    let rated = [2.0, 1.0, 3.0, 2.0, 60.0, 1.0, 4.0, 3.0, 1.0, 5.0, 2.0, 40.0];

    let (r, se_delta) = ratio_se(&ups, &rated);
    let n_rated: f64 = rated.iter().sum();
    let se_naive = (r * (1.0 - r) / n_rated).sqrt(); // pretends every rating is independent
    println!("rate {:.3} from {} ratings by {} users", r, n_rated, ups.len()); // rate 0.403 from 124 ratings by
12 users
    println!("naive se {:.3}  delta se {:.3}  ratio {:.2}", se_naive, se_delta, se_delta / se_naive);
    // naive se 0.044  delta se 0.108  ratio 2.46
    let per_user: Vec<f64> = ups.iter().zip(&rated).map(|(a, b)| a / b).collect();
    println!("mean of per-user rates {:.3}", mean(&per_user)); // 0.501, a different metric
}
```

4. Proxy metrics and Goodhart's law

Plain words

Paid by the line

Pay a writer by the line and you get long, padded articles. The line count was a reasonable hint of effort until it became the thing being paid for. That is Goodhart's law: once a number becomes a target, people (and optimizers) find ways to raise it that do not deliver what it was meant to measure. Models trained or tuned against a metric are extremely good at finding those ways.

For leaders

Assume every proxy will be gamed

Common AI proxies and how they fail when pushed:

- **Thumbs-up rate** rewards agreeable, flattering answers. Pushed hard, it selects for sycophancy.
- **Session length or messages per session** rise when the model is slow to resolve a task.
- **Judge score** inherits the judge's biases toward length and confident tone; see LLM as judge.
- **Harmful-compliance rate on a red-team set** falls to zero if the model learns to refuse everything that looks risky, harming benign users.

The defences: pair each proxy with a counter-metric that moves the other way when it is gamed (length with judged correctness, refusals with benign refusal rate), keep a slow, expensive measurement (human review, retention) as a periodic audit, and rotate or refresh proxies. Leading eval work treats Goodhart as the default condition to plan around.

Under the hood

Why optimization breaks proxies

Write the proxy as $M = V + \varepsilon$, true value plus error. Selecting or optimizing for high M preferentially picks items with high ε, so the gap between M and V grows with optimization pressure. With a heavy-tailed error, the strongest optimization eventually selects almost purely on error. That is the same mechanism as reward-model overoptimization in preferences and reward models, and the remedy is the same: measure V independently on a sample and stop pushing when the proxy and the independent measure diverge.

5. Offline and online metrics

Plain words

The driving test and the commute

A driving test checks skills on a set route with an examiner. The daily commute is real traffic, real hurry, real weather. Passing the test predicts safe commuting, but not perfectly. Offline evals are the driving test: fixed questions, graded the same way each time. Online metrics are the commute: what actually happens with real users. You need both, and you need to know how well one predicts the other.

For leaders

Use offline to filter, online to decide

Offline evals are cheap, repeatable and safe, so they gate which candidates reach users. Online tests are expensive and slow, but they measure the outcome. A model change should name the offline evals it was selected on and the online metric it is expected to move, and the team should track how often offline wins turn into online wins. The offline–online gap shows how to calibrate the conversion rate and what lives in the gap: distribution shift, user adaptation, latency, and the interface.

Under the hood

Making offline and online comparable

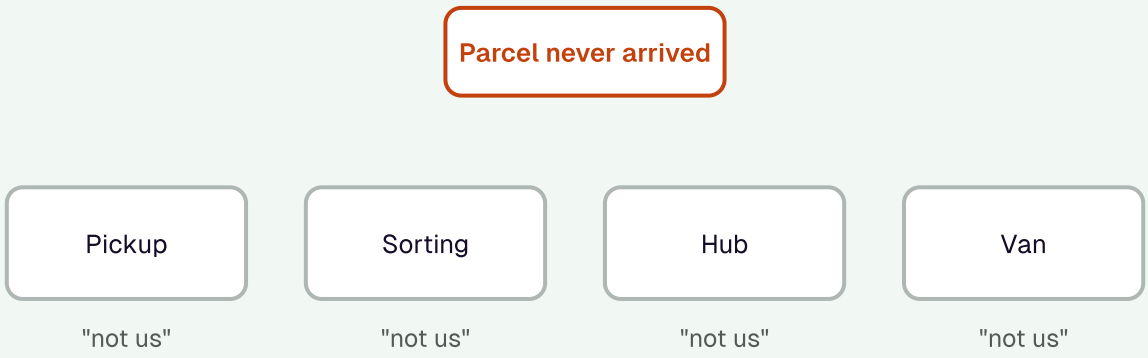
Build offline sets from logged production traffic (sampled by the same rules as the online population, privacy-filtered, and refreshed), grade them with a validated judge, and weight items by production frequency. Then each launch yields a pair (offline delta, online delta), and a regression across launches gives the calibration slope and its uncertainty. A slope near zero means the offline metric is not a useful filter for that product, whatever its face validity.

6. The instrumentation layer

Diagram

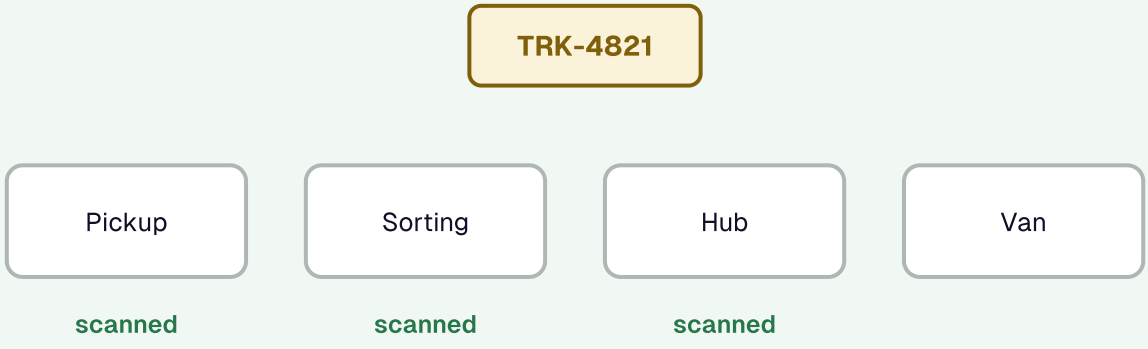
A tracking number for every answer

Why a shared ID is the most valuable field in your logs.



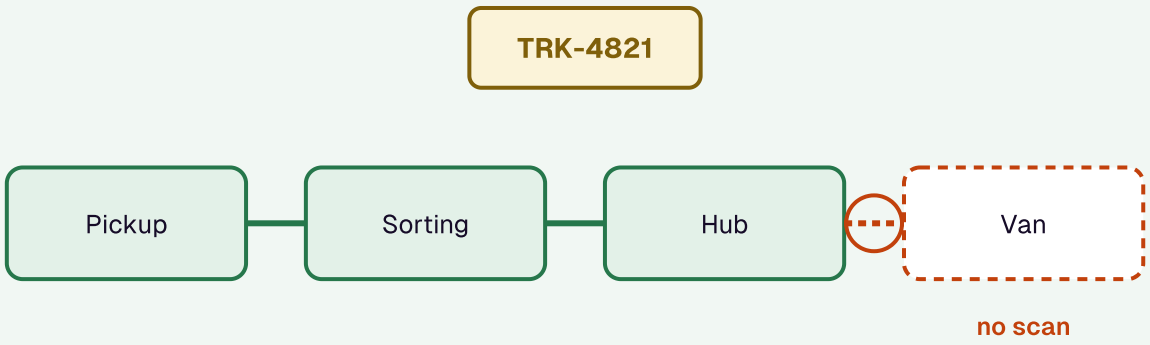
Each depot keeps its own records. None of them can be matched.

A parcel goes missing. Every depot keeps a log, and every depot says it was not them. Without a way to match one depot's record to the next, there is no way to find out where it disappeared. This is what debugging feels like when an AI product's logs cannot be joined.



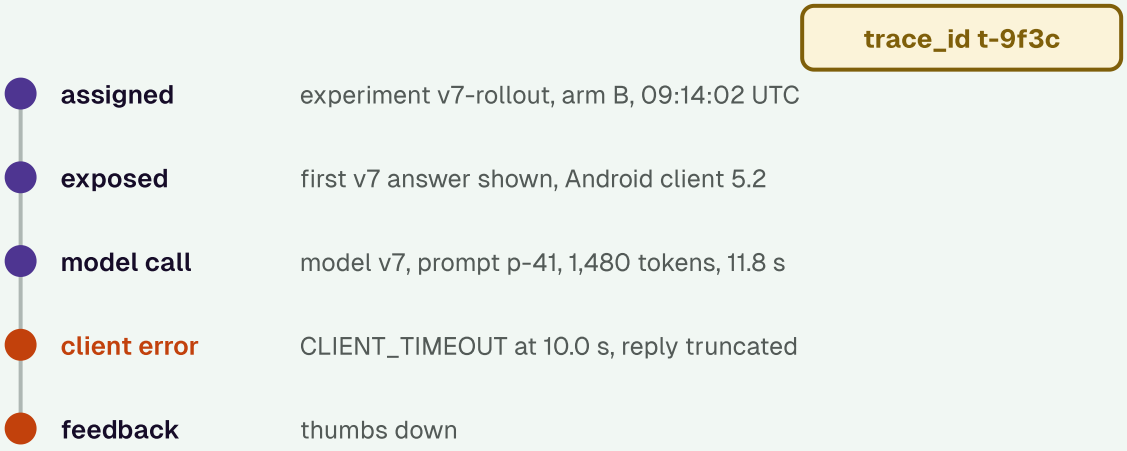
One number, printed once, scanned everywhere.

Now give the parcel a tracking number. Every depot scans it and writes down the time. The number costs almost nothing, and it turns four separate logs into one story about one parcel.

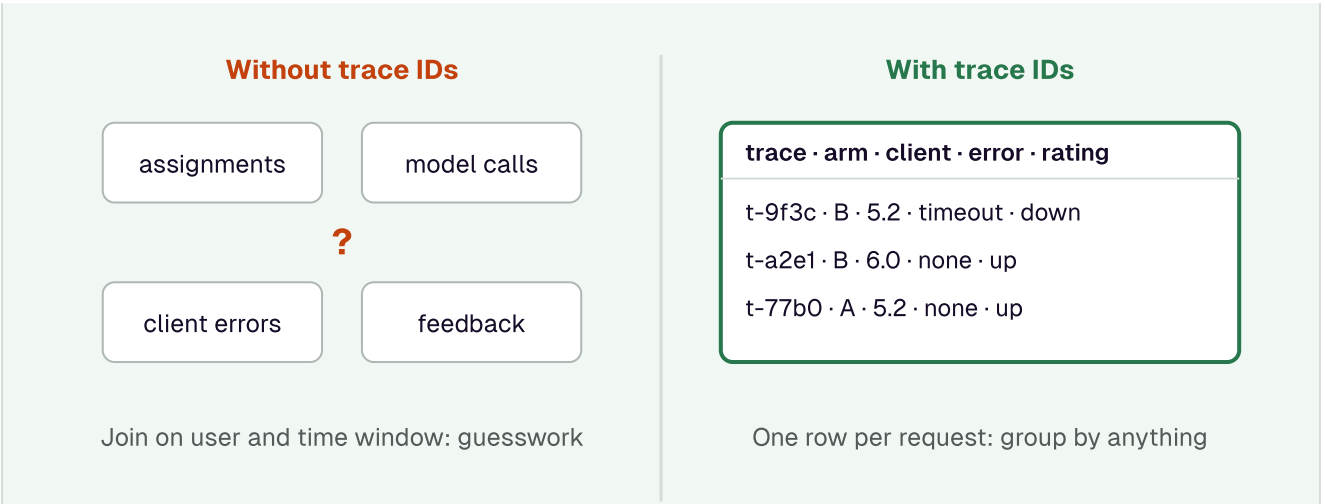


Lost between the hub and the van. Ask the hub's loading dock.

The trail shows the parcel reached the hub at 13:40 and was never scanned onto a van. The search shrinks from four depots to one loading dock. Missing events are evidence too, but only when you can tell which events belong together.



An AI request is a parcel with more depots. The trace ID is its tracking number: generated once at the start and copied onto every event, from assignment to the model call to the user's rating. Here it shows a thumbs-down that came from a client timeout on a long answer, not from a bad answer.



Without a shared ID, the assignment table, model logs, client errors and feedback live in separate piles, joined by guesses about users and time windows. With it, every request becomes one row you can slice by arm, client version, error code or model version. The rest of this section is about which columns that row needs.

Plain words

Write down who, what, when, and which version

To explain any change later, your records need to answer four questions for every interaction: who was it (and which group were they in), what did they actually see, when did it happen, and which version of everything produced it. Leave one out and some questions become unanswerable forever, because you cannot go back and log the past.

For leaders

The logging contract for experiments

- **Assignment logging.** Every time a unit is bucketed: unit ID, experiment, arm, timestamp. This is what the SRM check counts.
- **Exposure logging.** When the user actually **saw** the difference, not when they were assigned. Assignment happens at login; exposure happens when the first v7 answer renders. Analysing assigned-but-unexposed users dilutes the effect, and exposure that differs by arm causes SRM.
- **Trace and request IDs** that join model calls, tool calls, client events and feedback.
- **Versions on every event**: model, prompt template, system prompt, decoding config, safety filter, client app. Without them, a regression after a deploy cannot be attributed.
- **Event schemas** registered and versioned, so a renamed field does not silently zero a metric.

- **Timestamps in UTC** with both client and server time, so day boundaries and latency are computed consistently.

Budget for this as part of the feature, not as an afterthought. A launch whose logging cannot answer the questions in the experiment brief is not ready to launch.

Under the hood

Schemas, time, and sampling bias

A minimal event for an AI request might carry the fields below; everything an analysis needs to group by must be on the event or joinable through an ID.

json

```
{
  "event": "model_response",  "schema_version": 4,
  "trace_id": "t-9f3c",  "request_id": "r-1187",  "user_id_hash": "u-55ad",
  "experiment": {"id": "v7-rollout", "arm": "B", "assigned_at": "2026-10-12T09:14:02Z"},
  "exposed": true,
  "model": "assistant-v7",  "prompt_version": "p-41",  "decoding": {"temperature": 0.7},
  "client": {"platform": "android", "version": "5.2"},  "region": "eu-west",
  "ts_server": "2026-10-12T09:15:40.120Z",  "ts_client": "2026-10-12T11:15:39.870+02:00",
  "latency_ms": 11800,  "output_tokens": 1480,  "error_code": "CLIENT_TIMEOUT"
}
```

Time. Store server time in UTC as the source of truth; keep client time for latency and offline queues, but expect clock skew and late-arriving events. Define the analysis "day" explicitly (UTC, or the user's local day) and use it everywhere, or you get partial days: a test that starts at 17:00 Pacific (00:00 UTC) has a full first UTC day but a first local day only seven hours long.

Log sampling. Full traces are expensive, so teams sample. Sampling is safe only if it is independent of the outcome and of the arm. Two classic biases: **tail sampling** that keeps all errors but 1% of successes inflates error rates unless you reweight by the inverse sampling probability; and sampling decided per request after the response returns can depend on latency. Decide sampling at the trace's start by hashing the trace ID, log the sampling rate on the event, and keep metric computation on unsampled counters where possible.

7. Data-quality checks

Plain words

Weigh the scale before you weigh the flour

Before baking, you check the kitchen scale reads zero with nothing on it. Experiments need the same habit. Run a test where both groups get exactly the same thing and make sure the system reports no difference. Check that groups are the right size. Watch for data that suddenly goes missing. These checks are boring, and they catch the most expensive mistakes.

For leaders

Checks that should run automatically

- **A/A tests** on the platform: about 5% of metrics should come out significant at $p < 0.05$, and p-values should be roughly uniform. Far more means the variance computation is wrong.
- **SRM** on every experiment, overall and by day and platform (A/B tests).
- **Missing-data monitors**: event volume per source, per client version, per hour, with alerts on drops. A silent client logging failure looks exactly like users leaving.
- **Metric definitions as code**: one reviewed, versioned definition per metric, used by every dashboard and every experiment, with tests. Two teams computing "active user" differently is how launches get contradictory reads.

Under the hood

What a failed A/A test tells you

Run many A/A comparisons (re-randomize historical data into fake arms, which is cheap) and look at the distribution of p-values per metric. Too many small p-values usually means the variance ignores clustering, such as per-request metrics with user randomization, or that heavy-tailed outliers dominate; p-values piled near 1 mean the variance is overestimated. Both are bugs in the analysis, and both would otherwise distort every real experiment on that metric.

8. Debugging a surprising result: a worked investigation

A worked example with illustrative numbers. Assistant v7 is in a 50/50 test against v6. After a week, thumbs-up rate in treatment is 67% against 71% in control, a statistically clear drop, while offline evals had v7 ahead by 3 points. The investigation, in the order the triage chapter recommends:

1. **Trust first.** SRM on exposed users: p = 0.41, fine. Exposure rate 97% in both arms. Feedback events joined to 99.8% of traces in both arms. The data is not broken.
2. **Numerator and denominator.** Rating rate per exposed user is unchanged (12% versus 12%), so this is not a change in who rates.
3. **Slice by pre-planned segments.** By region: the same small drop everywhere. By client: 92% of the drop is on Android client 5.2, which is 18% of traffic.

4. **Join traces for that slice.** Error codes show CLIENT_TIMEOUT on 18% of v7 responses on 5.2, versus 2% for v6. v7 answers are 40% longer, and the 5.2 client has a hard 10-second timeout that truncates the reply.

5. **Check the alternative explanations.** Judge-scored quality of completed v7 answers is up 1.2 points on every client. Prompt version unchanged. The model is better; the old client cuts it off.

6. **Decide.** Raise the client timeout (or stream the response) on 5.2, keep v7, rerun on 5.2 users to confirm. Add "client timeout rate" as a guardrail for future model launches.

Every step needed a specific field. The widget below lets you remove fields and see which explanations become unknowable.

Interactive

Can you explain the drop? (shown at its default setting)

Toggle which fields your logs carry. Each candidate root cause is confirmed, ruled out, or stays unknowable, depending on whether the fields it needs were logged. Start from a typical dashboard and find the smallest set that explains the drop.

Scenario. Assistant v7 is in a 50/50 test against v6. After a week, thumbs-up rate is 67% in treatment versus 71% in control. Offline evals had v7 ahead. Leadership wants to know whether to roll back.

PRESETS

FIELDS LOGGED ON EACH EVENT

Model version | Prompt version | Exposure | Client version | Region | Latency | Error codes | Trace IDs

Confirmed	Ruled out	Unknowable
0	1	8

Unknowable

v7 writes worse answers

Needs Model version + Trace IDs. Missing: **Trace IDs**.

Unknowable

A prompt-template change shipped the same week

Needs Prompt version. Missing: **Prompt version**.

Unknowable

The drop comes from users assigned to v7 who never saw it

Needs Exposure. Missing: **Exposure**.

Unknowable

A logging bug loses feedback events in treatment

Needs Exposure + Trace IDs. Missing: **Exposure, Trace IDs**.

Unknowable

Fewer people rate, and the ones who do are grumpier

Needs Exposure. Missing: **Exposure**.

Unknowable

A regional outage in one data center

Needs Region + Error codes. Missing: **Error codes**.

Unknowable

v7 is slower everywhere, so people give up before rating

Needs Latency + Client version. Missing: **Client version**.

Unknowable

The old Android app (5.2) times out on v7's longer answers

Needs Client version + Latency + Error codes. Missing: **Client version, Error codes**.

Ruled out

A seasonal dip that hit everyone that week

Events tagged with v6 stay flat at 71% over the same days, so the dip is specific to v7.

The true cause stays hidden. 8 candidates are unknowable with these fields. Think about which fields would show where the drop happens and what failed there, then which ones rule out the other stories.

Eval lens

An online metric is a measurement instrument, with the same questions as an offline eval: is it valid for what we care about, is it reliable run to run, and what is its noise (measurement)? The instrumentation layer plays the role of the eval harness in building a harness: record every input, version and output so any number can be traced back to the items that produced it. A metric you cannot decompose to individual traces is as untrustworthy as a benchmark score you cannot decompose to individual items.

Principal asks

Which tier is this metric in, and who agreed? How do we know this driver predicts the north star under intervention, not just in correlation? What are the numerator and denominator, and did the treatment change the denominator? Is the variance computed at the randomization unit? What counter-metric catches gaming? Where is exposure logged, and does it fire identically in both arms? Can every event be joined by trace ID to model, prompt and client versions? If this metric dropped 4% tomorrow, which fields would we use to explain it, and do we log them today?

Common mistakes

Treating messages per session as engagement when it also measures failure. Reporting a rate without its denominator. Computing per-request standard errors in a user-randomized test. Analysing assigned users instead of exposed ones. Logging the model version but not the prompt or client version. Mixing local and UTC days. Tail-sampling logs and forgetting to reweight. Defining the same metric in three dashboards three ways. Promoting a diagnostic to a target.

Part VI · Chapter 47

Designing experiments for AI systems: the whole guide in one design

Every earlier chapter supplies one piece of a trustworthy decision about an AI system: the statistics of Part I, the evaluation craft of Part II, the understanding of what training changed from Part IV, the judgment of Part V, and the design discipline of this part. This capstone puts them together. It covers offline evals treated as real experiments, online tests for model changes, interleaving, human evaluation, and pre-registration, then works one realistic launch from question to decision and ends with a single checklist that points each item back to the chapter that covers it.

1. Offline evals are experiments

Plain words

Same exam, same day, same marker

To find out which of two students is better at maths, you give both the same exam, on the same day, marked by the same teacher. If one got an easier paper or a kinder marker, you would learn nothing. An offline eval is that exam for models: the same questions, the same settings, the same grader, so the only thing that differs is the model. And because models answer a little differently each time, you let each one sit the exam a few times.

For leaders

Treat an eval run like a lab experiment

- **Fixed items**: the eval set is frozen and versioned, and both models answer every item. Changing items between runs makes comparisons meaningless.

- **Paired comparison**: compare per item, not just two averages. Pairing removes item difficulty from the noise and often cuts the required set size severalfold (statistics for eval sets).
- **Seeds and sampling settings** recorded and repeated. A result that flips with the seed is not a result (leading eval work).
- **One change at a time**, or a design that separates them. A run that changes the model, the prompt and the grader at once cannot say which one helped.
- **Clean items**: checked for contamination against training data (contamination).

Under the hood

The model behind a paired eval

Write the score of model m on item i with seed s as $y_{mis} = \mu_m + b_i + \gamma_{mi} + \varepsilon_{mis}$: a model effect, an item difficulty shared by both models, a model-by-item interaction (some items suit one model more than the other), and sampling noise. The paired difference $d_i = \bar{y}_{Ai\cdot} - \bar{y}_{Bi\cdot}$ cancels b_i, so

$$\mathrm{Var}(\bar{d}) = \frac{1}{n}\left(\underbrace{\mathrm{Var}(\gamma_{Ai} - \gamma_{Bi})}_{\sigma^2_{\text{model}\times\text{item}}} + \frac{2\sigma^2_\varepsilon}{S}\right)$$

n items, S seeds per item and model. Item difficulty σ^2_b drops out entirely. More seeds shrink only the sampling term; only more items shrink the model-by-item term.

That decomposition tells you where to spend: if the seed term dominates, rerun; if the item term dominates, add items. Cluster the standard error by the unit that is actually independent (a conversation, a repository, a document) as in intervals, and run everything through a harness that records versions and seeds (building a harness).

2. Online A/B tests for model changes

Plain words

A new chef on half the tables

Testing a new model on real users is like letting a new chef cook for half the tables while the old chef cooks for the rest. You keep each table with the same chef all evening, because switching mid-meal

would confuse everyone. You watch whether diners are happier, but also whether the food comes out slower or costs more, because a better dish that takes an hour is not better.

For leaders

What is different about testing models

- **Randomize by user**, not by request. Users notice tone and capability, conversations carry memory, and per-request switching mixes the two models inside one conversation.
- **Cost and latency are first-class guardrails.** A model change moves both, and they can matter more than quality for a free tier.
- **Safety is a guardrail with low base rates.** Violations at 0.1% need very large samples to detect increases; pair the online guardrail with targeted offline red-team evals (safety).
- **Quality needs a measurement.** Thumbs are sparse and biased. Most teams run an LLM judge on a sample of production traffic from both arms, validated against human labels (LLM as judge).
- **Expect novelty.** A new model voice draws attention for a few days. Run at least two full weeks (A/B tests).

Under the hood

Pitfalls specific to LLM traffic

Shared caches: a prompt or response cache shared across arms lets control serve treatment outputs; key caches by model version. **Capacity**: a treatment model on a smaller serving pool may queue under load, so latency differences partly reflect provisioning, which will change at 100%. **Carryover**: users who saw treatment and leave the test take learned behaviour (longer prompts, new tasks) with them, which matters for re-randomized follow-ups. **Interference**: shared workspaces or agents acting on shared resources violate independence; randomize at the workspace level. **Triggering**: if the change only affects some requests (say, those that call a tool), log a counterfactual trigger in control and analyse triggered users, with variance at the user level (choosing metrics).

3. Interleaving for ranking and search

Plain words

Two playlists shuffled into one

Two friends each suggest songs for a party. Instead of playing one friend's list on Friday and the other's on Saturday, you shuffle both lists together and see whose songs people skip less. Every guest hears both, on the same night, so the weather, the crowd and the mood are identical for both

lists. Interleaving does this for search results: mix two rankers' results into one list and count whose results get clicked.

For leaders

Fast, sensitive, and narrow

Interleaving answers "which ranker do users prefer?" with far less traffic than an A/B test, because each user compares both rankers directly and between-user differences drop out. It suits retrieval for RAG, search, recommendations and tool selection. Its limit is scope: it measures preference within a results list, not downstream outcomes like task success, retention or cost. Use it to screen many ranking candidates quickly, then confirm the winner with an A/B test on the product metric.

Under the hood

Team-draft interleaving

In team-draft interleaving (Radlinski, Kurup and Joachims, CIKM 2008), the two rankers act like captains picking teams: a coin decides who picks first in each round, each picks its highest-ranked result not already chosen, and every slot remembers which team placed it. Clicks credit the team that placed the clicked result; a query is a win for whichever ranker earned more clicks, and the test is whether wins exceed 50% over queries (a sign test, clustered by user). Chapelle, Joachims, Radlinski and Yue (TOIS 2012) validated interleaving at scale on two commercial search engines, comparing it with editorial judgments and studying its statistical efficiency. For generative systems, the analogue is a side-by-side preference shown to a sample of users, which inherits the position-bias issues covered next.

4. Human evaluation design

Plain words

A blind taste test

In a fair taste test, tasters do not know which cola is which, the cups come in random order, and you first check that tasters agree with each other on a few practice cups. If they cannot tell the difference consistently, their verdict on the real test means little. Human evaluation of models needs the same three things: hide which model wrote what, randomize the order, and measure how often raters agree.

For leaders

Ratings are a measurement system to manage

- **Blinding**: raters never see model names, and outputs are normalised for tell-tale formatting where possible.
- **Randomized order**: left/right position randomized per item, because both people and judges prefer the first or second option at measurable rates.
- **Calibration**: a rubric with worked examples, a calibration session, and gold items mixed in to catch drift and inattention.
- **Agreement**: report inter-rater agreement on overlapping items. It is the ceiling on any judge or reward model trained on these labels (LLM as judge).
- **Rater effects**: some raters are harsher. Assign items so each rater sees both models' outputs, and analyse with rater as a factor.

Public leaderboards built on blinded pairwise votes, such as Chatbot Arena (Chiang et al., 2024), show the design at scale, and also its limits: the voters and prompts are whoever shows up, which is a sampling question (who is in your sample?).

Under the hood

Agreement and the rater as a cluster

Cohen's kappa corrects raw agreement p_o for chance agreement p_e:

$$\kappa = \frac{p_o - p_e}{1 - p_e}$$

For more than two raters or missing ratings, Krippendorff's alpha generalises the same idea.

Ratings from one rater are correlated, so a study with 2,000 ratings from 5 raters has far fewer than 2,000 independent observations for questions that involve rater variation. Fit a mixed model with item and rater random effects, or bootstrap over raters and items together. Plan the design so model and position are crossed with rater: every rater sees both models in both positions, which makes rater harshness and position bias estimable instead of confounded with the model effect.

5. Pre-registration and analysis plans

Plain words

Sealing your prediction in an envelope

A magician who writes a prediction, seals it in an envelope and hands it to the audience before the trick is convincing. One who announces the prediction after seeing the card is not. Pre-registration is the sealed envelope: write the plan for analysing the data before you see the data, and share it.

For leaders

It protects the team from itself

With dozens of metrics, segments and analysis choices, some combination will show a win by chance. Pre-registration, popularised in science partly in response to replication failures (Nosek et al., PNAS 2018, "The preregistration revolution"), makes the choices before the data can influence them. In product teams it is lightweight: the experiment brief plus an analysis plan, timestamped in the experiment system. Deviations are allowed and recorded; what is not allowed is presenting an exploratory finding as if it were the planned test.

Under the hood

What an analysis plan specifies

1. **Estimand**: the population, the unit, the metric definition (code reference), and the contrast (difference, ratio, non-inferiority margin).
2. **Test and variance**: e.g. difference in means with delta-method variance at the user level; CUPED covariate named.
3. **Error control**: α, sidedness, correction across the primary and guardrails, and the look schedule with its spending function.
4. **Exclusions**: bots, internal users, users with no exposure; decided by rules that cannot depend on arm.
5. **Segments**: the few pre-planned ones, with interaction tests.
6. **Decision rule**: what each outcome leads to, and what would change our mind.

6. A full worked design: moving the free tier to a cheaper model

A realistic launch, worked end to end. The numbers are illustrative; the structure is the point.

Step	Decision for this launch	Chapter
Question	Can the free tier move from model v7 to v8, a distilled model that costs about 38% less per request, without hurting users?	distillation
Hypothesis	Non-inferiority: weekly task success on v8 is no more than 1 point below v7. Secondary: cost per request falls by at least 25%.	A/B tests
Offline evidence	1,500 frozen production-sampled items, 3 seeds, paired: v8 – v7 = –0.6 points, 95% CI –1.7 to +0.5. Not enough to rule out a 1-point loss, so test online.	eval stats, intervals
Population	Logged-in free-tier users on supported clients; excludes internal accounts and known bots by pre-set rules.	sampling
Unit and assignment	User, by hash(user ID, fresh salt) into 50/50 after a 1% then 10% safety ramp.	randomization
Primary metric	Share of active users with at least one judge-graded successful task per week. Judge validated against humans (kappa 0.71 vs human-human 0.74).	metrics, judges
Guardrails	p95 latency; safety violation rate; benign refusal rate; client error rate; task success in the five smallest supported languages (distillation tends to lose long-tail skill).	metrics, safety
Power	Base 62%, margin 1 point, one-sided α = 0.05, 80% power: 29,133 users per arm, or 17,480 with CUPED on pre-period success (variance kept 0.6). At 4,000 new eligible users per arm per day: 5 days, so run 14 for weekly cycles and novelty.	A/B tests
Logging	Assignment and exposure (first v8 response rendered), trace IDs across model call, tool calls, client events and feedback; model, prompt, decoding and client versions on every event.	metrics
Analysis	Difference in user-level success with CUPED, delta-method variance; SRM check before anything else; group-sequential monitoring of safety guardrails only, primary read once at day 14.	A/B tests
Decision rule	Ship if the one-sided 95% lower bound on the success difference is above –1 point, cost falls at least 25%, and no	leading

Step	Decision for this launch	Chapter
	guardrail is breached. Keep a 2% holdout for one quarter.	
What would change our mind	A breach in any small-language guardrail, a safety increase beyond tolerance, or an effect that decays across exposure cohorts.	triage

Plain words

Proving "no worse", not "better"

This launch is not trying to make answers better. It is trying to save money without making them worse. That changes the question from "is it better?" to "is it close enough?" You set in advance how much worse is still acceptable (here, 1 point), and you ship only if the data rules out anything worse than that. "We did not see a difference" is not enough, because a small test never sees differences.

For leaders

The margin is a business decision

The 1-point margin is the number leadership must own: it says how much quality the company will trade for a 38% cost cut. Set it from the economics and user research before the test, not after seeing the estimate. A smaller margin needs quadratically more users: halving it to 0.5 points quadruples the sample. The decision rule also requires the cost saving to be real at test traffic levels, since the whole point of the launch is cost.

Under the hood

Non-inferiority sample size and decision

$$H_0 : \Delta \le -m \quad \text{vs} \quad H_1 : \Delta > -m, \qquad n_{\text{per arm}} = \frac{(z_{1-\alpha} + z_{1-\beta})^2 \, 2p(1-p)\,(1-\rho^2)}{m^2}$$

Assumes the true difference is zero. m is the margin; $(1-\rho^2)$ is the variance kept by CUPED. Reject H_0 (ship) when $\hat{\Delta} - z_{1-\alpha}\,\mathrm{SE} > -m$.

With an observed difference of −0.3 points and standard error 0.35 points, the one-sided lower bound is $-0.3 - 1.645 \times 0.35 = -0.88$, above −1, so the rule says ship.

python

```
import math
from statistics import NormalDist
```

```
Z = NormalDist().inv_cdf                    # standard normal quantile

def noninferiority_n(p, margin, alpha=0.05, power=0.8, variance_kept=1.0):
    """Users per arm to show the new model is at most `margin` worse (one-sided test).

    Assumes the true difference is zero. variance_kept below 1 models CUPED (1 - rho^2)."""
    z = Z(1 - alpha) + Z(power)
    return math.ceil(z * z * 2 * p * (1 - p) * variance_kept / margin ** 2)

def decide(diff, se, margin, alpha=0.05):
    """Ship only if the one-sided lower confidence bound sits above -margin."""
    lower = diff - Z(1 - alpha) * se
    return lower, lower > -margin

n = noninferiority_n(0.62, 0.01)                    # 62% weekly task success, 1-point margin
n_cuped = noninferiority_n(0.62, 0.01, variance_kept=0.6)
print(n, n_cuped)                                   # 29133 17480
days = math.ceil(n_cuped / 4_000)                   # 4,000 new eligible users per arm per day
print(days, 7 * math.ceil(days / 7))                # 5 7  (round up to whole weeks)
lower, ship = decide(diff=-0.003, se=0.0035, margin=0.01)
print(f"{lower:.4f} {'ship' if ship else 'hold'}")  # -0.0088 ship
```

rust

```
// Rust's standard library has no normal quantile, so we inline Acklam's rational
// approximation (relative error about 1e-9), which agrees with Python's NormalDist here.
fn z(p: f64) -> f64 {
    let a = [-3.969683028665376e1, 2.209460984245205e2, -2.759285104469687e2, 1.383577518672690e2,
-3.066479806614716e1, 2.506628277459239];
    let b = [-5.447609879822406e1, 1.615858368580409e2, -1.556989798598866e2, 6.680131188771972e1,
-1.328068155288572e1];
    let c = [-7.784894002430293e-3, -3.223964580411365e-1, -2.400758277161838, -2.549732539343734,
4.374664141464968, 2.938163982698783];
    let d = [7.784695709041462e-3, 3.224671290700398e-1, 2.445134137142996, 3.754408661907416];
    let tail = |q: f64| (((((c[0] * q + c[1]) * q + c[2]) * q + c[3]) * q + c[4]) * q + c[5]) / ((((d[0] * q +
d[1]) * q + d[2]) * q + d[3]) * q + 1.0);
    if p < 0.02425 {
        tail((-2.0 * p.ln()).sqrt())
    } else if p > 1.0 - 0.02425 {
        -tail((-2.0 * (1.0 - p).ln()).sqrt())
    } else {
        let q = p - 0.5;
        let r = q * q;
        (((((a[0] * r + a[1]) * r + a[2]) * r + a[3]) * r + a[4]) * r + a[5]) * q / (((((b[0] * r + b[1]) * r +
b[2]) * r + b[3]) * r + b[4]) * r + 1.0)
    }
}

/// Users per arm to show the new model is at most `margin` worse (one-sided test).
/// Assumes the true difference is zero. variance_kept below 1 models CUPED (1 - rho^2).
fn noninferiority_n(p: f64, margin: f64, alpha: f64, power: f64, variance_kept: f64) -> u64 {
    let zz = z(1.0 - alpha) + z(power);
    (zz * zz * 2.0 * p * (1.0 - p) * variance_kept / margin.powi(2)).ceil() as u64
}

/// Ship only if the one-sided lower confidence bound sits above -margin.
fn decide(diff: f64, se: f64, margin: f64, alpha: f64) -> (f64, bool) {
    let lower = diff - z(1.0 - alpha) * se;
    (lower, lower > -margin)
```

```
}

fn main() {
    let n = noninferiority_n(0.62, 0.01, 0.05, 0.8, 1.0); // 62% weekly task success, 1-point margin
    let n_cuped = noninferiority_n(0.62, 0.01, 0.05, 0.8, 0.6);
    println!("{} {}", n, n_cuped); // 29133 17480
    let days = (n_cuped as f64 / 4_000.0).ceil() as u64; // 4,000 new eligible users per arm per day
    println!("{} {}", days, 7 * ((days as f64) / 7.0).ceil() as u64); // 5 7  (round up to whole weeks)
    let (lower, ship) = decide(-0.003, 0.0035, 0.01, 0.05);
    println!("{:.4} {}", lower, if ship { "ship" } else { "hold" }); // -0.0088 ship
}
```

7. The design reviewer

Pick the options a team proposed. The reviewer flags each problem in one line and points to the chapter that covers it. Start from the default draft, which has several of the most common mistakes, and fix it.

Interactive

Experiment design reviewer (shown at its default setting)

A proposed online test of a new assistant model. Choose what the team plans for each item; the grade and flags update as you go.

Who is in the test?

- **All eligible users of the tier**
- Beta-program volunteers
- Employees only
- Last month's most active users

How are users assigned?

- **Hash of user ID, fresh salt**
- EU gets the new model, US keeps the old
- New model on odd days
- Users opt in from settings

Unit of randomization

- **User**
- Session
- Request

Primary metric

- Pre-registered driver plus guardrails
- **Thumbs-up rate only**
- Whichever moves most, chosen afterwards
- Offline benchmark score

Analysis

- Variance at the randomization unit (delta method)
- **t-test treating every request as independent**
- Compare the dashboard lines

Sample size

- **Power analysis for the MDE, whole weeks**
- Whatever arrives in 3 days
- Run until it is significant

When will they look?

Once, at the planned end | Weekly, pre-planned sequential boundary | **Daily, stop at the first p < 0.05**

What is logged?

Assignment, exposure, versions, trace IDs | **Assignment only** | Aggregate dashboards only

D

4 major, 0 minor (score 8 of 16). This test cannot support a launch decision as designed. Each major flag alone can produce a confident wrong answer.

MAJOR	**Primary metric:** Thumbs-up rate only. Sparse, favours flattery, and nothing guards cost, latency or safety. → Choosing metrics
MAJOR	**Analysis:** t-test treating every request as independent. Requests from one user are correlated, so standard errors are too small and false wins follow. → Choosing metrics
MAJOR	**Looks:** Daily, stop at the first p < 0.05. Peeking: 14 daily looks push the false positive rate above 20%. → A/B tests end to end
MAJOR	**Logging:** Assignment only. No exposure or versions: effects are diluted and a surprise cannot be explained. → Choosing metrics
OK	**Sample:** All eligible users of the tier. Matches the population the decision applies to. → Who is in your sample?
OK	**Assignment:** Hash of user ID, fresh salt. Random, reproducible, and independent of earlier tests. → Randomization
OK	**Unit:** User. Consistent experience, and retention can be measured. → Randomization
OK	**Sample size:** Power analysis for the MDE, whole weeks. Sized to detect the smallest effect that would change the decision. → A/B tests end to end

8. The end-to-end checklist

One list for any experiment on an AI system, offline or online. Each item points to the chapter that explains it.

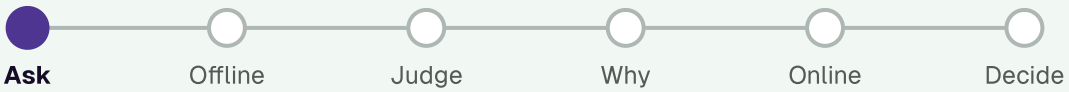
Stage	Check	Chapter
Design	The question is written as a decision, with the rule for each outcome.	A/B tests
Design	The causal claim needs an experiment; observational data would be confounded.	causation
Design	Population and sampling frame match where the decision applies.	sampling
Design	Unit of randomization chosen for consistency and independence; assignment is random and logged.	randomization
Design	Analysis plan pre-registered: estimand, test, variance, looks, segments.	this chapter
Measurement	Metric is valid and reliable for the construct; tier (north star, driver, guardrail, diagnostic) agreed.	measurement, metrics
Measurement	Ratio metrics show numerator and denominator; variance at the randomization unit.	metrics
Measurement	Exposure, versions and trace IDs logged; A/A test passes.	metrics
Evaluation	Offline items frozen, clean of contamination, paired across models, multiple seeds.	eval stats, contamination
Evaluation	Intervals reported, clustered by the independent unit.	intervals
Evaluation	LLM judge validated against humans; position and length bias controlled.	judges
Evaluation	Offline-to-online relationship calibrated for this product.	offline–online gap
Training	Known side effects of the training change (length, refusals, long-tail loss, reward hacking) have guardrails.	reward models, distillation, safety
Statistics	Powered for the MDE; duration in whole weeks.	eval stats, A/B tests

Stage	Check	Chapter
Statistics	SRM checked before results; peeking controlled by the plan.	A/B tests
Statistics	Multiple metrics and segments corrected or labelled exploratory.	eval stats
Decision	Launch table applied; surprising results triaged before celebration.	A/B tests, triage
Decision	Result reviewed with the questions a principal asks; holdout kept where warranted.	leading

Diagram

One question through the whole guide

The free-tier launch above, walked station by station. Numbers are illustrative.

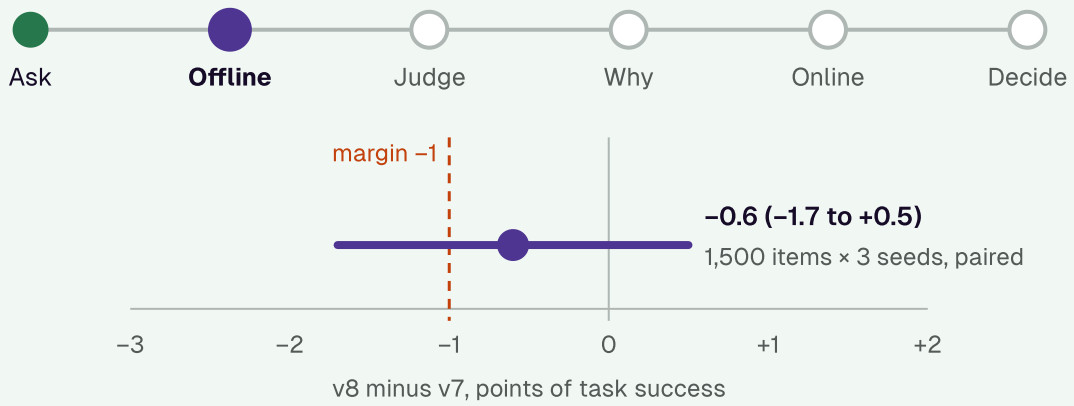


Question
Can the free tier move from v7 to v8, which costs 38% less?

Decision rule
Switch if task success is no more than 1 point worse, cost falls 25% or more, and no guardrail breaks.

design: A/B tests, this chapter

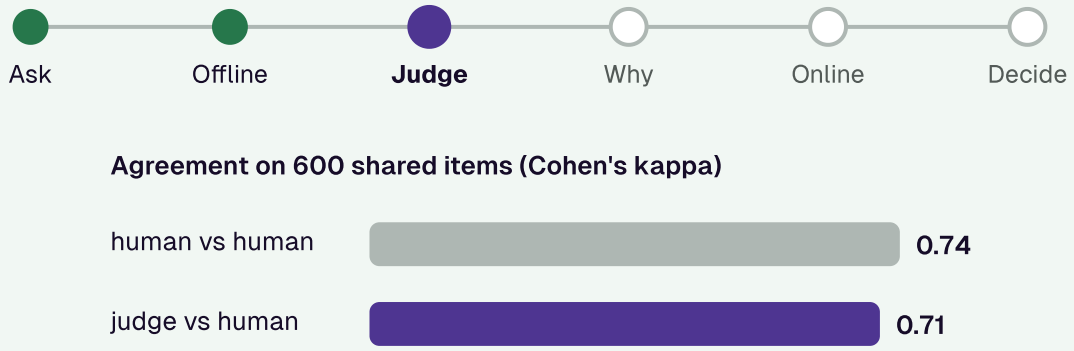
Everything starts with a decision, not a curiosity. The team writes the question, the margin they will accept, and what they will do with each answer. Because v8 is meant to save money, the question is "no worse than 1 point", a non-inferiority test.



The interval crosses the margin: offline cannot settle it.

Part I–II: eval stats, intervals

The team runs both models on 1,500 frozen items drawn from production, three seeds each, and compares item by item. v8 is 0.6 points behind, but the interval reaches −1.7, past the margin. Offline evidence narrows the question; it cannot answer it here.



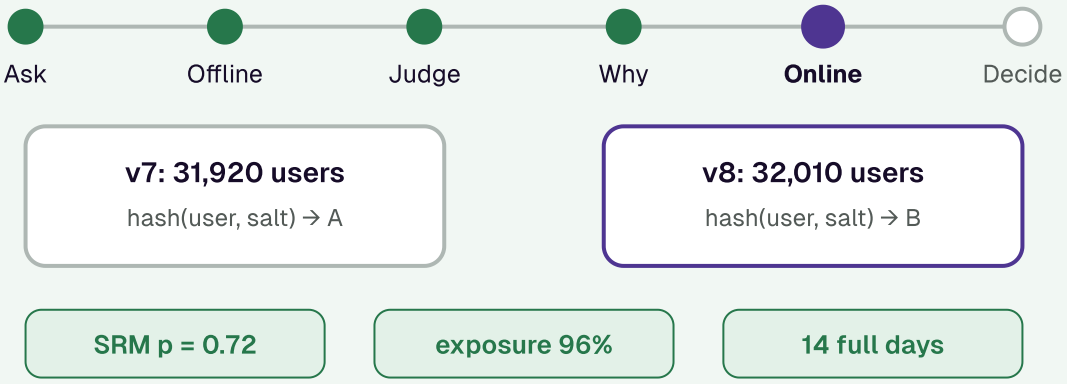
Close to the human ceiling: fit to grade production traffic.

Part II: LLM as judge

Task success online will be graded by an LLM judge, so the judge is checked first. On 600 items labelled by two people, the judge agrees with humans almost as often as humans agree with each other, with position bias controlled by swapping order.

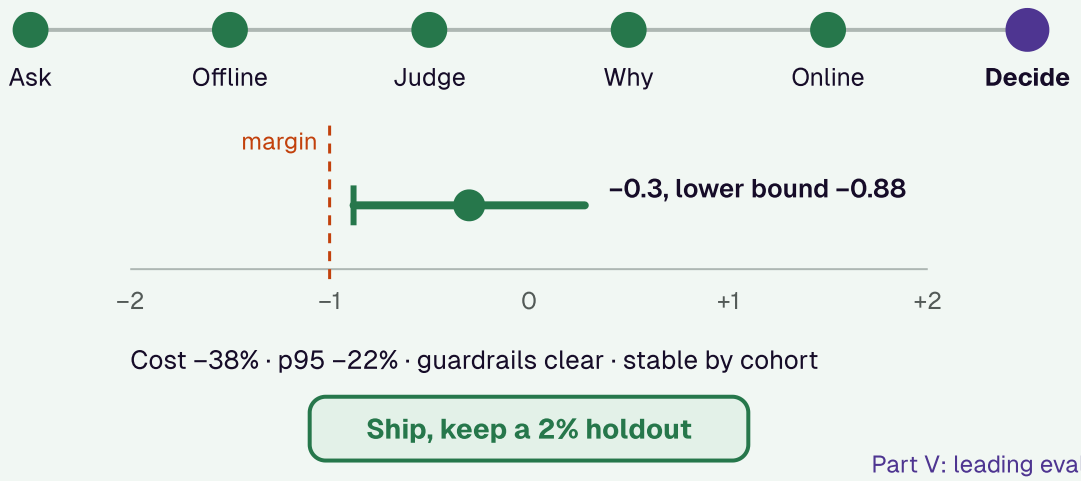
Part IV: distillation, safety

Knowing how v8 was made predicts where it might break. Distillation tends to lose rare skills first, so the team adds a guardrail on task success in the five smallest languages, alongside safety and refusal rates. The training chapters turn into a list of things to measure.



Part VI: A/B tests, metrics

Free-tier users are split by a hash of their ID for two full weeks, with exposure logged when the first answer renders and a trace ID on every event. Before anyone looks at task success, the split passes the SRM check and exposure is equal in both arms.



Cost −38% · p95 −22% · guardrails clear · stable by cohort

Ship, keep a 2% holdout

Part V: leading eval work

At day 14 the primary metric is read once. The lower bound sits above the margin, cost and latency fall, no guardrail breaks, and the effect is steady across exposure cohorts. The rule written on day 0 says ship, with a holdout to watch the long run. Part V's reviewer questions have answers on file.

Eval lens

The deepest lesson of the guide is that every number about a model is the output of an experiment, whether or not anyone designed it as one. A benchmark score has a sample (the items), a measurement (the grader), a comparison (the baseline), and noise (seeds, items, raters). An online metric adds assignment and exposure. Asking the design questions of every number, offline or online, is what separates an eval program from a leaderboard.

Principal asks

What decision does this experiment inform, and what result would change it? Is this a superiority or a non-inferiority question, and who set the margin? What did the offline eval establish, with what interval, and why is that not enough? What did training change that we should guard against? Who is sampled, how are they assigned, and is the analysis at the same unit? Which numbers are pre-registered, and what is exploratory? When will we look, and what stops the test early? If the result surprises us, which logs will explain it?

Common mistakes

Testing a cost-saving model for superiority and calling a flat result "no worse". Shipping on an offline win without an online test, or running an online test without offline screening. Randomizing model changes per request. Using an unvalidated judge as the primary metric. Unblinded human evals with fixed left/right order. Changing the analysis after seeing the data without saying so. Forgetting that the guardrails need power too.

Appendix

Reference

Every term in the guide, defined in a sentence or two.

Appendix · Chapter 48

Glossary

Every term used in the guide, defined in a sentence or two, with a link to the chapter that explains it properly.

524 terms. Type to filter.

6ND	The approximate training compute of a dense transformer: 6 FLOPs per parameter per training token (2 forward, 4 backward). → section
A/A test	An experiment in which both arms get the identical product. About 5% of metrics should come out significant at $p < 0.05$; many more reveals broken variance estimates, assignment, or logging. → section
Ablation	An experiment that changes one component while holding everything else fixed, to measure that component's effect; at a frontier lab, run compute-matched at several scales. → section
Abstention	Declining to answer, or saying "I don't know", when confidence is too low. Useful only if the grading or reward gives it more credit than a likely-wrong guess. → section
A/B test	A randomized controlled experiment on live users: units are split at random between the current version (A) and a change (B), and the difference in outcomes estimates the change's causal effect. → section
Accuracy	(TP + TN) / n. Misleading when one class is rare, because it mostly measures the common class. → section
Activation checkpointing	Discarding intermediate activations in the forward pass and recomputing them during backward to save memory at the cost of extra compute. → section

Activation function	The fixed non-linear function applied to a neuron's weighted sum (ReLU, tanh, GELU, SwiGLU). Without it, any stack of layers collapses into a single linear map. → section
Active parameters	In a mixture-of-experts model, the parameters actually used for one token; they set compute per token, while total parameters set memory. → section
AdamW	The standard LLM optimiser: Adam's per-parameter step scaling from running averages of the gradient and squared gradient, plus decoupled weight decay. → section
Adapter merging	Folding a trained LoRA update into the base weights (W + (alpha/r)BA) so serving has no extra cost. Inexact when the base is quantised. → section
Adaptive thinking	A mode in which the model decides per request whether and how much to think, steered by an effort setting. Used by current Claude models. → section
Advantage	How much better an action turned out than expected from that state: $A_t = R_t - V(s_t)$. → section
Alignment tax	Capability regressions caused by alignment training, first documented on public NLP benchmarks in InstructGPT. → section
Alpha spending function	A rule, introduced by Lan and DeMets (1983), for how much of the type I error budget is used up by each fraction of the planned sample, so interim looks need not be scheduled exactly. → section
Always-valid p-value	A p-value that stays valid however often it is checked, such as one built from a mixture sequential probability ratio test; the price is lower power at any fixed sample size. → section
Analysis plan	The pre-specified estimand, test, variance method, covariates, error control and look schedule, exclusions, segments, and decision rule for an experiment. → section
Annealing	Decaying the learning rate toward zero at the end of pretraining, often on a higher-quality data mix. It produces a visible drop in loss. → section

Arg max	The position (or input) at which a quantity is largest, rather than the largest value itself. → section
Aspect ratio (model shape)	The ratio of width to depth; published frontier decoders sit around 100 to 130 hidden units per layer, and loss depends only weakly on it at fixed size. → section
Assignment logging	Recording each time a unit is bucketed into an experiment arm, with unit ID, experiment, arm, and timestamp. It is what SRM checks count. → section
Attack success rate	The fraction of attack attempts that elicit the targeted unsafe behaviour; best reported as a function of attacker effort. → section
Attention	A layer that computes, for each position, a softmax-weighted average of other positions' values, with weights from how well a query matches each key (Bahdanau et al. 2014). → section
Attention head	One of several attention computations run in parallel with their own projections. Heads specialise, for example copying the previous token or linking a pronoun to its noun. → section
Attention sink	The tendency of many heads to put most of their weight on the first token when nothing else is relevant, effectively a no-op option (Xiao et al. 2023). → section
Attenuation	Noise in a predictor biases its estimated slope toward zero by the reliability factor λ, and noise in two variables shrinks their correlation by $\sqrt{(\lambda_X \lambda_Y)}$ (Spearman, 1904). → section
Attenuation bias	The pull of a regression slope toward zero when the predictor is measured with noise, as offline deltas are. → section
Autoregressive generation	Producing text one token at a time, each conditioned on all tokens so far: predict, pick, append, repeat. One forward pass per generated token. → section
AWQ	Activation-aware weight quantization: protects the weights most important to activations when quantizing to low bit widths (Lin et al., 2024). → section

Back-door criterion	A set Z identifies the causal effect of X on Y if no member of Z descends from X and Z blocks every back-door path; then $P(y \mid do(x)) = \sum_z P(y \mid x, z) P(z)$. → section
Back-door path	A path between treatment X and outcome Y that starts with an arrow pointing into X. Open back-door paths carry non-causal association and must be blocked by adjustment. → section
Backpropagation	The algorithm that computes every gradient in a network by applying the chain rule backward from the loss, reusing each intermediate result. Costs about twice a forward pass. → section
Backtest	Leave-one-out prediction of each past release from all others, used to estimate how well an offline suite predicts the next release. → section
Balanced accuracy	The mean of recall and specificity. 0.5 is chance at any prevalence. → section
Baseline	A quantity subtracted from the return in a policy gradient to reduce variance without adding bias, usually the value estimate. → section
Base model	A model after pretraining only. It continues any document in whatever style the document suggests and has no built-in notion of answering a user. → section
Batch normalisation	Normalising each feature using the mean and variance over the current batch (Ioffe and Szegedy 2015); central to CNNs, awkward for language models. → section
BCa bootstrap	Bias-corrected and accelerated bootstrap: shifts the percentile interval using a bias term and a jackknife skewness term. Better on skewed statistics. → section
Benchmark saturation	The state where top models approach a benchmark's ceiling, so remaining differences are dominated by label noise and the benchmark stops separating models. → section
Benjamini–Hochberg	A step-up procedure that controls the false discovery rate: reject the k smallest p values, where k is the largest rank with p(k) ≤ (k/m)·q. → section
Berkson's paradox	A spurious negative association produced by selecting on a collider, such as food and location quality among surviving restaurants (Berkson

	1946). → section
Best-of-N	Generate N responses and return the one a reward model or verifier scores highest. → section
bf16	Brain floating point: a 16-bit format with the same exponent range as fp32 but fewer mantissa bits. The standard training precision. → section
Bias	A systematic error that does not shrink with more data. The opposite of noise. → section
Bias (neuron)	A learned constant added to a neuron's weighted sum, shifting where it switches on independently of the inputs. Unrelated to statistical bias. → section
Big data paradox	The larger a non-random dataset, the more confidently wrong its estimates: error bars shrink with n while bias set by the data defect correlation does not. → section
Bits per byte	Total loss in bits divided by the text length in bytes. Lets you compare models with different tokenizers. → section
Blinding	Hiding which system produced an output from raters (and ideally from analysts) so expectations cannot influence the judgments. → section
Block bootstrap	Resampling contiguous chunks of time-ordered data so the ordering and drift survive. → section
Blocking	Grouping similar units (by tier, language, prompt) and randomizing within each group, which removes chance imbalance on the blocking variable and cuts variance by about the share it explains. Also called stratified randomization. → section
Bonferroni correction	Testing each of m hypotheses at α/m. Simple and conservative on large suites. → section
Bootstrap	Resample your data with replacement many times to see how much an estimate would wobble. → section
Bradley–Terry model	Models the probability that one item beats another as the logistic sigmoid of their score difference. The basis of reward model training and DPO. → section

Brier score	Mean squared error between predicted probabilities and 0/1 outcomes. A proper scoring rule; lower is better. → section
Byte-pair encoding (BPE)	A tokenizer training method that starts from characters or bytes and repeatedly merges the most frequent adjacent pair until the vocabulary reaches a target size. → section
Cache key	A hash over everything that can change a model output (task, prompt text, model id, sampling parameters, harness and judge versions), used to reuse results safely. → section
Calibrated probability	A probability that matches observed frequency: of the cases a model calls 70% likely, about 70% turn out true. Required for the cost formula $t = c_{FP}/(c_{FP} + c_{FN})$ to pick the right threshold. → section
Calibration	Whether predicted probabilities match observed frequencies: of items given 70% confidence, about 70% should be positive. → section
Canary string	A unique random token embedded in an eval set so that it can be filtered from training data and detected if a model has memorised the set. → section
Catastrophic forgetting	Loss of previously learned abilities when a model is trained further on narrow data. Detected with a fixed regression suite compared against the parent model. → section
Causal mask	The matrix that sets attention scores to future positions to $-\infty$, so each token can only attend to itself and earlier tokens. It lets one training pass make a prediction at every position. → section
CDF	Cumulative distribution function: the share of values at or below each point. Rises from 0 to 1. → section
Censoring	Knowing only that a value exceeds some limit, such as a run that hit the time cap or a user still active when data ends. Dropping or capping censored values biases summaries downward. → section
Central limit theorem	Averages tend toward a bell shape regardless of the raw data's shape, with variance shrinking as one over n. → section
Chain of thought (CoT)	Intermediate reasoning written before the final answer, elicited by worked examples (Wei et al. 2022) or a cue such as "Let's think step by

	step" (Kojima et al. 2022). Each written token adds a forward pass of computation and a line of scratchpad. → section
Chain rule	The derivative of a composition is the product of the local derivatives: if L depends on y which depends on w, then $\partial L/\partial w = (\partial L/\partial y)(\partial y/\partial w)$. → section
Chat template	The exact layout of special tokens and role markers used to format a conversation. Training, serving, and evaluation must all use the same one. → section
Chinchilla scaling	Hoffmann et al. (2022): for a fixed training budget, grow parameters and tokens together, about 20 tokens per parameter. → section
Choice (decision type)	A question type that returns one of the supplied options plus a probability for each option and a confidence value. → section
Clip-higher	Using a larger upper clipping bound than lower bound in the PPO ratio so low-probability tokens can be promoted faster, which helps preserve exploration. → section
Clopper–Pearson interval	An exact, conservative interval for a proportion built from the binomial distribution. Useful for very small n. → section
Cluster bootstrap	Resampling whole clusters (sessions, users) instead of rows, so the interval respects dependence within clusters. → section
Cluster randomization	Randomizing whole groups (orgs, regions, workspaces) instead of individuals, usually to contain spillover; power depends on the number of clusters. → section
Cluster sampling	Sampling whole groups (streets, conversations, repositories) and measuring every unit inside them. Cheaper, but variance is inflated by the design effect $1 + (m - 1)\rho$. → section
CNN (convolutional neural network)	A network built from convolution layers that apply one small shared filter at every position of a grid; the standard design for images before vision transformers. → section
Cochran's Q	Tests whether three or more models differ on the same binary tasks; the paired extension of McNemar. Says whether they differ, not which. → section

Cohen's kappa	Agreement between two raters corrected for the agreement expected by chance: (p_o – p_e) / (1 – p_e). → section
Cold-start SFT	A small supervised fine-tuning stage on curated long chain-of-thought examples before reasoning RL, used in DeepSeek-R1 to improve readability and stability. → section
Collider	A variable caused by two others, $X \to C \leftarrow Y$. Conditioning on it (filtering, stratifying, adding it as a control) creates a spurious association between X and Y. → section
Compute-matched comparison	Comparing two methods at the same training FLOPs or tokens, so a gain reflects the method rather than extra compute. → section
Compute-optimal	The split of a fixed training compute budget between model size and data that minimises loss. → section
Conditional probability P(A | B)	The probability of A given that B is known to have happened. The vertical bar reads 'given'. → section
Confidence interval	A range produced by a procedure that captures the truth a stated share of the time. → section
Confidence target	A grading rule that tells the model the threshold at which answering pays off, for example "answer only if more than t confident; wrong answers cost t/(1–t) points" (Kalai et al. 2025). → section
Confounder	A variable that causes both the treatment and the outcome, creating an association between them that is not the treatment's effect. Temperature for ice cream sales and drownings. → section
Confusion matrix	The 2×2 table of true positives, false positives, false negatives, and true negatives at a given threshold. Every classification metric is a ratio of its cells. → section
Constitutional AI	Training with a written set of principles: the model critiques and revises its own outputs, and AI-generated preference labels replace human harmlessness labels (Bai et al., 2022). → section
Contamination	Benchmark items, or close paraphrases of them, appearing in a model's training data, which inflates its score on that benchmark. → section

Contextual calibration	Estimate a prompt's built-in label bias by scoring a content-free input such as "N/A", then divide it out (Zhao et al. 2021). → section
Continuous batching	A serving technique that adds new requests to a running batch as others finish, keeping accelerators busy. → section
Convenience sample	A sample of whoever is easiest to reach, such as a team's own prompts or nearby colleagues. Its inclusion probabilities are unknown. → section
Convolution	Sliding one small kernel of weights across an input and taking a weighted sum at every position; encodes locality, weight sharing, and translation equivariance. → section
Correlation (Pearson's r)	Covariance divided by the product of the two standard deviations, between −1 and +1. Measures how well a straight line describes how two variables move together; says nothing about why. → section
Cosine schedule	A learning-rate schedule that decays from the peak to a minimum following half a cosine wave after warmup. → section
Cost-based threshold	The probability above which acting is cheaper than not acting, set by the costs of false positives and false negatives rather than by the model. → section
Coverage	How often an interval method actually contains the true value. A '95%' method can cover far less in practice. → section
Coverage bias	Bias from a sampling frame that omits part of the population, such as an email poll missing everyone without email. → section
Coverage–risk curve	Error rate on answered items as a function of the share of items answered (coverage). → section
Critical batch size	The batch size beyond which larger batches give little reduction in the number of training steps. Estimated by the gradient noise scale. → section
Critical point	Any point where the gradient is zero: a minimum, a maximum, or a saddle. The Hessian's eigenvalue signs tell them apart. → section
Cross-entropy loss	The average negative log probability a model assigns to the token that actually came next. The pretraining and SFT objective. → section

CUPED	Controlled-experiment Using Pre-Experiment Data (Deng et al., 2013): subtract θ times each unit's centred pre-period covariate from its outcome, cutting variance by a factor of 1 – ρ² without bias. → section
DAG (causal diagram)	Directed acyclic graph: nodes for variables, arrows from cause to effect, no loops. Used to decide which variables to adjust for. → section
Dangerous-capability evaluation	Testing whether a model could meaningfully assist with severe harms or act autonomously in risky ways, used to trigger safeguards under frontier safety policies. → section
DAPO	An RL recipe (Yu et al., 2025) adding clip-higher, dynamic sampling of informative groups, token-level loss, and overlong reward shaping to GRPO-style training. → section
Data defect correlation	Meng's $\rho_{R,G}$: the correlation between being recorded and the value measured. The sample mean's error equals $\rho\sqrt{(1-f)/f}\,\sigma$, so a tiny ρ ruins huge samples (Meng 2018). → section
Data mixture	The proportions of each data source (web, code, math, books, languages, synthetic) in a training run. Tuned with small proxy models. → section
Decision cascade	A design where a fast decision model handles every case, acts when confident, and escalates an uncertain band to an LLM judge or a person. → section
Decision model	A model that takes unstructured input plus a question with a fixed set of options and returns a typed answer with a probability distribution in one forward pass, generating no text. Examples: OpenAI's Decisions API, Laya, TypeSafe's Jev. → section
Decode	Generating output tokens one at a time; usually memory-bandwidth-bound. → section
Decontamination	Removing training documents that overlap with evaluation items, usually by long n-gram matching, before training. → section
Degradation problem	Deeper plain networks reaching higher training error than shallower ones, an optimisation failure rather than overfitting, which residual connections fixed. → section

Deliberative alignment	Training a model to recall and reason over a written safety specification before answering (Guan et al., 2024). → section
Delta method	Approximates the variance of a function of estimates, such as a ratio of means, from their variances and covariance; the standard way to get error bars on ratio metrics randomized by user. → section
Depth	The number of layers a signal passes through in sequence; more depth allows more steps of composition but adds latency and makes training harder. → section
Design effect	How much clustering inflates variance: deff = 1 + (m̄ – 1)·ρ. The naive standard error is too small by a factor of about sqrt(deff). → section
Diagnostic metric	A metric that explains why something moved, such as tokens per answer or tool calls per task. Useful for debugging, never a launch criterion. → section
Difference-in-differences	Estimates an effect as the treated group's change minus the comparison group's change over the same period, which removes stable differences between groups and shared trends. → section
Direction accuracy	The share of past releases where offline and online deltas agreed in sign. The number that matters for ship decisions. → section
Discordant pair	A task where one model passed and the other failed. The only tasks that carry information about which model is better. → section
do-operator	$do(X=x)$ denotes setting X by intervention rather than observing it; $P(y \mid do(x))$ is the causal quantity a randomized experiment estimates. → section
DoRA	Weight-Decomposed Low-Rank Adaptation (Liu et al., 2024). Splits each weight into magnitude and direction and applies LoRA to the direction. → section
DPO	Direct Preference Optimization (Rafailov et al. 2023): optimizes the KL-regularised RLHF objective with a classification loss on preference pairs, without a reward model or RL loop. → section

Dr. GRPO	A GRPO variant (Liu et al., 2025) that drops per-response length normalisation and standard-deviation scaling to remove a length bias and a difficulty bias. → section
Driver metric	A faster, more sensitive metric that is known from past experiments to move the north star, and is used as the primary metric of most tests. → section
d-separation	Graphical rule for whether a set of variables blocks every path between two nodes: chains and forks are blocked by conditioning on the middle, colliders are opened by it. → section
ECE (decision models)	Expected calibration error: the bin-weighted average gap between stated probability and observed rate. Depends on binning and is noisy on small samples; does not transfer between tasks. → section
Effective sample size	The number of independent observations your data is worth after clustering: N / deff. → section
Effect size	How big the difference is, as opposed to how confident you are that it exists. → section
Embedding	The vector a token is turned into by looking up its row in a learned table $E \in \mathbb{R}^{V \times d}$. Tokens used in similar ways end up with nearby vectors. → section
Emergent ability	A capability that appears to arise suddenly with scale. Often an artifact of all-or-nothing metrics; smooth metrics usually show gradual improvement. → section
Entropy collapse	The policy's output distribution becoming nearly deterministic during RL, so sampled responses stop differing and the learning signal disappears. → section
Eta squared	Effect size for ANOVA: the share of total variance explained by group membership. Report it beside F. → section
Eval harness	The software that sends tasks to a model, collects and scores outputs, and records results. The measuring instrument for every eval number. → section

Evidence-first prompting	Asking the judge to quote or reason about evidence before giving its verdict, so the verdict is conditioned on it. Reasoning written after the verdict cannot change it. → section
Expected calibration error (ECE)	Weighted average gap between confidence and accuracy across confidence bins. Depends on the number of bins. → section
Expected cost	(c_FP·FP + c_FN·FN) / n: the average cost per decision once the two errors are priced. → section
Expected value (E[X])	The long-run average of a random quantity; in practice estimated by averaging a sample. → section
Exploding gradient	Gradients growing geometrically through many layers or time steps because each step multiplies them by a factor above one; causes loss spikes and divergence. → section
Exposure logging	Recording the moment a user actually experienced the change (for example when the first new-model answer rendered), as opposed to when they were assigned. → section
F1 score	Harmonic mean of precision and recall, 2PR / (P + R). High only when both are high; ignores true negatives. → section
False discovery rate	The expected share of declared discoveries that are false. Controlled by Benjamini–Hochberg. → section
False negative (FN)	A positive case the model missed. → section
False positive (FP)	A negative case the model wrongly flagged: a false alarm. → section
False positive rate (FPR)	FP / (FP + TN): the share of negatives wrongly flagged. The x-axis of an ROC curve. → section
Family-wise error rate	The probability of at least one false positive across a set of tests. Controlled by Bonferroni. → section
F-beta score	Weighted harmonic mean of precision and recall; β > 1 weights recall more, β < 1 weights precision more. → section
Few-shot prompting	Putting a handful of worked examples in the prompt so that the most likely continuation follows the same pattern. Works on base models without any fine-tuning. → section

Flaky task	A task whose score varies across runs at fixed settings. Quarantined and reported separately rather than averaged into the headline. → section
Flip rate	The share of items whose verdict changes when only the order of options or responses is reversed. An estimate of how much of a verdict is position. → section
Forward KL	KL(teacher || student). Mode covering: penalises the student for missing any place the teacher puts mass. → section
Forward pass	Running inputs through the network layer by layer to produce an output. An LLM does one full forward pass per generated token, costing about 2 operations per parameter. → section
FP8 training	Training with 8-bit floating-point matrix multiplies for higher throughput and lower memory, with careful scaling to avoid overflow. Used at scale in DeepSeek-V3. → section
Framing effect	A change in a judge's verdicts caused by wording, label words, format or order rather than by the item judged. Survey research's "forbid" versus "not allow" is the classic human example. → section
Friedman test	A rank-based test for three or more models scored on the same tasks with continuous or ordinal scores. → section
FSDP	Fully Sharded Data Parallel, PyTorch's implementation of ZeRO-style sharding of model state across GPUs. → section
Full jitter	A retry backoff drawn uniformly between zero and an exponentially growing cap, so retries from a failure burst do not arrive in lockstep. → section
GAE	Generalized advantage estimation: an exponentially weighted sum of one-step temporal-difference errors, with λ trading bias for variance. → section
GAT (graph attention network)	A GNN that weights each neighbour's message with learned attention scores instead of fixed averages (Veličković et al. 2018). → section

GCN (graph convolutional network)	The GNN layer of Kipf and Welling (2017): a degree-normalised average over each node's neighbours and itself, followed by a shared weight matrix. → section
GELU	Gaussian error linear unit, $x\,\Phi(x)$: a smooth version of ReLU used in BERT and GPT-2. → section
Geometric mean	The exponential of the average log. On lognormal data it equals the median. → section
Global maximum	The highest value of the loss over the whole region considered; of no use in training except as a place to avoid. → section
Global minimum	The lowest value of the loss anywhere in parameter space. Large networks have many global or near-global minima. → section
GNN (graph neural network)	A network for graph-structured data in which each node repeatedly updates its representation from messages sent by the nodes it is linked to. → section
Goodhart's law	When a measure becomes a target, it stops being a good measure; optimisation pressure finds ways to raise the metric without the underlying quality. → section
GPTQ	A post-training weight quantization method that uses approximate second-order information to minimise layer output error (Frantar et al., 2023). → section
Gradient	The vector of partial derivatives of the loss with respect to every parameter. It points in the direction of steepest increase, so training steps the opposite way. → section
Gradient check	Comparing backprop's gradient with a finite-difference estimate, $(\mathcal{L}(\theta+\epsilon)-\mathcal{L}(\theta-\epsilon))/2\epsilon$, to catch bugs. → section
Gradient clipping	Rescaling the gradient when its norm exceeds a threshold (commonly 1.0) to prevent unstable updates. → section
Gradient descent	Repeatedly update the parameters by a small step against the gradient, $\theta \leftarrow \theta - \eta \nabla \mathcal{L}(\theta)$, so the loss goes down. With mini-batch gradients it becomes SGD; Adam and its variants change the step's size and direction. → section

GraphSAGE	A GNN that samples a fixed number of neighbours and learns aggregation functions, so it scales to huge graphs and to nodes unseen in training (Hamilton et al. 2017). → section
Greedy decoding	Always choosing the single most probable next token; deterministic. → section
Group advantage	In GRPO, a response's reward minus its group's mean reward, usually divided by the group's standard deviation. Zero for every response when all rewards in the group are equal. → section
Grouped-query attention (GQA)	Attention in which several query heads share each key/value head, shrinking the KV cache. → section
Group sequential design	A plan with a fixed number of interim analyses and critical values chosen so the overall error rate is alpha, such as Pocock or O'Brien–Fleming boundaries. → section
GRPO	Group Relative Policy Optimization (Shao et al., 2024): PPO-style RL that samples a group of responses per prompt and uses the group's mean and standard deviation of reward as the baseline, removing the critic network. → section
Guardrail metric	A metric that must not get worse whatever else improves, such as latency, cost, or safety violations; trust guardrails such as sample ratio decide whether the test can be believed at all. → section
Hallucination	Fluent, plausible output that is false or unsupported, such as an invented citation or a wrong date stated confidently. → section
Hat ($\hat{\theta}$, $\hat{p}$)	Marks an estimate computed from data, as opposed to the true unknown value. → section
Hessian	The matrix of second derivatives of the loss. Its eigenvalues are the curvatures along its eigenvectors and classify critical points. → section
Heteroscedasticity	Spread that changes with the value. Common in latency: slow things vary more. → section
Hidden layer	Any layer between the input and the output, whose values are internal features rather than something you feed in or read out. → section

Holdout	A small group kept on the old version after a launch, often for a quarter, to measure long-run and cumulative effects. → section
Horvitz–Thompson estimator	Unbiased estimator of a population total that weights each sampled unit by the inverse of its inclusion probability. → section
Human ceiling	The agreement two humans reach on the same labels. No judge or classifier can meaningfully exceed it against human labels. → section
ICC	Intraclass correlation: the share of variance that lives between clusters rather than within them. → section
Implicit reward	In DPO, $\beta \log \pi_\theta(y \mid x) / \pi_{\text{ref}}(y \mid x)$, the reward the policy implicitly defines. → section
Induction head	An attention head pattern that finds an earlier occurrence of the current token and predicts the token that followed it, completing repeated sequences (Olsson et al. 2022). → section
Inductive bias	The assumptions about the data that an architecture or learning method builds in, such as locality in a convolution or order in a recurrent network; strong bias learns from less data but caps flexibility. → section
Instruct / chat model	A base model further trained (SFT, then preference training such as RLHF or DPO) to continue conversation-formatted documents in the role of a helpful assistant. → section
InstructGPT	OpenAI's 2022 SFT, reward model, PPO recipe (Ouyang et al. 2022), the template for most later RLHF pipelines. → section
Instrumental variable	A variable that shifts treatment, is as good as random, and affects the outcome only through treatment. The Wald estimator divides its effect on the outcome by its effect on treatment, giving the effect for compliers. → section
Intention to treat	Analysing every unit in the arm it was assigned to, whether or not it used the treatment. It keeps the protection randomization gives; filtering after assignment does not. → section
Inter-annotator agreement	How often independent labelers make the same choice on the same item. The practical ceiling on reward model accuracy. → section

Interference	When one unit's treatment changes another unit's outcome, such as through shared GPU capacity, shared documents, or a marketplace. Also called spillover; it violates SUTVA. → section
Interleaving	An online comparison for rankers in which results from two systems are merged into one list and user clicks are credited to the system that contributed each result. Each user compares both, so it needs far less traffic than an A/B test. → section
Inter-rater agreement	How often independent raters agree on the same items, usually reported as raw agreement plus a chance-corrected statistic such as Cohen's kappa or Krippendorff's alpha. → section
Interrupted time series	Fits the pre-launch trend and seasonality of a metric and reads the effect as the post-launch gap from that forecast. Anything else that changed on the same date is indistinguishable from the treatment. → section
Interval scale	Equal steps with an arbitrary zero (Celsius, Elo-style ratings). Differences and means are meaningful; ratios are not. → section
Intraclass correlation (ICC)	The share of variance that lies between clusters (tasks, sessions, users) rather than within them. Drives the design effect. → section
Intrinsic dimension	The number of free directions needed to reach most of full fine-tuning performance. Low for pretrained language models, which is why low-rank updates work. → section
Inverse scaling in test-time compute	Tasks where longer reasoning lowers accuracy, for example counting with distractors or regression with spurious features (Gema et al. 2025). → section
IPO	Identity Preference Optimization (Azar et al. 2023): a squared loss toward a finite target margin, avoiding DPO's overfitting to deterministic preferences. → section
IQR	Interquartile range: the middle 50% of the data, p75 minus p25. Outliers cannot move it. → section
Jackknife	Recomputing a statistic with each observation left out in turn; used to estimate bias, variance, and the BCa acceleration. → section

Jailbreak	An input crafted to get a model to do something its safety training is meant to prevent. → section
k3 KL estimator	The per-sample estimator r − log r − 1, with r = π_ref/π_θ, used in GRPO. It is unbiased for KL(π_θ‖π_ref) and never negative. → section
Kaplan–Meier estimator	Estimates the survival curve from censored durations as the product, over event times, of the share of at-risk units that survive each event (Kaplan and Meier, 1958). → section
Kendall's tau	Rank correlation computed from concordant and discordant pairs, $\tau = (C - D)/(C + D)$. More stable than Spearman with few releases. → section
Kill criteria	A result, stated in advance, that would end a research project. → section
Kish effective sample size	$(\sum w_i)^2 / \sum w_i^2$: the number of unweighted random observations with the same precision as a weighted sample. → section
KL penalty	A term $\beta\,\mathrm{KL}(\pi_\theta \| \pi_{\mathrm{ref}})$ that keeps the trained policy close to the reference; β sets its strength. → section
Knowledge distillation	Training a student model to match a teacher's output distribution, not just the correct label (Hinton et al., 2015). → section
KTO	Kahneman–Tversky Optimization (Ethayarajh et al. 2024): preference optimization from unpaired desirable and undesirable examples. → section
Kurtosis	How heavy the tails are. High kurtosis means extremes are more common than a bell predicts. → section
KV cache	Stored attention keys and values for every previous token, reused at each decoding step; its size grows with layers, KV heads, context length, and concurrent sequences. → section
Ladder of evidence	Ordering of causal evidence from anecdote, raw correlation and adjusted comparison through natural experiments to randomized and replicated experiments. → section
Layer	A set of neurons that read the same inputs; computed as one matrix multiply plus an element-wise activation. Layers that are neither input nor output are hidden layers. → section

Layer normalisation	Normalising each example's feature vector by its own mean and variance (Ba et al. 2016), so behaviour is identical in training and inference and for any batch size. → section
Learning rate	The step size η in gradient descent. Too small wastes compute; too large overshoots and can diverge. On a quadratic with curvature λ, steps converge only if $\eta < 2/\lambda$. Usually warmed up and decayed over a run. → section
Length bias	The tendency of raters and reward models to prefer longer responses regardless of quality. → section
Length-biased sampling	Sampling units that are active at a random moment, which picks long-lived units (long sessions, slow jobs) in proportion to their duration. → section
Levels of measurement	Stevens's (1946) classification of scales as nominal, ordinal, interval, or ratio, defined by which transformations leave the information unchanged. → section
Likelihood displacement	A DPO failure where both chosen and rejected log-likelihoods fall during training and probability moves to responses neither label mentioned. → section
Likelihood scoring	Evaluating by comparing the log-probabilities a model assigns to fixed answer options, instead of grading text the model generates. → section
Likert scale	A rating scale with ordered response options such as 1 (strongly disagree) to 5 (strongly agree). Whether single items may be averaged is a long-running debate. → section
LLM as judge	Using a language model to grade or compare model outputs. A classifier that must itself be validated against humans. → section
Local maximum	A point higher than every nearby point: zero gradient and every direction curves downward. Gradient descent moves away from it. → section
Local minimum	A point lower than every nearby point: the gradient is zero and every direction curves upward (all Hessian eigenvalues positive). It need not be the lowest point overall. → section

Logits	The raw, unnormalised scores a model outputs for each vocabulary item before softmax turns them into probabilities. → section
Lognormal	The distribution of a quantity whose log is normal. Arises from multiplicative effects; the shape of latency and token counts. → section
Logprob scoring	Reading the probability of the verdict tokens instead of the sampled word, for example a renormalised P(yes) or a probability-weighted score as in G-Eval. → section
Log-rank test	Compares two or more survival curves using all event times; loses power when the curves cross. → section
LoRA	Low-Rank Adaptation (Hu et al., 2021). Learns a low-rank update BA added to a frozen weight, scaled by alpha/r. B starts at zero so training begins at the base model. → section
Loss landscape	The loss viewed as a surface over parameter space: each point is a setting of every weight, and its height is how wrong the model is with those weights. → section
Loss masking	Setting the label of non-target tokens (system, user, tool output) to an ignore value such as -100 so they don't contribute to the loss. → section
Loss spike	A sudden jump in training loss during a large run, usually from optimisation instability. Often handled by rolling back and skipping batches. → section
LSTM	Long short-term memory (Hochreiter and Schmidhuber 1997): an RNN with gates and an additively updated cell state so information and gradients survive many steps. → section
Macro average	Unweighted mean of a metric across classes, so every class counts equally. → section
MAD	Median absolute deviation: the median distance from the median. Multiply by 1.4826 to compare with the sd. → section
maj@k (self-consistency)	Sample k responses and return the most common final answer (Wang et al., 2022). → section
Mann–Whitney test	A rank-based test of whether values from one group tend to exceed values from another; its effect size is P(B > A) + ½P(tie). It compares

	medians only if the two shapes match. → section
Mann–Whitney U	A rank-based comparison of two groups that does not assume interval data. Its common-language effect is P(random x > random y). → section
Matching	Pairing each treated unit with similar untreated units on pre-treatment covariates; assumes no unmeasured confounders and enough overlap. → section
Matthews correlation coefficient (MCC)	Correlation between predictions and labels using all four confusion-matrix cells, from –1 to +1. A strong single-number summary under imbalance. → section
McNemar test	A test on discordant pairs: under no difference, each is a fair coin flip. Use the exact binomial version on small eval sets. → section
Mean	The arithmetic average: add everything, divide by n. Right for totals, misleading on skewed data. → section
Measurement error	The gap between a recorded value and the true one. Classical error inflates variance and attenuates slopes; error correlated with treatment or outcome causes bias. → section
Median	The middle value when sorted. The robust centre for skewed data like latency. → section
Mediator	A variable on the causal path from treatment to outcome ($X \to M \to Y$). Controlling for it removes the part of the effect that flows through it. → section
Memoryless	How long you have already waited tells you nothing about the remaining wait. True of the exponential and geometric. → section
Message passing	The general GNN template (Gilmer et al. 2017): compute messages along edges, aggregate them at each node with a permutation-invariant function, then update the node. → section
Metric directionality	Whether a metric's movement reliably means better or worse for users; messages per session, for example, can rise for good or bad reasons. → section
Metric sensitivity	How often a metric moves detectably when a change matters, determined by the effect size relative to the metric's noise. → section

MFU	Model FLOPs utilisation: useful training FLOPs per second divided by the hardware's peak. Large runs typically reach 35 to 50 percent. → section
Micro average	A metric computed from counts pooled across classes. In single-label multiclass, micro-F1 equals accuracy. → section
Mid-training	A stage between pretraining and fine-tuning that continues the same loss on curated data, often while the learning rate decays: high-quality text, long documents for context extension, or boosted code and math. → section
MinHash deduplication	A near-duplicate detection method that estimates Jaccard similarity between documents' n-gram sets using the minimum of random hash functions. → section
Minimum detectable effect	The smallest true difference an eval design can detect at a chosen power, usually 80%. Report it next to any null result. → section
Minimum detectable effect (MDE)	The smallest true effect a test has a stated power (usually 80%) to detect at its significance level. Required sample size scales with one over the MDE squared. → section
Min-K% Prob	A membership-inference signal that averages the lowest-probability tokens of a text; seen text tends to score higher (Shi et al., 2023). → section
Min-p sampling	Keeping only tokens whose probability is at least a fraction p_min of the top token's probability, so the cutoff adapts to model confidence. → section
Mixture	Two or more populations sharing an axis, often visible as two humps. Split them before summarising. → section
Mixture of experts (MoE)	An architecture where each token is routed to a few of many expert sub-networks, so total parameters can be large while per-token compute stays small. → section
MLP (feed-forward) layer	The per-position sublayer $W_2\,\phi(W_1 x)$ in each transformer block. It holds about two thirds of a dense model's parameters and behaves partly like a key-value memory of associations. → section

Mode	The most common value. Useful for categories and for spotting two humps. → section
Model collapse	Progressive loss of distribution tails and quality when models are trained recursively on their own outputs (Shumailov et al., 2024). Mitigated by keeping real data in the mix. → section
Model soup	A model made by averaging the weights of several fine-tunes of the same base (Wortsman et al., 2022). → section
Momentum	An optimizer term that keeps a running average of past gradients, so steps build speed along consistent directions and damp oscillations. → section
MRP (multilevel regression and post-stratification)	Fits a multilevel model of the outcome by demographic cell, then weights cell predictions by population counts. Rescued the heavily skewed Xbox poll of 2012 (Wang et al. 2015). → section
Multi-LoRA serving	Serving many adapters on one shared base model and batching their requests together, as in S-LoRA and Punica. → section
Multiple comparisons	Running many tests at once inflates the number of false positives; with 40 benchmarks at α = 0.05, about 2 will look significant by luck. → section
muP (maximal update parameterisation)	A way of scaling initialisation and per-layer learning rates with width so the best hyperparameters stay roughly constant, allowing tuning on a small proxy (Yang et al. 2022). → section
Negative binomial	A count distribution with variance above the mean, arising when the event rate varies across units. Fits tool calls per session. → section
Neural architecture search (NAS)	Automatically proposing and evaluating architectures with search or reinforcement learning; effective at small scale, rare at frontier scale because each candidate must be trained. → section
Neuron (artificial)	The basic unit of a neural network: a weighted sum of its inputs plus a bias, passed through an activation function, $y = \phi(\mathbf{w}^\top \mathbf{x} + b)$. → section

Next-token prediction	The single task a language model is trained on and performs: given a sequence of tokens, output a probability for every possible next token. Generation repeats it in a loop. → section
Neyman allocation	Stratified allocation with stratum sample sizes proportional to stratum size times stratum standard deviation, which minimises variance for a fixed total n. → section
NF4	4-bit NormalFloat: a 4-bit data type whose levels sit at quantiles of a normal distribution, suited to normally distributed weights. → section
Nominal scale	Numbers or words used only as names (club, model family, error category). Counts, proportions, and the mode are meaningful; averages are not. → section
Non-inferiority test	A test of whether a change is no worse than a pre-set margin, H0: Δ ≤ −m. The right framing for launches meant to save cost or latency. → section
Nonresponse bias	Bias that arises when sampled units that do not respond differ on the outcome from those that do. → section
Normal approximation (Wald interval)	The symmetric interval p ± z·sqrt(p(1−p)/n). Under-covers badly near 0 or 1 and can exceed 100%. → section
North star metric	The top-level metric that captures long-run value for a product, such as weekly users with a successful task. Usually too slow to test against directly, so experiments use drivers that predict it. → section
noul (Laya)	Laya's yes/no decision type, returning the probability that the answer is yes. Equivalent to a Decisions API predicate. → section
Novelty effect	A temporary boost because a change is new, which fades as users get used to it. A short test overstates the long-run effect. → section
Offline to online gap	The difference between how a model change moves an offline eval score and how it moves the production metric it is meant to predict. → section
Off-policy data	Training data generated by a different model than the one being trained. → section

Online / iterative DPO	Repeated rounds of sampling from the current policy, labeling pairs with a reward model or judge, and training with the DPO loss. → section
On-policy distillation	The student generates text and the teacher scores each token, so the student learns to correct its own mistakes (Agarwal et al., 2024). → section
Operating point	The specific threshold, and resulting precision and recall, at which a classifier is actually deployed. → section
Optional stopping	Letting the decision to stop data collection depend on the results seen so far. Without a sequential design it inflates false positives. → section
Option-ID selection bias	A preference for particular option labels such as "A" in multiple-choice prompts, traced mainly to token priors (Chujie Zheng et al. 2023). → section
Ordinal regression	A model (cumulative logit or probit) that treats an ordinal outcome as a latent continuous score cut by thresholds; handles covariates and clustering without assuming equal gaps. → section
Ordinal scale	Values with a meaningful order but unequal or unknown gaps (finishing place, 1–5 rating). Medians, percentiles, and rank tests are meaningful. → section
ORPO	Odds Ratio Preference Optimization (Hong et al. 2024): single-stage SFT plus an odds-ratio preference penalty, with no reference model. → section
Outcome reward model	A reward model that scores only the final response. → section
Outcome reward model (ORM)	A reward model, or verifier, that scores only the final result of a full response. → section
Overall evaluation criterion (OEC)	The single metric, or agreed weighted combination, that defines success for a product and that experiments are ultimately judged against; also called the north star metric. → section
Overdispersion	Count data whose variance exceeds its mean, so Poisson intervals are too narrow. Typical of tool calls and token counts. → section
Over-refusal	Refusing a benign request, often because it superficially resembles a harmful one. The false positive of safety behaviour. → section

Over-smoothing	Node representations in a deep GNN converging to nearly the same value because every layer averages over neighbours, erasing the differences between nodes. → section
Over-squashing	Information from a node's exponentially growing k-hop neighbourhood being compressed through a few edges into fixed-size vectors, which limits long-range reasoning in GNNs. → section
Over-training	Training a smaller model on far more tokens than compute-optimal, accepting slightly higher loss for much cheaper inference. → section
Packing	Concatenating several short training examples into one fixed-length sequence to avoid wasted padding. Needs per-document attention masks so examples can't see each other. → section
Paired	The same tasks were given to both models, so comparisons can use within-task differences. → section
Paired bootstrap	Resampling task indices and reading both models at each sampled task, to get an interval on the difference between them. → section
Paired comparison	Comparing two models task by task on the same tasks, which removes task difficulty from the noise and gives far more power than comparing two averages. → section
Parallel trends	The assumption behind difference-in-differences: without treatment, the treated and comparison groups would have changed by the same amount. Checked, partially, by plotting the pre-period. → section
Parameter	Any learned number in a model: every weight and every bias. "A 70B model" has about 70 billion; at 2 bytes each they need about 140 GB of memory. → section
Parameter-efficient fine-tuning (PEFT)	Methods that freeze the pretrained weights and train a small number of added or selected parameters. → section
pass@k	The probability that at least one of k sampled responses is correct, estimated without bias from n ≥ k samples as 1 – C(n−c, k)/C(n, k). → section
pass^k	The probability that all k attempts pass. A reliability measure used for agents, popularised by τ-bench. → section

Pass-rate gap	The difference between a judge's pass rate and humans' on the same items. A generous judge inflates every absolute number while still ranking models correctly. → section
PDF	Probability density function: how common values are around each point. The familiar humped curve. → section
Peeking	Checking a running test repeatedly and stopping the first time it looks significant. With 20 looks at a nominal 5% level, the false positive rate rises to about 25%. → section
Percentile bootstrap	An interval taken from the quantiles of statistics recomputed on resamples drawn with replacement. → section
Percentile / quantile	The value below which a given share of the data falls. p95 is the 0.95 quantile. → section
Perceptron	Rosenblatt's 1958 single-layer classifier: a weighted sum of inputs compared with a threshold, trained by nudging weights after mistakes. → section
Perplexity	The exponential of cross-entropy loss in nats. Roughly the number of equally likely choices the model feels it faces at each token. → section
Persona	The character a chat model is trained to play (the assistant). It is a learned pattern of text, not a separate agent with goals. → section
Perturbation test	Re-running a benchmark with names, numbers, or option order changed. A large drop relative to a clean control suggests memorisation. → section
Pipeline parallelism	Placing different layers on different GPUs and streaming micro-batches through them. → section
Policy	In RL for language models, the model's next-token distribution $\pi_\theta(a_t \mid s_t)$. → section
Policy gradient	An estimator of the gradient of expected reward using samples from the policy, weighting each action's log-probability gradient by its return or advantage. → section
Polysemantic neuron	A unit that responds to several unrelated concepts, common in LLMs because features are stored in superposition across many units. →

	section
Position bias	A pairwise judge's tendency to prefer the response shown in a particular position. Controlled by judging both orderings. → section
Position bias (decision prompts)	A judge's preference for whichever option appears in a particular slot, usually first. Measured by swapping order; mitigated by averaging both orders. → section
Post-stratification	Reweighting respondents so each group's share matches its known population share. Fixes bias from the wrong group mix but not bias within groups. → section
Potential outcomes	The framework that gives each unit an outcome under treatment, Y(1), and under control, Y(0); the causal effect is their difference, of which only one is ever observed. → section
Power	The chance of detecting a real effect of a given size, if it exists. → section
Power law	A distribution with no typical size, straight on log-log axes. Its mean may not exist; report percentiles only. → section
PPO	Proximal Policy Optimization (Schulman et al. 2017): a policy gradient method that clips the probability ratio so each batch can only move the policy a limited amount. → section
PR-AUC (average precision)	Area under the precision–recall curve, usually computed as average precision. Its chance baseline equals the prevalence. → section
PR curve	Precision plotted against recall as the threshold sweeps. Sensitive to prevalence. → section
Precision	TP / (TP + FP): of the items flagged, the share that were truly positive. → section
Predicate (Decisions API)	A question type that returns the probability that a stated condition is true. Equivalent to Laya's `noul`. → section
Preference pair	A prompt with a preferred (chosen) and dispreferred (rejected) response, written (x, y_w, y_l). → section

Prefill	Processing the input prompt in parallel before generation; usually compute-bound. → section
Pre-norm	Placing normalisation at the input of each residual branch, leaving the residual stream untouched; more stable at depth than the original post-norm transformer. → section
Pre-registration	Writing down the hypothesis, primary metric, analysis plan, and decision rule, timestamped, before seeing the data, so exploratory findings cannot be passed off as planned tests. → section
Pretraining	The first and most expensive training stage: next-token prediction over trillions of tokens of text and code. Most knowledge and raw capability comes from here. → section
Prevalence	The share of items that are truly positive. Precision and PR-AUC depend on it; recall, FPR, and ROC-AUC do not. → section
Primacy effect	A temporary dip, also called change aversion, because users are used to the old way. A short test understates or reverses the long-run effect. → section
Prior	What you believed before seeing this data. With small samples it does real work in a Bayesian estimate. → section
Probability ratio	$\pi_\theta(a \mid s)/\pi_{\theta_{\text{old}}}(a \mid s)$, how much an update changed an action's probability. PPO clips it to $[1-\epsilon, 1+\epsilon]$. → section
Probability sampling	Any design in which every population unit has a known, positive chance of selection, which makes unbiased estimates and honest error bars possible. → section
Process reward model	A reward model that scores each intermediate step of a reasoning chain (Lightman et al. 2023). → section
Process reward model (PRM)	A reward model that scores each intermediate reasoning step rather than only the final answer. → section
Product (Π)	Capital pi: multiply the terms that follow. → section
Prompt injection	Instructions hidden in content a model reads, such as a web page, document, or tool output, that try to override the user's or developer's instructions. → section

Propensity score	The probability of receiving treatment given observed covariates, e(x) = P(T=1 | X=x). Matching or weighting on it balances measured covariates only. → section
Proper scoring rule	A score minimised by reporting your true belief. Brier score is one; ECE is not. → section
Proxy metric	An easy-to-measure stand-in for a target that is slow, costly, or rare (thumbs-up for satisfaction, judge score for expert judgment). Optimizing it can drive it apart from the target. → section
p value	How often you would see a result this extreme if there were genuinely no effect. Not the probability the null is true. → section
QLoRA	LoRA trained on top of a base model stored in 4-bit NF4, with double quantisation and paged optimisers (Dettmers et al., 2023). Cuts memory enough to fine-tune large models on one GPU. → section
QQ plot	A plot of your data's quantiles against a reference distribution's. A straight line means a good match. → section
Quality classifier	A small model that scores documents for usefulness (for example educational value) so only the best slice of a web crawl is kept. → section
Quantization	Storing weights, and sometimes activations or the KV cache, in fewer bits (for example 8 or 4) to cut memory and raise speed, with some accuracy risk. → section
Quasi-experiment	A study that estimates a causal effect without random assignment by exploiting a situation that mimics it (a cutoff, a lottery, a staggered launch), at the price of an untestable assumption. → section
Query, key, value	Three learned projections of each position's vector. The query says what a position is looking for, the key what it offers as a match, the value what it hands over when attended to. → section
Raking	Iterative proportional fitting of weights to match several known population margins in turn, used when their joint distribution is unknown. → section

Ramp	Raising a change's traffic share in stages (for example 1%, 5%, 50%, 100%) so that crashes and guardrail breaches are caught while few users are exposed. → section
Random assignment	Deciding who gets a treatment by a chance mechanism (a coin flip or a hash of the user ID), so the groups are balanced on average on every trait, measured or not. → section
Rank (LoRA)	The inner dimension r of the LoRA matrices. The update can express at most r independent directions; trainable parameters per matrix are r times (rows + columns). → section
Rater calibration	Training and checking human raters against a rubric, worked examples, and gold items so their judgments are consistent with each other and stable over time. → section
Ratio metric	A metric defined as a sum over a sum, such as thumbs-up over rated responses. Its variance must be computed at the randomization unit, for example with the delta method, and its numerator and denominator reported separately. → section
Ratio scale	Equal steps and a true zero (time, tokens, cost), so ratios, logs, and geometric means are meaningful. → section
Reasoning effort	A coarse setting (for example low, medium, high) that steers how much a model thinks. Exact token counts vary per request. → section
Reasoning model	A model trained, usually with reinforcement learning on verifiable rewards, to write long chains of thought before answering. Examples: OpenAI o1 (Sept 2024), DeepSeek-R1 (Jan 2025). → section
Recall	TP / (TP + FN): of the truly positive items, the share that were flagged. Also called sensitivity or true positive rate. → section
Receptive field	The region of the input that can influence one output unit; it grows with each stacked convolution or pooling layer. → section
Red teaming	Deliberately searching for inputs that make a model fail or behave harmfully, by humans or automated attacker models. → section
Reference model	A frozen copy of the starting model used to measure and penalize how far the trained policy drifts. → section

Regression discontinuity	Compares units just above and just below a cutoff that decides treatment; valid if units cannot precisely control which side they land on. The estimate applies near the cutoff. → section
Regression to the mean	Units picked for an extreme measurement tend to be less extreme next time, because part of the extreme was luck. Expected bounce is (1 – ρ) times the distance from the mean, with ρ the test–retest correlation. → section
Regression triage	A fixed order of checks (is it real, where is it, what else changed, errors, replay, read outputs) for deciding whether an eval drop is noise, measurement, infrastructure, or model. → section
REINFORCE	The basic Monte Carlo policy gradient algorithm (Williams 1992). → section
Rejection sampling fine-tuning	Sample many responses per prompt, keep those that pass a verifier or score highest with a reward model, and fine-tune on them. → section
Reliability	How consistently a measurement gives the same result when repeated: test–retest correlation, inter-rater agreement, or ICC. Necessary for validity, not sufficient. → section
Reliability diagram	A plot of observed positive rate against stated probability, binned. Points on the diagonal mean calibrated; points below it mean overconfident. → section
ReLU	Rectified linear unit, $\max(0, z)$: keeps positive inputs and zeroes negative ones. Each ReLU unit adds one hinge, so ReLU networks are piecewise-linear. → section
Replay	Mixing general or earlier training data into a narrow fine-tuning run to reduce forgetting. → section
Residual connection	A skip path that adds a layer's input to its output, $h + F(h)$, giving gradients a direct route back and letting a layer default to the identity (He et al. 2015). → section
Residual stream	The running vector at each position that every sublayer reads from and adds to. Nothing overwrites it, so later layers can read what earlier layers wrote. → section

Responsible Scaling Policy	Anthropic's framework tying capability thresholds (AI Safety Levels) to required security and deployment safeguards; similar frameworks include OpenAI's Preparedness Framework and Google DeepMind's Frontier Safety Framework. → section
Retry inflation	Upward bias in a pass rate when retried tasks effectively get more than one draw at the answer, making them scored like pass@k rather than pass@1. → section
Reversal curse	The finding that models trained on "A is B" often fail to answer "what is B?" with A (Berglund et al. 2023): stored knowledge is directional. → section
Reverse causation	An association read in the wrong causal direction: the outcome causes the supposed cause, as wind turns windmills. → section
Reverse KL	KL(student || teacher). Mode seeking: penalises the student for putting mass where the teacher doesn't. Used in on-policy distillation. → section
RewardBench	A public benchmark (Lambert et al. 2024; RewardBench 2 in 2025) of preference pairs across chat, safety, and reasoning for comparing reward models. → section
Reward hacking	A policy earning reward through behavior the designer did not intend, such as padding answers for a length-biased reward model. → section
Reward model	A network trained on human or AI comparisons to output a scalar score predicting which response is preferred. The target for RLHF and the selector for best-of-N. → section
Reward overoptimization	Optimizing a proxy reward past the point where true quality peaks, so the proxy keeps rising while real quality falls. Quantified by Gao et al. (2022). → section
RLAIF	Reinforcement learning from AI feedback: preference labels produced by a model guided by written principles instead of by people. → section
RLCD	Reinforcement Learning for Calibrated Decisions: TypeSafe's name for its training method, which it says optimises for honest probabilities. Not described in detail; Laya describes its training as RL against strictly proper scoring rules. → section

RLHF	Reinforcement learning from human feedback: training a policy to maximize a learned reward model's score with a KL penalty to a reference model. → section
RLOO	REINFORCE leave-one-out: each response's baseline is the mean reward of the other responses for the same prompt, giving an unbiased advantage estimate. → section
RLVR	Reinforcement learning with verifiable rewards: RL where a program (answer checker, unit tests, format check) computes the reward instead of a learned reward model. → section
RMSNorm	A cheaper layer-normalisation variant that rescales by the root mean square of the features without subtracting the mean; used in Llama-family models. → section
RNN (recurrent neural network)	A network that reads a sequence one step at a time, updating a hidden state with the same weights at every step. → section
Robust	Resistant to outliers. The median is robust, the mean is not. → section
ROC-AUC	Area under the ROC curve; the probability a random positive scores higher than a random negative. 0.5 is chance, 1.0 is perfect. → section
ROC curve	Recall plotted against false positive rate as the threshold sweeps from high to low. Each threshold is one point. → section
Rollout	A response sampled from the current policy during RL training, then scored. → section
Rotary position embedding (RoPE)	A way to give attention a sense of order by rotating query and key vectors by an angle proportional to position, so their dot product depends on relative distance. → section
Router (thinking modes)	A fast classifier that sends each request to an instant model or a thinking model. GPT-5 launched in August 2025 with a real-time router of this kind. A router is a decision model whose errors can be measured. → section
Rubric decomposition	Replacing one holistic judgment with separate, observable yes/no checks combined by an explicit rule. CheckEval (Lee et al. 2025) reported large gains in agreement across evaluator models. → section

Rule of three	If zero events occur in n independent trials, an approximate 95% upper bound on the event rate is 3/n. → section
Saddle point	A critical point (zero gradient) that curves up in some directions and down in others, like a mountain pass. In high dimensions most critical points are saddles; they slow gradient descent rather than trap it. → section
Sample ratio mismatch	Arm sizes that differ from the planned split by more than chance, detected with a chi-square test; it signals broken assignment or logging, and results should not be read until it is explained. → section
Sample ratio mismatch (SRM)	Observed arm sizes that differ from the configured split by more than chance allows, tested with a chi-square goodness-of-fit test. It means something adds or drops users unevenly, and results should not be read until it is explained. → section
Sampling distribution	The spread of results you would get from repeating the whole experiment on fresh samples. → section
Sampling frame	The list or mechanism from which a sample is actually drawn, such as an email list or a log table. Anyone outside it has zero chance of selection. → section
Scaled dot-product attention	$\mathbf{softmax}(QK^\top/\sqrt{d_k}+M)V$. Dividing by $\sqrt{d_k}$ keeps scores from growing with vector width. → section
Scaling law	An empirical power-law relationship between loss and model size, data, or compute, used to predict large runs from small ones. → section
Score (decision type)	A question type over ordered levels that returns per-level probabilities and the probability-weighted level index, which can fall between levels. The Decisions API allows 2 to 10 levels. → section
Scratchpad	The tokens a model writes before its answer, which later tokens can read. It turns a fixed-depth network into a loop with working memory. → section
Sealed holdout	An evaluation set owned and run by someone independent of the training team, used only for final decisions so nobody can optimise toward it. → section

Seed variance	How much a final score changes when the same training recipe is rerun with a different random seed. → section
Selection bias	A difference between treated and comparison groups that exists before treatment, so the naive comparison mixes the effect with who ended up in each group. Opt-in betas are the classic case. → section
Selective prediction	Letting a model abstain on low-confidence cases and measuring error only on the cases it answers. → section
Self-attention	The mechanism that lets each position gather information from other positions: it scores its query against every key, softmaxes the scores, and takes the weighted sum of values. → section
Self-consistency	Sample several reasoning chains at temperature above zero and return the most common final answer (Wang et al. 2022). Combines sampling diversity with voting. → section
Self-correction	Asking a model to review and fix its own answer. Without new external information it often fails to improve reasoning answers and can make them worse (Huang et al. 2023). → section
Self-Instruct	Generating new instruction data by prompting a model with seed examples, then filtering (Wang et al., 2023). → section
Self-preference bias	A judge scoring outputs from its own model family more favourably. → section
Self-selection	Units choose their own treatment, so whatever drives the choice (engagement, motivation) also differs between the groups. → section
Self-selection bias	Bias from letting units choose whether to take part, as in opt-in polls, app-store reviews and thumbs feedback. → section
Sequence-level distillation	Training the student on complete outputs generated by the teacher (Kim and Rush, 2016). Works across tokenizers and through APIs. → section
Sequential testing	Testing designs that allow repeated looks at accumulating data while controlling the overall false positive rate, through planned boundaries or always-valid p-values. → section

Shrinkage (offline to online)	One minus the fitted slope of online delta on offline delta. A shrinkage of 0.4 means online moves are about 60% of offline moves. → section
SimPO	Simple Preference Optimization (Meng et al. 2024): reference-free, length-normalized implicit reward with a target margin. → section
Simpson's paradox	A trend that holds in every subgroup reverses or vanishes when the subgroups are pooled, because the groups being compared have different subgroup mixes. → section
Skew	Asymmetry, named for the side the tail is on. Mean divided by median above 1.2 signals right skew. → section
Skew ratio	Mean divided by median. Above about 1.2, report quantiles instead of the mean. → section
Slice contribution	A slice's share of the total score change, $n_s \bar{d}_s / N$. Rank slices by this rather than by percentage change. → section
Softmax	The function that turns logits into probabilities: exponentiate each score and divide by the sum. Temperature divides the logits first. → section
Soft targets	The teacher's full probability distribution over outputs, often softened with a temperature above 1 to expose how it ranks wrong answers. → section
Specificity	TN / (TN + FP): of the truly negative items, the share left alone. Recall for the negative class; one minus the false positive rate. → section
Speculative decoding	A small draft model proposes several tokens that the large model verifies in one pass, producing the same output distribution faster (Leviathan et al., 2023). → section
Standard deviation	Typical distance of a single measurement from the mean. Does not shrink with more data. → section
Standard error	Typical distance of your estimate from the truth. Shrinks as one over the square root of n. → section
Standardised mean difference	The difference in a covariate's mean between arms divided by the pooled standard deviation; a common balance check flags |SMD| above 0.1. → section

Standardization (direct)	Comparing groups by applying each group's subgroup rates to one shared reference mix, so differences in mix no longer drive the comparison. → section
STaR	Self-Taught Reasoner (Zelikman et al., 2022). Iteratively fine-tunes on the model's own reasoning traces that reach correct answers. → section
State-space model (SSM)	A sequence layer built on a linear recurrence $h_t = A\,h_{t-1} + B\,x_t$; trains in parallel and runs at constant memory per token. Mamba (Gu and Dao 2023) makes the parameters input-dependent. → section
Statistical power	The probability that a test detects a real effect of a given size. Small eval sets have low power for small improvements. → section
Stochastic gradient descent (SGD)	Gradient descent using the gradient of a small random mini-batch instead of the whole dataset: far cheaper per step, unbiased, and noisy, with noise that shrinks as the batch grows. → section
Stratified sampling	Dividing the population into groups (strata) and sampling at random within each, guaranteeing every group is represented and usually reducing variance. → section
Subscript (x_i)	A small index below a symbol giving a position: x_3 is the third value. It is a seat number, not multiplication. → section
Summarised reasoning	A model-written summary of hidden thinking tokens, shown instead of the raw chain of thought. Vendors bill for the full thinking, not the summary. → section
Summation (Σ)	Capital sigma: add up the terms that follow, with the index running from the value below to the value above. → section
Superficial alignment hypothesis	The claim from LIMA (Zhou et al., 2023) that a model's knowledge comes almost entirely from pretraining and alignment mostly teaches which format and style to use. Holds for style, less so for hard skills. → section
Supervised fine-tuning (SFT)	Training a pretrained model on example prompt and response pairs with the next-token loss, counting only the response tokens. Teaches format and behaviour more than knowledge. → section

Surrogate endpoint	Medicine's term for a proxy outcome. A surrogate correlated with the target can still mislead if the treatment moves it through a path that does not reach the target. → section
Survival analysis	Methods such as Kaplan-Meier that handle censored data correctly, for example requests cut off by a timeout. → section
Survivorship bias	Drawing conclusions only from units that survived a filter, such as bombers that returned or users still active, while ignoring those that did not. → section
SUTVA	Stable unit treatment value assumption: each unit's outcome depends only on its own treatment (no interference) and each treatment comes in one version. → section
SwiGLU	A gated activation, $(\mathrm{SiLU}(\mathbf{x}W) \odot \mathbf{x}V)W_2$, used in the MLP blocks of PaLM, LLaMA and most recent open LLMs; it adds a third weight matrix per block. → section
Switchback design	An experiment that flips the whole system between treatment and control in time blocks, used when units share capacity or a market; blocks must outlast carryover. → section
Sycophancy	A model telling users what they want to hear, such as agreeing with incorrect claims or abandoning a correct answer under pushback. → section
Synthetic control	Builds a weighted blend of untreated units that tracks the treated unit before treatment and uses it as the counterfactual afterwards (Abadie, Diamond and Hainmueller, 2010). → section
System 1 / System 2	Kahneman's shorthand for fast, automatic thinking versus slow, deliberate thinking. For language models, a loose analogy for instant answers versus thinking modes; the network is the same, only the computation spent differs. → section
Systematic sampling	Taking every k-th unit from an ordered list after a random start. Fails when the list has a period matching k. → section
System One model	TypeSafe AI's name for its class of fast, typed, probabilistic decision models, introduced with Jev in September 2026. → section

System prompt	Text placed at the start of a chat, in its own role, that sets instructions, persona or context. To the model it is ordinary tokens that training taught it to weight heavily. → section
Task vector	The difference between fine-tuned and base weights. Task vectors can be added or subtracted to combine or remove skills (Ilharco et al., 2023). → section
Teacher forcing	Training each step on the reference's previous tokens rather than the model's own. Causes exposure bias, because at inference the model conditions on its own outputs. → section
Team-draft interleaving	An interleaving method (Radlinski, Kurup and Joachims, 2008) where two rankers take turns, in coin-flip order, picking their best unchosen result, and clicks are credited to the team that placed each result. → section
Temperature	A divisor applied to logits before softmax. Below 1 sharpens the distribution toward the top token; above 1 flattens it. → section
Temperature scaling	Dividing logits by a single learned constant on held-out data to fix calibration without changing rankings. → section
Temperature vs reasoning	Temperature reshapes each next-token distribution and adds no computation. Reasoning adds tokens of computation before the answer and so changes the distribution itself. They are independent knobs. → section
Tensor parallelism	Splitting individual weight matrices across GPUs so each computes part of every layer. Needs fast interconnect. → section
Test–retest reliability	The correlation between repeated measurements of the same items under the same conditions, such as an eval run twice with different seeds or judge samples. → section
Test-time compute	Extra computation spent when answering, such as longer reasoning, more samples, or search, to improve accuracy without retraining. → section
Test-time compute curve	Accuracy plotted against thinking tokens or compute per answer. Typically near-linear on a log axis over a range, then flat, while latency and cost grow linearly. → section

Thinking budget	A limit or target for how many tokens a model may spend reasoning before it answers, set as a token count (Gemini 2.5 `thinkingBudget`, Claude's legacy `budget_tokens`) or as a coarse effort level. → section
Threshold	The score cutoff above which a classifier predicts positive. Moving it trades false positives for false negatives. → section
TIES-Merging	A merging method that trims small weight changes, resolves sign conflicts, and averages only agreeing entries (Yadav et al., 2023). → section
Time-split benchmark	A benchmark that only counts items created after a model's training cutoff, so they cannot have been memorised. → section
Time-window bias	Bias from sampling an atypical period, such as launch week, a holiday or one time zone's office hours. → section
Token	The unit a language model reads and writes: a chunk of text produced by the tokenizer, often a word piece. English averages roughly four characters per token for common tokenizers. → section
Top-k sampling	Sampling only from the k most probable next tokens. → section
Top-p (nucleus) sampling	Sampling from the smallest set of most probable tokens whose cumulative probability reaches p (Holtzman et al., 2020). → section
Top-two-box	The share of ratings in the top two points of a scale, such as 4 or 5 out of 5. A more honest summary of ordinal ratings than the mean. → section
Trace ID	An identifier created once at the start of a request and copied onto every event it produces (model calls, tool calls, client events, feedback) so they can be joined. → section
Transformer	An architecture of stacked self-attention and MLP blocks with residual connections and normalisation, processing all positions in parallel (Vaswani et al. 2017). → section
Transformer block	The repeated unit of a transformer: an attention sublayer then an MLP sublayer, each adding its output to the residual stream. Large models stack 32 to over 100 of them. → section

Triggered analysis	Analysing only the units that reached the point where treatment and control differ, with the trigger logged in both arms, to avoid diluting the effect with unaffected users. → section
Trimmed mean	The mean after dropping a fixed share from each end. Rarely right for a one-sided tail. → section
True negative (TN)	A negative case the model correctly left alone. → section
True positive (TP)	A positive case the model correctly flagged. → section
Twyman's law	"Any figure that looks interesting or different is usually wrong." In experimentation: treat a surprisingly large result as a likely bug until logging, SRM, and metric definitions check out. → section
Typed output	An answer constrained to values defined in advance (a label from a list, a level on a scale, a probability), so it never needs parsing. → section
Unembedding	The final matrix W_U that maps the last residual vector to one logit per vocabulary token. → section
Unit of randomization	The thing the coin flip decides about: request, session, user, or cluster. The variance calculation must account for it, or error bars come out too narrow. → section
Universal approximation theorem	A network with one hidden layer and a non-polynomial activation can approximate any continuous function on a bounded region arbitrarily well, given enough units. It guarantees existence only: not how many units, not that training finds them, not generalization. → section
Unsafe compliance	Answering a genuinely harmful request. The false negative of safety behaviour. → section
Validity	Whether a measurement captures what it claims to: content (covers the domain), criterion (predicts a trusted outcome), and construct (measures the concept in its name). → section
Value model (critic)	A network that predicts expected future reward from each state, used to compute advantages in PPO. → section
Vanishing and exploding gradients	When many layers multiply local derivatives smaller (or larger) than 1, gradients for early layers shrink toward zero (or blow up). Residual

	connections, normalization and careful initialization counter it. → section
Vanishing gradient	Gradients shrinking geometrically as they flow back through many layers or time steps, because each step multiplies them by a factor below one; early layers stop learning. → section
Variance	Standard deviation squared. Adds cleanly across independent sources. → section
Variance components	Splitting score variance into task-to-task and run-to-run parts, which tells you whether to add tasks or add samples per task. → section
Variance reduction	Techniques such as CUPED, regression adjustment, stratification, or capping that shrink the noise in a metric so the same test detects smaller effects. → section
Verbalised confidence	A probability the model writes as text ("80% sure"). For RLHF-tuned models often better calibrated than token probabilities (Tian et al. 2023), but still worth checking. → section
Verbalized confidence	A confidence the model states in words ("I'm about 70% sure"), as opposed to one read from its token probabilities. Both can be checked for calibration. → section
Verbosity bias	A judge's tendency to prefer longer answers regardless of quality, documented for LLM judges by Zheng et al. (2023). → section
Verifier	A program that checks whether a model's output is correct, used as the reward in RLVR or to select among samples at inference time. → section
Vocabulary	The fixed set of tokens a model can read and write, typically 50,000 to 250,000 entries, set by the tokenizer before pretraining. → section
Warmup	Ramping the learning rate up from near zero over the first steps of training, because early gradients are large and erratic. → section
Weight	A learned number that says how strongly one input or unit influences the next unit; positive weights push the output up, negative ones push it down. → section
Weighted average	Mean of a per-class metric weighted by class support. Can hide poor performance on rare classes. → section

Width	The size of each layer's hidden vector (the model dimension *d* in a transformer); parameters grow with the square of width. → section
Wilcoxon signed-rank test	The paired counterpart of Mann–Whitney: ranks the absolute per-item differences and compares the sums for positive and negative differences. → section
Wilson interval	A confidence interval for a proportion that stays inside 0 to 1 and keeps close to its stated coverage near the boundaries. The default for pass rates. → section
Winner's curse	The upward bias in the observed score of whichever variant you selected as best; with many variants and noisy scores, the winner looks better than it is. → section
Winsorize	Cap extreme values at a percentile rather than removing them. Justified only for confirmed measurement errors. → section
WSD schedule	Warmup-stable-decay: hold the learning rate constant for most of training and decay only at the end, so decayed checkpoints can be branched off at any point. → section
x-bar ($\bar{x}$)	The average of the values x_1 to x_n. → section
ZeRO	Zero Redundancy Optimizer: sharding optimiser states (stage 1), gradients (stage 2), and weights (stage 3) across data-parallel GPUs instead of replicating them. → section
Zero-variance group	A prompt where every sampled response gets the same reward; it contributes no policy-gradient signal in group-baseline methods. → section

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Built from three earlier study guides plus new training chapters. Numbers in worked examples are illustrative unless a source is named. Source and build script live in the accompanying repository.